1866 1922 1943 1973 2001 2010

A COLLECTION OF CLASSIC PAPERS

A Timeline of Animal Breeding

The papers that changed history, from Mendel to the H matrix

compiled by @deltagplus

INTRODUCTION

From phenotypic selection to the H matrix

Until the eighteenth century, the choice of breeding animals was based on their appearance and on the experience of the breeders. In England, from 1760 onward, Robert Bakewell began to apply selection systematically, mating close relatives to fix types and recording the performance of sheep and cattle. In 1822, Coates's Herd Book, the first herd book of the Shorthorn breed, was published, and pedigree acquired commercial value. At that time, however, there was still no theory capable of explaining how traits are transmitted from one generation to the next.

In 1866, Gregor Mendel published the results of crosses between pea varieties and showed that inheritance occurs through discrete factors that segregate during gamete formation and recombine in predictable proportions. The work received little attention until 1900, when it was rediscovered. During the same period, Francis Galton and Karl Pearson studied continuously varying traits with statistical methods, and in 1886 Galton described the regression of offspring toward the population mean. In the first two decades of the twentieth century, Mendelians and biometricians held interpretations of the inheritance of these traits that were considered incompatible. In 1908, Hardy and Weinberg independently showed that genotype frequencies remain constant under random mating.

In 1918, Ronald Fisher showed that the joint action of many genes of small effect, each segregating according to Mendel's laws, produces the continuous variation and the correlations between relatives observed by the biometricians. This work introduced the term variance and the partition of genetic variation into additive, dominance and epistatic components. In 1922, Sewall Wright defined the coefficients of inbreeding and relationship. These papers established the theoretical basis of quantitative genetics.

The application of this theory to animal breeding began with Jay Lush at Iowa State College in the 1930s. Lush formalized the concepts of heritability, breeding value and response to selection and showed how to combine the performance of an animal with that of its relatives to estimate its breeding value. In 1943, Lanoy Hazel generalized this principle in the selection index, which combines several traits weighted by their economic values. In 1950, James Rendel and Alan Robertson divided genetic gain into four selection paths and showed that most of the progress came from the choice of the parents of future bulls. The expansion of artificial insemination during the same period increased the number of daughters per bull and established the progeny test as the main selection tool in dairy cattle.

The selection index assumes known means and variances and the comparison of animals in a common environment, conditions that do not hold in field data, where the daughters of a bull are distributed across different herds and years. In 1949, Charles Henderson, who had also studied under Lush, presented the equations that would later be called the mixed model equations, which estimate environmental effects and predict breeding values simultaneously. In 1953, Henderson proposed methods for estimating variance components from unbalanced data, and in 1959 the mixed model equations were formally published. In 1973, Henderson consolidated the theory of the best linear unbiased predictor (BLUP) applied to sire evaluation. In 1976, he presented a method to obtain the inverse of the relationship matrix directly, which, together with advances in computing, made the evaluation of large populations feasible.

In the 1980s, the sire model was progressively replaced by the animal model, which includes every individual in the pedigree in the equations and uses all the relationship information. In the same period, models with maternal effects and with multiple traits were developed, the estimation of variance components by restricted maximum likelihood (REML), proposed in 1971, became established, and the Bayesian approach was introduced. In the 1990s, random regression models began to be applied to data with repeated measurements over the life of the animal, such as test day milk records.

From the 1980s onward, there were attempts to use molecular markers to identify genes of large effect and to incorporate this information into genetic evaluations. In 1989, Fernando and Grossman included a marked locus in the mixed model equations. The practical results of this strategy were limited, since most traits of economic importance are controlled by a large number of genes. In 1997, Nejati Javaremi, Smith and Gibson proposed measuring relationships directly from shared alleles. In 2001, Meuwissen, Hayes and Goddard proposed estimating the effects of markers distributed across the whole genome simultaneously, which gave rise to genomic selection. With the commercial availability of SNP panels, VanRaden presented in 2008 the genomic relationship matrix, or G matrix, which measures the realized relationship between individuals. In 2009, the first official genomic evaluations of dairy cattle were published in the United States.

The first genomic evaluations were carried out in several steps and used only the genotyped animals, which represented a fraction of the population. In 2009 and 2010, Legarra, Aguilar and Misztal and, independently, Christensen and Lund showed how to combine the A and G matrices into a single relationship matrix, the H matrix. This development gave rise to single step GBLUP, which evaluates genotyped and nongenotyped animals jointly. In 2014, the APY algorithm made it possible to apply the method to populations with millions of genotyped animals, and in 2015 metafounders offered a solution to the compatibility between the A and G matrices. Today, single step GBLUP is the method used in the main animal breeding programs.

This collection brings together the papers associated with each of these stages. The criterion for selection was the contribution of each paper to the solution of a specific

methodological problem. The papers are arranged in chronological order and divided into six blocks. Each block is preceded by a text that describes the context of the period, and each paper is preceded by a page that presents the problem it addressed and the aspects that deserve attention while reading.

Reading the original texts poses its own difficulties. Notation varies between authors, some texts were scanned from old copies, and many concepts appear in a preliminary form, prior to the formulation found in textbooks. Reading the originals nevertheless makes it possible to understand the origin and the rationale of the methods used today, since the construction of the H matrix brings together contributions from Wright, Lush and Henderson.

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BLOCK 1

1

The theoretical foundations

1866 to 1922

This block brings together the papers that established the theoretical basis for the transmission of traits between generations. In 1866, Mendel showed that inheritance occurs through discrete factors. In 1908, Hardy explained the stability of genotype frequencies in populations under random mating. In 1918, Fisher showed that continuous variation can be explained by the action of many genes of small effect. In 1922, Wright defined the coefficients of inbreeding and relationship. The papers in the following blocks rest on these four.

Mendel (1866)

Hardy (1908)

Fisher (1918)

Wright (1922)

Mendel (1866)

> *Mendel, G. 1866. Versuche über Pflanzenhybriden. Verhandlungen des naturforschenden Vereines in Brünn, vol. 4, pp. 3 to 47. English translation, Experiments in Plant Hybridization.*

The problem

In the nineteenth century the prevailing hypothesis was blending inheritance, according to which the traits of the parents would combine irreversibly in the offspring. Under this hypothesis, variation should disappear within a few generations, which is not observed. Mendel crossed pea varieties that differed in seven traits and counted the offspring over several generations. He showed that each trait is controlled by a pair of factors that do not blend, that separate during gamete formation and that recombine at random. This is the origin of the 3 to 1 and 9 to 3 to 3 to 1 ratios.

Points to note while reading

Note the experimental design. Mendel chose pure lines and well defined traits and counted thousands of plants. Note also that Mendel already reasons in terms of probability and combinations, which decades later made it possible to connect his theory to statistics. The version included here is the English translation published by the Electronic Scholarly Publishing Project.

Mendel, Gregor. 1866. Versuche über Plflanzen-hybriden. *Verhandlungen des naturforschenden Ver-eines in Brünn, Bd. IV für das Jahr 1865,* Abhand-lungen, 3–47.

EXPERIMENTS IN PLANT HYBRIDIZATION (1865)

GREGOR MENDEL

Read at the February 8th, and March 8th, 1865, meetings
of the Brünn Natural History Society

INTRODUCTORY REMARKS

EXPERIENCE OF ARTIFICIAL FERTILIZATION, such as is effected with ornamental plants in order to obtain new variations in color, has led to the experiments which will here be discussed. The striking regularity with which the same hybrid forms always reappeared whenever fertilization took place between the same species induced further experiments to be undertaken, the object of which was to follow up the developments of the hybrids in their progeny.

To this object numerous careful observers, such as Kölreuter, Gärtner, Herbert, Lecoq, Wichura and others, have devoted a part of their lives with inexhaustible perseverance. Gärtner especially in his work *Die Bastarderzeugung im Pflanzenreiche* [The Production of Hybrids in the Vegetable Kingdom], has recorded very valuable observations; and quite recently Wichura published the results of some profound investigations into the hybrids of the Willow. That, so far, no generally applicable law governing the formation and development of hybrids has been successfully formulated can hardly be wondered at by anyone who is acquainted with the extent of the task, and can appreciate the difficulties with which experiments of this class have to contend. A final decision can only be arrived at when we shall have before us the results of *detailed experiments* made on plants belonging to the most diverse orders.

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Those who survey the work done in this department will arrive at the conviction that among all the numerous experiments made, not one has been carried out to such an extent and in such a way as to make it possible to determine the number of different forms under which the offspring of the hybrids appear, or to arrange these forms with certainty according to their separate generations, or definitely to ascertain their statistical relations.

It requires indeed some courage to undertake a labor of such far–reaching extent; this appears, however, to be the only right way by which we can finally reach the solution of a question the importance of which cannot be overestimated in connection with the history of the evolution of organic forms.

The paper now presented records the results of such a detailed experiment. This experiment was practically confined to a small plant group, and is now, after eight years' pursuit, concluded in all essentials. Whether the plan upon which the separate experiments were conducted and carried out was the best suited to attain the desired end is left to the friendly decision of the reader.

SELECTION OF THE EXPERIMENTAL PLANTS

The value and utility of any experiment are determined by the fitness of the material to the purpose for which it is used, and thus in the case before us it cannot be immaterial what plants are subjected to experiment and in what manner such experiment is conducted.

The selection of the plant group which shall serve for experiments of this kind must be made with all possible care if it be desired to avoid from the outset every risk of questionable results.

The experimental plants must necessarily:

1. Possess constant differentiating characteristics
2. The hybrids of such plants must, during the flowering period, be protected from the influence of all foreign pollen, or be easily capable of such protection.

The hybrids and their offspring should suffer no marked disturbance in their fertility in the successive generations.

Accidental impregnation by foreign pollen, if it occurred during the experiments and were not recognized, would lead to entirely erroneous conclusions. Reduced fertility or entire sterility of certain forms, such as occurs in the offspring of many hybrids, would render

the experiments very difficult or entirely frustrate them. In order to discover the relations in which the hybrid forms stand towards each other and also towards their progenitors it appears to be necessary that all member of the series developed in each successive generations should be, *without exception*, subjected to observation.

At the very outset special attention was devoted to the *Leguminosae* on account of their peculiar floral structure. Experiments which were made with several members of this family led to the result that the genus *Pisum* was found to possess the necessary qualifications.

Some thoroughly distinct forms of this genus possess characters which are constant, and easily and certainly recognizable, and when their hybrids are mutually crossed they yield perfectly fertile progeny. Furthermore, a disturbance through foreign pollen cannot easily occur, since the fertilizing organs are closely packed inside the keel and the anthers burst within the bud, so that the stigma becomes covered with pollen even before the flower opens. This circumstance is especially important. As additional advantages worth mentioning, there may be cited the easy culture of these plants in the open ground and in pots, and also their relatively short period of growth. Artificial fertilization is certainly a somewhat elaborate process, but nearly always succeeds. For this purpose the bud is opened before it is perfectly developed, the keel is removed, and each stamen carefully extracted by means of forceps, after which the stigma can at once be dusted over with the foreign pollen.

In all, thirty–four more or less distinct varieties of Peas were obtained from several seedsmen and subjected to a two year's trial. In the case of one variety there were noticed, among a larger number of plants all alike, a few forms which were markedly different. These, however, did not vary in the following year, and agreed entirely with another variety obtained from the same seedsman; the seeds were therefore doubtless merely accidentally mixed. All the other varieties yielded perfectly constant and similar offspring; at any rate, no essential difference was observed during two trial years. For fertilization twenty–two of these were selected and cultivated during the whole period of the experiments. They remained constant without any exception.

Their systematic classification is difficult and uncertain. If we adopt the strictest definition of a species, according to which only those individuals belong to a species which under precisely the same circumstances display precisely similar characters, no two of these varieties could be referred to one species. According to the opinion of

experts, however, the majority belong to the species *Pisum sativum*; while the rest are regarded and classed, some as sub–species of *P. sativum*, and some as independent species, such as *P. quadratum*, *P. saccharatum*, and *P. umbellatum*. The positions, however, which may be assigned to them in a classificatory system are quite immaterial for the purposes of the experiments in question. It has so far been found to be just as impossible to draw a sharp line between the hybrids of species and varieties as between species and varieties themselves.

DIVISION AND ARRANGEMENT OF THE EXPERIMENTS

If two plants which differ constantly in one or several characters be crossed, numerous experiments have demonstrated that the common characters are transmitted unchanged to the hybrids and their progeny; but each pair of differentiating characters, on the other hand, unite in the hybrid to form a new character, which in the progeny of the hybrid is usually variable. The object of the experiment was to observe these variations in the case of each pair of differentiating characters, and to deduce the law according to which they appear in successive generations. The experiment resolves itself therefore into just as many separate experiments are there are constantly differentiating characters presented in the experimental plants.

The various forms of Peas selected for crossing showed differences in length and color of the stem; in the size and form of the leaves; in the position, color, size of the flowers; in the length of the flower stalk; in the color, form, and size of the pods; in the form and size of the seeds; and in the color of the seed–coats and of the albumen (endosperm). Some of the characters noted do not permit of a sharp and certain separation, since the difference is of a “more or less” nature, which is often difficult to define. Such characters could not be utilized for the separate experiments; these could only be applied to characters which stand out clearly and definitely in the plants. Lastly, the result must show whether they, in their entirety, observe a regular behavior in their hybrid unions, and whether from these facts any conclusion can be reached regarding those characters which possess a subordinate significance in the type.

The characters which were selected for experiment relate:

1. To the *difference in the form of the ripe seeds*. These are either round or roundish, the depressions, if any, occur

on the surface, being always only shallow; or they are irregularly angular and deeply wrinkled (*P. quadratum*).

2. To the *difference in the color of the seed albumen* (endosperm). The albumen of the ripe seeds is either pale yellow, bright yellow and orange colored, or it possesses a more or less intense green tint. This difference of color is easily seen in the seeds as their coats are transparent.
3. To the *difference in the color of the seed–coat*. This is either white, with which character white flowers are constantly correlated; or it is gray, gray–brown, leather–brown, with or without violet spotting, in which case the color of the standards is violet, that of the wings purple, and the stem in the axils of the leaves is of a reddish tint. The gray seed–coats become dark brown in boiling water.
4. To the *difference in the form of the ripe pods.* These are either simply inflated, not contracted in places; or they are deeply constricted between the seeds and more or less wrinkled (*P. saccharatum*).
5. To the *difference in the color of the unripe pods.* They are either light to dark green, or vividly yellow, in which coloring the stalks, leaf–veins, and calyx participate.*
6. To the *difference in the position of the flowers*. They are either axial, that is, distributed along the main stem; or they are terminal, that is, bunched at the top of the stem and arranged almost in a false umbel; in this case the upper part of the stem is more or less widened in section (*P. umbellatum*).
7. To the *difference in the length of the stem.* The length of the stem is very various in some forms; it is, however, a constant character for each, in so far that healthy plants, grown in the same soil, are only subject to unimportant variations in this character. In experiments with this character, in order to be able to discriminate with certainty, the long axis of 6 to 7 ft. was always crossed with the short one of ¾ ft. to 1½ ft.

* One species possesses a beautifully brownish–red colored pod, which when ripening turns to violet and blue. Trials with this character were only begun last year.

Each two of the differentiating characters enumerated above were united by cross–fertilization. There were made for the

1st	experiment	60	fertilizations on	15	plants.
2nd	experiment	58	fertilizations on	10	plants.
3rd	experiment	35	fertilizations on	10	plants.
4th	experiment	40	fertilizations on	10	plants.
5th	experiment	23	fertilizations on	5	plants.
6th	experiment	34	fertilizations on	10	plants.
7th	experiment	37	fertilizations on	10	plants.

From a larger number of plants of the same variety only the most vigorous were chosen for fertilization. Weakly plants always afford uncertain results, because even in the first generation of hybrids, and still more so in the subsequent ones, many of the offspring either entirely fail to flower or only form a few and inferior seeds.

Furthermore, in all the experiments reciprocal crossings were effected in such a way that each of the two varieties which in one set of fertilizations served as seed–bearer in the other set was used as the pollen plant.

The plants were grown in garden beds, a few also in pots, and were maintained in their natural upright position by means of sticks, branches of trees, and strings stretched between. For each experiment a number of pot plants were placed during the blooming period in a greenhouse, to serve as control plants for the main experiment in the open as regards possible disturbance by insects. Among the insects which visit Peas the beetle *Buchus pisi* might be detrimental to the experiments should it appear in numbers. The female of this species is known to lay the eggs in the flower, and in so doing opens the keel; upon the tarsi of one specimen, which was caught in a flower, some pollen grains could clearly be seen under a lens. Mention must also be made of a circumstance which possibly might lead to the introduction of foreign pollen. It occurs, for instance, in some rare cases that certain parts of an otherwise normally developed flower wither, resulting in a partial exposure of the fertilizing organs. A defective development of the keel has also been observed, owing to which the stigma and anthers remained partially covered. It also sometimes happens that the pollen does not reach full perfection. In this event there occurs a gradual lengthening of the pistil during the blooming period, until the stigmatic tip protrudes at the point of the keel. This remarkable appearance has also been observed in hybrids of *Phaseolus* and *Lathyrus.*

The risk of false impregnation by foreign pollen is, however, a very slight one with *Pisum*, and is quite incapable of disturbing the general result. Among more than 10,000 plants which were carefully examined there were only a very few cases where an indubitable false impregnation had occurred. Since in the greenhouse such a case was never remarked, it may well be supposed that *Brucus pisi*, and possibly also the described abnormalities in the floral structure, were to blame.

THE FORMS OF THE HYBRIDS

Experiments which in previous years were made with ornamental plants have already afforded evidence that the hybrids, as a rule, are not exactly intermediate between the parental species. With some of the more striking characters, those, for instance, which relate to the form and size of the leaves, the pubescence of the several parts, etc., the intermediate, indeed, is nearly always to be seen; in other cases, however, one of the two parental characters is so preponderant that it is difficult, or quite impossible, to detect the other in the hybrid.

This is precisely the case with the Pea hybrids. In the case of each of the seven crosses the hybrid–character resembles that of one of the parental forms so closely that the other either escapes observation completely or cannot be detected with certainty. This circumstance is of great importance in the determination and classification of the forms under which the offspring of the hybrids appear. Henceforth in this paper those characters which are transmitted entire, or almost unchanged in the hybridization, and therefore in themselves constitute the characters of the hybrid, are termed the *dominant*, and those which become latent in the process *recessive*. The expression *recessive* has been chosen because the characters thereby designated withdraw or entirely disappear in the hybrids, but nevertheless reappear unchanged in their progeny, as will be demonstrated later on.

It was furthermore shown by the whole of the experiments that it is perfectly immaterial whether the dominant character belongs to the seed plant or to the pollen plant; the form of the hybrid remains identical in both cases. This interesting fact was also emphasized by Gärtner, with the remark that even the most practiced expert is not in a position to determine in a hybrid which of the two parental species was the seed or the pollen plant.

Of the differentiating characters which were used in the experiments the following are dominant:

1. The round or roundish form of the seed with or without shallow depressions.
2. The yellow coloring of the seed albumen.
3. The gray, gray–brown, or leather brown color of the seed–coat, in association with violet–red blossoms and reddish spots in the leaf axils.
4. The simply inflated form of the pod.
5. The green coloring of the unripe pod in association with the same color of the stems, the leaf–veins and the calyx.
6. The distribution of the flowers along the stem.
7. The greater length of stem.

With regard to this last character it must be stated that the longer of the two parental stems is usually exceeded by the hybrid, a fact which is possibly only attributable to the greater luxuriance which appears in all parts of plants when stems of very different lengths are crossed. Thus, for instance, in repeated experiments, stems of 1 ft. and 6 ft. in length yielded without exception hybrids which varied in length between 6 ft. and 7½ ft.

The hybrid seeds in the experiments with seed–coat are often more spotted, and the spots sometimes coalesce into small bluish–violet patches. The spotting also frequently appears even when it is absent as a parental character.

The hybrid forms of the seed–shape and of the [color of the] albumen are developed immediately after the artificial fertilization by the mere influence of the foreign pollen. They can, therefore, be observed even in the first year of experiment, whilst all the other characters naturally only appear in the following year in such plants as have been raised from the crossed seed.

THE FIRST GENERATION FROM THE HYBRIDS

In this generation there reappear, together with the dominant characters, also the recessive ones with their peculiarities fully developed, and this occurs in the definitely expressed average proportion of three to one, so that among each four plants of this generation three display the dominant character and one the recessive.

This relates without exception to all the characters which were investigated in the experiments. The angular wrinkled form of the seed, the green color of the albumen, the white color of the seed–coats and the flowers, the constrictions of the pods, the yellow color of the unripe pod, of the stalk, of the calyx, and of the leaf venation, the umbel–like form of the inflorescence, and the dwarfed stem, all reappear in the numerical proportion given, without any essential alteration. *Transitional forms were not observed in any experiment*.

Since the hybrids resulting from reciprocal crosses are formed alike and present no appreciable difference in their subsequent development, consequently these results can be reckoned together in each experiment. The relative numbers which were obtained for each pair of differentiating characters are as follows:

Expt 1: Form of seed. — From 253 hybrids 7,324 seeds were obtained in the second trial year. Among them were 5,474 round or roundish ones and 1,850 angular wrinkled ones. Therefrom the ratio 2.96:1 is deduced.

Expt 2: Color of albumen. — 258 plants yielded 8,023 seeds, 6,022 yellow, and 2,001 green; their ratio, therefore, is as 3.01:1.

In these two experiments each pod yielded usually both kinds of seed. In well–developed pods which contained on the average six to nine seeds, it often happened that all the seeds were round (Expt. 1) or all yellow (Expt. 2); on the other hand there were never observed more than five wrinkled or five green ones on one pod. It appears to make no difference whether the pods are developed early or later in the hybrid or whether they spring from the main axis or from a lateral one. In some few plants only a few seeds developed in the first formed pods, and these possessed exclusively one of the two characters, but in the subsequently developed pods the normal proportions were maintained nevertheless.

As in separate pods, so did the distribution of the characters vary in separate plants. By way of illustration the first ten individuals from both series of experiments may serve.

	Experiment 1 Form of the Seed		Experiment 2 Color of the Albumen	
Plants	**round**	**wrinkled**	**yellow**	**green**
1	45	12	25	11
2	27	8	32	7
3	24	7	14	5
4	19	10	70	27
5	32	11	24	13
6	26	6	20	6
7	88	24	32	13
8	22	10	44	9
9	28	6	50	14
10	25	7	44	18

As extremes in the distribution of the two seed characters in one plant, there were observed in Expt. 1 an instance of 43 round and only two angular, and another of 14 round and 15 angular seeds. In Expt. 2 there was a case of 32 yellow and only one green seed, but also one of 20 yellow and 19 green.

These two experiments are important for the determination of the average ratios, because with a smaller number of experimental plants they show that very considerable fluctuations may occur. In counting the seeds, also, especially in Expt. 2, some care is requisite, since in some of the seeds of many plants the green color of the albumen is less developed, and at first may be easily overlooked. The cause of this partial disappearance of the green coloring has no connection with the hybrid–character of the plants, as it likewise occurs in the parental variety. This peculiarity is also confined to the individual and is not inherited by the offspring. In luxuriant plants this appearance was frequently noted. Seeds which are damaged by insects during their development often vary in color and form, but with a little practice in sorting, errors are easily avoided. It is almost superfluous to mention that the pods must remain on the plants until they are thoroughly ripened and have become dried, since it is only then that the shape and color of the seed are fully developed.

Expt. 3: Color of the seed–coats. — Among 929 plants, 705 bore violet–red flowers and gray–brown seed–coats; 224 had white flowers and white seed–coats, giving the proportion 3.15:1.

Expt. 4: Form of pods. — Of 1,181 plants, 882 had them simply inflated, and in 299 they were constricted. Resulting ratio, 2.95:1.

Expt. 5: Color of the unripe pods. — The number of trial plants was 580, of which 428 had green pods and 152 yellow ones. Consequently these stand in the ratio of 2.82:1.

Expt. 6: Position of flowers. — Among 858 cases 651 had inflorescences axial and 207 terminal. Ratio, 3.14:1.

Expt. 7: Length of stem. — Out of 1,064 plants, in 787 cases the stem was long, and in 277 short. Hence a mutual ratio of 2.84:1. In this experiment the dwarfed plants were carefully lifted and transferred to a special bed. This precaution was necessary, as otherwise they would have perished through being overgrown by their tall relatives. Even in their quite young state they can be easily picked out by their compact growth and thick dark–green foliage.

If now the results of the whole of the experiments be brought together, there is found, as between the number of forms with the dominant and recessive characters, an average ratio of 2.98:1, or 3:1.

The dominant character can have here a *double signification* — viz. that of a parental character or a hybrid–character. In which of the two significations it appears in each separate case can only be determined by the following generation. As a parental character it must pass over unchanged to the whole of the offspring; as a hybrid–character, on the other hand, it must maintain the same behavior as in the first generation.

THE SECOND GENERATION FROM THE HYBRIDS

Those forms which in the first generation exhibit the recessive character do not further vary in the second generation as regards this character; they remain constant in their offspring.

It is otherwise with those which possess the dominant character in the first generation [bred from the hybrids — i.e., the F_2 in modern terminology]. Of these *two*–thirds yield offspring which display the dominant and recessive characters in the proportion of three to one, and thereby show exactly the same ratio as the hybrid forms, while only *one*–third remains with the dominant character constant.

The separate experiments yielded the following results:

Expt. 1: Among 565 plants which were raised from round seeds of the first generation, 193 yielded round seeds only, and remained therefore constant in this character; 372, however, gave both round and wrinkled seeds, in the proportion of 3:1. The number of the hybrids, therefore, as compared with the constants is 1.93:1.

Expt. 2: Of 519 plants which were raised from seeds whose albumen was of yellow color in the first generation, 166 yielded exclusively yellow, while 353 yielded yellow and green seeds in the proportion of 3:1. There resulted, therefore, a division into hybrid and constant forms in the proportion of 2.13:1.

For each separate trial in the following experiments 100 plants were selected which displayed the dominant character in the first generation, and in order to ascertain the significance of this, ten seeds of each were cultivated.

Expt. 3: The offspring of 36 plants yielded exclusively gray–brown seed–coats, while of the offspring of 64 plants some had gray–brown and some had white.

Expt. 4: The offspring of 29 plants had only simply inflated pods; of the offspring of 71, on the other hand, some had inflated and some constricted.

Expt. 5: The offspring of 40 plants had only green pods; of the offspring of 60 plants some had green, some yellow ones.

Expt. 6.: The offspring of 33 plants had only axial flowers; of the offspring of 67, on the other hand, some had axial and some terminal flowers.

Expt. 7: The offspring of 28 plants inherited the long axis, of those of 72 plants some the long and some the short axis.

In each of these experiments a certain number of the plants came constant with the dominant character. For the determination of the proportion in which the separation of the forms with the constantly

persistent character results, the two first experiments are especially important, since in these a larger number of plants can be compared. The ratios 1.93:1 and 2.13:1 gave together almost exactly the average ratio of 2:1. Experiment 6 gave a quite concordant result; in the others the ratio varies more or less, as was only to be expected in view of the smaller number of 100 trial plants. Experiment 5, which shows the greatest departure, was repeated, and then in lieu of the ratio of 60:40, that of 65:35 resulted. *The average ratio of 2 to 1 appears, therefore, as fixed with certainty*. It is therefore demonstrated that, of those forms which possess the dominant character in the first generation, two–thirds have the hybrid–character, while one–third remains constant with the dominant character.

The ratio 3:1, in accordance with which the distribution of the dominant and recessive characters results in the first generation, resolves itself therefore in all experiments into the ratio of 2:1:1, if the dominant character be differentiated according to its significance as a hybrid–character or as a parental one. Since the members of the first generation spring directly from the seed of the hybrids, it is now clear that the hybrids form seeds having one or other of the two differentiating characters, and of these one–half develop again the hybrid form, while the other half yield plants which remain constant and receive the dominant or the recessive characters in equal numbers.

THE SUBSEQUENT GENERATIONS FROM THE HYBRIDS

The proportions in which the descendants of the hybrids develop and split up in the first and second generations presumably hold good for all subsequent progeny. Experiments 1 and 2 have already been carried through six generations; 3 and 7 through five; and 4, 5, and 6 through four; these experiments being continued from the third generation with a small number of plants, and no departure from the rule has been perceptible. The offspring of the hybrids separated in each generation in the ratio of 2:1:1 into hybrids and constant forms.

If ***A*** be taken as denoting one of the two constant characters, for instance the dominant, ***a***, the recessive, and ***Aa*** the hybrid form in which both are conjoined, the expression

$$\boldsymbol{A} + 2\boldsymbol{Aa} + \boldsymbol{a}$$

shows the terms in the series for the progeny of the hybrids of two differentiating characters.

The observation made by Gärtner, Kölreuter, and others, that hybrids are inclined to revert to the parental forms, is also confirmed by the experiments described. It is seen that the number of the hybrids which arise from one fertilization, as compared with the number of forms which become constant, and their progeny from generation to generation, is continually diminishing, but that nevertheless they could not entirely disappear. If an average equality of fertility in all plants in all generations be assumed, and if, furthermore, each hybrid forms seed of which one–half yields hybrids again, while the other half is constant to both characters in equal proportions, the ratio of numbers for the offspring in each generation is seen by the following summary, in which ***A*** and ***a*** denote again the two parental characters, and ***Aa*** the hybrid forms. For brevity's sake it may be assumed that each plant in each generation furnishes only four seeds.

				Ratios				
Generation	***A***	***Aa***	***a***	***A***	:	***Aa***	:	***a***
1	1	2	1	1	:	2	:	1
2	6	4	6	3	:	2	:	3
3	28	8	28	7	:	2	:	7
4	120	16	120	15	:	2	:	15
5	496	32	496	31	:	2	:	31
n				$2^n - 1$	:	2	:	$2^n - 1$

In the tenth generation, for instance, $2^n - 1 = 1{,}023$. There result, therefore, in each 2,048 plants which arise in this generation 1,023 with the constant dominant character, 1,023 with the recessive character, and only two hybrids.

THE OFFSPRING OF HYBRIDS IN WHICH SEVERAL DIFFERENTIATING CHARACTERS ARE ASSOCIATED.

In the experiments above described plants were used which differed only on one essential character. The next task consisted in ascertaining whether the law of development discovered in these applied to each pair of differentiating characters when several diverse characters are united in the hybrid by crossing.

As regards the form of the hybrids in these cases, the experiments showed throughout that this invariably more nearly approaches to that one of the two parental plants which possesses the greater number of

dominant characters. If, for instance, the seed plant has a short stem, terminal white flowers, and simply inflated pods; the pollen plant, on the other hand, a long stem, violet–red flowers distributed along the stem, and constricted pods; the hybrid resembles the seed parent only in the form of the pod; in the other characters it agrees with the pollen parent. Should one of the two parental types possess only dominant characters, then the hybrid is scarcely or not at all distinguishable from it.

Two experiments were made with a considerable number of plants. In the first experiment the parental plants differed in the form of the seed and in the color of the albumen; in the second in the form of the seed, in the color of the albumen, and in the color of the seed–coats. Experiments with seed characters give the result in the simplest and most certain way.

In order to facilitate study of the data in these experiments, the different characters of the seed plant will be indicated by ***A***, ***B***, ***C***, those of the pollen plant by ***a***, ***b***, ***c***, and the hybrid forms of the characters by ***Aa***, ***Bb***, and ***Cc***.

Expt. 1. —	***AB***, seed parents	***ab***, pollen parents
	A, form round	***a***, form wrinkled
	B, albumen yellow	***b***, albumen green

The fertilized seeds appeared round and yellow like those of the seed parents. The plants raised therefrom yielded seeds of four sorts, which frequently presented themselves in one pod. In all, 556 seeds were yielded by 15 plants, and of these there were:

315	round and yellow,
101	wrinkled and yellow,
108	round and green,
32	wrinkled and green.

All were sown the following year. Eleven of the round yellow seeds did not yield plants, and three plants did not form seeds. Among the rest:

38	had round yellow seeds	***AB***
65	round yellow and green seeds	***ABb***
60	round yellow and wrinkled yellow seeds	***AaB***
138	round yellow and green, wrinkled yellow and green seeds	***AaBb***

From the wrinkled yellow seeds 96 resulting plants bore seed, of which:

28	had only wrinkled yellow seeds	***aB***
68	wrinkled yellow and green seeds	***aBb***

From 108 round green seeds 102 resulting plants fruited, of which:

35	had only round green seeds	***Ab***
67	round and wrinkled green seeds	***Aab***

The wrinkled green seeds yielded 30 plants which bore seeds all of like character; they remained constant ***ab***.

The offspring of the hybrids appeared therefore under nine different forms, some of them in very unequal numbers. When these are collected and coordinated we find:

38	plants with the sign	***AB***
35	" " "	***Ab***
28	" " "	***aB***
30	" " "	***ab***
65	" " "	***ABb***
68	" " "	***aBb***
60	" " "	***AaB***
67	" " "	***Aab***
138	" " "	***AaBb***

The whole of the forms may be classed into three essentially different groups. The first includes those with the signs ***AB***, ***Ab***, ***aB***, and ***ab***: they possess only constant characters and do not vary again in the next generation. Each of these forms is represented on the average 33 times. The second group includes the signs ***ABb***, ***aBb***, ***AaB***, ***Aab***: these are constant in one character and hybrid in another, and vary in the next generation only as regards the hybrid–character. Each of these appears on any average 65 times. The form ***AaBb*** occurs 138 times: it is hybrid in both characters, and behaves exactly as do the hybrids from which it is derived.

If the numbers in which the forms belonging to these classes appear be compared, the ratios of 1:2:4 are unmistakably evident. The numbers 33, 65, 138 present very fair approximations to the ratio numbers of 33, 66, 132.

The development series consists, therefore, of nine classes, of which four appear therein always once and are constant in both characters; the forms ***AB***, ***ab***, resemble the parental forms, the two others present combinations between the conjoined characters ***A***, ***a***, ***B***, ***b***, which combinations are likewise possibly constant. Four classes appear always twice, and are constant in one character and hybrid in the other. One class appears four times, and is hybrid in both

characters. Consequently, the offspring of the hybrids, if two kinds of differentiating characters are combined therein, are represented by the expression

$$\boldsymbol{AB} + \boldsymbol{Ab} + \boldsymbol{aB} + \boldsymbol{ab} + 2\boldsymbol{ABb} + 2\boldsymbol{aBb} + 2\boldsymbol{AaB} + 2\boldsymbol{Aab} + 4\boldsymbol{AaBb}$$

This expression is indisputably a combination series in which the two expressions for the characters ***A*** and ***a***, ***B*** and ***b*** are combined. We arrive at the full number of the classes of the series by the combination of the expressions:

$$\boldsymbol{A} + 2\boldsymbol{Aa} + \boldsymbol{a}$$
$$\boldsymbol{B} + 2\boldsymbol{Bb} + \boldsymbol{b}$$

Expt. 2. —	***ABC***, seed parents	***abc***, pollen parents
	A, form round	***a***, form wrinkled
	B, albumen yellow	***b***, albumen green
	C, seed coat grey–brown	***c***, seed coat white

This experiment was made in precisely the same way as the previous one. Among all the experiments it demanded the most time and trouble. From 24 hybrids 687 seeds were obtained in all: these were all either spotted, gray–brown or gray–green, round or wrinkled. From these in the following year 639 plants fruited, and as further investigation showed, there were among them:

8	plants	***ABC***	22	plants	***ABCc***	45	plants	***ABbCc***
14	"	***Abc***	17	"	***AbCc***	36	"	***aBbCc***
9	"	***AbC***	25	"	***aBCc***	38	"	***AaBCc***
11	"	***Abc***	20	"	***abCc***	40	"	***AabCc***
8	"	***aBC***	15	"	***ABbC***	49	"	***AaBbC***
10	"	***aBc***	18	"	***Abbc***	48	"	***AaBbc***
10	"	***abC***	19	"	***aBbC***			
7	"	***abc***	24	"	***aBbc***			
			14	"	***AaBC***	78	"	***AaBbCc***
			18	"	***AaBc***			
			20	"	***AabC***			
			16	"	***Aabc***			

The whole expression contains 27 terms. Of these eight are constant in all characters, and each appears on the average ten times; twelve are constant in two characters, and hybrid in the third; each appears on the average 19 times; six are constant in one character and hybrid in the other two; each appears on the average 43 times. One form appears 78 times and is hybrid in all of the characters. The ratios 10:19:43:78 agree so closely with the ratios 10:20:40:80, or 1:2:4:8 that this last undoubtedly represents the true value.

The development of the hybrids when the original parents differ in three characters results therefore according to the following expression:

ABC + ***ABc*** + ***AbC*** + ***Abc*** + ***aBC*** + ***aBc*** + ***abC*** + ***abc*** + 2***ABCc*** + 2***AbCc*** + 2***aBCc*** + 2***abCc*** + 2***ABbC*** + 2***ABbc*** + 2***aBbC*** + 2***aBbc*** + 2***AaBC*** + 2***AaBc*** + 2***AabC*** + 2***Aabc*** + 4***ABbCc*** + 4***aBbCc*** + 4***AaBCc*** + 4***AabCc*** + 4***AaBbC*** + 4***AaBbc*** + 8***AaBbCc***.

Here also is involved a combination series in which the expressions for the characters ***A*** and ***a***, ***B*** and ***b***, ***C*** and ***c***, are united. The expressions:

A + 2***Aa*** + ***a***
B + 2***Bb*** + ***b***
C + 2***Cc*** + ***c***

give all the classes of the series. The constant combinations which occur therein agree with all combinations which are possible between the characters ***A***, ***B***, ***C***, ***a***, ***b***, ***c***; two thereof, ***ABC*** and ***abc***, resemble the two original parental stocks.

In addition, further experiments were made with a smaller number of experimental plants in which the remaining characters by twos and threes were united as hybrids: all yielded approximately the same results. There is therefore no doubt that for the whole of the characters involved in the experiments the principle applies *that the offspring of the hybrids in which several essentially different characters are combined exhibit the terms of a series of combinations, in which the developmental series for each pair of differentiating characters are united*. It is demonstrated at the same time that *the relation of each pair of different characters in hybrid union is independent of the other differences in the two original parental stocks*.

If *n* represent the number of the differentiating characters in the two original stocks, 3^n gives the number of terms of the combination series, 4^n the number of individuals which belong to the series, and 2^n the number of unions which remain constant. The series therefore contains, if the original stocks differ in four characters, $3^4 = 81$ classes, $4^4 = 256$ individuals, and $2^4 = 16$ constant forms: or, which is the same, among each 256 offspring of the hybrids are 81 different combinations, 16 of which are constant.

All constant combinations which in Peas are possible by the combination of the said seven differentiating characters were actually obtained by repeated crossing. Their number is given by $2^7 = 128$.

Thereby is simultaneously given the practical proof *that the constant characters which appear in the several varieties of a group of plants may be obtained in all the associations which are possible according to the laws of combination, by means of repeated artificial fertilization.*

As regards the flowering time of the hybrids, the experiments are not yet concluded. It can, however, already be stated that the time stands almost exactly between those of the seed and pollen parents, and that the constitution of the hybrids with respect to this character probably follows the rule ascertained in the case of the other characters. The forms which are selected for experiments of this class must have a difference of at least 20 days from the middle flowering period of one to that of the other; furthermore, the seeds when sown must all be placed at the same depth in the earth, so that they may germinate simultaneously. Also, during the whole flowering period, the more important variations in temperature must be taken into account, and the partial hastening or delaying of the flowering which may result there from. It is clear that this experiment presents many difficulties to be overcome and necessitates great attention.

If we endeavor to collate in a brief form the results arrived at, we find that those differentiating characters, which admit of easy and certain recognition in the experimental plants, *all behave exactly alike in their hybrid associations*. The offspring of the hybrids of each pair of differentiating characters are, one–half, hybrid again, while the other half are constant in equal proportions having the characters of the seed and pollen parents respectively. If several differentiating characters are combined by cross–fertilization in a hybrid, the resulting offspring form the terms of a combination series in which the combination series for each pair of differentiating characters are united.

The uniformity of behavior shown by the whole of the characters submitted to experiment permits, and fully justifies, the acceptance of the principle that a similar relation exists in the other characters which appear less sharply defined in plants, and therefore could not be included in the separate experiments. An experiment with peduncles of different lengths gave on the whole a fairly satisfactory results, although the differentiation and serial arrangement of the forms could not be effected with that certainty which is indispensable for correct experiment.

THE REPRODUCTIVE CELLS OF THE HYBRIDS

The results of the previously described experiments led to further experiments, the results of which appear fitted to afford some conclusions as regards the composition of the egg and pollen cells of hybrids. An important clue is afforded in *Pisum* by the circumstance that among the progeny of the hybrids constant forms appear, and that this occurs, too, in respect of all combinations of the associated characters. So far as experience goes, we find it in every case confirmed that constant progeny can only be formed when the egg cells and the fertilizing pollen are of like character, so that both are provided with the material for creating quite similar individuals, as is the case with the normal fertilization of pure species. We must therefore regard it as certain that exactly similar factors must be at work also in the production of the constant forms in the hybrid plants. Since the various constant forms are produced in one plant, or even in one flower of a plant, the conclusion appears logical that in the ovaries of the hybrids there are formed as many sorts of egg cells, and in the anthers as many sorts of pollen cells, as there are possible constant combination forms, and that these egg and pollen cells agree in their internal compositions with those of the separate forms.

In point of fact it is possible to demonstrate theoretically that this hypothesis would fully suffice to account for the development of the hybrids in the separate generations, if we might at the same time assume that the various kinds of egg and pollen cells were formed in the hybrids on the average in equal numbers. In order to bring these assumptions to an experimental proof, the following experiments were designed. Two forms which were constantly different in the form of the seed and the color of the albumen were united by fertilization.

If the differentiating characters are again indicated as **A**, **B**, **a**, **b**, we have:

AB,	Seed parents	***ab***,	Pollen parents
A,	form round	***a***,	form wrinkled
B,	albumen yellow	***b***,	albumen green

The artificially fertilized seeds were sown together with several seeds of both original stocks, and the most vigorous examples were chosen for the reciprocal crossing. There were fertilized:

1. The hybrids with the pollen of ***AB***
2. The hybrids with the pollen of ***ab***
3. ***AB*** with the pollen of the hybrids.

4. ***ab*** with the pollen of the hybrids.

For each of these four experiments the whole of the flowers on three plants were fertilized. If the above theory be correct, there must be developed on the hybrids egg and pollen cells of the forms ***AB***, ***Ab***, ***aB***, ***ab***, and there would be combined:

1. The egg cells ***AB***, ***Ab***, ***aB***, ***ab*** with the pollen cells ***AB***.
2. The egg cells ***AB***, ***Ab***, ***aB***, ***ab*** with the pollen cells ***ab***.
3. The egg cells ***AB*** with the pollen cells ***AB***, ***Ab***, ***aB***, and ***ab***.
4. The egg cells ***ab*** with the pollen cells ***AB***, ***Ab***, ***aB***, and ***ab***.

From each of these experiments there could then result only the following forms:

1. ***AB***, ***ABb***, ***AaB***, ***AaBb***
2. ***AaBb***, ***Aab***, ***aBb***, ***ab***
3. ***AB***, ***ABb***, ***AaB***, ***AaBb***
4. ***AaBb***, ***Aab***, ***aBb***, ***ab***

If, furthermore, the several forms of the egg and pollen cells of the hybrids were produced on an average in equal numbers, then in each experiment the said four combinations should stand in the same ratio to each other. A perfect agreement in the numerical relations was, however, not to be expected since in each fertilization, even in normal cases, some egg cells remain undeveloped or subsequently die, and many even of the well–formed seeds fail to germinate when sown. The above assumption is also limited in so far that while it demands the formation of an equal number of the various sorts of egg and pollen cells, it does not require that this should apply to each separate hybrid with mathematical exactness.

The *first and second* experiments had primarily the object of proving the composition of the hybrid egg cells, while the *third and fourth* experiments were to decide that of the pollen cells. As is shown by the above demonstration the first and third experiments and the second and fourth experiments should produce precisely the same combinations, and even in the second year the result should be partially visible in the form and color of the artificially fertilized seed. In the first and third experiments the dominant characters of form and color, ***A*** and ***B***, appear in each union, and are also partly constant and

partly in hybrid union with the recessive characters ***a*** and ***b***, for which reason they must impress their peculiarity upon the whole of the seeds. All seeds should therefore appear round and yellow, if the theory be justified. In the second and fourth experiments, on the other hand, one union is hybrid in form and in color, and consequently the seeds are round and yellow; another is hybrid in form, but constant in the recessive character of color, whence the seeds are round and green; the third is constant in the recessive character of form but hybrid in color, consequently the seeds are wrinkled and yellow; the fourth is constant in both recessive characters, so that the seeds are wrinkled and green. In both these experiments there were consequently four sorts of seed to be expected; namely, round and yellow, round and green, wrinkled and yellow, wrinkled and green.

The crop fulfilled these expectations perfectly. There were obtained in the

1st Experiment, 98 exclusively round yellow seeds;

3rd " 94 exclusively round yellow seeds

In the 2nd Experiment, 31 round and yellow, 26 round and green, 22 wrinkled and yellow, 26 wrinkled and green seeds.

In the 4th Experiment, 24 round and yellow, 25 round and green, 22 wrinkled and yellow, 27 wrinkled and green.

There could scarcely be now any doubt of the success of the experiment; the next generation must afford the final proof. From the seed sown there resulted for the first experiment 90 plants, and for the third 87 plants which fruited: these yielded for the

1st Exp.	**3rd Exp.**		
20	25	round yellow seeds	***AB***
23	19	round yellow and green seeds	***ABb***
25	22	round and wrinkled yellow seeds	***AaB***
22	21	round and wrinkled green and yellow seeds	***AaBb***

In the second and fourth experiments the round and yellow seeds yielded plants with round and wrinkled yellow and green seeds, ***AaBb***.

From the round green seeds plants resulted with round and wrinkled green seeds, ***Aab***.

The wrinkled yellow seeds gave plants with wrinkled yellow and green seeds, ***aBb***.

From the wrinkled green seeds plants were raised which yielded again only wrinkled and green seeds, ***ab***.

Although in these two experiments likewise some seeds did not germinate, the figures arrived at already in the previous year were not affected thereby, since each kind of seed gave plants which, as regards their seed, were like each other and different from the others. There resulted therefore from the

2nd. Expt.	**4th Expt.**	
31	24	plants of the form ***AaBb***
26	25	plants of the form ***Aab***
27	22	plants of the form ***aBb***
26	27	plants of the form ***ab***

In all the experiments, therefore, there appeared all the forms which the proposed theory demands, and they came in nearly equal numbers.

In a further experiment the characters of *flower–color and length of stem* were experimented upon, and selection was so made that in the third year of the experiment each character ought to appear in *half* of all the plants if the above theory were correct. ***A***, ***B***, ***a***, ***b*** serve again as indicating the various characters.

A, violet–red flowers.	***a***, white flowers.
B, axis long.	***b***, axis short.

The form ***Ab*** was fertilized with ***ab***, which produced the hybrid ***Aab***. Furthermore, ***aB*** was also fertilized with ***ab***, whence the hybrid ***aBb***. In the second year, for further fertilization, the hybrid ***Aab*** was used as seed parent, and hybrid ***aBb*** as pollen parent.

Seed parent, ***Aab***.	Pollen parent, **aBb**.
Possible egg cells, ***Ab, ab***.	Pollen cells, ***aB, ab***.

From the fertilisation between the possible egg and pollen cells four combinations should result, namely:

$$AaBb + aBb + Aab + ab$$

From this it is perceived that, according to the above theory, in the third year of the experiment out of all the plants,

Half	should	have	violet–red flowers	(***Aa***),	Classes	1, 3
"	"	"	white flowers	(***a***)	"	2, 4
"	"	"	a long axis	(***Bb***)	"	1, 2
"	"	"	a short axis	(***b***)	"	3, 4

From 45 fertilizations of the second year 187 seeds resulted, of which only 166 reached the flowering stage in the third year. Among these the separate classes appeared in the numbers following:

Class	Flower Color	Stem	
1	violet–red	long	47 times
2	white	long	40 times
3	violet–red	short	38 times
4	white	short	41 times

There subsequently appeared,

The	violet–red flower color	(***Aa***)	in	85 plants
"	white flower–color	(***a***)	in	81 plants
"	long stem	(***Bb***)	in	87 plants
"	short stem	(***b***)	in	79 plants

The theory adduced is therefore satisfactorily confirmed in this experiment also.

For the characters of *form of pod*, *color of pod*, and *position of flowers*, experiments were also made on a small scale and results obtained in perfect agreement. All combinations, which were possible through the union of the differentiating characters duly appeared, and in nearly equal numbers.

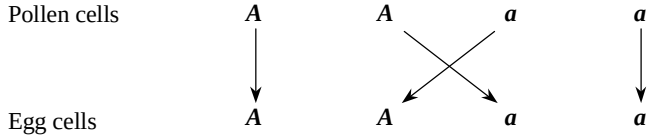
Experimentally, therefore, the theory is confirmed that *the pea hybrids form egg and pollen cells which, in their constitution, represent in equal numbers all constant forms which result from the combination of the characters united in fertilization.*

The difference of the forms among the progeny of the hybrids, as well as the respective ratios of the numbers in which they are observed, find a sufficient explanation in the principle above deduced. The simplest case is afforded by the developmental series of *each pair of differentiating characters*. This series is represented by the expression ***A***+2***Aa***+***a***, in which ***A*** and ***a*** signify the forms with constant differentiating characters, and ***Aa*** the hybrid form of both. It includes in three different classes four individuals. In the formation of these, pollen and egg cells of the form ***A*** and ***a*** take part on the average equally in the fertilization; hence each form [occurs] twice, since four individuals are formed. There participate consequently in the fertilization

The pollen cells: ***A*** + ***A*** + ***a*** + ***a***,
The egg cells: ***A*** + ***A*** + ***a*** + ***a***.

It remains, therefore, purely a matter of chance which of the two sorts of pollen will become united with each separate egg cell. According, however, to the law of probability, it will always happen, on the average of many cases, that each pollen form ***A*** and ***a*** will unite

equally often with each egg cell form ***A*** and ***a***, consequently one of the two pollen cells ***A*** in the fertilization will meet with the egg cell ***A*** and the other with the egg cell ***a***, and so likewise one pollen cell ***a*** will unite with an egg cell ***A***, and the other with the egg cell ***a***.



The result of the fertilization may be made clear by putting the signs for the conjoined egg and pollen cells in the form of fractions, those for the pollen cells above and those for the egg cells below the line. We then have

$$\frac{A}{A} + \frac{A}{a} + \frac{a}{A} + \frac{a}{a} .$$

In the first and fourth term the egg and pollen cells are of like kind, consequently the product of their union must be constant, namely ***A*** and ***a***; in the second and third, on the other hand, there again results a union of the two differentiating characters of the stocks, consequently the forms resulting from these fertilizations are identical with those of the hybrid from which they sprang. *There occurs accordingly a repeated hybridization*. This explains the striking fact that the hybrids are able to produce, besides the two parental forms, offspring which are like themselves ***A/a*** and ***a/A*** both give the same union ***Aa***, since, as already remarked above, it makes no difference in the result of fertilization to which of the two characters the pollen or egg cells belong. We may write then

$$\frac{A}{A} + \frac{A}{a} + \frac{a}{A} + \frac{a}{a} = A + 2Aa + a .$$

This represents the average result of the self–fertilization of the hybrids when two differentiating characters are united in them. In individual flowers and in individual plants, however, the ratios in which the forms of the series are produced may suffer not inconsiderable fluctuations. Apart from the fact that the numbers in which both sorts of egg cells occur in the seed vessels can only be regarded as equal on the average, it remains purely a matter of chance which of the two sorts of pollen may fertilize each separate egg cell. For this reason the separate values must necessarily be subject to

fluctuations, and there are even extreme cases possible, as were described earlier in connection with the experiments on the forms of the seed and the color of the albumen. The true ratios of the numbers can only be ascertained by an average deduced from the sum of as many single values as possible; the greater the number the more are merely chance effects eliminated.

The developmental series for hybrids in which two kinds of differentiating characters are united contains among 16 individuals nine different forms, viz.:

$$\boldsymbol{AB} + \boldsymbol{Ab} + \boldsymbol{aB} + \boldsymbol{ab} + 2\boldsymbol{ABb} + 2\boldsymbol{aBb} + 2\boldsymbol{AaB} + 2\boldsymbol{Aab} + 4\boldsymbol{AaBb}.$$

Between the differentiating characters of the original stocks ***Aa*** and ***Bb***, four constant combinations are possible, and consequently the hybrids produce the corresponding four forms of egg and pollen cells: ***AB***, ***Ab***, ***aB***, ***ab***, and each of these will on the average figure four times in the fertilization, since 16 individuals are included in the series. Therefore, the participators in the fertilization are

Pollen cells: ***AB*** + ***AB*** + ***AB*** + ***AB*** + ***Ab*** + ***Ab*** + ***Ab*** + ***Ab*** + ***aB*** + ***aB*** + ***aB*** + ***aB*** + ***ab*** + ***ab*** + ***ab*** + ***ab***.

Egg cells: ***AB*** + ***AB*** + ***AB*** + ***AB*** + ***Ab*** + ***Ab*** + ***Ab*** + ***Ab*** + ***aB*** + ***aB*** + ***aB*** + ***aB*** + ***ab*** + ***ab*** + ***ab*** + ***ab***.

In the process of fertilization each pollen form unites on an average equally often with each egg cell form, so that each of the four pollen cells ***AB*** unites once with one of the forms of egg cell ***AB***, ***Ab***, ***aB***, ***ab***. In precisely the same way the rest of the pollen cells of the forms ***Ab***, ***aB***, ***ab***, unite with all the other egg cells. We obtain therefore:

$$\frac{\boldsymbol{AB}}{\boldsymbol{AB}} + \frac{\boldsymbol{AB}}{\boldsymbol{Ab}} + \frac{\boldsymbol{AB}}{\boldsymbol{aB}} + \frac{\boldsymbol{AB}}{\boldsymbol{ab}} + \frac{\boldsymbol{Ab}}{\boldsymbol{AB}} + \frac{\boldsymbol{Ab}}{\boldsymbol{Ab}} + \frac{\boldsymbol{Ab}}{\boldsymbol{aB}} + \frac{\boldsymbol{Ab}}{\boldsymbol{ab}} +$$

$$\frac{\boldsymbol{aB}}{\boldsymbol{AB}} + \frac{\boldsymbol{aB}}{\boldsymbol{Ab}} + \frac{\boldsymbol{aB}}{\boldsymbol{aB}} + \frac{\boldsymbol{aB}}{\boldsymbol{ab}} + \frac{\boldsymbol{ab}}{\boldsymbol{AB}} + \frac{\boldsymbol{ab}}{\boldsymbol{Ab}} + \frac{\boldsymbol{ab}}{\boldsymbol{aB}} + \frac{\boldsymbol{ab}}{\boldsymbol{ab}},$$

or

$$\boldsymbol{AB} + \boldsymbol{ABb} + \boldsymbol{AaB} + \boldsymbol{AaBb} + \boldsymbol{ABb} + \boldsymbol{Ab} + \boldsymbol{AaBb} + \boldsymbol{Aab} +$$
$$\boldsymbol{AaB} + \boldsymbol{AaBb} + \boldsymbol{aB} + \boldsymbol{aBb} + \boldsymbol{AaBb} + \boldsymbol{Aab} + \boldsymbol{aBb} + \boldsymbol{ab} =$$

$$\boldsymbol{AB} + \boldsymbol{Ab} + \boldsymbol{aB} + \boldsymbol{ab} + 2\boldsymbol{ABb} + 2\boldsymbol{aBb} + 2\boldsymbol{AaB} + 2\boldsymbol{Aab} + 4\boldsymbol{AaBb}$$

In precisely similar fashion is the developmental series of hybrids exhibited when *three kinds of differentiating characters* are conjoined in them. The hybrids form eight various kinds of egg and pollen cells: ***ABC***, ***ABc***, ***AbC***, ***Abc***, ***aBC***, ***aBc***, ***abC***, ***abc***, and each pollen form unites itself again on the average once with each form of egg cell.

The law of combination of different characters which governs the development of the hybrids finds therefore its *foundation and explanation* in the principle enunciated, that the hybrids produce egg cells and pollen cells which in equal numbers represent all constant forms which result from the combinations of the characters brought together in fertilization.

EXPERIMENTS WITH HYBRIDS OF OTHER SPECIES OF PLANTS

It must be the object of further experiments to ascertain whether the law of development discovered for *Pisum* applies also to the hybrids of other plants. To this end several experiments were recently commenced. Two minor experiments with species of *Phaseolus* have been completed, and may be here mentioned.

An experiment with *Phaseolus vulgaris* and *Phaseolus nanus* gave results in perfect agreement. *Ph. nanus* had together with the dwarf axis, simply inflated, green pods. *Ph. vulgaris* had, on the other hand, an axis 10 ft. to 12 ft. high, and yellow colored pods, constricted when ripe. The ratios of the numbers in which the different forms appeared in the separate generations were the same as with *Pisum*. Also the development of the constant combinations resulted according to the law of simple combination of characters, exactly as in the case of *Pisum*. There were obtained:

Constant Combinations	**Axis**	**Color of the unripe pods**	**Form of the unripe pods**
1	long	green	inflated
2	"	"	constricted
3	"	yellow	inflated
4	"	"	constricted
5	short	green	inflated
6	"	"	constricted
7	"	yellow	inflated
8	"	"	constricted

The green color of the pod, the inflated forms, and the long axis were, as in *Pisum*, dominant characters.

Another experiment with two very different species of *Phaseolus* had only a partial result. *Phaseolus nanus*, L., served as seed parent, a perfectly constant species, with white flowers in short recemes and small white seeds in straight, inflated, smooth pods; as pollen parent was used *Ph. multiflorus*, W., with tall winding stem, purple–red flowers in very long racemes, rough, sickle–shaped crooked pods, and large seeds which bore black flecks and splashes on a peach–blood–red ground.

The hybrids had the greatest similarity to the pollen parent, but the flowers appeared less intensely colored. Their fertility was very limited; from 17 plants, which together developed many hundreds of flowers, only 49 seeds in all were obtained. These were of medium size, and were flecked and splashed similarly to those of *Ph. multiflorus*, while the ground color was not materially different. The next year 44 plants were raised from these seeds, of which only 31 reached the flowering stage. The characters of *Ph. nanus*, which had been altogether latent in the hybrids, reappeared in various combinations; their ratio, however, with relation to the dominant plants was necessarily very fluctuating owing to the small number of trial plants. With certain characters, as in those of the axis and the form of pod, it was, however, as in the case of *Pisum*, almost exactly 1:3.

Insignificant as the results of this experiment may be as regards the determination of the relative numbers in which the various forms appeared, it presents, on the other hand, the phenomenon of a remarkable change of color in the flowers and seed of the hybrids. In *Pisum* it is known that the characters of the flower– and seed–color present themselves unchanged in the first and further generations, and that the offspring of the hybrids display exclusively the one or the other of the characters of the original stocks. It is otherwise in the experiment we are considering. The white flowers and the seed–color of *Ph. nanus* appeared, it is true, at once in the first generation in one fairly fertile example, but the remaining 30 plants developed flower–colors which were of various grades of purple–red to pale violet. The coloring of the seed–coat was no less varied than that of the flowers. No plant could rank as fully fertile; many produced no fruit at all; others only yielded fruits from the flowers last produced, which did not ripen. From 15 plants only were well–developed seeds obtained. The greatest disposition to infertility was seen in the forms with preponderantly red flowers, since out of 16 of these only four yielded

ripe seeds. Three of these had a similar seed pattern to *Ph. multiflorus*, but with a more or less pale ground color; the fourth plant yielded only one seed of plain brown tint. The forms with preponderantly violet–colored flowers had dark brown, black–brown, and quite black seeds.

The experiment was continued through two more generations under similar unfavorable circumstances, since even among the offspring of fairly fertile plants there came again some which were less fertile and even quite sterile. Other flower–and seed–colors than those cited did not subsequently present themselves. The forms which in the first generation contained one or more of the recessive characters remained, as regards these, constant without exception. Also of those plants which possessed violet flowers and brown or black seed, some did not vary again in these respects in the next generation; the majority, however, yielded together with offspring exactly like themselves, some which displayed white flowers and white seed–coats. The red flowering plants remained so slightly fertile that nothing can be said with certainty as regards their further development.

Despite the many disturbing factors with which the observations had to contend, it is nevertheless seen by this experiment that the development of the hybrids, with regard to those characters which concern the form of the plants, follows the same laws as in *Pisum*. With regard to the color characters, it certainly appears difficult to perceive a substantial agreement. Apart from the fact that from the union of a white and a purple–red coloring a whole series of colors results, from purple to pale violet and white, the circumstance is a striking one that among 31 flowering plants only one received the recessive character of the white color, while in *Pisum* this occurs on the average in every fourth plant.

Even these enigmatic results, however, might probably be explained by the law governing *Pisum* if we might assume that the color of the flowers and seeds of *Ph. multiflorus* is a combination of two or more entirely independent colors, which individually act like any other constant character in the plant. If the flower–color $\boldsymbol{A}$ were a combination of the individual characters $\boldsymbol{A_1}$+ $\boldsymbol{A_2}$ + ... which produce the total impression of a purple coloration, then by fertilization with the differentiating character, white color, $\boldsymbol{a}$, there would be produced the hybrid unions $\boldsymbol{A_1a}$ + $\boldsymbol{A_2a}$ +..... and so would it be with the corresponding coloring of the seed–coats. According to the above assumptions, each of these hybrid color unions would be independent,

and would consequently develop quite independently from the others. It is then easily seen that from the combination of the separate developmental series a complete color–series must result. If, for instance, $A = A_1 + A_2$, then the hybrids A_1a and A_2a form the developmental series:

$$A_1 + 2\,A_1a + a$$
$$A_2 + 2\,A_2a + a$$

The members of this series can enter into nine different combinations, and each of these denotes another color:

$1\,A_1A_2$	$2\,A_1aA_2$	$1\,A_2a$
$2\,A_1A_2a$	$4\,A_1aA_2a$	$2\,A_2aa$
$1\,A_1a$	$2\,A_1aa$	$1\,aa$

The figures prescribed for the separate combinations also indicate how many plants with the corresponding coloring belong to the series. Since the total is 16, the whole of the colors are on the average distributed over each 16 plants, but, as the series itself indicated, in unequal proportions.

Should the color development really happen in this way, we could offer an explanation of the case above described, namely that of the white flowers and seed–coat color only appeared once among 31 plants of the first generation. This coloring appears only once in the series, and could therefore also only be developed once in the average in each 16, and with three color characters only once even in 64 plants.

It must, nevertheless, not be forgotten that the explanation here attempted is based on a mere hypothesis, only supported by the very imperfect result of the experiment just described. It would, however, be well worth while to follow up the development of color in hybrids by similar experiments, since it is probable that in this way we might learn the significance of the extraordinary variety in the *coloring of our ornamental flowers.*

So far, little at present is known with certainty beyond the fact that the color of the flowers in most ornamental plants is an extremely variable character. The opinion has often been expressed that the stability of the species is greatly disturbed or entirely upset by cultivation, and consequently there is an inclination to regard the development of cultivated forms as a matter of chance devoid of rules; the coloring of ornamental plants is indeed usually cited as an example of great instability. It is, however, not clear why the simple

transference into garden soil should result in such a thorough and persistent revolution in the plant organism. No one will seriously maintain that in the open country the development of plants is ruled by other laws than in the garden bed. Here, as there, changes of type must take place if the conditions of life be altered, and the species possesses the capacity of fitting itself to its new environment. It is willingly granted that by cultivation the origination of new varieties is favored, and that by man's labor many varieties are acquired which, under natural conditions, would be lost; but nothing justifies the assumption that the tendency to formation of varieties is so extraordinarily increased that the species speedily lose all stability, and their offspring diverge into an endless series of extremely variable forms. Were the change in the conditions the sole cause of variability we might expect that those cultivated plants which are grown for centuries under almost identical conditions would again attain constancy. This, as is well known, is not the case since it is precisely under such circumstances that not only the most varied but also the most variable forms are found. It is only the *Leguminosae*, like *Pisum*, *Phaseolus*, *Lens*, whose organs of fertilization are protected by the keel, which constitute a noteworthy exception. Even here there have arisen numerous varieties during a cultural period of more than 1,000 years under most various conditions; these maintain, however, under unchanging environments a stability as great as that of species growing wild.

It is more than probable that as regards the variability of cultivated plants there exists a factor which so far has received little attention. Various experiments force us to the conclusion that our cultivated plants, with few exceptions, are *members of various hybrid series*, whose further development in conformity with law is varied and interrupted by frequent crossings *inter se*. The circumstance must not be overlooked that cultivated plants are mostly grown in great numbers and close together, affording the most favorable conditions for reciprocal fertilization between the varieties present and species itself. The probability of this is supported by the fact that among the great array of variable forms solitary examples are always found, which in one character or another remain constant, if only foreign influence be carefully excluded. These forms behave precisely as do those which are known to be members of the compound hybrid series. Also with the most susceptible of all characters, that of color, it cannot escape the careful observer that in the separate forms the inclination to vary is displayed in very different degrees. Among plants which arise from *one* spontaneous fertilization there are often

some who offspring vary widely in the constitution and arrangement of the colors, while that of others shows little deviation, and among a greater number solitary examples occur which transmit the color of the flowers unchanged to their offspring. The cultivated species of *Dianthus* afford an instructive example of this. A white–flowered example of *Dianthus caryophyllus*, which itself was derived from a white–flowered variety, was shut up during its blooming period in a greenhouse; the numerous seeds obtained therefrom yielded plants entirely white–flowered like itself. A similar result was obtained from a sub–species, with red flowers somewhat flushed with violet, and one with flowers white, striped with red. Many others, on the other hand, which were similarly protected, yielded progeny which were more or less variously colored and marked.

Whoever studies the coloration which results in ornamental plants from similar fertilization can hardly escape the conviction that here also the development follows a definite law which possibly finds its expression *in the combination of several independent color characters.*

CONCLUDING REMARKS

It can hardly fail to be of interest to compare the observations made regarding *Pisum* with the results arrived at by the two authorities in this branch of knowledge, Kölreuter and Gärtner, in their investigations. According to the opinion of both, the hybrids in outward appearance present either a form intermediate between the original species, or they closely resemble either the one or the other type, and sometimes can hardly be discriminated from it. From their seeds usually arise, if the fertilization was effected by their own pollen, various forms which differ from the normal type. As a rule, the majority of individuals obtained by one fertilization maintain the hybrid form, while some few others come more like the seed parent, and one or other individual approaches the pollen parent. This, however, is not the case with hybrids without exception. Sometimes the offspring have more nearly approached, some the one and some the other of the two original stocks, or they all incline more to one or the other side; while in other cases *they remain perfectly like the hybrid* and continue constant in their offspring. The hybrids of varieties behave like hybrids of species, but they possess greater

variability of form and more pronounced tendency to revert to the original types.

With regard to the *form* of the hybrids and their development, as a rule an agreement with the observations made in *Pisum* is unmistakable. It is otherwise with the exceptional cases cited. Gärtner confesses even that the exact determination whether a form bears a greater resemblance to one or to the other of the two original species often involved great difficulty, so much depending upon the subjective point of view of the observer. Another circumstance could, however, contribute to render the results fluctuating and uncertain, despite the most careful observation and differentiation. For the experiments, plants were mostly used which rank as good species and are differentiated by a large number of characters. In addition to the sharply defined characters, where it is a question of greatly or less similarity, those characters must also be taken into account which are often difficult to define in words, but yet suffice, as every plant specialist knows, to give the forms a peculiar appearance. If it be accepted that the development of hybrids follows the law which is valid for *Pisum*, the series in each separate experiment must contain very many forms, since the number of terms, as is known, increases with the number of the differentiating characters as the powers of three. With a relatively small number of experimental plants the results therefore could only be approximately right, and in single cases might fluctuate considerably. If, for instance, the two original stocks differ in seven characters, and 100 – 200 plants were raised from the seeds of their hybrids to determine the grade of relationship of the offspring, we can easily see how uncertain the decision must become since for seven differentiating characters the combination series contains 16,384 individuals under 2,187 various forms; now one and then another relationship could assert its predominance, just according as chance presented this or that form to the observer in a majority of cases.

If, furthermore, there appear among the differentiating characters at the same time *dominant* characters, which are transmitted entire or nearly unchanged to the hybrids, then in the terms of the developmental series that one of the two original parents which possesses the majority of dominant characters must always be predominant. In the experiment described relative to *Pisum*, in which three kinds of differentiating characters were concerned, all the dominant characters belonged to the seed parent. Although the terms of the series in their internal composition approach both original parents equally, yet in this experiment the type of the seed parent

obtained so great a preponderance that out of each 64 plants of the first generation 54 exactly resembled it, or only differed in one character. It is seen how rash it must be under such circumstances to draw from the external resemblances of hybrids conclusions as to their internal nature.

Gärtner mentions that in those cases where the development was regular among the offspring of the hybrids the two original species were not reproduced, but only a few individuals which approached them. With very extended developmental series it could not in fact be otherwise. For seven differentiating characters, for instance, among more than 16,000 individuals — offspring of the hybrids — each of the two original species would occur only once. It is therefore hardly possible that these should appear at all among a small number of experimental plants; with some probability, however, we might reckon upon the appearance in the series of a few forms which approach them.

We meet with an *essential difference* in those hybrids which remain constant in their progeny and propagate themselves as truly as the pure species. According to Gärtner, to this class belong the *remarkably fertile hybrids Aquilegia atropurpurea canadensis*, *Lavatera pseudolbia thuringiaca*, *Geum urbanorivale*, and some *Dianthus* hybrids; and, according to Wichura, the hybrids of the Willow family. For the history of the evolution of plants this circumstance is of special importance, since constant hybrids acquire the status of new species. The correctness of the facts is guaranteed by eminent observers, and cannot be doubted. Gärtner had an opportunity of following up *Dianthus Armeria deltoides* to the tenth generation, since it regularly propagated itself in the garden.

With *Pisum* it was shown by experiment that the hybrids form egg and pollen cells of *different* kinds, and that herein lies the reason of the variability of their offspring. In other hybrids, likewise, whose offspring behave similarly we may assume a like cause; for those, on the other hand, which remain constant the assumption appears justifiable that their reproductive cells are all alike and agree with the foundation–cell of the hybrid. In the opinion of renowned physiologists, for the purpose of propagation one pollen cell and one egg cells unite in Phanerogams* into a single cell, which is capable by

* In Pisum it is placed beyond doubt that for the formation of the new embryo a perfect union of the elements of both reproductive cells must take place. How could we otherwise explain that among the offspring of the hybrids both original types reappear in equal numbers and with all their peculiarities? If the influence of the egg cell upon the pollen cell were only external, if it fulfilled the role of a nurse

assimilation and formation of new cells to become an independent organism. This development follows a constant law, which is founded on the material composition and arrangement of the elements which meet in the cell in a vivifying union. If the reproductive cells be of the same kind and agree with the foundation cell of the mother plant, then the development of the new individual will follow the same law which rules the mother plant. If it chance that an egg cell unites with a *dissimilar* pollen cell, we must then assume that between those elements of both cells, which determine opposite characters some sort of compromise is effected. The resulting compound cell becomes the foundation of the hybrid organism the development of which necessarily follows a different scheme from that obtaining in each of the two original species. If the compromise be taken to be a complete one, in the sense, namely, that the hybrid embryo is formed from two similar cells, in which the differences are *entirely and permanently accommodated* together, the further result follows that the hybrids, like any other stable plant species, reproduce themselves truly in their offspring. The reproductive cells which are formed in their seed vessels and anthers are of one kind, and agree with the fundamental compound cell.

With regard to those hybrids whose progeny is *variable* we may perhaps assume that between the differentiating elements of the egg and pollen cells there also occurs a compromise, in so far that the formation of a cell as the foundation of the hybrid becomes possible; but, nevertheless, the arrangement between the conflicting elements is only temporary and does not endure throughout the life of the hybrid plant. Since in the habit of the plant no changes are perceptible during the whole period of vegetation, we must further assume that it is only possible for the differentiating elements to liberate themselves from the enforced union when the fertilizing cells are developed. In the formation of these cells all existing elements participate in an entirely free and equal arrangement, by which it is only the differentiating ones which mutually separate themselves. In this way the production would be rendered possible of as many sorts of egg and pollen cells as there are combinations possible of the formative elements.

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only, then the result of each fertilization could be no other than that the developed hybrid should exactly resemble the pollen parent, or at any rate do so very closely. This the experiments so far have in nowise confirmed. An evident proof of the complete union of the contents of both cells is afforded by the experience gained on all sides that it is immaterial, as regards the form of the hybrid, which of the original species is the seed parent or which the pollen parent.

The attribution attempted here of the essential difference in the development of hybrids to a *permanent or temporary union* of the differing cell elements can, of course, only claim the value of an hypothesis for which the lack of definite data offers a wide scope. Some justification of the opinion expressed lies in the evidence afforded by *Pisum* that the behavior of each pair of differentiating characters in hybrid union is independent of the other differences between the two original plants, and, further, that the hybrid produces just so many kinds of egg and pollen cells as there are possible constant combination forms. The differentiating characters of two plants can finally, however, only depend upon differences in the composition and grouping of the elements which exist in the foundation–cells of the same in vital interaction.

Even the validity of the law formulated for *Pisum* requires still to be confirmed, and a repetition of the more important experiments is consequently must to be desired, that, for instance, relating to the composition of the hybrid fertilizing cells. A differential may easily escape the single observer, which although at the outset may appear to be unimportant, yet accumulate to such an extent that it must not be ignored in the total result. Whether the variable hybrids of other plant species observe an entire agreement must also be first decided experimentally. In the meantime we may assume that in material points an essential difference can scarcely occur, since the unity in the developmental plant of organic life is beyond question.

In conclusion, the experiments carried out by Kölreuter, Gärtner, and others with respect to *the transformation of one species into another by artificial fertilization* merit special mention. Particular importance has been attached to these experiments and Gärtner reckons them “among the most difficult of all in hybridization.”

If a species ***A*** is to be transformed into a species ***B***, both must be united by fertilization and the resulting hybrids then be fertilized with the pollen of ***B***; then, out of the various offspring resulting, that form would be selected which stood in nearest relation to ***B*** and once more be fertilized with ***B*** pollen, and so continuously until finally a form is arrived at which is like ***B*** and constant in its progeny. By this process the species ***A*** would change into the species ***B***. Gärtner alone has effected 30 such experiments with plants of genera *Aquilegia*, *Dianthus*, *Geum*, *Lavatera*, *Lynchnis*, *Malva*, *Nicotiana*, and *Oenothera*. The period of transformation was not alike for all species. While with some a triple fertilization sufficed, with others this had to be repeated five or six times, and even in the same species fluctuations were observed in various experiments. Gärtner ascribes

this difference to the circumstance that "the specific power by which a species, during reproduction, effects the change and transformation of the maternal type varies considerably in different plants, and that, consequently, the periods with which the one species is changed into the other must also vary, as also the number of generations, so that the transformation in some species is perfected in more, and in others in fewer generations". Further, the same observer remarks "that in these transformation experiments a good deal depends upon which type and which individual be chosen for further transformation".

If it may be assumed that in these experiments the constitution of the forms resulted in a similar way to that of *Pisum*, the entire process of transformation would find a fairly simple explanation. The hybrid forms as many kinds of egg cells as there are constant combinations possible of the characters conjoined therein, and one of these is always of the same kind as that of the fertilizing pollen cells. Consequently there always exists the possibility with all such experiments that even from the second fertilization there may result a constant form identical with that of the pollen parent. Whether this really be obtained depends in each separate case upon the number of the experimental plants, as well as upon the number of differentiating characters which are united by the fertilization. Let us, for instance, assume that the plants selected for experiment differed in three characters, and the species ***ABC*** is to be transformed into the other species ***abc*** by repeated fertilization with the pollen of the latter; the hybrids resulting from the first cross form eight different kinds of egg cells, namely:

ABC, ***ABc***, ***AbC***, ***aBC***, ***Abc***, ***aBc***, ***abC***, ***abc***

These in the second year of experiment are united again with the pollen cells ***abc***, and we obtain the series

AaBbCc + ***AaBbc*** + ***AabCc*** + ***aBbCc*** + ***Aabc*** + ***aBbc*** + ***abCc*** + ***abc***

Since the form abc occurs once in the series of eight terms, it is consequently little likely that it would be missing among the experimental plants, even were these raised in a smaller number, and the transformation would be perfected already by a second fertilization. If by chance it did not appear, then the fertilization must be repeated with one of those forms nearest akin, ***Aabc***, ***aBbc***, ***abCc***. It is perceived that such an experiment must extend the farther *the smaller the number of experimental plants and the larger the number of differentiating characters* in the two original species; and that, furthermore, in the same species there can easily occur a delay of one

or even of two generations such as Gärtner observed. The transformation of widely divergent species could generally only be completed in five or six years of experiment, since the number of different egg cells which are formed in the hybrid increases as the powers of two with the number of differentiating characters.

Gärtner found by repeated experiments that the *respective period of transformation* varies in many species, so that frequently a species ***A*** can be transformed into a species ***B*** a generation sooner than can species ***B*** into species ***A***. He deduces therefrom that Kölreuter's opinion can hardly be maintained that "the two natures in hybrids are perfectly in equilibrium". Experiments which in this connection were carried out with two species of *Pisum* demonstrated that as regards the choice of the fittest individuals for the purpose of further fertilization it may make a great difference which of two species is transformed into the other. The two experimental plants differed in five characters, while at the same time those of species ***A*** were all dominant and those of species ***B*** all recessive. For mutual transformation ***A*** was fertilized with pollen of ***B***, and ***B*** with pollen of ***A***, and this was repeated with both hybrids the following year. With the first experiment, ***B/A***, there were 87 plants available in the third year of experiment for selection of the individuals for further crossing, and these were *of the possible 32 forms*; with the second experiment, ***A/B***, 73 plants resulted, which *agreed throughout perfectly in habit with the pollen parent*; in their internal composition, however, they must have been just as varied as the forms in the other experiment. A definite selection was consequently only possible with the first experiment; with the second the selection had to be made at random, merely. Of the latter only a portion of the flowers were crossed with the ***A*** pollen, the others were left to fertilize themselves. Among each five plants which were selected in both experiments for fertilization there agreed, as the following year's culture showed, with the pollen parent:

1st Experiment	**2nd Experiment**			
2 plants	——	in	all	characters
3 plants	——	"	4	"
——	2 plants	"	3	"
——	2 plants	"	2	"
——	1 plant	"	1	character

In the first experiment, therefore, the transformation was completed; in the second, which was not continued further, two more fertilizations would probably have been required.

Although the case may not frequently occur in which the dominant characters belong exclusively to one or the other of the original parent plants, it will always make a difference *which* of the two possesses the majority of dominants. If the pollen parent has the majority, then the selection of forms for further crossing will afford a less degree of certainty than in the reverse case, which must imply a delay in the period of transformation, provided that the experiment is only considered as completed when a form is arrived at which not only exactly resembles the pollen parent in form, but also remains as constant in its progeny.

Gärtner, by the results of theses transformation experiments, was led to oppose the opinion of those naturalists who dispute the stability of plant species and believe in a continuous evolution of vegetation. He perceives in the complete transformation of one species into another an indubitable proof that species are fixed with limits beyond which they cannot change. Although this opinion cannot be unconditionally accepted we find on the other hand in Gärtner's experiments a noteworthy confirmation of that supposition regarding variability of cultivated plants which has already been expressed.

Among the experimental species there were cultivated plants, such as *Aquilegia atropurpurea* and *canadensis*, *Dianthus caryophyllus*, *chinensis*, and *japonicus*, *Nicotiana rustica* and *paniculata*, and hybrids between these species lost none of their stability after four or five generations.

Hardy (1908)

Hardy, G.H. 1908. Mendelian proportions in a mixed population. Science, vol. 28, no. 706, pp. 49 and 50.

The problem

Soon after the rediscovery of Mendel, an argument was raised against the new theory. If an allele is dominant, the trait it determines should spread until it reaches three quarters of the population. Since this was not observed, some concluded that Mendel's laws did not apply to populations. In a two page letter, the mathematician G.H. Hardy showed that, under random mating, genotype frequencies remain constant from one generation to the next. Wilhelm Weinberg independently reached the same result in the same year.

Points to note while reading

The Hardy and Weinberg equilibrium is the starting point of population genetics. Selection, migration, mutation and drift are interpreted as forces that move a population away from this equilibrium. Note that the demonstration uses only elementary algebra.

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School of Economics and Political Science, to which he was appointed in 1903, retains the readership in geography, to which, under its then title, he was appointed in 1902.

DISCUSSION AND CORRESPONDENCE

MENDELIAN PROPORTIONS IN A MIXED POPULATION

TO THE EDITOR OF SCIENCE: I am reluctant to intrude in a discussion concerning matters of which I have no expert knowledge, and I should have expected the very simple point which I wish to make to have been familiar to biologists. However, some remarks of Mr. Udny Yule, to which Mr. R. C. Punnett has called my attention, suggest that it may still be worth making.

In the *Proceedings of the Royal Society of Medicine* (Vol. I., p. 165) Mr. Yule is reported to have suggested, as a criticism of the Mendelian position, that if brachydactyly is dominant "in the course of time one would expect, in the absence of counteracting factors, to get three brachydactylous persons to one normal."

It is not difficult to prove, however, that such an expectation would be quite groundless. Suppose that Aa is a pair of Mendelian characters, A being dominant, and that in any given generation the numbers of pure dominants (AA), heterozygotes (Aa), and pure recessives (aa) are as $p:2q:r$. Finally, suppose that the numbers are fairly large, so that the mating may be regarded as random, that the sexes are evenly distributed among the three varieties, and that all are equally fertile. A little mathematics of the multiplication-table type is enough to show that in the next generation the numbers will be as

$$(p+q)^2 : 2(p+q)(q+r) : (q+r)^2,$$

or as $p_1 : 2q_1 : r_1$, say.

The interesting question is—in what circumstances will this distribution be the same as that in the generation before? It is easy to see that the condition for this is $q^2 = pr$. And since $q_1^2 = p_1 r_1$, whatever the values of p, q and r may be, the distribution will in any case continue unchanged after the second generation.

Suppose, to take a definite instance, that A is brachydactyly, and that we start from a population of pure brachydactylous and pure normal persons, say in the ratio of 1:10,000. Then $p = 1$, $q = 0$, $r = 10{,}000$ and $p_1 = 1$, $q_1 = 10{,}000$, $r_1 = 100{,}000{,}000$. If brachydactyly is dominant, the proportion of brachydactylous persons in the second generation is 20,001:100,020,001, or practically 2:10,000, twice that in the first generation; and this proportion will afterwards have no tendency whatever to increase. If, on the other hand, brachydactyly were recessive, the proportion in the second generation would be 1:100,020,001, or practically 1:100,000,000, and this proportion would afterwards have no tendency to decrease.

In a word, there is not the slightest foundation for the idea that a dominant character should show a tendency to spread over a whole population, or that a recessive should tend to die out.

I ought perhaps to add a few words on the effect of the small deviations from the theoretical proportions which will, of course, occur in every generation. Such a distribution as $p_1 : 2q_1 : r_1$, which satisfies the condition $q_1^2 = p_1 r_1$, we may call a *stable* distribution. In actual fact we shall obtain in the second generation not $p_1 : 2q_1 : r_1$ but a slightly different distribution $p_1' : 2q_1' : r_1'$, which is not "stable." This should, according to theory, give us in the third generation a "stable" distribution $p_2 : 2q_2 : r_2$, also differing slightly from $p_1 : 2q_1 : r_1$; and so on. The sense in which the distribution $p_1 : 2q_1 : r_1$ is "stable" is this, that if we allow for the effect of casual deviations in any subsequent generation, we should, according to theory, obtain at the next generation a new "stable" distribution differing but slightly from the original distribution.

I have, of course, considered only the very simplest hypotheses possible. Hypotheses other that that of purely random mating will give different results, and, of course, if, as appears to be the case sometimes, the character is not independent of that of sex, or

has an influence on fertility, the whole question may be greatly complicated. But such complications seem to be irrelevant to the simple issue raised by Mr. Yule's remarks.

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April 5, 1908

P. S. I understand from Mr. Punnett that he has submitted the substance of what I have said above to Mr. Yule, and that the latter would accept it as a satisfactory answer to the difficulty that he raised. The "stability" of the particular ratio 1:2:1 is recognized by Professor Karl Pearson (*Phil. Trans. Roy. Soc.* (A), vol. 203, p. 60).

PURE CULTURES FOR LEGUME INOCULATION

IN the 1907 Report of the Biologist of the North Carolina Agricultural Experiment Station, Dr. F. L. Stevens and Mr. J. C. Temple report some work upon cultures of the nodule-forming organisms of legumes. The cultures used were obtained from the United States Department of Agriculture. The investigators have presented their data in such a manner that the value of pure cultures for inoculating legumes appears questionable and their conclusions emphasize their attitude of disapproval. In carefully reviewing their report, a very brief outline of which appeared in SCIENCE, Vol. 26, 1907, p. 311, I have been impressed with the fact that the inferences drawn by the casual reader would almost certainly be unwarrantably antagonistic to the use of pure cultures for inoculating legumes. The investigators' objections to the actions of cultures supplied by this department are briefly as follows:

A considerable number of the cultures hermetically sealed in glass were sterile at the time they were examined by Dr. Stevens and Mr. Temple. The misconception in regard to the viability of cultures distributed by the department at the present time could have been prevented by the insertion of a footnote explaining that since July, 1906, small bottles with wax seals have been substituted for small tubes hermetically sealed in the flame of a blast lamp. It is surprising to me that four out of seven of the old-style cultures examined by Dr. Stevens should have been sterile, as my own investigations previous to adopting this method for distribution indicated that about one half of one per cent. of the cultures sealed in this way in routine work would be injured or sterilized by the heat of sealing. The law of chance must perhaps be invoked to explain the discrepancy in our figures. It must be remembered, however, that the cultures spoken of at this time are the old-style liquid cultures, and that the cultures distributed since July, 1906, are not open to criticism of this sort.

It is surprising to me also to learn that during the multiplication period conducted in the practical manner outlined for use on the farm such great contamination should have become manifest. Two years ago I had small samples of these gross cultures prepared on the farm returned to me by farmers in various parts of the country for examination, the sample being taken and mailed to me at the time the culture was applied to the seed. This, of course, allowed for greater development of contaminations than would have taken place at the time the culture was applied to the seed. Even with this handicap about two per cent. of the cultures received from the farmers were apparently pure, and if contaminated the contamination was evidently very slight indeed. About sixty per cent. were contaminated, but not excessively so, it being easy in all of these cases to isolate large numbers of *Pseudomonas radicicola*. The remainder were in rather bad condition, although I doubt if ten per cent. of the entire number received were so seriously contaminated as to be worthless.

The description of the pot experiments conducted by Dr. Stevens and Mr. Temple is confusing. In the first place, the sterilizing of soil by heating is well known to injure the soil seriously, and, regardless of the condition of the nodule-forming bacteria introduced, it is an open question whether soil sterilized by heating would allow nodule formation until a normal bacteriologic flora and normal soil conditions generally had been reestablished. It is impossible to determine whether any

Fisher (1918)

> *Fisher, R.A. 1918. The correlation between relatives on the supposition of Mendelian inheritance. Transactions of the Royal Society of Edinburgh, vol. 52, pp. 399 to 433.*

The problem

In the early twentieth century, Mendelians and biometricians disagreed about the inheritance of continuous traits. The biometricians measured correlations between relatives for height and weight and believed that this continuous variation could not arise from discrete factors. Fisher showed that many genes of small effect, each obeying Mendel's laws, produce exactly these correlations. To do so he introduced the term variance and separated genetic variation into an additive part, a dominance part and a part due to interaction between loci.

Points to note while reading

This is the founding text of quantitative genetics. While reading, focus on the idea of dividing the variance into components and on how the expected correlation between relatives depends on these parts. The notion of additive breeding value, used by every breeding program today, originates here.

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XV.—The Correlation between Relatives on the Supposition of Mendelian Inheritance.

By R. A. Fisher, B.A. *Communicated by* Professor J. Arthur Thomson. (With Four Figures in Text.)

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Several attempts have already been made to interpret the well-established results of biometry in accordance with the Mendelian scheme of inheritance. It is here attempted to ascertain the biometrical properties of a population of a more general type than has hitherto been examined, inheritance in which follows this scheme. It is hoped that in this way it will be possible to make a more exact analysis of the causes of human variability. The great body of available statistics show us that the deviations of a human measurement from its mean follow very closely the Normal Law of Errors, and, therefore, that the variability may be uniformly measured by the standard deviation corresponding to the square root of the mean square error. When there are two independent causes of variability capable of producing in an otherwise uniform population distributions with standard deviations σ_1 and σ_2, it is found that the distribution, when both causes act together, has a standard deviation $\sqrt{\sigma_1^2+\sigma_2^2}$. It is therefore desirable in analysing the causes of variability to deal with the square of the standard deviation as the measure of variability. We shall term this quantity the Variance of the normal population to which it refers, and we may now ascribe to the constituent causes fractions or percentages of the total variance which they together produce. It is desirable on the one hand that the elementary ideas at the basis of the calculus of correlations should be clearly understood, and easily expressed in ordinary language, and on the other that loose phrases about the "percentage of causation,"

which obscure the essential distinction between the individual and the population, should be carefully avoided.

Speaking always of normal populations, when the coefficient of correlation between father and son, in stature let us say, is r, it follows that for the group of sons of fathers of any given height the variance is a fraction, $1-r^2$, of the variance of sons in general. Thus if the correlation is ·5, we have accounted by reference to the height of the father for one quarter of the variance of the sons. For the remaining three quarters we must account by some other cause. If the two parents are independent, a second quarter may be ascribed to the mother. If father and mother, as usually happens, are positively correlated, a less amount must be added to obtain the joint contribution of the two parents, since some of the mother's contribution will in this case have been already included with the father's. In a similar way each of the ancestors makes an independent contribution, but the total amount of variance to be ascribed to the measurements of ancestors, including parents, cannot greatly exceed one half of the total. We may know this by considering the difference between brothers of the same fraternity: of these the whole ancestry is identical, so that we may expect them to resemble one another rather more than persons whose ancestry, identical in respect of height, consists of different persons. For stature the coefficient of correlation between brothers is about ·54, which we may interpret* by saying that 54 per cent. of their variance is accounted for by ancestry alone, and that 46 per cent. must have some other explanation.

It is not sufficient to ascribe this last residue to the effects of environment. Numerous investigations by GALTON and PEARSON have shown that all measurable environment has much less effect on such measurements as stature. Further, the facts collected by GALTON respecting identical twins show that in this case, where the essential nature is the same, the variance is far less. The simplest hypothesis, and the one which we shall examine, is that such features as stature are determined by a large number of Mendelian factors, and that the large variance among children of the same parents is due to the segregation of those factors in respect to which the parents are heterozygous. Upon this hypothesis we will attempt to determine how much more of the variance, in different measurable features, beyond that which is indicated by the fraternal correlation, is due to innate and heritable factors.

In 1903 KARL PEARSON devoted to a first examination of this hypothesis the

* The correlation is determined from the measurements of n individuals, $x_1, x_2, \ldots x_n$, and of their brothers, $y_1, y_2 \ldots, y_r$; let us suppose that each pair of brothers is a random sample of two from an infinite fraternity, that is to say from all the sons which a pair of parents might conceivably have produced, and that the variance of each such fraternity is V, while that of the sons in general is σ. Then the mean value of $(x-y)^2$ will be 2V, since each brother contributes the variance V. But expanding the expression, we find the mean value of both x^2 and y^2 is σ^2, while that of xy is $r\sigma^2$, where r is the fraternal correlation. Hence $2V=2\sigma^2(1-r)$, or $\frac{V}{\sigma^2}=1-r$. Taking the values ·5066 and ·2804 for the parental and marital correlations, we find that the *heights of the parents alone* account for 40·10 per cent. of the variance of the children, whereas the *total effect of ancestry*, deduced from the fraternal correlation, is 54·33 per cent.

twelfth of his Mathematical Contributions to the Theory of Evolution ("On a Generalised Theory of Alternative Inheritance, with special reference to Mendel's Laws," *Phil. Trans.*, vol. cciii, A, pp. 53–87. The subject had been previously opened by UDNY YULE, *New Phytologist*, vol. i). For a population of n equally important Mendelian pairs, the dominant and recessive phases being present in equal numbers, and the different factors combining their effects by simple addition, he found that the correlation coefficients worked out uniformly too low. The parental correlations were $\frac{1}{3}$ and the fraternal $\frac{5}{12}$.*

These low values, as was pointed out by YULE at the Conference on Genetics in 1906 (Horticultural Society's Report), could be satisfactorily explained as due to the assumption of complete dominance. It is true that dominance is a very general Mendelian phenomenon, but it is purely somatic, and if better agreements can be obtained without assuming it in an extreme and rigorous sense, we are justified in testing a wider hypothesis. YULE, although dealing with by no means the most general case, obtained results which are formally almost general. He shows the similarity of the effects of dominance and of environment in reducing the correlations between relatives, but states that they are identical, an assertion to which, as I shall show, there is a remarkable exception, which enables us, as far as existing statistics allow, to separate them and to estimate how much of the total variance is due to dominance and how much to arbitrary outside causes.

In the following investigation we find it unnecessary to assume that the different Mendelian factors are of equal importance, and we allow the different phases of each to occur in any proportions consistent with the conditions of mating. The heterozygote is from the first assumed to have any value between those of the dominant and the recessive, or even outside this range, which terms therefore lose their polarity, and become merely the means of distinguishing one pure phase from the other. In order to proceed from the simple to the complex we assume at first random mating, the independence of the different factors, and that the factors are sufficiently numerous to allow us to neglect certain small quantities.

* The case of the fraternal correlations has been unfortunately complicated by the belief that the correlation on a Mendelian hypothesis would depend on the number of the fraternity. In a family, for instance, in which four Mendelian types are liable to occur in equal numbers, it was assumed that of a family of four, one would be of each type; in a family of eight, two of each type; and so on. If this were the case, then in such families, one being of the type A would make it less likely, in small families impossible, for a second to be of this type. If, as was Mendel's hypothesis, the different qualities were carried by different gametes, each brother would have an independent and equal chance of each of the four possibilities. Thus the formulæ giving the fraternal correlations in terms of the number of the fraternity give values too small. The right value on Mendel's theory is that for an infinite fraternity. As PEARSON suggested in the same paper, "probably the most correct way of looking at any fraternal correlation table would be to suppose it a random sample of all pairs of brothers which would be obtained by giving a large, or even indefinitely large, fertility to each pair, for what we actually do is to take families of varying size and take as many pairs of brothers as they provide." In spite of this, the same confusing supposition appears in a paper by SNOW "On the Determination of the Chief Correlations between Collaterals in the Case of a Simple Mendelian Population Mating at Random" (E. C. SNOW, B.A., *Proc. Roy. Soc.*, June 1910); and in one by JOHN BROWNLEE, "The Significance of the Correlation Coefficient when applied to Mendelian Distributions" (*Proc. Roy. Soc. Edin.*, Jan. 1910).

1. Let us suppose that the difference caused by a single Mendelian factor is represented in its three phases by the difference of the quantities a, d, $-a$, and that these phases exist in any population with relative frequency P, 2Q, R, where $P + 2Q + R = 1$.

Then a population in which this factor is the only cause of variability has its mean at

$$m = Pa + 2Qd - Ra,$$

so that

$$P(a - m) + 2Q(d - m) - R(a + m) = 0.$$

Let now

$$P(a - m)^2 + 2Q(d - m)^2 + R(a + m)^2 = \alpha^2 \quad . \quad . \quad . \quad . \quad . \quad (I)$$

α^2 then is the variance due to this factor, for it is easily seen that when two such factors are combined at random, the mean square deviation from the new mean is equal to the sum of the values of α^2 for the two factors separately. In general the mean square deviation due to a number of such factors associated at random will be written

$$\sigma^2 = \Sigma\alpha^2 \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (II)$$

To justify our statement that α^2 is the contribution which a single factor makes to the total variance, it is only necessary to show that when the number of such factors is large the distributions will take the normal form.

If now we write

$$\mu_3 = P(a - m)^3 + 2Q(d - m)^3 - R(a + m)^3$$
$$\mu_4 = P(a - m)^4 + 2Q(d - m)^4 + R(a + m)^4,$$

and if M_3 and M_4 are the third and fourth moments of the population, the variance of which is due solely to the random combination of such factors, it is easy to see that

$$M_3 = \Sigma\mu_3$$
$$M_4 - 3\sigma^4 = \Sigma(\mu_4 - 3\alpha^4).$$

Now the departure from normality of the population may be measured by means of the two ratios

$$\beta_1 = \frac{M_3^2}{\sigma^6} \quad \text{and} \quad \beta_2 = \frac{M_4}{\sigma^4}.$$

The first of these is

$$(\Sigma\mu_3)^2/(\Sigma\alpha^2)^3,$$

and is of the order $\frac{1}{n}$, where n is the number of factors concerned, while the second differs from its Gaussian value 3 also by a quantity of the order $\frac{1}{n}$.

2. If there are a great number of different factors, so that σ is large compared to every separate α, we may investigate the proportions in which the different phases occur in a selected array of individuals. Since the deviation of an individual is simply due to a random combination of the deviations of separate factors, we must expect a given array of deviation, let us say x, to contain the phases of each factor

in rather different proportions to those in which they exist in the whole population. The latter will be represented now by $\bar{P}, 2\bar{Q}, \bar{R}$, while $P, 2Q, R$ stand for the proportions in some particular array under consideration.

Consider a population which is the same in every respect as the one we are dealing with save that all its members have one particular factor in the heterozygous phase, and let us modify it by choosing of each array a proportion $\bar{P}$ which are to become dominants and to increase by $a-d$, and a proportion $\bar{R}$ which become recessive and diminish by $a+d$: the mean is thereby moved to the extent $m-d$.

Of those which after this modification find themselves in the array with deviation x, the dominants formerly had a deviation $x-a+m$, the heterozygates $x-d+m$, and the recessives $x+a+m$, and since the variance of the original population was σ^2-a^2, the frequencies of these three types are in the ratio

$$\bar{P}e^{-\frac{(x-a+m)^2}{2(\sigma^2-a^2)}} : 2\bar{Q}e^{-\frac{(x-d+m)^2}{2(\sigma^2-a^2)}} : \bar{R}e^{-\frac{(x+a+m)^2}{2(\sigma^2-a^2)}},$$

or, when σ is great compared to a, so that $\frac{a^2}{\sigma^2}$ may be neglected,

$$\left.\begin{aligned} P &= \bar{P}\left[1+\frac{x}{\sigma^2}(a-m)\right] \\ Q &= \bar{Q}\left[1+\frac{x}{\sigma^2}(d-m)\right] \\ R &= \bar{R}\left[1-\frac{x}{\sigma^2}(a+m)\right] \end{aligned}\right\} \quad \ldots\ldots\ldots \quad \text{(III)}$$

giving the proportions in which the phases occur in the array of deviation x.

3. Hence the members of this array mating at random will have offspring distributed in the three phases in the proportion

$$\bar{P}^2\left[1+\frac{x}{\sigma^2}(a-m)\right]+\bar{P}\bar{Q}\left[2+\frac{x}{\sigma^2}(a-m+d-m)\right]+\bar{Q}^2\left[1+\frac{x}{\sigma^2}(d-m)\right],$$

$$\bar{P}\bar{Q}\left[2+\frac{x}{\sigma^2}(a-m+d-m)\right]+2\bar{Q}^2\left[1+\frac{x}{\sigma^2}(d-m)\right]+\bar{P}\bar{R}\left[2-\frac{x}{\sigma^2}(2m)\right]+\bar{Q}\bar{R}\left[2+\frac{x}{\sigma^2}(d-m-a-m)\right],$$

$$\bar{Q}^2\left[1+\frac{x}{\sigma^2}(d-m)\right]+\bar{Q}\bar{R}\left[2+\frac{x}{\sigma^2}(d-m-a-m)\right]+\bar{R}^2\left[1-\frac{x}{\sigma^2}(a+m)\right],$$

and therefore the deviation of the mean of the offspring is

$$2d(\bar{P}\bar{R}-\bar{Q}^2)+\frac{x}{\sigma^2}\left[\bar{P}\bar{Q}(a-d)^2+2\bar{P}\bar{R}(a^2-d^2)+\bar{Q}\bar{R}(a+d)^2+(\bar{P}\bar{R}-\bar{Q}^2)d(d-m)\right].$$

Omitting the terms in $(\bar{P}\bar{R}-\bar{Q}^2)$, which for random mating is zero, the regression due to a single factor is

$$\frac{x}{\sigma^2}\left[\bar{P}\bar{Q}(a-d)^2+2\bar{P}\bar{R}(a^2-d^2)+\bar{Q}\bar{R}(a+d)^2\right] \quad \ldots\ldots\ldots \quad \text{(IV)}$$

4. To interpret this expression, consider what is involved in taking a, d, $-a$ as representing the three phases of a factor. Genetically the heterozygote is intermediate between the dominant and the recessive, somatically it differs from their

mean by d. The steps from recessive to heterozygote and from heterozygote to dominant are genetically identical, and may change from one to the other in passing from father to son. Somatically the steps are of different importance, and the soma to some extent disguises the true genetic nature. There is in dominance a certain latency. We may say that the somatic effects of identical genetic changes are not additive, and for this reason the genetic similarity of relations is partly obscured in the statistical aggregate. A similar deviation from the addition of superimposed effects may occur between different Mendelian factors. We may use the term Epistacy to describe such deviation, which although potentially more complicated, has similar statistical effects to dominance. If the two sexes are considered as Mendelian alternatives, the fact that other Mendelian factors affect them to different extents may be regarded as an example of epistacy.

The contributions of imperfectly additive genetic factors divide themselves for statistical purposes into two parts: an additive part which reflects the genetic nature without distortion, and gives rise to the correlations which one obtains; and a residue which acts in much the same way as an arbitrary error introduced into the measurements. Thus, if for a, d, $-a$ we substitute the linear series

$$c+b,\ c,\ c-b,$$

and choose b and c in such a way that

$$\mathrm{P}(c+b-a)^2+2\mathrm{Q}(c-d)^2+\mathrm{R}(c-b+a)^2$$

is a minimum, we find for this minimum value δ^2,

$$\delta^2=\frac{4\mathrm{PQR}d^2}{\mathrm{PQ}+2\mathrm{PR}+\mathrm{QR}},$$

which is the contribution to the variance of the irregular behaviour of the soma; and for the contribution of the additive part, β^2, where

$$\beta^2=\mathrm{P}(c+b-m)^2+2\mathrm{Q}(c-m)^2+\mathrm{R}(c-b-m)^2,$$

we obtain

$$\beta^2=2b^2(\mathrm{PQ}+2\mathrm{PR}+\mathrm{QR}),$$

and since

$$b=a+\frac{\mathrm{Q}(\mathrm{P}-\mathrm{R})d}{\mathrm{PQ}+2\mathrm{PR}+\mathrm{QR}},$$

we have

$$\beta^2=2a^2(\mathrm{PQ}+2\mathrm{PR}+\mathrm{QR})-4\mathrm{Q}(\mathrm{P}-\mathrm{R})ad+\frac{2\mathrm{Q}^2(\mathrm{P}-\mathrm{R})^2d^2}{\mathrm{PQ}+2\mathrm{PR}+\mathrm{QR}}.$$

5. These expressions may be much simplified by using the equation

$$\mathrm{Q}^2=\mathrm{PR},$$

for then

$$\delta^2=4\mathrm{Q}^2d^2 \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (\mathrm{V})$$

$$\beta^2=2a^2\mathrm{Q}^2-4\mathrm{Q}(\mathrm{P}-\mathrm{R})ad+2\mathrm{Q}(\mathrm{P}-\mathrm{R})^2d^2 \quad . \quad . \quad . \quad . \quad . \quad (\mathrm{VI})$$

which appears in the regression in Article 3 (IV),

and

$$a^2=2a^2\mathrm{Q}-4\mathrm{Q}(\mathrm{P}-\mathrm{R})ad+2\mathrm{Q}(\mathrm{P}+\mathrm{R})d^2 \quad . \quad . \quad . \quad . \quad (\mathrm{VII})$$

In general

$$a^2 = \beta^2 + \delta^2,$$

and if

$$\sigma^2 = \Sigma a^2 \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (\text{VIII})$$

$$\tau^2 = \Sigma \beta^2 \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (\text{IX})$$

and

$$\epsilon^2 = \Sigma \delta^2 \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (\text{X})$$

then

$$\sigma^2 = \tau^2 + \epsilon^2.$$

The regression due to a single factor of the mean of the offspring of parents of a given array is

$$\frac{x^2}{\sigma^2} \cdot \frac{\beta^2}{2},$$

and adding up the effects of all factors we find

$$\frac{x}{\sigma^2} \quad \frac{\tau^2}{2},$$

so that the parental correlation for a static population mating at random is simply

$$\frac{1}{2} \quad \frac{\tau^2}{\sigma^2} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (\text{XI})$$

We may regard this formula otherwise. The correlation between the actual somatic measurements such as a, d, $-a$, and the representative linear quantities $c+b$, c, $c-b$ is $\frac{\tau}{\sigma}$. Thus the correlation of parent and child is made up of three factors, two of them representing the relations between the real and the representative measurements, and the third the correlation between the representative measurements of the two relatives. Thus the effect of dominance is simply to reduce certain relationship correlations in the ratio $\frac{\tau^2}{\sigma^2}$.

The values of the correlations between the representative measurements for random mating, which may be called the genetic correlations, are given in the accompanying table :—

Generations.	Half 2nd Cousin.	Half 1st Cousin.	Half Brother.	Ancestral Line.	Brother.	1st Cousin.	2nd Cousin.
Own	$1/64$	$1/16$	$1/4$	1	$1/2$	$1/8$	$1/32$
Father's	$1/128$	$1/32$	$1/8$	$1/2$	$1/4$	$1/16$	$1/64$
Grandfather's . . .	$1/256$	$1/64$	$1/16$	$1/4$	$1/8$	$1/32$	$1/128$
Great-grandfather's . .	$1/512$	$1/128$	$1/32$	$1/8$	$1/16$	$1/64$	$1/256$
Great-great-grandfather's .	$1/1024$	$1/256$	$1/64$	$1/16$	$1/32$	$1/128$	$1/512$

6. The above reasoning as to the effects of dominance applies without modification to the ancestral line, but in a special class of collaterals requires reconsideration. The reason is that the deviations from linearity are now themselves correlated. In other words, a father who is heterozygote instead of recessive may have offspring

who show a similar variation; but they may also be changed from heterozygote to dominant. In the case of siblings, however, whichever change takes place in one is more likely to occur in the other.

Thus, writing i, j, k for the deviations

$$a-m,\ d-m,\ -(a+m),$$

so that

$$iP+2jQ+kR=0 \quad . \quad . \quad .

types of mating down to the second generation. The three Mendelian phases will yield six types of mating, and ordinary cousinships are therefore connected by one of six types of sibship. The especially interesting case of double cousins, in which two members of one sibship mate with two members of another, can occur in twenty-one distinct ways, since any pair of the six types of sibship may be taken. The proportionate numbers of the three Mendelian phases in the children produced by the random matings of such pairs of sibships is given in the accompanying table :—

Type of sibship.	1.0.0	1. 1.0	0.1.0	1.2.1	0. 1.1	0.0.1
Frequency . .	p^4	$4p^3q$	$2p^2q^2$	$4p^2q^2$	$4pq^3$	$q4$
p^4	1.0.0	3. 1.0	1.1.0	1.1.0	1. 3.0	0.1.0
$4p^3q$	3.1.0	9. 6.1	3.4.1	3.4.1	3.10.3	0.3.1
$2p^2q^2$	1.1.0	3. 4.1	1.2.1	1.2.1	1. 4.3	0.1.1
$4p^2q^2$	1.1.0	3. 4.1	1.2.1	1.2.1	1. 4.3	0.1.1
$4pq^3$	1.3.0	3.10.1	1.4.3	1.4.3	1. 6.9	0.1.3
q^4	0.1.0	0. 3.1	0.1.1	0.1.1	0. 1.3	0.0.1
	$p.q.0$	$\frac{3p}{4}.\frac{p+3q}{4}.\frac{q}{4}$	$\frac{p}{2}.\frac{1}{2}.\frac{q}{2}$	$\frac{p}{2}.\frac{1}{2}.\frac{q}{2}$	$\frac{p}{4}.\frac{3p+q}{4}.\frac{3q}{4}$	$0.p.q$

The lowest line gives the proportions of the phases in the whole cousinship whose connecting sibship is of each of the six types.

If we pick out all possible pairs of uncle (or aunt) and nephew (or niece) we obtain the table

$p^3(p+\frac{1}{2}q)$	$\frac{1}{2}p^2q(3p+q)$	$\frac{1}{2}p^2q^2$
$\frac{1}{2}p^2q(3p+q)$	$\frac{1}{2}pq(p^2+6pq+q^2)$	$\frac{1}{2}pq^2(p+3q)$
$\frac{1}{2}p^2q^2$	$\frac{1}{2}pq^2(p+3q)$	$q^2(\frac{1}{2}p+q)$

the quadratic from which reduces exactly to $\frac{1}{4}\beta^2$, showing that when mating is at random the avuncular correlation is exactly one half of the paternal.

From the twenty-one types of double cousinship pairs may be picked, the proportions of which are shown in the table :—

$p^2(p+\frac{1}{4}q)^2$	$\frac{3}{2}p^2q(p+\frac{1}{4}q)$	$\frac{9}{16}p^2q^2$
$\frac{3}{2}p^2q(p+\frac{1}{4}q)$	$\frac{1}{2}pq(p^2+\frac{1}{2}\frac{3}{2}pq+q^2)$	$\frac{3}{2}pq^2(\frac{1}{4}p+q)$
$\frac{9}{16}p^2q^2$	$\frac{3}{2}pq^2(\frac{1}{4}p+q)$	$q^2(\frac{1}{4}p+q)^2$

which agrees with the table given by SNOW for ordinary first cousins. I cannot explain this divergence, unless it be that SNOW is in error, my values for ordinary first cousins leading to less than half this value for the correlation. Simplifying the quadratic in i, j, k, which is most easily done in this case by comparison with the avuncular table, we find for the correlation of double cousins

$$\frac{1}{4\sigma^2}(\tau^2+\tfrac{1}{4}\epsilon^2),$$

showing that double cousins, like brothers, show some similarity in the distribution

of deviations due to dominance, and that with these cousins the correlation will in general be rather higher than it is for uncle and nephew.

For ordinary first cousins I find the following table of the distribution of random pairs drawn from the six types of ordinary cousinship :—

$\frac{1}{4}p^3(4p+q)$	$\frac{1}{4}p^2q^2(7p+q)$	$\frac{3}{4}p^2q^2$
$\frac{1}{4}p^2q(7p+q)$	$\frac{1}{4}pq(p^2+14pq+q^2)$	$\frac{1}{4}pq^2(p+7q)$
$\frac{3}{4}p^2q^2$	$\frac{1}{4}pq^2(p+7q)$	$\frac{1}{4}q^3(p+4q)$

which yields the correlation $\frac{1}{8}\frac{\tau^2}{\sigma^2}$.

In a similar way the more distant kin may be investigated, but since for them reliable data have not yet been published, the table already given of genetic correlations will be a sufficient guide.

8. Before extending the above results to the more difficult conditions of assortative mating, it is desirable to show how our methods may be developed so as to include the statistical feature to which we have applied the term Epistacy. The combination of two Mendelian factors gives rise to nine distinct phases, and there is no biological reason for supposing that nine such distinct measurements should be exactly represented by the nine deviations formed by adding i, j, or k to i', j', or k'. If we suppose that i, j, k, i', j', k' have been so chosen as to represent the nine actual types with the least square error, we have now to deal with additional quantities, which we may term

$$\begin{matrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{matrix}$$

connected by the six equations, five of which are independent.

$$\begin{aligned} p^2e_{11}+2pqe_{21}+q^2e_{31}&=0 & p'^2e_{11}+2p'q'e_{12}+q'^2e_{13}&=0 \\ p^2e_{12}+2pqe_{22}+q^2e_{32}&=0 & p'^2e_{21}+2p'q'e_{22}+q'^2e_{23}&=0 \\ p^2e_{13}+2pqe_{23}+q^2e_{33}&=0 & p'^2e_{31}+2p'q'e_{32}+q'^2e_{33}&=0. \end{aligned}$$

This is a complete representation of any such deviations from linearity as may exist between two factors. Such dual epistacy, as we may term it, is the only kind of which we shall treat. More complex connections could doubtless exist, but the number of unknowns introduced by dual epistacy alone, four, is more than can be determined by existing data. In addition it is very improbable that any statistical effect, of a nature other than that which we are considering, is actually produced by more complex somatic connections.

The full association table between two relatives, when we are considering two distinct Mendelian factors, consists of eighty-one cells, and the quadratic expression to which it leads now involves the nine epistatic deviations. A remarkable simplification is, however, possible, since each quantity, such as e_{21}, which refers to a partially or

wholly heterozygous individual, is related to two other quantities, such as e_{11} and e_{31}, by just the same equation as that by which j is related to i and k, and occurs in the 9×9 table with corresponding coefficients. The elimination of the five deviations e_{21}, e_{12}, e_{32}, e_{23}, e_{22} is therefore effected by rewriting the 9×9 table as a 4×4 table, derived from the quadratic in i and k corresponding to the relationship considered.

Thus the variance, found by squaring the individual variations, is derived from the 3×3 table

p^2	—	—
—	$2pq$	—
—	—	q^2

which yields the 2×2 table

$\frac{p^2}{2q}(p+2q)$	$\frac{1}{2}pq$
$\frac{1}{2}pq$	$\frac{q^2}{2p}(2p+q)$

and the quadratic in e_{11}, e_{13}, e_{31}, e_{33}

$$\frac{1}{4pqp'q'}\Big[(p+2q)(p'+2q')p^2p'^2e_{11}{}^2 + 3 \text{ similar terms} + 2p^2q^2p'^2(p'+2q')e_{11}e_{31} + 3 \text{ similar terms}$$
$$+ 2p^2q^2p'^2q'^2(e_{11}e_{33} + e_{13}e_{31})\Big],$$

which also takes the form

$$\frac{1}{4pqp'q'}\Big[(p^2p'^2e_{11} - p^2q'^2e_{13} - q^2p'^2e_{31} + q^2q'^2e_{33})^2 + 2pqp'^2(pe_{11} + qe_{31})^2 + 3 \text{ similar terms}\Big].$$

The parental table

$p^3/4q$	$-\frac{1}{4}pq$
$-\frac{1}{4}pq$	$q^3/4p$

yields

$$\frac{1}{16pqp'q'}\Big[p^2p'^2e_{11} - p^2q'^2e_{13} - q^2p'^2e_{31} + q^2q'^2e_{33}\Big]^2,$$

and the fraternal table

$p^2/4q$	—
—	$q^2/4p$

leads us to the simple expression

$$\frac{1}{16pqp'q'}\Big[p^3p'^3e_{11}{}^2 + p^3q'^3e_{13}{}^2 + q^3p'^3e_{31}{}^2 + q^3q'^3e_{33}{}^2\Big].$$

For uncles and cousins we obtain respectively $\frac{1}{4}$ and $\frac{1}{16}$ of the parental contribution, while for double cousins the table

$\frac{p^2}{16q}(2p+q)$	$-\frac{1}{16}pq$
$-\frac{1}{16}pq$	$\frac{q^2}{16p}(p+2q)$

and a quadratic similar to that for the variance.

9. With assortative mating all these coefficients will be modified. There will be association between similar phases of different factors, so that they cannot be treated separately. There will also be an increase in the variance.

We must determine the nature of the association between different factors, and ascertain how it is related to the degree of assortative mating necessary to maintain it. Then we shall be able to investigate the statistical effects of this association on the variance of the population and on the correlations.

If μ be the marital correlation, then in a population with variance V the frequency of individuals in the range dx is

$$\frac{1}{\sqrt{2\pi \mathrm{V}}} e^{-\frac{x^2}{2\mathrm{V}}} dx = \mathrm{M},$$

and the frequency in the range dy is

$$\frac{1}{\sqrt{2\pi \mathrm{V}}} e^{-\frac{y^2}{2\mathrm{V}}} dy = \mathrm{N};$$

but the frequency of matings between these two groups is not simply MN, as would be the case if there were no marital correlation, but

$$\frac{1}{2\pi \mathrm{V} \sqrt{1-\mu^2}} e^{-\frac{1}{1-\mu^2} \frac{x^2 - 2\mu xy + y^2}{2\mathrm{V}}} dx\, dy,$$

which is equal to

$$\frac{\mathrm{MN}}{\sqrt{1-\mu^2}} e^{-\frac{\mu^2 x^2 - 2\mu xy + \mu^2 y^2}{2\mathrm{V}(1-\mu^2)}}.$$

In studying the effect of assortative mating we shall require to know the frequency of matings between two groups, each with a variance nearly equal to that of the whole population, but centred about means a and b. The frequencies of such groups in any ranges dx, dy can be written down, and if the chance of any mating depends only on x and y, the frequency of mating between these two groups can be expressed as a double integral. If M and N are the frequencies in the two groups, the frequency of mating between them is found to be

$$\mathrm{MN} e^{\frac{\mu a b}{\mathrm{V}}}.$$

10. We shall apply this expression first to determine the equilibrium value of the frequencies of the three phases of a single factor. Of the six types of mating which are possible, all save two yield offspring of the same genetic phase as their parents. With the inbreeding of the pure forms D × D and R × R obviously no change is made, and the same is true of the crosses D × H and R × H, for each of these yields the pure form and the heterozygote in equal numbers. On the other hand, in the cross D × R we have a dominant and a recessive replaced in the next generation by two heterozygotes, while in the cross H × H half of the offspring return to the homozygous condition. For equilibrium the second type of mating

must be twice as frequent as the first, and if I, J, and K are the means of the distributions of the three phases,

$$4Q^2e^{\frac{\mu J^2}{V}} = 4PRe^{\frac{\mu IK}{V}}.$$

Since $\frac{J^2}{V}$ and $\frac{IK}{V}$ are small quantities, we shall neglect their squares, and obtain the equation

$$PR - Q^2 = Q^2\mu\frac{J^2 - IK}{V} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (XIII)$$

If, as before, the two types of gamete are in the ratio $p : q$, the frequencies of the three phases are expressed by the equations

$$\left.\begin{aligned} P &= p^2 + p^2q^2\mu\frac{J^2 - IK}{V} \\ Q &= pq - p^2q^2\mu\frac{J^2 - IK}{V} \\ R &= q^2 + p^2q^2\mu\frac{J^2 - IK}{V} \end{aligned}\right\} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (XIV)$$

It is evident that

$$PI + 2QJ + RK = 0 \quad . \quad . \quad . \quad . \quad . \quad . \quad (XV)$$

and this enables us, whenever necessary, to eliminate J, and to treat only I and K as unknowns. These can only be found when the system of association between different factors has been ascertained. It will be observed that the changes produced in P, Q, and R are small quantities of the second order : in transforming the quantity

$$p^2q^2\mu\frac{J^2 - IK}{V}$$

we may write $-(p^2I + q^2K)$ for $2pqJ$, leading to the form

$$\frac{\mu}{4V}(p^2I - q^2K)^2,$$

which will be found more useful than the other.

11. The nine possible combinations of two factors will not now occur in the simple proportions PP′, 2PQ′, etc., as is the case when there is no association ; but whatever the nature of the association may be, we shall represent it by introducing new quantities, which by analogy we may expect to be small of the second order, defined so that the frequency of the type

$$DD' \text{ is } PP'(1 + f_{11}),$$

that of

$$DH' \text{ is } 2PQ'(1 + f_{12}),$$

and that of

$$DR' \text{ is } PR'(1 + f_{13}),$$

and so on.

Formally, we have introduced nine such new unknowns for each pair of factors, but since, for instance, the sum of the above three quantities must be P, we have the six equations

$$\left.\begin{aligned} P'f_{11} + 2Q'f_{12} + R'f_{13} &= 0 \qquad Pf_{11} + 2Qf_{21} + Rf_{31} = 0 \\ P'f_{21} + 2Q'f_{22} + R'f_{23} &= 0 \qquad Pf_{12} + 2Qf_{22} + Rf_{32} = 0 \\ P'f_{31} + 2Q'f_{32} + R'f_{33} &= 0 \qquad Pf_{13} + 2Qf_{23} + Rf_{33} = 0 \end{aligned}\right\} \quad . \quad . \quad . \quad (XVI)$$

five of which are independent. The unknowns are thus reduced to four, and we shall use $f_{11}, f_{13}, f_{31}, f_{33}$, since any involving a 2 in the suffix can easily be eliminated.

We have further

$$\left.\begin{aligned} \mathrm{I} &= i + \Sigma(\mathrm{P}'i'f_{11} + 2\mathrm{Q}'j'f_{12} + \mathrm{R}'k'f_{13}) \\ \mathrm{J} &= j + \Sigma(\mathrm{P}'i'f_{21} + 2\mathrm{Q}'j'f_{22} + \mathrm{R}'k'f_{23}) \\ \mathrm{K} &= k + \Sigma(\mathrm{P}'i'f_{31} + 2\mathrm{Q}'j'f_{32} + \mathrm{R}'k'f_{33}) \end{aligned}\right\} \quad . \quad . \quad . \quad . \quad . \quad . \quad \text{(XVII)}$$

in which the summation is extended over all the factors except that one to which i, j, k refer. Since we are assuming the factors to be very numerous, after substituting their values for the f's we may without error extend the summation over all the factors. The variance defined as the mean square deviation may be evaluated in terms of the f's

$$\mathrm{V} = \Sigma(\mathrm{P}i^2 + 2\mathrm{Q}j^2 + \mathrm{R}k^2) + 2\Sigma\{\mathrm{PP}'(1 + f_{11})ii' + 8 \text{ other terms}\},$$

which reduces to

$$\Sigma(\mathrm{P}i^2 + 2\mathrm{Q}j^2 + \mathrm{R}k^2) + 2\Sigma\{\mathrm{PP}'ii'f_{11} + 8 \text{ other terms}\},$$

so that

$$\mathrm{V} = \Sigma(\mathrm{P}i\mathrm{I} + 2\mathrm{Q}j\mathrm{J} + \mathrm{R}k\mathrm{K}) \quad . \quad . \quad . \quad . \quad . \quad . \quad \text{(XVIII)}$$

12. We can only advance beyond these purely formal relations to an actual evaluation of our unknowns by considering the equilibrium of the different phase combinations. There are forty-five possible matings of the nine types, but since we need only consider the equilibrium of the four homozygous conditions, we need only pick out the terms, ten in each case, which give rise to them. The method will be exactly the same as we used for a single factor. Thus the matings $\mathrm{DD}' \times \mathrm{DD}'$ have the frequency

$$\mathrm{PP}' \,.\, \mathrm{PP}' \,.\, (1 + f_{11})(1 + f_{11}) e^{\frac{\mu(\mathrm{I}+\mathrm{I}')^2}{\mathrm{V}}},$$

which for our purpose is equal to

$$\mathrm{P}^2\mathrm{P}'^2\left[1 + 2f_{11} + \frac{\mu}{\mathrm{V}}(\mathrm{I} + \mathrm{I}')^2\right].$$

Collecting now all the matings which yield DD', we have for equilibrium

$$\left.\begin{aligned} &\mathrm{P}^2\mathrm{P}'^2\left[1 + 2f_{11} + \tfrac{\mu}{\mathrm{V}}(\mathrm{I} + \mathrm{I}')^2\right] + 2\mathrm{P}^2\mathrm{P}'\mathrm{Q}'\left[1 + f_{11} + f_{12} + \tfrac{\mu}{\mathrm{V}}(\mathrm{I} + \mathrm{I}')(\mathrm{I} + \mathrm{J}')\right] \\ &+ 2\mathrm{PQP}'^2\left[1 + f_{11} + f_{21} + \tfrac{\mu}{\mathrm{V}}(\mathrm{I} + \mathrm{I}')(\mathrm{J} + \mathrm{I}')\right] + 2\mathrm{PQP}'\mathrm{Q}'\left[1 + f_{11} + f_{22} + \tfrac{\mu}{\mathrm{V}}(\mathrm{I} + \mathrm{I}')(\mathrm{J} + \mathrm{J}')\right] \\ &+ 2\mathrm{PQP}'\mathrm{Q}'\left[1 + f_{12} + f_{21} + \tfrac{\mu}{\mathrm{V}}(\mathrm{I} + \mathrm{J}')(\mathrm{J} + \mathrm{I}')\right] + \mathrm{P}^2\mathrm{Q}'^2\left[1 + 2f_{12} + \tfrac{\mu}{\mathrm{V}}(\mathrm{I} + \mathrm{J}')^2\right] \\ &+ \mathrm{Q}^2\mathrm{P}'^2\left[1 + 2f_{21} + \tfrac{\mu}{\mathrm{V}}(\mathrm{J} + \mathrm{I}')^2\right] + 2\mathrm{PQQ}'^2\left[1 + f_{12} + f_{22} + \tfrac{\mu}{\mathrm{V}}(\mathrm{I} + \mathrm{J}')(\mathrm{J} + \mathrm{J}')\right] \\ &+ 2\mathrm{Q}^2\mathrm{P}'\mathrm{Q}'\left[1 + f_{21} + f_{22} + \tfrac{\mu}{\mathrm{V}}(\mathrm{J} + \mathrm{I}')(\mathrm{J} + \mathrm{J}')\right] + \mathrm{Q}^2\mathrm{Q}'^2\left[1 + 2f_{22} + \tfrac{\mu}{\mathrm{V}}(\mathrm{J} + \mathrm{J}')^2\right] \\ &= \mathrm{PP}'(1 + f_{11}) \end{aligned}\right\} \quad \text{(XIX)}$$

Now since

$$(\mathrm{P} + \mathrm{Q})^2(\mathrm{P}' + \mathrm{Q}')^2 - \mathrm{PP}'(\mathrm{P} + 2\mathrm{Q} + \mathrm{R})(\mathrm{P}' + 2\mathrm{Q}' + \mathrm{R}') = (\mathrm{Q}^2 - \mathrm{PR})\mathrm{P}' + (\mathrm{Q}'^2 - \mathrm{P}'\mathrm{R}')\mathrm{P} + (\mathrm{Q}^2 - \mathrm{PR})(\mathrm{Q}'^2 - \mathrm{P}'\mathrm{R}')$$

the terms involving only P and Q, reduce (XIII) to the second order of small quantities,

$$-\frac{\mu}{V}[P'Q^2(J^2-IK)+P'Q'^2(J'^2-I'K')]=-\frac{\mu}{4V}[p'^2(IP-KR)^2+p^2(I'P'-K'R')^2].$$

Also collecting the terms in I and J, we find

$$\frac{\mu}{V}[(P'+Q')(IP+JQ)+(P+Q)(I'P'+J'Q')]^2,$$

which yields on eliminating J,

$$\frac{\mu}{4V}[p'(IP-KR)+p(I'P'-K'R')]^2,$$

while the result of collecting and transforming the terms in f is

$$\tfrac{1}{2}pp'[PP'f_{11}-PR'f_{13}-P'Rf_{31}+RR'f_{33}].$$

Hence, if the frequency of the type DD′ is unchanged

$$\frac{\mu}{2V}pp'(IP-KR)(I'P'-K'R')+\tfrac{1}{2}pp'[PP'f_{11}-PR'f_{13}-P'Rf_{31}+RR'f_{33}]=PP'f_{11} \quad . \quad . \quad . \quad \text{(XIX, } a\text{)}$$

Now the corresponding equations for the types DR′, RD′, R′D′ may be obtained simply by substituting K for I, R for P, and *vice versa*, as required; and each such change merely reverses the sign of the left-hand side, substituting q or q' for p or p' as a factor.

Combining the four equations

$$\frac{\mu}{2V}(IP-KR)(I'P'-K'R')=\tfrac{1}{2}[PP'f_{11}-PR'f_{13}-RP'f_{31}+RR'f_{33}] \quad . \quad . \quad . \quad \text{(XX)}$$

so that the set of four equations

$$\frac{\mu}{V}(IP-KR)(I'P'-K'R')=pp'f_{11}=-pq'f_{12}=-qp'f_{31}=qq'f_{33} \quad . \quad . \quad . \quad \text{(XXI)}$$

gives the whole of the conditions of equilibrium.

13. Substituting now in (XVII), which we may rewrite,

$$I=i+\Sigma[P'(i'-j')f_{11}-R'(j'-k')f_{13}]$$
$$K=k+\Sigma[P'(i'-j')f_{31}-R'(j'-k')f_{33}]$$

we have

$$IP-KR=iP-kR+\Sigma\frac{\mu}{V}(IP-KR)(I'P'-K'R')[p'(i'-j')+q'(j'-k')]=iP-kR+A(IP-KR),$$

where

$$A(1-A)=\frac{\mu}{V}\Sigma(i'P'-k'R')[p'(i'-j')+q'(j'-k')]$$

$$=\frac{\mu}{V}\Sigma\beta^2, \text{ since } \beta^2=\frac{(iP-kR)^2}{2Q}$$

or

$$A(1-A)=\mu\frac{r^2}{V} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad \text{(XXII)}$$

It would seem that there is an ambiguity in the value of A, so that the same amount of assortative mating would suffice to maintain two different degrees of association: we have, however, not yet ascertained the value of V. Since this also depends upon A, the form of the quadratic is changed, and it will be seen that the ambiguity disappears.

Supposing A determinate, we may determine the association coefficients f for

$$\left.\begin{aligned} p^2p'^2f_{11} &= \frac{\mu}{(1-A)^2}\frac{(iP-kR)(i'P'-k'R')}{V}pp' \\ p^2q'^2f_{12} &= -\frac{\mu}{(1-A)^2}\frac{(iP-kR)(i'P'-k'R')}{V}pq' \end{aligned}\right\} \quad . \quad . \quad . \quad . \quad (XXIII)$$

Hence

$$\begin{aligned} I &= i + \frac{\mu}{(1-A)^2}\frac{iP-kR}{pV}\Sigma[p'(i'-j')+q'(j'-k')](i'P'-k'R') \\ &= i + \frac{\mu}{(1-A)^2}\frac{iP-kR}{p}\cdot\frac{\tau^2}{V}, \end{aligned}$$

and so

$$\left.\begin{aligned} I &= i + \frac{A}{1-A}\frac{iP-kR}{p} \\ K &= k - \frac{A}{1-A}\frac{iP-kR}{q} \\ J &= j - \frac{A}{1-A}\frac{p-q}{2pq}(iP-kR) \end{aligned}\right\} \quad . \quad . \quad . \quad . \quad . \quad (XXIV)$$

Similarly

and

So that the sense in which the mean value of the heterozygote is changed by assortative mating depends only on whether p or q is greater. In spite of perfect dominance, the mean value of the heterozygote will be different from that of the dominant phase.

The value of the variance deduced from the expression

$$V = \Sigma(PiI + 2QjJ + RkK)$$

reduces to a similar form. For evidently

$$V = \Sigma a^2 + \frac{A}{1-A}\cdot\Sigma(iP-kR)[p(i-j)+q(j-k)].$$

Hence

$$V = \sigma^2 + \frac{A}{1-A}\tau^2 \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (XXV)$$

Therefore the equation for A finally takes the form

$$\mu\tau^2 = VA(1-A) = A(1-A)\sigma^2 + A^2\tau^2,$$

and may be otherwise written

$$A^2\epsilon^2 - A\sigma^2 + \mu\tau^2 = 0 \quad . \quad . \quad . \quad . \quad . \quad . \quad (XXVI)$$

Now, since the left-hand side is negative when A = 1, there can be only one root less than unity. Since, moreover,

$$(\mu - A^2)\tau^2 = (A - A^2)\sigma^2 \quad . \quad . \quad . \quad . \quad . \quad . \quad (XXVI, a)$$

it is evident that this root is less than μ, and approaches that value in the limiting case when there is no dominance.

A third form of this equation is of importance, for

$$\left.\begin{aligned} \frac{A}{\mu} &= \frac{\tau^2}{\sigma^2 - A\epsilon^2} = \frac{\tau^2 + \frac{A}{1-A}\tau^2}{\sigma^2 + \frac{A}{1-A}\tau^2} \end{aligned}\right\} \quad . \quad . \quad . \quad . \quad . \quad (XXVI, b)$$

which is the ratio of the variance without and with the deviations due to dominance.

14. *Multiple Allelomorphism.*—The possibility that each factor contains more than two allelomorphs makes it necessary to extend our analysis to cover the inheritance of features influenced by such polymorphic factors. In doing this we abandon the strictly Mendelian mode of inheritance, and treat of GALTON's "particulate inheritance" in almost its full generality. Since, however, well-authenticated cases of multiple allelomorphism have been brought to light by the Mendelian method of research, this generalised conception of inheritance may well be treated as an extension of the classical Mendelism, which we have so far investigated.

If a factor have a large number, n, of allelomorphs, there will be n homozygous phases, each of which is associated with a certain deviation of the measurement under consideration from its mean value. These deviations will be written i_1, i_2, . . . i_n, and the deviations of the heterozygous phases, of which there are $\frac{1}{2}n(n-1)$, will be written j_{12}, j_{13}, j_{23}, and so on. Let the n kinds of gametes exist with frequencies proportional to p, q, r, s, and so on, then when the mating is random the homozygous phases must occur with frequencies proportional to p^2, q^2, r^2, . . . , and the heterozygous phases to $2pq$, $2pr$, $2qr$, . . .

Hence, our measurements being from the mean,

$$p^2i_1 + q^2i_2 + r^2i_3 + \ldots + 2pqj_{12} + 2prj_{13} + \ldots = 0 \quad . \quad . \quad . \quad . \quad \text{(XII*)}$$

As before, we define a^2 by the equation

$$p^2i_1^2 + q^2i_2^2 + r^2i_3^2 + \ldots + 2pqj_{12}^2 + 2prj_{13}^2 + \ldots = a^2 \quad . \quad . \quad . \quad . \quad \text{(I*)}$$

and choosing l, m, n, . . . , so that

$$p^2(2l - i_1)^2 + q^2(2m - i_2)^2 + \ldots\ 2pq(l + m - j_{12})^2 + 2pr(l + n - j_{13})^2 + \ldots$$

is a minimum, we define β^2 by

$$4l^2p^2 + 4m^2q^2 + \ldots\ 2pq(l + m)^2 + 2pr(l + n)^2 \ldots = \beta^2,$$

the condition being fulfilled if

$$l = pi_1 + qj_{12} + rj_{13} + \ldots,$$
$$m = pj_{12} + qi_2 + rj_{23} + \ldots,$$

and so on.

Now

$$\beta^2 = \mathrm{S}(4l^2p^2) + \mathrm{S}(2pq\overline{l + m}^2),$$
$$= \mathrm{S}(2p(1 + p)l^2) + \mathrm{S}(4pqlm),$$

and since

$$pl + qm + rn + \ldots = 0,$$
$$\beta^2 = \mathrm{S}(2pl^2),$$

which may now be written as a quadratic in i and j, represented by the typical terms

$$2p^2i_1^2 + 4p^2qi_1j_{12} + 2pq(p + q)j_{12}^2 + 4pqrj_{12}j_{13}.$$

Now we can construct an association table for parent and child as in Article 6, though it is now more complicated, since the j's cannot be eliminated by equation (XII*), and its true representation lies in four dimensions; the quadratic in i and j derived

from it is, however, exactly one half of that obtained above, so that the contribution of a single factor to the parental product moment is $\frac{1}{2}\beta^2$. Hence the parental correlation is

$$\frac{1}{2}\frac{\tau^2}{\sigma^2},$$

where τ and σ retain their previous meanings.

Moreover, from the fraternal table we may obtain a quadratic expression having for its typical terms

$$\tfrac{1}{4}p^2(1+p)^2 i_1^2 + \tfrac{1}{2}p^3q^2 i_1 i_2 + p^2q(1+p) i_1 j_{12} + p^2qr i_1 j_{13}$$
$$\tfrac{1}{2}pq(1+p+q+2pq) j_{12}^2 + pqr(1+2p) j_{12} j_{13} + 2pqrs j_{12} j_{34},$$

which, when simplified by removing one quarter of the square of the expression in (XII*) becomes

$$\tfrac{1}{4}p^2(1+2p) i_1^2 + p^2q i_1 j_{12} + \tfrac{1}{2}pq(1+p+q) j_{12}^2 + pqr j_{12} j_{13},$$

or, simply,

$$\tfrac{1}{4}(\alpha^2+\beta^2).$$

Here, again, the introduction of multiple allelomorphism does not affect the simplicity of our results; the correlation between the dominance deviations of siblings is still exactly $\frac{1}{4}$, and the fraternal correlation is diminished by dominance to exactly one half the extent suffered by the parental correlation. The dominance ratio plays the same part as it did before, although its interpretation is now more complex. The fraternal correlation may be written, as in Article 6,

$$\frac{1}{2\sigma^2}(\tau^2+\tfrac{1}{2}\epsilon^2).$$

15. *Homogamy and Multiple Allelomorphism.*—The proportions of these different phases which are in equilibrium when mating is assortative must now be determined. As in Article 10, let $I_1, I_2, \ldots$ be the mean deviations of the homozygous phases, and $J_{12}, J_{13}, \ldots$ those of the heterozygous phases. Let the frequency of the first homozygous phase be written as $p^2(1+f_{11})$, and the others in the same way. Then, since p is the frequency of the first kind of gamete,

$$pf_{11} + qf_{12} + rf_{13} + \ldots = 0,$$

and

$$pf_{12} + qf_{22} + rf_{23} + \ldots = 0,$$

and so on.

Let

$$pI_1 + qJ_{12} + rJ_{13} + \ldots = L,$$
$$pJ_{12} + qI_2 + rJ_{23} + \ldots = M,$$

and so on, then L, M, . . . represent the mean deviations of individuals giving rise to gametes of the different kinds; hence, by Article 9,

$$2pq(1+f_{12}) = 2pqe^{\frac{\mu}{V}\cdot LM},$$

that is,

$$f_{12} = \mu/V \,.\, LM \quad . \quad . \quad .

those of the other factor from 1 to n, then, as the phase (12) of the first factor occurs with frequency $2pq(1+f_{12})$, and of the second factor, with frequency $2p'q'(1+f'_{12})$, we shall write the frequency with which these two phases coincide in one individual as $4pqp'q'(1+f'_{12\,.\,12})$, or as $4pqp'q'(1+f_{12})\ (1+f'_{12})\ (1+f_{12\,.\,12})$, so that

$$f'_{12\,.\,12}=f_{12\,.\,12}+f_{12}+f'_{12}.$$

The proportional increase of frequency of the gametic combination (1 . 1) is

$$\begin{aligned}&pp'f'_{11\,.\,11}+pq'f'_{11\,.\,12}+pr'f'_{11\,.\,13}+\ \ldots\\&+qp'f'_{12\,.\,11}+qq'f'_{12\,.\,12}+qr'f'_{12\,.\,13}+\ \ldots\end{aligned}$$

and so on.

By virtue of the equations connecting the f's of a single factor, this expression, which we shall term F_{11}, has the same value, whether written with dashed or undashed f's.

Individuals having the constitution (12 . 12) may be formed by the union either of gametes (1 . 1) and (2 . 2), or of gametes (1 . 2) and (2 . 1); hence the equations of equilibrium are of the form

$$\begin{aligned}2f'_{12\,.\,12}&=\mathrm{F}_{11}+\mathrm{F}_{22}+\frac{\mu}{\mathrm{V}}(\mathrm{L}+\mathrm{L}')(\mathrm{M}+\mathrm{M}')\\&\quad+\mathrm{F}_{12}+\mathrm{F}_{21}+\frac{\mu}{\mathrm{V}}(\mathrm{L}+\mathrm{M}')(\mathrm{M}+\mathrm{L}'),\end{aligned}$$

but

$$\begin{aligned}2f_{12\,.\,12}&=2f'_{12\,.\,12}-2f_{12}-2f'_{12}\\&=2f'_{12\,.\,12}-\frac{2\mu}{\mathrm{V}}(\mathrm{LM}+\mathrm{L}'\mathrm{M}'),\end{aligned}$$

therefore

$$2f_{12\,.\,12}=\mathrm{F}_{11}+\mathrm{F}_{22}+\mathrm{F}_{12}+\mathrm{F}_{21}+\frac{\mu}{\mathrm{V}}(\mathrm{L}+\mathrm{M})(\mathrm{L}'+\mathrm{M}')\qquad(\mathrm{XIX}^*)$$

By analogy with Article 12, the solution

$$f_{12\,.\,12}=\frac{\mu}{\mathrm{V}}(\mathrm{L}+\mathrm{M})(\mathrm{L}'+\mathrm{M}')$$

suggests itself, and on trial it leads to

$$\mathrm{F}_{11}=\frac{\mu}{\mathrm{V}}\mathrm{L}\mathrm{L}',$$

and is thereby verified.

Hence we may evaluate L, L', . . . , for

$$\begin{aligned}&\mathrm{L}-p\mathrm{I}_1+q\mathrm{J}_{12}+r\mathrm{J}_{13}+\ \ldots\\&\quad=l+\Sigma\{p'^2i'(pf_{11\,.\,11}+qf_{12\,.\,11}+\ \ldots)+2p'q'j'_{12}(pf_{11\,.\,12}+qf_{12\,.\,12}+\ \ldots)+\ \ldots\end{aligned}$$

but

$$pf_{11\,.\,11}+qf_{12\,.\,11}+\ \ldots=\frac{\mu}{\mathrm{V}}\mathrm{L}(\mathrm{L}'+\mathrm{M}'),$$

therefore

$$\begin{aligned}\mathrm{L}&=l+\frac{\mu}{\mathrm{V}}\mathrm{L}\Sigma\{p'^2i'(\mathrm{L}'+\mathrm{L}')+2p'q'j'_{12}(\mathrm{L}'+\mathrm{M}')+\ \ldots\}\\&=l+\frac{\mu}{\mathrm{V}}\mathrm{L}\Sigma(2p'i'\mathrm{L}'+2q'm'\mathrm{M}'\ \ldots).\end{aligned}$$

Let

$$L = l + AL,$$

then

$$L = \frac{l}{1 - A},$$

and

$$A = \frac{\mu}{V}\Sigma(2p'l'L' + 2q'm'M' + \ldots),$$

therefore

$$A(1 - A) = \frac{\mu}{V}\Sigma(2p'l'^2 + 2q'm'^2 + \ldots)$$

$$= \frac{\mu}{V}\Sigma\beta'^2,$$

therefore

$$A(1 - A) = \frac{\mu}{V}\tau^2 \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad (XXIII^*)$$

so that the association constant, A, appearing now in the constant ratio $l : L$, plays exactly the same part in the generalised analysis as it did in the simpler case.

It may now be easily shown that the mean deviations, I and J, may be calculated from the equations

$$\left.\begin{aligned} I_1 &= i_1 + \frac{2Al}{1 - A} \\ \text{and} \qquad J_{12} &= j_{12} + \frac{A}{1 - A}(l + m), \end{aligned}\right\} \quad \ldots\ldots\ldots\ldots \quad (XXIV^*)$$

and that the variance reduces, as before, to

$$\sigma^2 + \frac{\Lambda}{1 - A}\tau^2 \quad \ldots\ldots\ldots\ldots \quad (XXV^*)$$

16. *Coupling.*—In much modern Mendelian work coupling plays an important part, although the results of different investigators do not seem as yet to converge upon any one uniform scheme of coupling. The type found by MORGAN in the American Fruit Fly (*Drosophila*) is, however, of peculiar simplicity, and may be found to be the general type of the phenomenon.

An individual heterozygous in two factors may owe its origin to the union of either of two pairs of gametes, either $(1 . 1) \times (2 . 2)$ or $(1 . 2) \times (2 . 1)$; when coupling occurs, the gametes given off by such an individual, of all these four types, do not appear in equal numbers, preference being given to the two types from which the individual took its origin. Thus in a typical case these two types might each occur in 28 per cent. of the gametes, and the other two types in 22 per cent. Coupling of this type is reversible, and occurs with equal intensity whichever of the two pairs are supplied by the grandparents. We may have any intensity from zero, when each type of gamete contributes 25 per cent., to complete coupling, when only the two original types of gamete are formed, and the segregation takes place as if only one factor were in action.

The above analysis of polymorphic factors enables us to compare these two extreme cases; for there are 9 phase combinations of a pair of dimorphic factors, or, if we

separate the two kinds of double heterozygote, 10, which, apart from inheritance, can be interpreted as the 4 homozygous and the 6 heterozygous phases of a tetramorphic factor. The 4 gametic types of this factor are the 4 gametic combinations (1 . 1), (1 . 2), (2 . 1), (2 . 2).

The mean deviations associated with these 4 gametic types are $\mathrm{L}+\mathrm{L}'$, $\mathrm{M}+\mathrm{M}'$, . . ., and we therefore write

$$\mathfrak{L}=\mathrm{L}+\mathrm{L}', \quad \mathfrak{M}=\mathrm{L}+\mathrm{M}', \quad \mathfrak{N}=\mathrm{M}+\mathrm{L}', \quad \mathfrak{O}=\mathrm{M}+\mathrm{M}'.$$

Further, if these gametic types occur with frequency,

$$\mathfrak{p}=pp'\left(1+\frac{\mu}{\mathrm{V}}\mathrm{LL}'\right) \quad \mathfrak{q}=pq'\left(1+\frac{\mu}{\mathrm{V}}\mathrm{LM}'\right)$$

$$\mathfrak{r}=qp'\left(1+\frac{\mu}{\mathrm{V}}\mathrm{ML}'\right) \quad \mathfrak{s}=qq'\left(1+\frac{\mu}{\mathrm{V}}\mathrm{MM}'\right),$$

it is clear that the frequencies with which the homozygous phases occur, such as

$$p^2p'^2(1+f''_{11\,.\,11})=p^2p'^2\left\{1+\frac{\mu}{\mathrm{V}}(\mathrm{L}^2+\mathrm{L}'^2+4\mathrm{LL}')\right\}$$

$$\mathfrak{p}^2\left\{1+\frac{\mu}{\mathrm{V}}(\mathrm{L}+\mathrm{L}')^2\right\}=\mathfrak{p}^2\left(1+\frac{\mu}{\mathrm{V}}\mathfrak{L}^2\right),$$

are exactly those produced, if there really were a single tetramorphic factor.

In the same way the phases heterozygous in one factor also agree, for

$$2p^2p'q'(1+f''_{11\,.\,12})=2p^2p'q'\left\{1+\frac{\mu}{\mathrm{V}}\mathrm{L}^2+\mathrm{L}'\mathrm{M}'+2\mathrm{L}(\mathrm{L}'+\mathrm{M}')\right\}$$

$$=2\mathfrak{pq}\left\{1+\frac{\mu}{\mathrm{V}}(\mathrm{L}+\mathrm{L}')(\mathrm{L}+\mathrm{M}')\right\}=2\mathfrak{pq}\left(1+\frac{\mu}{\mathrm{V}}\mathfrak{L}\mathfrak{M}\right).$$

Finally, taking half the double heterozygotes,

$$2pqp'q'(1+f''_{12\,.\,12})=2pqp'q'\left\{1+\frac{\mu}{\mathrm{V}}(\mathrm{LM}+\mathrm{L}'\mathrm{M}'+(\mathrm{L}+\mathrm{M})(\mathrm{L}'+\mathrm{M}'))\right\}$$

$$2\mathfrak{ps}\left\{1+\frac{\mu}{\mathrm{V}}(\mathrm{L}+\mathrm{L}')(\mathrm{M}+\mathrm{M}')\right\}-2\mathfrak{ps}\left(1+\frac{\mu}{\mathrm{V}}\mathfrak{L}\mathfrak{O}\right),$$

or, equally,

$$2\mathfrak{qr}\left\{1+\frac{\mu}{\mathrm{V}}(\mathrm{L}+\mathrm{M}')(\mathrm{M}+\mathrm{L}')\right\}-2\mathfrak{qr}\left(1+\frac{\mu}{\mathrm{V}}\mathfrak{M}\mathfrak{N}\right).$$

From this it appears that a pair of factors is analytically replaceable by a single factor if the phase frequencies be chosen rightly; but the only difference in the inheritance in these two systems is that in the one case there is no coupling, and in the other coupling is complete. It would appear, therefore, that coupling is without influence upon the statistical properties of the population.

17. The effects both of dominance and of environment may be taken into account in calculating the coefficient of correlation; if we call x the actual height of the individual, y what his height would have been under some standard environment, and z what his height would have been if in addition, without altering the extent to which different factors are associated, each phase is given its representative value of

Article 5. Then, since we are using the term environment formally for arbitrary external causes independent of heredity, the mean x of a group so chosen that $y=t$ for each member will be simply t, but the mean y of a group so chosen that $x=t$ for each member will be c_1t, where c_1 is a constant equal to the ratio of the variance with environment absolutely uniform to that when difference of environment also makes its contribution. Similarly for the group $z=t$, the mean value of y is t, but for the group $y=t$ the mean z is c_2t, where

$$c_2=\frac{\tau^2}{\sigma^2-\mathrm{A}\epsilon^2} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad \text{(XXVII)}$$

Now, we may find the parental and grandparental correlations from the fact that the mean z of any sibship is the mean z of its parents; but we shall obtain very different results in these as in other cases, according to the interpretation which we put upon the observed correlation between parents. For, in the first place, this correlation may be simply the result of conscious selection. If the correlation for height stood alone this would be the most natural interpretation. But it is found that there is an *independent* association of the length of the forearm *: if it is due to selection it must be quite unconscious, and, as Professor PEARSON points out, the facts may be explained if to some extent fertility is dependent upon genetic similarity. Thus there are two possible interpretations of marital correlations. One regards the association of the apparent characteristics as primary: there must, then, be a less intense association of the genotype y, and still less of z. The other regards the association as primarily in y or z, and as appearing somewhat masked by environmental effects in the observed correlation. In the first place, let us suppose the observed correlation in x to be primary.

Then if μ is the correlation for x, $c_1\mu$ will be that for y, and this must be written for μ in the applications of the preceding paragraphs. Hence

$$\mathrm{A}=c_1c_2\mu,$$

and μ, $c_1\mu$ and A are the marital correlations for x, y, and z.

Since the mean z of a sibship is equal to the mean z of its parents, we may calculate the parental and grandparental correlations thus:—For group chosen so that $x=t$: mean y, $\bar{y}=c_1t$; mean z, $\bar{z}=c_1c_2t$; $\bar{x}$ of mate is μt; $\bar{z}$ of mate is $c_1c_2\mu t$. Therefore $\bar{z}$ of children is

$$c_1c_2\frac{1+\mu}{2}.$$

Hence, since there is no association except of z between parents and child, the parental correlation coefficient is

$$c_1c_2\frac{1+\mu}{2}.$$

Now, since we know the mean z of the children to be

$$c_1c_2\frac{1+\mu}{2}t,$$

* PEARSON and LEE, "On the Laws of Inheritance in Man," *Biometrika*, ii, 374.

the mean z of their mates is

$$c_1 c_2 \frac{1+\mu}{2} \mathrm{A} t,$$

and the grandparental correlation coefficient will be

$$c_1 c_2 \frac{1+\mu}{2} \frac{1+\mathrm{A}}{2}.$$

Similarly, that for the $(n+1)^{\text{th}}$ parent will be

$$c_1 c_2 \frac{1+\mu}{2} \left(\frac{1+\mathrm{A}}{2}\right)^n,$$

giving the Law of Ancestral Heredity as a necessary consequence of the factorial mode of inheritance.

18. If we suppose, on the other hand, that the association is essentially in y, the coefficient of correlation between y of husband and y of wife must be $\frac{\mu}{c_1}$ in order to yield an apparent correlation μ. Also

$$c_2 = \frac{\tau^2}{\sigma^2 - \mathrm{A}\epsilon^2},$$

and

$$\mathrm{A} = \frac{\mu}{c_1} c_2.$$

The parental correlation found as before is now

$$\frac{c_1 c_2 + \mathrm{A} c_1}{2},$$

and the higher ancestors are given by the general form

$$\frac{c_1 c_2 + \mathrm{A} c_1}{2} \left(\frac{1+\mathrm{A}}{2}\right)^n,$$

although A is now differently related to c_1, c_2, and μ.

In the third case, where the essential connection is between z of husband and z of wife—and this is a possible case if the association is wholly due to selective fertility or to the selection of other features affected by the same factors—the equation between the correlations for y and z is changed, for now the marital correlation for y is equal to $\mathrm{A}c_2$ when we retain the definition

$$c_2 = \frac{\tau^2}{\sigma^2 - \mathrm{A}\epsilon^2}.$$

Hence also

$$\mu = \mathrm{A} c_1 c_2,$$

and the correlation coefficients in the ancestral line take the general form

$$c_1 c_2 \left(\frac{1+\mathrm{A}}{2}\right)^{n+1}.$$

19. On the first of these theories a knowledge of the marital and the parental correlations should be sufficient to determine $c_1 c_2$, and thence to deduce the constant ratio of the ancestral coefficients.

Thus for three human measurements :—

	Stature.	Span.	Forearm.
μ	·2804	·1989	·1977
p	·5066	·4541	·4180
c_1c_2	·7913	·7575	·6980
A	·2219	·1507	·1377
$\frac{1}{2}(1+\mathrm{A})$ . . .	·6109	·5753	·5689

These figures are deduced from those given by PEARSON and LEE (*loc. cit.*), neglecting sex distinctions, which are there found to be insignificant, and taking the weighted means.

These values for $\frac{1}{2}(1+\mathrm{A})$ agree very satisfactorily with the two ratios of the ancestral correlations which have been obtained, ·6167 for eye colour in man, and ·6602 for coat colour in horses. It is evident that if we also knew the ratio of the ancestral correlations for these features, we could make a direct determination of A and ascertain to what extent it is the cause and to what extent an effect of the observed marital correlation.

20. The correlations for sibs, double cousins, and more distant relations of the same type, in which all the ancestors of a certain degree are common, may be found by considering the variance of the group of collaterals descended from such ancestors. The variance of a sibship, for example, depends, apart from environment, only upon the number of factors in which the parents are heterozygous, and since the proportion of heterozygotes is only diminished by a quantity of the second order, the mean variance of the sibships must be taken for our purposes to have the value appropriate to random mating,

$$\tfrac{1}{2}\tau^2+\tfrac{3}{4}\epsilon^2=\frac{\mathrm{V}}{4}[2c_2(1-\Lambda)+3(1-c_2)]$$

plus the quantity $\frac{\mathrm{V}}{c_1}-\mathrm{V}$ due to environment. But the variance of the population is V/c_1; and the ratio of the two variances must be $1-f$, where f is the fraternal correlation. Hence

$$f=\frac{c_1}{4}(1+c_2+2c_2\Lambda).$$

In the same way, the variance for a group of double cousins is unaffected by selective mating, and we find the correlation coefficient for double cousins to be

$$\frac{c_1}{16}(1+3c_2+12c_2\mathrm{A}),$$

showing how the effect of selective mating increases for the more distant kin.

On the first hypothesis, then, we must write,

$$\mu=\frac{\mathrm{A}}{c_1c_2},$$

$$p=c_1c_2\frac{1+\mu}{2},$$

and

$$f=\frac{c_1}{4}(1+c_2(1+2\Lambda)).$$

21. We shall use this formula for the fraternal correlation to estimate the relative importance of dominance and environment in the data derived from the figures given by PEARSON and LEE.

Assuming as the observed correlations

	Stature.	Span.	Cubit.
μ	·2804	·1989	·1977
p	·5066	·4541	·4180
f	·5433	·5351	·4619

we obtain as before

c_1c_2	·7913	·7575	·6980
A	·2219	·1507	·1377

and calculating c_1 from the formula

$$c_1 = 4f - c_1c_2(1 + 2A),$$

we obtain the three values

1·031 1·155 ·957

with a standard error of ·072, and a mean of 1·048.

This relatively large standard error, due principally to our comparative ignorance of the fraternal correlations (errors in μ have scarcely any effect, and those in p are relatively unimportant), prevents us from making on a basis of these results a close estimate of the contributions to the total variance of the factors under consideration.

Remembering that c_1 is intrinsically less than unity, the second value is inexplicably high, whilst the first and third are consistent with any value sufficiently near to unity. The mean of these results is materially greater than unity, and therefore gives no support to the supposition that there is any cause of variance in these growth features other than genetic differences. If this is so, we should put $c_1 = 1$, and compare the observed values of f with those calculated from the formula

$$4f = 1 + c_2(1 + 2A).$$

With their standard errors we obtain

	Stature.	Span.	Cubit.	Standard Error.
Observed . .	·5433	·5351	·4619	·016
Calculated . .	·5356	·4964	·4726	·008
Difference . .	−·0077	−·0387	+·0107	·018

The exceptional difference in the fraternal correlations for span might, perhaps, be due to the effects of epistacy, or it may be that the terms which we have neglected, which depend upon the finiteness of the number of factors, have some influence. It is more likely, as we shall see, that the assumption of direct sexual selection is

not justified for this feature. Accepting the above results for stature, we may ascribe the following percentages of the total variance to their respective causes :—

Ancestry		54 per cent.
Variance of sibship :		
$\frac{1}{2}\tau^2$	31 per cent.	
$\frac{3}{4}\epsilon^2$	15 „	
Other causes	...	
		46 „
		100 per cent.

Again it may be divided :

Genotypes (σ^2) :		
Essential genotypes (τ^2)	62 per cent.	
Dominance deviations (ϵ^2)	21 „	
		83 per cent.
Association of factors by homogamy		17 „
Other causes		...
		100 per cent.

These determinations are subject, as we have seen, to considerable errors of random sampling, but our figures are sufficient to show that, on this hypothesis, it is very unlikely that so much as 5 per cent. of the total variance is due to causes not heritable, especially as every irregularity of inheritance would, in the above analysis, appear as such a cause.

It is important to see that the large effect ascribed to dominance can really be produced by ordinary Mendelian factors. The dominance ratio, $\frac{\epsilon^2}{\sigma^2}$, which may be determined from the correlations, has its numerator and denominator composed of elements, δ^2 and α^2, belonging to the individual factors. We may thereby ascertain certain limitations to which our factors must be subject if they are successfully to interpret the existing results. The values of the dominance ratio in these three cases are found to be :

	Stature.	Span.	Cubit.	Standard Error.
Dominance ratio	·253	·274	·336	·045

22. The correlations for uncles and cousins, still assuming that the association of factors is due to a direct selection of the feature x, may be obtained by the methods of Article 14, using the two series already obtained : that for ancestors

$$c_1 c_2 \frac{1+\mu}{2}\left(\frac{1+A}{2}\right)^n,$$

and that for collaterals, like sibs and double cousins, which have all their ancestors of a certain degree in common,

$$\tfrac{1}{4}c_1[1+c_2(1+2A)],$$
$$\tfrac{1}{8}c_1[1+3c_2(1+4A)],$$

and so on.

Thus if a group be chosen so that $x = t$,

$\bar{y}$ of group is $c_1 t$,
$\bar{z}$ of group is $c_1 c_2 t$,
$\bar{z}$ of sibs is $c_1 c_2 \frac{1+A}{2} t$,

also

$\bar{y}$ of sibs is $\frac{1}{4} c_1 [1 + c_2(1+2A)]t$,
$\bar{y}$ of sibs mates is $\frac{1}{4} c_1 [1 + c_2(1+2A)] c_1 \mu t$,
$\bar{z}$ of sibs mates is $\frac{1}{4} c_1 [1 + c_2(1+2A)] A t$.

Hence

$\bar{z}$ of nephews is $\frac{1}{8} c_1 [2c_2(1+A) + \{1 + c_2(1+2A)\} A] t$,

giving the correlation

$$c_1 c_2 \left(\frac{1+A}{2}\right)^2 + \tfrac{1}{8} c_1 A (1 - c_2).$$

Again for cousins, if a group be chosen so that $x = t$, we have

$$\bar{y} \text{ of uncles is } \left[c_1 c_2 \left(\frac{1+A}{2}\right)^2 + \tfrac{1}{8} c_1 A (1 - c_2) \right] t,$$

$$\bar{z} \text{ of uncles is } c_1 c_2 \left(\frac{1+A}{2}\right)^2,$$

and

$$\bar{z} \text{ of uncles mates is } \left[c_1 c_2 \left(\frac{1+A}{2}\right)^2 + \tfrac{1}{8} c_1 A (1 - c_2) \right] A t,$$

hence

$$\bar{z} \text{ of cousins is } \left[c_1 c_2 \left(\frac{1+A}{2}\right)^3 + \tfrac{1}{16} c_1 A^2 (1 - c_2) \right] t,$$

giving the correlation

$$c_1 c_2 \left(\frac{1+A}{2}\right)^3 + \tfrac{1}{16} c_1 A^2 (1 - c_2).$$

The formulæ show that these two correlations should differ little from those for grandparent and great-grandparent, using the values already found, and putting $c_1 = 1$ we have

	Stature.	Span.	Cubit.
Grandparent . . .	·3095	·2612	·2378
Great-grandparent . .	·1891	·1503	·1353
Uncle	·3011	·2553	·2311
Cousin	·1809	·1445	·1288

23. On the third supposition, that the marital correlation is due primarily to an association in the essential genotype z, we obtain results in some respects more intelligible and in accordance with our existing knowledge.

From the fundamental equations

$$\mu = c_1 c_2 A,$$
$$p = \tfrac{1}{2}(c_1 c_2 + \mu),$$

we may deduce

$$c_1 c_2 = 2p - \mu,$$
$$A = \mu/(2p - \mu),$$

whence the following table is calculated :—

	Stature.	Span.	Cubit.	Standard Error.
μ . .	·2804	·1989	·1977	·0304
p . .	·5066	·4541	·4180	·0115
f . .	·5433	·5351	·4619	·0160
$c_1 c_2$ . .	·7328	·7093	·6383	·038
A . .	·3826	·2804	·3097	·028
$\frac{1}{2}(1+A)$.	·6913	·6402	·6549	·014

and making use of the fraternal correlations to separate c_1 and c_2, by the equations

$$f = \tfrac{1}{4}c_1[1 + c_2(1 + 2A)],$$

or

$$c_1 = 4f - 2p - \mu,$$

we obtain

c_1 . .	·8796	1·0333	·8139	·078
c_2 . .	·8331	·6864	·7842	·077
$\frac{\epsilon^2}{\sigma^2}$ . .	·2450	·3883	·2850	·105

The standard error for the dominance ratio is now very high, since the latter is proportional to the difference $f - p$. If we assume a known value for c_1, and calculate the dominance ratio from p and μ only, the standard error falls nearly to its value in Article 18.

The three values for the ratio of the ancestral correlations ·691, ·640, ·655 are now higher than that obtained from observations of eye colour, and are more similar to the value ·660 obtained for the coat colour of horses. Without knowing the marital correlations in these cases, it is not possible to press the comparison further. It would seem unlikely that the conscious choice of a mate is less influenced by eye colour than by growth features, even by stature. But it is not at all unlikely that eye colour is but slightly correlated with other features, while the growth features we know to be highly correlated, so that a relatively slight selection in a number of the latter might produce a closer correlation in each of them than a relatively intense selection of eye colour.

The value of c_1 for span is still greater than unity, 1·033, but no longer unreasonably so, since the standard error is about ·078. If we were considering span alone the evidence would be strongly in favour of our third hypothesis. A remarkable confirmation of this is that PEARSON and LEE (*loc. cit.*, p. 375), considering organic and marital correlations alone, show that the observed correlations could be accounted for by the following direct selection coefficients :—

Stature.	Span.	Cubit.
·2374	·0053	·1043

Naturally these cannot be taken as final, since there are a large number of other features, which may be connected with these and at the same time may be subject to sexual selection. The correlations of cross assortative mating are in fact smaller than they would be if direct selection to this extent were actually taking place. The influence of other features prevents us from determining what proportion of the observed association is due to direct selection, but if inheritance in these growth features is capable of representation on a Mendelian scheme—and our results have gone far to show that this is likely—it would be possible to distinguish the two parts by comparing the parental and fraternal correlations with those for grandparents and other kindred.

On our present supposition that the association is primarily in z, and for the case

of span this seems likely, the correlations for uncle and cousin will be the same as those for grandparent and great-grandparent, being given by the formulæ

$$c_1c_2\left(\frac{1+A}{2}\right)^2 \quad \text{and} \quad c_1c_2\left(\frac{1+A}{2}\right)^3,$$

leading to the numbers

	Stature.	Span.	Cubit.
Grandparent . . .	·3502	·2907	·2737
Great-grandparent . .	·2421	·1861	·1793

24. Neither these nor the similar table for the first hypothesis accord ill with the value obtained for uncle and nephew, ·265, from measurements of eye colour. It may, however, be thought that neither of them give high enough value for cousins. Certainly they do not approach some of the values found by Miss ELDERTON in her memoir on the resemblance of first cousins (*Eugenics Laboratory Memoirs*, iv). Series are there found to give correlations over ·5, and the mean correlation for the measured features is ·336. From special considerations this is reduced to ·270, but if the similarity of first cousins is due to inheritance, it must certainly be less than that between uncle and nephew. No theory of inheritance could make the correlation for cousins larger than or even so large as that for the nearer relationship.

It will be of interest finally to interpret our results on the assumption that the figures quoted (Article 20) represent actual coefficients of selection. Manifestly it would be better to obtain the value of A experimentally from the ratio of the ancestral correlations, using the collateral correlations to determine what are the marital correlations for y. For the present we must neglect the possibility of an independent selection in y; and although we know that the figures are not final, we shall write s, the coefficient of selection, equal to ·2374, ·0053, and ·1043 in our three cases.

Further, let

$$A = c_1c_2s + \frac{\mu - s}{c_1c_2},$$

so that

$$2p = c_1c_2(1+s) + \mu - s,$$

whence we deduce

	Stature.	Span.	Cubit.
c_1c_2	·7841	·7108	·6725
A	·2410	·2761	·2090
$\frac{1}{2}(1+A)$ . . .	·6205	·6381	·6045

the values of A being now in much closer agreement for the three features. Further, from the fraternal correlation we have

c_1	1·0112	1·0370	·8940

with a mean at ·9821.

Again, for the dominance ratio

·2763 ·3880 ·2940 ·3194 (mean),

leaving a trifle under 2 per cent. for causes not heritable, but requiring high values about ·32 for the dominance ratio.

25. *The Interpretation of the Statistical Effects of Dominance.*—The results which we have obtained, although subject to large probable errors and to theoretical reservations which render an exact estimate of these errors impossible, suggest that the ratio $\frac{\epsilon^2}{\sigma^2}$, the statistical measure of the extent of dominance, has values of about ·25 to ·38. In his initial memoir on this subject KARL PEARSON has shown that, under the restricted conditions there considered, this ratio should be exactly $\frac{1}{3}$. Subsequently UDNY YULE (Conference on Genetics) pointed out that the parental correlation could be raised from the low values reached in that memoir to values more in accordance with the available figures by the partial or total abandonment of the assumption of dominance. To this view Professor PEARSON subsequently gave his approval; but it does not seem to have been observed that if lower values are required—and our analysis tends to show that they are not—the statistical effects are governed not only by the physical ratio $\frac{d}{a}$, but by the proportions in which the three Mendelian phases are present. This effect is an important one, and very considerably modifies the conclusions which we should draw from any observed value of the dominance ratio.

The fraction $\frac{\delta^2}{a^2}$, of which the numerator and denominator are the contributions of a single factor to ϵ^2 and σ^2, is equal, as we have seen (Article 5, equations V–VII) to

$$\frac{2pqd^2}{(p+q)^2a^2-2(p^2-q^2)ad+(p^2+q^2)d^2},$$

and depends wholly upon the two ratios $\frac{d}{a}$ and $\frac{p}{q}$. We may therefore represent the variations of this function by drawing the curves for which it has a series of constant values upon a plane, each point on which is specified by a pair of particular values for these two ratios. The accompanying diagram (fig. 1, p. 430) shows such a series of curves, using $\frac{d}{a}$ and $\log\frac{p}{q}$ as co-ordinates. The logarithm is chosen as a variable, because equal intensity of selection will affect this quantity to an equal extent, whatever may be its value; it also possesses the great advantage of showing reciprocal values of $\frac{p}{q}$ in symmetrical positions.

It will be seen that ·3 is not by any means the highest value possible: when $d=a$, and when $\frac{p}{q}$ is very great, any value up to unity may appear; but high values are confined to this restricted region. When $\frac{d}{a}$ is less than ·3 the ratio is never greater than ·05, and we cannot get values as high as ·15 unless $\frac{d}{a}$ be as great as ·5. On the

other hand, all values down to zero are consistent with complete dominance, provided that the values of $\frac{p}{q}$ are sufficiently small.

We know practically nothing about the frequency distribution of these two ratios. The conditions under which Mendelian factors arise, disappear, or become modified are unknown. It has been suggested that they invariably arise as recessive mutations in a dominant population. In that case $\frac{p}{q}$ would initially be very high, and could only be lowered if by further mutation, and later by selection, the recessive phase became more frequent. These factors would, however, have little individual weight if better balanced factors were present, until $\frac{p}{q}$ had been lowered to about 10. In face of these theories it cannot be taken for granted that the distribution of these ratios is a simple one. It is natural, though possibly not permissible, to think of their distributions as independent. We may profitably consider further the case in which the distribution is symmetrical, in which the factor of known a and d is equally likely to be more frequent in the dominant as in the recessive phase.

For this case we combine the numerators and denominators of the two fractions

$$\frac{2pqd^2}{(p+q)^2a^2-2(p^2-q^2)ad+(p^2+q^2)d^2} \text{ and } \frac{2pqd^2}{(p+q)^2a^2+2(p^2-q^2)ad+(p^2+q^2)d^2},$$

and obtain the joint contribution

$$\frac{2pqd}{(p+q)^2a^2+(p^2+q^2)d^2},$$

the curves for which are shown in fig. 2, representing the combined effect of two similar factors, having their phases in inverse proportions. It will be seen that complete dominance does not preclude the possibility of low value for the dominance ratio: the latter might fall below ·02 if the greater part of the variance were contributed by factors having the ratio between p and q as high as 100 to 1. This ratio is exceedingly high; for such a factor only one individual in 10,000 would be a recessive. We may compare the frequency of deaf mutism with which about one child in 4000 of normal parents is said to be afflicted. It would be surprising if more equal proportions were not more common, and if this were so, they would have by far the greater weight.

The fact that the same intensity of selection affects the logarithm of $\frac{p}{q}$ equally, whatever its value may be, suggests that this function may be distributed approximately according to the law of errors. This is a natural extension of the assumption of symmetry, and is subject to the same reservations. For instance, a factor in which the dominant phase is the commonest would seem less likely to suffer severe selection than one in which the recessive phase outnumbers the other. But if symmetry be granted, our choice of a variable justifies the consideration of a normal distribution.

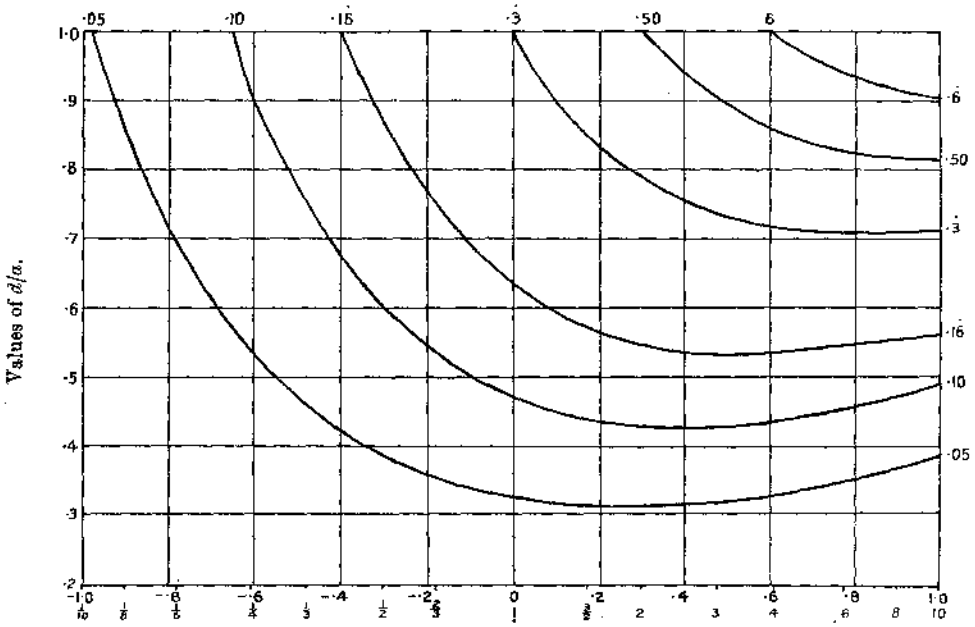


FIG. 1.—Values of $\log_{10}(p/q)$ (upper figures) and of p/q (lower figures).

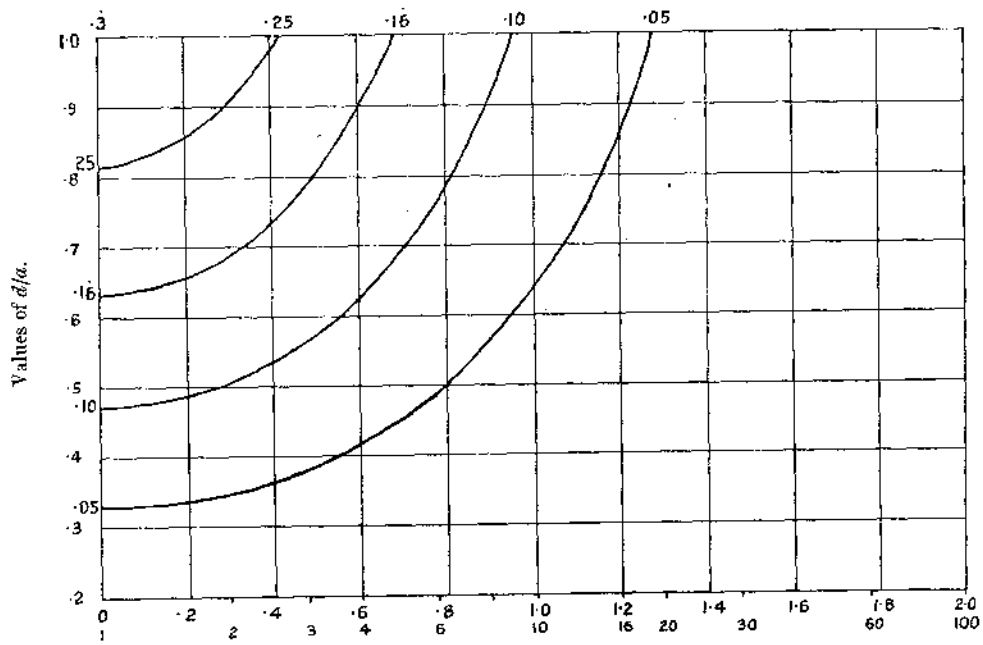


FIG. 2.—Values of $\log_{10}(p/q)$ (upper figures) and of p/q (lower figures).

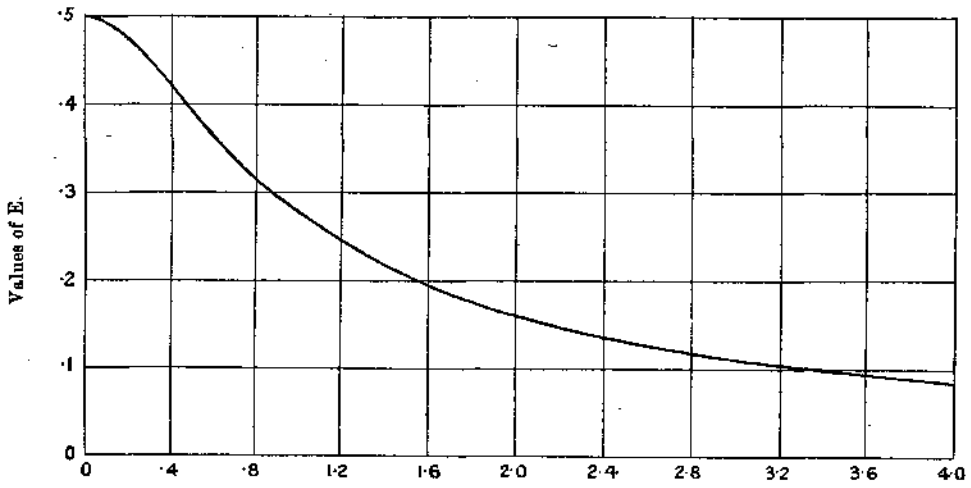


FIG. 3.—Values of $\sigma \log_{10} e = \cdot 4343\sigma$.

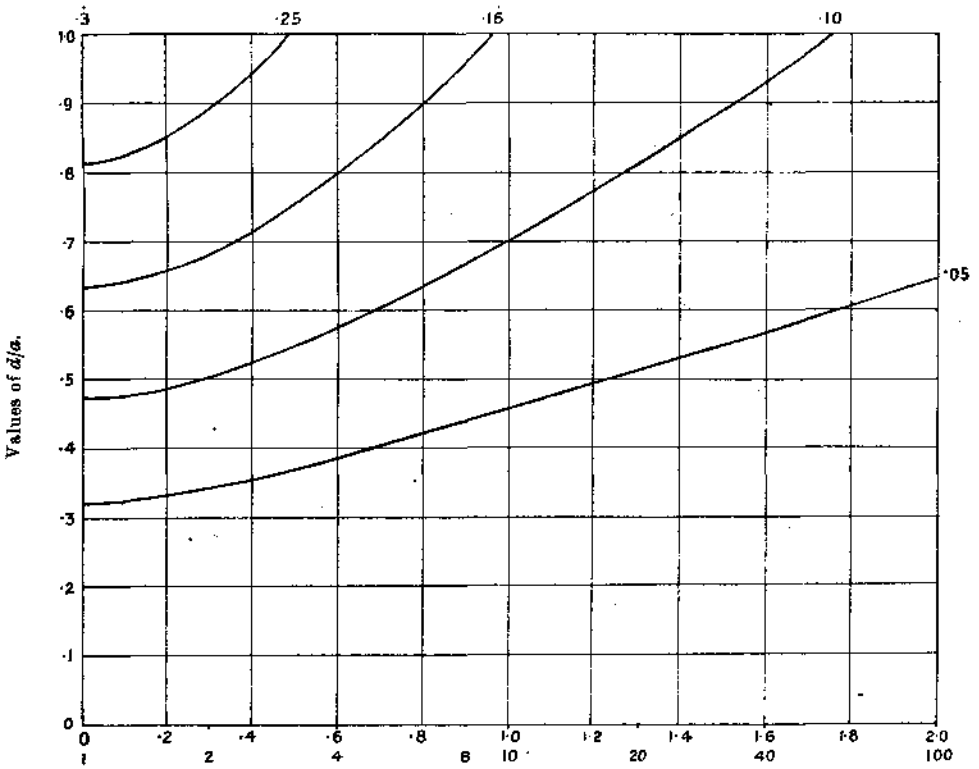


FIG. 4.—Values of $\log_{10}$ of standard ratio (upper figures) and of standard ratio (lower figures).

Writing ξ for $\log_e \frac{p}{q}$ and σ for the standard deviation of ξ, we have

$$p = e^{\frac{1}{2}\xi}/2\cosh\tfrac{1}{2}\xi, \quad q = e^{-\frac{1}{2}\xi}/2\cosh\tfrac{1}{2}\xi, \quad \text{and} \quad 2pq = \tfrac{1}{2}\operatorname{sech}^2\tfrac{1}{2}\xi.$$

Hence we have to evaluate

$$\mathrm{E} = \frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty} \tfrac{1}{2}\operatorname{sech}^2\tfrac{1}{2}\xi \,.\, e^{-\xi^2/2\sigma^2} d\xi = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \tfrac{1}{2}\operatorname{sech}^2\frac{\sigma\xi}{2} e^{-\frac{1}{2}\xi^2} d\xi \quad . \quad . \quad \text{(XXVIII)}$$

and the dominance ratio derived from the whole group is

$$\frac{\mathrm{E}d^2}{a^2 + (1-\mathrm{E})d^2}.$$

E is a function of σ only, which decreases steadily from its value $\frac{1}{2}$ when $\sigma = 0$, approaching when σ is large to the function $\frac{2}{\sigma\sqrt{2\pi}}$. The function $\left(16 + 16\sigma^2 + \frac{\pi^2}{4}\sigma^4\right)^{-\frac{1}{4}}$ osculates it at the origin, and appears on trial to represent it effectively to three significant figures. This function has been used for calculating the form of the accompanying curves. Fig. 3 shows the course of the function E. Fig. 4 gives the curves comparable to those of figs. 1 and 2, showing the value of the dominance ratio for different values $\frac{d}{a}$ and σ. If the assumptions upon which this diagram is based are justified, we are now advanced some way towards the interpretation of an observed dominance ratio. A ratio of ·25 gives us a lower limit of about ·8 for $\frac{d}{a}$, and no upper limit. If the possibility of superdominance ($d > a$) is excluded, then the ratio of the phases must be so distributed that the standard ratio e^{σ} is not greater than about 3:1. A greater value of the standard ratio would make the effect of dominance too small; a smaller value could be counteracted by a slight reduction of $\frac{d}{a}$. We have therefore no reason to infer from our dominance ratios that dominance is incomplete. We may speak of it as having at least four-fifths of its full value, but we can set no upper limit to it.

26. Throughout this work it has been necessary not to introduce any avoidable complications, and for this reason the possibilities of Epistacy have only been touched upon, and small quantities of the second order have been steadily ignored. In spite of this, it is believed that the statistical properties of any feature determined by a large number of Mendelian factors have been successfully elucidated. Due allowance has been made for the factors differing in the magnitude of their effects, and in their degree of dominance, for the possibility of Multiple Allelomorphism, and of one important type of Coupling. The effect of the dominance in the individual factors has been seen to express itself in a single Dominance Ratio. Further, the effect of marital correlation has been fully examined, and the relation between this association and the coefficient of marital correlation has been made clear.

By means of the fraternal correlation it is possible to ascertain the dominance

ratio and so distinguish dominance from all non-genetic causes, such as environment, which might tend to lower the correlations: this is due to the similarity in siblings of the effects of dominance which causes the fraternal correlation to exceed the parental. The fact that this excess of the fraternal correlation is very generally observed is itself evidence in favour of the hypothesis of cumulative factors. On this hypothesis it is possible to calculate the numerical influence not only of dominance, but of the total genetic and non-genetic causes of variability. An examination of the best available figures for human measurements shows that there is little or no indication of non-genetic causes. The closest scrutiny is invited on this point, not only on account of the practical importance of the predominant influence of natural inheritance, but because the significance of the fraternal correlation in this connection has not previously been realised.

Some ambiguity still remains as to the causes of marital correlations: our numerical conclusions are considerably affected according as this is assumed to be of purely somatic or purely genetic origin. It is striking that the indications of the present analysis are in close agreement with the conclusions of PEARSON and LEE as to the genetic origin of a part of the marital correlation, drawn from the effect of the correlation of one organ with another in causing the selection of one organ to involve the selection of another. This difficulty will, it is hoped, be resolved when accurate determinations are available of the ratio of the grandparental to the parental correlation. From this ratio the degree of genetic association may be immediately obtained, which will make our analysis of the Variance as precise as the probable errors will allow.

In general, the hypothesis of cumulative Mendelian factors seems to fit the facts very accurately. The only marked discrepancy from existing published work lies in the correlation for first cousins. SNOW, owing apparently to an error, would make this as high as the avuncular correlation; in our opinion it should differ by little from that of the great-grandparent. The values found by Miss ELDERTON are certainly extremely high, but until we have a record of complete cousinships measured accurately and without selection, it will not be possible to obtain satisfactory numerical evidence on this question. As with cousins, so we may hope that more extensive measurements will gradually lead to values for the other relationship correlations with smaller standard errors. Especially would more accurate determinations of the fraternal correlation make our conclusions more exact.

Finally, it is a pleasure to acknowledge my indebtedness to Major LEONARD DARWIN, at whose suggestion this inquiry was first undertaken, and to whose kindness and advice it owes its completion.

Wright (1922)

> *Wright, S. 1922. Coefficients of inbreeding and relationship. The American Naturalist, vol. 56, no. 645, pp. 330 to 338.*

The problem

Mating between relatives was common practice among breeders, but there was no measure of the degree of inbreeding of an animal or of the relationship between two animals. Wright used his method of path coefficients to define the coefficient of inbreeding and the coefficient of relationship. He showed how to calculate these coefficients by summing the contributions of each path linking two individuals to a common ancestor, and applied the method to famous Shorthorn bulls.

Points to note while reading

The formula that sums powers of one half along the paths of the pedigree is the origin of the relationship matrix A, which appears in all of Henderson's equations and which, many decades later, would be combined with the genomic matrix to form the H matrix. Note the examples with real animals, which show the level of inbreeding in the breeds of the time.

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COEFFICIENTS OF INBREEDING AND RELATIONSHIP

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In the breeding of domestic animals consanguineous matings are frequently made. Occasionally matings are made between very close relatives—sire and daughter, brother and sister, etc.—but as a rule such close inbreeding is avoided and there is instead an attempt to concentrate the blood of some noteworthy individual by what is known as line breeding. No regular system of mating such as might be followed with laboratory animals is practicable as a rule.

The importance of having a coefficient by means of which the degree of inbreeding may be expressed has been brought out by Pearl[1] in a number of papers published between 1913 and 1917. His coefficient is based on the smaller number of ancestors in each generation back of an inbred individual, as compared with the maximum possible number. A separate coefficient is obtained for each generation by the formula

$$Z_n = 100\left(1 - \frac{q_{n+1}}{p_{n+1}}\right) = 100\left(1 - \frac{q_{n-1}}{2^{n+1}}\right)$$

where $q_{n+1}/2^{n+1}$ is the ratio of actual to maximum possible ancestors in the $n+1$st generation. By finding the ratio of a summation of these coefficients to a similar summation for the maximum possible inbreeding in higher animals, *viz.*, brother-sister mating, he obtains a single coefficient for the whole pedigree.

This coefficient has the defect, as Pearl himself pointed

[1] American Naturalist, 1917, 51: 545–559; 51: 636–639.

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out, that it may come out the same for systems of breeding which we know are radically different as far as the effects of inbreeding are concerned. For example, in the continuous mating of double first cousins, an individual has two parents, four grandparents, four great grandparents and four in every generation, back to the beginning of the system. Exactly the same is true of an individual produced by crossing different lines, in each of which brother-sister mating has been followed. Yet in the first the individual will be homozygous in all factors if the system has been in progress sufficiently long; in the second he will be heterozygous in a maximum number of respects.

In order to overcome this objection Pearl has devised a partial inbreeding index which is intended to express the percentage of the inbreeding which is due to relationship between the sire and dam, inbreeding being measured as above described. A coefficient of relationship is used in this connection. These coefficients have been discussed by Ellinger[2] who suggests certain alterations and extensions by means of which the total inbreeding coefficient, a total relationship coefficient and a total relationship-inbreeding index for a given pedigree can be compared on the same scale.

An inbreeding coefficient to be of most value should measure as directly as possible the effects to be expected on the average from the system of mating in the given pedigree.

There are two classes of effects which are ascribed to inbreeding: First, a decline in all elements of vigor, as weight, fertility, vitality, etc., and second, an increase in uniformity within the inbred stock, correlated with which is an increase in prepotency in outside crosses. Both of these kinds of effects have ample experimental support as average (not necessarily unavoidable) consequences of inbreeding. The best explanation of the decrease in vigor is dependent on the view that Mendelian

[2] AMERICAN NATURALIST, 1920, 54: 540–545.

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factors unfavorable to vigor in any respect are more frequently recessive than dominant, a situation which is the logical consequence of the two propositions that mutations are more likely to injure than improve the complex adjustments within an organism and that injurious dominant mutations will be relatively promptly weeded out, leaving the recessive ones to accumulate, especially if they happen to be linked with favorable dominant factors. On this view it may readily be shown that the decrease in vigor on starting inbreeding in a previously random-bred stock should be directly proportional to the increase in the percentage of homozygosis. Numerous experiments with plants and lower animals are in harmony with this view. Extensive experiments with guinea-pigs conducted by the Bureau of Animal Industry are in close quantitative agreement. As for the other effects of inbreeding, fixation of characters and increased prepotency, these are of course in direct proportion to the percentage of homozygosis. Thus, if we can calculate the percentage of homozygosis which would follow on the average from a given system of mating, we can at once form the most natural coefficient of inbreeding. The writer[3] has recently pointed out a method of calculating this percentage of homozygosis which is applicable to the irregular systems of mating found in actual pedigrees as well as to regular systems. This method, it may be said, gives results widely different from Pearl's coefficient, in many cases even as regards the relative degree of inbreeding of two animals.

Taking the typical case in which there are an equal number of dominant and recessive genes (*A* and *a*) in the population, the random-bred stock will be composed of 25 per cent. *AA*, 50 per cent. *Aa* and 25 per cent. *aa*. Close inbreeding will tend to convert the proportions to 50 per cent. *AA*, 50 per cent. *aa*, a change from 50 per cent. homozygosis to 100 per cent. homozygosis. For a natural coefficient of inbreeding, we want a scale which

[3] *Genetics*, 1921, 6: 111–178.

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runs from 0 to 1, while the percentage of homozygosis is running from 50 per cent. to 100 per cent. The formula $2h-1$, where h is the proportion of complete homozygosis, gives the required value. This can also be written $1-2p$ where p is the proportion of heterozygosis. In the above-mentioned paper it was shown that the coefficient of correlation between uniting egg and sperm is expressed by this same formula, $f = 1-2p$. We can thus obtain the coefficient of inbreeding f_b for a given individual B, by the use of the methods there outlined.

The symbol r_{bc}, for the coefficient of the correlation between B and C, may be used as a coefficient of relationship. It has the value 0 in the case of two random individuals, .50 for brothers in a random stock and approaches 1.00 for individuals belonging to a closely inbred subline of the general population.

In the general case in which dominants and recessives are not equally numerous, the composition of the random-bred stock is of the form $x^2\ AA$, $2xy\ Aa$, $y^2\ aa$. The percentage of homozygosis is here greater than 50 per cent. The rate of increase, however, under a given system of mating, is always exactly proportional to that in the case of equality. The coefficient is thus of general application.

If an individual is inbred, his sire and dam are connected in the pedigree by lines of descent from a common ancestor or ancestors. The coefficient of inbreeding is obtained by a summation of coefficients for every line by which the parents are connected, each line tracing back from the sire to a common ancestor and thence forward to the dam, and passing through no individual more than once. The same ancestor may of course be involved in more than one line.

The path coefficient, for the path, sire (S) to offspring (O), is given by the formula $p_{o.s} = \frac{1}{2}\sqrt{(1+f_s)/(1+f_o)}$, where f_s and f_o are the coefficients of inbreeding for sire

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and offspring, respectively. The coefficient for the path, dam to offspring, is similar.

In the case of sire's sire (G) and individual, we have $p_{o.g} = p_{o.s}\ p_{s.g} = \frac{1}{4}\sqrt{(1+f_g)/(1+f_o)}$, and for any ancestor ($A$) we have for the coefficient pertaining to a given line of descent $p_{o.a} = (\frac{1}{2})^n\sqrt{(1+f_a)/(1+f_o)}$, where n is the number of generations between them in this line.

The correlation between two individuals (r_{bc}) is obtained by a summation of the coefficients for all connecting paths.

Thus

$$r_{bc} = \Sigma p_{ba}p_{ca} = \Sigma\left(\frac{1}{2}\right)^{n+n'}\frac{1+b_a}{\sqrt{(1+b_b)(1+b_c)}},$$

where n and n' are the number of generations in the paths from A to B and from A to C, respectively.

The formula for the correlation between uniting gametes, which is also the required coefficient of inbreeding, is

$$f_o = \tfrac{1}{2}r_{sd}\sqrt{(1+f_s)(1+f_d)},$$

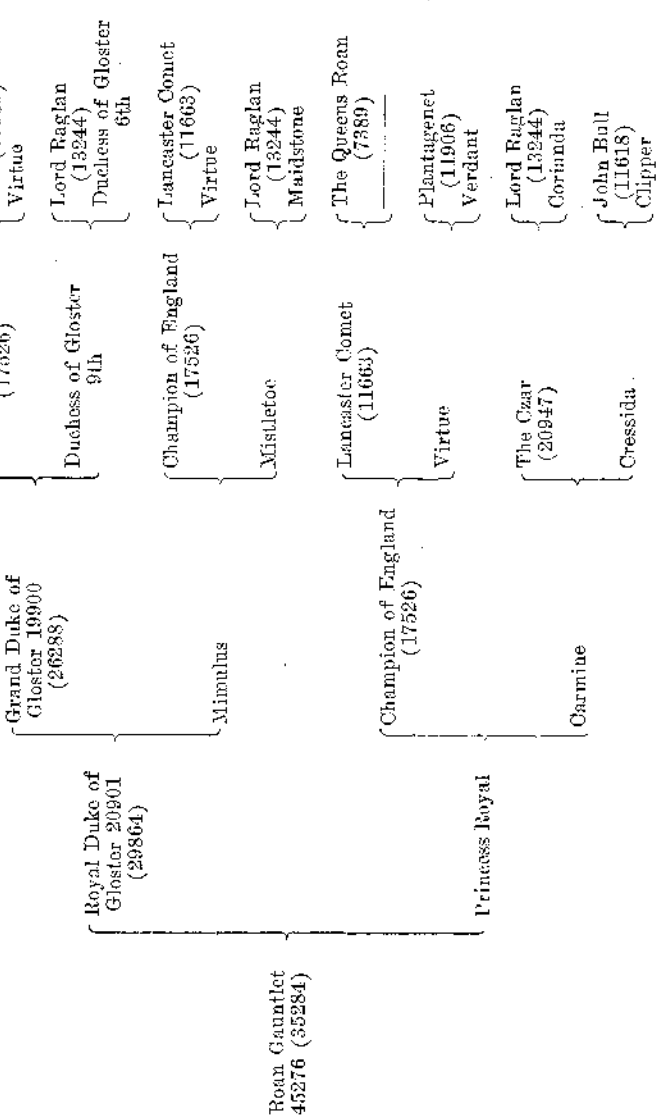
where r_{sd} is the correlation between sire and dam and f_s and f_d are coefficients of inbreeding of sire and dam. Substituting the value of r_{sd} we obtain

$$f_o = \Sigma\,(\tfrac{1}{2})^{n+n'+1}(1+f_a).$$

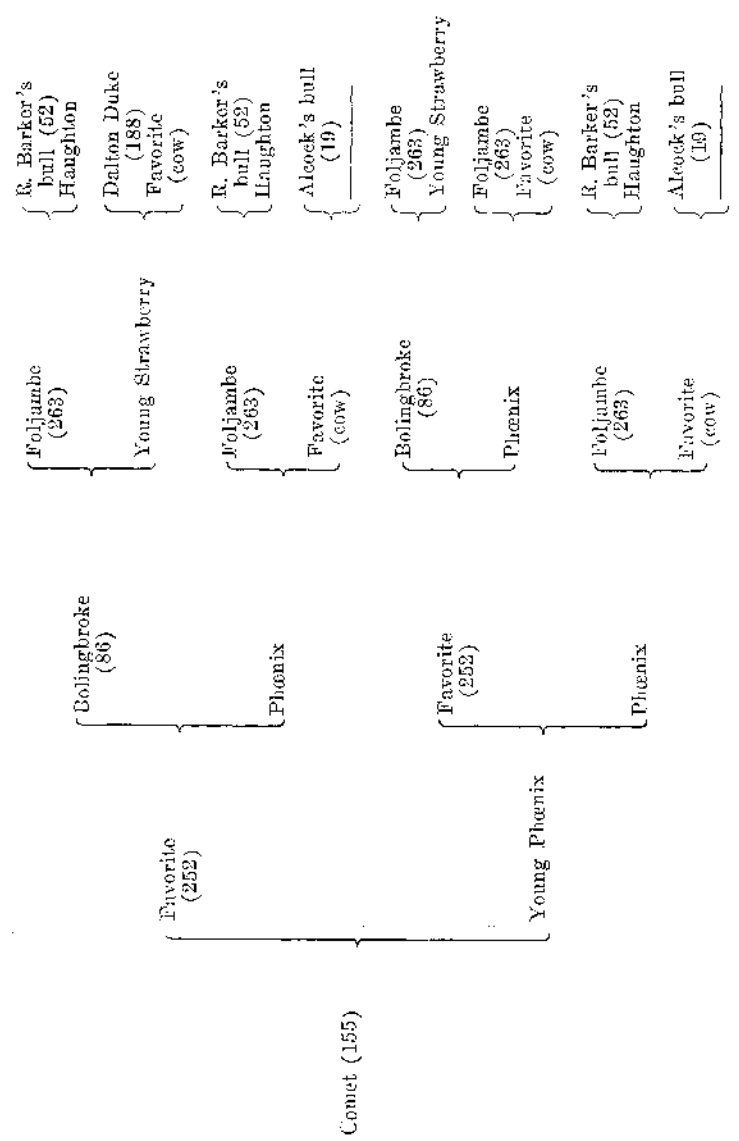
If the ancestor (A) is not inbred, the component for the given path is simply $(\frac{1}{2})^{n+n'+1}$ where n and n' are the number of generations from sire and dam respectively to the ancestor in question. If the common ancestor is inbred himself, his coefficient of inbreeding (f_a) must be worked out from his pedigree.

This formula gives the departure from the amount of homozygosis under random mating toward complete homozygosis. The percentage of homozygosis (assuming 50 per cent. under random mating) is $\frac{1}{2}(1+f_o) \times 100$.

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By this means the inbreeding in an actual pedigree, however irregular the system of mating, can be compared accurately with that under any regular system of mating.

As an illustration, take the pedigree of Roan Gauntlet, a famous Shorthorn sire, bred by Amos Cruickshank. This bull traces back in every line to a mating of Champion of England with a daughter or granddaughter of Lord Raglan. For the present purpose we will assume that these bulls were not at all inbred themselves and not related to each other. Since the sire traces twice to Champion of England and twice to Lord Raglan and the dam once to each bull, there are in all four lines by which the sire and dam are connected.

Individual	Common Ancestors of Sire and Dam	f_a	n	n'	$(\frac{1}{2})^{n+n'+1} \times (1 + b_a)$
Roan Gauntlet 45,276 (35,284)	Champion of England (17,526)	0	2	1	.062500
			2		.062500
	Lord Raglan (13,244)	0	3	3	.007812
			3		.007813
					.140625

The coefficient of inbreeding comes out 14.1 per cent., a rather low figure when compared to such systems as brother-sister mating (one generation 25 per cent., two generations 37.5 per cent., three generations 50 per cent., ten generations 88.6 per cent.) or parent-offspring mating, (one generation 25 per cent., two generations 37.5 per cent., three generations 43.8 per cent., approaching 50 per cent. as a limit).

As an example of closer inbreeding, take the pedigree of Charles Collings' bull, Comet. The sire was the bull Favorite and the dam was from a mating of Favorite with his own dam. As Favorite was himself inbred to some extent, it is necessary to calculate first his own coefficient of inbreeding.

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Individual	Common Ancestors of Sire and Dam	f_a	n	n'	$(\frac{1}{2})^{n+n'+1} \times (1 + f_a)$
Favorite (252)	Foljambe (263).....	0	1	1	.1250
	Favorite (cow).....	0	2	1	.0625
					.1875
Comet (115)	Favorite (252).....	.1875	0	1	.2969
	Phœnix.....	0	1	1	.1250
	Foljambe.....	0	2	2	.0312
	Favorite (cow).....	0	3	2	.0156
					.4687

In the case of Comet, Foljambe and Favorite (cow) each appears twice in the pedigree of the sire and three times in the pedigree of the dam. However, only those pedigree paths which connect sire and dam and which do not pass through the same animal twice are counted. The listing of Favorite (252) and Phœnix as common ancestors eliminates all but one path in each case as regards Foljambe and Favorite cow. The remaining paths are those due to the common descent of Bolingbroke, the sire's sire and Phœnix as the dam's dam from the above two animals.

By tracing the pedigrees back to the beginning of the herd book, the coefficients of inbreeding are slightly increased. This meant going back to the seventh generation for one common ancestor of the sire and dam of Favorite. The coefficient in the case of Favorite becomes .192 instead of .188 and that of Comet .471 instead of .469. Remote common ancestors in general have little effect on the coefficient. It will be noticed that Comet has a degree of inbreeding almost equal to three generations of brother-sister mating or an indefinite amount of sire-daughter mating where the sire is not himself inbred.

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BLOCK 2

2

From phenotype to breeding value

1933 to 1950

This block brings together the papers that turned the theory of quantitative genetics into selection procedures. Between 1933 and 1949, Lush defined heritability and breeding value and established how to weight the information from an animal and from its relatives. In 1943, Hazel and coworkers estimated genetic and environmental correlations, and Hazel presented the selection index for multiple traits. In 1950, Rendel and Robertson described the paths through which genetic gain is achieved in a breeding program. The concepts presented in this block remain in use.

Lush (1933)

Lush (1935)

Lush (1949)

Hazel, Baker and Reinmiller (1943)

Hazel (1943)

Rendel and Robertson (1950)

Lush (1933)

Lush, J.L. 1933. The bull index problem in the light of modern genetics. Journal of Dairy Science, vol. 16, no. 6, pp. 501 to 522.

The problem

In the 1920s and 1930s, several indexes were proposed to rank dairy bulls by the production of their daughters compared with that of their dams. Each index used a different formula, and there was considerable debate about which was correct. Lush analyzed these indexes in the light of Mendelian genetics. He showed that the daughter average must be regressed toward the population mean, because part of the observed difference is due to environment and chance, and that this regression depends on the number of daughters.

Points to note while reading

The paper presents, in simplified form, the principle that an estimate based on little information should remain closer to the mean. This shrinkage is exactly what BLUP does automatically, four decades later. Note also the clear distinction Lush makes between the phenotype of a cow and the breeding value she transmits.

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THE BULL INDEX PROBLEM IN THE LIGHT OF MODERN GENETICS*

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There is general agreement that information about a sire's offspring is important in estimating that sire's real breeding worth. There is not the same agreement on how to use that information. The purpose of the present article is to clarify the situation by calling attention to certain genetic principles which underlie the bull index problem.

GENETIC PRINCIPLES

1. *Inheritance is Mendelian in the broadest sense of the word.* This includes such modifications as linkage and sex-linkage. Inheritance is by particles or units which are present in pairs and which maintain their identity and later segregate out unchanged, instead of blending with each other as was thought in pre-Mendelian days. That there is in the higher organisms *any* inheritance which is not Mendelian in this broad sense of the word seems unlikely. Certainly there cannot be much. There may be a few minor exceptions such as plastid inheritance in plants or polyploidy but those things are rare or unknown in animals.

2. *The genes are not adaptively modified by their environment.* This statement denies the inheritance of acquired characters and like many other negative statements may be impossible of complete and rigorous proof. However, the many extensive and carefully conducted experiments which have failed to detect the inheritance of adaptive modifications constitute impressive evidence. In order not to be dogmatic, one should perhaps keep an open mind for new evidence on this point but it is almost incredible that environment should be able to modify the genes in an adaptive way to a degree important in practical animal breeding, and yet that such modifications should have escaped detection in the many experiments planned expressly to test that point.

3. *Observed yields are affected by environment.* Strictly speaking, the question of whether a characteristic is hereditary or environmental has no

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meaning because the genes cannot possibly produce the characteristic without the proper environment, and even in the proper environment the characteristic cannot develop unless the necessary genes are present. The characteristic is the end result of complicated interactions of genes among themselves and with their environments. There is no such thing as a disembodied gene or set of genes for a fat production of 800 pounds, for example. We may speak of a cow as being a 800 pound producer but it should never be forgotten that this applies to a particular set of conditions. Alter the conditions and the production will alter. The inheritance which would be for 800 pounds of fat under one set of conditions might just as well be called inheritance for 600 pounds under another set of conditions.

4. *The number of genes affecting each characteristic is large.* This is almost inevitably true for characteristics like milk production which are dependent on the combined functioning of many organs and organ systems and which might be raised or lowered by the altering of any one of many quite different physiological processes. The exact number of genes affecting milk and fat production matters little if it is much more than a half dozen pairs because then it will be physically impossible to learn the exact and complete genetic formula of any bull or cow.

The existence of a large number of genes and the general absence of intensive inbreeding within breeds has several other consequences:

A. No animals have *exactly* the genes the breeders desire but some have more nearly the ideal than others do. This makes it impossible to improve a herd or breed in all respects at once, simply by continually grading it up to a perfect individual, since the latter does not exist.

B. Entirely homozygous animals are so extraordinarily rare that the search for those to be used continually as sires is doomed to but partial success at most.

C. A high degree of homozygosity and the possession of a high proportion of the desired genes are uncorrelated, or nearly so. Indeed there is some evidence to show that on the whole heterozygosity rather than homozygosity is correlated with individual excellence although not necessarily with breeding worth.

5. *Gene frequency* (the proportion which a desired gene constitutes of all the genes which occupy that locus in the whole breed) is changed at a rate which would be appreciable within a breeder's lifetime only by selection; that is, by letting individuals possessing the desired genes leave more offspring than are produced by those lacking the desired genes. Mutation is so rare an event that the practical breeder need not take it into account. Random survival or extinction of genes is too weak a force to be important in a practical breeder's lifetime except in extreme inbreeding systems.

6. The *homozygosity* of a breed or other group is changed to any appreciable extent only as a result of changes in gene frequency, or (and much more powerfully) by some form of inbreeding or its opposite, the crossing of distinct strains.

7. *Genes often exhibit dominance.* This is not universal but it seems to be the usual situation, at least among genes for distinct differences in color and gross anatomy. The evidence is not so clear that either dominance or the lack of it is the usual condition among genes which have for their main effects quantitative change in a characteristic already present in some degree. There is no inherent tendency for dominant genes or recessive genes to replace each other in a population, except that undesired dominants are more exposed to the effects of selection than undesired recessives are. This has led to the general condition that undesired genes tend to be recessive and desired ones dominant, particularly if the traits which those genes affect have been the object of selection for many generations.

8. *Genes interact with each other.* Many genetic factors require the presence and cooperation of others in order to manifest their effects. These are known in genetics under various terms, such as inhibiting factors, complementary factors, epistatic factors, etc. They are most nearly summed up to the practical animal breeder in the term "nicking," although this term is ill-defined and often is used to include other phenomena such as rare Mendelian recombinations. The extensive inbreeding experiments so far carried out on animals and on corn indicate that there may be a considerable amount of such complex interaction among factors affecting size, growth rates, yield and such characteristics, which logically might be expected to be inherited in about as complicated a way as is milk production. There is not yet any very satisfactory basis for estimating whether such complex interactions among genes are the rule or are rare exceptions. Much of the gene interaction must be simply additive or multiplicative or else the observed daughter-dam and half-sister correlations could scarcely be as large as most actual studies have found them.

APPLICATIONS TO THE PROBLEM OF SIRE INDEXES

1. *Offspring average midway between parents.* The genetic fact which overshadows all others in its importance in the construction of sire indexes is that the genotypes of the offspring *average* midway between those of the two parents but any individual offspring may vary widely from that average. This is illustrated in the following example, where each capital letter represents a plus factor and the dashes indicate the unknown but probably very large number of factors omitted from the problem:

		PHENOTYPIC VALUES	
		a. No dominance. Each plus factor = 100	b. Complete dominance. Each factor pair = 150 if a plus is present
Sire:	*AABbCCDdee—Nn*	700	750
Dam:	*AaBbccDdEe—nn*	400	600
Offspring:			
Maximum	*AABBCcDDEe—Nn*	900	900
Minimum	*AabbCcddee———nn*	200	300
Average		550	675

The figures at the right show what the phenotypic values would be with certain arbitrarily assigned numbers to represent the effect of each plus factor. In both columns the average of all offspring is *midway* between the two parental values and in both the minimum offspring is below the poorer parent and the maximum offspring exceeds the higher parent.

2. *The consequences of dominance.* Those who take the trouble to juggle with such formulae as these in all possible Mendelian combinations will quickly find that *in the absence of dominance* the average phenotype as well as genotype of the offspring will be midway between the parents, *whatever the Mendelian formulae of the parents.*

If dominance exists the offspring still average midway between the parents wherever the average amount of heterozygosis among the offspring is the same as that of the parents. Where the offspring are more heterozygous than the parents (as in the first generation of a cross between two breeds) their average will be above the mean of the parents and might even exceed the higher parent (using "above" to mean in the direction of the dominant genes). Where the offspring are less heterozygous than the parents (*e.g.*, in intense inbreeding) they will average lower (*i.e.*, in the recessive direction) than the mean of the parents. For all practical purposes heterozygosis in a population *neither increases nor decreases* noticeably from one generation to another unless violent outbreeding at the one extreme or intensive inbreeding at the other are being practiced. Hence *in a population* bred as purebred dairy cattle are, the existence of dominance, even if complete, does not modify the general principle that offspring average midway between their parents.

The effects of dominance may be illustrated in another way by following the consequences of each of the six kinds of matings possible where a gene has only one allelomorph. Let $w = AA - aa$, that is, the difference between homozygous dominant and homozygous recessive phenotypes. Let the proportions of the two allelomorphic genes in the population be q *A* and $(1-q)a$.

Type of mating	Frequency of this mating in a population breeding at random	Average phenotype of offspring compared with midpoint of parents, if dominance is complete
1. $AA \times AA$	q^4	Equal
2. $AA \times Aa$	$4q^3(1-q)$	Equal
3. $AA \times aa$	$2q^2(1-q)^2$	Offspring higher by $w/2$
4. $Aa \times Aa$	$4q^2(1-q)^2$	Offspring lower by $w/4$
5. $Aa \times aa$	$4q(1-q)^3$	Equal
6. $aa \times aa$	$(1-q)^4$	Equal

The principle that offspring average midway between their parents, even though dominance is complete, is violated only by matings of types 3 and 4. In a population breeding at random, matings of type 4 are exactly twice as frequent as matings of type 3. The deviations of these two types of matings from the general rule exactly cancel each other. Close inbreeding will decrease the matings of type 3 more than those of type 4 (hence in part the decline which is a general consequence of inbreeding) while the crossing of distinctly unrelated lines will increase matings of type 3 much more than those of type 4. Other deviations from strictly random matings are of slight importance in altering the frequency of the different kinds of matings. Selection for the dominant diminishes matings of type 6 most and those of types 3 and 5 next. If the trait selected is influenced by many genes, such selection will have only a very faint tendency to alter the proportions of the types of matings, so far as one pair of genes is concerned.

Dominance does modify the consequences of *individual matings* slightly. If the parents are both phenotypic extremes in the same direction, their offspring will usually average a little nearer the mean of the race than the parents do. Thus if both parents are selected because they show the effects of an unusual number of dominant factors, they will have more than the normal proportion of gene pairs where the mating is of the $Aa \times Aa$ type and fewer matings of the $AA \times aa$ type. Conversely, if both parents are selected because they show the effects of few dominant factors, the opportunity for $Aa \times Aa$ matings will be reduced more than the opportunity for $AA \times aa$ matings.

Practically this effect will be slight and almost indistinguishable from the effect of random environmental variations which also lead to regression toward the mean of the race. (The "race" includes the interbreeding population which might in nature be a whole species or a geographically isolated portion of a species but in practical dairy cattle breeding would be the breed, since our pure breeding system isolates each breed from the rest of its species).

3. *The effects of environment.* Since the production of a cow depends upon the environment to which she is exposed as well as upon the genes she

has, any attempt to estimate whether her heredity was for high or low production will be subject to some error from mistaking the effects of environment for the effects of genes.

The larger the variation in environment to which the cows have been exposed, the larger will be the amount of error in estimating the cow's inheritance from her known production. Obviously, the ideal way to avoid this is to expose all cows to the same environment. It is equally obvious that this can never be done perfectly and cannot even be approximated when the comparison is between records made in herds under different management and at different ages.

(a) *Corrections to standard conditions.* The next best thing to having the records made under uniform environment is to estimate what each record would have been under standard conditions. Of course correction factors do not exactly allow for the effect which a difference in conditions had on all individual records. Perhaps it is true on the average that milking three times a day, instead of two, increases production twenty per cent, yet not all individual cows will respond exactly this much to the extra milking. A set of twice-a-day records corrected to a thrice-a-day basis will differ individually at least a little from the records which would have been obtained if these same cows had actually been milked three times a day, while every other environmental condition remained rigidly the same. Yet if the condition for which correction is being made is at all important, the use of a well-based correction factor to estimate what the record would have been under standard conditions, will remove much of the effects of differences in environment which would otherwise bias any conclusions to be drawn from the data. Such corrections should be used in spite of the fact that occasionally a "corrected" record will be more in error than if it had not been corrected at all.

Practical considerations limit the number of corrections which can be made. Ordinarily only a little accuracy would be gained by correcting for other than the most important circumstances such as age, times per day milked and (if such can be managed) a general estimate of the conditions of feeding and management. The preceding dry period may need to be taken into account if it was unusually short. Lactations incomplete through no fault of the cow or lactations terminated very early by recurrent pregnancy would need correction to standard length.

The choice of the conditions which shall be used as "standard" should be guided by the general principle that animals should be tested under conditions the same as those under which they are to be used. Thus the recent Holstein-Friesian decision to correct to thrice-a-day milking as a standard instead of four times is sound since few cows will be milked four times a day commercially while many are milked three times daily for purely commercial reasons. The standard conditions, including the ration fed, should

be good enough that few cows are seriously limited by those conditions. Beyond that level there is little use in forcing the cows. Indeed there is the certainty that some cows will respond to such forcing more than others without there being a corresponding difference in what they could produce under the commercial conditions for which after all they are being bred.

(b) *The effect of unknown differences in herd environment.* Cows tested in the same herd will tend to have been exposed to the same environment and management. If this environment is any better or any worse than is thought when the corrections for it are made, all the records from that herd will thereby be biased in the same direction. Increasing the number of records from that herd does not correct for this. This source of error is in all records and in general is more important the less we know about the management of the different herds.

This is perhaps the most difficult part of the problem which confronts one who attempts to rate large numbers of sires on the basis of published official records or C. T. A. records alone. A certain amount of information is gained by such estimates but one would often be mistaken in individual cases and the accuracy of the estimates could not rise to a very high level. Increasing the number of tested daughters beyond 3 or 4 under such conditions adds but little to the accuracy of the estimate of the sire's *genotype*, although it may add materially to the accuracy of an estimate of what other daughters *in that same herd* will produce.

(c) *The effect of random environment.* Random environment which affects one cow but not her stable mates (and much of which will be unknown even to the breeder himself) has as its principal effect (as in the case of dominance) a regression toward the mean of the breed.

The effects of random environment may be minimized by increasing the number of daughters included in the progeny test and by seeing that the records used are not selected records but are the records made nearest a given age or are selected by some other method not based on the magnitude of the record itself.

The effects of random environment on the record of any one cow are most effectively minimized by using an average of all her records, excluding only those *definitely known* to have been abnormal (as from abortion or other definite illness). Because of its effects on the standard deviations this averaging is not always permissible in statistical studies. That reason naturally has no bearing on individual selections made in breeding practice.

4. *Differences in the merit of the dams.* If each herd could be regarded as a random sample of the cows in the breed, the average merit of the herds would not be exactly the same, just as samples from the same source are not exactly alike, yet the differences would not be important. In such a condition and with the simplest assumptions about the genetic situation, the records of three daughters out of untested dams would furnish as much infor-

mation as the records of two daughters out of tested dams. With a more complicated genetic situation (as no doubt exists) the ratio would not be exactly this but the general principle would still be true that the errors introduced by the absence of the dams' records could be balanced by an increase (perhaps *about* a 50% increase) in the number of daughters tested.

But it seems certain that herds really do differ from each other more than if they were random samples of the breed. The error thus introduced into a sire index by neglecting the records of the dams of his daughters cannot be overcome by increasing the number of daughters tested. Instead it is a steady bias which tends, as the number of daughters increases, to stabilize itself not at zero but at a fixed proportion (depending on how the index is constructed) of the difference between the average merit of the dams in that herd and the average merit of the breed. Any index which does not take the records of the dams into account rests implicitly on the assumption that the dams are average cows of the breed in merit.

5. *The effect of "nicking."* Although we do not know how general this form of gene interaction is, we do know that its effect is to make estimates of the sire's breeding value less reliable than if all genes acted either in an additive or a multiplicative way to produce their effects, almost regardless of what other genes were already present. On the whole the effects of nicking will be similar to those of random environment and of dominance, but the misleading effects of nicking will not be as steadily overcome by increasing the number of offspring in the sample tested.

One more practical point about "nicking" does deserve mention. The more important it is, the more one would be anxious to continue to use a good bull on cows closely related to those with which he was mated when he produced the offspring which proved him. Any bull apparently proved poor by his first offspring but which because of excellent pedigree, appearance, or for any other reason was still thought worth continuing in use should be used on cows as unrelated as possible to those whose offspring apparently proved him to be a poor sire. If nicking is important the matter of line-breeding and staying within certain family lines is considerably more important than if nicking is really a rare and comparatively unimportant phenomenon.

6. *Number of daughters.* Each daughter receives from her sire a sample half of his inheritance. Since this is a sampling process, it may happen that no one daughter receives from her sire a sample which is exactly typical of him. As the number of daughters is increased, the average of what they have received from their sire corresponds more and more closely to being exactly equal to half the genes he had. This follows the well-known laws of random sampling (provided all daughters or at least a truly random sample of them are tested without selection). The error in estimating the sire's genotype from a knowledge of what he has transmitted to his daugh-

ters can be reduced to an unimportant amount by increasing the number of daughters tested.

The error from Mendelian sampling is such that (if it alone is considered) one daughter is as accurate an indication of a bull's worth as either his dam or his sire is. Three daughters are as accurate as full knowledge of both his sire and dam. More than three daughters give more information than could possibly come from the completest kind of study of his pedigree.

Increasing the number of daughters decreases the error from random environment as effectively as it decreases the errors from the sampling nature of inheritance.

Increasing the number of daughters decreases the error from dominance and from "nicking" but not quite as regularly and uniformly (especially with the latter) as in the case of inheritance sampling.

Increasing the number of daughters does *not* decrease the error from unknown or uncorrected differences in herd environment, or from differences in the average merit of the dams.

Therefore the answer to the question of how many daughters are necessary to prove a sire depends upon how sure of his "proof" one wants to be and how much can be gained by testing more daughters. If the herd environment or the average merit of the dams are important and have not been taken into account correctly, the accuracy of the sire rating will remain low under any circumstances, and not much is gained by increasing the number of tested daughters past three or four. On the other hand, if herd environment and average merit of the dams are correctly discounted and if random environment is important, the sire rating may be considerably increased in accuracy by increasing the number of daughters even past eight or ten. Five is rather widely used as a minimum number where the dams' records are also taken into account and this seems to the writer worth adopting to avoid confusion wherever a formal definition of a minimum must be made. Increasing the number of daughters never detracts from the accuracy of the sire estimate and under some conditions may increase it materially. Hence such an increase is always to be desired in individual cases. In the present state of our knowledge it seems to the writer undesirable to *publish* any sire index based on fewer than five daughters (more than that if the dams are not taken into account). Nevertheless even the first daughter provides at least a faint indication of a sire's worth.

7. *Multiplicative and additive factors.* The three kinds of dairy records, milk produced, fat produced and test, cannot be independent. One is the product of the other two. But which one? The "Mount Hope Index" expresses the view that milk produced and test are independent. The bulk of the evidence tends more toward the energy basis of measuring yield

as advocated by Gaines and co-workers at the Illinois Station. According to this view the inherited differences are differences in the rate at which the cow can transform energy. If the energy appears in the form of fat it automatically cannot appear as solids not fat. Other inherited (and environmental) factors determine whether the test shall be high or low.

It is possible that neither extreme view is absolutely correct. The author's tentative solution of the problem (admittedly imperfect but usable) is to regard the total fat and the test as independent, with the amount of milk being a function of those two. This is almost the same as the energy yield basis since most of the variation in energy yield is a result of variation in the fat yield.

If all factors affecting test, milk and fat were strictly multiplicative in their action the difficulty could be avoided by using logarithms instead of actual numbers for the observed tests and yields. The index would come out practically the same regardless of whether it was figured for test, milk yield or fat yield. Since most people are unfamiliar with logarithms and it is unlikely that *all* these relations are multiplicative, it seems best to apply the index to the two kinds of records which are most likely to be independent and to figure the third from the two resulting indexes. Those who use "fat-corrected-milk" would naturally apply the index to those figures and calculate a separate index for test if they were interested in that.

PROPOSED INDEXES

Space forbids a detailed review of any but the most satisfactory or widely publicized of the indexes so far proposed. No one is entirely satisfactory and yet the weaknesses of one are not exactly the same as the weaknesses of another. With one exception, all are based on the daughter average as a starting point. They differ in the use they make of the increase or decrease of daughters over dams. The exception is the proposal to rate sires solely on the increase or decrease of their daughters over the dams. That proposal has so many theoretical and practical unsoundnesses that it no longer seems worth discussion.

The others may all be expressed in the equation: Sire index = Daughter average + k (Daughters' average minus Dams' average). The indexes differ in the value they assign to k.

1. *The daughter average.* When k is zero, the index becomes simply the daughter average. This is theoretically correct[1] if one wishes to predict the performance of future daughters out of the *same kind of cows* and reared and tested under the *same* herd environment as the daughters whose production is already known. This is just like any other case in which an experiment is repeated. The most probable result is the same as that of the

[1] Wright, Sewall. On the evaluation of dairy sires. Proc. Amer. Soc. An. Prod. 1931: 71–78. 1932.

first experiment. Because these conditions are so largely fulfilled in actual data, those who have sought to compare from actual data the predicting value of various indexes have in general obtained favorable results with the daughter average as compared with the various indexes against which they have tested it.

The special weakness of the daughter average as a sire index is that it makes no allowance for differences in the merit of the cows to which different bulls are bred. This has two consequences. First, the error thus introduced into a sire estimate tends toward a constant value instead of tending toward zero as the number of tested daughters increases. Second, even if the dams were random samples of their breed, something like fifty per cent more tested daughters out of untested dams would be required for the same degree of accuracy as if the dams had been tested too.

Errors due to unknown or improperly corrected herd environment are relatively larger in the daughter average than in those indexes which also take into account the increase or decrease of daughters over dams tested under similar conditions.

In addition to its simplicity the daughter average has the advantage that it utilizes the information about all the daughters of a bull regardless of whether their dams were tested. Moreover it is harder to "fake" by selection of the records used than is an index involving a comparison of daughters and dams.

The daughter average will necessarily be less variable than the actual records of the cows. This follows the well-known statistical relation between the standard deviation of an average and the standard deviation of its constituents. Where

$$A = \frac{X_1 + X_2 + \cdots + X_n}{n},$$

and ${}^{\sigma}X_1 = {}^{\sigma}X_2 = {}^{\sigma}X_3 = \cdots = {}^{\sigma}X_n$ approximately, and

$${}^{r}X_1X_2 = {}^{r}X_1X_3 = {}^{r}X_2X_3 = \cdots = {}^{r}X_mX_n$$

approximately, this may be expressed as follows:

$${}^{\sigma}A = {}^{\sigma}X\sqrt{\frac{1 + (n-1)\,{}^{r}XX_n}{n}}$$

With a correlation of + .30 between paternal half sisters (about what has been found in official records), ${}^{\sigma}A$ when n is 5 or more, would be only about 5/9 to 2/3 as large as ${}^{\sigma}X$, the exact fraction depending on the size of n. With a smaller correlation than + .30 the decrease in the variability of the average would be even more extreme.

Thus a daughter average of 800 pounds of fat for a sire may be as rare and hard to obtain as an individual record of 1000 pounds of fat for a cow. The use of the daughter average as an index will require some kind of a mental translation to allow for the smaller variability of the sire index

when comparing bulls with cows. This is not a defect in principle but it is an inconvenience whenever it is necessary to compare the merits of female with those of male ancestors. This diminished variability is more extreme with the daughter average than with the indexes which also take into account the difference between daughters and dams.

LaMaster's proposal[2] to index bulls by the percentage which the daughters' production is of the average production of all cows tested in the same classes differs from the daughter average only in that he avoids the necessity of correcting for age, and for the management conditions by assuming that those things are equal for all cows grouped in any one class. The dams' records might also be expressed as percentages of the breed average, thus making it possible to consider increase or decrease of daughters, too. However, LaMaster's "production factor" does not combine in one index the daughter performance and its increase or decrease over dam's performance. LaMaster's "production factor" is subject to the objection that the value of the sires would not without an additional computation be expressed in pounds of milk or fat as the production records of the female ancestors usually are, although this is only a minor inconvenience in its application and does not invalidate its principles which are the same as those of the daughter average. La Master's "production factor" has the practically important advantage that its expression as a percentage gives the reader a quick picture of whether the index is relatively high or low without it being necessary for him to carry in mind actual class averages or measures of variability.

2. *The equal parent index.* Since the offspring tend to average midway between the parents, inheriting equally from them both, the simplest index which would include those facts is to add to the daughter average the increase of the daughters over their dams. That is, k in the general equation has a value of one. This is the index proposed by Nils Hansson in Sweden nearly twenty years ago. T. E. Woodward used it in 1922 on Guernsey fat records. It is the same index applied by Yapp in 1924 to "fat-corrected milk." The much publicized "Mount Hope Index" becomes simply this when it is freed from a misconception about the effects of dominance. (The index at present used by the Holstein-Friesian Association in its Herd Improvement Registry data under the name of "Mount Hope Index" has in fact been freed from this misconception about the consequences of dominance which distinguished the original Mount Hope Index from the general expression of this same idea.) That the offspring average midway between the parents is fundamentally the same idea as that introduced into genetic literature many years ago by J. Schmidt of the Carlsberg Laboratory under the term "diallel crossing," except that Schmidt's material permitted mat-

[2] LaMaster, J. P. Production factors applied to Jerseys. The Jersey Bulletin 51 (22): 763–764. 1932.

ing each of several different males to the same females. Because the females were mated to each of the males, he avoided the necessity of assuming anything about the genotypic values of the females. His findings were that the genotypic values of the sires differed by twice as much as the difference between the phenotypic averages of their progeny. R. A. Fisher in his book on "The Genetical Theory of Natural Selection" endorses as a sound general principle this finding from the experiments on diallel crossing. Wright (*loc. cit.*) shows that this index is a statistically valid means of estimating the sire's genotype for production but subject of course to a sampling error. The idea is foreshadowed in much of Galton's work on human inheritance where he studied the "parental mid-point," although most of his studies were on traits which could be manifested by both sexes and therefore he was not in much need of an actual "index." The idea of the equal parent index is such a simple one that it is probably as impossible as it is unnecessary to trace its origin or to couple it with the name of any one man. It seems to the writer that the term "equal parent index" is simple enough, accurately descriptive enough and free enough from implications about priority that this term might well come into general use. (Since this was written Professor V. A. Rice has in several articles urged the use of the term "intermediate index" which also seems to the writer accurately descriptive and free from implications about priority. Probably the name used is not important except that confusion should be avoided. Presumably the name ultimately in common use will be decided by the "survival of the fittest"!)

The advantage of the equal parent index over the daughter average is that the former removes the errors (except the effects of random environment on the dams' records) arising from differences in the productiveness of the cows to which each bull was bred. Also the range of the equal parent index is more nearly (though perhaps not exactly) like that of cows' actual records. This makes translation from actual records of cows to indexes of bulls and back again in pedigree study less necessary.

The weaknesses of the equal parent index are (1) that it is more easily "faked" by selecting the lowest records of the dams and the highest record of the daughters in case the one computing it deliberately seeks to deceive, (2) that if the herd environment of dams is different from that of the daughters, and if corrections for that are not entirely accurate, the effects are put down as effects of the sire's genotype and (3) that it doesn't use information about tested daughters out of dams which for one reason or another were not tested.

Weakness No. 1 might become important if the index should come to have much commercial value but of course does not exist for a man who is interested in finding out for his own use how good his own bull is. No. 2 will not often be an important source of error where the records were made

in the same herd, but the possibility should always be kept in mind that it may have been important in any one case. If the daughters' herd environment is correlated with that of the dams as highly as +.25, herd environment will be relatively a less serious source of error in the equal parent index than in the daughter average. If much culling of cows on the basis of their production records has been practiced, there will be some tendency for the dams to be only the selected survivors of their generation, while the daughters represent their whole generation. This will introduce a slight bias because (on account of this culling) the dams' records will average a little above the dams' genotypes, even when the records are fairly selected, whereas daughters' records will not be thus biased. As for objection No. 3, there would probably be little error introduced by assuming that the untested dams were equal in merit to the tested ones. Whatever *bias* there is in this assumption would tend to give the sire a lower index since the untested dams might be expected on the whole to average a little lower than the tested ones if they differ systematically at all. Those who may think that such an assumption (*i.e.*, that the untested dams would average the same as the tested ones) is a very rash one may be reminded that the use of the daughter average as an index rests upon the assumption that each group of dams is an average sample of the cows of the breed.

3. *Norton's index.* H. W. Norton, Jr., has tentatively suggested an index outwardly similar to the equal parent index, except that it takes into account the average amount of regression toward the mean of the breed which has occurred in the past, without bothering to inquire into the causes of that regression. Mr. Norton has calculated from a large number of Holstein-Friesian Advanced Registry records the average production by daughters out of cows of each level of production. This he calls the "expectancy for daughters" from dams of such a level. His index differs from the equal parent index only in that he substitutes this "expectancy" in place of the average record of the dams. Thus if a sire is mated to dams averaging 850 pounds, and produces daughters which average 810 pounds, Mr. Norton would turn to his expectancy table which shows that, in the breed as a whole, dams averaging 850 pounds have produced daughters averaging 718 pounds. This would be the "expectancy" for daughters from the particular group of dams. This particular sire's daughters exceeded their "expectancy" by 92 pounds. Adding this difference to the average of the daughters would get for this bull an index of 902 pounds.

This proposal has the merit of discounting the regression actually observed, so that in this respect the index figure will be directly comparable with cows' records. Likewise it has the advantage (for teaching purposes at least) that to most people it will seem to be based less on theory and more on actual data with a minimum amount of deduction.

The "expectancy" figure corresponds closely, except for a little curvilinearity at the extremes, to the equation: Expectancy $= \frac{\text{Dam}}{3} + \frac{2}{3}$ breed average. Thus, Sire index = Daughter + $\frac{1}{3}$ (Daughter-Dam) + $\frac{2}{3}$ Daughter-breed average). This index is therefore a mixture of about one-third equal parent index and two-thirds daughter average, the latter having had its range expanded so that it would be about the same as the range of cows' actual records. Apparently this index does not discount the error from genuine herd differentiation in the dams' merits as completely as the equal parent index does. Also it seems more vulnerable to errors from the daughter's herd environment being better (or worse) than is thought when their records are being corrected to standard conditions. On the other hand, because it takes into account the actual regression, including whatever there may be from dominance, nicking, random environment, and selection of dams, it probably is a little fairer to unusually good bulls mated to unusually good cows and to unusually poor bulls mated to poor cows, giving them in both cases an index rather fairly comparable with the actual records of cows. It might be especially suitable for use with official records where there has been considerable selection of cows to be tested, although no index can discount completely the effects of such selection.

An index of this type may deserve further study. But before such an index could be applied to other data than Holstein-Friesian Advanced Registry it would be necessary to prepare similar "expectancy" tables for each population concerned.

4. *Indexes with values of less than one for k.* Studies af actual data in an attempt to derive from them empirically an index which will predict with the minimum standard error of estimate the breeding value of a sire, may be expected generally to result in a formula which is a compromise between the daughter average and the equal parent index, that is, with k having a positive value between zero and 1.0. For example Turner's study of Guernsey sires (Missouri Research Bulletin 79, especially page 37) finds a value for k of 3/17. This general result is to be expected because future daughters of a bull would usually be reared in the same herd and out of the same kind of dams as his first daughters. The exact point of compromise (that is, the exact value of k) would largely depend upon the extent to which the dams of the early daughters of a sire really differed in merit from the dams of the later daughters, and on the extent to which the first daughters and the later daughters were in the same herd.

5. *Prospects of deriving a satisfactory index by simple statistical treatment of existing data.* There seems to be no prospect of settling the question of which index is the most accurate by appealing directly to actual data such as exist in official records or C. T. A. records. It seems impossi-

ble to exclude from such data all the effects of selection on the dams and daughters tested, the records used, and which sires are given opportunity to sire large numbers of daughters. More important still, such data are affected by the differentiation from herd to herd both in management and in the average merit of the cows. Even in single large herds the management and average merit of the cows vary from time to time.

The complexity of the problem is shown by figure 1, which is a graphic scheme for showing why correlations may be high or low under different circumstances. The square of a correlation coefficient measures the portion of the variance in one variable which disappears when the other is held constant. Thus squared correlation coefficients between an index and the sire's genotype (the thing it is intended to measure) may be pictured as fractions of which the denominator is all the variance in the index for a large population of bulls and the numerator is that portion of the variance which genuinely results from differences in the bulls' genotypes. Fraction

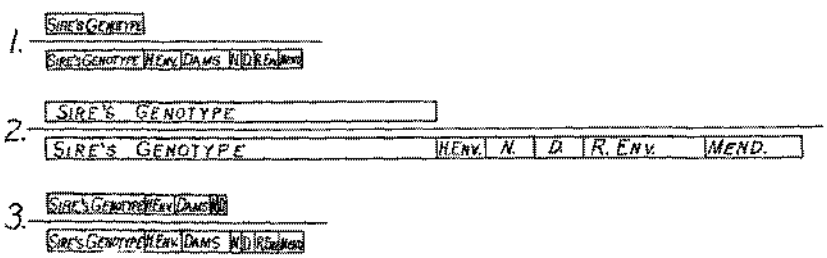


FIG. 1. Scheme for showing why certain indexes are more accurate than others. Each squared correlation is represented as a fraction, the denominator being the total variance in the index and the numerator being that portion of the variance arising from the thing which is to be predicted.—1. The squared correlation between the daughter average and the sire's genotype. 2. The squared correlation between the equal parent index and the sire's genotype. 3. The squared correlation between the average of a sire's first daughters and the average of his later daughters. Compare 1 and 2 or 1 and 3.

No. 1 represents this for the correlation between sire's genotype and daughter average, No. 2 represents the same thing for the equal parent index and No. 3 shows why the correlation between a sire's first daughters and his later daughters is so high in most studies of official records and C. T. A. data. The proportions of the different segments of the denominator are but rough approximations, the data to measure those proportions being so lacking in some cases that the proportions pictured scarcely deserve to be called more than guesses. The abbreviations are:

H.Env. = Herd environment or conditions of management which differ from herd to herd but which affect alike all the cows in the herd. So far as those effects are removed by the use of correction factors, this element would disappear from the

problem. Hence it will be more important in some groups of data than in others.

Dams = Differences in the average merit of the dams to which the sire was bred.

N. = Effects of complex gene interaction or "nicking."

D. = Effects of dominance.

R.Env. = Random environment which affects one cow but not her stable mates and is not removed by appropriate correction factors.

Mend. = Effects of Mendelism or chance in the sampling of the parental inheritance when the gametes are produced.

Comparing No. 1 and No. 2, the contribution of the sire's genotype to both numerator and denominator is much increased in the latter, in accordance with the principle from diallel crossing that the difference between the offspring is only half the difference between the sires' genotypes if the groups of dams are of equal merit. Since these fractions represent squares, the denominator of No. 2 is four times as long as that of No. 1 in all its constituents except the merit of the dams, which does not appear in No. 2 at all, and the herd environment which, if identical for daughters and dams, will be of the same size in No. 1 and No. 2. If herd environment is not the same for daughters and dams, it will contribute more to No. 2 than to No. 1, though not four times as much if the correlation between daughters' herd environment and dams' herd environment is as high as +.25. The absolute size of random environment is increased about five times in No. 2. In comparing No. 2 with No. 1, the range in absolute units is greatly increased, the effect of differences in the average merit of the dams is eliminated and relatively the effects of inadequately corrected herd environment become less, while the effects of random environment become a little larger.

Increasing the number of daughters tested decreases R.Env. and Mend. steadily in proportion to the number. N. and D. are also decreased by this but not so regularly nor effectively. H.Env. and Dams are not thus reduced. Figure 1 is drawn according to the writer's estimate of the proportion of variance from each source in indexes based upon about 5 to 8 daughters and on Advanced registry data where no corrections could be made except for age, times milked per day, and date of beginning of next pregnancy.

No. 3 shows graphically why the first daughters of a sire are usually so closely correlated with his subsequent daughters. Much of H.Env. and Dams and even some of the N. and D. is repeated in the numerator. The numerator is therefore far from representing the sire's genotype alone.

It is the writer's present opinion that the most profitable research work which can be done with official records or similar data at the present stage of our knowledge is to measure more accurately than in the rough approximations shown in figure 1, the relative importance of the different causes of

variance in production records. Perhaps if such an analysis is carried very far it may be advisable to sub-divide some of these terms or even to add terms for cross correlation between them as, for example, if sires with the best genotypes tend to be used in those herds where environment is best and *vice versa*. These diagrams may appear complicated but they must be simplicity itself compared to an accurate detailed picture of all the interactions of genes, environment and breeders' practices which together cause a given set of actual records to be what they are.

In this connection, critical reference should be made to two fallacies which (by unintentional inference, of course) have come into the literature on this subject largely as a result of the work done by those who sought to check the accuracy of indexes by a direct appeal to the official records.

The first is the inference that the daughter's record is in some way a "genotypic" measure of the sire whereas the dam's record is a "phenotypic" expression of the dam's genotype and therefore not so reliable. If daughters' and dams' records are both used without selection, the daughter's record of course stands in exactly the same relation to her genotype as the dam's record stands to the dam's genotype. In estimating the dam's genotype from her record there is always the possibility of error from mistaking the effects of unknown or inadequately corrected environment for the effects of genes. In estimating the sire's genotype from a daughter's record there is exactly the same possibility of error plus the additional sampling error of Mendelian inheritance whereby this daughter may happen to have received better (or worse) inheritance from him than he would have transmitted to an infinite number of daughters. The only respect in which the daughter average can become more "genotypic" than the dam's record is the purely statistical fact that increasing the number of daughters decreases the errors from random environment and from Mendelian inheritance. If they are decreased enough by having many daughters in the average, the combined error from them may become less than the error from random environment in a dam's record. But if the number of dams is increased also, the error from random environment diminishes there, too. For example if we have 10 tested daughters out of 10 tested dams, the average of the dams' records is a more accurate index of their average genotype for production than the daughter average is of the sire's genotype because the latter still contains at least a small amount of Mendelian sampling error not in the former. Most of the confusion on this subject arises out of failure to distinguish clearly between fundamental biological or statistical relations of single records and the purely statistical consequences of averaging these records in various ways.

The second fallacy arises when the cows mated to one bull are divided into groups on the basis of their known records and the average production

of the daughters of each group is calculated. This is legitimate when each cow is considered as an individual and one wishes to find the regression of individual daughters toward the mean of the breed. It is not legitimate to infer that this is the same situation as if one bull were mated to three separate herds of the productive levels represented by these dam's averages. The manner of division into the three groups insures that the effects of poor random environment will be much more prominent in the low group and that there will be many more effects of good random environment in the high group than if the cows had been assembled on any other basis than the *ex post facto* knowledge of their previous records. Much of the random environment is thus turned into biased environment. Naturally the regression toward the mean of the herd is extreme. If one really wished to pursue this sort of investigation far the legitimate procedure would be to make the division as is done, then to discard the dams' records on the basis of which the division was made, and then to use as the productivity of the three groups of dams, the average of their records in the next lactation of each. These averages of following lactations would be legitimately comparable with separate herds of high, medium and low production. Naturally data suitable for such a study and free from pronounced selection are not available now, although perhaps they may be when the Herd Improvement Tests have been in operation longer.

The most efficient statistical method for exploring the average relation between dam and daughter among the daughters of a single sire is to calculate a daughter-dam correlation on the whole population, then correct that for heterogeneity on account of the different sires represented.[3] The calculation is easy but the interpretation is difficult because when the sire is "held constant," so, too, is the herd environment and part of the variance in dam's merit, thus leaving random environment a relatively important rôle to play. It therefore seems impossible to assess the sire's importance by comparing the daughter-dam correlation in the breed as a whole with the average daughter-dam correlation among the daughters of individual sires, since so many other things besides the sire differ in the two correlations.

To the extent that the investigations of actual records purport to show that the dams' records are unimportant, they have succeeded only in showing that those records (and the daughters' records, too!) are importantly

[3] Richey, F. D. and Willier, J. G. A statistical study of the relation between seed-ear characters and productiveness in corn. U. S. D. A. Department Bulletin 1321. 1925. See especially pages 7 and 8. The method here given would not take into account the degrees of freedom lost by treating each sire as a unit but that is important only in establishing the statistical significance of the resulting correlations, and may be had from Fisher's method of the "analysis of co-variance" (Statistical Methods for Research Workers, 4th edition).

affected by environment. To the extent that in some of these investigations the sire seems more important than the dams, it is only that the sire is being credited with many of the effects of herd environment and with many of the herd differences in the dam's merit too! No doubt there is a smattering of sex-linkage in the problem, too; but that can hardly be large enough to be detected in view of the magnitude of the other sources of error in the present data.

PRACTICAL CONSIDERATIONS

1. The index should be as simple as is consistent with accuracy and usefulness. The daughter average is the simplest of the usable indexes but the equal parent index is almost as simple and is considerably more accurate if herds differ much in the average merits of their cows.

2. For convenience in use the bull index should be expressed in the same terms and with nearly the same range or standard deviation as the actual records of the cows, so that in comparing bulls and cows there would be little need for a mental translation from one kind of unit to another. The daughter average has a much narrower range than the cows' records and narrower than any of the other usable indexes. This could be overcome roughly (if it were thought worth the effort) by increasing the daughter average by some fifty to seventy per cent and subtracting the breed average from that figure. The range of the equal parent index would probably not be very different from the range of the actual records, although that would vary slightly with the number of daughters tested as well as with the correlations between herd mates, between half sisters and between daughter and dam in the data used.

3. The index should be available on as many bulls as possible and for each bull it is desirable to use all possible information. This is the strongest feature of the daughter average as an index. There should be a place for using information about the sire's sisters, dam and other female collateral relatives and ancestors. None of the present indexes do that.

4. The records used should be absolutely unselected or at least selected by some method which is not at all influenced by the magnitude of the records. This applies both to the selection of daughters to be tested and to the selection of records used when a cow or her daughter have more than one record. One of the strong points of the daughter average is that it is more difficult to load the dice by selecting high records in one place and low records in another. The plans with the least bias seem to be the use of (1) either the very first record which either animal has made, (2) the record made nearest maturity regardless of the magnitude of that record, or (3) records made in the same year, the choice of years used to be based upon one of the preceding plans and not to be influenced at all by the magnitude of either record. The first rule has the advantage of making the

final index figure available while the sire is younger than under the other procedure. The use of the record nearest maturity has the advantage of avoiding some of the individual errors in age correction factors but may contain errors from some permanent damages which may have been done to the producing ability of the cow while she was still young. For example, one or more quarters of her udder may have been destroyed by infection and this would be unfairly charged against her hereditary worth unless another correction were made for that fact. The third plan goes far toward keeping differences in herd environment from affecting the daughter-dam differences.

If indexes become popular and acquire any considerable commercial value, the matter of insuring that the records used shall not be selected by some biased methods, will become quite important. This will be more important if the index eventually adopted is one that takes into account the dam's record than if it is the daughter average alone.

5. *Index recommended.* The writer recommends the equal parent index for general use since it most nearly complies with practical considerations and is soundest in principle. Its greatest vulnerability is in the possibility that daughters and dams may not have been tested under the same conditions and that it may be distorted so much by deliberately biased selection of the records used. A minor weakness is its failure to utilize information about daughters out of untested dams. Information about daughters of untested dams can be used (but with somewhat less confidence) by assuming that their dams would have averaged the same as the tested dams. The equal parent index can be applied to any other characteristic which can be measured or scored with usable accuracy in the cows or their daughters.

This is not to say that the daughter average is useless as an index. Far from it. The daughter average is the first step on which the usable sire indexes are all built. But it neglects differences in the merit of the dams and is rather sharply affected by herd environment. There seems no reason for not correcting those two defects wherever the increase or decrease of daughters over dams is known so that such a correction can be made.

SUMMARY

The genetic principles affecting the validity of bull indexes are reviewed.

All the indexes accurate enough to deserve further consideration start with the average production of the daughters as a basis. They differ only in the use and emphasis which they make of the difference between daughters and dams.

Differences in herd environment affect all indexes since they enter into the daughter average which is the base of all indexes. They are less important in the equal parent index than in the daughter average if the cor-

relation between herd environment of daughter and of dam is as high as 25.

Differences in the average genetic merit of the cows to which the bull was mated are neglected in the daughter average but are discounted in the equal parent index.

Errors due to random environment and to the part played by chance in inheritance are reduced by increasing the number of daughters tested. Errors due to dominance and to "nicking" are also thus reduced but not so effectively. Errors due to herd environment and to differences in the average merit of the dams are biased and do not trend toward zero as the number of daughters tested is increased.

No index will give infallible results. The sources of error cannot be absolutely eliminated. All that can be done is to minimize them as much as possible. The more one knows about the conditions under which the records were made, the more he can reduce these errors. Even if all other possibilities of error were overcome, there would still remain the sampling error intrinsic in the Mendelian nature of inheritance.

The writer recommends the equal parent index (the daughter average plus the increase of daughters over dams) as soundest in principle, simple enough for field use, freest from systematic error, and having a range not very different from that of the actual records of cows.

Lush (1935)

Lush, J.L. 1935. Progeny test and individual performance as indicators of an animal's breeding value. Journal of Dairy Science, vol. 18, no. 1, pp. 1 to 19.

The problem

The choice of breeding animals can be based on the performance of the animal itself or on the performance of its progeny. Lush used Wright's path coefficients to compare the two sources of information quantitatively. He calculated how many offspring are needed for the progeny test to be as reliable as the animal's own record and showed that the answer depends on the heritability of the trait.

Points to note while reading

Note the path diagram and how Lush derives the accuracy of each source of information. For traits of low heritability, a few offspring already outweigh the animal's own record. This reasoning still underlies the progeny test and the decision about how much information is needed to rely on an evaluation.

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PROGENY TEST AND INDIVIDUAL PERFORMANCE AS INDICATORS OF AN ANIMAL'S BREEDING VALUE

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It has long been known by breeders that sometimes a few offspring indicate an animal's real breeding value more accurately than the animal's own appearance or performance does. The rediscovery of Mendelism and the discovery a little later (by Bateson and others) of complementary genes and of other epistatic relations among non-allelomorphic genes, provided a simple explanation for this. Occasionally there are cases where a single offspring furnishes more convincing proof of its parent's genotype than could ever be obtained by studying the parent itself. So far as color is concerned, this would be the case with a purebred red calf born in one of the black breeds of cattle.

These are special cases, deserving attention to be sure, but perhaps they are more significant as *exceptions* to a general rule than as *examples* of one? Some writers on animal breeding topics have considered them to be examples. For instance we are sometimes told that a progeny test is "genotypic" and therefore dependable, whereas a cow's own record is "phenotypic" and therefore not dependable. But no fundamental distinction is really involved, since the progeny test itself consists of some kind of an average of "phenotypic" records of the offspring. To suppose that the process of averaging would purge these phenotypic records of all their error and would bring out only the pure gold of genotypic truth seems to require too much faith in the potency of arithmetic!

The present paper presents the results of an inquiry into the biometrical relations which determine the relative accuracy of progeny test and parent's own performance as indicators of the parent's real breeding value. The complete answer contains many variables but most of these have little influence. An answer accurate enough for practical purposes can be presented in usably simple form. For convenience the problem is phrased in terms of dairy cattle breeding thus: Is a cow's own record or the average record

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of n of her daughters the more dependable indicator of the cow's own breeding value?

METHODS USED AND DEFINITIONS

These biometrical relations were explored by Wright's method of path coefficients (10). Figure 1 shows those relations. The small letters indicate path coefficients which measure (provided the cause and effect relationships are portrayed correctly in the diagram) the importance of the causes of variation in the effect. The path coefficient (h) from heredity (H) to observed characteristic or yield (C) is the ratio of the standard deviation of C due to H to the total standard deviation of C. The path coefficients h and e may be used to express the relative importance of heredity and environment on production records in a particular population of data. Because variation is measured by squares of deviations h and e will not add up to unity. Where there is no correlation between heredity and environment, the squared path coefficients, h^2 and e^2, will come near to expressing "the relative importance of heredity and environment" in the form in which that idea is ordinarily encountered in practical problems. A definite numerical value for h^2 is a description of the population from which it was derived and may not, without other knowledge, be safely applied to populations where the variations in environment or in heredity may be quite different. For prediction purposes, path coefficients are regression coefficients expressed on a scale in which each standard deviation is one unit.

Most of the terms in Figure 1 are self-explanatory. Phenotype is used here to mean what the outwardly observed record would be if it were possible to keep all environmental circumstances rigidly controlled. It differs from the actually observed record in that the latter has been modified by the variations in environment to which the various phenotypes were exposed. Non-additive effects of environment on the phenotype (such as concern Hogben so much in his recent "Nature and Nurture" (5)) may be minimized by expressing the records on scales which show in the most nearly additive way the effects of variations in environment and in heredity. For example there is some evidence to indicate (but perhaps not enough to prove conclusively?) that dairy production records should be expressed in logarithms of the yield instead of in pounds if the most efficient use is to be made of the information in those records. However, for the present question, non-additive combination effects of phenotype and environment may be considered as included in the environment, since they tend to decrease the correlation between records and genotypes.

"Genotype" is obtained by assigning each gene such a value that the simple sum of all the genes in each individual will give the "genotype" for that individual subject to the "least squares rule," that (1) the average

genotypic value of the population shall coincide with its average phenotypic value and (2) the sum of the squares of the individual deviations of "genotype" from "phenotype" shall be a minimum.* This "genotype" seems to be the same as Fisher's "expected value" of which he says (4, page 32), using human stature for illustration: "This expected value will not necessarily represent the real stature, though it may be a good approximation of it, but its statistical properties will be more intimately involved in the inheritance of real stature than the properties of that variate itself."

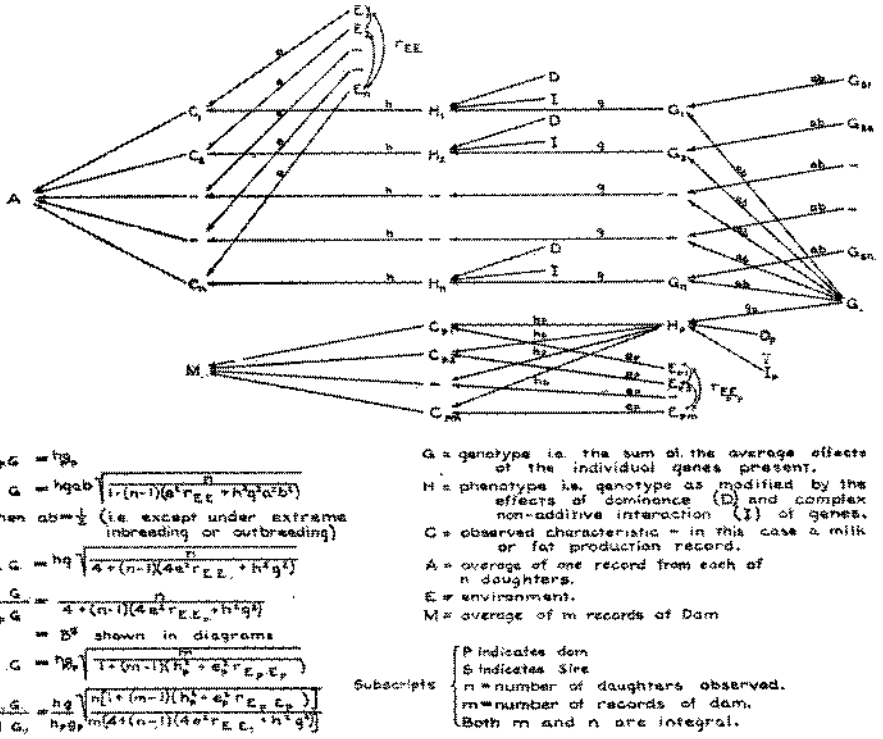


FIG. 1. PATH COEFFICIENT DIAGRAM SHOWING BIOMETRIC RELATIONS BETWEEN PARENT'S RECORDS AND PARENT'S BREEDING VALUE (G) AND THE BIOMETRIC RELATIONS BETWEEN THE PROGENY AVERAGE AND THE PARENT'S BREEDING VALUE.

These genotypic values will be perfectly correlated with Turner's definition (9, page 45) of "transmitting abilities," in so far as dominance is the only cause of discrepancy between genotype and phenotype, but Turner's definition makes no provision for non-additive combination effects of genes which are not allelomorphic to each other. For convenience in this article such non-additive combination effects of non-allelomorphic genes will be

*I am indebted to Dr. Sewall Wright for this method of expressing as much as possible of hereditary differences in terms which contribute to the correlations between relatives.

called epistasis or epistatic effects, although this is a slightly broader meaning than that for which "epistatic" was first used in genetics. In the usage of practical animal breeders, "nicking" most nearly has the same meaning but many breeders also apply the term, "nicking," to phenomena which are nothing but unusual Mendelian recombinations of genes each of which may have nearly additive effects.

"Genotype" as used here is perfectly correlated with breeding value if the latter term is understood to express the average merit of the offspring produced by mating this animal to mates which are a truly representative sample of their breed. If the mates are not a truly representative sample and if there is much epistatic interaction or if there is much dominance and mating is far from random, it must sometimes happen that an individual will produce offspring of higher (or lower) merit in one kind of herd than it would in other herds where the most frequent gene-combinations are quite different. In such cases an animal's breeding value as measured by actual records of progeny, will deviate from what its breeding value would be if its mates were truly representative of their whole breed. However such offspring will not transmit such dominance deviations or epistatic deviations, since those deviations are functions of the *gene combination* rather than of the individual genes considered separately. The *combination* is not transmitted but segregates at the reduction division and the genes reassemble in other combinations in the next generation. Hence "genotype" as used here to express the sum of the average effects of the constituent genes *is* "breeding value" if one considers mating the animal to representative animals of its breed, though it may deviate from "breeding value" if the latter term is used to mean the contribution which this animal could make to descendants produced by breeding it to some isolated and non-typical sub-group of the breed.

"Phenotype" is used here as a convenient intermediate step to show the modification of the genotype by dominance or epistatic relations, which are usually considered genetic phenomena although really they are only consequences of the chemistry and physiology by which the genes produce their effects and are not involved in the mechanism by which genes are transmitted. By this device we can express separately the proportion (h^2) of the observed variance which is hereditary in the broader sense of that word (*i.e.*, including the effects of dominance and epistasis as "hereditary") and the proportion (g^2) of the hereditary (in that broad sense) variance which can be expressed by this additive scheme for representing genotype. The proportion of the observed variance which is hereditary in the narrow sense of that word (*i.e.*, which would contribute directly to parent-offspring correlations) would be h^2g^2.

Non-additive effects of gene combinations are separated into those due to dominance (D) and all others (I). This is done merely because the sta-

tistical consequences of dominance have been explored (2, 3, 12) and can be stated in definite formulae. The statistical consequences of epistatic interactions are not so well known and must be quite complex in some cases, but on the whole probably are much the same as those of dominance, *i.e.*, to lower the correlation between genotype and outward characteristic and hence to lower the observed correlations between relatives, but to affect the correlations between some kinds of relatives more than those between other kinds. So far as dominance and epistasis merely lower the parent-offspring correlation (*i.e.*, diminish the value of g) their effects nearly cancel out of the present problem since they weaken the correlation between daughter average and dam's genotype almost as much as they do the correlation between the dam's average and her genotype. However so far as dominance and epistasis lower the correlation between sisters less than they do that between parent and offspring, they actually prevent the progeny test from becoming relatively as accurate an indicator of the dam's genotype as it would if there were no dominance or epistasis.

The importance of epistasis in practical breeding problems is still a matter of conjecture, with some workers thinking it very unimportant and others speculating that it may often be highly important, especially in herds already improved to a point considerably above the average level of the breed. Wherever the parent-offspring correlation is high under nearly random mating (after having been corrected for the effects of common environment), it seems likely that epistasis is unimportant. However a low parent-offspring correlation under those circumstances does not prove that epistasis is important, since such a low correlation might also result from h being small, *i.e.*, from random variations in environment having important effects on the records.

The importance of dominance depends primarily on whether complete dominance is the rule, how abundant the dominant genes are in contrast to their recessive allelomorphs, and somewhat on the deviations from random mating. In a population mating at random with dominance complete and with no epistasis, g^2 for the effects of each gene pair $= \frac{2(1-q)}{2-q}$ or conversely $d^2 = \frac{q}{2-q}$ = the portion of variance in H due to dominance. (The letter q here represents the frequency of the dominant genes among all the genes occupying that locus in the whole breed.) This d^2 which Fisher calls "the dominance ratio" (3) is thought by him to be typically about $\frac{1}{3}$ for most characteristics. Wright (12, pages 137–139) for other reasons, which appear to him more convincing, believes that this ratio is more typically about $\frac{1}{5}$, even if complete dominance is the almost universal rule among all genes. Moreover he questions (13) whether such complete dominance is the rule, especially among modifying genes each of which has

a comparatively small effect. Such modifying genes probably furnish much of the variance in quantitative characteristics which, like milk production, are vitally necessary in some degree but which may vary widely among different individuals in the breed.

Whether complete dominance is the rule or not, it is obvious that dominance will not often be the major obstacle in estimating an animal's genotype from its own appearance or performance. Even if dominance is universal and Fisher's estimate of d^2 is correct, g will generally have a value of about .82, while if Wright's view is correct g will usually be nearer .90. Both these figures are based on the assumption that epistasis is too small to be important. If epistasis is important, d^2 and g^2 would still retain about the same relation to each other, though both would be smaller. There will be occasional characteristics (of which lethals are notable examples) where dominance is complete and the recessive gene is so rare that the "dominance ratio" approaches close to unity. In those cases dominance may be the major source of the discrepancies between genotype and phenotype. However in such traits the recessive individuals are rare and each such trait will contribute but little to the total variance of the population. For example, in the United States in black breeds of cattle like the Angus, Galloway, or Holstein-Friesian, the gene for black must be around ten to fifteen times as frequent as its recessive allelomorph. Hence the "dominance ratio" for this particular trait will be well above .80. But the gene for red is so rare that less than 1% of the purebred calves of these breeds are born red. Hence this trait is a comparatively unimportant source of loss to breeders. Dominance has been discussed here more than its importance justifies, largely because it has received more attention than it deserves in the writings on applied genetics.

SOLUTION UNDER SIMPLEST CONDITIONS

Figure 1 shows the relations and correlations as they would be under random mating. The correlation between one record of the dam and the dam's genotype will be $h_p g_p$ while the correlation between one record of one daughter and the dam's genotype will be hgab. When m unselected records of the dam are averaged together the correlation between that average and the dam's genotype becomes

$$h_p g_p \sqrt{\frac{m}{1+(m-1)(h_p^2+e_p^2 r_{E_p E_p})}}$$

which approaches

$$g_p \sqrt{\frac{h_p^2}{h_p^2+e_p^2 r_{E_p E_p}}}$$

as a limit as m becomes indefinitely large. If $e_p^2 r_{E_p E_p}$ = zero (the most favorable condition for high values of r_{MG}), this approaches $r_{MG} = g_p$ as a

limit. Increasing the number of records observed thus tends to eliminate errors arising from random uncorrected variations in environment but does not tend to eliminate errors arising from dominance or epistasis or consistently biased and uncorrected environment.

On the other hand, the average record of n daughters each tested once is correlated with the dam's genotype thus:

$$hgab\sqrt{\frac{n}{1+(n-1)(h^2g^2a^2b^2+e^2r_{EE})}}$$

which approaches

$$\sqrt{\frac{h^2g^2a^2b^2}{h^2g^2a^2b^2+e^2r_{EE}}}$$

as n becomes indefinitely large. This approaches unity when e^2r_{EE} is zero. In other words, increasing n tends to cancel errors arising from the imperfections of both h and g, whereas increasing m tends to cancel the imperfections of h_p but does nothing to correct for the fact that g_p is less than unity. However the progeny test involves ab which arises from the intervening generation of Mendelian segregation and appears not at all in the correlation between dam's record and dam's genotype.

Thus a daughter's record usually is a less accurate (on account of ab) indicator of her dam's genotype than the dam's own record is, but the errors in daughters' records tend to be more completely cancelled by the averaging process than is the case with errors in the dam's record. Hence if g and g_p were quite small, it *might* be possible for n to be so large that its cancellation of errors arising from dominance or epistasis would more than make up for the Mendelian sampling errors which affect the progeny average but not the dam's own records.

The problem of the relative accuracy of progeny test and dam's own record as indicators of the dam's genotype therefore becomes the quantitative one of balancing the errors introduced by ab against the greater effectiveness of the progeny average in eliminating errors introduced by dominance and epistasis. Unfortunately for the simplicity of the answer, e^2r_{EE} and $e_p^2r_{E_pE_p}$ cannot safely be considered zero and their existence sets serious limits on the effectiveness of averaging as a means of eliminating errors, either from the daughter average or from the average of the dam's own records.

The value of ab will be nearly .5 even with moderately large departures from random mating (11, pages 118–119). Substituting that value for ab, the ratio (B) of r_{AG} to r_{MG} becomes

$$B=\frac{hg}{h_pg_p}\sqrt{\frac{n}{m}}\sqrt{\frac{1+(m-1)(h_p^2+e_p^2r_{E_pE_p})}{4+(n-1)(h^2g^2+4e^2r_{EE})}} \qquad (1)$$

If dam and daughters are produced by the same kind of a breeding system $g = g_p$ and if dams and daughters are equally typical samples of their generations and if the records used to represent the productiveness of dam and of daughters are chosen by a method equally fair to both, $h = h_p$ and the first part of this expression cancels. Assuming (as the case most favorable to the progeny test) that $m = 1$ and squaring to simplify the expression we have

$$B^2 = \frac{n}{4 + (n-1)(h^2g^2 + 4e^2r_{EE})} \quad \ldots\ldots\ldots\

The double line in figure 2 shows a plane parallel to the base at the level $B^2 = 1.0$. Points above this level denote conditions under which the progeny test is more accurate than the dam's own record. When e^2r_{EE} = zero, and h^2g^2 is as low as .10, the progeny test surpasses the cow's own record when $n = 5$, but n must exceed 6 when $h^2g^2 = .4$ and must exceed 11 when $h^2g^2 = .7$. This brings out the important points that the progeny test

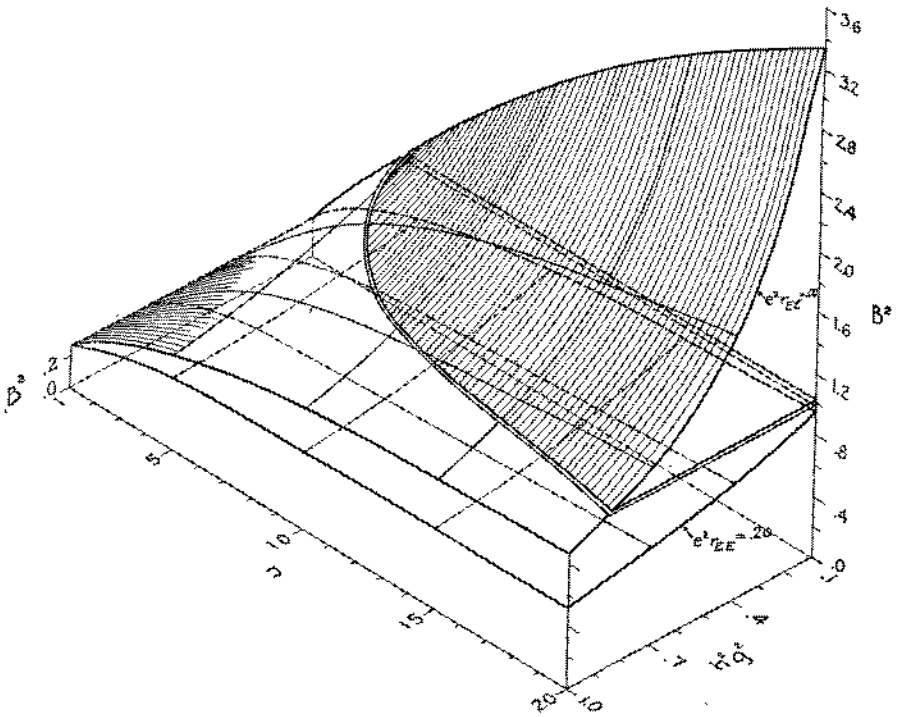


FIG. 2. The relative accuracy of progeny test and the individual's own performance when the number (n) of offspring in the progeny test varies and for traits which are highly hereditary (h^2g^2 is large) or slightly hereditary (h^2g^2 is small). The upper surface shows this condition when the offspring resemble each other only because they are half-sibs. The surface at $e^2r_{EE} = .20$ shows the condition when the offspring resemble each other enough for other reasons to bring about a correlation of +.20 between them on that account alone.

is not very dependable where the daughters resemble each other for other reasons than that they are out of the same dam[1] and that the superiority of the progeny test over the dam's own record is most marked where h^2g^2 is smallest, *i.e.*, where the characteristic is least hereditary in the narrow sense

[1] This same point was shown in more detail by the present writer in an article (8) directed primarily at how r_{AG} varies with increasing n. In that article e^2 was used to include all the causes for half sisters resembling each other except that they are by the same parent.

of that word, and therefore where neither indicator is highly accurate! Values of h^2g^2 less than .10 are not shown in either figure 2 or figure 3. For such values B^2 would become very large if e^2r_{EE} were also very small.

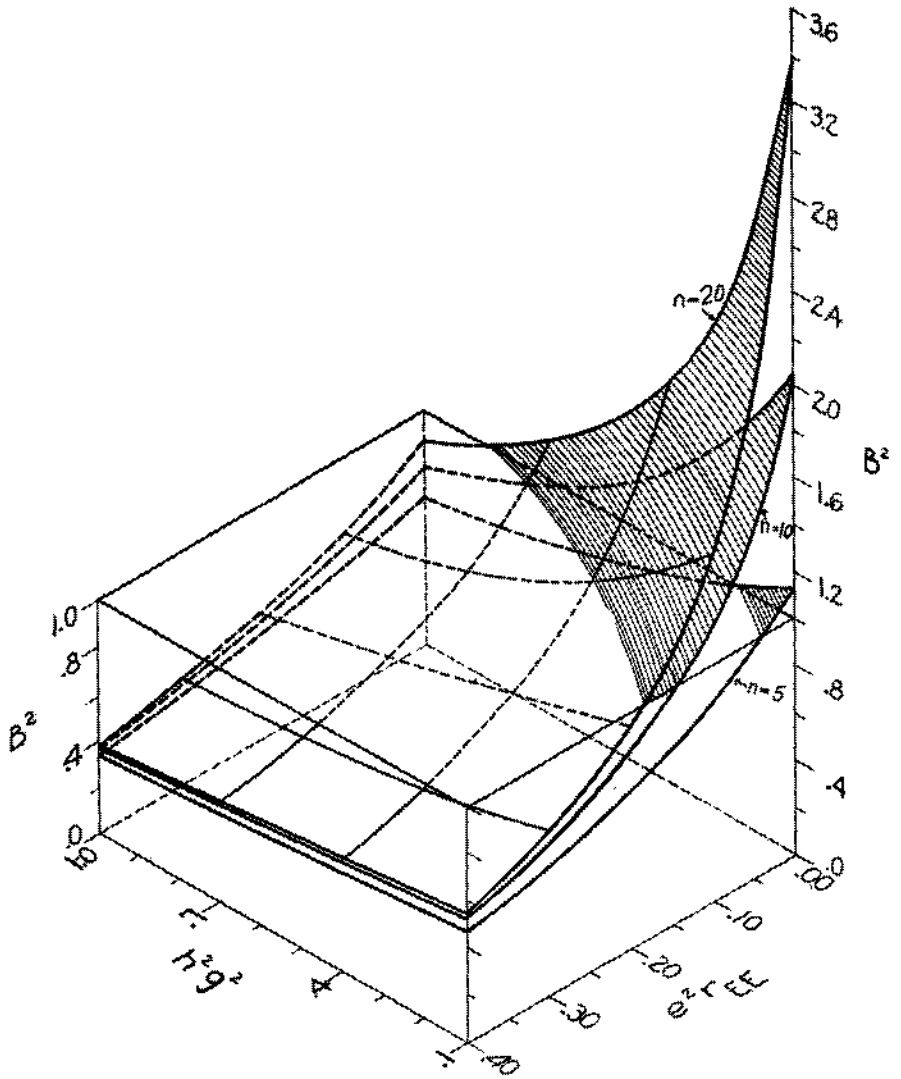


FIG. 3. The relative accuracy of progeny test and individual's own performance for three selected values of n (20, 10 and 5) for traits in which the importance of heredity varies and for populations in which the other reasons for offspring resembling each other vary from zero to a correlation of +.40 between the offspring for other reasons than that they are out of the same dam.

Figure 3 shows how h^2g^2 and e^2r_{EE} affect B^2 for three selected values of n (5, 10, and 20). The shading shows the regions where B^2 rises above 1.0.

The limiting effect of e^2r_{EE} on the accuracy of the progeny test is vividly shown in this figure. Only rarely can the accuracy of the progeny test in a whole population average as high as that of the parent's own record, if the correlation between herd mates due to herd environment is as high as .10 and if h^2g^2 is as large as .10.

CONDITIONS WHICH MAY AFFECT THE VALIDITY OF THE SIMPLE SOLUTION

Figures 2 and 3 show the situation when m is one, each daughter is represented by one record only, breeding is at random, the sires are of unknown breeding value, dams and daughters are produced by the same kind of breeding system, and dams are just as representative of their generation as daughters are of theirs. Now these conditions will never be fulfilled exactly, not even in experimental data. It is important to glance for a moment at the way deviations from these conditions will affect figures 2 and 3, although only deviations from the last condition can alter those figures materially in the direction of making the progeny test more reliable.

When m is more than one—and it usually will be if n is as large as two—the dam's record will be more reliable, in accordance with the formula in equation (1). When each daughter is represented by an average of several records, the effect is that of increasing h and making the progeny test more reliable but this is usually more than offset by the increase in m. Any dam who has several daughters, each with more than one record, is almost certain to have had many records of her own, so that the average number of records per daughter will nearly always be less than m. Hence the use of an average of all available records will tend to make both indicators more reliable but will usually increase that of the dam more than that of the progeny.

Deviations from random mating may be of two kinds so far as they affect this problem. First there may be a correlation between the mates of a cow, irrespective of whether they are like or unlike her. A positive correlation of this kind has the same effect as an increase in e^2r_{EE} and therefore lowers the reliability of the progeny test. A negative correlation of this kind (*i.e.*, some degree of success in deliberately seeing to it that each cow was mated both to very good bulls and to an equal number of very poor bulls) would increase the accuracy of the progeny test slightly since it would eliminate some of the chance existing under truly random breeding, that one cow would happen to be mated only to good bulls, another only to poor bulls, etc. However such a negative correlation cannot be high if n is large, because if a cow is mated to three or more bulls those bulls cannot each one be exactly opposite to every other one! Moreover it is hard to imagine that such a general negative correlation could exist under any circumstances except a breeding experiment where it was made

a cardinal point to get from each cow daughters by poor bulls and daughters by good bulls.

The second kind of deviation from random mating would be a general correlation between dams and sires so that it would generally be true through the whole breed that good cows were usually mated to good bulls, poor cows to poor bulls, etc. This could come about either through inbreeding or through deliberate assortative mating based on individual performance or the performance of near relatives. It is not unreasonable, either from the genetic principles involved or from the published studies of dairy data to suppose that this correlation within dairy breeds might well be as large as +.15 or +.20. So long as all breeders are working for high production, this correlation could hardly rise much above +.25 or +.30 without the practice of extreme inbreeding. The effect of such a general resemblance between sire and dam would be to increase the accuracy of the progeny test slightly, since each offspring would then tend to be like the dam, not only because of receiving half its inheritance from her but also because the other half of its inheritance came from a sire whose inheritance was more like that of the dam than if he had been chosen at random from all sires of the breed. A general correlation of +.25 between the genotypes of sires and dams would modify equation 2 enough that the progeny test might possibly be more accurate than the dam's own record while n is as small as 3, but this would be true only if e^2r_{EE} were zero and h^2g^2 were very small, and there were no other correlation between sires. Since it is not often that all three of these conditions will be fulfilled, it seems unlikely that such deviations from random mating as exist in actual dairy cattle populations will often alter figures 2 and 3 enough to make the progeny test in a whole population average as accurate as the cow's own record when the cow has as few as four daughters. It should be mentioned that in plants where self-fertilization is possible, the resultant perfect correlation between genotypes of sire and dam is quite attainable and makes it possible for a progeny test based on as few as two offspring to be as accurate as the parent's own record. This is doubtless the most important fundamental reason that plant breeders make so much use of the progeny test, particularly when working with self-fertilized organisms like wheat.

Sometimes the merits of the different sires will be partly known. If their genotypes were perfectly known and there were no correlation between sires and dams, formula 2 would become:

$$B^2 = \frac{n}{4 + (n-2)h^2g^2 + 4(n-1)e^2r_{EE}} \qquad (3)$$

Thus even such extreme knowledge as this would have but a trifling effect on figures 2 and 3. If mating is not at random, such information about the sires is quite useful in that it would cancel and prevent errors which would

otherwise be introduced by any such correlation between sires as that discussed in the next to the last paragraph. Knowledge of the sires therefore prevents a correlation between sires from causing the accuracy of the progeny test to fall far below the levels pictured in figures 2 and 3, but cannot of itself raise those levels much higher than is shown. If there is a general correlation between sires and dams and the merit of the sires is known from other evidence, then such knowledge will help to predict the dam's genotype but such help is because of that direct correlation between the genotypes of sire and dam, and not because of evidence from the progeny themselves.

Where dam and daughters have been produced by different breeding systems, the values of g for dam and daughter may not be identical and hence will not cancel out of equation 1 as has been assumed so far. The most extreme example of this, so far as concerns dominance, would be the case where the progeny test is conducted by breeding the cows to homozygous recessive bulls. In such a case the deceiving effects of dominance would become zero for the daughters and a correspondingly higher percentage of the total variance among them than among the dams would be truly genetic in the narrow sense of that word. Naturally such a test could not be made for a cow on account of her short life and limited reproductiveness, but a corresponding test of a bull might be practical if the information gained were worth the cost of getting it. However dominance is usually only a minor cause for actual records deviating from breeding value. The assembling and maintaining of a herd of recessive cows for such tests would be expensive, even if possible at all. Moreover such a plan does not appear to offer any help for avoiding deception by epistatic effects. These considerations make it seem quite unlikely that such a testing plan would often yield information worth the cost of getting it. If the sire and dam belonged to different breeds, there would be increased probability of important misleading epistatic effects which would probably preclude using for testers individuals of a breed developed for other purposes, as for example the use of beef-bred cows for testing the breeding value of a dairy bull for milk and fat production.

Some of the most brilliant research in genetics, particularly in the genetics of Drosophila and of maize, owes its success to the devising and use of tester strains so designed as to eliminate (except for Mendelian sampling errors) the effects not only of dominance but also of epistatic interactions, thus making the truly genetic portion of the variance (h^2g^2) among the offspring much higher than was the case among their parents. However the devising of such a tester strain has for its prerequisite a reasonably accurate hypothesis as to the number and kind of genes involved and (at least so far as concerns epistasis) of the precise way in which the effect of the gene combination differs from the sum of the average separate effects

of the genes concerned. The limited reproductive rate of farm animals, their comparatively long interval between generations and their expensiveness make it well nigh a lifetime's work to create such a tester strain, even if the investigator were wise enough or lucky enough to have chosen the correct working hypothesis at his first attempt and even if the genetic situation were simple. Hence it appears unlikely that much use can be made of such tester strains in the breeding of farm animals, however useful and even necessary such tester strains may be in research on fundamental principles of genetics.

The assumption that daughter and dam are equally representative of their generation and that their records were selected by methods equally fair to both will often be untrue especially in semi-commercial data, and is the only one of these assumptions likely ever to be important enough in a whole population to make much change in figures 2 and 3. The effect of such differences between daughters and dams is to make h in figure 1 unequal to h_p. If they are unequal these two terms cannot be cancelled out of equation 1 as has been inferred hitherto.

If but one record is used to represent each animal and if that record is selected on some such *ex post facto* basis as the choice of the highest record after having seen them all, such a selection deliberately picks out that record made under the most favorable combination of unknown conditions of environment and health. Such selection automatically tends to give to the cow having many lactations a figure less truly representative of her phenotypic producing ability than would be the case with a cow which had but one or two lactations from among which to select. Since the dam usually has more records than the daughter, the net result is to make the dam's record usually less typical of her than the daughter's record is typical of her. This tends to make the progeny test more accurate than is pictured in figures 2 and 3. It should be pointed out that this comes not from any intrinsic superiority of the progeny test, but rather from the bias introduced by the choice of records where variable numbers are available. Studies of the actual correlations between records of the same cow in different lactations in the same herd have generally given values ranging from about +.30 to +.60. Correlations as low as this give ample room for selections of the largest record for each cow to cause such selected records to deviate far from each cow's phenotype, especially if some cows have only one or two records and others have 4 or 5, from among which to select.

Probably more generally important is the fact that the dams are often a more highly selected group than the daughters are. Usually all of the progeny which were tested at all are included in the progeny test. If they were very poor producers, they may have been culled shortly afterward but their records remain for use in the progeny test. The conditions may be quite otherwise among the dams. Not all cows which are tested succeed

themselves in becoming dams of tested daughters. The dams being studied in any particular set of data are to some extent only the more favored survivors from among the group tested contemporarily with them. In studying correlations between the records of those dams and any measure of the subsequent performance of their progeny, one is by no means making a comparison between phenotypic and genotypic manifestations of production. Instead he is comparing the utility of those phenotypic differences still remaining among the dams after this selection has been practiced, with the utility of various averages of phenotypic differences among the less highly selected group of daughters. This seems to be a major part of the explanation for the practical findings of Jull (6, 7) and Copeland (1) that the ancestor's production record is of relatively limited value in breeding selections. In Jull's data all birds which laid less than 200 eggs were culled before they had a chance to become dams. What would have happened, if all hens or a strictly random sample of those hatched in each generation had been progeny-tested, of course can only be surmised. Copeland's data came from many herds and at various dates, so that selection was not by such uniform rules as in Jull's experimental data. Nevertheless Copeland's tables 1 and 5 show that there had been stronger selection among the dams than among either their daughters or granddaughters, else the extreme regression (about 50 to 70 pounds of fat) from the dams' average to the daughters' and granddaughters' averages would not have occurred.

Naturally such studies as those of Jull and Copeland (many others might be mentioned but these two are selected as typical of the more careful among recent ones) correspond closely to the conditions practical breeders actually face. Depending on the amount of selection involved and on how highly hereditary the trait really is, it may sometimes be true that such variation as remains among a highly selected group of dams may be less useful as an indication of the hereditary differences between those dams than the variation which exists among the relatively unselected daughters of those dams, even when there are as few as two daughters per dam. The point to be made here is that when this actually is the case, it is not because of any inherent biological superiority of the progeny test but rather it is because in the case of the dams we are comparing what *further selection* could do among a group of survivors on which considerable selection has already been practiced, with all which selection could do among a group of daughters where very little selection has yet been practiced. The practical utility of such comparisons is beyond argument for the breeder who must actually use, in some combination or other, some data from the more heavily selected population and some data from the less selected population. There is, however, some danger of confusion as to *the utility of the initial selections* in the populations from which the dams first come, if such studies are interpreted as meaning that an individual's own record is in-

herently less representative of its breeding value than the average records of as few as two or three of its offspring.

GENERAL CONSIDERATIONS AND DISCUSSION

The general conclusion to be drawn from all these considerations is that only under rare and unlikely combinations of conditions would a progeny test based on as few as four-daughters-average in an unselected population as accurate an indicator of a dam's breeding value as the dam's own performance. If the trait is highly hereditary, in the narrow sense of that word, or if the progeny resemble each other very much for reasons of having been exposed to a common environment, many more than four daughters may be required or it may even be quite impossible for the progeny test to average more accurate in a whole population than the dam's own performance, no matter how many daughters there are.

The superiority of the progeny test is greatest for traits which are least hereditary in the narrow sense of that word. For traits which are very faintly hereditary and for which the offspring do not resemble each other for reasons of having been under common environment, the progeny test can become several times as accurate as the dam's own record, but for that to be true the trait must be so slightly hereditary that neither indicator can be highly accurate!

The progeny test is needed most where one sex cannot express the trait, as in milk and fat production in dairy cattle, egg production in poultry, prolificacy in swine, etc. Since the biometrical relations between sire's genotype and progeny average are the same as those between dam's genotype and progeny average (except for sex-linked inheritance which must be a small part of the total inheritance in mammals) the foregoing considerations apply also to the progeny test of the sire. However, for traits which the sire cannot express himself, there is nothing but a pedigree estimate of the sire against which to compare the accuracy of a progeny test of him. Since a progeny test surpasses even the best pedigree estimate when there are more than three progeny (provided the offspring do not resemble each other very much for any other reason except that they are by the same sire), the progeny test in such cases is much more useful and more urgently needed for the sire than for the dam. However such progeny tests will rarely tell as much about the sire as the available information correctly used will tell about a dam of nearly equal age. Enthusiasm over some such slogan as the current one that "The next best thing to a proved sire is the son of a proved sire" should not cause us to forget that such a son is one generation of Mendelian segregation away from this sire and that half of his inheritance (a little more than half if the probably small amount of sex-linked inheritance in mammals is also considered) comes from his dam

whose breeding value may usually be estimated more closely than that of his proved sire, if the available information is fully and fairly used.

Although the progeny test will not often be *superior* to the dam's own record (where no differences of selection are involved), it will be noticed from equation 2 that $B = .5$ even when n and m are only one. Therefore even the most fragmentary progeny test is worth much as an indicator of the dam's genotype. Naturally, as in any other prediction, both indicators should be used where both are available. The principles of multiple correlation govern the amount of attention to be paid to each indicator where both are used. However the formulae for the relative amount of attention to be given to the two records are not only complex but also involve a term (which is usually indeterminable) for the degree to which the dam's record and her daughters' records resemble each other because of being made under common environment. The ratio between the standard regression coefficients will be in the same direction from unity as is the ratio between the primary correlations (B in the earlier algebraic discussion) but will be more extreme. Perhaps as good an approximate rule as any would be to give a daughter's production nearly half as much attention as the cow's own record where there is only one record for each, but to give between $\frac{3}{5}$ and $\frac{4}{5}$ as much attention to the daughter average as to the cow's own record where there are at least two or three offspring. This rule would be approximately correct if daughters and dams had been equally selected from their generations. Where the dams have already been highly selected, more importance than this should be given to the progeny test in making further selections but it does not seem possible to develop any simple general rule for this, since such a rule would include terms for the intensity of selection among dams and among daughters and also a term for the extent to which the trait is really hereditary.

In actual practice selection will also be based in part on pedigree. Chronologically pedigree becomes available first, then the individual's own performance and the progeny test comes latest. As some selection is practiced on each basis, the possibilities of further gains by additional selections on the same basis rapidly diminish unless new information becomes available. The sampling nature of Mendelian inheritance sets a lower limit on the usefulness of pedigree than is inherently necessary for either of the other two bases of selection. Hence the very early selections on the basis of pedigree come near exhausting the possibilities in that direction, although increased knowledge of the performance of parents and other ancestors or collateral relatives does make it profitable occasionally to revise an earlier pedigree estimate. On the other hand, knowledge of the individual's own performance can continue to increase at a practically important rate as long as it lives and knowledge of its progeny can increase

as long as previously untested progeny come on test and (at a slower rate) as long as progeny already tested continue to be tested at later ages.

This leads to the general picture that pedigree occupies first place only for such selections as must be made before the other two criteria are available. It is very distinctly in third place after a few such early selections have been made. Individual performance occupies first place after it becomes available and until considerable use has been made of it. Then progeny test is the most useful basis for further selections but when it is much used for such selections, the possibilities in that tend to be exhausted and the individual's own performance might again assume first place. Among mature animals under practical conditions individual performance and progeny test therefore will vary or alternate as most useful for further selections, according to the amount of new knowledge becoming available about each and according to how nearly the possibilities for selection on the basis of the existing knowledge about each have already been exhausted.

SUMMARY

1. The biometrical relations governing the relative accuracy of progeny test and of the parent's own performance as indicators of the parent's breeding value are presented and discussed.

2. A solution under the simplest conditions is presented algebraically and graphically. Under those conditions there must be at least five offspring before the progeny test in a whole population will usually be a more accurate indicator of the parent's breeding value than the parent's own performance.

3. Most deviations from those simplest conditions have only slight effects on the solution. However any general resemblance between the offspring for any other reason than that they are half-sibs through the parent in question sets serious limits on the accuracy of the progeny test. On the other hand if the parents or the records used to represent them are more highly selected than the offspring or their records, the progeny test may become relatively more accurate than under the simple conditions for which the algebraic solution is given.

4. The progeny test is needed most for traits which cannot be expressed in one sex and for traits which are but slightly hereditary.

5. The bases for estimating breeding value are pedigree, own performance, and progeny test. As fast as some selection is practiced on one of these bases, the possibilities for further progress by additional selection on the same basis rapidly diminish and correspondingly increased attention should be given to one of the other bases.

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The problem

During the 1930s and 1940s, the concept of heritability came to be used in all livestock species, but with different definitions and estimation methods. In this paper, presented at the International Congress of Genetics in 1948, Lush organizes the subject. He separates heritability in the broad sense from heritability in the narrow sense, explains estimation methods based on the resemblance between relatives and gathers the available estimates for several traits in cattle, pigs, sheep and poultry.

Points to note while reading

The definition of heritability in the narrow sense, linked to the average effects of genes, is the one still used to predict the response to selection. Note the estimation methods, such as the regression of offspring on dam and the correlation between sibs, and compare the values of the time with those used today.

HERITABILITY OF QUANTITATIVE CHARACTERS IN FARM ANIMALS

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ANY distinction in principle between qualitative and quantitative characters disappeared long ago, although in the early days of Mendelism it was often conjectured that they might be inherited according to fundamentally different laws.

If it is still convenient to call some characters qualitative and others quantitative, it is only to denote that the former naturally have a discontinuous and the latter a continuous distribution, or that the former are not easily measured on a familiar metrical scale. Colors are an example. Differences between colors can be measured in terms of length of light waves, hue, brilliance, etc., but most of us find it difficult to compare those measurements with our own visual impressions.

Most quantitative characters are affected by many pairs of genes and also importantly by environmental variations. It is rarely possible to identify the pertinent genes in a Mendelian way or to map the chromosome position of any of them. Fortunately this inability to identify and describe the genes individually is almost no handicap to the breeder of economic plants or animals. What he would actually do if he knew the details about all the genes which affect a quantitative character in that population differs little from what he will do if he merely knows how heritable it is and whether much or little of the hereditary variance comes from dominance or overdominance, and from epistatic interactions between the genes.

Definition. — Heritability is the fraction of the observed or phenotypic variance which was caused by differences between the genes or the genotypes of the individuals. It is used in both a narrow and a broad sense. In the broad sense it refers to the functioning of the whole genotype as a unit and is used in contrast with environmental. But the genotype is not transmitted as a unit. Instead its constituent genes segregate and come together in new combinations. The genes may interact with each other in non-additive ways, so that in certain combinations they have effects quite different from their average effects in

that population. These differences between their actual effects in each combination and their average effects in the whole population are called dominance deviations or epistatic deviations and are transmitted only in part, if at all. The narrow definition of heritability includes only the average effects of the genes. These will be called here the genic differences. This narrow meaning of heritability is used when the main emphasis is on expressing what fraction of the phenotypic differences between parents may reasonably be expected to be recovered in their offspring.

Like all other aspects of genetics, heritability concerns *differences* between individuals rather than the absolute values of the individual phenotypes. When the importance of various causes of differences is to be assessed, the differences are best expressed in terms of their squares; *i. e.* in terms of *variance*, although sometimes it may be convenient to use correlation and regression when obtaining the estimates from actual data.

If the causes of phenotypic variations are measured in terms of their effects on the phenotype, we can write: $p = h + e + j$ where p is the phenotype of the individual, *h* is heredity in the broad sense, *e* is environment, and *j* represents the joint effects or nonlinear interactions of heredity and environment which cannot be divided fairly between them. All are expressed as deviations from the population average. Since *h* itself is the sum of the additively genetic or genic deviations (*g*), plus the dominance deviations (*d*), plus the epistatic deviations (*i*), we can write: $p = g + d + i + e + j$. Then the variance of the individual phenotypes in the population is:

$$\sigma_p^2 = \sigma_g^2 + \sigma_d^2 + \sigma_i^2 + \sigma_e^2 + \sigma_j^2$$

The covariance terms are zero, except as any covariance which may exist between *e* and *h* is by definition put with σ_j^2 and the variance from nonlinear interactions of the genes with each other is by definition put in σ_i^2.

The precise definition of heritability in the narrow sense is:

$$\frac{\sigma_g^2}{\sigma_g^2 + \sigma_d^2 + \sigma_i^2 + \sigma_e^2 + \sigma_j^2}$$

and in the broad sense it is:

$$\frac{\sigma_g^2 + \sigma_d^2 + \sigma_i^2}{\sigma_g^2 + \sigma_d^2 + \sigma_i^2 + \sigma_e^2 + \sigma_j^2}$$

The two definitions differ only in what they do with the variance due to dominance and to epistasis.

Separating the phenotypic variance into its constituent parts is statistically similar to fitting a straight line to data which may not in fact lie on a straight line. It is also like fitting a plane to a surface which actually is irregularly curved and has valleys and peaks. The genic variance in breeding problems corresponds to what the statisticians call variance due to regression. The rest of the phenotypic variance corresponds to variance due to deviations from the straight line or from the plane. The separation of that remainder into portions due to dominance, to epistasis, to environment, or to interactions between environment and epistasis, corresponds to the separation the statistician would make if he analyzed his deviations from regression according to their relation to some third, fourth, or fifth variables, as is sometimes done in multiple curvilinear correlation.

The distinction between hereditary and environmental becomes blurred in such cases as when the heredity of the parents influences the kind of environment they provide for their offspring. An example is milk production among the mammals. The sow who produces a large amount of milk provides a more favorable environment for her pigs than the one who produces only a little. This difference between the pigs in one litter and those in another is environmental as far as the pigs are concerned, but that environment itself is partly conditioned by genes in the dams. Those genes will in part be transmitted to the pigs. Even more extreme examples occur in man in what is generally called social or cultural inheritance. In farm animals the variance from these things which are environmental on the individual but are partly caused by genes in their parents can generally be grouped satisfactorily enough under the term, maternal effect, since the sire almost never participates in rearing the young.

Heritability applies to a particular character in a particular population. Since it is a ratio, its value can change as either the numerator or the denominator changes. Confusion about this aspect of heritability can make it appear contradictory. For example, if a character can be influenced strongly by environmental variations, its heritability would be low in a population in which the environment varies widely. But in another population in which the physical control is so rigid that the environmental variations are very slight, the same character would appear highly hereditary. Or two different populations might be kept under equally variable environments but one of them might be almost

entirely homozygous, as an inbred line tends to become, while in the other population the frequencies of several important genes might be near .5, or the population might consist of several races or of several highly inbred but unrelated lines differing in their means. Then heritability of the same character would be almost zero in the first population but very high in the second. Such of this confusion as results automatically from the mathematical properties of ratios can be avoided if the genic variance or the hereditary variance is expressed in absolute terms, rather than as ratios, but that would not get around the fact that the genic or hereditary variance in a given character actually may be larger in some populations than in others.

Confusion may also arise when it is not clear whether the heritability being mentioned is that of differences between individuals or of differences between the averages of groups such as families, progenies, breeds, species, etc. The basic unit for estimates of heritability concerns differences between individuals. Heritability of individual differences can be converted into heritability of differences between group averages merely by multiplying by $\frac{1+(n-1)r}{1+(n-1)t}$ where r is the correlation between hereditary values and t is the correlation between phenotypic values of individual members of the same group and each group contains n individuals. This starts out as 1.0 when $n=1$ and tends toward $\frac{r}{t}$ times the heritability of individual differences as n becomes indefinitely large.

A clear distinction must be drawn between the *facts* concerning heritability and *estimates* of heritability which are derived from the available data. The population has certain biological parameters but our estimates of these are subject to sampling errors because we derive them from a finite sample of data. Also our estimates are subject to errors of interpretation. The magnitude of purely sampling errors can be estimated by appropriate statistical methods. Various common possibilities of errors in interpretation will be mentioned in connection with the methods of estimating heritability.

The Importance of Knowing Heritability. — To know whether heritability is high or low is important when making efficient breeding plans. When heritability in the narrow sense is high, reliance should be placed mainly on mass selection. As heritability becomes lower the plan should put less emphasis on mass selection and more upon pedigrees, sib tests, and progeny tests. These latter are primarily aids

to mass selection, intended to correct the errors made by using that when heritability is low.

If the epistatic variance is a large fraction, more reliance is placed on selection between families and on the linebreeding type of inbreeding in which it is sought to recover the particular combination of genes which made some ancestor outstandingly successful.

If overdominance is prominent the breeding plan turns toward the making of inbred lines, then testing these in crosses with each other, and then multiplying the ones which cross most favorably, so that those crosses can be produced commercially.

If variance due to interactions between heredity and environment is large, the ideal breeding plan tends more toward producing a separate variety for each ecological niche which is large enough to justify the costs of that.

Methods of Estimating Heritability. — All methods of estimating heritability rest in one way or another on measuring the extent to which related individuals are more like each other than unrelated ones. It is a question of the extent to which phenotypic likeness parallels genetic likeness. In statistical terms it is the regression of genetic differences on phenotypic differences. The more promising methods include: the study of isogenic lines, regression of offspring on parental phenotypes, seeking to develop a high line and a low line by selecting in opposite directions from the same initial population, resemblances between full sibs, and resemblances between half sibs. Resemblances between remote relatives offer little help in estimating heritability because the degree of expected genetic likeness is so small that fantastically large numbers would be necessary to detect with statistical significance such differences as one would have any right to expect.

Pitfalls of Interpretation. — The correlation between relatives is likely to have an environmental component. If so, part of the decrease in variance between relatives, as compared with the variance between unrelated individuals, occurs because the environmental differences between relatives are less. This environmental component can largely be prevented by designing the experiments properly. Or one can estimate that component in data collected from breeders' herds and subtract it before estimating heritability, but that offers many possibilities for mistakes.

Variance due to dominance affects the resemblance between some kinds of relatives more than that between others. This is a minor obstacle in practically all cases except when using resemblances between

full sibs or when using data from populations of partially inbred lines.

Part of the epistatic variance is included in the resemblances between all kinds of relatives. This portion is not clearly worked out in all cases. It is very small for epistatic effects requiring the simultaneous presence of many genes but is noticeable for the two-gene and three-gene effects. It rises curvilinearly with inbreeding. Most methods of estimation include only a little of the epistatic variance except from the simpler combinations.

That part of the variance which was caused in the offspring by environmental differences due to genetic peculiarities of their parents or other ancestors will appear more strongly in the correlations between some kinds of relatives than of others. In most farm animals it is likely to be important only in resemblances between dam and offspring or between maternal sibs.

Corrections for non-randomness of the mating system are unimportant in most data collected from breeders' herds unless breeders have for some time differed widely in their ideals. Such corrections do become important in data based on experiments involving many partially inbred lines. The amount of inbreeding can be computed from the pedigrees, but the amount and direction of any phenotypic assortive mating is difficult to estimate.

The data may pertain to selected animals. This will bias correlations but not regressions on the selected variable, provided the selection was primarily on the individual's own phenotype rather than on the phenotypes of its relatives. Hence the regression of unselected offspring on selected parents can be used. Little if anything can be done with data in which the offspring also were selected.

Isogenic Lines. — Isogenic lines are rare in farm animals, the only examples being identical twins and highly inbred lines. Identical twins occur too rarely to be of much use except in man and in cattle. If inbred lines are to be almost isogenic, they need to have inbreeding coefficients well above 90 %. Such have not yet been reached with any animals other than guinea pigs and mice, unless perhaps some lines of poultry may be inbred that highly.

The principle involved is that all variance (B) between members of the same isogenic line is environmental, whereas the variance (A + B) between two unrelated individuals chosen at random from that population contains both genetic and environmental variance. Therefore

$\frac{A}{A+B}$ would be an estimate of heritability in the broad sense if the environment varied as much between members of the same line as between individuals from different lines. That it might do in a well-designed experiment with inbred lines or with the clones which are possible in asexually propagated plants. But both members of a pair of identical twins would be subject to many of the same peculiarities of environment, so that B computed from them would underestimate the environmental variance. Consequently this method overestimates heritability to the extent that the common environment of identical twins is important. In man identical twins reared apart from soon after birth have been studied to overcome this difficulty arising from common environment. This method is being used at Wiad and in New Zealand with twins in cattle. It does not avoid similarity of pre-natal environment but that seems unlikely to be important for most economic characters which are measured only when maturity is approached. Getting around the difficulty with common environment has been attempted by computing C as the variance within pairs of fraternal twins and then doubling $\frac{C-B}{A+B}$ to get an estimate of heritability. But C—B really contains $\frac{\sigma_g^2}{2}+\frac{3}{4}\sigma_d^2$ + more than $^3/_4$ of σ_i^2. Therefore doubling it gives an overestimate of heritability in the broad sense by $\frac{\sigma_d^2}{2}$ and more than $^1/_2$ of σ_i^2. This is reasonably accurate, however, if dominance and epistasis are unimportantly small.

Also considering B as an estimate of the environmental variance throws all of the interactions between heredity and environment into the genetic portion because the effects of environmental variations within isogenic lines are measured only on a uniform genetic background. The interaction variance comes from differences which occur only when both genotype and environment vary but which are so non-linear that they cannot fairly be ascribed to either.

An example of using inbred lines for estimating heritability is WRIGHT's study of the amount of spotting in guinea pigs. The line had been inbred so long that the parent-offspring correlation was practically zero, although that correlation was above .2 in the control stock. Therefore the inbred line could be assumed to be practically isogenic with respect to this character. Since the members of the inbred line were mostly from different dams and the colony was on an ex-

perimental basis with conditions for it and for the control stock kept as nearly standard as practical, the variance in spotting within the inbred line would be a fair measure of the total extent to which environmental variations affected spotting. Comparing the variance within the inbred line and the variance within a control stock yielded the estimate that heritability in the broad sense was about .42.

Regression on Mid-parent. — That the regression of offspring on mid-parent could be used as a measure of heritability was pointed out as long ago as GALTON's time. Applied to the data of PEARSON and LEE on measurements in man it yields figures of .79 for stature, .76 for span and .70 for length of forearm.

This regression of offspring on mid-parent gives an estimate of heritability a little larger than the narrow one but distinctly smaller than the broad one. It includes the genic variance and that due to the genetic portion of whatever the parental environment causes, plus something less than half of the epistatic variance. It includes no dominance variance and nothing from environment if that was adequately randomized; *i. e.* if the environments of parent and offspring were uncorrelated. No correction is needed for mating system. Like the method of isogenic lines, but unlike the others which follow, it needs no multiplication by a factor for the incompleteness of the genetic resemblance. Therefore its sampling errors are not multiplied correspondingly. It is not applicable to characters expressed in one sex only nor to characters which can be measured only by destroying the individual. If the parents are the survivors of some more or less intense selection which was based entirely on their own phenotypes, this regression will still give an unbiased estimate although its fiducial limits are wider than if the parents had been unselected. But if the selection of the parents was based also on other considerations, such as the merit of their sibs or other relatives, and if those other considerations really were correlated with their transmitting ability, then the regression will give an underestimate of heritability.

A good recent example of heritability, as deduced from regression on the mid-parent, is in BERGE's data on inheritance of vertebrae number in swine. He found a regression of .74 in spite of the fact that vertebrae number is a threshold characteristic which took only four different values. Based on SHEPARD's correction, about .06 of the remaining .26 of the variance can be ascribed to this coarseness of classification. Little room is left for truly environmental effects or for accidents of development in embryology, if there is any dominance or epistasis at

all. Perhaps this is not surprising since this character is a skeletal one formed early in embryology. Tangible environmental differences would hardly be expected to affect it, although it would not have been surprising if accidents of development in early embryology had played a bigger role than they appear to do.

Selection Experiments. — Estimating heritability from experiments in which a high and a low line are derived from an initial population by mass selection in opposite directions is really a special way of using regression of the offspring on the mid-parent. Among the classical examples with animals are the following. CASTLE and PHILLIPS' selection for the hooded pattern in rats yielded about .32 for the heritability, as computed from their tables 14 and 20. The selections of GOODALE and of MACARTHUR for size and for white spotting in mice show selection to have been highly successful, whence we may conclude that heritability was high, but they selected largely on progeny tests or sib tests and they did not report selection differentials so that heritability could be computed. The current work at the Illinois Station by KRIDER *et al.* in selecting for high and low rates of gain in swine yields heritability estimates which are of the order of $^1/_6$ to $^1/_5$. The Alabama selections for economy of gain in swine, reported by DICKERSON and GRIMES, indicated that heritability was about .24. The principle of the method is that the difference produced between the means of the two lines is divided by the total amount by which the phenotypes of the parents in all the generations exceeded the mean of the generation in which they were born. More detail about the genetic and statistical principles involved may be seen in Student's analysis of the Illinois data on high and low oil in corn. The only real difference in principle between this selection method and observing the regression of offspring on mid-parent on a non-experimental population is that epistatic variance may contribute much to the latter but can contribute little to the former except in the first and second generations of selection. If the data from the first two generations in such selection experiments are discarded, this method comes nearer than any other to excluding epistatic variance from the estimate of heritability in the narrow sense. The difference between progress in the first and second generation and progress in later generations appears to be the best indication we have of whether epistasis really is important.

Resemblance between Parent and Offspring. — In a random bred population the correlation between parent and offspring includes half of the genic variance and something less than a fourth of the epistatic

variance. It may include something from environment if the environment of the offspring is partly determined by the individual characters of the parents, or if there is a correlation between the environments of parent and offspring. No variance due to dominance is included. In planned experiments the environmental correlations may be reduced to zero by proper experimental design. However, if the data come from non-experimental sources as from breeders' herds or from several different locations or from over a period long enough for time trends to have been important, similarity of environment between parent and offspring is likely to contribute something to their resemblance.

When estimating heritability from the parent-offspring resemblance the steps are: first, subtract the environmental contribution; second, multiply the remainder by two; third, correct for non-randomness in the mating system. The first step is usually much the more important and more difficult to do accurately. The second step is simple and easy. In most non-experimental material the deviations from random mating are so slight that the third step is comparatively unimportant but it becomes important if the estimate is made from experiments in which inbred lines are being formed. Occasionally it will become important in data from a group of herds whose breeders held widely divergent ideals. The third step is to divide by $1 + m$ if the non-randomness was of the inbreeding kind, or by $1 + r_{SD}$ if it was phenotypic assortive mating. Here m is the relationship between mates and r_{SD} is the phenotypic correlation between mates. In principle, the offspring are used merely to indicate the genic values of their dams. Really the method merely asks how much difference can be expected between their offspring per unit of phenotypic difference between those dams. In slightly more technical language, it asks for the regression of genic differences among those dams on the phenotypic differences among them. If the regression is on an intra-sire basis, all of the information is relative to such differences as exist between females which get mated to the same male. The method bypasses any question about the differences between females which get mated to different sires. Those are simply left unanalyzed.

So far as the environmental component in the resemblance of dam and offspring comes from each such pair having been kept in the same herd but the general environment varying from one herd to another, making the analysis on an intra-sire basis automatically eliminates that environmental component if, as is usual, all the mates and daughters of a sire were reared, kept and tested in the same herd. If the daughters

of dams who did unusually well were given better than average treatment merely because they were daughters of those dams, while the daughters of dams who did poorly were given poor treatment because their dams had not been promising, this would still leave some environmental contribution in the likeness between daughter and dam, even in data analyzed on an intra-sire basis. This possibility may need checking although it seems unlikely often to be important.

An example of using intra-sire regression of offspring is in the study by LUSH *et al.* of lactation yields of milk and butterfat in dairy cattle. This yielded estimates of .25 to .30 for heritability of differences in fat production and .33 for differences in milk production between cows mated to the same bull. All the dams had at least two records each. Therefore repeatability as well as heritability could be studied. From .52 to .60 of the variance between single records, each made by different cows mated to the same bull, was due to temporary environmental circumstances which changed from one lactation to the next. From .25 to .33 was heritable in the sense yielded by the intra-sire regression method. This leaves .10 to .15 which would include that due to dominance, the remainder of that due to epistasis and all of that due to permanent effects of environment on the dam. That this remainder is small and surely included quite a little from permanent effects of environment, indicates that epistasis and dominance are of small importance for butterfat production. The definition of permanent effects of environment here includes any circumstances which extended over two lactations, even though they may not have lasted longer.

Intra-dam regression of offspring on sire would have the same advantages and in addition would not include anything from maternal environment. However, it isn't often that one gets extensive data of this kind where the phenotype is manifest in the sires and many maternal half sibs are available. Also the sires are likely to have been selected more intensively. If so, the phenotypic variation among them would be much smaller and the estimate subject to a larger sampling error.

Use of Full Sibs. — The phenotypic resemblance between full sibs in a random mating population contains half the genic variance, one fourth of the dominance variance, something less than one fourth of the epistatic variance, and the variance from common environment. The procedure when estimating heritability is: First, subtract the environmental portion. Second, double the remainder. Third, correct for any non-randomness of the mating system.

The main difference between the full-sib and the parent-offspring

resemblance is that full sibs have in common one fourth of the dominance variance and are almost certain to have more similar environment. That the observed correlations between full sibs are generally higher than those between parent and offspring can be interpreted as evidence either for dominance or for a larger environmental component, or for any mixture of the two. The two resemblances also differ to a minor degree in that selection is likely to have made the parent-offspring correlation lower but assortive mating would have made it a bit higher than the full-sib correlation. The estimate obtained from the full-sib correlation will include the genic variance, plus half the dominance variance, plus less than half of the epistatic variance, plus or minus twice the error made in subtracting the environmental variance, plus or minus any of the usually small errors in correcting for deviations from random mating. The importance of correcting accurately for the environmental term is obvious. For example, in GOWEN's study of milk and test in Jersey cattle no allowance was made for an environmental component in the correlation between daughter and dam. Environmental differences between herds would have made such a contribution. The excess of full-sib resemblance over parent-offspring resemblance he alternatively interpreted as an environmental component or as a result of dominance. On the hypothesis that there was no dominance he found heritability of test to be .83 (his Table 18) and of milk to be .56 or .53 according to whether the departures from random mating were caused by phenotypic assortive mating or by mating relatives together. But if herd-to-herd differences in environment contributed .10 to the daughter-dam correlation for test and .25 to that for milk [1] then he would have found .64 and about .09 or .08. Or if the environmental components were only .05 and .20 respectively, he would have found .71 and about .19 or .18. These figures differ widely. Obviously it is highly important to discount the environmental contribution accurately, although that is often difficult. Any errors made in the discounting are doubled.

In such data as dairy records gathered from many herds, much of this discounting for environment can be done automatically by doing the analysis on an intra-herd or "Stalldurchschnitt" basis but even then the maternal sib resemblance could still contain a small environmental component.

Use of Half Sibs. — The procedure is as follows: First, subtract the environmental component from their observed resemblance. Second,

[1] Other work indicates these values to be reasonable, although they certainly need more verifying.

multiply the result by four. Third, correct for any deviations from random mating. This estimate would include the genic variance and a small portion of the epistatic but nothing from dominance. If the half sibs were paternal it would include nothing from the maternal environment, but if they were maternal it would include four times that component. It would include nothing from the other environment if the environmental differences between half-sibs were as large as those between non-sibs. That would need to be verified in each case. For example, if the data were collected from many breeders' herds and each sire's offspring were wholly in one herd, any differences in average environment from one herd to the other would produce differences between the progenies of sires but not differences within them.

Automatic correction for this circumstance can be made by making such analyses on an intra-herd basis. Even when that is done, time trends in the environment pertaining to a herd may have been important so that, for example, the offspring of a sire tested mostly in 1935 were subjected to a different environment from offspring of another sire tested in that same herd in 1940. This possibility can be largely avoided if the analysis is made on an intra-season as well as an intra-herd basis. That is feasible where in all years the offspring of several different sires are present. It wouldn't yield much information if most of the data come from herds so small that all contemporaries were offspring of the same sire.

The multiplication by four automatically multiplies the sampling error, as well as any error from not having subtracted the correct amount for the environmental component. This multiplication by four is a distinct disadvantage of using half sibs, as compared with the full-sib or parent-offspring methods where multiplication is by two, or with the isogenic lines or regression on mid-parent where no multiplication is made. But sometimes the amount of data concerning half sibs is so much larger that the sampling error will be reduced enough to more than offset this handicap of the half-sib method.

Basing the estimates on resemblances between sibs of any kind does not require any use of the parental phenotypes. Sib resemblances can therefore be used for characters which require destroying the individuals in the process of measuring them, as many carcass characters do. Also the estimates are almost wholly free from the effects of selection among the parents. Only the tiny effects which such selection would have in reducing the variability of the next generation are involved.

Often the half-sib method is most informative if combined with the full-sib one and if both maternal and paternal half sibs are available. The correlation between maternal half sibs minus that between paternal half sibs would permit measuring the importance of maternal environment, provided one were sure that the usually higher contemporaneity of the paternal half sibs did not introduce a larger environmental component into their resemblance. If the herd-to-herd environmental components could be removed first, the excess of the full-sib correlation over the sum of the maternal and paternal half-sib correlations would yield an estimate of the dominance component. But the size of the sampling errors and the uncertainty about removing the environmental component exactly make the dependability of such an estimate dubious.

CHAPMAN'S analysis of variations in the ovaries of the rats in a colony at Wisconsin is an excellent example of the interpretations which can be made from correlations between full sibs and between both kinds of half sibs in a controlled experiment, where there were also aunt-niece resemblances and unrelated rats reared by the same dam. Among individuals born in the same month and weighing the same, he found .25 of the variance genic, .03 due to dominance, practically none from epistasis, .06 from similar contemporary pre-natal maternal environment but none from pre-natal environment alike for different litters from the same dam, and about .09 from post-natal environment alike for contemporaries.

These methods have the characteristics of intra-class correlations. They depend on the number of degrees of freedom within sires. The reason for this can be seen readily if we consider an extreme example of estimating the likeness between paternal half sibs from data which include only the daughters of two sires but each sire has 100 daughters. That gives 198 degrees of freedom for estimating the differences between daughters of the same sire but only one degree of freedom for estimating differences between sires. The result is not nearly as dependable as if there were two daughters from each of 100 sires, in which case there would be 100 degrees of freedom for estimating the variance between daughters of the same sire and 99 degrees of freedom for estimating the variance between sires. In the former case the two sires could easily have happened to be very nearly alike genetically and hence the resemblance between paternal half sibs would appear small. The two sires could just as easily have happened to be unusually far apart genetically. In that case the difference between sires would appear far larger than is really typical of sires from that population.

An example of the use of paternal half sibs is the work of HAZEL *et al.* who were studying the heritability of differences in growth rate in swine at various ages. They found heritabilities of .15, .28, and .17 respectively for the first, second and third 56-day periods after birth. They found the most convenient technique to be an analysis of variance and covariance between full sibs, between paternal half sibs, and between non-sibs, but it is the same in principle as a comparison of correlations between full sibs and between paternal half sibs. Also they separated the correlation between the performance of the same pig in two different periods of its life into a genetic part and an environmental part, but that is aside from our main topic here.

The Use of Diallel Crossing. — Where a diallel crossing experiment has been conducted, the data are well adapted to an analysis of differences between full sibs, between maternal half sibs, and between paternal half sibs from many dams. As an example consider the data of HAZEL and LAMOREUX on sexual maturity and body weight in the fowl. With n sires each mated to k different hens in the first period and another n sires mated to the very same hens in the second period, and with each dam producing t daughters to each sire, the structure of the experiment is as follows:

	Degrees of Freedom
Between periods	1
Between blocks (each block being the k hens mated to the same two sires)	n—1
Between the two sires used on the same hens	n—1
Between hens mated to the same sire	n(k—1)
Interaction between hen and sire	n(k—1)
Between full sibs	2nk(t—1)

The excess of the interaction term over that between full sibs consists of some dominance and some epistatic variance. It can be considered a partial test of the importance of "nicking". The amount the variance between hens mated to the same sire exceeds the interaction term is a basis for estimating the variance from maternal environment plus that from genic differences between the dams. The excess of that between sires over that between hens mated to the same sire permits an estimate of the genic differences between the sires, plus a small amount from epistatic variance. Variance due to sex-linked genes would appear in

this term also. Using it is equivalent to estimating heritability from the correlation between paternal half sibs.

In this particular example, HAZEL and LAMOREUX found comparatively little evidence of a real interaction big enough to be important. However their population was substantially random bred and under those circumstances they could have expected to pick up only a little of the epistatic variance plus a quarter of the dominance variance in their interaction component. Using males and females from different inbred or partially inbred lines in a diallel crossing scheme would provide a more sensitive test for the epistatic or overdominance portions of the variance. It remains possible that epistatic variance or overdominance are important for these two characteristics but the data do not suggest that positively. The maternal environment had an important effect on body weight but not on sexual maturity. There was no evidence that sex-linked genes influenced either character.

Threshold Characteristics. — That the restriction of the vertebrae number to only four classes in BERGE's study would lead to some underestimate of heritability was already mentioned. This situation is much more extreme in characters which can take only two forms, as is the case with mortality when the individuals are classed only as dead or alive. An example of analyzing such data is given by LUSH *et al.* on inheritance of resistance to mortality in the domestic fowl.

Since such data are binomial, the variance and the mean are correlated. This requires translating the estimates obtained for variance on the percentage scale into terms of variance on the probit scale if the analysis and interpretation are to be consistent. If the variance between family means is small compared to the individual variance within families, as was so in this case, if the numbers per family are moderate or small, and especially if the numbers per family vary considerably, transformation of the primary data to some other scale, such as the arc sine transformation, seems unproductive of worth-while results.

In this particular case the maternal effect was extremely small, the estimate being little above zero. This indicates that dominance is not important, that transmission of the causative agent through the egg is not important in determining the result, and probably that there is little epistasis involved. The data yielded consistent estimates of heritability around 16 % on the probit scale. Heritability on the observed percentage scale would be $\frac{z^2}{p(1-p)}$ times that where p is the mean fraction dying and z is the ordinate of the normal curve at the point

below which p of the area lies. Heritability on the observed percentage scale would therefore fall toward zero as the percentage mortality declines.

The genetics of the case is such that mass selection is rather effective when mortality is well above 30 % but declines rapidly in effectiveness as the incidence of mortality declines further. The sib test as a basis of selection acquires increasing importance as mortality becomes less but it, too, begins to lose its effectiveness as mortality becomes still lower. To reduce mortality to much lower than 10 percent, especially if much of it is genuinely accidental, requires other means, such as progeny-testing on susceptible stocks, or techniques for raising the average incidence without changing its basis.

The analysis in this case of mortality in the fowl was based in principle on comparing full sib and paternal half-sib resemblances. Maternal half sibs were not available since one hen was mated to the same male throughout the breeding season. The analysis of such a threshold characteristic requires moderately large numbers of progeny from each sire and preferably also from each dam.

Important Unsolved Problems. — Several things about the techniques of estimating heritability are still unsatisfactory. Chief among these are how to get a more dependable estimate of the importance of epistatic variance and how to estimate the importance of overdominance. The more important epistasis and overdominance are, the more the optimum breeding plan swings toward selection on a family basis and there is correspondingly less room for emphasis on individual selection.

A clearer distinction between variance which is purely environmental and that which is environmental for the individual but due to genetic characteristics of its parents or other relatives is to be desired. In most of the examples mentioned here, that which is caused by environmental differences which in turn were partly caused by the genes of the parents has been treated either as purely environmental or it has gone into what was called the maternal effect, even though most of that effect may be from circumstances which are purely environmental in origin but are alike for full sibs.

Most of the numerical estimates for heritability of economically important characters have been surprisingly high in view of the fact that many of these characters are known to be readily and strongly influenced by changes in nutrition and by other environmental circumstances. Most of these estimates have been between .2 and .4, but some have been even higher than that. Heritability of skeletal dimensions or

peculiarities has generally been nearer to .7 but then those are hard to influence much, even by rather drastic changes in nutrition. Accessory evidence which is not very decisive but deserves some credence, makes one suppose that breeders generally have been selecting rather intensely for these things for many generations. If breeders have been selecting intensely and if heritability is as high as these estimates, the breed average should have been improving rapidly for many generations and should still be doing so. But the actual evidence does not indicate improvement that rapid. Admittedly the evidence on the actual rate of improvement is scanty. Also an unknown fraction of the changes in the population means as, for example, part of the one or two pounds of fat per cow per year, which is the rate at which average dairy production has been increasing, may have been caused by gradual changes in environment. Yet, even with this much uncertainty, it seems to me that an important amount of contradiction remains. This is a challange and demands a solution. I have no solution in which I have strong faith, but the following possibilities are mentioned as worth examining.

Perhaps the breeders have not been and are not yet practicing as intense selection as we think. Only a few fragmentary studies of this point have been published. For the most part they indicate that for some reason or another the actual average selection differential is smaller than most of us think. Examples are SEATH's study on dairy cattle and BARTRAM's on litter size in swine. It could be that the selection differential is so low that there really is no contradiction, but I hardly think so.

Possibly most of the heritability we have been finding is from epistatic effects. This is supported by the extremely high heritability which seems to come from estimates based on isogenic lines, as compared with those based on resemblance between ordinary relatives, but the evidence from isogenic lines is extremely scanty. If most of the hereditary variance were epistatic, much of the actual selection would merely repair the regression caused by segregation of the epistatic combinations each generation. Also heritability in the broad sense would have to be extremely high to give such high correlations as actually occur between half sibs and more distant relatives. I feel that this is a part of the explanation but doubt that it can be by any means the whole of it. The more direct tests for epistasis have nearly all given small or doubtful indications.

Overdominance is a possible explanation but, if it were very important, the estimates based on full sibs should be much higher than

those based on parent and offspring. This has not generally been the case.

To a considerable extent the same genes may affect two characters which appear to be entirely different. We might then be selecting strongly for some genes when we select for characteristic *A*, but selecting strongly against those same genes when we select for characteristic *B* at a later stage, thus undoing the earlier selection. This amounts to the same thing as overdominance if the merit of the individual as a whole is considered, but we would never see that if we only studied each characteristic by itself. There is need for more study of the truly genetic correlations between different characteristics. Animal husbandry emphasis in the past on *balance* and the repeated warnings against "*einseitige Zuchtwahl*" are a part of that pattern but the evidence on which those warnings are based seems scanty indeed.

At any rate the general picture is that most of the economically important characteristics are heritable to such a degree that this can be found with statistical significance whenever we look for it, even on moderately small amounts of data. The possibilities of improving farm animals by suitable genetic methods are real and important. It looks as if more extensive analyses of the heritability of many characters would point the emphasis more toward making distinct families and selecting between them than has been customary in the past. However, such making of distinct families and selecting between them requires time and numbers. It also involves more inbreeding than can be practiced without some decline in merit. There is indeed abundant room for research concerning more economical and efficient ways of combining and using these facts.

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Hazel, Baker and Reinmiller (1943)

Hazel, L.N., Baker, M.L., and Reinmiller, C.F. 1943. Genetic and environmental correlations between the growth rates of pigs at different ages. Journal of Animal Science, vol. 2, no. 2, pp. 118 to 128.

The problem

When animals are selected for more than one trait, or for the same trait at different ages, the genetic association between the traits must be known. Hazel and coworkers showed how to separate the observed correlation between two measurements into a genetic part and an environmental part, using the analysis of covariance between and within families. The data were the weight gains of pigs over several growth periods.

Points to note while reading

The genetic correlation is an essential element of Hazel's selection index, published in the same year, and of all the multiple trait models that followed. Note that the covariance is partitioned in the same way as the variance, and that genetic and environmental correlations can differ considerably from the phenotypic correlation.

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GENETIC AND ENVIRONMENTAL CORRELATIONS BETWEEN THE GROWTH RATES OF PIGS AT DIFFERENT AGES[1]

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Introduction

EARLY growth rate in pigs is correlated with growth in subsequent periods. Presumably the association is due to the effects of genetic and/or environmental factors which act over relatively long periods of time. The extent to which growth genes with persistent effects operate in determining hereditary variations in growth rate is of interest in swine breeding research. Information of this kind is required to attain maximum accuracy in measuring breeding value and hence to make most efficient use of selection.

For example, if genes with effects of short duration were responsible for most of the hereditary variation, growth rate in one stage would be relatively independent of that in another stage, depending upon the duration of each stage and the interval between stages. According to this possibility most of the observed association between growth rates in the different stages must be due to the effects of persistent environmental circumstances. Consequently separate consideration might be given the different stages in measuring hereditary growth rate for the entire period.

Conversely, if growth genes with persistent effects are responsible for most of the hereditary variation, they would contribute to the observed correlation between gain in different stages. Growth rate in each period would contain information concerning hereditary growth rate in other stages. In fact, growth rate for a short period where hereditary variations are relatively large might even be a more accurate indication of hereditary growth rate over the entire period than growth rate from birth to market age. Consequently information about the kind of genes which influence growth rate is useful in planning a breeding program designed to increase the economic value of swine.

The authors have investigated the genetic and environmental relations which contribute to the gross correlations between growth rate in three 56-

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day periods from birth to 168 days and have attempted to apply the results in measuring hereditary growth rate with maximum accuracy.

Analytical Method

Each of the main sources which may cause variation in growth rate represents the sum of the effects of a large number of agencies which act simultaneously and more or less continuously during the growing period, so that the specific effect of any one source is not directly measurable. However each source may be visualized as a quantitative variable, the actual values for which correspond to the effects which that source has upon the growth rate of individual pigs. Accordingly, let growth rate for each individual pig be the sum of the three variables: (1) A genetic source (G), composed of the *average* effects of all genes which influence growth rate; (2) litter environment (L), comprising the effects of all environmental factors which influence the litter as a whole; and (3) a residual source (E), including the effects of dominance, epistasis and environmental factors peculiar to individual pigs. Then growth rate in the ith period is defined by the equation

$$X_i = G_i + L_i + E_i \qquad (1),$$

where each variable is considered as a deviation from the general mean. Thus variation in growth rate is the consequence of differences in one or more of the three sources. The variance $[V(\ )]$ in growth rate in the ith period is

$$V(X_i) = V(G_i) + V(L_i) + V(E_i) \qquad (2),$$

provided the three sources are statistically independent.

The growing period can be divided into any number of more or less arbitrary intervals, the gain in each period being considered as a distinct quantitative trait with its own sources of variation. Then the covariance $[\text{cov}\ (\ )]$ between the growth rate in the ith and jth periods is

$$\text{cov}\ (X_iX_j) = \text{cov}\ (G_iG_j) + \text{cov}\ (L_iL_j) + \text{cov}\ (E_iE_j) \qquad (3),$$

provided each source in the ith period is statistically independent of the other two in the jth period. The correlation between the effects of a source in different periods can be written from equations 2 and 3. For example, the correlation between hereditary growth rate (the genetic correlation) in the ith and jth periods is $\dfrac{\text{cov}\ (G_iG_j)}{\sqrt{V(G_i) \cdot V(G_j)}}$. Since these correlations are of particular interest, estimating the variances in equation 2 for each period and the covariances in equation 3 for each combination of periods is essential to the solution of the present problem.

The first step is calculating from the data the components of variance and

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covariance attributed to (1) differences between sires, (2) differences between litters by the same sire and (3) differences between litter mates. The form of the analysis used is presented in table 1, along with the theoretical composition of the variance and covariance. The theoretical composition given in table 1 exemplifies the effect which a limited number has upon the sample variance and covariance as compared to those expected in a population of infinite size. The components of variance are additive, $V(P)+V(M)+V(S)$ being the variance expected among pigs from different litters and by different sires. The same is true for the covariance.

In a population mated at random, the percentages of the additively genetic

TABLE 1.—THEORETICAL COMPOSITION OF MEAN SQUARES AND COVARIANCES

Source	Degrees of freedom	Mean square	Covariance
Total	$nmk-1$	$V(P_t)+\frac{k(nm-1)}{nmk-1}V(M_t)$ $+\frac{mk(n-1)}{nmk-1}V(S_t)$	$\operatorname{cov}(P_tP_j)+\frac{k(nm-1)}{nmk-1}\operatorname{cov}(M_tM_j)$ $+\frac{mk(n-1)}{nmk-1}\operatorname{cov}(S_tS_j)$
Between sires	$n-1$	$V(P_t)+kV(M_t)+mkV(S_t)$	$\operatorname{cov}(P_tP_j)+k\operatorname{cov}(M_tM_j)$ $+mk\operatorname{cov}(S_tS_j)$
Between litters by the same sire	$n(m-1)$	$V(P_t)+kV(M_t)$	$\operatorname{cov}(P_tP_j)+k\operatorname{cov}(M_tM_j)$
Within litters	$mn(k-1)$	$V(P_t)$	$\operatorname{cov}(P_tP_j)$

variance represented in $V(P)$, $V(M)$ and $V(S)$ are 50, 25 and 25, respectively. In the partially inbred lines from which the data were taken, most of the sires used each season were from different lines and were mated to females in the same line. Consequently the genetic contribution to $V(S)$ is larger, while that to $V(M)$ and to $V(P)$ is smaller than expected in a random mating system. According to formulae developed by Dickerson (1941) and to the inbreeding which has taken place, it seems probable that about 40 per cent of the additively genetic variance is represented in $V(S)$ and in $V(P)$, while $V(M)$ contains the remaining 20 per cent.[3] The genetic covariance is distributed in the same manner as the additively genetic variance. Since all

[3] This distribution of the genetic variance is an estimate based on the specific amount of inbreeding which has occurred in the North Platte swine herd. The exact distribution of the variance due to dominance deviations and epistatic (non-additive) gene interactions depends upon several factors the importance of which cannot be measured (the prevalence of dominance and epistasis, gene frequency, the degree of dominance and the particular types of non-additive interactions). According to Fisher (1918) and Wright (1939), most of the variances from these sources appears in $V(P)$.

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litters were treated as nearly alike as was practical, $V(S)$ and cov (S_iS_j) may be assumed entirely due to genetic differences between sires.

Based on the above assumptions, estimates of the components of variance (equation 2) and of covariance (equation 3) were derived as follows:

Variance	*Covariance*
$V(G) = 2.5V(S)$	cov $(G_iG_j) = 2.5$ cov (S_iS_j)
$V(L) = V(M) - 0.5V(S)$	cov (L_iL_j) = cov $(M_iM_j) - 0.5$ cov (S_iS_j) (4).
$V(E) = V(P) - V(S)$	cov (E_iE_j) = cov (P_iP_j) − cov (S_iS_j)

Results

A description of the data and the general management of the swine herd from which the data were collected is given elsewhere (Baker, Hazel and Reinmiller, 1943).

The results of the analysis of variance and covariance for the gain in the three 56-day periods are summarized in table 2. The gains in the three periods

TABLE 2.—ANALYSIS OF VARIANCE AND COVARIANCE FOR GAINS IN THREE 56-DAY PERIODS

Source	Degrees of freedom	Mean squares			Covariances		
		X_1	X_2	X_3	X_1X_2	X_1X_3	X_2X_3
Intraseason total	986	38.8	156.0	242.1	34.0	15.6	106.3
Between sires	54	131.8	574.7	669.7	154.2	86.0	331.4
Between litters by the same sire	197	93.7	287.3	408.7	80.1	40.7	139.5
Within litters	735	17.3	90.1	166.0	12.8	3.7	80.9

are designated X_1, X_2, and X_3 according to the age at which they occur. These data were analyzed on an intraseasonal basis (the differences due to the eight seasonal means being eliminated) so that the degrees of freedom for the intraseason total and for between sires in the same season were reduced by seven.

Agencies affecting the litter as a unit caused highly significant variation in gains for each period, as evidenced by the much larger size of the mean square between litters by the same sire as compared to that between litter mates. Some variation, but less extreme, was caused by differences between sires. The mean square between sires in the same season (using the mean square between litters by the same sire for error variance) was significant in the second and third periods and was almost significant ($F = 1.40$) in the first.

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The gross correlations between gains in the three periods are $r_{12} = .437$, $r_{13} = .161$ and $r_{23} = .547$ (from table 2). Since these three correlations are highly significant, one or more of the main sources of variation must contribute significantly to the covariation between periods. The present analytical problem is that of reducing these correlations into the relations existing between the genotypic and environmental effects which are responsible for the observed variation in the different periods.

The variance for each period and the covariance for each combination of periods in table 2 was analyzed into the three components of variation or

TABLE 3.—COMPONENTS OF VARIANCE AND COVARIANCE ATTRIBUTED TO DIFFERENCES (1) BETWEEN SIRES (S), (2) BETWEEN LITTERS BY THE SAME SIRE (M) AND (3) BETWEEN LITTER MATES (P)

Period	$V(S_i)$	$V(M_i)$	$V(P_i)$	Combination of periods	cov (S_iS_j)	cov (M_iM_j)	cov (P_iP_j)
1	2.38	19.98	17.27	1:2	4.62	17.56	12.75
2	17.93	51.18	90.09	1:3	2.83	9.63	3.71
3	16.28	63.23	166.01	2:3	11.98	15.25	80.92

covariation indicated in table 1. Although the exact values for these components of the variance and covariance are of little interest in themselves they are presented in table 3 as an essential step in the analysis. In these data, the number of pigs per litter (k) was 3.838, and the number of litters per sire (m) was 4.177.

The components in table 3 were used to derive the hereditary and environmental variances and covariances as indicated in equation 4. These latter are given in table 4 for the three periods. The first figure in each row is the variance for the source and period indicated by the row and column. Figures other than the first in each row are covariances. For example, $V(G_1) = 5.95$ and cov $(G_1G_3) = 11.55$. The phenotypic variances and covariances correspond to those for the intraseason total shown in table 2. Since each phenotypic estimate is the sum of the genetic and the two environmental components, it is slightly larger than the corresponding estimate in table 2 because of the restriction imposed by limited sample size.

Figure 1 presents a diagram of the genetic and environmental sources presumed to cause variation in gain in each period and of the relations between these sources which cause the observed correlation between gain in different periods. The quantitative importance of each source in causing variation is represented by path coefficients[4] while the relations between

[4] Specifically, the path coefficient is that fraction of the standard deviation of the dependent variable for which the designated factor is directly responsible (Wright, 1934).

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each source in different periods is represented by correlation coefficients. The values for the correlations and path coefficients in figure 1 were calculated from the estimates given in table 4. The curved lines represent correlation

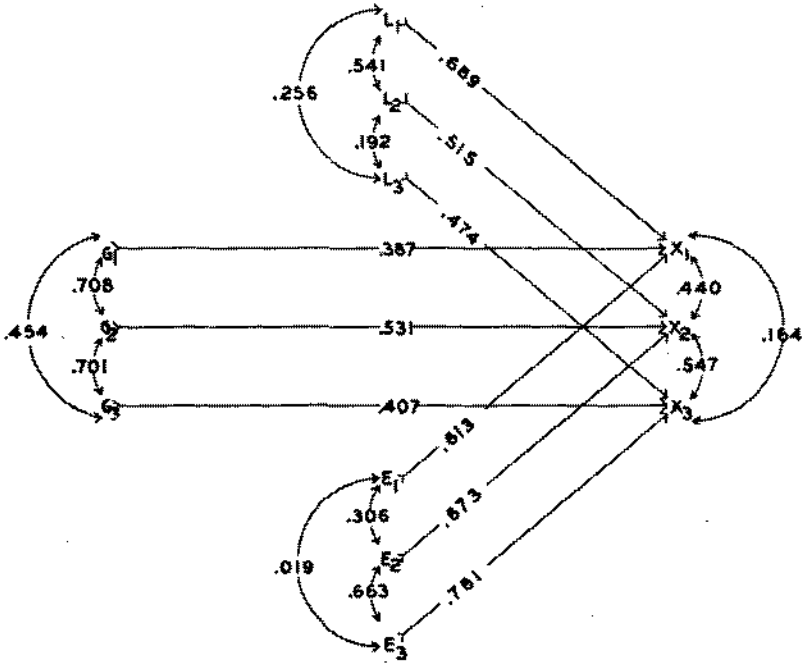


Figure 1. Path coefficient diagram to illustrate the genetic and environmental relations presumed to underly the observed correlation between gain in the three periods. X_1, X_2 and X_3 = gain in periods 1, 2 and 3, respectively. L_1, L_2 and L_3 = litter effect in the three periods. G_1, G_2 and G_3 = genotype for gain in the three periods. E_1, E_2 and E_3 = dominance and epistatic deviations and environment peculiar to individual pigs in the three periods. The straight lines represent path coefficients. The curved lines represent correlation coefficients.

coefficients. Each was calculated by dividing the appropriate covariance by the square root of the product of the two corresponding variances from table 4. For example, the genetic correlation is

$$r_{G_1G_2} = \frac{\text{cov}\,(G_1G_2)}{\sqrt{V(G_1)\cdot V(G_2)}} = \frac{11.55}{\sqrt{5.95\cdot 44.82}} = .708.$$

The straight lines represent path coefficients, the direction of influence being indicated by the arrow. Each path coefficient was calculated by dividing the square root of the variance for a given source by the square root of

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the phenotypic variance from table 4. For example, the path from G_1 to $X_1 = \sqrt{\frac{V(G_1)}{V(X_1)}} = \sqrt{\frac{5.95}{39.63}} = .387$. Each of the gross correlations in figure 1 is the sum of three chains of causal relations, each chain being the product of one correlation and two path coefficients. For example,

$$r_{X_1X_3} = (.689)(.256)(.474) + (.387)(.454)(.407) + (.613)(.019)(.781) = .164.$$

The gross correlations in figure 1 differ only slightly from those calculated

TABLE 4.—GENOTYPIC AND ENVIRONMENTAL VARIANCES AND COVARIANCES FOR THE THREE PERIODS

Source	Period	Period 1	Period 2	Period 3
Genotype	G_1	5.95	11.55	7.07
	G_2		44.82	29.95
	G_3			40.70
Litter environment	L_1	18.79	15.25	8.22
	L_2		42.22	9.26
	L_3			55.09
Environment peculiar to the individual	E_1	14.89	8.13	0.88
	E_2		72.16	68.94
	E_3			149.73
Phenotype	X_1	39.63	34.93	16.17
	X_2		159.20	108.15
	X_3			245.52

from table 2 (.440, .164 and .547 as compared to .437, .161 and .547, respectively).

Several points in figure 1 appear sufficiently important to deserve comment. First, heredity plays a relatively minor role in causing variations in growth rate, particularly in the first and third periods. The heritability of gain (the path coefficients squared) is .15, .28 and .17 in the three periods, leaving 85, 72 and 83 per cent of the observed variance to be attributed to environmental circumstances. Second, the genetic correlations are generally larger than the corresponding environmental correlations. Consequently heredity exerts a less *important* but more *constant* influence upon growth over the entire period than do environmental circumstances. Third, the genetic correlations between the gains in the two adjacent periods are considerably higher than that between the periods separated by 56 days.

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Presumably these points would be true regardless of the number or length of arbitrary periods used.

The importance of litter environment in causing variation in growth rate decreases with age, whereas that of environment peculiar to individual pigs increases. The correlation between E_1 and E_3 is very low in view of the relationship between E_1 and E_2 and between E_2 and E_3. Evidently the environmental factors which cause variation in the growth rate of individual pigs early in life persist into the second period but not into the third. Since $r_{E_2E_3} = .663$, the factors which begin operating in the second period are quite important in the third. This is not true for the environmental factors which influence the litter as a whole. Since $r_{L_1L_3}$ is larger than $r_{L_2L_3}$, the effects which operate in the first period must be more important in causing variation in the third period than in the second. The differences in these correlations may be due to sampling errors; hence their biological significance is questionable.

Application

The permanent improvement accomplished by selection is in proportion to the accuracy with which genetic differences among animals can be recognized. Since genotypes cannot be evaluated directly, selection must be practiced for some estimate of the genotype based on measurable characteristics. Consequently the correlation between the genotype and the variable used as a basis of selection provides a logical basis for comparing the efficiency of different measures of genetic merit.

As indicated previously, the gains in different periods may vary in the extent to which they represent hereditary growth rate over the entire period of 168 days. Growth rate over the entire period is the sum of the gains in the three 56-day periods. Thus hereditary growth rate for the entire period is

$$G_t = G_1 + G_2 + G_3.$$

Consequently (from table 4),

$\text{cov}(G_tX_1) = V(G_1) + \text{cov}(G_1G_2) + \text{cov}(G_1G_3) = 5.95 + 11.55 + 7.07 = 24.57$;

$\text{cov}(G_tX_2) = 86.32$;

$\text{cov}(G_tX_3) = 77.72$;

and $V(G_t) = 24.57 + 86.32 + 77.72 = 188.61$;

whence

$$r_{G_tX_1} = \frac{\text{cov}(G_tX_1)}{\sqrt{V(G_t) \cdot V(X_1)}} = .284;$$

$r_{G_tX_2} = .498$;

and $r_{G_tX_3} = .361$.

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Thus the correlations indicate that of the three 56-day periods the second is the most efficient in measuring hereditary growth rate for the 168-day period.

Perhaps the most logical measure of G_t is

$$X_t = X_1 + X_2 + X_3,$$

where X_t is the gain for the entire period. Since

$$\text{cov}\,(G_t X_t) = V(G_t),$$

then

$$r_{G_t X_t} = \frac{V(G_t)}{\sqrt{V(G_t) \cdot V(X_t)}} = \sqrt{\frac{188.61}{762.85}} = .497.$$

An additional measure is the multiple regression equation

$$I = b_1 X_1 + b_2 X_2 + b_3 X_3,$$

where the b's are regression coefficients so chosen that the sum of squares of the deviations of I from G_t are a minimum. The simultaneous equations to be solved are

$$b_1 V(X_1) + b_2 \text{ cov } (X_1 X_2) + b_3 \text{ cov } (X_1 X_3) = \text{cov } (G_t X_1),$$
$$b_1 \text{ cov } (X_1 X_2) + b_2 V(X_2) + b_3 \text{ cov } (X_2 X_3) = \text{cov } (G_t X_2),$$
$$b_1 \text{ cov } (X_1 X_3) + b_2 \text{ cov } (X_2 X_3) + b_3 V(X_3) = \text{cov } (G_t X_3).$$

The estimates required for the solution of these equations are given above or in table 4. The multiple regression equation turns out to be

$$I = .064 X_1 + .130 X_2 + .038 X_3,$$

and

$$R_{G_t \cdot X_1 X_2 X_3} = r_{G_t I} = .516.$$

Recapitulating, the multiple regression equation gives the most accurate estimate of G_t, followed rather closely by gain in the second period and by gain over the entire period.

In the above comparisons it was assumed that change in any period was equally desirable. In practice, gain in one period may have more economic value than that in another period because of the extent to which it reflects greater economy of food utilization or more desirable carcass composition. This situation can be met by redefining G_t so that

$$G_t = a_1 G_1 + a_2 G_2 + a_3 G_3,$$

where the a's represent the economic values for the corresponding genotypes.

Discussion

From the practical view, the subdivision of the growing period for the purpose of measuring hereditary growth rate more accurately is of questionable value. The multiple correlation (.516) based on gains in the three 56-day periods is only 4 per cent larger than that based on gain over the entire period (.497). However, a subdivision might easily be justified if improved

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growth rate in one period were considered more desirable than in another. In this event, similar analyses for swine herds differing in genetic background and under different systems of management would be needed to test the general applicability of these results.

This study yields some rather pertinent results in regard to the selection of boars as it is generally practiced. Weaning weight (usually at about 56 days) is often used as a measure of growth rate in making initial selections. Final selections are based on weight at about 180 days of age, after the normal growth of many boars is disturbed by sexual development. Weight at 56 days is probably not a good measure of hereditary growth rate since gain from birth to 56 days is a poor measure ($r_{G_1X_1} = .284$). Gain from 56 to 112 days ($r_{G_1X_2} = .498$) is probably a more accurate measure in boars than gain over the entire period ($r_{G_1X_3} = .497$, for barrows and gilts) because it occurs before puberty. Consequently the use of gain from 56 to 112 days in selecting boar pigs is a promising alternative to the usual method. Most of the initial selection at 56 days could be eliminated so that some good individuals would be saved. Castration at 112 days is little more difficult than at earlier ages, and all boars except those to be used could be castrated then.

Several methods of calculating growth rate have been proposed by different investigators. Gain in live weight (in pounds) during constant age periods was used in the present study because of its simplicity and directness, because gain in the different periods is directly additive and because it corresponds to the measures most widely used in swine breeding. The results reported herein may not hold for other methods of calculating growth rate.

Summary

The gross correlations between the gains in three 56-day periods from birth to 168 days were analyzed. The genetic and environmental relations presumed to underly these correlations are shown in figure 1.

The genetic variance constituted only a small fraction of the observed variance in each of the three periods (15, 28 and 17 per cent, respectively). However, the genetic correlations were larger than the corresponding environmental correlations, indicating that genes with persistent effects were responsible for much of the genetic variation. Consequently heredity has a less important but more constant influence upon growth rate than either of the environmental sources.

The genetic correlations between the gain in adjacent periods were considerably larger than that for the two periods separated by 56 days.

Five possible measures of hereditary growth rate for the 168-day period were compared. A multiple regression equation based on gain in each 56-day period was about 4 per cent more accurate than gain from 56 to 112 days or

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gain from birth to 168 days. Gain from birth to 56 days and from 112 to 168 days were not efficient measures.

The possibility of using gain from 56 to 112 days as a measure of hereditary growth rate in selecting boar pigs is suggested.

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Hazel (1943)

Hazel, L.N. 1943. The genetic basis for constructing selection indexes. Genetics, vol. 28, no. 6, pp. 476 to 490.

The problem

Breeding programs rarely aim at a single trait. Hazel showed how to combine several measurements into a single index that maximizes progress in total economic value. The weight of each measurement is obtained from a system of equations that uses the phenotypic and genetic variances and covariances and the economic value of each trait.

Points to note while reading

The index equations are, in essence, the same ones Henderson would later solve with unknown fixed effects. Note the concept of the aggregate genotype, which is the breeding objective expressed in monetary terms, and the distinction between what is measured and what one wants to improve. The economic indexes used today follow this logic.

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THE GENETIC BASIS FOR CONSTRUCTING SELECTION INDEXES[1]

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"The key is man's power of accumulative selection: nature gives successive variations; man adds them up in certain directions useful to him."—Darwin, p. 35, sixth edition of *The Origin of Species*. 1920.

INTRODUCTION

THE idea of a yardstick or selection index for measuring the net merit of breeding animals is probably almost as old as the art of animal breeding itself. In practice several or many traits influence an animal's practical value, although they do so in varying degrees. The information regarding different traits may vary widely, some coming from an animal's relatives and some from the animal's own performance for traits which are expressed once or repeatedly during its lifetime. LUSH (1935) emphasized that permanent improvement from phenotypic selection is proportional to the additively genetic (heritable) fraction of the observed variance and that this varies for different traits. DOBZHANSKY (1937) suggested "that most, and possibly all, genes have manifold effects." These factors make wise selection a complicated and uncertain procedure; in addition fluctuating, vague, and sometimes erroneous ideals often cause the improvement resulting from selection to be much less than could be achieved if these obstacles were overcome.

In the initial stages of breeding investigations conducted by the REGIONAL SWINE BREEDING LABORATORY and cooperating state experiment stations, an arbitrary method of selecting breeding animals had to be adopted. In the meantime the theoretical aspects of the problem were investigated while data were being collected. While many fundamental genetic problems are still incompletely solved (particularly as regards the prevalence of dominance, epistasis and pleiotropic effects of genes in quantitative inheritance, the nature of heterosis, and the interaction of genotype and environment), the accumulation of genetic knowledge justifies an exploration of this problem. According to formulas presented by HAZEL and LUSH (1943), selection for an index which gives proper weight to each trait is more efficient than selection for one trait at a time or for several traits with an independent culling level for each trait. The principles of constructing and using selection indexes which permit the attainment of maximum genetic progress are given in the present paper. Three selection indexes were constructed (and compared as to efficiency) from data taken on the Iowa Station swine herd from the fall of 1937 through the spring of 1940. Such indexes are subject to revision when the accuracy of

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the statistics upon which they are based can be increased by analyzing additional data.

Selection indexes, constructed with attention to the genetic and economic bases for the various traits, should be valuable in livestock breeding programs. GALTON'S "Law of Regression," presented before the Mendelian nature of inheritance was clear, represents an early step in index construction. The sire index, widely used in selecting sires for butterfat and milk production, is a practical example of an index based on one trait but using information about several relatives. SMITH (1937) developed an index designed for the selection of plant lines, using FISHER'S concept of discriminant functions to derive a linear equation based on observable characteristics as the best available guide to the genetic value of each line.

ANALYTICAL METHOD

The net genetic improvement which can be brought about by selecting among a group of animals is the sum of the genetic gains made for the several traits which have economic importance. It is logical to weight the gain made for each trait ($\overline{G}_i$) by the relative economic value of that trait (a_i). Thus the average genetic superiority of a selected group over the group from which it was chosen is

$$\overline{H} = a_1\overline{G}_1 + a_2\overline{G}_2 + \cdots + a_n\overline{G}_n. \tag{1}$$

The relative economic value for each trait depends upon the amount by which profit may be expected to increase for each unit of improvement in that trait. Good approximations to relative economic values often can be obtained from long-time price averages and cost-of-production figures. As an example, WINTERS (1940) found that the average price of wool per pound was 3.4 times that of lamb per pound. If additional feed or labor costs are associated with increased production for either trait, the increased cost per unit should be discounted when calculating relative economic values. These values may vary from breed to breed or from region to region within the same breed. They may change, even while a breeding program is in progress, if permanent shifts in market demand occur.

Animals vary in breeding value, as in phenotype, for each of the several traits. The aggregate value of an animal is the sum of its several genotypes (assuming a distinct genotype for each economic trait), each genotype being weighted according to the relative economic value of that trait. An animal's genotype for a given trait may be defined as the sum of the average (strictly additive) effects of its genes which influence that trait. Thus the aggregate genotype of an animal is

$$H = a_1G_1 + a_2G_2 + \cdots + a_nG_n. \tag{2}$$

Environmental factors, dominance, and epistasis may make phenotypic performance unlike the genotype for that trait; hence animals having the highest values for H cannot be recognized directly with perfect accuracy. Selection for improved breeding value therefore must be practiced indirectly by select-

ing directly for a correlated variable (I) based on the phenotypic performance of each animal for the several traits. When selection is practiced on a large population, the genetic average of the selected group minus that of the original population represents the genetic gain from selection ($\overline{H}$ in equation 1), being

$$\overline{H} = (i)R_{IH}\sigma_H. \tag{3}$$

Here ī is the average superiority of the indexes (in standard deviation units) for the selected group as compared to the whole group, R_{IH} is the correlation between H and I, while σ_H is the standard deviation of H in the whole group.

The selection differential (ī) depends upon the proportion which can be culled, being limited by the percentage needed for replacements for a particular species. The standard deviation of breeding value,

$$\sigma_H = \sqrt{a_1^2\sigma_{G_1}^2 + a_2^2\sigma_{G_2}^2 + \cdots + 2a_1a_2\sigma_{G_1}\sigma_{G_2}r_{G_1G_2} + \cdots},$$

depends upon gene frequency and to some extent upon the mating system. These can be changed only a little by the breeder. Consequently the opportunity for increasing the progress expected from selection lies in making R_{IH} as large as possible. Accordingly I is defined as

$$I = b_1X_1 + b_2X_2 + \cdots + b_nX_n, \tag{4}$$

where the X's represent phenotypic performance for the several traits and the b's are multiple regression coefficients chosen so as to make R_{IH} as large as possible. These regression coefficients may be calculated from n simultaneous equations

$$\begin{aligned}
&\beta_1 + \beta_2r_{X_1X_2} + \cdots + \beta_nr_{X_1X_n} = r_{X_1H}\\
&\beta_1r_{X_1X_2} + \beta_2 + \cdots + \beta_nr_{X_2X_n} = r_{X_2H}\\
&\qquad\cdots\cdots\cdots\cdots\cdots\cdots\\
&\beta_1r_{X_1X_n} + \beta_2r_{X_2X_n} + \cdots + \beta_n = r_{X_nH}
\end{aligned} \tag{5}$$

where $\beta_i = b_i\dfrac{\sigma_{X_i}}{\sigma_H}$ and r_{X_iH} is the correlation between H and the i-th phenotypic measurement.

The simultaneous equations can be solved only if estimates of the various correlations can be calculated. The usual methods of interclass and intraclass correlation are generally sufficient to calculate the phenotypic correlations ($r_{X_iX_j}$). WRIGHT'S (1934) method of path coefficients is convenient for calculating the more complex correlations between H and phenotypic performance (r_{X_iH}). The path coefficient diagram (fig. 1) indicates the various relations between H and the phenotypic measurement for each trait (X_i). This correlation is the sum of the various paths from X_i to H, as follows,

$$r_{X_iH} = r_{G_iX_i}\{d_1r_{G_1G_i} + d_2r_{G_2G_i} + \cdots + d_nr_{G_nG_i}\}, \tag{6}$$

where $$d_i = a_i \frac{\sigma_{G_i}}{\sigma_H}.$$

Therefore we must have estimates of genetic variability for each trait (σ_{G_i}), the correlation between genotype and phenotypic performance for each trait ($r_{X_iG_i}$), and the correlation between genotypes for different traits ($r_{G_iG_j}$) to solve the simultaneous equations.

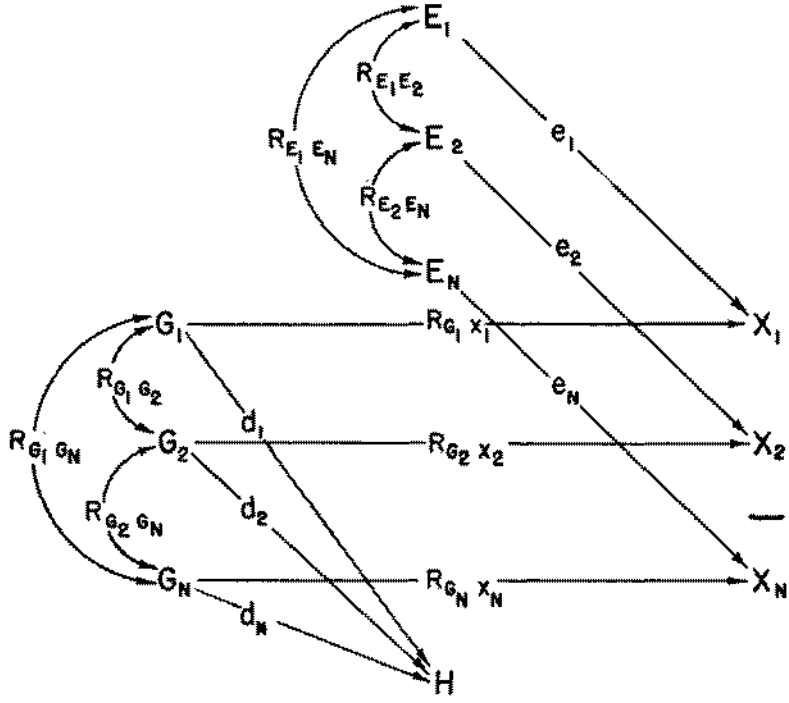


FIGURE 1. Path coefficient diagram showing the relation between phenotypic measurements (X_i) and the aggregate genotype (H). For further explanation of symbols see text.

Let an observed trait (X) be the sum of the average effects of the genes (G)[3] which an animal inherits, plus the combined effects of environment, dominance and epistasis (E); thus

(7) $$X = G + E.$$

Then the observed variance is

(8) $$\sigma_X^2 = \sigma_G^2 + \sigma_E^2,$$

[3] We have defined the genotype as the sum of the average effects of genes because WRIGHT (1921, 1935) has shown that only the average (additive) effects contribute to permanent gain from selection. The effects of environment, dominance, and epistasis may logically be grouped together since they act similarly to mask the genotype.

if G and E are uncorrelated, as would be the case unless particular efforts were made to give the better genotypes better or worse than average treatment. The correlation between the genotype and phenotype for the same trait is

$$r_{G_iX_i} = \frac{\sigma_{G_i}}{\sigma_{X_i}} = g_i, \tag{9}$$

where that trait is measured on the animal itself. The term g_i is the square root of the heritable (additively genetic) fraction of the observed variance.

DARWIN observed the importance of correlated variation, as evidenced by the statement, "Hence if man goes on selecting, and thus augmenting, any peculiarity, he will almost certainly modify unintentionally other parts of the structure, owing to the mysterious laws of correlation." The genetic basis for this statement is evident if genes have manifold effects. Many cases of pleiotropic genes have been reported for laboratory animals, although they have received little attention in domestic animals. Linkage and non-random mating systems may also cause correlated variation; however, their effects would be less permanent and consequently less important in selection. Repeated crossing over ultimately makes the coupling and repulsion heterozygotes equally numerous, while the transient nature of breeding herds prevents sustained departure from random mating in most cases. An additional and usually much more important cause of correlated variation within an interbreeding population lies in the environmental circumstances peculiar to each animal, particularly for traits which develop during the same, or in adjacent, periods of time. Thus if trait I is correlated with trait J in the same animal, I may serve as an indicator either of the animal's genotype for J or of the environmental circumstances to which that animal was exposed when traits I and J were both being developed.

Statistically an observed phenotypic correlation may be analyzed into its constituents, a genetic correlation ($r_{G_iG_j}$) and an environmental correlation ($r_{E_iE_j}$), as indicated in figure 1. The observed phenotypic correlation between two traits measured on the same animal is

$$r_{X_iX_j} = g_ig_jr_{G_iG_j} + e_ie_jr_{E_iE_j}, \quad \text{where} \quad e_i = \frac{\sigma_{E_i}}{\sigma_{X_i}}. \tag{10}$$

Sex-limited traits and traits such as carcass merit cannot be measured directly on all breeding animals. Sometimes selection must be practiced before each animal's performance for every trait is known. For these reasons there is a possibility of increasing genetic progress in a breeding program by using information about the performance of relatives. Examples of how r_{X_iH} may be calculated are given in the subsequent section where X_i is (1) the individual's own performance, (2) the performance of the individual's dam, and (3) the average perfromance of a group of relatives of which the individual is a member.

APPLICATION

Three selection indexes for young boars and gilts were constructed from data taken on the Iowa Station swine herd from the fall of 1937 through the spring of 1940. The history of the herd was given by BYWATERS (1937) and more recently by WHATLEY (1942). LUSH (1940) showed that the economic return from the swine enterprise depended largely upon three traits—growth rate, market suitability, and size of litter—the relative economic values of which were ½, 1, and 2, respectively, in the units used in this study. Growth rate was measured by weight at 180 days of age. Market suitability was measured by a numerical score given by each of several judges. The details of these measurements were given by WHATLEY for 180-day weight and by STONAKER and LUSH (1942) for score. The measure of litter size was that suggested by LUSH and MOLLN (1942),

$$P = n_0 + n_{21} + n_{56} + W_{21}/10 + W_{56}/30,$$

the n's and W's referring to the number of pigs and weight of the litter, respectively, at the ages designated by the subscripts. Young boars and gilts are not old enough to have produced litters themselves at the time first selections must be made; consequently the estimate of productivity for each young animal is based upon the litter in which it was born.

The aggregate breeding value of an animal is therefore defined as

$$H = \tfrac{1}{3}G_W + G_S + 2G_P,$$

G_W, G_S and G_P referring to the genotypes for 180-day weight, market score and productivity, respectively. The following statistics are needed to construct one or more of the three indexes:

A. Phenotypic constants
   1. Standard deviation for each of the three traits
   2. Phenotypic correlation between each pair of traits
   3. Phenotypic correlations between the traits of relatives

B. Genetic constants
   1. Heritable fraction of the variance in each trait
   2. Genetic correlation between each pair of traits.

Differences due to season and line of breeding were eliminated in calculating these statistics, as selection is usually practiced between animals born in the same season and within the same line or other interbreeding population. The phenotypic constants shown in table 1 were calculated by the usual methods of variance and covariance analysis (SNEDECOR 1940).

Several procedures were outlined by LUSH (1941) for estimating heritability (g_1^2) in farm livestock. Intrasire regressions of offspring on dam were considered preferable to correlation coefficients in the present case because of the rather intense selection which had been practiced in the Iowa Station swine

TABLE 1

Phenotypic constants necessary to construct one or more of the three indexes.

TRAIT	STANDARD DEVIATION	DEGREES OF FREEDOM
180-day weight (in pounds)	31.86	1513
Market score (in points)	4.78	1513
Productivity (in points)	9.72	282
CORRELATION BETWEEN TRAITS	**CORRELATION COEFFICIENT**	**STANDARD ERROR**
Pig's own weight and score	0.614	0.016
Pig's own weight and productivity of its dam	−.024	.059
Pig's own score and productivity of its dam	−.081	.059
Weight of litter mates	.314	.027
Score of litter mates	.204	.026
Weight and score of litter mates	.186	.026

herd for the traits under consideration. The intensity of selection was evidenced by the higher averages and lower variability for the dams as compared to their offspring. The intrasire regressions of offspring on dam are shown in table 2.

TABLE 2

Regression of each trait for the offspring on each of the three traits for the dam.

OFFSPRING'S TRAIT	DAM'S TRAIT	SYMBOL*	REGRESSION COEFFICIENT	DEGREES OF FREEDOM
Weight	Weight	$b_{W_2W_1}$	0.149	232
Score	Score	$b_{S_2S_1}$	.049	189
Productivity	Productivity	$b_{P_2P_1}$	.081	82
Weight	Score	$b_{W_2S_1}$	.153	189
Score	Weight	$b_{S_2W_1}$	.013	232
Weight	Productivity	$b_{W_2P_1}$	−.067	204
Productivity	Weight	$b_{P_2W_1}$	.004	52
Score	Productivity	$b_{S_2P_1}$	−.024	204
Productivity	Score	$b_{P_2S_1}$	.041	52

* The sub-subscripts 1 and 2 refer to dam and offspring, respectively, the regressions being those of offspring's trait on dam's trait in each case.

Values of g_i, calculated from the formula,[4] $g_i = \sqrt{2b_{i_2i_1}}$, are given in table 3 for the three traits.

[4] This formula was derived from the relation,

$$b_{i_2i_1} = g_i^2 t + e_i^2 r_{EE}.$$

The genetic relationship between dam and offspring (t) was taken as 0.5, and the environmental correlation (r_{EE}) was assumed zero, since no particular efforts had been made to give parent and offspring similar or dissimilar treatment.

TABLE 3

Genetic constants derived from intrasire regressions of offspring on dam.

STATISTIC	SYMBOL	ESTIMATE
$\sqrt{\text{heritability of weight}}$	g_W	0.546
$\sqrt{\text{heritability of score}}$	g_S	.313
$\sqrt{\text{heritability of productivity}}$	g_P	.402
Genetic correlation between weight and score	$r_{G_WG_S}$	.519
Genetic correlation between weight and productivity	$r_{G_WG_P}$	.0±
Genetic correlation between score and productivity	$r_{G_SG_P}$	.0±

Procedures for estimating the genetic correlations ($r_{G_iG_j}$) have not been developed previously. Equation 10 is not suitable for this purpose because no direct method is available for separating the genetic and environmental correlations for two traits measured upon the same animal. For example, any environmental accident such as differential exposure to parasites or infection which affected growth and plumpness would either raise or lower the growth rate and score of the same animal. Thus the actually observed correlation between the two traits on the same animal conceivably could be due wholly to such environmental circumstances, or to genes which affect both traits, or to a mixture of the two causes in any proportion. To measure the genetic correlations by themselves it was necessary to correlate one trait in one animal with the other in a relative. The formula,[5]

$$r_{G_iG_j} = \sqrt{\frac{b_{i_2i_1} \cdot b_{j_2i_1}}{b_{i_2i_1} \cdot b_{j_2j_1}}} = \sqrt{\frac{(\text{cov } I_2J_1)(\text{cov } J_2I_1)}{(\text{cov } I_2I_1)(\text{cov } J_2J_1)}},$$

was adopted because it appeared to be unbiased by selection and to utilize most of the available information. Estimates of the genetic correlations between each pair of the three traits are given in table 3. Consistent estimates could not be derived for $r_{G_WG_P}$ and $r_{G_SG_P}$, since in each case the two regressions required in the numerator differed in sign. Since these regressions are small and not significantly different from zero, it seemed more accurate to assign both $r_{G_WG_P}$ and $r_{G_SG_P}$ values of zero than to attempt an alternative method of calculating them.

The genetic constants probably include a small fraction of the epistatic deviations and exclude the average differences between groups of dams to

[5] Selection for the independent variable does not bias the corresponding regression coefficient (EISENHART 1939). The quantity $\sqrt{b_{i_2i_1} \cdot b_{j_2i_1}}$ consequently can be taken as an estimate of the correlation between two traits of dam and offspring in an unselected population. The biometric relation,

$$r_{i_1j_1} = r_{i_2i_1} = g_i g_j r_{G_iG_j}$$

permits the derivation of the formula given above for $r_{G_iG_j}$. The two regressions (or correlations) actually provide two independent estimates of $r_{G_iG_j}$ in unselected populations and an arithmetic average of the two estimates would be less biased by sampling errors than the geometric average used here.

which sires were mated. Since some inbreeding had occurred, they may be slightly nearer zero than would be expected in a non-inbred population. If they are biased, and this bias is equal and in the same direction for the different genetic constants, the index will be no less accurate for selecting breeding animals.

Weight at 180 days of age and market score are available for each animal before breeding age. The first index constructed was based on these two traits. Values for r_{WH} and r_{SH} were calculated by substituting the statistics from tables 1 and 3 into formula 6 as follows:

$$r_{WH} = \frac{0.546}{\sigma_H}\{5.803(1) + (1.496)(.519)\} = 3.593/\sigma_H$$

$$r_{SH} = \frac{0.313}{\sigma_H}\{5.803(.519) + 1.496(1)\} = 1.409/\sigma_H.$$

The two simultaneous equations are:

$$\beta_W + 0.614\beta_S = 3.593/\sigma_H$$
$$0.614\beta_W + \beta_S = 1.409/\sigma_H.$$

Solving, we get $\beta_W = 4.381/\sigma_H$ and $\beta_S = -1.282/\sigma_H$.
Since $\beta_i = b_i\sigma_{Xi}/\sigma_H$, the first index is

$$I_1 = 0.137W - 0.268S,$$

where W and S represent the pig's own 180-day weight and market score, respectively. If the index can thus be made more convenient for use or for keeping records, it may be multiplied by any constant or any constant may be added to it without influencing its relative accuracy, since such procedure does not change R_{IH}.

A second index was constructed by using the productivity of the dam as a measure of each pig's productivity in the index, the lapse of one generation being compensated by multiplying r_{PH} by one-half as follows:

$$r_{PH} = \frac{0.402}{2\sigma_H}(7.802) = 1.566/\sigma_H.$$

The three simultaneous equations were solved as before, the second index being

$$I_2 = 0.136W - 0.232S + 0.164P.$$

The third index was designed to include information about the average weight and score of the litter in which each pig was born, in addition to the three traits in the second index. The procedure used was to consider the average weight ($\bar{W}$) and score ($\bar{S}$) of the litter as a fourth and fifth variable, expressing the necessary correlations in terms of the correlations given in table 1 and as a function of the number of pigs per litter (k). This latter step was necessary because the number upon which the average is based influences the

variability of the average and the amount of information in the average concerning each pig's breeding value. The additional phenotypic correlations in terms of k and the correlations in table 1 are

$$r_{w\bar{w}} = \sqrt{\frac{1 + (k-1).314}{k}},$$

$$r_{s\bar{s}} = \sqrt{\frac{1 + (k-1).204}{k}},$$

$$r_{w\bar{s}} = \frac{.614 + (k-1).186}{\sqrt{k[1 + (k-1).204]}},$$

$$r_{s\bar{w}} = \frac{.614 + (k-1).186}{\sqrt{k[1 + (k-1).314]}},$$

$$r_{p\bar{w}} = -.024\sqrt{\frac{k}{1 + (k-1).314}}$$

$$r_{p\bar{s}} = -.081\sqrt{\frac{k}{1 + (k-1).204}}, \quad \text{and}$$

$$r_{\bar{w}\bar{s}} = \frac{.614 + (k-1).186}{\sqrt{[1 + (k-1).314][1 + (k-1).204]}}.$$

The additional correlations between H and the litter averages are

$$r_{\bar{w}H} = \frac{3.593[1 + (k-1).5]}{\sigma_H\sqrt{k[1 + (k-1).314]}} \quad \text{and}$$

$$r_{\bar{s}H} = \frac{1.409[1 + (k-1).5]}{\sigma_H\sqrt{k[1 + (k-1).204]}}.$$

The five simultaneous equations were solved as before, giving the following partial regression coefficients:

$$b_W = .098$$

$$b_S = -.165$$

$$b_P = \frac{0.358\{0.164(k-1) + 0.016(k-1)^2\}}{2.18\{0.998(k-1) + 0.899(k-1)^2\}}$$

$$b_{\bar{w}} = \frac{k\{0.270 + (0.004)(k-1)\}}{7.14\{3.273(k-1) + 0.324(k-1)^2\}}$$

$$b_{\bar{s}} = \frac{-k\{0.070 + (0.02)(k-1)\}}{1.07\{0.491(k-1) + 0.049(k-1)^2\}}.$$

The absolute values of the three latter coefficients are given in table 4 for different values of k. The number of pigs in the litter has so little effect on

TABLE 4

Partial regression coefficients for dam's productivity, average weight of the litter and average score of the litter for different numbers of pigs in the litter, in the third index.

NUMBER OF PIGS IN THE LITTER (k)	b_P	$b_{\bar{W}}$	$b_{\bar{S}}$
1	0.164	0.038	−0.067
2	.165	.059	− .113
3	.165	.072	− .148
4	.166	.081	− .175
5	.166	.088	− .197
6	.167	.093	− .215
7	.167	.098	− .230
8	.167	.101	− .243
9	.168	.104	− .254
10	.168	.106	− .264
11	.168	.109	− .273
12	.169	.110	− .280
13	.169	.112	− .287
14	.169	.113	− .293
15	.169	.115	− .299

b_P that this could be taken as a constant (0.166 or 0.167) with very little error.

The amount of genetic progress expected when a given index is used in making selections is proportional to R_{IH} (see formula 3). Hence these values provide a basis for choosing an index which is easy and simple to use yet which is of nearly maximum accuracy. For example the three indexes previously constructed may be compared as follows:

$$R_{I_1H} = \sqrt{\beta_W r_{WH} + \beta_S r_{SH}} = 0.363$$

$$R_{I_2H} = 0.395 \quad \text{and}$$

$$R_{I_3H} = 0.404, \quad \text{for} \quad k = 5.$$

The second and third indexes are 8.8 and 11.3 percent, respectively, more efficient than the first. Since the time and effort expended in keeping records is but a small fraction of the total labor connected with a breeding program, the second index would almost certainly be preferable to the first. The third might also be chosen over the second, since genetic progress could be increased still further, and the extra labor would be only that of computing and using the litter averages from data already taken.

DISCUSSION

FISHER'S (1930) "fundamental theorem of natural selection" and WRIGHT'S (1931) emphasis that genetic change depends upon genetic variability and selection intensity indicate that the factors which are important in natural selection also hold for selection as practiced by man. In equation 3, σ_H is a

measure of genetic variability and $\bar{i}$ a measure of selection intensity. The breeder in applying artificial selection to an animal population has the opportunity of increasing the accuracy of his selections (increasing R_{IH}) considerably over what may hold in the "trial-and-error" methods of natural selection. The breeder has additional opportunities of increasing $\bar{i}$ (within limits) and of increasing genetic variability by the intentional control of population size (inbreeding) and migration (outcrossing) which have not been considered in the present study. WRIGHT (1940) has examined the conditions under which the supplementary use of these latter methods may be advantageous.

For the special case where the traits are uncorrelated, R_{IH} is a maximum when each regression coefficient is equal to (or proportional to) the product of the relative economic value and heritability for each trait ($b_i = a_i g_i^2$). The correlations which may exist between traits complicate the calculation of the partial regression coefficients, just as correlations between the independent variables do in any multiple regression analysis. When the phenotypic correlation is large as compared to the genetic correlation, the regression coefficient for a trait with little economic importance or slight heritable variation may be negative, because its function in the index then becomes mainly that of indicating the environment for a more important and more highly heritable trait. An example of this was seen in the negative regression coefficients for score in the three indexes calculated previously. However, selection for the animals having the highest indexes would create some improvement for score because of the positive genetic correlation between score and weight.

An index constructed from data taken on a herd in one locality may not be widely applicable. The reasons for this are:

1. Relative economic values for a trait may vary with the particular locality or nature of the enterprise.

2. The genetic constitution of herds may differ, especially where they are under distinctly non-random mating systems such as intense inbreeding.

3. Different managerial practices may cause the standard deviations for the traits to vary in different herds. The standard deviations for subjective traits such as market conformation measured by judging or by scores may vary because different judges will vary the range over which they spread their scores.

4. Few herds are large enough to provide data sufficient to make the sampling errors of the genetic constants small.

The best way to test whether or not selection indexes can be standardized and recommended for general use seems to be to compare several indexes constructed from data taken on different herds.

The data in the present study were sufficiently numerous to provide accurate estimates of the phenotypic constants in table 1. They were less satisfactory with regard to the reliability of the genetic constants given in table 3. Some idea of the general accuracy of these figures can be obtained by comparing them to similar estimates by other investigators. WHATLEY (1942) used several methods to estimate heritability of 180-day weight in the Iowa Station swine herd through 1938, concluding that "at least 30 percent and possibly

more than 40 percent of the individual variance" was hereditary. WHATLEY and NELSON (1942) estimated that 180-day weight in the Oklahoma Duroc herd was 23 percent heritable, while BAKER and coworkers (1943) found a figure of 25 percent for 168-day weight in the Nebraska (North Platte substation) Duroc herd. These estimates help to substantiate the figure of 30 percent for the heritability of 180-day weight found in this study. WHATLEY and NELSON also estimated that market score was about 33 percent heritable, while STONAKER and LUSH (1942) obtained an estimate of 20 percent for the Iowa Station herd from data which included that used in this study. These estimates indicate that our estimate of 10 percent for the heritability of score may be too small. Previous reports of the heritability of productivity have not been made, but LUSH and MOLLN (1942) found that the correlation between litters by the same sow was between 0.15 and 0.20 for the items which are included in productivity. They quoted a number of investigations which substantiated their findings. In general these results substantiate our figure of 16 percent if most of the permanent differences in these items between sows are hereditary.

Other attempts to estimate genetic correlations have not been made. The genetic correlation of 0.52 between 180-day weight and score indicates that about half of the genes which influence one trait also influence the other (assuming equal gene frequency and equal effects of the genes). However most of the observed correlation of 0.61 was due to an environmental correlation. Although an effort was made to score all pigs at a constant weight of 225 pounds, part of this correlation was probably due to a subjective tendency on the part of the judges to assign scores in accordance with differences in age or weights of the pigs. The genetic correlations between productivity and weight or score may have been either positive or negative so far as the evidence from the present data indicate. They appeared to be small; hence assigning them a value of zero is unlikely to have caused serious errors in the indexes.

From the studies of heritability which have been made for economic traits in different farm animals, it seems that the best indexes which can be constructed will be far from perfect. The confusing effects of environment, dominance, and epistasis in masking genotypes cause the progress in the present case to be less than half of what might be made if genotypes could be recognized precisely. Thus the indexes constructed for swine permit from 36.1 to 40.4 percent as much gain as could be made with a perfect index (where $R_{IH}=1$), which is the limit of what could be achieved if the exact Mendelian composition of every animal were known. These indexes could be improved somewhat by more perfect control of the environment, by the wise use of corrections for known environmental circumstances, by more accurately measuring differences in phenotypes, and by including the performance of additional relatives in the index; however, the use of these methods is limited by practical considerations. Although R_{IH} is likely to increase with the age of the animal (as more becomes known about its phenotype and as its progeny becomes observable) so that more gain can be made from selecting within a group of

older animals, the length of generation will also increase. DICKERSON and HAZEL (1942) have shown that the interval between generations in some cases is increased by progeny testing more than enough to offset the increased accuracy of selection, the net result of more emphasis on the progeny test then being a decrease in the *annual* rate of genetic improvement. While these considerations do not indicate much possibility of phenomenally rapid improvement in animal populations from selection alone, the progress which can be made with properly constructed indexes is considerably greater than can be expected when the ideals toward which selection is directed are confused or erroneous.

CONCLUSIONS

The genetic gain which can be made by selecting for several traits simultaneously within a group of animals is the product of (1) the selection differential, (2) the multiple correlation between aggregate breeding value and the selection index, and (3) genetic variability. The first of these may be very small due to the breeder's carelessness, procrastination, etc., and is limited by the rate of reproduction for each species, while the third is relatively beyond man's control; hence the greatest opportunity of increasing the progress from selection is by insuring that the second is as large as possible.

A multiple correlation method of constructing selection indexes having maximum accuracy was presented. The following constants must be known in order to solve the simultaneous equations:

1. Relative economic values for the different traits
2. Phenotypic constants
   a. Standard deviations for each trait
   b. Correlation between each pair of traits
3. Genetic constants
   a. Heritability of each trait
   b. Genetic correlations between each pair of traits

Examples of the construction of selection indexes for young boars and gilts were presented from data taken on the Iowa Station swine herd using (1) 180-day weight and market score of the individual animal, (2) the two previous traits and productivity of the dam, and (3) the three previous traits and the average weight and score of the litter of which each pig is a member.

The progress which can be made by using the above indexes varied from 36 to 40 percent of that which could be made with a perfect index. The loss is due to the confusing effects of environmental circumstances, dominance, and epistasis, all of which can make phenotypes unlike genotypes.

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Rendel and Robertson (1950)

> *Rendel, J.M., and Robertson, A. 1950. Estimation of genetic gain in milk yield by selection in a closed herd of dairy cattle. Journal of Genetics, vol. 50, no. 1, pp. 1 to 8.*

The problem

Rendel and Robertson estimated the genetic progress possible in a closed dairy herd and divided genetic gain into four paths, which are sires of bulls, sires of cows, dams of bulls and dams of cows. For each path, the gain depends on the intensity of selection, the accuracy and the generation interval. They concluded that the possible progress was about one percent per year and that the selection of cows contributed very little.

Points to note while reading

The formula for annual genetic gain, which sums the progress of the four paths and divides it by the sum of the generation intervals, is still the standard way of comparing breeding programs. Note the weight of the generation interval, which explains why genomic selection, by shortening this interval, doubled the rate of genetic gain in dairy cattle.

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ESTIMATION OF GENETIC GAIN IN MILK YIELD BY SELECTION IN A CLOSED HERD OF DAIRY CATTLE

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(With One Text-figure)

INTRODUCTION

The simplest and most direct way of breeding farm animals is by mass selection, that is, by choosing breeding stock purely on the basis of their own phenotype and mating the chosen animals at random. In dairy cattle the value of this practice is reduced, since such selection cannot be practised on males. It has therefore been suggested that the bull should be judged on the performance of his progeny in order to make up for the fact that he has no measurable phenotype himself. In this way progress would be hastened. The advocates of this view can point to evidence in historical documents and claim that the cattle of centuries ago were not greatly different from our own in milk yield, which argues the inefficiency of any mass selection which may have taken place since then. However, such evidence, though interesting historically, does not carry much scientific weight; and it seems desirable to examine the method of mass selection, first, from the theoretical point of view in order to estimate its effect under the most favourable circumstances and, secondly, from the practical aspect in considering how much selection is actually practised in herds and how much genetic gain can be expected therefrom.

PREVIOUS ESTIMATES OF GENETIC GAIN FOR DAIRY HERDS

It is obvious that the direct observation of the variation of the average yield of a herd gives little indication of the genetic improvement achieved. There are many reasons why the yield of a herd should increase with time—the owner will increase his skill at managing his herd with practice, and there has been a great improvement in methods of feeding dairy cattle since the 1920's. In the reverse direction, there is the influence of two wars to be reckoned with, and the recovery from these provides two further periods of general increase in milk yields.

At first sight it would seem possible to measure the effect of such extraneous changes on yields by considering the yield of the same cow in successive years. After subtracting environmental trends so determined from the total, the genetic changes are left. This method is used by Lörtscher (1937) and with slightly more sophisticated statistics by Nelson (1943). There is, however, a logical error in this approach which vitiates the conclusions drawn. The records of the same cows in successive years will vary systematically for two reasons: first, changes in management whose effect we wish to evaluate, and secondly, increase in age of the cow. Before comparison of records in successive years is made, therefore, all yields must be reduced to their equivalent at a standard age. To do this, correction factors must be used; if these are calculated from the data provided by the herd under review, they will include the changes due to either environmental or genetic

trends. Suppose, for example, feeding is being steadily improved and yields are rising for this reason; suppose also that yields are rising because of selection leading to genetic gains; we shall then have the picture given in Fig. 1.

In the figure, the position is simplified by considering only two lactations. Point *A* is the level of first lactations in year 0, and *B* is the level of second lactations of the same cows in year 1. The genetic and management levels are assumed to be improving at the rate of g and m per year. *BE* is the apparent age increase, which is, in fact, made up of the true age effect a and the management improvement m. The first lactations in year 1 will be at the point *C*. We can actually measure the increase in yield of a given cow with time, $a+m$, and the increase in the mean yield of the herd, $g+m$. If we calculate the

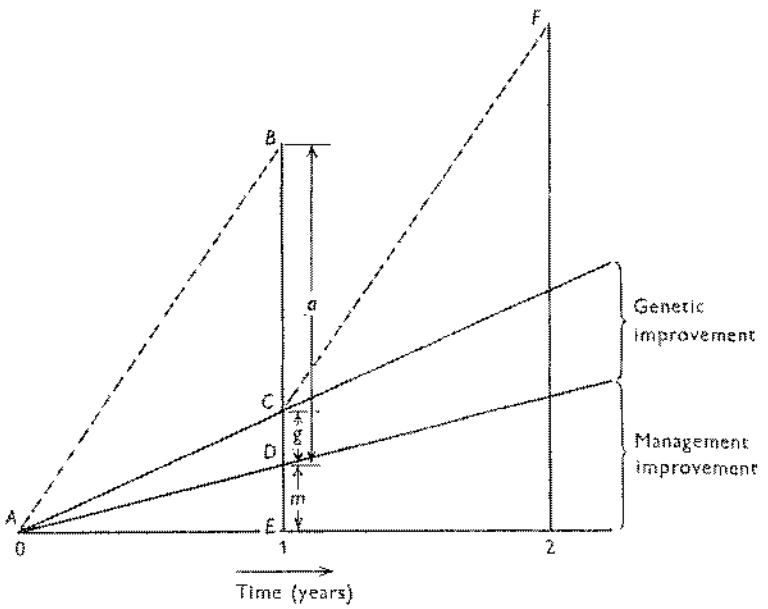


Fig. 1.

age correction factor from $a+m$, assuming it is equal to a, then when the conversion to standard yield is made both m and a are taken out, leaving no apparent effect of environment. If, on the other hand, the factor is calculated by comparing first and second lactations made in the same year a slope corresponding to $a-g$ is being measured, and the correction is then reduced by the amount of any genetic trend, so that the environmental effect is made to include the genetic. In this case, the effects of selection must be allowed for. Algebraically we can say that $g+m$ and $m+a$ can be measured, from which we can deduce $a-g$, which can also be estimated without first knowing $g+m$ and $m+a$, but none of the three entities can be known separately.

Correction factors may be taken from estimates made in other herds. These will have against them the objection already outlined and also doubt as to their suitability. The effect of age on yield is not independent of management. A cow which is matured early by heavy feeding will not show as marked an age effect as a cow reared on a lower plane of nutrition (Bonnier, Hansson & Skjervold, 1948), so that correction factors must be fitted to the management of the herd in question if they are to be appropriate. If, for instance, we have a quick-maturing herd, either for genetic or management reasons, use of corrections

derived from slower-maturing animals in this method will suggest that management is getting worse (cows' corrected first records will be above the true value), and consequently the genetic level improving, even though there is really no change in either. It may be said that such errors will be small, but the age effect is large compared with the other effects, and any errors in its estimation will be cumulative. Suppose the factor converting first to second lactation is 0·005 too high, then first lactations when converted will be 0·5% higher than second (other things being equal), the environment will appear to be deteriorating at 0·5% per year, and the genetic level will be correspondingly increased. Since the actual trends are of the order of 0·75% per year, errors of this size are of some importance, and are likely to be exceeded in most estimates in practice.

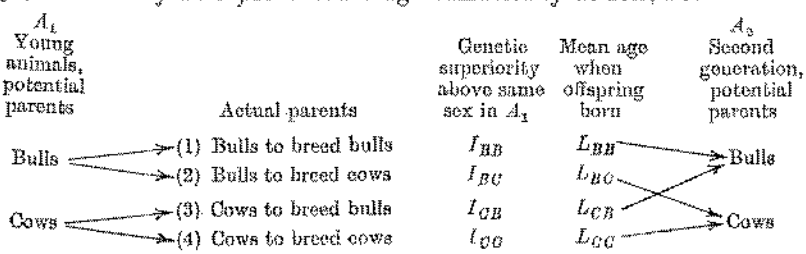
It must be concluded that such indirect estimates of the genetic gains under selection are useless. The alternative is to estimate the genetic gains directly from the selection applied. This is the method used here.

IMPROVEMENT TO BE EXPECTED FROM DIRECT SELECTION UNDER OPTIMUM CONDITIONS

The application of the principles of genetics to the problems arising in breeding farm animals for economical production are of comparatively recent origin. They are based on the theoretical considerations of Wright, Fisher and Haldane. The detailed application of their findings to animal breeding has been mostly due to Lush and his co-workers. The general principles of this new approach are given by Lush in his book *Animal Breeding Plans*. The basic concept, that of 'heritability', is a statistical one, relating the phenotype of an animal to its genotypic value. In its narrow sense, as we shall apply it here to individuals, it can be defined as the regression of genotypic value on phenotype. The heritability of milk yield based on one lactation only is 0·25 in cows, and it is this value we shall use here. Values close to this have been obtained in many investigations, both in America and this country. The statement that the heritability of milk yield is one-quarter, put in non-statistical language, means that the difference in the breeding value of any two cows is, on the average, one-quarter of the difference between their records.

The expression for the estimated rate of genetic improvement for a known selection method has been given by Dickerson & Hazel (1944) in their paper on the progeny test. It may be desirable here to give a derivation of this expression as applied to our particular case.

The process of breeding and selection may be envisaged as follows. Young animals (heifers and young bulls) are taken into the breeding herd on the basis of their parents' performance and may themselves be regarded as the potential parents of the next generation. The decision as to which of these animals actually will pass their genes on is made on the basis of some measurement of their own genotype—by own performance in cows and by progeny tests in bulls. Genes may be transmitted to the next generation in four ways, which may be represented diagrammatically as follows:



Owing to the smaller number of males required, I_{CB} will necessarily be greater than I_{CC} and so on. In addition some of the classes of actual parents are mutually inclusive. For instance, a heifer calf from a cow of group 3 would be taken into the herd but not a bull calf from a cow of group 4.

The average rate of improvement may be obtained as follows. Let B and C be the mean genetic level of young animals born in a given year and taken into the breeding herd as potential parents. Then if ΔG is the genetic improvement per year, the mean genotype of sires of such bulls is $B-L_{BB}\Delta G+I_{BB}$ and of dams of such bulls $C-L_{CB}\Delta G+I_{CB}$, so that we have an equation

$$B=\tfrac{1}{2}(B-L_{BB}\Delta G+I_{BB}+C-L_{CB}\Delta G+I_{CB}),$$

and similarly

$$C=\tfrac{1}{2}(B-L_{BC}\Delta G+I_{BC}+C-L_{CC}\Delta G+I_{CC}).$$

On adding the two equations, B and C vanish and we are left with

$$\Delta G=\frac{\Sigma I}{\Sigma L}.$$

This gives the rate of genetic improvement to be expected from a selection programme in terms of the selection differentials and the generation lengths. In our case I_{BB} and I_{BC} are zero, since there is no selection amongst bulls.

The phenotype selection differential I_P measures the difference between the mean of the parents and the mean of all animals of the same sex in the population from which they were drawn. To convert this into a genetic measure, we must multiply by the heritability h^2. Then

$$I_G=I_P h^2.$$

If we assume that the phenotype is distributed normally, then we can express I_P as $\sigma_P \bar{i}$, where $\bar{i}$ is the superiority in terms of standard deviations and is obtainable in terms of the proportion selected, from tables (Pearson, 1931; Fisher & Yates, 1938), and σ_P is the phenotypic standard deviation. But also

$$\sigma_G=h\sigma_P,$$

where σ_G is the standard deviation of genotypic value.

Thus

$$\begin{aligned} I_G &= \bar{i}h^2\sigma_P \\ &= \bar{i}h\sigma_G. \end{aligned}$$

The latter is the most convenient form, as for a given proportion selected the only variable is h.

Let us consider a herd of about 100 cows into which no breeding stock is introduced and in which there is no progeny testing and bulls are chosen solely on the milk records of their dam. The selection is being practised only in choosing cows to breed from. It is therefore necessary to calculate the possible selection differential in selecting cows to breed cows and selecting cows to breed bulls. As far as the former is concerned, let the proportion of cows in the first lactation be f_1, and so on, and let the natural mortality rates be 1/6 in the first three lactations and 1/3 for the fourth and after. Assume that we select on the first lactation only and reject all calves born from cows that are themselves culled. Let α be the number of mature heifers produced per cow in the herd per year.

Then the replacements available $=\alpha\,(f_2+f_2+f_3+f_4+\ldots)$, the first f_2 being calves born to cows, subsequently accepted into the herd, in the first lactation. Then

$$f_1=\alpha\,(2f_2+f_3+f_4+\ldots).$$

But we know from the mortality figures that at equilibrium

$$f_3=\tfrac{5}{6}f_2, f_4=\tfrac{5}{6}f_3, f_5=\tfrac{5}{6}f_4,$$

so that the later proportions can be expressed in terms of f_2, giving

$$f_1=\alpha\times 4{\cdot}916 f_2.$$

Of this f_1, one-sixth will die during the first lactation leaving $\frac{5}{6}\,(\alpha\times 4{\cdot}916 f_2)$, from which f_2 are to be selected.

The percentage culled is then $\dfrac{4{\cdot}097\alpha-1}{4{\cdot}097\alpha}=1-\dfrac{0{\cdot}244}{\alpha}$. An accurate value of α is not available, but it may be assumed to lie between 0·35 and 0·40. This gives for percentage culled 39·0 and 30·3 respectively.

For milk yield, we may assume that $\sigma_P=0{\cdot}20\,\overline{Y}$ and $\sigma_G=0{\cdot}10\,\overline{Y}$, where $\overline{Y}$ is the average yield of the herd. Then the expected genetic superiority of the selected cows above their contemporaries will be $0{\cdot}032\,\overline{Y}$ and $0{\cdot}024\,\overline{Y}$ respectively. Calculation assuming culling at the end of the second lactation shows roughly the same results as on the first.

We may assume that the bulls are bred from the top 5 % of the herd (Lush, 1946, p. 147). The genetic superiority of their dams would then be $0{\cdot}103\,\overline{Y}$. Assuming a sum of generation lengths of about 13 years, which seems to be about the minimum possible, we have for the expected genetic improvement

$$\Delta G=\frac{0{\cdot}135\,\overline{Y}}{13}=0{\cdot}010\,\overline{Y} \quad \text{if} \quad \alpha=0{\cdot}4$$

and

$$\Delta G=0{\cdot}009\,\overline{Y} \quad \text{for} \quad \alpha=0{\cdot}35.$$

Thus, under the optimum conditions for mass selection the expected rate is 1·0 % per year. Of this, about one-quarter comes from the early culling of heifers and the rest from the selection of the dams of bulls.

THE ESTIMATING OF GENETIC IMPROVEMENT FROM THE SELECTION PRACTISED IN A HERD

Just as the rate of genetic improvement was estimated from the calculated superiority of parents over their contemporaries in the previous section, an estimation of the actual rate of genetic improvement in a herd can be obtained from the selection actually practised. Records of a herd founded in 1903 which is still in existence have been kindly made available to us. Although some breeding stock has been introduced during that time, about two-thirds of the animals born up to 1933 were sired by home-bred bulls. Thus it is possible to calculated the selection actually practised both in breeding cows and bulls. Selection was not all done at a given age as in our theoretical example, but there was a continual selection at all ages. The amount of selection was calculated as follows.

The records had, for other purposes, been divided into seven groups of sixty cows, the cows in each group being contemporaries. A calculation was therefore made within each group of the first lactation records of all cows having daughters in the milking herd, weighted according to the number of such daughters. This was compared with the average of all first lactations in the group. This gives the 'apparent' phenotypic selection differential as measured on first lactation and, on multiplying by 0·25, gives the 'apparent' genotypic

selection differential as shown in Table 1. This, however, does not represent the true selection differential if there has been selection in later lactations. Column 1 shows the actual amount of culling on first lactation as measured by the difference between the first lactations of animals which also had a second, and the whole group. The lack of correspondence between the two columns is a reflexion of the degree to which parenthood was independent of culling. Animals culled after their first lactation frequently left daughters in the herd and are therefore part of the selected breeding stock, whereas animals kept in the herd did not all have daughters. If all the selection was actually on the first lactation, then column 2 should be one-quarter of column 1. The fact that it is more than one-quarter indicates that some culling had also gone on in later lactations. The genetic improvement can then better be calculated on all the records available. The heritability then becomes dependent on the number of records, and the formula for genetic superiority is

$$I_{CC} = \Sigma n h^2_l (Y_l - \overline{Y}),$$

Table 1. *Estimation of selection differential from culling practised*

Period	Culling on 1st lactation* (gal.)	Genetic superiority of dams: Based on 1st lact. (gal.)	Genetic superiority of dams: Based on 1st–4th lact. (gal.)	No. of daughters in herd
1	+0·4	+6·1	+8·5	67
2	−7·2	+2·0	−3·5	44
3	+19·5	+3·5	+6·6	54
4	+2·7	+9·7	+16·2	30
5	+38·9	+0·7	+2·0	19
6	+34·0	+10·3	+17·6	24
7	+35·0	−1·4	−2·3	37
Total	+16·0	+4·8	+6·3	275

* Means of 1st lactations of cows having 2nd lactations − mean of all 1st lactations.

Table 2. *Selection differential on dams of bulls*

	Estimated genetic superiority of dams (based on 1st–4th lactations) (gal.)	No. of bulls	No. of daughters
Total	+38·1	67	289
Bull *AQ*	+101·4	1	65
Total less *AQ*	+17·5	66	224

where h^2_l is heritability based on l lactations, $\overline{Y}$ is the mean of the first lactations of the whole group, Y_l is the mean of the lactations (corrected to first lactation) and n the number of daughters for each cow. Column 3 shows the values of I_{CC}, thus calculated, for each of the groups. The average value shows an increase of about one-third over that calculated on first lactations only. This value for I_{CC} should be compared with that theoretically possible as calculated above, which was, depending on the value of n, around 2·5–3 % of the herd average. This gives in this herd, where $\overline{Y}$ is 450 gal., a possible superiority of 12–14 gal. The fact that the actual figure is below this is due to many reasons —involuntary culling as, for example, for reactions to the tuberculin test, played some part, and in some stages selection was inefficient in the sense that a cow was culled after her first lactation, but nevertheless her heifer calf came into the herd.

In a similar way, the improvement due to selection of dams of bulls can be calculated. In this case each dam was weighted according to the number of daughters which her son had in the herd, as this varied considerably. This estimate is very uncertain, as one bull from an outstanding dam had sixty-five daughters in the herd. Exclusion of him from the

calculation reduces I_{CB} from 38 to 17. The results are shown in Table 2 and are calculated on all the available records of the dam. Using the figure for I_{CB} which includes the bull *AQ*, we have a value of $I_{CC}+I_{CB}$ of 44·1 gal.

The generation lengths were also calculated for the herd with the following results:

L_{BB}	34 months
L_{BC}	36 months
L_{CB}	44 months
L_{CC}	47 months

giving a total for ΣL of 13½ years, and an estimate for the genetic improvement per year of $\Delta G = 44{\cdot}1/13{\cdot}5 = 3{\cdot}3$ gal./year.

The generation interval is exceptionally short. Lush has estimated the mean generation interval (i.e. $\frac{1}{4}\Sigma L$) in dairy cattle at 4–4½ years. An approximate check on British herd books gave an estimate of ΣL for Friesians and Shorthorns of 18·4 and 19·1 years respectively.

This estimate of the rate of genetic improvement of the herd has several snags. The first and most obvious one is in the applicability of heritability data derived from other herds and breeds of cattle to this particular herd. An estimate of heritability within the herd itself is in agreement with the figure used, but the standard error of the estimate is fairly large. Another source of error is in the sampling error of the estimate, both statistical and Mendelian. The value of the method lies not so much in the absolute value of the advance calculated as in the comparison of it with the optimum possible advance. In this case we have a value of I_{CC} of 6·3 gal. compared with a possible of 12–14 gal. and of I_{CB} of 38 gal. compared with a possible of 50 gal.

Discussion

The maximum possible rate of genetic improvement in a closed herd would seem to be of the order of 1 % of the average yield per year. In the herd whose records have been analysed, the probable improvement achieved by selection is 0·7 % per year, although the actual value could vary between fairly wide limits. The rate of improvement in normal pedigree herds would be expected to be less than this for several reasons:

(*a*) The rate of improvement depends on the mean generation length, which in the herd in question was just over 3 years. An examination of the Shorthorn and Friesian herd books shows that in those breeds the mean generation length was over 4½ years. This would reduce the maximum improvement expected to about 0·6 % per year.

(*b*) Most herds are not closed and bulls are brought in. Knowledge of their breeding value would be less accurate than if they were home-bred, e.g. the management factor is not so well known.

(*c*) Selection will also be practised for other factors, e.g. type and butterfat percentage. This will reduce the selection differential on yield.

All these factors will reduce the rate of genetic improvement of breeds taken as a whole.

Practically all of the improvement expected comes from the selection of dams of bulls—only one-quarter comes from the selection of dams of cows. The contribution on the latter

side could be improved by anything which would increase the average number of heifer calves born in the lifetime of a cow. Amongst these would be:

(*a*) Increased longevity, although this would also increase the mean generation length.

(*b*) Sex determination, giving a larger selection differential in producing replacements.

(*c*) Superovulation and transplantation of ova of good cows into host cows with essentially the same effect as (*b*).

There is no immediate prospect of either (*b*) or (*c*) being applied commercially on a large scale.

It is often argued that the selection for productivity alone is liable to decrease the vigour and longevity of the animals produced. The evidence at present available on this point suggests that although there may be some such negative relationship between yield and longevity, it is not of very great effect. In addition, the cows are being naturally selected for longevity, in that after the first selection they are kept in the herd as long as they will live.

Progeny testing has been suggested as a method of increasing the rate of improvement. It has been shown (Dickerson & Hazel, 1944) that in a closed herd of 120 cows, it does not represent a useful method, as the increase in generation length more than offsets the selection differential. However, the use of artificial insemination in connexion with progeny testing makes it much more effective, and in a paper to be published shortly it will be shown that with a unit of size 2000 cows, it should be possible to obtain a rate of genetic improvement of about 1·8% of the average yield per year.

Summary

1. Methods of estimating the rate of genetic gain achieved in animal improvement are discussed and criticized.

2. A theoretical method of estimation of the maximum possible genetic gain of milk yield under direct selection is presented for a closed herd of dairy cattle. The maximum possible gain is approximately 1% of the average yield.

3. The method can also be used for the estimation of genetic gain from the selection practised in an actual herd. An estimated gain of milk yield of 0·7% per year is given for one herd.

We should like to take this opportunity of thanking the owner of the herd used in this discussion for giving us access to the records over a very long period.

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BLOCK 3

3

Henderson and mixed models

1949 to 1976

The selection index assumes that animals are compared in a common environment and that means and variances are known. In field data these conditions do not hold, since the daughters of a bull produce in different herds and years and the animals being evaluated have already undergone selection. This block brings together the papers in which Henderson developed, between 1949 and 1976, the mixed model equations and BLUP, along with the methods for estimating variance components, including REML, the model for maternal effects and the method for obtaining the inverse of the relationship matrix, which made the adoption of the animal model possible.

Henderson (1949)

Henderson (1953)

Henderson, Kempthorne, Searle and von Krosigk (1959)

Willham (1963)

Patterson and Thompson (1971)

Henderson (1973)

Henderson (1975)

Henderson (1976)

Henderson and Quaas (1976)

BLOCK 3 PAPER 10

Henderson (1949)

> *Henderson, C.R. 1949. Estimation of changes in herd environment. Journal of Dairy Science, vol. 32, p. 706. Published abstract and original manuscript.*

The problem

Herd production averages change from year to year, partly because of genetic improvement and partly because of changes in management. Separating the two is difficult, because culled cows are not a random sample. In this abstract and the accompanying manuscript, Henderson proposes estimating changes in environment while accounting for the repeatability of lactations and for culling. The system of equations he sets up is the first version of the mixed model equations.

Points to note while reading

The published abstract is only a few lines long and appears on the first page, alongside other abstracts from the meeting. The typed manuscript that follows shows the equations and a numerical example solved by hand. Note that year effects and cow effects are estimated at the same time, which is the central idea of everything Henderson would do afterward.

have ranged from 110 to 168 lb., averaging 145 lb.

With the exception of two lighter weight calves, normal delivery has been possible. Four have been delivered by Caesarean section. Of these none has survived more than a few hours. All other calves were dismembered. As a result, the future reproductive capacity of these cows has been impaired greatly. During the ninth month of pregnancy, there appears to be a complete absence of all physical changes normally occurring.

Pedigree analyses indicate that all calves manifesting this anomaly are homozygous for an autosomal recessive gene. It is concluded tentatively that prolonged gestation is caused by an hormonal imbalance between the fetus and mother, when the fetus is of the mutant genotype. This unique genetic material should prove valuable for certain physiological studies.

P3 Estimation of Changes in Herd Environment. C. R. HENDERSON, Cornell University, Ithaca, N. Y.

Accurate appraisals of the results of breeding programs and most efficient estimates of breeding values of individuals whose own records and whose relatives' records were made in several different years require quantitative measures of the effects of changing herd environment. Least squares or modified least squares methods for obtaining such measures give biased estimates when records of cows culled from the herd are either above or below the herd average. This bias results from the lack of perfect repeatability of records. In contrast, the method of maximum likelihood automatically takes into account incomplete repeatability and annual culling levels and utilizes all of the records in such a way as to obtain the most precise estimates possible of the yearly environmental effects.

The maximum likelihood method has been utilized to obtain annual correction factors for several New York dairy herds. The method is illustrated with data from one of these herds and less accurate but less laborious modifications are described. Examples are given of the use of the correction factors for estimating the genetic improvement in the herd, predicting the breeding values of cows and evaluating sire proofs. Application of the method to computation of age correction factors also is discussed.

P4 The Number of Proved Sons Necessary to Evaluate the Transmitting Ability of a Sire. W. E. WASHBON AND W. J. TYLER, West Virginia University, Morgantown.

One hunderd seventy-four Holstein sires with eight or more D.H.I.A. proved sons were studied to determine the least number of proved sons necessary to estimate most accurately the performance of those to be proved later. The data included average butterfat production of the proved sons' daughters, average difference of daughters' production as compared to their dams and per cent of proved sons that maintained or increased butterfat production in the herds in which they were used. Averages of the first three to ten proved sons, respectively, were compared with averages of the following three, five and ten sons.

Highly significant correlations ($r = 0.35$ to 0.65) indicated that the average butterfat production of the daugthers of the first three proved sons was nearly as accurate as data on more sons in estimating the average butterfat production of the daughters of the next three, five or ten proved sons of a sire.

Similarly, the significant correlations ($r = 0.24$ to 0.40) for the sons' daughters' increase or decrease in butterfat production from their dams indicated that data on the first three or four proved sons were nearly as accurate as data on a larger number in predicting what might be expected from the next three, five or ten proved sons in this respect.

For per cent of sons improving production the correlations were significant when the first four, five and six sons were compared with the next ten sons ($r = 0.30$).

A sire's future granddaughters' butterfat production and its difference from their dams' production apparently can be estimated nearly as well from the performance of the first three or four sons as from a larger number. The per cent of a sire's future proved sons that likely will improve production is more reliable if the prediction is based upon the performance of at least the first four or five proved sons.

P5 Calf Mortality, Sex Ratio and Incidence of Twinning in Two University of Minnesota Herds. K. MILLER AND L. GILMORE, Minn. Agr. Expt. Station, St. Paul.

The University calf records at Grand Rapids and St. Paul have been analyzed with respect to prenatal and postnatal mortality for the first 6 mo. and for other information such as the sex ratio and incidence of multiple births.

At University Farm for the 12 years 1934 to 1945, inclusive, 50 of the 592 births, among Guernseys, Holstein-Fresians and Jerseys, were born dead. Ninty-one died during the first month, 36 more by the end of the sixth month

P.L. Cattle - Breeds + Breeding, General

ESTIMATION OF CHANGES IN HERD ENVIRONMENT

C. R. Henderson
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26 JUN 1951 J. DONALD.

One of the important problems in dairy cattle breeding is the estimation of year to year fluctuations or trends in environmental factors influencing milk production. An examination of herd averages with respect to changes from one year to another tells nothing at all about the causes of these changes. Increased production may be the result of improved feeding and management, of improved genetic merit, or of some combination of changed environment and genotypes. Consequently, if the effectiveness of a breeding program is to be appraised, it is necessary that the observed changes in production be separated into environmental and genetic components. Also most effective selection among individuals who made their records and whose relatives made their records in different years requires that these records be corrected for any year to year differences in environment.

Accounting for environmental changes is particularly important in appraising bull proofs. Since the dams' records are usually made several years earlier than are the daughters' records, any very appreciable change in environment can seriously bias the bull proof. Attempts have been made to get around this difficulty by using only concurrent records in cases where there is reason to suspect that there have been some rather marked changes in feeding and management practices between the time when most of the dams made their records and the time when most of the daughters made their records. The difficulty with this method is that many of the dams leave the herd before their daughters make their records, and consequently a great deal of information must be sacrificed. One way which might be suggested to obviate partially the difficulty due to small numbers is to utilize records of all cows who were in production in both of the years under question. For example, a dam made a record in 1946 and her daughter a record in 1949 and we wish to take into account possible changes in environment between these two years. We could assemble the records of all cows in the herd who made records in both 1946 and 1949 and compute the difference between the average production of this selected group of cows in each of the two years under question.

This method for correcting daughter-dam comparisons seems to offer a logical method for solving the general problem of estimating changes in herd environment. That is, the change from year 1 to year 2 would be the difference between the average of these two years with respect to the selected group of cows who made records in both of these years. Similarly the change from year 2 to year 3 would be the difference in average production for these two years in a second selected group of cows who made records in both years 2 and 3. This procedure could be carried on for any number of years desired. It is obvious that this is not the most efficient possible method of estimating changes, since the change from year 1 to year 3 can be estimated not only by comparison of the year 1 versus year 2 with the year 2 versus year 3 estimates, but also by comparisons within the group of cows with records in both years 1 and 3. It can be seen that there are many possible combinations of comparisons and the problem arises as to how best to weight them.

The method of least squares is a computational procedure which enables one properly to weight the various comparisons. Although it will be shown that this method may yield seriously biased estimates, it is described here because it is frequently used in the solution of problems like this one involving unequal subclass numbers in factorial classifications and is therefore familiar to some workers. Also the maximum likelihood method presented in this paper closely resembles the least squares procedure. A model applicable to the least squares method is as follows:

$y_{ij}=\mu+a_i+c_j+e_{ij}$, where y_{ij} is the record made by the jth cow in the ith year, μ is the population mean, a_i is the environmental effect of the ith year, c_j is the real producing ability of the jth cow, and e_{ij} is a random error associated with the record of the jth cow made in the ith year. The errors are assumed to be uncorrelated, and in most cases they are assumed to be normally distributed. Essentially all this model says is that the record of a cow can be at least approximately expressed as the sum of the population mean, a cow effect, an environmental effect peculiar to a particular year, and an error which is presumably the sum of many different environmental effects impinging upon a particular cow in a particular year. The real producing ability of the cow is a function of her genotype and of permanent environmental factors peculiar to her. The problem is to assign values to μ, a_i, and c_j in such a way as to minimize the residual sum of squares. The solution to the following set of simultaneous equations accomplishes this objective.

$$n_{..}\hat{\mu}+\Sigma n_{i.}\hat{a}_i+\Sigma n_{.j}\hat{c}_j=\Sigma\Sigma y_{ij}$$

$$n_{i.}\hat{\mu}+n_{i.}\hat{a}_i+\Sigma n_{ij}\hat{c}_j=\Sigma y_{ij} \quad \text{(One such equation for each of the } a_i\text{)}$$

$$n_{.j}\hat{\mu}+\Sigma n_{ij}\hat{a}_i+n_{.j}\hat{c}_j=\Sigma y_{ij} \quad \text{(One such equation for each } c_j\text{)}, \text{ where } n_{ij}=1$$

if the jth cow made a record in the ith year and =0 if the jth cow did not make a record in the ith year, $n_{..}$ = the total number of records, $n_{i.}$ = the number of records made in the ith year, and $n_{.j}$ = the number of records made by the jth cow. The solution to these equations is not as formidable as first appears for $\hat{\mu}+\hat{c}_j=\frac{1}{n_{.j}}(\Sigma y_{ij}-\Sigma n_{ij}\hat{a}_i)$. Utilizing this fact, the equations can be reduced to ones involving only the $\hat{a}_i$. Then after imposing the restriction $\Sigma\hat{a}_i=0$, the $\hat{a}_i$ can be solved.

As was stated above, the least squares estimate, and the same can be said of comparisons of yearly averages of cows selected because they had records in two specified years, can give seriously biased results. The bias is of this sort. If the cows which are culled from the herd have average records below the herd mean, the environment will appear to be getting worse and worse if in fact there has been no change in environment. Similarly if the environment has been improving, improvement in environment appears less than it really has been. If the environment has been getting worse, the deterioration in environment will appear worse than it actually has been. Conversely, if the cows culled from the herd are of above average production, the environment will seem to have improved more or to have deteriorated less than is actually the case.

The reason for the bias in the least squares method is the incomplete repeatability of dairy records. It is a well known fact that a group of cows selected because their records were above the herd average during the past one or more years are likely to produce less far above the herd average

next year. Conversely cows selected for below average production are likely to be less far below the herd average in their succeeding lactations. The reason for this is that cows with above average production, more frequently than not, receive an above average temporary environmental contribution. Since this temporary environmental factor does not carry over to succeeding lactations, these later records are likely to fall below the earlier ones. The crucial point is that least squares and similar methods are essentially year to year comparisons of records of the survivors of each year's culling. Therefore, if the survivors of culling made above herd average records prior to the culling of certain of their mates, we should expect these survivors' records to be less in succeeding years and consequently to make it appear that the environment is becoming poorer from year to year.

What is needed then is some method by which knowledge of repeatability of dairy records and of the year to year culling levels can be utilized to adjust the estimates of the yearly environmental effects. The method of maximum likelihood now to be described does all of this automatically in a set of simultaneous equations which are little more complex than the least squares equations. In addition the method effects faster genetic progress through selection on own performance than does any other method for utilizing annual production records.

Now let us modify the model used for the least squares analysis as follows:

$y_{ijk} = \mu + a_i + b_j + c_{jk} + e_{ijk}$, where y_{ijk} is the record made in the ith year by the kth cow of the jth group of cows, μ is the population average, a_i is the environmental effect of the ith year, b_j is the average real producing ability of the jth group of cows, c_{jk} is the real producing ability of the kth cow of the jth group, and e_{ijk} is a random environmental effect peculiar to the individual record. The new element in this model as compared to the least squares model is the b_j. It represents the average real producing ability of a particular group of cows. It might, for example, represent the daughters of a particular bull or a set of cows born within a specified period. Now let us make the additional assumptions that the c_{jk} are normally and independently distributed with mean 0 and variance σ_c^2, that the e_{ijk} are normally and independently distributed with mean 0 and variance σ_e^2, and that the c's and e's are uncorrelated. These assumptions mean that the cows of a particular group are randomly drawn from a normal population whose mean is $\mu + b_j$ and whose variance is σ_c^2. Furthermore temporary environment is not correlated with real producing ability.

With the assumptions specified above we can now write the joint distribution of the y_{ijk} and the c_{jk} and can proceed to compute values of μ, a_i, b_j, and c_{jk} which will maximize the probability of obtaining the

sample of records actually at hand. The distribution is

$$L = \prod_{ijk} \frac{1}{\sqrt{2\pi\sigma_e^2}} e^{-\frac{1}{2\sigma_e^2}(y_{ijk}-\mu-a_i-b_j-c_{jk})^2} \prod_{jk} \frac{1}{\sqrt{2\pi\sigma_c^2}} e^{-\frac{c_{jk}^2}{2\sigma_c^2}}$$

The maximizing values are the solution to the following set of simultaneous equations:

$$n_{...}\tilde{\mu}+\Sigma n_{i..}\tilde{a}_i+\Sigma n_{.j.}\tilde{b}_j+\Sigma\Sigma n_{.jk}\tilde{c}_{jk}=\Sigma\Sigma\Sigma y_{ijk}$$

$$n_{i..}(\tilde{\mu}+\tilde{a}_i)+\Sigma n_{ij.}\tilde{b}_j+\Sigma\Sigma n_{ijk}\tilde{c}_{jk}=\Sigma\Sigma y_{ijk}, \text{ and similarly for all a's.}$$

$$n_{.j.}(\tilde{\mu}+\tilde{b}_j)+\Sigma n_{ij.}\tilde{a}_i+\Sigma n_{.jk}\tilde{c}_{jk}=\Sigma\Sigma y_{ijk}, \text{ and similarly for all b's.}$$

$$n_{.jk}(\tilde{\mu}+\tilde{b}_j)+\Sigma n_{ijk}\tilde{a}_i+(n_{.jk}+\frac{\sigma_e^2}{\sigma_c^2})\tilde{c}_{jk}=\Sigma y_{ijk}, \text{ and similarly for all c's.}$$

n_{ijk}=1 if the jk^{th} cow has a record in the ith year and =0 otherwise.

A dot in the subscript denotes summation over that particular subscript. That is, $n_{...}$ refers to the total number of records, $n_{i..}$ to the number of records in the ith year, $n_{.j.}$ to the number of records made by the cows of the jth group, $n_{.jk}$ to the number of records made by the kth cow of the jth group, and $n_{ij.}$ to the number of records made by the jth group in the ith year.

It will be noted that $\frac{\sigma_e^2}{\sigma_c^2}$ appears in certain coefficients. This ratio is closely related to repeatability of dairy records as defined by Lush, and concerning the size of which several studies have been made. If we denote repeatability by $r=\frac{\sigma_c^2}{\sigma_c^2+\sigma_e^2}$, then $\frac{\sigma_e^2}{\sigma_c^2}$ can be written as $\frac{1-r}{r}$ For example, if we take r=.40, $\frac{\sigma_e^2}{\sigma_c^2}=1.5$.

Solving for $\tilde{c}_{jk}$ in the above equations we get

$$\tilde{c}_{jk}=\frac{\sigma_c^2}{n_{.jk}\sigma_c^2+\sigma_e^2}(\Sigma y_{ijk}-n_{.jk}\tilde{\mu}-n_{.jk}\tilde{b}_j-\Sigma n_{ijk}\tilde{a}_i).$$

Consequently we can substitute these expressions for $\tilde{c}_{jk}$ in the μ, a, and b equations and thereby eliminate the $\tilde{c}_{jk}$. Then we have a set of equations in which $\tilde{\mu}+\tilde{b}_j$ can be expressed in terms of the $\tilde{a}_i$ and certain observed records

in which by proper substitution the equations can be reduced to one involving only the $\tilde{a}_i$.

Let us look at the expression for $\tilde{c}_{jk}$ and now assume that μ is known and that all a_i and b_j=0. This would be the case if we assume that there have been no changes in the herd environment nor in the genetic merit of the herd and that the herd average is known without error. Then,

$$\tilde{c}_{jk} = \frac{\sigma_c^2}{n_{.jk}\sigma_c^2+\sigma_e^2} (\Sigma_i y_{ijk} - n_{.jk}\mu)$$

$$= \frac{n_{.jk} r}{1+(n_{.jk}-1)r} (\bar{y}_{.jk}-\mu)$$

This last expression is exactly equivalent to the method derived by Lush for estimating the real producing ability of a cow. That is, his method is in fact the maximum likelihood method under the assumptions listed above. The method described here utilizes these same principles but permits the assumption of changing herd environment and genotype.

Several different modifications of the model described above are possible. For example, linear or curvilinear regressions could be fit to freshening date and to date of birth. In any case the general procedures are the same. Also if there is some question regarding the applicability of standard age correction factors to a particular herd, the age of the cow at time of freshening can be worked into the model.

Summary and Conclusions

A maximum likelihood solution to the problem of estimating changes in herd environment has been presented. The method utilizes in the most efficient possible way all of the records in a herd, gives estimates which are free from the bias inherent in least squares or similar methods, and automatically yields estimates of the real producing abilities of each of the cows.

The computational procedure is not particularly difficult for those familiar with least squares analyses. A number of computational short-cuts have been developed and have been described in a numerical example which can be obtained by writing the author at Cornell University.

PL CATTLE; BREEDS & BREEDING

NUMERICAL EXAMPLE OF ESTIMATION OF CHANGES IN HERD ENVIRONMENT

P. DON. 26 JUN 1951

C. R. Henderson
Cornell University

It should be emphasized that the amount of data in the following example is too small to give accurate estimates of changes in herd environment. The numbers of records have been kept low in order to simplify the presentation.

Table 1 shows the mature equivalent records of 35 cows for the years 1946, 1947 and 1948. The cows have been divided into four groups depending upon their date of birth. These groups include a group born before 1944, a group born in 1944, a group born in 1946, and a group born in 1946.

The equations for estimating the changes in environment by means of the method of maximum likelihood can be simplified somewhat in comparison to those given in the paper presented at the 1949 meeting of the American Dairy Science Association. This simplification involves combining μ and b_j into a single parameter. Then the equations are as follows:

$n_{i..}a_i + \sum_j n_{ij.}(\mu+b_j) + \sum_{jk} n_{ijk}c_{jk} = \sum_{jk} y_{ijk}$, and similarly for the other a_i equations.

$\sum_i n_{ij.}a_i + n_{.j.}(\mu+b_j) + \sum_k n_{.jk}c_{jk} = \sum_{ik} y_{ijk}$, and similarly for the other $\mu+b_j$ equations.

$\sum_i n_{ijk}a_i + n_{.jk}(\mu+b_j) + (n_{.jk} + \sigma_e^2/\sigma_c^2)c_{jk} = \sum_i y_{ijk}$, and similarly for the other c_{jk} equations.

a_1, a_2, and a_3 = the environmental effects peculiar to cows freshening in 1946, 1947, and 1948 respectively.

$\mu+b_1$, $\mu+b_2$, $\mu+b_3$, and $\mu+b_4$ = the permanent producing abilities of the sub-populations born before 1944, in 1944, in 1945, and in 1946 respectively.

TABLE 1

MATURE EQUIVALENT RECORDS

Cow	Year of Birth	Year Cow Freshened 1946	1947	1948	Cow Sum
	Before				
1	1944	343	492	311	1146
2		437	544	482	1463
3		390	355		745
4		264			264
5		410			410
6		384			384
7		405	230		635
8		324	395	347	1066
9		447	369	393	1209
10		381	425	392	1198
11		367	397	317	1081
12		282			282
13		395	370	246	1011
14		344	477		821
15		381	339	342	1062
16		333	299	380	1012
17		263			263
18		376	393	349	1118
19		412	322	291	1025
20		381	272		653
21	1944	389	320	326	1035
22		441	378	274	1093
23		384	410	353	1147
24		514	388	453	1355
25		421	332	332	1085
26	1945		273		273
27			385	265	650
28			343	355	698
29			233	245	478
30			346	323	669
31	1946			254	254
32				233	233
33				304	304
34				427	427
35				372	372
Freshening Year Sum		9468	9087	8366	26921

The cows actually born in any specified year are considered to be a random sample of cows from some infinite sub-population of cows which might have been born in that year in the herd being studied. The sub-population is assumed to be normally distributed with mean $\mu+b_j$ and variance σ_c^2.

c_{11} = the permanent producing ability of the first cow born before 1944, her permanent producing ability being expressed as a deviation from the mean of her sub-population (cows born before 1944). In what is to follow c_{11} will be changed to c_1 to correspond to the cow numbers presented in Table 1. Similarly c_{12} will be changed to c_2, etc.

The individual records are assumed to be subject to a random error, e_{ijk}, which is normally distributed with mean 0 and variance σ_e^2.

In this example, the equation pertaining to a_1 is

$$25a_1+20(\mu+b_1)+5(\mu+b_2)+c_1+c_2+\ldots+c_{25}= 9468.$$

25, the coefficient of a_1, arises from the fact that there were 25 records made by cows freshening in 1946. 20, the coefficient of $\mu+b_1$, comes from the fact that of records made by cows freshening in 1946 there were 20 records made by cows who were born before 1944. 5, the coefficient of $\mu+b_2$, arises from the fact that of records made by cows freshening in 1946, 5 records were made by cows born in 1944. 1, the coefficient of c_1, c_2, etc., comes from the fact that of records made by cows freshening in 1946, the cows involved were 1, 2, ..., 25, each with one record. The right member of the equation, 9468, is the sum of all records started in 1946. The equations pertaining to a_2 and a_3 are obtained in a similar manner and are presented in Table 2.

The equation pertaining to $\mu+b_1$ is

$$20a_1+15a_2+11a_3+46(\mu+b_1)+3c_1+3c_2+\ldots+2c_{20}= 16,848$$

20, the coefficient of a_1, is the number of records made by cows born before 1944 and in which the freshening date was 1944.6. 15, the coefficient of a_2, is the number of records made by this same group of cows in which the freshening date was 1945.1. 11, the coefficient of a_3, is the number of records made

by the same group of cows in which the freshening date was 1948. 46, the coefficient of $\mu+b_1$, is the total number of records made by cows born before 1944. 3, the coefficient of c_1, is the total number of records made by the first cow born before 1944, and similarly for the coefficients of the other c_{1k}. 16,848, the right member of the equation, is the sum of all records made by cows born before 1944. The equations pertaining to $\mu+b_2$, $\mu+b_3$, and $\mu+b_4$ are set up in a similar manner and are presented in Table 2.

The equation pertaining to c_1 is

$$a_1+a_2+a_3+3(\mu+b_1)+4.5c_1 = 1146$$

1, the coefficient of a_1, a_2, and a_3, results from the fact that cow 1 made records in which the freshening dates were 1944, 1945, and 1948. 3, the coefficient of $\mu+b_1$, results from the fact that cow 1 made 3 records and is a member of the sub-population of cows born before 1944. 4.5, the coefficient of c_1, is the sum of 3, the total number of records made by cow 1, and 1.5, the value of σ_e^2/σ_c^2 used in this example. σ_e^2/σ_c^2 is equal to $(1-r)/r$, where r is a repeatability of butterfat records. Equations for the other c_{jk} are set up in a similar manner.

Table 2 presents the full set of maximum likelihood equations in tabular form. The equations pertaining to a_i and $\mu+b_j$ are read vertically. The column headings denote the equations and the row headings denote the unknowns (parameter estimates). The entries in the table are the appropriate coefficients of the unknowns. The last row contains the right members of the equations. For example, the equation pertaining to $\mu+b_3$ is

$$5a_2+4a_3+9(\mu+b_3)+c_{26}+2c_{27}+2c_{28}+2c_{29}+2c_{30} = 2768.$$

The equations pertaining to the c_{jk} are read horizontally, the column headings denoting the unknowns and the right members. The row headings denote the equations. In the next to the extreme right column is a heading, c_{jk}. This means that the only c_{jk} unknown in the equation is the one pertaining to the jk^{th} cow. For example, the equation for cow 30 is

$$a_2+a_3+2(\mu+b_3)+3.5c_{30} = 669.$$

TABLE 2

MAXIMUM LIKELIHOOD EQUATIONS

	a_1	a_2	a_3	$\mu+b_1$	$\mu+b_2$	$\mu+b_3$	$\mu+b_4$	c_{jk}	Right Member
a_1	25	0	0	20	5	0	0	-	-
a_2	0	25	0	15	5	5	0	-	-
a_3	0	0	25	11	5	4	5	-	-
$\mu+b_1$	20	15	11	46	0	0	0	-	-
$\mu+b_2$	5	5	5	0	15	0	0	-	-
$\mu+b_3$	0	5	4	0	0	9	0	-	-
$\mu+b_4$	0	0	5	0	0	0	5	-	-
c_1	1	1	1	3	0	0	0	4.5	1146
c_2	1	1	1	3	0	0	0	4.5	1463
c_3	1	1	0	2	0	0	0	3.5	745
c_4	1	0	0	1	0	0	0	2.5	264
c_5	1	0	0	1	0	0	0	2.5	410
c_6	1	0	0	1	0	0	0	2.5	384
c_7	1	1	0	2	0	0	0	3.5	635
c_8	1	1	1	3	0	0	0	4.5	1066
c_9	1	1	1	3	0	0	0	4.5	1209
c_{10}	1	1	1	3	0	0	0	4.5	1198
c_{11}	1	1	1	3	0	0	0	4.5	1081
c_{12}	1	0	0	1	0	0	0	2.5	282
c_{13}	1	1	1	3	0	0	0	4.5	1011
c_{14}	1	1	0	2	0	0	0	3.5	821
c_{15}	1	1	1	3	0	0	0	4.5	1062
c_{16}	1	1	1	3	0	0	0	4.5	1012
c_{17}	1	0	0	1	0	0	0	2.5	253
c_{18}	1	1	1	3	0	0	0	4.5	1118
c_{19}	1	1	1	3	0	0	0	4.5	1025
c_{20}	1	1	0	2	0	0	0	3.5	653
c_{21}	1	1	1	0	3	0	0	4.5	1035
c_{22}	1	1	1	0	3	0	0	4.5	1093
c_{23}	1	1	1	0	3	0	0	4.5	1147
c_{24}	1	1	1	0	3	0	0	4.5	1355
c_{25}	1	1	1	0	3	0	0	4.5	1085
c_{26}	0	1	0	0	0	1	0	2.5	273
c_{27}	0	1	1	0	0	2	0	3.5	650
c_{28}	0	1	1	0	0	2	0	3.5	698
c_{29}	0	1	1	0	0	2	0	3.5	478
c_{30}	0	1	1	0	0	2	0	3.5	669
c_{31}	0	0	1	0	0	0	1	2.5	264
c_{32}	0	0	1	0	0	0	1	2.5	233
c_{33}	0	0	1	0	0	0	1	2.5	304
c_{34}	0	0	1	0	0	0	1	2.5	427
c_{35}	0	0	1	0	0	0	1	2.5	372
Right Member	9468	9087	8366	16,848	5715	2768	1590	-	-

The solution to the equations in Table 2 is a formidable task as they stand for there are 42 unknowns. It is, however, quite easy to reduce these equations to 7 in number by making use of the fact that c_{jk} can be expressed in terms of the a_i, $\mu+b_j$, and the right member of the c_{jk} equation. For example,

$$c_{30} = \frac{1}{3.5}\left[669-a_2-a_3-2(\mu+b_3)\right] .$$

The computational procedure for making such substitutions is facilitated by preparing a table like that of Table 3. The entries in this table not in parentheses are the same as in the lower portion of Table 2. The entries in parentheses are obtained by dividing each of the entries not in parentheses in a particular row by the entry in that row under the column headed c_{jk}. For example, in the first row of the table, the entry in parentheses under a_1 is $1/4.5 = .2222$. The entries under a_2 and a_3 in this row are also $1/4.5$. The entry under $\mu+b_1$ is $3/4.5 = .6667$. The entry under "Right Member" is $1146/4.5 = 254.67$.

Table 4 can now be prepared. It is a new set of equations in which the c_{jk} have been expressed in terms of the other parameters. In these equations the column headings denote the unknowns while the row headings denote the equations. The right column entries denote the right members of the equations. This table is prepared by using information in Tables 2 and 3. For example, the entry in the upper left hand corner of Table 4 is equal to $25-1(.2222)-1(.2222)-1(.2857)-\ldots-1(.2222) = 18.30$. That is, this entry is equal to the original coefficient of a_1 in the a_1 equation minus the cross products of the entries in the a_1 column of Table 3. The entry in the a_2 column of the a_1 row of Table 4 is equal to $0-1(.2222)-1(.2222)-1(.2857)-1(.2857)-\ldots-1(.2222)-1(.2222) = -4.70$. This entry is simply the coefficient of a_2 in the a_1 equation of Table 2 minus the cross products between the entries not in parentheses in column a_1 and entries in parentheses in column a_2 of Table 3. The entry in row a_1, column a_3 of Table 4 is equal to $0-1(.2222)-1(.2222)-\ldots-1(.2222)-1(.2222) = -3.56$. The entry in the $\mu+b_1$ column

TABLE 3

Cow	a_1	a_2	a_3	$\mu+b_1$	$\mu+b_2$	$\mu+b_3$	$\mu+b_4$	c_{jk}	Right Member	
1	1(.2222)	1(.2222)	1(.2222)	3(.6667)				4.5	1146	(254.67)
2	1(.2222)	1(.2222)	1(.2222)	3(.6667)				4.5	1463	(325.11)
3	1(.2857)	1(.2857)		2(.5714)				3.5	745	(212.86)
4	1(.4000)			1(.4000)				2.5	264	(105.60)
5	1(.4000)			1(.4000)				2.5	410	(164.00)
6	1(.4000)			1(.4000)				2.5	384	(153.60)
7	1(.2857)	1(.2857)		2(.5714)				3.5	635	(181.43)
8	1(.2222)	1(.2222)	1(.2222)	3(.6667)				4.5	1066	(236.89)
9	1(.2222)	1(.2222)	1(.2222)	3(.6667)				4.5	1209	(268.67)
10	1(.2222)	1(.2222)	1(.2222)	3(.6667)				4.5	1198	(266.22)
11	1(.2222)	1(.2222)	1(.2222)	3(.6667)				4.5	1081	(240.22)
12	1(.4000)			1(.4000)				2.5	282	(112.80)
13	1(.2222)	1(.2222)	1(.2222)	3(.6667)				4.5	1011	(224.67)
14	1(.2857)	1(.2857)		2(.5714)				3.5	821	(234.57)
15	1(.2222)	1(.2222)	1(.2222)	3(.6667)				4.5	1062	(236.00)
16	1(.2222)	1(.2222)	1(.2222)	3(.6667)				4.5	1012	(224.89)
17	1(.4000)			1(.4000)				2.5	263	(105.20)
18	1(.2222)	1(.2222)	1(.2222)	3(.6667)				4.5	1118	(248.44)
19	1(.2222)	1(.2222)	1(.2222)	3(.6667)				4.5	1025	(227.78)
20	1(.2857)	1(.2857)		2(.5714)				3.5	653	(186.57)
21	1(.2222)	1(.2222)	1(.2222)		3(.6667)			4.5	1035	(230.00)
22	1(.2222)	1(.2222)	1(.2222)		3(.6667)			4.5	1093	(242.89)
23	1(.2222)	1(.2222)	1(.2222)		3(.6667)			4.5	1147	(254.89)
24	1(.2222)	1(.2222)	1(.2222)		3(.6667)			4.5	1355	(301.11)
25	1(.2222)	1(.2222)	1(.2222)		3(.6667)			4.5	1085	(241.11)
26		1(.4000)				1(.4000)		2.5	273	(109.20)
27		1(.2857)	1(.2857)			2(.5714)		3.5	650	(185.71)
28		1(.2857)	1(.2857)			2(.5714)		3.5	698	(199.43)
29		1(.2857)	1(.2857)			2(.5714)		3.5	478	(136.57)
30		1(.2857)	1(.2857)			2(.5714)		3.5	669	(191.14)
31			1(.4000)				1(.4000)	2.5	254	(101.60)
32			1(.4000)				1(.4000)	2.5	233	(93.20)
33			1(.4000)				1(.4000)	2.5	304	(121.60)
34			1(.4000)				1(.4000)	2.5	427	(170.80)
35			1(.4000)				1(.4000)	2.5	372	(148.80)

TABLE 4

	a_1	a_2	a_3	$\mu+b_1$	$\mu+b_2$	$\mu+b_3$	$\mu+b_4$	Right Member
a_1	18.30	-4.70	-3.56	8.38	1.67	0	0	3987.81
a_2	-4.70	18.76	-4.70	5.38	1.67	2.31	0	3425.96
a_3	-3.56	-4.70	18.30	3.67	1.67	1.71	3.00	2993.59
$\mu+b_1$	8.38	5.38	3.67	17.43	0	0	0	6315.26
$\mu+b_2$	1.67	1.67	1.67	0	5.00	0	0	1905.00
$\mu+b_3$	0	2.31	1.71	0	0	4.03	0	1233.10
$\mu+b_4$	0	0	3.00	0	0	0	3.00	954.00

of the a_1 row of Table 4 is 20-1(.6667)-1(.6667)-1(.5714)-...-1(.6667)-1 (.5714)=8.38. In a similar manner the entries in the $\mu+b_2$, $\mu+b_3$, and $\mu+b_4$ columns of the b_1 row are obtained. The right member of Table 4 in the a_1 row is 9468-1(254.67)-1(325.11)-...-1(301.11)-1(241.11)=3987.81. In a similar manner each of the other entries of Table 4 can be computed. Since the entries are symmetric about the upper left hand to lower right hand diagonal, it is necessary to compute only the diagonal elements and those to the right of it. The entries to the left of the diagonal can be copied from those already computed for those to the right.

There is a useful check on the accuracy of the entries in Table 4. In each row the sum of the entries under the a_i is equal to the sum of the entries under the $\mu+b_j$. For example in the first row 18.30-4.70-3.56=8.38 +1.67 except for rounding error. A similar check exists for the entries in the right hand column of Table 4. The sum of the entries in the a_i rows is equal to the sum of the entries in the $\mu+b_j$ rows except for rounding errors. That is, 3987.81+3425.96+2993.59=6315.26+1905.00+1233.10+954.00.

Although the equations have now been reduced to 7 in number it is possible to reduce them still further by making use of the fact that $\mu+b_j$ can be expressed in terms of the a_j and the right members. Making use of this fact the equations can be reduced to 3 in number. Table 5 facilitates these substitutions.

The entries in Table 5 correspond to the last four rows of Table 4. The entries <u>in parentheses</u> in Table 5 are the entries <u>not in parentheses</u> divided by the entry under $\mu+b_j$ for that particular row. For example the entry in parentheses in the first column and first row is 8.38/17.43=.4808.

Now Table 6 is set up. It is a set of equations in which both the c_{jk} and $\mu+b_j$ have been expressed in terms of the a_i and the right members. The procedure for doing this is just like that used in preparing Table 4. For example the entry in the a_1 row and a_1 column of Table 6 is 18.30-8.38 (.4808)-1.67(.3340) = 13.71. The entry in the a_2 column of the a_1 row of

Table 6 is -4.70-8.38(.3087)-1.67(.3340)= -7.84. The entry in the right member of the first row is 3987.81-.4808(6315.26)-.3340(1905.00) = 315.16. A useful check on the accuracy of the computations of Table 6 is the fact that the entries in each row sum to 0 over the first 3 columns except for rounding error. For example 13.71-7.84-5.88 = -.01. The right members also add to 0 except for rounding errors. That is, 315.16+133.36-449.88 = -1.36.

TABLE 5

$\mu+b_j$	a_1		a_2		a_3		$\mu+b_j$	Right Member
1	8.38	(.4808)	5.38	(.3087)	3.67	(.2106)	17.43	6315.26
2	1.67	(.3340)	1.67	(.3340)	1.67	(.3340)	5.00	1905.00
3	0	(0)	2.31	(.5732)	1.71	(.4243)	4.03	1233.10
4	0	(0)	0	(0)	3.00	(1.0000)	3.00	954.00

TABLE 6

	a_1	a_2	a_3	Right Member
a_1	13.71	-7.84	-5.88	315.16
a_2	-7.84	15.22	-7.37	133.36
a_3	-5.88	7.37	13.24	-449.88

There is no solution to the equations of Table 6 as they stand inasmuch as the sum of any two of the equations is equal to the third equation with signs changed. This difficulty can easily be got around by imposing the restriction that the sum of the a_i=0, that is $a_1+a_2+a_3 = 0$. This is a perfectly logical restriction inasmuch as the yearly environmental effects are thereby expressed as deviations about the mean of such effects. Making use of this rostriction, $a_3 = -a_1-a_2$. Consequently the equations of Table 6 can be reduced to those of Table 7. For example, the coefficient of a_1 in the a_1 equation of Table 7 is equal to 13.71-(-5.88)=19.59. The coefficient of a_2 in the a_1 equation is -7.84-(-5.88)= -1.96. The a_3 equation of Table 6 can be deleted since it is redundant.

TABLE 7

	a_1	a_2	Right Member
a_1	19.59	-1.96	315.16
a_2	-.47	22.59	133.36

The solution to the equations of Table 7 is $\widetilde{a}_1=17$ and $\widetilde{a}_2=6$. Consequently $\widetilde{a}_3=-17-6=-23$. What these figures mean is, that on the basis of the available data, cows freshening in 1946 receive an average environment which results in butterfat production of 17 pounds above the average of the three years studied. Consequently, records made in that year should be corrected by subtracting 17 pounds from each of them. In a like manner records made in 1947 should be corrected by subtracting 6 pounds, and records of cows freshening in 1948 should be corrected by adding 23 pounds.

We can now proceed to obtain the maximum likelihood estimates for the $\mu+b_j$ and the c_{jk}. The former are obtained by substituting the values just obtained for the $\widetilde{a}_i$ in the last four equations of Table 4 and solving for the $\mu+b_j$. For example,

$$\widetilde{\mu+b}_1=\frac{1}{17.43}\left[6315.26-17(8.38)-6(5.38)+23(3.67)\right]=357.$$

In a like manner the value of the other $\widetilde{\mu+b}_j$ are 381, 312, and 341. Finally if we substitute in the c_{jk} equations of Table 2 the computed values for the $\widetilde{a}_i$ and $\widetilde{\mu+b}_j$, we can obtain the estimates of c_{jk}. For example,

$$\widetilde{c}_1=\frac{1}{4.5}\left[1146-1(17)-1(6)-1(-23)-3(357)\right]=17$$

Then the estimate of the real producing ability of the first cow is

$\widetilde{\mu+b}_1+\widetilde{c}_1=357+17=374$, and similarly for the other cows.

Henderson (1953)

Henderson, C.R. 1953. Estimation of variance and covariance components. Biometrics, vol. 9, no. 2, pp. 226 to 252.

The problem

Estimating heritability, genetic correlations and the indexes themselves requires estimates of variance components. With field data, in which the number of daughters per bull and per herd varies widely, classical analysis of variance cannot be used. Henderson presented three methods for unbalanced data, which became known as Henderson's Methods 1, 2 and 3.

Points to note while reading

Method 3, based on fitting constants, was the standard in animal breeding for about thirty years, until it was replaced by REML. Note that the methods equate sums of squares to their expected values, and that Henderson already recognizes the limitations of this approach when fixed and random effects are present at the same time.

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ESTIMATION OF VARIANCE AND COVARIANCE COMPONENTS*

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INTRODUCTION

The theory of variance component analysis has been discussed recently by Crump (1946, 1951) and by Eisenhart (1947). These papers and, indeed, most of the published works on estimating variance components deal with the one-way classification, with "nested" classifications, and with factorial classifications having equal subclass numbers. Also most papers on this subject are concerned with what Eisenhart (1947) has called Model II; that is, all elements of the linear model save μ are regarded as random variables. In the above cases, estimation of variance components is usually accomplished by computing the mean squares in the standard analysis of variance, equating these mean squares to their expectations, and solving for the unknown variances. These techniques are described in many statistical textbooks.

Unfortunately, research workers in some of those fields in which much use is made of variance component estimates are unable to obtain data which have the above described characteristics. This is particularly true in those fields in which survey data must be used or where, even in a well-planned experiment, the subclasses are of quite unequal size due, for example, to differences in litter numbers. Also,

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Model II is sometimes not appropriate. Instead the data more appropriately correspond to what Eisenhart called the Mixed Model. For example, the data may represent several different years, and the year effects should be regarded as fixed rather than as random variables.

It is the purpose of this paper to describe some methods for estimating variance components in the non-orthogonal case and to illustrate the methods with a small sample of butterfat records made by cows resulting from an artificial breeding program. The three methods described are:

1. Compute sums of squares as in the standard analysis of variance of corresponding orthogonal data. Equate these sums of squares to their expectations obtained under the assumption of Model II and solve for the unknown variances.
2. Obtain least squares estimates of fixed effects, "correct" the data according to these estimates of the fixed effects, and then using the corrected data in place of the original data, proceed as in Method 1.
3. Compute mean squares by a conventional least squares analysis of non-orthogonal data (method of fitting constants, weighted squares of means, e.g.). Equate these mean squares to their expectations and solve for the unknown variances.

These three methods henceforth called Method 1, Method 2, and Method 3 vary greatly in computational labor. Method 1 is the simplest. Method 2 in many cases is only slightly more difficult. Method 3 is usually much the most laborious. Method 1, however, leads to biased estimates if certain elements of the model are fixed or if some of them are correlated. Estimates obtained by Method 2 are free of the first of these biases, but not of the second. Method 3 yields unbiased estimates, but the computations required may be prohibitive. The relative sizes of the sampling variances of estimates obtained by these three methods are not known.

DESCRIPTION AND ILLUSTRATION OF METHODS OF ESTIMATION

The Data

In New York State most artificial breeding of dairy cows is accomplished with semen supplied by the New York Artificial Breeders' Cooperative, Inc. This cooperative organization has approximately 60 bulls in service. The operations of the organization are conducted in such a manner that it is largely a matter of chance to which bull's semen a particular cow is bred. This fact as well as the large number

of daughters sired by each bull make the production records of these daughters particularly suitable for studying the genetic differences among bulls and for estimating the magnitudes of other sources of variation in milk production records. Good estimates of these variances are needed in designing efficient testing and selection programs.

The difficulties in estimating the pertinent variance components are typical of those faced by research workers in animal breeding and in other fields as well. The difficulties in the present example are due to the following causes:

1. Several years' data are involved and time trends are known to be important.
2. The two major classifications of the data are sire and herd. The number of sires exceeds 100 and the number of different herds exceeds 2000.
3. The number of observations per herd-sire subclass varies; the majority being 0.

We have estimated from these data the pertinent variances. Both Method 1 and Method 2 have been employed and have yielded estimates essentially the same. A small sample of records is presented in this paper and the three methods of estimation are illustrated.

Table 1 shows the number of first lactation butterfat records in each of the year $\times$ herd $\times$ sire subclasses and also the sum of the records for each of these subclasses.

TABLE 1

Herd	Sire	Year				Total
		1	2	3	4	
1	1	3–1414	2– 981			5–2395
1	2		4–1766	2– 862		6–2628
1	3				5–1609	5–1609
2	1	1– 404	3–1270			4–1674
2	2			5–2109		5–2109
2	3			4–1563	2– 740	6–2303
3	1		3–1705			3–1705
3	2		4–2310	2–1134		6–3444
4	1	3–1113	5–1951			8–3064
4	3			3–1291	6–2457	9–3748
Total		7–2931	21–9983	16–6959	13–4806	57–24679

The Linear Model

Let y_{hijk} denote the record made in the h-th year by the k-th daughter of the j-th sire in the i-th herd. Suppose that the appropriate linear model representing these observations is

$$y_{hijk} = \mu + a_h + h_i + s_j + (hs)_{ij} + e_{hijk}$$

$h = 1, \cdots, p \qquad k = 1, \cdots, n_{hij}$

$i = 1, \cdots, q \qquad \sum_h \sum_i \sum_j n_{hij} = N$

$j = 1, \cdots, r \qquad$ Total number of filled subclasses $= s$

μ is common to all observations. a_h is common to all observations in the h-th year, h_i to all observations in the i-th herd, and s_j to all records made by daughters of the j-th sire; $(hs)_{ij}$ is peculiar to all records made by the daughters of the j-th sire in the i-th herd. Peculiar to each record is a random element e_{hijk} which is assumed to have mean zero and variance σ_e^2. The assumptions made concerning the other elements of the model are described for each estimation method.

*Method 1**

Method 1 can be used only if it is assumed that, except for μ, all elements of the model are uncorrelated variables with means zero and variances σ_a^2, σ_h^2, σ_s^2, σ_{hs}^2, or σ_e^2. This is, of course, the Eisenhart Model II.

The following quantities are computed:

$$T = \sum_h \sum_i \sum_j \sum_k y_{hijk}^2 \qquad H = \sum_i \frac{y_{\cdot i \cdot \cdot}^2}{n_{\cdot i \cdot \cdot}}$$

$$A = \sum_h \frac{y_{h \cdot \cdot \cdot}^2}{n_{h \cdot \cdot}} \qquad S = \sum_j \frac{y_{\cdot \cdot j \cdot}^2}{n_{\cdot \cdot j}}$$

$$HS = \sum_i \sum_j \frac{y_{\cdot ij \cdot}^2}{n_{\cdot ij}} \qquad CF = \frac{y_{\cdot \cdot \cdot \cdot}^2}{N}$$

Dots in the subscripts denote summation. For example,

$$y_{h \cdot \cdot \cdot} = \sum_i \sum_j \sum_k y_{hijk} .$$

Next the expectations of the above quantities are computed. Under the assumptions of Model II, the coefficients of μ^2 and the variances in these expectations are as shown in Table 2.

*This method was first suggested to me by Dr. S. Lee Crump.

TABLE 2

	μ^2	σ_a^2	σ_h^2	σ_r^2	σ_{hs}^2	σ_e^2
T	N	N	N	N	N	N
A	N	N	K_1	K_2	K_3	p
HS	N	K_4	N	N	N	s
H	N	K_5	N	K_6	K_6	q
S	N	K_7	K_8	N	K_8	r
CF	N	K_9	K_{10}	K_{11}	K_{12}	1

N, p, q, r, s in the above table were defined in the statement of the linear model. $K_1, K_2, \cdots, K_{12}$ must be computed as follows:

$$K_1 = \sum_h \frac{\sum_i n_{hi\cdot}^2}{n_{h\cdot\cdot}} \qquad K_7 = \sum_j \frac{\sum_h n_{h\cdot j}^2}{n_{\cdot\cdot j}}$$

$$K_2 = \sum_h \frac{\sum_j n_{h\cdot j}^2}{n_{h\cdot\cdot}} \qquad K_8 = \sum_j \frac{\sum_i n_{\cdot ij}^2}{n_{\cdot\cdot j}}$$

$$K_3 = \sum_h \frac{\sum_i \sum_j n_{hij}^2}{n_{h\cdot\cdot}} \qquad K_9 = \sum_h n_{h\cdot\cdot}^2/N$$

$$K_4 = \sum_i \sum_j \frac{\sum_h n_{hij}^2}{n_{\cdot ij}} \qquad K_{10} = \sum_i n_{\cdot i\cdot}^2/N$$

$$K_5 = \sum_i \frac{\sum_h n_{hi\cdot}^2}{n_{\cdot i\cdot}} \qquad K_{11} = \sum_j n_{\cdot\cdot j}^2/N$$

$$K_6 = \sum_i \frac{\sum_j n_{\cdot ij}^2}{n_{\cdot i\cdot}} \qquad K_{12} = \sum_i \sum_j n_{\cdot ij}^2/N$$

If the data were orthogonal, the sums of squares in the analysis of variance would be

$$\text{Among Years} = A - CF$$
$$\text{Among Herds} = H - CF$$
$$\text{Among Sires} = S - CF$$
$$\text{Herds} \times \text{Sires} = HS - H - S + CF$$
$$\text{Error} = T - A - HS + CF$$

If these same quantities are computed in spite of the non-orthogonality and are equated to their expectations, unbiased estimates of the

variances can be obtained by solving the resulting equations. The necessary expectations are derived from Table 2. To illustrate, E (Among Years) $= E(A - CF) = E(A) - E(CF)$.

Computation of the K's is facilitated by constructing from Table 1 the following two-way tables of subclass numbers (Tables 3, 4, 5).

TABLE 3

Herd	Year				Total
	1	2	3	4	
1	3	6	2	5	16
2	1	3	9	2	15
3	0	7	2	0	9
4	3	5	3	6	17
Total	7	21	16	13	57

TABLE 4

Sire	Year				Total
	1	2	3	4	
1	7	13	0	0	20
2	0	8	9	0	17
3	0	0	7	13	20
Total	7	21	16	13	57

TABLE 5

Herd	Sire			Total
	1	2	3	
1	5	6	5	16
2	4	5	6	15
3	3	6	0	9
4	8	0	9	17
Total	20	17	20	57

Also certain totals are computed from Table 1.

Year	Herd	Sire
1. 2931	1. 6632	1. 8838
2. 9983	2. 6086	2. 8181
3. 6959	3. 5149	3. 7660
4. 4806	4. 6812	
Total 24,679	Total 24,679	Total 24,679

Using the above totals and the totals in Table 1,

$$A = \frac{2931^2}{7} + \cdots + \frac{4806^2}{13} = 10{,}776{,}451$$

$$HS = \frac{2395^2}{5} + \cdots + \frac{3748^2}{9} = 10{,}970{,}369$$

$$H = \frac{6632^2}{16} + \cdots + \frac{6812^2}{17} = 10{,}893{,}666$$

$$S = \frac{8838^2}{20} + \frac{8181^2}{17} + \frac{7660^2}{20} = 10{,}776{,}278$$

$$CF = \frac{24{,}679^2}{57} = 10{,}685{,}141$$

The expectations of these quantities are presented in Table 6. The computations of these entries proceed as follows:

From Table 3, $K_1 = \dfrac{3^2 + 1^2 + 3^2}{7} + \cdots + \dfrac{5^2 + 2^2 + 6^2}{13} = 19.51$

From Table 4, $K_2 = \dfrac{7^2}{7} + \dfrac{13^2 + 8^2}{21} + \dfrac{9^2 + 7^2}{16} + \dfrac{13^2}{13} = 39.22$

From Table 1, $K_3 = \dfrac{3^2 + 1^2 + 3^2}{7} + \cdots + \dfrac{5^2 + 2^2 + 6^2}{13} = 15.10$

From Table 1, $K_4 = \dfrac{3^2 + 2^2}{5} + \cdots + \dfrac{3^2 + 6^2}{9} = 37.35$

From Table 3, $K_5 = \dfrac{3^2 + 6^2 + 2^2 + 5^2}{16} + \cdots + \dfrac{3^2 + 5^2 + 3^2 + 6^2}{17} = 21.49$

From Table 5, $K_6 = \dfrac{5^2 + 6^2 + 5^2}{16} + \cdots + \dfrac{8^2 + 9^2}{17} = 24.04$

From Table 4, $K_7 = \dfrac{7^2 + 13^2}{20} + \dfrac{8^2 + 9^2}{17} + \dfrac{7^2 + 13^2}{20} = 30.33$

From Table 5, $K_8 = \dfrac{5^2 + 4^2 + 3^2 + 8^2}{20} + \cdots + \dfrac{5^2 + 6^2 + 9^2}{20} = 18.51$

From Table 3, $K_9 = \dfrac{7^2 + 21^2 + 16^2 + 13^2}{57} = 16.05$

From Table 3, $K_{10} = \dfrac{16^2 + 15^2 + 9^2 + 17^2}{57} = 14.93$

From Table 4, $K_{11} = \dfrac{20^2 + 17^2 + 20^2}{57} = 19.11$

From Table 1, $K_{12} = \dfrac{5^2 + 6^2 + \cdots + 9^2}{57} = 6.19$

TABLE 6

	μ^2	σ_a^2	σ_h^2	σ_s^2	σ_{hs}^2	σ_e^2	
T	57	57.	57.	57.	57.	57	11,124,007
A	57	57.	19.51	39.22	15.10	4	10,776,451
HS	57	37.35	57.	57.	57.	10	10,970,369
H	57	21.49	57.	24.04	24.04	4	10,893,666
S	57	30.33	18.51	57.	18.51	3	10,776,278
CF	57	16.05	14.93	19.11	6.19	1	10,685,141

The equations to be solved are presented in Table 7. The first equation reads: $40.95\ \sigma_a^2 + 4.58\ \sigma_h^2 + 20.11\ \sigma_s^2 + 8.91\ \sigma_{hs}^2 + 3\ \sigma_e^2 = 91{,}310$.

TABLE 7

	σ_a^2	σ_h^2	σ_s^2	σ_{hs}^2	σ_e^2	
$A - CF$	40.95	4.58	20.11	8.91	3	91,310
$H - CF$	5.44	42.07	4.93	17.85	3	208,525
$S - CF$	14.28	3.58	37.89	12.32	2	91,137
$HS - H - S + CF$	1.58	−3.58	−4.93	20.64	4	−14,434
$T - A - HS + CF$	−21.30	−4.58	−20.11	−8.91	44	62,328

The solution to these equations is $\sigma_a^2 = 763$, $\sigma_h^2 = 4531$, $\sigma_s^2 = 1587$, $\sigma_{he}^2 = -164$, $\sigma_e^2 = 2950$. If σ_{hs}^2 is set equal to 0, the solution is $\sigma_a^2 = 756$, $\sigma_h^2 = 4468$, $\sigma_s^2 = 1542$, $\sigma_e^2 = 2952$. These estimates, of course, have no practical value for p, q, r, and s are much too small for accurate estimation of the corresponding variances. The illustration of their computation does, however, show that even with many different classes the computations are relatively simple. We have successfully adapted most of these computations to International Business Machines operations.

A difficulty with Method 1 is that it may be inappropriate to regard the year effects as random variables. If these effects actually are fixed, the estimates of σ_h^2 , σ_s^2 , and σ_{hs}^2 are biased. The estimate of σ_e^2 may or may not be biased depending on how it is estimated. This estimate is biased if obtained from the equations of Table 7. If, however, σ_e^2 had been estimated from

$$T - \sum_k \sum_i \sum_j \frac{y_{hij\cdot}^2}{n_{hij}},$$

the within year $\times$ herd $\times$ sire subclass sum of squares, the estimate would be unbiased regardless of the assumptions concerning the a_k .

It might be well at this point to state briefly a convenient procedure for finding the expected values of quantities like H, S, etc. Substitute for the y's their corresponding linear models, and then remembering the assumptions concerning the elements of the model proceed to write out the expectations. For example,

$$E \sum_i \sum_j \frac{y_{\cdot ij\cdot}^2}{n_{\cdot ij}} = \sum_i \sum_j E \frac{y_{\cdot ij\cdot}^2}{n_{\cdot ij}}$$

$$= \sum_i \sum_j E[n_{\cdot ij}\mu + n_{1ij}a_1$$

$$+ \cdots + n_{pij}a_p + n_{\cdot ij}h_j + n_{\cdot ij}s_j + n_{\cdot ij}(hs)_{ij}$$

$$+ \sum_k \sum_k e_{kijk}]^2/n_{\cdot ij}$$

$$= \sum_i \sum_j E[n_{\cdot ij}^2\mu^2 + n_{1ij}^2a_1^2$$

$$+ \cdots + n_{pij}^2a_p^2 + n_{\cdot ij}^2h_i^2 + n_{\cdot ij}^2s_i^2 + n_{\cdot ij}^2(hs)_{ij}^2$$

$$+ \sum_k \sum_k e_{kijk}^2 + \text{cross products all having zero expectation}]/n_{\cdot ij}$$

$$= \sum_i \sum_j [n_{\cdot ij}^2 \mu^2 + n_{1ij}^2 \sigma_a^2 + \cdots + n_{hij}^2 \sigma_a^2 + n_{\cdot ij}^2 \sigma_b^2 + n_{\cdot ij}^2 \sigma_s^2 + n_{\cdot ij}^2 \sigma_{bs}^2 + \sum_k \sum_k \sigma_e^2]/n_{\cdot ij}$$

$$= N\mu^2 + \sum_i \sum_j \frac{\sum_k n_{kij}^2}{n_{\cdot ij}} \sigma_a^2 + N(\sigma_b^2 + \sigma_s^2 + \sigma_{bs}^2) + s\sigma_e^2$$

Method 2

The bias in estimating variance components due to the assumption that fixed elements of the model are random variables can be eliminated by using Method 2. At the same time the relative simplicity of Method 1 can be retained. Method 2 involves estimating the fixed effects by least squares, correcting the data in accordance with these estimates, and then applying Method 1 to the "corrected" data.

This method was used by Hazel and Terrill (1945) on data which were orthogonal except for the fixed effects. Their estimates were biased for they assumed for computational purposes that, except for fixed effects, the expectations of sums of squares of corrected data are the same as the expectations of the corresponding sums of squares of the uncorrected data. Method 2 enables one to appraise this bias and to correct for it.

Before we apply this method to our example, let us consider the general case. Suppose the linear model is

$$(1) \qquad y_a = \sum_{i=1}^{p} b_i x_{ia} + e_a \qquad a = 1, \cdots, N$$

The x's are known. The e's are uncorrelated with mean = 0 and variance = σ_e^2.

If the b's are all fixed, the least squares equations for estimating them are as shown in (2). The b's are, in fact, not all fixed in the variance components estimation problem, but they can be estimated by least squares as a matter of expediency.

$$(2) \qquad \begin{aligned} \sum_{i=1}^{p} C_{1i}\hat{b}_i &= Y_1 \qquad C_{ij} = \sum_{a=1}^{N} x_{ia}x_{ja} \\ \sum_{i=1}^{p} C_{2i}\hat{b}_i &= Y_2 \qquad Y_i = \sum_{a=1}^{N} x_{ia}y_a \\ &\vdots \qquad\qquad \vdots \\ \sum_{i=1}^{p} C_{pi}\hat{b}_i &= Y_p \end{aligned}$$

It is sometimes necessary to impose one or more linear restrictions on the estimates in order to obtain a solution to equations (2).

Now suppose that $b_1, \cdots, b_s$ are fixed and also that for all $i = 1, \cdots, s$

$$E(\hat{b}_i - b_i)^2 = K_i\sigma_e^2 \tag{3}$$

It is not true that all least squares estimates have this property. For example, in our butterfat production example described in Method 1

$$E(\hat{\mu} - \mu)^2 \neq K_\mu\sigma_e^2$$

Instead

$$E(\hat{\mu} - \mu)^2 = \frac{1}{p}\sigma_a^2 + \frac{1}{q}\sigma_b^2 + \frac{1}{r}\sigma_s^2 + \lambda\sigma_{bs}^2 + K_\mu\sigma_e^2$$

Method 2 applies only to correcting data by least squares estimates for which (3) applies. It is not difficult to determine which $\hat{b}$'s qualify.

Now the data are corrected as follows (in practice only certain linear functions of the observations need to be corrected):

$$z_u = y_u - \sum_{i=1}^{s} \hat{b}_i x_{iu} \tag{4}$$

Suppose that for $i, j = s+1, \cdots, r \leq p$ all $C_{ij} = 0$ when $i \neq j$. Let

$$Z_i = \sum_u x_{iu} z_u .$$

Then compute (5). Note that

$$Z_u = Y_u - \sum_{i=1}^{s} \hat{b}_i C_{ui}$$

$$\sum_{i=s+1}^{r} Z_i^2/C_{ii} \tag{5}$$

It is found that, except for σ_e^2, the expectation of (5) is the same as the expectation of (6) with $b_1, \cdots, b_s$ assumed $= 0$.

$$\sum_{i=s+1}^{r} Y_i^2/C_{ii} \tag{6}$$

The coefficient of σ_e^2 in the expectation of (5) is increased over that of (6) by the quantity.

$$\sum_{i=1}^{s}\sum_{j=1}^{s} C^{ij}P_{ij} , \qquad \text{where} \tag{7}$$

C^{ij} are elements of the matrix inverse to the matrix of C_{ij} $(i, j = 1, \cdots, p)$, and

$$P_{uv} = \sum_{i=s+1}^{p} C_{iu}C_{iv}/C_{ii}$$

Computation of (7) is simple if s is small and if least squares equations (2) can be rewritten as (8). This can be done in many cases.

$$\sum_{j=1}^{p} C_{ij}b_j = Y_i \qquad i = 1, \cdots, s \tag{8}$$

$$\sum_{j=1}^{s} C_{ij}b_j + C_{ii}b_i = Y_i \qquad i = s+1, \cdots, p$$

Note in these equations that for all $i, j = s+1, \cdots, p$, $C_{ij} = 0$ when $i \neq j$. When equations (8) prevail, C^{ij} and $\hat{b}_i$ $(i, j = 1, \cdots, s)$ can be computed from equations (9).

$$\sum_{j=1}^{s} C'_{ij}\hat{b}_j = Y'_i \qquad (i = 1, \cdots, s) \tag{9}$$

In equations (9)

$$C'_{uv} = C_{uv} - \sum_{i=s+1}^{p} C_{iu}C_{iv}/C_{ii}$$

$$Y'_u = Y_u - \sum_{i=s+1}^{p} C_{iu}Y_i/C_{ii}$$

The least squares estimates of $b_1, \cdots, b_s$ are the solution to equations (9), and C^{ij} $(i, j = 1, \cdots, s)$ are the elements of the matrix inverse to the matrix of coefficients in (9).

Let us illustrate Method 2 with the data of Table 1. We shall now assume that the a's are fixed. First the least squares estimates of the a's are computed. This is done most simply by estimating them jointly with $d_{ij} = \mu + h_i + s_j + (hs)_{ij}$. Thus the equations are reduced to the form of equations (8). Looking at Table 1 these equations are

$$7\hat{a}_1 + 3\hat{d}_{11} + \hat{d}_{21} + 3\hat{d}_{41} = 2931$$

and similarly for the other "a" equations

$$3\hat{a}_1 + 2\hat{a}_2 + 5\hat{d}_{13} = 2395$$

and similarly for the other "d" equations
Then by (9) these equations reduce to the ones shown in Table 10.

TABLE 10

$\hat{a}_1$	$\hat{a}_2$	$\hat{a}_3$	$\hat{a}_4$	
3.825	−3.825	0.	0.	− 73.500
−3.825	6.492	−2.667	0.	101.500
0.	−2.667	6.000	−3.333	41.333
0.	0.	−3.333	3.333	− 69.333

One restriction must be imposed before a solution is obtainable. A convenient one is $\hat{a}_4 = 0$. Then the solution is $\hat{a}_1 = 12.08$, $\hat{a}_2 = 31.30$, $\hat{a}_3 = 20.80$, $\hat{a}_4 = 0$.

Inverting the matrix of coefficients of Table 10 with fourth row and column deleted, the C^{ij} pertaining to the a's are obtained. These are presented in Table 11.

TABLE 11

a_1	a_2	a_3
.936488	.675000	.300000
.675000	.675000	.300000
.300000	.300000	.300000

Now the data can be corrected for the $\hat{a}$'s. For example, the corrected total for the subclass pertaining to herd 1 × sire 1 is 2395 − 3(12.08) − 2(31.30) = 2296.16. The corrected subclass and class totals are shown in Table 12.

TABLE 12

Herd	Sire			Total
	1	2	3	
1	2296.16	2461.20	1609.00	6366.36
2	1568.02	2005.00	2219.80	5792.82
3	1611.10	3277.20		4888.30
4	2871.26		3685.60	6556.86
Total	8346.54	7743.40	7514.40	23,604.34

Using the totals of Table 12 in conjunction with the subclass and class numbers of Tables 1 and 4, the corrected sums of squares are computed. Thus, $H' = (6366.36)^2/16 + \cdots + (6556.86)^2/17$. These quantities are:

$$HS' = 10{,}016{,}791 \qquad S' = 9{,}833{,}620$$

$$H' = 9{,}954{,}295 \qquad CF' = 9{,}774{,}822$$

Next the amounts by which the coefficients of σ_e^2 are increased in the corrected as compared to the uncorrected sums of squares are needed. Using (7), the P_{ij} pertaining to HS' are computed. Looking at Table 1,

$$P_{11} = \frac{3^2}{5} + \frac{1^2}{4} + \frac{3^2}{8} = 3.175$$

$$P_{12} = \frac{3(2)}{5} + \frac{1(3)}{4} + \frac{3(5)}{8} = 3.825$$

Table 13 presents the complete set of P's for HS'.

TABLE 13

a_1	a_2	a_3
3.175	3.825	0.
3.825	14.508	2.667
0.	2.667	10.000

The sum of products of corresponding entries in Table 11 and Table 13 is

$$.936438(3.175) + \cdots + .300000(10.000) = 22.53$$

Therefore, the coefficient of σ_e^2 in $E(HS') = 10 + 22.53 = 32.53$.

The P_{ij} for H' can be computed by reference to Table 3 thus

$$P_{11} = \frac{3^2}{16} + \frac{1^2}{15} + \frac{3^2}{17} = 1.159$$

$$P_{12} = \frac{3(6)}{16} + \frac{1(3)}{15} + \frac{3(5)}{17} = 2.207$$

Table 14 presents the complete set of P_{ij} for H'.

TABLE 14

a_1	a_2	a_3
1.159	2.207	1.504
2.207	9.765	4.988
1.504	4.988	6.024

Multiplying these values by those of Table 11, the addition to σ_e^2 in $E(H') = 16.54$.

The P_{ij} for S' are shown in Table 15. Referring to Table 4,

$$P_{11} = \frac{7^2}{20}, \qquad P_{12} = \frac{7(13)}{20}, \qquad \text{etc.}$$

TABLE 15

a_1	a_2	a_3
2.450	4.550	0.
4.550	12.215	4.235
0.	4.235	7.215

Then the addition to σ_e^2 in $E(S') = 2.450\ (.936438) + \cdots = 21.39$.
Finally the P_{ij} for CF' are

$$P_{11} = \frac{7^2}{57}, \qquad P_{12} = \frac{7(21)}{57}, \qquad \text{etc. as shown in Table 16.}$$

TABLE 16

a_1	a_2	a_3
.860	2.579	1.965
2.579	7.737	5.895
1.965	5.895	4.491

Multiplying and summing corresponding entries of Tables 11 and 16, the addition to the coefficient of σ_e^2 in $E(CF') = 15.57$.

Table 17 shows the corrected sums of squares and their expectations.

TABLE 17

	μ^2	σ_h^2	σ_s^2	σ_{hs}^2	σ_e^2	
HS'	57.	57.	57.	57.	32.53	10,016,791
H'	57.	57.	24.04	24.04	20.54	9,954,295
S'	57.	18.51	57.	18.51	24.39	9,833,620
CF'	57.	14.93	19.11	6.19	16.57	9,774,822

The equations to be solved are presented in Table 18

TABLE 18

	σ_h^2	σ_s^2	σ_{hs}^2	σ_e^2	
$H' - CF'$	42.07	4.93	17.85	3.97	179,473
$S' - CF'$	3.58	37.80	12.32	7.82	58,798
$HS' - H' - S' + CF'$	−3.58	−4.93	20.64	4.17	3,698

The estimate of σ_e^2 can be obtained readily from the residual sum of squares after estimating the a's and d's, that is from

$$\sum_h \sum_i \sum_j \sum_k y_{hijk}^2 - \text{Reduction } (a_h , d_{ij}).$$

$$\sum_h \sum_i \sum_j \sum_k y_{hijk}^2 = 11,124,007$$

$$\text{Reduction } (a_h , d_{ij}) = 12.08\ (-73.500) + 31.30\ (101.500) + 20.80\ (41.333) + HS = 3149 + 10,970,369 = 10,973,518$$

The residual sum of squares is therefore 11,124,007 − 10,973,518 = 150,489, with expectation 44 σ_e^2 . Consequently $\hat{\sigma}_e^2 = 150,489/44 = 3,420$. Substituting $\sigma_e^2 = 3,420$ in the equations of Table 18 and solving, the estimates of the variances are $\sigma_h^2 = 3792$, $\sigma_s^2 = 409$, $\sigma_{hs}^2 = 243$. It is not surprising that these estimates are different from the estimates obtained by Method 1. The sampling variances must be extremely large in both cases.

Adapting Method 2 to a model with covariates is easy to accomplish. In some problems this would simplify the computations. For example, if many years were involved, the number of fixed elements in the model could be reduced by fitting a quadratic or cubic to years instead of estimating individual yearly effects as was done in this example. If the years exhibit no trend, the simplest procedure is to regard a's as random variables and then to apply Method 1.

Method 3

When it is computationally feasible, Method 3 is the most satisfactory of the three methods for estimating variance components. For one thing it gets around the difficulty of fixed elements in the model. For another, it yields unbiased estimates even though certain elements of the model are correlated. The manner in which interference by these correlations is eliminated is described subsequently.

Unfortunately Method 3 is not likely to be computationally feasible in the non-orthogonal case unless the number of different classes is small or unless the design incorporates planned non-orthogonality and consequently the mean squares of the analysis of variance can be computed without solving least squares equations. In these two cases the expectations of the mean squares are easy to compute. For example, the analysis of the balanced incomplete block design is simple and so is the writing of the expectations of the mean squares. The basic facts needed for employing Method 3 are stated below.

Let the linear model describing y_a , the a-th observation be

$$y_a = \sum_{i=1}^{p} b_i x_{ia} + e_a \tag{10}$$

The x's are known. The e's have mean zero, are uncorrelated, and have common variance σ_e^2 . For the present we shall not specify which b's are fixed and which distributed.

Now if b_{q+1} , $\cdots$, b_p are set $= 0$, the least squares estimates of b_1 , $\cdots$, b_q are the solution to equations (11).

$$\begin{aligned} &\hat{b}_1 C_{11} + \hat{b}_2 C_{12} + \cdots + \hat{b}_q C_{1q} = Y_1 \\ &\hat{b}_1 C_{21} + \hat{b}_2 C_{22} + \cdots + \hat{b}_q C_{2q} = Y_2 \\ &\text{etc.} \end{aligned} \tag{11}$$

In equation (11)

$$C_{ij} = \sum_{a=1}^{N} x_{ia} x_{ja}$$

$$Y_i = \sum_{a=1}^{N} x_{ia} y_a$$

The reduction in sum of squares due to $\hat{b}_1$, $\cdots$, $\hat{b}_q$ is

$$R(b_1 , \cdots , b_q) = \sum_{i=1}^{q} \hat{b}_i Y_i \tag{12}$$

But since

$$\hat{b}_i = \sum_{j=1}^{q} C^{ij} Y_j ,$$

where C^{ij} are elements of the matrix inverse to the C_{ij} matrix ($i, j = 1, \cdots, q$),

$$R(b_1 , \cdots , b_q) = \sum_{i=1}^{q} \sum_{j=1}^{q} C^{ij} Y_i Y_j \tag{13}$$

Using (13), the expectation of $R(b_1 , \cdots , b_q)$ is easy to write, but not necessarily easy to compute.

$$E[R(b_1 , \cdots , b_q)] = \sum_{i=1}^{q} \sum_{j=1}^{q} C^{ij} E(Y_i Y_j) \tag{14}$$

Use will be made of the fact that (14) can be written

$$E[R(b_1 , \cdots , b_q)] = \sum_{i=1}^{q} \sum_{j=1}^{p} C_{ij} E(b_i b_j) + \sum_{i=q+1}^{p} \sum_{j=1}^{q} C_{ij} E(b_i b_j) + \sum_{i=q+1}^{p} \sum_{j=q+1}^{p} \lambda_{ij} E(b_i b_j) + q' \sigma_e^2 , \tag{15}$$

where

$$\lambda_{uv} = \sum_{i=1}^{q} \sum_{j=1}^{q} C^{ij} [C_{iu} C_{jv} + C_{iv} C_{ju}] \qquad \text{when } u \neq v,$$

and $$\lambda_{uu} = \sum_{i=1}^{q} \sum_{j=1}^{q} C^{ij} C_{iu} C_{ju} \qquad \text{(See 14)}.$$

In most variance components problems $E b_i b_j = 0$ $(i \neq j)$. Thus only the λ_{ij} need to be computed. q' refers to the number of independent equations in (11).

It is easy to verify that the expectation of the uncorrected total sum of squares is

$$E \sum_{\alpha=1}^{N} y_\alpha^2 = \sum_{i=1}^{p} \sum_{j=1}^{p} C_{ij} E(b_i b_j) + N \sigma_e^2 . \tag{16}$$

Now it becomes clear why the residual mean square has expectation σ_e^2 regardless of the assumptions concerning the b's. Making use of (15) it is seen that

$$E[R(b_1 , \cdots , b_p)] = \sum_{i=1}^{p} \sum_{j=1}^{p} C_{ij} E(b_i b_j) + p' \sigma_e^2 . \tag{17}$$

Therefore, the expectation of the residual sum of squares is

$$(18) \qquad E[\sum_a y_a^2 - R(b_1 , \cdots , b_p)]$$

$$= \left[\sum_{i=1}^{p} \sum_{j=1}^{p} C_{ij} E(b_i b_j) + N\sigma_e^2 \right] - \left[\sum_{i=1}^{p} \sum_{j=1}^{p} C_{ij} E(b_i b_j) + p'\sigma_e^2 \right]$$

$$= (N - p')\sigma_e^2$$

Suppose that $b_{q+1} , \cdots , b_p$ are independently distributed with means = 0 and common variance σ^2. This variance can be estimated by equating $R(b_1 , \cdots , b_p) - R(b_1 , \cdots , b_q)$ to its expectation. The expectation of this difference is seen by reference to (15) and (17) to be

$$(19) \qquad \sum_{i=q+1}^{p} \sum_{j=q+1}^{p} (C_{ij} - \lambda_{ij}) E(b_i b_j) + (p' - q')\sigma_e^2$$

$$= \sum_{i=q+1}^{p} (C_{ii} - \lambda_{ii})\sigma^2 + (p' - q')\sigma_e^2$$

Then using the estimate of σ_e^2 arising from (18) an unbiased estimate of σ^2 can be obtained by equating $R(b_1 , \cdots , b_p) - R(b_1 , \cdots , b_q)$ to (19). It will be noted that the assumptions made concerning $b_1 , \cdots , b_q$ are of no consequence.

Now we shall illustrate Method 3 with our data of Table 1. If one were to carry out the usual tests of hypotheses by least squares, the following sums of squares would be computed.

Among Years = R(years, herd × sire subclasses) − R(herd × sire subclasses)

Among Herds = R(years, herds, sires) − R(years, sires)

Among Sires = R(years, herds, sires) − R(years, herds)

Herds × Sires = R(years, herd × sire subclasses) − R(years, herds, sires)

Residual = $\sum_h \sum_i \sum_j \sum_k y_{hijk}^2 - R$(years, herd × sire subclasses)

The last four of these quantities can also be used to estimate σ_h^2 , σ_s^2 , σ_{hs}^2 , and σ_e^2. If years were regarded as random variables, the first would be used to estimate σ_e^2 . Our present assumption is that the year effects are fixed, however.

According to (15) we need not be concerned with μ and the a's in the expectations since the expectation of each of the above reductions

contains the following quantity,

$$N\mu^2 + 2\sum_h n_{h\cdot\cdot}\mu a_h + \sum_h n_{h\cdot\cdot}a_h^2 .$$

This expression vanishes in the sums of squares, which are differences between two reductions.

Aside from μ and a_h terms, the expectations of the pertinent reductions are those shown in Table 19.

TABLE 19

	σ_h^2	σ_s^2	σ_{hs}^2	σ_e^2
$\sum_h\sum_i\sum_j\sum_k y_{hijk}^2$	N	N	N	N
R(years, herd × sire subclasses)	N	N	N	$p+s-1$
R(years, herds, sires)	N	N	K_1	$p+q+r-2$
R(years, herds)	N	K_2	K_3	$p+q-1$
R(years, sires)	K_4	N	K_5	$p+r-1$

The entries other than the K's in Table 19 are derived from (15) and (16). The K's are computed by (14). The expectations of the sums of squares of the analysis of variance are presented in Table 20.

TABLE 20

S.S.	σ_h^2	σ_s^2	σ_{hs}^2	σ_e^2
Among Herds	$N-K_4$	0	K_1-K_5	$q-1$
Among Sires	0	$N-K_2$	K_3-K_5	$r-1$
Herds × Sires	0	0	$N-K_1$	$s-q-r+1$
Residual	0	0	0	$N-p-s+1$

The computations of the various reductions proceed as follows in our example. R(years, herd × sire subclasses) = 10,973,518 as was shown in the description of Method 2. To obtain R(years, herds, sires) the equations of Table 21 need to be solved. In these equations μ is estimated jointly with each a_h, while h_4 and s_3 are set = 0. These three or some other set of three restrictions on the estimates are necessary.

TABLE 21

	a_1	a_2	a_3	a_4	h_1	h_2	h_3	s_1	s_2	
a_1	7	0	0	0	3	1	0	7	0	2931
a_2	0	21	0	0	6	3	7	13	8	9983
a_3	0	0	16	0	2	9	2	0	9	6959
a_4	0	0	0	13	5	2	0	0	0	4806
h_1	3	6	2	5	16	0	0	5	6	6632
h_2	1	3	9	2	0	15	0	4	5	6086
h_3	0	7	2	0	0	0	9	3	6	5149
s_1	7	13	0	0	5	4	3	20	0	8838
s_2	0	8	9	0	6	5	6	0	17	8181

The solution is

$$a_1 = 414.77 \qquad h_1 = 6.13 \qquad s_1 = 2.48$$
$$a_2 = 419.83 \qquad h_2 = -8.15 \qquad s_2 = 15.09$$
$$a_3 = 412.39 \qquad h_3 = 143.05$$
$$a_4 = 368.59$$

$$R(\text{years, herds, sires}) = 414.77(2931) + \cdots + 15.09(8181)$$
$$= 10{,}921{,}107$$

In order to compute $R(\text{years, herds})$ the s_1 and s_2 rows and columns of Table 21 are deleted and the resulting equations solved. The solution is

$$a_1 = 414.76 \qquad h_1 = 11.35$$
$$a_2 = 422.99 \qquad h_2 = -6.35$$
$$a_3 = 418.32 \qquad h_3 = 150.16$$
$$a_4 = 366.30$$

$$R(\text{years, herds}) = 414.76(2931) + \cdots + 150.16(5149)$$
$$= 10{,}919{,}698$$

The reduction due to years and sires requires solution of the equations of Table 21 with the h_1 , h_2 , h_3 rows and columns deleted. The solution is

$$a_1 = 425.46 \qquad a_3 = 407.72$$
$$a_2 = 461.13 \qquad a_4 = 369.69$$

$$s_1 = -6.75 \qquad s_2 = 48.38$$

$$R(\text{years, sires}) = 425.46(2931) + \cdots + 48.38(8181)$$

$$= 10{,}800{,}679$$

The computations of the K's in Table 19 require inversions of certain matrices. To obtain K_1, the inverse of the matrix of coefficients in Table 21 is needed. This inverse matrix is presented in Table 22. The entries to the left of the diagonal are omitted since the matrix is symmetric.

TABLE 22

	a_1	a_2	a_3	a_4	h_1	h_2	h_3	s_1	s_2
a_1	.64297	.41685	.15893	.03469	−.07895	−.02813	−.05580	−.46226	−.22447
a_2		.42381	.16668	.03183	−.06608	−.04169	−.09206	−.38257	−.21929
a_3			.19176	.02719	−.03647	−.08558	−.04440	−.13108	−.12625
a_4				.10925	−.00501	−.04759	−.04686	−.00004	.02410
h_1					.14349	.06382	.09075	.00884	−.05104
h_2						.14976	.07769	−.02062	−.02907
h_3							.23943	.00581	−.07214
s_1								.46163	.25050
s_2									.28088

Now we need the coefficients of σ^2_{hs} in the expectations of squares and products of the right members of the equations of Table 21. These computations are facilitated by setting up Table 23.

TABLE 23

Herd × sire subclasses

Right members	11	12	13	21	22	23	31	32	41	43
$y_{1\cdot\cdot}$	3	0	0	1	0	0	0	0	3	0
$y_{2\cdot\cdot}$	2	4	0	3	0	0	3	4	5	0
$y_{3\cdot\cdot}$	0	2	0	0	5	4	0	2	0	3
$y_{4\cdot\cdot}$	0	0	5	0	0	2	0	0	0	6
$y_{\cdot 1\cdot}$	5	6	5	0	0	0	0	0	0	0
$y_{\cdot 2\cdot}$	0	0	0	4	5	6	0	0	0	0
$y_{\cdot 3\cdot}$	0	0	0	0	0	0	3	6	0	0
$y_{\cdot\cdot 1}$	5	0	0	4	0	0	3	0	8	0
$y_{\cdot\cdot 2}$	0	6	0	0	5	0	0	6	0	0

The coefficients of $\sigma^2_{i_s}$ in the squares and products of right members are the squares or products of appropriate rows in Table 23. For example, the coefficient of $\sigma^2_{1_3}$ in $E\,y_1^2\ldots$ is $3^2 + 1^2 + 3^2 = 19$. That in $E(y_1\ldots y_2\ldots)$ is $3(2) + 1(3) + 3(5) = 24$. The complete set of coefficients is presented in Table 24, with entries to the left of the diagonal omitted due to the symmetry of the matrix.

TABLE 24

	a_1	a_2	a_3	a_4	h_1	h_2	h_3	s_1	s_2
a_1	19	24	0	0	15	4	0	43	0
a_2		79	16	0	34	12	33	71	48
a_3			58	26	12	49	12	0	49
a_4				65	25	12	0	0	0
h_1					86	0	0	25	36
h_2						77	0	16	25
h_3							45	9	36
s_1								114	0
s_2									97

Multiplying and summing corresponding entries of Tables 22 and 24,

$$K_1 = 19(.64297) + 2(24)(.41685) + \cdots + 97(.28088)$$

$$= 38.29$$

Calculation of K_2 requires the inverse of the matrix of coefficients of Table 21 with s_1 and s_2 columns and rows deleted. This inverse is presented in Table 25.

TABLE 25

	a_1	a_2	a_3	a_4	h_1	h_2	h_3
a_1	.17524	.03588	.03770	.03027	−.06049	−.04552	−.03629
a_2		.10581	.05361	.03375	−.06365	−.06022	−.09421
a_3			.13358	.03635	−.05523	−.09823	−.07138
a_4				.10523	−.05576	−.04461	−.03433
h_1					.12204	.05734	.06178
h_2						.14663	.06867
h_3							.20025

Also needed in the computation of K_2 are the coefficients of σ_e^2 in the expectations of squares and products of right members of the least squares equations. Table 26 facilitates this computation.

TABLE 26

Right members	Sires		
	1	2	3
$y_{1\cdot\cdot\cdot}$	7	0	0
$y_{2\cdot\cdot\cdot}$	13	8	0
$y_{3\cdot\cdot\cdot}$	0	9	7
$y_{4\cdot\cdot\cdot}$	0	0	13
$y_{\cdot 1\cdot\cdot}$	5	6	5
$y_{\cdot 2\cdot\cdot}$	4	5	6
$y_{\cdot 3\cdot\cdot}$	3	6	0

The coefficient of σ_e^2 in $E(y_1^2\cdot\cdot\cdot)$ is $7^2 = 49$, in $E(y_1\cdot\cdot\cdot y_2\cdot\cdot\cdot)$ is $7(13) = 91$, etc. The complete set is shown in Table 27.

TABLE 27

	a_1	a_2	a_3	a_4	b_1	b_2	b_3
a_1	49	91	0	0	35	28	21
a_2		233	72	0	113	92	87
a_3			130	91	89	87	54
a_4				169	65	78	0
b_1					86	80	51
b_2						77	42
b_3							45

$$\text{Now } K_2 = 49\ (.17524) + 2(91)\ (.03588) + \cdots + 45\ (.20025)$$
$$= 42.29$$

K_3 is obtained from Table 24 and 25, thus

$$K_3 = 19\ (.17524) + 2\ (24)\ (.03588) + \cdots + 45\ (.20025)$$
$$= 31.71.$$

In a similar manner K_4 is found to be 22.57 and K_5 to be 22.57 (the equality of K_4 and K_5 is only a coincidence.)

Table 28 presents the pertinent reductions and their expectations (excluding μ and a_h terms)

TABLE 28

	σ_h^2	σ_s^2	σ_{hs}^2	σ_e^2	
$\Sigma\Sigma\Sigma\Sigma\ y_{hsijk}^2$	57	57	57	57	11,124,007
R(years, herd × sire subclasses)	57	57	57	13	10,973,518
R(years, herds, sires)	57	57	38.29	9	10,921,107
R(years, herds)	57	42.29	31.71	7	10,919,698
R(years, sires)	22.57	57	22.57	6	10,800,679

Then the equations to be solved are those of Table 29.

TABLE 29

	σ_h^2	σ_s^2	σ_{hs}^2	σ_e^2	
Among Herds	34.43	0	15.72	3	120,438
Among Sires	0	14.71	6.58	2	1,409
Herds × Sires	0	0	18.71	4	52,411
Residual	0	0	0	44	150,489

The solution to these equations is $\sigma_h^2 = 2255$, $\sigma_s^2 = -1295$, $\sigma_{hs}^2 = 2070$, and $\sigma_e^2 = 3420$.

ESTIMATION OF COMPONENTS OF COVARIANCE

The same general principles described in Methods 1, 2, and 3 for estimating variances can be employed to estimate covariances. To illustrate, suppose an observation is made on each of the progeny resulting from single crosses among inbred lines. If y_{ijk} is the observation on the k-th progeny of the i-th male line by the j-th female line cross, a model which might reasonably be assumed is

$$y_{ijk} = \mu + g_i + g_j + m_j + s_{ij} + e_{ijk}$$

where g_i is the general combining ability of the i-th line, g_j is the general combining ability of the j-th line, m_j is the maternal ability (exclusive of the genes transmitted to the progeny) of the j-th line, s_{ij} is peculiar to crosses $i \times j$ and of $j \times i$, and e_{ijk} is a random error. Suppose further that the elements of the model fit Eisenhart's Model II except that $E(g_i m_i) = \sigma_{gm}$.

The problem is to estimate the variances and σ_{gm}. If Method 3 is used, unbiased estimates of the variances are obtained. If Method 1 is used, the estimates are biased due to the presence of σ_{gm}. If the least squares estimates of g_i and m_i are computed, an unbiased estimate of σ_{gm} can be derived from $\sum_i \hat{g}_i \hat{m}_i$, the expectation of which is $(p-1)\sigma_{gm} + \sum_i k_i \sigma_e^2$, where p is the number of lines and $k_i \sigma_e^2$ is the covariance between $\hat{g}_i$ and $\hat{m}_i$, assuming that g_i and m_i are fixed.

A more frequently occurring type of covariance estimation problem in animal breeding arises in connection with estimation of genetic and phenotypic correlations. Observations are taken on two or more traits in some population. The following linear models characterize such observations on two traits.

$$y_a = \sum_{i=1}^{p} b_i x_{ia} + e_a \tag{20}$$

$$y'_a = \sum_{i=1}^{p} b'_i x_{ia} + e'_a$$

b_i and b'_i are fixed for $i = 1, \cdots, q$

b_i and b'_i are random variables for $i = q+1, \cdots, p$

So far as the random variables are concerned,

$$Eb_i = Eb'_i = 0 \qquad E(e'_a)^2 = \sigma_{e'}^2$$

$$E(b_i)^2 = \sigma_i^2 \qquad E(b_i b'_i) = \sigma_{ii'}$$

$$E(b'_i)^2 = \sigma_{i'}^2 \qquad E(e_a e'_a) = \sigma_{ee'}$$

$$E(e_a)^2 = \sigma_e^2 \qquad \text{All other covariances} = 0$$

Now in place of least squares reductions like

$$\sum_i \hat{b}_i Y_i \qquad \text{or} \qquad \sum_i Y_i^2/C_{ii}$$

we substitute

$$\sum_i \hat{b}_{i'} Y_i \qquad \text{or} \qquad \sum_i Y_i Y_{i'}/C_{ii} ,$$

where $Y_{i'}$, refers to a right member of the least squares equations for the second trait.

Then the expectations of these reductions are exactly the same as described for estimation of variance components except that $\sigma_{ii'}$, is substituted for σ_i^2 and $\sigma_{ee'}$, is substituted for σ_e^2. Therefore, any of the

three methods for estimating variances can be used equally well for estimating covariances when (20) is the model.

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Henderson, Kempthorne, Searle and von Krosigk (1959)

Henderson, C.R., Kempthorne, O., Searle, S.R., and von Krosigk, C.M. 1959. The estimation of environmental and genetic trends from records subject to culling. Biometrics, vol. 15, no. 2, pp. 192 to 218.

The problem

This is the paper in which the mixed model equations were formally published. The problem is the same as in 1949, the estimation of environmental and genetic trends from records of cows subject to culling. The authors show that the solutions of the equations maximize the joint density of the observations and the random effects, so that the estimates of the fixed effects and the predictions of the random effects come from a single system.

Points to note while reading

Note the structure of the equations, with the block of fixed effects, the block of random effects and the inverse of the variance matrix added to the random block. This form is exactly the one used today in every genetic evaluation program, including single step.

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THE ESTIMATION OF ENVIRONMENTAL AND GENETIC TRENDS FROM RECORDS SUBJECT TO CULLING*

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INTRODUCTION

A very common problem which arises in animal or plant breeding research is that of assessing the gain which has resulted from a selection program carried on over a number of years. To be specific, let us suppose that we have a closed dairy herd which has been maintained over a number of years with selection being practiced. The records available for assessing any genetic improvement consist of production records of cows in the various years and can be represented by a two-way classification, cow by year. At first it might be thought that such a two-way classification could be analyzed by the method of fitting constants [Yates, 1934]. Applications of this technique have, however, led to the apparent conclusion that the environment gradually deteriorated over the period of years, as indicated by the fact that the constants fitted for years tend to decrease year by year.

Henderson [1949] pointed out that a least squares procedure in which the cow effects are regarded as fixed leads to biased estimates. Lush and Shrode [1950] gave a simple explanation of the biases arising in the estimation of age correction factors; similar considerations apply to the estimation of year effects. The present paper is the combined

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result of work in this problem carried out over the past few years independently at Iowa and Cornell.

During the summer of 1957 the two of us at Iowa (Kempthorne and von Krosigk), after preparing a paper for this journal, had correspondence and personal discussion with the two at Cornell (Henderson and Searle), from which it was felt that a combined paper would best suit our present state of knowledge of the problem. A method for estimating environmental trend when repeatability is assumed known was outlined by Henderson in mimeographed material that has been circulating for some years; a method not requiring an assumed value for repeatability and now being proposed by two of us (K. and K.) appeared at first to differ, but when one of us (S.) showed that for given repeatability these two methods are equivalent, it seemed more suitable to present all the work in one publication rather than two separate ones. This paper therefore presents both methods and the manner in which they are equivalent, attributing the various sections to their appropriate authors.

ORIGIN OF THE BIAS IN ROUTINE LEAST SQUARES
(Kempthorne and von Krosigk)

Lush and Shrode [1950] have shown how bias enters into the estimation of age correction factors due to culling. Similar arguments apply when estimating year effects. These admit of easy explanation in the simple situation in which a cow's first-year record x_{i1}, and her second-year record x_{i2}, conform to a bivariate normal distribution, with a strictly additive difference γ between first and second year records, based on the model

$$x_{i1} = \mu + c_i + e_{i1}, \qquad x_{i2} = \mu + \gamma + c_i + e_{i2}.$$

The e's are errors arising from differing environments among the same animal's records from year to year, and from inaccuracies of measurement, and c_i is common to all records of cow i. The c's and e's are random variables with zero means and variances σ_c^2 and σ_e^2 respectively, and all covariances are zero. The expectation with regard to the errors is over a hypothetical set of repetitions in the particular year with any particular cow. The expectation with regard to the cow effects is over the population of cows which could have entered the records, it being hypothesized that the particular set entering the records is a random sample of the possibilities that could have arisen. The fact that cows in a herd will probably be somewhat related would vitiate the assumption that the covariances between any two c's is zero. We shall, however, in this paper not extend the argument to take care of this situation.

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It should be emphasized that we are supposing no age or lactation-number effect, on the assumption that certain correction factors obtained apart from the data to be analyzed are appropriate. The validity of this assumption will be discussed at the end of the paper.

With this model the herd average for the first year (mean value of x) is μ and that for the second year (mean value of y) is $\mu + \gamma$. x and y both have variances $\sigma_a^2 + \sigma_e^2$ and the covariance between them is σ_a^2, so that based on a bivariate normal distribution the mean value of y given x is

$$\mu + \gamma + r(x - \mu)$$

where $r = \sigma_a^2/(\sigma_a^2 + \sigma_e^2)$, a ratio known in animal breeding work as repeatability [Lush, 1949]. If we take the expectation of this conditional mean over all the cows who had a first record we obtain of course $\mu + \gamma$, the population mean for y. But the culling has the effect that the expectation of x over the cows retained in the herd after the first record is equal to a number say μ', which would not in general be equal to μ, unless the culling were either based only on some attributes statistically independent of x, or were a peculiar type of balanced culling which did not affect the mean. The usual thing is probably some sort of truncation selection, though probably based on an index of which x is one component rather than on x alone. It therefore follows that the mean of the second-year records of cows which are retained is

$$\mu + \gamma + r(\mu' - \mu).$$

In this simple case the method of least squares gives an estimate of the year difference γ, as the average of second-year records minus the average of first-year records of those cows that had a second-year record, and this estimate has expected value,

$$\mu + \gamma + r(\mu' - \mu) - \mu' = \gamma - (1 - r)(\mu' - \mu).$$

This is the paired comparison method of estimation (method B of Lush and Shrode); with the usual type of selection μ' will be greater than μ, so this estimate of the year difference is biased downward to the extent

$$(1 - r)(\mu' - \mu).$$

With the gross comparison method (method A of Lush and Shrode) the year difference is estimated by the difference between the mean of all second-year records and the mean of all first-year records; such an estimate has expectation

$$\mu + \gamma + r(\mu' - \mu) - \mu,$$

and hence is biased upward by an amount $r(\mu' - \mu)$.

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METHOD I (HENDERSON)

Suppose the linear model for a cow's record is

$$y_{ikt} = \mu + d_k + g_t + c_{it} + e_{ikt}$$

where y_{ikt} is the record in the kth year made by the ith cow of the tth group of cows in a herd. These groups might for example represent daughters of a bull, sets of cows born within a specified period, or groups of cows that enter the herd together. μ is the population average, d_k is the environmental effect of the kth year, g_t is the mean real producing ability of the tth group of cows, c_{it} is the real producing ability of the ith cow of the tth group, and e_{ikt} is a random environmental effect peculiar to the individual record. We will assume that the c_{it} are normally and independently distributed with zero means and variance σ_c^2, that the e_{ikt} are normally and independently distributed with zero means and variance σ_e^2, and that the c's and e's are uncorrelated. These assumptions imply that the cows of a particular group are randomly drawn from a normal population with mean $\mu + g_t$ and variance σ_c^2. Furthermore, temporary environment is not correlated with real producing ability. The problem is to estimate differences among the d's and g's assuming that repeatability is known.

The following method for maximum likelihood estimation of fixed elements of mixed linear models has been derived by one of us (H.).

Let the mixed linear model be

$$y = X\beta + Zu + e$$

where β is a vector of fixed effects, while u and e are independent vectors of variables that are normally distributed with zero means and variance-covariance matrices $D\sigma^2$ and $R\sigma^2$ respectively. Then y has a multivariate normal distribution with means $X\beta$ and variance-covariance matrix $(R + ZDZ')\sigma^2$, and the m.l. estimator of β, say $\hat{\beta}$, is the solution to

$$X'(R + ZDZ')^{-1}X\hat{\beta} = X'(R + ZDZ')^{-1}y \qquad (1)$$

assuming that the coefficient matrix is non-singular and that β is estimable. The difficulty in applying this method is that $R + ZDZ'$ is, in practice, often large and non-diagonal.

Now the same estimator can be obtained by maximizing for variation in β and u the joint density function of y and u. This function is

$$f(y, u) = g(y|u)h(u)$$

$$= \text{Const. } \exp\left[-\frac{1}{2\sigma^2}(y - X\beta - Zu)'R^{-1}(y - X\beta - Zu) - \frac{1}{2\sigma^2}u'D^{-1}u\right].$$

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Differentiating this with respect to β and u and equating to zero gives the following equations:

$$X'R^{-1}X\tilde{\beta} + X'R^{-1}Z\tilde{u} = X'R^{-1}y$$

$$Z'R^{-1}X\tilde{\beta} + (Z'R^{-1}Z + D^{-1})\tilde{u} = Z'R^{-1}y.$$

These equations except for D^{-1} are identical to those for the m.l. estimation of β and u regarding u as fixed. In many problems the equations are easy to write since R and D are diagonal.

To prove that $\tilde{\beta}$ and $\hat{\beta}$ are identical we eliminate $\tilde{u}$ obtaining

$$X'WX\tilde{\beta} = X'Wy$$

where

$$W = R^{-1} - R^{-1}Z(Z'R^{-1}Z + D^{-1})^{-1}Z'R^{-1}.$$

Now, if it can be shown that $W = (R + ZDZ')^{-1}$ these equations are identical to those given in (1) and then $\tilde{\beta} = \hat{\beta}$. We show this by proving that $(R + ZDZ')W = I$.

$$\begin{aligned}(R + ZDZ')W &= (R + ZDZ')[R^{-1} - R^{-1}Z(Z'R^{-1}Z + D^{-1})^{-1}Z'R^{-1}] \\ &= I + ZDZ'R^{-1} - Z(Z'R^{-1}Z + D^{-1})^{-1}Z'R^{-1} \\ &\qquad - ZDZ'R^{-1}Z(Z'R^{-1}Z + D^{-1})^{-1}Z'R^{-1} \\ &= I + ZDZ'R^{-1} - Z(I + DZ'R^{-1}Z)(Z'R^{-1}Z + D^{-1})^{-1}Z'R^{-1} \\ &= I + ZDZ'R^{-1} - ZD(D^{-1} + Z'R^{-1}Z)(Z'R^{-1}Z + D^{-1})^{-1}Z'R^{-1} \\ &= I + ZDZ'R^{-1} - ZDZ'R^{-1} \\ &= I, \text{ thus completing the proof.}\end{aligned}$$

In applying this method to the present problem μ, d's, and g's correspond to β, and c's correspond to u. The joint distribution of the y_{ikt} and c_{it} is

$$\prod_{ikt}\left\{\frac{1}{\sqrt{2\pi\sigma_e^2}}\exp\frac{-1}{2\sigma_e^2}(y_{ikt} - \mu - d_k - g_t - c_{it})^2\right\}$$

$$\times\left\{\prod_{it}\frac{1}{\sqrt{2\pi\sigma_a^2}}\exp\frac{-c_{it}^2}{2\sigma_a^2}\right\}.$$

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The maximizing values for variations in μ, g's, d's, and c's are the solutions to the following set of equations:

$$n_{..}\hat{\mu} + \sum_k n_{.k.}\hat{d}_k + \sum_t n_{..t}\hat{g}_t + \sum_i \sum_t n_{i.t}\hat{c}_{it} = y_{...}, \qquad (2)$$

$$n_{.k.}\hat{\mu} + n_{.k.}\hat{d}_k + \sum_t n_{.kt}\hat{g}_t + \sum_i \sum_t n_{ikt}\hat{c}_{it} = y_{.k.}, \qquad (3)$$

and analogous equations for other years (k);

$$n_{..t}\hat{\mu} + \sum_k n_{.kt}\hat{d}_k + n_{..t}\hat{g}_t + \sum_i n_{i.t}\hat{c}_{it} = y_{..t}, \qquad (4)$$

and analogous equations for other groups (t);

$$n_{i.t}\hat{\mu} + \sum_k n_{ikt}\hat{d}_k + n_{i.t}\hat{g}_t + \left(n_{i.t} + \frac{1-r}{r}\right)\hat{c}_{it} = y_{i.t}, \qquad (5)$$

and analogous equations for other cows (i, t).

$n_{ikt} = 1$ if the ith cow has a record in the kth year, and is zero otherwise. A dot in the subscript denotes summation over that particular subscript.

It will be noted that the term $(1 - r)/r$ appears in the coefficient of the $\hat{c}$'s in equations (5). This arises from the ratio σ_s^2/σ_e^2 and the definition of repeatability, namely $\sigma_s^2/(\sigma_s^2 + \sigma_e^2)$. As used here repeatability is not intra-herd as Lush [1949] uses it, but rather intra-group, (within g_t groups) and therefore depends on the choice of groups.

Solving equations (5) gives

$$\hat{c}_{it} = \frac{n_{i.t}r}{1 + (n_{i.t} - 1)r}(\bar{y}_{i.t} - \hat{\mu} - \hat{g}_t - \bar{\hat{d}}).*$$

$\bar{\hat{d}}$ is the mean of $\hat{d}_k$ associated with that particular cow. Using these expressions we can eliminate $\hat{c}_{it}$ from equations (2), (3), and (4). These reduced equations are of the form

$$\sum_k m_{k'k}\hat{d}_k + \sum_t m_{k't}(\hat{\mu} + \hat{g}_t) = z_{k'}. \qquad (6)$$

and analogous equations for other years (k');

$$\sum_k m_{kt}\hat{d}_k + m_{.t}(\hat{\mu} + \hat{g}_t) = z_{.t} \qquad (7)$$

and analogous equations for other groups (t).

The m's of these equations are functions of the n's, and the z's are

*Animal breeders will be interested in the fact that $\hat{c}_{it}$ is Lush's [1949] "most probable producing ability" modified by correcting the records for years and then expressing the result as a deviation from the group mean.

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functions of n's and y's of equations (2), (3), (4), (5). As a check on the computations it should be noted that

$$\sum_k m_{k'k} = \sum_t m_{k't} \quad \text{for each year } (k')$$

$$\sum_k m_{kt} = m_{.t} \quad \text{for each group } (t)$$

$$\sum_{k'} z_{k'.} = \sum_t z_{.t} .$$

It is shown later that these equations are equivalent to the m.l. equations of Method II when r is assumed known.

Now from (7)

$$\hat{\mu} + \hat{g}_t = \frac{1}{m_{.t}} (z_{.t} - \sum_k m_{kt} \hat{d}_k).$$

Substituting this in (6) we get equations in $\hat{d}_k$ as follows

$$\sum_k w_{kk'} \hat{d}_k = v_{k'} \tag{8}$$

and analogous equations for other years (k'). The fact that

$$\sum_k w_{kk'} = 0 \quad \text{for each year } (k')$$

and

$$\sum_{k'} v_{k'} = 0$$

can be used as a computational check. Because of this linear relationship among the coefficients (8) does not have a unique solution. Consequently, we impose one constraint on the estimators, a convenient one being $\hat{d}_f = 0$, where f refers to the final year. To solve (8) with this constraint we delete the last equation and the last unknown of the remaining equations. Then $\hat{d}_k$ is the m.l. estimator of $d_k - d_f$, and $\hat{\mu} + \hat{g}_t$ is the m.l. estimator of $\mu + g_t + d_f$. Therefore $\hat{d}_k - \hat{d}_{k'}$ is the m.l. estimator of $d_k - d_{k'}$ and $(\hat{\mu} + \hat{g}_t) - (\hat{\mu} + \hat{g}_{t'})$ is the m.l. estimator of $g_t - g_{t'}$.

Use of an incorrect value of repeatability biases the estimates of environmental and genetic trends. For example, if too large a value of r is used and if cows that were culled had lower records than their contemporaries, the estimate of environmental trend is biased downward.

If r is known, the sampling variance-covariance matrix of $\hat{\mu} + \hat{g}_t$ and $\hat{d}_k$, when $\hat{d}_f = 0$, is σ_e^2 times the inverse of the coefficients of (6) and (7) with the fth row and column deleted. The upper left submatrix

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of this inverse can be obtained by inverting the coefficients of (8) with the last row and last column deleted.

The following methods for testing hypotheses concerning the d's and g's are appropriate if r is assumed known.

The denominator sum of squares in the F tests is

$$\sum_i \sum_k \sum_t y_{ikt}^2 - \sum_i \sum_t \frac{r}{[1 + (n_{i.t} - 1)r]} y_{i.t}^2 - \sum_t z_{.t}^2/m_{.t} \text{ [from (7)]} - \sum_k \hat{d}_k v_k \text{ [from (8)]}.$$

This has degrees of freedom = number of records − number of years − number of groups + 1. The numerator sum of squares for testing the hypothesis that all d_k are equal is

$$\sum_k \hat{d}_k v_k \text{ from (8)}.$$

The numerator sum of squares for testing the hypothesis that all g_t are equal is

$$\sum_t z_{.t}^2/m_{.t} \text{ [from (7)]} + \sum_k \hat{d}_k v_k \text{ [from (8)]} - \sum_k \tilde{d}_k z_{k.}$$

where $\tilde{d}_k$ are solutions to equations (6) after first deleting $(\hat{\mu} + \hat{g}_t)$.

A NUMERICAL EXAMPLE OF METHOD I (HENDERSON)

Suppose we have the following records, classified according to group, cow in group, and year of freshening.

Group	Cow	Year Fresh 1	2	3	Total
1	1	450	420	410	1280
	2	500	470	500	1470
	3	410	430		840
	4	390			390
2	1		400	430	830
	2		430	380	810
	3		380		380
3	1			500	500
	2			460	460
Total		1750	2530	2680	6960

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We shall take a value of 1/2 for r; then as in (5), $(1 - r)/r = 1$ is added to $n_{i..}$ to form the diagonal coefficients of these equations, (5). Then the complete set of equations to be solved, namely those in (2) through (5) is

$$\begin{bmatrix}
16 & 4 & 6 & 6 & 9 & 5 & 2 & 3 & 3 & 2 & 1 & 2 & 2 & 1 & 1 & 1 \\
4 & 4 & 0 & 0 & 4 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\
6 & 0 & 6 & 0 & 3 & 3 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 \\
6 & 0 & 0 & 6 & 2 & 2 & 2 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\
9 & 4 & 3 & 2 & 9 & 0 & 0 & 3 & 3 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\
5 & 0 & 3 & 2 & 0 & 5 & 0 & 0 & 0 & 0 & 0 & 2 & 2 & 1 & 0 & 0 \\
2 & 0 & 0 & 2 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\
3 & 1 & 1 & 1 & 3 & 0 & 0 & 4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
3 & 1 & 1 & 1 & 3 & 0 & 0 & 0 & 4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
2 & 1 & 1 & 0 & 2 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\
2 & 0 & 1 & 1 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 \\
2 & 0 & 1 & 1 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 \\
1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\
1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\
1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2
\end{bmatrix}
\begin{bmatrix}
\hat{\mu} \\ \hat{d}_1 \\ \hat{d}_2 \\ \hat{d}_3 \\ \hat{g}_1 \\ \hat{g}_2 \\ \hat{g}_3 \\ \hat{c}_{11} \\ \hat{c}_{12} \\ \hat{c}_{13} \\ \hat{c}_{14} \\ \hat{c}_{21} \\ \hat{c}_{22} \\ \hat{c}_{23} \\ \hat{c}_{31} \\ \hat{c}_{32}
\end{bmatrix}
=
\begin{bmatrix}
6960 \\ 1750 \\ 2530 \\ 2680 \\ 3980 \\ 2020 \\ 960 \\ 1280 \\ 1470 \\ 840 \\ 390 \\ 830 \\ 810 \\ 380 \\ 500 \\ 460
\end{bmatrix}.$$

The $\hat{c}$'s are eliminated to obtain (6) and (7) as illustrated by the coefficient of $\hat{\mu} + \hat{g}_1$ in the second equation of (6), which is

$$3 - \frac{1(3)}{4} - \frac{1(3)}{4} - \frac{1(2)}{3} = \frac{5}{6}.$$

The right hand member of the first equation reduces to

$$1750 - \frac{1(1280)}{4} - \frac{1(1470)}{4} - \cdots - \frac{0(460)}{2} = \frac{3525}{6}.$$

Multiplying all equations by 6 to eliminate fractions gives the following equations (6) and (7):

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$$\begin{bmatrix} 16 & -5 & -3 & 8 & 0 & 0 \\ -5 & 24 & -7 & 5 & 7 & 0 \\ -3 & -7 & 23 & 3 & 4 & 6 \\ 8 & 5 & 3 & 16 & 0 & 0 \\ 0 & 7 & 4 & 0 & 11 & 0 \\ 0 & 0 & 6 & 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} \hat{d}_1 \\ \hat{d}_2 \\ \hat{d}_3 \\ \hat{\mu} + \hat{\phi}_1 \\ \hat{\mu} + \hat{\phi}_2 \\ \hat{\mu} + \hat{\phi}_3 \end{bmatrix} = \begin{bmatrix} 3525 \\ 4955 \\ 5795 \\ 6975 \\ 4420 \\ 2880 \end{bmatrix}.$$

Note that the sum of the first three equations equals the sum of the last three.

Now we eliminate $\hat{\mu} + \hat{\phi}_t$ to obtain (8). For example, the coefficient of $\hat{d}_2$ in the first equation is

$$-5 - \frac{8(5)}{16} - \frac{0(7)}{11} - \frac{0(0)}{6} = \frac{-15}{2}$$

and the first right hand member is

$$3525 - \frac{8(6975)}{16} - \frac{0(4420)}{11} - \frac{0(2880)}{6} = \frac{75}{2}.$$

Then multiplying each equation by 528 we get

$$\begin{bmatrix} 6336 & -3960 & -2376 \\ -3960 & 9495 & -5535 \\ -2376 & -5535 & 7911 \end{bmatrix} \begin{bmatrix} \hat{d}_1 \\ \hat{d}_2 \\ \hat{d}_3 \end{bmatrix} = \begin{bmatrix} 19{,}800 \\ -19{,}755 \\ -45 \end{bmatrix}.$$

These are equations (8). Note that their sum is zero.

Imposing the restriction $\hat{d}_3 = 0$, the solution is

$$\hat{d}_1 = 2.468, \qquad \hat{d}_2 = -1.051$$

and substituting in the second set of equations and solving for $\hat{\mu} + \hat{\phi}$ we get

$$\hat{\mu} + \hat{\phi}_1 = 435.032$$

$$\hat{\mu} + \hat{\phi}_2 = 402.487$$

$$\hat{\mu} + \hat{\phi}_3 = 480.000.$$

METHOD II (KEMPTHORNE AND VON KROSIGK)

The discussion of the origin of the bias in the method of fitting

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constants leads to an obvious estimation procedure for the case of two years. We use the following notation:

$\bar{x}$ = average of all first-year records,
$\bar{x}_1$ = average of first-year records of cows which are retained in the herd,
$\bar{y}$ = mean of second-year records (of cows retained, of course).

The regression line of the joint distribution of the y's and x's is estimated unbiasedly even if selection of any sort is made with regard to the x's, provided only that the true regression, or relationship to x of conditional mean of y for given x, is linear throughout the entire range.* This happens automatically if the joint distribution of y and x in the unselected data is bivariate normal. It follows that the line

$$y - \bar{y} = \hat{r}(x - \bar{x}_1)$$

is an unbiased estimate of the line

$$y - (\mu + \gamma) = r(x - \mu),$$

where

$$\hat{r} = \frac{\sum (y - \bar{y})(x - \bar{x}_1)}{\sum (x - \bar{x}_1)^2},$$

summations being over cows with both first- and second-year records. It follows then that

$$\bar{y} + \hat{r}(\bar{x} - \bar{x}_1)$$

is an unbiased estimate of

$$\mu + \gamma,$$

since $\bar{x}$ is an unbiased estimate of μ with errors independent of the errors of estimation of r. Hence an unbiased estimate of γ is

$$\bar{y} + \hat{r}(\bar{x} - \bar{x}_1) - \bar{x} = (\bar{y} - \bar{x}) - \hat{r}(\bar{x}_1 - \bar{x}).$$

We shall find the variance conditionally on the x's for which second records exist. We have

$$\hat{\gamma} = \bar{y} + \hat{r}(\bar{x} - \bar{x}_1) - \bar{x} = (\bar{y} - \hat{r}\bar{x}_1) - (1 - \hat{r})\bar{x},$$

$$V(\hat{\gamma}) = V(\bar{y} - \hat{r}\bar{x}_1) + V[(1 - \hat{r})\bar{x}] - 2 \operatorname{Cov} [\bar{y} - \hat{r}\bar{x}_1 , (1 - \hat{r})\bar{x}]$$

$$= \sigma^2_{y|x}\left\{\frac{1}{n_2} + \frac{\bar{x}_1^2}{S}\right\} + (1 - r)^2 \frac{\sigma_x^2}{n_1} + [\mu^2 + V(\bar{x})] \frac{\sigma^2_{y|x}}{S}$$

$$+ 2\mu \operatorname{Cov} (\hat{r}, \bar{y} - \hat{r}\bar{x}_1),$$

*A bias would occur if culling were based also on some attribute correlated with both x and y. This would occur if selection were based on dam's performance for example. The bias can be seen to be of the order of $r_1^2(1 - r)$ where r_1 is the correlation of the attribute with x and y. This will be small in many cases, and for the particular case discussed here.

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with $S = \sum (x - \bar{x}_1)^2$ and n_1, n_2 equal to the total number of first and second records respectively.

$$\therefore\ V(\hat{\gamma}) = \sigma^2_{y|x}\left(\frac{1}{n_2} + \frac{\bar{x}_1^2}{S}\right) + (1 - r)^2 \frac{\sigma_x^2}{n_1} + [\mu^2 + V(\bar{x})] \frac{\sigma^2_{y|x}}{S} - 2\mu\bar{x}_1 \frac{\sigma^2_{y|x}}{S}.$$

An unbiased estimate of this is given by

$$V(\hat{\gamma}) = s^2_{y|x}\left(\frac{1}{n_2} + \frac{\bar{x}_1^2}{S}\right) + (1 - r)^2 \frac{s_x^2}{n_1} + \frac{s^2_{y|x}s_x^2}{n_1 S} + \bar{x}^2 \frac{s^2_{y|x}}{S} - 2\bar{x}\bar{x}_1 \frac{s^2_{y|x}}{S}$$

$$= (1 - r)^2 \frac{s_x^2}{n_1} + s^2_{y|x}\left\{\frac{1}{n_2} + \frac{(\bar{x}_1 - \bar{x})^2}{S}\right\} + \frac{s^2_{y|x}s_x^2}{n_1 S},$$

with s_x^2 = mean square among first records
and $s^2_{y|x}$ = mean square about regression of second on first records and is an unbiased estimate of $\sigma^2_{y|x}$.

The unconditional variance would depend also on Cov $(\bar{x}, \bar{x}_1)$ which would be based on the culling procedure.

The procedure given above leads to an unbiased estimate of the year effect but the estimate is not efficient in the sense of exhausting the information even in the data from two years with successive records, because we have estimated r only by the regression of second-year records on first-year records. Under the assumptions the mean square of totality of first-year records is an estimate of σ^2, while the mean square of the second-year records about the fitted line is an estimate of $\sigma^2(1 - r^2)$. So we could obtain an estimate of r,* apart from sign, purely from these two mean squares. One problem of fitting is therefore to get the estimate of r which exhausts the data. Also data will extend over more than 2 years for many cows.

For these reasons and because a solution can be obtained by the method of maximum likelihood, we shall now use this method to attack the general case. The material given above is to be regarded as a first approximation to an exhaustive solution and indicates the sort of thing that can happen when one deals with data culled on the basis of part of the actual data. The approximate solution given above will also be of use in obtaining the maximum likelihood solution, because this solution will be obtainable only by iterative solution of non-linear equations.

As a framework for analysis of the general situation, suppose we have t records of a sample of the population of individuals and that there is no systematic environmental effect and no culling. Then it

*This is independent of the regression estimate because the sum of squares of deviations of y about its regression on x is independent of that regression and of the x's.

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is reasonable to regard the jth record of the ith individual as being given by the equation

$$y_{ij} = \mu + c_i + e_{ij}$$

in which the portion c_i is common to all records of the ith cow. Also we suppose that the c_i's are normally and independently distributed with mean zero and variance σ_c^2, that the e_{ij} are normally and independently distributed with mean zero and variance σ_e^2, and that the e_{ij} and c_i are independent. Then we have, with V denoting variance, and Cov denoting covariance,

$$V(y_{ij}) = \sigma_c^2 + \sigma_e^2 = \sigma^2$$

$$\text{Cov}\,(y_{ij}\,,\, y_{ij'}) = \sigma_c^2 = r\sigma^2,$$

with

$$r = \frac{\sigma_c^2}{\sigma_c^2 + \sigma_e^2}.$$

The variance-covariance matrix of the t records of an individual is therefore

$$\sigma^2\begin{bmatrix} 1 & & & r \\ & \cdot & \cdot & \\ & & & \cdot \\ r & & & 1 \end{bmatrix}$$

i.e. σ^2 times a matrix with unity on the diagonal and r in every off-diagonal position. Let us now consider the regression of each record on the preceding ones. We have the following relationships, in which z_{ij} equals $y_{ij} - \mu$:

$$z_{i2} = rz_{i1} + e_{i2.1}\,, \qquad V(e_{i2.1}) = \sigma^2(1 - r^2)$$

$$z_{i3} = \frac{r}{1+r}(z_{i1} + z_{i2}) + e_{i3.12}\,, \qquad V(e_{i3.12}) = \sigma^2\left(1 - \frac{2r^2}{1+r}\right)$$

$$z_{i4} = \frac{r}{1+2r}(z_{i1} + z_{i2} + z_{i3}) + e_{i4.123}\,, \qquad V(e_{i4.123}) = \sigma^2\left(1 - \frac{3r^2}{1+2r}\right)$$

$$z_{i5} = \frac{r}{1+3r}(z_{i1} + z_{i2} + z_{i3} + z_{i4}) + e_{i5.1234}\,,$$

$$V(e_{i5.1234}) = \sigma^2\left(1 - \frac{4r^2}{1+3r}\right)$$

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and so on. These regression equations may be obtained by the use of the method of least squares. As an example, to find the regression of z_{i4} on z_{i1}, z_{i2}, and z_{i3} we have to fit the equation

$$z_{i4} = \beta_1 z_{i1} + \beta_2 z_{i2} + \beta_3 z_{i3}$$

which gives the normal equations

$$\begin{aligned}
\beta_1\sigma^2 + \beta_2 r\sigma^2 + \beta_3 r\sigma^2 &= r\sigma^2 \\
\beta_1 r\sigma^2 + \beta_2\sigma^2 + \beta_3 r\sigma^2 &= r\sigma^2 \\
\beta_1 r\sigma^2 + \beta_2 r\sigma^2 + \beta_3\sigma^2 &= r\sigma^2
\end{aligned}$$

to which the solution is

$$\beta_1 = \beta_2 = \beta_3 = \frac{r}{1+2r}.$$

The residual variance is the total variance of z_{i4} minus the sum of products of regression coefficients and right-hand sides of the normal equations. In addition, from the general regression theorem that the residual from a regression is uncorrelated with the "independent" variables that are included in the regression we know that

$e_{i2.1}$ is independent of z_{i1},
$e_{i3.12}$ is independent of z_{i1} and z_{i2} and hence of $e_{i2.1}$
$e_{i4.123}$ is independent of z_{i1}, and z_{i3} and hence of $e_{i3.12}$ and so on.

The joint distribution of the y_{ij}, $j = 1, 2, \cdots, t$ with i fixed is therefore expressible as

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{z_{i1}^2}{2\sigma^2}\right] \times \frac{1}{\sqrt{2\pi\sigma^2}\sqrt{1-r^2}} \exp\left[-\frac{1}{2}\frac{e_{i2.1}^2}{\sigma^2(1-r^2)}\right]$$
$$\times \frac{1}{\sqrt{2\pi\sigma^2}\sqrt{1-\frac{2r^2}{1+r}}} \exp\left[-\frac{e_{i3.12}^2}{2\sigma^2\left(1-\frac{2r^2}{1+r}\right)}\right] \times \cdots$$

We now apply these elementary distributional results, which are probably widely known even if not stated explicitly somewhere in the literature, to the problem at hand.

We suppose that the first records of an individual are not subject to culling* at all and that individuals are culled after the first record on any basis. The existence of a second record usually depends in some way on the value of the first record, of the third record on the value of

*Any such culling is more properly to be regarded purely as selection of incoming animals and the effects of such selection will be included in genetic parameters to be estimated.

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the first two records. We suppose that the group of individuals contributing first records in year t deviates from the population mean systematically by an amount g_t , and that records made in the kth year deviate from the population mean systematically by an amount d_k . We can therefore write down the joint distribution of the totality of records as follows. Let y_{ijkt} denote the jth record of individual i which was made in year k, this individual having entered the herd in year t. Then the distribution of the y_{ijkt} is

$$F_1 \times F_{2|1} \times F_{3|12} \times \cdots ,$$

where

$$F_1 = \prod_{(1)} \frac{1}{\sqrt{2\pi\sigma^2}} \exp -\left[\frac{(y_{i1kt} - \mu - g_t - d_k)^2}{2\sigma^2}\right],$$

$\prod_{(1)}$ denoting the product over all first records, and

$$F_{2|1} = \prod_{(2)} \frac{1}{\sqrt{2\pi\sigma^2}\sqrt{1-r^2}} \cdot \exp -\left[\frac{\{(y_{i2kt} - \mu - g_t - d_k) - r(y_{i1k't} - \mu - g_t - d_{k'})\}^2}{2\sigma^2(1-r^2)}\right],$$

$\prod_{(2)}$ denoting the product over all second records, k' denoting the year of the first record corresponding to the second record y_{ijkt} ,

and

$$F_{3|12} = \prod_{(3)} \frac{1}{\sqrt{2\pi\sigma^2}\sqrt{1-\frac{2r^2}{1+r}}} \cdot \exp\left[-\left\{(y_{i3kt} - \mu - g_t - d_k) - \frac{r}{1+r}(y_{i2k't} - \mu - g_t - d_{k'}) - \frac{r}{1+r}(y_{i1k''t} - \mu - g_t - d_{k''})\right\}^2 \Big/ 2\sigma^2\left(1 - \frac{2r^2}{1+r}\right)\right],$$

$\prod_{(3)}$ denoting the product over all third records, and k', k'' denoting the years in which the first and second records corresponding to the third record y_{i3kt} were made, and so on.

This joint distribution takes account of the fact that culling on prior records to an unknown extent was made. It incorporates, superficially at least, all the ingredients which one would like, namely

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μ, the mean in an arbitrary base population;
g_t, the amount by which the group of individuals entering the herd in year t deviate from the base population;
d_k, the amount by which the records made in year k deviate from those made in an arbitrary base year;
r, the repeatability;
σ^2, the variance of records of a population of unculled individuals.

The only regrettable feature of the joint distribution is the assumption that σ^2 does not change, though if repeatability is not high and culling and selection not intense, one would not expect the genetic part of σ^2 to change appreciably over a period of many years. We have no easy alternative to assuming the environmental component remains constant.* Without this assumption the notion of repeatability would, of course, break down, at least partially.

The logarithm of the likelihood is equal, apart from a constant, to

$$\log L = \sum_{(1)} \left[-\frac{1}{2} \log \sigma^2 - \frac{(y_{i1kt} - \mu - g_t - d_k)^2}{2\sigma^2} \right]$$

$$+ \sum_{(2)} \left[-\frac{1}{2} \log \sigma^2 - \frac{1}{2} \log (1 - r^2) \right.$$

$$\left. - \frac{\{(y_{i2kt} - \mu - g_t - d_k) - r(y_{i1k't} - \mu - g_t - d_{k'})\}^2}{2\sigma^2(1 - r^2)} \right]$$

$$+ \sum_{(3)} \left[-\frac{1}{2} \log \sigma^2 - \frac{1}{2} \log \left(1 - \frac{2r^2}{1 + r}\right) \right.$$

$$- \left\{ (y_{i3kt} - \mu - g_t - d_k) - \frac{r}{1 + r} (y_{i2k't} - \mu - g_t - d_{k'}) \right.$$

$$\left. \left. - \frac{r}{1 + r} (y_{i1k''t} - \mu - g_t - d_{k''}) \right\}^2 \Big/ 2\sigma^2 \left(1 - \frac{2r^2}{1 + r}\right) \right]$$

$$+ \text{etc.},$$

where $\sum_{(1)}$ denotes summation over all first records, $\sum_{(2)}$ summation over all second records, and so on. An alternative expression for $\log L$ is more suitable for visualizing the likelihood and the problem of maximizing this likelihood. Let N denote the total number of records available, n_1 the total number of first records, n_2 the number of second records, and so on. Also let $\sum_{(i)}$ denote summation over individuals

*The extent to which this assumption is false could be examined by fitting a more general model.

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with only a first record, $\sum_{(2)}$ summation over individuals with only first and second records, and so on. Then

$$\log L = -\frac{N}{2}\log\sigma^2 - \frac{n_2}{2}\log(1-r^2) - \frac{n_3}{2}\log\left(1-\frac{2r^2}{1+r}\right)$$
$$-\frac{n_4}{2}\log\left(1-\frac{3r^2}{1+2r}\right) - \cdots \sum_{(1)}\frac{(y_{i1k\cdot t}-\mu-g_t-d_k)^2}{2\sigma^2}$$
$$-\sum_{(2)}\left[\frac{\{(y_{i2kt}-\mu-g_t-d_k)-r(y_{i1k'\cdot t}-\mu-g_t-d_{k'\cdot})\}^2}{2\sigma^2(1-r^2)}\right.$$
$$\left.+\frac{(y_{i1k't}-\mu-g_t-d_{k'\cdot})^2}{2\sigma^2}\right]$$
$$-\sum_{(3)}\left[\left\{(y_{i3kt}-\mu-g_t-d_k)-\frac{r}{1+r}(y_{i2k'\cdot t}-\mu-g_t-d_{k'\cdot})\right.\right.$$
$$\left.\left.-\frac{r}{1+r}(y_{i1k''\cdot t}-\mu-g_t-d_{k''\cdot})\right\}^2\Big/2\sigma^2\left(1-\frac{2r^3}{1+r}\right)\right.$$
$$+\frac{\{(y_{i2k't}-\mu-g_t-d_{k'\cdot})-r(y_{i1k''\cdot t}-\mu-g_t-d_{k''\cdot})\}^2}{2\sigma^2(1-r^2)}$$
$$\left.+\frac{(y_{i1k't}-\mu-g_t-d_{k'\cdot})^2}{2\sigma^2}\right]+\cdots \text{ etc.}$$

With records covering s years we shall have $(2s+3)$ parameters to fit, of which one g and one d may be chosen arbitrarily. We may as well carry on the mathematical manipulations symmetrically in all the g's and d's, and impose necessary conditions at as late a stage as possible. We find

$$\frac{\partial \log L}{\partial g_p} = \sum_{(1)_p}\frac{(y_{i1kp}-\mu-g_p-d_k)}{\sigma^2}$$
$$+\sum_{(2)_p}\frac{(y_{i2kp}-\mu-g_p-d_k)-r(y_{i1k'p}-\mu-g_p-d_{k'\cdot})}{\sigma^2(1+r)}$$
$$+\sum_{(3)_p}\left\{(y_{i3kp}-\mu-g_p-d_k)-\frac{r}{1+r}(y_{i2k'p}-\mu-g_p-d_{k'\cdot})\right.$$
$$\left.-\frac{r}{1+r}(y_{i1k''p}-\mu-g_p-d_{k''\cdot})\right\}\Big/\sigma^2(1+2r)+\text{etc.},$$

where $\sum_{(2)_p}$ for instance means summation over all individuals of group p (i.e. entering the herd in year p) with first and second records only.

The derivative with regard to a particular d effect, d_q, say, is not as easily reduced to manageable proportions. We find

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$$\frac{\partial L}{\partial d_q} = \sum_{(q)}{}' (y_{i1q\iota} - \mu - g_\iota - d_q)/\sigma^2$$

$$+ \sum_{(qq')}{}' \{(y_{i2q\iota} - \mu - g_\iota - d_q) - r(y_{i1q'\iota} - \mu - g_\iota - d_{q'})\}/\sigma^2(1 - r^2)$$

$$- \sum_{(q'q)}{}' \{(y_{i2q'\iota} - \mu - g_\iota - d_{q'}) - r(y_{i1q\iota} - \mu - g_\iota - d_q)\}/\sigma^2(1 - r^2)$$

$$+ \sum_{(qq'q'')}{}' \left\{(y_{i3q\iota} - \mu - g_\iota - d_q) - \frac{r}{1+r}(y_{i2q'\iota} - \mu - g_\iota - d_{q'}) - \frac{r}{1+r}(y_{i1q''\iota} - \mu - g_\iota - d_{q''})\right\} \Big/ \sigma^2\left(1 - \frac{2r^2}{1+r}\right)$$

$$- \sum_{(q'qq'')}{}' \left\{(y_{i3q'\iota} - \mu - g_\iota - d_{q'}) - \frac{r}{1+r}(y_{i2q\iota} - \mu - g_\iota - d_q) - \frac{r}{1+r}(y_{i1q''\iota} - \mu - g_\iota - d_{q''})\right\} \frac{r}{1+r} \Big/ \sigma^2\left(1 - \frac{2r^2}{1+r}\right)$$

$$- \sum_{(q'q''q)}{}' \left\{(y_{i3q''\iota} - \mu - g_\iota - d_{q''}) - \frac{r}{1+r}(y_{i2q'\iota} - \mu - g_\iota - d_{q'}) - \frac{r}{1+r}(y_{i1q\iota} - \mu - g_\iota - d_q)\right\} \frac{r}{1+r} \Big/ \sigma^2\left(1 - \frac{2r^2}{1+r}\right)$$

$$+ \sum_{(qq'q''q''')}{}' \left\{(y_{i4q\iota} - \mu - g_\iota - d_q) - \frac{r}{1+2r}(y_{i3q'\iota} - \mu - g_\iota - d_{q'}) - \frac{r}{1+2r}(y_{i2q''\iota} - \mu - g_\iota - d_{q''}) - \frac{r}{1+2r}(y_{i1q'''\iota} - \mu - g_\iota - d_{q'''})\right\} \Big/ \sigma^2\left(1 - \frac{3r^2}{1+2r}\right)$$

$$- \sum_{(q'qq''q''')}{}' \left\{(y_{i4q'\iota} - \mu - g_\iota - d_{q'}) - \frac{r}{1+2r}(y_{i3q\iota} - \mu - g_\iota - d_q) - \frac{r}{1+2r}(y_{i2q''\iota} - \mu - g_\iota - d_{q''}) - \frac{r}{1+2r}(y_{i1q'''\iota} - \mu - g_\iota - d_{q'''})\right\} \frac{r}{1+2r} \Big/ \sigma^2\left(1 - \frac{3r^2}{1+2r}\right)$$

$$- \sum_{(q'q''qq''')}{}' \left\{(y_{i4q'\iota} - \mu - g_\iota - d_{q'}) - \frac{r}{1+2r}(y_{i3q''\iota} - \mu - g_\iota - d_{q''}) - \frac{r}{1+2r}(y_{i2q\iota} - \mu - g_\iota - d_q)\right.$$

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$$- \frac{r}{1+2r}(y_{i1q'''t} - \mu - g_t - d_{q'''})\Big\} \frac{r}{1+2r} \Big/ \sigma^2\left(1 - \frac{3r^2}{1+2r}\right)$$

$$- \sum_{(q'q''q'''q)}{}' \Big\{(y_{i4q't} - \mu - g_t - d_{q'}) - \frac{r}{1+2r}(y_{i2q''t} - \mu - g_t - d_{q''})$$

$$- \frac{r}{1+2r}(y_{i2q'''t} - \mu - g_t - d_{q'''})$$

$$- \frac{r}{1+2r}(y_{i1qt} - \mu - g_t - d_q)\Big\} \frac{1}{1+2r} \Big/ \sigma^2\left(1 - \frac{3r^2}{1+2r}\right) + \text{etc.}$$

where, as an example of the summation notation, $\sum'_{(q'q''qq'''')}$ means summation over the function in the brackets for those individuals whose third record is in year q, and q', q'', q'''' denote the years in which the first, second, and fourth record respectively of each such individual were made. These derivatives are indeed formidable but the method of obtaining the maximum likelihood estimates will be iterative and one will make rough estimates of the parameters which will be used to compute the first derivatives numerically. Adjustments to the rough estimates will be obtained by solution of linear equations. Alternatively if one can determine the best estimate of r, the estimates of μ, g_t , d_k are obtained as the solution of linear equations.

The derivative of $\log L$ with respect to μ is the sum of the derivatives with respect to the g's or the d's.

In addition we have

$$\frac{\partial \log L}{\partial \sigma^2} = -\frac{N}{2\sigma^2} + \sum_{(1)} (y_{i1kt} - \mu - g_t - d_k)^2/2\sigma^4$$

$$+ \sum_{(2)} \{(y_{i2kt} - \mu - g_t - d_k) - r(y_{i1k't} - \mu - g_t - d_{k'})\}^2/2\sigma^4(1 - r^2)$$

$$+ \sum_{(3)} \Big\{(y_{i3kt} - \mu - g_t - d_k) - \frac{r}{1+r}(y_{i2k't} - \mu - g_t - d_{k'})$$

$$- \frac{r}{1+r}(y_{i1k''t} - \mu - g_t - d_{k''})\Big\}^2 \Big/ 2\sigma^4\left(1 - \frac{3r^2}{1+r}\right) + \text{etc.}$$

It is easy to evaluate σ^2 by equating this derivative to zero if one has values for μ, the g's, the d's, and r. The derivative with respect to r is not as simple, for we have

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$$\frac{\partial \log L}{\partial r} = \frac{n_2 r}{(1-r^2)} + \frac{n_3 r(2+r)}{(1-r)(1+r)(1+2r)}$$

$$+ \frac{n_4 3r(2+2r)}{2(1-r)(1+2r)(1+3r)} + \frac{n_5 4r(2+3r)}{2(1-r)(1+3r)(1+4r)} + \cdots$$

$$- \sum_{(2)} \frac{(e_{i2kt} - re_{i1k't})^2}{2\sigma^2} \frac{2r}{(1-r^2)^2}$$

$$- \sum_{(3)} \frac{\left(e_{i3kt} - \frac{r}{1+r} e_{i2k't} - \frac{r}{1+r} e_{i1k''t}\right)^2}{2\sigma^2} \frac{2r(2+r)}{(1+2r)^2(1-r)^2}$$

$$- \sum_{(4)} \frac{\left(e_{i4kt} - \frac{r}{1+2r} e_{i3k't} - \frac{r}{1+2r} e_{i2k''t} - \frac{r}{1+2r} e_{i1k'''t}\right)^2}{2\sigma^2}$$

$$\cdot \frac{3r(2+2r)}{(1+3r)^2(1-r)^2} - \text{etc.}$$

$$+ \sum_{(2)} \frac{(e_{i2kt} - re_{i1k't})e_{i1k't}}{\sigma^2(1-r)(1+r)}$$

$$+ \sum_{(3)} \frac{\left(e_{i3kt} - \frac{r}{1+r} e_{i2k't} - \frac{r}{1+r} e_{i1k''t}\right)(e_{i2k't} + e_{i1k''t})}{\sigma^2(1-r)(1+r)(1+2r)}$$

$$+ \sum_{(4)} \left(e_{i4kt} - \frac{r}{1+2r} e_{i3k't} - \frac{r}{1+2r} e_{i2k''t} - \frac{r}{1+2r} e_{i1k'''t}\right)$$

$$\cdot (e_{i3k't} + e_{i2k''t} + e_{i1k'''t})/\sigma^2(1-r)(1+2r)(1+3r) + \text{etc.}$$

where $e_{ijkt} = y_{ijkt} - \mu - g_t - d_k$.

The maximum likelihood estimates for μ, g_t, d_k, r, σ^2 are the values for these quantities which make the derivatives equal to zero. The equations obtained by setting the derivatives equal to zero are non-linear in the parameters so the method of solution followed will be iterative. In a situation with estimation of parameters $\theta_1, \theta_2, \cdots, \theta_r$ a general procedure is to expand the quantities $\partial \log L/\partial\theta_i$ by a Taylor series around guessed values, ignoring terms of higher order than quadratic, giving for θ_1, for instance,

$$\left.\frac{\partial \log L}{\partial \theta_1}\right|_{\hat{\theta}} = \left.\frac{\partial \log L}{\partial \theta_1}\right|_{\theta_0} + (\hat{\theta}_1 - \theta_{10}) \left.\frac{\partial^2 \log L}{\partial \theta_1^2}\right|_{\theta_0}$$

$$+ (\hat{\theta}_2 - \theta_{20}) \left.\frac{\partial^2 \log L}{\partial \theta_1 \partial \theta_2}\right|_{\theta_0} + \cdots (\hat{\theta}_r - \theta_{r0}) \left.\frac{\partial^2 \log L}{\partial \theta_1 \partial \theta_r}\right|_{\theta_0}.$$

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In this equation $\underline{\hat{\theta}}$ is one arbitrary point $(\hat{\theta}_1, \hat{\theta}_2, \cdots, \hat{\theta}_r)$ in r-dimensional space and $\underline{\theta}_0$ or $(\theta_{10}, \theta_{20}, \cdots, \theta_{r0})$ is another arbitrary point, and all derivatives on the right-hand side are evaluated at $(\theta_{10}, \theta_{20}, \cdots, \theta_{r0})$. The maximum likelihood estimates are those for which the left-hand sides are zero, so by putting these left-hand sides equal to zero and solving for $\hat{\theta}_1 - \theta_{10}, \hat{\theta}_2 - \theta_{20}, \cdots$, we get an approximation to the maximum likelihood estimates. The closer the original $\theta_{10}, \theta_{20}, \cdots, \theta_{r0}$ are to the maximum likelihood values, the smaller will be the steps from θ_{i0} to $\hat{\theta}_i$ and when these steps are negligible we take $\hat{\theta}_i$ or θ_{i0} to be the solution.

A modification of the purely mathematical procedure for finding the maximum of a function is to use in a statistical situation the expected values of the second derivatives rather than the actual derivatives. It was in fact in these terms that the iterative method of maximum likelihood was first presented. As a general rule the authors prefer to use actual derivatives but in the present case it can be seen there are considerable advantages in using the expected values of the second derivatives.

The problem is strictly a computational one from this point on. It does not seem appropriate to pursue the matter further herein, particularly as we have had no experience with a set of data of the dimensions that ordinarily arise. We shall therefore close this section with some remarks on the situation, leaving the computational details to a later publication.

It is clear that if we possess a good estimate of r (see below, however) the problem is strictly one of weighted least squares with known relative weights. This suggests that one should first estimate r in as good a way as possible, and then use weighted least squares on linear functions of the records. One would then presumably go through at least one cycle of the iteration which would result in estimates and estimated variances and covariances of the estimates.

The extent and direction of bias resulting from estimating r without taking genetic trend into account is an important and unsolved problem. Another question lies in the propriety of applying an r value for one herd to another herd when there is no reason to suppose that r is constant.

A PARTIAL EXAMINATION OF A SET OF DATA

(KEMPTHORNE AND VON KROSIGK)

The basic ideas of the estimation procedure of Method II have been explored with the data of the Iowa Board of Control herd at Woodward for the period 1940 to 1954. The aspects we have examined are the

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consistency of the regression equations of a record on preceding records and the estimation of r.

Considering only first and second records made in successive years, we examined the homogeneity of the regressions between starting groups, which are defined as the groups of cows entering the records in each year. The analysis of variance is given in Table 1.

TABLE 1

TEST FOR HOMOGENEITY OF WITHIN STARTING YEAR REGRESSIONS

Source	d.f.	S.S.	M.S.
Among groups	14	38,210,685	
Common regression	1	257,323	
Additional due to separate regressions	13	52,514	4,040
Residual	205	1,067,441	5,207
Total	233	39,587,963	

There is no evidence at all for different regressions so we take, as the estimate from this portion of the data, r equal to 0.501 with a standard error of 0.071.

For the portion of the data with first, second, and third records in three successive years, we found that the within-group partial regression of third record on first record was $0.278 \pm .115$ and of the second record on first record was 0.501 ± 0.133. The difference of these two coefficients leads to a 't' value equal to 1.1, which in no way contradicts the hypothesis of equality of the two coefficients. The estimate forcing the two partial regression coefficients to be equal was 0.378 ± 0.071.

For the portion of the data with four records in successive years the multiple regression of fourth record on first, second, and third gave partial regression coefficients as follows:

$$\begin{aligned} \text{on first:} &\quad 0.145 \pm 0.218 \\ \text{on second:} &\quad 0.268 \pm 0.184 \\ \text{on third:} &\quad 0.228 \pm 0.140 \end{aligned}$$

The F value for differences of these was 0.07, which does not contradict the hypothesis of equality. The estimate forcing the three partial coefficients to be the same was 0.218 ± 0.066.

These three estimates are independent. They can best be represented in tabular form. Table 2 shows the estimates, the resulting estimates

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TABLE 2
ESTIMATES OF REPEATABILITY

Source	Estimate	$\hat{r}$	Weight
$\hat{r}$	0.501 ± .071	0.501 ± .071	198
$r/(1+r)$	0.378 ± .071	0.608 ± .184	29
$r/(1+2r)$	0.218 ± .066	0.387 ± .208	23

of r, and the weights. The best estimate of r from the regression coefficients is 0.503 ± .063. This estimate has been used to obtain estimates of yearly environmental trends from first and second records only. The computations are summarized in Table 3. The sum of the estimates of the yearly environmental trends gives an estimate of 31 ± 98 pounds of butterfat for the total change in environment.

TABLE 3
ESTIMATED YEARLY ENVIRONMENTAL EFFECTS

Year	n_1	n_2	$\bar{x}$	$\bar{x}_1$	$\bar{y}$	$(\bar{y}-\bar{x})-\hat{r}(\bar{x}_1-\bar{x})$
1940	26	20	360.8	363.7	422.0	59.7 ± 18.0
1941	14	11	382.7	403.6	431.6	38.4 ± 24.3
1942	22	22	449.3	449.3	428.1	−21.2 ± 17.6
1943	13	9	373.1	414.6	414.1	20.1 ± 26.7
1944	6	3	423.4	413.3	355.0	−63.3 ± 44.8
1945	13	10	352.2	344.2	335.8	−12.4 ± 25.4
1946	5	2	363.4	327.0	363.0	17.9 ± 54.2
1947	37	31	354.5	357.7	345.2	−10.9 ± 14.6
1948	21	16	385.1	411.7	388.3	−10.2 ± 20.0
1949	31	23	388.3	415.7	441.0	38.9 ± 16.8
1950	31	22	411.6	447.3	372.3	−57.3 ± 17.2
1951	28	18	361.4	389.8	375.3	−0.4 ± 18.7
1952	39	19	376.2	401.0	444.2	55.5 ± 17.8
1953	40	27	464.0	473.1	444.6	−24.0 ± 15.3

To illustrate the importance of the assumption that age effects have been eliminated suppose all of the cows in the above example freshened for the first time at two years and two months. Then if these first records were multiplied by, say, 1.25 instead of 1.28 the estimate of the total environmental change would be increased by approximately 120 pounds. Of course, if records at all ages were used any effect of age being incorrectly discounted would be damped down. However, first and second records will always comprise the major share of the data.

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If a term were added to the model for ages so that the effect of ages could be estimated simultaneously with the other factors, one would have more confidence in the estimates. However, it can be seen, as pointed out by Rendel and Robertson [1950], that if ages and starting dates were both classified with the same accuracy they would be perfectly confounded and there would not be a unique solution to the equations. It does not seem logically defensible to classify the records by years for start of lactation and by, say, months for ages in order to break down the perfect correlation between the two.

EQUIVALENCE OF THE METHODS (SEARLE)

Method II uses four subscripts on y, y_{ijkt} being the record made in year k by the ith cow of the group of cows whose first records were made in year t, its being the cow's jth record. Thus j is no more than an ordinal indicator of which record of the cow y_{ijkt} is and in terms of the elements of the model it plays no part. In the case where a cow's second record is always made in the year immediately following that of her first record, and her third record always comes in the year following that of her second, and so on, $j = k + 1 - t$. The presence of the j as a subscript to y merely emphasizes that records may not occur in concurrent years—when they do, as in many situations, the j is redundant because of the above relationship. In either case the model of Method I is applicable:

$$y_{ikt} = \mu + d_k + g_t + c_{it} + e_{ikt} .$$

In this model the random variables are the c's and e's. For the cow (i, t) having $n_{i.t}$ records it is convenient to define the following column vectors: (i) $\mathbf{y}_{it}$, her $n_{i.t}$ records, (ii) $\mathbf{g}_{it}$, a vector of $n_{i.t}$ 1's, and (iii) $\mathbf{d}_{it}$ and $\mathbf{e}_{it}$ as the d's and e's appropriate to her $n_{i.t}$ records. Then the likelihood of the sample of y's for all cows is

$$\prod_{i,t} \left\{ \frac{1}{(\sqrt{2\pi})^{n_{i.t}} \mid A_{it} \mid^{1/2}} \exp -1/2(c_{it}\mathbf{g}'_{it} + \mathbf{e}'_{it}) A_{it}^{-1}(c_{it}\mathbf{g}_{it} + \mathbf{e}_{it}) \right\}$$

where A_{it} is a square matrix of order $n_{i.t}$, all diagonal terms being $\sigma^2 = \sigma_c^2 + \sigma_e^2$, and all non-diagonal terms being $r\sigma^2$, where r is repeatability. This is the variance-covariance matrix of the $n_{i.t}$ records of the cow (i, t). Due to its special form the likelihood can be expanded and be shown to be equal to

$$L = \frac{\frac{1}{(2\pi\sigma_e^2)^{n_{...}/2}} \exp\left[-\frac{\sum_i \sum_j \sum_t e_{ijt}^2}{2\sigma_e^2} \right] \frac{1}{(2\pi\sigma_c^2)^{n_{...}/2}} \exp\left[-\frac{\sum_{it} c_{it}^2}{2\sigma_c^2} \right]}{\prod_{it} [1 + (n_{i.t} - 1)r] \prod_{it} \left(\frac{1}{2\pi\sigma_e^2(1-r)[1 + (n_{i.t} - 1)r]} \right)^{1/2} \exp -\frac{[(1-r)c_{it} - r \sum_k e_{ikt}]^2}{2\sigma_e^2(1-r)[1 + (n_{i.t} - 1)r]}}$$

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where M is the number of cows. This expression is the likelihood of the sample of y's. It is therefore the likelihood used in Method II, although it was there written as the product over all cows of the conditional likelihoods

$$F_1 F_{2|1} F_{3|2,1} \cdots$$

equal to

$$L_1(y_{i1\iota}) L_2(y_{i2\iota} \mid y_{i1\iota}) L_3(y_{i3\iota} \mid y_{i2\iota}, y_{i1\iota}) \cdots .$$

With the substitution

$$e_{ik\iota} = y_{ik\iota} - \mu - d_k - g_\iota - c_{i\iota}$$

the numerator of L as given above is the joint distribution function which has been maximized in Method I with respect to μ, the d's, g's, and the c's. In Method II, L has been maximized with respect to μ, the d's, g's, and r. Therefore the Method I equations with the following three amendments will yield the Method II equations:

(i) Include the equations arising from maximizing L with respect to r.
(ii) Subtract from the equations arising from the differentiation with respect to μ, the d's, and g's, those terms arising from differentiating the logarithm of the denominator of L with respect to these same parameters.
(iii) Delete the equations arising from maximizing L with respect to the $c_{i\iota}$'s.

In the situation where r is assumed known and is not to be estimated the equations of (i) will not occur.

As an example of the terms in (ii), consider the equation arising from differentiation with respect to μ. In Method I this will come from equating to zero the expression

$$\frac{\partial}{\partial \mu}\left\{ -\frac{\sum_i \sum_k \sum_\iota (y_{ik\iota} - \mu - d_k - g_\iota - c_{i\iota})^2}{2\sigma_e^2} \right\}$$

and this is

$$-\frac{\sum_i \sum_k \sum_\iota (y_{ik\iota} - \hat{\mu} - \hat{d}_k - \hat{g}_\iota - \hat{c}_{i\iota})(-1)}{(1-r)\sigma_e^2} .$$

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In Method II there will be added to this expression the following

$$\frac{\partial}{\partial \mu}\left\{\sum_{ij}\frac{[(1-r)c_{ij}-r\sum_{k}(y_{ikt}-\mu-d_k-g_t-c_{ij})]^2}{2\sigma_e^2(1-r)[1+(n_{i.t}-1)r]}\right\}$$

which equals

$$\sum_{ij}\frac{n_{i.t}r\{1+(n_{i.t}-1)c_{ij}-r[y_{i.t}-n_{i.t}(\hat{\mu}+\hat{g}_t)-\sum_{k}n_{ikt}\hat{d}_k]\}}{r(1-r)\sigma^2[1+(n_{i.t}-1)r]}.$$

In the difference the terms in $\hat{c}_{ij}$ are

$$\sum_{ij}\frac{\hat{c}_{ij}}{(1-r)\sigma^2}(-n_{i.t}+n_{i.t})\equiv 0,$$

i.e. the $\hat{c}$'s are eliminated from the equation; and this will be found to be true for the equations corresponding to the differentiations with respect to the d's and the g's. Thus together with (iii) we see that the equations of Method II are those of Method I after the $\hat{c}$'s have been eliminated, namely equations (6) and (7) of Method I.

Hence we have shown that when the repeatability r is assumed known, the two methods are equivalent. That is, maximizing the likelihood with respect to μ, the d_k's, and the g_t's (as in Method II), gives the same equations as result from eliminating the $\hat{c}_{ij}$'s from the equations obtained by maximizing the joint distribution function of the y's and c's, with respect to μ, the d_k's, the g_t's, and the c_{ij}'s (as in Method I). Thus when r is assumed known, the equations of Method II are simply those of Method I with the $\hat{c}$'s eliminated; and hence the two methods give the same estimates of estimable linear functions of the fixed effects μ, d_k, and g_t. Furthermore, since it has been proved in Method II that these estimates are unbiased by selection, such unbiasedness also applies to Method I.

SUMMARY

Two methods for the separation of genetic and environmental time trends are presented and compared. The reason the classical least squares approach to this problem yields biased estimates is described in standard statistical terminology. The first method is illustrated with a small numerical example and a partial examination of a set of data is used to illustrate some of the assumptions and the estimation of parameters in Method II.

The methodology is presented and discussed in terms of dairy records. However, the same technique would apply to many other cases, often without the complication of age-correction factors.

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Willham (1963)

Willham, R.L. 1963. The covariance between relatives for characters composed of components contributed by related individuals. Biometrics, vol. 19, no. 1, pp. 18 to 27.

The problem

The weaning weight of a calf depends on its own genes, but it also depends on the maternal ability of the cow, which in turn has a genetic basis. Willham showed how to write the covariance between relatives when the phenotype is made up of a direct genetic effect and a maternal genetic effect, including a covariance between the two.

Points to note while reading

This is the model underlying the evaluation of traits with maternal effects, such as birth and weaning weights in beef cattle. Note how the covariance between relatives gains new terms when the dam contributes to the phenotype of her offspring, and why this complicates the estimation of variance components.

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THE COVARIANCE BETWEEN RELATIVES FOR CHARACTERS COMPOSED OF COMPONENTS CONTRIBUTED BY RELATED INDIVIDUALS[1]

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SUMMARY

For some species in which family members are dependent on or are in close proximity to one another, a character may be composed of several conceptual components, some being contributed by related individuals. A general expression for the genotypic covariance between relatives is synthesized for such characters when mating is random. The relationships between individuals contributing to a measure of the character are important in defining the genotypic variance and the response to selection. A character of an offspring influenced by a maternal effect illustrates the expression.

INTRODUCTION

Sometimes a character is divisible into meaningful component characters as was done by Falconer [1960] for litter size in mice. Each component character is conceivably influenced by a genotypic value and an environmental value. For species in which family members depend on or are in close proximity to one another, the component characters may be contributed by related individuals. Growth during the suckling period in mammals is affected both by the offspring whose growth is measured and by his dam who furnishes the developmental environment during the period. The value observed when the character, growth, is measured on an offspring is the phenotypic value of that offspring. But the character is composed of at least two distinct component characters; offspring growth and a maternal effect contributed by a related individual, the dam. While the maternal contribution is strictly environmental relative to the offspring, genotypic differences among dams for the maternal effect are expressed in their offspring.

This paper describes the genotypic variance when a character is the

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sum of several component characters contributed by related individuals. This is done by synthesizing from existing theory an expression for the covariance between relatives. The covariance is expressed in terms of additive, dominance, and epistatic components of genotypic variance and covariance for the component characters and the relationships between the individuals involved when mating is random. The phenotypic value of an offspring that is influenced by a maternal effect illustrates the expression. This example was chosen since the present generalizations evolved from works of Dickerson [1947], Koch and Clark [1955] and Kempthorne [1955]. These authors examined the composition of the covariance between specific relatives when a maternal effect was considered.

THEORY

Let P_x represent the phenotypic value of individual X. For simplicity assume P_x is the sum of two component characters; one being influenced by the genotypic value of X and the other by the genotypic value of an individual related to X say W. Denote the two components of the character as o and m symbolizing the offspring component and the maternal effect. Let

$$P_x = G_{ox} + E_{ox} + G_{mw} + E_{mw}$$

where G_{ox} and E_{ox} are the genotypic and environmental values respectively of individual X for component o, and the terms G_{mw} and E_{mw} are the genotypic and environmental values of individual W for component m as expressed in P_x. The latter two terms are defined in terms of the variance of P_x that is causally attributable to them. Fig. 1 illustrates such a model in terms of a path coefficient diagram when W is the dam of X and the (G')'s denote additive genetic values rather than genotypic values.

The genotypic variance of such phenotypic values is best examined by synthesizing a general expression for the genotypic covariance between relatives, since the variance depends on the relationship between the individuals contributing to the phenotypic value. The genotypic covariance between P_x and P_y (where for P_y individuals Y and Z contribute components o and m respectively) for any relationship between X and Y is

$$\begin{aligned}\mathrm{Cov}\,(G_{ox}\,,\,G_{oy}) + \mathrm{Cov}\,(G_{ox}\,,\,G_{mz}) \\ + \mathrm{Cov}\,(G_{mw}\,,\,G_{oy}) + \mathrm{Cov}\,(G_{mw}\,,\,G_{mz}).\end{aligned}$$

Each covariance can be evaluated in terms of either the genotypic variance for the particular component or the genotypic covariance be-

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tween two components and the relationship between the two individuals involved. Thus, the first covariance between X and Y for component o is given by Kempthorne [1957] as

$$2p_{xy}\sigma^2_{A_o} + u_{xy}\sigma^2_{D_o} + \sum_{r,s} (2p_{xy})^r (u_{xy})^s \sigma^2_{(A^rD^s)_o} \qquad 2 \leq r + s \leq N$$

where $\sigma^2_{A_o}$, $\sigma^2_{D_o}$, and $\sigma^2_{(A^rD^s)_o}$ are the additive, dominance, and epistatic components of genotypic variance for component o. For the epistatic terms, r and s refer to the number of loci involved in the interaction and N refers to the number of loci segregating for component o. The coefficient, $2p_{xy}$, is Wright's coefficient of relationship with no inbreeding or twice Malecot's "*coefficient de parenté*". The coefficient, u_{xy} , represents the probability that the two genes at a locus in individual X are identical by descent with the two genes at that locus in individual Y. The value of u_{xy} is zero unless the two individuals are related by two lines of descent such as full sibs and double first cousins.

The second covariance, Cov $(G_{ox}$, $G_{mz})$, is the genotypic covariance between X and Z for component o expressed in P_x and component m expressed in P_y . Mode and Robinson [1959] indicated that the genotypic covariance between characters or in this case components of a character can be partitioned in an analogous manner to the genotypic variance. Thus, the Cov $(G_{ox}$, $G_{mz})$ is

$$2p_{xz}\sigma_{A_oA_m} + u_{xz}\sigma_{D_oD_m} + \sum_{r,s} (2p_{xz})^r (u_{xz})^s \sigma_{(A^rD^s)_o(A^rD^s)_m} \qquad 2 \leq r + s \leq M$$

where $\sigma_{A_oA_m}$, $\sigma_{D_oD_m}$, and $\sigma_{(A^rD^s)_o(A^rD^s)_m}$ represent the additive, dominance, and epistatic components of genotypic covariance between o and m respectively. M represents the number of loci segregating that affect both o and m.

Any number of such covariances can be evaluated and the results summed to give the total genotypic covariance between the phenotypic value of two related individuals. Thus the genotypic covariance between P_x and P_y when characters o and m are considered is

$$\begin{aligned} &\sum_{r,s} (2p_{xy})^r (u_{xy})^s \sigma^2_{(A^rD^s)_o} \\ &+ \sum_{r,s} (2p_{xz})^r (u_{xz})^s \sigma_{(A^rD^s)_o(A^rD^s)_m} \\ &+ \sum_{r,s} (2p_{wy})^r (u_{wy})^s \sigma_{(A^rD^s)_o(A^rD^s)_m} \\ &+ \sum_{r,s} (2p_{wz})^r (u_{wz})^s \sigma^2_{(A^rD^s)_m} \end{aligned} \qquad (1)$$

where $r + s$ goes from 1 to the number of loci segregating. The two genotypic covariance terms can be added. But the two terms are

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subsequently given separately since the two relationships, X with Z and Y with W, may differ. For additive genetic effects only, this covariance is

$$2p_{xy}\sigma^2_{A_o} + [2p_{xz} + 2p_{wy}]\sigma_{A_oA_m} + 2p_{wz}\sigma^2_{A_m} .$$

This result for specific relatives when maternal effects were considered was given by Dickerson [1947] and Koch and Clark [1955] while Kempthorne [1955] included dominance in a one locus case.

If there are k components of a character contributing to a phenotypic value there will be k variance terms and $k(k-1)/2$ different covariance terms in (1). The covariance (1) depends on four relationships rather than the usual one between X and Y when one character is considered. If there are k components of a character and each is contributed by a different individual, the covariance between the phenotypic values of two individuals depends on k^2 relationships. These relationships are in part responsible for the interesting properties of the problem.

APPLICATION

Some properties of expression (1) are general for such characters while other properties depend on the specific problem. The application of expression (1) to the problem of a maternal effect is given in Table 1. The compositions of the covariances between several common relatives appear in terms of additive and dominance components of genotypic variance and covariance. The covariances can be extended to include epistasis using expression (1) since the coefficients of the variances and covariances appearing in the body of the table are the values of $2p$ and u for the four relationships involved in each of the covariances between relatives. For example, when X and Y are offspring and dam as in row three of Table 1; $2p_{xy} = \frac{1}{2}$, $2p_{xz} = \frac{1}{4}$ since Z is the granddam of X, $2p_{wy} = u_{wy} = 1$ since W and Y are the same individual, and $2p_{yz} = \frac{1}{2}$ since Z is the dam of W. Using these values and (1) the coefficient for $\sigma_{A_oA_m}$ is $[(1)^2 + (\frac{1}{2})^2] = 17/16$.

The genotypic components of covariance between o and m in the total genotypic variance of P_x (row one of Table 1) depend on the relationship between X and W which are the individuals contributing to the phenotypic value. When X and W are offspring and dam as for a maternal effect, only additive genetic and additive epistatic components of covariance appear in the total genotypic variance of P_x. This results because the value of u between offspring and dam is zero and thus there is no covariance between dominance deviations or epistatic terms containing dominance.

The fraction of the selection differential that would be realized if

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TABLE 1

THE COMPOSITION OF SEVERAL GENOTYPIC COVARIANCES BETWEEN RELATIVES WHEN MATERNAL EFFECTS ARE CONSIDERED. THE FIGURES IN THE TABLE ARE THE COEFFICIENTS OF THE COMPONENTS OF COVARIANCE GIVEN AT THE HEADS OF THE COLUMNS.

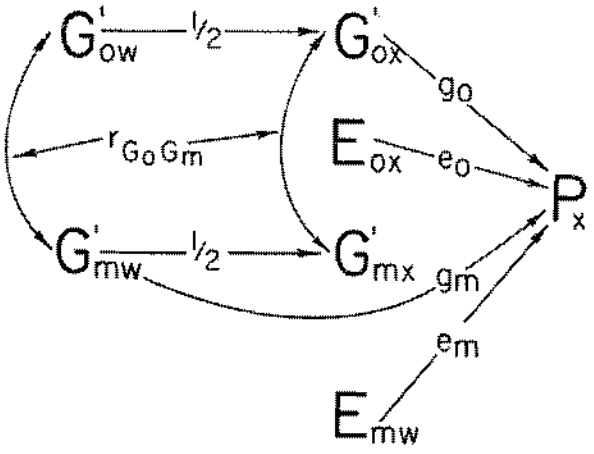
Covariances	*Components*					
	$\sigma^2_{A_o}$	$\sigma^2_{D_o}$	$\sigma_{A_oA_m}$	$\sigma_{D_oD_m}$	$\sigma^2_{A_m}$	$\sigma^2_{D_m}$
Genotypic variance of P_x	1	1	1/2 + 1/2	0 + 0	1	1
Cov $(G'_{ox} + G'_{mx}, P_x)$	1	0	1 + 1/2	0 + 0	1/2	0
Dam-Offspring	1/2	0	1 + 1/4	1 + 0	1/2	0
Sire-Offspring	1/2	0	0 + 1/4	0 + 0	0	0
Full-Sib	1/2	1/4	1/2 + 1/2	0 + 0	1	1
Paternal-Half-Sib	1/4	0	0 + 0	0 + 0	0	0
Maternal-Half-Sib	1/4	0	1/2 + 1/2	0 + 0	1	1
Double-First-Cousins						
Sires Full-Sibs and Dams Full-Sibs	1/4	1/8	1/4 + 1/4	0 + 0	1/2	1/4
Both opposite sexes are Full-Sibs	1/4	1/8	1/4 + 1/4	0 + 0	0	0
Single-First-Cousins						
Sires Full-Sibs	1/8	0	0 + 0	0 + 0	0	0
Dams Full-Sibs	1/8	0	1/4 + 1/4	0 + 0	1/2	1/4
A Sire and a Dam are Full-Sibs	1/8	0	1/4 + 0	0 + 0	0	0

selection were on P_x was defined by Dickerson [1947] as the regression of the transmitting ability (additive genetic value) of X on the phenotype of X. This regression is shown in terms of path coefficients for a maternal effect in Figure 1. Although an individual has but one genotype, Figure 1 clearly shows that individual X has two additive genetic values (G'), one for component o that is expressed in P_x and one for component m that will be expressed only if X subsequently becomes a dam. The numerator of this regression is the covariance between the sum of the additive genetic values of X and the phenotype of X. Using expression (1) the Cov $(G'_{ox} + G'_{mx}, P_x)$ is

$$\sigma^2_{A_o} + [1 + 2p_{ox}]\sigma_{A_oA_m} + 2p_{ox}\sigma^2_{A_m}$$

where $2p_{ox} = 1$. Here again the relationships between the individuals contributing to the phenotypic value are important. Components of the character such as m contributed by individuals other than X receive

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$$\beta(G'_{ox} + G'_{mx})P_x = g_o^2 + 3/2\, g_o\, r_{G_oG_m}\, g_m + 1/2\, g_m^2$$

FIGURE 1

A PATH COEFFICIENT DIAGRAM DESCRIBING A PHENOTYPIC VALUE INFLUENCED BY A MATERNAL EFFECT (AFTER DICKERSON 1947)

attention relative to the contributor's relationship with X when selection is based on P_x. The sign of the covariance is important in determining the magnitude of this covariance.

A few of the problems involved in the separation of additive genetic or genotypic variance and covariance from the other sources of variance using covariances between relatives will be sketched for maternal effects with the aid of Table 1. The covariances can be roughly divided into two groups. The first contains covariances based on the transmitted effects of males. The other contains those based on both the direct (maternal) and transmitted effects of females. Comparable covariances of the first group have been used to separate the transmitted from the direct effects found in the second group. Several collateral relationships have identical composition with no maternal effect but differ when a maternal effect is included. An example in Table 1 is that of maternal and paternal half sibs. Also the difference between the covariance between full sibs and twice the covariance between paternal half sibs has been extensively used to indicate the variance attributable to a common litter environment. Dickerson [1947] used the fact that this difference was negative to suggest an antagonism between components o and m for some traits of swine. Differences between such covariances include not only functions of the maternal variance but

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also the covariance between o and m. The sign of this covariance may obscure the magnitude of the maternal variance. For the most common collateral relatives the coefficients of the additive genetic covariance and the additive genetic maternal variance are identical making separation even more difficult. Full and maternal sibs have the same dam, consequently all the genotypic variance and at least part of the environmental variance for the maternal effect appear in the covariances. Using relationships among individuals whose dams are related such as maternal cousins reduces the environmental contribution to the maternal variance. When a sire of one cousin is the full sib of the other's dam, covariance between o and m is present but not the maternal variance (see Table 1). Cousin relationships thus appear well suited to the study of genetic maternal performance and have been used to some extent.

The composition of parent-offspring covariances differ for the parental sex. The dam-offspring covariance contains the covariance between the dominance deviations of o and m which appear in no other covariance nor the variance, as was noted by Kempthorne [1955]. In general, when X and Y bear the same relationship to each other as X and W which are the individuals contributing to P_c, the covariance between X and Y is likely to be the most complex. The dam-offspring covariance can be negative when $\sigma_{A_oA_m}$ is negative due to the large coefficient for the covariance term, $(1 + \frac{1}{4})$. Summing the sire-offspring and dam-offspring covariances gives some idea of the covariance between additive genetic values of X and the phenotypic value of X. Koch and Clark [1955] used this sum to predict selection response for characters in beef cattle. Since for such characters more than one relationship is involved in most covariances between relatives, dividing by a particular relationship (say $\frac{1}{4}$ for maternal half-sibs) to measure a portion of the genotypic variance cannot be done.

Table 2 gives four hypothetical examples to illustrate numerically how the correlations between the common relatives are affected by a maternal effect. The values taken for the parameters not involving the maternal effect are the same in all four examples ($\sigma^2_{A_o} = 32$, $\sigma^2_{D_o} = 4$, $\sigma^2_E = 160$, and $\sigma_{E_oE_m} = 0$). All components for $(r + s) > 2$ have been set equal to zero. Example I has no maternal effect and is given for comparison. Example II has a maternal effect without a correlation between o and m ($\sigma^2_{A_m} = 2$, $\sigma^2_{D_m} = 1$, $\sigma^2_{E_m} = 20$ and $\sigma_{A_oA_m} = \sigma_{D_oD_m} = 0$). Example III is like II except there is a correlation of $+1$ between o and m ($\sigma_{A_oA_m} = 8$ and $\sigma_{D_oD_m} = 2$). Example IV is like II except there is correlation of -1 between o and m. Maternal variance is roughly one-eighth the importance of the offspring variance, but in example III and

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TABLE 2
THE CORRELATION BETWEEN RELATIVES
FOR FOUR EXAMPLES (SEE TEXT)

Relationship	*Examples* I	II	III	IV
Variance of P_x	196	219	227	211
Regression of $(G'_{ox} + G'_{mx})$ on P_x	.164	.151	.198	.100
Dam-Offspring	.082	.078	.128	.024[a]
Sire-Offspring	.082	.073	.079	.066
Full-Sib[b]	.087	.183	.211	.152
Paternal-Half-Sib	.041	.037	.035	.038
Maternal-Half-Sib[c]	.041	.096	.128	.062
Double-First-Cousin	.042	.043	.059	.026

[a] $.024 = \frac{5}{211} = \frac{1/2\ (32) + 5/4\ (-8) + (-2) + 1/2\ (2)}{32 + 4 + 160 - 8 + 2 + 1 + 20}$

[b] Full-sibs contain all of $\sigma^2_{E_m}$ approximating the common environment of full-sibs especially for species having litters.

[c] Maternal-half-sibs contain $1/2\sigma^2_{E_m}$ approximating the permanent environmental effects common to offspring of a dam.

IV the genotypic correlation between o and m is high. Although untrustworthy for broad generalizations these examples numerically illustrate the covariances in Table 1. The examples fail to include the complex possibilities of environmental correlations that might exist in actual data.

DISCUSSION

The goal of most quantitative genetic studies is to describe the kind and relative amount of genotypic variance for the character examined. Practical problems center around predicting progress from selection. Subdivision of a character into components can help to elucidate the genetics involved. But the possible complexity of the genotypic variance when a character is composed of components is evident from expression (1) and from the example of a maternal effect. The genotypic variance of P_x or the covariance of an individual with himself, for two components setting all terms for $(r + s) > 2$ equal to zero, is

$$\sigma^2_{A_o} + [2p_{xm} + 2p_{wx}]\sigma_{A_oA_m} + \sigma^2_{A_m}$$

where $2p_{xx} = 2p_{ww} = 1$. Description of even additive genetic variances and covariances prove difficult with only two components. This

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results because the components are measured jointly and usually only a few different sorts of relationship are included in a set of data. When the components are uncorrelated and one is of primary interest, the other may be considered as a random environmental effect. However, when the components are correlated the phenotypic value should be considered as a joint expression. Also problems of environmental covariances between relatives exist and may be of different magnitude depending on management of even the same organism.

Phenotypic values composed of components contributed by ancestral relations such as maternal effects deserve special attention. Possibly a maternal effect is influenced by a maternal effect. Then the granddam of the offspring would exert a direct effect through the dam on the phenotypic value of her daughter's offspring. This is plausible if early environment of the dam influences her subsequent performance as a mother. The general consequence of such direct ancestral effects on selective improvement would be to increase the inertia of the population. That is, the population would resist change since the present level of the population depends on several generations simultaneously. Further, negative correlations among the characters would reduce the phenotypic variance and tend to hold a population close to a particular value, perhaps an optimum.

Several single locus examples of maternal effects suggest interaction between the genotype of dam and offspring. The second degree statistics become difficult since not all combinations of dam and offspring genotypes are possible. That is, AA females produce no aa offspring. This introduces correlations among interaction terms involving allelic genes possessed by different but related individuals.

Numerous phenotypic values are recognized as being composed of several component characters. For example, phenotypic values of family size require at least three as indicated by Falconer [1960] for litter size. Certainly in humans, family effects are real for some measurements. The expression of the offspring genotype depends on the home environment which has a genetic base in the parents, grandparents, collateral relatives and in fact the whole family heritage. Possibly dividing characters into components attributable to different causes and examining the variation will aid in interpreting the genotypic variance of the more complex quantitative characters.

ACKNOWLEDGMENTS

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Patterson and Thompson (1971)

Patterson, H.D., and Thompson, R. 1971. Recovery of inter-block information when block sizes are unequal. Biometrika, vol. 58, no. 3, pp. 545 to 554.

The problem

Ordinary maximum likelihood underestimates variances because it does not account for the degrees of freedom used to estimate the fixed effects. Working with field trials in incomplete blocks, Patterson and Thompson proposed maximizing the likelihood of only the part of the data that does not depend on the fixed effects. This gave rise to restricted maximum likelihood, or REML.

Points to note while reading

The context of the paper is agricultural experimentation with plants, and the connection with animal breeding is not obvious on first reading. Note the idea of working with contrasts of the data that are free of fixed effects. Over time, REML became the standard method for estimating variance components in animal models.

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Printed in Great Britain

Recovery of inter-block information when block sizes are unequal

BY H. D. PATTERSON AND R. THOMPSON
University of Edinburgh

SUMMARY

A method is proposed for estimating intra-block and inter-block weights in the analysis of incomplete block designs with block sizes not necessarily equal. The method consists of maximizing the likelihood, not of all the data, but of a set of selected error contrasts. When block sizes are equal results are identical with those obtained by the method of Nelder (1968) for generally balanced designs. Although mainly concerned with incomplete block designs the paper also gives in outline an extension of the modified maximum likelihood procedure to designs with a more complicated block structure.

1. INTRODUCTION

In this paper we consider the estimation of weights to be used in the recovery of inter-block information in incomplete block designs with possibly unequal block sizes. The problem can also be thought of as one of estimating constants and components of variance from data arranged in a general two-way classification when the effects of one classification are regarded as fixed and the effects of the second classification are regarded as random.

Nelder (1968) described the efficient estimation of weights in generally balanced designs, in which the blocks are usually, although not always, of equal size. Lack of balance resulting from unequal block sizes is, however, common in some experimental work, for example in animal breeding experiments. The maximum likelihood procedure described by Hartley & Rao (1967) can be used but does not give the same estimates as Nelder's method in the balanced case.

As will be shown, the two methods in effect use the same weighted sums of squares of residuals but assign different expectations. In the maximum likelihood approach, expectations are taken over a conditional distribution with the treatment effects fixed at their estimated values. In contrast Nelder uses unconditional expectations.

The difference between the two methods is analogous to the well-known difference between two methods of estimating the variance σ^2 of a normal distribution, given a random sample of n values. Both methods use the same total sum of squares of deviations. But whereas one method equates the sum of squares to $(n-1)\sigma^2$ the other equates the sum of squares to $n\sigma^2$. The former method gives an unbiased estimate of σ^2; the latter maximizes the likelihood of the sample.

Another method for unbalanced designs has been described by Cunningham & Henderson (1968). When corrected as described by Thompson (1969) this method allows for errors in the estimation of treatment effects but in general the estimates are not efficient.

The method proposed in the present paper is a modified maximum likelihood procedure, more efficient than the Cunningham and Henderson method and giving the same results as Nelder's method in the analysis of balanced designs. The contrasts among yields are divided into two sets: (i) contrasts between treatment totals; and (ii) contrasts with zero

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expectation, i.e. error contrasts. The method consists of maximizing the joint likelihood of all possible contrasts in set (ii). Contrasts in set (i) are excluded from the likelihood function on the grounds that, as long as treatment effects are regarded as unknown, fixed, as opposed to random, and without restraints, no contrast in set (i) can provide any information on error.

The modified maximum likelihood method was considered by Patterson (1964) in a components-of-variance problem arising in the analysis of rotation experiments. This paper was, however, primarily concerned with evaluating the efficiencies of simpler methods and gave no details beyond formulae for asymptotic variances.

2. The model

We suppose that the incomplete block design has t treatments and n units, plots, in b blocks of possibly unequal size and that the $n \times 1$ vector of yields, $\mathbf{y}$, can be represented by the model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\alpha} + \boldsymbol{\epsilon}, \tag{1}$$

where $\mathbf{X}$ is an $n \times t$ matrix of rank t determined by the allocation of treatments to units, $\boldsymbol{\alpha}$ is a $t \times 1$ vector of treatment parameters and $\boldsymbol{\epsilon}$ is a random variable normally distributed with mean zero and variance given by

$$\mathbf{V} = \mathbf{H}\sigma^2, \quad \mathbf{H} = \mathbf{Z}\boldsymbol{\Gamma}\mathbf{Z}' + \mathbf{I}. \tag{2}$$

Further $\boldsymbol{\Gamma} = \gamma\mathbf{I}$, $\mathbf{Z}$ is an $n \times b$ matrix with elements Z_{ij} equal to 1 when unit i is in block j ($i = 1, \ldots, n$; $j = 1, \ldots, b$) and equal to 0 otherwise, and γ and σ^2 are unknown scalars. A model with more general $\mathbf{Z}$ and $\boldsymbol{\Gamma}$ is considered in §10.

The problem is to estimate $\boldsymbol{\alpha}$, γ and σ^2. Sometimes reparameterization may be possible. Provided γ is not negative, $\gamma\sigma^2$ and σ^2 can be regarded as components of variance σ_b^2 and σ_e^2, say. When blocks sizes are all equal to k we may require to estimate functions

$$V_1 = (k\gamma + 1)\sigma^2, \quad V_2 = \sigma^2.$$

This is, in fact, what Nelder (1968) does.

The matrix $\mathbf{H}$ is essentially nonsingular. The inverse $\mathbf{H}^{-1}$ can be written in the form

$$\mathbf{H}^{-1} = \mathbf{I} - \mathbf{Z}(\mathbf{Z}'\mathbf{Z} + \boldsymbol{\Gamma}^{-1})^{-1}\mathbf{Z}'. \tag{3}$$

Thus $\mathbf{H}^{-1}$ exists if $(\mathbf{Z}'\mathbf{Z} + \boldsymbol{\Gamma}^{-1})^{-1}$ exists; $\mathbf{Z}'\mathbf{Z}$ is diagonal with elements k_j, where k_j is the number of plots in block j. Clearly $\mathbf{Z}'\mathbf{Z} + \boldsymbol{\Gamma}^{-1}$ is singular if $\gamma = -k_j^{-1}$ for some j. But this would imply that the variance of the mean for block j is zero. A smaller value of γ implies a negative variance. We can, therefore, reasonably impose the condition that $\gamma > -1/k_{\max}$, where $k_{\max}$ is the largest number of plots in a single block.

Other conditions must also be satisfied for estimation of γ and σ^2 to be possible. These conditions will be considered in § 6.

Error model (2) also implies that the correlation between two units in the same block is independent of block size. This model is often used in animal experiments with blocks consisting of genetically related animals; usually the relationships within a block can be assumed to be independent of the number of animals in the block. In some other applications, for example field plot experiments, it might be more appropriate to specify a correlation that varies from block to block with the largest blocks showing the smallest correlations.

Estimates of $\boldsymbol{\alpha}$, γ and σ^2 will be denoted by $\hat{\boldsymbol{\alpha}}$, $\hat{\gamma}$ and $\hat{\sigma}^2$. We will also use the circumflex to denote functions of $\hat{\gamma}$, for example $\hat{\boldsymbol{\Gamma}} = \hat{\gamma}\mathbf{I}$ and $\hat{\mathbf{H}} = \mathbf{Z}\hat{\boldsymbol{\Gamma}}\mathbf{Z}' + \mathbf{I}$.

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3. Modified maximum likelihood method

The logarithm of the likelihood function of $\mathbf{y}$ is given by

$$L = \text{const} - \tfrac{1}{2}\log|\mathbf{H}| - \tfrac{1}{2}n\log\sigma^2 - \frac{1}{2\sigma^2}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha})'\mathbf{H}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha}). \tag{4}$$

Hartley & Rao (1967) estimate $\boldsymbol{\alpha}$, γ and σ^2 by maximizing L. In the present paper we divide the data into two parts, with separate logarithmic likelihoods L' and L'', estimate γ and σ^2 by maximizing L' and estimate $\boldsymbol{\alpha}$ by maximizing L''.

The two parts can be represented by transformed yields $\mathbf{Sy}$ and $\mathbf{Qy}$ with the following properties.

(i) The matrix $\mathbf{S}$ is of rank $n-t$ and $\mathbf{Q}$ is a matrix of rank t.

(ii) The two parts are statistically independent, i.e. $\text{cov}(\mathbf{Sy}, \mathbf{Qy}) = 0$.
This condition is met if

$$\mathbf{SHQ}' = 0, \tag{5}$$

(iii) The matrix $\mathbf{S}$ is chosen so that

$$E(\mathbf{Sy}) = 0, \quad \text{i.e. } \mathbf{SX} = 0. \tag{6}$$

(iv) The matrix $\mathbf{QX}$ is of rank t, so that every linear function of the elements of $\mathbf{Qy}$ estimates a linear function of the elements of $\boldsymbol{\alpha}$.

It follows from (i) and (ii) that the likelihood of $\mathbf{y}$ is the product of the likelihoods of $\mathbf{Sy}$ and $\mathbf{Qy}$, i.e.

$$L = L' + L''. \tag{7}$$

Suitable matrices $\mathbf{S}$ and $\mathbf{Q}$ are given by

$$\mathbf{S} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \tag{8}$$

$$\mathbf{Q} = \mathbf{X}'\mathbf{H}^{-1}. \tag{9}$$

The matrix $\mathbf{S}$ is symmetric, idempotent, of rank $n-t$ and independent of γ. The elements of $\mathbf{Sy}$ are deviations of yields from treatment means. An estimate of γ is required for the transformation $\mathbf{Qy}$.

4. Estimation of γ and σ^2

First, we estimate γ and σ^2 by maximizing L', the logarithmic likelihood of $\mathbf{Sy}$. The variance matrix $\mathbf{SHS}\sigma^2$ is singular but suitable expressions for L' are available in terms of either a generalized inverse $(\mathbf{SHS})^{-g}$ or the latent roots of $\mathbf{SHS}$.

In deriving these expressions we use an $n \times (n-t)$ matrix $\mathbf{P}$ whose columns are orthogonal vectors of both $\mathbf{S}$ and $\mathbf{SHS}$. As $\mathbf{S}$ is idempotent and symmetric, it can be expressed in the form $\mathbf{AA}'$, where $\mathbf{A}$ is an $n \times (n-t)$ matrix such that $\mathbf{A}'\mathbf{A} = \mathbf{I}$. Now let $\mathbf{B}$ be an orthogonal matrix such that $\mathbf{B}'\mathbf{A}'\mathbf{HAB}$ is diagonal.

The required matrix $\mathbf{P}$ is given by $\mathbf{AB}$. It has the following properties: (i) $\mathbf{P}'\mathbf{P} = \mathbf{I}$, (ii) $\mathbf{PP}' = \mathbf{S}$ and (iii) $\mathbf{P}'\mathbf{HP} = \text{diag}(\xi_s)$, a diagonal matrix with elements ξ_s $(s = 1, \ldots, n-t)$. We note from (i) and (ii) that $\mathbf{P}'\mathbf{S} = \mathbf{P}'$. Hence $\mathbf{P}'\mathbf{y}$ can be derived from $\mathbf{Sy}$ by the transformation $\mathbf{P}'(\mathbf{Sy})$. Also $\mathbf{Sy}$ can be derived from $\mathbf{P}'\mathbf{y}$ by the transformation $\mathbf{P}(\mathbf{P}'\mathbf{y})$. It follows that the likelihood of $\mathbf{Sy}$ is also the likelihood of $\mathbf{P}'\mathbf{y}$.

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Property (iii) shows that

$$\mathbf{P}'\mathbf{HS} = \text{diag}\,(\xi_s)\,\mathbf{P}', \tag{10}$$

i.e. the ξ_s are the nonzero latent roots of **HS**. We can now write **SHS** in the spectral form

$$\mathbf{SHS} = \mathbf{P}\,\text{diag}\,(\xi_s)\,\mathbf{P}' \tag{11}$$

and define a generalized inverse $(\mathbf{SHS})^{-g}$ such that

$$(\mathbf{SHS})^{-g} = \mathbf{P}\,\text{diag}\,(\xi_s^{-1})\,\mathbf{P}'. \tag{12}$$

The product of $(\mathbf{SHS})^{-g}$ with **SHS** is **S**. It is also worth noting that $(\mathbf{SHS})^{-g}$ is unchanged on multiplication by **S**.

The elements of $\mathbf{P}'\mathbf{y}$ are contrasts $u_s\,(s = 1, \ldots, n-t)$ with variances $\xi_s\sigma^2$. Hence the required log likelihood function is given by

$$L' = \text{const} - \tfrac{1}{2}\sum_s \log \xi_s - \tfrac{1}{2}(n-t)\log\sigma^2 - R/(2\sigma^2), \tag{13}$$

where R is the weighted sum of squares of the u_s given by

$$R = \Sigma\,(u_s^2/\xi_s) = \mathbf{y}'(\mathbf{SHS})^{-g}\,\mathbf{y}. \tag{14}$$

The estimates $\hat{\gamma}$ and $\hat{\sigma}^2$ maximizing L' are obtained by solving the equations

$$\frac{\partial L'}{\partial \gamma} = -\tfrac{1}{2}E + \frac{B}{2\sigma^2} = 0, \tag{15}$$

$$\frac{\partial L'}{\partial \sigma^2} = -\frac{1}{2}\frac{n-t}{\sigma^2} + \frac{R}{2\sigma^4} = 0, \tag{16}$$

where

$$B = \sum_s u_s^2 d_s/\xi_s^2 = \mathbf{y}'(\mathbf{SHS})^{-g}\,\mathbf{SDS}(\mathbf{SHS})^{-g}\,\mathbf{y}, \tag{17}$$

$$E = \sum_s (d_s/\xi_s) = \text{tr}\,\{(\mathbf{SHS})^{-g}\,\mathbf{SDS}\}. \tag{18}$$

Further d_s denotes $\partial\xi_s/\partial\gamma$ and **D** denotes $\partial\mathbf{H}/\partial\gamma$.

The expected values of $\partial L'/\partial\gamma$ and $\partial L'/\partial\sigma^2$ for fixed γ are both zero. Solution of equations (15) and (16) consists therefore of equating B and R to their expectations in the conditional distribution with γ fixed.

The information matrix is

$$\frac{1}{2}\begin{bmatrix} f_{11} & f_{12}/\sigma^2 \\ f_{12}/\sigma^2 & f_{22}/\sigma^4 \end{bmatrix},$$

where

$$f_{11} = \sum_s (d_s/\xi_s)^2 = \text{tr}\,\{(\mathbf{SHS})^{-g}\,\mathbf{SDS}\}^2, \tag{19}$$

$$f_{12} = E, \tag{20}$$

$$f_{22} = n-t. \tag{21}$$

So far the results are appropriate when **H** is any symmetric $n \times n$ matrix of rank n such that $\text{rank}(\mathbf{SHS}) = \text{rank}(\mathbf{S}) = n-t$. For practical purposes, however, and for comparison with other methods of estimation, we require simpler expressions for B, R, E and the elements of the information matrix. When **H** takes the particular form specified in (2) suitable expressions can be obtained by substituting the appropriate differentials **D** and d_s and a simplified form of $(\mathbf{SHS})^{-g}$.

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The differential $\mathbf{D}$ is simply $\mathbf{ZZ'}$ and the d_s are given by the latent roots of $\mathbf{ZZ'S}$, or $\mathbf{SZZ'S}$. The latter can be demonstrated by substituting for $\mathbf{H}$ in equation (11) and rearranging to give

$$\mathbf{SZZ'S} = \mathbf{P}\,\mathrm{diag}\,(\lambda_s)\,\mathbf{P'}, \tag{22}$$

where the λ_s are such that

$$\xi_s = \lambda_s\gamma + 1. \tag{23}$$

Hence $d_s = \lambda_s$.

We now express $(\mathbf{SHS})^{-g}$ in terms of a $b \times b$ matrix. First, we note that the factors of $\mathbf{Z'SZ\Gamma Z'S}$ include both the $b \times b$ matrix $\mathbf{Z'SZ}$ and the $n \times n$ matrix $\mathbf{SZ\Gamma Z'S} = \mathbf{S(H-I)\,S}$. Hence

$$(\mathbf{Z'SZ})\,\mathbf{\Gamma Z'S} + \mathbf{Z'S} = \mathbf{Z'S(SHS)},$$

so that

$$\mathbf{\Gamma Z'(SHS)}^{-g} = \mathbf{W}^{-1}\mathbf{Z'S}, \tag{24}$$

where

$$\mathbf{W} = \mathbf{Z'SZ} + \mathbf{\Gamma}^{-1}. \tag{25}$$

Premultiplying both sides of (24) by $\mathbf{SZ}$, we obtain

$$(\mathbf{SHS} - \mathbf{S})\,(\mathbf{SHS})^{-g} = \mathbf{SZW}^{-1}\mathbf{Z'S},$$

so that

$$(\mathbf{SHS})^{-g} = \mathbf{S} - \mathbf{SZW}^{-1}\mathbf{Z'S}. \tag{26}$$

The quantities B, R, E and f_{11} can now be expressed in terms of $\mathbf{W}$, a vector

$$\boldsymbol{\beta} = \mathbf{\Gamma Z'(SHS)}^{-g}\,\mathbf{y} = \mathbf{W}^{-1}\mathbf{Z'Sy} \tag{27}$$

and a matrix

$$\mathbf{U} = \mathbf{Z'(SHS)}^{-g}\mathbf{Z} = \mathbf{\Gamma}^{-1} - \mathbf{\Gamma}^{-1}\mathbf{W}^{-1}\mathbf{\Gamma}^{-1}. \tag{28}$$

The required expressions for B and R can be obtained by direct substitution. They are

$$B = \boldsymbol{\beta'\beta}/\gamma^2, \tag{29}$$

$$R = \mathbf{y'Sy} - \mathbf{y'SZ}\boldsymbol{\beta}. \tag{30}$$

Also, since $\mathrm{tr}\,\{(\mathbf{SHS})^{-g}\mathbf{ZZ'S}\}$ is equivalent to $\mathrm{tr}\,\{\mathbf{Z'(SHS)}^{-g}\mathbf{Z}\}$, E is given by

$$E = f_{12} = \mathrm{tr}\,(\mathbf{U}). \tag{31}$$

Similarly,

$$f_{11} = \mathrm{tr}\,(\mathbf{U}^2). \tag{32}$$

In practice, of course, we use $\hat{\mathbf{H}}$, $\hat{B}$, $\hat{\boldsymbol{\beta}}$, $\hat{R}$, etc., given by the expressions for $\mathbf{H}$, B, $\boldsymbol{\beta}$, R, etc., but with γ replaced by $\hat{\gamma}$.

These results depend on $\mathbf{W}$ being nonsingular; in fact $\mathbf{W}$ is always nonsingular under the conditions imposed on $\mathbf{H}$ in §2 as the largest λ_s is necessarily not larger than $k_{\max}$.

5. Estimation of $\boldsymbol{\alpha}$

The estimate of $\boldsymbol{\alpha}$ is obtained by maximizing

$$L'' = \mathrm{const} - \tfrac{1}{2}\log|\mathbf{X'H}^{-1}\mathbf{X}| - \tfrac{1}{2}t\log\sigma^2 - \frac{1}{2\sigma^2}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha})'\,\mathbf{H}^{-1}\mathbf{X}(\mathbf{X'H}^{-1}\mathbf{X})^{-1}\mathbf{X'H}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha}). \tag{33}$$

The estimate is given by

$$\hat{\boldsymbol{\alpha}} = (\mathbf{X'}\hat{\mathbf{H}}^{-1}\mathbf{X})^{-1}\mathbf{X'}\hat{\mathbf{H}}^{-1}\mathbf{y}. \tag{34}$$

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An equivalent procedure is to minimize the weighted sum of squares

$$(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha})'\hat{\mathbf{H}}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha})$$

with fixed $\hat{\gamma}$.

6. Practical solution of equations (15) and (16)

Equations (15) and (16) can be solved by Fisher's iterative method. We start with a preliminary estimate, $\hat{\gamma}_0$ say, of γ. Substitute this estimate for γ in (25) to give $\hat{\mathbf{W}}$, and hence determine $\hat{\boldsymbol{\beta}}$, $\hat{\mathbf{U}}$, $\hat{B}$, $\hat{R}$, $\hat{f}_{12}$ and $\hat{f}_{11}$ equations from (27) to (32). Then an approximation to $\hat{\sigma}^2$ is given by

$$\hat{\sigma}^2 = \hat{f}^{12}\hat{B} + \hat{f}^{22}\hat{R}, \tag{35}$$

and a closer approximation to $\hat{\gamma}$ is given by

$$\hat{\gamma} = \hat{\gamma}_0 + (\hat{f}^{11}\hat{B} + \hat{f}^{12}\hat{R})/\hat{\sigma}^2. \tag{36}$$

In these expressions $\hat{f}^{11}$, $\hat{f}^{12}$ and $\hat{f}^{22}$ are the elements of the inverse matrix $\hat{\mathbf{F}}^{-1}$, where

$$\hat{\mathbf{F}} = \begin{bmatrix} \hat{f}_{11} & \hat{f}_{12} \\ \hat{f}_{12} & \hat{f}_{22} \end{bmatrix}.$$

If $\hat{\gamma}_0$ is small it may be more accurate numerically not to calculate $\hat{\boldsymbol{\beta}}$ explicitly, but to use instead

$$\hat{\boldsymbol{\beta}}/\hat{\gamma} = (\hat{\gamma}_0\mathbf{Z}'\mathbf{S}\mathbf{Z} + \mathbf{I})^{-1}\mathbf{Z}'\mathbf{S}\mathbf{y}.$$

A preliminary estimate of σ^2 is not required. Also $\boldsymbol{\alpha}$ need not be estimated until $\hat{\gamma}$ has been determined. A convenient expression for $\hat{\boldsymbol{\alpha}}$ in terms of $\hat{\gamma}$ can be obtained by noting that $\hat{\boldsymbol{\alpha}}$ and $\hat{\boldsymbol{\beta}}$ in (34) and (27) satisfy the equations

$$\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\alpha}} + \mathbf{X}'\mathbf{Z}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}, \tag{37}$$

$$\mathbf{Z}'\mathbf{X}\hat{\boldsymbol{\alpha}} + (\mathbf{Z}'\mathbf{Z} + \hat{\boldsymbol{\Gamma}}^{-1})\hat{\boldsymbol{\beta}} = \mathbf{Z}'\mathbf{y}. \tag{38}$$

Hence

$$\hat{\boldsymbol{\alpha}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{y}-\mathbf{Z}\hat{\boldsymbol{\beta}}). \tag{39}$$

Equations (37) and (38) were suggested by Henderson (Henderson, Kempthorne, Searle & von Krosigk, 1959) as the basis of a practical procedure for estimating $\boldsymbol{\alpha}$ when γ is known.

When b is large an alternative procedure may be preferable, requiring inversion of a $t \times t$ matrix and a diagonal $b \times b$ matrix instead of the nondiagonal $\mathbf{W}$.

Equation (34) is used first to estimate $\boldsymbol{\alpha}$. Equation (3) provides a convenient expression for $\hat{\mathbf{H}}^{-1}$, involving inversion of a diagonal matrix only. Then

$$\hat{\boldsymbol{\beta}} = (\mathbf{Z}'\mathbf{Z} + \hat{\boldsymbol{\Gamma}}^{-1})^{-1}\mathbf{Z}'(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\alpha}}), \tag{40}$$

a rearrangement of equation (38). The value of $\hat{B}$ is calculated as before and $\hat{R}$ is given by

$$\hat{R} = \mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{X}\hat{\boldsymbol{\alpha}} - \mathbf{y}'\mathbf{Z}\hat{\boldsymbol{\beta}}. \tag{41}$$

The matrix $\hat{\mathbf{W}}^{-1}$, required in the calculation of $\hat{f}_{11}$ and $\hat{f}_{12}$, can be conveniently expressed in terms of the matrices $(\mathbf{Z}'\mathbf{Z} + \hat{\boldsymbol{\Gamma}}^{-1})^{-1}$ and $(\mathbf{X}'\mathbf{H}^{-1}\mathbf{X})^{-1}$ already used in the estimation of $\boldsymbol{\alpha}$. The expression is

$$\hat{\mathbf{W}}^{-1} = (\mathbf{Z}'\mathbf{Z} + \hat{\boldsymbol{\Gamma}}^{-1})^{-1} + (\mathbf{Z}'\mathbf{Z} + \boldsymbol{\Gamma}^{-1})^{-1}\mathbf{Z}'\mathbf{X}(\mathbf{X}'\hat{\mathbf{H}}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z} + \hat{\boldsymbol{\Gamma}}^{-1})^{-1}. \tag{42}$$

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Solution of equations (15) and (16) breaks down if $\mathbf{F}$ is singular. We distinguish three cases. When $f_{22} = 0$, i.e. $n = t$, neither γ nor σ^2 can be estimated as the treatment contrasts account for all $n-1$ degrees of freedom. When $f_{11} = 0$ but $f_{22} \neq 0$, σ^2 can be estimated but not γ. This situation arises when some treatment comparison is totally confounded with every block comparison. Each column of $\mathbf{Z}$ is then given by a linear combination of the columns of $\mathbf{X}$ so that $\mathbf{SZ} = 0$ and $\mathbf{W}^{-1} = \boldsymbol{\Gamma}$, and hence $\mathbf{U} = 0$. The matrix $\mathbf{F}$ is also singular when treatment contrasts account for all intra-block degrees of freedom but not for all inter-block degrees of freedom. In this case $f_{11} = (n-t)/(\gamma+1)^2$, $f_{12} = (n-t)/(\gamma+1); (\gamma+1)\sigma^2$ can be estimated but not the individual γ and σ^2.

A further complication is that $\mathbf{X}'\mathbf{X}$ may be singular This can be dealt with by replacing $(\mathbf{X}'\mathbf{X})^{-1}$ and $(\mathbf{X}'\mathbf{H}^{-1}\mathbf{X})^{-1}$ in (8), (34), (39) and (42) by generalized inverses and making t the rank of $\mathbf{X}$ instead of the number of treatments.

7. Relationship with other methods

We now consider the relationship between I, the method described by Hartley & Rao (1967), II, the method proposed by Cunningham & Henderson (1968) and later modified by Thompson (1969) and III, the method of the present paper. These three methods give different estimates of γ but use essentially the same methods of estimating $\boldsymbol{\alpha}$ and σ^2 once $\hat{\gamma}$ has been determined.

Hartley & Rao (1967) show that the unconditional maximum likelihood estimates are obtained by solving the following equations for $\boldsymbol{\alpha}$, σ^2 and γ:

$$\frac{1}{\sigma^2}(\mathbf{X}'\mathbf{H}^{-1}\mathbf{y} - \mathbf{X}'\mathbf{H}^{-1}\mathbf{X}\boldsymbol{\alpha}) = 0, \tag{43}$$

$$-\tfrac{1}{2}\operatorname{tr}(\mathbf{H}^{-1}\mathbf{Z}\mathbf{Z}') + \frac{1}{2\sigma^2}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha})'\mathbf{H}^{-1}\mathbf{Z}\mathbf{Z}'\mathbf{H}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha}) = 0, \tag{44}$$

$$-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha})'\mathbf{H}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha}) = 0. \tag{45}$$

The numerical terms $(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha})'\mathbf{H}^{-1}\mathbf{Z}\mathbf{Z}'\mathbf{H}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha})$ and $(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha})'\mathbf{H}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha})$ are equivalent to B and R. This can be demonstrated as follows. Consider the expressions for L, L' and L'' given in (4), (13) and (33). As $L = L' + L''$, we have

$$(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha})'\mathbf{H}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha}) = R + (\mathbf{y}-\mathbf{X}\boldsymbol{\alpha})'\mathbf{H}^{-1}\mathbf{X}(\mathbf{X}'\mathbf{H}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{H}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha}). \tag{46}$$

Substitution of the solution for $\boldsymbol{\alpha}$ given in §5, or the solution given by equation (43), shows that the second term on the right hand side of (46) is zero. Hence

$$R = (\mathbf{y}-\mathbf{X}\boldsymbol{\alpha})'\mathbf{H}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha}). \tag{47}$$

Differentiation of both sides of equation (47) with respect to γ gives

$$B = (\mathbf{y}-\mathbf{X}\boldsymbol{\alpha})'\mathbf{H}^{-1}\mathbf{Z}\mathbf{Z}'\mathbf{H}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\alpha}). \tag{48}$$

Thus the Hartley–Rao method consists of equating B and R to their expected values in the conditional distribution with both γ and $\boldsymbol{\alpha}$ fixed. It differs from method III in that in the latter the expected values take account of errors in the estimation of $\boldsymbol{\alpha}$.

Method II differs from method III in that the sum of squares $\boldsymbol{\beta}'\mathbf{Z}'\mathbf{S}\mathbf{y}$ is used instead of B.

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Thus $\boldsymbol{\beta}'\mathbf{Z}'\mathbf{Sy}$ and R are equated to their expected values in the conditional distribution with fixed γ. As in method III the estimates of γ and σ^2 are wholly derived from $\mathbf{Sy}$. Method II does not, however, maximize the likelihood of $\mathbf{Sy}$ except in special cases when $\boldsymbol{\beta}$ is proportional to $\mathbf{Z}'\mathbf{Sy}$, for example in complete block designs and in symmetric balanced incomplete block designs.

8. EQUAL BLOCK SIZES

Nelder (1968) proposed a method for estimating stratum variances in a general class of balanced designs. We now show that application of this method to incomplete block designs with blocks of equal size k gives the same results as method III.

The method consists of equating the sums of squares of residuals,

$$(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\alpha}})'\left(\frac{\mathbf{ZZ}'}{k}-\frac{\mathbf{11}'}{n}\right)(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\alpha}}), \tag{49}$$

$$(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\alpha}})'\left(\mathbf{I}-\frac{\mathbf{ZZ}'}{k}\right)(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\alpha}}) \tag{50}$$

to their expectations, where $\mathbf{1}$ is a unit vector.

Equations (48) and (47) show that $\hat{B}$ and $\hat{R}$, the quantities used in method III, are also sums of squares of residuals. When block sizes are all equal to k the expressions for $\hat{B}$ and $\hat{R}$ simplify to

$$\hat{B} = (\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\alpha}})'\,\mathbf{ZZ}'(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\alpha}})/(\hat{\gamma}k+1)^2,$$

$$\hat{R} = (\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\alpha}})'\,(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\alpha}})-\hat{\gamma}\hat{B};$$

$\mathbf{1}'(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\alpha}})$, the sum of residuals, is zero. Equating $\hat{B}$ and $\hat{R}$ to their expectations is therefore equivalent to equating the expressions (49) and (50) to their expectations, i.e. the method proposed in the present paper gives the same results as Nelder's (1968) method.

Table 1. *Data used by Cunningham & Henderson* (1968)

	Treatment 1		Treatment 2		Totals
Block 1	3, 2,	5	2, 3,	5	10
2	2, 3, 5, 6, 7,	23	8, 8, 9	25	48
3	3	3	4, 4, 3, 2, 5	18	21
Totals		31		48	79

Table 2. *Estimation of* σ^2, γ *from the data of Table* 1. *Initial estimate of* $\gamma = 1$

	Cycle 1	2	3	4
B	6·3194	2·7584	2·8537	2·8505
R	42·7221	40·2145	40·2981	40·2952
$\hat{f}^{11}$	0·8155	1·8424	1·7817	1·7836
$10\hat{f}^{12}$	−0·8530	−1·2824	−1·2611	−1·2618
$10\hat{f}^{22}$	0·7142	0·7143	0·7143	0·7143
$\hat{\sigma}^2$	2·5123	2·5186	2·5185	2·5185
$\hat{\gamma}$	1·6006	1·5708	1·5718	1·5718

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9. Example

We have applied the new method, method III, to the data of the example discussed by Cunningham & Henderson (1968) and Thompson (1969). The data are in Table 1. Table 2 gives the results of four cycles of the iterative procedure, starting with $\hat{\gamma}_0 = 1$. Convergence is rapid. Table 3 compares the results with those given by methods I and II and by the analysis of variance method described by Cunningham & Henderson (1968).

Table 3. *Comparison of estimates of* σ^2 *and* γ

Method	Estimate of σ^2	Estimate of γ
Analysis of variance	2·5237	1·7479
Method I	2·3518	1·0652
Method II	2·4822	1·8028
Method III	2·5185	1·5718

10. More general results

Hartley & Rao (1967) extended the unconditional maximum likelihood method to a general class of designs with c block factors. The class includes for example split-plot designs and row-and-column designs. With these designs the model given by equation (1) still applies but the variance matrix of $\boldsymbol{\epsilon}$ is now $\mathbf{H}\sigma^2$, where

$$\mathbf{H} = \left(\mathbf{I} + \sum_{p=1}^{c} \mathbf{Z}_p \mathbf{Z}_p' \gamma_p\right). \tag{51}$$

We have to estimate $c+1$ parameters, σ^2 and γ_p $(p = 1, \ldots, c)$.

Each plot is at exactly one level of each block factor. For example, if a design is arranged in rows and columns each plot is in exactly one row and one column. Element (i,j) of $\mathbf{Z}_p$ is 1 when plot i is at level j of block factor p $(j = 1, \ldots, b_p; p = 1, \ldots, c)$; otherwise the element is 0. The matrices $\mathbf{Z}_p'\mathbf{Z}_p$ are diagonal.

The modified maximum likelihood method of the present paper can also be applied to this more general class of designs. We again divide the data into two parts with logarithmic likelihoods L' and L'', estimate γ_p and σ^2 by maximizing L', and estimate $\boldsymbol{\alpha}$ by maximizing L''.

With suitable redefinition of $\boldsymbol{\Gamma}$ and $\mathbf{Z}$, results closely follow those already given for the simpler model. Only the main modifications will be described here.

We redefine $\boldsymbol{\Gamma}$ as a diagonal $b \times b$ matrix, i.e. $\boldsymbol{\Gamma} = \mathrm{diag}\,(\gamma_p \mathbf{I}_p)$ $(p = 1, \ldots, c)$, where $b = \Sigma b_p$ and $\mathbf{I}_p$ is the $b_p \times b_p$ identity matrix; $\mathbf{Z}$ is the partitioned matrix $(\mathbf{Z}_1 \vdots \ldots \vdots \mathbf{Z}_c)$. With these definitions $\mathbf{H}$ can again be written in the form $\mathbf{Z}\boldsymbol{\Gamma}\mathbf{Z}' + \mathbf{I}$. The vector $\boldsymbol{\beta}$ and the matrix $\mathbf{U}$ defined by (27) and (28) are also partitioned. The $b_p \times 1$ subvectors of $\boldsymbol{\beta}$ will be denoted by $\boldsymbol{\beta}_p$ and the $b_p \times b_q$ submatrices of $\mathbf{U}$ will be denoted by $\mathbf{U}_{pq}$ $(p, q = 1, \ldots, c)$.

The estimates $\hat{\gamma}_p$ and $\hat{\sigma}^2$ are obtained by solving the c equations

$$-\tfrac{1}{2}E_p + \frac{1}{2\sigma^2}B_p = 0 \quad (p = 1, \ldots, c),$$

together with equation (16), where

$$B_p = \mathbf{y}'(\mathbf{SHS})^{-g}\mathbf{Z}_p\mathbf{Z}_p'(\mathbf{SHS})^{-g}\mathbf{y},$$
$$E_p = \mathrm{tr}\,\{(\mathbf{SHS})^{-g}\mathbf{Z}_p\mathbf{Z}_p'\mathbf{S}\}.$$

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Working expressions for B_p and E_p are

$$B_p = \boldsymbol{\beta}_p' \boldsymbol{\beta}_p / \gamma_p^2, \quad E_p = \mathrm{tr}\,(\mathbf{U}_{pp}).$$

The information matrix is

$$\tfrac{1}{2}\begin{bmatrix} \mathrm{tr}\,(\mathbf{U}_{11}^2) & \mathrm{tr}\,(\mathbf{U}_{12}\mathbf{U}_{21}) & \dots & \mathrm{tr}\,(\mathbf{U}_{1c}\mathbf{U}_{c1}) & \mathrm{tr}\,(\mathbf{U}_{11})/\sigma^2 \\ \mathrm{tr}\,(\mathbf{U}_{12}\mathbf{U}_{21}) & \mathrm{tr}\,(\mathbf{U}_{22}^2) & \dots & \mathrm{tr}\,(\mathbf{U}_{2c}\mathbf{U}_{c2}) & \mathrm{tr}\,(\mathbf{U}_{22})/\sigma^2 \\ . & . & \dots & . & . \\ \mathrm{tr}\,(\mathbf{U}_{1c}\mathbf{U}_{c1}) & \mathrm{tr}\,(\mathbf{U}_{2c}\mathbf{U}_{c2}) & \dots & \mathrm{tr}\,(\mathbf{U}_{cc})^2 & \mathrm{tr}\,(\mathbf{U}_{cc})/\sigma^2 \\ \mathrm{tr}\,(\mathbf{U}_{11})/\sigma^2 & \mathrm{tr}\,(\mathbf{U}_{22})/\sigma^2 & \dots & \mathrm{tr}\,(\mathbf{U}_{cc})/\sigma^2 & (n-t)/\sigma^4 \end{bmatrix}.$$

Generalization of the iterative solution described in § 6 presents little difficulty. Singularities in the information matrix again indicate that one or more parameters cannot be estimated. For example, we cannot estimate γ_q when $\mathbf{U}_{qq} = 0$.

Further generality can be achieved by defining $\mathbf{Z}_p\mathbf{Z}_p'$ as any real symmetric matrix with $\mathbf{Z}_p'\mathbf{Z}_p$ not necessarily diagonal. We can also deal with variance matrices of the form $\mathbf{V} = (\mathbf{J} + \mathbf{Z}\boldsymbol{\Gamma}\mathbf{Z}')\sigma^2$, where $\mathbf{J}$ is symmetric. When $\mathbf{J}$ is idempotent and such that $\mathbf{JZ} = \mathbf{Z}$, $\mathbf{JX} = \mathbf{X}$ and $\mathbf{J} \neq \mathbf{I}$, generalized inverses are required in place of $\mathbf{H}^{-1}$, $(\mathbf{X}'\mathbf{X})^{-1}$ and $(\mathbf{X}'\mathbf{H}^{-1}\mathbf{X})^{-1}$. When $\mathbf{J}$ is not idempotent a preliminary transformation $\mathbf{Ty}$ can be used with $\mathbf{T}$ such that $\mathbf{TJT}'$ is idempotent.

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Some key words: Analysis of incomplete block designs; Components of variance; Designs with unequal block sizes; Recovery of inter-block information; Maximum likelihood.

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Henderson (1973)

Henderson, C.R. 1973. Sire evaluation and genetic trends. In Proceedings of the Animal Breeding and Genetics Symposium in Honor of Dr. Jay L. Lush. American Society of Animal Science, pp. 10 to 41.

The problem

Written for the symposium in honor of Lush, this is the text in which Henderson organizes the entire theory of sire evaluation. He distinguishes three situations, according to what is known about the distribution of the data, and arrives at the best linear unbiased predictor, BLUP, as the solution for the real case in which the means are unknown. The paper also deals with genetic groups, genetic trends and how data from many herds can be used at the same time.

Points to note while reading

This is the classic paper on BLUP applied to the sire model. Note the definitions of BP, BLP and BLUP and how each requires less prior knowledge. The appendixes present numerical examples that help in understanding the equations.

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SIRE EVALUATION AND GENETIC TRENDS

C. R. Henderson[1]

Introduction

Sire evaluation can logically be formulated as a process of prediction of future progeny of a sire produced by matings with specified females and making their records in some specified environment. Sewall Wright (1931) over 40 years ago suggested three types of prediction that might be of interest: (1) progeny of a particular mating, (2) future daughters in the same herd but out of a new sample of dams, (3) daughters out of a random sample of dams of the breed. Dr. Lush as early as 1931 had elucidated the principles of sire evaluation, Lush (1931, 1933).

As was pointed out by Lehman (1961), two types of selection problems have been studied by statisticians. These are Model I and Model II selection, analagous to the corresponding models of analysis of variance. In Model I the candidates for selection are fixed, for example, choices are to be made among treatments, a random sample of observations having been taken on each fixed treatment. No really unified theory has been developed for this type of selection. In contrast, Model II selection involves candidates that are regarded as a random sample from some specified population. Model II represents the classical selection problem in animal breeding and had essentially been solved by Wright and Lush early in the 1930's. Smith's (1936) application to plant breeding and, in particular, Hazel's (1943) application to animal breeding formalized the techniques.

A third type of selection that has apparently been overlooked by both statisticians and animal breeders might well be called mixed model selection following the analogy of the mixed model of analysis of variance. In this case the candidates for selection are randomly drawn from more than one population, and the merit of each is the sum of the sub-population mean and the value of the particular random variable associated with that animal. Apparently the first statement of this problem, at least in connection with sire evaluation, was Henderson's (1966) suggestion that in order to account for genetic trend and for different selection policies of AI studs and of dairymen's choice of sires for natural service, bulls to be evaluated be divided into groups and the evaluation be the sum of the estimate of the group and the selection index type evaluation of the deviation of the individual sire from this mean. This idea has been incorporated into the Cornell method of sire evaluation now in use in the Northeast.

The present paper includes a review of Model II prediction methods that have been available for many years. This is followed by a discussion of the problems of unknown fixed effects, large bodies of multiply classified data with unequal numbers, mixed model selection, and culling and selection bias. The motivation for my contribution to these latter problems came first from my association with Drs. Hazel, Lush, Kempthorne, and Mood while I was a graduate student at Iowa State, and second from being confronted with these problems for many years at Cornell University. I should like also to thank my fellow graduate students for discussing these problems with me and helping clarify my thinking. The same should be said of my own graduate students at Cornell and in particular Shayle Searle whose contribution to my understanding of linear methods has been very substantial. Much of the presentation that follows is in matrix notation, and for this I offer no apology as this has rapidly become an essential tool of any serious student of animal breeding.

Classes of Selection Problems Relative to Information Available

The selection problem and some solutions will be presented for situations with decreasing levels of knowledge of the populations of which the candidates for selection are members.

1. The form of the joint distribution of records and of the genetic values to be predicted is known, and in addition the numerical values of the parameters of the distribution are known. It has been proved that the conditional

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mean of genetic values, given the records, has optimum properties. This method will be called Best Prediction (BP).

2. The form of the distribution is not known or certain parameters are not known, but we do know the means of the records, the means of the genetic values, and variances and covariances of records and genetic values. That is, we know the first and second moments of the distribution. In that case Best Linear Prediction (BLP) can be used. This involves finding that linear function of the records which minimizes the average of squared errors of prediction. In case the distribution is normal, BLP is also BP.

3. The problem is the same as for BLP, but now we do not know the means. That is, only the second central moments are known. In this case we use Best Linear Unbiased Prediction (BLUP). In this method we find that linear function of the records which has the same expectation (mean) as the genetic value to be predicted and which in the class of such functions minimizes the average of squared errors.

4. The last class of selection problems with which we are concerned is the one in which neither first nor second moments are known and we wish to use linear prediction methods. Of course we never, in fact, really know parameter values, but we may have good prior estimates of them. In the BP case we have good estimates of all parameters. In the BLP case we have good estimates of the first and second moments. In the BLUP case we have good estimates of the second central moments. Now if we have no prior estimates of either first or second moments, we need to estimate them from the same data that are to be used for prediction. This is essentially an unsolved problem, but recent work by Hartley and Rao (1967) and extended by us may offer some hope. A method has been discovered for the normal distribution to find joint ML estimates of the means, the variances, and the conditional means of the genetic values. Recall that the true conditional means are optimum for the normal case. It appears logical to substitute ML estimates of the first and second moments for parameter values in these expressions.

Model II Prediction by Minimum Square Error (Best Prediction)

We observe a set of records say, y, where $y' = [y_1 \ldots y_n]$ is a row vector of n elements. Jointly distributed with y is a non-observable vector, w, which we wish to predict. For example, y' = [record on sire, record on dam, record on female progeny]

w' = [future progeny of this sire and dam, future progeny of the female progeny with record].

Let the predictor of the ith element of w be

$$\hat{w}_i = f(y),$$

some function, not necessarily linear, of the records. The question is what is the "best" function to use. To answer this question it is required to define "best". One way is to say we wish to minimize the error of prediction, $\hat{w}_i - w_i$, in conceptual repetitions of sampling and predicting. But how do we measure error? Logically it could be

$$E \mid \hat{w}_i - w_i \mid^p,$$

the expected value of some power of the absolute value of the deviation. Then what p should be chosen? The most common choice is 2, probably because it leads to mathematical simplicity and to other desirable properties. The method which minimizes $E(\hat{w}_i - w_i)^2$ is called Minimum Mean Square Error Prediction or sometimes Best Prediction (BP). Hereafter we shall call it BP. It can be proved (see Cochran, 1951 or Rao, 1965) that BP of w_i is

$$\hat{w}_i = E(w_i \mid y),$$

that is, the conditional mean of w_i, given y. Unfortunately, however, in order to compute this we must know the form of the joint distribution of [y,w], we must be able from that to derive the conditional distribution, and finally the values of the parameters that appear in this conditional distribution must be known. One or more of these three requirements is likely to be unfulfilled in practice.

Some of the properties of BP in addition to the one used in its derivation are stated below.

1. $E(\hat{w}_i) = E(w_i)$. Consequently, BP is an unbiased predictor even though unbiasedness was not imposed in its derivation.

2. $Var(\hat{w} - w)$ = average conditional Var (w|y). By Var of a vector we mean the variance-covariance matrix of elements of the vector. We see that the variance of the prediction errors is the same as the variance of w in the conditional distribution (fixed y), averaged over the range of values of y. Also $Var(\hat{w}) = Cov(\hat{w},w)$.

3. Of *all* functions of y, $\hat{w}_i = E(w_i \mid y)$

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maximizes the correlation between $\hat{w}_i$ and w_i, where $\hat{w}_i$ is the ith element of w. This correlation is $\sigma_{\hat{w}i}/\sigma_{wi}$.

4. k samples of [y,w] are drawn. We wish to predict w_i, the ith element of w, so that if we select $s<k$ of the w_i by truncation on the predictions, the w_i associated with the s highest predictions will have the largest possible mean. BP has this property.

Best Linear Prediction (BLP)

Often in practice we do not know the distribution of (y,w). What can be done in that case? One possibility is to restrict ourselves to *linear* predictors. This greatly simplifies the problem.

Now we compute $\hat{w}_i = b_i' y + k_i$, where b_i and k_i are a vector and a scalar respectively that will minimize $E(\hat{w}_i - w_i)^2$. We call this method Best Linear Prediction (BLP). The solution is

$$\hat{w}_i = \alpha_i + c_i' V^{-1} (y-\theta), \qquad (1)$$

where α_i = mean of w_i,

$c_i = Cov(y,w_i)$, the ith column of the matrix of covariances between y and w, say C,

$V = Var(y)$,

θ = mean of y.

V is of course a symmetric matrix, and θ is a column vector. We can also write BLP of w_i as

$$\hat{w}_i = \alpha_i + b_i'(y-\theta),$$

where b_i is the solution to

$$V b_i = c_i. \qquad (2)$$

Equation (1) follows simply from (2), for note that

$$b_i = V^{-1} c_i \text{ or } b_i' = c_i' V^{-1}.$$

This is a statement in matrix notation of the equations to be solved in selection index methods to find weighting factors. A simple generalization to BLP of the vector, w, is

$$\hat{w} = \alpha + C'V^{-1}(y-\theta), \qquad (3)$$

where α = the column vector of means of the vector, w

$C = Cov(y,w)$.

An example of the use of this result is the evaluation of all animals in a herd by using all available records. Then y represents these records, w represents the genetic merits of all animals to be evaluated, V is the variance-covariance matrix of records, and C is the set of covariances between records and genetic merits.

Now it is obvious that BLP requires only the first and second moments of the distribution of (y,w), a much less stringent requirement than BP. Some of the properties of BLP are:

1. $\hat{w}$ is unbiased, that is, $E(\hat{w}) = E(w)$. This is true even though unbiasedness was not required in the derivation.
2. $Var(\hat{w}) = Cov(\hat{w},w) = C'V^{-1}C$, and $E(\hat{w}-w)^2 = Var(\hat{w}-w) = G-C'V^{-1}C$, where $G = Var(w)$.
3. Of all *linear* functions of y, BLP of w_i maximizes the correlation between the predictor and the predictand.
4. BLP of a linear function of w, say of $m'w$, is that same linear function of BLP of w. That is,

BLP of $m'w = m'\hat{w}$, where $\hat{w}$ is BLP of w.

This last property was proved by Henderson in the early 1950's, see Karam, Chapman and Pope (1953) for an application to selection index. This property also has value in evaluation of line crosses, Henderson (1952).

5. When [y,w] are jointly normally distributed, BLP is also BP. In other words the best predictor is linear in the normal case.

Dr. Lush clearly was using BLP methods long before the formalization of selection index. It was not always clear whether or not he invoked normality, but certainly his work laid the foundations for selection index in animal breeding. His proof of its optimum property, maximization of genetic progress through truncation selection on a linear index, did invoke normality. Hazel's (1943) rationale for the method depended upon E(w|y) being linear in y, a property of the multivariate normal. The first major contribution of Dr. Hazel to selection index methods, was, I believe, his description of the aggregate genotype of the animal in terms of a linear function of genotypic values for each of several traits. In the notation of the present paper, the definition is $m'u$, and this is treated as the predictand ("dependent variable") in the regression type equations for finding weights on records.

The equations to be solved in that case are

$$V b = C m. \qquad (4)$$

Hazel's second major contribution to selection index was his derivation of methods to estimate the elements of C m. In his notation these are functions of heritabilities and genetic correlations.

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BLP represents, it seems to me, most of the present thinking, teaching, and application of selection index principles in animal breeding. It has been extremely useful in helping to predict the relative merits of alternative selection programs and to gain perspective on the importance of various sources of information in evaluating animals. It was, for example, the basis of all serious attempts at evaluating dairy sires objectively prior to the wide use of artificial insemination.

Some Serious Defects of BLP

Conventional selection index (BLP) is totally inadequate to handle most problems of sire evaluation, and, in fact, most real life selection problems today. Some of the difficulties are enumerated below.

1. The means of w and y are required. For example, in AI sire evaluation we must know year, season, and group means even though no prior data exist on new groups and years. A solution to this problem is presented in a later section, "Best Linear Unbiased Prediction".

2. In some cases the required variances and covariances are unknown. No satisfactory solution to this problem has been developed, and perhaps none is really possible.

3. As a consequence of AI multiply classified data with very large numbers of potential subclasses and extremely unbalanced data, problems arise with which conventional selection index is totally incapable of dealing.

4. From a pedagogical viewpoint the justification for selection index that invokes the argument of maximization of genetic progress through truncation on a linear index (Lush, 1948) is not valid when the candidates for selection have unequal information. A different proof involving maximization of the probability of a correct ranking has been derived, Henderson (1963). Normality is required in this proof.

Best Linear Unbiased Prediction (BLUP)

Suppose now that $\alpha = E(w)$ and $\theta = E(y)$ are unknown, but V and C are known. In that case we cannot use BLP (selection index) as in (1). The usual procedure in animal breeding applications has been to estimate α and θ in some way and then to substitute these estimates for the parameter values. The properties of such a method depended upon what estimates were used. Henderson (1963) presented a straightforward method based on the following principles.

Compute $\hat{w}_i = b_i'y + k_i$ such that

$$E(\hat{w}_i) = E(w_i)$$

$E(\hat{w}_i - w_i)^2$ is minimum.

That is, we find the minimum mean square error predictor in the class of unbiased predictors.

A general statement of the mean of y is

$$\theta = X\beta.$$

Further, if the problem is to be solved, the mean of $w = \alpha$ must be estimable from the data $= y$.

Suppose $E(w) = P\beta$.

Then if $p_i\beta$ is estimable we can find an unbiased predictor of w_i, where p_i is the ith row of P. It is easily proved that k_i always $= 0$, and b_i is the solution to

$$\begin{bmatrix} V & X \\ X' & 0 \end{bmatrix} \begin{bmatrix} b_i \\ \gamma \end{bmatrix} = \begin{bmatrix} c_i \\ p_i' \end{bmatrix}. \tag{5}$$

Solving for b_i we find that

$$\hat{w}_i = p_i (X'V^{-1}X)^{-}X'V^{-1}y + c_i' V^{-1} [y - X(X'V^{-1}X)^{-}X'V^{-1}y]. \tag{6}$$

This is a particularly interesting result because

$(X'V^{-1}X)^{-}X'V^{-1}y = \hat{\beta}$ = some solution to the generalized least squares (GLS) equations,

$$X'V^{-1}X\hat{\beta} = X'V^{-1}y. \tag{7}$$

$(X'V^{-1}X)^{-}$ denotes some generalized inverse of $X'V^{-1}X$.

Then it is easy to see that the predictor in (6) is

$$\begin{aligned} \hat{w}_i &= p_i\hat{\beta} + c_i'V^{-1}(y - X\hat{\beta}) \\ &= \hat{\alpha}_i + c_i'V^{-1}(y - \hat{\theta}). \end{aligned} \tag{8}$$

That is, the best linear unbiased predictor is BLP with GLS estimates of α_i and θ substituted for the corresponding parameters. Thus we have proved that the practice of substitution of estimates of means in the selection index is a valid procedure, and in fact is BLUP provided the GLS estimators are used.

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It must be recognized that for a particular set of values of β some linear function of y that is a biased predictor of w_i will have smaller mean square error of prediction than will BLUP. The difficulty, of course, with using a biased predictor is that since β is unknown, we might actually choose biased functions that have *larger* mean square error than BLUP. Further, most users would, I think, be reluctant deliberately to bias comparisons between different groups, for example to underevaluate young sires as compared to older ones, as was the case during the last years of application of the herdmate method to Northeast data.

Re-writing Model so That Non-observable Random Variables Have Mean Zero

It is more convenient in subsequent sections to deal with non-observable random variables that have mean zero. Consequently we define for the BLUP procedure

$$w = P\beta + u,$$

where $E(u) = 0$,
$Var(u) = G$,
$Cov(y,u) = C$.
Then as before $E(w) = P\beta$,
$Var(w) = G$,
$Cov(y,w) = C$.

Some Important Properties of BLUP

1. In the class of linear unbiased predictors, BLUP maximizes the correlation between the predictor and the predictand.
2. The BLUP of $m'w$ is $m'\hat{w}$, where $\hat{w}$ is BLUP of w.
3. Under normality
   a. $E(u|\hat{u}) = \hat{u}$.
   b. $Var(\hat{u}-u) = Var(u|\hat{u})$.
   c. $\hat{w}$ is ML estimator and best linear unbiased estimator of conditional mean of w.
   d. In the class of linear predictors with means zero, $\hat{u}$ maximizes the probability of a correct ranking of the elements of u.
4. Some variances and covariances

$$Var(\hat{u}) = Cov(\hat{u},u) = C'V^{-1}C - C'V^{-1}X(X'V^{-1}X)^{-}X'V^{-1}C$$

$$Var(\hat{u}-u) = G - Var(\hat{u}), \text{ where } G = Var(u)$$

$$Cov(P\hat{\beta}, \hat{u}) = \text{null}$$

$$Var(P\hat{\beta}) = P(X'V^{-1}X)^{-}P'$$

$$Cov(P\hat{\beta}, u) = P(X'V^{-1}X)^{-}X'V^{-1}C$$

$$Cov(P\hat{\beta}, \hat{u}-u) = -Cov(P\hat{\beta}, u)$$

$$\begin{aligned} Var(\hat{w}-w) &= Var(P\hat{\beta} + \hat{u} - P\beta - u) \\ &= P(X'V^{-1}X)^{-}P' - P(X'V^{-1}X)^{-}X'V^{-1}C - C'V^{-1}X(X'V^{-1}X)^{-}P' \\ &\quad + G - C'V^{-1}C + C'V^{-1}X(X'V^{-1}X)^{-}X'V^{-1}C. \end{aligned}$$

Problems With Mixed Model Selection

By mixed model selection we mean that selection is among candidates that are regarded as a random sample of variables with unknown means not equal to zero. The classical selection index problem assumes that the means of the genetic values and of y are known. Thus selection index has optimum properties. When the mean of y is unknown, but the mean of w is zero, BLUP has optimum properties in the class of linear predictors with mean zero. The maximum probability of a correct ranking is the justification for calling the method optimum. But if the mean of w is not zero, selection then is among elements of $u + P\beta$. I have not been able to prove that BLUP is the optimum procedure for purposes of ranking. If I understand correctly some of the work on Model I selection, it appears that GLS is optimum for ranking elements of β. If that is so, and knowing that BLUP is optimum for ranking elements of u, it may be true that BLUP, which ranks by a combination of the two, may be optimum. This, however, has not been proved.

The Problem of Selection With Large Data Sets and Unequal Subclass Numbers.

The foregoing linear predictions, BLP (selection index) and BLUP (selection index modified for unknown β), are feasible only when the order of V is small enough that it can be inverted by whatever computing facilities are available. In most instances this is likely to be approximately 200. This obviously rules out sire evaluation in the AI era, involving as it does thousands if not millions of records. Fortunately a seemingly very different method can be used in this situation, and in fact it does yield BLUP. This method was originally called Maximum Likelihood, which it is in a certain sense, but it is actually more general than ML because it does not depend upon normality. For purposes of this paper I shall call it the Mixed Model Solution (MMS). Before describing the method, its power is illustrated by its applica-

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tion to sire evaluation in the Northeast. The model assumed is

$$y_{ijk\ell} = h_i + g_j + s_{jk} + e_{ijk\ell},$$

where h refers to herd-year-season, g refers to group, and s refers to sire within group. Only first lactation records of AI progeny are used. Nevertheless in the Holstein breed the magnitude of the problem is indicated by the fact that there are approximately 85,000 herd-year-seasons, 33 groups, 1,400 sires, and 450,000 total records. Thus V is of order 450,000 and a selection index type solution is clearly impossible. But with careful choice of computing strategy the time required to apply MMS is no greater than for the superficially much more simple herdmate method that was first used in the Northeast.

I was first confronted with the problem of multiply classified data with unequal subclass numbers during my Ph.D. program at Iowa State. By that time (1947) the use of least squares for estimation and hypothesis testing had been developed, particularly by Yates (1934), and had been used to some extent in animal breeding research. As was true in the case of selection index development, Iowa State had also pioneered in least squares applications. A. E. Brandt's Ph.D. thesis dealt with the problem (Brandt, 1933) as did an early publication of Snedecor (1934). Later W. G. Cochran's teaching and consulting work at Ames became apparent in this field. Notable examples of this were R. H. Nelson's Ph.D. thesis dealing with the effect of inbreeding on fat yield and the series of publications from the U.S. Sheep Experiment Station at Dubois by Hazel and Terrill (see Hazel and Terrill, 1945). Dr. Cochran had left Iowa State by the time I returned for Ph.D. work, but during my last year Oscar Kempthorne joined the statistics group and greatly influenced my thinking.

My thesis problem dealt with a study of several years' data arising from single crosses between 12 inbred lines of swine. Only about half of the potential reciprocal crosses had been made, the numbers of progeny per cross varied greatly, year effects were obvious, and levels of inbreeding of the dams varied substantially. The problem was (1) to estimate the relative importance of general, specific, and maternal combining abilities, and (2) to predict the merits of the lines for top crossing and the merits of single crosses for commercial application. The first was a variance components estimation problem and motivated the development of my estimation methods, Henderson (1953). The second was a selection index problem with unknown β and in the context of unequal numbers. Originally least squares estimation seemed to me the logical solution to the selection problem. After completing this type of analysis it became obvious to me, as it should have at the outset, that least squares prediction should make little sense to anyone who had studied with Dr. Lush. The problem is that least squares prediction is analagous to evaluation of animals without taking into account the amount of information available on each. The inevitable consequence is that a prediction with large sampling variance is almost certain either to over-evaluate seriously or to under-evaluate seriously the predictand. The only solution to this problem that occurred to me at that time was what I called "regressed least squares". Let $\tilde{u}_i$ = L S prediction of $\tilde{u}_i$. Then the regressed least squares predictor is

$$\frac{\mathrm{Cov}(\tilde{u}_i, u_i)}{\mathrm{Var}(\tilde{u}_i)}\tilde{u}_i.$$

This predictor is not BLUP, and a further problem which I did not appreciate at the time is, that because of frequent non-estimability in least squares, the predictor is not necessarily invariant to the solution to the L S equations. A much better method for this unequal numbers case is described in the next section.

Mixed Model Solution (MMS) to the Unequal Numbers Problem

Due to a lucky misinterpretation of maximum likelihood I discovered in early 1949 a method involving only a slight modification of least squares that is, in fact, BLUP. The idea came from a deceptively simple problem assigned by Mood to his mathematical statistics class. The problem was:

"Given an IQ score of 130, what is the ML estimate of that individual's true IQ? We know that the mean of both true IQ and of IQ test scores assumed normally distributed is 100, the variance of test error is 25, and the variance of true IQ is 225." This seemed to be identical to the problem of finding "most probable producing ability" (MPPA) as taught by Dr. Lush. The answer surely then must be

$$100 + \frac{225}{225 + 25}(130 - 100) = 127.$$

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This indeed was the correct answer but Mood did not accept the methodology. Instead he suggested that we maximize the joint distribution of true IQ and IQ test score for variations in true IQ. Dr. Lush acted surprised and unconvinced when we told him that his MPPA was ML. It turned out that his doubts were justified, for the function maximized was not actually a likelihood function. In a later edition of Mood's book, Mood and Graybill (1963), the problem was completely re-stated. However, in 1949 I used Mood's idea to derive a general method for estimating β and predicting u. The generalization follows:

Let a general mixed linear model be

$$y = X\beta + Z u + e, \qquad (9)$$

where y is the observation vector,

X is a known, fixed, n × p matrix with rank = $r \leq \min(n,p)$,

β is an unknown, fixed vector,

Z is a known, fixed matrix,

u and e are non-observable random vector variables with null means and $\mathrm{Var}\begin{bmatrix} u \\ e \end{bmatrix} = \begin{bmatrix} G & 0 \\ 0 & R \end{bmatrix}$

G and R are known and non-singular. R is an identity matrix times a scalar in most animal breeding applications. G is often a diagonal matrix. Now when (u,e) are normally distributed, the joint density of (y,u) is maximized for variations in β and u by the solution to the following equations

$$\begin{bmatrix} X'R^{-1}X & X'R^{-1}Z \\ Z'R^{-1}X & Z'R^{-1}Z+G^{-1} \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} X'R^{-1}y \\ Z'R^{-1}y \end{bmatrix}. \qquad (10)$$

It was noted that if R=I these are regular LS equations, except for the G^{-1} term, under the assumption that u is fixed. Further, if G is diagonal, the only modification of LS is the addition of variance ratios to the diagonal coefficients of $\hat{u}$. At the time these equations were derived I did not know precisely what were the properties of the solution, but there were certain clues to possible desirable ones. For one thing, it was noted that in simple cases $\hat{u}$ were selection index criteria for u but with $\hat{\beta}$ of (10) substituted for β. For another, it was found that application of the method to the balanced incomplete block design gave the same estimates of treatment differences as Yates' (1940) recovery of inter-block information method. Since it was known that Yates used the principle of best linear unbiased estimation (BLUE), this suggested that MMS yielded BLUE of estimable functions of β.

The first brief publication of MMS was Henderson (1950). Its application to evaluation of linecrosses was reported at the Iowa State Heterosis Conference, Henderson (1952). An application to estimation of intra-herd genetic and environmental trends was presented at meetings of the American Dairy Science Association, Henderson (1949). We also used the method in the early 50's to evaluate AI sires according to a very simple model,

$$y_{ijk} = a_i + s_j + e_{ijk},$$

where a_i refers to year of freshening, and s_j to sire value, regarded as random. These results were used internally only, since daughter-dam comparisons still were regarded as somewhat sacred in the dairy industry.

In 1956 I managed to prove with help from Searle that $\tilde{\beta}$ of (10) is in fact $\hat{\beta}$, a solution to GLS equations. This result does not require normality, but in the normal case $\hat{\beta}$ is ML, as I had originally thought. The formal proof of the properties of $\hat{\beta}$ was published in 1959 by Henderson, Kempthorne, Searle, and von Krosigk (1959). It was also confirmed in 1956 that $\hat{u}$, the solution in (10), is the selection index criterion with $\hat{\beta}$ substituted for β. This result was published by Henderson (1963). We now summarize these results of MMS as well as some others that were discovered later.

1. $K'\hat{\beta}$ is BLUE of the set of estimable linear functions, $K'\beta$.

2. $\hat{u}$ is BLUP with all properties described in the section, "Best Linear Unbiased Prediction".

3. $\hat{u}$ is unique regardless of whether the coefficient matrix of (10) has full rank.

4. $K'\hat{\beta} + M'\hat{u}$ is BLUP of $K'\beta + M'u$ provided $K'\beta$ is estimable.

5. Variances and covariances

Let some g-inverse of the coefficient matrix of (10) be

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}$$

Then $\mathrm{Var}(K'\hat{\beta}) = K'C_{11}K$, assuming that $K'\beta$ is estimable.

$\mathrm{Cov}(K'\hat{\beta}, \hat{u}) = \text{null}$

$\mathrm{Cov}(K'\hat{\beta}, u) = -K'C_{12}$.

$\mathrm{Cov}(K'\hat{\beta}, \hat{u}-u) = K'C_{12}$.

$\mathrm{Var}(\hat{u}) = \mathrm{Cov}(\hat{u},u) = G - C_{22}$.

$\mathrm{Var}(\hat{u}-u) = C_{22}$.

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6. Model for solutions.

It is often informative to express the predictions as linear functions of the elements of the linear model for y. We define these expressions as models for the predictors. The model for $K'\hat{\beta} = K'\beta - K'C_{12}G^{-1}u + K'(C_{11}X' + C_{12}Z')R^{-1}e$, provided $K'\beta$ is estimable. Model for $\hat{u} = (I - C_{22}G^{-1})u + (C_{21}X' + C_{22}Z')R^{-1}e$.

Prediction When Variances Are Unknown

The linear methods described above (BLUP and the MMS method to compute BLUP) assume that β is unknown but that the required variances and covariances are known. In many selection and evaluation situations good estimates are available from prior data, and these estimates can probably legitimately be regarded as parameter values. But it may be true that no prior estimates exist, or even if they do, we may suspect that our population may be different from that for which the estimates were made.

Little success has attended efforts to find optimum selection methods when the variance must be estimated from the same data that are to be used in computing selection criteria. We know that the conditional mean is optimum for selection, but as described in the section on BP, this requires knowledge of the form of the conditional distribution and the numerical values of the parameters of that distribution. We know that in the normal distribution, the conditional mean is

$\hat{w} = \alpha + C'V^{-1}(y - \theta)$, see (3).

In the mixed model and under normality the ML estimate of the conditional mean of w is $\hat{w} = \hat{u} + P\hat{\beta}$, where $\hat{u}$ and $\hat{\beta}$ are solutions to (10), $E(w) = P\beta$, and G, R are known.

Then, because of the invariance property of ML estimates, the substitution of ML estimates, if available, for α, C, V, θ in (3) or ML estimates of G, R in (10) would result in ML estimates of the required conditional means. Unfortunately ML estimation of variance components in the unequal subclass numbers case has been regarded as probably impossible. Hartley and Rao (1967) presented a method, involving very difficult numerical analysis methods, for such estimation.

Due to a suggestion by Glen Ufford, I have recently been able to derive an estimation method based on Hartley and Rao's fundamental work that appears to be computationally feasible for some problems. The method is based on the mixed model equations (10).

The model now will be assumed to contain no correlated variables, and the random variables of the model (except for the error term) refer to classifications and interactions. That is

$$Z = [Z_1 \vdots \cdots \vdots Z_s],$$

$$u' = [u_1' \vdots \cdots \vdots u_s'],$$

$$G = \begin{bmatrix} I\,\sigma_1^2 & & 0 \\ & \ddots & \\ 0 & & I\,\sigma_s^2 \end{bmatrix}$$

$$R = I\,\sigma_e^2 .$$

The number of columns in Z_i = the number of elements in the vector u_i is q_i.

Every row of Z_i (i=1,...,s) contains one "1" and $q_i - 1$ zeros.

Now the problem is to estimate $\sigma_1^2, \ldots, \sigma_s^2, \sigma_e^2$ by ML.

In the variance component model, equations (10) simplify to

$$\begin{bmatrix} X'X & X'Z_1 & \cdots & X'Z_s \\ Z_1'X & Z_1'Z_1 + I\frac{\sigma_e^2}{\sigma_1^2} & \cdots & Z_1'Z_s \\ \vdots & \vdots & & \vdots \\ Z_s'X & Z_s'Z_1 & \cdots & Z_s'Z_s + I\frac{\sigma_e^2}{\sigma_s^2} \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{u}_1 \\ \vdots \\ \hat{u}_s \end{bmatrix} = \begin{bmatrix} X'y \\ Z_1'y \\ \vdots \\ Z_s'y \end{bmatrix} \qquad (10a)$$

Now let

$$\begin{bmatrix} Z_1'Z_1 + I\frac{\sigma_e^2}{\sigma_1^2} & \cdots & Z_1'Z_s \\ \vdots & & \vdots \\ Z_s'Z_1 & \cdots & Z_s'Z_s + I\frac{\sigma_e^2}{\sigma_s^2} \end{bmatrix}^{-1} = \begin{bmatrix} T_{11} & T_{12} & \cdots & T_{1s} \\ T_{12}' & T_{22} & \cdots & T_{2s} \\ \vdots & \vdots & & \vdots \\ T_{1s}' & T_{2s}' & \cdots & T_{ss} \end{bmatrix} \qquad (10b)$$

Then the ML estimates are as follows

$\widehat{K'\beta} = K'\hat{\beta}$, where $\hat{\beta}$ is any solution to (10a) and $K'\beta$ is estimable.

$\hat{\sigma}_e^2 = (y'y - \hat{\beta}'X'y - \hat{u}'Z'y)/n.$

$\hat{\sigma}_i^2 = (\hat{u}_i'\hat{u}_i + \hat{\sigma}_e^2 \operatorname{tr} T_{ii})/q_i.$

These solutions require, of course, an iterative solution. The starting point is to substitute first estimates of σ_e^2/σ_i^2 in (10b) and solve for $\hat{\beta}$, $\hat{u}$. Then these results are used to find second estimates of σ_e^2, σ_i^2, which in turn are substituted in (10b) to find second estimates of $\hat{\beta}$, $\hat{u}$. The process is repeated until the solution converges.

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Selection and Culling Bias

Data used in sire evaluation, and in fact in most animal breeding research and application, are field collected from herds in which culling and selection have been practiced. This must surely raise the question as to whether the estimates and predictions arising from such data will be biased. For example, suppose $n_1 + n_2$ dairy cows make first records in some common environmental situation. Then n_1 of them are chosen to produce second records in a second environmental situation. Assuming that appropriate *a priori* corrections can be made for age, what is the best estimate of the difference between the two environments? Let y_{11}, y_{12} be the means of the n_1 cows, and y_{21} be the mean of the n_2 cows. Then the LS estimate of the difference between environments is $y_{12} - y_{11}$. Knowing, however, that y_{12} is likely to be larger than y_{21} and knowing that repeatability of records is less than one, we should expect y_{12} to be less than y_{11} if the environments were equal. Consequently the estimate of environment 2 − environment 1 is biased downward. Lush and Shrode (1950) discussed this problem in connection with the paired comparison method of age factor estimation. Alan Robertson and I became concerned with this problem when we were at Iowa State and hoped to estimate genetic and environmental trends in Dr. Lush's Holstein herd. We got no further with the problem at that time than to resolve that least squares estimation would give biased estimates, the bias being in the direction of underestimating environmental trend and overestimating genetic trend. When I later discovered the MMS method it occurred to me that this might solve the problem. I did a small scale simulation study using the model,

$$y_{ijk\ell} = g_i + c_{ij} + a_k + e_{ijk\ell},$$

where g referred to genetic group (date of birth grouping, e.g.),

c referred to cows in groups,

a referred to year that lactation was started.

The c and e elements were regarded as normally and independently distributed.

The simulation results did indeed indicate that culling bias was removed by MMS, so this method was reported at the 1949 American Dairy Science Association meetings, Henderson (1949). A more detailed discussion of this problem was published later, Henderson *et al.* (1959). Both of these papers lacked generality in that they did not prove what conditions are required for unbiasedness. A first attempt at generalization was Henderson (1965) in which a rule was stated that insured unbiasedness of estimates of estimable linear functions of β using MMS. Since then methods have been derived for adjusting for bias due to selection not only on records used in the analysis but also selection on the basis of prior knowledge (not based on records in the analysis), and bias due to non-random errors. These results have not been published prior to this paper but were presented in seminars and symposia at Iowa State University, Texas A&M University, University of Nebraska, and the 1971 A.D.S.A. meetings.

Two possible approaches to the selection bias problem have been proposed. One is to suppose that the subclass numbers have some sort of distribution and are correlated with certain random elements of the model. For example, it might be logical to assume that the number of progeny a sire has in AI is positively correlated with his transmitting ability for milk yield. Harville (1968) used this notion in an attempt to eliminate selection bias from variance components estimates. A second approach, which I have found easier to apply, is to define a set of linear functions that describe the selection process. For example, in the culling illustration above, the linear function could be

$$E(y_{11} - y_{21}) \neq 0.$$

An abbreviated description of the method follows.

Suppose that in repeated sampling one of the following inequalities exists.

$$E(L'y) \neq L'X\beta$$
$$\text{or } E(L'u) \neq 0$$
$$\text{or } E(L'e) \neq 0.$$

In the absence of selection these expectations would be $L'X\beta$, 0, 0 respectively.

In each of these cases L is the selection matrix. Note that we do not need to specify the value of the expectation other than to state that it is different from what it would be if there were no selection or culling. Also, given one of the inequalities, the means, variances and covariances are different in the conditional distribution from those stated under (9). If we assume normality, we can write these parameters under the conditional distribution. For example,

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$$E[u|EL'y = s] = GZ'L(L'VL)^{-1}(s - L'X\beta)$$

$$E[y|EL'y = s] = X\beta + VL(L'VL)^{-1}(s - L'X\beta).$$

Next we proceed to derive BLUP of $K'\beta + M'u$ under the conditional distribution, remembering that E(u) will not necessarily be 0 in the conditional distribution. It turns out that only simple modifications of (10) are required.

Selection on y

The BLUP equations when selection is on y are:

$$\begin{bmatrix} X'R^{-1}X & X'R^{-1}Z & X'L \\ Z'R^{-1}X & Z'R^{-1}Z+G^{-1} & 0 \\ L'X & 0 & L'VL \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{u} \\ \hat{l} \end{bmatrix} = \begin{bmatrix} X'R^{-1}y \\ Z'R^{-1}y \\ L'y \end{bmatrix} \quad (11)$$

Note that if $L'X$ is null, the solution to $\hat{\beta}$, u is the same as in the no selection case. The condition, $L'X$ null, implies selection within levels of fixed effects. When $L'X \neq 0$, it is not necessarily true that u is predictable, for remember that $E(u) \neq 0$ in the conditional model. To test for predictability note that if [X:VL] is not full column rank, it is always true that T_1 and T_2 can be found such that

$$[X:VL]\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \text{null matrix}$$

where the number of cols. in $\begin{bmatrix} T_1 \\ T_2 \end{bmatrix}$ is the number of cols. of [X:VL] minus the rank of that matrix. Then for $k'\beta + m'u$ to be predictable it must be true that

$$[k' : m'GZ'L]\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \text{null vector.}$$

Selection on u

If selection is on u, that is if $E(L'u) \neq 0$, the following modifications of MMS (10) are required to obtain BLUP.

$$\begin{bmatrix} X'R^{-1}X & X'R^{-1} & 0 \\ Z'R^{-1}X & Z'R^{-1}Z+G^{-1} & -L \\ 0 & -L & L'GL \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{u} \\ \hat{f} \end{bmatrix} = \begin{bmatrix} X'R^{-1}y \\ Z'R^{-1}y \\ 0 \end{bmatrix} \quad (12)$$

No obvious form of L would result in regular MMS giving unbiased predictions. To test for predictability of $k'\beta + m'u$ note that if [X:ZGL] is not full column rank, it is always true that T_1, T_2 can be found such that

$$(X:ZGL)\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \text{null matrix.}$$

Then for predictability of $k'\beta + m'u$ it must be true that

$$(k' : m'GL)\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \text{null vector.}$$

If L is unknown, but assumed to exist, or if the modified equations are too difficult, we can sometimes obtain unbiased, but not minimum variance predictors by proceeding as though u were fixed. That is, solve,

$$\begin{bmatrix} X'R^{-1}X & X'R^{-1}Z \\ Z'R^{-1}X & Z'R^{-1}Z \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} X'R^{-1}y \\ Z'R^{-1}y \end{bmatrix} \quad (13)$$

Then $k'\hat{\beta} + m'u$ is an unbiased predictor of $k'\beta + m'u$ if the following conditions hold.

$$[X : Z]\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \text{null matrix.}$$

$$[k' : m']\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \text{null vector.}$$

A better method than (13) can be used if

$u' = [u_1' : u_2']$,

selection is on u_1, and

$\text{Cov}(u_1, u_2) = \text{null}$.

Now we solve (14) for $\hat{\beta}, \hat{u}_1, \hat{u}_2$.

$$\begin{bmatrix} X'R^{-1}X & X'R^{-1}Z_1 & X'R^{-1}Z_2 \\ Z_1'R^{-1}X & Z_1'R^{-1}Z_1 & Z_1'R^{-1}Z_2 \\ Z_2'R^{-1}X & Z_2'R^{-1}Z_1 & Z_2'R^{-1}Z_2+G_{22}^{-1} \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{u}_1 \\ \hat{u}_2 \end{bmatrix} = \begin{bmatrix} X'R^{-1}y \\ Z_1'R^{-1}y \\ Z_2'R^{-1}y \end{bmatrix} \quad (14)$$

Then $k'\hat{\beta} + m_1'\hat{u}_1 + m_2'\hat{u}_2$ is an unbiased predictor of $k'\beta + m_1'u_1 + m_2'u_2$ if

$$[X : Z_1]\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \text{null matrix,}$$

and $$[k' : m_1']\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \text{null vector.}$$

This means that any function of u_2 alone can be predicted. Examples of selection on y and

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on u are presented in the section, "Application of MMS to Sire Evaluation".

Selection on e

If $L'e \neq 0$, the modifications of (10) required for BLUP are

$$\begin{bmatrix} X'R^{-1}X & X'R^{-1}Z & X'L \\ Z'R^{-1}X & Z'R^{-1}Z+G^{-1} & Z'L \\ L'X & L'Z & L'RL \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{u} \\ \hat{\gamma} \end{bmatrix} = \begin{bmatrix} X'R^{-1}y \\ Z'R^{-1}y \\ L'y \end{bmatrix} . \quad (15)$$

In this type of selection MMS (10) would give unbiased predictors only if selection were within levels of both fixed and random effects, which would result in $L'X$, $L'Z$ both equal to null matrices. The condition required for predictability is

$k'\hat{\beta} + m'\hat{u}$ is a predictor of $k'\beta + m'u$ if

$$[X : RL] \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \text{null matrix, and } [k' : m'0] \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \text{null vector.}$$

Application of Mixed Model Solution (MMS) to Sire Evaluation

It seems safe to conclude that most sire evaluation problems at present, and certainly those of the future, involve field collected data with all of the problems inherent in such data, including (1) multiply-classified with unequal subclass numbers, (2) records subject to culling and selection, (3) unknown, fixed factors, (4) sires from different subpopulations. The preceding sections represent, I believe, most of what is now known concerning prediction for purposes of selection. If this is true, it becomes clear that prediction by best linear unbiased procedures is the method of choice. Further, the mixed model solution provides a powerful computing tool that is straightforward and relatively easy to apply in the modern computing field. It is logical to assume that computers of the near future will be much faster than the present ones, and that some remarkable developments in expanded computer memory are almost certain. If we do agree that BLUP is the logical choice of a sire evaluation method, this is only the starting point, for two key questions must be answered.

1. What data should be used?
2. What linear model should be assumed?

Another still more important question also must be considered in the long run, and this is what data should be collected in the first place. We shall assume here that this last question has been decided for us, at least with respect to how we evaluate sires today with records already available. Once the two questions above have been answered, the computations to be done follow immediately from MMS. I shall make no attempt here to state what is a reasonable model, but rather will give a few examples to illustrate the flexibility of the method. It should be pointed out, however, that the data chosen partially determine the model, and conversely, the model chosen may to some extent influence what data should be used. It also needs to be stated that the more complete is the model the less are the chances for bias to result, but adding unimportant elements to the model results in greater computational cost and larger sampling variances. My philosophy in general would be to use as simple a model as possible that will account for the important sources of variation.

The Herdmate Method as Approximate BLUP

It is interesting in retrospect to see what model and data set to which BLUP is applied yield the same results as the herdmate method. In fact, when the method was proposed, Henderson, Carter, and Godfrey (1954), most of the principles of BLUP were already available, but computing facilities were totally inadequate to utilize the method. The herdmate method was a compromise that was computationally feasible. For purposes of this discussion we shall restrict ourselves to use of only first lactation records on the sires to be evaluated, AI only from the outset in the Cornell system. Let the model for the kth AI progeny record of the jth sire in the ith herd be

$$y_{ijk} = \mu + \frac{1}{2} g_i + v_i + s_j + e_{ijk}$$

Let the model for the jth herdmate record in the ith herd be

$$x_{ij} = \mu + g_i + v_i + \epsilon_{ij}$$

g represents herd genetic value,
v represents herd environmental value,
s represents sire.

All elements of the two models except μ are regarded as random variables with mean zero and mutually uncorrelated. The relative magnitudes of the variances are

$$\sigma_g^2 = .1,\ \sigma_v^2 = .4,\ \sigma_s^2 + \sigma_e^2 = .575,\ \sigma_\epsilon^2 = .5$$

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$\sigma_g^2/(\sigma_g^2 + \sigma_w^2) = .2$ = intra-herd heritability and this implies under certain assumptions an intra-sire regression of daughter on true herdmate average = .9, Mao, Henderson and Miller (1972), which was our estimate of this parameter, and an adjustment that was used in our herdmate method.

$\sigma_e^2/(\sigma_g^2 + \sigma_w^2) = 1$, which was our estimate when we first used the herdmate method, and which was used in the adjustment of herdmate average for numbers. Later estimates indicate that this ratio is actually considerably greater than 1.

Now if we assume that each AI progeny has only non-AI herdmates and apply BLUP procedures, the prediction of a sire based on one progeny only is

$$k\left(y_{ijk} - \frac{.9n}{n+1} x_{ij} - \frac{.1n+1}{n+1} \mu\right),$$

where k depends upon n and σ_s^2, and n = number of herdmates. Note that the $.9n/(n+1)$ weight on x is the product of the $n/(n+1)$ adjustment of herdmates for numbers and .9, the intra-sire regression of progeny on herdmates. Then if we average equally for each sire all the quantities in the parentheses and adjust for number of progeny we have essentially the herdmate evaluation. A better prediction would be to weight the individual deviations, and the appropriate weighting would depend upon the n's. This was considered too laborious when we started our herdmate evaluations using only a tabulator and a card calculating punch (IBM 602A). The point to this discussion of the herdmate method is to emphasize the rather drastic limitations of the model. Some of its deficiencies in the present situation are:

1. Many AI progeny in some herds.
2. The sires are not randomly assigned to herds.
3. The sires are not all from the same population, as is implied by adjustment of all of them to a common mean.
4. The herd-year-season effects are not uncorrelated. Genetic or environmental trend would invalidate this assumption.

The Northeastern Sire Comparison Method

The next illustration of application of BLUP to sire evaluation is the Northeastern Sire Comparison, not because we claim that it has an ideal model or data set, but rather because it has been used now for more than two years and consequently we know a good deal about its behavior.

By 1966 problems with the herdmate method as used in the Northeast were becoming apparent. The two most serious ones were (1) evaluations of sires that continued in service over a long period of time after being returned to service following a sampling test generally tended to decline, and (2) sires of later generations appeared to be underevaluated relative to older sires. Both of these difficulties could be explained by an upward trend in the genetic merit of herdmates. A method for evaluating sires that would eliminate this source of bias as well as certain others was reported by Henderson (1966). This involved the application of MMS to a rather elaborate model. Paul Miller tried out this method in his Ph.D. thesis at Cornell and did indeed confirm that genetic trend was one of the apparent causes of the difficulty with the herdmate method. The computational problems associated with my 1966 suggestion were, however, too great for routine application on our computing facilities. Consequently Miller, Lentz and I designed a more simple model, and this has proved to be quite feasible on our IBM 360/model 40. In fact, the time required is less than our old herdmate method, the major time required for the latter being maintenance of the herdmate file.

The model used is

$$y_{ijk\ell} = h_i + g_j + s_{jk} + e_{ijk\ell}$$

The records are first lactation records produced by AI progeny and adjusted for the usual factors such as age and number of days in milk (if the cow leaves the herd still in milk and prior to 305 days production).

h refers to herd-year-season.

g refers to group, where the group is defined as a set of sires from the same stud and entering service during the same period of time.

s refers to sires within a group. The breeding values of sires in the same group are regarded as a random sample from the same population. The e's are regarded as random variables mutually uncorrelated and uncorrelated with any other variables in the model.

Although it is logical to regard the h_i as random variables, we carry out the computations as though they were fixed. The reason for this is that we suspect that

$$\sum_i h_i n_{ijk}/n_{.jk}$$

differs from one sire to another. That is, the average merit of herds associated with sires

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differs. This represents L'u type of selection, and the bias due to it can be eliminated by computing as though h were fixed. When herd size is relatively large and/or σ_e^2/σ_h^2 is relatively small, the increase in prediction error variance is small due to this strategy. But in situations with small herd size I would be inclined to accept some bias in order to reduce variance.

The computing strategy for our procedure has been developed to the extent that we have a very fast operating program. Some of the features are

1. Group equations are ignored during the input of herd data.
2. h (herd-year-season) is absorbed at the end of each individual herd-year-season and the non-zero only coefficients and right hand sides contributed by that HYS are written on tape.
3. At the end of a breed, L S equations involving sires only are formed by sorting and summing the tape output.
4. Group equations and coefficients are formed by adding appropriate sire coefficients and right hand sides.
5. σ_e^2/σ_s^2 is added to the diagonal coefficients of the sire equations.
6. A Lagrange multiplier equation is added.
7. The equations are solved by Gauss-Seidel iteration using as input only non-zero coefficients. Advantage is taken of the facts that

$$\sum_k \hat{s}_{jk} = 0, \text{ for all } j \ ,$$

and the Lagrange multiplier solution is 0.

8. The evaluation of the jk^{th} sire is

$$\hat{g}_j + \hat{s}_{jk}.$$

The number of sires to be evaluated may be so large that the foregoing method is not feasible, for example, if natural service sires are included. We have designed a strategy for this situation that has been tried and found to be very efficient. It involves absorbing natural service sires into their respective groups and into HYS (or in herds if one is willing to use additional computer time). Then HYS (or herd) is absorbed into AI sires and NS groups. The data on HYS (or H) and NS sires are saved. At the end of the breed data input, the group equations for AI sires are formed, a Lagrange multiplier is added, and the resulting equations are solved. This leads to almost BLUP of AI sires and groups and to a good approximation to NS groups. Then by a back solution the one or more pieces of data for NS sires are corrected for HYS (or herd) and NS group. Then these are combined and NS sire variables predicted by

$$\text{adj. sum}/\left(n + \frac{\sigma_e^2}{\sigma_s^2}\right), \text{ where n = number of progeny.}$$

Evaluation Using Repeated Records

It may be desirable in some dairying regions to use all records on the daughters. This is true particularly if herd size is small and restriction to first records will severely limit the number of herdmate records available. We assume the simple repeatability model, that is

$$\text{Var}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \end{bmatrix} = \begin{bmatrix} 1 & r & r & \cdots \\ r & 1 & r & \cdots \\ r & r & 1 & \cdots \\ \vdots & \vdots & \vdots & \end{bmatrix} \sigma_y^2 \ ,$$

where y_1, y_2, etc. refer to records on the same cow, and r = repeatability.

An expansion of the Northeastern model for this situation is straightforward.

$$y_{ijk\ell m} = h_{ij} + g_k + s_{k\ell} + p_{ik\ell m} + e_{ijk\ell m}$$

h_{ij} refers to the jth year-season in the ith herd

g, s are as before

$p_{ik\ell m}$ refers to the r.p.a. $= g_k + s_{k\ell}$ of the mth daughter of the $k\ell^{th}$ sire in the ith herd.

s, p, e are regarded as independent random variables.

Under a simple additive genetic and repeatability model

$$\sigma_e^2 = (1-r)\,\sigma_y^2, \ \sigma_s^2 = \tfrac{1}{2} h^2 \sigma_y^2,$$

$$\sigma_p^2 = (r - \tfrac{1}{4}h^2)\sigma_y^2$$

Assuming that we absorb no sires, the data are entered sorted by cows within herd. At the end of data entry for each cow absorb that into sires and HYS using as the diagonal coefficient $n + \sigma_e^2/\sigma_p^2$, where n is the number of records on the cow. At the end of the herd absorb, HYS. This will require inverting a non-diagonal matrix, the coefficients of HYS in the HYS equations. The order of the matrix is the number of YS in that herd. Then proceed as in the Northeastern method. If the predicted breeding values of cows are wanted, they can be computed by

$$\hat{g}_k + \hat{s}_{k\ell} + \frac{\frac{3}{4}h^2}{r - \frac{1}{4}h^2}\,\hat{p}_{ik\ell m}$$

The repeated records model can also be used in the absorbed sire method. This is now being

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programmed at Cornell to enable BLUP to be used in the natural service sire and repeated records case.

Multiple Trait Evaluation of Sires

For purposes of illustration of sire evaluation for multiple traits we shall use an expansion of the NE model for one trait.

$$y_{ijk\ell m} = h_{ij} + g_{ik} + s_{ik\ell} + e_{ijk\ell m}$$

The i subscript denotes trait number.

h, g, s refer to HYS, group, and sire respectively.

We compute as though HYS were fixed.

$$\text{Var}\begin{bmatrix} s_{1k\ell} \\ s_{2k\ell} \\ \vdots \end{bmatrix} = G_s, \qquad \text{Var}\begin{bmatrix} e_{1jk\ell m} \\ e_{2jk\ell m} \\ \vdots \end{bmatrix} = R_p .$$

All other covariances are zero. Under a simple, additively genetic model,

$G_s = \frac{1}{4}$ genetic variance-covariance of traits.

R_p = phenotypic variance covariance matrix minus G_s.

Then G and R of (10) are block diagonal matrices. For purposes of computing the coefficient matrix, R^{-1} is needed. Also if some progeny have data missing on some traits, subsets of R must be inverted. For example, if the number of traits is 3, possible matrices needed are

$$R^{-1} = \begin{bmatrix} \begin{pmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{pmatrix}^{-1} & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} p_{11} & 0 & p_{13} \\ 0 & 0 & 0 \\ p_{31} & 0 & p_{33} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & \begin{pmatrix} r_{22} & r_{23} \\ r_{32} & r_{33} \end{pmatrix}^{-1} \\ 0 & \end{bmatrix}$$

$$\begin{bmatrix} r_{11}^{-1} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & r_{22}^{-1} & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & r_{33}^{-1} \end{bmatrix}, \quad \text{where}$$

$$\begin{bmatrix} p_{11} & p_{13} \\ p_{31} & p_{33} \end{bmatrix} = \begin{bmatrix} r_{11} & r_{13} \\ r_{31} & r_{33} \end{bmatrix}^{-1}$$ The matrices

correspond to cows with records on all traits, traits 1 and 2, ..., trait 3 only. Define these matrices as P_{123}, P_{12}, ..., P_3. Now suppose a sire has a daughter with records on traits 1 and 3 in a particular HYS. Call these records y_1, y_3. Then the contribution of these records to the coefficients and RHS of the mixed model equations is

$$P_{13}\begin{bmatrix} s_{1k\ell} \\ 0 \\ s_{2k\ell} \end{bmatrix}, \quad P_{13}\begin{bmatrix} y_1 \\ 0 \\ y_3 \end{bmatrix}$$

Sum the data for all sires in a HYS. Let the coefficient matrix for sire x sire be S_α and the RHS be t_α. Then we have as the partial coefficient matrix and RHS

$$\begin{bmatrix} S_1 & & & 0 & S_1 \\ & S_2 & & & S_2 \\ & & \ddots & & \vdots \\ 0 & & & & \vdots \\ & & & S_n & S_n \\ S_1 & S_2 & \cdots & S_n & \Sigma S \end{bmatrix}, \quad \begin{bmatrix} t_1 \\ t_2 \\ \vdots \\ t_n \\ \Sigma t \end{bmatrix}$$

Now absorb HYS. Then the ith diagonal submatrices are

$$S_i - S_i(\Sigma S)^{-1} S_i,$$

the ij^{th} off-diagonal submatrix is

$$-S_i(\Sigma S)^{-1} S_j, \text{ and the } i^{th} \text{ RHS is}$$

$$t_i - S_i(\Sigma S)^{-1}(\Sigma t).$$

At the end of data accumulation we have three equations for each sire. Now sum these to obtain 3 equations for each group. Finally add G_s^{-1} to the diagonal submatrices of sire equations and solve. If there are no data on say the third trait for a particular sire, the equation for the third trait for that sire will be simply

$$q'\begin{bmatrix} s_{1k\ell} \\ s_{2k\ell} \\ s_{3k\ell} \end{bmatrix} = 0, \text{ where } q' \text{ is the third row of } G^{-1}.$$

This yields BLUP of the third trait of that sire.

If no data on say the third trait are available for an entire group, we cannot estimate that group's value for the third trait, and it must be recognized that the rank of the coefficient matrix of the equations to be solved is reduced by one.

It should be clear that the above method for multiple trait evaluation applies to a repeated records model in which the variance of first, second, etc. records are unequal, and the covariances between different pairs of records are unequal. That is, the simple repeatability

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model is not valid. Then first record is trait 1, the second is trait 2, etc.

The predicted aggregate genetic merit now is

$$\sum_i a_i (\hat{g}_{ik} + \hat{s}_{ik}),$$

where a_i are economic values for the different traits.

Accounting for Relationships Among Sires to be Evaluated

Some increase in accuracy of evaluation can be effected by taking into account the relationships among sires. In fact, if it can be assumed that all of these sires are out of random sample of dams from that same population, we would not need to do any grouping. In that case the Northeastern Sire Comparison model would be

$$y_{ijk} = h_i + s_j + e_{ijk}.$$

After absorbing HYS the equations could be written

$$P\hat{s} = r.$$

P will have rank one less than its order. The vector, s, refers to sires. Now modify these equations as follows

$$(P + \frac{\sigma_e^2}{\sigma_s^2} A^{-1})\hat{s} = r,$$

where A is the numerator relationship matrix of sires. Then $\hat{s}$ is BLUP of the sires to be evaluated.

Using Records of Progeny of Daughters of the Sire to be Evaluated

It is possible to expand all of the above methods to include progeny records of the daughters of the sires to be evaluated. This would mean also that these maternal grandsires would be evaluated by both their progeny and their grandprogeny. We illustrate the method in the repeated records model, which now becomes for a progeny record with known maternal grandsire,

$$y_{ijk\ell mno} = h_{ij} + \frac{1}{2} g_k + \frac{1}{2} s_{k\ell} + g_m + s_{mn} + p_{ik\ell mno} + e_{ijk\ell mno}$$

g_k refers to group of grandsire and g_m to group of sire. Under a simple additively genetic and repeatability model

$$\sigma_s^2 = \frac{1}{4} h^2 \sigma_y^2$$

$$\sigma_p^2 = (r - \frac{5}{16} h^2)\sigma_y^2$$

$$\sigma_e^2 = (1-r)\sigma_y^2.$$

The model for a progeny record with maternal grandsire unknown is the same as described in the section, "Evaluation Using Repeated Records". Note that σ_p^2 in the latter model is $(r - h^2/4)\sigma_y^2$. This difference in σ_p^2 would result in a slightly different addition to the diagonal of daughter equations in the absorption of daughters. The difference is so small that probably it is not worth taking into account. If only first records are used, p and e are combined, and the variance of this sum is $(1 - 5\,h^2/16)\sigma_y^2$ under a simple model.

Inclusion of c^2 in the Model

It is thought by some that it is important in sire evaluation to take into account the possibility that progeny of the same sire in the same environmental subset (herd, e.g.) have a common environmental component. If such components can be regarded as having a distribution with known second moments, this can be incorporated into the MMS procedures. As a simple example let us suppose that in the N E model there is an environmental component that is common to a sire's first lactation progeny in the same herd. A model for this could be

$$y_{ijk\ell m} = h_{ij} + g_k + s_{k\ell} + e_{ik\ell} + e_{ijk\ell m}$$

i refers to herd, j to yr-season

$e_{ik\ell}$ is a random component affecting records of all progeny of the k, ℓ^{th} sire in the ith herd. This implies that the environmental correlation is the same for half-sibs in different year-seasons as those in the same year-seasons. This, of course, is not necessarily a valid assumption. The e term could be regarded as an interaction.

Now we proceed as in the NE method. That is, we absorb HYS at the end of each such data set, but with this difference. We accumulate the sire coefficients and right hand sides over all YS of the herd. The resulting equations are then

$$Q\,s = r.$$

Now, taking c into account, the equations are

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$$\begin{bmatrix} Q & Q \\ Q & Q+\frac{\sigma_e^2}{\sigma_c^2} I \end{bmatrix} \begin{bmatrix} s \\ c \end{bmatrix} = \begin{bmatrix} r \\ r \end{bmatrix}$$

Next absorb c to obtain coefficients and RHS for sires,

$$[I - Q(Q+\frac{\sigma_e^2}{\sigma_c^2})^{-1}]\, Q,$$

and $[I - Q(Q+\frac{\sigma_e^2}{\sigma_c^2})^{-1}]\, r$, respectively.

This coefficient matrix and r.h.s. are written on tape and we proceed as in the NE method.

Evaluation Using All Records

This method should give smaller prediction errors than any other because it utilizes all relationships. However, it is suitable only for small numbers of animals. Its greatest usefulness is probably for closed herds.

Let the model for the records on a single trait be

$$y = X\beta + Zp + e$$

p is a vector variable of real producing abilities.

e is a vector variable of temporary environmental errors.

$\text{Var}(p) = B\,\sigma_e^2$,

where $B = \dfrac{h^2 A_{11} + (r-h^2)I}{1-r}$,

h^2 = heritability, r = repeatability, and A_{11} is the numerator relationship matrix of animals producing the records.

σ_e^2 is the variance of every element of e = $(1-r)\sigma_y^2$.

This model assumes that the only cause of covariance between records of related animals is additive genetic variance. It also assumes that variance of unselected first records is the same as that for second, etc. Also it assumes no adjacency effects between records on the same animal. Other more complicated models would require more elaborate techniques. If there are some animals with two or more records, set up and solve these equations.

$$\begin{bmatrix} X'X & X'ZB \\ BZ'X & BZ'ZB+B \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ v \end{bmatrix} = \begin{bmatrix} X'y \\ BZ'y \end{bmatrix}$$

Thus $K'\hat{\beta}$ is BLUE of $K'\beta$, provided this function is estimable.

BLUP of p(r.p.a.) = $\hat{p} = B\,v$.

BLUP of a.g.v. of animals making records is

$$\frac{h^2}{1-r} A_{11}\, v.$$

BLUP of a.g.v. of animals not making records but related to those that did make records, sires or potential sires, for example, is

$$\frac{h^2}{1-r} A'_{12}\, v,$$

where A'_{12} is numerator relationship between animals not making records and those that did make records.

If there are only single unselected records used and we are interested only in predicting additive genetic values, solve these equations

$$\begin{bmatrix} X'X & X'S_{11} \\ S_{11}X & S_{11}^2 + S_{11} \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ w \end{bmatrix} = \begin{bmatrix} X'y \\ S_{11}y \end{bmatrix}$$

where $S = h^2 A/(1-h^2)$, and A is numerator relationship matrix.

Then BLUP of a.g.v. of animals with records is

$$S_{11}\, w,$$

and for animals without records is

$$S'_{12}\, w.$$

It appears that this method could be used to obtain joint estimates of environmental trend, genetic trend, and age effects. Previous methods, except one suggested by Henderson (1966) that used the dams to measure genetic trend of the females' contributions, could estimate satisfactorily only two of the three factors: genetic trend, environmental trend, and age. In the present method, time periods and ages could be included in β, and the predictions of a.g.v. ordered by date of birth could be used to estimate genetic trend.

Some Potential Biases in Sire Evaluation

Since most data available for sire evaluation do not arise from well designed experiments but rather from field data, many potential

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hazards suggest themselves and probably others have not even been thought about. Some of them will now be appraised and suggestions made concerning eliminating bias due to them.

Failure to Take Into Account Progeny Numbers Varying from One Sire to Another

At least in AI sire evaluation the numbers of tested progeny differ markedly, ranging from 1 to more than 10,000. Properly accounting for these differences is a property of selection index and not doing so is a property of least squares. Consequences of not adjusting include:

a. Prediction error variances are needlessly large. To illustrate, let the model be a very simple one

$$y_{ij} = s_i + e_{ij},$$

with $\sigma_e^2/\sigma_s^2 = \alpha$.

The L.S. prediction is $\tilde{s}_i = \bar{y}_{i.}$. This has prediction error variance σ_e^2/n. In contrast, the selection index prediction is $n\bar{y}_{i.}/(n+\alpha)$ with prediction error variance $\sigma_e^2/(n+\alpha)$. If $\alpha = 15$, the range of variances when n goes from 1 to ∞ is from σ_e^2 to 0 in L.S. and from $\sigma_e^2/16$ to 0 for selection index.

b. Sires with the very highest evaluations are most likely to be those with small numbers of progeny. Consequently, with high selection intensity sires with small numbers have too high probability of being selected. These results are a consequence of the variances of the *predictors* (in contrast to variance of prediction errors described above). The variance for the L.S. predictor is $(n+\alpha)\sigma_e^2/n\alpha$ and for S.I. predictor is $n\sigma_e^2/\alpha(n+\alpha)$. The range of values as n goes from 1 to ∞ when $\alpha = 15$ is $16\sigma_e^2/15$ to $\sigma_e^2/15$ for L.S. and $\sigma_e^2/240$ to $\sigma_e^2/15$ for S.I. Note that the variance of the predictor decreases with increasing n in L.S., but increases in BLUP.

c. Biases in estimates of fixed effects can result.

Correlation between Initial Progeny Test and Future Progeny Numbers

No biases result if BLUP is used, but estimates are biased if L.S. is used.

Correlation between Culling Level and Mean of First Records

If progeny groups having high first records are culled less intensively than groups having low first records, biases result depending upon how multiple records are used.

a. If only first records are used, no bias results.

b. If evaluation is based on mean of all records ignoring what daughter made each record, low first record sires are over-evaluated relative to high.

c. If evaluation is based on mean of progeny means, high first record sires are over-evaluated relative to low.

d. If BLUP techniques are used, high and low first record sires are compared fairly. The present U.S.D.A. adjustment for number of records also eliminates bias due to culling.

Average Level of Herd Effects Associated with Sire A is Higher than the Average Associated with Sire B

If herds are regarded as random and BLUP techniques are used, sire A is over-evaluated relative to sire B. If BLUP computation is carried out regarding herds as fixed, no bias results.

Sires with Similar Initial Progeny Tests Tend to be Associated in the Same Herds to a Greater Extent Than Sires With Dissimilar Initial Proofs

If BLUP is used, no bias results from this selection on sires provided their initial proofs are used in the evaluation. This is equivalent to L'y selection with L'X = null. If initial proofs are not used, this is equivalent to selection of L'u type. This is discussed in the next section.

Some Knowledge of Values of Sires Exists External to Records to be Used in the Evaluation

Based on this knowledge similar sires tend to be associated in the same herds. Some ways of eliminating this source of bias are:

a. Use BLUP modified for L'u selection provided L' can be written.

b. If sires can be assigned logically to two or more fixed groups based on this prior knowledge, bias is reduced or eliminated by BLUP.

c. If a non-diagonal variance-covariance matrix of the s_i can be written, use this in the BLUP equations.

d. If no knowledge of how the selection was made exists, L.S. eliminates this bias, but introduces other undesirable features already described. This is rather a measure of last resort.

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Sires to be Evaluated Cannot Logically be Regarded as a Random Sample From a Common Population

The herdmate method, recently discontinued at Cornell, and a similar method used by USDA make this assumption. Some reasons why this assumption proved false in the Northeast and probably would if tested elsewhere, are:

a. Sires of different generations in a situation in which genetic progress is occurring are not from the same population.

b. Contemporary sires of different studs differ because of differing selection practices.

c. Sires used in most organized AI studs are markedly superior to those used in natural service.

Our most recent estimate in the Northeast indicates that the most recently tested Holstein sire of a large stud have progeny approximately 800 lb. above contemporary progeny of NS sires. When we first studied this in the early 50's, the difference was 0. Both populations have increased, but AI much faster.

A sire with many progeny is little affected by failure to group or by being placed in the wrong group for his evaluation approaches his own progeny mean adjusted for herd effects.

Estimation of Genetic and Environmental Trends

Using any of the models of the preceding section, the resulting estimates can be used to estimate genetic trends.

Assume that the grouping is done by time periods within some other classification, e.g., studs and NS or for all AI *vs.* all NS. Then within each stud the trend in $\hat{g}_i$ is an estimate of the trend in genetic merit of the unweighted average of the sires of that subpopulation. An estimate of the trend in merit of the resulting progeny attributable to sires would be $\sum_i \sum_j n_{ijk}(\hat{g}_i+\hat{s}_{ij})/n_{..k}$ for all k, where the k subscript denotes some period of time.

The MMS method provides a powerful tool for estimating environmental and genetic trends. The mixed linear model is completely general with respect to what terms can be included. It also permits a judicious choice of which elements are fixed and which are random. Finally it provides a technique for determining whether selection and culling will bias the estimates. This last feature is particularly important because it would be in a selection situation that one would be particularly interested in estimating trends.

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APPENDIX A
Numerical Examples of General Prediction Methods

Simple examples are presented of the basic prediction methods presented in the foregoing report, namely, BP, BLP, BLUP, and the Mixed Model Solution to BLUP.

BP

Suppose $y' = [y_1 \; y_2]$ and w, a scalar variable to be predicted, have the joint density function,

$$f(y,w) = 2(ay_1 + by_2 + cw)/(a+b+c),$$

where y_1, y_2, w all are in the range 0 to 1.
Then BP of w is E(w|y).
The conditional distribution of u is

$$h(w|y) = f(y,w)/g(y).$$

Now $g(y) = \int_0^1 f(y,w)\,dw$

$$= 2(ay_1 + by_2 + c/2)/(a+b+c).$$

Consequently $h(w|y) = (ay_1 + by_2 + cw)/(ay_1 + by_2 + c/2)$,

and $\hat{w} = E(w|y) = \int_0^1 wh(w|y)dw$

$$= (3\,ay_1 + 3\,by_2 + 2c)/(6\,ay_1 + 6\,by_2 + 3c).$$

This clearly is not a linear index. Also in order to compute BP the values of a, b, c must be known.

BLP

Suppose we have the following data on 5 sires, the first 2 of which are members of group 1 and the last 3 are members of group 2. Further, progeny are located in 2 different herds.

Number of progeny

	Group 1		Group 2		
Herd \ Sire	1	2	3	4	5
1	5	4	0	10	1
2	0	2	5	1	5

Progeny totals

51	36	–	74	9
–	25	31	10	54

The linear model is assumed to be

$$y_{ijk\ell} = h_i + g_j + s_{jk} + e_{ijk\ell}$$

Suppose we have the following parameter values.

$$\mathrm{Var}(s) = .8\,I, \quad \mathrm{Var}(e) = 8\,I$$
$$[h_1, h_2, g_1, g_2] = [2, 0, 7, 9]$$

To find BLP we can use subclass means rather than individual observations, thus reducing the number of "observations" from 33 to 8. The variance of a subclass mean is $.8 + 8/n_{ijk}$. The covariance between subclass means relating to the same sire is .8. All others are zero. Using these rules:

$$V = \begin{bmatrix} 2.4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2.8 & 0 & 0 & .8 & 0 & 0 & 0 \\ 0 & 0 & 1.6 & 0 & 0 & 0 & .8 & 0 \\ 0 & 0 & 0 & 8.8 & 0 & 0 & 0 & .8 \\ 0 & .8 & 0 & 0 & 4.8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2.4 & 0 & 0 \\ 0 & 0 & .8 & 0 & 0 & 0 & 8.8 & 0 \\ 0 & 0 & 0 & .8 & 0 & 0 & 0 & 2.4 \end{bmatrix}$$

Suppose we want to predict

$$g_i + s_{ij} + \frac{1}{2}(h_1 + h_2)$$

Then the covariance between a subclass mean and the s pertaining to that subclass is .8 and all other covariances are 0.
Using this rule:

$$C = \begin{bmatrix} .8 & 0 & 0 & 0 & 0 \\ 0 & .8 & 0 & 0 & 0 \\ 0 & 0 & 0 & .8 & 0 \\ 0 & 0 & 0 & 0 & .8 \\ 0 & .8 & 0 & 0 & 0 \\ 0 & 0 & .8 & 0 & 0 \\ 0 & 0 & 0 & .8 & 0 \\ 0 & 0 & 0 & 0 & .8 \end{bmatrix}$$

Then the weights to apply to $\bar{y}_{ijk\cdot} - g_i - h_j$ according to (3) are

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$$C'V^{-1} = \begin{bmatrix} .33333 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & .25 & 0 & 0 & .125 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & .33333 & 0 & 0 \\ 0 & 0 & .47619 & 0 & 0 & 0 & .04762 & 0 \\ 0 & 0 & 0 & .0625 & 0 & 0 & 0 & .3125 \end{bmatrix}$$

$$\bar{y}_{ijk} - g_i - h_j = \begin{pmatrix} 10.2 \\ 9 \\ 7.4 \\ 9 \\ 12.5 \\ 6.2 \\ 10 \\ 10.8 \end{pmatrix} - \begin{pmatrix} 9 \\ 9 \\ 11 \\ 11 \\ 7 \\ 9 \\ 9 \\ 9 \end{pmatrix} = \begin{pmatrix} 1.2 \\ 0 \\ -3.6 \\ -2 \\ 5.5 \\ -2.8 \\ 1 \\ 1.8 \end{pmatrix}$$

Then the predictions of $g_i + s_{ij} + \frac{1}{2}h_1 + \frac{1}{2}h_2$ are

$$\begin{bmatrix} g_1 + \frac{1}{2}(h_1 + h_2) \\ \vdots \\ g_2 + \frac{1}{2}(h_1 + h_2) \end{bmatrix} + C'V^{-1} \begin{bmatrix} \bar{y}_{111.} - h_1 - g_1 \\ \vdots \\ \bar{y}_{225.} - h_2 - g_2 \end{bmatrix}$$

$$= \begin{pmatrix} 8 \\ 8 \\ 10 \\ 10 \\ 10 \end{pmatrix} + \begin{pmatrix} .4 \\ .6875 \\ -.9333 \\ -1.6667 \\ .4375 \end{pmatrix} \cong \begin{pmatrix} 8.4 \\ 8.7 \\ 9.1 \\ 8.3 \\ 10.4 \end{pmatrix} \qquad (22)$$

BLUP

Suppose now in the example above h and g are unknown. Using the method of (5),

$$\beta' = [h_1 \;\; h_2 \;\; g_1 \;\; g_2]$$

$$X = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}, \text{ and } P = \begin{bmatrix} 1/2 & 1/2 & 1 & 0 \\ 1/2 & 1/2 & 1 & 0 \\ 1/2 & 1/2 & 0 & 1 \\ 1/2 & 1/2 & 0 & 1 \\ 1/2 & 1/2 & 0 & 1 \end{bmatrix}$$

Then the weights, B', to apply to $\bar{y}_{ijk}$ are the solution to

$$\begin{bmatrix} V & X \\ X' & 0 \end{bmatrix} \begin{bmatrix} B \\ T \end{bmatrix} = \begin{bmatrix} C \\ P' \end{bmatrix}$$

The solution to B' is

$$\begin{bmatrix} .60582 & .16803 & -.22537 & -.04787 & .22695 & .10315 & .04891 & .11998 \\ .26501 & .42462 & -.15617 & -.03287 & .31097 & .07213 & .05386 & .08306 \\ .02293 & .03669 & .38068 & .05973 & -.05982 & .45688 & -.00091 & .10365 \\ -.07027 & -.03244 & .55345 & .00126 & .05271 & .17965 & .08961 & .17863 \\ .01806 & .02890 & .33635 & .31460 & -.04696 & .12434 & .00314 & .41942 \end{bmatrix}$$

Then B' $\begin{bmatrix} \bar{y}_{111.} \\ \cdot \\ \cdot \\ \cdot \\ \bar{y}_{225.} \end{bmatrix} = \begin{bmatrix} 10.85 \\ 10.64 \\ 7.12 \\ 8.19 \\ 8.73 \end{bmatrix}$.

Applying the method of substituting $\hat{\beta}$ for β in the BLP procedure, we first find GLS of β.

$$X'V^{-1}X = \begin{bmatrix} .62548 & -.06443 & .29167 & .26935 \\ -.06443 & .47366 & .06250 & .34673 \\ .29167 & .06250 & .35417 & 0 \\ .26935 & .34673 & 0 & .61608 \end{bmatrix}$$

$$X'V^{-1} \begin{bmatrix} \bar{y}_{111.} \\ \cdot \\ \cdot \\ \cdot \\ \bar{y}_{225.} \end{bmatrix} = \begin{bmatrix} 4.69062 \\ 3.91771 \\ 3.60625 \\ 5.00208 \end{bmatrix}$$

It should be noted that $X'V^{-1}X$ has rank, 3, rather than 4, its order. Consequently no unique solution exists for $\hat{\beta}$. Nevertheless we can use any solution since $X\beta$ and $P\beta$ are estimable, and consequently $X\hat{\beta}$ and $P\hat{\beta}$ are invariant to the solution. Let us take the solution for which $\hat{g}_2 = 0$. This is $(\hat{h}_1 \; \hat{h}_2 \; \hat{g}_1 \; \hat{g}_2) = [7.139, 8.881, 2.736, 0]$.

Then $\begin{bmatrix} \bar{y}_{111.} \\ \cdot \\ \cdot \\ \cdot \\ \bar{y}_{225.} \end{bmatrix} - \begin{bmatrix} \hat{h}_1 + \hat{g}_1 \\ \cdot \\ \cdot \\ \cdot \\ \hat{h}_2 + \hat{g}_2 \end{bmatrix} = \begin{pmatrix} .325 \\ -.874 \\ .261 \\ 1.861 \\ .883 \\ -2.681 \\ 1.119 \\ 1.919 \end{pmatrix}$,

and $C'V^{-1}$ (corrected $\bar{y}_{ijk}$) $= \begin{bmatrix} .108 \\ -.108 \\ -.893 \\ .178 \\ .716 \end{bmatrix}$.

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Then the predictions of $g_i + s_j + \frac{1}{2}(h_1 + h_2)$ are

$$\begin{bmatrix} 10.746 \\ 10.746 \\ 8.010 \\ 8.010 \\ 8.010 \end{bmatrix} + \begin{pmatrix} .108 \\ -.108 \\ -.893 \\ .178 \\ .716 \end{pmatrix} = \begin{pmatrix} 10.854 \\ 10.638 \\ 7.117 \\ 8.188 \\ 8.726 \end{pmatrix}, \quad (23)$$

the same solution as before.

BLUP Using Mixed Model Solution Assuming Variances are Known

Now $\beta' = (h_1 \;\; h_2 \;\; g_1 \;\; g_2)$

$u' = [s_{11} \;\; s_{12} \;\; s_{21} \;\; s_{22} \;\; s_{23}]$

$$Z = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad R = 8 \begin{bmatrix} 1/5 & & & & \\ & 1/4 & & & 0 \\ & & \ddots & & \\ & 0 & & \ddots & \\ & & & & 1/5 \end{bmatrix}$$

$$G = .8\,I$$

Then the equations of (10) become

$$\frac{1}{8}\begin{bmatrix} 20 & 0 & 9 & 11 & 5 & 4 & 0 & 10 & 1 \\ 0 & 13 & 2 & 11 & 0 & 2 & 5 & 1 & 5 \\ 9 & 2 & 11 & 0 & 3 & 6 & 0 & 0 & 0 \\ 11 & 11 & 0 & 22 & 0 & 0 & 5 & 11 & 0 \\ 5 & 0 & 3 & 0 & 15 & 0 & 0 & 0 & 0 \\ 4 & 2 & 6 & 0 & 0 & 16 & 0 & 0 & 0 \\ 0 & 5 & 0 & 5 & 0 & 0 & 15 & 0 & 0 \\ 10 & 1 & 0 & 11 & 0 & 0 & 0 & 21 & 0 \\ 1 & 5 & 0 & 6 & 0 & 0 & 0 & 0 & 16 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ g_1 \\ g_2 \\ s_{11} \\ s_{12} \\ s_{23} \\ s_{24} \\ s_{25} \end{bmatrix} = \frac{1}{8} \begin{bmatrix} 170 \\ 130 \\ 115 \\ 178 \\ 51 \\ 61 \\ 31 \\ 84 \\ 63 \end{bmatrix} \quad (24)$$

The coefficient matrix has rank one less than order, so no unique solution exists to $\hat{h}$ and $\hat{g}$. But $\hat{s}$ is unique. A solution is [7.139 8.881 2.736 0 .108 −.108 −.893 .178 .716]. Note that this is the same solution to $\hat{\beta}$ and $\hat{u}$ as by the other two BLUP methods.

A g-inverse of the coefficient matrix provides the variances of prediction errors.

Unknown Variances and Maximum Likelihood

The mixed model equations are the same as in (24), except that the last 5 diagonal coefficients are

$$[5 + \hat{\sigma}_e^2/\hat{\sigma}_s^2, \ldots, 6 + \hat{\sigma}_e^2/\hat{\sigma}_s^2].$$

We assume that $y'y = 2837$.

By an iterative process a solution was found to be

$$\hat{\sigma}_e^2 = 6.56234, \quad \hat{\sigma}_s^2 = .83613$$

$$[\hat{h}_1 \;\; \hat{h}_2 \;\; \hat{g}_1 \;\; \hat{g}_2] = [7.09381 \;\; 8.89221 \;\; 2.77282 \;\; 0]$$

$$\hat{s}' = [.12973 \;\; -.12973 \;\; -1.04768 \;\; .22122 \;\; .82646]$$

To prove that this is a solution,

$$6.56234/.83613 = 7.84847.$$

Adding this ratio to 5, ..., 11 to form the last diagonal coefficients and solving the mixed model equations, we obtain the above results for $\hat{\beta}$, $\hat{u}$.

Further,

$$[2837 - 7.09381\,(170) - \ldots - .82646\,(63)]/33 = 6.56234$$

$$[.12973^2 + \ldots + .82646^2 + 6.5234\,(\tfrac{1}{12.84847} + \ldots + \tfrac{1}{13.84847})]/5 = .83613 .$$

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APPENDIX B
Examples of Selection Bias

Selection on y

Suppose that we have the following sets of progeny on 4 sires.

Sire	No. of progeny
1	5 and 100
2	4
3	2
4	5

Let those 5 progeny means be denoted by $y_{11}, y_{12}, y_{21}, y_{31}, y_{41}$. Sires 1 and 2 are regarded as a sample from group 1, and sires 3 and 4 as a sample from group 2. Suppose $\sigma_e^2/\sigma_s^2 = 5$.

Then the BLUP equations are

$$\begin{bmatrix} 109 & 0 & 105 & 4 & 0 & 0 \\ 0 & 7 & 0 & 0 & 2 & 5 \\ 105 & 0 & 110 & 0 & 0 & 0 \\ 4 & 0 & 0 & 9 & 0 & 0 \\ 0 & 2 & 0 & 0 & 7 & 0 \\ 0 & 5 & 0 & 0 & 0 & 10 \end{bmatrix} \begin{bmatrix} \hat{g}_1 \\ \hat{g}_2 \\ \hat{s}_1 \\ \hat{s}_2 \\ \hat{s}_3 \\ \hat{s}_4 \end{bmatrix} = \begin{bmatrix} y_{11} + y_{12} + y_{21} \\ y_{31} + y_{41} \\ y_{11} + y_{12} \\ y_{21} \\ y_{31} \\ y_{41} \end{bmatrix}$$

Then the predictors of g + s turn out to be

$$\begin{bmatrix} \hat{g}_1 + \hat{s}_1 \\ \hat{g}_1 + \hat{s}_2 \\ \hat{g}_2 + \hat{s}_3 \\ \hat{g}_2 + \hat{s}_4 \end{bmatrix} = \begin{bmatrix} .0469 & .9386 & .0145 & 0 & 0 \\ .0181 & .3610 & .6209 & 0 & 0 \\ 0 & 0 & 0 & .5455 & .4545 \\ 0 & 0 & 0 & .1818 & .8182 \end{bmatrix} \begin{bmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{31} \\ y_{41} \end{bmatrix}$$

Now what if we assume that

$$E\begin{bmatrix} y_{11} - y_{21} \\ y_{11} - y_{31} \\ y_{11} - y_{41} \end{bmatrix} \neq \begin{bmatrix} 0 \\ g_1 - g_2 \\ g_1 - g_2 \end{bmatrix} ?$$

That is $L' = \begin{bmatrix} 1 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & -1 \end{bmatrix}$.

This would imply that sire 1 was selected to have additional progeny because of the relative merits of his original 5 progeny.

Now $L'X = L' \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & -1 \\ 1 & -1 \end{bmatrix} \neq$ null.

Consequently the predictions are biased in regular BLUP.

$$L'VL = L' \begin{bmatrix} .4 & .2 & 0 & 0 & 0 \\ .2 & .21 & 0 & 0 & 0 \\ 0 & 0 & .45 & 0 & 0 \\ 0 & 0 & 0 & .7 & 0 \\ 0 & 0 & 0 & 0 & .4 \end{bmatrix} L\sigma_e^2$$

$$= \begin{bmatrix} .4 & .2 & -.45 & 0 & 0 \\ .4 & .3 & 0 & -.7 & 0 \\ .4 & .2 & 0 & 0 & -.4 \end{bmatrix} L\sigma_e^2 = \begin{bmatrix} .85 & .4 & .4 \\ .4 & 1.1 & .4 \\ .4 & .4 & .8 \end{bmatrix} \sigma_e^2$$

Then the modified BLUP equations, described in (11) are

$$\begin{bmatrix} 109 & 0 & 105 & 4 & 0 & 0 & 0 & 1 & 1 \\ 0 & 7 & 0 & 0 & 2 & 5 & 0 & 1 & 1 \\ 105 & 0 & 110 & 0 & 0 & 0 & 0 & 0 & 0 \\ 4 & 0 & 0 & 9 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 7 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 & 10 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & .85 & .4 & .4 \\ 1 & 1 & 0 & 0 & 0 & 0 & .4 & 1.1 & .4 \\ 1 & 1 & 0 & 0 & 0 & 0 & .4 & .4 & .8 \end{bmatrix} \begin{bmatrix} \hat{g}_1 \\ \hat{g}_2 \\ \hat{s}_1 \\ \hat{s}_2 \\ \hat{s}_3 \\ \hat{s}_4 \\ \hat{t}_1 \\ \hat{t}_2 \\ \hat{t}_3 \end{bmatrix} = \begin{bmatrix} y_{11} + y_{12} + y_{21} \\ y_{31} + y_{41} \\ y_{11} + y_{12} \\ y_{21} \\ y_{31} \\ y_{41} \\ y_{11} \quad y_{21} \\ y_{21} \quad y_{31} \\ y_{31} \quad y_{41} \end{bmatrix}$$

The predictors coming from the solution to these equations are

$$\begin{bmatrix} \hat{g}_1 + \hat{s}_1 \\ \hat{g}_1 + \hat{s}_2 \\ \hat{g}_2 + \hat{s}_3 \\ \hat{g}_2 + \hat{s}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ -.5555 & 1.1111 & .4444 & 0 & 0 \\ 1.3131 & -1.7172 & .4041 & .5455 & .4545 \\ .9192 & -1.2020 & .2828 & .1818 & .8182 \end{bmatrix} \begin{bmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{31} \\ y_{41} \end{bmatrix}$$

To demonstrate that these predictors are unbiased note that

$$E\left[\begin{pmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{31} \\ y_{41} \end{pmatrix} L'y\right] = \begin{bmatrix} g_1 \\ g_1 \\ g_1 \\ g_2 \\ g_2 \end{bmatrix} + \begin{bmatrix} .4 & .4 & .4 \\ .2 & .2 & .2 \\ .45 & 0 & 0 \\ 0 & .7 & 0 \\ 0 & 0 & .4 \end{bmatrix} (L'VL)^{-} \begin{bmatrix} y_{11} - g_1 \\ y_{12} - g_1 \\ y_{21} - g_1 \\ y_{31} - g_2 \\ y_{41} - g_2 \end{bmatrix}$$

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Then

$$E\left[\begin{pmatrix} \hat{g}_1 + \hat{s}_1 \\ \hat{g}_1 + \hat{s}_2 \\ \hat{g}_2 + \hat{s}_3 \\ \hat{g}_2 + \hat{s}_4 \end{pmatrix} L'y\right] = \begin{bmatrix} g_1 \\ g_1 \\ g_2 \\ g_2 \end{bmatrix} + \begin{bmatrix} .2 & .2 & .2 & .2 \\ -.2 & 0 & 0 & 0 \\ 0 & .2 & 0 & 0 \\ 0 & 0 & -.2 & -.2 \end{bmatrix} (L'VL)^{-1} \begin{bmatrix} y_{11} - g_1 \\ y_{12} - g_1 \\ y_{21} - g_2 \\ y_{22} - g_2 \end{bmatrix}$$

Then

$$\begin{bmatrix} \hat{\mu} + \hat{s}_1 \\ \hat{\mu} + \hat{s}_2 \\ \hat{\mu} + \hat{s}_3 \\ \hat{\mu} + \hat{s}_4 \end{bmatrix} = \begin{bmatrix} .6255 & .1115 & .1793 & .0837 \\ .1394 & .5684 & .1992 & .0930 \\ .0717 & .0638 & .8167 & .0478 \\ .1673 & .1488 & .2390 & .4449 \end{bmatrix} \begin{bmatrix} \bar{y}_1. \\ \bar{y}_2. \\ \bar{y}_3. \\ \bar{y}_4. \end{bmatrix}$$

It is easy to prove that

$$E\left[\begin{pmatrix} g_1 + s_1 \\ g_1 + s_2 \\ g_2 + s_3 \\ g_2 + s_4 \end{pmatrix} L'y\right]$$ has the same expectation.

Now suppose it is suspected that sires 1 and 2 were selected differently from sires 3 and 4. This can be expressed as

$$E(s_1 + s_2 - s_3 - s_4) \neq 0.$$

That is L' of L'u is $[1 \;\; 1 \;\; -1 \;\; -1]$.

Using the method of (12) the modified BLUP equations are

Selection on u

Suppose we have the following data on sires and the model assumed is

$$y_{ij} = \mu + s_i + e_{ij},$$

where $\text{Var}\begin{pmatrix} s \\ e \end{pmatrix} = \begin{bmatrix} \frac{1}{10} I & 0 \\ 0 & I \end{bmatrix} \sigma^2$.

Sire	n_i
1	10
2	8
3	25
4	5

$$\begin{bmatrix} 48 & 10 & 8 & 25 & 5 & 0 \\ 10 & 20 & 0 & 0 & 0 & -1 \\ 8 & 0 & 18 & 0 & 0 & -1 \\ 25 & 0 & 0 & 35 & 0 & 1 \\ 5 & 0 & 0 & 0 & 15 & 1 \\ 0 & -1 & -1 & 1 & 1 & .4 \end{bmatrix} \begin{bmatrix} \hat{\mu} \\ \hat{s}_1 \\ \hat{s}_2 \\ \hat{s}_3 \\ \hat{s}_4 \\ \hat{t} \end{bmatrix} = \begin{bmatrix} y_{..} \\ y_{1.} \\ y_{2.} \\ y_{3.} \\ y_{4.} \\ 0 \end{bmatrix}$$

The solution is

$$\begin{bmatrix} \hat{\mu} + \hat{s}_1 \\ \hat{\mu} + \hat{s}_2 \\ \hat{\mu} + \hat{s}_3 \\ \hat{\mu} + \hat{s}_4 \end{bmatrix} = \begin{bmatrix} .7647 & .2353 & 0 & 0 \\ .2941 & .7059 & 0 & 0 \\ 0 & 0 & .9091 & .0909 \\ 0 & 0 & .4545 & .5455 \end{bmatrix} \begin{bmatrix} \bar{y}_1. \\ \bar{y}_2. \\ \bar{y}_3. \\ \bar{y}_4. \end{bmatrix}$$

Then in the no selection model BLUP of $\mu + s_i$ is

$$\begin{bmatrix} \hat{\mu} + \hat{s}_1 \\ \vdots \\ \hat{\mu} + \hat{s}_4 \end{bmatrix},$$

where $\hat{\mu}, \hat{s}_i$ are solution to

$$\begin{bmatrix} 48 & 10 & 8 & 25 & 5 \\ 10 & 20 & 0 & 0 & 0 \\ 8 & 0 & 18 & 0 & 0 \\ 25 & 0 & 0 & 35 & 0 \\ 5 & 0 & 0 & 0 & 15 \end{bmatrix} \begin{bmatrix} \hat{\mu} \\ s_1 \\ s_2 \\ s_3 \\ s_4 \end{bmatrix} = \begin{bmatrix} y_{..} \\ y_{1.} \\ y_{2.} \\ y_{3.} \\ y_{4.} \end{bmatrix}$$

Note that this is considerably different from the no selection solution. It is interesting to note that the selection bias can be eliminated in this example by regarding sires 1 and 2 as a sample from a population different from that of sires 3 and 4. The BLUP equations become

$$\begin{bmatrix} 18 & 0 & 10 & 8 & 0 & 0 \\ 0 & 30 & 0 & 0 & 25 & 5 \\ 10 & 0 & 20 & 0 & 0 & 0 \\ 8 & 0 & 0 & 18 & 0 & 0 \\ 0 & 25 & 0 & 0 & 35 & 0 \\ 0 & 5 & 0 & 0 & 0 & 15 \end{bmatrix} \begin{bmatrix} \hat{g}_1 \\ \hat{g}_2 \\ \hat{s}_1 \\ \hat{s}_2 \\ \hat{s}_3 \\ \hat{s}_4 \end{bmatrix} = \begin{bmatrix} y_{1.} + y_{2.} \\ y_{3.} + y_{4.} \\ y_{1.} \\ y_{2.} \\ y_{3.} \\ y_{4.} \end{bmatrix}$$

The solution to $\hat{g} + \hat{s}$ is exactly the same as that for $\hat{\mu} + \hat{s}$ in the selection model.

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APPENDIX C
Repeated Records

Suppose we have the following records in 2 herds.

Group	Sire	Progeny	Herd 1 Years 1	2	3	Herd 2 Years 1	2
1	1	1	14	16	13		
1	1	2	12	–	–		
1	2	3	15	10	–		
1	2	4	–	18	13		
1	2	5	–	–	16		
1	2	6	–	12	11		
1	3	7	–	20	19		
2	4	8	15	14	–		
2	4	9	–	–	21		
1	1	10				12	14
1	1	11				16	–
1	3	12				13	18
1	3	13				22	–
2	4	14				–	13
2	4	15				15	16
2	4	16				17	13
2	5	17				18	20
2	5	18				–	14

We assume that $h^2 = .25$, $r = .45$. Then $\sigma_c^2 = .55\,\sigma_y^2$, $\sigma_s^2 = .0625\,\sigma_y^2$, $\sigma_p^2 = .3875\,\sigma_y^2$. Thus we add $.55/.0625 = 8.8$ to the diagonals of the s coefficients and $.55/.3875 = 1.41935$ to the diagonals of the p coefficients of the regular least squares equations.

Some of these equations are

Group 1: $19\,g_1 + 7s_1 + 7s_2 + 5s_3 + 3\,hy_{11} + 5hy_{12} + 5hy_{13} + 4hy_{21} + 2hy_{22} + 3c_1 + c_2 + 2\,c_3 + 2\,c_4 + c_5 + 2\,c_6 + 2\,c_7 + 2\,c_{10} + c_{11} + 2\,c_{13} + c_{14} = 284.$

Sire 1: $7\,g_1 + 15.8\,s_1 + 2hy_{11} + hy_{12} + hy_{13} + 2hy_{21} + hy_{22} + 3\,c_1 + c_2 + 2\,c_{10} + c_{11} = 97.$

Herd y_{11}: $3\,g_1 + g_2 + 2\,s_1 + s_2 + s_4 + 4hy_{11} + c_1 + c_2 + c_3 + c_8 = 56.$

Cow$_1$: $3\,g_1 + 3\,s_1 + hy_{11} + hy_{12} + hy_{13} + 4.41935\,c_1 = 43.$

A solution is

$\hat{g}' = [\ 8496 \quad 0]$

$\hat{s}' = [\ .3372 \quad .3582 \quad .6954 \quad .1150 \quad .1150]$

$\hat{hy}' = [15.2834 \quad 16.1284 \quad 15.8178 \quad 16.2423 \quad 16.0414]$

$\hat{c}' = [\ .2645 \quad 1.0232 \quad 1.3149 \quad .4297 \quad .5745 \quad -1.9099 \quad 2.1531$
$.7863 \quad 2.1895 \quad 1.1435 \quad .3906 \quad .2852 \quad 2.4436 \quad 1.2886$
$.3082 \quad .6006 \quad 1.6045 \quad -.3913]$

Note that

$\hat{s}_1 + \hat{s}_2 + \hat{s}_3 = 0$, and $\hat{s}_4 + \hat{s}_5 = 0$.

Also sum of $\hat{p}$ of progeny of

i^{th} sire $= \dfrac{\sigma_p^2}{\sigma_s^2} = \hat{s}_i$ for all i.

The relative merits of the sires is predicted by $\hat{g} + \hat{s}$.

For example, sire 1 = $-.85 - .34 = -1.19$

sire 5 = $0 + .12 = .12$

The relative r.p.a. of the cows are predicted by $\hat{g} + \hat{s} + \hat{p}$.

For example, cow 1 = $-.85 - .34 - .26 = -1.45$

cow 18 = $0 + .12 - .89 = -.77$

The relative breeding values of cows are predicted by

$$\hat{g} + \hat{s} + \frac{\frac{3}{4}h^2}{r - \frac{1}{4}h^2}\,\hat{p} = \hat{g} + \hat{s} + .4839\,\hat{p}$$

For example, cow 1 = $-.85 - .34 - .13 = -1.32$.

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APPENDIX D
Multiple Trait Evaluation

Suppose we have the following data involving 2 traits.

Group	Sire	Progeny	Herd 1 Traits 1	Herd 1 Traits 2	Herd 2 Traits 1	Herd 2 Traits 2
1	1	1	8	7		
1	1	2	10	3		
1	1	3	5			
1	2	4	9	12		
1	2	5	–	13		
2	3	6	10	5		
2	3	7	8	–		
2	4	8	6	3		
1	1	9			7	8
1	1	10			5	
2	3	11			12	6
2	3	12			9	3
2	4	13			8	7
2	4	14			–	9

$$\text{Var}\begin{bmatrix} s_{1k\ell} \\ s_{2k\ell} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \quad \text{Var}\begin{bmatrix} e_{1jk\ell m} \\ e_{2jk\ell m} \end{bmatrix} = \begin{bmatrix} 17 & 9 \\ 9 & 13 \end{bmatrix}$$

$$P_{12} = \begin{bmatrix} .092857 & -.064286 \\ -.064286 & .121429 \end{bmatrix},$$

$$P_1 = \begin{bmatrix} .058824 & 0 \\ 0 & 0 \end{bmatrix},$$

$$P_2 = \begin{bmatrix} 0 & 0 \\ 0 & .076923 \end{bmatrix}$$

$$G_s^{-1} = \begin{bmatrix} .6 & -.2 \\ -.2 & .4 \end{bmatrix}$$

To illustrate

$$S_2 \text{ in herd 1} = P_{12} + P_2 = \begin{bmatrix} .092857 & -.064286 \\ -.064286 & .198352 \end{bmatrix}$$

$$t_2 \text{ in herd 1} = P_{12}\begin{bmatrix} 9 \\ 12 \end{bmatrix} + P_2\begin{bmatrix} 0 \\ 13 \end{bmatrix} = \begin{bmatrix} .06429 \\ 1.87857 \end{bmatrix}$$

A solution is

$$[g_{11}\ g_{12}\ g_{21}\ g_{22}] = [-1.08 \quad 3.64 \quad 0 \quad 0]$$

$$[s_{11}\ s_{12} \ldots s_{41}\ s_{42}]$$

$$= [\ .34 \quad -1.48 \quad .34 \quad 1.48 \quad .23 \quad -.55 \quad -.23 \quad .55]$$

$$[h_{11}\ h_{12}\ h_{21}\ h_{22}]$$

$$= [9.03 \quad 4.40 \quad 9.08 \quad 5.89]$$

The second subscript refers to trait number. For example, –.34 is the prediction of trait one on sire one.

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APPENDIX E
Related Sires

Suppose that we have data on progeny of related sires as follows. Sires 1 and 2 are unrelated, Sires 3 and 4 are sons of 1, and sires 5, 6, 7 are sons of 2.

Then $A = \begin{bmatrix} 1 & 0 & ½ & ½ & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & ½ & ½ & ½ \\ ½ & 0 & 1 & ¼ & 0 & 0 & 0 \\ ½ & 0 & ¼ & 1 & 0 & 0 & 0 \\ 0 & ½ & 0 & 0 & 1 & ¼ & ¼ \\ 0 & ½ & 0 & 0 & ¼ & 1 & ¼ \\ 0 & ½ & 0 & 0 & ¼ & ¼ & 1 \end{bmatrix}$

Suppose $\sigma_e^2/\sigma_s^2 = 10$.

Number of Progeny

Sire	Herd 1	Herd 2	Herd 3	$n_{i.}$	$y_{i..}$
1	2	0	4	6	70
2	3	2	0	5	81
3	5	4	9	18	233
4	3	1	4	8	135
5	0	2	5	7	102
6	0	0	8	8	114
7	2	3	5	10	123
$n_{.j}$	15	12	35		
$y_{.j.}$	210	129	519		

The mixed model equations are

$$\frac{1}{6}\begin{bmatrix} 136 & 0 & -40 & -40 & 0 & 0 & 0 & 12 & 0 & 24 \\ 0 & 150 & 0 & 0 & -40 & -40 & -40 & 18 & 12 & 0 \\ -40 & 0 & 188 & 0 & 0 & 0 & 0 & 30 & 24 & 54 \\ -40 & 0 & 0 & 128 & 0 & 0 & 0 & 18 & 6 & 24 \\ 0 & -40 & 0 & 0 & 122 & 0 & 0 & 0 & 12 & 30 \\ 0 & -40 & 0 & 0 & 0 & 128 & 0 & 0 & 0 & 48 \\ 0 & -40 & 0 & 0 & 0 & 0 & 140 & 12 & 18 & 30 \\ 12 & 18 & 30 & 18 & 0 & 0 & 12 & 90 & 0 & 0 \\ 0 & 12 & 24 & 6 & 12 & 0 & 18 & 0 & 72 & 0 \\ 24 & 0 & 54 & 24 & 30 & 48 & 30 & 0 & 0 & 210 \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \\ s_6 \\ s_7 \\ h_1 \\ h_2 \\ h_3 \end{bmatrix} = \begin{bmatrix} 70 \\ 81 \\ 233 \\ 135 \\ 102 \\ 114 \\ 123 \\ 210 \\ 129 \\ 519 \end{bmatrix}$$

The solution is

$$\hat{s}' = [-.736 \quad .744 \quad -.630 \quad .815 \quad .524 \quad -.033 \quad -.308]$$

$$\hat{h}' = [14.037 \quad 10.758 \quad 14.958]$$

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APPENDIX F

Using Records of Progeny of Daughters of Sires to be Evaluated

The following progeny and grand-progeny records are available.

Grand-sire	Herd 1 Sire 1	Herd 1 Sire 5	Herd 1 Sire 6	Herd 1 Sire 7	Herd 2 Sire 4	Herd 2 Sire 5	Herd 2 Sire 6	Herd 2 Sire 7
1	–	12	21, 15, 9	–	–	13, 18	11	–
2	–	12	–	18	–	15, 14, 17	–	22, 12
3	13, 20 15, 17	–	–	15	19		13, 12	14
4	16, 19	10, 12	–	–	–	13	15, 14	–

Sires 1 and 2 are members of group 1, sires 3 and 4 of group 2, and sires 5, 6, 7 of group 3.

Suppose $h^2 = \frac{16}{41}$.

Since there is only one record per sire, σ_e^2 and σ_p^2 are combined. Then the new

$$\sigma_e^2 = (1 - \frac{5}{16} h^2)\sigma_y^2 = \frac{36}{41} \sigma_y^2$$

$$\sigma_s^2 = \frac{1}{4} h^2 \sigma_y^2 = \frac{4}{41} \sigma_y^2.$$

Consequently $\sigma_e^2/\sigma_s^2 = 9$, to be added to diagonals of sire equations.
Some of the mixed model equations are:

Group 1: $10.5\, g_1 + 3.5\, g_2 + 7\, g_3 + 8.75\, s_1 + 1.75\, s_2 + 2.5\, s_3 + s_4 + 3.5\, s_5 + 2\, s_6 + 1.5\, s_7 + 9\, h_1 + 5\, h_2 = 223.50$

Sire 1: $8.75\, g_1 + 3.5\, g_2 + 3.5\, g_3 + 17.75\, s_1 + 2.5\, s_3 + s_4 + 1.5\, s_5 + 2\, s_6 + 8\, h_1 + 2.5\, h_2 = 168.50$

Herd 2: $5\, g_1 + 3.5\, g_2 + 14\, g_3 + 2.5\, s_1 + 2.5\, s_2 + 2\, s_3 + 1.5\, s_4 + 6\, s_5 + 5\, s_6 + 3\, s_7 + 15\, h_2 = 222$

A solution is

$\hat{g}' = [13.391 \quad 9.420 \quad 8.851]$
$\hat{s}' = [-.155 \quad .155 \quad -.068 \quad .068 \quad -.492 \quad -.168 \quad .660]$
$\hat{h}' = [-1.087 \quad 0]$

Note that $\Sigma \hat{s} = 0$ within groups.

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APPENDIX G

Inclusion of c^2 (Interaction) in Model

We have the following data

Group	Sire	Herd-yr. 11	12	13	24	25
1	1	14, 12			12, 16	
1	2	15	18, 12	16		
1	3		20		13, 22	
2	4	15			17, 15	13
2	5				18	14

Suppose first that c^2 pertains to daughters of a sire in the same herd-year. This is equivalent to interaction between sire and herd-year.

Suppose $\sigma_e^2/\sigma_s^2 = 15$ and $\sigma_e^2/\sigma_{shy}^2 = 30$.

Then some of the equations to be solved are as follows.

Group 1: $11\,g_1 + 4\,s_1 + 4\,s_2 + 3\,s_3 + 3\,hy_{11} + 3\,hy_{12} + hy_{13} + 4\,hy_{24} + 2\,shy_{1,11} + 2\,shy_{1,24} + shy_{2,11} + 2\,shy_{2,12} + shy_{2,13} + shy_{3,12} + 2\,shy_{3,24} = 170$

Sire 1: $4\,g_1 + 19\,s_1 + 2\,hy_{11} + 2\,hy_{24} + 2\,hy_{1,11} + 2\,shy_{1,24} = 54$

hy_{11}: $3\,g_1 + g_2 + 2\,s_1 + s_2 + s_4 + 4\,hy_{11} + 2\,shy_{1,11} + shy_{2,11} + shy_{4,11} = 56$

$shy_{1,11}$: $2\,g_1 + 2\,s_1 + 2\,hy_{11} + 32\,shy_{1,11} = 26$

A solution to the equations is

$\hat{g}' = [-1.746 \quad 0]$

$\hat{s}' = [-.210 \quad -.195 \quad .405 \quad -.080 \quad .080]$

$\hat{hy}' = [15.492 \quad 18.439 \quad 19.511 \quad 17.100 \quad 13.500]$

Table of $\widehat{shy}$

Sire	Herd-year 11	12	13	24	25
1	−.033			−.071	
2	.047	−.094	−.051		
3		.094		.109	
4	−.013		.051	−.064	−.014
5				.026	.014

Note that $\sum_{jk} \widehat{shy}_{ijk} = \frac{15}{30}\,\hat{s}_i$

$\sum_{i} \widehat{shy}_{ijk} = 0$

$\Sigma\,\hat{s} = 0$ within each group.

Now suppose that c^2 pertains to daughters of a sire in the same herd (rather than herd-year).

Suppose $\sigma_e^2/\sigma_s^2 = 15$, $\sigma_e^2/\sigma_{sh}^2 = 30$.

Now some of the equations are:

Group 1: $11\,g_1 + 4\,s_1 + 4\,s_2 + 3\,s_3 + 3\,hy_{11} + 3\,hy_{12} + hy_{13} + 4\,hy_{24} + 2\,sh_{11} + 2\,sh_{12} + 4\,sh_{21} + sh_{31} + 2\,sh_{32} = 170$

Sire 1: $4\,g_1 + 19\,s_1 + 2\,hy_{11} + 2\,hy_{24} + 2\,sh_{11} + 2\,sh_{12} = 54$

hy_{11}: $3\,g_1 + g_2 + 2s_1 + s_2 + s_4 + 4\,hy_{11} + 2\,sh_{11} + sh_{21} + sh_{41} = 56$

sh_{21}: $4\,g_1 + 4\,s_2 + hy_{11} + 2\,hy_{12} + hy_{13} + 34\,sh_{21} = 61$

A solution is

$\hat{g}' = [-1.716 \quad 0]$

$\hat{s}' = [-.215 \quad -.189 \quad .404 \quad -.078 \quad .078]$

$\hat{hy}' = [15.493 \quad 18.405 \quad 19.520 \quad 17.086 \quad 13.518$

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Table of $\widehat{sh}$

Sire	Herd 1	Herd 2
1	−.035	−.072
2	−.094	—
3	.094	.108
4	.035	−.074
5	—	.039

Note that $\sum_j \widehat{sh}_{ij} = \frac{15}{30}\,\hat{s}_i$

$\sum_i \widehat{sh}_{ij} = 0$

$\Sigma\ \hat{s} = 0$ within each group.

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APPENDIX H

Using All Records

Suppose we have the following records on 5 cows.

	Period		
Cow	1	2	3
1	15	13	19
2	10	20	–
3	–	14	16
4	–	17	–
5	–	–	12

X, referring to periods, =

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Z referring to cows =

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Suppose A =

$$\begin{bmatrix} 1 & \frac{1}{4} & \frac{1}{4} & \frac{1}{16} & \frac{1}{16} \\ \frac{1}{4} & 1 & \frac{1}{4} & \frac{1}{16} & \frac{1}{16} \\ \frac{1}{4} & \frac{1}{4} & 1 & \frac{1}{16} & \frac{1}{16} \\ \frac{1}{16} & \frac{1}{16} & \frac{1}{16} & 1 & \frac{1}{4} \\ \frac{1}{16} & \frac{1}{16} & \frac{1}{16} & \frac{1}{4} & 1 \end{bmatrix},$$

$$X'X = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix},$$

$$X'Z = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

$h^2 = .25, \quad r = .50.$

$$Z'Z = \begin{bmatrix} 3 & & & & \\ & 2 & & & \\ & & 2 & 0 & \\ & & 0 & 1 & \\ & & & & 1 \end{bmatrix}, \quad X'y = \begin{bmatrix} 25 \\ 64 \\ 47 \end{bmatrix}, \quad Z'y = \begin{bmatrix} 47 \\ 30 \\ 30 \\ 17 \\ 12 \end{bmatrix}$$

Then B =

$$\begin{bmatrix} 1 & \frac{1}{8} & \frac{1}{8} & \frac{1}{32} & \frac{1}{32} \\ \frac{1}{8} & 1 & \frac{1}{8} & \frac{1}{32} & \frac{1}{32} \\ \frac{1}{8} & \frac{1}{8} & 1 & \frac{1}{32} & \frac{1}{32} \\ \frac{1}{32} & \frac{1}{32} & \frac{1}{32} & 1 & \frac{1}{8} \\ \frac{1}{32} & \frac{1}{32} & \frac{1}{32} & \frac{1}{8} & 1 \end{bmatrix}$$

Then

$$\begin{bmatrix} X'X & X'ZB \\ BZ'X & BZ'ZB+B \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{v} \end{bmatrix} = \begin{bmatrix} X'y \\ BZ'y \end{bmatrix}$$

becomes

$$\begin{bmatrix} 2 & 0 & 0 & 1.125 & 1.125 & .25 & .0625 & .0625 \\ 0 & 4 & 0 & 1.28125 & 1.28125 & 1.28125 & 1.09375 & .21875 \\ 0 & 0 & 3 & 1.15625 & .28125 & 1.15625 & .1875 & 1.0625 \\ 1.125 & 1.28125 & 1.15625 & 4.06445 & .7832 & .7832 & .17578 & .17578 \\ 1.125 & 1.28125 & .28125 & .7832 & 3.05008 & .67383 & .14844 & .14844 \\ .25 & 1.28125 & 1.15625 & .7832 & .67383 & 3.08008 & .14844 & .14844 \\ .0625 & 1.09375 & .1875 & .17570 & .14844 & .14844 & 2.02246 & .38184 \\ .0625 & .21875 & 1.0625 & .17578 & .14844 & .14844 & .38184 & 2.02246 \end{bmatrix} \begin{bmatrix} \hat{\mu} \\ \hat{p}_2 \\ \hat{p}_3 \\ \hat{v}_1 \\ \hat{v}_2 \\ \hat{v}_3 \\ \hat{v}_4 \\ \hat{v}_5 \end{bmatrix} = \begin{bmatrix} 25 \\ 64 \\ 47 \\ 55.406 \\ 40.531 \\ 40.531 \\ 21.844 \\ 17.469 \end{bmatrix}$$

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The solution is

$$\hat{p}' = [11.519 \quad 15.466 \quad 16.178]$$

$$\hat{v}' = [.959 \quad 1.009 \quad -.685 \quad .882 \quad -2.164]$$

Then $B\hat{v}$ gives the predicted real producing abilities of these 5 cows.

The predictions of a.g.v.'s are

$$\frac{.25}{1-.50} A\hat{v} = \frac{1}{2} A\hat{v}.$$

Henderson (1975)

Henderson, C.R. 1975. Best linear unbiased estimation and prediction under a selection model. Biometrics, vol. 31, no. 2, pp. 423 to 447.

The problem

The animals that come to have records have already undergone selection, and this could bias the evaluations. Henderson showed under which conditions BLUP remains unbiased even in the presence of selection. He also presented in an organized form the properties of the mixed model equations and the formulas for the variances of prediction errors.

Points to note while reading

Note the conditions that guarantee the absence of bias, which require selection to have been based on linear functions of the data included in the model. This is why an evaluation must use all the data and the whole pedigree, a point later taken up by Kennedy, Schaeffer and Sorensen.

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BEST LINEAR UNBIASED ESTIMATION AND PREDICTION UNDER A SELECTION MODEL

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SUMMARY

Mixed linear models are assumed in most animal breeding applications. Convenient methods for computing BLUE of the estimable linear functions of the fixed elements of the model and for computing best linear unbiased predictions of the random elements of the model have been available. Most data available to animal breeders, however, do not meet the usual requirements of random sampling, the problem being that the data arise either from selection experiments or from breeders' herds which are undergoing selection. Consequently, the usual methods are likely to yield biased estimates and predictions. Methods for dealing with such data are presented in this paper.

1. INTRODUCTION

Data available to animal breeding research workers for estimation of genetic and environmental parameters and to practitioners for making selection decisions almost invariably are provided by herds in which selection has been practiced. As a consequence, the usual assumptions of random sampling invoked for estimation and prediction are seldom valid. For example, production records in dairy cows have been used to estimate genetic and environmental trends. It is obvious from knowledge of the industry that the mean of first lactation records on cows that also produce a second record is higher than the mean of first records on contemporaries that do not produce a second record. This should suggest caution in applying the usual linear model methods to these data. Henderson [1949] and Henderson *et al.* [1959] presented methods for obtaining unbiased estimators in such cases. Similarly, Lush and Shrode [1950] described the bias in estimation of age effects due to culling on previous production. A similar problem occurs in evaluation of sires used in artificial breeding programs. Progeny tests are made on large numbers of sires. Then, only those with the best progeny are continued in service to produce many more progeny with records. Further, there is a tendency among those retained for the number of subsequent progeny to be positively correlated with past progeny production.

Two different possibilities for the solution of problems of this sort suggest themselves: (1) invoking a joint distribution of certain random variables of the model and the design matrix and (2) writing a distribution conditional on selected random variables. The first of these methods has proved to be extremely difficult, but some useful results can be attained using the second, as described in this paper. These are based on deriving best linear unbiased estimators and predictors under a model conditional on selection of certain linear functions of random variables jointly distributed with the random variables of the usual linear model. For example, in the cow culling case the selection variable could be the difference between means of first records on cows continuing in production and those that are culled.

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It is obvious that the subclass numbers are unequal in data which have arisen from selection situations, and in fact there are usually many empty subclasses. Professor Snedecor helped many research workers deal with the problems of unequal subclass numbers through his teaching and his book with its examples of strains of mice crossclassified with isolations of typhoid bacilli (Snedecor [1946]).

This paper presents some general results on best linear unbiased estimation and prediction from mixed linear models conditional on a selected vector variable. The methods are an extension of some of the author's mixed model techniques for the unconditional model. These latter methods are summarized in the next section.

2. BEST LINEAR UNBIASED PREDICTION

2.1 A mixed linear model

A mixed linear model is assumed in many genetic applications and can be represented as follows.

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \mathbf{e} \tag{1}$$

where $\mathbf{y}$ is an $n \times 1$ observation vector, $\mathbf{X}$ is a known, $n \times p$ matrix, $\boldsymbol{\beta}$ is an unknown, fixed vector, and $\mathbf{Z}$ is a known, $n \times q$ matrix. $\mathbf{u}$ and $\mathbf{e}$ are nonobservable random vectors with null means and

$$\text{Var}\begin{bmatrix}\mathbf{u}\\ \mathbf{e}\end{bmatrix} = \begin{bmatrix}\mathbf{G} & \mathbf{0}\\ \mathbf{0} & \mathbf{R}\end{bmatrix}\sigma^2$$

where σ^2 is a scalar, possibly unknown, and $\mathbf{G}$ and $\mathbf{R}$ are both nonsingular.

Now, given a sample vector, $\mathbf{y}$, we wish to do one or more of the following:

1. Estimate some estimable linear function of $\boldsymbol{\beta}$.
2. Test hypotheses regarding $\boldsymbol{\beta}$.
3. Estimate $\mathbf{G}$, $\mathbf{R}$, and σ^2.
4. Predict $\mathbf{u}$ or some linear functions of $\mathbf{u}$.
5. Predict linear functions of $\boldsymbol{\beta}$ and $\mathbf{u}$ jointly.

The animal breeding research worker is usually concerned with the first three, and the practitioner with the last two of these. The research worker is concerned also with developing better methods for the practitioner to accomplish the last two.

2.2 Best linear prediction

It is known that if $\boldsymbol{\beta}$, $\mathbf{G}$, and $\mathbf{R}$ are known, the best *linear* predictor of $\mathbf{k}'\boldsymbol{\beta} + \mathbf{m}'\mathbf{u}$ is

$$\mathbf{k}'\boldsymbol{\beta} + \mathbf{m}'\mathbf{G}\mathbf{Z}'\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \tag{2}$$

where $\mathbf{V} = \mathbf{R} + \mathbf{Z}\mathbf{G}\mathbf{Z}'$. In the multivariate normal case, (2) is of course $E(\mathbf{k}'\boldsymbol{\beta} + \mathbf{m}'\mathbf{u} \mid \mathbf{y})$ and consequently is the *best* predictor. Whether or not normality is implied, (2) is the usual selection index used in animal breeding.

2.3 Best linear unbiased prediction

Now if $\boldsymbol{\beta}$ is unknown, as is usually the case, the method of (2) cannot be used. A modification invoking unbiasedness can be employed, however. By unbiased we mean that $E(\text{predictor}) = E(\mathbf{k}'\boldsymbol{\beta} + \mathbf{m}'\mathbf{u}) = \mathbf{k}'\boldsymbol{\beta}$. Henderson [1963] showed that the best linear unbiased predictor (BLUP) of $\mathbf{k}'\boldsymbol{\beta} + \mathbf{m}'\mathbf{u}$ is

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$$\mathbf{k}'\hat{\boldsymbol{\beta}} + \mathbf{m}'\mathbf{G}\mathbf{Z}'\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) \tag{3}$$

where $\hat{\boldsymbol{\beta}}$ is any solution to (4), the generalized least squares (GLS) equations,

$$\mathbf{X}'\mathbf{V}^{-1}\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{V}^{-1}\mathbf{y}. \tag{4}$$

The difficulty with this method is that $\mathbf{V}$ is often a matrix so large that its inversion is very costly. An alternative method was suggested by Henderson [1950]. The prediction of $\mathbf{k}'\boldsymbol{\beta} + \mathbf{m}'\mathbf{u}$ is $\mathbf{k}'\hat{\boldsymbol{\beta}} + \mathbf{m}'\hat{\mathbf{u}}$ where $\hat{\boldsymbol{\beta}}$ and $\hat{\mathbf{u}}$ are any solution to (5);

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \end{bmatrix}. \tag{5}$$

Henderson *et al.* [1959] proved that $\hat{\boldsymbol{\beta}}$ of (5) is a solution to (4), the GLS equations. The proof involved showing that

$$\mathbf{R}^{-1} - \mathbf{R}^{-1}\mathbf{Z}(\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1})^{-1}\mathbf{Z}'\mathbf{R}^{-1} = \mathbf{V}^{-1}. \tag{6}$$

Henderson [1963] proved that $\hat{\mathbf{u}}$ of (5) is equal to $\mathbf{G}\mathbf{Z}'\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$ of (3). This was done by showing that

$$(\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1})^{-1}\mathbf{Z}'\mathbf{R}^{-1} = \mathbf{G}\mathbf{Z}'\mathbf{V}^{-1} \tag{7}$$

and by noting that $\mathbf{X}\hat{\boldsymbol{\beta}}$ of (5) is the GLS estimator of $\mathbf{X}\boldsymbol{\beta}$.

The obvious advantage computationally of (5) over (3) is that neither $\mathbf{V}$ nor its inverse is required. Of course $\mathbf{R}$ has the same dimensions as $\mathbf{V}$, but it usually is an identity matrix. Further, $\mathbf{G}$ is often diagonal, and $\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1}$ is either diagonal or has a large diagonal submatrix.

2.4 Distributional properties of $\hat{\boldsymbol{\beta}}$ *and* $\hat{\mathbf{u}}$

Let some symmetric g-inverse of the coefficient matrix of (5) be

$$\begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{12}' & \mathbf{C}_{22} \end{bmatrix}. \tag{8}$$

$\mathbf{C}_{22}$ is always unique, but $\mathbf{C}_{11}$ and $\mathbf{C}_{12}$ are unique only if $\mathbf{X}$ has full column rank. Now assuming that $\mathbf{K}'\boldsymbol{\beta}$ is estimable,

$$\begin{aligned}
&\text{Var}\,(\mathbf{K}'\hat{\boldsymbol{\beta}}) = \mathbf{K}'\mathbf{C}_{11}\mathbf{K}\sigma^2, \\
&\text{Cov}\,(\mathbf{K}'\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}}') = \mathbf{0}, \\
&\text{Cov}\,(\mathbf{K}'\hat{\boldsymbol{\beta}}, \mathbf{u}') = -\mathbf{K}'\mathbf{C}_{12}\sigma^2, \\
&\text{Cov}\,(\mathbf{K}'\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}}' - \mathbf{u}') = \mathbf{K}'\mathbf{C}_{12}\sigma^2, \\
&\text{Var}\,(\hat{\mathbf{u}}) = (\mathbf{G} - \mathbf{C}_{22})\sigma^2, \\
&\text{Cov}\,(\hat{\mathbf{u}}, \mathbf{u}') = \text{Var}\,(\hat{\mathbf{u}}), \\
&\text{Var}\,(\hat{\mathbf{u}} - \mathbf{u}) = \mathbf{C}_{22}\sigma^2.
\end{aligned}$$

Accordingly,

$$\text{Var}\,(\mathbf{K}'\hat{\boldsymbol{\beta}} + \mathbf{M}'\hat{\mathbf{u}} - \mathbf{K}'\boldsymbol{\beta} - \mathbf{M}'\mathbf{u}) = [\mathbf{K}' \vdots \mathbf{M}'] \begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{12}' & \mathbf{C}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{K} \\ \mathbf{M} \end{bmatrix} \sigma^2.$$

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The proof of these results for the full rank case are presented in Appendix A.

3. EXAMPLES OF BLUP IN THE NO SELECTION MODEL

Suppose we have records on progeny of two sires in two herds with progeny numbers as follows.

	Herd	
Sire	1	2
1	5	10
2	5	0

We assume that the model for the kth progeny of the ith sire in the jth herd is

$$y_{ijk} = \mu + s_i + h_j + e_{ijk} \tag{9}$$

where $\mathbf{e}' = [e_{111}, \cdots, e_{213}]$, $E(\mathbf{e}) = 0$, and $\mathrm{Var}(\mathbf{e}) = \mathbf{I}\sigma^2$.

We wish to predict $s_1 - s_2$. The best linear unbiased predictor depends upon the distribution of $\mathbf{s}' = [s_1 \quad s_2]$ and $\mathbf{h}' = [h_1 \quad h_2]$. Various assumptions are described below.

3.1 Sires and herds fixed

The regular least squares equations are

$$\begin{bmatrix} 20 & 15 & 5 & 10 & 10 \\ 15 & 15 & 0 & 5 & 10 \\ 5 & 0 & 5 & 5 & 0 \\ 10 & 5 & 5 & 10 & 0 \\ 10 & 10 & 0 & 0 & 10 \end{bmatrix} \begin{bmatrix} \hat{\mu} \\ \hat{s}_1 \\ \hat{s}_2 \\ \hat{h}_1 \\ \hat{h}_2 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 5 \\ 5 & 10 & 0 \\ 0 & 0 & 5 \\ 5 & 0 & 5 \\ 0 & 10 & 0 \end{bmatrix} \bar{\mathbf{y}} \tag{10}$$

where $\bar{\mathbf{y}}' = [\bar{y}_{11.} \quad \bar{y}_{12.} \quad \bar{y}_{21.}]$. For any solution to (10),

$$\hat{s}_1 - \hat{s}_2 = [1 \quad 0 \quad -1]\bar{\mathbf{y}} \tag{11}$$

with variance $= 0.4\sigma^2$.

3.2 Sires fixed and herds random

$E(\mathbf{h}) = \mathbf{0}$ and $\mathrm{Var}(\mathbf{h}) = 0.5\mathbf{I}\sigma^2$. Then the BLUP equations according to (5) are the same as (10) except that a diagonal matrix with diagonal coefficients

$$[0 \quad 0 \quad 0 \quad 2 \quad 2]$$

is added to the coefficient matrix of (10). Now for any solution,

$$\hat{s}_1 - \hat{s}_2 = [11 \quad 2 \quad -13]\bar{\mathbf{y}}/13 \tag{12}$$

with variance $= 24\sigma^2/65$, which is less than $0.4\sigma^2$. Therefore, if in fact $\mathbf{h}$ were random but computations were done as in (10), the prediction error variance would have been unnecessarily large.

3.3 Sires random and herds fixed

$E(\mathbf{s}) = \mathbf{0}$ and $\mathrm{Var}(\mathbf{s}) = 0.1\mathbf{I}\sigma^2$. Now the BLUP equations are like those of (10) except that a diagonal matrix with diagonal elements

$$[0 \quad 10 \quad 10 \quad 0 \quad 0]$$

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is added to the coefficient matrix. The resulting solution is

$$\hat{s}_1 - \hat{s}_2 = [1 \quad 0 \quad -1]\mathbf{y}/3 \tag{13}$$

with prediction error variance $= 2\sigma^2/15$. We illustrate calculation of this variance from a g-inverse of the coefficient matrix. A g-inverse is

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 5 & 1 & -3 & -5 \\ 0 & 1 & 5 & -3 & -1 \\ 0 & -3 & -3 & 9 & 3 \\ 0 & -5 & -1 & 3 & 11 \end{bmatrix} \frac{1}{60}$$

and $\text{Var}(\hat{s}_1 - \hat{s}_2 - s_1 + s_2) = \sigma_e[5 - (2)(1) + 5]/60 = 2\sigma^2/15$.

3.4 Sires and herds random

Now the diagonal matrix with diagonals

$$[0 \quad 10 \quad 10 \quad 2 \quad 2]$$

is added to the coefficient matrix of (10). The solution to $\hat{s}_1 - \hat{s}_2$ is

$$[11 \quad 2 \quad -13]\mathbf{y}/37 \tag{14}$$

with prediction error variance $= 24\sigma^2/185$.

4. EXAMPLES OF EFFECT OF SELECTION

The example of section 3 is used to illustrate the consequences of selection. Note that sire 2 has no progeny in herd 2. Suppose we know that the progeny in herd 1 were produced earlier than those in herd 2 and we suspect that the reason sire 2 was not used in herd 2 was that his progeny were inferior to those of sire 1 in herd 1. If this is true, is the prediction of $s_1 - s_2$ biased as a consequence? In the absence of selection,

$$\begin{aligned} E(\bar{y}_{11.} - \bar{y}_{21.}) &= E(s_1 - s_2) \\ &= s_1 - s_2 \text{ in the fixed sire model} \\ &= 0 \text{ in the random sire model.} \end{aligned}$$

Suppose that due to selection this expectation is $s_1 - s_2 + \alpha$ in the fixed sire case or is α in the random sire case. Now assuming a multivariate normal distribution we examine the bias in prediction of $s_1 - s_2$ under the various models and suggest some unbiased predictors.

4.1 Fixed sires and herds

Conditional means, given that $E(\bar{y}_{11.} - \bar{y}_{21.}) = s_1 - s_2 + \alpha$, are

$$E\begin{bmatrix} \bar{y}_{11.} \\ \bar{y}_{12.} \\ \bar{y}_{21.} \\ s_1 - s_2 \end{bmatrix} = \begin{bmatrix} \mu + s_1 + h_1 \\ \mu + s_1 + h_2 \\ \mu + s_2 + h_1 \\ s_1 - s_2 \end{bmatrix} + \begin{bmatrix} 0.2 \\ 0 \\ -0.2 \\ 0 \end{bmatrix} t$$

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where $t = [\mathrm{Var}(\bar{y}_{11.} - \bar{y}_{21.})/\sigma^2]^{-1}\alpha$. Consequently, $\hat{s}_1 - \hat{s}_2$ of (11) has expectation $s_1 - s_2 + .4t$, a biased predictor. Does an unbiased linear predictor exist? Let a predictor be $\mathbf{b}'\bar{\mathbf{y}}$. For unbiasedness, the following conditions are sufficient:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0.2 & 0 & -0.2 \end{bmatrix} \mathbf{b} = \begin{bmatrix} 0 \\ 1 \\ -01 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

No solution exists to $\mathbf{b}$ so no unbiased predictor exists.

4.2 Sires fixed and herds random

Now the conditional means given $E(\bar{y}_{11.} - \bar{y}_{21.}) = s_1 - s_2 + \alpha$ are

$$E\begin{bmatrix} \bar{y}_{11.} \\ \bar{y}_{12.} \\ \bar{y}_{21.} \\ s_1 - s_2 \end{bmatrix} = \begin{bmatrix} \mu + s_1 \\ \mu + s_1 \\ \mu + s_2 \\ s_1 - s_2 \end{bmatrix} + \begin{bmatrix} 0.2 \\ 0 \\ -0.2 \\ 0 \end{bmatrix} t.$$

Then the expectation of $\hat{s}_1 - \hat{s}_2$ from (12) is $s_1 - s_2 + 24t/65$, a biased predictor. If $\mathbf{b}'\bar{\mathbf{y}}$ is to be an unbiased predictor, a sufficient condition is

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0.2 & 0 & -0.2 \end{bmatrix} \mathbf{b} = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}.$$

A unique solution exists. It is $\mathbf{b}' = [-1 \quad 2 \quad -1]$. Thus, a single unbiased predictor exists.

4.3 Sires random and herds fixed

The conditional means given $E(\bar{y}_{11.} - \bar{y}_{21.}) = \alpha$ are

$$E\begin{bmatrix} \bar{y}_{11.} \\ \bar{y}_{12.} \\ \bar{y}_{21.} \\ s_1 - s_2 \end{bmatrix} = \begin{bmatrix} \mu + h_1 \\ \mu + h_2 \\ \mu + h_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.3 \\ 0.1 \\ -0.3 \\ 0.2 \end{bmatrix} t.$$

From the solution in (13), $E(\hat{s}_1 - \hat{s}_2) = 0.2t$, an unbiased predictor. Note that if $s_1 - s_2$ had been predicted as though $\mathbf{s}$ were fixed as in (11), the expectation of the predictor would be $0.6t$. That is, the predictor of the difference is too large. It is easy to show that (13) is the only unbiased predictor.

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4.4 Sires and herds random

The conditional means given $E(\bar{y}_{11.} - \bar{y}_{21.}) = \alpha$ are

$$E\begin{pmatrix} \bar{y}_{11.} \\ \bar{y}_{12.} \\ \bar{y}_{21.} \\ s_1 - s_2 \end{pmatrix} = \begin{pmatrix} \mu \\ \mu \\ \mu \\ 0 \end{pmatrix} + \begin{pmatrix} 0.3 \\ 0.1 \\ -0.3 \\ 0.2 \end{pmatrix} t.$$

From (14), $E(\hat{s}_1 - \hat{s}_2) = .2t$, an unbiased predictor. This is not the only one, the complete set being $\mathbf{b}'\mathbf{y}$ where $\mathbf{b}' = [1 + 2a \vdots -1 - 3a \vdots a]$. The solution of (14) has minimum variance as we shall see in the next section.

4.5 Selection on s

Suppose that instead of production in herd 1 the choice of sire to be used in herd 2 was based on some prior knowledge of the sires regarded as random in the unconditional distribution.

$$E(s_1 - s_2) = \alpha \neq 0.$$

Then the conditional means are

$$E\begin{pmatrix} \bar{y}_{11.} \\ \bar{y}_{12.} \\ \bar{y}_{21.} \\ s_1 - s_2 \end{pmatrix} = \begin{pmatrix} 0.1 \\ 0.1 \\ -0.1 \\ 0.2 \end{pmatrix} t$$

where $t = (2\sigma_s^2/\sigma^2)^{-1}\alpha$. Assuming herds random, the BLUP solution for the unconditional distribution of (14) gives $E(\hat{s}_1 - \hat{s}_2) = 0.0703t$, which is biased.

If instead we use the solution in (12) in which sires are regarded as fixed, $E(\hat{s}_1 - \hat{s}_2) = 0.2t$, an unbiased predictor. A general unbiased predictor is $[a \vdots 1 - a \vdots -1]\hat{\mathbf{y}}$.

5. BLUP IN A SELECTION MODEL

In this section, we derive the general solution to best linear unbiased prediction and estimation in a selection model. The examples of section 4 can all be described as a particular case of a model described in section 5.1.

5.1 A mixed model with selection

The unconditional model for $\mathbf{y}$ is the same as (1) but with the additional condition that the form of distribution is multivariate normal. We also invoke an additional vector variable $\mathbf{w}$, jointly normally distributed with $\mathbf{y}$, $\mathbf{u}$, and $\mathbf{e}$. The following additional parameters exist.

$$E(\mathbf{w}) = \mathbf{d}. \tag{15}$$

$$\text{Cov}\left(\begin{pmatrix} \mathbf{y} \\ \mathbf{u} \\ \mathbf{e} \end{pmatrix}, \mathbf{w}'\right) = \begin{pmatrix} \mathbf{B} \\ \mathbf{B}_u \\ \mathbf{B}_e \end{pmatrix}\sigma^2 \tag{16}$$

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where $\mathbf{B} = \mathbf{Z}\mathbf{B}_u + \mathbf{B}_e$.

$$\mathrm{Var}(\mathbf{w}) = \mathbf{H}\sigma^2. \tag{17}$$

Now suppose that some sort of selection on $\mathbf{w}$ has occurred such that in the selection model

$$\begin{aligned} E(\mathbf{w}) &= \mathbf{s} \neq \mathbf{d}, \\ \mathrm{Var}\,(\mathbf{w}) &= \mathbf{H}_s\sigma^2 \neq \mathbf{H}\sigma^2. \end{aligned} \tag{18}$$

For example, in the model of section 4.4,

$$\mathbf{w} = \bar{y}_{11\cdot} - \bar{y}_{21\cdot} ,$$

$$E(\mathbf{w}) = \mathbf{s} \neq 0,$$

$$\mathbf{H} = \mathrm{Var}\,(\bar{y}_{11\cdot} - \bar{y}_{21\cdot})/\sigma^2 = 0.6,$$

$$\mathbf{B}_u = \frac{1}{\sigma^2}\mathrm{Cov}\left[\begin{pmatrix} s_1 \\ s_2 \\ h_1 \\ h_2 \end{pmatrix}, (\bar{y}_{11\cdot} - \bar{y}_{21\cdot})\right] = \begin{pmatrix} 0.1 \\ -0.1 \\ 0 \\ 0 \end{pmatrix},$$

$$\mathbf{B}_e = \frac{1}{\sigma^2}\mathrm{Cov}\left[\begin{pmatrix} \bar{e}_{11\cdot} \\ \bar{e}_{12\cdot} \\ \bar{e}_{21\cdot} \end{pmatrix}, (\bar{y}_{11\cdot} - \bar{y}_{21\cdot})\right] = \begin{pmatrix} 0.2 \\ 0 \\ -0.2 \end{pmatrix},$$

$$\mathbf{B} = \frac{1}{\sigma^2}\mathrm{Cov}\left[\begin{pmatrix} \bar{y}_{11\cdot} \\ \bar{y}_{12\cdot} \\ \bar{y}_{21\cdot} \end{pmatrix}, (\bar{y}_{11\cdot} - \bar{y}_{21\cdot})\right] = \begin{pmatrix} 0.3 \\ 0.1 \\ -0.3 \end{pmatrix}.$$

Given these parameters of (1), (15), (16), (17), and (18), we can write the remaining parameters of the conditional (selection) model. For this purpose, we use a result due to Pearson [1903]. $[\mathbf{v}_1' \vdots \mathbf{v}_2']$ is normally distributed with mean $= [\boldsymbol{\mu}_1' \vdots \boldsymbol{\mu}_2']$ and variance

$$\begin{pmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{12}' & \mathbf{C}_{22} \end{pmatrix}.$$

Then, if $\mathbf{v}_2$ is selected such that $E(\mathbf{v}_2) = \boldsymbol{\mu}_2 + \mathbf{k}$ and $\mathrm{Var}(\mathbf{v}_2) = \mathbf{C}_s$, the parameters of the conditional distribution are

$$\begin{aligned} E(\mathbf{v}_1) &= \boldsymbol{\mu}_1 + \mathbf{C}_{12}\mathbf{C}_{22}^{-1}\mathbf{k} \\ \mathrm{Var}\begin{pmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \end{pmatrix} &= \begin{pmatrix} \mathbf{C}_{11} - \mathbf{C}_{12}\mathbf{C}_0\mathbf{C}_{12}' & \vdots & \mathbf{C}_{12}\mathbf{C}_{22}^{-1}\mathbf{C}_s \\ \cdots & \vdots & \cdots \\ \mathbf{C}_s\mathbf{C}_{22}^{-1}\mathbf{C}_{12}' & \vdots & \mathbf{C}_s \end{pmatrix} \end{aligned} \tag{19}$$

where $\mathbf{C}_0 = \mathbf{C}_{22}^{-1}(\mathbf{C}_{22} - \mathbf{C}_s)\mathbf{C}_{22}^{-1}$. Note that if $\mathbf{v}_2$ is fixed and consequently $\mathbf{C}_s = \mathbf{0}$, we have the well-known result $\mathrm{Var}(\mathbf{v}_1) = \mathbf{C}_{11} - \mathbf{C}_{12}\mathbf{C}_{22}^{-1}\mathbf{C}_{12}'$. Applying the result in (19) to our mixed model, the parameters of the conditional distribution are

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$$E\begin{pmatrix}\mathbf{y}\\ \mathbf{u}\\ \mathbf{w}\end{pmatrix} = \begin{pmatrix}\mathbf{X}\boldsymbol{\beta} + \mathbf{B}\mathbf{t}\\ \mathbf{B}_u\mathbf{t}\\ \mathbf{s}\end{pmatrix} \tag{20}$$

where $\mathbf{t} = \mathbf{H}^{-1}(\mathbf{s} - \mathbf{d})$,

$$\operatorname{Var}\begin{pmatrix}\mathbf{y}\\ \mathbf{u}\\ \mathbf{w}\end{pmatrix} = \left(\begin{array}{c:c:c} \mathbf{V} - \mathbf{B}\mathbf{H}_0\mathbf{B}' & \mathbf{Z}\mathbf{G} - \mathbf{B}\mathbf{H}_0\mathbf{B}_u' & \mathbf{B}\mathbf{H}^{-1}\mathbf{H}_s \\ \hdashline \mathbf{G}\mathbf{Z}' - \mathbf{B}_u\mathbf{H}_0\mathbf{B}' & \mathbf{G} - \mathbf{B}_u\mathbf{H}_0\mathbf{B}_u' & \mathbf{B}_u\mathbf{H}^{-1}\mathbf{H}_s \\ \hdashline \mathbf{H}_s\mathbf{H}^{-1}\mathbf{B}' & \mathbf{H}_s\mathbf{H}^{-1}\mathbf{B}_u' & \mathbf{H}_s \end{array}\right) \tag{21}$$

where $\mathbf{H}_0 = \mathbf{H}^{-1}(\mathbf{H} - \mathbf{H}_s)\mathbf{H}^{-1}$.

5.2 Best linear unbiased prediction

We wish to predict

$$\mathbf{k}'\boldsymbol{\beta} + \mathbf{m}'\mathbf{u} + \mathbf{f}'(\mathbf{w} - \mathbf{d})$$

by $\mathbf{b}'\mathbf{y}$ such that the predictor is unbiased and has minimum variance of prediction error under the conditional model.

$$E[\mathbf{k}'\boldsymbol{\beta} + \mathbf{m}'\mathbf{u} + \mathbf{f}'(\mathbf{w} - \mathbf{d})] = \mathbf{k}'\boldsymbol{\beta} + \mathbf{m}'\mathbf{B}_u\mathbf{t} + \mathbf{f}'\mathbf{H}\mathbf{t}$$

$$E(\mathbf{b}'\mathbf{y}) = \mathbf{b}'\mathbf{X}\boldsymbol{\beta} + \mathbf{b}'\mathbf{B}\mathbf{t}$$

For unbiasedness, it is required that $\mathbf{X}'\mathbf{b} = \mathbf{k}$ and $\mathbf{B}'\mathbf{b} = \mathbf{B}_u'\mathbf{m} + \mathbf{H}\mathbf{f}$. The variance of the error of prediction is

$$\operatorname{Var}(\mathbf{b}'\mathbf{y} - \mathbf{m}'\mathbf{u} - \mathbf{f}'\mathbf{w}) = \{\mathbf{b}'(\mathbf{V} - \mathbf{B}\mathbf{H}_0\mathbf{B}')\mathbf{b} - 2\mathbf{b}'(\mathbf{Z}\mathbf{G} - \mathbf{B}\mathbf{H}_0\mathbf{B}_u')\mathbf{m} - 2\mathbf{b}'\mathbf{B}\mathbf{H}^{-1}\mathbf{H}_s\mathbf{f} + \operatorname{Var}(\mathbf{m}'\mathbf{u} + \mathbf{f}'\mathbf{w})\}\sigma^2.$$

This is minimized subject to unbiasedness by $\mathbf{b}$, the solution if it exists to (22),

$$\left(\begin{array}{c:c:c} \mathbf{V} - \mathbf{B}\mathbf{H}_0\mathbf{B}' & \mathbf{X} & \mathbf{B} \\ \hdashline \mathbf{X}' & \mathbf{0} & \mathbf{0} \\ \hdashline \mathbf{B}' & \mathbf{0} & \mathbf{0} \end{array}\right)\begin{pmatrix}\mathbf{b}\\ \boldsymbol{\theta}\\ \boldsymbol{\phi}\end{pmatrix} = \begin{pmatrix}\mathbf{Z}\mathbf{G}\mathbf{m} - \mathbf{B}\mathbf{H}_0\mathbf{B}_u'\mathbf{m} + \mathbf{B}\mathbf{H}^{-1}\mathbf{H}_s\mathbf{f}\\ \mathbf{k}\\ \mathbf{B}_u'\mathbf{m} + \mathbf{H}\mathbf{f}\end{pmatrix}. \tag{22}$$

From the third equation of (22), $\mathbf{B}'\mathbf{b} = \mathbf{B}_u'\mathbf{m} + \mathbf{H}\mathbf{f}$. Substituting this in the first equation of (22), we get

$$\begin{pmatrix}\mathbf{V} & \mathbf{X} & \mathbf{B}\\ \mathbf{X}' & \mathbf{0} & \mathbf{0}\\ \mathbf{B}' & \mathbf{0} & \mathbf{0}\end{pmatrix}\begin{pmatrix}\mathbf{b}\\ \boldsymbol{\theta}\\ \boldsymbol{\phi}\end{pmatrix} = \begin{pmatrix}\mathbf{Z}\mathbf{G}\mathbf{m} + \mathbf{B}\mathbf{f}\\ \mathbf{k}\\ \mathbf{B}_u'\mathbf{m} + \mathbf{H}\mathbf{f}\end{pmatrix}. \tag{23}$$

This is a fortunate result because the predictor does not require knowledge of $\mathbf{s}$ and $\mathbf{H}_s$. Provided a solution exists, it is

$$\mathbf{b}'\mathbf{y} = \mathbf{k}'\hat{\boldsymbol{\beta}} + (\mathbf{m}'\mathbf{B}_u + \mathbf{f}'\mathbf{H})\hat{\mathbf{t}} + (\mathbf{m}'\mathbf{G}\mathbf{Z}' + \mathbf{f}'\mathbf{B}')\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{B}\hat{\mathbf{t}}) \tag{24}$$

where $\hat{\boldsymbol{\beta}}$ and $\hat{\mathbf{t}}$ are any solution to (25),

$$\left(\begin{array}{c:c} \mathbf{X}'\mathbf{V}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{V}^{-1}\mathbf{B} \\ \hdashline \mathbf{B}'\mathbf{V}^{-1}\mathbf{X} & \mathbf{B}'\mathbf{V}^{-1}\mathbf{B} \end{array}\right)\begin{pmatrix}\hat{\boldsymbol{\beta}}\\ \hat{\mathbf{t}}\end{pmatrix} = \begin{pmatrix}\mathbf{X}'\mathbf{V}^{-1}\mathbf{y}\\ \mathbf{B}'\mathbf{V}^{-1}\mathbf{y}\end{pmatrix}. \tag{25}$$

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From the second equation of (25),

$$\mathbf{B}'\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{B}\hat{\mathbf{t}}) = 0.$$

Substituting this in (24), we obtain

$$\mathbf{b}'\mathbf{y} = \mathbf{k}'\hat{\boldsymbol{\beta}} + (\mathbf{m}'\mathbf{B}_s + \mathbf{f}'\mathbf{H})\hat{\mathbf{t}} + (\mathbf{m}'\mathbf{G}\mathbf{Z}'\mathbf{V}^{-1})(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{B}\hat{\mathbf{t}}). \tag{26}$$

Now from the matrix identities in (6) and (7),

$$\mathbf{b}'\mathbf{y} = \mathbf{k}'\hat{\boldsymbol{\beta}} + (\mathbf{m}'\mathbf{B}_u + \mathbf{f}'\mathbf{H})\hat{\mathbf{t}} + \mathbf{m}'\hat{\mathbf{v}} \tag{27}$$

where $\hat{\boldsymbol{\beta}}$, $\hat{\mathbf{t}}$, and $\hat{\mathbf{v}}$ are any solution to (28),

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{B} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{B} \\ \mathbf{B}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{B}'\mathbf{R}^{-1}\mathbf{Z} & \mathbf{B}'\mathbf{R}^{-1}\mathbf{B} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{v}} \\ \hat{\mathbf{t}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{B}'\mathbf{R}^{-1}\mathbf{y} \end{bmatrix}. \tag{28}$$

The advantages of the method of (27) and (28) over (24) and (25) are the same as described for the advantages of (5) over (3) and (4).

A further modification of (28) that yields a direct solution to $\hat{\mathbf{u}}$, the best linear unbiased predictor of $\mathbf{u}$, can be accomplished as follows. Premultiply each side of equation (28) by

$$\mathbf{P} = \begin{bmatrix} \mathbf{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_u' & \mathbf{I} \end{bmatrix}. \tag{29}$$

This gives

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{B} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{B} \\ \mathbf{B}_e'\mathbf{R}^{-1}\mathbf{X} & \mathbf{B}_e'\mathbf{R}^{-1}\mathbf{Z} - \mathbf{B}_u\mathbf{G}^{-1} & \mathbf{B}_e'\mathbf{R}^{-1}\mathbf{B} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{v}} \\ \hat{\mathbf{t}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{B}_e'\mathbf{R}^{-1}\mathbf{y} \end{bmatrix}. \tag{30}$$

Now insert $\mathbf{P}'(\mathbf{P}')^{-1} = \mathbf{I}$ between the coefficient matrix and $[\hat{\boldsymbol{\beta}}' \quad \hat{\mathbf{v}}' \quad \hat{\mathbf{t}}']'$ of (30) and simplify. This gives

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{B}_e \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{B}_e - \mathbf{G}^{-1}\mathbf{B}_u \\ \mathbf{B}_e'\mathbf{R}^{-1}\mathbf{X} & \mathbf{B}_e'\mathbf{R}^{-1}\mathbf{Z} - \mathbf{B}_u'\mathbf{G}^{-1} & \mathbf{B}_e'\mathbf{R}^{-1}\mathbf{B}_e + \mathbf{B}_u'\mathbf{G}^{-1}\mathbf{B}_u \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{v}} + \mathbf{B}_u\hat{\mathbf{t}} \\ \hat{\mathbf{t}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{B}_e'\mathbf{R}^{-1}\mathbf{y} \end{bmatrix}. \tag{31}$$

Now define $\hat{\mathbf{v}} + \mathbf{B}_u\hat{\mathbf{t}}$ as $\hat{\mathbf{u}}$ and substitute this in (27) and (31). This is a logical definition since $E(\hat{\mathbf{u}}) = \mathbf{B}_u\mathbf{t}$. Now (27) can be written as

$$\begin{aligned} \mathbf{b}'\mathbf{y} &= \mathbf{k}'\hat{\boldsymbol{\beta}} + \mathbf{m}'(\hat{\mathbf{v}} + \mathbf{B}_u\hat{\mathbf{t}}) + \mathbf{f}'\mathbf{H}\hat{\mathbf{t}} \\ &= \mathbf{k}'\hat{\boldsymbol{\beta}} + \mathbf{m}'\hat{\mathbf{u}} + \mathbf{f}'\mathbf{H}\hat{\mathbf{t}} \end{aligned} \tag{32}$$

where $\hat{\boldsymbol{\beta}}$, $\hat{\mathbf{u}}$, and $\hat{\mathbf{t}}$ are solutions to (31) with $\hat{\mathbf{u}} = \hat{\mathbf{v}} + B_u\hat{\mathbf{t}}$.

5.3 Estimability

In the no selection case, the solution to $\hat{\mathbf{u}}$ is always unique, but $\boldsymbol{\beta}$ is estimable only if $\mathbf{X}$ has full column rank. In the selection case, however, $\mathbf{u}$ is not necessarily predictable. The requirement for this is that $\mathbf{B}_u\mathbf{t}$ be estimable. Of course, even if this is not true, there

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may be certain linear functions of interest, say $\mathbf{m}'\mathbf{u}$, that are predictable. This will be true provided $\mathbf{m}'\mathbf{B}_u\mathbf{t}$ is estimable. One way to check estimability is to determine if $[\mathbf{k}' \vdots \mathbf{m}' \vdots \mathbf{f}'\mathbf{H}]'E[\hat{\boldsymbol{\beta}}' \quad \hat{\mathbf{u}}' \quad \hat{\mathbf{t}}']' = \mathbf{k}'\boldsymbol{\beta} + (\mathbf{m}'\mathbf{B}_u + \mathbf{f}'\mathbf{H})\mathbf{t}$ where

$$E\begin{pmatrix}\hat{\boldsymbol{\beta}}\\ \hat{\mathbf{u}}\\ \hat{\mathbf{t}}\end{pmatrix} = \begin{pmatrix}\mathbf{X}'\mathbf{R}^{-1}\mathbf{X}\\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X}\\ \mathbf{B}_s'\mathbf{R}^{-1}\mathbf{X}\end{pmatrix}\boldsymbol{\beta} + \begin{pmatrix}\mathbf{X}'\mathbf{R}^{-1}\mathbf{B}\\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{B}\\ \mathbf{B}_s'\mathbf{R}^{-1}\mathbf{B}\end{pmatrix}\mathbf{t}$$

premultiplied by the g-inverse of the coefficient matrix of (31) used in the solution to $[\hat{\boldsymbol{\beta}}' \vdots \hat{\mathbf{u}}' \vdots \hat{\mathbf{t}}']$.

5.4 Distributional properties of estimators and predictors

Let some symmetric g-inverse of the coefficient matrix in (31) be

$$\begin{pmatrix}\mathbf{C}_{11} & \mathbf{C}_{12} & \mathbf{C}_{13}\\ \mathbf{C}_{12}' & \mathbf{C}_{22} & \mathbf{C}_{23}\\ \mathbf{C}_{13}' & \mathbf{C}_{23}' & \mathbf{C}_{33}\end{pmatrix}. \tag{33}$$

Then, $\mathrm{Var}(\mathbf{K}'\hat{\boldsymbol{\beta}} + \mathbf{M}'\hat{\mathbf{u}} + \mathbf{F}'\hat{\mathbf{t}})/\sigma^2$ is

$$[\mathbf{K}' \vdots \mathbf{M}' \vdots \mathbf{F}'] \cdot \begin{pmatrix}\mathbf{C}_{11} & \mathbf{C}_{12}\mathbf{B}_u' & \mathbf{C}_{13}\\ \mathbf{B}_u\mathbf{C}_{13}' & \mathbf{G} - \mathbf{C}_{22} + \mathbf{C}_{23}\mathbf{B}_u' + \mathbf{B}_u\mathbf{C}_{23}' - \mathbf{B}_u\mathbf{H}_0\mathbf{B}_u' & \mathbf{B}_u\mathbf{C}_{33} - \mathbf{B}_u\mathbf{H}_0\\ \mathbf{C}_{13}' & \mathbf{C}_{33}\mathbf{B}_u' - \mathbf{H}_0\mathbf{B}_u' & \mathbf{C}_{33} - \mathbf{H}_0\end{pmatrix}\begin{pmatrix}\mathbf{K}\\ \mathbf{M}\\ \mathbf{F}\end{pmatrix}. \tag{34}$$

$\mathrm{Cov}\,([\mathbf{K}'\hat{\boldsymbol{\beta}} + \mathbf{M}'\hat{\mathbf{u}} + \mathbf{F}'\hat{\mathbf{t}}], [\mathbf{u}' \vdots \mathbf{w}'])/\sigma^2$

$$= [\mathbf{K}' \vdots \mathbf{M}' \vdots \mathbf{F}']\begin{pmatrix}-\mathbf{C}_{12} + \mathbf{C}_{13}\mathbf{B}_u' & \mathbf{0}\\ \mathbf{G} - \mathbf{C}_{22} + \mathbf{C}_{23}\mathbf{B}_u' - \mathbf{B}_u\mathbf{H}_0\mathbf{B}_u' & \mathbf{B}_u\mathbf{H}^{-1}\mathbf{H}_s\\ -\mathbf{C}_{23}' + \mathbf{C}_{33}\mathbf{B}_u' - \mathbf{H}_0\mathbf{B}_u' & \mathbf{H}^{-1}\mathbf{H}_s\end{pmatrix}. \tag{35}$$

Note that in contrast to the no selection case $\mathrm{Var}(\hat{\mathbf{u}}) \neq \mathrm{Cov}(\hat{\mathbf{u}}, \mathbf{u}')$. From (21),

$$\frac{1}{\sigma^2}\mathrm{Var}\begin{pmatrix}\mathbf{u}\\ \mathbf{w}\end{pmatrix} = \begin{pmatrix}\mathbf{G} - \mathbf{B}_u\mathbf{H}_0\mathbf{B}_u' & \mathbf{B}_u\mathbf{H}^{-1}\mathbf{H}_s\\ \mathbf{H}_s\mathbf{H}^{-1}\mathbf{B}_u' & \mathbf{H}_s\end{pmatrix}. \tag{36}$$

Combining the results of (34), (35), and (36), we obtain the results of (37) for variances of prediction errors where $\mathbf{H}\hat{\mathbf{t}}$ is the predictor of $\mathbf{w} - \mathbf{d}$ in the full rank case. The variance of error of prediction of $\mathbf{K}'\boldsymbol{\beta} + \mathbf{M}'\mathbf{u} + \mathbf{F}'(\mathbf{w} - \mathbf{d})$ is

$$[\mathbf{K}' \vdots \mathbf{M}' \vdots \mathbf{F}']\begin{pmatrix}\mathbf{C}_{11} & \mathbf{C}_{12} & \mathbf{C}_{13}\mathbf{H}\\ \mathbf{C}_{12}' & \mathbf{C}_{22} & \mathbf{C}_{23}\mathbf{H}\\ \mathbf{H}\mathbf{C}_{13}' & \mathbf{H}\mathbf{C}_{23}' & \mathbf{H}\mathbf{C}_{33}\mathbf{H} - \mathbf{H}\end{pmatrix}\begin{pmatrix}\mathbf{K}\\ \mathbf{M}\\ \mathbf{F}\end{pmatrix}\sigma^2. \tag{37}$$

From (31) and (37), we now have the fortunate result that best linear unbiased predictors and the prediction error variances require no knowledge of the intensity and not even

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the direction of selection. In contrast, it can be seen that the correlation between $\mathbf{u}_i$ and $\hat{\mathbf{u}}_i$ cannot be computed unless $\mathbf{H}_i$ is known. The results in (34) and (35) are proved for the full rank case in Appendix B.

6. SOME APPLICATIONS

6.1 Selection on $\mathbf{y}$

The most common type of selection involves selection on $\mathbf{y}$. For this case, let $\mathbf{w} = \mathbf{L}'\mathbf{y}$ where $\mathbf{L}'$ has full row rank. Then

$$\mathbf{B}_u = \text{Cov}(\mathbf{u}, \mathbf{y}'\mathbf{L})/\sigma^2 = \mathbf{G}\mathbf{Z}'\mathbf{L},$$

$$\mathbf{B}_e = \text{Cov}(\mathbf{e}, \mathbf{y}'\mathbf{L})/\sigma^2 = \mathbf{R}\mathbf{L},$$

$$\mathbf{B} = \text{Cov}(\mathbf{y}, \mathbf{y}'\mathbf{L})/\sigma^2 = \mathbf{V}\mathbf{L},$$

$$\mathbf{H} = \text{Var}(\mathbf{L}'\mathbf{y})/\sigma^2 = \mathbf{L}'\mathbf{V}\mathbf{L}.$$

Substituting these in (31), the BLUP equations for the selection model are

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} & \mathbf{X}'\mathbf{L} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} & \mathbf{0} \\ \mathbf{L}'\mathbf{X} & \mathbf{0} & \mathbf{L}'\mathbf{V}\mathbf{L} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \\ \hat{\mathbf{t}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{L}'\mathbf{y} \end{bmatrix}. \tag{38}$$

It should be noted in particular that if $\mathbf{L}'\mathbf{X} = \mathbf{0}$, $\hat{\boldsymbol{\beta}}$ and $\hat{\mathbf{u}}$ are any solution to the BLUP equations in the no selection model. [See Eq. (5)].

We illustrate $\mathbf{L}'\mathbf{y}$ type of selection with a sire example. Suppose we have the following progeny from 4 sires in 2 herds.

		Herd	
Group	Sire	1	2
1	1	5	20
	2	3	0
2	3	4	0
	4	6	0

The model assumed is

$$y_{ijkl} = g_i + s_{ij} + h_k + e_{ijkl}$$

where $\mathbf{g}$ is fixed; $\mathbf{s}$, $\mathbf{h}$, and $\mathbf{e}$ are normally distributed with null means, and

$$\text{Var}\begin{bmatrix} \mathbf{s} \\ \mathbf{h} \\ \mathbf{e} \end{bmatrix} = \begin{bmatrix} 0.1\mathbf{I} & 0 & 0 \\ 0 & 0.5\mathbf{I} & 0 \\ 0 & 0 & \mathbf{I} \end{bmatrix} \sigma^2.$$

Now we assume that sire 1 was selected to have progeny in herd 2 because his progeny were better in herd 1 than were those of the other sires. In terms of the model,

$$E(g_1 + s_1 + h_1 + \bar{e}_{111\cdot}) > E(g_1 + s_2 + h_1 + \bar{e}_{121\cdot})$$

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or $E[(s_1 + \bar{e}_{111.}) - (s_2 + \bar{e}_{121.})] > 0$,

$$E(g_1 + s_1 + h_1 + \bar{e}_{111.}) > E(g_2 + s_3 + h_1 + \bar{e}_{231.})$$

or $E[(s_1 + \bar{e}_{111.}) - (s_3 + \bar{e}_{231.})] > (g_2 - g_1)$.

Also $E[(s_1 + \bar{e}_{111.}) - (s_4 + \bar{e}_{241.})] > (g_2 - g_1)$.

Rewrite the model as

$$\bar{y}_{ijk.} = g_i + s_{ij} + h_k + \bar{e}_{ijk.}$$

where **s** and **h** have the same distributional properties as before and

$$\text{Var}(\bar{\mathbf{e}}) = \begin{pmatrix} 5 & & & & & \\ & & & & 0 & \\ & & 20 & & & \\ & & & 3 & & \\ & & & & 4 & \\ & 0 & & & & \\ & & & & & 6 \end{pmatrix}^{-1} \sigma^2.$$

Now we can describe the selection as

$$\mathbf{L}'\bar{\mathbf{y}} = \begin{pmatrix} 1 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \bar{y}_{111.} \\ \vdots \\ \bar{y}_{241.} \end{pmatrix}.$$

$$\mathbf{V} = \begin{pmatrix} 0.8 & 0.1 & 0.5 & 0.5 & 0.5 \\ 0.1 & 0.65 & 0 & 0 & 0 \\ 0.5 & 0 & 0.933\ldots & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 & 0.85 & 0.5 \\ 0.5 & 0 & 0.5 & 0.5 & 0.766\ldots \end{pmatrix}.$$

$$\mathbf{L}'\mathbf{V}\mathbf{L} = \begin{pmatrix} 0.733\ldots & 0.3 & 0.3 \\ 0.3 & 0.65 & 0.3 \\ 0.3 & 0.3 & 0.566\ldots \end{pmatrix}.$$

$$\mathbf{L}'\mathbf{X} = \mathbf{L}' \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & -1 \\ 1 & -1 \end{pmatrix}.$$

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Then the BLUP equations according to (38) are

$$\begin{bmatrix} 28 & 0 & 25 & 3 & 0 & 0 & 8 & 20 & 0 & 1 & 1 \\ 0 & 10 & 0 & 0 & 4 & 6 & 10 & 0 & 0 & -1 & -1 \\ 25 & 0 & 35 & 0 & 0 & 0 & 5 & 20 & 0 & 0 & 0 \\ 3 & 0 & 0 & 13 & 0 & 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 14 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 & 16 & 6 & 0 & 0 & 0 & 0 \\ 8 & 10 & 5 & 3 & 4 & 6 & 20 & 0 & 0 & 0 & 0 \\ 20 & 0 & 20 & 0 & 0 & 0 & 0 & 22 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.733\ldots & 0.3 & 0.3 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0.3 & 0.65 & 0.3 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0.3 & 0.3 & 0.566\ldots \end{bmatrix} \begin{bmatrix} g_1 \\ g_2 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \\ h_1 \\ h_2 \\ t_1 \\ t_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} 5 & 20 & 3 & 0 & 0 \\ 0 & 0 & 0 & 4 & 6 \\ 5 & 20 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 6 \\ 5 & 0 & 3 & 4 & 6 \\ 0 & 20 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & -1 \end{bmatrix} \bar{\mathbf{y}}. \tag{39}$$

The sires are evaluated by $\hat{g}_i + \hat{s}_{i'}$. The solution is

$$\begin{bmatrix} \hat{g}_1 + \hat{s}_1 \\ \hat{g}_1 + \hat{s}_2 \\ \hat{g}_2 + \hat{s}_3 \\ \hat{g}_2 + \hat{s}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ -0.3846 & 1.1538 & 0.2308 & 0 & 0 \\ 0.6653 & -0.9148 & 0.2495 & 0.5946 & 0.4054 \\ 0.5821 & -0.8004 & 0.2183 & 0.2703 & 0.7297 \end{bmatrix} \bar{\mathbf{y}}. \tag{40}$$

To check that this solution is unbiased, note that

$$E(\bar{\mathbf{y}}) = \begin{bmatrix} g_1 \\ g_1 \\ g_1 \\ g_2 \\ g_2 \end{bmatrix} + \begin{bmatrix} 0.3 & 0.3 & 0.3 \\ 0.1 & 0.1 & 0.1 \\ -0.433\ldots & 0 & 0 \\ 0 & -0.35 & 0 \\ 0 & 0 & -0.266\ldots \end{bmatrix} \mathbf{t}.$$

Using this result and the solution of (40),

$$E\begin{bmatrix} \hat{g}_1 + \hat{s}_1 \\ \hat{g}_1 + \hat{s}_2 \\ \hat{g}_2 + \hat{s}_3 \\ \hat{g}_2 + \hat{s}_4 \end{bmatrix} = \begin{bmatrix} g_1 \\ g_1 \\ g_2 \\ g_2 \end{bmatrix} + \begin{bmatrix} 0.1 & 0.1 & 0.1 \\ -0.1 & 0 & 0 \\ 0 & -0.1 & 0 \\ 0 & 0 & -0.1 \end{bmatrix} \mathbf{t},$$

which is unbiased because this is the expectation of $\mathbf{g} + \mathbf{s}$. Observe that if the model had been

$$y_{ijk} = \mu + s_i + h_j + e_{ijk},$$

$\mathbf{X}' = [1 \cdots 1]$ and $\mathbf{L}'\mathbf{X} = \mathbf{0}$. Consequently, regular BLUP would be unbiased.

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6.2 Selection on $\mathbf{u}$

In some cases selection may have occurred on $\mathbf{u}$. In this case, let $\mathbf{w} = \mathbf{L}'\mathbf{u}$. Then

$$\mathbf{B}_u = \text{Cov}(\mathbf{u}, \mathbf{u}'\mathbf{L})/\sigma^2 = \mathbf{GL},$$
$$\mathbf{B}_e = \text{Cov}(\mathbf{e}, \mathbf{u}'\mathbf{L})/\sigma^2 = \mathbf{0},$$
$$\mathbf{B} = \text{Cov}(\mathbf{y}, \mathbf{u}'\mathbf{L})/\sigma^2 = \mathbf{ZGL},$$
$$\mathbf{H} = \text{Var}(\mathbf{L}'\mathbf{u})/\sigma^2 = \mathbf{L}'\mathbf{GL}.$$

Substituting in (31), we get as the BLUP equations

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} & \mathbf{0} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} & -\mathbf{L} \\ \mathbf{0} & -\mathbf{L}' & \mathbf{L}'\mathbf{GL} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \\ \hat{\mathbf{t}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{0} \end{bmatrix}. \tag{41}$$

As an example of this, suppose we have progeny on 6 sires numbering 5, 8, 3, 2, 4, and 2. The model is

$$y_{ij} = \mu + s_i + e_{ij}$$

where μ is fixed, s_i and e_{ij} are normally distributed with means = 0,

$$\text{Var}(\mathbf{s}) = \frac{1}{9}\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0.5 & 0 & 0 & 0 \\ 0 & 0.5 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}\sigma^2,$$

$\text{Var}(\mathbf{e}) = \mathbf{I}\sigma^2$, and $\text{Cov}(\mathbf{s}, \mathbf{e}') = \mathbf{0}$. $\text{Var}(\mathbf{s})$ might imply an additive genetic model with all sires unrelated except 2 and 3 which are full sibs. Now suppose we have prior information on the sires that makes us believe that $s_1 + s_2 > s_3 + s_4$ and $s_1 + s_2 > s_5 + s_6$. Then,

$$\mathbf{L}'\mathbf{u} = \begin{bmatrix} 1 & 1 & -1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 & -1 & -1 \end{bmatrix}\mathbf{u}.$$

The equations of (41) become

$$\begin{bmatrix} 24 & 5 & 8 & 3 & 2 & 4 & 2 & 0 & 0 \\ 5 & 14 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \\ 8 & 0 & 20 & -6 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & -6 & 15 & 0 & 0 & 0 & 1 & 0 \\ 2 & 0 & 0 & 0 & 11 & 0 & 0 & 1 & 0 \\ 4 & 0 & 0 & 0 & 0 & 13 & 0 & 0 & 1 \\ 2 & 0 & 0 & 0 & 0 & 0 & 11 & 0 & 1 \\ 0 & -1 & -1 & 1 & 1 & 0 & 0 & 0.33\ldots & 0.166\ldots \\ 0 & -1 & -1 & 0 & 0 & 1 & 1 & 0.166\ldots & 0.44\ldots \end{bmatrix} \begin{bmatrix} \hat{\mu} \\ \hat{s}_1 \\ \hat{s}_2 \\ \hat{s}_3 \\ \hat{s}_4 \\ \hat{s}_5 \\ \hat{s}_6 \\ \hat{t}_1 \\ \hat{t}_2 \end{bmatrix} = \begin{bmatrix} 5 & 8 & 3 & 2 & 4 & 2 \\ 5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \bar{y}$$

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where $\bar{\mathbf{y}}' = [\bar{y}_1 \cdots \bar{y}_6]$. The solution is approximately

$$\begin{pmatrix} \hat{\mu} \\ \hat{s}_1 \\ \hat{s}_2 \\ \hat{s}_3 \\ \hat{s}_4 \\ \hat{s}_5 \\ \hat{s}_6 \end{pmatrix} = \begin{pmatrix} 0.134 & 0.180 & 0.148 & 0.150 & 0.244 & 0.144 \\ 0.594 & 0.165 & -0.195 & -0.226 & -0.213 & -0.125 \\ 0.082 & 0.519 & -0.043 & -0.138 & -0.264 & -0.156 \\ -0.213 & 0.101 & 0.388 & 0.143 & -0.263 & -0.156 \\ -0.323 & -0.129 & 0.292 & 0.466 & -0.192 & -0.114 \\ -0.095 & -0.221 & -0.162 & -0.125 & 0.493 & 0.110 \\ -0.088 & -0.228 & -0.165 & -0.120 & 0.264 & 0.338 \end{pmatrix} \bar{\mathbf{y}}.$$

$$E(\bar{\mathbf{y}}) = \begin{pmatrix} \mu \\ \mu \\ \mu \\ \mu \\ \mu \\ \mu \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 0.5 & 1 \\ -0.5 & 0.5 \\ -1 & 0 \\ 0 & -1 \\ 0 & -1 \end{pmatrix} \mathbf{t}/9.$$

Then,

$$E\begin{pmatrix} \hat{\mu} \\ \hat{s}_1 \\ \hat{s}_2 \\ \hat{s}_3 \\ \hat{s}_4 \\ \hat{s}_5 \\ \hat{s}_6 \end{pmatrix} = \begin{pmatrix} \mu \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 1 \\ 0.5 & 1 \\ -0.5 & 0.5 \\ -1 & 0 \\ 0 & -1 \\ 0 & -1 \end{pmatrix} \mathbf{t}/9,$$

which is the expectation of

$$\begin{pmatrix} \mu \\ \mathbf{s} \end{pmatrix}.$$

In some applications, equations (41) may be difficult to write or to solve. An unbiased but not minimum prediction error variance method is available. Suppose selection has the form $\mathbf{L}_1'\mathbf{u}_1$ where $\mathbf{u}' = [\mathbf{u}_1' \vdots \mathbf{u}_2']$ and

$$\mathbf{G} = \begin{pmatrix} \mathbf{G}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{G}_{22} \end{pmatrix}.$$

Then it is easy to prove that a solution to equations of the form of (5) with $\mathbf{u}_1$ regarded as fixed yield these results: 1) $\hat{\mathbf{u}}_2$ is a unique, unbiased predictor of $\mathbf{u}_2$ and 2) $\mathbf{k}'\hat{\boldsymbol{\beta}} + \mathbf{m}_1'\hat{\mathbf{u}}_1$ is a unique, unbiased predictor of $\mathbf{k}'\boldsymbol{\beta} + \mathbf{m}_1'\mathbf{u}_1$ provided this is an estimable function under a fixed $\mathbf{u}_1$ model.

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6.3 Selection on **e**

In this case, let $\mathbf{w} = \mathbf{L}'\mathbf{e}$. Then,

$$\mathbf{B}_u = \text{Cov}\,(\mathbf{u}, \mathbf{e}'\mathbf{L})/\sigma^2 = \mathbf{0},$$
$$\mathbf{B}_e = \text{Cov}\,(\mathbf{e}, \mathbf{e}'\mathbf{L})/\sigma^2 = \mathbf{RL},$$
$$\mathbf{B} = \text{Cov}\,(\mathbf{y}, \mathbf{e}'\mathbf{L})/\sigma^2 = \mathbf{RL},$$
$$\mathbf{H} = \text{Var}\,(\mathbf{L}'\mathbf{e})/\sigma^2 = \mathbf{L}'\mathbf{RL}.$$

Substituting these in (31), we get

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} & \mathbf{X}'\mathbf{L} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} & \mathbf{Z}'\mathbf{L} \\ \mathbf{L}'\mathbf{X} & \mathbf{L}'\mathbf{Z} & \mathbf{L}'\mathbf{RL} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \\ \hat{\mathbf{t}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{L}'\mathbf{y} \end{bmatrix}. \tag{42}$$

As an example, suppose that 4 sires have progeny numbering 5, 10, 5, and 4. The model is

$$y_{ij} = \mu + s_i + e_{ij}$$

where **s** and **e** are normally distributed with means zero and

$$\text{Var}\begin{bmatrix} \mathbf{s} \\ \mathbf{e} \end{bmatrix} = \begin{bmatrix} .1\mathbf{I} & 0 \\ 0 & \mathbf{I} \end{bmatrix}\sigma^2.$$

We suspect that the $\bar{e}_i$ associated with sires 1 and 2 are better than those for 3 and 4 because of differential selection of mates. Then,

$$\mathbf{L}'\bar{\mathbf{e}} = [1 \quad 1 \quad -1 \quad -1]\bar{\mathbf{e}}$$

where $\bar{\mathbf{e}}' = [\bar{e}_1 \cdots \bar{e}_4]$. Then the equations according to (42) are

$$\begin{bmatrix} 24 & 5 & 10 & 5 & 4 & 0 \\ 5 & 15 & 0 & 0 & 0 & 1 \\ 10 & 0 & 20 & 0 & 0 & 1 \\ 5 & 0 & 0 & 15 & 0 & -1 \\ 4 & 0 & 0 & 0 & 14 & -1 \\ 0 & 1 & 1 & -1 & -1 & 0.75 \end{bmatrix} \begin{bmatrix} \hat{\mu} \\ \hat{s}_1 \\ \hat{s}_2 \\ \hat{s}_3 \\ \hat{s}_4 \\ \hat{l} \end{bmatrix} = \begin{bmatrix} 5 & 10 & 5 & 4 \\ 5 & 0 & 0 & 0 \\ 0 & 10 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 4 \\ 1 & 1 & -1 & -1 \end{bmatrix} \bar{\mathbf{y}}$$

where $\bar{\mathbf{y}}' = [\bar{y}_1 \cdots \bar{y}_4]$. The solution is approximately

$$\begin{bmatrix} \hat{\mu} \\ \hat{s}_1 \\ \hat{s}_2 \\ \hat{s}_3 \\ \hat{s}_4 \end{bmatrix} = \begin{bmatrix} 0.251 & 0.361 & 0.211 & 0.177 \\ 0.153 & -0.198 & 0.013 & 0.032 \\ -0.198 & 0.261 & -0.043 & -0.020 \\ 0.013 & -0.043 & 0.180 & -0.150 \\ 0.032 & -0.020 & -0.150 & 0.138 \end{bmatrix} \bar{\mathbf{y}},$$

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$$E(\bar{\mathbf{y}}) = \begin{bmatrix} \mu \\ \mu \\ \mu \\ \mu \end{bmatrix} + \begin{bmatrix} 0.2 \\ 0.1 \\ -0.2 \\ -0.25 \end{bmatrix} \mathrm{t}.$$

Then,

$$E\begin{bmatrix} \hat{\mu} \\ \hat{\delta}_1 \\ \hat{\delta}_2 \\ \hat{\delta}_3 \\ \hat{\delta}_4 \end{bmatrix} = \begin{bmatrix} \mu \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \mathrm{t},$$

as it should be for unbiasedness.

6.4 Sequential selection

The foregoing methods apply to sequential selection. To illustrate, suppose we wish to select for genetic merit of trait 1. We observe traits 1 and 2 on several animals. A selection index for trait 1 is constructed and a certain fraction of animals is selected. Later trait 3 is observed in the survivors of the initial selection. What is the best predictor of trait 3 and what are its distributional properties?

Suppose the model for traits 1, 2, and 3 is

$$\mathbf{y} = \mathbf{I}\mathbf{g} + \mathbf{I}\mathbf{e}$$

where $\mathbf{y}$, $\mathbf{g}$, and $\mathbf{e}$ are column vectors of order 3; $E(\mathbf{g}) = E(\mathbf{e}) = \mathbf{0}$;

$$\mathrm{Var}\,(\mathbf{e}) = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix} \sigma^2;$$

$$\mathrm{Var}\,(\mathbf{g}) = \begin{bmatrix} 5 & 2 & -1 \\ 2 & 6 & 3 \\ -1 & 3 & 7 \end{bmatrix} \sigma^2.$$

Note that $E(\mathbf{y}) = \mathbf{0}$, a usual assumption in selection index applications but not a very realistic one.

Now, given a sample $[\bar{y}_1 \quad \bar{y}_2]$, the best predictor of $[g_1 \quad g_2]'$ is

$$\begin{bmatrix} \hat{g}_1 \\ \hat{g}_2 \end{bmatrix} = (\mathbf{R}_2^{-1} + \mathbf{G}_2^{-1})^{-1}\mathbf{R}_2^{-1}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \frac{1}{79}\begin{bmatrix} 48 & 11 \\ 14 & 46 \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

where $\mathbf{R}_2$ and $\mathbf{G}_2$ refer to 2×2 submatrices of $\mathbf{R}$ and $\mathbf{G}$. Now we select on the basis of

$$\frac{1}{79}[48 \quad 11]\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

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such that the expectation of this function is not **0**. That is, $\mathbf{L}' = [48 \quad 11]/79$. Clearly $\mathbf{L}'\mathbf{X}$ does not exist since $\mathbf{X}$ does not exist. Consequently, BLUP of $\mathbf{g}$, given $[y_1 \quad y_2 \quad y_3]$, is the same solution as in the unconditional model. That is,

$$\begin{bmatrix} \hat{g}_1 \\ \hat{g}_2 \\ \hat{g}_3 \end{bmatrix} = (\mathbf{R}^{-1} + \mathbf{G}^{-1})^{-1}\mathbf{R}^{-1}\mathbf{y}$$

$$= \begin{bmatrix} 1.2983 & -0.0305 & 0.4203 \\ -0.0305 & 2.1508 & 1.3661 \\ 0.4203 & 1.3661 & 1.9559 \end{bmatrix} \begin{bmatrix} 0.6154 & 0.3846 & -0.4615 \\ 0.3846 & 0.6154 & -0.5385 \\ -0.4615 & -0.5385 & 0.8462 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

The variance of the prediction error of g_1 is $1.2983\sigma^2$ from (37). Let us examine the correlation between $\hat{g}_1$ and g_1. For this, we need $\mathbf{B}_*$ and $\mathbf{H}_0$.

$$\mathbf{B}_* = \text{Cov}\,(g_1, \mathbf{L}'\mathbf{y})/\sigma^2 = [5 \quad 2 \quad -1] \begin{bmatrix} 48 \\ 11 \\ 0 \end{bmatrix} \frac{1}{79} = 3.3165;$$

$$\mathbf{H} = \text{Var}\,(\mathbf{L}'\mathbf{y})/\sigma^2 = 262/79 = 3.3165.$$

Let $\mathbf{H}_* = k\mathbf{H}$ where k is in the range 0 to 1. Then $\mathbf{H}_0 = \mathbf{H}^{-1}(\mathbf{H} - \mathbf{H}_*)\mathbf{H}^{-1} = 0.3015(1 - k)$. From (34),

$$\text{Var}\,(\hat{g}_1)/\sigma = 5 - 1.2983 - (3.3165)^2(0.3015)(1 - k)$$

$$= 0.3853 + 3.3165k$$

This also is $\text{Cov}(\hat{g}_1, g_1)/\sigma^2$.

$$\text{Var}\,(g_1)/\sigma^2 = 5 - 3.3165(1 - k)$$

$$= 1.6835 + 3.3165k$$

Therefore,

$$r_{\hat{g}_1, g_1} = \sqrt{\frac{0.3853 + 3.3165k}{1.6835 + 3.3165k}}.$$

As would be expected, the more intense selection is in stage 1; that is, the smaller k is, the smaller the correlation is.

It should be noted that the results of this paper do not necessarily apply to more than two stage selection because the selected variates are not normally distributed and consequently the results of (21) do not apply.

7. BIAS DUE TO INCORRECT VARIANCES

It should be recognized that the equations of (31) give unbiased estimates and predictions under the assumption that the correct $\mathbf{R}$, $\mathbf{G}$, $\mathbf{B}_*$, and $\mathbf{B}_*$ are used. Statements of the bias due to using some other values can be written but are hard to interpret. In the case of selection on $\mathbf{y}$ and with $\mathbf{L}'\mathbf{X} = \mathbf{0}$, some fairly simple results can be stated.

Suppose that $\mathbf{R} = \mathbf{I}$ as is often the case, and we use an estimate of $\mathbf{G}$, say $\hat{\mathbf{G}}$. Then the BLUP equations for estimation of $\boldsymbol{\beta}$ and prediction of $\mathbf{u}$ are

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$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{X} & \mathbf{Z}'\mathbf{Z} + \hat{\mathbf{G}}^{-1} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{Z}'\mathbf{y} \end{bmatrix}. \tag{43}$$

Now under the distribution conditional on $\mathbf{L}'\mathbf{y}$,

$$E(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta} + (\mathbf{I} + \mathbf{Z}\mathbf{G}\mathbf{Z}')\mathbf{L}\hat{\mathbf{t}},$$

$$E(\mathbf{u}) = \mathbf{G}\mathbf{Z}'\mathbf{L}\hat{\mathbf{t}}.$$

Assume that $\mathbf{X}$ is full rank. Then, an inverse of the coefficient matrix of (43) is

$$\begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{12}' & \mathbf{C}_{22} \end{bmatrix}.$$

$$E(\hat{\boldsymbol{\beta}}) = (\mathbf{C}_{11}\mathbf{X}' + \mathbf{C}_{12}\mathbf{Z}')[\mathbf{X}\boldsymbol{\beta} + (\mathbf{I} + \mathbf{Z}\mathbf{G}\mathbf{Z}')\mathbf{L}\mathbf{t}].$$

Making use of $\mathbf{L}'\mathbf{X} = \mathbf{0}$, this simplifies to

$$\boldsymbol{\beta} + \mathbf{C}_{12}(\mathbf{G}^{-1} - \hat{\mathbf{G}}^{-1})\mathbf{G}\mathbf{Z}'\mathbf{L}\mathbf{t}$$

which, of course, reduces to $\boldsymbol{\beta}$ if $\mathbf{G} = \hat{\mathbf{G}}$ or if no selection has been practiced. Note further that if regular least squares were used, $\hat{\mathbf{G}}^{-1} = \mathbf{0}$ and the bias is $\mathbf{C}_{12}\mathbf{Z}'\mathbf{L}\mathbf{t}$, assuming in this last case that $\boldsymbol{\beta}$ and $\mathbf{u}$ are estimable under a fixed $\mathbf{u}$ model. Similarly,

$$E(\hat{\mathbf{u}}) = \mathbf{G}\mathbf{Z}'\mathbf{L}\mathbf{t} + \mathbf{C}_{22}(\mathbf{G}^{-1} - \hat{\mathbf{G}}^{-1})\mathbf{G}\mathbf{Z}'\mathbf{L}\mathbf{t}$$

which reduces to the correct expectation, $\mathbf{G}\mathbf{Z}'\mathbf{L}\mathbf{t}$, if $\mathbf{G} = \hat{\mathbf{G}}$ or if no selection had been practiced.

MEILLEUR ESTIMATEUR LINEAIRE SANS BIAIS (BLUE) ET PREDICTEUR SOUS UN MODELE DE SELECTION

RESUME

Les modèles linéaires mixtes sont choisis dans la plupart des applications sur lignées animales. Sont disponibles des méthodes de calcul:

—des BLUE des fonctions linéaires estimables des éléments fixés du modèle,

—des prédicteurs linéaires sans biais des éléments aléatoires du modèle.

La plupart des données relatives aux lignées animales ne satisfont cependant pas aux conditions usuelles de l'échantillonnage aléatoire. Le problème est que les données proviennent soit d'une expérimentation de sélection, soit de troupeaux qui subissent une sélection.

Par conséquent, les méthodes usuelles donnent vraisemblablement des estimateurs et des prédicteurs biaisés. On présente, dans ce papier, des méthodes pour traiter de telles données.

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APPENDIX A

Derivation of Variance-Covariance Matrix of Estimators, Predictors, and Prediction Errors in the No Selection Model.

Let a regular inverse of the coefficient matrix of (5) be

$$\begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{12}' & \mathbf{C}_{22} \end{bmatrix}.$$

Then

$$\begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{C}_{11} & \vdots & \mathbf{C}_{12} \\ \cdots & \cdots & \cdots \\ \mathbf{C}_{12}' & \vdots & \mathbf{C}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{Q}_1' \\ \mathbf{Q}_2' \end{bmatrix} \bar{\mathbf{y}} \text{ , say.}$$

The following identity is useful,

$$\begin{bmatrix} \mathbf{Q}_1' \\ \mathbf{Q}_2' \end{bmatrix} [\mathbf{X} \vdots \mathbf{Z}] = \begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{12}' & \mathbf{C}_{22} \end{bmatrix} \left(\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \vdots & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} \\ \cdots & \cdots & \cdots \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \vdots & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ 0 & \mathbf{G}^{-1} \end{bmatrix} \right)$$

$$= \begin{bmatrix} \mathbf{I} & 0 \\ 0 & \mathbf{I} \end{bmatrix} - \begin{bmatrix} 0 & \mathbf{C}_{12}\mathbf{G}^{-1} \\ 0 & \mathbf{C}_{22}\mathbf{G}^{-1} \end{bmatrix}.$$

That is,

$$\begin{aligned} \mathbf{Q}_1'\mathbf{X} &= \mathbf{I}, \\ \mathbf{Q}_1'\mathbf{Z} &= -\mathbf{C}_{12}\mathbf{G}^{-1}, \\ \mathbf{Q}_2'\mathbf{X} &= 0, \\ \mathbf{Q}_2'\mathbf{Z} &= \mathbf{I} - \mathbf{C}_{22}\mathbf{G}^{-1}. \end{aligned}$$

Using these results,

$$\begin{aligned} \text{Var } (\hat{\boldsymbol{\beta}})/\sigma^2 &= \mathbf{Q}_1'(\mathbf{R} + \mathbf{Z}\mathbf{G}\mathbf{Z}')\mathbf{Q}_1 \\ &= \mathbf{Q}_1'\mathbf{X}\mathbf{C}_{11} + \mathbf{Q}_1'\mathbf{Z}\mathbf{C}_{12}' + \mathbf{Q}_1'\mathbf{Z}\mathbf{G}\mathbf{Z}'\mathbf{Q}_1 \\ &= \mathbf{I}\mathbf{C}_{11} - \mathbf{C}_{12}\mathbf{G}^{-1}\mathbf{C}_{12}' + \mathbf{C}_{12}'\mathbf{G}^{-1}\mathbf{G}\mathbf{G}^{-1}\mathbf{C}_{12} \\ &= \mathbf{C}_{11}. \\ \text{Cov } (\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}}')/\sigma^2 &= \mathbf{Q}_1'(\mathbf{R} + \mathbf{Z}\mathbf{G}\mathbf{Z}')\mathbf{Q}_2 \\ &= \mathbf{Q}_1'\mathbf{X}\mathbf{C}_{12} + \mathbf{Q}_1'\mathbf{Z}\mathbf{C}_{22} + \mathbf{Q}_1'\mathbf{Z}\mathbf{G}\mathbf{Z}'\mathbf{Q}_2 \\ &= \mathbf{I}\mathbf{C}_{12} - \mathbf{C}_{12}\mathbf{G}^{-1}\mathbf{C}_{22} - \mathbf{C}_{12}\mathbf{G}^{-1}\mathbf{G}(\mathbf{I} - \mathbf{G}^{-1}\mathbf{C}_{22}) \\ &= 0. \end{aligned}$$

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$$\begin{aligned}
\text{Cov}\,(\hat{\boldsymbol{\beta}}, \mathbf{u}')/\sigma^2 &= \mathbf{Q}_1'\mathbf{Z}\mathbf{G} \\
&= -\mathbf{C}_{12}\mathbf{G}^{-1}\mathbf{G} \\
&= -\mathbf{C}_{12} . \\
\text{Cov}\,(\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}}' - \mathbf{u}')/\sigma^2 &= \mathbf{0} - (-\mathbf{C}_{12}) \\
&= \mathbf{C}_{12} . \\
\text{Var}\,(\hat{\mathbf{u}})/\sigma^2 &= \mathbf{Q}_2'(\mathbf{R} + \mathbf{Z}\mathbf{G}\mathbf{Z}')\mathbf{Q}_2 \\
&= \mathbf{Q}_2'\mathbf{X}\mathbf{C}_{12} + \mathbf{Q}_2'\mathbf{Z}\mathbf{C}_{22} + \mathbf{Q}_2'\mathbf{Z}\mathbf{G}\mathbf{Z}'\mathbf{Q}_2 \\
&= \mathbf{0} + (\mathbf{I} - \mathbf{C}_{22}\mathbf{G}^{-1})\mathbf{C}_{22} + (\mathbf{I} - \mathbf{C}_{22}\mathbf{G}^{-1})\mathbf{G}(\mathbf{I} - \mathbf{G}^{-1}\mathbf{C}_{22}) \\
&= \mathbf{G} - \mathbf{C}_{22} . \\
\text{Cov}\,(\hat{\mathbf{u}}, \mathbf{u})/\sigma^2 &= \mathbf{Q}_2'\mathbf{Z}\mathbf{G} \\
&= (\mathbf{I} - \mathbf{C}_{22}\mathbf{G}^{-1})\mathbf{G} \\
&= \mathbf{G} - \mathbf{C}_{22} . \\
\text{Var}\,(\hat{\mathbf{u}} - \mathbf{u})/\sigma^2 &= \mathbf{G} - \mathbf{C}_{22} - 2(\mathbf{G} - \mathbf{C}_{22}) + \mathbf{G} \\
&= \mathbf{C}_{22} .
\end{aligned}$$

APPENDIX B

Derivation of Variance-Covariance Matrix of Estimators, Predictors, and Prediction Error in Selection Model.

Let the regular inverse of the coefficient matrix of (25) be

$$\begin{bmatrix} \mathbf{T}_{11} & \mathbf{T}_{13} \\ \mathbf{T}_{13}' & \mathbf{T}_{33} \end{bmatrix} . \tag{B1}$$

Then,

$$\begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{t}} \end{bmatrix} = \begin{bmatrix} \mathbf{T}_{11} & \mathbf{T}_{13} \\ \mathbf{T}_{13}' & \mathbf{T}_{33} \end{bmatrix} \begin{bmatrix} \mathbf{X}'\mathbf{V}^{-1}\mathbf{y} \\ \mathbf{B}'\mathbf{V}^{-1}\mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{Q}_1' \\ \mathbf{Q}_3' \end{bmatrix} y, \quad \text{say.}$$

The following identity is useful,

$$\begin{bmatrix} \mathbf{Q}_1' \\ \mathbf{Q}_3' \end{bmatrix} [\mathbf{X} \vdots \mathbf{B}] = \begin{bmatrix} \mathbf{T}_{11} & \mathbf{T}_{13} \\ \mathbf{T}_{13}' & \mathbf{T}_{33} \end{bmatrix} \begin{bmatrix} \mathbf{X}'\mathbf{V}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{V}^{-1}\mathbf{B} \\ \mathbf{B}'\mathbf{V}^{-1}\mathbf{X} & \mathbf{B}'\mathbf{V}^{-1}\mathbf{B} \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} .$$

Thus, $\mathbf{Q}_1'\mathbf{X} = \mathbf{I}$, $\mathbf{Q}_1'\mathbf{B} = \mathbf{0}$, $\mathbf{Q}_3'\mathbf{X} = \mathbf{0}$, and $\mathbf{Q}_3'\mathbf{B} = \mathbf{I}$.

$$\begin{aligned}
\text{Var}\,(\hat{\boldsymbol{\beta}})/\sigma^2 &= \mathbf{Q}_1'(\mathbf{V} - \mathbf{B}\mathbf{H}_0\mathbf{B}')\mathbf{Q}_1 \\
&= \mathbf{Q}_1'\mathbf{X}\mathbf{T}_{11} + \mathbf{Q}_1'\mathbf{B}\mathbf{T}_{13}' - \mathbf{Q}_1'\mathbf{B}\mathbf{H}_0\mathbf{B}'\mathbf{Q}_1 \\
&= \mathbf{I}\mathbf{T}_{11} + \mathbf{0} - \mathbf{0} \\
&= \mathbf{T}_{11} .
\end{aligned}$$

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$$\begin{aligned}
\text{Cov } (\hat{\beta}, \hat{t})/\sigma^2 &= \mathbf{Q}_1'(\mathbf{V} - \mathbf{B}\mathbf{H}_0\mathbf{B}')\mathbf{Q}_3 \\
&= \mathbf{Q}_1'\mathbf{X}\mathbf{T}_{13} + \mathbf{Q}_1'\mathbf{B}\mathbf{T}_{33} - \mathbf{Q}_1'\mathbf{B}\mathbf{H}_0\mathbf{B}'\mathbf{Q}_3 \\
&= \mathbf{I}\mathbf{T}_{13} + 0 - 0 \\
&= \mathbf{T}_{13} . \\
\text{Cov } (\hat{\beta}, \mathbf{w}')/\sigma^2 &= \mathbf{Q}_1'\mathbf{B}\mathbf{H}^{-1}\mathbf{H}_2 \\
&= 0. \\
\text{Var } (\hat{t})/\sigma^2 &= \mathbf{Q}_3'(\mathbf{V} - \mathbf{B}\mathbf{H}_0\mathbf{B}')\mathbf{Q}_3 \\
&= \mathbf{Q}_3'\mathbf{X}\mathbf{T}_{13} + \mathbf{Q}_3'\mathbf{B}\mathbf{T}_{33} - \mathbf{Q}_3'\mathbf{B}\mathbf{H}_0\mathbf{B}'\mathbf{Q}_3 \\
&= 0 + \mathbf{T}_{33} - \mathbf{H}_0 \\
&= \mathbf{T}_{33} - \mathbf{H}_0 . \\
\text{Cov } (\hat{t}, \mathbf{w}')/\sigma^2 &= \mathbf{Q}_3'\mathbf{B}\mathbf{H}^{-1}\mathbf{H}_2 \\
&= \mathbf{H}^{-1}\mathbf{H}_2 .
\end{aligned}$$

Let the regular inverse of the coefficient matrix of (28) be

$$\begin{pmatrix} \mathbf{T}_{11} & \mathbf{T}_{12} & \mathbf{T}_{13} \\ \mathbf{T}_{12}' & \mathbf{T}_{22} & \mathbf{T}_{23} \\ \mathbf{T}_{13}' & \mathbf{T}_{23}' & \mathbf{T}_{33} \end{pmatrix}. \tag{B2}$$

From partitioned matrix inversion methods and the identity of (6), it is clear that $\mathbf{T}_{11}$, $\mathbf{T}_{13}$, and $\mathbf{T}_{33}$ of (B2) are identical to submatrices of (B1) with the same identification. From (B2) and (28),

$$\begin{pmatrix} \hat{\beta} \\ \hat{\nu} \\ \hat{t} \end{pmatrix} = \begin{pmatrix} \mathbf{T}_{11} & \mathbf{T}_{12} & \mathbf{T}_{13} \\ \mathbf{T}_{12}' & \mathbf{T}_{22} & \mathbf{T}_{23} \\ \mathbf{T}_{13}' & \mathbf{T}_{23}' & \mathbf{T}_{33} \end{pmatrix} \begin{pmatrix} \mathbf{X}'\mathbf{R}^{-1} \\ \mathbf{Z}'\mathbf{R}^{-1} \\ \mathbf{B}'\mathbf{R}^{-1} \end{pmatrix} \mathbf{y} = \begin{pmatrix} \mathbf{Q}_1' \\ \mathbf{Q}_2' \\ \mathbf{Q}_3' \end{pmatrix} \mathbf{y}, \quad \text{say.}$$

$$\begin{pmatrix} \mathbf{Q}_1' \\ \mathbf{Q}_2' \\ \mathbf{Q}_3' \end{pmatrix} [\mathbf{X} \vdots \mathbf{Z} \vdots \mathbf{B}] = \begin{pmatrix} \mathbf{T}_{11} & \mathbf{T}_{12} & \mathbf{T}_{13} \\ \mathbf{T}_{12}' & \mathbf{T}_{22} & \mathbf{T}_{23} \\ \mathbf{T}_{13}' & \mathbf{T}_{23}' & \mathbf{T}_{33} \end{pmatrix} \left\{ \begin{pmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{B} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{B} \\ \mathbf{B}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{B}'\mathbf{R}^{-1}\mathbf{Z} & \mathbf{B}'\mathbf{R}^{-1}\mathbf{B} \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{G}^{-1} & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\}$$

$$= \begin{pmatrix} \mathbf{I} & 0 & 0 \\ 0 & \mathbf{I} & 0 \\ 0 & 0 & \mathbf{I} \end{pmatrix} - \begin{pmatrix} 0 & \mathbf{T}_{12}\mathbf{G}^{-1} & 0 \\ 0 & \mathbf{T}_{22}\mathbf{G}^{-1} & 0 \\ 0 & \mathbf{T}_{23}'\mathbf{G}^{-1} & 0 \end{pmatrix}$$

That is, $\mathbf{Q}_1'\mathbf{X} = \mathbf{I}$, $\mathbf{Q}_1'\mathbf{Z} = -\mathbf{T}_{12}\mathbf{G}^{-1}$, $\mathbf{Q}_1'\mathbf{B} = 0$, $\mathbf{Q}_2'\mathbf{X} = 0$, $\mathbf{Q}_2'\mathbf{Z} = \mathbf{I} - \mathbf{T}_{22}\mathbf{G}^{-1}$, $\mathbf{Q}_2'\mathbf{B} = 0$, $\mathbf{Q}_3'\mathbf{X} = 0$, $\mathbf{Q}_3'\mathbf{Z} = -\mathbf{T}_{23}'\mathbf{G}^{-1}$, and $\mathbf{Q}_3'\mathbf{B} = \mathbf{I}$. These identities are used extensively.

$$\begin{aligned}
\text{Cov } (\hat{\beta}, \hat{\nu}')/\sigma^2 &= \mathbf{Q}_1'(\mathbf{R} + \mathbf{Z}\mathbf{G}\mathbf{Z}' - \mathbf{B}\mathbf{H}_0\mathbf{B}')\mathbf{Q}_2 \\
&= \mathbf{Q}_1'\mathbf{X}\mathbf{T}_{12} + \mathbf{Q}_1'\mathbf{Z}\mathbf{T}_{22} + \mathbf{Q}_1'\mathbf{B}\mathbf{T}_{23}' + \mathbf{Q}_1'\mathbf{Z}\mathbf{G}\mathbf{Z}'\mathbf{Q}_2 + \mathbf{Q}_1'\mathbf{B}\mathbf{H}_0\mathbf{B}'\mathbf{Q}_2 \\
&= \mathbf{I}\mathbf{T}_{12} - \mathbf{T}_{12}\mathbf{G}^{-1}\mathbf{T}_{22} + 0(\mathbf{T}_{23}') - \mathbf{T}_{12}\mathbf{G}^{-1}\mathbf{G}(\mathbf{I} - \mathbf{G}^{-1}\mathbf{T}_{22}) - 0(\mathbf{H}_0\mathbf{B}')\mathbf{Q}_2 \\
&= 0.
\end{aligned}$$

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$$\begin{aligned}\text{Cov}\ (\hat{\beta}, u')/\sigma^2 &= Q_1(ZG - BH_0B_u') \\ &= -T_{12}G^{-1}G - 0(H_0B_u') \\ &= -T_{12}\ .\end{aligned}$$

$$\begin{aligned}\text{Var}\ (\hat{v})/\sigma^2 &= Q_2'(R + ZGZ' - BH_0B')Q_2 \\ &= Q_2'XT_{12} + Q_2'ZT_{22} + Q_2'BT_{23}' + Q_2'ZGZ'Q_2 - Q_2'BH_0B'Q_2 \\ &= (0)(T_{12}) + (I - T_{22}G^{-1})T_{22} + (0)(T_{23}') \\ &\quad + (I - T_{22}G^{-1})G(I - G^{-1}T_{22}) + (0)(H_0)(0) \\ &= G - T_{22}\ .\end{aligned}$$

$$\begin{aligned}\text{Cov}\ (\hat{v}, \hat{t}')/\sigma^2 &= Q_2'(R + ZGZ' - BH_0B')Q_3 \\ &= Q_2'XT_{13} + Q_2'ZT_{23} + Q_2'BT_{33} + Q_2'ZGZ'Q_3 - Q_2'BH_0B'Q_3 \\ &= 0(T_{13}) + (I - T_{22}G^{-1})T_{23} + 0(T_{33}) \\ &\quad + (I - T_{22}G^{-1})G(-G^{-1}T_{23}) - (0)(H_0B')Q_3 \\ &= 0.\end{aligned}$$

$$\begin{aligned}\text{Cov}\ (\hat{v}, u')/\sigma^2 &= Q_2'(ZG - BH_0B_u') \\ &= (I - T_{22}G^{-1})G - (0)(H_0B_u') \\ &= G - T_{22}\end{aligned}$$

$$\begin{aligned}\text{Cov}\ (\hat{v}, w')/\sigma^2 &= Q_2'BH^{-1}H_s \\ &= 0.\end{aligned}$$

$$\begin{aligned}\text{Cov}\ (\hat{t}, u')/\sigma^2 &= Q_3'(ZG - BH_0B_u') \\ &= -T_{23}' - H_0B_u'.\end{aligned}$$

Now utilizing $\hat{u} = \hat{v} + B_u\hat{t}$,

$$\begin{aligned}\text{Cov}\ (\hat{\beta}, \hat{u}') &= \text{Cov}\ (\hat{\beta}, \hat{v}') + \text{Cov}\ (\hat{\beta}, \hat{t}'B_u') \\ &= T_{13}B_u'\sigma^2.\end{aligned}$$

$$\begin{aligned}\text{Var}\ (\hat{u}) &= \text{Var}\ (\hat{v} + B_u\hat{t}) \\ &= (G - T_{22} + B_uT_{33}B_u' - B_uH_0B_u')\sigma^2.\end{aligned}$$

$$\begin{aligned}\text{Cov}\ (\hat{u}, \hat{t}') &= \text{Cov}\ (\hat{v}, \hat{t}') + B_u\ \text{Var}\ (\hat{t}) \\ &= (B_uT_{33} - B_uH_0)\sigma^2.\end{aligned}$$

$$\begin{aligned}\text{Cov}\ (\hat{u}, u') &= \text{Cov}\ (\hat{v}, u') + \text{Cov}\ (B_u\hat{t}, u') \\ &= (G - T_{22} - B_uT_{23}' - B_uH_0B_u')\sigma^2.\end{aligned}$$

$$\begin{aligned}\text{Cov}\ (\hat{u}', w') &= \text{Cov}\ (\hat{v}, w') + \text{Cov}\ (B_u\hat{t}, w') \\ &= B_uH^{-1}H_s\sigma^2.\end{aligned}$$

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Now we look at the elements in the inverse of the coefficient matrix in (31),

$$\begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} & \mathbf{C}_{13} \\ \mathbf{C}_{12}' & \mathbf{C}_{22} & \mathbf{C}_{23} \\ \mathbf{C}_{13}' & \mathbf{C}_{23}' & \mathbf{C}_{33} \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & -\mathbf{B}_u \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{T}_{11} & \mathbf{T}_{12} & \mathbf{T}_{13} \\ \mathbf{T}_{12}' & \mathbf{T}_{22} & \mathbf{T}_{23} \\ \mathbf{T}_{13}' & \mathbf{T}_{23}' & \mathbf{T}_{33} \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_u' & \mathbf{I} \end{bmatrix}^{-1}$$

$$= \left[\begin{array}{c:c:c} \mathbf{T}_{11} & \mathbf{T}_{12} + \mathbf{T}_{13}\mathbf{B}_u' & \mathbf{T}_{13} \\ \hdashline \mathbf{T}_{12}' + \mathbf{B}_u\mathbf{T}_{13}' & \mathbf{T}_{22} + \mathbf{B}_u\mathbf{T}_{23}' + \mathbf{T}_{23}\mathbf{B}_u' + \mathbf{B}_u\mathbf{T}_{33}\mathbf{B}_u' & \mathbf{T}_{23} + \mathbf{B}_u\mathbf{T}_{33} \\ \hdashline \mathbf{T}_{13}' & \mathbf{T}_{23}' + \mathbf{T}_{33}\mathbf{B}_u' & \mathbf{T}_{33} \end{array}\right].$$

Using these relationships between the elements of $\mathbf{C}$ and of $\mathbf{T}$ applied to the preceding variances and covariances of this appendix, we obtain

$$\frac{1}{\sigma^2}\operatorname{Cov}\left(\begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \\ \hat{\mathbf{t}} \end{bmatrix}, \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \\ \hat{\mathbf{t}} \\ \mathbf{u} \\ \mathbf{w} \end{bmatrix}'\right) = \left[\begin{array}{c:c:c} \mathbf{C}_{11} & \mathbf{C}_{13}\mathbf{B}_u' & \\ \hdashline \mathbf{B}_u\mathbf{C}_{13}' & \mathbf{G} - \mathbf{C}_{22} + \mathbf{C}_{23}\mathbf{B}_u' + \mathbf{B}_u\mathbf{C}_{23}' - \mathbf{B}_u\mathbf{H}_0\mathbf{B}_u' & \\ \hdashline \mathbf{C}_{13}' & \mathbf{C}_{33}\mathbf{B}_u' - \mathbf{H}_0\mathbf{B}_u' & \end{array}\right.$$

$$\left.\begin{array}{c:c:c} \mathbf{C}_{13} & -\mathbf{C}_{12} + \mathbf{C}_{13}\mathbf{B}_u' & \mathbf{0} \\ \hdashline \mathbf{B}_u\mathbf{C}_{33} - \mathbf{B}_u\mathbf{H}_0 & \mathbf{G} - \mathbf{C}_{22} + \mathbf{C}_{23}\mathbf{B}_u' - \mathbf{B}_u\mathbf{H}_0\mathbf{B}_u' & \mathbf{B}_u\mathbf{H}^{-1}\mathbf{H}_s \\ \hdashline \mathbf{C}_{33} - \mathbf{H}_0 & -\mathbf{C}_{23}' + \mathbf{C}_{33}\mathbf{B}_u' - \mathbf{H}_0\mathbf{B}_u' & \mathbf{H}^{-1}\mathbf{H}_s \end{array}\right].$$

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Henderson (1976)

Henderson, C.R. 1976. A simple method for computing the inverse of a numerator relationship matrix used in prediction of breeding values. Biometrics, vol. 32, no. 1, pp. 69 to 83.

The problem

The mixed model equations require the inverse of the relationship matrix A. For thousands of animals, inverting A directly was impossible with the computers of the time. Henderson discovered that the inverse can be built directly from the list of parents, with simple rules that add small contributions for each animal and its parents. The inverse is sparse and can be built for a pedigree of any size.

Points to note while reading

This is one of Henderson's papers with the greatest practical impact, because it made the animal model feasible. Note the rules for building the inverse, which depend on how many parents are known. In the same year, Quaas extended the method to include inbreeding. The same rules are still used, including within the H matrix.

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A SIMPLE METHOD FOR COMPUTING THE INVERSE OF A NUMERATOR RELATIONSHIP MATRIX USED IN PREDICTION OF BREEDING VALUES

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SUMMARY

The inverse of a numerator relationship matrix is needed for best linear unbiased prediction of breeding values. The purpose of this paper to is present a rapid and simple method for computation of the elements of this inverse without computing the relationship matrix itself. The method is particularly useful in non-inbred populations but is much faster than the conventional method in the presence of inbreeding.

1. INTRODUCTION

Most methods for predicting breeding values use records on certain close relatives of the animals to be predicted and ignore the information which could be provided by other relatives. Henderson [1952, 1973] suggested methods for utilizing all relationships in intra-herd female selection and inter-herd sire evaluation. These methods required either inversion of the numerator relationship matrix, in most cases a very large one, or a transformation of the best linear unbiased prediction (BLUP) equations by pre- and post-multiplying by the relationship matrix. The former procedure seemed impractical because of the large size of the matrix, and the latter had the distinct disadvantage of reducing the efficiency of an iterative solution. This paper presents a very simple and most unexpected method for finding the inverse of the relationship matrix directly from pedigree information without computing the matrix whose inverse is wanted. The method derives from the fact that a vector of additive genetic values can be written as the product of an easily computed triangular matrix and a vector of uncorrelated random variables with equal variances.

2. PREDICTION BY BEST LINEAR UNBIASED PREDICTION

The general principles of prediction of breeding values by best linear unbiased prediction methods (BLUP) have been stated by Henderson [1963, 1973].

Let a general mixed model for such prediction be

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}_1\mathbf{u} + \mathbf{Z}_2\mathbf{a} + \mathbf{e} \tag{1}$$

where $\mathbf{X}$ is a known matrix, β is an unknown fixed vector, $\mathbf{Z}_1$ is a known matrix, $\mathbf{u}$ is a nonobservable random vector variable, $\mathbf{Z}_2$ is a known matrix, $\mathbf{a}$ is a nonobservable random vector representing additive genetic merits and $\mathbf{e}$ is a nonobservable random vector of "errors." We assume that $\mathbf{u}$, $\mathbf{a}$ and $\mathbf{e}$ all have null means and that

$$\text{Var}(\mathbf{u}) = \mathbf{G}\sigma^2,$$

$$\text{Var}(\mathbf{a}) = k\mathbf{A}\sigma^2,$$

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where k is a scalar and

$$\text{Var}(\mathbf{e}) = \mathbf{I}\sigma^2.$$

$\mathbf{G}$ and $\mathbf{A}$ are known and nonsingular, and $\mathbf{A}$ is the numerator relationship matrix defined below. It is assumed that $\mathbf{u}$, $\mathbf{a}$ and $\mathbf{e}$ are mutually uncorrelated. σ^2 is a scalar, possibly unknown.

The following relates the notation of Henderson [1963] to the present paper.

Henderson [1963]	Present Paper
$\mathbf{Z}$	$[\mathbf{Z}_1 \vdots \mathbf{Z}_2]$
$\mathbf{u}'$	$[\mathbf{u}' \vdots \mathbf{a}']$
$\mathbf{G}$	$\begin{bmatrix} \mathbf{G} & \mathbf{0} \\ \mathbf{0} & k\mathbf{A} \end{bmatrix}$

Now, BLUP of $\mathbf{a}$ is $\hat{\mathbf{a}}$, any solution to

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z}_1 & \mathbf{X}'\mathbf{Z}_2 \\ \mathbf{Z}_1'\mathbf{X} & \mathbf{Z}_1'\mathbf{Z}_1 + \mathbf{G}^{-1} & \mathbf{Z}_1'\mathbf{Z}_2 \\ \mathbf{Z}_2'\mathbf{X} & \mathbf{Z}_2'\mathbf{Z}_1 & \mathbf{Z}_2'\mathbf{Z}_2 + k^{-1}\mathbf{A}^{-1} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \\ \hat{\mathbf{a}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{Z}_1'\mathbf{y} \\ \mathbf{Z}_2'\mathbf{y} \end{bmatrix}. \tag{2}$$

The solution to $\hat{\boldsymbol{\beta}}$ is not necessarily unique, but $\hat{\mathbf{u}}$ and $\hat{\mathbf{a}}$ are unique regardless of the solution to $\hat{\boldsymbol{\beta}}$.

In some important applications, either $[\hat{\boldsymbol{\beta}}' \vdots \hat{\mathbf{u}}']$ has relatively few elements or

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z}_1 \\ \mathbf{Z}_1'\mathbf{X} & \mathbf{Z}_1'\mathbf{Z}_1 + \mathbf{G}^{-1} \end{bmatrix}$$

has a large submatrix that is easy to invert (is diagonal, for example) and consequently most or all of the elements of $[\hat{\boldsymbol{\beta}}' \vdots \hat{\mathbf{u}}']$ can be absorbed easily. For example, in the Cornell method of sire evaluation used in the Northeast (Henderson [1973]), $\boldsymbol{\beta}$ represents a few groups and a very large number of fixed HYS (herd-year-season) effects. There is no $\mathbf{u}$ in this model. If, on the other hand, HYS were regarded as random variables with variance $= c\mathbf{I}\sigma^2$, these would form the $\mathbf{u}$ vector and groups would be the $\boldsymbol{\beta}$ vector. In either case, HYS can be absorbed easily leaving a set of equations with a few fixed groups and a very large number of sires (approximately 1,500 Holsteins). Presently, the interrelationships among sires are ignored. That is, $\mathbf{A}$ is regarded as having the value $\mathbf{I}$. The resulting equations are well-suited to an iterative solution because of many zero coefficients and, in particular, because the diagonal coefficients are large relative to the off-diagonal ones.

The prohibitive cost of inverting $\mathbf{A}$ has dictated in the past that the reduction in prediction error variance by using $\mathbf{A}^{-1}$ not be utilized. A simple and fast method for finding this matrix is presented in this paper with the thought that animal breeders can now utilize this additional information to increase the accuracy of selection. The method is particularly efficient for noninbred populations. Henderson [1975a, 1975b] reported in detail how to incorporate $\mathbf{A}^{-1}$ in both interherd evaluation of sires and interherd evaluation of males and females.

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3. THE NUMERATOR RELATIONSHIP MATRIX

The numerator relationship matrix, **A**, is a matrix which has the following properties:

(i) Is symmetric.

(ii) $a_{ii} = 1 + f_i$ where a_{ii} is the ith diagonal of **A**, and f_i is the inbreeding coefficient of the ith animal, Wright [1922].

(iii) $a_{ij} = r_{ij}\sqrt{a_{ii}a_{jj}}$ where r_{ij} is Wright's [1922] coefficient of relationship between animals i and j. He derived his results by his path coefficient method and presented details of the computation of r and f. Also see Malécot [1948].

3.1 A Recursive Method for Computing **A**.

The path coefficient method for computing **A** is not very well-suited to computerization, but a well-known recursive method is.

The animals for which $\mathbf{A}^{-1}$ is to be computed are identified by $1, 2, \cdots, n$. They must be ordered so that parents precede their progeny, and $1, \cdots, b$ must be a "base" population of animals regarded as noninbred and unrelated. Then the upper left b^2 submatrix of **A** is **I**. From that starting point, the upper left submatrix is expanded by order one in succession until the n^2 matrix is formed.

(i) If both parents of the tth individual are known, say p and q,

$$a_{it} = a_{ti} = .5(a_{ip} + a_{iq}) \quad \text{for} \quad i = 1, \cdots, t-1; \tag{3}$$

$$a_{tt} = 1 + .5a_{pq}\,. \tag{4}$$

(ii) If one parent only is known, say p,

$$a_{it} = a_{ti} = .5a_{ip} \quad \text{for} \quad i = 1, \cdots, t-1; \tag{5}$$

$$a_{tt} = 1. \tag{6}$$

(iii) If neither parent is known,

$$a_{ti} = a_{it} = 0 \quad \text{for} \quad i = 1, \cdots, t-1; \tag{7}$$

$$a_{tt} = 1. \tag{8}$$

We illustrate these and subsequent computations with the following example involving inbreeding.

TABLE 1

PARENTS OF INDIVIDUALS

Individual	Parents
1	Unknown
2	Unknown
3	1 and unknown
4	1 and 2
5	3 and 4
6	1 and 4
7	5 and 6

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Using the method described in (3) through (8), **A** is presented in Table 2.
For example,

$$a_{56} = (.5 + .625)/2 = .5625;$$

$$a_{77} = 1 + (.5625/2) = 1.28125.$$

3.2 A Recursive Method for Computing **L**.

Another recursive method involving computation of a triangular matrix can also be used. Let **L** be a lower triangular matrix such that $\mathbf{LL}' = \mathbf{A}$.

The n animals are ordered as described in the recursive computation of **A**. The upper b^2 submatrix of **L** is **I**. Then each of the remaining rows of **L** is computed in order by the following method.

(i) If both parents of t are known, say p and q where $p < q$,

$$\begin{aligned} l_{ti} &= .5(l_{pi} + l_{qi}) \quad \text{for} \quad j = 1, \cdots, p \\ &= .5l_{qi} \quad \text{for} \quad j = p + 1, \cdots, q \\ &= 0 \quad \text{for} \quad j = q + 1, \cdots, t - 1, \end{aligned} \tag{9}$$

a case which appears only if $q < t - 1$;

$$l_{tt} = \sqrt{1 + .5 \sum_{j=1}^{p} l_{pj} l_{qj} - \sum_{j=1}^{q} l_{tj}^2}. \tag{10}$$

Note that

$$.5 \sum_{j=1}^{p} l_{pj} l_{qj} = f_t \tag{11}$$

and

$$l_{tt} = \sqrt{.5 - .25(f_p + f_q)}. \tag{12}$$

TABLE 2

NUMERATOR RELATIONSHIP MATRIX

	1	2	3	4	5	6	7
1	1.00000	0.00000	0.50000	0.50000	0.50000	0.75000	0.62500
2	0.00000	1.00000	0.00000	0.50000	0.25000	0.25000	0.25000
3	0.50000	0.00000	1.00000	0.25000	0.62500	0.37500	0.50000
4	0.50000	0.50000	0.25000	1.00000	0.62500	0.75000	0.68750
5	0.50000	0.25000	0.62500	0.62500	1.12500	0.56250	0.84375
6	0.75000	0.25000	0.37500	0.75000	0.56250	1.25000	0.90625
7	0.62500	0.25000	0.50000	0.68750	0.84375	0.90625	1.28125

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(ii) If only one parent is known, say p,

$$l_{tj} = .5l_{pj} \quad \text{for} \quad j = 1, \cdots, p$$
$$= 0 \quad \text{for} \quad j = p+1, \cdots, t-1, \tag{13}$$

a case which appears only if $p < t - 1$;

$$l_{tt} = \sqrt{1 - \sum_{j=1}^{p} l_{tj}^2} = \sqrt{.75 - .25f_p}\,. \tag{14}$$

(iii) If neither parent is known,

$$l_{tj} = 0 \quad \text{for} \quad j = 1, \cdots, t-1; \tag{15}$$

$$l_{tt} = 1. \tag{16}$$

Using the methods of (9) through (16), **L** is presented in Table 3. To illustrate

$l_{53} = .5(.866025 + 0) = .433013;$

$l_{55} = \{1 + .5[(.5)^2 + 0(.5) + .866025(0)] - (.5)^2 - (.25)^2 - (.433013)^2 - (.353553)^2\}^{1/2}$
$= \sqrt{.5}.$

It is easy to verify that $\mathbf{LL}' = \mathbf{A}$ in this example.

4. INVERSION OF NUMERATOR RELATIONSHIP MATRIX

4.1 Conventional Method.

Most computing centers have one or more matrix inversion routines based on various modifications of Gauss elimination methods. These work well for small matrices, say in the order of 2 to 150, the upper limit depending upon the memory capacity of the machine. They are very slow if auxiliary storage is required, and even for matrices stored in memory

TABLE 3

TRIANGULAR MATRIX FOR COMPUTING **A**

	1	2	3	4	5	6	7
1	1.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
2	0.000000	1.000000	0.000000	0.000000	0.000000	0.000000	0.000000
3	0.500000	0.000000	0.866025	0.000000	0.000000	0.000000	0.000000
4	0.500000	0.500000	0.000000	0.707107	0.000000	0.000000	0.000000
5	0.500000	0.250000	0.433013	0.353553	0.707107	0.000000	0.000000
6	0.750000	0.250000	0.000000	0.353553	0.000000	0.707107	0.000000
7	0.625000	0.250000	0.216506	0.353553	0.353553	0.353553	0.637378

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the time required is approximately proportional to the cube of the order. Consequently, these conventional routines are not suitable for the problem of utilizing **A** matrices in most animal breeding applications that involve hundreds or thousands of animals.

4.2 A Simple Method Using **L**.

Note that $\mathbf{L} = \mathbf{TD}$ where **D** is a diagonal matrix with diagonal elements identical to those of **L** and **T** is a matrix computed by exactly the method described for **L** in Section 3.2 except that all diagonal elements of $\mathbf{T} = 1$. Then

$$\mathbf{A}^{-1} = (\mathbf{LL}')^{-1} = (\mathbf{TDDT}')^{-1} = (\mathbf{T}^{-1})'(\mathbf{D}^{-1})^2\mathbf{T}^{-1}. \tag{17}$$

$\mathbf{T}^{-1}$ is easy to compute due to the recursive computation of **T**. It is a special case of the following result.

Let $\mathbf{P}_0$ be an n^2 lower triangular matrix. Let

$$\mathbf{P}_1 = \begin{bmatrix} \mathbf{P}_0 & \mathbf{0} \\ \mathbf{a}'\mathbf{P}_0 & 1 \end{bmatrix} \tag{18}$$

where $\mathbf{a}'$ is a $1 \times n$ vector. Then

$$\mathbf{P}_1^{-1} = \begin{bmatrix} \mathbf{P}_0^{-1} & \mathbf{0} \\ -\mathbf{a}' & 1 \end{bmatrix} \tag{19}$$

as can be proved by showing that

$$\begin{bmatrix} \mathbf{P}_0 & \mathbf{0} \\ \mathbf{a}'\mathbf{P}_0 & 1 \end{bmatrix}\begin{bmatrix} \mathbf{P}_0^{-1} & \mathbf{0} \\ -\mathbf{a}' & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{P}_0^{-1} & \mathbf{0} \\ -\mathbf{a}' & 1 \end{bmatrix}\begin{bmatrix} \mathbf{P}_0 & \mathbf{0} \\ \mathbf{a}'\mathbf{P}_0 & 1 \end{bmatrix} = \mathbf{I}.$$

T is derived by a series of computations like (18).

$$\mathbf{T}_{b+1} = \begin{bmatrix} \mathbf{T}_b & \mathbf{0} \\ \mathbf{a}_b'\mathbf{T}_b & 1 \end{bmatrix}$$

$$\mathbf{T}_{b+2} = \begin{bmatrix} \mathbf{T}_{b+1} & \mathbf{0} \\ \mathbf{a}'_{b+1}\mathbf{T}_{b+1} & 1 \end{bmatrix}$$

$$\vdots$$

$$\mathbf{T}_n = \begin{bmatrix} \mathbf{T}_{n-1} & \mathbf{0} \\ \mathbf{a}'_{n-1}\mathbf{T}_{n-1} & 1 \end{bmatrix}$$

where $\mathbf{a}_i'$ is a vector of length $i - 1$ with all elements $= 0$ except for .5 corresponding to known parents of the ith animal. Then by (19), $\mathbf{T}^{-1}$ is a lower triangular matrix with all 1's in the diagonal, and with the only nonzero elements to the left of the diagonal of the ith row being $-.5$ for columns corresponding to known parents. The **T** matrix for the example follows. Then using the result for the inverse, $\mathbf{T}^{-1}$ follows.

To illustrate computation of $\mathbf{T}^{-1}$, the fifth row has $-.5$ in columns 3 and 4 corresponding to the fact that 3 and 4 are the parents of 5.

As a consequence of $\mathbf{A}^{-1}$ having the form of (17), simple rules can be stated for computing this matrix.

(i) Define $(\mathbf{D}^{-1})^2$ as **B**.

(ii) Set $\mathbf{A}^{-1} = \mathbf{B}$.

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TABLE 4
T MATRIX

	1	2	3	4	5	6	7
1	1.000	0.000	0.000	0.000	0.000	0.000	0.000
2	0.000	1.000	0.000	0.000	0.000	0.000	0.000
3	0.500	0.000	1.000	0.000	0.000	0.000	0.000
4	0.500	0.500	0.000	1.000	0.000	0.000	0.000
5	0.500	0.250	0.500	0.500	1.000	0.000	0.000
6	0.750	0.250	0.000	0.500	0.000	1.000	0.000
7	0.625	0.250	0.250	0.500	0.500	0.500	1.000

(iii) Add the following contributions to elements of $\mathbf{A}^{-1}$ from each of the n animals with one or both parents known.

(a) If both parents of the ith animal are known, say p and q, add

$-.5b_{ii}$ to (p, i), (i, p), (q, i), and (i, q) elements;

$.25b_{ii}$ to (p, p), (p, q), (q, p), and (q, q) elements.

(b) If only one parent is known, say p, add

$-.5b_{ii}$ to (p, i) and (i, p) elements;

$.25b_{ii}$ to (p, p) element.

TABLE 5
$\mathbf{T}^{-1}$

	1	2	3	4	5	6	7
1	1.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	1.0	0.0	0.0	0.0	0.0	0.0
3	-.5	0.0	1.0	0.0	0.0	0.0	0.0
4	-.5	-.5	0.0	1.0	0.0	0.0	0.0
5	0.0	0.0	-.5	-.5	1.0	0.0	0.0
6	-.5	0.0	0.0	-.5	0.0	1.0	0.0
7	0.0	0.0	0.0	0.0	-.5	-.5	1.0

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It can be seen that the only nontrivial computation required to find $\mathbf{A}^{-1}$ is $\mathbf{L}$ whose diagonal elements are needed. $\mathbf{A}$ and $\mathbf{T}$ are not required. When n is so large that $\mathbf{L}$ cannot be stored in computer memory, serious consideration needs to be given to development of efficient strategies for employing auxiliary storage. One possibility is to store only nonzero elements, identified by row and column numbers[1].

Now we illustrate the simple method for writing $\mathbf{A}^{-1}$. The diagonal elements of $(\mathbf{D}^{-1})^{\frac{1}{2}} = \mathbf{B}$ from Table 3 are [1, 1, 4/3, 2, 2, 2, 2.461534]. These become the diagonals of $\mathbf{A}^{-1}$ as a starting point. Further contributions of the seven animals are as follows:

1 and 2: None
3: $-2/3$ to (1, 3) and (3, 1) elements
1/3 to (1, 1) element
4: -1 to (1, 4), (4, 1), (2, 4), (4, 2)
.5 to (1, 1), (1, 2), (2, 1), (2, 2)
5: -1 to (3, 5), (5, 3), (4, 5), (5, 4)
.5 to (3, 3), (3, 4), (4, 3), (4, 4)
6: -1.230767 to (5, 7), (7, 5), (6, 7), (7, 6)
.615304 to (5, 5), (5, 6), (6, 5), (6, 6)

Summing these contributions, $\mathbf{A}^{-1}$ has the values shown in Table 6.

4.3 Further Simplification in a Noninbred Population.

The major cost of computing $\mathbf{A}^{-1}$ as described in Section 4.2 is findingt he diagonals of $\mathbf{L}$. In a noninbred population, this is no longer a problem since they can be found without computing $\mathbf{L}$. This is true because the diagonals can take on only three possible values; namely,

$\sqrt{.5}$ if both parents are known;
$\sqrt{.75}$ if only one parent is known;
1 if neither parent is known.

Thus, the diagonal elements of $\mathbf{B}$ are either 2, 4/3, or 1. Consequently, $\mathbf{A}^{-1}$ can be computed by searching a list of the n individuals and adding the following quantities to $\mathbf{A}^{-1}$ pre-set to null.

(i) If both parents of the ith individual are known, say p and q, add
2 to (i, i) element;
-1 to (p, i), (i, p), (q, i), and (i, q) elements;
.5 to (p, p), (p, q), (q, p), and (q, q) elements.
(ii) If only one parent is known, say p, add
4/3 to (i, i) element;
$-2/3$ to (p, i) and (i, p) elements;
1/3 to (p, p) element.
(iii) If neither parent is known, add
1 to (i, i) element.

An efficient method for large n is to write on dise or tape each of the above contributions to $\mathbf{A}^{-1}$, identified by row and column. These can then be sorted by columns within rows and summed. We have used this method for evaluation of about 3,000 sires. Only approximately three minutes were required by an IBM 370/135 to generate $\mathbf{A}^{-1}$ from a list of these sires and their parents.

[1] My colleague, Dr. R. L. Quaas, has found a method for rapid computation of the diagonals of L and A using minimal computer memory. Consequently, the methods of this paper can be used easily in inbred populations.

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TABLE 6

$\mathbf{A}^{-1}$

	1	2	3	4	5	6	7
1	2.333333	0.500000	-.666667	-.500000	0.000000	-1.000000	0.000000
2	0.500000	1.500000	0.000000	-1.000000	0.000000	0.000000	0.000000
3	-.666667	0.000000	1.833333	0.500000	-1.000000	0.000000	0.000000
4	-.500000	-1.000000	0.500000	3.000000	-1.000000	-1.000000	0.000000
5	0.000000	0.000000	-1.000000	-1.000000	2.615384	0.615384	-1.230767
6	-1.000000	0.000000	0.000000	-1.000000	0.615384	2.615384	-1.230767
7	0.000000	0.000000	0.000000	0.000000	-1.230767	-1.230767	2.461534

The computation of $\mathbf{A}^{-1}$ for a noninbred population described in Table 7 is illustrated below. $\mathbf{L}$ for this population is shown in Table 8.
Note that the diagonals are [1, 1, $\sqrt{.5}$, $\sqrt{.5}$, $\sqrt{.5}$, $\sqrt{.75}$, $\sqrt{.75}$, $\sqrt{.75}$, $\sqrt{.5}$] as they should be according to knowledge of the parents. The contributions to elements of $\mathbf{A}^{-1}$ by each individual follows.

1: 1 to (1, 1)
2: 1 to (2, 2)
3: 2 to (3, 3)
−1 to (1, 3), (3, 1), (2, 3), (3, 2)
.5 to (1, 1), (1, 2), (2, 1), (2, 2)

TABLE 7

PARENTS OF NONINBRED INDIVIDUALS

Individual	Parents
1	Unknown
2	Unknown
3	1 and 2
4	1 and 2
5	1 and 2
6	1 and unknown
7	1 and unknown
8	2 and unknown
9	7 and 8

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TABLE 8

L FOR NONINBRED POPULATION

	1	2	3	4	5	6	7	8	9
1	1.000000								
2	0.000000	1.000000							
3	0.500000	0.500000	0.707107						
4	0.500000	0.500000	0.000000	0.707107					
5	0.500000	0.500000	0.000000	0.000000	0.707107				
6	0.500000	0.000000	0.000000	0.000000	0.000000	0.866025			
7	0.500000	0.000000	0.000000	0.000000	0.000000	0.000000	0.866025		
8	0.000000	0.500000	0.000000	0.000000	0.000000	0.000000	0.000000	0.866025	
9	0.250000	0.250000	0.000000	0.000000	0.000000	0.000000	0.433012	0.433012	0.707107

4: 2 to (4, 4)
−1 to (1, 4), (4, 1), (2, 4), (4, 2)
.5 to (1, 1), (1, 2), (2, 1), (2, 2)

5: 2 to (5, 5)
−1 to (1, 5), (5, 1), (2, 5), (5, 2)
.5 to (1, 1), (1, 2), (2, 1), (2, 2)

6: 4/3 to (6, 6)
−2/3 to (1, 6), (6, 1)
1/3 to (1, 1)

7: 4/3 to (7, 7)
−2/3 to (1, 7), (7, 1)
1/3 to (1, 1)

8: 4/3 to (8, 8)
−2/3 to (2, 8), (8, 2)
1/3 to (2, 2)

9: 2 to (9, 9)
−1 to (7, 9), (9, 7), (8, 9), (9, 8)
.5 to (7, 7), (7, 8), (8, 7), (8, 8)

Adding these contributions, we get $\mathbf{A}^{-1}$ as shown in Table 9.

5. INVERSION OF A SUBMATRIX OF A

It must be remembered that in the foregoing methods $\mathbf{A}^{-1}$ is computed by including all of the "base population" animals. We may, in fact, not want to include some of these in our $\mathbf{A}$ matrix. In that case, the simple method for finding $\mathbf{A}^{-1}$ cannot be used. We can, however, obtain the desired result from partitioned matrix methods. Suppose $\mathbf{A}$ is $(p + q)^2$ and $\mathbf{A}^{-1}$ can be computed by the simple method, but we want the inverse of only the lower q^2 submatrix. Let

$$\mathbf{A}^{-1} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{12}' & \mathbf{A}_{22} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{12}' & \mathbf{B}_{22} \end{bmatrix}$$

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TABLE 9

$\mathbf{A}^{-1}$ FOR NONINBRED POPULATION

$$\frac{1}{6}\begin{bmatrix} 19 & 9 & -6 & -6 & -6 & -4 & -4 & 0 & 0 \\ 9 & 17 & -6 & -6 & -6 & 0 & 0 & -4 & 0 \\ -6 & -6 & 12 & 0 & 0 & 0 & 0 & 0 & 0 \\ -6 & -6 & 0 & 12 & 0 & 0 & 0 & 0 & 0 \\ -6 & -6 & 0 & 0 & 12 & 0 & 0 & 0 & 0 \\ -4 & 0 & 0 & 0 & 0 & 8 & 0 & 0 & 0 \\ -4 & 0 & 0 & 0 & 0 & 0 & 11 & 3 & -6 \\ 0 & -4 & 0 & 0 & 0 & 0 & 3 & 11 & -6 \\ 0 & 0 & 0 & 0 & 0 & 0 & -6 & -6 & 12 \end{bmatrix}$$

where $\mathbf{B}_{11}$ and $\mathbf{A}_{11}$ are p^2 and $\mathbf{B}_{22}$ and $\mathbf{A}_{22}$ are q^2. Then, by rules for inversion of a partitioned matrix, it is seen that

$$\mathbf{A}_{22}^{-1} = \mathbf{B}_{22} - \mathbf{B}_{12}'\mathbf{B}_{11}^{-1}\mathbf{B}_{12} . \tag{20}$$

This is an easy computation if p is small or if $\mathbf{B}_{11}$ is diagonal or patterned in some way that makes its inversion easy. However, the resulting matrix is likely to be less sparse than the entire $\mathbf{A}^{-1}$ matrix.

Suppose we want the inverse of the lower 7×7 submatrix of $\mathbf{A}$ in our noninbred population. Using the method of (20), this is

$$\frac{1}{6}\begin{bmatrix} 12 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 12 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 12 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 11 & 3 & -6 \\ 0 & 0 & 0 & 0 & 3 & 11 & -6 \\ 0 & 0 & 0 & 0 & -6 & -6 & 12 \end{bmatrix} - \frac{1}{6}\begin{bmatrix} -6 & -6 \\ -6 & -6 \\ -6 & -6 \\ -4 & 0 \\ -4 & 0 \\ 0 & -4 \\ 0 & 0 \end{bmatrix} \left[\frac{1}{6}\begin{bmatrix} 19 & 9 \\ 9 & 17 \end{bmatrix}\right]^{-1}$$

$$\cdot \frac{1}{6}\begin{bmatrix} -6 & -6 & -6 & -4 & -4 & 0 & 0 \\ -6 & -6 & -6 & 0 & 0 & -4 & 0 \end{bmatrix}.$$

The result is shown in Table 10. Note that this is less sparse than the inverse of Table 9.

6. EXAMPLE IN SIRE EVALUATION

A model for sire evaluat on might be

$$y_{ijk} = s_i + h_j + e_{ijk}$$

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TABLE 10

INVERSE OF LOWER 7 × 7 SUBMATRIX OF A

1.5537	-.4463	-.4463	-.1322	-.1322	-.1653	0.0000
-.4463	1.5537	-.4463	-.1322	-.1322	-.1653	0.0000
-.4463	-.4463	1.5537	-.1322	-.1322	-.1653	0.0000
-.1322	-.1322	-.1322	1.1460	-.1873	0.0992	0.0000
-.1322	-.1322	-.1322	-.1873	1.6460	0.5992	-1.0000
-.1653	-.1653	-.1653	0.0992	0.5992	1.6240	-1.0000
0.0000	0.0000	0.0000	0.0000	-1.0000	-1.0000	2.0000

where s_i represents the value of the ith sire and h_j represents the value of the jth herd-year-season. The random vector $\mathbf{s}$ is regarded as having null means and variance-covariance matrix $\mathbf{A}\sigma_s^2/15$. The random vector $\mathbf{e}$ has null means and variance-covariance matrix $\mathbf{I}\sigma_e^2$. $\mathbf{s}$ and $\mathbf{e}$ are uncorrelated. $\mathbf{h}$ is regarded as fixed. The implication of Var $(\mathbf{s}) = \mathbf{A}\sigma_s^2/15$ is that we have an additive genetic model with heritability, $\mathbf{h}^2 = .25$. Then in a noninbred population, $\sigma_s^2 = .25h^2\sigma_y^2$ and $\sigma_e^2 = (1 - .25h^2)\sigma_y^2$. If h^2 is assumed to be .25, $\sigma_e^2/\sigma_s^2 = 15$.

Suppose six sires have progeny numbers distributed as shown in Table 11.

$$y_{i\cdot\cdot} = [0, 1210, 1070, 920, 700, 7120]$$

$$y_{\cdot j\cdot} = [1980, 2480, 1780, 1350, 1740, 1690]$$

TABLE 11

PROGENY NUMBERS

	Herd-Year-Season					
Sire	1	2	3	4	5	6
1	0	0	0	0	0	0
2	3	2	5	0	0	1
3	0	1	2	3	4	2
4	4	3	3	1	0	0
5	2	0	1	3	5	3
6	8	12	7	10	8	12

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Sires 3 and 4 are sons of sire 1, and sires 5 and 6 are sons of sire 2. All dams are unknown. Sires 1 and 2 are noninbred and unrelated. Then, $\mathbf{A}$ is as shown in Table 12. Inverting $\mathbf{A}$ and multiplying by 15, we obtain $(\mathbf{A}/15)^{-1}$ as in Table 13.

Now if we set up least squares equations and then absorb $\hat{\mathbf{h}}$, the equations are

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 24.8660 & -5.3562 & -6.4248 & -5.6373 & -7.4477 \\ 0 & -5.3562 & 8.8039 & -.7778 & -1.8725 & -.7974 \\ 0 & -6.4248 & -.7778 & 10.0294 & -.6765 & -2.1503 \\ 0 & -5.6373 & -1.8725 & -.6765 & 9.0000 & -.8137 \\ 0 & -7.4477 & -.7974 & -2.1503 & -.8137 & 11.2092 \end{bmatrix} \begin{bmatrix} \hat{s}_1 \\ \hat{s}_2 \\ \hat{s}_3 \\ \hat{s}_4 \\ \hat{s}_5 \\ \hat{s}_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 1103.07 \\ -3.30 \\ -100.98 \\ -335.29 \\ -663.50 \end{bmatrix} \tag{21}$$

or $\mathbf{C}\hat{\mathbf{s}} = \mathbf{r}$, say. Then the BLUP equations are

$$(\mathbf{C} + 15\mathbf{A}^{-1})\hat{\mathbf{s}} = \mathbf{r}. \tag{22}$$

The solution is

$$\hat{\mathbf{s}}' \simeq [1.38,\ 17.97,\ 3.28,\ 0.18,\ -1.97,\ -11.17].$$

$[\mathbf{C} + 15\ \mathbf{A}^{-1}]^{-1}\sigma^2$ is the variance-covariance matrix of prediction errors; that is, Var $(\hat{\mathbf{s}} - \mathbf{s})$. This is approximately

$$\begin{bmatrix} .059 & .011 & .024 & .024 & .008 & .009 \\ .011 & .039 & .014 & .015 & .023 & .024 \\ .024 & .014 & .047 & .013 & .011 & .010 \\ .024 & .015 & .013 & .046 & .010 & .012 \\ .008 & .023 & .011 & .010 & .048 & .015 \\ .009 & .024 & .010 & .012 & .015 & .047 \end{bmatrix} \sigma^2. \tag{23}$$

If we had used 15 $\mathbf{I}$ rather than 15 $\mathbf{A}^{-1}$, the equations would have been

TABLE 12

$\mathbf{A}$

1.00	0.00	0.50	0.50	0.00	0.00
0.00	1.00	0.00	0.00	0.50	0.50
0.50	0.00	1.00	0.25	0.00	0.00
0.50	0.00	0.25	1.00	0.00	0.00
0.00	0.50	0.00	0.00	1.00	0.25
0.00	0.50	0.00	0.00	0.25	1.00

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TABLE 13
$15\mathbf{A}^{-1}$

25	0	−10	−10	0	0
0	25	0	0	−10	−10
−10	0	20	0	0	0
−10	0	0	20	0	0
0	−10	0	0	20	0
0	−10	0	0	0	20

$$(\mathbf{C} + 15\mathbf{I})\hat{\mathbf{s}} = \mathbf{r}. \tag{24}$$

Under the assumption that $\mathbf{G} = \mathbf{A}/15$ rather than $\mathbf{I}/15$ that was used in (24), the prediction error variance-covariance matrix is

$$\{[\mathbf{C} + 15]^{-1} + [(\mathbf{C} + 15\mathbf{I})^{-1}15\mathbf{I}(\tfrac{1}{16}\mathbf{A} - \tfrac{1}{16}\mathbf{I})15\mathbf{I}(\mathbf{C} + 15\mathbf{I})^{-1}]\}\sigma^2.$$

This turns out to be approximately

$$\begin{bmatrix} .067 & .009 & .024 & .024 & .005 & .005 \\ .009 & .040 & .014 & .015 & .023 & .024 \\ .024 & .014 & .047 & .013 & .011 & .010 \\ .024 & .015 & .013 & .046 & .010 & .012 \\ .005 & .023 & .011 & .010 & .052 & .018 \\ .005 & .024 & .010 & .012 & .018 & 0.50 \end{bmatrix}\sigma^2. \tag{25}$$

As would be expected, the diagonals of (23) are less than those of (25) but the gain in accuracy is not great. This along with the cost of computing $\mathbf{A}^{-1}$ has resulted in little use having been made of $\mathbf{A}$ in sire evaluation. Now with the very easy method for computing $\mathbf{A}^{-1}$ available as described in this paper, particularly in a noninbred population, it appears that advantage should be taken of the small but real reduction in prediction error variance through use of all relationships.

UNE METHODE SIMPLE POUR LE CALCUL DE L'INVERSE D'UNE MATRICE DE NUMERATEURS DE COEFFICIENTS DE PARENTE UTILISEE DANS LA PREDICTION DES VALEURS GENETIQUES

RÉSUMÉ

L'inverse d'une matrice de numérateurs de coefficients de parenté est nécessaire pour obtenir la meilleure prédiction linéaire sans biais des valeurs génétiques. Le but de ce papier est de présenter une méthode

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simple et rapide pour le calcul des éléments de cet inverse sans calculer la matrice de parenté elle-même. La methodé est particulièrement utile dans les populations non consanguines mais est beaucoup plus rapide que la méthode conventionnelle en présence de consanguinité.

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Key Words: Prediction of breeding values, Recursive computation of relationships, Relationship matrix as product of triangular matrices, Use of relationships in sire evaluation.

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Henderson and Quaas (1976)

Henderson, C.R., and Quaas, R.L. 1976. Multiple trait evaluation using relatives' records. Journal of Animal Science, vol. 43, no. 6, pp. 1188 to 1197.

The problem

The selection index already combined several traits, but it did not deal with environmental effects or with data from many relatives. Henderson and Quaas extended BLUP to several traits at once, using the records of the animal and of all its relatives. They showed how to transform the traits to simplify the computations and illustrated the method with an example.

Points to note while reading

Note the covariance structure, which combines the relationship matrix with the matrix of genetic covariances between traits. Multiple trait BLUP makes it possible to use correlated traits to improve prediction and to correct the bias caused by sequential selection, and it underlies modern multiple trait evaluations.

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MULTIPLE TRAIT EVALUATION USING RELATIVES' RECORDS

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SUMMARY

Methods have been derived for best linear unbiased prediction of breeding values for multiple traits utilizing records on large numbers of relatives as well as the individual's own records. The method is based upon an extension of Henderson's method for efficient utilization of relatives' records in single trait evaluation. Examples of the method are given for three traits in beef cattle.

(Key Words: Best Linear Unbiased Prediction, Relatives' Records, Records on Correlated Traits.)

INTRODUCTION

Selection index methods have been used to solve two problems in animal breeding: a) optimum weighting of records on the individual and its relatives in single trait selection, and b) phenotypic selection on multiple traits. Dairy cattle breeders have made the most use of the first of these applications. Beef cattle, swine, and sheep breeders have been more concerned with multiple trait selection. A further application, selection for multiple traits using relatives' records, has been difficult, particularly if the number of relatives is large. Henderson (1963, 1973) showed how, in theory, genetic relationships among all of the animals that are to be evaluated can be incorporated in best linear unbiased prediction (BLUP) of their merits. The method requires finding the inverse of the numerator relationship matrix of these animals. If the number of animals is large, computing this matrix and then inverting it was a formidable, if not impossible, task. Recently, however, a method has been discovered for computing the inverse directly from pedigree information with much less effort than is required in finding the relationship matrix itself (Henderson, 1975a, 1976). A simple extension of these methods, finding the inverse of the numerator relationship matrix and using it in BLUP equations, now makes multiple trait evaluation using many relatives' records computationally feasible. The purpose of this paper is to present this method and to illustrate it with an example from beef cattle breeding. If all fixed effects are known from prior data, the BLUP solution reduces to the selection index solution, but the former is much easier computationally when all relationships are used.

PREDICTION BY LINEAR MODEL METHODS

Henderson (1963) has formulated a general mixed linear model as

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \mathbf{e}$$

where $\mathbf{y}$ is the vector of observations adjusted if possible for fixed factors that have been estimated from prior data; β is a fixed vector; $\mathbf{X}$ is a known matrix relating elements of β to each $\mathbf{y}$; $\mathbf{u}$ is an unknown random vector, some or all of whose elements may represent breeding values; $\mathbf{Z}$ is a known matrix relating elements of $\mathbf{u}$ to elements of $\mathbf{y}$; $\text{Var}(\mathbf{u}) = \mathbf{G}\sigma^2$ where $\mathbf{G}$ is a known nonsingular matrix and σ^2 is a scalar, possibly unknown; $\text{Var}(\mathbf{e}) = \mathbf{R}\sigma^2$ where $\mathbf{R}$ is a known, nonsingular matrix; and $\text{Cov}(\mathbf{u},\mathbf{e}') = 0$.

Then, the best linear unbiased estimator (BLUE) of any set of estimable function, $\mathbf{K}'\beta$, is $\mathbf{K}'\hat{\beta}$ where $\hat{\beta}$ is any solution to [1]. The best linear unbiased predictor (BLUP) of $\mathbf{u}$ is $\hat{\mathbf{u}}$, the solution to [1]. Also, BLUP of $\mathbf{m}'\mathbf{u} = \mathbf{m}'\hat{\mathbf{u}}$.

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z}+\mathbf{G}^{-1} \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \end{bmatrix}. \quad [1]$$

The sampling variances can be obtained from any g-inverse of the coefficient matrix of [1]. Let some g-inverse be

$$\begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}'_{12} & \mathbf{C}_{22} \end{bmatrix} \quad [2]$$

[1] Department of Animal Science.

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Then,

$$\text{Var}(\mathbf{K}'\hat{\boldsymbol{\beta}}) = \mathbf{K}'\mathbf{C}_{11}\mathbf{K}\sigma^2 \text{ provided } \mathbf{K}'\boldsymbol{\beta} \text{ is estimable;}$$

$$\text{Cov}(\mathbf{K}'\hat{\boldsymbol{\beta}},\hat{\mathbf{u}}') = \mathbf{0};$$

$$\text{Cov}(\mathbf{K}'\hat{\boldsymbol{\beta}},\hat{\mathbf{u}}'-\mathbf{u}') = \mathbf{C}_{12}\sigma^2;$$

$$\text{Var}(\hat{\mathbf{u}}) = \text{Cov}(\hat{\mathbf{u}},\mathbf{u}') = (\mathbf{G}-\mathbf{C}_{22})\sigma^2;$$

$$\text{Var}(\hat{\mathbf{u}}-\mathbf{u}) = \mathbf{C}_{22}\sigma^2.$$

The methods presented in the following sections are special cases of this general model and method.

REGULAR SELECTION INDEX USING ONLY RECORDS ON THE INDIVIDUAL

The linear model assumed for the ith animal considered for selection is

$$\mathbf{y}_i = \mathbf{u}_i + \mathbf{e}_i$$

where $\mathbf{y}_i' = [y_{i1} \cdots y_{iq}]$ is the vector of records on traits $1, \ldots, q$ on the ith animal. Similarly, $\mathbf{u}_i' = [u_{i1} \ldots u_{iq}]$ and $\mathbf{e}_i' = [e_{i1} \ldots e_{iq}]$. $E(\mathbf{y}_i) = \mathbf{0}$; $\text{Var}(\mathbf{u}_i) = \mathbf{G}_0$; $\text{Var}(\mathbf{e}_i) = \mathbf{R}_0$; $\text{Cov}(\mathbf{u}_i,\mathbf{e}_i') = \mathbf{0}$. Then, $\text{Var}(\mathbf{y}_i) = \mathbf{G}_0 + \mathbf{R}_0 = \mathbf{V}_0$, say.

Defining the ij^{th} element of G_0, R_0 and V_0 as g_{ij}, r_{ij} and v_{ij}, the following statements clarify the meaning of these matrices

g_{ii} = additive genetic variance of ith trait
r_{ii} = environmental variance of ith trait
v_{ii} = phenotypic variance of ith trait
g_{ii}/v_{ii} = heritability of ith trait
$g_{ij}/\sqrt{g_{ii}g_{jj}}$ = genetic correlation between traits i and j
$r_{ij}/\sqrt{r_{ii}r_{jj}}$ = environmental correlation between traits i and j
$v_{ij}/\sqrt{v_{ii}v_{jj}}$ = phenotypic correlation between traits i and j.

Suppose we want to select on the basis of the best linear prediction of $\mathbf{m}'\mathbf{u}_i$ where m is a vector of economic weights. By best linear predictor is meant that in the class of linear functions of $\mathbf{y}_i$, the predictor chosen, say g_i, minimizes $E(g_i-\mathbf{m}'\mathbf{u}_i)^2$. This is equivalent to maximizing the correlations between g_i and $\mathbf{m}'\mathbf{u}_i$. Also, in the multivariate normal case, g_i is the expectation of $\mathbf{m}'\mathbf{u}_i$, given $\mathbf{y}_i$.

By regular selection index methods, utilizing records on the individual only,

$$g_i = \mathbf{b}'\mathbf{y}_i$$

where b is the solution to

$$\mathbf{V}_0\mathbf{b} = \mathbf{G}_0\mathbf{m}. \quad [3]$$

The same result can be obtained from the general method [1] by noting that $\mathbf{X}\boldsymbol{\beta}$ does not exist, $\mathbf{Z} = \mathbf{I}$, $\mathbf{R} = \mathbf{R}_0$, and $\mathbf{G} = \mathbf{G}_0$. Then the equations become

$$(\mathbf{R}_0^{-1} + \mathbf{G}_0^{-1})\hat{\mathbf{u}}_i = \mathbf{R}_0^{-1}\mathbf{y}_i \quad [4]$$

and $g_i = \mathbf{m}'\hat{\mathbf{u}}_i$. This method has no obvious advantage over [3], but it serves as an introduction to an efficient method for more complicated problems.

BLUP USING RELATIVES' RECORDS

Suppose now that we have n related animals each with records on q traits and that the records have unknown means = $\mathbf{X}\boldsymbol{\beta}$. If we order the elements of both y, the data vector, and u, the vector of breeding values, by animals within traits and assume an additive genetic model,

$$\text{Var}(\mathbf{u}) = \mathbf{G} = \begin{bmatrix} g_{11}\mathbf{A} & g_{12}\mathbf{A} & \cdots & g_{1q}\mathbf{A} \\ g_{12}\mathbf{A} & g_{22}\mathbf{A} & \cdots & g_{2q}\mathbf{A} \\ \vdots & \vdots & & \vdots \\ g_{1q}\mathbf{A} & g_{2q}\mathbf{A} & \cdots & g_{qq}\mathbf{A} \end{bmatrix} = \mathbf{G}_0 * \mathbf{A} \quad [5]$$

A is the nxn numerator relationship matrix of the animals considered for selection, * refers to the right direct production operation (Searle, 1966), and g_{ij} is the ij^{th} element of G_0.

$$\text{Var}(\mathbf{e}) = \mathbf{R} = \begin{bmatrix} r_{11}\mathbf{I} & r_{12}\mathbf{I} & \cdots & r_{1q}\mathbf{I} \\ r_{12}\mathbf{I} & r_{22}\mathbf{I} & \cdots & r_{2q}\mathbf{I} \\ \vdots & \vdots & & \vdots \\ r_{1q}\mathbf{I} & r_{2q}\mathbf{I} & \cdots & r_{qq}\mathbf{I} \end{bmatrix} = \mathbf{R}_0 * \mathbf{I} \quad [6]$$

where r_{ij} is the ij^{th} element of $\mathbf{R}_0$, and I is an nxn identity matrix. We partition $\boldsymbol{\beta}'$ into $[\boldsymbol{\beta}'_1 \cdots \boldsymbol{\beta}'_q]$ corresponding to the q different traits. Similarly, we partition X into

$$\begin{bmatrix} \mathbf{X}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{X}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{X}_q \end{bmatrix}.$$

Thus, $\mathbf{X}_i$ relates the n records on the ith trait to the $\boldsymbol{\beta}_i$ subvector. It is of order nxp_i where p_i denotes the number of elements in $\boldsymbol{\beta}_i$. Z is an

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nqxnq identity matrix. Then, the best linear unbiased predictors (BLUP) can be found by solving

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{12}' & C_{22} \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} t_1 \\ t_2 \end{bmatrix} \quad [7]$$

$$C_{11} = X'R^{-1}X =$$

$$\begin{bmatrix} r^{11}X_1'X_1 & r^{12}X_1'X_2 & \cdots & r^{1q}X_1'X_q \\ r^{12}X_2'X_1 & r^{22}X_2'X_2 & \cdots & r^{2q}X_2'X_q \\ \vdots & \vdots & & \vdots \\ r^{1q}X_q'X_1 & r^{2q}X_q'X_2 & \cdots & r^{qq}X_q'X_q \end{bmatrix} \quad [8]$$

where r^{ij} is the ij^{th} element of R_0^{-1},

$$C_{12} = X'R^{-1}Z = X'R^{-1} =$$

$$\begin{bmatrix} r^{11}X_1' & r^{12}X_1' & \cdots & r^{1q}X_1' \\ r^{12}X_2' & r^{22}X_2' & \cdots & r^{2q}X_2' \\ \vdots & \vdots & & \vdots \\ r^{1q}X_q' & r^{2q}X_q' & \cdots & r^{qq}X_q' \end{bmatrix}, \quad [9]$$

and

$$C_{22} = Z'R^{-1}Z+G^{-1} = R_0^{-1} * I+G_0^{-1} * A^{-1} =$$

$$\begin{bmatrix} r^{11}I+g^{11}A^{-1} & r^{12}I+g^{12}A^{-1} & \cdots & r^{1q}I+g^{1q}A^{-1} \\ r^{12}I+g^{12}A^{-1} & r^{22}I+g^{22}A^{-1} & \cdots & r^{2q}I+g^{2q}A^{-1} \\ \vdots & \vdots & & \vdots \\ r^{1q}I+g^{1q}A^{-1} & r^{2q}I+g^{2q}A^{-1} & \cdots & r^{qq}I+g^{qq}A^{-1} \end{bmatrix} \quad [10]$$

where g^{ij} is the ij^{th} element of G_0^{-1}. A^{-1} can be computed rapidly by the methods described by Henderson (1975a, 1976).

$$t_1 = X'R^{-1}y = \begin{bmatrix} \sum_{j=1}^{q} r^{1j}X_1'y_j \\ \sum_{j=1}^{q} r^{2j}X_2'y_j \\ \vdots \\ \sum_{j=1}^{q} r^{qj}X_q'y_j \end{bmatrix} \quad [11]$$

where t_1 has q subvectors, the length of the ith being p_i, and

$$t_2 = Z'R^{-1}y = R^{-1}y = \begin{bmatrix} \sum_{j=1}^{q} r^{1j}y_j \\ \sum_{j=1}^{q} r^{2j}y_j \\ \vdots \\ \sum_{j=1}^{q} r^{qj}y_j \end{bmatrix} \quad [12]$$

where t_2 has q subvectors each of length n.

We shall illustrate these methods with three traits in beef cattle: birth weight, weaning weight, and feedlot gain. From the publication of Petty and Cartwright (1966), we have the following parameter estimates from summaries of a large number of publications.

$$G_0 = \begin{bmatrix} 28.6 & 73.77 & .50 \\ 73.77 & 566 & 2.29 \\ .50 & 2.29 & .0276 \end{bmatrix};$$

$$R_0 = \begin{bmatrix} 36.3 & 67.43 & .06 \\ 67.43 & 1454 & -.53 \\ .06 & -.53 & .0254 \end{bmatrix}.$$

Suppose we have data on five animals as in table 1. The fixed effects in the model are a common mean for birth weight, and season of birth for weaning weight and feedlot gain. Consequently,

$$\beta = [\beta_{11} \quad \beta_{21} \quad \beta_{22} \quad \beta_{31} \quad \beta_{32}]$$

and

$$X = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

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TABLE 1. DATA ON FIVE ANIMALS

Animal	Season of birth	Birth weight	Weaning weight	Feedlot gain
1	1	61	362	1.96
2	1	72	401	2.05
3	2	68	350	1.81
4	2	78	410	2.01
5	2	65	340	1.74

Now let us suppose that animals 1, 2 and 3 are half-sibs, and 4 and 5 are half-sibs unrelated to the first set. Then,

$$\mathbf{A} = \begin{bmatrix} 1 & .25 & .25 & 0 & 0 \\ .25 & 1 & .25 & 0 & 0 \\ .25 & .25 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & .25 \\ 0 & 0 & 0 & .25 & 1 \end{bmatrix},$$

and

$$\mathbf{A}^{-1} = \begin{bmatrix} 1.1111 & -.2222 & -.2222 & 0 & 0 \\ -.2222 & 1.1111 & -.2222 & 0 & 0 \\ -.2222 & -.2222 & 1.1111 & 0 & 0 \\ 0 & 0 & 0 & 1.0667 & -.2667 \\ 0 & 0 & 0 & -.2667 & 1.0667 \end{bmatrix}.$$

$$\mathbf{R}_0^{-1} = \begin{bmatrix} .030405 & -.001447 & -.102022 \\ -.001447 & .000762 & .019317 \\ -.102022 & .019317 & 40.014145 \end{bmatrix};$$

$$\mathbf{G}_0^{-1} = \begin{bmatrix} .059630 & -.005120 & -.655442 \\ -.005120 & .003101 & -.164391 \\ -.655442 & -.164391 & 61.745515 \end{bmatrix};$$

Then, by the rules of [8], [9], [10], [11] and [12],

$\mathbf{C}_{11} =$

$$\begin{bmatrix} .15203 & -.00289 & -.00434 & -.20404 & -.30607 \\ -.00289 & .00152 & .0 & .03863 & .0 \\ -.00434 & .0 & .00229 & .0 & .05795 \\ -.20404 & .03863 & .0 & 80.02829 & .0 \\ -.30607 & .0 & .05795 & .0 & 120.04243 \end{bmatrix};$$

$\mathbf{C}_{12} =$

$$\begin{bmatrix} .03041 & -.00145 & .0 & -.10202 & .0 \\ .03041 & -.00145 & .0 & -.10202 & .0 \\ .03041 & .0 & -.00145 & .0 & -.10202 \\ .03041 & .0 & -.00145 & .0 & -.10202 \\ .03041 & .0 & -.00145 & .0 & -.10202 \\ -.00145 & .00076 & .0 & .01932 & .0 \\ -.00145 & .00076 & .0 & .01932 & .0 \\ -.00145 & .0 & .00076 & .0 & .01932 \\ -.00145 & .0 & .00076 & .0 & .01932 \\ -.00145 & .0 & .00076 & .0 & .01932 \\ -.10202 & .01932 & .0 & 40.01414 & .01932 \\ -.10202 & .01932 & .0 & 40.01414 & .01932 \\ -.10202 & .0 & .01932 & .0 & 40.01414 \\ -.10202 & .0 & .01932 & .0 & 40.01414 \\ -.10202 & .0 & .01932 & .0 & 40.01414 \end{bmatrix};$$

$\mathbf{C}_{22}$ is presented in equation [13].

$\mathbf{t}_1' =$

[6.7868 .4663 .4601 161.6266 222.2007].

The three subvectors of $\mathbf{t}_2'$ are

[1.1309 1.3997 1.3764 1.5732 1.3068],

[.2254 .2409 .2032 .2783 .1986],

and

[79.1971 82.4295 72.2491 80.3907 69.5610].

Then, the solution is $\hat{\beta}' =$
[68.910 387.459 363.331 2.032 1.840].

The three subvectors of $\hat{\mathbf{u}}'$ are

[−3.341 .862 −.808 4.320 −1.581],

[−10.693 2.943 −4.798 17.890 −7.433],

and

[−.0658 .0164 −.0262 −.1200 −.0570].

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$$
\begin{bmatrix}
.09666 & -.01325 & -.01325 & .0 & .0 & -.00714 & .00114 & .00114 & .0 & .0 & -.83029 & .14565 & .14565 & .0 & .0 \\
-.01325 & .09666 & -.01325 & .0 & .0 & .00114 & -.00714 & .00114 & .0 & .0 & .14565 & -.83029 & .14565 & .0 & .0 \\
-.01325 & -.01325 & .09666 & .0 & .0 & .00114 & .00114 & -.00714 & .0 & .0 & .14565 & .14565 & -.83029 & .0 & .0 \\
.0 & .0 & .0 & .09401 & -.01590 & .0 & .0 & .0 & -.00691 & .00137 & .0 & .0 & .0 & -.80116 & .17478 \\
.0 & .0 & .0 & -.01590 & .09401 & .0 & .0 & .0 & .00137 & -.00691 & .0 & .0 & .0 & .17478 & -.80116 \\
-.00714 & .00114 & .00114 & .0 & .0 & .00421 & -.00069 & -.00069 & .0 & .0 & -.16334 & .03653 & .03653 & .0 & .0 \\
.00114 & -.00714 & .00114 & .0 & .0 & -.00069 & .00421 & -.00069 & .0 & .0 & .03653 & -.16334 & .03653 & .0 & .0 \\
.00114 & .00114 & -.00714 & .0 & .0 & -.00069 & -.00069 & .00421 & .0 & .0 & .03653 & .03653 & -.16334 & .0 & .0 \\
.0 & .0 & .0 & -.00691 & .00137 & .0 & .0 & .0 & .00407 & -.00083 & .0 & .0 & .0 & -.15603 & .04834 \\
.0 & .0 & .0 & .00137 & -.00691 & .0 & .0 & .0 & -.00083 & .00407 & .0 & .0 & .0 & .04384 & -.15603 \\
-.83029 & .14565 & .14565 & .0 & .0 & -.16334 & .03653 & .03653 & .0 & .0 & 108.62027 & -13.72123 & -13.72123 & .0 & .0 \\
.14565 & -.83029 & .14565 & .0 & .0 & .03653 & -.16334 & .03653 & .0 & .0 & -13.72123 & 108.62027 & -13.72123 & .0 & .0 \\
.14565 & .14565 & -.83029 & .0 & .0 & .03653 & .03653 & -.16334 & .0 & .0 & -13.72123 & -13.72123 & 108.62027 & .0 & .0 \\
.0 & .0 & .0 & -.80116 & .17478 & .0 & .0 & .0 & -.15603 & .04384 & .0 & .0 & .0 & 105.87603 & -16.46547 \\
.0 & .0 & .0 & .17478 & -.80116 & .0 & .0 & .0 & .04384 & -.15603 & .0 & .0 & .0 & -16.46547 & 105.87603
\end{bmatrix} \quad [13]
$$

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Now suppose the economic weights on the three traits are a_1, a_2 and a_3, respectively. Then BLUP of the total merit of animal 1 is

$-3.341a_1 - 10.693a_2 - .0658a_3$ and similarly for the other four animals.

MISSING RECORDS

When records are missing, $\mathbf{Z}$ is an identity matrix with rows pertaining to missing records deleted. Then, $\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z}$ is

$$\begin{bmatrix} \mathbf{D}_{11} & \mathbf{D}_{12} & \cdots & \mathbf{D}_{1q} \\ \mathbf{D}_{12} & \mathbf{D}_{22} & \cdots & \mathbf{D}_{2q} \\ \vdots & \vdots & & \vdots \\ \mathbf{D}_{1q} & \mathbf{D}_{2q} & \cdots & \mathbf{D}_{qq} \end{bmatrix} \quad [14]$$

Each $\mathbf{D}_{ij}$ is diagonal and has the form

$$\begin{bmatrix} h_{1ij} & 0 & \cdots & 0 \\ 0 & h_{2ij} & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & h_{nij} \end{bmatrix} \quad [15]$$

where h_{kij} is the ij^{th} element of $\mathbf{H}_k$, a matrix peculiar to the kth animal. It is computed as follows:

a. If the animal has no records missing, $\mathbf{H}_k = \mathbf{R}_0^{-1}$.
b. If the animal has all records missing, $\mathbf{H}_k = \mathbf{0}$.
c. Otherwise, find the regular inverse of the elements of $\mathbf{R}_0$ pertaining to records that are present and fill out the remaining elements with zeros for the other elements of $\mathbf{H}_k$.

For example, if q = 4 and the second and fourth traits are missing,

$$\mathbf{H}_k = \begin{bmatrix} p_{11} & 0 & p_{13} & 0 \\ 0 & 0 & 0 & 0 \\ p_{13} & 0 & p_{33} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

where

$$\begin{bmatrix} p_{11} & p_{13} \\ p_{13} & p_{33} \end{bmatrix} = \begin{bmatrix} r_{11} & r_{13} \\ r_{13} & r_{33} \end{bmatrix}^{-1}.$$

Now, regard $\mathbf{X}_i$ as having dimension $n \times p_i$ by setting rows of $\mathbf{X}_j$ to 0 pertaining to missing records. Similarly, expand $\mathbf{y}$ to length nq, inserting 0 for each missing record. Then, the BLUP equations are

$$\begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}'_{12} & \mathbf{C}_{22} \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{t}_1 \\ \mathbf{t}_2 \end{bmatrix} \quad [16]$$

where

$$\mathbf{C}_{11} = \begin{bmatrix} \mathbf{X}_1'\mathbf{D}_{11}\mathbf{X}_1 & \mathbf{X}_1'\mathbf{D}_{12}\mathbf{X}_2 & \cdots & \mathbf{X}_1'\mathbf{D}_{1q}\mathbf{X}_q \\ \mathbf{X}_2'\mathbf{D}_{12}\mathbf{X}_1 & \mathbf{X}_2'\mathbf{D}_{12}\mathbf{X}_2 & \cdots & \mathbf{X}_2'\mathbf{D}_{2q}\mathbf{X}_q \\ \vdots & \vdots & & \vdots \\ \mathbf{X}_q'\mathbf{D}_{1q}\mathbf{X}_q & \mathbf{X}_2'\mathbf{D}_{2q}\mathbf{X}_q & \cdots & \mathbf{X}_q'\mathbf{D}_{qq}\mathbf{X}_q \end{bmatrix}; \quad [17]$$

$$\mathbf{C}_{12} = \begin{bmatrix} \mathbf{X}_1'\mathbf{D}_{11} & \mathbf{X}_1'\mathbf{D}_{12} & \cdots & \mathbf{X}_1'\mathbf{D}_q \\ \mathbf{X}_2'\mathbf{D}_{12} & \mathbf{X}_2'\mathbf{D}_{22} & \cdots & \mathbf{X}_2'\mathbf{D}_q \\ \vdots & \vdots & & \vdots \\ \mathbf{X}_q'\mathbf{D}_{1q} & \mathbf{X}_q'\mathbf{D}_{2q} & \cdots & \mathbf{X}_q'\mathbf{D}_{qq} \end{bmatrix}; \quad [18]$$

$$\mathbf{C}_{22} = \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} = \begin{bmatrix} \mathbf{D}_{11}+g^{11}\mathbf{A}^{-1} & \mathbf{D}_{12}+g^{12}\mathbf{A}^{-1} & \cdots & \mathbf{D}_{1q}+g^{1q}\mathbf{A}^{-1} \\ \mathbf{D}_{12}+g^{12}\mathbf{A}^{-1} & \mathbf{D}_{22}+g^{22}\mathbf{A}^{-1} & \cdots & \mathbf{D}_{2q}+g^{2q}\mathbf{A}^{-1} \\ \vdots & \vdots & & \vdots \\ \mathbf{D}_{1q}+g^{1q}\mathbf{A}^{-1} & \mathbf{D}_{2q}+g^{2q}\mathbf{A}^{-1} & \cdots & \mathbf{D}_{qq}+g^{qq}\mathbf{A}^{-1} \end{bmatrix}; \quad [19]$$

$$\mathbf{t}_j = \begin{bmatrix} \sum_{j=1}^{q} \mathbf{X}_1'\mathbf{D}_{1j}\mathbf{y}_j \\ \sum_{j=1}^{q} \mathbf{X}_2'\mathbf{D}_{2j}\mathbf{y}_j \\ \vdots \\ \sum_{j=1}^{q} \mathbf{X}_q'\mathbf{D}_{qj}\mathbf{y}_j \end{bmatrix}; \quad [20]$$

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$$t_2 = \begin{bmatrix} \sum_{j=1}^{q} D_{1j}y_j \\ \sum_{j=1}^{q} D_{2j}y_j \\ \vdots \\ \sum_{j=1}^{q} D_{qj}y_j \end{bmatrix} . \qquad [21]$$

We illustrate the missing records method with the same example as before, except now we pretend that birth weight is missing on animal 5 and feedlot gains are missing on animals 3 and 4. First we compute the five required $\mathbf{H}_k$'s. For animals 1 and 2, these are

$$\mathbf{R}_0^{-1} = \begin{bmatrix} .030405 & -.001447 & -.102022 \\ -.001447 & .00762 & .019317 \\ -.102022 & .019317 & 40.014145 \end{bmatrix} .$$

For 3 and 4,

$$\begin{bmatrix} \begin{bmatrix} 36.3 & 67.43 \\ 67.43 & 1454 \end{bmatrix}^{-1} & \begin{matrix} 0 \\ 0 \end{matrix} \\ \begin{matrix} 0 & 0 \end{matrix} & 0 \end{bmatrix} =$$

$$\begin{bmatrix} .030145 & -.001398 & 0 \\ -.001398 & .000753 & 0 \\ 0 & 0 & 0 \end{bmatrix} .$$

For 5,

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & \begin{bmatrix} 1454 & -.53 \\ -.53 & .0254 \end{bmatrix}^{-1} & \end{bmatrix} =$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & .000693 & .014461 \\ 0 & .014461 & 39.671821 \end{bmatrix} .$$

Then, $\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z}$ is shown in equation [22].
By the rules of [17], [18], [19], [20] and [21],

$\mathbf{C}_{11} =$

$$\begin{bmatrix} .12110 & -.00289 & -.00280 & -.20404 & .0 \\ -.00289 & .00152 & .0 & .03863 & .0 \\ -.00280 & .0 & .00220 & .0 & .01446 \\ -.20404 & .03863 & .0 & 80.02829 & .0 \\ .0 & .0 & .01446 & .0 & 39.67182 \end{bmatrix} ;$$

$\mathbf{C}'_{12} =$

$$\begin{bmatrix} .03041 & -.00145 & .0 & -.10202 & .0 \\ .03041 & -.00145 & .0 & -.10202 & .0 \\ .03015 & .0 & -.00140 & .0 & .0 \\ .03015 & .0 & -.00140 & .0 & .0 \\ .0 & .0 & .0 & .0 & .0 \\ -.00145 & .00076 & .0 & .01932 & .0 \\ -.00145 & .00076 & .0 & .01932 & .0 \\ -.00140 & .0 & .00075 & .0 & .0 \\ -.00140 & .0 & .00075 & .0 & .0 \\ .0 & .0 & .00069 & .0 & .01446 \\ -.10202 & .01932 & .0 & 40.01414 & .0 \\ -.10202 & .01932 & .0 & 40.01414 & .0 \\ .0 & .0 & .0 & .0 & .0 \\ .0 & .0 & .0 & .0 & .0 \\ .0 & .0 & .01446 & .0 & 39.67182 \end{bmatrix} ;$$

$\mathbf{C}_{22}$ is presented in equation [23].

$\mathbf{t}_1' =$

[5.8692 .4663 .6287 161.6266 73.9457].

The three subvectors of $\mathbf{t}_2'$ are

[1.1389 1.3997 1.5606 1.7780 0],

[.2254 .2409 .1683 .1995 .2608],

and

[79.1971 82.4295 0 0 73.9457].

The solution is $\hat{\beta} =$

[69.720 389.143 362.368 2.0379 1.7413];

The three subvectors of $\hat{\mathbf{u}}'$ are

[−3.7673 .4351 −1.1566 3.7217 .0190],

[−11.5752 2.0611 −4.8214 15.1516 −3.2053],

and

[−.0715 .0108 −.0251 .0798 −.0083].

APPROXIMATION TO REDUCE COMPUTATIONAL LABOR

The method of the previous section leads to BLUE of $\mathbf{X}\beta$ and BLUP of $\mathbf{u}$, provided either no selection has occurred or selection has been

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[22]

.030405	0	0	0	0	0	-.001447	0	0	0	0	0	-.102022	0	0	0	0	0	0	0	0	0	0
0	.030405	0	0	0	0	0	-.001447	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	.030145	0	0	0	0	0	-.001398	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	.030145	0	0	0	0	0	-.001398	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	.030145	0	0	0	0	0	-.001398	0	0	0	0	0	0	0	0	0	0	0	0
-.001447	0	0	0	0	0	.000762	0	0	0	0	0	.019317	0	0	0	0	0	0	0	0	0	0
0	-.001447	0	0	0	0	0	.000762	0	0	0	0	0	.019317	0	0	0	0	0	0	0	0	0
0	0	-.001398	0	0	0	0	0	.000753	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	-.001398	-.001398	0	0	0	0	0	.000753	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	.000693	0	0	0	0	0	0	0	0	0	0	0	.014461
-.102022	0	0	0	0	0	.019317	0	0	0	0	0	40.014145	0	0	0	0	0	0	0	0	0	0
0	-.102022	0	0	0	0	0	.019317	0	0	0	0	0	40.014145	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	.014461	0	0	0	0	0	0	0	0	0	0	39.671821

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$$
\begin{bmatrix}
.09666 & -.01325 & -.01325 & .0 & .0 & -.00714 & .00114 & .00114 & .0 & .0 & -.83029 & .14565 & .14565 & .0 & .0 \\
-.01325 & .09666 & -.01325 & .0 & .0 & .00114 & -.00714 & .00114 & .0 & .0 & .14565 & -.83029 & .14565 & .0 & .0 \\
-.01325 & -.01325 & .09640 & .0 & .0 & .00114 & .00114 & -.00709 & .0 & .0 & .14565 & .14565 & -.72827 & .0 & .0 \\
.0 & .0 & .0 & .09375 & -.01590 & .0 & .0 & .0 & -.00686 & .00137 & .0 & .0 & .0 & -.69914 & .17478 \\
.0 & .0 & .0 & -.01590 & .06361 & .0 & .0 & .0 & .00137 & -.00546 & .0 & .0 & .0 & .17478 & -.69914 \\
-.00714 & .00114 & .00114 & .0 & .0 & .00421 & -.00069 & -.00069 & .0 & .0 & -.16334 & .03653 & .03653 & .0 & .0 \\
.00114 & -.00714 & .00114 & .0 & .0 & -.00069 & .00421 & -.00069 & .0 & .0 & .03653 & -.16334 & .03653 & .0 & .0 \\
.00114 & .00114 & -.00709 & .0 & .0 & -.00069 & -.00069 & .00420 & .0 & .0 & .03653 & .03653 & -.18266 & .0 & .0 \\
.0 & .0 & .0 & -.00686 & .00137 & .0 & .0 & .0 & .00406 & -.00083 & .0 & .0 & .0 & -.17535 & .04384 \\
.0 & .0 & .0 & .00137 & -.00546 & .0 & .0 & .0 & -.00083 & .00400 & .0 & .0 & .0 & .04384 & -.16089 \\
-.83029 & .14565 & .14565 & .0 & .0 & -.16334 & .03653 & .03653 & .0 & .0 & 108.62027 & -13.72123 & -13.72123 & .0 & .0 \\
.14565 & -.83029 & .14565 & .0 & .0 & .03653 & -.16334 & .03653 & .0 & .0 & -13.72123 & 108.62027 & -13.72123 & .0 & .0 \\
.14565 & .14565 & -.72827 & .0 & .0 & .03653 & .03653 & -.18266 & .0 & .0 & -13.72123 & -13.72123 & 68.60613 & .0 & .0 \\
.0 & .0 & .0 & -.69914 & .17478 & .0 & .0 & .0 & -.17535 & .04384 & .0 & .0 & .0 & 65.86188 & -16.46547 \\
.0 & .0 & .0 & .17478 & -.69914 & .0 & .0 & .0 & .04384 & -.16089 & .0 & .0 & .0 & -16.46547 & 105.53370
\end{bmatrix} \qquad [23]
$$

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within fixed effects. See Henderson (1975b).

Other unbiased estimators of $X\beta$ could be obtained and these used to adjust y. The resulting solutions to u would be unbiased but would not have minimum prediction errors. For example, the saving in computational labor by using regular least squares to estimate $X\beta$ might be enough to warrant accepting larger prediction errors.

CONCLUSIONS

The methods described in this paper would seem to make practical the incorporation of many relatives' records in multiple trait evaluation by BLUP. This does not imply, however, that such methods should always be used. One needs to balance accuracy of prediction against computational labor.

Relatives' records are of greatest use when heritabilities are low and in particular when the trait cannot be observed in the individual that is a candidate for selection. General rules that indicate how much the use of correlated traits contributes to accuracy are not known for the unknown mean and unequal information case. For a particular data set, however, the procedures in this paper present straightforward methods for determining prediction error variances using all data or only partial data (by pretending data are missing).

Finally, it should be stated that if the variances and covariances used in the computations are substantially different from the true values, it is possible for the prediction error to be larger than if correlated traits had not been used at all.

LITERATURE CITED

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BLOCK 4

4

The animal model and its extensions

1980 to 2004

Direct computation of the inverse of A and increased computing capacity made it possible to include every animal in the pedigree in the mixed model equations. The animal model uses information from all relatives, accounts for selection and nonrandom mating, and allows males and females to be compared on the same scale. This block presents the application of the animal model in beef cattle, its theoretical basis and its genetic properties. It also includes the Bayesian approach, which later provided the basis for part of the genomic methods, the first inclusion of molecular markers in BLUP and random regression models.

- Quaas and Pollak (1980)
- Gianola and Fernando (1986)
- Henderson (1988)
- Kennedy, Schaeffer and Sorensen (1988)
- Fernando and Grossman (1989)
- Jamrozik, Schaeffer and Dekkers (1997)
- Schaeffer (2004)

Quaas and Pollak (1980)

Quaas, R.L., and Pollak, E.J. 1980. Mixed model methodology for farm and ranch beef cattle testing programs. Journal of Animal Science, vol. 51, no. 6, pp. 1277 to 1287.

The problem

Until then, BLUP had been used mainly in the progeny testing of dairy bulls. In beef cattle, the data come from farms and ranches, with bulls and cows evaluated on their own records and on those of their relatives. Quaas and Pollak applied the multiple trait animal model to these programs and proposed the reduced animal model, which lowers the number of equations by absorbing the animals without progeny.

Points to note while reading

Note how the animal model allows males and females to be evaluated on the same scale and uses all the relationship information. The reduced animal model was important at a time of limited computing capacity. It is no longer needed today, but the logic of separating parents from nonparents remains useful for understanding the model.

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MIXED MODEL METHODOLOGY FOR FARM AND RANCH BEEF CATTLE TESTING PROGRAMS

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Summary

Although mixed model methods for multiple traits offer a means for efficient utilization of beef cattle performance test data, excessive computational requirements have precluded routine application. This paper describes how these procedures can be applied to weaning weight and postweaning gain records to estimate breeding values for weaning weight (direct and maternal) and yearling weight. Computational simplifications are emphasized. These include the use of an equivalent model which allows a great reduction in the number of equations that need to be solved. Advantages of these procedures for a genetic evaluation program are discussed.

(**Key Words:** Mixed Model Methodology, Best Linear Unbiased Prediction, Estimated Breeding Values, Performance Testing Programs, Genetic Evaluation.)

Introduction

Henderson first discussed what has come to be known as best linear unbiased prediction (BLUP) 30 years ago (Henderson, 1949). Since then, BLUP theory has been developed and extended to apply to a variety of prediction problems (e.g., Henderson, 1973; Harville, 1976). The impetus for this development and the major application of the methodology has been the genetic evaluation of dairy animals. It is now used world-wide for this purpose.

As Willham (1979) comments, "beef breeders are becoming increasingly aware of the power of the analysis method. . ." Many of the national sire evaluation programs in the United States and Canada utilize BLUP procedures. The power of the method, however, has not been fully realized because it has been applied only to progeny test data. It has not been applied to the performance data collected and processed by breed organizations for within-herd use. This has been due, at least in part, to the real or imagined computational cost involved. The history of BLUP application, however, has been that more and more of the theory becomes applicable because of the steady increase in computer power and the discovery of computing short cuts. A notable example of the latter was the discovery that the inverse of a matrix of relationships is easier (and faster) to compute than are the relationships themselves (Henderson, 1976). This leads to the surprising result that it is easier to incorporate information on all relatives into evaluations than it is to use for example, only information on close relatives.

A next logical step is to incorporate not only relatives' records for a single trait, but records on correlated traits, as well. This could be done, for example, in the evaluations of weaning and yearling weights of beef cattle. Henderson and Quaas (1976) showed how this can be achieved through the solution of a set of equations that exceeds the number of records in magnitude. The primary objective of this paper is to demonstrate, by the concept of equivalent models, how the computational problem can be reduced to manageable proportions. Advantages of using mixed model methodology to summarize performance test information are discussed.

BLUP with the Full Model

The Full Model. It is assumed that the data consist of weaning weight and postweaning gain records as opposed to weaning and yearling weights. This avoids a part-whole relationship. The model assumed for these records is:

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$$\begin{bmatrix} y_w \\ y_g \end{bmatrix} = \begin{bmatrix} X_w & 0 \\ 0 & X_g \end{bmatrix} \begin{bmatrix} \beta_w \\ \beta_g \end{bmatrix} + \begin{bmatrix} Z_w & 0 & Z_m & Z_p \\ 0 & Z_g & 0 & 0 \end{bmatrix} \begin{bmatrix} u_w \\ u_g \\ u_m \\ u_p \end{bmatrix} + \begin{bmatrix} e_w \\ e_g \end{bmatrix}, \quad (1)$$

where

$\mathbf{y}_w(\mathbf{y}_g)$ = vector of weaning weight (gain) records,

$\beta_w(\beta_g)$ = vector of fixed effects affecting weaning weight (gain) records,

$\mathbf{X}_w(\mathbf{X}_g)$ = known incidence matrix relating weaning weight (gain) records to fixed effects,

$\mathbf{u}_w(\mathbf{u}_g)$ = random vector representing direct breeding values for weaning weight (gain),

$\mathbf{u}_m$ = random vector representing maternal breeding values for weaning weight,

$\mathbf{u}_p$ = random vector representing permanent maternal environmental effects on weaning weights,

$\mathbf{Z}_w(\mathbf{Z}_g)$ = known incidence matrix relating weaning weight (gain) records to elements of $\mathbf{u}_w(\mathbf{u}_g)$,

$\mathbf{Z}_m(\mathbf{Z}_p)$ = known incidence matrix relating weaning weight records to elements of $\mathbf{u}_m(\mathbf{u}_p)$ corresponding to the dams of the animals making the records and,

$\mathbf{e}_w(\mathbf{e}_g)$ = random vector representing environmental factors peculiar to each weaning weight (gain) record.

Several points about the model should be made. It is an "animal model" in contrast to a sire model. That is, the equation for a record contains a term for the breeding value of the animal making the record. This allows for the possibility that sires and(or) dams may have records. It also means that evaluations will be obtained for animals that are not sires (or dams). A less desirable consequence is that the number of random elements to be predicted, at least one for each record, becomes exceedingly large.

A second point regarding the model pertains to the Z matrices. An incidence matrix usually has one nonzero element, a 1, in each row and at least one nonzero element in each column. In model (1), however, the Z matrices very well may have columns of zeros. For example, if an animal is represented in $\mathbf{u}_w(\mathbf{u}_g)$ but does not have a record in $\mathbf{y}_w(\mathbf{y}_g)$, the corresponding column of $\mathbf{Z}_w(\mathbf{Z}_g)$ will be null. A similar situation will exist for $\mathbf{Z}_m$. In fact, the only non-null columns of $\mathbf{Z}_m$ will correspond to cows with progeny weaning weights. Allowing for such incidence matrices facilitates expressing the mixed model equations when one wants to include random effects not in the model (Henderson, 1977), such as the maternal breeding values of sires. Extra elements may also be included to simplify some computations. For example, the inverse of a relationship matrix will be needed, and this is easy to compute if all common ancestors are included (Henderson, 1976).

Final specification of the model is completed by the assumptions regarding first and second moments of the random terms, which are that all random vectors have null means and that the u vectors are uncorrelated with the e vectors. Furthermore, for $\mathbf{u}' = (\mathbf{u}_w' \ \mathbf{u}_g' \ \mathbf{u}_m')$, $\text{var}(\mathbf{u}) = \mathbf{G} = \mathbf{G}_0 * \mathbf{A}$, where A is the numerator relationship matrix for all animals to be evaluated and $\mathbf{G}_0$ is the 3 × 3 matrix of additive genetic variances and covariances with elements g_{ij}. The * denotes the direct product operator; that is,

$$\mathbf{G} = \begin{bmatrix} g_{11}\mathbf{A} & g_{12}\mathbf{A} & g_{13}\mathbf{A} \\ g_{21}\mathbf{A} & g_{22}\mathbf{A} & g_{23}\mathbf{A} \\ g_{31}\mathbf{A} & g_{32}\mathbf{A} & g_{33}\mathbf{A} \end{bmatrix}$$

This representation implies that every animal is represented in each of the subvectors of u. Thus, G is larger than otherwise would be necessary but allows computation of $\mathbf{G}^{-1}$ as $\mathbf{G}_0^{-1} * \mathbf{A}^{-1}$. $\mathbf{G}^{-1}$ would be impossible to compute

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by straight inversion techniques. It is assumed that $\mathbf{u}_p$ is uncorrelated with the other u vectors and that var $(\mathbf{u}_p) = r_{33}\mathbf{I}$. Finally, it is also assumed that if every animal has records on both traits, var$(\mathbf{e}) = \mathbf{R}_o * \mathbf{I} = \mathbf{R}$, where $\mathbf{e}' = (\mathbf{e}_w' \ \mathbf{e}_g')$ and $\mathbf{R}_o$ is the 2 × 2 variance-covariance matrix of temporary environmental effects influencing weaning weights and gains. For the more realistic situation in which some animals will have only a single record, appropriate rows and columns of **R** are deleted. Suppose, for example, there are three animals with the following records:

Animal	Traits recorded
1	1, 2
2	1
3	2

Then,

$$\mathbf{R} = \begin{bmatrix} r_{11} & 0 & r_{12} & 0 \\ 0 & r_{11} & 0 & 0 \\ r_{12} & 0 & r_{22} & 0 \\ 0 & 0 & 0 & r_{22} \end{bmatrix}$$

and

$$\mathbf{R}^{-1} = \begin{bmatrix} r^{11} & 0 & r^{12} & 0 \\ 0 & r_{11}^{-1} & 0 & 0 \\ r^{12} & 0 & r^{22} & 0 \\ 0 & 0 & 0 & r_{22}^{-1} \end{bmatrix},$$

where

$$\begin{bmatrix} r^{11} & r^{12} \\ r^{12} & r^{22} \end{bmatrix} = \mathbf{R}_o^{-1}.$$

It is assumed that r_{33} and the elements of $\mathbf{R}_o$ and $\mathbf{G}_o$ are known, which in practice means that good estimates are available.

The Mixed Model Equations. The BLUP of the elements of the u vectors are the solutions to a set of mixed model equations. These equations corresponding to model (1) are similar to those presented by Henderson and Quaas (1976) and modified to include the maternal effects $\mathbf{u}_m$ and $\mathbf{u}_p$. These equations are:

$$\begin{bmatrix} \mathbf{X}_w'\mathbf{R}^{11}\mathbf{X}_w & \mathbf{X}_w'\mathbf{R}^{12}\mathbf{X}_g & \mathbf{X}_w'\mathbf{R}^{11}\mathbf{Z}_w & \mathbf{X}_w'\mathbf{R}^{12}\mathbf{Z}_g & \mathbf{X}_w'\mathbf{R}^{11}\mathbf{Z}_m & \mathbf{X}_w'\mathbf{R}^{11}\mathbf{Z}_p \\ \mathbf{X}_g'\mathbf{R}^{21}\mathbf{X}_w & \mathbf{X}_g'\mathbf{R}^{22}\mathbf{X}_g & \mathbf{X}_g'\mathbf{R}^{21}\mathbf{Z}_w & \mathbf{X}_g'\mathbf{R}^{22}\mathbf{Z}_g & \mathbf{X}_g'\mathbf{R}^{21}\mathbf{Z}_m & \mathbf{X}_g'\mathbf{R}^{21}\mathbf{Z}_p \\ \mathbf{Z}_w'\mathbf{R}^{11}\mathbf{X}_w & \mathbf{Z}_w'\mathbf{R}^{12}\mathbf{X}_g & \mathbf{Z}_w'\mathbf{R}^{11}\mathbf{Z}_w + g^{11}\mathbf{A}^{-1} & \mathbf{Z}_w'\mathbf{R}^{12}\mathbf{Z}_g + g^{12}\mathbf{A}^{-1} & \mathbf{Z}_w'\mathbf{R}^{11}\mathbf{Z}_m + g^{13}\mathbf{A}^{-1} & \mathbf{Z}_w'\mathbf{R}^{11}\mathbf{Z}_p \\ \mathbf{Z}_g'\mathbf{R}^{21}\mathbf{X}_w & \mathbf{Z}_g'\mathbf{R}^{22}\mathbf{X}_g & \mathbf{Z}_g'\mathbf{R}^{21}\mathbf{Z}_w + g^{21}\mathbf{A}^{-1} & \mathbf{Z}_g'\mathbf{R}^{22}\mathbf{Z}_g + g^{22}\mathbf{A}^{-1} & \mathbf{Z}_g'\mathbf{R}^{21}\mathbf{Z}_m + g^{23}\mathbf{A}^{-1} & \mathbf{Z}_g'\mathbf{R}^{21}\mathbf{Z}_p \\ \mathbf{Z}_m'\mathbf{R}^{11}\mathbf{X}_w & \mathbf{Z}_m'\mathbf{R}^{12}\mathbf{X}_g & \mathbf{Z}_m'\mathbf{R}^{11}\mathbf{Z}_w + g^{31}\mathbf{A}^{-1} & \mathbf{Z}_m'\mathbf{R}^{12}\mathbf{Z}_g + g^{32}\mathbf{A}^{-1} & \mathbf{Z}_m'\mathbf{R}^{11}\mathbf{Z}_m + g^{33}\mathbf{A}^{-1} & \mathbf{Z}_m'\mathbf{R}^{11}\mathbf{Z}_p \\ \mathbf{Z}_p'\mathbf{R}^{11}\mathbf{X}_w & \mathbf{Z}_p'\mathbf{R}^{12}\mathbf{X}_g & \mathbf{Z}_p'\mathbf{R}^{11}\mathbf{Z}_w & \mathbf{Z}_p'\mathbf{R}^{12}\mathbf{Z}_g & \mathbf{Z}_p'\mathbf{R}^{11}\mathbf{Z}_m & \mathbf{Z}_p'\mathbf{R}^{11}\mathbf{Z}_p + r_{33}^{-1}\mathbf{I} \end{bmatrix} \begin{bmatrix} \hat{\beta}_w \\ \hat{\beta}_g \\ \hat{\mathbf{u}}_w \\ \hat{\mathbf{u}}_g \\ \hat{\mathbf{u}}_m \\ \hat{\mathbf{u}}_p \end{bmatrix} = \begin{bmatrix} \mathbf{X}_w'(\mathbf{R}^{11}\mathbf{y}_w + \mathbf{R}^{12}\mathbf{y}_g) \\ \mathbf{X}_g'(\mathbf{R}^{21}\mathbf{y}_w + \mathbf{R}^{22}\mathbf{y}_g) \\ \mathbf{Z}_w'(\mathbf{R}^{11}\mathbf{y}_w + \mathbf{R}^{12}\mathbf{y}_g) \\ \mathbf{Z}_g'(\mathbf{R}^{21}\mathbf{y}_w + \mathbf{R}^{22}\mathbf{y}_g) \\ \mathbf{Z}_m'(\mathbf{R}^{11}\mathbf{y}_w + \mathbf{R}^{12}\mathbf{y}_g) \\ \mathbf{Z}_p'(\mathbf{R}^{11}\mathbf{y}_w + \mathbf{R}^{12}\mathbf{y}_g) \end{bmatrix} \quad (2)$$

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where

$$\begin{bmatrix} R^{11} & R^{12} \\ R^{21} & R^{22} \end{bmatrix} = \begin{bmatrix} \text{var}(e_1) & \text{cov}(e_1, e_2') \\ \text{cov}(e_2, e_1') & \text{var}(e_2) \end{bmatrix}^{-1}.$$

These are, to say the least, a formidable set of equations even in the shorthand of matrix representation. The number of equations, somewhat less than four times the number of animals to be evaluated, provides motivation to reduce the magnitude of computations.

One might be willing to assume that some off-diagonal elements of G_0 or R_0 are null. This does not reduce the number of equations, but it does reduce the number of nonzero elements of the coefficient matrix. Therefore, the equations can be formed and solved iteratively more quickly. However, even with $r_{12} = g_{12} = g_{13} = 0$, Pollak and Quaas (1981a) found these equations to be poorly behaved for iteration. This approach reduces accuracy and can be justified only if the loss in accuracy is small and the saving in computational time makes the procedure feasible.

A second approach is to eliminate some of the equations by absorption. To be most useful, large numbers of equations must be readily absorbed and back solutions easily computed for the absorbed equations. Therefore, which equations are candidates for absorption? First, the $\hat{u}_p$ equations can be easily absorbed as the equations are formed if the data are sorted by dam. This reduces the problem by a factor equal to the number of dams, which helps some but not much.

For models involving breeding values for two traits, Slanger (1977) has shown how one of the equations for most of the animals can be eliminated. Presumably, this procedure could be generalized for more complicated models — e.g., (1) — but the task does not appear trivial. Furthermore, except under special circumstances, this procedure is not exact (Slanger, 1979).

The major reduction can be made by noting the structure of A^{-1}. The only nonzero off-diagonal elements are between parent and progeny and between mates (Henderson, 1976). Therefore, if the animals are ordered so that all animals with progeny precede those with no progeny, the structure of A^{-1} is :

$$\begin{bmatrix} A_{sd}^{-1} + .25M'DM & -.5M'D \\ -.5DM & D \end{bmatrix},$$

where A_{sd}^{-1} is the inverse of the matrix of relationships among sires and dams, M is an incidence matrix relating parents to progeny and D is diagonal. This pattern to A^{-1} means that there are large diagonal blocks in any of the submatrices of the coefficient matrix involving A^{-1}. This structure can be exploited to eliminate all equations in (2) pertaining to animals with no progeny. The remaining equations, numbering approximately three times the number of sires and dams, can be solved for BLUE of estimable functions of the fixed effects and BLUP of the breeding values of the sires and dams.

Furthermore, close inspection of (2) reveals that the solutions for a nonparent depend only upon its own records, the fixed solutions and the solutions of its parents. This means that solutions for any nonparent can be computed from a set of three equations.

The concept of equivalent models will be used to arrive at rules for forming the reduced equations directly.

BLUP with an Equivalent Model

Henderson has alluded to the concept of equivalent models in several papers, e.g., inclusion of a herd × sire interaction in the model to account for an environmental correlation among paternal half-sibs in the same herd (Henderson, 1974). Equivalent models are defined as models that generate the same first and second moments of the observations. The concept is useful, because the way the model is written affects computations utilizing mixed model equations.

Suppose, for example, that

$$E(y) = X\beta$$

and

$$\text{var}(y) = Z_1 V_1 Z_1' + Z_2 V_2 Z_2' + R_1.$$

Then two equivalent linear models are

$$y = X\beta + Z_1 u_1 + Z_2 u_2 + e_1 . \qquad (3)$$

with

$$\text{var} \begin{bmatrix} u_1 \\ u_2 \\ e_1 \end{bmatrix} = \begin{bmatrix} V_1 & 0 & 0 \\ 0 & V_2 & 0 \\ 0 & 0 & R_1 \end{bmatrix},$$

and

$$y = X\beta + Z_1 u_1 + e_2 , \qquad (4)$$

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with $\text{var}\begin{bmatrix} u_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} V_1 & 0 \\ 0 & R_2 \end{bmatrix}$, where $R_2 = Z_2 V_2 Z_2' + R_1$.

The mixed model equations corresponding to (3) and (4) are

$$\begin{bmatrix} X'R_1^{-1}X & X'R_1^{-1}Z_1 & X'R_1^{-1}Z_2 \\ Z_1'R_1^{-1}X & Z_1'R_1^{-1}Z_1 + V_1^{-1} & Z_1'R_1^{-1}Z_2 \\ Z_2'R_1^{-1}X & Z_2'R_1^{-1}Z_1 & Z_2'R_1^{-1}Z_2 + V_2^{-1} \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{u}_1 \\ \hat{u}_2 \end{bmatrix} = \begin{bmatrix} X'R_1^{-1}y \\ Z_1'R_1^{-1}y \\ Z_2'R_1^{-1}y \end{bmatrix}, \quad (5)$$

and

$$\begin{bmatrix} X'R_2^{-1}X & X'R_2^{-1}Z_1 \\ Z_1'R_2X & Z_1'R_2^{-1}Z_1 + V_1^{-1} \end{bmatrix} \begin{bmatrix} \tilde{\beta} \\ \tilde{u}_1 \end{bmatrix} = \begin{bmatrix} X'R_2^{-1}y \\ Z_1'R_2^{-1}y \end{bmatrix}, \quad (6)$$

respectively. It is readily shown that $\hat{u}_1 = \tilde{u}_1$ and $k'\hat{\beta} = k'\tilde{\beta}$, where $k'\beta$ is an estimable function. To do so, absorb the $\hat{u}_2$ equations in (5) and note that $R_2^{-1} = R_1^{-1} - R_1^{-1}Z_2(Z_2'R_1^{-1}Z_2 + V_2^{-1})^{-1}Z_2'R_1^{-1}$. Hence, if one is only interested in BLUE of $k'\beta$ and BLUP of u_1, it does not matter whether computations are made with (5) or (6). If, for example, R_1 and V_2 are diagnoal, (5) has the advantage of being easier to form because the only nondiagonal inverse involved is V_1^{-1}. On the other hand, (6) has the advantage of involving fewer equations. In many applications, the model is written as (3), but because $Z_2'R_1^{-1}Z_2 + V_2^{-1}$ is diagonal the $\hat{u}_2$ equations are absorbed to give (6). Hence, writing the model in the form of model (3) serves only to facilitate formation of (6). If (6) is solved for $\tilde{\beta}$ and $\tilde{u}_1$, then $\tilde{u}_2 = (Z_2'R_1^{-1}Z_2 + V_2^{-1})^{-1}Z_2'R_1^{-1}(y - X\tilde{\beta} - Z_1\tilde{u}_1)$. If $Z_2'R_1^{-1}Z_2 + V_1^{-1}$ is easy to invert, these back solutions are readily computed.

The difference between models (3) and (4) is in the composition of the error term. In general, any random effects can be put into the error term as long as they are not correlated to the random effects remaining in the model. The mixed model equations formed from the reduced equivalent model will be exactly the same as those obtained by absorbing equations from the full set of equations. To be useful, however, the error variance-covariance matrix must be easily inverted. It will now be shown how this can be accomplished for model (1).

Equivalent Model for Records of a Single Animal. The equations for records of a single animal may be denoted as

$$y_w^c = \mu_w^c + u_w^c + u_m^d + u_p^d + e_w^c,$$

$$y_g^c = \mu_g^c + u_g^c + e_g^c,$$

where y_w^c and y_g^c are the weaning weight and gain records of the calf (c), u_w^c and u_g^c are the calf's breeding values (direct) for the two traits, u_m^d and u_p^d are the maternal breeding value and permanent environmental effects of calf's dam (d) and e_w^c and e_g^c are the environmental effects associated with the calf's records. μ_w^c and μ_g^c represent the linear function of fixed effects affecting the calf's records. Then, assuming that c and d are noninbred,

$$\text{var}\begin{bmatrix} y_w^c \\ y_g^c \end{bmatrix} = \begin{bmatrix} g_{11} + g_{13} + g_{33} + r_{33} + r_{11} & g_{12} + \tfrac{1}{2}g_{23} + r_{12} \\ g_{12} + \tfrac{1}{2}g_{23} + r_{12} & g_{22} + r_{22} \end{bmatrix}$$

To arrive at an equivalent model that does not involve u_w^c and u_g^c, write

$$\begin{aligned} y_w^c &= \mu_w^c + (\tfrac{1}{2}u_w^s + \tfrac{1}{2}u_w^d + e_w^c) + u_m^d + u_p^d + e_w^c \\ &= \mu_w^c + \tfrac{1}{2}u_w^s + \tfrac{1}{2}u_w^d + u_m^d + u_p^d + (e_w^c + e_w^c), \\ y_g^c &= \mu_g^c + (\tfrac{1}{2}u_g^s + \tfrac{1}{2}u_g^d + e_g^c) + e_g^c \\ &= \mu_g^c + \tfrac{1}{2}u_g^s + \tfrac{1}{2}u_g^d + (e_g^c + e_g^c), \end{aligned}$$

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where u_w^s and u_g^s are the breeding values for the calf's sire (s) and dam (d) and ϵ_w^c and ϵ_g^c represent Mendelian sampling in the formation of the calf's genotype. Note that ϵ_w^c and ϵ_g^c are uncorrelated to other terms of the model but not with each other. Written this way, the model includes a term for an "average" gamete from each parent ($\frac{1}{2}u^s$ and $\frac{1}{2}u^d$) plus a term (ϵ^c) for the deviation from "average" of the particular gametes received by the calf. Hence, this might be termed a gametic model as opposed to a genotypic model. With "error" defined as $\epsilon^c + e^c$, we have

$$\text{var}\begin{bmatrix}\epsilon_w^c + e_w^c \\ \epsilon_g^c + e_w^c\end{bmatrix} = \begin{bmatrix}\frac{1}{2}g_{11} + r_{11} & \frac{1}{2}g_{12} + r_{12} \\ \frac{1}{2}g_{12} + r_{12} & \frac{1}{2}g_{22} + r_{22}\end{bmatrix},$$

and the two models, for records on a single animal, are equivalent.

The next question is whether they are equivalent for records on different animals. Suppose we have records of both a dam and her calf. Look at the gametic models for gain.

$$y_g^d = \mu_g^d + \tfrac{1}{2}u_g^{mgs} + \tfrac{1}{2}u_g^{mgd} + (\epsilon_g^d + e_g^d), \quad \text{and}$$

$$y_g^c = \mu_g^c + \tfrac{1}{2}u_g^s + \tfrac{1}{2}u_g^d + (\epsilon_g^c + e_g),$$

where mgs (mgd) refers to the calf's maternal grandsire (granddam), i.e., the dam's sire (dam). In this case the error for the dam's record ($\epsilon_g^d + e_g^d$) is correlated to $\frac{1}{2}u_g^d$ of the calf's record. Hence, for the gametic model to be equivalent to the genotypic model for records of parents and their progeny, specification of covariances among some errors and some elements of the u vectors is necessary. If one ignores this possibility and computes with the gametic model, records will be given improper weights. A parent's own record, for example, will be treated as that of a full-sib. While the gametic model is not equivalent for records of ancestors and descendents, it is equivalent for collateral relatives.

The advantage of the gametic model is that the number of sires and dams is likely to be smaller than the number of animals with records when the data span several years. To utilize this advantage, use both models: the genotypic model for records of parents and the gametic model for records of animals with no recorded progeny.

Equivalent Model for All Records. Matrix representation of the equivalent model is the same as that of the genotypic model (1), but with some terms defined differently. Those that do not change are $\mathbf{X}_w$, $\mathbf{X}_g$, β_w, β_g, $\mathbf{Z}_p$ and $\mathbf{u}_p$. The rest change. The vectors $\mathbf{u}_w$, $\mathbf{u}_g$ and $\mathbf{u}_m$ contain elements pertaining only to sires and dams, and all columns to $\mathbf{Z}_m$ pertaining to nonparents are deleted. $\mathbf{Z}_w$ ($\mathbf{Z}_g$) still relates records to elements of $\mathbf{u}_w$ ($\mathbf{u}_g$), each row corresponding to a particular record. If the record is a parent's, the row contains a single "1" corresponding to the position of the animal's breeding value in $\mathbf{u}_w$ ($\mathbf{u}_g$). If the animal is not a parent, however, the row contains two "½'s" corresponding to the positions of its parents' breeding values in $\mathbf{u}_w$ ($\mathbf{u}_g$).

The variance of u has the same representation as before, but the relationship matrix A pertains only to sires and dams. The "error" variance-covariance matrix also has the same structure as before in that errors are uncorrelated between animals but not between records for the same animal. Now there is a different within-animal error structure depending upon whether the animal is a parent or not. For records of parents, it is $\mathbf{R}_o$, as before, while for nonparents it is $\mathbf{P}_o$, where

$$\mathbf{R}_o = \begin{bmatrix}r_{11} & r_{12} \\ r_{12} & r_{22}\end{bmatrix}$$

and

$$\mathbf{P}_o = \begin{bmatrix}\rho_{11} & \rho_{12} \\ \rho_{12} & \rho_{22}\end{bmatrix} = \begin{bmatrix}\frac{1}{2}g_{11} + r_{11} & \frac{1}{2}g_{12} + r_{12} \\ \frac{1}{2}g_{12} + r_{12} & \frac{1}{2}g_{22} + r_{22}\end{bmatrix}.$$

Hence, $\mathbf{R}$ has nonzero elements in the same places as before, but if these pertain to records of nonparents, they are ρ's instead of r's; the same holds for $\mathbf{R}^{-1}$.

To illustrate what these Z matrices look like, we will use an example the pedigree given in figure 1. The Z matrices are given in table 1.

Forming the Mixed Model Equations. The mixed model equations formed from the equivalent model have the same representation as before (2), but some matrices that were formerly diagonal (e.g., $\mathbf{Z}'_w\mathbf{R}^{11}\mathbf{Z}_w$), are no longer so because some rows of Z's now contain two nonzero elements. However, the equations can be built in the same way as before. One proceeds through the records one at a time, adding the "contributions" to the coefficient

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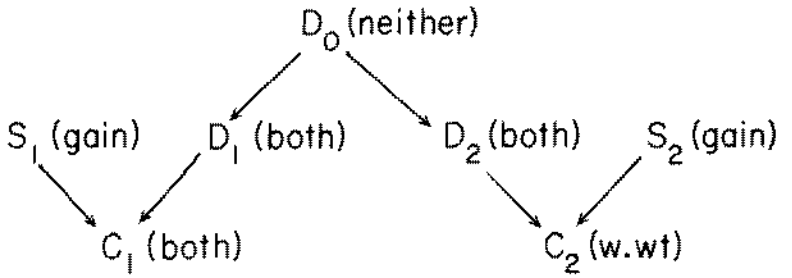


Figure 1. Example pedigree. Records available on each animal are indicated in parentheses.

matrix and right-hand sides for each record. These contributions differ depending upon which records are available for the animal and whether the animal is a parent or not. Contributions for the "least-squares" part of the coefficient matrix are presented in table 2. $G_0^{-1} * A^{-1}$ must then be added to the appropriate block of the coefficient matrix and r_{33}^{-1} added to the diagonals of the u_p equations. The operations are similar to those involved in the formation of A^{-1} from a list of sires and dams, and so, with slight modifications, the entire coefficient matrix can be formed at once. If data are sorted by dam, it also is possible to absorb the u_p equations immediately.

Iterative Solutions. Although the number of equations that need be solved has been reduced considerably, a large set remains. These will most likely have to be solved iteratively. The algorithm most commonly used is Gauss-Seidel (Van Norton, 1959).

Denoting the equations as $Cb = w$, this algorithm takes the form

$$b_i^{(k+1)} = c_{ii}^{-1}\left[w_i - \sum_{j<i} c_{ij} b_j^{(k+1)} - \sum_{j>i} c_{ij} b_j^{(k)}\right], \quad (7)$$

where $b_j^{(k)}$ is the value obtained for the jth

TABLE 1. Z MATRICES CORRESPONDING TO EXAMPLE IN FIGURE 1

Observation vector	Order of records	Order of elements of u					Matrix
		S_1	S_2	D_0	D_1	D_2	
y_w	D_1	0	0	0	1	0	$= Z_w$
	D_2	0	0	0	0	1	
	C_1	.5	0	0	.5	0	
	C_2	0	.5	0	0	.5	
y_w	D_1	0	0	1	0	0	$= Z_m$
	D_2	0	0	1	0	0	
	C_1	0	0	0	1	0	
	C_2	0	0	0	0	1	
y_w	D_1			1	0	0	$= Z_p$
	D_2			1	0	0	
	C_1			0	1	0	
	C_2			0	0	1	
y_g	S_1	1	0	0	0	0	$= Z_g$
	S_2	0	1	0	0	0	
	D_1	0	0	0	1	0	
	D_2	0	0	0	0	1	
	C_1	.5	0	0	.5	0	

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TABLE 2. CONTRIBUTIONS TO COEFFICIENT MATRIX[a] AND RIGHT-HAND SIDES (RHS) BY VARIOUS TYPES OF RECORDS

Parent record		Nonparent record	
Contribution[b]	Location	Contribution[b]	Location
δ_{11}	$\mu_w \times \mu_w$	δ_{11}	$\mu_w \times \mu_w$
δ_{11}	$\mu_w \times u_w$	$.5\delta_{11}$	$\mu_w \times u_w^s; \mu_w \times u_w^d$
δ_{11}	$\mu_w \times u_m^d$	δ_{11}	$\mu_w \times u_m^d$
δ_{11}	$\mu_w \times u_p^d$	δ_{11}	$\mu_w \times u_p^d$
δ_{11}	$u_w \times u_w$	$.25\delta_{11}$	$u_w^s \times u_w^s; u_w^s \times u_w^d; u_w^d \times u_w^d$
δ_{11}	$u_w \times u_m^d$	$.5\delta_{11}$	$u_w^s \times u_m^d; u_w^d \times u_m^d$
δ_{11}	$u_w \times u_p^d$	$.5\delta_{11}$	$u_w^s \times u_p^d; u_w^d \times u_p^d$
δ_{11}	$u_m^d \times u_m^d$	δ_{11}	$u_m^d \times u_m^d$
δ_{11}	$u_m^d \times u_p^d$	δ_{11}	$u_m^d \times u_p^d$
δ_{11}	$u_p^d \times u_p^d$	δ_{11}	$u_p^d \times u_p^d$
δ_{12}	$\mu_w \times \mu_g$	δ_{12}	$\mu_w \times \mu_g$
δ_{12}	$\mu_w \times u_g$	$.5\delta_{12}$	$\mu_w \times u_g^s; \mu_w \times u_g^d$
δ_{12}	$\mu_g \times u_w$	$.5\delta_{12}$	$\mu_g \times u_w^s; \mu_g \times u_w^d$
δ_{12}	$\mu_g \times u_m^d$	δ_{12}	$\mu_g \times u_m^d$
δ_{12}	$\mu_g \times u_p^d$	δ_{12}	$\mu_g \times u_p^d$
δ_{12}	$u_w \times u_g$	$.25\delta_{12}$	$u_w^s \times u_g^s; u_w^s \times u_g^d; u_w^d \times u_g^s; u_w^d \times u_g^d$
δ_{12}	$u_g \times u_m^d$	$.5\delta_{12}$	$u_g^s \times u_m^d; u_g^d \times u_m^d$
δ_{12}	$u_g \times u_p^d$	$.5\delta_{12}$	$u_g^s \times u_p^d; u_g^d \times u_p^d$
δ_{22}	$\mu_g \times \mu_g$	δ_{22}	$\mu_g \times \mu_g$
δ_{22}	$\mu_g \times u_g$	$.5\delta_{22}$	$\mu_g \times u_g^s; \mu_g \times u_g^d$
δ_{22}	$u_g \times u_g$	$.25\delta_{22}$	$u_g^s \times u_g^s; u_g^s \times u_g^d; u_g^d \times u_g^d$
θ_1	μ_w RHS	θ_1	μ_w RHS
θ_1	u_w RHS	$.5\theta_1$	u_w^s RHS; u_w^d RHS
θ_1	u_m^d RHS	θ_1	u_m^d RHS
θ_1	u_p^d RHS	θ_1	u_p^d RHS
θ_2	μ_g RHS	θ_2	μ_g RHS
θ_2	u_g RHS	$.5\theta_2$	u_g^s RHS; u_g^dRHS

[a]Only contributions on or above the diagonal are given; sires precede dams in the u vectors.

[b]$\theta_i = \delta_{i1} y_w + \delta_{i2} y_g$; δ's defined in table 3.

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TABLE 3. DEFINITIONS OF δ'S IN TABLE 2

	Records available								
	Both records			Weaning weight only			Gain only		
Type of record	δ_{11}	δ_{12}	δ_{22}	δ_{11}	δ_{12}	δ_{22}	δ_{11}	δ_{12}	δ_{22}
Parent	r^{11}	r^{12}	r^{22}	r_{11}^{-1}	0	0	0	0	r_{22}^{-1}
Nonparent	ρ^{11}	ρ^{12}	ρ^{22}	ρ_{11}^{-1}	0	0	0	0	ρ_{22}^{-1}

element of the solution vector at the kth iterate. If $\mathbf{C}$ is symmetric positive definite, the procedure is guaranteed to converge. It is not guaranteed to converge very rapidly. It works best when the diagonal elements are large in relation to the off-diagonals. This, unfortunately, is not a property of the multiple trait equtions. The situation can be ameliorated if the equations can be partitioned (possibly with reordering) as

$$\begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} & \cdots & \mathbf{C}_{1n} \\ \mathbf{C}_{21} & \mathbf{C}_{22} & \cdots & \mathbf{C}_{2n} \\ \cdot & \cdot & \cdot & \\ \cdot & \cdot & \cdot & \\ \cdot & \cdot & \cdot & \\ \mathbf{C}_{n1} & \mathbf{C}_{n2} & \cdots & \mathbf{C}_{nn} \end{bmatrix} \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \cdot \\ \cdot \\ \cdot \\ \mathbf{b}_n \end{bmatrix} = \begin{bmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \cdot \\ \cdot \\ \cdot \\ \mathbf{w}_n \end{bmatrix},$$

such that the dominant coefficients in any equations are contained in the diagonal blocks. The computing algorithm is then taken to be

$$\mathbf{b}_i^{(k+1)} = \mathbf{C}_{ii}^{-1}\left[\mathbf{w}_i - \sum_{j<i}\mathbf{C}_{ij}\mathbf{b}_j^{(k+1)} - \sum_{j>i}\mathbf{C}_{ij}\mathbf{b}_j^{(k)}\right]. \quad (8)$$

This is known as block iteration (Vargas, 1962) and the multiple trait equations are well suited for this approach. Suppose one has set up the equations with the $\hat{\mathbf{u}}_p$ equations absorbed. A logical partitioning is to let $\mathbf{b}_1$ refer to the fixed effects and $\mathbf{b}_2$ through $\mathbf{b}_n$ refer to the solutions for a particular animal, i.e., $\mathbf{b}_{i+1}' = (\hat{u}_w^i\ \hat{u}_g^i\ \hat{u}_m^i)$. The reduction in rounds to convergence can be dramatic. Working with a special case of the multiple traits equations, L. D. Van Vleck (*personal communication*) obtained convergence after four rounds using (8); by contrast, he obtained convergence after 50 rounds with (7). Even a more modest reduction will justify the expense of inverting the diagonal blocks.

The speed of convergence also depends upon how "good" the starting values are. Therefore, when processing records annually, one should use the previous year's solutions as starting values. For single trait equations, R. W. Everett (*personal communcation*) found that this procedure reduced the number of rounds required even when the stopping criterion (maximum change in any solution) was halved.

Back-solutions for Calves. Define

$$\mathbf{y}^c = \begin{bmatrix} y_w^c - \hat{\mu}_w^c - \hat{u}_m^d - \hat{u}_p^d \\ y_g^c - \hat{\mu}_g \\ 0 \end{bmatrix}$$

with the convention that if the animal has no weaning weight or gain record, the corresponding element of $\mathbf{y}^c$ is null. Also, define the matrix $\mathbf{R}_o^c$ as

$$\begin{bmatrix} r^{11} & r^{12} & 0 \\ r^{21} & r^{22} & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} r_{11}^{-1} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & r_{22}^{-1} & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

or null, depending upon whether the animal has records for both traits, only weaning weight or only gain, or no records. Then,

$$\hat{\mathbf{u}}^c = (\mathbf{R}_o + a^{cc}\mathbf{G}_o^{-1})^{-1}\left[\mathbf{R}_o^c\mathbf{y}^c - \mathbf{G}_o^{-1}(a^{cs}\hat{\mathbf{u}}^s + a^{cd}\hat{\mathbf{u}}^d)\right],$$

where the a's are elements from $\mathbf{A}^{-1}$ (for all animals) and $\mathbf{u}^s$ and $\mathbf{u}^d$ are vectors of solutions for the animals sire and dam. If there is no inbreeding, $a^{cs} = a^{cd} = -.5a^{cc} = -1$. Therefore,

$$\hat{\mathbf{u}}^c = \mathbf{B}^c\mathbf{y}^c + (\mathbf{I} - \mathbf{B}^c)(.5\hat{\mathbf{u}}^s + .5\hat{\mathbf{u}}^d) \quad (9.1)$$

$$= \mathbf{B}^c(\mathbf{y}^c - .5\hat{\mathbf{u}}^s - .5\hat{\mathbf{u}}^d) + .5(\hat{\mathbf{u}}^s + \hat{\mathbf{u}}^d), \quad (9.2)$$

where $\mathbf{B}^c = (\mathbf{R}_o^c + 2\mathbf{G}_o^{-1})^{-1}\mathbf{R}_o^c$.

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Since there are only four possibilities for $\mathbf{B}^c$, depending upon which records are available for the animal, the value need not be computed anew for each animal. The two terms in (9.2) have meaning. Recall the representation: $\mathbf{u}^c = .5\mathbf{u}^s + .5\mathbf{u}^d + \epsilon^c$. The first term is BLUP of ϵ^c; the latter, BLUP of $.5(\mathbf{u}^s + \mathbf{u}^d)$. Or, equivalently, the latter term predicts the average breeding value of future full sibs while the first predicts the difference of this animal from those sibs.

Accounting for Inbreeding. Three modifications need be made if there is inbreeding. First, $\mathbf{A}^{-1}$ must be computed by an algorithm that accounts for inbreeding (see, for example, Quaas, 1976). Second, $\mathbf{P}_o$, the within-animal covariance matrix of errors, will depend upon the inbreeding of sires and dams and these will be different for each calf (nonparent). That is,

$$\begin{aligned}\mathbf{P}_o &= \text{var } \epsilon^c + \text{var } \mathbf{e}^c \\ &= .5\,(1\text{-}f)\,\mathbf{G}_1 + \mathbf{R}_o,\end{aligned}$$

where $\mathbf{G}_1$ is the leading 2×2 block of $\mathbf{G}_o$ and f is the average of the inbreeding coefficients for the calf's parents. These inbreeding coefficients for parents are a byproduct of the algorithm for computing $\mathbf{A}^{-1}$ (Quaas, 1976), so their calculation is not a problem. To use them, however, requires computing $\mathbf{P}_o^{-1}$ for each calf with inbred parents, which may not be worthwhile for the levels of inbreeding found in most beef cattle populations.

Finally, computing back-solutions requires that $\mathbf{B}^c$ of (9.1) or (9.2) be defined as $(\mathbf{R}_o^c + a^{cc}\mathbf{G}_o^{-1})^{-1}\,\mathbf{R}_o^c$, where $a^{cc} = 2/(1\text{-}f)$.

Discussion

By these procedures, animals are evaluated (breeding values estimated) for weaning weight (direct and maternal) and for postweaning gain. From these evaluations, others can be constructed. For example, a yearling weight evaluation can be computed as $\hat{\mathbf{u}}_w + 160\ \hat{\mathbf{u}}_g$. Most probable producing ability of dams for weaning weight is $.5\hat{\mathbf{u}}_w + \hat{\mathbf{u}}_m + \hat{\mathbf{u}}_p$. In some current programs, bulls are evaluated for maternal performance on the basis of the weaning weights of their daughter's calves. This is an evaluation of the bull as a maternal grandsire. Such an evaluation can be computed as $.25\ \hat{\mathbf{u}}_w + .5\hat{\mathbf{u}}_m$. We do not recommend that all of these be computed, but they illustrate what can be done.

These evaluations computed by BLUP procedures with all available data have several advantages over the use of weight ratios. The reason for using ratios is to remove yearly environmental differences, which they do, although not efficiently nor completely. They also, however, remove genetic trend. In contrast, the use of $\mathbf{A}^{-1}$ in the mixed model procedure accounts for genetic trend (Henderson, 1975a; Pollak and Quaas 1981a). This means that fair comparisons among generations can be made. From the point of view of the breeder, it also means that the evaluations will reflect progress made by an effective selection program. It is possible to provide breeders an estimate of the genetic trends in their herds. This can be computed as the mean evaluation of the parents (weighted by number of progeny for sires) of each year's calf crop.

Additionally, yearling weight ratios can be affected by culling at weaning if computed only for selected animals. The ratios for all animals will be lower than they would have been had lighter animals not been culled at weaning. This means that ratios do not provide fair comparisons among animals in different contemporary groups when selection intensity varies. It also is disconcerting for the breeder who culls heavily on the basis of weaning weights that one-half of his selected animals receive yearling weight ratios of less than 100. In contrast again, multiple trait mixed model procedures account for the effects of sequential culling (Henderson 1975b; Pollak and Quaas, 1981b).

The procedures outlined are not simple, especially by comparison to the calculation of adjusted weight ratios. They are, we feel, practicable at a modern data processing center. The use of the procedures requires two types of information — pedigrees and performance data — which are routinely processed by breed associations. Hence, implementation would integrate two of the major functions of breed organizations.

Another reason for implementation is suggested by Willham (1979). Sire selection is the most important selection decision facing breeders and it cannot be based solely on a progeny test. Bulls are selected primarily on the basis of their own performance and that of their ancestors and collateral relatives. So, to be most useful, an evaluation program of potential herd sires should incorporate this information. The immediate application of the procedures de-

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scribed herein is to accomplish this on a within-herd basis, but the long-range goal is to extend this across herds. Willham (1979) predicts that in the future, the distinction between within-herd performance testing and national sire evaluation programs will diminish. One step toward this end is to apply available technology to the performance test data currently collected.

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Gianola and Fernando (1986)

Gianola, D., and Fernando, R.L. 1986. Bayesian methods in animal breeding theory. Journal of Animal Science, vol. 63, no. 1, pp. 217 to 244.

The problem

Henderson's methods were built within classical statistics, but several questions, such as the use of prior information, the estimation of variances and the analysis of traits that do not follow a normal distribution, were difficult to handle in that framework. Gianola and Fernando proposed Bayesian inference as a general framework for animal breeding. They showed that BLUP corresponds to the mean of the posterior distribution under normality and indicated how to treat more complex problems.

Points to note while reading

The Bayesian approach gained ground in the 1990s with Gibbs sampling and became the basis of genomic methods such as BayesA, BayesB and BayesCπ. Note the interpretation of BLUP as a Bayesian estimate and the discussion of informative and noninformative prior distributions.

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BAYESIAN METHODS IN ANIMAL BREEDING THEORY[1,2]

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ABSTRACT

This paper proposes the Bayesian approach as a conceptual strategy to solve problems arising in animal breeding theory. General elements of Bayesian inference, e.g., prior and posterior distributions, informative vs noninformative priors, likelihood functions, finite samples, "memory" properties and integration of nuisance parameters are illustrated with animal breeding examples. Shrinkage estimators are discussed from a Bayesian viewpoint. A general framework for estimation of breeding value with or without selection or assortative mating, including the situation where variances and covariances are unknown, is presented. Selection indexes, best linear unbiased prediction, nonlinear merit functions, nonlinear models and estimation of genetic parameters are discussed from a Bayesian perspective.

(Key Words: Bayesian Theory, Animal Breeding Methods, Estimation, Prediction.)

Introduction

It is useful to start with a definition of what is meant by "animal breeding theory" and by "Bayesian viewpoint." Animal breeding theory deals with the formulation and validation of mathematical (primarily statistical) models aimed at developing procedures for selecting and mating individuals so that performance is optimal in some sense. Kendall and Buckland (1971) give the following definitions:

Bayesian inference: "A form of inference which regards parameters as being random variables possessed of prior distributions reflecting the accumulated state of knowledge."

Bayes' estimation: "The estimation of population parameters by the use of methods of inverse probability."

The development of statistical principles in animal breeding has proceeded in a somewhat disorderly manner. For example, there are at present no less than 10 methods of estimation of genetic parameters from unbalanced data. Seemingly, none of these methods was developed with a view towards the use to be made of the estimates. If the idea is to use them in functions of the data aimed to rank candidates for selection so as to maximize expected genetic progress, unbiasedness, minimum variance or asymptotic normality are perhaps irrelevant. Similarly, maximization of likelihood does not necessarily give guidance on how to rank optimally breeds of cattle. The objectives of this paper is to propose the Bayesian approach as a conceptual strategy to solve animal breeding problems, to illustrate how well-known results (some of which were obtained from noncontentious applications of Bayes' theorem) can be retrieved from a Bayesian perspective, and to suggest possible areas of research in which this methodology can lead to fruitful results.

Bayes Theorem

General. Consider an unobservable vector θ and data $\mathbf{y}$, and their joint density $f(\theta, \mathbf{y})$. From standard probability theory

$$f(\theta, \mathbf{y}) = f(\mathbf{y}|\theta) \cdot f(\theta) \qquad [1]$$

and

$$f(\theta, \mathbf{y}) = f(\theta|\mathbf{y}) \cdot f(\mathbf{y}), \qquad [2]$$

where $f(\theta)$ and $f(\mathbf{y})$ are the marginal densities of θ and $\mathbf{y}$, respectively. From [1] and [2]

[1] Invitational paper presented at a symp. on "New Statistical Methods in Animal Breeding and Genetics," 76th Annu. Meet. of the Amer. Soc. of Anim. Sci., Univ. of Missouri, Columbia, August 9, 1984.

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$$f(\theta|\mathbf{y}) = f(\mathbf{y}|\theta)\cdot f(\theta)/f(\mathbf{y}). \qquad [3]$$

Note that

$$f(\mathbf{y}) = \int_{R_\theta} f(\mathbf{y}, \theta)\, d\theta$$

$$= \int_{R_\theta} f(\mathbf{y}|\theta)\, f(\theta)\, d\theta = \underset{\theta}{E}[f(\mathbf{y}|\theta)], \qquad [4]$$

where $\underset{\theta}{E}$ indicates averaging with respect to the distribution of θ (Box and Tiao, 1973). It is clear that $f(\mathbf{y})$ is not a function of θ. Hence [3] can be written as

$$f(\theta|\mathbf{y}) \propto f(\mathbf{y}|\theta)\cdot f(\theta), \qquad [5]$$

which is Bayes theorem. In the Bayesian terminology, $f(\theta)$ is known as the prior density of θ, which reflects the relative uncertainty about the possible values of θ before the data vector $\mathbf{y}$ is realized. The density $f(\mathbf{y}|\theta)$ is the likelihood function, which represents the contribution of $\mathbf{y}$ to knowledge about θ (Lindley, 1971; Zellner, 1971; Box and Tiao, 1973). Finally, $f(\theta|\mathbf{y})$ is the posterior density which subsumes the state of uncertainty about θ given previous knowledge, data or non-data based (Zellner, 1971), and the data $\mathbf{y}$. Inferences about θ are made from the posterior density and the solution to the statistical problem is completed with

$$\text{Prob}(\theta \epsilon R|\mathbf{y}) = \int_R f(\theta|\mathbf{y})\, d\theta, \qquad [6]$$

where R is a region of the space of θ. Fixing the probability in [6] at, say, $1 - \alpha$, for a given α, it is possible to obtain an interval for θ such that its probability content is $1 - \alpha$. Note that [6] is conceptually (and numerically in many instances) different from the usual "confidence interval" where repeated sampling of $\mathbf{y}$ is envisaged.

Bayes theorem can be extended to make inferences about yet unobserved data points (Zellner, 1971). Let $\tilde{\mathbf{y}}$ be a vector of unobserved data so

$$f(\tilde{\mathbf{y}}, \theta|\mathbf{y}) = f(\tilde{\mathbf{y}}|\theta, \mathbf{y})\cdot f(\theta|\mathbf{y}) \qquad [7a]$$

is the joint "predictive" density function (Lindley, 1971; Zellner, 1971). The marginal "predictive" density function of $\tilde{\mathbf{y}}$ is then

$$f(\tilde{\mathbf{y}}|\mathbf{y}) = \int_{R_\theta} f(\tilde{\mathbf{y}}, \theta|\mathbf{y})\, d\theta. \qquad [7b]$$

The expressions in [5], [7a] and [7b] provide a general framework for problems in "estimation of parameters" and "prediction." These are features of the more general inferential problem solved by [6].

Remarks. It is apparent in [5] that θ is regarded as random. In the Bayesian framework there is no distinction between "fixed" and "random" variables. This distinction, usually required in animal breeding (e.g., Henderson, 1953, 1973) is often troublesome as put aptly by Lefort (1981; translated from French):

> "Fixed or random? To this question relative to the nature of a parameter, the genetic value of a sire, for example, geneticists often respond in an intuitive manner without clearly stating the logical reasons for the choice. In some instances, a parameter is considered random in a first set of calculations and fixed in a second one, and the geneticist does not seem bothered by this inconsistency . . . I would like to contribute some elements of response and to show how the Bayesian approach to statistics permits us to justify and improve in several situations, the usual methods employed by geneticists."

Loosely speaking, a "fixed" effect is, in a Bayesian sense, a random variable on which prior knowledge is diffuse or vague, i.e., a priori the investigator is indifferent to its likely value. While this conceptualization of a fixed effect is useful from a Bayesian perspective in retrieving results derived using "classical" statistical arguments (Zellner, 1971; Box and Tiao, 1973), the idea that prior knowledge is vague is not always defensible. Suppose a Jersey cow is to be mated to a Shorthorn or to a Chianina bull. Most animal breeders would be prepared to bet before the calf is born that the first mating would lead to a less difficult parturition than the second one. Similarly, in studying patterns of development of bulls versus steers, it would seem naive to state a priori that these two types of animals are likely to follow the same growth curve. The above illustrate the Bayesian justification for treating fixed effects as random variables. While the items in question may not necessarily represent a random sample from a distribution, the prior density reflects randomness stemming from relative uncertainty about the values of the parameters before the data $\mathbf{y}$ are collected.

Prior knowledge is always used in arriving at decisions. Berger (1980) stated

> ". . . if significant prior information does exist, then ignoring it for the sake of objectivity seems ill advised. Nevertheless, the difficulties

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> in accurately specifying prior information (and convincing others that this prior information is reasonable) suggests that it is probably wise to use prior information sparingly in inference problems."

Zellner (1971) distinguished between two types of prior information: data based and non-data based. The first type may be information contained in past data that are relevant to the problem at hand, and that have been obtained in a scientific manner. For example, heritability estimates from previous data sets are used to calculate genetic evaluations using linear (Henderson, 1973) or nonlinear methods (Gianola and Foulley, 1983; Smith, 1983; DeLorenzo, 1983; Smith and Allaire, 1985). Non-data based prior information may be the result of theoretical considerations or of personal beliefs. In the latter case, the posterior inferences made by person A may differ from those made by B unless the information in the likelihood function overwhelms the prior (Box and Tiao, 1973).

The use of priors based on data or on theoretical grounds (e.g., the Mendelian process), as opposed to personal priors, is of general acceptance in statistical practice. There are some other types of noncontroversial, non-data based priors that arise in animal breeding theory. For example, in estimation of variance components from a paternal half-sib family structure, the model imposes the natural restrictions

$$\sigma_s^2 \geq 0;\ \sigma_e^2 > 0;\ \sigma_e^2/\sigma_s^2 \geq 3,$$

where σ_s^2 and σ_e^2 are the "sire" and "residual" components of variance, respectively. The space of

$$\theta = [\sigma_e^2, \sigma_s^2]$$

is then restricted from $\{0 < \sigma_e^2 < \infty;\ 0 \leq \sigma_s^2 < \infty\}$ to $\{0 < \sigma_e^2 < \infty;\ 0 \leq \sigma_s^2 \leq \sigma_e^2/3\}$. A method of estimation which ignores these restrictions may lead to ridiculous estimates of heritability. The situation becomes further complicated in the multivariate case where, as shown by Hill and Thompson (1978), estimates of genetic parameters derived from the analysis of variance can lead to sizable probabilities of nonpositive definiteness of estimated genetic covariance matrices; if these matrices are then used in the construction of selection indexes, absurd results may be obtained. Another instance where non-data based priors arise is in categorical responses: the estimated percentage of a sire's progeny in a particular category should not exceed 1 nor be less than 0 (Gianola, 1980a,b, 1982; Quaas and Van Vleck, 1980). The use of non-stochastic prior information in linear models is reviewed by Toutenburg (1982).

The likelihood function formalizes the contribution of data to knowledge about θ. Suppose S unrelated sires are sampled at random from a conceptual population of sires, and that their transmitting abilities, represented by the vector $\theta' = [s_1, \ldots, s_S]$, are distributed, a priori, as $N(\upsilon, \sigma_s^2)$. We are interested in making inferences about θ given progeny data. The model for the average of the progeny of the ith sire is

$$\bar{y}_{i.} = \mu + s_i + \bar{e}_{i.};\ i = 1, \ldots, S, \qquad [8]$$

where μ is a known constant and $\bar{e}_{i.} \sim N(0, \sigma_e^2/n_i)$ is the average of the residuals associated with the records of the n_i progeny of the ith sire. Because μ is assumed known, one can write

$$w_i = \bar{y}_i - \mu = s_i + \bar{e}_{i.}.$$

Clearly, $E(w_i|s_i) = s_i$ and $Var(w_i|s_i) = \sigma_e^2/n_i$. Because the sires are unrelated, the prior density in [5] is

$$f(\theta) = \prod_{i=1}^{S} \frac{1}{\sqrt{2\pi\sigma_s^2}} \exp\left\{-\frac{(s_i - \upsilon)^2}{2\sigma_s^2}\right\} \qquad [9]$$

and the likelihood function, with $Cov(\bar{e}_{i.}, \bar{e}_{j.}) = 0$, $i \neq j$, is

$$f(w_1, \ldots, w_S|\theta) = \prod_{i=1}^{S} \sqrt{\frac{n_i}{2\pi\sigma_e^2}} \exp\left\{-\frac{n_i(w_i - s_i)^2}{2\sigma_e^2}\right\}. \qquad [10]$$

In view of the assumption of independence

$$f(s_i|w_1, \ldots, w_S) = f(s_i|w_i);\quad i = 1, \ldots, S \qquad [11]$$

and using [9] and [10] in [11] we obtain the posterior density of the transmitting ability of the ith sire

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$$f(s_i|w_i) \propto \exp\left[-\frac{1}{2}\left(\frac{\sigma_s^2 + \sigma_e^2/n_i}{\sigma_s^2\sigma_e^2/n_i}\right) \times \left(s_i - \frac{w_i\sigma_s^2 + \upsilon\sigma_e^2/n_i}{\sigma_s^2 + \sigma_e^2/n_i}\right)^2\right]. \quad [12]$$

Note that this is a normal density with mean

$$E(s_i|w_i) = \frac{w_i\sigma_s^2 + \upsilon\sigma_e^2/n_i}{\sigma_s^2 + \sigma_e^2/n_i} \quad [13a]$$

$$= \frac{\upsilon(\sigma_s^2)^{-1} + w_i(\sigma_e^2/n_i)^{-1}}{(\sigma_s^2)^{-1} + (\sigma_e^2/n_i)^{-1}}. \quad [13b]$$

Clearly, the posterior mean is a weighted average of the prior mean of the transmitting abilities (υ) and of the sample mean (w_i), the weights being the reciprocal of their variances. Put

$$a = (\sigma_s^2)^{-1}/[(\sigma_s^2)^{-1} + (\sigma_e^2/n_i)^{-1}],$$

so [13b] can be written as

$$E(s_i|w_i) = a\upsilon + (1 - a)\, w_i. \quad [13c]$$

Adding and subtracting υ, and rearranging

$$E(s_i|w_i) = \upsilon + (1 - a)(w_i - \upsilon)$$

$$= \upsilon + (1 - a)(\bar{y}_i - \mu - \upsilon), \quad [14]$$

which is the familiar result for the conditional mean in a bivariate normal distribution. Further, upon rearrangement

$$(1 - a) = n_i/(n_i + \frac{\sigma_e^2}{\sigma_s^2}) \quad [15]$$

and, in the particular case that $\upsilon = 0$, [14] becomes

$$E(s_i|w_i) = \frac{n_i}{n_i + \frac{\sigma_e^2}{\sigma_s^2}}(\bar{y}_{i.} - \mu). \quad [16]$$

This is the classical selection index evaluation of transmitting ability of a sire via progeny testing (e.g., Van Vleck, 1979). The Bayesian treatment highlights the fact that [14] or [16] are weighted contributions of prior knowledge and of data as shown in [13b]. This was noted by Robertson (1955) and Dempfle (1977). The variance of the posterior distribution is from [12]

$$Var(s_i|w_i) = \frac{\sigma_s^2\sigma_e^2/n_i}{\sigma_s^2 + \sigma_e^2/n_i}$$

$$= \frac{1}{(\sigma_s^2)^{-1} + (\sigma_e^2/n_i)^{-1}} \quad [17]$$

or, in a more familiar form

$$Var(s_i|w_i) = \sigma_e^2(n_i + \frac{\sigma_e^2}{\sigma_s^2})^{-1} \quad [18a]$$

$$= \sigma_s^2(1 - \frac{n_i}{n_i + \frac{\sigma_e^2}{\sigma_s^2}}) \quad [18b]$$

with

$$Corr^2[s_i, E(s_i|w_i)] = \frac{n_i}{n_i + \sigma_e^2/\sigma_s^2}. \quad [19]$$

It is of interest to examine the behavior of the posterior distribution as n_i increases. From [12], it is clear that the posterior density would tend towards [10]. This illustrates that the prior tends to be overwhelmed or dominated by the likelihood as the amount of data, e.g., the number of progeny per sire, increases (Box and Tiao, 1973). In other words, the contribution from prior knowledge is relatively more important when the information is scant than when it is plentiful. In the limit, when $n_i \to \infty$,

$$E(s_i|w_i) \to w_i \quad [20]$$

and

$$Var(s_i|w_i) \to 0. \quad [21]$$

The Bayesian approach does not rely on the abstract concept of repeated sampling of data required by classical procedures. In these, the properties of point and of interval estimators are based on data that we might have got if the experiment were repeated an infinite number of times, and not on the data that are at hand. Lindley (1971) illustrates this problem as follows: suppose we wish to obtain an unbiased estimator, $t(y)$, of the parameter θ. Then

$$\int_R t(y)\, f(y|\theta)\, dy = \theta \quad [22]$$

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needs to be satisfied, where R is the sampling space of the data $\mathbf{y}$. The properties of the estimator $t(\mathbf{y})$ require reference not only to the $\mathbf{y}$ at hand but to all possible realizations of this vector! Consider now the problem of interval estimation of θ. In sampling theory θ is an unknown fixed quantity and the confidence interval is random: over repeated sampling of $\mathbf{y}$, the interval is expected to contain θ a certain proportion of the time (Zellner, 1971; Box and Tiao, 1973). However, given the data at hand, θ is either inside or outside of the interval.

In the Bayesian system, the inferences are made directly from the posterior distribution in [5], i.e., only reference to the data at hand and to the prior density is needed. A number of point estimates can be derived to summarize features of the posterior distribution, particularly when the latter is analytically intractable, and probability statements about θ can be made via [6] without recourse to imaginary data. Returning to the progeny test example, if one wishes to make probability statements about the true transmitting ability of sire i given w_i, it suffices to refer

$$z = \frac{s_i - E(s_i|w_i)}{\sqrt{Var(s_i|w_i)}} \qquad [22]$$

to a N(0, 1) distribution and to compute, for any fixed q_1 and q_2

$$\text{Prob}\{q_1 < s_i < q_2|w_i\} = \Phi(z_2) - \Phi(z_1), \qquad [23]$$

where $\Phi(.)$ is the standard normal distribution function.

Bayes theorem provides a precise solution to the problem of finite sample sizes. For each data set, no matter how small or large, there is an exact posterior distribution from which inferences can be made, e.g., the posterior of θ given $\mathbf{y}_1$ is different from that of θ given $\mathbf{y}_1$ and $\mathbf{y}_2$. Many statistical procedures widely advocated in animal breeding such as maximum likelihood (Henderson, 1973; Thompson, 1973, 1977, 1982; Schaeffer et al., 1978; Rothschild et al., 1979; Meyer, 1983; Meyer and Thompson, 1984) have only asymptotic justification, that is, their properties are well defined only when samples are large. Are these properties reached in typical animal breeding situations? How applicable are these methods to experimental settings where scarcity of data is the rule rather than the exception? Because when samples are large the posterior tends towards the likelihood function, a Bayesian justification for maximum likelihood procedures is provided. However, this is not so when sample sizes are finite, in which case only [5] furnishes a conceptual solution to the problem.

There is an asymptotic Bayesian theory (Zellner, 1971; Cox and Hinkley, 1974) with large sample results similar to those of maximum likelihood. This theory is particularly useful when the exact posterior density is either difficult to obtain or to integrate. Write [5] as

$$f(\theta|\mathbf{y}) \propto \exp\{\ln[f(\mathbf{y}|\theta)]\} \cdot f(\theta) \qquad [24]$$

and using a Taylor series, expand both factors in [24] about $\hat{\theta}$, the maximum likelihood estimate of θ. Retaining only the leading term of the combined expansion, we have after some rearrangement

$$f(\theta|\mathbf{y}) \propto \exp\left\{-\tfrac{1}{2}(\theta - \hat{\theta})' \times \left[-\frac{\partial^2 \ln f(\mathbf{y}|\theta)}{\partial\theta\,\partial\theta'}\right]_{\theta=\hat{\theta}} (\theta - \hat{\theta})\right\}, \qquad [25]$$

which is a multivariate normal density with mean vector

$$E(\theta|\mathbf{y}) = \hat{\theta} \qquad [26a]$$

and variance-covariance matrix

$$Var(\theta|\mathbf{y}) = \left[-\frac{\partial^2 \ln f(\mathbf{y}|\theta)}{\partial\theta\,\partial\theta'}\right]^{-1}_{\theta=\hat{\theta}}. \qquad [26b]$$

If [25] is a singular multivariate normal density, then the unique inverse in [26b] is replaced by a generalized inverse (Rao, 1973). The result in [25] proves the "asymptotic domination" of the likelihood function, i.e., as sample size increases, the posterior tends towards a multivariate normal density centered at the maximum likelihood estimate.

Returning to the progeny test example, given the likelihood function in [10] it is easy to establish that

$$\hat{\theta}_i = w_i = \bar{y}_{i.} - \mu \qquad [27a]$$

and

$$\frac{\partial^2 \ln f(w_i|\theta_i)}{(\partial\theta_i)^2} = -\frac{n_i}{\sigma_e^2} \qquad [27b]$$

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so, asymptotically, $E(s_i|w_i) = w_i$ and $Var(s_i|w_i) = \sigma_e^2/n_i$.

The Bayesian setting provides a precise description of the process by which new knowledge is combined with previous knowledge (Zellner, 1971; Box and Tiao, 1973). Suppose a data set $\mathbf{y}_1$ is obtained, so using [5]

$$f(\theta|\mathbf{y}_1) \propto f(\mathbf{y}_1|\theta)\cdot f(\theta). \quad [28]$$

Before collecting additional data, the state of knowledge is then represented by [28]. A new data set $\mathbf{y}_2$, not necessarily independent of $\mathbf{y}_1$ is obtained. Then

$$f(\theta|\mathbf{y}_1,\mathbf{y}_2) \propto f(\mathbf{y}_2|\mathbf{y}_1,\theta)\cdot f(\theta|\mathbf{y}_1) \quad [29a]$$

$$\propto f(\mathbf{y}_2|\mathbf{y}_1,\theta)\cdot f(\mathbf{y}_1|\theta)\cdot f(\theta) \quad [29b]$$

$$\propto f(\mathbf{y}_2,\mathbf{y}_1|\theta)\cdot f(\theta). \quad [29c]$$

This shows that given data $\mathbf{y}_1$ and $\mathbf{y}_2$, the posterior density can be obtained in several ways. For example, in [29a] and [29b] an analysis based on $\mathbf{y}_1$ is used to construct a distribution which serves as a prior for the second experimental stage. Alternatively, $\mathbf{y}_1$ and $\mathbf{y}_2$ can be merged to produce a joint likelihood function which, when combined with the initial prior as in [29c], yield $f(\theta|\mathbf{y}_1,\mathbf{y}_2)$. Note in [29c] that if $\mathbf{y}_1$ and $\mathbf{y}_2$ are conditionally independent, then

$$f(\theta|\mathbf{y}_1,\mathbf{y}_2) \propto f(\mathbf{y}_2|\theta)\cdot f(\mathbf{y}_1|\theta)\cdot f(\theta). \quad [30]$$

In the context of the progeny testing example, suppose the ith sire has two independent batches of progeny consisting of $n_{i,1}$ and $n_{i,2}$ records, respectively; let $v = 0$. Merging the two batches of data as in [29c], we obtain from [16]

$$E(s_i|w_{i,1},w_{i,2}) = \frac{n_{i,1} + n_{i,2}}{n_{i,1} + n_{i,2} + \sigma_e^2/\sigma_s^2}\times \left(\frac{n_{i,1}\bar{y}_{i,1} + n_{i,2}\bar{y}_{i,2}}{n_{i,1} + n_{i,2}} - \mu\right), \quad [31]$$

where $\bar{y}_{i,1}$ and $\bar{y}_{i,2}$ are the progeny averages in batches 1 and 2, respectively. From [18a]

$$Var(s_i|w_{i,1},w_{i,2}) = \sigma_e^2\left(n_{i,1} + n_{i,2} + \frac{\sigma_e^2}{\sigma_s^2}\right)^{-1}. \quad [32]$$

If we use [29a], the prior distribution for the second batch is the posterior density given the first batch of progeny. The parameters of this prior distribution are then as in [16] and [18a] with a subscript included to denote the first batch. The posterior mean after the two batches is, using [13b], [16] and [18a] and putting $\lambda = \sigma_e^2/\sigma_s^2$,

$$E(s_i|w_{i,1},w_{i,2}) = \frac{\frac{n_{i,1}}{n_{i,1} + \lambda}(\bar{y}_{i,1} - \mu)(n_{i,1} + \lambda)/\sigma_e^2 + (\bar{y}_{i,2} - \mu)n_{i,2}/\sigma_e^2}{(n_{i,1} + \lambda)/\sigma_e^2 + n_{i,2}/\sigma_e^2}, \quad [33]$$

If we use [29a], the prior distribution for the second batch is the posterior density given the first batch of progeny. The parameters of this prior distribution are then as in [16] and [18a] with a subscript included to denote the first batch. The posterior mean after the two batches is, using [13b], [16] and [18a] and putting $\lambda = \sigma_e^2/\sigma_s^2$, as shown in [33], which, after algebra, is identical to [31]. Similarly from [17]

$$Var(s_i|w_{i,1},w_{i,2}) = \frac{1}{(n_{i,1} + \lambda)/\sigma_e^2 + n_{i,2}/\sigma_e^2}, \quad [34]$$

which is precisely [32].

In many instances the distribution of the data $\mathbf{y}$ depends not only on a set of parameters of interest (θ_1, say), but on the other incidental or "nuisance" parameters (θ_2). The joint posterior density is then from [5]

$$f(\theta_1,\theta_2|\mathbf{y}) \propto f(\mathbf{y}|\theta_1,\theta_2)\cdot f(\theta_1,\theta_2), \quad [35]$$

where $f(\theta_1,\theta_2)$ is the joint prior density of θ_1 and θ_2. Because the interest centers on inferences about θ_1 in the "absence of" θ_2, the nuisance parameters can be eliminated by integrating the joint posterior density with respect to θ_2. We have

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$$f(\theta_1|\mathbf{y}) = \int_{R_{\theta_2}} f(\theta_1,\theta_2|\mathbf{y})\, d\theta_2 \quad [36a]$$

$$= \int_{R_{\theta_2}} f(\theta_1|\theta_2,\mathbf{y})\, f(\theta_2|\mathbf{y})\, d\theta_2, \quad [36b]$$

where R_{θ_2} indicates the space of θ_2 (Zellner, 1971; Box and Tiao, 1973). The form in [36b] is particularly useful in studies of robustness (Box and Tiao, 1962). Suppose, for example, that θ_2 contains the elements of a genetic variance-covariance matrix and that the main interest is on the components of θ_1, a vector of breeding values. The conditional density

$$f(\theta_1|\theta_2 = \mathbf{k},\mathbf{y})$$

would describe the inferences that would be made when it is assumed that $\theta_2 = \mathbf{k}$, and $f(\theta_2|\mathbf{y})$ reflects the "plausibility of such an assumption" (Box and Tiao, 1973); note in [36b] that $f(\theta_2|\mathbf{y})$ acts as a weight function multiplying the conditional density of θ_1. Hence, $f(\theta_1|\mathbf{y})$ describes the overall inference which can be made when appropriate weights are given to the possible values of θ_2, posterior to data.

Consider hypothetically the problem of predicting the breeding value of an individual (θ_i) from a phenotypic measurement (y_i). Take, a priori, $\theta_i \sim$ N.I.I.D. (0, h^2), $E(y_i|\theta_i) = \theta_i$, and $Var(y_i|\theta_i) = 1 - h^2$, where h^2 is heritability. The conditional density $f(\theta_i|h^2 = k,y_i)$ is normal with mean

$$E(\theta_i|h^2 = k,y_i) = ky_i \quad [37a]$$

and

$$Var(\theta_i|h^2 = k,y_i) = k(1 - k). \quad [37b]$$

Now, because there is only one data point, we can argue that

$$f(h^2|y_i) \propto f(h^2), \quad [38]$$

where $f(h^2)$ and $f(h^2|y_i)$ are the prior and marginal posterior densities of heritability, respectively. Further suppose that, a priori, h^2 can take only 3 values (0, .10, .20) with Prob $\{h^2 = 0\}$ = .10, Prob $\{h^2 = .10\}$ = .40 and Prob $\{h^2 = .20\}$ = .50. Heritability is a nuisance parameter in this problem as we are interested in marginal inferences about breeding values. Then, using [36b] we can "integrate out" h^2 to obtain

$$f(\theta_i|y_i) = f(\theta_i|h^2 = 0,y_i) \times .10 + f(\theta_i|h^2 = .10,y_i) \times .40 + f(\theta_i|h^2 = .20,y_i) \times .50. \quad [39]$$

Then

$$E(\theta_i|y_i) = .10E(\theta_i|h^2 = 0,y_i) + .40E(\theta_i|h^2 = .10,y_i) + .50E(\theta_i|h^2 = .20,y_i) = .14y_i. \quad [40]$$

Note that .14 is the prior mean of the distribution of possible values of heritability. Similarly,

$$Var(\theta_i|y_i) = \underset{h^2}{E}\,[Var(\theta_i|h^2 = k, y_i)] + \underset{h^2}{Var}\,[E(\theta_i|h^2 = k, y_i)]. \quad [41]$$

The first term in [41] is the weighted average of $h^2(1 - h^2)$, which is approximately .12. The second term, from [37a] is y_i^2 Var(h^2), with Var(h^2) = .0044. Hence

$$Var(\theta_i|y_i) \doteq .12 + .0044y_i^2. \quad [42]$$

Equations [40] and [42] illustrate that heritability has been effectively "integrated out" as these expressions no longer depend on h^2. However, both the average and the variance of the marginal posterior density of breeding values given data depend on y_i. This illustrates two points. First, when h^2 is not known, and one is forced to construct a prior distribution about its likely values, the posterior density of breeding values given data is not normal (from [25] it is asymptotically normal, however). Second, when h^2 is integrated out, inferences from the marginal posterior density of breeding values are not necessarily sharper than those made from [37a] and [37b]. Because the marginal density of y_i is N(0, 1), values of y_i larger in absolute value than 3 are unlikely. For y_i = 3, [42] yields

$$Var(\theta_i|y_i = 3) \doteq .1569,$$

which is smaller than [37b] when h^2 = .20, but larger when h^2 = 0 or .10. However, inferences from $f(\theta_i|y_i)$ are more realistic than those made from the conditional density $f(\theta_i|h^2 = k, y_i)$

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because uncertainty about the value of h^2 is not taken into account in the latter. This approach can be generalized, at least conceptually, to a much more realistic setting, e.g., multiple traits, unequal information, related animals, by incorporating the uncertainty about elements of a genetic variance-covariance matrix, say, via a prior distribution.

Box and Tiao (1973) warn that the integration of nuisance parameters should be done in a judicious manner. For example, breeders might be interested in examining how the posterior distribution of breeding values changes with their genetic variance-covariance matrix (θ_2). If $f(\theta_1|\theta_2, \mathbf{y})$ is very sensitive to changes in θ_2, is it important to examine $f(\theta_2|\mathbf{y})$ carefully. If this density is rather flat, that is when there is little information from data or prior knowledge about the genetic covariance matrix, then in addition to reporting $f(\theta_1|\mathbf{y})$ it would be useful to illustrate how $f(\theta_1|\theta_2, \mathbf{y})$ changes with θ_2 over a range of "likely" values of θ_2. If, on the contrary $f(\theta_2|\mathbf{y})$ is sharp, i.e., when most of the density is in the neighborhood of its mode ($\hat{\theta}_2$, say), then

$$f(\theta_1|\mathbf{y}) \doteq f(\theta_1|\theta_2 = \hat{\theta}_2, \mathbf{y}). \qquad [43]$$

This, in our opinion gives justification to a procedure usually employed in animal breeding: if $\mathbf{y}$ gives "sharp" information about the nuisance vector θ_2 (e.g., heritabilities and genetic correlations) then modal or maximum likelihood-type estimates can be used in lieu of true values in $f(\theta_1|\theta_2, \mathbf{y})$. As shown in [43], this would be approximately equivalent to making inferences from the marginal distribution of breeding values given data.

There are situations in which knowledge about a problem is either vague or nonexistent. For example, a gene may be transferred by recombinant DNA techniques from species A to two different strains of species B with the question of interest being the pleiotropic effects of this gene upon two different genetic backgrounds. Because a foreign gene is tantamount to a mutation, its overall effect might turn out to be disruptive. However, investigators may feel uncertain about likely values of these effects before data are collected. In this case, Bayesian inference resorts to vague, diffuse or noninformative priors, i.e., prior densities that reflect ignorance rather than knowledge (Jeffreys, 1961). If θ is the effect of this gene, a vague prior could be represented as

$$f(\theta) \propto \text{constant}; -\infty < \theta < \infty. \qquad [44]$$

This prior density is not proper because its integral over all possible values of θ does not converge. Improper priors are controversial but most, if not all, results obtained using classical statistical arguments can be derived via Bayes theorem using improper or noninformative priors (Zellner, 1971; Box and Tiao, 1973); the posterior density is always proper because [44] gets absorbed in the constant of integration associated with the posterior. In order to illustrate the fact that [44] can be used to represent ignorance, consider the situation in [9] in which a normal, informative, prior is used. Prior ignorance about the likely values of the transmitting abilities of sires can be viewed as the limiting case where $\sigma_s^2 \to \infty$; the prior density becomes flatter and flatter so prior variation in σ_i tends to [44]. However, this improper prior is not suitable for all parameters. If one is interested in inferences about a variance, σ^2, priors of the form of [44] turn out to be informative (Jeffreys, 1961). A more reasonable noninformative prior for a variance would be (Zeller, 1971)

$$f(\sigma^2) \propto \frac{1}{\sigma^2}; 0 < \sigma^2 < \infty. \qquad [45]$$

We contrast noninformative vs informative priors with an example in the domain of evaluation of cattle strains and breeds. Taylor (1976) argued that available strains can be regarded as belonging to a population of strains with something in common and, similarly, breeds can be viewed as sampled from a "hyperpopulation" (Lindley and Smith, 1972) of cattle; neither strains nor breeds need to be sampled at random from their respective conceptual populations. Let β and $\mathbf{s}$ be vectors representing breed and strain effects and suppose that populations are of infinite size. Take, a priori, the informative priors

$$\beta \sim N(1\beta, I\sigma_\beta^2) \qquad [46]$$

and

$$\mathbf{s} \sim N(1\mathbf{s}, I\sigma_s^2), \qquad [47]$$

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where σ^2_β and σ^2_s are variances between breeds and strains, respectively, and assume that β and s are independently distributed. An experiment is conducted by randomly sampling sires and dams within each of the strains. Let the conditional distribution of the data given β and s be normal with mean vector

$$E(\mathbf{y}|\beta,\mathbf{s}) = \mathbf{B}\beta + \mathbf{S}\mathbf{s} \qquad [48a]$$

and variance-covariance matrix

$$\text{Var}(\mathbf{y}|\beta,\mathbf{s}) = \mathbf{V}\sigma^2, \qquad [48b]$$

where σ^2 is a scalar assumed to be known. Variation in y given β and s reflects variability between and within families or equivalently, dispersion among individuals within strains. As in [12], because the prior and the likelihood function are both multivariate normal, the joint posterior density of β and s is also multivariate normal with mean as shown in [49], and variance-covariance matrix as in [50], with [49] and [50] following results of Lindley and Smith (1972). Observe that the posterior distribution of β and s can be written as $N(\mathbf{Qq},\mathbf{Q}\sigma^2)$, where Q is the matrix preceding the vector in [49], denoted as q. As pointed out by Dempfle (1977) in a vectorial generalization of [13b], the posterior mean in [49] can be written as a weighted average involving β, s and the sample means. Note in [49] that the posterior means are shrunk toward β and s depending on the ratio of σ^2 to σ^2_β and σ^2_s, respectively. If σ^2_β and σ^2_s are very large relative to σ^2, i.e., when there is considerable prior uncertainty about breed and strain effects, there is little shrinkage and the information from the sample dominates.

$$E\begin{bmatrix}\beta|\mathbf{y}\\ \mathbf{s}|\mathbf{y}\end{bmatrix} = \begin{bmatrix}\mathbf{B}'\mathbf{V}^{-1}\mathbf{B} + \mathbf{I}\frac{\sigma^2}{\sigma^2_\beta} & \mathbf{B}'\mathbf{V}^{-1}\mathbf{S}\\ \mathbf{S}'\mathbf{V}^{-1}\mathbf{B} & \mathbf{S}'\mathbf{V}^{-1}\mathbf{S} + \mathbf{I}\frac{\sigma^2}{\sigma^2_s}\end{bmatrix}^{-1}\begin{bmatrix}\mathbf{B}'\mathbf{V}^{-1}\mathbf{y} + \mathbf{1}\beta\frac{\sigma^2}{\sigma^2_\beta}\\ \mathbf{S}'\mathbf{V}^{-1}\mathbf{y} + \mathbf{1}\mathbf{s}\frac{\sigma^2}{\sigma^2_s}\end{bmatrix} \qquad [49]$$

$$\text{Var}\begin{bmatrix}\beta|\mathbf{y}\\ \mathbf{s}|\mathbf{y}\end{bmatrix} = \begin{bmatrix}\mathbf{B}'\mathbf{V}^{-1}\mathbf{B} + \mathbf{I}\frac{\sigma^2}{\sigma^2_\beta} & \mathbf{B}'\mathbf{V}^{-1}\mathbf{S}\\ \mathbf{S}'\mathbf{V}^{-1}\mathbf{B} & \mathbf{S}'\mathbf{V}^{-1}\mathbf{S} + \mathbf{I}\frac{\sigma^2}{\sigma^2_s}\end{bmatrix}^{-1}\sigma^2, \qquad [50]$$

When σ^2_β and σ^2_s tend to infinity, this corresponds to the situation when [46] and [47] are noninformative priors, and $E(\beta|\mathbf{y})$ and $E(\mathbf{s}|\mathbf{y})$ tend towards a generalized least-squares solution (Searle, 1971); in this case Q is a singular matrix. In the Bayesian world, if the prior is proper (Lindley and Smith, 1972) "estimability" problems do not arise. Now, to further illustrate, suppose that the problem consists only of evaluating breeds (i.e., ignore strains) so that the posterior mean is

$$E(\beta|\mathbf{y}) = (\mathbf{B}'\mathbf{V}^{-1}\mathbf{B} + \mathbf{I}\frac{\sigma^2}{\sigma^2_\beta})^{-1} \times (\mathbf{B}'\mathbf{V}^{-1}\mathbf{y} + \mathbf{1}\beta\frac{\sigma^2}{\sigma^2_\beta}). \qquad [51]$$

Because $\mathbf{y} = \mathbf{B}\beta + \epsilon$, where $\epsilon \sim N(0, \mathbf{V}\sigma^2)$, then from [51] and [46]

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$$\underset{y}{E}[E(\beta|y)] = (\mathbf{B}'\mathbf{V}^{-1}\mathbf{B} + \mathbf{I}\frac{\sigma^2}{\sigma^2_\beta})^{-1} \times (\mathbf{B}'\mathbf{V}^{-1}\mathbf{B}\mathbf{1}\beta + \mathbf{1}\beta\frac{\sigma^2}{\sigma^2_\beta}) = \mathbf{1}\beta, \quad [52]$$

which illustrates that $E(\beta|y)$ is "unbiased" for the prior mean (remember that unbiasedness is irrelevant in the Bayesian framework). Put now $\beta = 0$ and rearrange [51] as

$$(\mathbf{B}'\mathbf{V}^{-1}\mathbf{B} + \mathbf{I}\frac{\sigma^2}{\sigma^2_\beta})\,\beta^* = \mathbf{B}'\mathbf{V}^{-1}\mathbf{y}. \quad [53]$$

This is an extended form of Henderson's (1950, 1963, 1973, 1975) mixed model equations in which the usual "random" effects (u in his notation) have been absorbed into the "fixed" effects (β in this case), and with the generalized least-squares estimator of β ($\hat{\beta}$, say) shrunk towards zero. Solving for β^* and taking the expectation with respect to the marginal distribution of y

$$\underset{y}{E}(\beta^*) = (\mathbf{B}'\mathbf{V}^{-1}\mathbf{B} + \mathbf{I}\frac{\sigma^2}{\sigma^2_\beta})^{-1} \times \mathbf{B}'\mathbf{V}^{-1}\mathbf{B}\,E(\beta) \quad [54a]$$

$$= (\mathbf{B}'\mathbf{V}^{-1}\mathbf{B} + \mathbf{I}\frac{\sigma^2}{\sigma^2_\beta})^{-1} \times (\mathbf{B}'\mathbf{V}^{-1}\mathbf{B} + \mathbf{I}\frac{\sigma^2}{\sigma^2_\beta} - \mathbf{I}\frac{\sigma^2}{\sigma^2_\beta}) \times E(\beta) \quad [54b]$$

$$= E(\beta) - \frac{\sigma^2}{\sigma^2_\beta} \times (\mathbf{B}'\mathbf{V}^{-1}\mathbf{B} + \mathbf{I}\frac{\sigma^2}{\sigma^2_\beta})^{-1} \times E(\beta), \quad [54c]$$

which is biased for $E(\beta)$ unless this expectation is truly a null vector. Because in the "classical" treatment (e.g., Henderson, 1973) $E(\beta) = \beta$, i.e., β is fixed, then [54c] shows that β^* is biased for β. However, β^* might be "better" (in the non-Bayesian sense of mean-square error) than $\hat{\beta}$, the generalized least-squares estimator of β, an argument which has received considerable attention recently (Stein, 1956; Hoerl and Kennard, 1970a,b; Bibby and Toutenburg, 1977; Toutenberg, 1982). Consider in particular, the "ridge" type estimators where a constant k is used heuristically instead of σ^2/σ^2_β. This tends to make the elements of β^* "similar" to one another. However, unless this commonality is justified on the basis of an assumption like the one in [46], it is not sensible to shrink the estimates toward a common value (Lindley and Smith, 1972). For example, if β contains effects of breeds (β_1) and of herd-year-seasons (β_2), making β^*_1 "similar" to β^*_2, or vice-versa, is hardly defensible. In this case, different shrinkage factors for β_1 and β_2, say k_1 and k_2, might be justified. This acrobacy is not needed in the Bayesian formulation, as $\sigma^2_{\beta_1}$, $\sigma^2_{\beta_2}$ and σ^2 would provide the natural mechanism by which commonality or exchangeability (DeFinnetti, 1964) is exploited. Returning to [51], [52], [53] we see, from a Bayesian perspective, that it is not sensible to shrink β toward zero unless $E(\beta) = 0$. Because [51] is the Bayes rule for a quadratic loss function and because [46] is proper, then $E(\beta|y)$ in [51] is admissible, i.e., there is no solution with a smaller risk, so β^* and $\hat{\beta}$ are inadmissible (Wald, 1950; Lindley, 1971; Berger, 1980). However, β^* might represent an "improvement" over β provided that a rationale for the choice of k is offered. These considerations depend, of course, on knowing σ^2 and σ^2_β. However, from the argument of [43], modal-type estimates (provided that the likelihood is peaked) could be used in lieu of the true values.

Bayesian Estimation of Breeding Value

General Framework. The objective of a breeding program is to elicit, by selection, favorable trends in a merit function (T). While T is usually unobservable, e.g., "aggregate genetic value" in classical selection index theory (Smith, 1936; Hazel, 1943) or "producing ability" (Van Vleck, 1979), it may also involve phenotypic values (Smith and Allaire, 1985). It can be linear in the variables of interest, e.g., breeding values, but economic or biological considerations may dictate a non-linear form (Moav and Hill, 1966; Wilton et al., 1968; Dickerson, 1970; Harris, 1970; Henderson, 1977b; Allaire, 1980; Goddard, 1983). If changes in T reflect in some sense monetary utility occurring from variation in genetic or phenotypic inputs, the law of diminishing returns should be observed in the definition of T.

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The problem can be viewed as one of making inferences about a vector $\theta_1 = \mathbf{T}$ from data $\mathbf{y}$. Nuisance parameters θ_2 ("fixed" effects in the classical sense) and θ_3 (genetic and phenotypic variances and covariances) often need to be brought into the picture for a complete formulation of the problem and then "integrated out." In other settings such as in the Northeast sire comparison method (Henderson, 1973), θ_2 is an integral part of the merit function as the transmitting ability of a sire is a function of elements of both θ_1 and θ_2. The problem can be solved in general by writing the posterior density $f(\theta_1,\theta_2,\theta_3|\mathbf{y})$, and if needed, integrating out θ_2 and θ_3 from it. Then, joint (or marginal) probability statements about θ_1 (or its elements) are carried out using [6]. Suppose now that interest centers on $\theta_1' = [T_1, T_2, \ldots, T_m]$, e.g., a vector of breeding values, and that $k < m$ individuals are needed for breeding. Prior to the data, or when there is no justification for collecting further data, e.g., dairy bulls with "repeatabilities" close to 1, one would select the k elements of θ_1 with largest prior means. The same would be done posterior to the data, i.e., the k individuals selected would be those with the largest $E(T_i|\mathbf{y})$. Bulmer (1980), Goffinet (1983) and Fernando and Gianola (1984) proved that the above strategy maximizes expected genetic progress.

Henderson (1973) pointed out that $E(T_i|\mathbf{y})$, which he called "best" predictor: a) minimizes mean squared error of prediction, b) is unbiased for $E(T_i)$ and c) among all functions of $\mathbf{y}$ maximizes the correlation between predictor and predictand. These properties are of statistical interest but only (c) has a direct connection to the problem of improvement by selection. In the multivariate normal distribution (or when the conditional mean of T_i given $\mathbf{y}$ is linear), when candidates have the same amount of information, ranking with $E(T_i|\mathbf{y})$ maximizes expected response to truncation selection via maximization of the correlation between predictor and predictand; this is a well known result in quantitative genetics (e.g., Falconer, 1981). With unequal information, ranking individuals with $E(T_i|\mathbf{y})$ and retaining those with the largest values maximizes the mean of k selected individuals irrespective of the distribution. In order to prove this result, Bulmer (1980), Goffinet (1983) and Fernando and Gianola (1984) invoked hypothetical repetitions of $\mathbf{y}$. The Bayesian solution yields the same answer with the proviso that inferences are made from the data at hand without reference to imaginary or hypothetical data.

Nuisance Parameters. The calculation of $E(\mathbf{T}|\mathbf{y})$ requires integrating out or knowing the nuisance parameters θ_2 and θ_3. Henderson (1973) restricted attention to linear functions of $\mathbf{y}$ and for the situation where θ_3 (but not θ_2) is known, proposed his method of best linear unbiased prediction, which is now used worldwide as a standard procedure of sire evaluation. For the case where θ_3 is not known he wrote:

> "Little success has attended efforts to find optimum selection methods when the variance must be estimated from the same data that are to be used in computing selection criteria."

and argued that maximum likelihood procedures would provide a reasonable answer to the problem. Henderson's intuition is validated by the following Bayesian arguments which do not require normality nor linearity. First, incorporate uncertainty about the values of θ_1, θ_2 and θ_3 via prior distribution. Then, given the data $\mathbf{y}$

$$f(\theta_1,\theta_2,\theta_3|\mathbf{y}) \propto f(\mathbf{y}|\theta_1,\theta_2,\theta_3) \times f(\theta_1,\theta_2,\theta_3). \quad [55]$$

If interest centers on θ_1, a vector of breeding values, say, integrate out θ_2 and θ_3. Using [36b]

$$f(\theta_1|\mathbf{y}) = \int_{R_{\theta_2}}\int_{R_{\theta_3}} f(\theta_1|\theta_2,\theta_3,\mathbf{y}) \times f(\theta_2,\theta_3|\mathbf{y}) \times d\theta_2 d\theta_3. \quad [56]$$

Now, if the joint posterior $f(\theta_2,\theta_3|\mathbf{y})$ is peaked so that most of the density is at the mode $(\hat{\theta}_2,\hat{\theta}_3)$, the argument of [43] yields

$$f(\theta_1|\mathbf{y}) \doteq f(\theta_1|\theta_2 = \hat{\theta}_2, \theta_3 = \hat{\theta}_3, \mathbf{y}). \quad [57]$$

If flat priors for θ_2 and θ_3 are used, then $\hat{\theta}_2$ and $\hat{\theta}_3$ are precisely the maximum likelihood estimators of θ_2 and θ_3, the reason for this being that

$$f(\theta_2,\theta_3|\mathbf{y}) \propto f(\mathbf{y}|\theta_2,\theta_3) \quad [58]$$

so the mode of the joint posterior corresponds to the maximum of the likelihood function. An

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alternative approach to inferences about θ_1 (or θ_2) is the following. Suppose that interest is on θ_1 and write

$$f(\theta_1,\theta_2|\mathbf{y}) \doteq f(\theta_1,\theta_2|\theta_3 = \hat{\hat{\theta}}_3,\mathbf{y}). \quad [59]$$

where now $\hat{\hat{\theta}}_3$ is the mode of the marginal density of θ_3 given y. Calculating $\hat{\hat{\theta}}_3$ involves integrating θ_2 out of [58] and then maximizing $f(\theta_3|\mathbf{y})$. If a flat prior is used for θ_3, then under normality $\hat{\hat{\theta}}_3$ is the restricted maximum likelihood estimator of θ_3 (Harville, 1974, 1977). Further, if we are interested in marginal inferences about θ_1, then θ_2 should be integrated out of [59]. This leads to

$$f(\theta_1|\mathbf{y}) \doteq f(\theta_1|\theta_3 = \hat{\hat{\theta}}_3,\mathbf{y}). \quad [60]$$

We regard [60] as a better approximation to $f(\theta_1|\mathbf{y})$ than [57], because it formally integrates θ_2 out of [55] rather than using a modal estimator of θ_2. This integration is feasible under normality because marginal inferences about θ_1 (or θ_2) can be made directly from [59]. If we are interested in joint inferences about θ_1 and θ_2 (such as in the Northeast sire comparison method) then one would use [59]. Under normality, this is tantamount to obtaining the solution to Henderson's mixed model equations (Henderson, 1973) with θ_3 replaced by the restricted maximum likelihood estimate of θ_3 (provided that flat priors for θ_2 and θ_3 are used. We emphasize that the arguments in [57] and [60] apply to posterior modes rather than to maximum likelihood estimates of the nuisance parameters. We return to the problem of estimating θ_3 later but remind the reader that if θ_3 is a vector of variances, flat priors as in [45] do not represent ignorance adequately (Jeffreys, 1961). There may be estimators that are more sensible from a Bayesian viewpoint than those based upon likelihood ideas, particularly when sample sizes are small. While integrating θ_1 (or θ_2) out of [59] is easy under multivariate normality, this may not be true in general. In the domain of generalized linear models (Nelder and Wedderburn, 1972; McCullaugh and Nelder, 1983) it is tempting to mimic the computations that arise when the distribution is multivariate normal.

Selection and Assortative Mating. Field data from which genetic evaluations are to be calculated originate in herds or flocks in which selection is routinely practiced. The intensity of selection depends on the species in question, on its reproductive rate and on the number of traits that contribute real or imaginary value to returns from livestock production. Assortative mating (positive or negative) also is practiced in animal breeding (Allaire, 1980; Smith and Allaire, 1985). The question of how to estimate breeding value in a population undergoing selection and assortative mating is, therefore, of considerable practical importance. The problem of selection has been discussed by Henderson et al. (1959) and Henderson (1973, 1975, 1977b, 1982), particularly in the context of estimation of breeding value and fixed effects, and by Curnow (1961), Thompson (1973, 1977, 1982) and Robertson (1966, 1977) in the domain of estimation of genetic parameters. All these authors have worked in a multivariate normal setting using arguments based on conditioning on the type of selection that effectively occurred (Henderson, 1975), invoking "translation invariant" selection decisions (Henderson, 1982) or likelihood ideas (Curnow, 1961; Thompson, 1973, 1977, 1982). Henderson (1975) found that selection upon linear functions of the data that do not depend on the fixed effects leads to estimators and predictors that are unbiased by selection. In fact, these estimators and predictors are the same ones that would be obtained if it is assumed that selection has not occurred. The same applies if selection and assortative mating are simultaneously provided that the mating decisions are based on linear translation invariant functions; this consequence is somewhat hidden in the paper of Henderson (1975). Henderson (1982) observed that the "usual" unbiased estimators of variances and covariances do not retain this property when selection occurs. He speculated that minimum variance quadratic unbiased estimators and related iterative methods might retain some of their properties, e.g., consistency, in the selection model. Thompson (1977, 1982) used the term "unbiased estimating equations" to indicate that under normality, the likelihood function and associated equations have the same form when selection occurs than when the animals are selected strictly at random. This implies, at least in the multivariate normal setting, that "essentially parameters in the unselected population are being estimated" (Thompson, 1982). The parameters in the selected population can then be obtained, assuming normality, using well known theory (Robertson, 1977; Bulmer, 1980).

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In the Bayesian setting, it is simple to show that given certain conditions, inferences about breeding values and genetic parameters should be made from the posterior density constructed as if selection or assortative mating had not occurred. The result agrees with the developments based upon normality described above, and holds for any distribution. As before, let θ_1 be a vector of unknown merits and θ_2 and θ_3 be vectors of unknown nuisance parameters, e.g., "fixed effects" and variances and covariances, respectively. Selection and mating decisions are made upon available data $\mathbf{y}_o$, an $n_o \times 1$ vector, which may consist of observations on one or several traits. Depending on $\mathbf{y}_o$, further data $\mathbf{y}_i$ ($i = 1, \ldots, s$) are collected. If $\mathbf{y}_o$ falls in the ith discrete and disjoint partition (R_i) of the n_o-dimensional space, selection and mating decisions are made which lead to the realization of $\mathbf{y}_i$. Then $\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_s$ are mutually exclusive data sets because Prob $\{\mathbf{y}_o \in R_i \Omega \mathbf{y}_o \in R_j\} = 0$, $i \neq j$. In the absence of selection or assortative mating, it is useful to regard $\mathbf{y}_o, \mathbf{y}_1, \ldots, \mathbf{y}_s, \theta_1, \theta_2, \theta_3$ as having a joint distribution with density

$$f(\mathbf{y}_o, \mathbf{y}_1, \ldots, \mathbf{y}_s, \theta_1, \theta_2, \theta_3). \qquad [61]$$

The marginal density of $f(\mathbf{y}_o, \mathbf{y}_i, \theta_1, \theta_2, \theta_3)$ is obtained upon appropriate integration. Inferences should be based on

$$f(\theta_1, \theta_2, \theta_3 | \mathbf{y}_o, \mathbf{y}_i), \qquad [62]$$

and θ_2 and θ_3 can be integrated out as in [56a]. With $\hat{\theta}_3$ being a modal estimator, inferences are made as in [60]. Now, when selection and mating decisions are made, the joint density of $\mathbf{y}_o, \mathbf{y}_i, \theta_1, \theta_2, \theta_3$ is defined if $\mathbf{y}_o \in R_i$. Then we write

$$f(\mathbf{y}_o, \mathbf{y}_i, \theta_1, \theta_2, \theta_3 | \mathbf{y}_o \in R_i) = \frac{f(\mathbf{y}_o, \mathbf{y}_i, \theta_1, \theta_2, \theta_3)}{\text{Prob}\{\mathbf{y}_o \in R_i\}} \qquad [63]$$

because $\mathbf{y}_o \in R_i$ is a discrete event. Using conditional probability arguments

$$f(\mathbf{y}_o, \mathbf{y}_i | \mathbf{y}_o \in R_i) = \frac{f(\mathbf{y}_o, \mathbf{y}_i)}{\text{Prob}\{\mathbf{y}_o \in R_i\}}, \qquad [64]$$

where $f(\mathbf{y}_o, \mathbf{y}_i)$ is the joint density of $\mathbf{y}_o, \mathbf{y}_i$ as if selection had not occurred. Then, combining [63] and [64]

$$f(\theta_1, \theta_2, \theta_3 | \mathbf{y}_o, \mathbf{y}_i, \mathbf{y}_o \in R_i) = \frac{f(\mathbf{y}_o, \mathbf{y}_i, \theta_1, \theta_2, \theta_3)/\text{Prob}\{\mathbf{y}_o \in R_i\}}{f(\mathbf{y}_o, \mathbf{y}_i)/\text{Prob}\{\mathbf{y}_o \in R_i\}} \qquad [65a]$$

$$= f(\theta_1, \theta_2, \theta_3 | \mathbf{y}_o, \mathbf{y}_i), \qquad [65b]$$

which is precisely [62]. This proves that if selection and mating decisions are based upon $\mathbf{y}_o$ and, further, if $\mathbf{y}_o$ is in the data available for analysis, then inferences about breeding values, fixed effects and variances or covariances should be made from the posterior density as if selection had not occurred. This generalizes the results of Henderson (1975, 1982) and Thompson (1982), which depend upon normality. If the nuisance parameters θ_3 are unknown and if interest is on θ_1, then we use the argument of [60]: employing all data ($\mathbf{y}_o$, $\mathbf{y}_i$), estimate θ_3 with a modal estimator (restricted maximum likelihood if normality is assumed and flat priors are used) and make inferences from

$$f(\theta_1 | \mathbf{y}_o, \mathbf{y}_i, \theta_3 = \hat{\theta}_3).$$

Note that in the Bayesian setting, the requirement of "translation invariance" or equivalently $\mathbf{L}'\mathbf{X} = \mathbf{0}$, made by Henderson (1975, 1982) does not arise, the reason being that in the Bayesian approach θ_2 is regarded as a random variable. Apart from this distinction, [65b] is in full agreement with Henderson (1975) when normality is assumed. In Henderson's (1975) treatment of selection, linear selection decisions are required. However, the derivation of [65b] does not impose the restriction that selection and mating decisions based upon $\mathbf{y}_o$ need to be linear, even if normality is assumed. If one argues from a classical viewpoint, however, where θ_2 is regarded as fixed, then these nonlinear selection decisions would need to be translation invariant as pointed out by Goffinet (1983) and Fernando and Gianola (1984). In conclusion, the posterior distribution of breeding values and parameters is the same with or without selection or assortative mating. Hence, given data and prior knowledge, any decision rule based on a posterior distribution including $\mathbf{y}_o$ will be unaffected by nonrandom mating of individuals. The extension of the argument used to derive [65b] to several generations or stages of selection is straightforward.

Bayesian View of Linear Selection Indexes. The general problem is as follows (Smith,

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1936; Hazel, 1943; Henderson, 1963; Cunningham, 1969; Van Vleck, 1979): we wish to predict an unobservable vector $\mathbf{T} = \mathbf{A}\theta_1$ from a vector of records $\mathbf{y}$. The known matrix $\mathbf{A}$ contains relative weights for a number of different traits and θ_1 is a vector of breeding values or producing abilities. Further, $\mathbf{y}$ consists of records on the individuals themselves or on relatives, and the model describing this vector is

$$\mathbf{y} = \theta_2 + \mathbf{Z}\theta_1 + \mathbf{e}, \qquad [66]$$

where θ_2 is a known vector of constants specifying the locarion of the marginal distribution of $\mathbf{y}$, $\mathbf{Z}$ is a matrix relating θ_1 to $\mathbf{y}$, and $\mathbf{e}$ is a vector of residuals. Because of linearity one can write without loss of generality

$$\begin{bmatrix} \theta_1 \\ \mathbf{e} \\ \mathbf{y} \end{bmatrix} \sim N\left(\begin{bmatrix} \upsilon \\ \mathbf{0} \\ \theta_2 \end{bmatrix}, \begin{bmatrix} \mathbf{G} & \mathbf{0} & \mathbf{GZ}' \\ \mathbf{0} & \mathbf{R} & \mathbf{R} \\ \mathbf{ZG} & \mathbf{R} & \mathbf{V} \end{bmatrix} \right), \qquad [67]$$

where $\mathbf{V} = \mathbf{ZGZ}' + \mathbf{R}$, and $\mathbf{G}$ and $\mathbf{R}$ are known; υ is usually taken as null. The posterior density of θ_1 is

$$f(\theta_1|\mathbf{y}) \propto f(\mathbf{y}|\theta_1) \cdot f(\theta_1) \qquad [68]$$

and has a multivariate normal form because θ_1 is normal and the distribution of $\mathbf{y}$ given θ_1 is also normal. Because $\mathbf{T}$ is linear in θ_1, the posterior of $\mathbf{T}$ is also normal with mean vector

$$E(\mathbf{T}|\mathbf{y}) = \mathbf{A}\, E(\theta_1|\mathbf{y}) \qquad [69]$$

and covariance matrix

$$\mathrm{Var}(\mathbf{T}|\mathbf{y}) = \mathbf{A}\, \mathrm{Var}(\theta_1|\mathbf{y})\mathbf{A}'. \qquad [70]$$

Note that [69] states the well-known linear invariance property of selection indexes. From properties of the multivariate normal distribution (e.g., Searle, 1971)

$$E(\mathbf{T}|\mathbf{y}) = \mathbf{A}[\upsilon + \mathbf{GZ}'\mathbf{V}^{-1}(\mathbf{y} - \theta_2)] \qquad [71]$$

$$= \mathbf{A}\upsilon + \mathbf{AGZ}'\mathbf{V}^{-1}(\mathbf{y} - \theta_2). \qquad [72]$$

Put $\mathbf{AGZ}'\mathbf{V}^{-1} = \mathbf{B}'$, transpose and rearrange so as to obtain

$$\mathbf{VB} = \mathbf{ZGA}' \qquad [73a]$$

or

$$\mathbf{VB} = \mathrm{Cov}(\mathbf{y}, \theta_1')\mathbf{A}' \qquad [73b]$$

or

$$\mathbf{VB} = \mathrm{Cov}(\mathbf{y}, \mathbf{T}'). \qquad [73c]$$

Note that [73b] and [73c] are the selection index equations of Smith (1936) and Hazel (1943), and $\mathbf{B} = \mathbf{V}^{-1}\mathbf{ZGA}'$ is the matrix of selection index weights, given that $\mathbf{V}$ and $\mathbf{G}$ are known.

Observe that in order to compute $\mathbf{B}$ the inverse of $\mathbf{V}$ is required. Now, in the multivariate normal distribution

$$\mathrm{Var}(\theta_1|\mathbf{y}) = \mathbf{G} - \mathbf{GZ}'\mathbf{V}^{-1}\mathbf{ZG} \qquad [74]$$

$$= [\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1}]^{-1} \qquad [75]$$

with [75] obtained after some algebra. Thus

$$\mathrm{Var}(\mathbf{T}|\mathbf{y}) = \mathbf{A}[\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1}]\mathbf{A}'. \qquad [76]$$

When the order of $\mathbf{y}$ is much larger than that of θ_1 it is tedious to compute $\mathbf{B}$ with the Smith-Hazel equations of [73c]. The posterior density in [68] is

$$f(\theta_1|\mathbf{y}) \propto \exp\{-\tfrac{1}{2}(\mathbf{y} - \theta_2 - \mathbf{Z}\theta_1)'\mathbf{R}^{-1} \times (\mathbf{y} - \theta_2 - \mathbf{Z}\theta_1)\} \times \exp\{-\tfrac{1}{2}(\theta_1 - \upsilon)'\mathbf{G}^{-1} \times (\theta_1 - \upsilon)\}. \qquad [77]$$

Because this density is symmetric and unimodal, the mode and the mean are the same. Then, the posterior mean can be obtained by taking logs in [77], differentiating with respect to θ_1 and equating to zero. The resulting equations are

$$(\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1})\, E(\theta_1|\mathbf{y}) = \mathbf{Z}'\mathbf{R}^{-1}(\mathbf{y} - \theta_2) + \mathbf{G}^{-1}\upsilon. \qquad [78]$$

Solving [78] for $E(\theta_1|\mathbf{y})$ gives

$$E(\theta_1|\mathbf{y}) = (\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1})^{-1} \times [\mathbf{Z}'\mathbf{R}^{-1}(\mathbf{y} - \theta_2) + \mathbf{G}^{-1}\upsilon]. \qquad [79]$$

Then $E(\mathbf{T}|\mathbf{y})$ is computed using [79] in [69], with υ usually taken as a null vector. These two expressions provide an algorithm for computing the selection index solution, a result that seem-

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ingly fails to be recognized by many animal breeders in spite of papers by Henderson (1950, 1963, 1973, 1974). If the order of $\mathbf{y}$ is much larger than the order of θ_1, then [79] coupled with [69] are computationally easier than [73c]. In many instances [73c] is computationally not feasible; however, [79] will usually be amenable to an iterative solution. Inferences about $\mathbf{T}$, given data, are then completed by application of normal distribution theory to [79] (with [69]) and [75]. Apparently, the first one that treated selection indexes in a Bayesian framework was Solomon (1971).

Bayesian View of Best Linear Unbiased Prediction. The connection between "Bayes estimators" and best linear unbiased prediction (Henderson, 1973) was discussed by Ronningen (1971) and Dempfle (1977). The setting is as with selection indexes, except that θ_2 in [66] is structured as $\theta_2 = \mathbf{X}\beta$, where $\mathbf{X}$ is a known matrix relating β to $\mathbf{y}$, and β is unknown. This vector is assumed to follow a priori a multivariate normal distribution, i.e., $\beta \sim N(\alpha,\Gamma)$ which is independent of the distribution of θ_1. With $\mathbf{G}$ and $\mathbf{R}$ known, the joint posterior density of β and θ_1 is then

$$\begin{aligned} f(\theta_1,\beta) \propto \exp &\left\{-\tfrac{1}{2}(\mathbf{y}-\mathbf{X}\beta-\mathbf{Z}\theta_1)'\mathbf{R}^{-1}\right. \\ &\left.\times(\mathbf{y}-\mathbf{X}\beta-\mathbf{Z}\theta_1)\right\} \\ &\times \exp\left\{-\tfrac{1}{2}(\beta-\alpha)'\Gamma^{-1}\right. \\ &\left.\times(\beta-\alpha)\right\} \\ &\times \exp\left\{-\tfrac{1}{2}(\theta_1-\upsilon)'\mathbf{G}^{-1}\right. \\ &\left.\times(\theta_1-\upsilon)\right\}. \qquad [80] \end{aligned}$$

Because [80] is normal (apart from a proportionality factor), the mode and the mean are the same. Hence, the mean vector of the joint posterior distribution can be found by maximization rather than by integration. This is easily done by differentiating [80] with respect to β and θ_1, setting to zero and rearranging. As shown in Dempfle (1977), the equations satisfy

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X}+\Gamma^{-1} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z}+\mathbf{G}^{-1} \end{bmatrix} \times \begin{bmatrix} \mathrm{E}(\beta|\mathbf{y}) \\ \mathrm{E}(\theta_1|\mathbf{y}) \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y}+\Gamma^{-1}\alpha \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y}+\mathbf{G}^{-1}\upsilon \end{bmatrix}, \qquad [81]$$

which is an extended version of Henderson's (1950, 1973, 1975) mixed model equations. As in [49] and [50], we can write

$$\begin{bmatrix} \beta|\mathbf{y} \\ \theta_1|\mathbf{y} \end{bmatrix} \sim N(\mathbf{Qq},\mathbf{Q}), \qquad [82]$$

where $\mathbf{Q}$ is the inverse of the coefficient matrix in [81] and $\mathbf{q}$ is the vector in the right-hand side.

If $\upsilon = \mathbf{0}$ and if a flat prior is taken for the elements of β (so that $\Gamma^{-1} \to \mathbf{0}$), [81] yields the mixed model equations

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z}+\mathbf{G}^{-1} \end{bmatrix} \begin{bmatrix} \tilde{\beta} \\ \hat{\theta}_1 \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \end{bmatrix}. \qquad [83]$$

The well-known inadmissibility of least squares (Stein, 1956; Lindley, 1971; Lindley and Smith, 1972) is illustrated in [83]. In least-squares, $\mathbf{G}^{-1}$ is a null matrix and this implies ignoring prior knowledge about θ_1. Hence, inferences about θ_1 made via least-squares are less sharp than those obtained with [83]. However, [83] tends asymptotically to a set of least-squares equations as one would expect from the arguments leading to [26a] and [26b].

As a point of historical interest, [83] was obtained by maximization of [80] (but taking a flat prior for β) with respect to β and θ_1 (Henderson et al., 1959), and the resulting estimators were called "maximum likelihood." While [80] is not a likelihood function but a posterior density, it can be shown in the setting of [83] that

$$\mathrm{E}(\beta|\mathbf{y}) = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-}\mathbf{X}'\mathbf{V}^{-1}\mathbf{y} = \tilde{\beta}, \qquad [84a]$$

where $(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-}$ is the inverse of $\mathbf{X}'\mathbf{V}^{-1}\mathbf{X}$, and

$$\mathrm{E}(\theta_1|\mathbf{y}) = \mathbf{G}\mathbf{Z}'\mathbf{V}^{-1}[\mathbf{y}-\mathbf{X}\tilde{\beta}] = \hat{\theta}_1. \qquad [84b]$$

Note that $\tilde{\beta}$ and $\hat{\theta}_1$ are the maximum likelihood estimators of β and of $\mathrm{E}(\theta_1|\beta,\mathbf{y})$, the latter being the mean of a conditional distribution in which β is fixed. However, $\hat{\theta}_1$ is not the maximum likelihood estimator of θ_1; it is the θ_1-component of the mean vector of the joint

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posterior density of β and θ_1 given $\mathbf{y}$. Because of multivariate normality $\hat{\theta}_1$ as indicated in [84b], is also the mean vector of the posterior distribution of θ_1 given $\mathbf{y}$. Here, β has been integrated out, this being the reason for the invariance (with respect to β) property of $\hat{\theta}_1$ (Henderson, 1982).

There are potentially interesting applications of [81]. Consider, for example the prediction of breeding values of individuals without data (Henderson, 1977a) from observations on relatives with data. Partition the vector of breeding values as $\theta_1' = [\theta_{11}', \theta_{12}']$ where θ_{11} pertains to individuals that contribute information ($\mathbf{y}$) to the likelihood function. Observe that

$$f(\theta_{11},\theta_{12}|\mathbf{y}) \propto f(\mathbf{y}|\theta_{11},\theta_{12}) \times f(\theta_{11},\theta_{12}). \quad [85]$$

Application of [81] yields

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X}+\Gamma^{-1} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} & \mathbf{0} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z}+\mathbf{G}^{11} & \mathbf{G}^{12} \\ \mathbf{0} & \mathbf{G}^{21} & \mathbf{G}^{22} \end{bmatrix} \times \begin{bmatrix} \mathrm{E}(\beta|\mathbf{y}) \\ \mathrm{E}(\theta_{11}|\mathbf{y}) \\ \mathrm{E}(\theta_{12}|\mathbf{y}) \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} \Gamma^{-1}\alpha \\ \mathbf{G}^{11}\upsilon_{11}+\mathbf{G}^{12}\upsilon_{12} \\ \mathbf{G}^{21}\upsilon_{11}+\mathbf{G}^{22}\upsilon_{12} \end{bmatrix}, \quad [86]$$

where

$$\begin{bmatrix} \mathbf{G}^{11} & \mathbf{G}^{12} \\ \mathbf{G}^{21} & \mathbf{G}^{22} \end{bmatrix} = \begin{bmatrix} \mathbf{G}_{11} & \mathbf{G}_{12} \\ \mathbf{G}_{21} & \mathbf{G}_{22} \end{bmatrix}^{-1} \quad [87]$$

$\mathbf{G}_{11} = \mathrm{Var}(\theta_{11})$, $\mathbf{G}_{12} = \mathrm{Cov}(\theta_{11},\theta_{12})$, $\mathbf{G}_{22} = \mathrm{Cov}(\theta_{22})$, and υ_{11} and υ_{12} are the prior means of θ_{11} and θ_{12}. Under the usual assumptions ($\Gamma^{-1} \to \mathbf{0}$, $\upsilon_{11} = \mathbf{0}$, $\upsilon_{12} = 0$), [86] yields equations [7] in Henderson (1977a). Upon elimination of $\mathrm{E}(\theta_{12}|\mathbf{y})$ from [86], it is easy to establish that $\mathrm{E}(\theta_{11}|\mathbf{y})$ can also be computed from

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X}+\Gamma^{-1} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z}+\mathbf{G}_{11}^{-1} \end{bmatrix} \times \begin{bmatrix} \mathrm{E}(\beta|\mathbf{y}) \\ \mathrm{E}(\theta_{11}|\mathbf{y}) \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y}+\Gamma^{-1}\alpha \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y}+\mathbf{G}_{11}^{-1}\upsilon_{11} \end{bmatrix}. \quad [88]$$

Equations [81] can also be applied to the problem of updating breeding values given data that accrue sequentially over time. This has been dealt with by Martinez and Rothschild (1983) using complicated theory and algebra developed by Kalman (1960) and Sallas and Harville (1981). The Bayesian solution is extremely simple, at least conceptually. Suppose data $\mathbf{y}_i$ accrue at the ith stage to predict θ_i, a vector of breeding values to be evaluated at this stage. From data on previous stages, one can construct a prior distribution for θ_i which is an application of the "memory" property of Bayes theorem stated in [29a], [29c] and [30]. The final results would be the same as when all the data are analyzed jointly. This works when the prior is "conjugate," i.e., when the prior and the posterior are in the same family of distributions (Cox and Hinkley, 1974). In linear methodology the prior and the posterior are both multivariate normal so the requirement is satisfied. Assuming normality, we take a priori

$$\theta_i \sim \mathrm{N}(\upsilon_{i-1}, \mathbf{G}_{i-1}),$$

where υ_{i-1} and $\mathbf{G}_{i-1}$ are the posterior mean vector and covariance matrix of θ at stage $i-1$. Then we use these in lieu of υ and $\mathbf{G}$ in [81] with $\mathbf{y}$ replaced by $\mathbf{y}_i$. Candidates no longer relevant after stage i can be integrated out of the posterior distribution. Also, if there are some potential candidates (e.g., unborn animals) that do not contribute to the likelihood of the ith stage, these can be evaluated via [86]. A similar rationale can be employed for β, the vector of "fixed" effects. In some cases, e.g., herd-year-seasons, it is debatable whether or not knowledge from previous stages constitutes relevant prior information. However, β may include components such as sex or parity effects where data from previous stages might be useful (note here that the Bayesian approach provides a formal mechanism to use "correction factors" incorporating a measure of uncertainty about them). Because, a posteriori, β and θ become correlated, a joint prior density would need to be specified at stages other than 0.

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However, in view of the obvious computational difficulties that this would entail, it is likely that only marginal densities could be considered. Furthermore, in most data sets only approximations to the marginal prior densities could be accommodated. For example, in

$$\theta_i \sim N(v_{i-1}, G_{i-1}),$$

the matrix G_{i-1} is the sector corresponding to θ of the inverse of the coefficient matrix in [81] at stage $i-1$. Because usually [81] can be solved by iteration only, one may wish to compromise by taking G_{i-1} as a diagonal matrix containing either approximations or the exact values of the diagonal elements of the inverse needed to solve [81].

Another application of [81] is in incorporating information on artificial insemination (A.I.) sire proofs to within-herd prediction problems. Partition θ_1 in [81] as

$$\theta_1' = [\theta_H', \theta_A'],$$

where θ_H and θ_A are the transmitting abilities of the home-bred and A.I. sires, respectively. A priori, there may be reason to believe that θ_H and θ_A follow different normal distributions with distinct location and spread parameters; in particular, the distribution of θ_A might be much sharper than that of θ_H because the A.I. sires are evaluated in a large number of herds. Take

$$\begin{bmatrix} \theta_H \\ \theta_A \end{bmatrix} \sim N\left(\begin{bmatrix} v_H \\ v_A \end{bmatrix}, \begin{bmatrix} G_{HH} & G_{HA} \\ G_{AH} & G_{AA} \end{bmatrix} \right), \quad [89]$$

where G_{HA} is a function of the additive relationships between the H and A bulls, v_A is a vector of A.I. sire proofs and G_{AA} is the posterior dispersion (or prediction error covariance matrix, Henderson, 1975) of such proofs. The data in the herd are described by

$$\begin{bmatrix} y_H \\ y_A \end{bmatrix} = \begin{bmatrix} X_H & 0 \\ 0 & X_A \end{bmatrix} \begin{bmatrix} \beta_H \\ \beta_A \end{bmatrix} + \begin{bmatrix} Z_H & 0 \\ 0 & Z_A \end{bmatrix} \begin{bmatrix} u_H \\ u_A \end{bmatrix} + \begin{bmatrix} e_H \\ e_A \end{bmatrix}, \quad [90]$$

where y_H (y_A) are the progeny records of the H(A) sires. Given β_H, β_A, u_H, u_A

$$\text{Var} \begin{bmatrix} y_H | \beta_H, \beta_A, u_H, u_A \\ y_A | \beta_H, \beta_A, u_H, u_A \end{bmatrix} = \begin{bmatrix} R_{HH} & R_{HA} \\ R_{AH} & R_{HH} \end{bmatrix} \quad [91]$$

and note that the specification of [90] and [91] can potentially account for preferential treatment of the progeny of A.I. sires. Using [81], [92] resembles a set of multiple trait equations (Henderson and Quaas, 1976) and where G^{ij} and R^{ij} ($i = H,A$; $j = H,A$) are partitions of the inverses of the covariance matrices in [89] and [91], respectively. Usually, the Γ^{ij} matrices will be taken as null, the R^{ij} segments will be patterned and $v_H = 0$. The difference between [92] and a multiple trait evaluation is that v_A and G_{AA} here are a vector of sire proofs and its prediction error covariance matrix, respectively. In practice, as pointed out previously, G_{AA} could be replaced by a diagonal matrix with elements that are approximations to the exact values. Note in [92] that if G^{AA} is large (which corresponds to very small diagonal elements of the prior covariance matrix), then

$$\begin{bmatrix} X_H'R^{HH}X_H + \Gamma^{HH} & X_H'R^{HA}X_A + \Gamma^{HA} & X_H'R^{HH}Z_H & X_H'R^{HA}Z_A \\ & X_A'R^{AA}X_A + \Gamma^{AA} & X_A'R^{AH}Z_H & X_A'R^{AA}Z_A \\ & & Z_H'R^{HH}Z_H + G^{HH} & Z_H'R^{HA}Z_A + G^{HA} \\ \text{SYMMETRIC} & & & Z_A'R^{AA}Z_A + G^{AA} \end{bmatrix} \begin{bmatrix} E(\beta_H | y_H, y_A) \\ E(\beta_A | y_H, y_A) \\ E(u_H | y_H, y_A) \\ E(u_A | y_H, y_A) \end{bmatrix}$$

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$$= \begin{bmatrix} X_H'R^{HH}y_H + X_H'R^{HA}y_A + \Gamma^{HH}\alpha_H + \Gamma^{HA}\alpha_A \\ X_A'R^{AH}y_H + X_A'R^{AA}y_A + \Gamma^{AH}\alpha_H + \Gamma^{AA}\alpha_A \\ Z_H'R^{HH}y_H + Z_H'R^{HA}y_A + G^{HH}u_H + G^{HA}u_A \\ Z_A'R^{AH}y_H + Z_A'R^{AA}y_A + G^{AH}u_H + G^{AA}u_A \end{bmatrix}, \quad [92]$$

$$E(u_A|y_H,y_A) \doteq u_A.$$

Also, u_A will enter non-trivially in

$$E(u_H|y_H,y_A)$$

so our prior knowledge on A.I. proofs "helps" in sharpening inferences about the transmitting abilities of the home-bred sires.

Nonlinear Merit Functions. Consider first a single trait quadratic merit function, e.g.,

$$T_i = a_o + a_1\theta_i + a_{11}\theta_i^2,$$

where a_o, a_1, a_{11} are known economic weights, and θ_i is the breeding value for a particular trait of the ith individual ($i = 1, \ldots, m$). We have information on one or several traits represented by the vector $\mathbf{y}$, which has a joint distribution with θ_1 (the vector of breeding values), and with the nuisance vectors θ_2 (fixed effects) and θ_3 (variance and covariance components). As before, we select the $k < m$ individuals with the largest conditional means, $E(T_i|y)$, which can be calculated from the posterior density

$$f(T_i,\theta_2,\theta_3|y),$$

after integrating out θ_2 and θ_3. We have

$$\begin{aligned} E(T_i|y) &= a_o + a_1 E(\theta_i|y) \\ &\quad + a_{11}E(\theta_i^2|y) \\ &= a_o + a_1 E(\theta_i|y) \\ &\quad + a_{11}Var(\theta_i|y) \\ &\quad + a_{11}E^2(\theta_i|y). \end{aligned} \quad [93]$$

We evaluate θ_3 at $\hat{\hat{\theta}}_3$, a modal estimator, and then calculate $E(\theta_i|y)$ and $Var(\theta_i|y)$ from [60]. Under normality and using a flat prior for θ_3, $E(\theta_i|y)$ and $Var(\theta_i|y)$ can be computed from [81] and [82], with R and G evaluated at $\hat{\hat{\theta}}_3$, the restricted maximum likelihood estimator of θ_3. If we take a flat prior for β and put $u = 0$, then we use the mixed model equations of [83] with $R(\hat{\hat{\theta}}_3)$ and $G(\hat{\hat{\theta}}_3)$. Here,

$$E(\theta_i|y) = \tilde{\theta}_i,$$

the best linear unbiased predictor of θ_i. Smith and Allaire (1985), observed that Wilton et al. (1968) ignore a_{11} $Var(\theta_i|y)$. Harris et al. (1983) account for this but give formulae which are seldom feasible to compute, as the inverse of the variance-covariance matrix of y is required. Further, θ_3 is taken as known in both papers.

We emphasize that θ_1 in [83] is a component of the posterior mean (or mode under normality) of $f(\theta_1,\beta|y)$, and not the "maximum likelihood" estimator of θ_1. The misunderstanding of maximum likelihood arising in Wilton et al. (1968) is of historical interest. Based on a "maximum likelihood" derivation of [83] obtained by Henderson et al. (1959), the above authors argued that because of the invariance property of maximum likelihood estimators (MLE), the MLE of

$$T_i = a_o + a_1\theta_i + a_{11}\theta_i^2 \quad [94]$$

would be

$$\hat{T}_i = a_1 + a_1\tilde{\theta}_i + a_{11}\tilde{\theta}_i^2, \quad [95]$$

where $\tilde{\theta}_i$ is an appropriate element of [83]. Note, from [84b], that $\hat{T}_i$ in [95] is actually the MLE of

$$\begin{aligned} T_i^* &= a_o + a_1 E(\theta_i|y,\beta) \\ &\quad + E^2(\theta_i|y,\beta), \end{aligned} \quad [96]$$

where $E(\theta_i|y,\beta)$ is the mean of a conditional distribution in which β is fixed. The discrepancy between [93] and [95] is even more serious for higher-order merit functions, a point brought up by Van Vleck (1979) arguing in a different context. Although a powerful source of "bias,"

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the contribution of $\text{Var}(\theta_i|y)$ to [93] is expected to decrease as the information on θ_i increases.

Next in order of complexity would be a kth order polynomial on the breeding values:

$$T_i = \sum_{j=0}^{k} a_j \theta_i^j. \qquad [97]$$

Candidates would be ranked using

$$E(T_i|y) = \sum_{j=0}^{k} a_j E(\theta_i^j|y). \qquad [98]$$

For terms with $j \geq 2$, the relationship (Smith and Allaire, 1985)

$$\begin{aligned} E(\theta_i^j|y) = {} & (j-1)\,\text{Var}(\theta_i|y) \\ & \times E(\theta_i^{j-2}|y) \\ & + E(\theta_i|y) \\ & \times E(\theta_i^{j-1}|y), \end{aligned} \qquad [99]$$

holds under normality. For example, if the merit function is cubic ($k = 3$), the ranking rule would be

$$\begin{aligned} E(T_i|y) = {} & a_0 + a_1 E(\theta_i|y) \\ & + a_2\,\text{Var}(\theta_i|y) \\ & + a_2\,E^2(\theta_i|y) \\ & + a_3\,\{2\,\text{Var}(\theta_i|y) \\ & \times E(\theta_i|y) + E(\theta_i|y) \\ & \times [\text{Var}(\theta_i|y) \\ & + E^2(\theta_i|y)]\}. \end{aligned} \qquad [100]$$

Under the assumptions leading to [83] and putting

$$E(\theta_i|y) = \hat{\theta}_i,$$

[100] can be written after rearrangement as

$$\begin{aligned} E(T_i|y) = {} & a_0 + a_1\hat{\theta}_i + a_2\hat{\theta}_i^2 \\ & + a_3\hat{\theta}_i^3 + \text{Var}(\theta_i|y) \\ & \times (a_2 + 3a_3\hat{\theta}_i). \end{aligned} \qquad [101]$$

The "invariance property" discussed before would neglect the last term in [101] This can cause a serious "bias", particularly when the posterior dispersion (or prediction error variance) is not negligible. The approximation

$$\sum_{j=0}^{k} a_j \hat{\theta}_i^j$$

can be justified only when the amount of information on the ith candidate is large; in this case $\text{Var}(\theta_i|y)$ would be close to 0.

Suppose now the merit function is bilinear with the form

$$\begin{aligned} T_i = {} & a_0 + a_A\theta_i^A + a_{AA}(\theta_i^A)^2 + a_B\theta_i^B \\ & + a_{BB}(\theta_i^B)^2 + a_{AB}\theta_i^A\theta_i^B \end{aligned} \qquad [102]$$

where $a_0, \ldots, a_{AB}$ are economic weights and $\theta_i^A(\theta_i^B)$ is the breeding value of the ith animal for trait A(B). The ranking rule is

$$\begin{aligned} E(T_i|y) = {} & a_0 + a_A\,E(\theta_i^A|y) \\ & + a_{AA}\,\text{Var}(\theta_i^A|y) \\ & + a_{AA}E^2(\theta_i^A|y) \\ & + a_B E(\theta_i^B|y) \\ & + a_{BB}\,\text{Var}(\theta_i^B|y) \\ & + a_{BB}\,E^2(\theta_i^B|y) \\ & + a_{AB}\,E(\theta_i^A|y)\,E(\theta_i^B|y) \\ & + a_{AB}\,\text{Cov}(\theta_i^A,\theta_i^B|y). \end{aligned} \qquad [103]$$

Under multivariate normality, with $v_A = 0$, $v_B = 0$ and taking flat priors for θ_2 and θ_3, one can compute

$$\hat{\theta}_i^A = E(\theta_i^A|y)$$

and

$$\hat{\theta}_i^B = E(\theta_i^B|y)$$

solving the multiple trait mixed model equations, with **R** and **G** replaced by their multiple trait restricted maximum likelihood estimates. The terms

$$\text{Var}(\theta_i^A|y),\ \text{Var}(\theta_i^B|y)$$

and

$$\text{Cov}(\theta_i^A,\theta_i^B|y)$$

can be obtained (if computationally feasible) from an inverse of the coefficient matrix arising in these equations. Generalization of [102] to a multilinear function is straightforward, at least under normality (see, e.g., Henderson, 1977b).

More complex merit functions can be expanded about prior means using Taylor series

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(Moav and Hill, 1966). Let the merit function be

$$T_i = g(\theta_i^A, \theta_i^B, \ldots, \theta_i^N),$$

where θ_i^k is the breeding value of the ith individual for the kth trait. If the function is differentiable and if the linear term of an expansion about the prior means is retained

$$T_i \doteq g(v_1, v_2, \ldots, v_N) + \sum_{k=1}^{N} [g'_k(\theta_i^A, \theta_i^B, \ldots, \theta_i^N)]_{\theta=v} \times (\theta_i^k - v_k), \quad [104]$$

where $[g'_k(\theta_i^A, \theta_i^B, \ldots, \theta_i^N)]_{\theta=v}$ is the first derivative of T_i with respect to θ_i^k evaluated at $\theta = v$, the vector of prior means. Then, candidates would be ranked with

$$E(T_i|\mathbf{y}) \doteq g(v) + \sum_{k=1}^{N} [g'_k(\theta_i^A, \theta_i^B, \ldots, \theta_i^N)]_{\theta=v} \times [E(\theta_i^k|\mathbf{y}) - v_k)]. \quad [105]$$

Under multivariate normality and with flat priors for θ_2 and θ_3, $E(\theta_i^k|\mathbf{y})$ is the predicted breeding value for the kth trait of the ith individual obtained from a mixed model N-trait evaluation with variances and covariances replaced by their restricted maximum likelihood estimates. If one expands T_i about the posterior rather than the prior means, it is clear from [104] that

$$\hat{T}_i \doteq g[E(\theta_i^A|\mathbf{y}), \ldots, E(\theta_i^N|\mathbf{y})],$$

which is similar to [95]; when information is limited, this may be a poor approximation to $E(T_i|\mathbf{y})$. In this case it might be useful to retain higher-order terms in the expansion. We recommend using these expansions cautiously, and to exercise judgment on a case by case basis. Smith and Allaire (1985) have suggested approximations using Gaussian quadrature or smoothing via cubic splines.

Nonlinear Models. Sometimes the relationship between parameters $(\theta_1, \theta_2, \theta_3)$ and data $(\mathbf{y})$ is not linear. This occurs, for example, in analysis of growth functions and lactation curves (e.g., Gianola and Kachman, 1983; Kachman and Gianola, 1983), categorical responses (Thompson, 1979; Gianola and Foulley, 1982, 1983; DeLorenzo, 1983; Foulley et al., 1983; Gilmour, 1983; Foulley and Gianola, 1984; Harville and Mee, 1984), and in studies of endurance or of age of onset (Famula, 1981; Smith, 1983; Wolynetz and Binns, 1983) where additional complications due to censoring arise. We illustrate the application of the Bayesian framework to the nonlinear problem of estimating breeding value for a binary trait.

A commonly used model in the analysis of categorical responses (Dempster and Lerner, 1950; Curnow and Smith, 1975) postulates a normally distributed variable in a hypothetical scale. Gianola and Foulley (1983) model this conceptual variate as

$$y_i = \mathbf{x}_i'\beta + \mathbf{z}_i'\theta_1 + e_i; \quad i = 1, \ldots, K, \quad [106]$$

where y_i is an unobservable realized value in the hypothetical scale, β and θ_1 are as before, $\mathbf{x}_i'$ and $\mathbf{z}_i'$ are known instance vectors, and $e_i \sim N(0, 1)$ and K is the number of individuals, which is fixed by sampling. If $y_i > t$, where t is a fixed threshold, then the trait is observed, e.g., the animal dies; if $y_i < t$, the animal survives. The probability of survival is

$$\text{Prob}\{y_i < t|\beta,\theta_1) = \Phi(t - \mathbf{x}_i'\beta - \mathbf{z}_i'\theta_1) = 1 - \Phi(\mathbf{x}_i'\beta + \mathbf{z}_i'\theta_1 - t), \quad [107]$$

where $\Phi(.)$ is the standard normal distribution function. In the binary case, there is no loss of generality in setting $t = 0$, and the likelihood function, assuming conditional independence of individuals, is

$$L(\beta,\theta_1) \propto \prod_{i=1}^{K} [1 - \Phi(\mathbf{x}_i'\beta + \mathbf{z}_i'\theta_1)]^{\lambda_i} \times [\Phi(\mathbf{x}_i'\beta + \mathbf{z}_i'\theta_1)]^{1-\lambda_i}, \quad [108]$$

where λ_i is an indicator variable. If the animal survives, $\lambda_i = 1$; otherwise, $\lambda_i = 0$. A priori, take

$$\begin{bmatrix} \beta \\ \theta_1 \end{bmatrix} \sim N\left(\begin{bmatrix} \alpha \\ v \end{bmatrix}, \begin{bmatrix} \Gamma & \mathbf{0} \\ \mathbf{0} & \mathbf{A}\sigma^2_{\theta_1} \end{bmatrix} \right), \quad [109]$$

where $\mathbf{A}$ is a matrix of additive relationships, and

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$$\sigma^2_{\theta_1} = h^2/(1 - h^2);$$

the reason for this is that the residual variance is 1 so

$$h^2 = \sigma^2_{\theta_1}/(1 + \sigma^2_{\theta_1}).$$

With known heritability, taking $v = 0$ and putting a flat prior on β, the joint posterior density is then

$$\begin{aligned} f(\theta_1,\beta|\mathbf{Y},h^2) \propto \prod_{i=1}^{K} & [1 - \Phi(\mathbf{x}_i'\beta \\ & + \mathbf{z}_i'\theta_1)]^{\lambda_i} \\ & \times [\Phi(\mathbf{x}_i'\beta + \mathbf{z}_i'\theta_1)]^{1-\lambda_i} \\ & \times \exp\{-\tfrac{1}{2}\theta_1'\mathbf{A}^{-1}\theta_1 \\ & \times (1 - h^2)/h^2\}, \end{aligned} \qquad [110]$$

where $\mathbf{Y}$ indicates the data, in this case categorical responses.

We wish to make inferences about θ_1 "free" of β. Because [110] is not normal the integration of β out of [110] is technically complex. A possible solution (Gianola and Foulley, 1982, 1983), is to base inferences on the joint posterior density of θ_1 and β. Lindley and Smith (1972) suggested the mode of the joint posterior density of θ_1 and β as an approximation to

$$E(\theta_1,\beta|\mathbf{Y},h^2).$$

The posterior mode can be found (Gianola and Foulley, 1983) by iteration with

$$\begin{bmatrix} \mathbf{X}'\mathbf{W}^{[j]}\mathbf{X} & \mathbf{X}'\mathbf{W}^{[j]}\mathbf{Z} \\ \mathbf{Z}'\mathbf{W}^{[j]}\mathbf{X} & \mathbf{Z}'\mathbf{W}^{[j]}\mathbf{Z} + \mathbf{A}^{-1}\left(\frac{1-h^2}{h^2}\right) \end{bmatrix} \times$$

$$\begin{bmatrix} \tilde{\beta}^{[j+1]} \\ \tilde{\theta}_1^{[j+1]} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{W}^{[j]}\mathbf{y}^{[j]} \\ \mathbf{Z}'\mathbf{W}^{[j]}\mathbf{y}^{[j]} \end{bmatrix}, \qquad [111]$$

where $[j]$ is the jth iterate, $\mathbf{y}^{[j]}$ and $\mathbf{W}^{[j]}$ are shown in [112a] and [112b], and the ith element of $\mathbf{v}^{[j]}$ is

$$\mathbf{y}^{[j]} = \mathbf{X}\beta^{[j]} + \mathbf{Z}\theta_1^{[j]} + (\mathbf{W}^{[j]})^{-1}\mathbf{v}^{[j-1]} \qquad [112a]$$

$$\mathbf{W}^{[j]} = \text{Diag}\left\{\frac{\phi^2(\mathbf{x}_i'\beta^{[j]} + \mathbf{z}_i'\theta_i^{[j]})}{[1 - \Phi(\mathbf{x}_i'\beta^{[j]} + \mathbf{z}_i'\theta_i^{[j]}]\,\Phi(\mathbf{x}_i'\beta^{[j]} + \mathbf{z}_i'\theta_1^{[j]})}\right\}; i = 1, \ldots, K \qquad [112b]$$

$$v_i^{[j]} = -\frac{\phi(\mathbf{x}_i'\beta^{[j]} + \mathbf{z}_i'\theta_1^{[j]})}{1 - \Phi(\mathbf{x}_i'\beta^{[j]} + \mathbf{z}_i'\theta_i^{[j]})} \qquad [112c]$$

when the animal survives, or

$$v_i^{[j]} = \frac{\phi(\mathbf{x}_i'\beta^{[j]} + \mathbf{z}_i'\theta_1^{[j]})}{\Phi(\mathbf{x}_i'\beta^{[j]} + \mathbf{z}_i'\theta_1^{[j]})} \qquad [112d]$$

when it dies. With $\hat{\beta}$ and $\hat{\theta}_1$ being the modal values, probabilities of survival can then be estimated as

$$\Phi(\mathbf{x}_i'\hat{\beta} + \mathbf{z}_i'\hat{\theta}_1), \; i = 1, \ldots, K.$$

Harville and Mee (1984) note that this may not be a "good" estimator of such probabilities. Posterior modes are not known to have general invariance properties, except in the case of linear transformations under normality. A seemingly superior approach would be to derive the joint posterior of $\Phi(\mathbf{x}_i'\beta + \mathbf{z}_i'\theta_1)$ and then maximize it. Foulley et al. (1983), Gilmour (1983) and Harville and Mee (1984) discuss the situation where h^2 is not known.

Bayesian Estimation of Genetic Parameters

The nuisance vector, θ_3, the genetic and residual variances and covariances, needs to be known or integrated out in order to calculate the optimal ranking rule or posterior mean. When θ_3 is not known, we suggested basing inferences on [60], with θ_3 replaced by $\hat{\theta}_3$, a modal-type estimator obtained by maximization of $f(\theta_3|\mathbf{y})$. A less desirable course of

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action would be to base inferences on [57] where $\hat{\theta}_2$ and $\hat{\theta}_3$ are the modal values of

$$f(\theta_2,\theta_3|y).$$

We review here Bayesian approaches to estimation of genetic parameters, with emphasis on components of variance as these play a central role in animal breeding theory.

Box and Tiao (1973) wrote:

> "The sampling theory approach to the problem of variance components encounters a number of snags. These have bothered statisticians for many years, as is evidenced by the great variety of attempts which have been made to resolve the problems."

The above statement applies particularly well to the estimation of variance components with mixed linear models and unbalanced data, where a plethora of methods has been developed in the last 30 years (Searle, 1971, 1979; Henderson, 1984a). Early emphasis in animal breeding was on quadratic unbiased estimators (Henderson, 1953). A large number of quadratic unbiased estimators can be derived, but it is difficult to choose among them. Sampling variance of the estimators is usually not a useful criterion for comparison because it depends on true parameter values and is tedious to compute. The work of Rao (1971) and La Motte (1970, 1971) generated interest on minimum variance quadratic unbiased estimation (MIVQUE) of variance components. Searle (1979) and Henderson (1984a,b) give an account of this approach, and describe computing algorithms. MIVQUE theory and associated methods present problems: negative estimates of variance are not precluded and confidence intervals for variance components can include negative values. Further, if θ_3 is a nuisance vector in a problem where interest centers on inferences about breeding values, the MIVQUE properties of an estimator of θ_3 may be unimportant. The problem of negative estimates of variance (or non-positive definite covariance matrices in the multivariate case) is particularly pervasive. The model is often blamed for producing negative estimates (Smith and Murray, 1984), but this is not convincing because negative values do occur even when the model is correct (Lehman, 1983). More recently (Patterson and Thompson, 1971; Harville, 1977; Searle, 1979; Thompson, 1982; Meyer, 1983; Henderson, 1984a,b; Meyer and Thompson, 1984), emphasis has been on maximum likelihood (ML) and on restricted maximum likelihood (REML). Although these estimators are best asymptotic normal (Harville, 1977), they do not have "small sample" justification (Lehman, 1983). From an animal breeding viewpoint, the methods based on likelihood have been found appealing because they ". . . have considerable power to control bias due to selection" (Henderson, 1984b). Harville (1977) conjectured that ML or REML estimators derived under normality might be "reasonable" under other distributions.

Estimation of variance components using Bayesian methods has been dealt with by Tiao and Tan (1965, 1966), Hill (1965, 1967), Klotz et al. (1969), Lindley and Smith (1972), and reviewed by Harville (1977). Inferences about variances components would be based on

$$f(\theta_3|y) \propto f(y|\theta_3)\cdot f(\theta_3), \qquad [113]$$

where $f(\theta_3)$ is the prior density. When θ_3 contains variances, [113] is defined in the range $(0, \infty)$ for each of the elements of θ_3, so the problem of negative estimates never arises (Box and Tiao, 1973). Exact posterior distributions of variance components have been derived for some simple models and balanced data (Box and Tiao, 1973). Because the data are fixed in the posterior distribution, inferences are exact for any sample size. When sample sizes are small, the marginal posterior distribution of components other than the residual can be extremely skewed, even J-shaped. In this instance, standard errors of variance components estimates (and of related estimators of genetic parameters) are virtually uninformative, probably worthless. The models that usually arise in animal breeding are not likely to be amenable to the type of mathematical analysis described above. In many instances, only summary features of the posterior distribution, e.g., means, modes or medians, can be obtained and the choice may depend on computational feasibility.

Klotz et al. (1969) found that posterior means may yield "inefficient" (in a mean squared error sense) estimators of variance components and suggested that the mode or the median may give "better" estimators. Box and Tiao (1973) strongly argued, however, that mean squared error is usually of little relevance to inferences about variance components (or any other parameters), unless the act of

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choosing a particular value for a variance results in an economic penalty proportional to the squared distance between this estimate and the "true" value! Nevertheless, modal estimators of variance components (or of any other genetic parameters) can be justified from the arguments leading to [57] and [60], provided that the main concern is making inferences about breeding values.

Consider first the situation in which flat priors are used for θ_2 and θ_3. Under normality it is easy to integrate out θ_1 out of the joint posterior density and we arrive at [58] from which it is clear that the mode of the joint posterior density of θ_2 and θ_3 corresponds to the maximum likelihood estimators of the fixed effects and variance components, respectively. These can be computed using, e.g., the algorithm of Henderson (1973). The integration of θ_2 out of $f(\theta_2,\theta_3|\mathbf{y})$ leads to a posterior density whose mode is REML (Harville, 1974, 1977). In our view, this precisely points out why REML should be preferred to ML, as opposed to the arguments usually advanced, e.g., taking into account the estimation of θ_2, or the downwards bias of ML (Patterson and Thompson, 1974). Mean squared error comparisons between ML and REML (Corbeil and Searle, 1976) have given ambiguous answers.

Can ML or REML be improved in a mean squared error sense? Pukelsheim (1976), Anderson (1978) and Searle (1979) consider the dispersion-mean model, in which

$$\mathbf{y} \sim (\chi\theta_3, \mathbf{W}), \qquad [114]$$

where $\mathbf{y}$, χ, $\mathbf{W}$ are appropriately defined vectors and matrices. In particular, they show that a solution to

$$\chi'\mathbf{W}^{-}\chi\hat{\theta}_3 = \chi'\mathbf{W}^{-}\mathbf{y} \qquad [115]$$

yields MINQUE and REML by iteration (Searle, 1979), and a modification of the above gives ML. Using a "ridge-type" argument, it would be possible to shrink the elements of $\hat{\theta}_3$ as

$$(\chi'\mathbf{W}^{-}\chi + \mathbf{D})\,\hat{\theta}_3 = \chi'\mathbf{W}^{-}\mathbf{y} \qquad [116]$$

for an appropriately chosen diagonal matrix $\mathbf{D}$. As discussed previously, shrinkage for the sake of reducing mean squared error does not fit at all in the Bayesian framework. However, this may be appealing for workers that are concerned about mean squared error but not about small biases (Harville, 1977). A theoretically more sensible approach would be, as a first step, to use noninformative priors for the elements of θ_3. Because flat priors do not represent ignorance about variances adequately, using Jeffreys rules (1961) for a set of variance components might be a reasonable starting point. For example, in a model with only "fixed" effects, one would take a priori

$$f(\theta_2) \propto \text{constant};$$
$$-\infty < \theta_{2i} < \infty \text{ for all } i \qquad [117]$$

and

$$f(\sigma^2) \propto \sigma^{-2}; 0 < \sigma^2 < \infty \qquad [118]$$

with θ_2 and σ^2 independent. Under normality, and with independent observations, the likelihood function (n data points) is

$$f(\mathbf{y}|\theta_2,\sigma^2) \propto \frac{1}{\sigma^n}\exp \times \left\{-\frac{1}{2\sigma^2}(\mathbf{y}-\mathbf{X}\theta_2)'(\mathbf{y}-\mathbf{X}\theta_2)\right\} \qquad [119]$$

The joint posterior density is (Zellner, 1971)

$$f(\theta_2,\sigma^2|\mathbf{y}) \propto \frac{1}{\sigma^{n+2}}\exp\left\{-\frac{1}{2\sigma^2}[\upsilon s^2 + (\theta_2-\hat{\theta}_2)]\,\mathbf{X}'\mathbf{X}(\theta_2-\hat{\theta}_2)]\right\} \qquad [120]$$

as shown in [120], where $\hat{\theta}_2 = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$, υ = n-rank $(\mathbf{X})$ and s^2 is the usual residual mean square. The σ^2-component of the mode of this density (Harville, 1977) is

$$\hat{\sigma}^2 = \frac{\upsilon s^2}{n+2}, \qquad [121]$$

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which is biased downwards, especially when the order of θ_2 is large. The marginal posterior density of σ^2, i.e., with θ_2 integrated out, is

$$f(\sigma^2|y) \propto \left(\frac{1}{\sigma^2}\right)^{\frac{\upsilon+2}{2}} \exp\left\{-\frac{\upsilon s^2}{2\sigma^2}\right\} \quad [122]$$

and its mode is

$$\tilde{\sigma} = \upsilon s^2/(\upsilon + 2), \quad [123]$$

which has a slight downward bias; when rank (X) = 2, [123] is the maximum likelihood estimator of σ^2. Harville (1977) stated that [123] has a uniformly smaller mean squared error than the ML and REML estimators of σ^2, and suggested that the application of Jeffreys (1961) priors to a more general variance component setting might be "intriguing."

Another possibility would be to use informative prior distributions for θ_1, θ_2 and θ_3 along the lines of Lindley and Smith (1972), and to estimate these vectors from the mode of a joint posterior distribution. The modal values satisfy

$$\frac{\partial}{\partial\theta_i}[f(\theta_1,\theta_2,\theta_3|y)] = 0; \quad i = 1, \ldots, 3. \quad [124]$$

For simplicity, consider a mixed model with 2 variance components so

$$\theta_3' = [\sigma^2_{\theta_1}, \sigma^2].$$

The first component is the variance of breeding values and the second is a residual variance. A priori, take

$$f(\theta_1,\theta_2,\sigma^2_{\theta_1},\sigma^2) = f(\theta_2)\cdot f(\theta_1|\sigma^2_{\theta_1}) \times f(\sigma^2_{\theta_1})\cdot f(\sigma^2_e), \quad [125]$$

noting that the distribution of breeding values depends on $\sigma^2_{\theta_1}$. The densities are

$$f(\theta_2) \propto \text{constant} \quad [126a]$$

$$f(\sigma^2_e) \propto \left(\frac{1}{\sigma^2_e}\right)^{\frac{\upsilon_e+2}{2}} \exp\left\{-\frac{\upsilon_e s^2_e}{2\sigma^2_e}\right\} \quad [126b]$$

$$f(\theta_1|\sigma^2_{\theta_1}) \propto \frac{1}{(\sigma^2_{\theta_1})^{q/2}} \times \exp\left\{-\frac{1}{2\sigma^2_{\theta_1}}\theta_1'\mathbf{A}^{-1}\theta_1\right\} \quad [126c]$$

and

$$f(\sigma^2_{\theta_1}) \propto \left(\frac{1}{\sigma^2_{\theta_1}}\right)^{\frac{\upsilon_{\theta_1}+2}{2}} \exp\left\{-\frac{\upsilon_{\theta_1} s^2_{\theta_1}}{2\sigma^2_{\theta_1}}\right\}, \quad [126d]$$

where υ_e (υ_{θ_1}) is a precision parameter (equivalent to degrees of freedom) associated with s^2_e ($s^2_{\theta_1}$); the latter are the prior variances, and q is the number of elements in θ_1. If the likelihood function is multivariate normal, the joint posterior density is as shown in [127].

$$f(\theta_1,\theta_2,\sigma^2_{\theta_1},\sigma^2|y) \propto \left(\frac{1}{\sigma^2_e}\right)^{\frac{1}{2}(\upsilon_e+2+n)} \exp\left\{-\frac{1}{2\sigma^2_e}[\upsilon_e s^2_e + (y - X\theta_2 - Z\theta_1)'\mathbf{R}^{-1} \times (y - X\theta_2 - Z\theta_1)]\right\}\cdot \left(\frac{1}{\sigma^2_{\theta_1}}\right)^{\frac{1}{2}(\upsilon_{\theta_1}+2+q)} \exp \times \left\{-\frac{1}{2\sigma^2_{\theta_1}}[\upsilon_{\theta_1} s^2_{\theta_1} + \theta_1'\mathbf{A}^{-1}\theta_1]\right\}. \quad [127]$$

Now, from [124], observe that the modal values also satisfy

$$\frac{\partial}{\partial\theta_1}[f(\theta_1,\theta_2|\theta_3,y] = 0 \quad [128a]$$

and

$$\frac{\partial}{\partial\theta_2}[f(\theta_1,\theta_2|\theta_3,y] = 0 \quad [128b]$$

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$$\frac{\partial}{\partial\theta_3}[f(\theta_3|\theta_1,\theta_2,\mathbf{y}] = 0. \qquad [128c]$$

Equations [128a] and [128b] are

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{A}^{-1}\frac{\hat{\sigma}^2_e}{\hat{\sigma}^2_{\theta_1}} \end{bmatrix} \begin{bmatrix} \hat{\theta}_2 \\ \hat{\theta}_1 \end{bmatrix} =$$

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \end{bmatrix}, \qquad [129]$$

which are Henderson's mixed model equations (1973), evaluated at $\sigma^2_e = \hat{\sigma}^2_e$ and $\sigma^2_{\theta_1} = \hat{\sigma}^2_{\theta_1}$, the modal values of the variance components. These can be found by maximizing [127] with respect to σ^2_e and $\sigma^2_{\theta_1}$, and then evaluating the resulting expressions at $\hat{\theta}_1$ and $\hat{\theta}_2$. We have

$$\hat{\sigma}^2_e = \frac{\hat{\mathbf{e}}'\mathbf{R}^{-1}\hat{\mathbf{e}} + \upsilon_e s^2_e}{n + 2 + \upsilon_e} \qquad [130a]$$

and

$$\hat{\sigma}^2_{\theta_1} = \frac{\hat{\theta}_1\mathbf{A}^{-1}\hat{\theta}_1 + \upsilon_{\theta_1} s^2_{\theta_1}}{q + 2 + \upsilon_{\theta_1}}, \qquad [130b]$$

where $\hat{\mathbf{e}} = \mathbf{y} - \mathbf{X}\hat{\theta}_2 - \mathbf{Z}\hat{\theta}_1$. Equations [129] and [130a,b] require an iterative solution which will "typically" converge (Lindley and Smith, 1972). Starting with guesses σ^2_e and $\sigma^2_{\theta_1}$, calculate [129]; then $\hat{\theta}_1$ and $\hat{\theta}_2$ are used in [130a,b]. The cycle is repeated until the solutions stabilize. The estimators of variance are nonnegative because [130a,b] involve sums of squares and positive numbers. Further, there is no need to store the inverse of [129] to compute [130a,b]. In practice, [129] is amenable to an iterative solution and only $\hat{\theta}_1$ and $\hat{\mathbf{e}}$ are needed in [130a,b]; $\mathbf{R}$ is usually an identity matrix. Lindley and Smith (1972) warn that while it is possible to put $\upsilon_e = 0$, which from [126b] corresponds to Jeffreys (1961) noninformative prior, a value of $\upsilon_{\theta_1} = 0$ may lead to modal estimates in [130b] that are zero. Harville (1977) points out that the problem may stem from "dependencies" existing between components of θ_3 and θ_1 in [127]. The point merits exploration as a multiple-trait extension of [130a] and [130b] would be computationally feasible. For example, in a two-trait situation one could specify the prior densities for the variances as in [126b] and [126d], and the genetic and residual correlations could follow a uniform distribution between −1 and 1. Alternatively (Box and Tiao, 1973), one could use Jeffreys rules to construct a noninformative prior for a covariance matrix. In the two-trait case this would give

$$f(\sigma^2_{e_1},\sigma^2_{e_2},\sigma_{e_{12}},\sigma^2_{\theta_1},\sigma^2_{\theta_2},\sigma_{\theta_{12}}) \propto$$

$$\begin{vmatrix} \sigma^2_{e_1} & \sigma_{e_{12}} \\ \sigma_{e_{12}} & \sigma^2_{e_2} \end{vmatrix}^{3} \begin{vmatrix} \sigma^2_{\theta_1} & \sigma_{\theta_{12}} \\ \sigma_{\theta_{12}} & \sigma^2_{\theta_2} \end{vmatrix}, \qquad [131]$$

where $\sigma_{e_{12}}$ and $\sigma_{\theta_{12}}$ are environmental and genetic covariances, respectively.

An area of potential importance in animal breeding is the estimation of genetic parameters with nonlinear models. Working with growth curves, Kachman and Gianola (1984) derived a Bayesian estimator of variance and covariance components which is a generalization of REML. As an illustration of another approach which could be followed in the domain of categorical responses, we return to the situation in [110]. Uncertainty about h^2 can be introduced via a uniform prior

$$f(h^2) = \text{constant};$$
$$0 < h^2 < 1. \qquad [132]$$

Then, the posterior density is

$$f(\theta_1,\beta,h^2|\mathbf{Y}) \propto C\left(\frac{1-h^2}{h^2}\right)^{q/2} \exp$$
$$\times \left\{-\tfrac{1}{2}\theta'_1\mathbf{A}^{-1}\theta_1(1-h^2)/h^2\right\} \qquad [133]$$

where C is a function which does not depend on h^2. The h^2-component of the mode of [133] can be easily found by differentiation. The resulting estimator is

$$\hat{h}^2 = \frac{\hat{\theta}_1\mathbf{A}^{-1}\hat{\theta}_1/q}{1 + \hat{\theta}'_1\mathbf{A}^{-1}\hat{\theta}_1/q}, \qquad [134]$$

where $\hat{\theta}'_1\mathbf{A}^{-1}\hat{\theta}_1/q$ is an estimator of $\sigma^2_{\theta_1}$, and 1 is the residual variance; $\hat{\theta}_1$ can be calculated from [111] and this, jointly with [134], define an iterative algorithm. Although the estimator in [134] satisfies [132], it presents the same problems of [130a] and [130b] discussed previously.

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Henderson (1988)

> *Henderson, C.R. 1988. Theoretical basis and computational methods for a number of different animal models. Journal of Dairy Science, vol. 71, supplement 2, pp. 1 to 16.*

The problem

Late in his career, Henderson gathered the different types of animal model in a single text. He covers models with one or several traits, repeated records, maternal effects, nonadditive genetic effects and genetic groups. For each case, he presents the theoretical basis and the way to set up and solve the equations, including the reduced animal model.

Points to note while reading

The text works as a catalog of the models used in genetic evaluations and can be used as a reference for choosing the appropriate model for each data set. The copy is scanned and some pages are of uneven quality.

Theoretical Basis and Computational Methods for a Number of Different Animal Models

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ABSTRACT

The animal model is actually a set of many different models. Their common feature is that all animals in a herd or region or nation are evaluated jointly in contrast, for example, to separate male and female evaluations. An animal model can account for repeated records, multiple traits, nonadditive genetic effects, litter effects, and, in addition, a number of environmental effects, both fixed and random. Some of these possible models are described, some computational aspects are presented, and where possible, corresponding reduced animal model methods are presented.

INTRODUCTION

Best linear unbiased prediction under the animal model has rapidly become the method of choice of those responsible for evaluation of animals. Section 1.6 presents desirable properties of animal models.

1.1 Types of Animal Models

There is not a single animal model. This paper deals with the following types of data sets and associated models.

1. Single record, single trait, additive genetic effects only.
2. Repeated records, single trait, additive genetic effects, and simple repeatability model.
3. Multiple traits, single records per trait, additive genetic effects.
4. Single trait, repeated records treated as multiple traits, additive genetic effects.
5. Direct and maternal additive genetic effects, single trait, single records.
6. Nonadditive genetic model both single as well as repeated records, single traits.
7. Nonadditive genetic model, multiple traits, single records.
8. Single trait, multiple records, nonadditive and maternal genetic effects.

These are not all of the possibilities, perhaps the most complicated are multiple traits, repeated records, nonadditive, and maternal effects. Also, in the case of pigs and sheep, multiple births complicate the model and may require incorporating negative correlations among intralitter errors.

The common feature of all animal models is evaluation of breeding values or total genetic merits or prediction of future records of all individuals in the population of interest. This is in contrast to evaluation of sires, breeds, breed crosses, or line crosses.

1.2 Early History of Animal Models and the Reduced Animal Model

It is not certain where the term, animal model (AM) originated, but it is probably due to Quaas and Pollak (18) in their paper presenting the reduced animal model (RAM). Apparently the first application of BLUP to an animal model was by Henderson (7) in his paper describing estimation of genetic and environmental trends with culling on earlier records. Further details were presented by Henderson et al. (16). The notion of using the numerator relationship matrix in the analysis by BLUP under an animal model was first by Henderson (8). This was not computationally feasible until his presentation of the rapid method for computing A^{-1}, the inverse of the relationship matrix (12). A detailed description of intraherd evaluation under an animal model with repeated records was presented by Henderson (9). The application to multiple traits was described by Henderson and Quaas (17). The inclusion of maternal effects and the first presentation of RAM were in a paper by Quaas and Pollak (18).

1.3 Reduced Animal Model

The RAM is actually a computing method to obtain BLUP under the AM. It can be conceptualized in two different ways: 1) absorption breed-

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ing values of animals that have no progeny with records from the AM equations; or 2) writing an equivalent model with a shorter vector of breeding values (parents only) and accounting for the breeding values of progeny by modified Z matrix. Mendelian sampling and unknown parents affecting the variances of records of progeny are taken into account by adding appropriate amounts of genetic variance and covariance to the error vector variance or to the vector of permanent environmental effects. The second of these methods is usually much easier computationally. Unfortunately, it has not been possible to apply the method to nonadditive genetic models.

1.4 Some basic results useful for Animal Models

A basic linear model is:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \mathbf{e} \qquad [1.1]$$

where $\boldsymbol{\beta}$ is p × 1, fixed, and unknown; $\mathbf{u}$, $\mathbf{e}$ are random vectors, which under a no selection model have null expectations and $\text{var}(\mathbf{u}) = \mathbf{G}$, $\text{var}(\mathbf{e}) = \mathbf{R}$, and $\text{cov}(\mathbf{u}, \mathbf{e}') = \mathbf{0}$.

Matrices G and R are usually positive definite and sometimes are assumed to be known to proportionality. The matrix G is positive semidefinite if identical twins are present.

The matrix, X, is a fixed, known n × p matrix with rank ≤ min(n,p), and Z is a fixed, known matrix with possibly some null columns corresponding to elements of u on which there are no observations.

The mixed model equations for BLUE of estimable functions $\boldsymbol{\beta}$ and for BLUP of u are in [1.2]. The elements of u can contain additive genetic effects, nonadditive genetic effects, maternal effects, and permanent environmental effects. It can also contain nongenetic effects and interactions.

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} \end{bmatrix} \begin{bmatrix} \boldsymbol{\beta}^0 \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \end{bmatrix} \qquad [1.2]$$

The solution, $\boldsymbol{\beta}^0$, is not unique if rank (X) < p, but if $\mathbf{K}'\boldsymbol{\beta}$ is estimable, $\mathbf{K}'\boldsymbol{\beta}$ is unique and is BLUE for any solution, $\boldsymbol{\beta}^0$. In contrast, $\hat{\mathbf{u}}$ is unique and is BLUP. These results depend, of course, upon the assumption of a correct model and the elements of G and R known at least to proportionality.

Let some symmetric g-inverse of the coefficient matrix of [1.2] be

$$\begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{12}' & \mathbf{C}_{22} \end{bmatrix} \qquad [1.3]$$

Nonsymmetric g-inverses of nonsingular matrices exist, but symmetric g-inverses are easier to compute and can be half-stored in memory. Now for estimable $\mathbf{K}'\boldsymbol{\beta}$, $\text{var}(\mathbf{K}'\boldsymbol{\beta}^0) = \mathbf{K}'\mathbf{C}_{11}\mathbf{K}$, $\text{cov}(\mathbf{K}'\boldsymbol{\beta}^0, \hat{\mathbf{u}} - \mathbf{u}) = \mathbf{K}'\mathbf{C}_{12}$, $\text{var}(\hat{\mathbf{u}} - \mathbf{u}) = \mathbf{C}_{22}$, and $\text{var}(\hat{\mathbf{u}}) = \mathbf{G} - \mathbf{C}_{22}$. [1.4]

Sometimes prediction of elements, say $\mathbf{u}_0$, that are not included in u of [1.1] are desired. For example, one may wish to predict the producing abilities of possible matings of animals already evaluated. Also, one may wish to predict future records on traits not observed in a multiple trait evaluation. In this case we need BLUP of the future errors, say $\mathbf{e}_0$.

Let $\text{cov}(\mathbf{u}_0, \mathbf{u}') = \mathbf{C}_1$, $\text{cov}(\mathbf{u}_0, \mathbf{e}') = \mathbf{C}_2$, $\text{cov}(\mathbf{e}_0, \mathbf{e}') = \mathbf{C}_3$. Then $\hat{\mathbf{u}}_0 = \mathbf{C}_1\mathbf{G}^{-1}\hat{\mathbf{u}} + \mathbf{C}_2\mathbf{R}^{-1}\hat{\mathbf{e}}$, and:

$$\hat{\mathbf{e}}_0 = \mathbf{C}_3\mathbf{R}^{-1}\hat{\mathbf{e}}, \qquad [1.5]$$

where $\hat{\mathbf{e}} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}^0 - \mathbf{Z}\hat{\mathbf{u}}$.

Note that for prediction of $\mathbf{e}_0$, $\text{cov}(\mathbf{e}_0, \mathbf{u}')$ is not included. In all cases of which I am aware this is null. Prediction error variances for missing elements in u and e can be computed by methods described by Henderson (14).

1.5 Some Additional Problems

In succeeding sections of this paper no consideration is given to how to write $\mathbf{X}\boldsymbol{\beta}$ and other elements of u other than animal effects. The u of this paper refers to animal effects, such as genetic values, permanent environmental effects, and producing abilities. Grouping is a particularly interesting problem. In the animal model with all parents (except base population animal) identified and their records utilized no grouping is needed. The first mention of grouping may have been by Henderson (7). Further details regarding grouping are presented in Section 10.

Unequal genetic and environmental variances from herd to herd and from year-season to year-season probably should be assumed. If we knew such variances, it would be easy to incorporate these in G and R of mixed model equations.

Unfortunately, such parameters are unknown and seldom are there enough data within herds to estimate them accurately. It seems logical to take as the estimate some compromise between the estimate coming from the individual herd and the population estimate. Empirical Bayesian estimation does this [see Gianola et al. (4)]. After such estimates are obtained, consequences of ignoring differences among herds could then examined.

1.6 Properties of Best Linear Unbiased Predictors in the Animal Model

Given certain assumptions, selection by BLUP using the animal model has highly desirable properties. The assumptions are:

1. Correct model, including variances and covariances known at least to proportionality.
2. Multivariate normal distributions, implying a large number of loci. Actually quite a small number of loci gives good approximations to normality.
3. Selection and mating decisions are based on translation invariant functions of the data, and these data are used in the analysis.

Then BLUP has these properties:

1. Unbiased even when selection has occurred.
2. In the class of translation invariant selection functions, BLUP used as the selection criterion maximizes the expected value of the selected individuals provided a fixed number is selected from a fixed number of candidates (1, 3, 5).

If the variances and covariances are unknown, their REML estimates probably should be used in mixed model equations in place of $\mathbf{G}$ and $\mathbf{R}$ (4). A detailed description of translation invariant selection functions is by Henderson (13).

1.7. A Model Combining β and Elements of u Excluding Animal Effects

Often the vector $\mathbf{u}$, of [1.1] includes random variables other than animal effects. The mixed model equations for various animal models that follow in this paper can be simplified by writing [1.1] as

$$\mathbf{y} = \mathbf{T}\boldsymbol{\tau} + \mathbf{Z}\mathbf{u} + \mathbf{e} \qquad [1.6]$$

where $\boldsymbol{\tau}' = [\boldsymbol{\beta}', \mathbf{u}_0']$ and $\mathbf{u}_0$ are random variables other than animal effects. The associated incidence matrix is $\mathbf{T} = [\mathbf{X}:\mathbf{Z}_0]$. The u vector of [1.6] is the subvector of [1.1] pertaining to animal effects.

Now let "var" $(\tau) = \mathbf{H}$ and using the Bayesian notion of a flat prior on β,

$$\mathbf{H}^{-1} \to \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & [\mathrm{Var}(\mathbf{u}_0)]^{-1} \end{bmatrix}$$

Then the mixed model equations are:

$$\begin{bmatrix} \mathbf{T}'\mathbf{R}^{-1}\mathbf{T} + \mathbf{H}^{-1} & \mathbf{T}'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{T} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} \end{bmatrix} \begin{bmatrix} \boldsymbol{\tau}^0 \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{T}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \end{bmatrix} \qquad [1.7]$$

1.8 Some Precautions to Take in Writing Animal Models and Reduced Animal Models

Missing information can sometimes cause problems in animal models. This is true particularly when some dams are not identified in a maternal model. To check if there are troubles assume that there is no inbreeding or selection and that variances are homogenous. Then the variance of every record should be the same. This can be checked by examining the diagonals of $\mathbf{ZGZ}' + \mathbf{R}$. If this is done for a maternal model, the diagonals for animals with a record but unidentified dams are smaller than those for animals with identified dams. Section 6 describes how to modify $\mathbf{R}$ for this problem. If there is the slightest doubt concerning the effect of missing information, one should look at the diagonals of $\mathbf{ZGZ}' + \mathbf{R}$.

The reduced animal model is appropriate for additive genetic models but apparently not for nonadditive genetic models. As is well known, RAM includes setting up mixed model equations with additive genetic values only for parents with recorded progeny. The $\mathbf{Z}$ matrix pertaining to the descendants with records but who themselves have no recorded progeny contains .5 in the column pertaining to a for each identified parent and appropriate trait. This formulation by itself does not account for Mendelian sampling and for the additive genetic contribution of unknown parents to var(y) of progeny. Therefore it is required that either var(e) or var(p) pertaining to progeny must be increased by appropriate fractions of additive genetic variance and of additive genetic covariance in the case of multiple traits and maternal models. It is easy to derive these for single trait, no maternal effects models, but for more complicated cases it is difficult to state specifically general rules. A variety of examples is presented in subsequent sections, and these suggest how to

proceed in more complicated situations. Whenever there is doubt concerning whether the modification has been done properly, one should compute var(y) for both AM and RAM. These must be equal if RAM is to produce the same estimates and predictions as AM.

2. One Trait, Single Records, Additive Genetic and Simple Repeatability Model

The basic model is:

$$\mathbf{y} = \mathbf{T}\boldsymbol{\tau} + \mathbf{Z}\mathbf{a} + \boldsymbol{\epsilon} \qquad [2.1]$$

where a refers to additive genetic values and ϵ, to random error that includes p, permanent environmental effects, and e, temporary environmental effects.

$$\mathrm{E}(\mathbf{a}),\ \mathrm{E}(\boldsymbol{\epsilon}) \text{ are both null,}$$

$\mathrm{var}(\mathbf{a}) = \mathbf{A}\sigma_a^2$, $\mathrm{var}(\boldsymbol{\epsilon}) = \mathbf{I}\sigma_e^2$, $\mathrm{var}(\mathbf{p}) = \mathbf{I}\sigma_p^2$, and all subvectors are uncorrelated. Then $\boldsymbol{\epsilon} = \mathbf{e} + \mathbf{p}$ and $\mathrm{var}(\boldsymbol{\epsilon}) = \mathbf{I}\,(\sigma_e^2 + \sigma_p^2)$.

2.1 Animal Model

Now the mixed model equations are:

$$\begin{bmatrix} \mathbf{T}'\mathbf{T} + \mathbf{H}^{-1} & \mathbf{T}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{T} & \mathbf{Z}'\mathbf{Z} + \mathbf{A}^{-1}\alpha \end{bmatrix} \begin{bmatrix} \boldsymbol{\tau}^0 \\ \hat{\mathbf{a}} \end{bmatrix} = \begin{bmatrix} \mathbf{T}'\mathbf{y} \\ \mathbf{Z}'\mathbf{y} \end{bmatrix} \qquad [2.3]$$

where $\alpha = (\sigma_e^2 + \sigma_p^2)/\sigma_a^2$.

The Z matrix is I if all animals have a record, but is I with rows associated with missing records deleted otherwise.

If BLUP of p is desired this is:

$$\hat{\mathbf{p}} = \hat{\boldsymbol{\epsilon}}\sigma_p^2/(\sigma_e^2 + \sigma_p^2) \qquad [2.4]$$

for animals with a record and 0 otherwise.

$$\hat{\boldsymbol{\epsilon}} = \mathbf{y} - \mathbf{T}\boldsymbol{\tau}^0 - \mathbf{Z}\hat{\mathbf{a}} \qquad [2.5]$$

Illustration. In all examples that follow $\mathbf{T} = \mathbf{X}$ and $\boldsymbol{\tau} = \boldsymbol{\beta}$. That is, only animal effects and e comprise random variables. We have 6 animals related in this way. The relationship matrix for 1,2 is:

$$\begin{bmatrix} 1.25 & .625 \\ .625 & 1.0 \end{bmatrix}.$$

Animals 3 and 4 have parents 1 and 2, animal 5 has sire 1 and unknown dam, and animal 6 has unknown sire and dam 2. Then A for the six animals is:

$$\begin{bmatrix} 1.25 & .625 & .9375 & .9375 & .625 & .3125 \\ & 1.0 & .8125 & .8125 & .3125 & .5 \\ & & 1.3125 & .875 & .46875 & .40625 \\ & \text{symmetric} & & 1.3125 & .46875 & .40625 \\ & & & & 1.0 & .15625 \\ & & & & & 1.0 \end{bmatrix} \qquad [2.6]$$

Animals 2 to 6 have a record and these are [4, 5, 7, 8, 9]. We assumed that $\sigma_a^2 = 4$, $\sigma_p^2 = 2$, $\sigma_e^2 = 8$. Then $\alpha = (2 + 8)/4 = 2.5$. It was assumed that:

$$\mathbf{X}' = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 3 & 4 & 2 & 7 & 6 \end{bmatrix}.$$

This gives

$$\boldsymbol{\beta}^0 = (3.957, .613)',$$
$$\hat{\mathbf{a}} = (-.067, -.226, -.335, .146, -.080, .228)',$$
$$\hat{\boldsymbol{\epsilon}} = (-1.569, -1.074, 1.672, -.167, 1.138)',$$

and $\hat{\mathbf{p}}$ for animals 2 to 6 $= .2\hat{\boldsymbol{\epsilon}} = (-.314, -.215, .334, -.033, .228)'$.

$$\hat{p}_1 = 0.$$

2.2 Reduced Animal Model

The reduced animal model of Quaas and Pollak (18) is widely used for the assumptions of model [2.1]. Partition a as $[\mathbf{a}_p' : \mathbf{a}_o']$, where p refers to parents with one or more recorded progeny and o refers to offspring with a record who themselves have no recorded progeny. Order the data by records on parents followed by records on progeny. Then the model is:

$$\mathbf{y} = \mathbf{T}\boldsymbol{\tau} + \mathbf{Z}_r\mathbf{a}_p + \mathbf{v} \qquad [2.7]$$

$\mathbf{Z}_r = \begin{bmatrix} \mathbf{Z}_p \\ \mathbf{Z}_o \end{bmatrix}$, where $\mathbf{Z}_p$ is an identity matrix of order equal to the number of parents, provided each has a record. Otherwise delete rows from I corresponding to missing records. The submatrix, $\mathbf{Z}_o$, contains all o's except for .5 in row and column pertaining to a known parent. Then if both parents

of the ith progeny are known, the ith row of $\mathbf{Z}_r$ contains two .5's. If one parent is known, it contains a single .5. If neither parent is known, the row is null.

$$\text{Var}(\mathbf{v}) = \mathbf{I}(\sigma_p^2 + \sigma_e^2) + \sigma_a^2\mathbf{D} \equiv \mathbf{R}_r. \quad [2.8]$$

The matrix $\mathbf{D}$ is a diagonal with o's in the diagonal elements pertaining to parents with a record. The element pertaining to a progeny is $1 - .25\,(a_{ss} + a_{dd})$, where a_{ss}, a_{dd} refer to diagonals of A for its sire and dam. If the sire (dam) is unknown, set $a_{ss}(a_{dd})$ to 0.

The mixed model equations are:

$$\begin{bmatrix} \mathbf{T}'\mathbf{R}_r^{-1}\mathbf{T} & \mathbf{T}'\mathbf{R}_r^{-1}\mathbf{Z}_r \\ \mathbf{Z}_r'\mathbf{R}_r^{-1}\mathbf{T} & \mathbf{Z}_r'\mathbf{R}_r^{-1}\mathbf{Z}_r + \mathbf{A}_p^{-1}/\sigma_a^2 \end{bmatrix} \begin{bmatrix} \boldsymbol{\tau}^0 \\ \hat{\mathbf{a}}_p \end{bmatrix} = \begin{bmatrix} \mathbf{T}'\mathbf{R}_r^{-1}\mathbf{y} \\ \mathbf{Z}_r'\mathbf{R}_r^{-1}\mathbf{y} \end{bmatrix} \quad [2.9]$$

The relationship matrix for parents is $\mathbf{A}_p$. Now $\mathbf{K}'\boldsymbol{\beta}^0$ and $\hat{\mathbf{a}}_p$ are the same as the solution to [2.3].

To compute the entire vector, $\hat{\mathbf{a}}$, let $\mathbf{T}_1' = \mathbf{Z}_r$ written as though there are no missing records. $\mathbf{T}_2' = \sigma_a^2\mathbf{D}_a\mathbf{R}_r^{-1}$ where $\mathbf{D}_a$ is the matrix $\mathbf{D}$ of [2.8] augmented by a row of o's for each missing record in the appropriate location.

Then $\hat{\mathbf{a}} = \mathbf{T}_1'\hat{\mathbf{a}}_p + \mathbf{T}_2'\hat{\mathbf{v}}$, where:

$$\hat{\mathbf{v}} = \mathbf{y} - \mathbf{T}\boldsymbol{\tau}^0 - \mathbf{Z}_r\hat{\mathbf{a}}_p \quad [2.10]$$

To compute the prediction error covariance matrix for the entire a vector, let a g-inverse of the matrix of [2.9] be:

$$\begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{12}' & \mathbf{C}_{22} \end{bmatrix} = \begin{bmatrix} \mathbf{C}_1 \\ \mathbf{C}_2 \end{bmatrix} = \mathbf{C} \quad [2.11]$$

Then $\text{var}(\hat{\mathbf{a}} - \mathbf{a}) = \sigma_a^2\mathbf{A} - \mathbf{T}_1'(\sigma_a^2\mathbf{A}_p - \mathbf{C}_{22})\mathbf{T}_1 - \mathbf{T}_2'(\mathbf{R}_r - \mathbf{W}_r\mathbf{C}\mathbf{W}_r')\mathbf{T}_2 - \mathbf{T}_1'\mathbf{C}_2\mathbf{W}_r\mathbf{T}_2 - \mathbf{T}_2'\mathbf{W}_r'\mathbf{C}_2'\mathbf{T}_1$, where $\mathbf{W}_r = (\mathbf{X}:\mathbf{Z}_r)$. [2.12]

Illustration of Reduced Animal Model. Using the same data as in the AM illustration,

$$\mathbf{Z}_r = \begin{bmatrix} 0 & 1.0 \\ .5 & .5 \\ .5 & .5 \\ .5 & 0 \\ 0 & .5 \end{bmatrix}$$

$\mathbf{D} = \text{diag}\,(0, .4375, .4375, .6875, .75)$.

Then $\mathbf{R} = 10\mathbf{I} + 4\mathbf{D}$
$= \text{diag}(10.0, 11.75, 11.75, 12.75, 13.0)$.

Then $\boldsymbol{\beta}^0 = (3.957, .613)'$ as in AM.

$\hat{\mathbf{a}}_p = (-.067, -.226)$ as in AM.

Now to compute $\hat{\mathbf{a}}$,

$$\mathbf{T}_1' = \begin{bmatrix} 1 & 0 \\ \mathbf{Z}_r \end{bmatrix}, \mathbf{T}_2' = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ & & 4\mathbf{D}\,\mathbf{R}_r^{-1} & & \end{bmatrix}$$

$\hat{\mathbf{v}} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}^0 - \mathbf{Z}_r\hat{\mathbf{a}}_p = (-1.569, -1.261, 1.964, -.213, 1.479)'$. Then $\hat{\mathbf{a}} = \mathbf{T}_1'\hat{\mathbf{a}}_p + \mathbf{T}_2'\hat{\mathbf{v}}$ = same result as in AM.

2.3 Best Linear Unbiased Predictors of Animal with No Record and No Recorded Progeny

Note that in Section 2.2 the only animals with no record included are those with recorded progeny. The BLUP of a of an animal with no recorded progeny and no record itself if $.5(\hat{a}_s + \hat{a}_d)$. Of course $\hat{p}$ of such an animal is 0.

3. One Trait, Repeated Records, Additive Genetic and Simple Repeatability Model

We next consider the same model as in Section 2 except that there are repeated records on animals.

3.1 Basic Model

$$\mathbf{y} = \mathbf{T}\boldsymbol{\tau} + \mathbf{Z}\mathbf{a} + \mathbf{Z}\mathbf{p} + \mathbf{e} \quad [3.1]$$

$\text{Var}(\mathbf{a}) = \mathbf{A}\sigma_a^2$, $\text{var}(\mathbf{p}) = \mathbf{I}\sigma_p^2$, $\text{var}(\mathbf{e}) = \mathbf{I}\sigma_e^2$

$\text{Cov}(\mathbf{a},\mathbf{p}')$, $\text{cov}(\mathbf{a},\mathbf{e}')$, $\text{cov}(\mathbf{p},\mathbf{e}')$ are null.

Then $\mathbf{Z}$ has the following form. Let the number of records on the animals be n_1, n_2, etc. Then the first n_1 rows of $\mathbf{Z}$ are n_1 repetitions of the first row of $\mathbf{I}$ with order equal to the number of animals. The $n_1 + 1, \ldots, n_1 + n_2$ rows are n_2 repetitions of the second row of I, etc. The mixed model equations are:

$$\begin{bmatrix} \mathbf{T}'\mathbf{T} + \mathbf{H}^{-1}\sigma_e^2 & \mathbf{T}'\mathbf{Z} & \mathbf{T}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{T} & \mathbf{Z}'\mathbf{Z} + \mathbf{A}^{-1}\alpha_a & \mathbf{Z}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{T} & \mathbf{Z}'\mathbf{Z} & \mathbf{Z}'\mathbf{Z} + \mathbf{I}\alpha_p \end{bmatrix} \begin{bmatrix} \boldsymbol{\tau}^0 \\ \hat{\mathbf{a}} \\ \hat{\mathbf{p}} \end{bmatrix} = \begin{bmatrix} \mathbf{T}'\mathbf{y} \\ \mathbf{Z}'\mathbf{y} \\ \mathbf{Z}'\mathbf{y} \end{bmatrix} \quad [3.2]$$

$\alpha_a = \sigma_e^2/\sigma_a^2$, $\alpha_p = \sigma_e^2/\sigma_p^2$.

The solution to $\hat{p}_i = 0$ if the ith animal has no records. We can easily absorb $\hat{\mathbf{p}}$ because

$\mathbf{Z'Z} + \mathbf{I}\alpha_p$ is diagonal. Then having solved for $\hat{\mathbf{a}}$,

$$\hat{\mathbf{p}} = [\mathbf{Z'Z} + \mathbf{I}\alpha_p]^{-1}[\mathbf{Z'y} - \mathbf{Z'X}\tau^0 - \mathbf{Z'Z}\hat{\mathbf{a}}] \quad [3.3]$$

3.2 Reduced Animal Model

The model is:

$$\mathbf{y} = \mathbf{T}\tau + \mathbf{Z}_r\mathbf{a}_r + \mathbf{Z}s + \mathbf{e} \quad [3.4]$$

$\mathbf{Z}$ is the same as in [3.1] and $\mathbf{Z}_r$ is the same as in Section 2.2 except that rows are repeated as many times as the corresponding animal has records.

$$\text{Var}(\mathbf{s}) = \mathbf{I}\sigma_p^2 + \mathbf{D}\sigma_a^2 = \mathbf{S}$$

The matrix $\mathbf{D}$ is exactly the same as in Section 2.2 single record situations, but includes an element 0 for any parent with a missing record. Thus, Mendelian sampling and unknown parents are accounted for by additions to var(p) to form var(s).

The mixed model equations for RAM are:

$$\begin{bmatrix} \mathbf{T'R}^{-1}\mathbf{T} + \mathbf{H}^{-1} & \mathbf{T'R}^{-1}\mathbf{Z}_r & \mathbf{T'R}^{-1}\mathbf{Z} \\ \mathbf{Z}_r'\mathbf{R}^{-1}\mathbf{T} & \mathbf{Z}_r'\mathbf{R}^{-1}\mathbf{Z}_r + \mathbf{A}_p^{-1}/\sigma_a^2 & \mathbf{Z}_r'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z'R}^{-1}\mathbf{T} & \mathbf{Z'R}^{-1}\mathbf{Z}_r & \mathbf{Z'R}^{-1}\mathbf{Z} + \mathbf{S}^{-1} \end{bmatrix} \begin{bmatrix} \tau^0 \\ \hat{\mathbf{a}}_r \\ \hat{\mathbf{s}} \end{bmatrix} = \begin{bmatrix} \mathbf{T'R}^{-1}\mathbf{y} \\ \mathbf{Z}_r'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z'R}^{-1}\mathbf{y} \end{bmatrix} \quad [3.5]$$

The solution to $\hat{\mathbf{s}}$ of [3.5] is not the same as $\hat{\mathbf{p}}$ of [3.2] except for parents. It is easy to absorb $\hat{\mathbf{s}}$ because $\mathbf{Z'R}^{-1}\mathbf{Z} + \mathbf{S}^{-1}$ is diagonal.

To compute $\hat{\mathbf{a}}$, BLUP of the entire a vector, let $\mathbf{T}_2'$ be the same as in Section 2.2, that is, $\mathbf{Z}_r$ written as though every parent and progeny has a single record.

Let $\mathbf{T}_2'$ be $(\sigma_a^2\,\mathbf{DS}^{-1})$. Then:

$$\hat{\mathbf{a}} = \mathbf{T}_1'\hat{\mathbf{a}} + \mathbf{T}_2'\hat{\mathbf{s}} \quad [3.6]$$

To compute $\hat{\mathbf{p}}$ let $\mathbf{T}_3' = \sigma_p^2\mathbf{S}^{-1}$. Then $\hat{\mathbf{p}} = \mathbf{T}_3'\hat{\mathbf{s}}$.

Illustration.

Animal	Parents	No. of records	y
1	0 – 0	. .	. . .
2	0 – 0	3	3, 4, 5
3	0 – 0	2	6, 5
4	1 – 2	2	7, 6
5	1 – 3	3	5, 4, 3
6	1 – 0	1	2
7	0 – 3	2	8, 9
8	0 – 0	1	7

The relationship of base animals, 1, 2, 3, is

$$\begin{bmatrix} 1.0 & .5 & .75 \\ .5 & 1.0 & .75 \\ .75 & .75 & 1.25 \end{bmatrix}.$$

For purposes of illustration $\sigma_a^2 = 6$, $\sigma_p^2 = 4$, $\sigma_e^2 = 10$, Then $\alpha_a = 10/6$, $\alpha_p = 10/4$.

The value of X′ is:

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 3 & 4 & 2 & 5 & 3 & 7 & 8 & 4 & 2 & 3 & 6 & 2 & 5 & 9 \end{bmatrix}.$$

Then the regular mixed model equations give this solution:

$$\begin{aligned} \boldsymbol{\beta}^0 &= (4.295, .232)'. \\ \hat{\mathbf{a}} &= (-.342, -.238, .046, -.101, \\ &\quad -.370, -1.026, 1.083, .186)'. \\ \hat{\mathbf{p}} &= (0, -.410, .103, .253, \\ &\quad -.338, -.760, 1.028, .124)'. \end{aligned}$$

Reduced animal model solution

Let $\overline{\mathbf{Z}}_r$ = matrix for one record for each of the 8 animals. This is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ .5 & .5 & 0 \\ .5 & 0 & .5 \\ .5 & 0 & 0 \\ 0 & 0 & .5 \\ 0 & 0 & 0 \end{bmatrix}$$

Then $\mathbf{Z}_r$ is $\overline{\mathbf{Z}}_r$ with rows repeated for number of records for that animal. Thus,

$$\mathbf{Z}_r = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ & \cdot & \\ & \cdot & \\ & \cdot & \\ 0 & 0 & .5 \\ 0 & 0 & .5 \\ 0 & 0 & 0 \end{bmatrix}$$

To compute var(s), $\mathbf{D}$ = diag. (0, 0, 0, .5, .4375, .75, .6875, 1.0)

Then $\mathbf{S}$ = diag. (4, 4, 4, 7, 6.625, 8.5, 8.125, 10.0) These parameters lead to the solution:

$\hat{\boldsymbol{\beta}}''$ = same as in Section 3.1,
$\hat{\mathbf{a}}_p$ = same as first three elements of â in Section 3.1,
$\hat{\mathbf{s}}$ = (0, −.410, .103, .442, −.560, −1.614, 2.087, .310)'.

Then â from RAM is $\overline{\mathbf{Z}}_1\hat{\mathbf{a}}_p + \mathbf{T}_2'\hat{\mathbf{s}}$, where $\mathbf{T}_2' = 6\mathbf{D}\,\mathbf{S}^{-1}$. This gives the same result as in Section 3.1. Also, $\hat{\boldsymbol{\beta}}$ from RAM is $4\,\mathbf{S}^{-1}\hat{\boldsymbol{\beta}}$. This gives same result as in Section 3.1.

4. Multiple Traits, Single Records per Trait, Additive Genetic Model

Now we have b animals and t traits. There is either 1 or no record on a trait on an animal. Potentially there are bt records.

4.1 Animal Model

The model is:

$$\mathbf{y} = \mathbf{T}\boldsymbol{\tau} + \mathbf{Z}\mathbf{a} + \mathbf{e} \qquad [4.1]$$

The record vector, $\mathbf{y}$, is ordered traits in animals and so is $\mathbf{a}$. All bt elements of a are included. If all bt potential records are observed, $\mathbf{Z}$ is $\mathbf{I}$ with order, bt. Otherwise delete rows corresponding to missing records. Then $\mathbf{Z}$ has order n × bt, where n < bt.

Let $\mathbf{G}_o$ be a t × t matrix, the usual additive genetic variance-covariance matrix for an unselected, noninbred population.

Let $\mathbf{R}_o$ be the corresponding t × t error covariance matrix. Then $h_i^2 = g_{ii}/(g_{ii} + r_{ii})$, and genetic correlation between traits i and j is $g_{ij}/(g_{ii}g_{jj})^{.5}$.

Now $\mathbf{G} = \text{var}(\mathbf{a}) = \mathbf{A} \otimes \mathbf{G}_o$ where ⊗ denotes direct product. If all bt records are observed, $\mathbf{R}_t = \mathbf{I} \otimes \mathbf{R}_o$, where $\mathbf{I}$ has order equal to number of animals, b. If records are missing, delete rows and columns of $\mathbf{I} \otimes \mathbf{R}_o$ corresponding to missing records to produce $\mathbf{R}$.

The mixed model equations are:

$$\begin{bmatrix} \mathbf{T}'\mathbf{R}^{-1}\mathbf{T} + \mathbf{H}^{-1} & \mathbf{T}'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{T} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\tau}}^0 \\ \hat{\mathbf{a}} \end{bmatrix} = \begin{bmatrix} \mathbf{T}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \end{bmatrix} \qquad [4.2]$$

and â contains BLUP of breeding values of all animals and traits. If we wish to predict records on traits and animals not observed, we can do this using â and ê, where $\hat{\mathbf{e}} = \mathbf{y} - \mathbf{T}\hat{\boldsymbol{\tau}}^0 - \mathbf{Z}\hat{\mathbf{a}}$. The prediction of a record on the jth trait of the ath animal is $\hat{a}_{ij} + \hat{e}_{ij}$. The solution $\hat{a}_{ij}$ is found from the solution to [4.2]. To predict missing e_{ij} proceed as follows.

Let $\mathbf{R}_d$ be $\mathbf{R}_t$ defined above with columns deleted that correspond to missing records and rows deleted corresponding to records observed. The BLUP of missing e, say e_m is:

$$\hat{\mathbf{e}}_m = \mathbf{R}_d\mathbf{R}^{-1}\hat{\mathbf{e}} \qquad [4.3]$$

4.2 Reduced Animal Model

The model is:

$$\mathbf{y} = \mathbf{T}\boldsymbol{\tau} + \mathbf{Z}_r\mathbf{a}_r + \boldsymbol{\epsilon} \qquad [4.4]$$

and

$$\mathbf{Z}_r = \begin{bmatrix} \mathbf{Z}_p \\ \mathbf{Z}_o \end{bmatrix} \qquad [4.5]$$

where $\mathbf{Z}_p = \mathbf{I}$ with order = number of parents xt if all parents have records on all t traits. Otherwise delete rows of I corresponding to missing records.

$\mathbf{Z}_o$ contains number of rows equal to number of records made by progeny. Every row is null except for .5 under column pertaining to a parent and trait. Now $\mathbf{R}$ for progeny must be modified for Mendelian sampling and missing parents.

Let $\mathbf{H}_r$ be a block diagonal matrix with diagonal blocks of order txt. The blocks for parents are null. The blocks for progeny are

$$\mathbf{B}_i \# \mathbf{G}_o$$

where $\mathbf{B}_i = [1 - .25\,(a_{sd} + a_{dd})]\,\mathbf{J}$, and $\mathbf{J}$ is a t × t matrix of 1's, and # denotes Hadamard product.

Then the variance of errors if there are no missing data, is $\mathbf{H}_t + \mathbf{R}_t$.

Now delete rows and columns representing missing data to produce $\mathbf{R}_r$, the variance of errors in the sample. Then solve:

$$\begin{bmatrix} \mathbf{T}'\mathbf{R}_r^{-1}\mathbf{T} + \mathbf{H}^{-1} & \mathbf{T}'\mathbf{R}^{-1}\mathbf{Z}_r \\ \mathbf{Z}_r'\mathbf{R}_r^{-1}\mathbf{T} & \mathbf{Z}_r'\mathbf{R}_r^{-1}\mathbf{Z}_r + \mathbf{G}_p^{-1} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\tau}}^0 \\ \hat{\mathbf{a}}_p \end{bmatrix} = \begin{bmatrix} \mathbf{T}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}_r'\mathbf{R}^{-1}\mathbf{y} \end{bmatrix} \qquad [4.6]$$

The upper left submatrix of G that pertains to parents is $\mathbf{G}_p$. Now $\boldsymbol{\tau}^0$ is same as in AM, as is $\hat{\mathbf{a}}_p$.

To compute the entire $\hat{\mathbf{a}}$ vector, let $\mathbf{T}_1' = \mathbf{Z}_r$ written as though there are no missing data, $\mathbf{T}_2' = \mathbf{H}_d\mathbf{R}_r^{-1}$, where $\mathbf{H}_d$ is $\mathbf{H}_r$ with columns deleted for missing records. Then $\hat{\mathbf{a}} = \mathbf{T}_1'\hat{\mathbf{a}}_p + \mathbf{T}_2'\hat{\mathbf{e}}$, where $\hat{\mathbf{e}} = \mathbf{y} - \mathbf{T}\hat{\boldsymbol{\tau}}^0 - \mathbf{Z}_r\hat{\mathbf{a}}_p$. To compute the entire $\hat{\mathbf{e}}$ vector, including those for missing records, let $\mathbf{T}_3' = \mathbf{R}_c\mathbf{R}_r^{r-1}$, where $\mathbf{R}_c = \mathbf{R}_r$ with columns deleted for missing data. Then $\hat{\mathbf{e}} = \mathbf{T}_3'\hat{\mathbf{e}}$.

Illustration. We have these data:

	Records		
Animal	Trait		Parents
	1	2	
1	. . .	. . .	0–0
2	3	5	0–0
3	6	9	1–2
4	8	. . .	1–0
5	3	. . .	0–2
6	2	. . .	0–0

The relationship matrix for base animals, 1 and 2, is I. It is assumed that $\mathbf{G}_o = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$ and $\mathbf{R}_o = \begin{bmatrix} 5 & 2 \\ 2 & 6 \end{bmatrix}$, and

$$\mathbf{X}' = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 4 & 3 & 7 & 5 & 2 & 5 \end{bmatrix}$$

From the pattern of observations above,

$$\mathbf{Z} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$\mathbf{R}_r$ is a 12 × 12 matrix of block diagonal form with six diagonal blocks each being $\mathbf{R}_o$.

Then $\mathbf{R} = \mathbf{R}_r$ with rows (and columns) 1, 2, 8, 10, 12 deleted. This is:

$$\begin{bmatrix} 5 & 2 & 0 & 0 & 0 & 0 & 0 \\ & 6 & 0 & 0 & 0 & 0 & 0 \\ & & 5 & 2 & 0 & 0 & 0 \\ \text{Symmetric} & & & 6 & 0 & 0 & 0 \\ & & & & 5 & 0 & 0 \\ & & & & & 5 & 0 \\ & & & & & & 5 \end{bmatrix}$$

The resulting mixed model solution is $\boldsymbol{\beta}^0 = [1.204, .956]'$, $\hat{\mathbf{a}} = [.618, -.110, .101, .061, .624, -.017, .838, -.232, -.001, .048, -1.494, .498]'$, with ordering traits within animals.

Now we apply RAM to the solution. Let $\overline{\mathbf{Z}}_r$ be the incidence matrix when no records are missing. For our problem this is:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ .5 & 0 & .5 & 0 \\ 0 & .5 & 0 & .5 \\ .5 & 0 & 0 & 0 \\ 0 & .5 & 0 & 0 \\ 0 & 0 & .5 & 0 \\ 0 & 0 & 0 & .5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Columns represent in order trait 1 on animal 1, trait 2 on animal 1, trait 1 on animal 2, and trait 2 on animal 2. The rows represent all 6 animals with traits ordered in animals.

Then the actual $\mathbf{Z}_r$ matrix is $\overline{\mathbf{Z}}_r$ with rows 1, 2, 8, 10, 12 deleted. Var ($\boldsymbol{\epsilon}$) for no missing data is block diagonal with the 6 diagonal blocks being:

$$\begin{bmatrix} 5 & 2 \\ 2 & 6 \end{bmatrix}, \begin{bmatrix} 5 & 2 \\ 2 & 6 \end{bmatrix}, \begin{bmatrix} 6.5 & 1.5 \\ 1.5 & 7.0 \end{bmatrix}, \begin{bmatrix} 7.25 & 1.25 \\ 1.25 & 7.5 \end{bmatrix}, \begin{bmatrix} 7.25 & 1.25 \\ 1.25 & 7.5 \end{bmatrix}, \begin{bmatrix} 8 & 1 \\ 1 & 8 \end{bmatrix}.$$

Then $\mathbf{R}_r$ = the above matrix with rows (and columns) 1, 2, 8, 10, 12 deleted, and:

$$\mathbf{G}_p = \begin{bmatrix} 3.0 & -1.0 & 0 & 0 \\ & 2.0 & 0 & 0 \\ \text{Symmetric} & & 3.0 & -1.0 \\ & & & 2.0 \end{bmatrix}$$

Then the RAM mixed model equations give the solution

$\boldsymbol{\beta}^0$ = same as AM,
$\hat{\mathbf{a}}_p$ = same as first four elements of $\hat{\mathbf{a}}$ in AM solution.

To compute the entire $\hat{\mathbf{a}}$ vector,

$$\hat{\mathbf{a}} = \bar{\mathbf{Z}}_r\hat{\mathbf{a}}_p + \mathbf{T}_2'\,\hat{\mathbf{e}}$$

where $\mathbf{T}_2' = \mathbf{H}_a\mathbf{R}_r^{-1}$ and

$$\mathbf{H}_a = \begin{bmatrix} & & & & & & \\ & 1.5 & -.5 & 0 & 0 & 0 \\ & -.5 & 1.0 & 0 & 0 & 0 \\ & 0 & 0 & 2.25 & 0 & 0 \\ \mathbf{0}_{(8,7)} & 0 & 0 & -.75 & 0 & 0 \\ & 0 & 0 & 0 & 2.25 & 0 \\ & 0 & 0 & 0 & -.75 & 0 \\ & 0 & 0 & 0 & 0 & 3.0 \\ & 0 & 0 & 0 & 0 & -1.0 \end{bmatrix} \quad \mathbf{0}_{(4,7)}$$

This gives $\hat{\mathbf{a}}$ the same as in AM.

To compute the entire $\hat{\mathbf{e}}$ vector, including those for missing observations, $\hat{\mathbf{e}} = \mathbf{T}_3'\,\hat{\boldsymbol{\epsilon}}$, where $\mathbf{T}_3' = \mathbf{R}_c\mathbf{R}_r^{-1}$ and:

$$\mathbf{R}_c = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 5 & 2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 6 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

$\hat{\boldsymbol{\epsilon}}$ = [−.217, −.090, 1.568, 1.127, 1.706, −.167, −3.985]'.
$\hat{\mathbf{e}}$ = [0, 0, −.217, −.090, 1.303, 1.119, 1.177, .471, −.115, .046, −2.491, −.996]'.

The $\hat{\mathbf{e}}$ associated with observations are the same as $\hat{\mathbf{e}} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}^0 - \mathbf{Z}\hat{\mathbf{a}} - \mathbf{Z}\hat{\mathbf{p}}$ of the animal model.

5. Repeated Records Treated as Multiple Traits

The methods of Section 4 can be used for repeated records on a single trait, one model for which was described in Section 3. Thus, trait i could refer to the ith record (or parity). How does this relate to Section 3? These two models are equivalent provided the following is true regarding the model of Section 4:

$\mathbf{G}_o$ has all elements = σ_a^2.

$\mathbf{R}_o$ has $\sigma_p^2 + \sigma_e^2$ in diagonals and σ_p^2 in all off-diagonals.

This implies in the context of multiple traits that the genetic variance of each trait is the same and the genetic correlations between all pairs of traits is 1. Further, the environmental variance of all traits are equal and pairs of environmental correlations are equal. Finally the phenotypic variance of all traits are equal and all pairs of phenotypic correlations are equal.

Now $\mathbf{G}$ is singular, and the methods for this situation are required, Harville (6). Then the mixed model solution yields the same $\hat{\mathbf{a}}$ for all records (parities) on the same animal and these are the $\hat{\mathbf{a}}_i$ of the repeatability model. This, of course, is the assumption of the repeatability model of Section 2.

Clearly the methods of Section 2 are much easier than those of this section when the assumptions regarding $\mathbf{G}_o$ and $\mathbf{R}_o$ are true. The crucial question is whether it is logical to assume that this simple repeatability model is valid. We need better estimates of $\mathbf{G}_o$, $\mathbf{R}_o$ than are now available to resolve this issue. Most data available for analysis involve sequential culling, and therefore ANOVA type estimators are biased. It appears the REML estimators are unbiased by selection. Assume that we estimate $\mathbf{G}_o$ and $\mathbf{R}_o$. Then taking these estimates as parameter values we could evaluate increases in prediction error variances when the repeatability model is assumed. See Henderson (10) for a simple method to accomplish this. We could also evaluate bias due to use of the repeatability model when sequential culling has occurred. This would require simulation.

Illustration.

Records

	Parities		
Animals	1	2	3
1	5	3	4
2	2	4	6
3	3	7	2
4	4	1	5
5	6	3	2

$$\mathbf{A} = \begin{bmatrix} 1.0 & .5 & .75 & .5 & .25 \\ & 1.0 & .75 & .25 & .5 \\ & & 1.25 & .375 & .375 \\ & & & 1.0 & .125 \\ & & & & 1.0 \end{bmatrix}$$

$$\mathbf{X}' = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 4 & 3 & 5 & 3 & 6 & 4 & 7 & 3 & 2 & 4 & 5 & 6 & 4 \end{bmatrix}.$$

$\sigma_e^2 = 6$, $\sigma_a^2 = 3$, $\sigma_p^2 = 2$.

Regular mixed model solution regarding parities as repeated records are

$$\boldsymbol{\beta}^0 = [4.506, -1.90]',$$
$$\hat{\mathbf{a}} = [.061, .154, .234, -.197, .090]',$$
$$\hat{\mathbf{p}} = [.002, .019, .169, -.202, .011]'.$$

Multiple trait solution with

$$\mathbf{G}_0 = \begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix}, \mathbf{R}_0 = \begin{bmatrix} 8 & 2 & 2 \\ 2 & 8 & 2 \\ 2 & 2 & 8 \end{bmatrix}$$

$\boldsymbol{\beta}^0$ = same as before.

$$\hat{\mathbf{a}} = [.061, .061, .061, \ldots, .090, .090, .090]'$$

6. Additive Genetic and Maternal Model for Single Trait, Single Records, One or Zero Progeny Per Female

6.1 Animal Model

The model is:

$$\mathbf{y} = \mathbf{T}\boldsymbol{\tau} + \mathbf{Z}_{(a)}\mathbf{a} + \mathbf{Z}_{(m)}\mathbf{m} + \boldsymbol{\epsilon} \qquad [6.1]$$

where m are maternal effects.

$$\text{Var}\begin{bmatrix} \mathbf{a} \\ \mathbf{m} \end{bmatrix} = \begin{bmatrix} \mathbf{A}\sigma_a^2 & \mathbf{A}\sigma_{am} \\ \mathbf{A}\sigma_{am} & \mathbf{A}\sigma_m^2 \end{bmatrix} = \mathbf{G} \qquad [6.2]$$

$$\text{Var}(\boldsymbol{\epsilon}) = \mathbf{I}\sigma_e^2 + \mathbf{P}(\sigma_{am} + \sigma_m^2) = \mathbf{R} \qquad [6.3]$$

where $\mathbf{P}$ is a diagonal matrix with 0's in diagonals corresponding to animals with a record and a known dam and 1's corresponding to animals with a record and unknown dam. This forces the variance of every y in a noninbred, unselected population to be $\sigma_a^2 + \sigma_m^2 + \sigma_{am} + \sigma_e^2$ [See Quaas and Pollak (18)].

$\mathbf{Z}_{(a)}$ has the same form as Z of AM in Section 2.

Every row of $\mathbf{Z}_{(m)}$ is null except for a "1" under the column pertaining to a known dam. Then the mixed model equations are:

$$\begin{bmatrix} \mathbf{T}'\mathbf{R}^{-1}\mathbf{T} + \mathbf{H}^{-1} & \mathbf{T}'\mathbf{R}^{-1}\mathbf{Z}_{(a)} & \mathbf{T}'\mathbf{R}^{-1}\mathbf{Z}_{(m)} \\ \mathbf{Z}'_{(a)}\mathbf{R}^{-1}\mathbf{T} & \mathbf{Z}'_{(a)}\mathbf{R}^{-1}\mathbf{Z}_{(a)} + \mathbf{B}_{11} & \mathbf{Z}'_{(a)}\mathbf{R}^{-1}\mathbf{Z}_{(m)} + \mathbf{B}_{12} \\ \mathbf{Z}'_{(m)}\mathbf{R}^{-1}\mathbf{T} & \mathbf{Z}'_{(m)}\mathbf{R}^{-1}\mathbf{Z}_{(a)} + \mathbf{B}_{12} & \mathbf{Z}'_{(m)}\mathbf{R}^{-1}\mathbf{Z}_{(m)} + \mathbf{B}_{22} \end{bmatrix}$$
$$\begin{bmatrix} \boldsymbol{\tau}^0 \\ \hat{\mathbf{a}} \\ \hat{\mathbf{m}} \end{bmatrix} = \begin{bmatrix} \mathbf{T}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'_{(a)}\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'_{(m)}\mathbf{R}^{-1}\mathbf{y} \end{bmatrix}, \text{where } \mathbf{G}^{-1} = \begin{bmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}'_{12} & \mathbf{B}_{22} \end{bmatrix} \qquad [6.4]$$

6.2 Reduced Animal Model

The model is:

$$\mathbf{y} = \mathbf{T}\boldsymbol{\tau} + \mathbf{Z}_{(a)}\mathbf{a}_p + \mathbf{Z}_{(m)}\mathbf{m}_p + \mathbf{v} \qquad [6.5]$$

$\mathbf{Z}_{(a)}$ has the same form as $\mathbf{Z}_r$ in Section 2.2.

$\mathbf{Z}_{(m)}$ is formed from the columns of $\mathbf{Z}_{(m)}$ of Section 6.1 for parents. Thus, $\mathbf{Z}_{(a)}$ and $\mathbf{Z}_{(m)}$ have the same number of columns.

$$\text{Var}(\mathbf{v}) = \mathbf{R} \text{ of } [6.3] \text{ plus } \sigma_a^2\mathbf{D} = \mathbf{R}_r \qquad [6.6]$$

The matrix, D, is described following equation [2.8]. Let $\mathbf{Z}_r = [\mathbf{Z}_{(a)} : \mathbf{Z}_{(m)}]$. Then solve [6.7].

$$\begin{bmatrix} \mathbf{T}'\mathbf{R}_r^{-1}\mathbf{T} + \mathbf{H}^{-1} & \mathbf{T}'\mathbf{R}_r^{-1}\mathbf{Z}_r \\ \mathbf{Z}'_r\mathbf{R}_r^{-1}\mathbf{T} & \mathbf{Z}'_r\mathbf{R}_r^{-1}\mathbf{Z}_r + \mathbf{G}_r^{-1} \end{bmatrix} \begin{bmatrix} \boldsymbol{\tau}^0 \\ \hat{\mathbf{a}}_p \\ \hat{\mathbf{m}}_p \end{bmatrix} = \begin{bmatrix} \mathbf{T}'\mathbf{R}_r^{-1}\mathbf{y} \\ \mathbf{Z}'_r\mathbf{R}_r^{-1}\mathbf{y} \end{bmatrix} \qquad [6.7]$$

$$\mathbf{G}_r = \begin{bmatrix} \sigma_a^2\mathbf{A}_p & \sigma_{am}\mathbf{A}_p \\ \sigma_{am}\mathbf{A}_p & \sigma_m^2\mathbf{A}_p \end{bmatrix} \qquad [6.8]$$

where $\mathbf{A}_p$ is the relationship matrix for parents.

To compute all $\hat{\mathbf{a}}$ and $\hat{\mathbf{m}}$ write a $\mathbf{Z}_{(a)}$ matrix as though every animal has a record. Call this $\mathbf{T}'_1$.

Let $\mathbf{T}'_2 = \sigma_a^2\mathbf{D}\mathbf{R}_r^{-1}$ augmented with a row of O's for every animal with a missing record. Then:

$$\hat{\mathbf{a}} = \mathbf{T}'_1\hat{\mathbf{a}}_p + \mathbf{T}'_2\hat{\mathbf{v}} \qquad [6.9]$$

where $\hat{\mathbf{v}} = \mathbf{y} - \mathbf{T}\boldsymbol{\tau}^0 - \mathbf{Z}_r\begin{bmatrix} \hat{\mathbf{a}}_p \\ \hat{\mathbf{m}}_p \end{bmatrix}$

$$\text{Now } \hat{\mathbf{m}} = \mathbf{T}'_1\mathbf{m}_p + \mathbf{T}'_3\hat{\mathbf{v}} \qquad [6.10]$$

where $\mathbf{T}'_3 = \dfrac{\sigma_{am}}{\sigma_a^2}\mathbf{T}'_2$.

Illustration. We have the following records:

Animal	Parents	Records
1	0 – 0	. . .
2	0 – 0	. . .
3	1 – 0	3
4	1 – 0	5
5	1 – 2	7
6	1 – 3	9
7	1 – 4	2
8	1 – 0	8

Animals 1 and 2 are noninbred and unrelated. Note that females have no more than one tested progeny. If that were not the case, permanent environmental effects would be needed in the model. Maternal and direct permanent environmental effects are assumed to be uncorrelated.

$$\sigma_a^2 = 3,\ \sigma_m^2 = 2,\ \sigma_{am} = -1,\ \sigma_e^2 = 6.$$

$\mathbf{Z}_{(a)} = \mathbf{I}_{(8)}$ with first two rows deleted:

$$\mathbf{Z}_{(am)} = \begin{bmatrix} 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&0&0&0&0&0&0 \end{bmatrix}$$

$\mathbf{P} = \text{diag}\,(1\ 1\ 0\ 0\ 0\ 1)$.
The $\mathbf{R} = 6\mathbf{I} + (-1 + 2)\,\mathbf{P} = \text{diag}\,(7, 7, 6, 6, 6, 7)$

$$\mathbf{X}' = \begin{bmatrix} 1&1&1&1&1&1\\ 2&4&3&6&5&7 \end{bmatrix}$$

Then the mixed model solution is

$$\begin{aligned} \boldsymbol{\beta}^0 &= [2.014, .825],\\ \hat{\mathbf{a}} &= [-.052, .137, -.111, -.226,\\ &\quad .454, .290, -.822, .032]',\\ \hat{\mathbf{m}} &= [-.156, .411, .260, -.581,\\ &\quad -.010, -.072, -.141, -.097]'. \end{aligned}$$

The RAM solution proceeds as follows with parents being animals 1 to 4:

$$\mathbf{Z}_{(ra)} = \begin{bmatrix} 0&0&1&0\\ 0&0&0&1\\ .5&.5&0&0\\ .5&0&.5&0\\ .5&0&0&.5\\ .5&0&0&0 \end{bmatrix}.$$

$$\mathbf{Z}_{(rm)} = \begin{bmatrix} 0&0&0&0\\ 0&0&0&0\\ 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1\\ 0&0&0&0 \end{bmatrix}$$

$\mathbf{D} = \text{diag}(0, 0, .5, .5, .5, .75)$.

Then $\mathbf{R}r = \mathbf{R} + 3\mathbf{D} = \text{diag}\,(7, 7, 7.5, 7.5, 7.5, 9.25)$.

Parents are animals 1 to 4. Therefore, $\mathbf{G}_s = \mathbf{G}$ with rows (columns) 5 to 8 and 13 to 16 deleted.

This gives the solution

$\boldsymbol{\tau}^0$ = same as in AM.
$\hat{\mathbf{a}}_p$ = first four elements of $\hat{\mathbf{a}}$ of AM.
$\hat{\mathbf{m}}_p$ = first four elements of $\hat{\mathbf{m}}$ of AM.

To compute the entire $\hat{\mathbf{a}}$ vector:

$$\hat{\mathbf{a}} = \mathbf{T}_1' \hat{\mathbf{a}}_p + \mathbf{T}_2' \hat{\mathbf{v}}$$

where $\mathbf{T}_1'$ is $\mathbf{Z}_{(ra)}$ written as though no records are missing. This is

$$\begin{bmatrix} 1&0&0&0\\ 0&1&0&0\\ & \mathbf{Z}_{ra} & \end{bmatrix}$$

and:

$$\mathbf{T}_2' = \begin{bmatrix} 0&0&0&0&0&0\\ 0&0&0&0&0&0\\ & & 3\mathbf{D} & & & \end{bmatrix} \mathbf{R}_r^{-1}$$

Then $\hat{\mathbf{a}}$ is the same as in AM, and

$$\hat{\mathbf{m}} = \mathbf{T}_3' \hat{\mathbf{m}}_p + \mathbf{T}_3' \hat{\mathbf{v}},$$

where $\mathbf{T}_3' = (-1/3)\mathbf{T}_2'$.
This gives $\hat{\mathbf{m}}$ = same as in AM.

For the above computations:

$$\begin{aligned} \hat{\mathbf{v}} &= \mathbf{y} - \mathbf{X}\boldsymbol{\beta}^0 - \mathbf{Z}_r \begin{bmatrix} \hat{\mathbf{a}}_p \\ \hat{\mathbf{m}}_p \end{bmatrix}\\ &= [-.553, -.089, 2.057, 1.858, -3.419, .237]'. \end{aligned}$$

7. Nonadditive Genetic Model, Single Trait

We assume a noninbred population in linkage equilibrium so that the Cockerham (2) model applies. Let the model be:

$$\mathbf{y} = \mathbf{T}\boldsymbol{\tau} + \mathbf{Z}_{(g_1 + g_2 + \ldots + g_t + p)} + \mathbf{e} \quad [7.1]$$

The $\mathbf{g}_i$ refers to type of gene action, e.g., additive = **a**, dominance = **d**, additive by additive = **aa**, and **p** refers to permanent environmental effects.

$$\begin{aligned} \text{Var}(\mathbf{a}) &= \mathbf{A}\sigma_a^2 \qquad \text{Var}(\mathbf{aa}) = \mathbf{A}\#\mathbf{A}\sigma_{aa} \\ &\qquad\qquad \text{etc.} \\ \text{Var}(\mathbf{d}) &= \mathbf{D}\sigma_d^2 \qquad \text{Var}(\mathbf{p}) = \mathbf{I}\sigma_p^2 \end{aligned} \quad [7.2]$$

where # denotes Hadamard product.

$$\begin{aligned} &\text{Let } \text{var}(\mathbf{g}_i) = \mathbf{G}_i\sigma_i^2 \\ &\text{Cov}(\mathbf{g}_i, \mathbf{g}_j') = \mathbf{0} \text{ for all } i \neq j. \\ &\text{Cov}(\mathbf{g}_i, \mathbf{p}') = \mathbf{0} \text{ for all } i. \end{aligned}$$

We could set up mixed model equations that contain all of the $\mathbf{g}_i$ as well as **p**, but this could be an extremely large set of equations. Further, none of the $\mathbf{G}_i$ except for **A** has an easily computable inverse. Consequently, a better method is to let

$$\begin{aligned} \mathbf{t} &= \mathbf{g}_1 + \ldots + \mathbf{g}_t + \mathbf{p} \\ \text{Var}(\mathbf{t}) &= \mathbf{C} = \sum_{i=1}^{t} \mathbf{G}_i\sigma_i^2 + \mathbf{I}\sigma_p^2 \\ \text{Var}(\mathbf{e}) &= \mathbf{I}\sigma_e^2 \end{aligned} \quad [7.3]$$

Then the mixed model equations are

$$\begin{bmatrix} \mathbf{T}'\mathbf{T} + \mathbf{H}^{-1}\sigma_e^2 & \mathbf{T}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{T} & \mathbf{Z}'\mathbf{Z} + \mathbf{C}^{-1}\sigma_e^2 \end{bmatrix} \begin{bmatrix} \boldsymbol{\tau}^0 \\ \hat{\mathbf{t}} \end{bmatrix} = \begin{bmatrix} \mathbf{T}'\mathbf{y} \\ \mathbf{Z}'\mathbf{y} \end{bmatrix} \quad [7.4]$$

Z is as described for single records in Section 2 and for multiple records in Section 3.

One purpose of the nonadditive genetic model is to design mating strategies to maximize expected mean of progeny production.

Now $\hat{\mathbf{t}}$ is BLUP of future production. Consequently we could predict t for one progeny of each of the possible matings among available males and females that were evaluated by the solution to [7.4]. Let the vector of **t** of such potential matings be $\mathbf{t}_o$. Then compute

$$\begin{bmatrix} \mathbf{C}_{pp} & \mathbf{C}_{po} \\ \mathbf{C}_{po}' & \mathbf{C}_{oo} \end{bmatrix}$$

where $\mathbf{C}_{pp}$ is the **C** of [7.3]. Then $\hat{\mathbf{t}}_o' = \mathbf{C}_{po}'\mathbf{C}_{pp}^{-1}\hat{\mathbf{t}}_p$ where $\mathbf{t}_p$ is $\hat{\mathbf{t}}$ of [7.4].

One might do a preliminary selection of sires and dams based on their $\hat{\mathbf{a}}$. This would reduce the order of $\mathbf{C}_{po}$.

If BLUP of a is wanted, this is $\hat{\mathbf{a}} = \sigma_a^2\mathbf{A}\mathbf{C}^{-1}\hat{\mathbf{t}}$. [7.5]

Also, $\hat{\mathbf{d}} = \sigma_d^2\mathbf{D}\,\mathbf{C}^{-1}\hat{\mathbf{t}}$, etc.

The reduced animal model is not feasible for nonadditive genetic models because no simple $\mathbf{Z}_i$ exists.

Illustration with Repeated Records.

Animals	Parents	Records
1	0 – 0	. . .
2	0 – 0	2, 4, 6
3	1 – 2	9, 5
4	1 – 0	3, 2
5	0 – 2	6

A for base animals 1, 2 = **I**.

$$\begin{aligned} \text{Model: } \mathbf{y} &= \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{a} + \mathbf{Z}\mathbf{a}\mathbf{a} + \mathbf{Z}\mathbf{p} + \mathbf{e} \\ &= \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{t} + \mathbf{e} \end{aligned}$$

$\text{Var}(\mathbf{a}) = 4\mathbf{A}$. $\text{Var}(\mathbf{aa}) = 3\mathbf{A} \# \mathbf{A}$. $\text{Var}(\mathbf{p}) = 2\mathbf{I}$. $\text{Var}(\mathbf{e}) = 8\mathbf{I}$. Then $\text{var}(\mathbf{t}) = \text{Var}(\mathbf{a}) + \text{Var}(\mathbf{aa}) + \text{Var}(\mathbf{p}) =$

$$\mathbf{C} = \begin{bmatrix} 9.0 & 0 & 2.75 & 2.75 & 0 \\ & 9.0 & 2.75 & 0 & 2.75 \\ & & 9.0 & 1.1875 & 1.1875 \\ \text{symmetric} & & & 9.0 & 0 \\ & & & & 9.0 \end{bmatrix}$$

$$\mathbf{X}' = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 4 & 5 & 6 & 2 & 3 & 4 \end{bmatrix}$$

The mixed model solution is:

$$\begin{aligned} \boldsymbol{\beta}^0 &= [.8865, 1.0308] \\ \hat{\mathbf{t}} &= [-.1026 \quad .0789 \quad .2825 \quad -.6522 \quad .5190]'. \end{aligned}$$

Now

$$\begin{aligned} \hat{\mathbf{a}} &= 4\mathbf{A}\mathbf{C}^{-1}\hat{\mathbf{t}} \\ &= [-.0746, .1124, .0998, -.2707, .2330]', \\ \hat{\mathbf{a}}\mathbf{a} &= 3\mathbf{A}\#\mathbf{A}\mathbf{C}^{-1}\hat{\mathbf{t}} \\ &= [-.0280, .0097, .1017, -.2258, .1682]', \end{aligned}$$

and

$$\hat{\mathbf{p}} = 2\mathbf{IC}^{-1}\hat{\mathbf{t}}$$
$$= [0, -.0432, .0810, -.1556, .1179]'.$$

$$\hat{\mathbf{a}} + \hat{\mathbf{aa}} + \hat{\mathbf{p}} = \hat{\mathbf{t}}$$

8. Nonadditive Genetic Model, Multiple Traits, Single Records

We have k noninbred animals and m traits. The model is:

$$\mathbf{y} = \mathbf{T}\boldsymbol{\tau} + \mathbf{Zt} + \mathbf{e} \qquad [8.1]$$

where **t** denotes producing abilities and has order km with traits ordered within animals. Also the n<km records are ordered in this manner.

Let **t** be the sum of two or more types of genetic effects plus permanent environmental effects. We shall illustrate with $\mathbf{t} = \mathbf{a} + \mathbf{d} + \mathbf{aa} + \mathbf{p}$.

Let $\mathbf{G}_{(a)}$ be the additive genetic variance-covariance matrix and similarly for $\mathbf{G}_{(d)}$, $\mathbf{G}_{(aa)}$, $\mathbf{G}_{(p)}$. These matrices have order, m.

Then $\text{var}(\mathbf{a}) = \mathbf{A} \otimes \mathbf{G}_{(a)}$, $\text{var}(\mathbf{d}) = \mathbf{D} \otimes \mathbf{G}_{(d)}$, $\text{var}(\mathbf{aa}) = \mathbf{A} \# \mathbf{A} \otimes \mathbf{G}_{(aa)}$, $\text{var}(\mathbf{p}) = \mathbf{I} \otimes \mathbf{G}_{(p)}$, and $\text{var}(\mathbf{t}) = \mathbf{G} = \text{var}(\mathbf{a}) + \text{var}(\mathbf{d}) + \text{var}(\mathbf{aa}) + \text{var}(\mathbf{p})$. [8.2]

Let $\mathbf{R}_o$ be temporary environmental variance-covariance matrix. Then $\mathbf{R} = \mathbf{I} \otimes \mathbf{R}_o$, with rows and columns deleted for missing records.

Let **Z** be a km order identity matrix with rows deleted for missing records. Then the mixed model equations are:

$$\begin{bmatrix} \mathbf{T}'\mathbf{R}^{-1}\mathbf{T} + \mathbf{H}^{-1} & \mathbf{T}'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{T} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} \end{bmatrix} \begin{bmatrix} \boldsymbol{\tau}^0 \\ \mathbf{t} \end{bmatrix} = \begin{bmatrix} \mathbf{T}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \end{bmatrix} \qquad [8.3]$$

Then $\hat{\mathbf{a}} = \mathbf{A} \otimes \mathbf{G}_{(a)}\mathbf{G}^{-1}\hat{\mathbf{t}}$, etc. for $\mathbf{aa}, \hat{\mathbf{d}}, \hat{\mathbf{p}}$.

Illustration.

Records

Animals	Traits 1	2	3	Parents
1	3	5	6	0 – 0
2	8	9	. . .	0 – 0
3	7	. . .	6	1 – 2
4	. . .	5	. . .	1 – 2
5	. . .	4	7	1 0

A for animals 1, 2 = **I**.

There are two types of gene action, additive and additive × additive:

$$\mathbf{G}_{(a)} = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 5 \end{bmatrix}, \mathbf{G}_{(aa)} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix},$$

$$\mathbf{G}_{(p)} = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 4 & 5 \\ 2 & 5 & 6 \end{bmatrix}, \mathbf{R}_o = \begin{bmatrix} 8 & 3 & 2 \\ 3 & 9 & 4 \\ 2 & 4 & 7 \end{bmatrix}$$

Let $\mathbf{t} = \mathbf{a} + \mathbf{aa} + \mathbf{p}$.

Then $\mathbf{G} = \text{var}(\mathbf{t}) = \mathbf{A} \otimes \mathbf{G}_{(a)} + \mathbf{A}\#\mathbf{A} \otimes \mathbf{G}_{(aa)} + \mathbf{I} \otimes \mathbf{G}_{(p)}$.

$$\mathbf{X}' = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Some of the elements of **G** are

$$\begin{bmatrix} 8 & 4 & \cdots & 1.25 & .75 \\ 4 & 11 & \cdots & 2.75 & 1.5 \\ \vdots & \vdots & & \vdots & \vdots \\ 1.25 & 2.75 & \cdots & 11 & 9 \\ .75 & 1.5 & \cdots & 9 & 15 \end{bmatrix}$$

$$\mathbf{R} = \begin{bmatrix} 8 & 3 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ & 9 & 4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ & & 7 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ & & & 8 & 3 & 0 & 0 & 0 & 0 & 0 \\ & & & & 9 & 0 & 0 & 0 & 0 & 0 \\ & & & & & 8 & 2 & 0 & 0 & 0 \\ & \text{Symmetric} & & & & & 7 & 0 & 0 & 0 \\ & & & & & & & 9 & 0 & 0 \\ & & & & & & & & 9 & 4 \\ & & & & & & & & & 7 \end{bmatrix}$$

The mixed model solution is

$$\boldsymbol{\beta}^0 = (5.737, 5.783, 6.947)',$$
$$\hat{\mathbf{t}} = (-1.360, -.664, -.667, 1.255, 1.725, 1.290, .493, -.141, -.618, -.090, -.301, -.473, -.460, -.865, -.060)', \text{ ordered traits in animals.}$$
$$\text{Then } \hat{\mathbf{a}} = [-.486, -.448, -.263, .582, .640, .160, .182, .132, .194, -.005, -.011, -.105, -.388, -.513, .023]'$$

$$\hat{\mathbf{aa}} = [-.326, -.164, -.201, .312, .446, .278, .123, .012, -.174, -.031, -.076, -.100, -.125, -.265, .049]',$$
$$\hat{\mathbf{p}} = [-.548, -.052, -.203, .360, .639, .853, .188, -.285, -.054, -.214, -.268, .053, .086, .133]'.$$

Note that $\hat{\mathbf{a}} + \hat{\mathbf{aa}} + \hat{\mathbf{p}} = \hat{\mathbf{t}}$.

$$\hat{\mathbf{e}} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}^0 - \mathbf{Z}\hat{\mathbf{t}}.$$
$$= [-1.377, .119, .281, 1.008, 1.492, .769, -.330, -.482, -.918, .113]'$$

$\hat{\mathbf{e}}_o$ referring to missing data are calculated by $\mathbf{R}_d\mathbf{R}^{-1}\hat{\mathbf{e}}$, where $\mathbf{R}_d = \mathbf{I} \otimes \mathbf{R}_o$ with columns 6, 8, 10, 12, 13 corresponding to missing data deleted and the rows 1, 2, 3, 4, 5, 7, 9, 11, 14, 15 corresponding to observations deleted.

This gives $\hat{\mathbf{e}}_o = [.712, .027, -.161, -.214, -.239]'$

9. Nonadditive, Direct Genetic and Maternal Effects, One Trait, Multiple Records

We have a noninbred population in linkage equilibrium with zero or several records per animal on a single trait. The model is:

$$\mathbf{y} = \mathbf{T}\boldsymbol{\tau} + \mathbf{Z}_{(g)}\mathbf{g} + \mathbf{Z}_{(m)}\mathbf{m} + \boldsymbol{\epsilon} \qquad [9.1]$$

where g represents direct genetic effects and direct permanent environmental effects and m represents maternal genetic effects and maternal permanent environmental effects.

$$\mathbf{g} = \mathbf{g}_{(a)} + \mathbf{g}_{(d)} + \mathbf{g}_{(aa)} + \text{etc.} + \mathbf{g}_{(p)}, \qquad [9.2]$$
$$\mathbf{m} = \mathbf{m}_{(a)} + \mathbf{m}_{(d)} + \mathbf{m}_{(aa)} + \text{etc.} + \mathbf{m}_{(p)}.$$

$$\text{Var}\begin{bmatrix}\mathbf{g}_{(a)}\\ \mathbf{m}_{(a)}\end{bmatrix} = \mathbf{C}_{(a)} \otimes \mathbf{A}$$
$$\text{Var}\begin{bmatrix}\mathbf{g}_{(d)}\\ \mathbf{m}_{(d)}\end{bmatrix} = \mathbf{C}_{(d)} \otimes \mathbf{D} \qquad [9.3]$$
$$\text{Var}\begin{bmatrix}\mathbf{g}_{(p)}\\ \mathbf{d}_{(m)}\end{bmatrix} = \mathbf{C}_{(p)} \otimes \mathbf{I}$$

$\mathbf{C}_{(a)}$ is 2 × 2 variance-covariance matrix for direct additive and maternal additive effects.

$\mathbf{C}_{(d)}$ is same for direct and maternal dominance effects, etc. Then:

$$(\mathbf{G}) = \mathbf{C}_{(a)} \otimes \mathbf{A} + \mathbf{C}_{(d)} \otimes \mathbf{D} + \ldots + \mathbf{C}_{(p)} \otimes \mathbf{I} \qquad [9.4]$$

$$\text{Var}(\boldsymbol{\epsilon}) = \mathbf{I}\sigma_e^2 + \mathbf{P} - \mathbf{R} \qquad [9.5]$$

$\mathbf{P}$ is a diagonal matrix to account for unknown dams. Its diagonal elements are 0 for records on animals with known dams. For animals with an unknown dam it is

$$\sigma^2_{m(a)} + \sigma^2_{m(d)} + \ldots + \sigma^2_{m(p)} + \sigma_{(a)m(a)} + .5\sigma_{g(aa)m(aa)} + .25\sigma_{g(aaa)m(aaa)} + \ldots \qquad [9.6]$$

Note that the covariance term is twice the relationship between dam and progeny for a particular effect multiplied by $\sigma_{g(i)m(i)}$. These relationships are .5, .25, .125, for **a, aa, aaa,** respectively, and 0 for any effect involving dominance.

$\mathbf{Z}_{(g)}$ is same as $\mathbf{Z}_{(a)}$ of Section 6.1 and $\mathbf{Z}_{(m)}$ is the same as $\mathbf{Z}_{(m)}$ of that section.

Then the mixed model equations are:

$$\begin{bmatrix}\mathbf{T}'\mathbf{R}^{-1}\mathbf{T} + \mathbf{H}^{-1} & \mathbf{T}'\mathbf{R}^{-1}\mathbf{Z}_{(g)} & \mathbf{T}'\mathbf{R}^{-1}\mathbf{Z}_{(m)}\\ \mathbf{Z}'_{(g)}\mathbf{R}^{-1}\mathbf{T} & \mathbf{Z}'_{(g)}\mathbf{R}^{-1}\mathbf{Z}_{(g)} + \mathbf{B}_{11} & \mathbf{Z}'_{(g)}\mathbf{R}^{-1}\mathbf{Z}'_{(m)} + \mathbf{B}_{11}\\ \mathbf{Z}'_{(m)}\mathbf{R}^{-1}\mathbf{T} & \mathbf{Z}'_{(m)}\mathbf{R}^{-1}\mathbf{Z}_{(g)} + \mathbf{B}'_{12} & \mathbf{Z}'_{(m)}\mathbf{R}^{-1}\mathbf{Z}_{(m)} + \mathbf{B}_{22}\end{bmatrix}$$
$$\begin{bmatrix}\boldsymbol{\tau}^0\\ \hat{\mathbf{g}}\\ \hat{\mathbf{m}}\end{bmatrix} = \begin{bmatrix}\mathbf{T}'\mathbf{R}^{-1}\mathbf{y}\\ \mathbf{Z}'_{(g)}\mathbf{R}^{-1}\mathbf{y}\\ \mathbf{Z}'_{(m)}\mathbf{R}^{-1}\mathbf{y}\end{bmatrix} \qquad [9.7]$$

where $\mathbf{G}^{-1} = \begin{bmatrix}\mathbf{B}_{11} & \mathbf{B}_{12}\\ \mathbf{B}'_{12} & \mathbf{B}_{22}\end{bmatrix}$

If $\begin{bmatrix}\hat{\mathbf{g}}_{(a)}\\ \hat{\mathbf{m}}_{(a)}\end{bmatrix}$ is wanted, this can be computed by:

$$[\mathbf{C}_{(a)} \otimes \mathbf{A}]\,\mathbf{G}^{-1}\begin{bmatrix}\hat{\mathbf{g}}\\ \hat{\mathbf{m}}\end{bmatrix} \qquad [9.8]$$

and similarly for $\begin{bmatrix}\hat{\mathbf{g}}_{(d)}\\ \hat{\mathbf{m}}_{(d)}\end{bmatrix}$, etc.

Illustration.

Animals	Parents	Records
1	0 - 0	. . .
2	0 - 0	3, 5, 6
3	1 - 2	8, 7
4	1 - 2	9, 4
5	0 - 4	3

$$\mathbf{X}' = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 2 & 4 & 3 & 5 & 4 & 3 \end{bmatrix}$$

Types of gene action are additive, additive × additive, additive × additive × additive, dominance. In addition producing ability and maternal effects include permanent environment.

Thus,

$$\mathbf{g} = \mathbf{g}_{(a)} + \mathbf{g}_{(aa)} + \mathbf{g}_{(aaa)} + \mathbf{g}_{(d)} + \mathbf{g}_{(p)},$$
$$\mathbf{m} = \mathbf{m}_{(a)} + \mathbf{m}_{(aa)} + \mathbf{m}_{(aaa)} + \mathbf{m}_{(d)} + \mathbf{m}_{(p)}.$$

$$\mathbf{C}_{(a)} = \begin{bmatrix} 5 & 3 \\ 3 & 6 \end{bmatrix}, \mathbf{C}_{(aa)} = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix},$$

$$\mathbf{C}_{(aaa)} = \begin{bmatrix} 6 & 2 \\ 2 & 9 \end{bmatrix}, \mathbf{C}_{d} = \begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix},$$

$$\mathbf{C}_{(p)} = \begin{bmatrix} 5 & 3 \\ 3 & 4 \end{bmatrix}, \text{ and } \sigma_e^2 = 8.$$

The **C** parameters and relationships give:

$$\mathbf{G} = \begin{bmatrix} 23 & 0 & 4.25 & 4.25 & 1.594 & 9 & 0 & 2.25 & 2.25 & .906 \\ & 23 & 4.25 & 4.25 & 1.594 & 0 & 9 & 2.25 & 2.25 & .906 \\ & & 23 & 5 & 1.594 & 2.25 & 2.25 & 9 & 2 & .906 \\ & & & 23 & 4.25 & 2.25 & 2.25 & 2 & 9 & 2.25 \\ & & & & 23 & .906 & .906 & .906 & 2.25 & 9 \\ & & & & & 26 & 0 & 4.875 & 4.875 & 1.828 \\ & & & & & & 26 & 4.875 & 4.875 & 1.828 \\ & & & & & & & 26 & 5.875 & 1.828 \\ & & & & & & & & 26 & 4.875 \\ & & & & & & & & & 26 \end{bmatrix}$$

To compute **P**, note that the first three records involve animal 2 with an unknown dam. Therefore the first three elements of **P** are:

$$6 + 3 - 9 + 4 + 4 + 1(3) + .5(2) + .25(2) = 30.5.$$

Consequently **R** = diagonal (38.5, 38.5, 38.5, 8, 8, 8, 8, 8).

Then we have var(y) with all diagonal elements = 61.5. The mixed model solution is:

$$\boldsymbol{\beta}' = [-1.078, 2.068]',$$
$$\hat{\mathbf{g}} = [-.118, .532, 1.111, -1.443, -.778]',$$
$$\hat{\mathbf{m}} = [-.157, -.016, .294, -1.146, -.426]',$$

and

$$\begin{bmatrix} \hat{\mathbf{g}}_{(a)} \\ \hat{\mathbf{m}}_{(a)} \end{bmatrix} = \mathbf{C}_a \otimes \mathbf{AG}^{-1} \begin{bmatrix} \hat{\mathbf{g}} \\ \hat{\mathbf{m}} \end{bmatrix}$$

$= [-.082, .054, .140, -.250, -.220, -.102, -.042, .021, -.266, -.190]'$ and similarly for the other types of gene action and for $\hat{\mathbf{g}}_p$, $\hat{\mathbf{m}}_p$. Note that $\mathbf{g}_{(a)} + \mathbf{g}_{(aa)} + \mathbf{g}_{(aaa)} + \mathbf{g}_{(d)} + \mathbf{g}_{(p)} = \hat{\mathbf{g}}$ and similarly for $\hat{\mathbf{m}}$.

10. Grouping

One of the more controversial questions in the application of BLUP in either a sire model or an animal model is whether to include genetic groups and if so how to form groups. The original BLUP sire program at Cornell grouped by stud and by year of birth of the sire. At that time (1970), $\mathbf{A}^{-1}$ was not utilized and group solutions differed dramatically. Generally there was a pronounced upward trend in the solutions, indicating a genetic trend. Also studs differed greatly. With the introduction of $\mathbf{A}^{-1}$ in the equations (1974), the group solutions differed much less than before. Consequently grouping by studs was discontinued.

As further studies of selection progressed and in particular from the results of (11), it became apparent that under a certain set of assumptions no grouping should be done. These assumptions are multivariate normality, variance and covariance known to proportionality, selections made by translation invariant linear functions of the records, and these records are used in the analysis. This being true dairy cattle breeders have come, I think, to a consensus that grouping is needed only to account for unknown parents.

The grouping strategy used by Westell (19) and to be implemented by the USDA is compared with **M'u** type selection (15). These give identical results. A numerical example is presented in that paper and the general principles are presented.

Partition $\mathbf{a}'$ as $[\mathbf{a}_b' : \mathbf{a}_d']$ when $\mathbf{a}_b$ refers to additive genetic values of base population animals plus

"phantom" parents, and d to descendants. Partition **A** as:

$$\begin{bmatrix} \mathbf{A}_{bb} & \mathbf{A}_{bd} \\ \mathbf{A}_{bd} & \mathbf{A}_{dd} \end{bmatrix}$$

Let $\mathbf{M}_b'$ describe group into which an element of $\mathbf{a}_b$ is assigned. This matrix for example might look like:

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

Group 1 contains animals 1, 2, 3; group 2 contains animals 4 and 6; and group 3 contains animals 5 and 7.

Then the coefficients of $\overline{\mathbf{X}}_g$, the incidence matrix for groups when all animals have a single record is $[\mathbf{A}_{bb} : \mathbf{A}_{bd}]\mathbf{M}_b$.

Then the evaluation of the ith animal is $\hat{a}_i + \overline{\mathbf{X}}_{gi}\mathbf{g}^0$ where $\overline{\mathbf{X}}_{gi}\mathbf{g}^0$ is the contribution of **g** to the ith animal. Usually $\mathbf{g}^0$ is not unique so we need to consider estimable functions, for example, animal differences.

The same evaluation can be obtained from mixed model equations without groups, but modified for M'u type selection (11). Let $\mathbf{a}^0$ be some solution to these equations, with the understanding that $\mathbf{a}^0$ is usually not unique. But differences using the elements of $\mathbf{a}^0$ are the same as differences among groups plus additive values when grouping is done.

11. Conclusions

There are many different animal models to consider. Modelling is a crucial issue, as is efficiency of computation as exemplified by use of RAM when possible and by combining subvectors, e.g., **p** and **e** in single trait, one record situations and $\mathbf{d} = \Sigma\, \mathbf{d}_{(i)}$ and $\mathbf{m} = \Sigma_i\, \mathbf{m}_{(i)}$ as described in Section 9.

Care must be used in RAM and in maternal models, so that the diagonals of var(**y**) are equal in a noninbred population. This can be checked by computing $\mathbf{ZGZ}' + \mathbf{R} = \text{var}(\mathbf{y})$.

The animal model or its equivalent, RAM, would appear to be the method of choice for evaluation of breeding values and producing abilities, and for the prediction of future records because if all records are used, selection is by translation invariant functions, and **G** and **R** are known to proportionality, the predictions are BLUP and in the class of such selection functions maximize the expected mean of selected individuals.

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Kennedy, Schaeffer and Sorensen (1988)

Kennedy, B.W., Schaeffer, L.R., and Sorensen, D.A. 1988. Genetic properties of animal models. Journal of Dairy Science, vol. 71, supplement 2, pp. 17 to 26.

The problem

Kennedy, Schaeffer and Sorensen discuss the genetic properties of the animal model. They show that when all the data and the whole pedigree used in selection are included in the model, BLUP accounts for selection, nonrandom mating and the change in genetic means over time. They also discuss the effects of inbreeding and of Mendelian sampling.

Points to note while reading

Note the idea that the breeding value of an animal is the average of its parents plus a Mendelian sampling deviation, and how the animal model estimates this deviation. This paper explains why the genetic trends estimated by the animal model are reliable, provided the information is complete.

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Genetic Properties of Animal Models

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ABSTRACT

For traits controlled by a large number of loci acting additively, use of an animal model with the additive genetic relationship matrix can account for changes in genetic mean and variance due to chance or selection. Changes in genetic variance can result from inbreeding and linkage disequilibrium as a result of selection. Through the relationship matrix, breeding values are expressed as linear functions of breeding values of base population animals and subsequent Mendelian sampling contributions that are unaffected by selection and account for inbreeding. Correct definition of the base population is important. With finite numbers of loci, an animal model does not account for changes in genetic variance due to changes in gene frequency. Genotypic values can be expressed in terms of separate gametic contributions of each parent by gametic models, and a gametic relationship matrix can be constructed. With dominance, use of an animal model with the dominance relationship matrix leads to BLUP of dominance genetic merit if number of loci is sufficiently large and no inbreeding occurs. With inbreeding, inclusion of the dominance relationship matrix (even if modified to account for inbreeding contributions) does not account for inbreeding depression. However, effects of inbreeding depression might be accommodated by including the inbreeding coefficient as a covariate in the model.

INTRODUCTION

Statistical models applied to data obtained from animals attempt to describe biological processes and effects quantitatively. In practice, the model only can approximate biological reality. The goal is to fit a practical model that describes the biological situation as closely as possible. For animal breeding, increasing use is being made of animal models for prediction of breeding values, estimation of selection response, and estimation of genetic parameters such as heritability.

Henderson (8) notes that many different animal models can be applied to different biological situations and data structures. He has described the statistical bases and outlined computations for several different animal models (8). The purpose of this paper is to examine and illustrate genetic properties of these models and to discuss conditions under which they may represent either good or poor approximations for the true models.

ANIMAL MODELS

A common feature of all animal models is that they use the additive genetic relationship matrix (**A**) either directly or indirectly. Diagonal elements of **A** equal $1 + F_i$, where F_i is coefficient of inbreeding for animal i; off-diagonal elements equal the numerator of Wright's (22) coefficient of relationship between animals i and j. When multiplied by the additive genetic variance (σ_a^2), $\mathbf{A}\sigma_a^2$ describes the variance-covariance structure among additive genetic (breeding) values of animals (**a**). Also, with most animal models, average breeding value, E(**a**), is **0**.

In animal populations, however, genetic mean and variance seldom are constant but change as a consequence of finite population size or selection (4). In closed populations of finite size, inbreeding accumulates each generation (as does genetic drift) and affects both genetic means and variances. Genetic means and variances also are changed by selection. Genetic variances change because of changes in gene frequency and establishment of covariances between genotypic values of different loci (linkage disequilibrium) (1).

Mendelian Sampling

Kennedy and Sorensen (11) have shown that animal models by use of **A** or $\mathbf{A}^{-1}$ can accommodate some of the changes in genetic mean and

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variance. The breeding value of animal i (a_i) can be expressed as:

$$a_i = .5a_s + .5a_d + w_i \qquad [1]$$

where a_s and a_d are breeding values of the sire and dam of the animal, .5 is the regression of a_i on a_s or a_d, and w_i represents the Mendelian sampling contribution to a_i. The Mendelian sampling terms have an expected value of 0 and variance of $.5(1 - \bar{F})\sigma_a^2$, where $\bar{F}$ is the average inbreeding coefficient of the animal's parents. Selection affects distribution of a_s and a_d but not w_i, which is normal and independent of parental breeding values if the number of loci is large enough, strictly infinite (1, 2). If a parent of an animal is not known or not included in **A**, then half its breeding value is included in w_i, and distribution of w_i reflects Mendelian sampling terms plus the contribution of the unknown parent. If the unknown parents are unselected, then expected value of w_i still is 0. However, variance of w_i is σ_a^2 if both parents are not known or $(.75 - .25F_p)\sigma_a^2$ if only one parent is not known, assuming the unknown parents are not inbred.

The model for [1] can be extended back one generation such that a_i can be written as a function of grandparental breeding values and additional Mendelian sampling terms:

$$a_i = .25a_{pgs} + .25a_{pgd} + .25a_{mgs} + .25a_{mgd} + .5w_s + .5w_d + w_i$$

where pgs, pgd, mgs, and mgd identify paternal grandsire, paternal granddam, maternal grandsire, and maternal granddam and s and d identify sire and dam. Again, Mendelian sampling terms are unaffected by selection in the previous generation and are independent of ancestral breeding values. The process can be extended back to any group of ancestors or to a conceptual base population of animals, presumably unrelated, with unknown or unidentified parents.

Expression of A as TWT'

The vector of breeding values (**a**), therefore, can be expressed as the product of a matrix of appropriate regression coefficients (**T**) and a vector of ancestral breeding values and Mendelian sampling terms from subsequent generations (**w**):

$$\mathbf{a} = \mathbf{Tw}$$

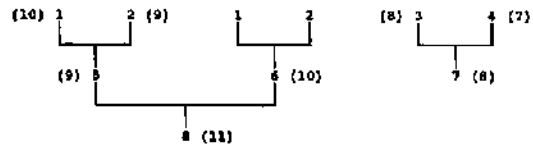


Figure 1. Pedigree and phenotypic records of animals from Kennedy and Sorensen (11) for illustration of an animal model; records are in parentheses beside animal number.

Accordingly,

$$\mathrm{Var}(\mathbf{a}) = \mathbf{A}\sigma_a^2 = \mathbf{T}[\mathrm{Var}(\mathbf{w})]\mathbf{T}'\sigma_a^2 = \mathbf{TWT}'\sigma_a^2$$

where **W** is a diagonal matrix with diagonal elements $.5(1 - \bar{F}_i)\sigma_a^2$, $(.75 - .25F_p)\sigma_a^2$, or σ_a^2 depending if both, one, or no parents are identified. This relationship has been used to derive simple computational methods for obtaining $\mathbf{A}^{-1}$ (6). For computation of **T**, ordering a such that parents precede offspring, usually is convenient.

Decomposition of **A** into **TWT'** is illustrated through Kennedy and Sorensen's (11) example pedigrees of eight animals (Figure 1). Animals 1 through 4 have unknown parents and are assumed unrelated. Both parents of animals 5 through 8 are known. For this example:

$$\mathbf{w}' = [a_1 \quad a_2 \quad a_3 \quad a_4 \quad m_5 \quad m_6 \quad m_7 \quad m_8]$$

where a_1, a_2, a_3, and a_4 are breeding values of ancestors with unknown parents and m_5, m_6, m_7, and m_8 are Mendelian sampling contributions to breeding values of their descendants. Numerically:

$$\mathbf{W} = \mathrm{diag}[1 \quad 1 \quad 1 \quad 1 \quad .5 \quad .5 \quad .5 \quad .5]$$

Off-diagonal elements of **W** are 0 because animals 1 to 4 are unrelated and subsequent Mendelian sampling terms are independent of each other and ancestral breeding values even if selection has occurred (1).

If animals are ordered genealogically, **T** is a lower triangular matrix with diagonal values of 1:

$$\begin{array}{ccc|cccccccc} s & d & i & & & & & & & & \\ & & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ & & 2 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ & & 3 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ & & 4 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 5 & .5 & .5 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 2 & 6 & .5 & .5 & 0 & 0 & 0 & 1 & 0 & 0 \\ 3 & 4 & 7 & 0 & 0 & .5 & .5 & 0 & 0 & 1 & 0 \\ 5 & 6 & 8 & .5 & .5 & 0 & 0 & .5 & .5 & 0 & 1 \end{array} = \mathbf{T}$$

Elements of **T** are:

$$t_{ij} = .5(t_{sj} + t_{dj})$$

for $j = 1, \ldots, i - 1$, if both parents are identified;

$$t_{ij} = .5(t_{s(d)j})$$

if only one parent is identified, and

$$t_{ij} = 0$$

if neither parent is identified; where s and d refer to the sire and dam of animal i. For example:

$$t_{81} = .5(t_{51} + t_{61}) = .5(.5 + .5) = .5$$

In addition, **T** may be computed using path coefficients. Examination of Figure 1 shows that genes can pass from animal 1 to animal 8 through two paths: $1 \rightarrow 5 \rightarrow 8$ and $1 \rightarrow 6 \rightarrow 8$. The value of t_{81} is the sum of probabilities over the two paths:

$$t_{81} = (.5)^2 + (.5)^2 = .5.$$

Only direct relationships are accounted for by **T**; collateral relationships are not in **T** although they are in **A**. For example, animals 5 and 6 are full siblings, however, $t_{65} = 0$.

Breeding values of the eight animals in the example can be expressed as:

$$\mathbf{a} = \mathbf{Tw} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ .5a_1 + .5a_2 + m_5 \\ .5a_1 + .5a_2 + m_6 \\ .5a_3 + .5a_4 + m_7 \\ .5a_1 + .5a_2 + .5m_5 + .5m_6 + m_8 \end{bmatrix}$$

For example, the breeding value of animal 8 is expressed as regressions on grandparental breeding values (a_1 and a_2), Mendelian sampling contributions to parental breeding values (m_5 and m_6), and the animal's own Mendelian sampling component (m_8). The variance of a_8 is:

$$\text{Var}(.5a_1 + .5a_2 + .5m_5 + .5m_6 + m_8)$$

If animals 1 and 2 are not inbred, not related, and not selected, then:

$$\text{Var}(a_8) = [.25 + .25 + .25(.5) + .25(.5) + .5]\sigma_a^2 = 1.25\sigma_a^2$$

The .25 reflects the contribution of inbreeding ($F_8 = .25$).

Under a model of infinite loci, changes in genetic variance from both inbreeding and selection subsequent to establishment of the base population (ancestors with unknown parents) clearly are accounted for by $\mathbf{A}\sigma_a^2 = \mathbf{TWT}'\sigma_a^2$. The constant σ_a^2 refers to additive genetic variance in the base population prior to subsequent selection and inbreeding. As shown, breeding values of all animals can be expressed as linear functions of breeding values of ancestral or base population animals and subsequent Mendelian sampling terms. If the number of loci are finite, however, changes in genetic variance from changes in gene frequency are not accounted for by $\mathbf{A}\sigma_a^2$ (see Finite Number of Loci).

Accounting for Changes in Genetic Mean Through A

How **A** can account for changes in genetic mean either from chance or selection can be illustrated with the example pedigrees and data in Figure 1. If the only fixed effect is the mean (μ) and each individual has a single record, the animal model equations are:

$$\begin{bmatrix} n & \mathbf{1}' \\ \mathbf{1} & \mathbf{I} + \mathbf{A}^{-1}\lambda \end{bmatrix} \begin{bmatrix} \hat{\mu} \\ \hat{\mathbf{a}} \end{bmatrix} = \begin{bmatrix} \mathbf{1}'\mathbf{y} \\ \mathbf{y} \end{bmatrix} \qquad [2]$$

where **y** is the vector of observations on n animals, **I** is an identity matrix, and $\lambda = \sigma_e^2/\sigma_a^2 = (1 - h^2)/h^2$ where σ_e^2 is the environmental variance and h^2 is heritability. Numerically, $\mathbf{A}^{-1}$ is:

$$\mathbf{A}^{-1} = \begin{bmatrix} 2 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1.5 & .5 & 0 & 0 & -1 & 0 \\ 0 & 0 & .5 & 1.5 & 0 & 0 & 1 & 0 \\ -1 & -1 & 0 & 0 & 2.5 & .5 & 0 & -1 \\ -1 & -1 & 0 & 0 & .5 & 2.5 & 0 & -1 \\ 0 & 0 & -1 & -1 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 0 & 2 \end{bmatrix}$$

If $h^2 = .5$, then [2] is:

$$\begin{bmatrix} 8 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 3 & 1 & 0 & 0 & -1 & -1 & 0 & 0 \\ 1 & 1 & 3 & 0 & 0 & -1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 2.5 & .5 & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & .5 & 2.5 & 0 & 0 & -1 & 0 \\ 1 & -1 & -1 & 0 & 0 & 3.5 & .5 & 0 & -1 \\ 1 & -1 & -1 & 0 & 0 & .5 & 3.5 & 0 & -1 \\ 1 & 0 & 0 & -1 & 1 & 0 & 0 & 3 & 0 \\ 1 & 0 & 0 & 1 & 1 & -1 & -1 & 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mu} \\ \hat{a}_1 \\ \hat{a}_2 \\ \hat{a}_3 \\ \hat{a}_4 \\ \hat{a}_5 \\ \hat{a}_6 \\ \hat{a}_7 \\ \hat{a}_8 \end{bmatrix} = \begin{bmatrix} 72 \\ 10 \\ 9 \\ 8 \\ 7 \\ 9 \\ 10 \\ 8 \\ 11 \end{bmatrix} \quad [3]$$

This system of equations makes adjustment for breeding values of mates, thus accounting for nonrandom mating (both assortative and disassortative). For example, consider the equation for animal 1, which can be expressed as:

$$\hat{a}_1 = [(10 - \hat{\mu}) + (\hat{a}_5 - .5\hat{a}_2) + (\hat{a}_6 - .5\hat{a}_2)]/3.$$

The contribution of each progeny ($\hat{a}_5$ and $\hat{a}_6$) is adjusted by half the estimated breeding value of the other parent or mate ($\hat{a}_2$). Also, the phenotypic record is adjusted for fixed effects, in this case $\hat{\mu}$.

Another property of this system of equations is illustrated through the equation for animal 8, which can be expressed as:

$$\hat{a}_8 = (\hat{a}_5 + \hat{a}_6)/2 + [11 - \hat{\mu} - (\hat{a}_5 + \hat{a}_6)/2]/3.$$

The first term is the predicted breeding value of animal 8 based on the average estimated breeding value of its parents ($\hat{a}_5$ and $\hat{a}_6$). The second term is an estimate of the Mendelian sampling contribution to $\hat{a}_8$. This term is the phenotypic observation adjusted for fixed effects and deviated from parental average (predicted breeding value) weighted by within-family heritability. Within-family heritability is:

$$.5(1 - \bar{F})h^2/[.5(1 - \bar{F})h^2 + (1 - h^2)]$$

where $\bar{F}$ is average inbreeding coefficient of the parents. For this example, within-family heritability is $.5(.5)/[.5(.5) + .5] = 1/3$. Even though animal 8 is inbred, there is no adjustment for inbreeding because its parents are not inbred.

Solution of [3] is:

$$\begin{bmatrix} \hat{\mu} \\ \hat{a}_1 \\ \hat{a}_2 \\ \hat{a}_3 \\ \hat{a}_4 \\ \hat{a}_5 \\ \hat{a}_6 \\ \hat{a}_7 \\ \hat{a}_8 \end{bmatrix} = \begin{bmatrix} 8.7059 \\ .8676 \\ .3676 \\ -.3676 \\ -.8676 \\ .6716 \\ 1.0049 \\ -.6471 \\ 1.3235 \end{bmatrix}$$

Animals 1 through 4 represent the base population; because they are unrelated and generations do not overlap, their average $\hat{a}$ is 0:

$$(\hat{a}_1 + \hat{a}_2 + \hat{a}_3 + \hat{a}_4)/4 = 0.$$

Implicit with this animal model is the assumption that expected breeding value of base population animals is 0. Expected breeding value of animals in subsequent generations is not 0, which is illustrated through the average $\hat{a}$ of the direct progeny of the base population animals:

$$(\hat{a}_5 + \hat{a}_6 + \hat{a}_7)/3 = .343$$

These changes can reflect both drift and selection. For example, animals 1 and 2, which had the highest phenotypic values, produced two offspring, and animals 3 and 4, with the lowest phenotypic values, produced only one offspring. The result of this selection, whether intentional or not, is reflected in the average breeding value of their offspring.

Therefore, animal models can be used to estimate response to selection (9, 11, 17, 19). Under an additive infinitesimal model, estimates of selection response from animal models using $A\sigma_a^2$ are unbiased and have smaller sampling variance than usual estimates based on least squares estimates of differences from controls. Also, $A\sigma_a^2$ can be used to account for drift variance in analysis of data from selection experiments (16).

Estimation of Genetic Variance

An assumption of the animal model for prediction of breeding values is that σ_a^2 in the base population is known, but in practice σ_a^2 must be estimated. Animal models can be used to obtain estimates of σ_a^2 that are unbiased by selection and inbreeding (18, 19) using either minimum variance quadratic unbiased estimation (MIVQUE), if good prior values of h^2 are available, or REML procedures. (Estimates of h^2 and σ_a^2 based on traditional sire models can be biased considerably if relationships beyond those due to a common sire are not accounted for and inbreeding is ignored.) Application of unbiased estimates of σ_a^2 obtained from the data yields unbiased estimates of breeding values and selection response over repeated sampling (19). Therefore, lack of exact knowledge of σ_a^2 in the base population is not a critical limitation to the use of animal models. Some additional error is introduced in estimates of breeding values and selection response but not necessarily bias over repeated sampling.

Base Population

Correct understanding and definition of the base population are critical for animal models. With A, animals without identified parents are treated as base population animals and are assumed to be sampled from a single population with average breeding value of 0 and common variance σ_a^2. Applying an animal model with disregard for definition of the base population animals to which all subsequent breeding values are tied can lead to errors. If base population animals actually were sampled from populations with different genetic means, such as when populations are mixed or new animals with unknown parents enter the population over time, then the model should account for the subpopulation structure and allow for different expected values of base population animals. This often can be accomplished through use of a judicious grouping strategy (21), and this problem has been discussed in greater detail by Henderson (8).

Finite Number of Loci

Because $A\sigma_a^2$ does not acknowledge changes in genetic variance associated with changes in gene frequency, an infinite number of loci has been assumed. However, simulation suggests that animal models provide good approximations for estimation of breeding values for traits controlled by small numbers of loci (even if selection has occurred) if the genetic model is additive (12).

Data of Maki-Tanila and Kennedy (12) found in Table 1 are used as illustration. Response to three generations of selection was estimated with an animal model for a trait controlled by two loci with a heritability of .5. Three frequencies of the favorable allele were considered (q = .1, .5, or .9). Animal model estimates of response averaged over 100 replicates were not significantly different from true response. However, selection response tended to be underestimated slightly if frequency of the favorable allele was low and overestimated if frequency was high. For low frequency, genetic variance assumed by the model would underestimate true genetic variance because σ_a^2 would increase as q increased (until q > .5). For high frequency, true genetic variance would be slightly underestimated as q approached fixation. Nonetheless, animal models appear to provide good approximations if loci are few even under selection (at least for short-term selection).

GAMETIC MODELS

Definition

For some situations, expressing breeding values in terms of separate gametic contributions of each parent is useful. Gametic models can be used to reduce computational requirements, to understand properties of animal models better, and to estimate genotypic values and variances under certain modes of inheritance.

If inheritance is additive, a_i can be expressed as the sum of the additive genetic contributions of gametes from its sire (g_s) and dam (g_d):

$$a_i = g_s + g_d \qquad [4]$$

TABLE 1. True response and estimated response with an animal model to three generations of selection for a trait controlled by two autosomal loci acting additively (heritability = .5).[1]

Frequency of favorable alleles	True genetic mean		Estimated genetic mean	
	$\bar{x}$	SE	$\bar{x}$	SE
.1	3.03	.29	2.95	.28
.5	2.30	.17	2.26	.17
.9	1.41	.08	1.59	.12

[1]From Maki-Tanila and Kennedy (12).

Without selection or linkage and with random union of parental gametes:

$$\text{Var}(a_i) = \text{Var}(g_s) + \text{Var}(g_d) = .5\sigma_a^2 + .5\sigma_a^2 = \sigma_a^2. \quad [5]$$

Each gametic value can be expressed in terms of its parental breeding value plus a Mendelian sampling contribution; i.e., the genetic value of a gamete obtained from its parent (g_p) is:

$$g_p = .5a_p + w_p \quad [6]$$

where a_p is parental breeding value and w_p is chance contribution from sampling of genes from parent to progeny.

$$\text{Var}(g_p) = \text{Var}(.5a_p) + \text{Var}(w_p)$$

and without selection:

$$\text{Var}(g_p) = .25\sigma_a^2 + .25(1 - F_p)\sigma_a^2 = .5(1 - .5F_p)\sigma_a^2.$$

Reduced Animal Model

Quaas and Pollak (13) developed a gametic or reduced animal model that used the right-hand side of [6] to reduce computational requirements for genetic evaluation. In their model:

$$a_i = g_s + g_d = .5a_s + .5a_d + w_i$$

where $w_i = w_s + w_d$ with:

$$\text{Var}(w_i) = .25(1 - F_s)\sigma_a^2 + .25(1 - F_d)\sigma_a^2 = .5(1 - \bar{F})\sigma_a^2.$$

Equations for breeding values of animals are expressed in terms of parental breeding values and a Mendelian sampling term that combines both sire and dam contributions. The vector of breeding values for all progeny with identified parents can be expressed as:

$$\mathbf{a} = \mathbf{Z}\mathbf{a}_p + \mathbf{w}$$

where $\mathbf{Z}$ is a matrix with one row for each progeny and one column for each parent. All elements of each row of $\mathbf{Z}$ are 0 except for elements corresponding to the animal's parents, in which case the elements are .5. Again, if number of loci is sufficiently large, distribution of Mendelian sampling terms are independent of each other and independent of parental breeding values. Therefore, under the reduced animal model:

$$\text{Var}(\mathbf{a}) = \mathbf{Z}\mathbf{A}_p\mathbf{Z}'\sigma_a^2 - \mathbf{W}$$

where $\mathbf{A}_p$ is the additive relationship matrix among parents of $\mathbf{a}$. If a parent of an animal is not identified, then $\mathbf{W}$ is modified to account for breeding value of the unidentified parent, as outlined previously.

Quaas and Pollak (13) used this model to reduce computational requirements because the order of the animal equations of the coefficient matrix [2] is only the order of parents, not the order of all animals. However, the reduced animal model also is of interest because it expresses breeding values in the same form as outlined for the full animal model to illustrate how use of the relationship matrix ties breeding values of individuals back to the base population.

Change in Genetic Variance from Selection

Kennedy and Sorensen (11) also used a gametic model to show change in genetic variance among progeny after selection of their parents on phenotype. They used [4] and [5] to show that gametic variance among progeny (g^*) after selection on parents of one sex, say sires, is:

$$\text{Var}(g_s^*) = .5(1 + .5h^2k_s)\sigma_a^2$$

where k_s is proportional change in phenotypic variance among selected sires. Random union of gametes leads to a genetic variance among progeny of selected sires and dams of:

$$\text{Var}(a^*) = (1 + .5h^2k)\sigma_a^2$$

where $k = .5(k_s + k_d)$. This is the same result derived by Bulmer (1) for change in genetic variance due to gametic or linkage disequilibrium from one generation of selection on phenotype. With directional or stabilizing selection, variance declines (negative k); with disruptive selection, variance increases (positive k).

Gametic Relationship Matrix

Gibson et al. (5) have proposed a gametic model for estimation of autosomally inherited genetic effects that are expressed from only one parent's gamete. In this model, a_i for these genes is as in [4], but genes from only one of the gametes

(sire or dam) are expressed in the phenotypic record of the animal. Genes from the other gamete are not expressed. Application of this model requires construction of the variance-covariance matrix among gametes (**G**) or its inverse ($\mathbf{G}^{-1}$).

Construction of **G** was outlined by Smith (15), who applied it to computing dominance genetic variances in inbred populations, and is illustrated in Figure 2. Resulting gametic relationships in **G** are in Table 2. The first gamete for each animal represents the gamete from its sire (subscript 1), and the second represents the gamete from its dam (subscript 2). Animals A and B are assumed unrelated and not inbred. The matrix is constructed recursively, similar to construction of **A**. All diagonal elements are 1. If X_iY_j represents row and column identification for an off-diagonal element in **G**, then:

$$X_iY_i = .5(X_iS_1 + X_iS_2)$$
$$X_iY_j = .5(X_iD_1 + X_iD_2)$$

where S and D are sire and dam of Y. For example:

$$E_1F_1 = .5(E_1E_1 + E_1E_2) = .5(1 + 1/8) = 9/16$$
$$E_1F_2 = .5(E_1B_1 + E_1B_2) = .5(1/4 + 1/4) = 1/4$$

Additive and dominance relationships can be computed from **G**. For example, the additive relationship between E and F is:

$$\underline{a}_{EF} = .5(E_1F_1 + E_1F_2 + E_2F_1 + E_2F_2)$$
$$= .5(9/16 + 1/4 + 9/16 + 0) = 11/16$$

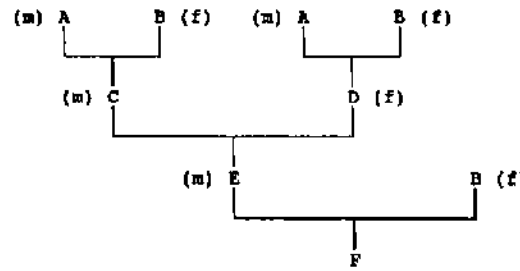


Figure 2. Example pedigree for construction of gametic relationship matrix; sex of animal is in parentheses (m = male, f = female).

and the dominance relationship is

$$\underline{d}_{EF} = (E_1F_1)(E_2F_2) + (E_1F_2)(E_2F_1)$$
$$= [(9/16)(0) + (1/4)(9/16)] = 9/64.$$

Construction of the dominance relationship matrix (**D**) from **G** can have useful applications for nonadditive genetic models.

Additional details on construction of **G** and $\mathbf{G}^{-1}$ have been discussed by Schaeffer and Kennedy (14).

NONADDITIVE GENETIC MODELS

Animal models can be applied to genetic evaluation or estimation of variance components for dominance and other nonadditive genetic effects (7, 8), maternal effects (8), and cytoplasmic effects (10). Kennedy (9) and Kennedy and Sorensen (11)

TABLE 2. Gametic relationships for animals A, B, C, D, E, and F from example pedigree in Figure 2.

Animal gamete	Animal gamete											
	A_1	A_2	B_1	B_2	C_1	C_2	D_1	D_2	E_1	E_2	F_1	F_2
A_1	1	0	0	0	1/2	0	1/2	0	1/4	1/4	1/4	0
A_2	0	1	0	0	1/2	0	1/2	0	1/4	1/4	1/4	0
B_1	0	0	1	0	0	1/2	0	0	1/4	0	1/8	1/2
B_2	0	0	0	1	0	1/2	0	0	1/4	0	1/8	1/2
C_1	1/2	1/2	0	0	1	0	1/2	0	1/2	1/4	3/8	0
C_2	0	0	1/2	1/2	0	1	0	0	1/2	0	1/4	1/2
D_1	1/2	1/2	0	0	1/2	0	1	0	1/4	1/2	3/8	0
D_2	0	0	0	0	0	0	0	1	0	1/2	1/4	0
E_1	1/4	1/4	1/8	1/8	1/2	1/2	1/4	0	1	1/8	9/16	1/4
E_2	1/4	1/4	0	0	1/4	0	1/2	1/2	1/8	1	9/16	0
F_1	1/4	1/4	1/8	1/8	3/8	1/4	3/8	1/4	9/16	9/16	1	1/8
F_2	0	0	1/2	1/2	0	1/2	0	0	1/4	0	1/8	1

have discussed genetic properties of animal models for situations in which nonadditive genetic effects are important.

Relationship Matrices

Henderson (7, 8) has outlined models for dominance and epistatic effects based on a population assumed to be noninbred and in linkage and Hardy-Weinberg equilibrium. In this case, Cockerham's (3) formulation of genetic covariance between relatives x and y is appropriate:

$$\mathrm{Cov}(G_x G_y) = \sum_{i=0} \sum_{j=0} (\underline{a}_{xy})^i (\underline{d}_{xy})^j \sigma^2_{ij} \text{ for } i + j > 0$$

where $\underline{a}_{xy}$ and $\underline{d}_{xy}$ are additive and dominance genetic relationships between x and y and i is number of nonallelic genes acting together with j allelic pairs. All nonadditive relationships can be generated as functions of **A**. From additive relationships between sires and dams, **D** can be computed. Higher order nonadditive relationship matrices can be generated as Hadamard products (denoted by #) of **A** or **D** (7). For example, the additive × additive relationship matrix is $\mathbf{A}\#\mathbf{A}\sigma^2_{aa}$. A sufficiently large number of loci (infinitesimal model) also is assumed with Henderson's (7, 8) nonadditive models.

Inbreeding

Most animal populations undergo some inbreeding. Also, selection often has been practiced such that the population is neither in Hardy-Weinberg nor linkage equilibrium, and assumptions of Henderson's (8) nonadditive genetic models are not valid. In addition, traits may be controlled by relatively few loci.

In inbred populations, **D** cannot be constructed simply from **A** as is the case for noninbred populations. Smith (15) proposed construction of **D** from **G** to account for inbreeding. Although **D** computed in this manner may represent inbreeding in the variance-covariance structure of dominance deviations, it does not account for inbreeding depression as illustrated by Maki-Tanila and Kennedy (12). After three generations of random mating for a trait controlled by two loci with favorable alleles completely dominant and heritability of .5, $\bar{F}$ was .14. Inbreeding depression occurred, and the genetic mean declined as expected (Table 3). However, estimates of the genetic mean from animal models reflected this decline only partially. Inclusion of **D** computed according to Smith (15) improved estimates of the genetic mean, but declines due to inbreeding depression still were not accounted for fully.

Although Maki-Tanila and Kennedy (12) examined a two locus model, failure of the animal model to account for inbreeding depression also would be expected for traits controlled by a large number of nonadditive loci if dominance is directional. Inbreeding depression is a consequence of an increased frequency of homozygous genotypes associated with increased coefficients of inbreeding. Unlike selection, for which changes in gene

TABLE 3. True response and estimated response with an animal model to three generations of selection for a trait controlled by autosomal loci with complete dominance and heritability of .5[1]

Number of loci	Type of selection	Frequency of favorable alleles	True genetic mean	Estimated genetic mean[2]	
				A only	A and D
2	Random	.5	−.41	−.14	−.20
		.9	−.75	−.17	−.47
	Directional	.1	3.23	1.36	2.74
		.5	1.00	1.12	1.09
		.9	.10	.67	.55
30	Directional	.1	7.46	2.58	6.17
		.9	−2.73	−.22	−.71

[1]Abridged from Maki-Tanila and Kennedy (12).

[2]**A** = Additive relationship matrix; **D** = dominance relationship matrix.

and genotypic frequency at any single locus will be small if number of loci is large, effects of inbreeding on genotypic frequency at a given locus are the same regardless of number of other loci influencing the trait. If a large number of loci show directional dominance, effects of inbreeding depression can still be large, and this should be accounted for, possibly through inclusion of F in the model as a covariate.

Similarly, inbreeding complicates epistatic relationships, and a fully general model with inbreeding and epistasis seems unmanageable (20). Whether ignoring inbreeding contributions to epistatic relationships can provide reasonable approximations has not been examined. Epistasis without dominance cannot cause inbreeding depression (4), but inbreeding depression is nonlinear with F if there is dominance × dominance interaction.

Selection

With a large number of loci, dominance deviations are normally distributed in unselected noninbred populations. Bulmer (2) asserts further that the distribution of dominance deviations is not affected by selection in previous generations and that dominance effects in unselected progeny of selected parents remain normally distributed and independent of additive effects. Therefore, if number of loci is sufficiently large, an animal model with D should provide best linear unbiased predictors of dominance genetic merit even for selected (but not inbred) populations.

With inbreeding and selection, inability of D to account for effects of inbreeding depression remains. In selected populations, confounding between increases in average additive genetic merit and F results in further complications. An animal model with inclusion of $A\sigma_a^2$, $D\sigma_d^2$ according to Smith (15), and F as a covariate should be examined with simulated data as a possible solution to the problem.

Maki-Tanila and Kennedy (12) examined effects of including $D\sigma_d^2$ computed as by Smith (15) but not F on estimates of genetic merit for traits controlled by 2 or 30 loci with complete dominance. Three generations of directional selection were practiced, and $\bar{F}$ was .14 at generation 3. Response depended on frequency of the favorable allele (Table 3). In addition, inbreeding depression occurred. At high frequency (.9), this resulted in either no increase in average genetic merit (2 loci) or a decrease in average genetic merit (30 loci) even though selection had been practiced. For both 2 and 30 loci, estimates of genetic means from animal model solutions were considerably biased (particularly when D was not included in the model) except when gene frequencies were intermediate (.5). Inclusion of D removed some but not all bias. Behavior of animal models in inbred and selected populations should be examined further if dominance is important for traits controlled by both large and small numbers of loci.

Bulmer's (1) selection model allows for additive and dominance effects but assumes no epistasis. Consequences of ignoring selection in formulating epistatic relationship matrices and applying these to animal model predictions of epistatic effects presumably would depend on the nature of the epistatic gene action, and these should be investigated.

CONCLUSIONS

Animal models have some clearly defined and useful genetic properties for prediction of genotypic values, estimation of selection response, and estimation of genetic parameters such as heritability in selected and inbred populations. They allow for complete incorporation of information on relatives, and their use should be encouraged. However, care must be exercised that the animal model used represents a good approximation of the true underlying genetic model, particularly if nonadditive genetic effects are important or number of loci influencing the trait is relatively small. Proposed animal models should be examined for robustness to various possible underlying genetic models, both theoretically and through computer simulation, because results can depend upon the assumptions of the model.

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Fernando and Grossman (1989)

Fernando, R.L., and Grossman, M. 1989. Marker assisted selection using best linear unbiased prediction. Genetics Selection Evolution, vol. 21, no. 4, pp. 467 to 477.

The problem

With the arrival of molecular markers came the idea of using a marker linked to a gene of large effect to improve selection. Fernando and Grossman showed how to include this marked locus in the mixed model equations, alongside the polygenic effect. To do so, they built a covariance matrix for the alleles at the marked locus and showed how to obtain its inverse recursively.

Points to note while reading

This is the first step toward placing molecular information in Henderson's equations. Note that the covariance matrix of the marked locus is built from gametes and recombination, and not from the expected pedigree relationship. It is an idea that anticipates the genomic matrix.

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Original article

Marker assisted selection using best linear unbiased prediction

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Summary – Best linear unbiased prediction (BLUP) is applied to a mixed linear model with additive effects for alleles at a market quantitative trait locus (MQTL) and additive effects for alleles at the remaining quantitative trait loci (QTL). A recursive algorithm is developed to obtain the covariance matrix of the effects of MQTL alleles. A simple method is presented to obtain its inverse. This approach allows simultaneous evaluation of fixed effects, effects of MQTL alleles, and effects of alleles at the remaining QTLs, using known relationships and phenotypic and marker information. The approach is sufficiently general to accommodate individuals with partial or no marker information. Extension of the approach to BLUP with multiple markers is discussed.

marker-assisted selection – best linear unbasied prediction – genetic marker

Résumé – Sélection assistée par un marqueur: utilisation du meilleur prédicteur linéaire sans biais (BLUP). *La méthode du BLUP (meilleure prédiction linéaire sans biais) est appliquée à un modèle linéaire mixte comprenant des effets additifs associé aux allèles d'un locus quantitatif flanqué d'un gène marqueur, et d'effets additifs pour les autres locus quantitatifs. Un algorithme récursif permet d'obtenir la matrice de covariances associée aux effets des allèles du locus marqué. Une méthode simple est aussi proposée pour calculer l'inverse de cette matrice. Cette approche permet d'évaluer simultanément les effets fixés, les effets des allèles du locus marqué, et les effets génétiques additifs de l'ensemble des autres locus, d'après les relations de parenté, les données phénotypiques et l'information sur les marqueurs. Cette approche est assez générale pour tenir compte de données incomplètes chez certains individus. On discute l'extension à un BLUP avec plusieurs marqueurs.*

sélection assistée par un marqueur – meilleure prédiction linéaire sans biais – marqueur génétique

INTRODUCTION

Genetic engineering techniques have produced a variety of molecular genetic markers with the potential to identify a large number of genetic polymorphisms (Soller

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and Beckmann, 1982; Smith and Simpson, 1986; Schumm *et al.*, 1988). Marker-assisted selection is one application of these techniques to animal and plant breeding. Information on marker loci that are linked to quantitative trait loci, together with phenotypic information, could be used to increase genetic progress by increasing accuracy of selection and by reducing generation interval (Soller, 1978; Smith and Simpson, 1986).

Geldermann (1975) proposed a least-squares procedure to estimate effects of marker alleles on quantitative traits. Based on selection index principles, Soller (1978) combined marker information and phenotypic information to obtain genetic evaluations. This method has been used to study the additional genetic progress expected from marker-assisted selection (Soller, 1978; Soller and Beckmann, 1983, Smith and Simpson, 1986). Because of the complex nature of animal breeding data, however, these methods may not be applicable directly to marker-assisted selection with field data.

Data from field-recorded populations are affected by non-genetic nuisance factors, such as age of animal, age of dam, management system, season of birth and herb. Also, non-random mating, selection and overlapping generations contribute to the complexity of the data. Best linear unbasied prediction (BLUP; Henderson, 1973, 1975, 1982) deals with these complications when predicting breeding values from phenotypic data. The objective of this paper is to present methodology for the application of BLUP to marker-assisted selection in animal breeding. Each methodological development is illustrated with a numerical example using a single hypothetical pedigree

METHODOLOGY

Consider a single polymorphic marker locus (ML), closely linked to a quantitative trait locus (QTL). Let M_i^p and M_i^m denote alleles at the ML that individual i inherited from its paternal (p) and its maternal (m) parent, and let Q_i^p and Q_i^m denote alleles at the market QTL (MQTL) linked to M_i^p and M_i^m, as shown below:

M_i^p	Q_i^p
M_i^m	Q_i^m

Let v_i^p and v_i^m be the additive effects of Q_i^p and Q_i^m. Additive effects of alleles at the remaining QTLs, unlinked to the ML, will be denoted by the residual additive effect u_i. Now, the additive effect for individual i, a_i, can be written as

$$a_i = v_i^p + v_i^m + u_i \tag{1}$$

The usual model to obtain BLUP if additive effects, given phenotypic information, is

$$y_i = \mathbf{x}_i'\beta + a_i + e_i \tag{2}$$

where y_i is the phenotypic value of individual i, $\mathbf{x}_i'$ is a vector of known constants, β is a vector of unknown fixed effects, and e_i is a random error. Using equ.(2), BLUP allows information from relatives to contribute to the predictor of a_i through the

covariance matrix of a_i values. Note that this covariance matrix depends on the type of genetic information available. When only relationship information ($\boldsymbol{r}$) is available, the covariance of a_i values is

$$\mathrm{Cov}(\mathbf{a}|\boldsymbol{r}) = \boldsymbol{G}_{a|r}$$

which is proportional to the numerator relationship matrix (*e.g.*, Henderson, 1976). When marker information ($\boldsymbol{m}$) is also available, the covariance matrix a_i values is

$$\mathrm{Cov}(\mathbf{a}|\boldsymbol{r},\boldsymbol{m}) = \boldsymbol{G}_{a|r,m}$$

It can be shown that $\boldsymbol{G}_{a|r} \neq \boldsymbol{G}_{a|r,m}$, in general. For example, the covariance between half-sibs that receive the same ML allele from their common parent is higher than the covariance between half-sibs that receive different ML alleles. This is because half-sibs receiving the same ML allele also receive the same MQTL allele with greater frequency than half-sibs receiving different ML alleles.

A. Marker model

To obtain BLUP with phenotypic and marker information, it is convenient to use

$$y_i = \mathbf{x}_i'\beta + v_i^p + v_i^m + u_i + e_i \tag{3}$$

which is equivalent to equ.(2). The covariance matrix of v_i values ($\boldsymbol{G}_v$) depends on relationship and marker information. The covariance matrix of u_i values ($\boldsymbol{G}_u$) depends only on relationship information and is proportional to the numerator relationship matrix (*e.g.*, Henderson, 1976). Given the covariance matrices $\boldsymbol{G}_v$ and $\boldsymbol{G}_u$, BLUPs of v_i and u_i values can be obtained using the mixed model equations (Henderson, 1973). The inverse of $\boldsymbol{G}_u$, which is required on the mixed model equations, usually is obtained using an algorithm given by Henderson (1976). A recursive algorithm to construct $\boldsymbol{G}_v$ is given in section B, and an algorithm to obtain its inverse is in section C.

B. Covariance matrix of MQTL effects

1. Theory. To construct $\boldsymbol{G}_v$, consider the covariance between additive effects of MQTL alleles. Without loss of generality, consider only paternal MQTL alleles. Suppose arbitrary individuals o and o' have sires s and s'. The MQTL alleles inherited by o and o' from their sires are Q_o^p and $Q_{o'}^p$, having additive effects v_o^p and $v_{o'}^p$. For paternal MQTL alleles in o and o', the covariance between their additive effects v_o^p and $v_{o'}^p$ is

$$\begin{aligned}\mathrm{Cov}(v_o^p, v_{o'}^p) &= \mathrm{Cov}(v_o^p, v_{o'}^p | Q_o^p \equiv Q_{o'}^p) \cdot P(Q_o^p \equiv Q_{o'}^p) \\ &= \mathrm{Var}(v_o^p) \cdot P(Q_o^p \equiv Q_{o'}^p)\end{aligned} \tag{4}$$

where $\mathrm{Var}(v_o^p) = \sigma_v^2$ is the additive variance of an MQTL allele and $P(Q_o^p \equiv Q_{o'}^p)$ is the probability that Q_o^p is identical by descent to $Q_{o'}^p$. For an arbitrary pair of individuals, one is not a direct descendent of other. If o is not a direct descendant of the o', Q_o^p can be identical by descent to $Q_{o'}^p$ in 2 mutually exclusive ways:

1) Q_o^p is identical by descent to the maternal MQTL allele of the sire of o' ($Q_{s'}^p$) and o' inherits $Q_{s'}^p$, or
2) Q_o^p is identical by descent to the paternal MQTL allele of the sire of o' ($Q_{s'}^m$) and o' inherits $Q_{s'}^m$.

If marker information is available, the conditional probability that o' inherits $Q^p_{s'}$, given that o' inherits $M^p_{s'}$, is $(1-r)$, where r is the recombination rate between the ML and the MQTL. Thus if o' inherits $M^p_{s'}$, the probability in equ.(4) can be calculated recursively as

$$P(Q^p_o \equiv Q^p_{o'}) = P(Q^p_o \equiv Q^p_{s'}) \cdot (1-r) + P(Q^p_o \equiv Q^m_{s'}) \cdot r \tag{5}$$

Similarly, if o' inherits $M^m_{s'}$

$$P(Q^p_o \equiv Q^p_{o'}) = P(Q^p_o \equiv Q^p_{s'}) \cdot r + P(Q^p_o \equiv Q^m_{s'}) \cdot (1-r) \tag{6}$$

If marker information is not available, so that it is not known whether o' inherits $M^p_{s'}$ or $M^m_{s'}$, 0.5 replaces r in equs.(5) and (6). This is because, in the absence of marker information, $Q^p_{s'}$ and $Q^m_{s'}$ have equal probability of being transmitted to o'.

The above development leads to a tabular method to construct $\boldsymbol{G}_v$, which is similar to the method used to construct the numerator relationship matrix (*e.g.* Henderson, 1976). Note that $\boldsymbol{G}_v$ has twice as many rows as individuals because each individual has 2 effects: 1 for the paternal and 1 for the maternal MQTL allele. The rows and columns of $\boldsymbol{G}_v$ should be ordered so that those corresponding to progeny follow those for their parents. Let the row indices of $\boldsymbol{G}_v$, corresponding to the effects of MQTL alleles of individual $o(v^p_o, v^m_o)$, be i^p_o, i^m_o; of its sire $s(v^p_s, v^m_s)$, be i^p_s, i^m_s; and of its dam $d(v^p_d, v^m_d)$, be i^p_d, i^m_d. Also, let element ij of $\boldsymbol{G}_v$ be g_{ij}. Then from equs.(4), (5) and (6), the elements of row i^p_o, below the diagonal, are obtained as

$$g_{i^p_o, j} = (1-\rho^p_o) g_{i^p_s, j} + \rho^p_o g_{i^m_s, j} \tag{7a}$$

for $j = 1 \ldots i^p_o - 1$, where $\rho^p_o = r$ if o inherits M^p_s or $\rho^p_o = (1-r)$ if o inherits M^m_s. Elements of column i^p_o, above the diagonal, are obtained from the corresponding row elements because $\boldsymbol{G}_v$ is symmetric. Similarly, elements of row i^m_o, below the diagonal, are obtained as

$$g_{i^m_o, j} = (1-\rho^m_o) g_{i^p_d, j} + \rho^m_o g_{i^m_d, j} \tag{7b}$$

for $j = 1 \ldots i^m_o - 1$, where $\rho^m_o = r$ if o inherits M^p_d and $\rho^m_o = (1-r)$ if o inherits M^m_d. Elements of column i^m_o, above the diagonal, are obtained from the corresponding row elements.

From equ.(4), the diagonal elements of $\boldsymbol{G}_v$ are equal to σ^2_v. If marker information cannot be used to determine which of the 2 marker alleles o are inherited from its sire or its dam, then 0.5 replaces ρ^p_o in equ.(7a) or ρ^m_o in (7b).

2. Numerical example. Consider the pedigree in Table I. To construct $\boldsymbol{G}_v$, rows and columns are arranged by individual and by paternal and maternal MQTL alleles within individual (Table II). For convenience, we will assume that $\sigma^2_v = 1$ and that $r = 0.1$. The first two individuals are assumed to be unrelated; thus the upper left 4×4 submatrix of $\boldsymbol{G}_v$ is the identity matrix. Elements on the diagonal are equal to $\sigma^2_v = 1$. Now, row elements below the diagonal can be obtained from equs.(7a) and (7b); column elements above the diagonal are obtained by symmetry. Each row element for v^p_3 is equal to $(1-r) = 0.9$ times the corresponding row element for v^p_1 plus $r = 0.1$ times the corresponding row element for v^m_1. Each row element for v^m_3 is equal to $r = 0.1$ times the corresponding row element for v^p_2 plus $(1-r) = 0.9$ times the corresponding row element for v^m_2. The ML allele inherited by 4 from

its sire is unknown. Thus, each row element for v_4^p is the mean ($r = 0.5$) of the corresponding row element for v_1^p and for v_1^m. Marker information is available for v_4^m, so that each row element for v_4^m is $(1 - r) = 0.9$ times the corresponding row element for v_3^p plus $r = 0.1$ times the corresponding row element for v_3^m.

Table I. Pedigree with marker information

Animal	*Sire*	*Dam*	*Marker inherited from*	
			Sire	*Dam*
1	–	–	–	–
2	–	–	–	–
3	1	2	M_1^p	M_2^m
4	1	3	–	M_3^p

Table II. Covariance matrix of MQTL effects: $\boldsymbol{G}_v$

	v_1^p	v_1^m	v_2^p	v_2^m	v_3^p	v_3^m	v_4^p	v_4^m
v_1^p	1.0	0.0	0.0	0.0	0.9	0.0	0.5	0.81
v_1^m	0.0	1.0	0.0	0.0	0.1	0.0	0.5	0.09
v_2^p	0.0	0.0	1.0	0.0	0.0	0.1	0.0	0.01
v_2^m	0.0	0.0	0.0	1.0	0.0	0.9	0.0	0.09
v_3^p	0.9	0.1	0.0	0.0	1.0	0.0	0.5	0.9
v_3^m	0.0	0.0	0.1	0.9	0.0	1.0	0.0	0.1
v_4^p	0.5	0.5	0.0	0.0	0.5	0.0	1.0	0.45
v_4^m	0.81	0.09	0.01	0.09	0.9	0.1	0.45	1.0

C. Algorithm for inverting $\boldsymbol{G}_v$

1. Theory. The approach taken here follows that by Quaas *et al.* (1984) and Quaas (1988) to invert the matrix of additive relationships. We define a linear model to relate the effect of the paternal MQTL allele of an individual (o) to effects of paternal and maternal MQTL alleles of its sire (s)

$$v_o^p = (1 - \rho_o^p)v_s^p + \rho_o^p v_s^m + \varepsilon_o^p \tag{8a}$$

where ε_o^p is a residual effect. Similarly, a linear model for effect of the maternal MQTL allele of o is

$$v_o^m = (1 - \rho_o^m)v_d^p + \rho_o^m v_d^m + \varepsilon_o^m \tag{8b}$$

It can be shown that the residuals ε_o^p in equ.(8a) and ε_o^m in (8b) have a diagonal covariance matrix ($\boldsymbol{G}_\varepsilon$; see Appendix). Now, the vector of effects of MQTL alleles ($\mathbf{v}$) can be written as

$$\boldsymbol{v} = \boldsymbol{P}\mathbf{v} + \boldsymbol{\varepsilon} \tag{9}$$

where $\boldsymbol{P}$ is a matrix with each row containing only two non-zero elements, if the parent is known or containing only zeros, if the parent is unknown; and where $\boldsymbol{\varepsilon}$ is a vector of residuals. For example, row i_o^p will have $(1-\rho_o^p)$ in column i_s^p and ρ_o^p in column i_s^m, if the sire of i is known. Similarly, row i_o^m will have $(1-\rho_o^m)$ in the column i_d^p and ρ_o^m in column i_d^m, if the dam of i is known.

To proceed, we need the diagonal elements of $\boldsymbol{G}_\varepsilon$. Consider, for example, the variance of ε_o^p. From equ.(8a), if the sire of o is known

$$\begin{aligned}\mathrm{Var}(v_o^p) = (1-\rho_o^p)^2 \cdot \mathrm{Var}(v_s^p) + (\rho_s^p)^2 \cdot \mathrm{Var}(v_s^m) \\ +2(1-\rho_o^p)\rho_o^p \cdot \mathrm{Cov}(v_s^p, v_s^m) + \mathrm{Var}(\varepsilon_o^p)\end{aligned}$$

because effects of MQTL alleles of sire s are uncorrelated with residuals of its offspring o (see Appendix). Hence

$$\begin{aligned}\mathrm{Var}(\varepsilon_o^p) = \mathrm{Var}(v_o^p) - (1-\rho_o^p)^2 \cdot \mathrm{Var}(v_s^p) - (\rho_o^p)^2 \cdot \mathrm{Var}(v_s^m) \\ -2(1-\rho_o^p)\rho_o^p \cdot \mathrm{Cov}(v_s^p, v_s^m)\end{aligned} \tag{10}$$

The covariance between the effects of paternal and maternal MQTL alleles can be written as

$$\mathrm{Cov}(v_s^p, v_s^m) = \mathrm{Var}(v_s^p) \cdot P(Q_s^p \equiv Q_s^m) = \mathrm{Var}(v_s^p) \cdot F_s \tag{11}$$

where F_s is the inbreeding of sire s. Now, equ.(10) can be written as

$$\mathrm{Var}(\varepsilon_o^p) = 2\sigma_v^2(1-\rho_o^p)\rho_o^p(1-F_s) \tag{12a}$$

because $\mathrm{Var}(v_o^p) = \mathrm{Var}(v_s^p) = \mathrm{Var}(v_s^m) = \sigma_v^2$, and where $(1-\rho_o^p)\rho_o^p = (1-r)r$ for ρ_o^p or for $\rho_o^p = (1-r)$. When the sire is not inbred: $\mathrm{Var}(\varepsilon_o^p) = 2\sigma_v^2(1-r)r$, if marker information is available; or $\mathrm{Var}(\varepsilon_o^p) = \sigma_v^2/2$, if marker information is not available. If the sire is not known, $\mathrm{Var}(\varepsilon_o^p) = \sigma_v^2$.

Similarly, if dam of o is known, the variance of ε_o^m is

$$\mathrm{Var}(\varepsilon_o^m) = 2\sigma_v^2(1-\rho_o^m)\rho_o^m(1-F_d) \tag{12b}$$

where $(1-\rho_o^m)\rho_o^m = (1-r)r$ for $\rho_o^m = r$ or for $\rho_o^m = (1-r)$ and where F_d is the inbreeding of dam d. When the dam is not inbred: $\mathrm{Var}(\varepsilon_o^m) = 2\sigma_v^2(1-r)r$, if marker information is available; or $\mathrm{Var}(\varepsilon_o^m) = \sigma_v^2/2$, if marker information is not available. If the dam is not known, $\mathrm{Var}(\varepsilon_o^m) = \sigma_v^2$.

Rearranging (9), $\boldsymbol{v}$ can be written as

$$\boldsymbol{v} = (\boldsymbol{I}-\boldsymbol{P})^{-1}\boldsymbol{\varepsilon} \tag{13}$$

for non-singular $(\boldsymbol{I}-\boldsymbol{P})$, and thus $\boldsymbol{G}_v$ can be written as

$$\boldsymbol{G}_v = (\boldsymbol{I}-\boldsymbol{P})^{-1}\boldsymbol{G}_\varepsilon(\boldsymbol{I}-\boldsymbol{P}')^{-1} \tag{14}$$

From equ.(14), it is clear that $\boldsymbol{G}_v^{-1}$ can be written as

$$\boldsymbol{G}_v^{-1} = (\boldsymbol{I}-\boldsymbol{P}')\boldsymbol{G}_\varepsilon^{-1}(\boldsymbol{I}-\boldsymbol{P}) \tag{15}$$

As shown earlier, $\boldsymbol{P}$ has a simple structure, with each row containing at most 2 non-zero elements, and $\boldsymbol{G}_\varepsilon^{-1}$ is diagonal.

To obtain the rules for inverting $\boldsymbol{G}_v^{-1}$, equ.(15) is written as

$$\boldsymbol{G}_v^{-1} = \boldsymbol{Q}\boldsymbol{G}_\varepsilon^{-1}\boldsymbol{Q}' \tag{16}$$

where $\boldsymbol{Q} = (\boldsymbol{I}-\boldsymbol{P}')$. Because $\boldsymbol{G}_\varepsilon$ is diagonal, equ.(16) can be written as

$$\boldsymbol{G}_v^{-1} = \sum_{j=1}^{2n} \boldsymbol{q}_j \, \boldsymbol{q}_j' d_j \tag{17}$$

where n is number of individuals in the pedigree, $\boldsymbol{q}_j$ is column j of $\boldsymbol{Q}$, and d_j is diagonal element j of $\boldsymbol{G}_\varepsilon^{-1}$. By definition of $\boldsymbol{Q}$, element j of $\boldsymbol{q}_j$ is unity. Further, $\boldsymbol{q}_j$ will have, at most, only 2 other non-zero elements; for $j = i_o^p$, element i_s^p equals $-(1-\rho_o^p)$ and element i_s^m equals $-\rho_o^p$, if the sire of o is known. Similarly, for $j = i_o^m$, element i_d^p equals $-(1-\rho_o^m)$ and element i_d^m equals $-\rho_o^m$, if the dam of o is known. Thus, given parent and marker information of an individual, the contributions to $\boldsymbol{G}_v^{-1}$, corresponding to effects of paternal and maternal MQTL alleles of the individual, are easily obtained.

Now, to obtain the inverse of $\boldsymbol{G}_v$: 1) calculate diagonals of $\boldsymbol{G}_\varepsilon$: when the parent is known, the diagonal is given by equ.(12a) or (12b), and when the parent is unknown, the diagonal is σ_v^2; 2) set $\boldsymbol{G}_v^{-1}$ to the null matrix; 3) for each offspring o, with sire s and dam d, add the following to the indicated elements of $\boldsymbol{G}_v^{-1}$:

if sire is known, add $(1-\rho_o^p)^2 d_{i_o^p}$ to diagonal element i_s^p, i_s^p;
$(1-\rho_o^p)\rho_o^p d_{i_o^p}$ to elements i_s^p, i_s^m and i_s^m, i_s^p;
$-(1-\rho_o^p) d_{i_o^p}$ to elements i_s^p, i_o^p and i_o^p, i_s^p;
$(\rho_o^p)^2 d_{i_o^p}$ to diagonal element i_s^m, i_s^m;
and $-\rho_o^p d_{i_o^p}$ to elements i_s^m, i_o^p and i_o^p, i_s^m;
if dam is known, add $(1-\rho_o^m)^2 d_{i_o^m}$ to diagonal element i_d^p, i_d^p;
$(1-\rho_o^m)\rho_o^m d_{i_o^m}$ to elements i_d^p, i_d^m and i_d^m, i_d^p;
$(1-\rho_o^m) d_{i_o^m}$ to elements i_d^p, i_o^m and i_o^m, i_d^p;
$(\rho_o^m)^2 d_{i_o^m}$ to diagonal element i_d^m, i_d^m;
and $-\rho_o^m d_{i_o^m}$ to elements i_d^m, i_o^m and i_o^m, i_d^m
and always add $d_{i_o^p}$ to element i_o^p, i_o^p;
and $d_{i_o^m}$ to element i_o^m, i_o^m

2. Numerical example. Consider the pedigree in Table I. To construct $\boldsymbol{G}_\varepsilon$, we again take $\sigma_v^2 = 1$ and $r = 0.1$. Because the parents of individuals 1 and 2 are not known, the first 4 elements on the diagonal of $\boldsymbol{G}_\varepsilon$ are $\sigma_v^2 = 1$. For individual 3, each parent is known and marker information is available. Thus, from equs.(12a) and (12b), the two diagonals of $\boldsymbol{G}_\varepsilon$ corresponding to effects of paternal and maternal MQTL alleles of individual 3 are $2(1-r)r = 0.18$. Each parent of individual 4 is also known, but the marker inherited from the sire is not known. Therefore, the diagonal of $\boldsymbol{G}_\varepsilon$ corresponding to v_4^p is 0.5, and that corresponding to v_4^m is $2(1-r)r = 0.18$.

The $\boldsymbol{P}$ matrix for this example is given in Table III. The first 4 rows of $\boldsymbol{P}$ are null because parents of the first 2 individuals are not known. The sire of individual 3 is 1, and M_1^p was transmitted to 3. Thus, the row corresponding to v_3^p has $(1-r) = 0.9$ in the column corresponding to v_1^p and $r = 0.1$ in the column corresponding to v_1^m. Similarly, the dam of individual 3 is 2, and M_2^m was transmitted to 3. Thus, the row corresponding to v_3^m has $r = 0.1$ in the column corresponding to v_2^p and $(1-r) = 0.9$ in the column corresponding to v_2^m. The sire of individual 4 is 1, but marker information is not available. Thus, the row corresponding to v_4^p has 0.5 in the columns corresponding to v_1^p and v_1^m. The dam of individual 4 is 3, and M_3^p

was transmitted to 4. Thus, the row corresponding to v_4^m has $(1-r)=0.9$ in the column corresponding to v_3^p and $r=0.1$ in the column corresponding to v_3^m.

The matrix $\boldsymbol{Q}=(\boldsymbol{I}-\boldsymbol{P}')$ is given in Table IV. The product $\boldsymbol{Q}\boldsymbol{G}_\varepsilon^{-1}\boldsymbol{Q}'$ is given in Table V. It can be verified that this is identical to the inverse of the matrix $\boldsymbol{G}_v$ in Table II.

Table III. $\boldsymbol{P}$ matrix

	v_1^p	v_1^m	v_2^p	v_2^m	v_3^p	v_3^m	v_4^p	v_4^m
v_1^p	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
v_1^m	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
v_2^p	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
v_2^m	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
v_3^p	0.9	0.1	0.0	0.0	0.0	0.0	0.0	0.0
v_3^m	0.0	0.0	0.1	0.9	0.0	0.0	0.0	0.0
v_4^p	0.5	0.5	0.0	0.0	0.0	0.0	0.0	0.0
v_4^m	0.0	0.0	0.0	0.0	0.9	0.1	0.0	0.0

Table IV. $\boldsymbol{Q}$ matrix

	v_1^p	v_1^m	v_2^p	v_2^m	v_3^p	v_3^m	v_4^p	v_4^m
v_1^p	1.0	0.0	0.0	0.0	−0.9	0.0	−0.5	0.0
v_1^m	0.0	1.0	0.0	0.0	−0.1	0.0	−0.5	0.0
v_2^p	0.0	0.0	1.0	0.0	0.0	−0.1	0.0	0.0
v_2^m	0.0	0.0	0.0	1.0	0.0	−0.9	0.0	0.0
v_3^p	0.0	0.0	0.0	0.0	1.0	0.0	0.0	−0.9
v_3^m	0.0	0.0	0.0	0.0	0.0	1.0	0.0	−0.1
v_4^p	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0
v_4^m	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0

Table V. $\boldsymbol{Q}\boldsymbol{G}_\varepsilon^{-1}\boldsymbol{Q}'$

	v_1^p	v_1^m	v_2^p	v_2^m	v_3^p	v_3^m	v_4^p	v_4^m
v_1^p	6.0	1.0	0.0	0.0	−5.0	0.0	−1.0	0.0
v_1^m	1.0	1.556	0.0	0.0	−0.556	0.0	−1.0	0.0
v_2^p	0.0	0.0	1.056	0.5	0.0	−0.556	0.0	0.0
v_2^m	0.0	0.0	0.5	5.5	0.0	−5.0	0.0	0.0
v_3^p	−5.0	−0.556	0.0	0.0	10.056	0.5	0.0	−5.0
v_3^m	0.0	0.0	−0.556	−5.0	0.5	5.611	0.0	−0.556
v_4^p	−1.0	−1.0	0.0	0.0	0.0	0.0	2.0	0.0
v_4^m	0.0	0.0	0.0	0.0	−5.0	−0.556	0.0	5.556

D. BLUP with multiple markers

If information on another marker locus linked to a QTL is available, the model can be expanded to include effects of alleles of this MQTL. This approach, however, results in $2n$ additional equations for each marker introduced into the analysis. Thus, for a large number of individuals (n) and a large number of MQTLs, solving the mixed model equations may not be feasible. An alternative would be to use equ.(2), with

$$a_i = \sum_k (v_{ki}^p + v_{ki}^m) + u_i \tag{18}$$

where v_{ki}^p and v_{ki}^m are effects of paternal and maternal alleles of the kth MQTL. The covariance matrix of effects of MQTL alleles at each locus ($\boldsymbol{G}_{v_k}$) can be constructed using the tabular method described in Section II.B. Then, assuming gametic equilibrium, the covariance of matrix a_i values ($\boldsymbol{G}_{a|r,m}$) can be obtained as

$$\boldsymbol{G}_{a|r,m} = \boldsymbol{Z}(\sum_k \boldsymbol{G}_{v_k})\boldsymbol{Z}' + \boldsymbol{G}_u \tag{19}$$

where $\boldsymbol{Z}$ is a $n \times 2n$ matrix with elements for row i containing a 1 corresponding to each of the paternal and maternal MQTL effects of individual i and zeros for the remaining elements. The problem with this approach, however, is that it could not be applied to large systems, unless a simple algorithm to invert $\boldsymbol{G}_{a|r,m}$ is available.

DISCUSSION

Results presented here are an application of BLUP to marker-assisted selection. This is a generalization of the method presented by Soller (1978) and Soller and Beckmann (1983). This generalization allows simultaneous evaluation of fixed effects, MQTL effects and the residual QTL effects, using known relationships and phenotypic and marker information. It is sufficiently general to accommodate individuals with partial or no marker information.

Several authors have calculated the additional genetic progress expected from marker-assisted selection (Soller, 1978, Soller and Beckmann, 1983; Smith and Simpson, 1986). Because the method presented here is a generalization of the method considered by these authors, their results give an indication of the advantage expected by using marker-assisted BLUP.

Application of this procedure requires knowledge of the recombination rate (r) between the marker and the MQTL and the variance of the additive effect of the MQTL alleles (σ_v^2). Assuming that effects of MQTL alleles are normally distributed, the model presented here could be used to estimate r and σ_v^2 by restricted maximum likelihood (REML; Patterson and Thompson, 1971). The robustness of REML estimation, with respect to the distribution of effects of MQTL alleles, needs to be examined.

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APPENDIX

Proof that G_ε is diagonal

Let o be an individual that is not a direct descendant of o'. From equs.(8a) and (8b), the additive effects of the MQTL alleles of o and o' are

$$v_o^z = (1-\rho_o^z)v_\xi^p + \rho_o^z v_\xi^m + \varepsilon_o^z \tag{A1}$$

and

$$v_{o'}^{z'} = (1-\rho_{o'}^{z'})v_{\xi'}^p + \rho_{o'}^{z'} v_{\xi'}^m + \varepsilon_{o'}^{z'} \tag{A2}$$

where z can take values p or m, $\xi = s$ when $z = p$, or $\xi = d$ when $z = m$. Similarly, z' can take values p or m, $\xi' = s'$ when $z' = p$, or $\xi' = d$ when $z' = m$. Note that for an arbitrary pair of individuals, ones is not direct descendant of the other. Therefore, to prove that $\boldsymbol{G}_\varepsilon$ is diagonal, it is sufficient to show thar the covariance between ε_o^z and $\varepsilon_{o'}^{z'}$ is null.

From equs.(A1) and (A2), the covariance between additive effects of MQTL alleles v_o^z and $v_{o'}^{z'}$ can be written as

$$\begin{aligned}\mathrm{Cov}(v_o^z, v_{o'}^{z'}) &= \mathrm{Cov}[v_o^z, (1-\rho_{o'}^{z'})v_{\xi'}^p + \rho_{o'}^{z'} v_{\xi'}^m + \varepsilon_{o'}^{z'}] \\ &= (1-\rho_{o'}^{z'})\mathrm{Cov}(v_o^z, v_{\xi'}^p) + \rho_{o'}^{z'}\mathrm{Cov}(v_o^z, v_{\xi'}^m) \\ &\quad + \mathrm{Cov}(v_o^z, \varepsilon_{o'}^{z'})\end{aligned} \tag{A3}$$

But, from equs.(7a) and (7b)

$$\mathrm{Cov}(v_o^z, v_{o'}^{z'}) = (1-\rho_{o'}^{z'})\mathrm{Cov}(v_o^z, v_{\xi'}^p) + \rho_{o'}^{z'}\mathrm{Cov}(v_o^z, v_{\xi'}^m) \tag{A4}$$

Thus, for equ.(A3) to equal (A4), the third term in equ.(A3), $\mathrm{Cov}(v_o^z, \varepsilon_{o'}^{z'})$, must be zero. The same reasoning can be used to show that $\mathrm{Cov}(v_\xi^p, \varepsilon_{o'}^{z'})$ and $\mathrm{Cov}(v_\xi^m \varepsilon_{o'}^{z'})$ are zero. Therefore, given that $\mathrm{Cov}(v_o^z, \varepsilon_{o'}^{z'})$, $\mathrm{Cov}(v_\xi^p, \varepsilon_{o'}^{z'})$ and $\mathrm{Cov}(v_\xi^m, \varepsilon_{o'}^{z'})$ are zero, $\mathrm{Cov}(\varepsilon_o^z, \varepsilon_{o'}^{z'})$ must be zero.

Further, taking o to be the a parent of o', the residual $(\varepsilon_{o'}^{z'})$ in equ.(A2) is uncorrelated with $v_{\xi'}^p$ and with $v_{\xi'}^m$ in equ.(A2), because $\mathrm{Cov}(v_o^z, \varepsilon_{o'}^{z'}) = 0$, as shown above. The result that be effect of each MQTL allele of a parent is uncorrelated with the residual $(\varepsilon_{o'}^{z'})$ of its offspring was used to obtain equ.(10)

Jamrozik, Schaeffer and Dekkers (1997)

Jamrozik, J., Schaeffer, L.R., and Dekkers, J.C.M. 1997. Genetic evaluation of dairy cattle using test day yields and random regression model. Journal of Dairy Science, vol. 80, no. 6, pp. 1217 to 1226.

The problem

Dairy cattle evaluations used 305 day yields, obtained from monthly test day records and adjusted by extension factors. This approach lost the shape of the lactation curve and the effect of each test date. Jamrozik, Schaeffer and Dekkers applied the random regression model directly to the individual test day records of more than 700,000 Canadian cows, estimating a lactation curve for each animal.

Points to note while reading

Note that each cow receives random regression coefficients, which describe how her breeding value changes over the course of lactation. The model yields both the breeding value for 305 days and values for persistency. This was the model adopted in Canada, and it served as the basis for evaluations in several countries.

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GENETICS AND BREEDING

Genetic Evaluation of Dairy Cattle Using Test Day Yields and Random Regression Model

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ABSTRACT

A model for analyzing test day records that contains both fixed and random regression coefficients was applied to the genetic evaluation of first lactation data for Canadian Holstein dairy cows. Data were 5.1 million test day records with milk, fat, and protein yields from calvings between 1988 and 1995 from herds in four regions of Canada. Each evaluated animal received five predictions for each trait representing the random regression coefficients. From these solutions, a range of estimated breeding values for various parts of the lactation could be calculated. Three genetic measures of persistency were compared. Bulls could be selected for both yields and persistency of their daughters in whatever combination was desirable. Test day analyses could result in monthly genetic evaluations for yield traits.
(**Key words**: random regression model, genetic evaluation, test day yields, persistency)

Abbreviation key: **OC** = official Canadian 305-d EBV, **RIP dip** = dip in bull EBV record in progress, **RRM** = random regression model, **T** = EBV for a complete 305-d lactation from TD yields, **TD** = test day.

INTRODUCTION

Test day (**TD**) yields (24-h measurements) for milk, fat, and protein are used to calculate yields for standard 305-d lactations. A TD model for genetic evaluation can account for factors that are specific to each TD, such as management groups within a herd on TD, day of the year (including weather conditions), and, for each cow, DIM, pregnancy status, medical treatments, and number of times milked on TD (3, 4). Many of these factors change for a cow from one TD to the next and would be difficult to model for 305-d yields. Shapes of lactation curves could differ by herd also.

The TD model of Ptak and Schaeffer (4) assumed a fixed lactation shape within age and season of calving subclasses, and the EBV measured genetic differences between animals in the height of curves. This TD model has been applied to SCS in Canada (6). Schaeffer and Dekkers (8) extended the TD model to allow the shape of the lactation curve to differ for each cow by the inclusion of random regression coefficients for each animal (1). The lactation curve for an individual cow is viewed as two sets of regressions on DIM. Fixed regressions for all cows belonging to the same subclass (for example, regions, age at calving, and season of calving) describe the general shape for that cow, and the random regressions for a cow describe the genetic deviations from the fixed regressions, which allows each cow to have a different shape of lactation curve on a genetic level. This model for TD yields is a random regression model (**RRM**).

Variance and covariance components for the RRM were estimated by Jamrozik and Schaeffer (2) from a data file of records for 6516 Canadian Holstein cows and 50,412 TD. A total of 45 parameters were estimated for a single-trait RRM for milk, fat, and protein yields during first lactation.

The objective of this study was to apply RRM to TD records for a large population of dairy cattle to obtain genetic evaluations for cows and bulls. The RRM analysis provides five solutions for each animal for each trait, but, from these solutions, EBV can be calculated for several quantities for partial lactations and for persistency. This paper describes how the RRM solutions can be utilized for selection.

MATERIALS AND METHODS

Data

There were 5.1 million TD records on Holstein cows from four milk recording organizations in Canada [Ontario, Québec, prairie provinces, (Manitoba, Saskatchewan, and Alberta), and British Columbia] for cows calving from 1988 through 1995. Data were

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restricted to first lactations, DIM on TD between 5 and 305 d, age at first calving between 18 and 48 mo, milk yields from 1.5 to 90 kg, and fat and protein percentages from 1.5 to 10.0%. There were TD records for 709,357 cows that were sired by 13,322 bulls; 567,493 herd-test date subclasses were represented. Including ancestors, the total number of animals evaluated was 1,096,272.

Model

The single-trait RRM for TD was

$$y_{ijkl} = HTD_i + \sum_{m=0}^{4} \beta_{km} z_{jlm} + \sum_{m=0}^{4} \alpha_{jm} z_{jlm} + p_j + e_{ijkl},$$

where y_{ijkl} is record l of cow j made on $DIMt_{jl}$ of first lactation for a cow belonging to subclass k of region by age at calving by season of calving; HTD_i is the fixed effect of herd-test date i; p_j is random permanent environmental effect associated with all TD yields of cow j within a lactation; e_{ijkl} is random residual effect; and β_{km} and α_{jm} are fixed and random regression coefficients, respectively. Let

$$\mathbf{z}_{jl} = (z_{jl0}\ z_{jl1}\ z_{jl2}\ z_{jl3}\ z_{jl4})' = (1\ c\ c^2\ d\ d^2)$$

where $c = t_{jl}/305$, and $d = \ln(1/c)$. Division by 305 reduces the range of the covariables.

The fixed regressions were estimated within 32 subclasses of region by age at calving by season of calving. Regions were defined by the four processing centers. Age groups were <25 mo, 25 to 29 mo, 30 to 34 mo, and >34 mo. Seasons were September through February and March through August. The numbers of TD records per subclass are given in Table 1.

A more complex model could include fixed or random regressions for each herd-year subclass to account for management effects on shape of lactation curves of cows within the herd. This possibility is an area for future research to improve the RRM.

In matrix notation, the model can be written as

$$\mathbf{y} = \mathbf{Xb} + \mathbf{ZQg} + \mathbf{Za} + \mathbf{Wp} + \mathbf{e},$$

where $\mathbf{b} = HTD_i$ and β_{km}, $\mathbf{a} = \alpha_{jm}$, $\mathbf{p}$ = vector of permanent environmental effects for cows with TD records, $\mathbf{e}$ = vector of residual effects, and $\mathbf{X}$, $\mathbf{Z}$, and $\mathbf{W}$ = incidence and covariable matrices. The vector $\mathbf{g}$ = phantom parent groups, and $\mathbf{Q}$ = relationship of animals to phantom parent groups. Assume that

$$\begin{pmatrix} \mathbf{a} \\ \mathbf{p} \\ \mathbf{e} \end{pmatrix} \sim \mathbf{N}(\mathbf{0}, \mathbf{V})$$

with

$$\mathbf{V} = \begin{pmatrix} \mathbf{G} \otimes \mathbf{A} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}\sigma_p^2 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{R} \end{pmatrix}$$

where

$$\mathbf{G} = V\ (\alpha_{j0},\ \alpha_{j1},\ \alpha_{j2},\ \alpha_{j3},\ \alpha_{j4})'$$

is the genetic covariance matrix of the random regression coefficients, assumed to be the same for all animals, $\mathbf{A}$ = additive genetic relationship matrix among the animals, $\otimes$ = Kronecker product function (10), $\mathbf{I}$ = identity matrix, σ_p^2 = variance of the permanent environmental effects assumed to be constant over DIM classes, and $\mathbf{R}$ = diagonal matrix with elements that depend on DIM, i.e.,

TABLE 1. Number of test day records per region, age, and season subclass.

		Region			
Age at calving (mo)	Season	British Columbia	Prairie[1]	Québec	Ontario
≤24	1	33,195	53,704	161,450	184,570
	2	31,338	56,159	84,587	155,964
25–29	1	128,924	190,978	778,018	649,200
	2	113,223	174,336	483,368	553,691
30–34	1	33,207	62,861	287,967	234,637
	2	30,773	61,067	165,635	182,510
≥34	1	6713	16,639	60,736	51,162
	2	4804	11,933	25,334	32,232

[1]Manitoba, Saskatchewan, and Alberta.

$$\mathbf{R} = \text{diag}\ \{\sigma^2_{e_k}\}$$

where k denotes DIM (1 to 305). Regression equations were used to derive the appropriate residual variance for a particular TD record, $\sigma^2_{e_k}$. The 29 different residual variances estimated by Jamrozik and Schaeffer (2) were fitted to a regression model with linear and quadratic terms for DIM. The resulting prediction equations were

for milk

$$\sigma^2_{e_k} = 5.775 - 0.030987k + 0.000079k^2,$$

for fat

$$\sigma^2_{e_k} = 1.616 - 0.009995k + 0.000023k^2,$$

and for protein

$$\sigma^2_{e_k} = 0.580 - 0.00247k + 0.000006k^2;$$

R^2 were 0.74, 0.92, and 0.51, respectively. Although the R^2 for protein was low, the effect of errors in the residual variances on EBV was small. Parameters for **G** and permanent environmental variances for milk, fat, and protein yields were given previously by Jamrozik and Schaeffer (2) and are shown in Table 2.

Six phantom parent groups were formed on the basis of one group for unknown sires and five groups for unknown dams by year of birth. In Canada, genetic evaluation of 305-d yields utilizes 32 phantom parent groups, and more groups will be needed with RRM when routine evaluations based on TD are adopted. Mixed model equations were developed that were similar to those of Quaas (5) that solved for group plus animal additive genetic effects directly. Iterations (9) were performed until the sum of squares of changes between two consecutive rounds, divided by the sum of squares of the latest solution, attained a value of $<1 \times 10^{-8}$, which required 320, 271, and 302 rounds of iteration for milk, fat, and protein yields, respectively.

TABLE 2. Estimates of genetic (co)variances for random regression coefficients and permanent environmental (pe) variances for milk, fat, and protein yields (kilograms).

Parameters		Milk	Fat × 10	Protein × 10
α_0	α_0	47.92	6.65	6.19
α_0	α_1	–3.81	–1.61	–1.05
α_0	α_2	–45.05	–5.14	–5.04
α_0	α_3	–34.27	–5.21	–4.61
α_0	α_4	5.89	0.97	0.83
α_1	α_1	1.76	0.65	0.42
α_1	α_2	4.72	1.42	1.02
α_1	α_3	–0.18	1.06	0.54
α_1	α_4	0.28	–0.18	–0.07
α_2	α_2	47.84	4.65	4.60
α_2	α_3	34.02	4.20	3.74
α_2	α_4	–6.15	–0.79	–0.69
α_3	α_3	34.02	5.17	4.05
α_3	α_4	–6.73	–1.03	–0.80
α_4	α_4	1.44	0.22	0.17
pe		5.79	0.75	0.50

Functions of Solutions

Let the sum of the group plus additive genetic solutions for animal i (for one trait) be represented as

$$\hat{\mathbf{a}}_i' = (\hat{a}_{0i}\ \hat{a}_{1i}\ \hat{a}_{2i}\ \hat{a}_{3i}\ \hat{a}_{4i}).$$

These are the EBV for the random regression coefficients. By themselves these EBV are not very useful for ranking or selecting animals, but functions of these EBV can provide useful information to producers for selection decisions. The EBV of animal i for yield on the DIM k, D_k, can be calculated as

$$D_k = \hat{a}_{0i} + \hat{a}_{1i}c_k + \hat{a}_{2i}c_k^2 + \hat{a}_{3i}d_k + \hat{a}_{4i}d_k^2,$$

where $c_k = k/305$ and $d_k = \ln(1/c_k)$. An EBV for a complete 305-d lactation from TD yields, **T**, is obtained by summing the EBV for each DIM. That is,

$$\begin{aligned} T &= \sum_{k=1}^{305} D_k \\ &= \sum_{k=1}^{305} (\hat{a}_{0i} + \hat{a}_{1i}c_k + \hat{a}_{2i}c_k^2 + \hat{a}_{3i}d_k + \hat{a}_{4i}d_k^2) \\ &= \left(\sum_{k=1}^{305} 1\right)\hat{a}_{0i} + \left(\sum_{k=1}^{305} c_k\right)\hat{a}_{1i} + \left(\sum_{k=1}^{305} c_k^2\right)\hat{a}_{2i} \\ &\quad + \left(\sum_{k=1}^{305} d_k\right)\hat{a}_{3i} + \left(\sum_{k=1}^{305} d_k^2\right)\hat{a}_{4i} \\ &= (305\ 153\ 102.17\ 301.22\ 581.23)\hat{\mathbf{a}}_i. \end{aligned}$$

The EBV for various partial lactation yields are obtained by varying the summation interval. Let C_m represent the EBV for yield in part m of lactation. Then,

$$C_1 = \sum_{k=1}^{100} D_k$$
$$= (100\ 16.56\ 3.64\ 208.29\ 519.12)\hat{\mathbf{a}}_i,$$
$$C_2 = \sum_{k=101}^{200} D_k$$
$$= (100\ 49.34\ 25.24\ 72.54\ 56.50)\hat{\mathbf{a}}_i,$$
$$C_3 = \sum_{k=201}^{305} D_k$$
$$= (105\ 87.10\ 73.29\ 20.39\ 5.50)\hat{\mathbf{a}}_i.$$

Note how the contributions of the five random regression coefficients change from the beginning to the end of the lactation. During the early lactation period, $\hat{a}_{3i}$ and $\hat{a}_{4i}$ are weighted more heavily than during late lactation, and $\hat{a}_{1i}$ and $\hat{a}_{2i}$ are weighted more heavily at the end of lactation. The intercept, $\hat{a}_{0i}$, has the same weight for every DIM.

Because the RRM allows the estimation of the shapes of lactation curves for individual cows, EBV for persistency can be easily derived from the analysis to provide additional information to breeders that was not possible with analyses of 305-d yields. Although EBV for persistency can be provided easily, whether to recommend selection for greater or lower persistency for improved economic returns is not known. Cows with high persistency may endure more stress during lactation, which could adversely affect reproduction or longevity. The ramifications of genetically changing the shape of lactation curves are not known.

The EBV for persistency were defined in three ways in this study. The first definition, P_1, was the additional genetic yield (gained or lost) from d 60 to 280 relative to an average animal having the same yield on d 60 (Figure 1). The shaded area of Figure 1 is approximated as the area of a triangle with a base equal to 220 d and a height of $(D_{280} - D_{60})$. Thus,

$$D_{280} = (1\ 0.92\ 0.84\ 0.09\ 0.01)\hat{\mathbf{a}}_i,$$
$$D_{60} = (1\ 0.20\ 0.04\ 1.63\ 2.64)\hat{\mathbf{a}}_i,$$
$$D_{280} - D_{60} = (0\ 0.72\ 0.80\ -1.54\ -2.64)\hat{\mathbf{a}}_i,$$
$$P_1 = (D_{280} - D_{60}) \times 110.0.$$

Note that $\hat{a}_{0i}$ has disappeared in the calculations, and, therefore, P_1 is independent of the intercept of the lactation curve. Higher values of P_1 indicate greater additional yield, which is indicated by the shape of the lactation curve. A similar quantity (not used in this study) could be calculated easily by summing the additional yields for each day from d 60 to 280, which would not require the approximation of a triangular shape, i.e.,

$$\sum_{k=60}^{280} (D_k - D_{60}).$$

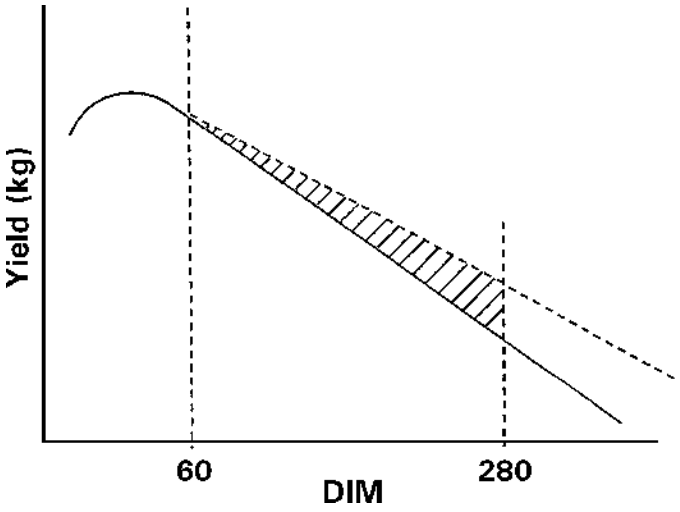


Figure 1. Typical lactation curve of dairy cows; shaded portion shows the additional yield from greater persistency.

The second definition of persistency, P_2, was based on the ratio of partial lactation yield from d 201 to 305 divided by partial lactation yield from d 1 to 100. Phenotypic means of the partial lactation yields, f_1 and f_3, for Ontario Holsteins, aged 25 to 29 mo and calving in September to February were added to the EBV before the ratio was calculated and then was deviated from the ratio of the phenotypic means of the cumulative yields. Thus,

$$P_2 = \left(\frac{C_3 + f_3}{C_1 + f_1} - \frac{f_3}{f_1}\right) \times 100.$$

The third definition of persistency, P_3, was based on the ratio of partial lactation yield from d 201 to 305 divided by total 305-d yield. Similarly, phenotypic means, f_3 and F, had to be added to the C_3 and T, respectively, based on the same group of cows used for P_2. Thus,

$$P_3 = \left(\frac{C_3 + f_3}{T + F} - \frac{f_3}{F}\right) \times 100.$$

The solutions, $\hat{\mathbf{a}}_i$, and the EBV for yields and persistency for 1730 sires with a minimum of 50 daughters were correlated with one another within milk,

fat, and protein yields. Official Canadian 305-d EBV (**OC**), based on all lactations per cow from January 1996, were also included.

RESULTS AND DISCUSSION

Fixed Regressions

Examples of the lactation curves for milk yield that were derived from the fixed regression coefficients are shown in Figures 2 and 3 for Québec and the prairie provinces, respectively, for cows calving during the period from March to August. Regional and age differences in shape of lactation curves are apparent. The lactation peak of Québec cows was higher; the decline after the peak was steeper than those of cows in the prairie provinces. Similar differences occurred among season groups and for fat and protein yields (not shown). In the future, as more TD records become available, the number of fixed regressions could be expanded to allow six 2-mo seasons; more age groups; an interaction of region, age, and season of calving with time; and separate lactation curves for herd and year subclasses.

In this study, covariables for the fixed and random regressions were represented by functions of the same form. However, the covariables for the fixed regressions could be different from those for the random regressions. Logically, the fixed regressions could accommodate five or more covariables, and the random regressions could perhaps be reduced to three or four covariables. Research in this area is warranted.

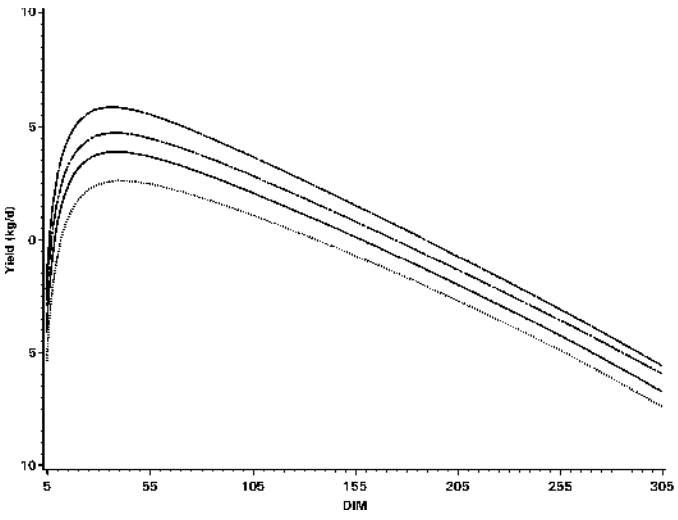


Figure 2. Daily milk yield curves deviated from the overall mean for Holsteins from Québec that calved during the period from March to August. Age groups are <25 mo (· · ·), 25 to 29 mo (—), 30 to 34 mo (–·–), and >34 mo (— —).

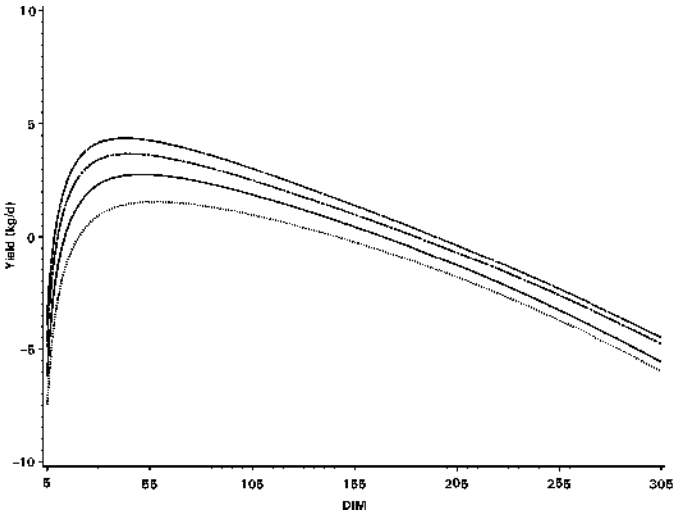


Figure 3. Daily milk yield curves deviated from the overall mean for Holsteins from the prairie provinces that calved during the period from March to August. Age groups are <25 mo (· · ·), 25 to 29 mo (—), 30 to 34 mo (–·–), and >34 mo (— —).

EBV

The random regression solutions for milk yield of 10 bulls are given in Table 3, and the EBV for 305-d yield, partial lactation yields, and persistency are given in Table 4. Variability was substantial among bulls in the regression coefficients and EBV. Hence, the shapes of lactation curves for bull and daughter groups were also variable.

Means and standard deviations of solutions and EBV for 1730 bulls with ≥50 daughters are given in Table 5. For all three traits, $\hat{a}_{0i}$ had the largest variability, followed by either $\hat{a}_{2i}$ or $\hat{a}_{3i}$, which corresponded to the intercept, and covariables of $(DIM/305)^2$ and ln(305/DIM). The EBV for 305-d yields are affected most by a_{0i}; modification was needed because of the shape of the lactation curve, primarily through a_{2i} and a_{3i}, which are positively correlated with each other, but both are negatively correlated with a_{0i}. These correlations are less than unity, and, therefore, significant changes in shape of lactation curves can be made genetically.

The variability of T was less than that of OC, which were based on all lactation records of a cow and on data from 1957 to the present; T was based on TD yields within first lactations only and from fewer years of data. The variability of P_1 was similar to the variability of C_1, C_2, or C_3, but the variability of the other measures of persistency were much lower because they were ratios of yields rather than yields themselves.

TABLE 3. Estimated breeding values of random regression coefficients for milk yield (kilograms) for 10 bulls from a random regression model.[1]

Bull	Daughters	Test day records	$\hat{a}_{0i}$	$\hat{a}_{1i}$	$\hat{a}_{2i}$	$\hat{a}_{3i}$	$\hat{a}_{4i}$
Aerostar	10.742	75,218	5.15	0.27	1.17	0.03	–0.35
Bandit	37	276	4.77	0.19	1.40	0.67	–0.35
Eagle	231	1795	5.30	–0.28	–0.31	0.14	–0.40
Felix	36	279	11.34	0.43	–3.87	–5.41	0.71
Lindy	8619	58,321	3.05	–0.60	–0.10	1.29	–0.57
Mascot	242	1335	8.61	–0.05	–1.85	–2.16	–0.03
Prelude	3421	15,341	–0.65	0.17	4.48	3.89	–0.85
Rockie	278	1719	9.40	–0.94	–5.77	–3.42	0.24
Sambo	77	600	14.85	–0.65	–9.26	–7.65	1.08
Valiant	658	4096	9.84	–0.61	–5.10	–3.72	0.28

[1]$\hat{a}_{0i}$ = Intercept, $\hat{a}_{1i}$ = coefficient of c = DIM/305, $\hat{a}_{2i}$ = coefficient of c^2, $\hat{a}_{3i}$ = coefficient of d = ln(1/c), and $\hat{a}_{4i}$ = coefficient of d^2.

TABLE 4. Estimated breeding values for 305-d lactation yield, partial lactation yields, and persistency for milk yield (kilograms) for 10 bulls from a random regression model.[1]

Bull	T	C_1	C_2	C_3	P_1	P_2	P_3
Aerostar	1539	349	541	649	220	12	2.24
Bandit	1628	445	551	632	126	9	1.73
Eagle	1353	347	496	510	42	8	1.31
Felix	1909	367	705	838	404	18	3.03
Lindy	887	268	334	284	–110	2	0.28
Mascot	1761	387	654	721	208	14	2.26
Prelude	964	323	291	350	–3	3	0.81
Rockie	1240	314	513	414	–74	5	0.58
Sambo	1810	411	726	674	110	11	1.56
Valiant	1429	327	571	532	49	9	1.29

[1]T = EBV for 305-d yields; C_m = EBV for partial lactation yields where m = 1 is from d 1 to 100, m = 2 is from d 101 to 200, and m = 3 is from d 201 to 305; and P_n = EBV for persistency where n = 1 is additional of gain or loss in yield (kilograms), n = 2 is the ratio of yield during the last 100 d of lactation to yield during the first 100 d of lactation deviated from the average, and n = 3 is the ratio of yield during the last 100 d to total 305-d yield deviated from the average.

TABLE 5. Means and standard deviations from the random regression model of EBV of 1730 bulls with 50 or more daughters.

EBV[1]	Milk yield		Fat yield		Protein yield	
	$\bar{X}$	SD	$\bar{X}$	SD	$\bar{X}$	SD
$\hat{a}_{0i}$	0.32	3.74	0.014	0.127	0.019	0.112
$\hat{a}_{1i}$	–0.05	0.56	0.003	0.034	–0.001	0.024
$\hat{a}_{2i}$	0.29	3.09	0.008	0.097	0.001	0.082
$\hat{a}_{3i}$	0.36	2.58	0.003	0.100	0.001	0.077
$\hat{a}_{4i}$	–0.12	0.52	–0.001	0.022	–0.002	0.016
T	158.45	579.96	5.695	20.696	4.919	15.993
C_1	44.25	172.22	1.591	7.275	1.055	4.544
C_2	56.61	214.19	1.838	6.941	1.850	5.805
C_3	57.59	226.86	2.266	7.804	2.014	6.675
P_1	–4.13	180.91	0.641	6.622	0.445	5.410
P_2	0.77	6.04	0.982	5.573	1.274	5.937
P_3	0.10	1.18	0.167	1.083	0.170	1.077
OC	210.55	799.20	8.055	29.253	6.932	23.930

[1]$\hat{a}_{ji}$ = EBV for the random regression coefficients; T = EBV for 305-d yields; C_m = EBV for partial lactation yields where m = 1 is for yield from d 1 to 100, m = 2 is for yield from d 101 to 200, and m = 3 is for yield from d 201 to 305; P_n = EBV for persistency where n = 1 is additional gain or loss in yield (kilograms), n = 2 is the ratio of yield during the last 100 d to yield during the first 100 d of lactation deviated from the average, and n = 3 is the ratio of yield during the last 100 d to total 305-d yield deviated from the average; and OC = official Canadian 305-d EBV from January 1996.

TABLE 6. Correlations between EBV for milk yield of 1730 bulls with 50 or more daughters from the random regression model.

EBV[1]	T	C_1	C_2	C_3	P_1	P_2	P_3	OC
$\hat{a}_{0i}$	0.71	0.54	0.80	0.66	0.08	0.46	0.32	0.65
$\hat{a}_{1i}$	–0.19	–0.41	–0.29	0.10	0.85	0.51	0.64	–0.14
$\hat{a}_{2i}$	–0.21	–0.14	–0.34	–0.11	0.17	–0.03	0.13	–0.17
$\hat{a}_{3i}$	–0.17	0.07	–0.27	–0.24	–0.33	–0.40	–0.28	–0.15
$\hat{a}_{4i}$	–0.06	–0.23	0.01	0.02	0.34	0.22	0.16	–0.06
T		0.90	0.99	0.94	0.10	0.55	0.49	0.94
C_1			0.87	0.72	–0.29	0.13	0.10	0.83
C_2				0.92	0.06	0.54	0.46	0.92
C_3					0.42	0.78	0.75	0.90
P_1						0.85	0.90	0.13
P_2							0.98	0.53
P_3								0.49

[1]$\hat{a}_{ji}$ = EBV for the random regression coefficients; T = EBV for 305-d yields; C_m = EBV for partial lactation yields where m = 1 is for yield from d 1 to 100, m = 2 is for yield from d 101 to 200, and m = 3 is for yield from d 201 to 305; P_n = EBV for persistency where n = 1 is additional gain or loss in yield (kilograms), n = 2 is the ratio of yield during the last 100 d to yield during the first 100 d of lactation deviated from the average, and n = 3 is the ratio of yield during the last 100 d to total 305-d yield deviated from the average; and OC = official Canadian 305-d EBV from January 1996.

Correlations

Correlations among EBV for production and persistency are given in Tables 6, 7, and 8 for milk, fat, and protein yields, respectively. Correlations were very similar for all three traits. Correlations of T with OC were 0.93 and 0.94. If the assumption is made that EBV from RRM are more accurate due to better accounting for environmental effects on TD yields and shapes of lactation curves, then the current animal model EBV are about 94% efficient. However, T were derived from TD data for first lactation from a few years, and OC were derived from 305-d data for all lactations over many more years. Thus, the correlations could be low due to differences in data. Ptak and Schaeffer (4) found correlations that were similar to those of this study when 305-d and TD data on the same cows were analyzed with both a TD model and an animal model; therefore, the low correlations may not be affected much by differences in data going into the EBV.

The correlation between C_2 and C_3 was greater than between C_1 and C_2, but both correlations were greater than the correlation between C_1 and C_3. Thus, partial lactation yields during the last 205 d of lactation were more similar than partial lactation yields during the first 100 d of lactation. Daily yields during the early part of lactation are more variable than daily yields after 100 d, and this period also includes the variability in peak yields.

TABLE 7. Correlations between EBV for fat yield of 1730 bulls with 50 or more daughters from the random regression model.

EBV[1]	T	C_1	C_2	C_3	P_1	P_2	P_3	OC
$\hat{a}_{0i}$	0.56	0.41	0.66	0.51	0.07	0.26	0.15	0.50
$\hat{a}_{1i}$	–0.02	–0.09	–0.17	0.17	0.50	0.34	0.50	0.01
$\hat{a}_{2i}$	0.00	0.01	–0.13	0.10	0.19	0.13	0.27	0.03
$\hat{a}_{3i}$	–0.03	0.13	–0.12	–0.10	–0.33	–0.28	–0.19	–0.01
$\hat{a}_{4i}$	0.02	–0.05	0.05	0.06	0.31	0.14	0.12	–0.02
T		0.91	0.98	0.93	0.06	0.30	0.32	0.93
C_1			0.86	0.71	–0.30	–0.11	–0.06	0.82
C_2				0.91	0.03	0.33	0.30	0.92
C_3					0.40	0.62	0.64	0.89
P_1						0.90	0.94	0.09
P_2							0.98	0.34
P_3								0.35

[1]$\hat{a}_{ji}$ = EBV for the random regression coefficients; T = EBV for 305-d yields; C_m = EBV for partial lactation yields where m = 1 is for yield from d 1 to 100, m = 2 is for yield from d 101 to 200, and m = 3 is for yield from d 201 to 305; P_n = EBV for persistency where n = 1 is additional gain or loss in yield (kilograms), n = 2 is the ratio of yield during the last 100 d to yield during the first 100 d of lactation deviated from the average, and n = 3 is the ratio of yield during the last 100 d to total 305-d yield deviated from the average; and OC = official Canadian 305-d EBV from January 1996.

TABLE 8. Correlations between EBV for protein yield of 1730 bulls with 50 or more daughters from the random regression model.

EBV[1]	T	C_1	C_2	C_3	P_1	P_2	P_3	OC
$\hat{a}_{0i}$	0.72	0.56	0.80	0.65	0.19	0.43	0.30	0.63
$\hat{a}_{1i}$	0.00	–0.09	–0.15	0.20	0.60	0.35	0.51	0.04
$\hat{a}_{2i}$	–0.27	–0.27	–0.38	–0.14	0.18	0.03	0.17	–0.19
$\hat{a}_{3i}$	–0.35	–0.15	–0.43	–0.37	–0.33	–0.39	–0.29	–0.27
$\hat{a}_{4i}$	0.20	0.09	0.23	0.21	0.29	0.22	0.17	0.12
T		0.87	0.99	0.95	0.32	0.57	0.53	0.93
C_1			0.84	0.68	–0.11	0.10	0.09	0.80
C_2				0.92	0.27	0.56	0.49	0.91
C_3					0.60	0.80	0.77	0.89
P_1						0.91	0.95	0.32
P_2							0.98	0.55
P_3								0.52

[1]$\hat{a}_{ji}$ = EBV for the random regression coefficients; T = EBV for 305-d yields; C_m = EBV for partial lactation yields where m = 1 is for yield from d 1 to 100, m = 2 is for yield from d 101 to 200, and m = 3 is for yield from d 201 to 305; P_n = EBV for persistency where n = 1 is additional gain or loss in yield (kilograms), n = 2 is the ratio of yield during the last 100 d to yield during the first 100 d of lactation deviated from the average, and n = 3 is the ratio of yield during the last 100 d to total 305-d yield deviated from the average; and OC = official Canadian 305-d EBV from January 1996.

The EBV for persistency differed in the correlations with EBV for 305-d yields, either from RRM or the official EBV. Persistency—defined as the additional gain or loss in yield between d 60 and 280 relative to an average cow, P_1—was much less correlated with 305-d EBV for all traits than the other two measures of persistency. The P_1 was, therefore, less affected by peak yield than the other measures and could be used with EBV for 305-d yield to change the shape of lactation curves by selection.

P_1 was highly correlated (0.85 to 0.95) with the other two measures of persistency. P_1 was negatively correlated with C_1, the EBV for partial lactation yield from d 1 to 100; P_1 had small correlations with C_2, the EBV for partial lactation yield from d 101 to 200; and P_1 was positively correlated with the EBV for partial lactation yield, C_3, from d 201 to 305. The other two EBV for persistency were positively and more highly correlated with all measures of partial lactation yield for milk and protein yields but were negatively correlated with C_1 for fat yields. P_2 and P_3 were highly correlated (0.98) with each other for all traits. Thus, any of the three measures of persistency could be used for selection to obtain similar results. However, selection on EBV for 305-d yields would indirectly improve P_2 and P_3 more readily than it would improve P_1. If the shape of lactation curve is to be changed, then perhaps use of P_1 for selection should be preferred. Selection of animals with high EBV for C_3 could also be used to change persistency.

The correlation of the same persistency measure between milk and fat yields averaged 0.70, between milk and protein yields averaged 0.90, and between fat and protein yields averaged 0.80. Thus, persistency has a similar meaning for milk, fat, and protein yields. A correlation of 0.90, however, allows bull EBV for persistency to be high for milk yield and low for protein yield or vice versa. Usually persistency is associated with milk yields only, and EBV for persistency for fat and protein yields may not be needed.

Persistency can be affected significantly by nutrition, hormones, and pregnancy. These factors have not yet been included in the RRM. At the moment, daughters of bulls are assumed to be randomly affected by these variables because information to the contrary is lacking. The effects, on a genetic basis, of selection for greater persistency on reproduction and longevity are unknown and need to be determined before solid recommendations can be made to producers.

CONCLUSIONS

Advantages of the RRM are that the environmental effects peculiar to each TD can be analyzed, the shape of the lactation curve is allowed to differ for each animal, the solutions allow calculation of EBV for partial lactation yields or persistency, and animals can be evaluated on the basis of a single TD yield. A disadvantage of RRM is an increased computational requirement because of the five correlated genetic effects per animal in the mixed model equations and because more TD records need to be processed compared with 305-d yields. However, computer technology is continually improving, and the ability to handle RRM should become easier over time.

Another potential disadvantage is that TD records are available in Canada only since 1988. What should be done about cows that appear as dams of currently active cows? Their EBV would be based solely on progeny and not on their own records. One possibility

would be to combine the EBV of dams based on 305-d yields in a multiple-trait model with EBV for 305-d yield, T, from RRM in a procedure similar to the multiple across-country evaluation method, MACE (7). This alternative could be discontinued after a sufficient number of years of TD records have accumulated.

The RRM does not rely on factors to project records to a 305-d basis prior to analysis. Usually, projection factors assume a fixed shape of lactation curve for all cows and tend to underestimate 305-d yields from early TD records on the more persistent cows and to overestimate yields on the less persistent cows. This phenomenon has had an adverse effect on early bull EBV for bulls that have daughters with very high (or very low) persistency. Bull EBV based on many short lactations tend to be underevaluated or overevaluated, but later, when those daughters have completed lactations, the bull EBV can increase or decrease substantially. This effect has been termed the record-in-progress dip (or **RIP dip**) by producers. The effect appears again when a large batch of second crop daughters begin to lactate. The effects of RIP dip should be less with RRM because the EBV for bulls would be initially influenced by the EBV of ancestors for the shape parameters. Thus, if the ancestors had more persistent progeny, then the persistency would be reflected in the EBV of the bull before the bull has any lactating daughters. In an analysis of a 305-d yield, the persistency of ancestors is unknown, and, hence, the RIP dip effect exists. Also, the RIP dip effect on second crop daughters would be moderated in the RRM by the completed lactations of the first crop daughters.

The biggest advantage of RRM in the future will be the multiple-trait versions of the model that will analyze milk, fat, and protein yields within each of the first three lactations. Thus, milk, fat, and protein yields need not be available for each TD. A multiple-trait analysis will allow greater flexibility to milk recording programs and will allow producers more options for submitting milk weights, with or without component testing. Inherent to any TD model, the intervals between TD no longer need to be close to 30 d, so that fewer tests may occur during a lactation.

A multiple-trait RRM will produce nine sets (three lactations times three traits) of five random regression coefficients per animal in the evaluation. The shapes of the lactation curves for different lactations of the same cow could differ. The problem of (co)variance estimation is complex for this model because of the number of correlated random effects. The genetic variance-covariance matrix alone would be of order 45. The problem can be reduced by using only three covariables for second and later lactations rather than five, and by using only three covariables for all fixed regressions. However, for first lactation cows perhaps a five covariable model should be retained for the random regressions. These possibilities are currently under study.

Alternatively, genetic evaluations based only on TD information for first lactation in a multiple-trait RRM would also reduce computational problems. Given the high correlation between 305-d evaluations during different lactations, the need to include later lactations in genetic evaluation can be questioned, even though later lactations do provide additional information. The effects of selection bias that are associated with later lactations would also be eliminated when first lactations only are used or all cows are required to have TD data for first lactation. The value of including later lactation data, compared with the cost of analyzing that data, needs to be assessed.

If computational efficiency for RRM can be optimized, then RRM lends itself nicely to more frequent calculation of EBV. If TD records are collected monthly, then EBV may be updated monthly also. In this case, the genetic evaluation software would be tightly linked to the database software. Whenever the database has a certain number of new TD records, say 50,000, then the genetic evaluation programs would start automatically and compute new EBV.

The TD records may be of different types (e.g., supervised 24-h samples, unsupervised weights, and single milking samples from either two times or three times a day milking). Each type of TD record may have a different residual variance, or accuracy, which should be accounted for in the RRM. Milk recording organizations need to record accurately the type of record that is going into the database. Also, other factors related to TD need to be better recorded, such as management groups within a herd, pregnancy status, and medical treatments, so they can be included in the model.

Various forms of TD models will likely be proposed in the next few years because of the intuitive appeal of analyzing the basic components of 305-d lactation yields, the TD yields. This paper has presented one potential TD model.

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Schaeffer (2004)

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The problem

After almost a decade of use in dairy cattle, random regression models spread to other traits measured several times over the life of an animal, such as growth, feed intake, survival and reproduction. In this review, Schaeffer describes the structure of a random regression model and discusses its past and potential applications.

Points to note while reading

Note the choice of functions used to describe the trajectories, such as Legendre polynomials and splines, and the discussion of the number of parameters. The review also shows how these models make it possible to study genotype by environment interaction by treating the environment as a continuous scale.

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Application of random regression models in animal breeding

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Abstract

Random regression models (RRM) have become common for the analysis of longitudinal data or repeated records on individuals over time. Applications in animal breeding research are emphasized while recognizing that RRM are used in many biological situations including human health. The best known application of RRM has been to genetic evaluation of dairy cattle using test day production records. The basic structure of a RRM is given. Other applications include growth traits in all species, genotype by environment interactions, and ad hoc proposals for the analysis of survival data and fertility data. RRM allow the researcher to study changes in genetic variability with time and allow selection of individuals to alter the general patterns of response over time.
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Keywords: Random regressions; Test day models; Growth; Survival; Fertility

1. Introduction

Random regression models (RRM) were introduced by Henderson (1982) and Laird and Ware (1982). Schaeffer and Dekkers (1994) suggested their use in dairy cattle breeding for the analysis of test day production records. Meyer and Hill (1997) showed the equivalence of covariance functions to RRM. RRM are intended for use on longitudinal data or 'repeated' records situations where observations for a trait are collected several times during the course of an animal's life (Hill and Brotherstone, 1999). The study of RRM has also increased over the last 8 years as evidenced by over 24 papers on RRM that were presented at the 7th World Congress of Genetics Applied to Livestock Production in August 2002.

There have been numerous studies about the application of RRM to the genetic evaluation of dairy cattle for test day milk production. Jensen (2001) discussed this particular application and the research problems that remain to be solved. More will be discussed in Section 3.

Areas of animal breeding that have already utilized RRM include conformation traits, body condition scores, feed intake, and heart girth measures in dairy cattle; weights and back fat thickness in swine and beef cattle; fork length and weights in rainbow trout; and litter size in swine. Other potential applications include wool yield in sheep; sperm production and quality in male reproduction of any species; lifetime milk production in dairy cattle;

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genotype by environment interactions; survival analyses; and female reproduction. RRM have also been used in human health studies and could be used in many biological situations. The objectives of this paper are to describe the structure of a typical RRM, and to focus on past and potential applications of RRM in animal breeding.

2. Structure of RRM

RRM have a basic structure that is similar in most applications. A simplified RRM for a single trait can be written as

$$y_{ijkn:t} = F_i + g(t)_j + r(a, x, m1)_k + r(\text{pe}, x, m2)_k + e_{ijkn:t},$$

where $y_{ijkn:t}$ is the nth observation on the kth animal at time t belonging to the ith fixed factor and the jth group; F_i is a fixed effect that is independent of the time scale for the observations, such as a cage effect, a location effect or a herd-test date effect; $g(t)_j$ is a function or functions that account for the phenotypic trajectory of the average observations across all animals belonging to the jth group; $r(a,\ x,\ m1)_k = \sum_{\ell=0}^{m_1} a_{k\ell} x_{ijk:l}$ is the notation adopted for a random regression function. In this case, a denotes the additive genetic effects of the kth animal, x is the vector of time covariates, and $m1$ is the order of the regression function. So that $x_{ijk:l}$ are the covariables related to time t, and a_{kl} are the animal additive genetic regression coefficients to be estimated; $r(\text{pe},\ x,\ m2)_k = \sum_{\ell=0}^{m_2} p_{k\ell} x_{ijk:l}$ is a similar random regression function for the permanent environmental (pe) effects of the kth animal; and $e_{ijkn:t}$ is a random residual effect with mean null and with possibly different variances for each t or functions of t.

The function, $g(t)_j$, can be either linear or nonlinear in t. Such a function is necessary in a RRM to account for the phenotypic relationship between y and the time covariables (or other types of covariables that could be used in a RRM). In a test day model, $g(t)_j$ accounts for different lactation curve shapes for groups of animals defined by years of birth, parity number, and age and season of calving within parities, for example. With growth data, $g(t)_j$ accounts for the growth curve of males or females of breed X or breed Y from young or old dams.

If the shape of the phenotypic relationship is not known or is nonlinear, then $g(t)_j$ could be a set of classification variables. Classification variables take up more degrees of freedom and require a large number of observations per level, but they do not force the user to explicitly define the shape of the trajectory. A mathematical function, on the other hand, does not use many degrees of freedom and gives a smooth trajectory over time regardless of the number of observations. The choice of classification variables or mathematical function is up to the researcher. If data are very numerous, and the mathematical function fits the data well, then either approach will generally lead to the same results. The phenotypic relationships, $g(t)_j$, are important to a RRM analysis and deserve care and effort in their correct specification.

The random regressions are intended to model the deviations around the phenotypic trajectories. The pattern of variation may be very different in shape or appearance from the phenotypic relationships, and may be more simple than $g(t)_j$. Orthogonal polynomials of standardized units of time have been recommended as covariables (Kirkpatrick et al., 1990). Orthogonal polynomials have computational advantages. The primary general advantage is the reduced correlations among the estimated coefficients. A standardized unit of time, w, ranges from −1 to +1, and is derived as

$$w = \frac{2(t - t_{\min})}{(t_{\max} - t_{\min})} - 1,$$

where $t_{\min}$ is the earliest date (or the youngest age) and $t_{\max}$ is the latest date (or oldest age) represented in the data. The order of the orthogonal polynomials would be m_1 and m_2, i.e. the highest power of polynomial. Note that m_1 and m_2 do not need to be equal, but often (for simplicity of computing) they are chosen to be the same. Meyer (2000) and Pool et al. (2000), for example, compared many RRM models with different orders of orthogonal polynomials for the genetic and pe effects. Several types of orthogonal polynomials are available, but Legendre polynomials have been utilized (Kirkpatrick et al., 1990). The first six Legendre polynomial functions of standardized units of time are given in Table 1. Thus, if w=−0.2, then the covariables that would go into the model (for

Table 1
The first six Legendre polynomial functions of standardized units of time, w

Order		For $w=-0.2$
0	$0.7071w^0$	0.7071
1	$1.2247w^1$	−0.2449
2	$-0.7906w^0+2.3717w^2$	−0.6957
3	$-2.8062w^1+4.6771w^3$	0.5238
4	$0.7955w^0-7.9550w^2+9.2808w^4$	0.4921
5	$4.3973w^1-20.5206w^3+18.4685w^5$	−0.7212

order equal to 5) are shown in the last column of Table 1. Covariables based upon orthogonal polynomials are small numbers that reduce problems with rounding errors, and they provide relatively small correlations between the estimated regression coefficients.

The residual variance should not be assumed to be constant from t_{min} to t_{max}. The residual effect is also known as a temporary environmental effect. Changes in residual variance might be predictable depending on the trajectory of the phenotypic data. For example, if RRM were being applied to growth data, weights may increase linearly with age, and the variance of weights may increase quadratically with age. Thus, the residual variance would be expected to increase in a similar manner as the phenotypic variance. Residual variances can be fit with a function of t, or assumed to have an autoregressive structure, or can be grouped into intervals having equal variance within the intervals. Research in this area is needed.

In matrix notation the RRM is

$$\mathbf{y} = \mathbf{Xb} + \mathbf{Z}_1\mathbf{a} + \mathbf{Z}_2\mathbf{p} + \mathbf{e},$$

where $\mathbf{b}$ contains F_i and $g(t)_j$ effects, $\mathbf{a}$ contains m_1+1 additive genetic regression coefficients for each animal, $\mathbf{p}$ contains m_2+1 permanent environmental regression coefficients for each animal with data, and $\mathbf{e}$ contains the temporary environmental effects. Also,

$$Var\begin{pmatrix}\mathbf{a}\\ \mathbf{p}\\ \mathbf{e}\end{pmatrix} = \begin{pmatrix}\mathbf{A}\otimes\mathbf{G} & 0 & 0\\ 0 & \mathbf{I}\otimes\mathbf{P} & 0\\ 0 & 0 & \end{pmatrix},$$

where $\mathbf{G}$ is the variance–covariance matrix of the additive genetic random regression coefficients of order m_1+1, $\mathbf{P}$ is the variance–covariance matrix of the permanent environmental random regression coefficients of order m_2+1; and $\mathbf{R}$ is a diagonal matrix of temporary environmental variances which could vary depending on t, or $\mathbf{R}$ could be block diagonal with an autocorrelation structure for each animal's records. The mixed model equations (MME) are represented as

$$\begin{pmatrix} X'R^{-1}X & X'R^{-1}Z_1 & X'R^{-1}Z_2\\ Z_1'R^{-1}X & Z_1'R^{-1}Z_1 + A^{-1}\otimes G^{-1} & Z_1'R^{-1}Z_2\\ Z_2'R^{-1}X & Z_2'R^{-1}Z_1 & Z_2'R^{-1}Z_2 + I\otimes P^{-1}\end{pmatrix}\begin{pmatrix}\hat{b}\\ \hat{a}\\ \hat{p}\end{pmatrix} = \begin{pmatrix}X'R^{-1}y\\ Z_1'R^{-1}y\\ Z_2'R^{-1}y\end{pmatrix}.$$

Assumptions about the distributions of $\mathbf{y}$ and other random variables are not necessary to derive best linear unbiased predictors (BLUP) (Goldberger, 1962; Henderson, 1984) or the MME, but when $\mathbf{y}$ is normally distributed then BLUP is also BLP if the model is correct and variances and covariances are known. In order to estimate the elements of $\mathbf{G}$, $\mathbf{P}$, and $\mathbf{R}$ via Bayesian methods or restricted maximum likelihood, then normality of the random variables must be assumed (see for example Jamrozik and Schaeffer, 1997). This paper will concentrate on the applications of RRM and not on the estimation of (co)variance parameters, nor on the computational details of estimating (co)variances or solving MMEs. Some of the applications in this paper have applied RRM to discrete data, and therefore, a BLUP analysis would not be optimum. However, the presentation of these ideas may stimulate others to find better solutions.

3. Application to test day records

Several reviews on the use of RRM for the analysis of test day records of dairy cattle have been given by Swalve (1995, 1998, 2000) and Jensen (2001). A detailed summary of the Canadian application of a test day model (TDM) was given by Schaeffer et al. (2000). Research still continues on determining the order of fit for the random regressions for additive genetic and permanent environmental effects that explains the most variation in the observations. Pool et al. (2000) con-

cluded that the additive genetic effects require a lower order of Legendre polynomials than the pe effects. The use of lactational shape functions for the covariables of time (i.e. days in milk (DIM)) have largely been abandoned because the estimated covariance matrices, **G** and **P**, usually have very high correlations between the coefficients which can hinder the estimation process. Orthogonal polynomials of time have much lower correlations among the coefficients and provide estimates of the covariance matrices that tend to be more robust over different data sets. There could be differences in computing properties between different types of orthogonal polynomials (e.g. Legendre vs. Chebyshev polynomials).

RRM have been used with multiple trait models to analyze test day records. Each lactation is treated as a separate trait, and each of milk, fat, and protein yields, and somatic cell scores are separate traits within each lactation. Feed intake, body condition scores, and heart girth also have been measured on test day and analyzed by multiple trait RRM (Liu, 1998; Jones et al., 1999; Gallo et al., 2001; Veerkamp et al., 2001). Only the first three lactations have been included in the Canadian TDM due to computational limitations, but research on different models that might be able to handle more lactations is ongoing. Jensen (2001), for example, suggested a multidimensional RRM for test day records which has been studied by Guo et al. (2002). The approach has a set of random regressions for parity number and a set of random regressions for DIM. Thus, persistency is assumed to be the same across parities. To circumvent this problem the covariables for parity number should be multiplied by those for DIM. If the order of the polynomials for parity number was 4 (i.e. five covariables) and the order of the polynomials for DIM was 4, then there would be a total of 25 covariables. Multiply this by four traits and there would be 100 coefficients per animal to estimate, and **G** would be of order 100. Lifetime production ability could be computed from these coefficients. The computing issues might be resolved by considering two traits at a time (e.g. milk and fat, milk and protein, and milk and SCS) rather than all four at once. Reduced rank models may be possible to reduce the number of coefficients to be estimated per animal.

An issue with all RRM analyses is to choose the appropriate range of time or ages for study. In the Canadian TDM, for example, only TD records between 5 and 305 DIM are included, but longer lactations of 330, 345, or 400 days have been chosen in some countries. Given that cows are producing milk longer, the need to restrict test day records between 5 and 305-days is unnecessary. All countries should allow test day records to go up to 400-days, but estimated breeding values (EBVs) may be expressed for any length of lactation that is suitable for that country. Changing the endpoint of the time interval results in changes to the (co)variance matrices, **G** and **P**, too. Thus, if every country agreed to a 400-days upper limit then comparisons of (co)variance matrices would become easier. Estimation of (co)variance matrices in RRM is sufficiently time consuming so that care should be taken in choosing the time or age range for a study.

Jamrozik et al. (2002) applied a multiple country RRM to first lactation test day records from Canada, Italy, New Zealand and Australia. The management systems for Canada and Italy were similar, but different from the pasture based systems in New Zealand and Australia. Hence the fixed lactation curves differed in the same manner. A RRM was used in order to deal with differences in lactation lengths between countries. New Zealand tended to have shorter lactations than the other countries. The joint country multiple trait model was used as a basis for the international comparison of dairy bulls. Extension of procedures to deal with over 20 countries would not be computationally possible unless those countries could be grouped into five or fewer groups. However, the main time limiting factor would be the huge number of cows and test day records to be processed. The study showed that an international RRM was possible, but may not be practical for Holsteins.

4. Applications similar to test day models

Besides dairy cattle, test day models have also been applied to dairy goats and dairy sheep. The following applications utilize the same basic structure of RRM as has been applied to test day yields of dairy cattle. Differences would be in the definition of subclasses for the fixed curves, the intervals of time, and the order of the random regressions

that might be used. A difficulty with goats and sheep is that the offspring are often allowed to suckle during the first week or more, so that test day records in early lactation are not readily available. This period could be critical for fitting the phenotypic trajectories.

Egg production in chickens also follows a lactation-like curve, and therefore, RRM can be applied to weekly or monthly egg production (Anang et al., 2000). The exact same issues would arise as with test day production. Similarly, annual wool production of sheep, litter size of sows per parity, and annual antler yield of deer or elk could also be analyzed by RRM similar to that of test day models.

Another application could be to semen fertility of dairy bulls where the observations can be the volume of semen, the percentage motility of the sperm, or the percentage of dead sperm, per ejaculate. Semen samples of dairy bulls are collected once or twice per week after the bulls have been returned to service. There would be many observations for bulls that were older than 6 years. Age at collection, handler, season of the year, interval of time between collections, and diet could be factors included in a model. RRM could be applied to male semen fertility in essentially any species of livestock.

In nutritional studies, blood samples of animals are often collected every 5–15 min to measure various substances in the blood in order to determine absorption of nutrients or the rate of release of enzymes. Hormones may have random spikes depending on when they are released in the blood and RRM may not deal with these appropriately. Due to costs not many animals are measured in this way. RRM could be used to model these changes, but usually there are not enough animals collected to do any kind of genetic study. The emphasis would be more on the dietary effects and the fixed ‘‘curves’’. The animal random regressions would be nuisance effects and would be a combination of genetic and pe effects.

5. Application to growth traits

The pattern of animal growth over time can be modeled by random regressions. In livestock species, growth is generally an economically important trait. Associated with growth are feed intake, feed efficiency, fat deposition, muscle development, bone length, degree of maturity, and body condition. Growth is slightly different from test day milk yields because body weights are cumulative over time. This would be analogous to accumulating daily milk yields of cows through the lactation rather than having individual test day yields on given days in the lactation. Accumulated weights have part–whole correlations from one weighing to the next, but will likely continue to be measured and analyzed as such.

One of the first applications of RRM to growth in pigs was made by Andersen and Pedersen (1996). In their study, pigs were weighed twice weekly from 30 to 115 kg live weight. Machines monitored individual feed intake even though animals were in pens of 12 individuals. Thus, pigs started the test at different ages and consequently were weighed at different days on test. Weight and weight gains were modeled as a function of time, but were also modeled as a function of feed intake from which a measure of feed efficiency was derived. That is, the genetic merit for growth was a function of the amount of feed intake. Usually growth and feed intake are highly correlated both phenotypically and genetically, so that the genetic variation in growth remaining after accounting for feed intake would be reduced. The fixed curves of the model, $g(t)_j$, were a fourth order polynomial of days on test (not orthogonal polynomials), while the order of random regressions was 2. Growth rate was fairly linear between 30 and 115 kg, but did decrease between 30 and 50 days on test, and further decreased between 50 and 80 days on test, for both gilts and castrated males. Rather than model weights against feed intake, a multiple trait RRM model having both weight and feed intake traits against time on test would be a better way to examine feed efficiency without reducing the genetic variation in weight. A multiple trait RRM would simultaneously account for the changes in genetic and residual variation in each trait while allowing both traits and the relationship between those traits to vary together with time. The general concept would be not to model one trait against another if they are genetically correlated.

Maternal genetic effects of growth traits are known to be important in beef cattle. Albuquerque and Meyer (2001) studied growth in Nelore cattle from birth to 630 days of age. The general RRM

structure was augmented to include random regressions for maternal genetic effects and maternal permanent environmental effects. Let r(ma, x, $m3$) and r(mp, x, $m4$) denote the random regressions on maternal genetic of order $m3$ and maternal pe effects of order $m4$, respectively. However, Albuquerque and Meyer (2001) assumed zero correlations between direct and maternal genetic effects at all time points in order to simplify computations. Different orders of fit for the random regressions were applied to three different data sets. Using their notation, one of the favored models was $k=6664$ which refers to the order (plus one) of the Legendre polynomials for direct genetic, maternal genetic, animal PE, and maternal PE effects, respectively, i.e. $k=(m1+1)(m3+1)(m2+1)(m4+1)$. Such a model has 77 (co)variances in addition to the residual variances to be estimated. Another favored model based on Bayesian Information Content (BIC) was $k=4463$ with 51 parameters to be estimated. With either model, maternal genetic variance increased from birth to around 115 days of age and decreased thereafter, while direct genetic variance increased throughout from birth to 630 days of age and was generally much larger than the maternal genetic variance. Residual variances were small and increased only slightly with age. The effect of zero correlations between direct and maternal genetic effects was not examined, but perhaps may not be too important in these particular data.

Besides Andersen and Pedersen (1996), RRM have been applied to growth traits by Schnyder et al. (2001), Meyer (1999, 2000), Magnabosco et al. (2000), Schenkel et al. (2002), McKay et al. (2002), Veerkamp and Thompson (1999) and Uribe et al. (2000). The key issues in application of RRM to growth traits are the number of times individuals need to be measured, at what times in their lives, and what will be the upper age range. The costs of collecting these measurements would also play a role in determining how often and when to measure growth. A RRM provides some freedom in this regard, and animals are generally weighed at all ages from birth to maturity. Some animals could have many weights recorded while other animals may have only a few. Fixed growth curves should be estimated for each sex, within years of birth, within breeds or breed crosses, and within different parities of dams. Much work remains in applications to growth.

Besides animals, RRM could be applied to growth of plants, such as crops (which grow quickly) or trees (which grow slowly). RRM could be used to model growth of bacterial populations grown under certain conditions. Similar to growth would be a decay function such as the degradation of nutrients in the gut as they were digested in various parts of the gastrointestinal tract. Application of RRM to growth traits is in itself a growing area of research.

6. Application to survival analyses

Survival is a trait that could be observed every day of an animal's life, but only the point at which an animal dies or is culled is usually recorded in the data. Survival analyses are typically undertaken using a non-linear proportional hazard model (Ducrocq, 1987, 1994; Ducrocq and Solkner, 1998). However, Veerkamp et al. (1999) proposed that a RRM could be used for survival analyses. At any given point in time an animal is either alive (1), or dead (0). The survival 'curve' for an individual animal is a straight line at the value 1 from birth until the animal is culled (i.e. removed from the population via death, accident or management decision), after which a value 0 is assigned and continues as a straight line to the end of the observation period. An average of the survival 'curves' of many animals gives a population survival curve of the expected survival rate at various ages. At the beginning of the observation period, usually birth or first calving, nearly all animals are still alive, so that the phenotypic variance is very close to zero. The phenotypic variance becomes larger with time until 50% of the population remains, after which the variance decreases towards zero again at the end of the observation period when nearly all animals have been culled.

Suppose the observation period for dairy cattle begins at first calving and ends 75 months later. Thus, every cow that has been culled has 75 monthly observations for survival. For example, if a cow was culled in month 41, then the observation vector for this cow would have the first 40 elements being 1, and the next 35 being 0. If a cow has not had an opportunity to be culled and was currently milking in month 40, then this cow would have an observation vector of length 40, with all elements equal to 1.

A RRM for survival might be as follows:

$$y_{ijklno:t} = (YS : Ht)_{ij} + (YSAP{:}t)_{ikl} + r(a, x, m1)_n + r(\mathrm{pe}, x, m2)_n + e_{ijklno:t},$$

where *YS* refers to the *i*th year-season of first calving; *H* refers to the *j*th herd of first calving; *A* refers to the *k*th age at first calving groups; *P* refers to the *l*th production level as a deviation from herd average in first lactation; *x* are the appropriate orthogonal polynomials of time, *t*, after first calving; *a* are the random regression coefficients for additive genetic value of animal *n*; pe are the random regression coefficients for permanent environmental effects of animal *n*; and $e_{ijklno:t}$ is a residual effect. The appropriate order of the orthogonal polynomials has yet to be determined.

The herd-time effect within year-season $(YS{:}Ht)_{ij}$, specifies a group of cows first calving in the same year-season, *i*, now located within herd *j* in time period *t*. This contemporary group would conceptually face the same management decision factors at that point in time. Cows could change herds and would be part of a different group of contemporaries born in the same year-season, but now in a different herd and time period. This effect also implies that herd management decisions are not consistent over years or seasons within years, and the change in size of the herd would also be considered with this effect. Rather than having *t* effects within each (*YS*:*H*), these effects could be modeled as random regressions with $mh+1$ coefficients per (*YS*:*H*). These effects would be similar to herd-test date effects in a test day model for production traits.

The $(YSAP{:}t)_{ikl}$ effects are the fixed survival curves for different ages at first calving and production levels. Consequently, the additive genetic random regression coefficients for animals would represent survival ability (involuntary culling) after accounting for production. The fixed curves are nested within year-season of first calving to allow for changes in population decision strategies over time. This would account for the general declining population sizes within a country.

This model allows the inclusion of animals that have not yet been culled (missing observations), and provides genetic evaluations of cows for survival, which were not initially available from the Survival Kit (Ducrocq and Solkner, 1998). Genetic evaluations (EBV) could be expressed as the percentage survival rate at a specific number of months after first calving (e.g. the one that gives the highest genetic variance). Alternatively, EBV could be reported as the number of months after first calving that results in 50% culling of a bull's progeny (i.e. half-life EBV). Research on RRM models for survival are just beginning.

The reviewers correctly pointed out that the observations on survival are binary (discrete) rather than continuous, and therefore, a BLUP analysis of these data will not be optimum. Perhaps the Survival Kit could be altered to include random regressions to cover the time dependent variables. Otherwise a generalized linear mixed model with its additional computational complexity could be applied.

7. Application to fertility in dairy cattle

Reproductive performance in dairy cattle is economically important, but typically has a low heritability. Reproductive performance includes the age at first estrus, the age at first service, the number of services to conception, age at conception, age at calving, the ease of calving, if the calf was stillborn, or if the cow aborted. These traits are observed before every parturition. Thus, there are observations on the same animal over a period of years. Genetic merit for reproductive performance could change over time as well. Let parity number be the unit of time being observed. Most cows have only three to four parities, but some cows can have as many as ten or more. Let the data files be limited to a maximum of ten parities and standardize the parity numbers from −1 (for first parity) to +1 (for tenth parity). A RRM for many of the fertility traits, say age at first estrus/days to first estrus after calving, for example, might be as follows:

$$y_{ijkmno:t} = (YSP{:}t)_{ij} + (YH{:}t)_{ik} + r(a,\ \mathrm{parity},\ m1)_m + r(\mathrm{pe},\ \mathrm{parity},\ m2)_m + r(\mathrm{ma},\ \mathrm{parity},\ m3)_n + r(\mathrm{mp},\ \mathrm{parity},\ m4)_n + e_{ijkmno:t}.$$

where $y_{ijkmno:t}$ is an observation on age at first estrus/days to first estrus after calving; *Y* refers to year of birth/calving; *S* refers to season of birth/calving, *H*

refers to herd of birth/calving; P refers to production level as a deviation from herd average for that lactation; the random regressions are as described for earlier models for additive genetic, maternal genetic, animal pe, and maternal pe effects; and $e_{ijkmno:t}$ is a residual effect.

A multiple trait RRM should be applied to all reproductive traits simultaneously. For some traits the sex of calf and service sire effects might be added to the model. Adjustment for production levels may be useful. Conformation traits like feet and legs could affect the ability of a cow to become pregnant; rump traits could affect calving ease and fertility, and body condition scores have been shown to have a relationship to fertility. Development of an appropriate model for each trait may require substantial work.

Estimation of the (co)variance matrices for the random regressions could be problematic because data would not be available on all seven parities for all animals. Data should be required to have first parity information for all cows. Analysis of cows with data on all seven parities would firstly give a relatively small number of cows, and secondly would be a group of cows that have had exceptional reproductive performance (otherwise they would have been culled). In order to obtain unbiased estimates, data on all cows will have to be used. Covariances between direct and maternal genetic effects may need to be assumed to be zero, as was made by Albuquerque and Meyer (2001) in their growth studies.

The results of such an analysis would be a different genetic value for each animal for each parity. An issue would be how to combine the genetic values of each parity. First lactation information should probably be weighted more heavily than that from later lactations Breeding values could be expressed relative to an optimal reproductive performance perspective. All of the traits analyzed could be combined into an economic fertility index.

Reproductive traits like number of inseminations, calving ease, and stillbirths are not continuous traits, and so BLUP analyses would not be optimum. Studies comparing these approximations to more suitable types of analyses are needed. If the approximations are computationally more feasible and lead to selection of the same animals as the theoretically correct methodology, then an optimal solution may be unnecessary.

The time variable in the above RRM was parity number which ranged from 1 to 10, so that there are only ten possible values of the x variable. This is not a problem for a RRM, conceptually or computationally. The time variable in a test day model is DIM which go from 5 to 305 days, or 301 possible values. Time is usually recorded in discrete units (days, weeks, months, years, etc.) even though time is considered to be continuous. Parity number could be substituted by age at calving (in months), but this variable may be highly related to one of the reproductive traits. Parity number was chosen for this model because reproductive performance of cows can change from one parity to the next, and the performance in previous parities could influence the reproductive performance in the current parity, and a RRM is a good way to study this variability.

8. Application to genotype by environment interaction studies

A genotype by environment interaction implies that animals would rank differently in different environments. If those environments can be ordered such as by a temperature–humidity index (THI), or by elevation, or by average herd production, then a RRM could be applied with regressions on these variables rather than on time or age. RRM are useful when the environments cover a range of different conditions rather than just a few categories. Ravagnolo and Misztal (2000) used the THI to obtain different heritabilities and different rankings of animals at different heat stress levels. Test day yields for two particular DIM classes were analyzed by the following model:

$$y_{ijklmn} = \mathrm{HTD}_i + A_j + D_k + F_l + r(a, f(i),\ 1)_m + r(\mathrm{pe}, f(i),\ 1)_m + e_{ijklmn},$$

where HTD_i is a herd-test date effect, A_j is an age at calving class, D_k is a DIM class effect, F_l is a milking frequency class, a and pe are the random animal additive genetic and permanent environmental effects, respectively, $f(i)$ is a heat stress value (ranging from 70 to 93 and averaging 78), and $r(a, f(i), 1)_m$ and $r(\mathrm{pe}, f(i), 1)_m$ are random regressions for animal additive genetic and permanent environmental linear effects,

respectively. This model differs from the general structure of RRM in that there were no fixed curves, $g(t)_j$ in the model for $f(i)$. However, because a HTD is associated with only one THI value the equations for $g(t)_j$ would be completely confounded with the HTD equations. The $f(i)$ values were not standardized and converted to orthogonal polynomials. Because only the linear component was considered, there was no need to use orthogonal polynomials. If higher order polynomials were included, then orthogonal polynomials should be considered. A common residual variance was assumed for all records. Only first parity cows were included. Heritability increased as THI increased, and correlations between predicted breeding values (i.e. $\hat{a}_{m,1}$ with $\hat{a}_{m,1}+\hat{a}_{m,2}\times f(i)$) decreased as THI increased.

Lawlor et al. (2002) used a RRM to model the change in genetic variances over time. The model implies that genetic values of animals also change with time and that re-ranking among cows calving in the same year might occur with time (if they could be re-observed), but cows are only observed in 1 year and do not have repeated records. Years 1981–1996 were standardized to values between −1 and 1. Production traits, productive life, and four linear type scores were analyzed jointly in a multiple trait model. Only first order Legendre polynomials were used with random regressions for herds, herd by sire interactions, and animal additive genetic effects. The model did not include any fixed curves for changes in traits over time. The authors related changes in variances and correlations to linkage disequilibrium and selection practices. Because there were no fixed curves considered in their model, changes in variances could be associated with changes in phenotypic means rather than due to linkage disequilibrium.

Strandberg et al. (2000) used a RRM to study genotype by environment interactions between populations of Nordic Red and White cattle. The independent variable was the herd-year average as a deviation from the overall year averages across the three countries (i.e. Finland, Norway, and Sweden). The model used was

$$y_{ij} = g(X_{ij}, 1) + r(s,\ X_{ij}, 1)_i + e_{ij},$$

where y_{ij} was either 305-days protein yield, days open, or interval from calving to last insemination; X_{ij} was a herd-year average deviation; $r(s, X_{ij}, 1)$ are sire random regressions of order 1, $g(X_{ij}, 1)$ is the fixed regression function on X_{ij}; and e_{ij} was a random residual effect assumed to have a common variance across countries. Nearly 15 million records were analyzed. Heritabilities of all traits increased as herd-year deviations increased. Considerable re-ranking of sires occurred in herd-year averages that were 1.5 S.D. or more units below the average, while re-rankings were not as severely affected by herd-year averages that were more than 1.5 S.D. units above the mean environment. By having a regression on a measure of the herd environment rather than a fixed herd effect, then the problem of lack of genetic connections between countries is circumvented, but perhaps other problems arise. The random regressions should also have considered higher order polynomials even if the phenotypic relationship was expected to be linear.

While there have been attempts to study genotype by environment interactions with RRM, improvements should be made to the models that have been used. Studies should include fixed curves to account for the phenotypic relationships, and random regressions on higher order orthogonal polynomials need to be explored. Because animals in these studies have only one observation (in one particular time interval or one environment), the hope is that a RRM can pull out a correlation or trend through the connections created by sires. A simulation study should be conducted to determine if a RRM can do this adequately.

9. Conclusions

Whenever there is a relationship between variables and time or age, then RRM are potentially applicable to that situation. A good RRM should contain a set of fixed curves that depict the phenotypic relationship between the variables for various groups of individuals, and random regressions for animal additive genetic effects and animal permanent environmental effects. There could be other random regression factors such as for herd effects or maternal genetic effects. Orthogonal polynomials are most appropriate for the covariates in RRM. Research is needed to determine the best order of polynomials to include in different applications. Hierarchical models (e.g. Schnyder et al., 2001) might

be used for RRM where the random regressions may not be linear functions of time or age. RRM have been discussed in applications to animal breeding, but other sciences (e.g. crops, social sciences, human health, biology) have benefited from such models. There will likely be many more applications of RRM in the future.

10. Notation

RRM, random regression model
EBV, estimated breeding value
DIM, days in milk
THI, temperature humidity index

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5

The genomic era and the G matrix

1997 to 2008

From the 1980s onward, the search for genes of large effect with molecular markers produced limited results for traits controlled by many loci. The alternative was the simultaneous use of markers distributed across the whole genome. This block brings together the proposal to measure relationships from shared alleles, from 1997, the paper that gave rise to genomic selection, from 2001, and the G matrix of VanRaden, from 2008. With the G matrix, genomic evaluation came to use Henderson's mixed model equations, with the A matrix replaced by the G matrix.

Nejati Javaremi, Smith and Gibson (1997)

Meuwissen, Hayes and Goddard (2001)

VanRaden (2008)

Nejati Javaremi, Smith and Gibson (1997)

Nejati-Javaremi, A., Smith, C., and Gibson, J.P. 1997. Effect of total allelic relationship on accuracy of evaluation and response to selection. Journal of Animal Science, vol. 75, no. 7, pp. 1738 to 1745.

The problem

Pedigree relationship is an expectation. Two full sibs share on average half of their genes, but in practice this value varies from one pair to another. Nejati Javaremi, Smith and Gibson proposed measuring relationship directly from the identity of alleles at many loci, the so called total allelic relationship. Through simulation, they showed that using this relationship in place of the expected relationship increases the accuracy of evaluations and the response to selection.

Points to note while reading

This paper is the direct precursor of the G matrix. Note how the relationship matrix is built from alleles and compare it with VanRaden's G matrix, eleven years later. At the time it was not possible to genotype a large number of markers at low cost, and the proposal could only be applied with the development of SNP panels.

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Effect of Total Allelic Relationship on Accuracy of Evaluation and Response to Selection

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ABSTRACT: Total allelic relationship (TA) as a possible alternative to the pedigree-derived additive genetic relationship (RA) is defined. The TA measures the actual allelic identity between individuals for loci segregating for the trait concerned. Its value was studied by simulation in populations of different family structure, different numbers of loci, different numbers of alleles per locus, and different heritability levels. The alternative types of relationship matrices were used in mixed model equations to derive best linear unbiased prediction estimates (EBV) of breeding values (BV). Accuracies of evaluations were calculated as correlations of EBV with true breeding values. In populations with random selection and mating, EBVTA derived using TA had higher accuracies than EBVRA derived using RA. In populations with selection, EBVTA was more accurate and resulted in higher responses than selection on EBVRA. We conclude that not accounting for variation in average measures of relationship and identity in state can be important sources of variance of prediction error, and taking account of them increases the accuracy of selection.

Key Words: Genetic Markers, Genetic Parameters, Genetic Gain, Best Linear Unbiased Prediction

J. Anim. Sci. 1997. 75:1738–1745

Introduction

In quantitative genetic evaluation, information from relatives is given different weights based on the degree of relatedness to the animal being evaluated. In estimating inbreeding and genetic relationship, an average relationship value is derived for all loci based on pedigree information.

Like most other biological means, there will be variation about the mean. But infinitesimal theory of quantitative genetics, which assumes that a trait is influenced by an infinite number of loci with free recombination, does not take such variation into account. Weir et al. (1980) found that average measures of inbreeding can be associated with substantial variances. The variance of true relationship about the average pedigree prediction has been investigated for several types of relationship (Franklin, 1977; Cockerham and Weir, 1983; Hill, 1993a,b; Rasmuson, 1993). The effect of accounting for variation in relationship on response to selection has not been investigated.

Best linear unbiased prediction of breeding values of animals, derived from mixed model equations (**MME**) of Henderson (1972), is commonly used in genetic selection of animals. The accuracy of such estimates are, however, affected by the amount of information available. Most of the information in many genetic evaluation programs is provided by the performance of relatives through use of the matrix of additive genetic relationship. By using a more accurate estimate of the relationship among relatives, the accuracy of estimation and consequently the rate of genetic response may be increased. These methods may be appropriate for incorporating quantitative trait loci (**QTL**), identified by molecular markers, into the genetic evaluation.

The objective of this study is to evaluate the effect of replacing average pedigree relationship by allelic relationship in genetic evaluation.

Materials and Methods

Theory

To derive the allelic relationship between two individuals at a given locus, *l*, each allele of the first individual is compared with the two alleles of the second individual; the average identity of the two

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alleles of one individual with the two alleles of the other individual is considered as the allelic identity at this locus:

$$TA_l = 2 \times \frac{\sum_{i=1}^{2}\sum_{j=1}^{2} I_{ij}}{4} = \frac{\sum_{i=1}^{2}\sum_{j=1}^{2} I_{ij}}{2} \qquad (1)$$

where I_{ij} is the identity of the ith allele of the first individual with the jth allele of the second (I_{ij} takes the value of 1 if the two alleles are identical and zero if they are not); TA is the total allelic relationship. The coefficient of 2 emphasizes that TA is twice the coefficient of relationship (Malécot, 1948) and is analogous to the numerator relationship (Wright, 1922). Total allelic identity of individuals x and y averaged over L loci affecting the trait is then

$$TA_{xy} = \frac{\sum_{l=1}^{L} TA_l}{L} = \frac{\sum_{l=1}^{L} \left(\frac{\sum_{i=1}^{2}\sum_{j=1}^{2} I_{lij}}{2}\right)}{L}. \qquad (2)$$

As an example, consider the following two individuals,

Individual 1					Individual 2				
A_2	B_1	C_1	D_1	E_2	A_1	B_1	C_2	D_1	E_1
A_2	B_1	C_2	D_1	E_2	A_2	B_2	C_2	D_2	E_1

mated to produce the following offspring:

Individual 3					Individual 4				
A_2	B_1	C_1	D_1	E_2	A_2	B_1	C_2	D_1	E_2
A_1	B_1	C_2	D_2	E_1	A_2	B_1	C_2	D_1	E_1

Individual 5					Individual 6				
A_2	B_1	C_2	D_1	E_2	A_2	B_1	C_2	D_1	E_2
A_1	B_2	C_2	D_1	E_1	A_2	B_1	C_2	D_1	E_1

The pedigree-derived additive genetic relationship matrix among these six individuals would be

	1	2	3	4	5	6
1	1	0	.5	.5	.5	.5
2	0	1	.5	.5	.5	.5
3	.5	.5	1	.5	.5	.5
4	.5	.5	.5	1	.5	.5
5	.5	.5	.5	.5	1	.5
6	.5	.5	.5	.5	.5	1

and the TA matrix would be

	1	2	3	4	5	6
1	1.8	.8	1.2	1.6	1.2	1.6
2	.8	1.4	1.0	1.2	1.2	1.2
3	1.2	1.0	1.2	1.2	1.0	1.2
4	1.6	1.2	1.2	1.8	1.4	1.8
5	1.2	1.2	1.0	1.4	1.4	1.4
6	1.6	1.2	1.2	1.8	1.4	1.8

The elements of TA matrix follow directly from application of Equation 2 to the information on genotype at the five loci. Although individuals 3 to 6 are all progeny of a single pair of parents, hence full-sibs, their total allelic relationship ranges from 1.0 to 1.8, in contrast to the value of .5 for all progeny estimated from pedigree information. Their parents have also a total allelic relationship of .8 with each other as compared with zero relationship in the pedigree method. Total allelic relationship would not change even if there were no pedigree relationship among these six individuals.

There are two principal differences between the method of total allelic relationship and the standard pedigree method of deriving relationships. In the pedigree method, all individuals of the base generation are assumed to be unrelated with zero inbreeding. In deriving total allelic relationship, individuals of the base generation have some degree of homozygosity for the defined loci (i.e., analogous to inbreeding in the biological sense) and hence allelic relationships with each other.

In the pedigree method, all progeny of a particular mating are assigned the same relationship and no variation is considered. The total allelic relationship accounts for variation in progeny genotypes.

Pedigree relationship assuming total allelic relationship is known in the base generation will be used to separate the possible effects of these two differences. If the true allelic relationship of the parents in the base generation were known, the pedigree derived relationship matrix would become

	1	2	3	4	5	6
1	1.8	.8	1.3	1.3	1.3	1.3
2	.8	1.4	1.1	1.1	1.1	1.1
3	1.3	1.1	1.4	1.2	1.2	1.2
4	1.3	1.1	1.2	1.4	1.2	1.2
5	1.3	1.1	1.2	1.2	1.4	1.2
6	1.3	1.1	1.2	1.2	1.2	1.4

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This approach will be used to explore the contribution of different sources of information to accuracy of relationship information in the simulation below.

Simulation Studies

Two simulation studies were undertaken: in the first study, parents of each generation were chosen randomly; in the second, parents were selected on various selection criteria.

No Selection. Populations of size 100 with three different family structures were simulated: 1) full-sib families of size 2 (**FS-2**), 2) full-sib families of size 20 (**FS-20**), and 3) half-sib (**HS**) families of size 20. Individuals of the base population (generation zero) were sampled at random from a large population with specified parameters. Each generation male and female parents of the next generation were chosen at random, with mating to produce the required family structure. Five, 50 or 100 unlinked QTL affected the trait each with two or five alleles at equal frequencies. Heritability was either .1 or .3. Allelic effects at each locus were sampled from a normal distribution with mean $\mu = 0$ and variance equal to the proportion of additive genetic variance expected of an average locus. Total additive genetic variance was $\sigma_a^2 = \Sigma_i^L \sigma_{a_i}^2$, where L is the number of loci. In simulating progeny, one allele per locus was randomly chosen from each parent, and a residual effect was added to the sum of genetic effects over all loci.

Different family structures were used to study the behavior of TA under different pedigree structures. For the FS-2 family structure, 50 males were each randomly mated to one of 50 females. Each mating resulted in one male and one female progeny, so 100 progeny were produced in total to replace the 100 parents. For the FS-20 family structure, five males and five females per generation were chosen randomly and 20 full-sibs were produced from each mating. For HS family structure, five males and 50 females per generation were chosen and mated randomly. Each male was mated to 10 females, and each mating resulted in two progeny. Each sire had 20 progeny, 10 pairs of full-sibs, being half-sibs to each other.

Selection. Populations of size 200 were simulated. Base generation individuals were sampled from a population in Hardy-Weinberg and gametic phase equilibria. Each generation, five of 100 males and 50 of 100 females were selected as parents. Each male was mated to 10 females, resulting in four progeny per family (two of each sex). The procedure was repeated for four generations, and 1,000 individuals were available for analysis at the end of each simulation. Each generation, individuals were evaluated for their breeding values using a linear mixed model (described below) incorporating all available pedigree information. The same base population was used for four selection strategies: 1) selection based on estimated breeding values (**EBV**) derived using TA (**EBVTA**), 2) selection based on EBV derived using pedigree-derived additive genetic relationship (**RA**) (**EBVRA**), 3) selection based on true breeding value (**BV**), and 4) phenotypic selection. The expected genetic mean at the base generation in each replicate of simulation was set to zero. The whole procedure was repeated 10 times for each of 5, 10, 25, 50, and 100 loci controlling the trait at each of three heritability levels of .1, .3, and .5, giving a total of 600 cases for simulation. Two alleles per locus at equal frequencies were used for all experiments. Allelic effects were sampled from a normal distribution with equal expected values across all loci; free recombination was assumed.

Model

The same linear model was used to describe each individual record in both experiments, and generations were discrete. The system of equations may be described, in matrix form, as

$$\mathbf{y} = \mathbf{Xb} + \mathbf{Zu} + \mathbf{e}$$

where $\mathbf{y}$ is the vector of phenotypic records on individuals, $\mathbf{b}$ is the vector of fixed effect of generations, $\mathbf{u}$ is the vector of random additive genetic effects of animals, $\mathbf{e}$ is a vector of random residual effects, $\mathbf{X}$ is an incidence matrix relating fixed effects of generations to the records, and $\mathbf{Z}$ is an incidence matrix relating random effects of animals to the records.

The expected values and (co)variances are

$$E(\mathbf{y}) = \mathbf{Xb}, \quad E(\mathbf{u}) = \mathbf{0}, \quad E(\mathbf{e}) = \mathbf{0}$$

$$Var(\mathbf{y}) = \mathbf{ZGZ'} + \mathbf{R}, \quad Var(\mathbf{u}) = \mathbf{G} = \mathbf{A}\sigma_a^2,$$
$$Var(\mathbf{e}) = \mathbf{R} = \mathbf{I}\sigma_e^2, \quad Cov(\mathbf{u},\mathbf{e}) = \mathbf{0}$$

where $\mathbf{I}$ is an identity matrix and $\mathbf{A}$ is the matrix of relationships among individuals.

The mixed model equations are

$$\begin{pmatrix} \mathbf{X'R^{-1}X} & \mathbf{X'R^{-1}Z} \\ \mathbf{Z'R^{-1}X} & \mathbf{Z'R^{-1}Z} + \mathbf{G^{-1}} \end{pmatrix} \begin{pmatrix} \hat{\mathbf{b}} \\ \hat{\mathbf{u}} \end{pmatrix} = \begin{pmatrix} \mathbf{X'R^{-1}y} \\ \mathbf{Z'R^{-1}y} \end{pmatrix}.$$

When using TA as the relationship matrix, it is possible to have genetically identical individuals at all the trait loci; the frequency increases as number of loci decreases. This causes $\mathbf{G}$ to be singular. Following Henderson (1984), the mixed model equations were changed to the following form to avoid inverting $\mathbf{G}$:

$$\begin{pmatrix} \mathbf{X'R^{-1}X} & \mathbf{X'R^{-1}Z} \\ \mathbf{GZ'R^{-1}X} & \mathbf{GZ'R^{-1}Z+I} \end{pmatrix} \begin{pmatrix} \hat{\mathbf{b}} \\ \hat{\mathbf{u}} \end{pmatrix} = \begin{pmatrix} \mathbf{X'R^{-1}y} \\ \mathbf{GZ'R^{-1}y} \end{pmatrix}$$

Breeding values were then estimated by solving the mixed model equations. Estimated breeding values

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Table 1. Correlations of estimated breeding values (EBV) with true breeding values in random mating populations. Trait with five loci and two alleles per locus[a]

h^2	Family structure	G	Phenotype		EBVRA		EBVTA		EBVRATB	
.1	2 Full sibs	0	.24	(.03)	.24	(.03)	.76	(.04)	.76	(.04)
		4	.34	(.03)	.42	(.04)	.94	(.01)	.43	(.03)
	20 Full sibs	0	.32	(.03)	.32	(.03)	.84	(.04)	.84	(.04)
		4	.27	(.05)	.37	(.10)	.94	(.02)	.35	(.10)
	20 Half sibs	0	.31	(.04)	.31	(.04)	.81	(.06)	.81	(.06)
		4	.29	(.04)	.43	(.06)	.92	(.03)	.43	(.06)
.3	2 Full sibs	0	.57	(.03)	.57	(.03)	.95	(.02)	.95	(.02)
		4	.53	(.03)	.64	(.03)	.98	(.00)	.64	(.02)
	20 Full sibs	0	.47	(.02)	.47	(.02)	.92	(.01)	.92	(.01)
		4	.41	(.05)	.52	(.07)	.98	(.00)	.51	(.07)
	20 Half sibs	0	.51	(.04)	.51	(.04)	.93	(.01)	.93	(.01)
		4	.47	(.03)	.57	(.03)	.98	(.00)	.57	(.03)

[a]Standard errors are in parentheses. All results based on 10 replicates. G = generations, EBVRA = EBV, additive relationship. EBVTA = EBV, total allelic relationship. EBVRATB = EBV, additive relationship, total allelic relationship in the base.

were derived using pedigree relationship (EBVRA), total allelic relationship (EBVTA), and pedigree relationship adjusted for base generation allelic relationship (EBVRATB). Individual QTL were assumed to be identified, but their effects on the genotype was not incorporated into calculating TA. Comparisons between methods were made on the basis of correlations with true breeding values of phenotypic values and of the three different EBV for the trait. Estimated breeding values were estimated using information from the current and all previous generations, but correlations of EBV with true breeding values were calculated within generations.

Results

No Selection

Tables 1 to 3 present correlations of the different types of estimated breeding values with true breeding values for the simulated data for the various combinations of family structure, numbers of loci, numbers of alleles per locus, and heritabilities. Standard errors of these correlations based on 10 replicates are also presented. Because the correlations followed well-behaved trends, only representative portions of the results are presented.

Correlations of BV with phenotypic records are a function of heritability and generally declined across generations as a result of a slight decrease in heritability due to increased inbreeding. Correlations of BV with EBVRA are a function of heritability and the amount of information available, which increases across generations. Because the reduction in heritability across generations was small, the overall result was an increased correlation. In the base generation, where no pedigree information is available, the correlation of EBVRA with BV was equal to that of phenotypic values with BV (Tables 1 to 3).

The correlation of BV with EBVTA was always greater than that with phenotypic values or with EBVRA. This is due to the fact that the alternative matrix of relationship (i.e., TA) uses information on records of relatives more accurately by accounting for variation in the additive genetic relationship. The correlation of BV with EBVRATB was identical to that with EBVTA in the base generation because the matrix of relationships for these two estimates of BV are the same in the base generation. In subsequent generations, correlation of the BV with EBVRATB declined toward that with EBVRA, showing that the adjustment for the base generation allelic relationship had little long-term effect.

The difference in the correlations of EBVTA and EBVRA with BV was greater when a small number of QTL were involved in the expression of the trait. This is expected due to the increasing variance of average relationship as number of loci affecting the trait decreases. However, even with 100 loci affecting the trait, the difference between the correlations of EBVTA and EBVRA with BV was still appreciable, generally being close to .2. The differences were similar across different family structures and for the two heritabilities. The advantage of EBVTA over EBVRA with traits of lower heritability decreased with an increase in the number of loci. This advantage increased over generations because of extra information provided by increased number of individuals. When heritability was low and more loci were affecting the trait, extra information provided by animals over several generations increased the gap between the correlations of BV with EBVTA and EBVRA. For example, in Table 3 with 100 loci, the difference was .39 – .30 = .09 in the base generation and .63 – .40 = .23 in the fourth generation. Increased number of alleles per locus (five vs two) reduced the advantages of EBVTA over EBVRA (Table 2 compared with Table 1). This trend was similar for different heritability levels.

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Table 2. Correlation of different estimated breeding values (EBV) with true breeding values in random mating populations. Trait with five loci and five alleles per locus[a]

h^2	Family structure	G	Phenotype		EBVRA		EBVTA		EBVRATB	
.1	2 Full sibs	0	.41	(.03)	.41	(.03)	.71	(.04)	.71	(.04)
		4	.37	(.03)	.49	(.03)	.91	(.00)	.49	(.03)
	20 Full sibs	0	.39	(.01)	.39	(.01)	.74	(.02)	.74	(.02)
		4	.36	(.03)	.51	(.06)	.87	(.03)	.50	(.07)
	20 Half sibs	0	.36	(.04)	.36	(.04)	.66	(.06)	.66	(.06)
		4	.41	(.02)	.57	(.02)	.89	(.01)	.57	(.02)
.3	2 Full sibs	0	.67	(.03)	.67	(.03)	.89	(.02)	.89	(.02)
		4	.67	(.03)	.74	(.03)	.97	(.00)	.74	(.03)
	20 Full sibs	0	.65	(.02)	.65	(.02)	.88	(.01)	.88	(.01)
		4	.52	(.03)	.66	(.03)	.96	(.01)	.67	(.03)
	20 Half sibs	0	.63	(.02)	.63	(.02)	.89	(.01)	.89	(.01)
		4	.58	(.04)	.69	(.03)	.96	(.00)	.69	(.03)

[a]Standard errors are in parentheses. All results based on 10 replicates. G = generations, EBVRA = EBV, additive relationship. EBVTA = EBV, total allelic relationship. EBVRATB = EBV, additive relationship, total allelic relationship in the base.

In general, EBVTA gave higher correlations with BV than any other EBV or phenotype. An increased number of loci or increased number of alleles per locus diminished the advantage of EBVTA over EBVRA.

Selection

The effects of number of loci controlling the trait showed a well-behaved trend, and results are therefore presented only for simulations with 5, 25, and 100 loci in Tables 4, 5, and 6, respectively. In each table, the genetic responses and their standard errors in each of four generations of selection are given for each of four selection strategies and three heritability levels. All populations in each table have the same genetic mean in the base generation.

Initial response to selection was higher, but long-term response (over four generations) was lower when a small number of loci (5 vs 25 and 100) controlled the expression of a trait. This was due to faster reduction of genetic variance across generations caused by fixation of alleles with few loci. With 100 loci, alleles were far from fixation at generation 4.

High heritability levels resulted in higher rates in genetic response when not selecting on BV directly. For example, cumulative response to four generations of phenotypic selection was 1.376 σ_g, 2.075 σ_g, and 2.303 σ_g for heritability levels of .1, .3, and .5, respectively (Table 4). Similar results were observed when selection was based on EBVRA. Although heritability has some effect on the rate of response to selection based on EBVTA, the effect is minimal compared with effect on EBVRA or phenotypic selection.

Responses to selection based on true breeding values show the highest possible gain from selection for this population structure. Response to selection

Table 3. Correlation of different estimated breeding values (EBV) with true breeding values in random mating populations. Trait with 100 loci and two alleles per locus[a]

h^2	Family structure	G	Phenotype		EBVRA		EBVTA		EBVRATB	
.1	2 Full sibs	0	.30	(.03)	.30	(.03)	.39	(.03)	.39	(.03)
		4	.29	(.02)	.40	(.04)	.63	(.03)	.40	(.04)
	20 Full sibs	0	.24	(.03)	.24	(.03)	.29	(.05)	.29	(.05)
		4	.24	(.02)	.40	(.07)	.55	(.05)	.41	(.07)
	20 Half sibs	0	.31	(.02)	.31	(.02)	.35	(.03)	.35	(.03)
		4	.24	(.04)	.45	(.06)	.63	(.03)	.45	(.06)
.3	2 Full sibs	0	.53	(.01)	.53	(.01)	.66	(.02)	.66	(.02)
		4	.51	(.03)	.64	(.02)	.84	(.01)	.64	(.02)
	20 Full sibs	0	.57	(.02)	.57	(.02)	.69	(.02)	.69	(.02)
		4	.47	(.03)	.58	(.05)	.81	(.02)	.57	(.06)
	20 Half sibs	0	.55	(.01)	.55	(.01)	.64	(.01)	.64	(.01)
		4	.56	(.02)	.65	(.02)	.84	(.01)	.64	(.02)

[a]Standard errors are in parentheses. All results based on 10 replicates. G = generations, EBVRA = EBV, additive relationship. EBVTA = EBV, total allelic relationship. EBVRATB = EBV, additive relationship, total allelic relationship in the base.

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Table 4. Genetic response in each of four generations of selection with five loci affecting the trait[a]

Heritability	Selection criterion	Generation 1		2		3		4	
.1	EBVTA	1.248	(.158)	.819	(.083)	.422	(.077)	.172	(.028)
	EBVRA	.419	(.097)	.642	(.099)	.315	(.055)	.281	(.051)
	Phenotype	.419	(.097)	.275	(.107)	.323	(.060)	.359	(.072)
	BV[b]	1.391	(.142)	.780	(.091)	.393	(.056)	.139	(.028)
.3	EBVTA	1.334	(.137)	.805	(.085)	.409	(.066)	.142	(.028)
	EBVRA	.806	(.175)	.609	(.116)	.410	(.055)	.340	(.056)
	Phenotype	.806	(.175)	.525	(.109)	.369	(.039)	.375	(.087)
.5	EBVTA	1.377	(.144)	.800	(.081)	.396	(.053)	.123	(.026)
	EBVRA	1.000	(.163)	.610	(.083)	.434	(.068)	.290	(.068)
	Phenotype	1.000	(.163)	.556	(.078)	.421	(.058)	.326	(.056)

[a]All values are in genetic SD units with SE in parentheses, based on 10 replicates. BV = breeding value. EBVRA = estimated BV, additive relationship. EBVTA = estimated BV, total allelic relationship.
[b]Values are the same for all heritability levels.

based on the other three criteria may be compared with this upper limit.

In this simulation, phenotypic selection resulted in the lowest response at lower heritability levels (Tables 4, 5, and 6). Selection based on EBVRA usually gave responses higher than that of phenotypic selection but lower than that of other methods of selection. Estimated breeding values using TA resulted in higher response than either phenotypic selection or selection based on EBVRA for any number of loci and at all heritability levels. Response to selection based on EBVTA with small numbers of loci and(or) higher heritability was close to the upper limit given by selection based on true breeding values. Even with the lowest heritability and 100 loci, EBVTA gave a 50% higher response than EBVRA, and selection on BV directly gave 131% higher response than EBVRA (Table 6, cumulative response over four generations).

Discussion

Total allelic relationship uses information from records of relatives more accurately because it accurately accounts for the variation in the additive genetic relationship among relatives and also because it uses information from pedigree-unrelated individuals, which leads to increased correlations of BV with EBVTA. This increase in accuracy of evaluation is due to accounting for identity in state, which also takes care of variation in average relationship. Failure of EBVRATB to maintain high accuracy after the base generation indicates that only accounting for the relationship in the base generation without considering variation in relationships in later generations will not have a lasting positive effect on the accuracy of evaluation.

Infinitesimal theory assumes free recombination between individual QTL. Even if a large number of

Table 5. Genetic response in each of four generations of selection with 25 loci affecting the trait[a]

Heritability	Selection criterion	Generation 1		2		3		4	
.1	EBVTA	1.080	(.090)	1.057	(.064)	.926	(.050)	.738	(.039)
	EBVRA	.463	(.094)	.555	(.108)	.480	(.058)	.483	(.063)
	Phenotype	.463	(.094)	.472	(.065)	.406	(.050)	.437	(.058)
	BV[b]	1.501	(.067)	1.042	(.035)	.972	(.044)	.832	(.020)
.3	EBVTA	1.376	(.066)	1.009	(.057)	.895	(.037)	.812	(.043)
	EBVRA	.845	(.086)	.665	(.070)	.700	(.080)	.563	(.059)
	Phenotype	.845	(.086)	.641	(.076)	.574	(.068)	.619	(.067)
.5	EBVTA	1.461	(.069)	1.038	(.028)	.932	(.041)	.840	(.043)
	EBVRA	1.117	(.085)	.840	(.049)	.656	(.053)	.748	(.045)
	Phenotype	1.117	(.085)	.880	(.054)	.682	(.070)	.743	(.063)

[a]All values are in genetic SD units with SE in parentheses, based on 10 replicates. BV = breeding value. EBVRA = estimated BV, additive relationship. EBVTA = estimated BV, total allelic relationship.
[b]Values are the same for all heritability levels.

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Table 6. Genetic response in each of four generations of selection with 100 loci affecting the trait[a]

Heritability	Selection criterion	Generation 1		2		3		4	
.1	EBVTA	.716	(.126)	.766	(.073)	.710	(.056)	.754	(.052)
	EBVRA	.409	(.027)	.533	(.055)	.478	(.076)	.558	(.087)
	Phenotype	.409	(.027)	.413	(.065)	.492	(.101)	.463	(.067)
	BV[b]	1.468	(.068)	1.081	(.065)	1.043	(.022)	.976	(.041)
.3	EBVTA	1.095	(.122)	.901	(.047)	.911	(.057)	.925	(.047)
	EBVRA	.751	(.082)	.825	(.100)	.642	(.067)	.678	(.076)
	Phenotype	.751	(.082)	.691	(.066)	.620	(.068)	.610	(.048)
.5	EBVTA	1.269	(.088)	.984	(.046)	.946	(.049)	.938	(.044)
	EBVRA	1.035	(.073)	.820	(.061)	.702	(.060)	.764	(.044)
	Phenotype	1.035	(.073)	.740	(.070)	.791	(.065)	.782	(.058)

[a]All values are in genetic SD units with SE in parentheses, based on 10 replicates. BV = breeding value. EBVRA = estimated BV, additive relationship. EBVTA = estimated BV, total allelic relationship.
[b]Values are the same for all heritability levels.

loci affect the trait of interest, linkage between loci due to residing on a limited number of chromosomes will cause the QTL to be passed in blocks of genes. In this study, there was considerable difference between EBVTA and EBVRA even with 100 nonlinked loci affecting the trait. In an organism with a genome of 30 Morgans in length, and considering an average of one recombination per one Morgan, approximately 60 blocks of genes will freely segregate.

The increase in the correlations with BV and response to selection was larger at lower heritability and when small numbers of loci are involved in expressing the trait, and this favors use of TA over RA for such traits. The greater increase in accuracy with small number of loci was due to higher variation in TA among relatives. Variation in relationships is an important source of variance of prediction error and can be used to increase the accuracy of selection. Although EBVRA is robust and has been used in practice to derive best linear unbiased prediction of breeding values, there is still room to improve the accuracy of selection by accounting for TA, which traces alleles that are identical in state.

The objective of this article was a partial exploration of the extent to which the approximations involved in deriving elements of the relationship matrix from pedigree information contribute to loss of accuracy in estimating breeding values. The definition of TA here assumed that genotypes at all QTL were known, but variation among QTL in their effects was not accounted for. Obviously, if QTL could be identified it would be possible to estimate the effects of these QTL and use this information directly in selection. The results here for relatively small numbers of loci (5 or 25) show that selection on EBVTA gives almost as much response as selection on breeding values directly. Even with 100 loci and the lowest heritability, selection on EBVTA improved response by 50% over conventional EBV. This strongly suggests that tracking genetic relationships at QTL will generally be much more important than estimating their effects. Only when a large number of loci are involved would having knowledge about QTL effects have the potential to substantially increase selection responses over EBVTA. In this case, however, obtaining accurate estimates of the many small QTL effects will be extremely difficult, and errors in these estimates might well result in little or no extra gain over using relationship information as in EBVTA.

Large numbers of polymorphic codominant genetic markers are now available. It should be possible to derive some approximation of TA using genetic markers. Use of such approximations will not yield responses as high as in this article, but extra response should still be possible, and this will be explored in a separate report. In general, it seems worthwhile when exploring the use of either QTL or marker information in selection programs to examine the extent to which estimates of QTL effect contribute information over the information that can be estimated to improve the accuracy of relationship matrix.

Implications

Even though pedigree relationship is used as a major source of information in genetic evaluation, it does not disclose all the available information. Accounting for identity by state and considering the variation in relationship can help significantly increase the accuracy of selection.

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Meuwissen, Hayes and Goddard (2001)

> *Meuwissen, T.H.E., Hayes, B.J., and Goddard, M.E. 2001. Prediction of total genetic value using genome-wide dense marker maps. Genetics, vol. 157, no. 4, pp. 1819 to 1829.*

The problem

Efforts to find genes of large effect had yielded little for animal breeding. Meuwissen, Hayes and Goddard proposed using a dense map of markers covering the whole genome and estimating the effects of all chromosome segments at the same time, without testing the significance of each one. With a training population that has phenotypes and genotypes, it would be possible to predict the breeding value of young animals from their genotypes alone.

Points to note while reading

This is the founding paper of genomic selection. Note the comparison between least squares, BLUP, BayesA and BayesB, and the accuracy of about 0.85 obtained by simulation. Note also the division of the data into a training population and a validation population, which became the standard in the field.

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Prediction of Total Genetic Value Using Genome-Wide Dense Marker Maps

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ABSTRACT

Recent advances in molecular genetic techniques will make dense marker maps available and genotyping many individuals for these markers feasible. Here we attempted to estimate the effects of ~50,000 marker haplotypes simultaneously from a limited number of phenotypic records. A genome of 1000 cM was simulated with a marker spacing of 1 cM. The markers surrounding every 1-cM region were combined into marker haplotypes. Due to finite population size (N_e = 100), the marker haplotypes were in linkage disequilibrium with the QTL located between the markers. Using least squares, all haplotype effects could not be estimated simultaneously. When only the biggest effects were included, they were overestimated and the accuracy of predicting genetic values of the offspring of the recorded animals was only 0.32. Best linear unbiased prediction of haplotype effects assumed equal variances associated to each 1-cM chromosomal segment, which yielded an accuracy of 0.73, although this assumption was far from true. Bayesian methods that assumed a prior distribution of the variance associated with each chromosome segment increased this accuracy to 0.85, even when the prior was not correct. It was concluded that selection on genetic values predicted from markers could substantially increase the rate of genetic gain in animals and plants, especially if combined with reproductive techniques to shorten the generation interval.

SELECTION for economically important quantitative traits in animals and plants is traditionally based on phenotypic records of the individual and its relatives. Estimated breeding values, based on this phenotypic data, are commonly calculated by best linear unbiased prediction (BLUP; Henderson 1984). One justification for molecular genetics research on livestock and crop species is the expectation that information at the DNA level will lead to faster genetic gain than that achieved based on phenotypic data only. The availability of a sparse map of genetic markers has resulted in the detection of some quantitative trait loci (QTL; Georges *et al.* 1995). The inclusion of marker information into BLUP breeding values was demonstrated by Fernando and Grossman (1989) and is predicted to yield 8–38% extra genetic gain (Meuwissen and Goddard 1996). However, the usefulness of information from a sparse marker map in outbreeding species is limited because the linkage phase between a marker and QTL must be established for every family in which the marker is to be used for selection.

The total number of single nucleotide polymorphisms (SNP) is estimated at many millions (Halushka *et al.* 1999), and the advent of DNA chip technology may make genotyping of many animals for many of these markers feasible (and perhaps even cost effective). However, the precision of mapping QTL by traditional linkage analysis is little improved by the use of a very dense marker map (Darvasi *et al.* 1993). Therefore, a different approach is needed to efficiently use all this marker information.

With a dense marker map some markers will be very close to the QTL and probably in linkage disequilibrium with it (*e.g.*, Hastbacka *et al.* 1992). Therefore, some marker alleles will be correlated with positive effects on the quantitative trait across all families and can be used for selection without the need to establish linkage phase in each family. Close markers can be combined into a haplotype. Chromosome segments that contain the same rare marker haplotypes are likely to be identical by descent (IBD) and hence carry the same QTL allele. Our approach is to estimate the effect on the quantitative trait of small chromosome segments defined by the haplotypes of marker alleles that they carry.

Quantitative traits are usually affected by many genes and consequently the benefit from marker-assisted selection is limited by the proportion of the genetic variance explained by the QTL. It would be desirable to utilize all QTL affecting the trait in marker-assisted selection. However, a dense marker map defines a very large number of chromosome segments and so there will be many effects to be estimated, probably more than there are phenotypic data points from which to estimate them.

The problem is essentially the same if we assume that

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the HUGO project and comparative mapping efforts will identify all 50,000 or so genes (*e.g.*, Aparicio 2000) in the cattle and pig genome. Hence, nearly all genes will be known, many with SNPs defined within them, and DNA chip technology will make it feasible to genotype animals for all these polymorphisms. However, when we come to estimate the allelic effects of all these genes on traits, we are again facing the estimation of very many effects in a data set of limited size, and we will not have enough degrees of freedom to fit all effects simultaneously by least squares (Lande and Thompson 1990).

In least-squares analyses, a stepwise approach can be adopted to tackle problems with insufficient degrees of freedom: genes are added to the model if they significantly improve the fit of the existing model. It seems, however, quite arbitrary to set the effects of loci to zero that are just below the significance threshold and include the full effect of those that are above this threshold. A better weighting of the information must be possible. Furthermore, selection of loci with the largest effects results in the selection of overpredicted effects. Lande and Thompson (1990) avoided this selection bias by using one-half of the data to select the loci with largest effects and using the other half to reestimate the effects. This splitting of the data set remains a suboptimal use of the information.

BLUP of allelic effects can be calculated even if there are more effects to be predicted than data points. If we assume that all loci or genes explain *a priori* an equal amount of variance (*i.e.*, the variance per locus is V_g/n, where V_g is the total genetic variance and n is the number of loci), we have only one variance to estimate. But having equal variances explained by all loci seems an unrealistic assumption.

In Bayesian statistics, parameters such as variance explained by the ith locus, V_{gi}, are assumed to come themselves from a prior distribution, $p(V_{gi})$. Hence, the variance can vary across loci, and combining of the information from the prior distribution and that of the data yields an estimate of V_{gi}. This Bayesian approach, where the variance due to each locus can vary, seems more realistic than assuming that the variance due to each locus is fixed at V_g/n, as is the case in the BLUP method.

The aim of this article is to compare least-squares, BLUP, and Bayesian analyses for their accuracy of predicting total breeding value of individuals in a situation where a limited number of recorded individuals are genotyped for many markers with many alleles per marker. Since the situation where the allelic effects of many known genes need to be estimated is very similar, we expect that the results will also hold for this situation.

METHODS

The simulated data: The alternative methods were compared by applying them to simulated data. To achieve a realistic distribution of QTL effects and gene frequencies, the simulated population was allowed to evolve until it reached an equilibrium between mutation and loss of genetic variance due to finite population size. Any particular population is likely to have experienced natural and artificial selection in the past, but this has been ignored for simplicity. The population was simulated with an effective population size of $N_e = 100$.

The genome was assumed to consist of only 10 chromosomes of 100 cM each. In the middle of each centimorgan, there was a QTL at which a mutation could occur and the QTL would become polymorphic. If the QTL is polymorphic and its allelic effects differ, it contributes to the genetic variance of the trait. At the beginning and end of every centimorgan of the chromosome a marker locus was situated; *i.e.*, there were 101 marker loci per chromosome, and every QTL was flanked by two marker loci at a distance of 0.5 cM.

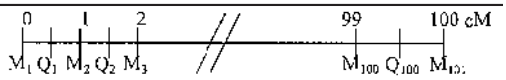
Mutations occurred randomly at the QTL at a rate of $m = 2.5 \times 10^{-5}$ per locus per generation. Since there are 1000 QTL, the mutation rate was 2.5×10^{-2} per haploid genome. For each new mutation the effect was drawn from a gamma distribution. Hayes and Goddard (2000) reviewed published estimates of QTL effects and concluded that their distribution resembled a gamma distribution with shape parameter $\beta = 0.4$. The gamma distribution with this shape was used here to simulate the effects of mutations at the QTL (see Table 1), which were assumed to be additive. Since the gamma distribution yields only positive effects, the sign of the QTL effect is sampled to be positive or negative with probability 0.5. The scale parameter of the gamma distribution was arbitrarily set to 1.66, which resulted in a genetic variance of 1 (see appendix). Because the environmental variance was also assumed to be 1, heritability was 0.5. The mutation variance added to the trait each generation was therefore $\sigma^2_m = 1000 \times m \times E(a^2) = 5 \times 10^{-3}$ environmental variance units (see appendix).

The mutation rate at the marker loci was 2.5×10^{-3} to allow a high probability of polymorphic marker loci, and every mutation at a marker locus resulted in a new unique marker allele. Hence, many marker loci were multiallelic. This might resemble the situation where several closely linked biallelic SNP markers are combined into one multiallelic marker haplotype.

To arrive at a mutation-drift balance, populations were simulated for 1000 generations at an effective size of 100. After these 1000 generations, the actual size of the populations was increased, to 200 (100 males and 100 females) in generation 1001, and to 2000 (20 half-sib families of size 100 each) in generations 1002 and 1003. The animals in generations 1001 and 1002 were marker genotyped and recorded for the trait. Phenotypic records were obtained by adding a normally distributed error term with variance 1 to the genetic value of the individuals. The 2000 animals of generation 1003 are assumed to be juveniles that did not (yet) have phenotypic records and their breeding values will be

TABLE 1

The parameters of the simulated genetic model

Map per chromosome[a]	
Number of chromosomes is the total number of morgans	10
Mutation rate of QTL	2.5×10^{-5}
Distribution of additive mutational effects	Gamma(1.66; 0.4)
Dominance of QTL effects	0
Mutation rate of marker loci	2.5×10^{-3}
Population structure	
Generations 1–1000	Ideal[b], $N = 100$
Generation 1001	Ideal[b], $N = 200$
Generation 1002	20 half-sib families, $N = 2000$
Generation 1003 and later	Ideal[b], $N = 2000$
Marker genotyping	Generations 1001 and later
Phenotypic recording	Generations 1001 and 1002

[a] M, marker position; Q, QTL position.

[b] Ideal denotes a population structure where the effective size equals the actual population size. This structure is simulated by giving every male (female) in generation $t - 1$ an equal probability of becoming the sire (dam) of animal i in generation t, which implies no selection and random mating of males and females.

estimated using marker information only. The statistical methods will be compared for their accuracy of predicting the true genetic values of the animals in generation 1003.

The data set for the estimation of the marker effects consisted of 200 and 2000 marker-genotyped and recorded animals in generations 1001 and 1002, respectively. Every animal was genotyped for 1010 (= 10 × 101) marker loci. It was assumed that the linkage phases of the tightly linked markers were known without error, but in practice they have to be estimated from the genotyping information and the family relationships between the animals. In the estimation procedures, the alleles of the marker loci that flank every centimorgan (possible QTL) of the genome are combined into one marker haplotype; *e.g.*, if the alleles of the flanking markers are 2 and 3 the haplotype allele will be denoted by 2_3. The simulation resulted in on average ~50 different haplotypes per 1-cM segment; *i.e.*, the total number of haplotype effects that needed to be estimated was ~50,000.

Least-squares estimation: Since we need to estimate 50,000 haplotype effects using 2200 phenotypic records, we cannot estimate all effects simultaneously by least squares, and some stepwise procedure for including the effects needs to be adopted. We used the following simple procedure here:

a. Perform single segment regression analyses for every segment, i, using the model

$$\boldsymbol{y} = \mu \mathbf{1}_n + \boldsymbol{X}_i \boldsymbol{g}_i + \boldsymbol{e},$$

where $\boldsymbol{y}$ is the data vector; μ is the overall mean; $\mathbf{1}_n$ is a vector of n ones; $\boldsymbol{g}_i$ represents the genetic effects of the haplotypes at the ith 1-cM segment; $\boldsymbol{X}_i$ is the design matrix for the ith segment; and $\boldsymbol{e}$ is the error deviation. The log-likelihood of the above model is calculated as $-0.5[n \ln(\sigma_e^2) + \boldsymbol{e}'\boldsymbol{e}/\sigma_e^2]$, where n is the number of records; $\boldsymbol{e}$ and σ_e^2 denote estimates of the error deviations and error variance, respectively, with $\sigma_e^2 = \boldsymbol{e}'\boldsymbol{e}/(n - \text{Rank}([\mathbf{1}_n \boldsymbol{X}_i]))$. These calculations yield a log-likelihood for every segment.

b. Plot the likelihood at every segment against the position of the segment. To have a likelihood peak we need one valley to the left and one valley to the right of the peak, and we required here that log-likelihood in the valleys was at least 14 units lower than that at the likelihood peak (which was found between the left and right valley). The 14 log-likelihood units are the natural log equivalent to a LOD score of 6 units (the use of the conventional LOD score of 3 yielded too many effects for simultaneous estimation). Note that the two lower likelihoods, which are exceeded by >14, are not necessarily adjacent to the position of the likelihood peak. These likelihood peaks imply a QTL segregating at the midpoint of the chromosome segment. There are usually several likelihood peaks per chromosome.

c. Estimate the effects of the haplotypes at the QTL positions simultaneously by the model

$$\boldsymbol{y} = \mu \mathbf{1}_n + \Sigma_i \boldsymbol{X}_i \boldsymbol{g}_i + \boldsymbol{e},$$

where summation Σ_i is over all QTL positions corresponding to a likelihood peak and $\boldsymbol{g}_i$ was estimated at the peak. All other haplotype effects are assumed to be zero. The overall mean is also arbitrarily set to zero, because its effect cannot be distinguished from that of the fixed haplotype effects.

In this way a complete array of estimates of haplotype effects was obtained.

BLUP estimation: BLUP estimation was by the model

$$\boldsymbol{y} = \mu\mathbf{1}_n + \Sigma_i \boldsymbol{X}_i\boldsymbol{g}_i + \boldsymbol{e},$$

where summation Σ_i is over all 1000 1-cM segments. The haplotype effects are assumed random here and their variance is $E(\sigma^2_{gi}) = 0.001$ (see APPENDIX). The error variance is also needed for BLUP estimation and is set equal to its true value of 1. The estimates of $\boldsymbol{g}_i$ are obtained from the mixed-model equations (HENDERSON 1984).

Bayesian estimation: method BayesA: With Bayesian estimation the data are modeled at two levels. First there is the model at the level of the data, and second there is a model at the level of the variances of the chromosome segments. The model at the level of the data is equal to that with BLUP estimation except that the variances of the segments are $\mathrm{Var}(g_{ij}) = \sigma^2_{gi}$, which differ for every segment and are estimated by the model for the variances of the segments. The latter estimation combines the information from the prior distribution of the variances and that from the data.

The prior distribution of variances of segments was the scaled inverted chi-square distribution, $\chi^{-2}(\nu, S)$, where S is a scale parameter and ν is the number of degrees of freedom. This is a convenient choice because when the information from this prior distribution is combined with the information from the data, the resulting posterior distribution is also a scaled inverted chi square (*e.g.*, WANG *et al.* 1993),

$$\mathrm{Post}(\sigma^2_{gi}|\boldsymbol{g}_i) = \chi^{-2}(\nu + n_i, S + \boldsymbol{g}_i'\boldsymbol{g}_i),$$

where n_i is the number of haplotype effects at segment i. This posterior distribution cannot be used directly for estimation because it is conditional on the unknown $\boldsymbol{g}_i$ effects. However, Gibbs sampling is based on posterior distributions conditional on all other effects, and hence it is used for the estimation of effects and variances here. The APPENDIX shows that the mean and variance of σ^2_{gi} are 0.001 and 1.675×10^{-4}, respectively. The scaled inverted chi-square distribution with $\nu = 4.012$ and $S = 0.0020$ has the same mean and variance, and this distribution was therefore used as the prior distribution of σ^2_{gi}.

When implementing the Gibbs sampler, the variances σ^2_{gi} were sampled from the above posterior distribution. For the error variance, σ^2_e, the prior distribution was $\chi^{-2}(-2, 0)$, which yields a uniformly distributed prior, *i.e.*, a flat prior, and the conditional posterior is

$$\mathrm{Post}(\sigma^2_e|\boldsymbol{e}_i) = \chi^{-2}(n - 2, \boldsymbol{e}_i'\boldsymbol{e}_i).$$

Given the error variance and the haplotype effects, the overall mean μ is sampled from the normal distribution,

$$N[\mathbf{1}_n'\boldsymbol{y} - \mathbf{1}_n'\boldsymbol{X}\boldsymbol{g}; \sigma^2_e/n],$$

where $\boldsymbol{X} = [\boldsymbol{X}_1\ \boldsymbol{X}_2\ \boldsymbol{X}_3\ \ldots]$ is the design matrix of all haplotype effects; and $\boldsymbol{g}$ is a vector of all haplotype effects. Finally, the haplotype effects, g_{ij}, were sampled from the normal distribution,

$$N[\boldsymbol{X}_{ij}'\boldsymbol{y} - \boldsymbol{X}_{ij}'\boldsymbol{X}\boldsymbol{g}_{(ij=0)} - \boldsymbol{X}_{ij}'\mathbf{1}_n\mu; \sigma^2_e/(\boldsymbol{X}_{ij}'\boldsymbol{X}_{ij} + \lambda_i)],$$

where $\boldsymbol{X}_{ij}$ is column of $\boldsymbol{X}$ of effect g_{ij}; $\boldsymbol{g}_{(ij=0)}$ equals $\boldsymbol{g}$ except that the effect of g_{ij} is set to zero; and $\lambda_i = \sigma^2_e/\sigma^2_{gi}$.

The Gibbs sampler was run for 10,000 cycles and by graphical inspection the first 1000 cycles were discarded as burn in. The samples of $\boldsymbol{g}$ from all later cycles were averaged to obtain an estimate of the haplotype effects $\boldsymbol{g}$.

Bayesian estimation: method BayesB: In reality, the distribution of genetic variances across loci is that there are many loci with no genetic variance (not segregating) and a few with genetic variance. However, the prior density of method BayesA does not have a density peak at $\sigma^2_{gi} = 0$; in fact its probability of $\sigma^2_{gi} = 0$ is infinitesimal. Method BayesB therefore uses a prior that has a high density, π, at $\sigma^2_{gi} = 0$ and has an inverted chi-square distribution for $\sigma^2_{gi} > 0$; *i.e.*, the prior distribution is

$$\begin{aligned} \sigma^2_{gi} &= 0 && \text{with probability } \pi, \\ \sigma^2_{gi} &\sim \chi^{-2}(\nu, S) && \text{with probability } (1 - \pi), \end{aligned}$$

where $\nu = 4.234$ and $S = 0.0429$ yield the mean and variance of σ^2_{gi} given that $\sigma^2_{gi} > 0$ (see APPENDIX).

In principle the Gibbs sampling algorithm of BayesA could also be used for BayesB; however, the Gibbs sampler will not move through the entire sampling space of method BayesB. This is because the sampling of $\sigma^2_{gi} = 0$ is not possible, if $\boldsymbol{g}_i'\boldsymbol{g}_i > 0$. On the other hand, the sampling of $\boldsymbol{g}_i = 0$ has an infinitesimal probability if $\sigma^2_{gi} > 0$. This problem is resolved by sampling σ^2_{gi} and $\boldsymbol{g}_i$ simultaneously from the distribution

$$p(\sigma^2_{gi}, \boldsymbol{g}_i|\boldsymbol{y}^*) = p(\sigma^2_{gi}|\boldsymbol{y}^*) \times p(\boldsymbol{g}_i|\sigma^2_{gi}, \boldsymbol{y}^*),$$

where $\boldsymbol{y}^*$ denotes the data $\boldsymbol{y}$ corrected for the mean and all other genetic effects except $\boldsymbol{g}_i$. The above indicates that we should sample σ^2_{gi} without conditioning on $\boldsymbol{g}_i$ (in contrast to BayesA) from $p(\sigma^2_{gi}|\boldsymbol{y}^*)$, and next sample $\boldsymbol{g}_i$ conditional on σ^2_{gi} and $\boldsymbol{y}^*$ as with BayesA from $p(\boldsymbol{g}_i|\sigma^2_{gi}, \boldsymbol{y}^*)$ (note that $\boldsymbol{g}_i = 0$ if $\sigma^2_{gi} = 0$). The distribution $p(\sigma^2_{gi}|\boldsymbol{y}^*)$ cannot be expressed in the form of a known distribution and therefore Gibbs sampling cannot be applied here. We used the following Metropolis-Hastings algorithm to sample from $p(\sigma^2_{gi}|\boldsymbol{y}^*)$, where the prior distribution, $p(\sigma^2_{gi})$, is used as the driver distribution to suggest updates for the Metropolis-Hastings chain (*e.g.*, GILKS *et al.* 1996):

1. Sample $\sigma_{g(\mathrm{new})}{}^2$ from the prior distribution $p(\sigma^2_{gi})$;
2. Replace the current σ^2_{gi} by $\sigma_{g(\mathrm{new})}{}^2$ with a probability of $\mathrm{Min}[p(\boldsymbol{y}^*|\sigma_{g(\mathrm{new})}{}^2)/p(\boldsymbol{y}^*|\sigma^2_{gi}); 1]$, and go to step 1, where $p(\boldsymbol{y}^*|\sigma^2_{gi})$ denotes the likelihood of the data given variance σ^2_{gi}. Note that this likelihood equals the posterior distribution, *i.e.*, where we really want to sample from, divided by the driver/prior distribution, which is as required by the independence sampling implementation of the Metropolis-Hastings algorithm (GILKS *et al.* 1996).

The calculation of the likelihood, $p(\boldsymbol{y}^*|\sigma^2_{gi})$, is described

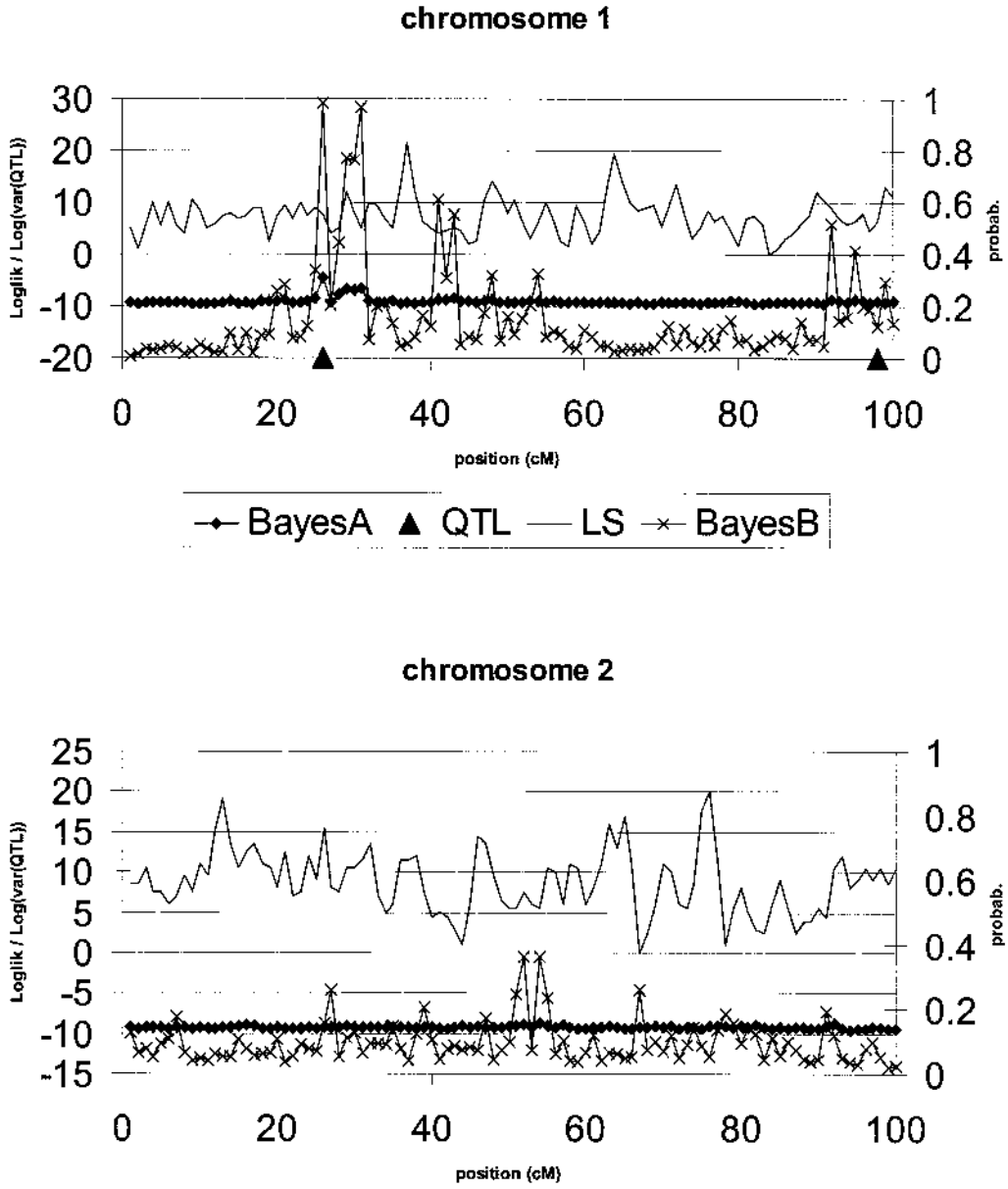


FIGURE 1.—Indicators of QTL positions at chromosomes 1 and 2 in the first replicate. BayesA, logarithm of estimated QTL variance; BayesB, posterior probability of having a QTL; LS, least-squares log-likelihood of having a single QTL as a deviation from the minimum LS-log-likelihood at the chromosome; QTL, position of the QTL that explains >2% of the total variance.

in the variance component estimation literature (*e.g.*, HENDERSON 1984). Now, the Monte Carlo Markov chain (MCMC) algorithm of BayesB consists of running a Gibbs chain as in BayesA, except that samples of σ^2_{gi} are obtained by running the above Metropolis-Hastings algorithm for 100 cycles instead of simply sampling σ^2_{gi} from an inverse chi-square distribution. The MCMC chain of BayesB was also run for 10,000 cycles and the first 1000 cycles were discarded as burn in.

Comparison of true and estimated breeding values: For all estimation methods, the estimated breeding values of the animals in generation 1003 were obtained from

$$\hat{u} = \mu \mathbf{1}_{2000} + \boldsymbol{X}_{(1003)}\boldsymbol{g},$$

where $\boldsymbol{X}_{(1003)}$ is the design matrix for generation 1003, *i.e.*, based on their marker genotyping, and $\boldsymbol{g}$ is the vector of estimates of the haplotype effects, which is obtained by least-squares, BLUP, or Bayesian estimation. Due to a recombination or a mutation, some new haplotypes occurred in generation 1003, whose effects had not been estimated in generations 1001 and 1002 and were assumed to equal zero. In the simulation, the true breeding values, $\boldsymbol{u}$, were also known in generation 1003, such that the correlation of true and estimated breeding values and the regression of true on estimated breeding values could be calculated. The latter regression coefficient equals 1 if the breeding value estimates are unbiased in the BLUP sense, *i.e.*, $E(u|\hat{u}) = \hat{u}$. The latter implies that if we select the animal with the highest $\hat{u}$, its true breeding value is expected to equal the estimate $\hat{u}$, so its breeding value is not over- or underpredicted.

RESULTS

Figure 1 shows the QTL detection results of the methods least squares (LS), BayesA, and BayesB for chromosomes 1 and 2 of one of the replicates. For LS the log-likelihoods of the individually fitted QTL are shown. For BayesA the logarithm of the estimates of σ^2_{gi} is shown, and for BayesB the posterior probability of $\sigma^2_{gi} > 0$. BLUP estimation did not provide an obvious parameter for QTL detection and was thus omitted from Figure 1. Chromosomes 1 and 2 are shown since they seemed to also characterize the results from the other chromosomes. Although there are more segregating QTL on the chromosomes, only QTL that explained >2% of the total genetic variance are shown; in fact, the QTL at positions 26 and 98 on chromosome 1 explained ~40 and 2% of the total genetic variance, respectively. Chromosome 2 was typical in that it contained no large QTL (all QTL explained <2% of genetic total genetic variance), which was the case for 5 out of 10 chromosomes in this replicate.

Although there is some correspondence between LS-likelihood peaks and the posterior probability peaks of BayesB, the LS-likelihood fluctuated much more than the posterior probability and thus yielded more false positive estimates of QTL effects (Figure 1). The method to detect QTL by LS, as described in METHODS, found QTL at positions 37 and 64 of chromosome 1 and at positions 13, 65, and 76 of chromosome 2 and found 18 QTL in total while there were 8 QTL with a $\sigma^2_{gi} > 2\%$ of the total genetic variance. The logarithm of σ^2_{gi} yielded a very flat line along most of the chromosomes and peaked only when there was a large QTL; *i.e.*, it seemed a conservative criterion for detecting QTL. The posterior probability of $\sigma^2_{gi} > 0$ of BayesB was in general a more sensitive criterion for detecting QTL, but still seemed to miss the QTL at position 98 of chromosome 1 although the posterior probability was somewhat increased in this region. In general, the really large QTL, say $\sigma^2_{gi} > 10\%$ of the total genetic variance, were detected accurately by BayesA and BayesB, but the smaller QTL were often not detected. Also, some false positive estimates were indicated by BayesB, although usually some smaller QTL were in the neighborhood of the indicated position, which may together with sampling error have increased the posterior probability of having a QTL.

When the estimates of the haplotype effects were used to estimate the true breeding values (TBV) of the animals in generation 1003, the correlations between true and estimated breeding values (EBV) were as given in Table 2. Because method BayesA required much more computer time and had poorer results than BayesB, it was investigated only in replicate 1. For the methods LS, BLUP, and BayesB, replicate 1 yielded correlations and regressions that were very similar to the averages shown in Table 2, except that BLUP yielded only $b_{TBV;EBV} = 0.844$ in this replicate.

TABLE 2

Comparing estimated *vs.* true breeding values in generation 1003

	$r_{TBV;EBV}$ + SE	$b_{TBV;EBV}$ + SE
LS	0.318 ± 0.018	0.285 ± 0.024
BLUP	0.732 ± 0.030	0.896 ± 0.045
BayesA	0.798	0.827
BayesB	0.848 + 0.012	0.946 + 0.018

Mean of five replicated simulations, except for BayesA which is based on one replicate. LS, least squares; BLUP, best linear unbiased prediction; BayesA, Bayesian method with inverse chi-square prior distribution; BayesB, Bayesian method where the prior density of having zero QTL effects was increased; $r_{TBV;EBV}$, correlation between estimated and true breeding values (equals accuracy of selection); $b_{TBV;EBV}$, regression of true on estimated breeding value.

The correlations of Table 2 reflect the accuracy of selection when selection is for the marker-based breeding value estimates. The accuracy of selection for LS predictions of TBV was rather low, which is probably due to the poor detection of QTL by LS (Figure 1), and because the estimation of the allelic effects of on average 15.4 detected marker haplotypes resulted in on average 872 equations. Hence, there were only 2200/872 = 2.5 d.f. per estimated effect, which would have resulted in large sampling errors and thus poor predictions of TBV. BLUP resulted in a reasonably high accuracy of selection, despite incorrectly assuming equal σ^2_{gi} for all loci. BayesA resulted in ~9% more accuracy than BLUP and the accuracy of BayesB exceeded that of BLUP by ~16%. The selection accuracies of BLUP, BayesA, and BayesB are very high for schemes where the animals have no performance information of their own. For comparison, a pedigree-based selection index would result in an accuracy of selection of ~0.4.

Table 2 also shows the regression of true breeding values on estimated breeding values. This regression should be 1 for methods that are unbiased in the BLUP sense (see METHODS). This regression coefficient was substantially <1 for LS, which indicates that the EBV would need to be regressed back to their mean to become an unbiased predictor of the TBV; *i.e.*, the estimated breeding values are too variable. In fact the LS-EBV were substantially more variable than the TBV (result not shown). The regression coefficient became closer to 1 if BLUP, BayesA, and, especially, BayesB were used, but was still somewhat <1. The small deviation of $b_{TBV;EBV}$ from 1 that remained may occur because the inverted chi-square distribution, which was used as a prior, is not equal to the simulated distribution of variances at the segregating loci, which is due to differences between gamma-distributed effects of mutations.

Table 3 shows correlations between TBV and EBV and regression coefficients of TBV on EBV when the estimation of marker haplotype effects was based on a

TABLE 3

Correlations between true and estimated breeding values when the number of phenotypic records is varied

	No. of phenotypic records		
	500	1000	2200
LS	0.124	0.204	0.318
BLUP	0.579	0.659	0.732
BayesB	0.708	0.787	0.848

reduced number of phenotypic records. The situation with 1000 or 500 records was obtained by deleting 1200 and 1700 records, respectively, of those animals of generation 1002, which had the smallest number of offspring in generation 1003; *i.e.*, the animals with most offspring were recorded. Table 3 does not show the results for method BayesA because of its computational costs and because it is expected to yield lower correlations than BayesB in any case. When the number of records was reduced from 2200 to 500, the correlation between TBV and EBV was reduced by 61, 21, and 17% for methods LS, BLUP, and BayesB, respectively. As expected, LS is least able to handle situations with few records and many effects to estimate and therefore showed a much larger reduction of the correlation between TBV and EBV as the number of records decreased. Method BayesB maintained a reasonably high correlation of 0.708 even when the estimation of the haplotype effects was based on only 500 phenotypic records.

Table 4 investigates the effects of having a less dense marker map, *i.e.*, where markers are spaced at every 2 or 4 cM. The situation with a marker distance of 2 cM was obtained by omitting every second marker from the original data set, and the situation with a 4-cM distance was obtained by again omitting every second marker. Note that the number of possible QTL positions remained at 100 per chromosome. For the BLUP analysis, σ^2_{gi} was increased from 0.0028 to 0.0056 and 0.0112, respectively. For BayesB the prior probability of having a QTL, $(1 - \pi)$, was increased from 0.053 to 0.106 and 0.212, respectively. For LS, the correlation between EBV and TBV increased as the distance between the markers increased. This is probably because the number of effects that are to be estimated is reduced when the number of markers is reduced, and the estimation of fewer effects reduces the shortage of degrees of freedom for LS estimation. BLUP and BayesB showed a small reduction in accuracy of ~4% when the marker spacing increased from 1 to 2 cM, and a 9–13% lower correlation when the spacing increased to 4 cM. This suggests that for the current population structure with $N_e = 100$, a marker spacing of 2 cM still yields sufficiently large linkage disequilibria between markers and QTL to predict the QTL effects. With the larger marker spacing of 4 cM, these linkage disequilibria are reduced.

TABLE 4

Correlations between true and estimated breeding values when the density of the marker map is varied and effective population size is 100

	Marker spacing (cM)		
	1	2	4
LS	0.318	0.354	0.363
BLUP	0.732	0.708	0.668
BayesB	0.848	0.810	0.737

DISCUSSION

Methods were presented for the estimation of allelic effects of marker or gene loci. The methods were compared in a situation where the allelic effects of small marker haplotypes surrounding 1-cM regions had to be estimated. Figure 1 indicated that BayesA and BayesB were able to predict the position of large QTL (say $\sigma^2_{gi} >$ 10% of the total genetic variance), but often did not identify the smaller QTL. However, the posterior probabilities of these smaller QTL were low but not zero for any of the possible QTL positions, so the small QTL still contributed to the prediction of total genetic values. The contributions of small QTL in the Bayesian methods were probably similar to their contribution in the BLUP prediction of total genetic values, where small and equal σ^2_{gi} are used for all loci. Because BLUP resulted in a reasonably high accuracy of predicting TBV, it seems that a correct positioning of QTL is not essential to achieve this. However, BayesA and especially BayesB did identify the positions of the largest QTL, which probably contributed to its increased accuracy of predicting TBV over BLUP.

The markers used in the simulations more closely resembled microsatellite markers than SNPs, which are biallelic and have a much lower mutation rate. However, three to five closely linked biallelic SNP markers may be pooled to obtain ~2^3 different haplotypes, which resemble the about seven alleles per marker that were used in the simulation. If the closely linked markers are within a region of ~0.25 cM, their recombination rate would resemble the mutation rate of the simulated markers (Table 1). The construction of haplotypes from the SNP markers, however, requires knowledge about the linkage phase of the markers. This requires at least two generations of typed individuals (*i.e.*, generations 1001 and 1002 here) and should be possible with high precision when the markers are closely linked; *i.e.*, (double) recombinations are very unlikely. In situations where the linkage phases are still uncertain, this uncertainty may be accounted for in the design matrix of the haplotype, $\boldsymbol{X}_i$, by having p and $(1 - p)$ at the elements

that belong to haplotypes A and B, instead of 0 and 1 (or 1 and 0). The information contributed by this record to estimate the difference between haplotypes A and B will be reduced, but this record still contributes to the differences between A (B) and all other haplotypes, such that the overall loss of information may be limited.

The accuracy with which breeding values can be predicted from the markers is limited by two factors. First, the linkage disequilibria between markers and QTL may be incomplete so that the marker haplotypes do not explain all the variance at the QTL. The results in Table 4 suggest that, in a population with $N_e = 100$, a marker spacing of 2 cM yields almost a maximum accuracy of prediction of the QTL effects. Because linkage disequilibria are a function of $N_e c$ (Sved 1971; Goddard 1991), Table 4 may more generally be interpreted as investigating the effect of $N_e c$ on accuracy of prediction of TBV; *i.e.*, the term "marker distance" could be replaced by $N_e c$, where c denotes distance between markers (in morgans). It follows from Table 4 that $N_e c$ should be <2 ($= 100 \times 0.02$) to achieve close to maximum accuracy of prediction of TBV given the information content of the current markers (the heterozygosity of markers was ~50%). With more informative markers, larger values of $N_e c$ may be used. Also, if estimation of the haplotype effects is restricted to a part of the population (*e.g.*, a few related families, the elites of a breeding scheme), the linkage disequilibrium within such a group may be much larger than expected based on $N_e c$ (Farnir *et al.* 2001) and prediction errors will be reduced within this part of the population. However, these estimates of haplotype effects may not be very useful to predict breeding values outside this part of the population.

The second effect that limits the accuracies of selection in Table 2 is the sampling error on the estimates of the haplotype effects. These sampling errors increase if the environmental variance divided by the number of genotyped and recorded animals increases. Hence, with a reduced heritability (h^2) such that ($1 - h^2$) becomes twice as large, the number of records needs to be doubled to achieve a similar accuracy. Table 3 shows that when the number of records increases from 1000 to 2200 the accuracy of predicting TBV still increases substantially. It seems that in these situations where very many effects are estimated from a limited number of records, a doubling of the number of records will yield, in practical situations, a substantial increase in accuracy (even when the number of records was already large). When haplotype effects were estimated from only 500 records, however, the accuracy of predicting TBV using BayesB was still much higher than could be expected from a pedigree-based selection index.

It may be expected that when going from a 10-M to a 30-M genome, the prediction of effects of individual chromosome segments becomes poorer because three times as many effects need to be predicted. However, in this case the EBV of an animal equals the sum of three times as many segments, such that prediction errors of individual segments are averaged out over three times as many effects. Hence, it may be expected that accuracies of selection with a 30-M genome will be similar to those of Table 2. When going from a 1-M genome (in preliminary tests of the programs) to 10-M genomes, we also found very similar accuracies of selection (although the 1-M results were more variable because sometimes there was very little genetic variance to be predicted).

The four methods of analysis used increasingly informative prior distributions for the σ^2_{gi}. LS performs badly because it greatly overestimates some haplotype effects and underestimates others. BLUP, although it uses a very simple prior, regresses estimates back toward zero, especially if there are few individuals carrying a particular haplotype. In the case of LS, better model selection methods can be used to determine which QTL effects should be included in the model; *e.g.*, start with the largest QTL and next include the second largest and so on until the QTL become too small to be included. However, the overestimation problems remain and in view of the huge number of possible QTL models and the poor results of LS in Table 2, it seems that more sophisticated methods are needed such as the use of prior distributions.

The BLUP method that was used here could also be improved upon. First, the total genetic variance could have been estimated by REML (residual maximum likelihood; Patterson and Thompson 1971) within a replicate and dividing the estimated genetic variance over the segments instead of the expected genetic variance (although, when the genome is sufficiently large, there will be little difference between these two values). The main problem with BLUP, however, remains, namely the really big QTL will be too heavily regressed back to zero. Second, the variances of the segments of QTL with large effects may be estimated by REML for use in the BLUP analysis. This leads to model selection problems as in the case of LS (which segments should be included in the analysis), and furthermore the estimation of many variances of chromosomal segments may be computationally as demanding as the estimation of these variances by BayesB.

It was assumed here that gene effects are additive, while some degree of dominance will probably occur in practice. In the presented models, only additive effects were fitted such that only the "average effects" of the genes (Falconer and Mackay 1996) are estimated, which is appropriate for the prediction of breeding values. This is probably satisfactory in many situations, except when prediction of dominant gene actions is important. In the latter case, dominance effects may be included in the model. Also, it was assumed here that to obtain the prior distribution of σ^2_{gi} mutation rate and the distribution of mutational effects were known. Although these parameters can be estimated in a metaanalysis (Hayes and Goddard 2001), they may still be dif-

ferent in specific situations. For example, the selection method may be different from that used in the populations of the metaanalysis, or the trait may be closely related to fitness, which renders most mutational effects negative. However, because the differences between BLUP, BayesA, and BayesB are relatively small (Table 2), the effect of using an incorrect prior distribution of σ^2_{gi} on the accuracy of selection seems to be small. The results of Table 2 indicate that (a) we need a proper prior distribution to avoid the overestimation problems of LS; and (b) the prior distribution should allow for small (or no) QTL effects with a high probability and for large QTL effects with a low probability, and the exact shape of the distribution seems of lesser importance (also since the inverted chi-square prior of BayesB did not perfectly agree with the gamma distribution of the QTL effects).

The simulation model assumes that every centimorgan contains a QTL that might affect the trait if a mutation occurred. However, in practice, only ~5% of these potential QTL were segregating. Thus the statistical model can be described as allowing that each marker bracket could contain a QTL. In reality, the QTL that can affect the trait will show some distribution across the genome and might be found in clusters close to each other; *i.e.*, some segments of 1 cM can contain more than one QTL. However, the effects of several closely linked QTL may be reasonably well approximated by one QTL with an increased genetic variance. Furthermore, the gamma distribution of QTL was based on QTL detection experiments (Hayes and Goddard 2001), which detect the effects of chromosomal segments rather than individual genes. In this view, the distribution of the effects of the 1-cM chromosomal segments may have been reasonably well reflected by the gamma distribution that was used here.

The main computation problem of the methods presented is that very many effects are fitted simultaneously, here 1000, such that the information matrix ($\boldsymbol{X'X}$) cannot be stored in the RAM memory of the computer. Therefore, the models were solved by the iteration on the data technique (Schaeffer and Kennedy 1986). This iterative technique, however, requires that, after the solutions of one haplotype effect are updated, the right-hand sides of all other effects are adjusted for the new solutions; *i.e.*, computer time increases approximately quadratically with the number of effects fitted. Method BayesA was most computer intensive and took ~2 weeks on a Pentium500 PC. Method BayesB required much less CPU, *i.e.*, ~1 day, because many effects have $\sigma^2_{gi} = 0$ in any cycle of the chain and thus do not enter the equations. In conclusion, the presented methods are computer intensive but are feasible on large computers even for genome sizes of 30 M or more.

The accuracies of selection in Table 2 are comparable to those obtained after a progeny test and are very high for animals without performance or progeny records.

TABLE 5

The correlation between estimated and true breeding values in generations 1003–1008, where the estimated breeding values are obtained from the BayesB marker estimates in generations 1001 and 1002

Generation	$r_{TBV;EBV}$
1003	0.848
1004	0.804
1005	0.768
1006	0.758
1007	0.734
1008	0.718

The generations 1004–1008 are obtained in the same way as 1003 from their parental generations.

Their use would dramatically increase the rate of genetic gain especially in traits where selection on phenotypic records is difficult, such as traits displayed only in females or after slaughter, disease resistance traits, or traits with low heritability. Further improvement could be had by combining high accuracies of selection with very short generation intervals to increase the number of selection cycles per unit of time. Georges and Massey (1991) and Haley and Visscher (1998) took this idea to the extreme in their "velogenetics" schemes for cattle, where oocytes were harvested from *in utero* calves (or obtained from *in vitro* meiosis of cultured cells), matured *in vitro*, fertilized, selected on the basis of their marker genotypes, and implanted in recipient cows (or cultured again), resulting in generation intervals of 6 months or less. Such a process could be repeated for several "generations" using method BayesB to predict the breeding values of the fertilized oocytes.

In velogenetics schemes the decline of the accuracy of selection over generations determines how often the haplotype effects need to be reestimated. This decline of accuracy is ~5% per generation between generations 1003 and 1005 (Table 5), and becomes smaller in later generations. This reduction is much larger than expected based on the recombination rate between a QTL and its nearest markers, *i.e.*, 0.5%. The latter indicates that more distant markers also contributed to the high accuracy of prediction of TBV. Or, putting it a different way, BayesB does not accurately predict the genetic value of individual 1-cM chromosome segments; instead it accurately predicts the total genetic value of larger chromosome segments (of, say, 4 cM or more), and therefore its accuracy reduces markedly as the large chromosome segments break up due to several generations of recombination. A larger effective population size will decrease the size of IBD chromosomal segments and will therefore improve the prediction of genetic value of small chromosome segments, provided that a sufficient number of phenotypic records is available. However, the accuracy of prediction of TBV in genera-

tion 1008, *i.e.*, six generations after the estimation of the haplotype effects, is still 0.718, which seems sufficiently high to make velogenetics breeding schemes successful. However, selection will cause changes of haplotype frequencies, which will make some haplotypes more important that were previously at low frequency, and thus less accurately estimated. Also, nonadditive genetic effects would cause a larger reduction in accuracy during generations 1003–1008 than shown in Table 5.

This study considered the selection of animals without phenotypic records. However, the results also apply to cases of genetic counseling, where the genetic risk of individuals for multifactorial diseases, *i.e.*, diseases that can (in part) be caused by many different mutations at different loci, is assessed. Alternatively, the response of individuals to various treatments can be predicted on the basis of marker information in situations where the responses to treatments depend on the genetic disposition of the individuals. Furthermore, well-chosen informative prior distributions in Bayesian estimation methods may prove useful in other examples of data mining where there are too few degrees of freedom for conventional statistical models.

CONCLUSIONS

1. By using a dense marker map covering all chromosomes, it is possible to accurately estimate the breeding value of animals that have no phenotypic record of their own and no progeny.
2. This requires the estimation of a large number of marker haplotype effects. Using least squares, all haplotype effects could not be estimated simultaneously. Even when only the largest effects were included, they were overestimated and the accuracy of predicting breeding value was low.
3. Methods that assumed a prior distribution for the variance associated with each chromosome segment gave more accurate predictions of breeding values even when the prior was not correct.
4. Selection on breeding values predicted from markers could substantially increase the rate of genetic gain in animals and plants especially if combined with reproductive techniques to shorten the generation interval.

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APPENDIX

Expectation and variance of genetic variance due to QTL in a mutation-drift model: The genetic variance due to an additive locus equals

$$\sigma^2_{gi} = 2p(1 - p)a^2$$

(Falconer and Mackay 1996), where $2p(1 - p)$ represents the heterozygosity, H, and a = one-half the difference between the performances of the two homozygotes. Because H and a^2 are independent,

$$E(\sigma^2_{gi}) = E(H) \times E(a^2).$$

$E(a^2) = E[(0.5(a_1 - a_2))^2] = 0.5\ E(a_1^2)$, where a_1 (a_2) = effect of homozygote carrying the first (second) mutation, and the above is because $E(a_1^2) = E(a_2^2)$ and $E(a_1 a_2) = 0$. The effect a_1 was sampled from the

gamma(1.66; 0.4) distribution, where the shape parameter (0.4) was estimated by Hayes and Goddard (2001) and the scale parameter is chosen such that the total genetic variance equals 1 (see next paragraph). The second moment of this gamma distribution is $E(a_1^2) = 0.203$, *i.e.*, $E(a^2) = 0.102$.

The expected heterozygosity of QTL is

$$E(H) = 4N_e m/(4N_e m + 1) = 0.0099$$

(Lynch and Walsh 1998), where m is the mutation rate (2.5×10^{-5}) per haploid locus per generation, and N_e is the effective population size (100). Hence, $E(\sigma_{gi}^2) = 0.102 \times 0.0099 = 0.001$. So 1000 QTL result in an expected total genetic variance of 1. Thus, an environmental variance of 1 yields a heritability of 0.5. The variance generated by new mutations each generation equals $2 \times 1000 \times m \times E(a^2) = 5 \times 10^{-3}$ environmental variance units.

The above E(H) is unconditional on whether the locus is segregating or not, as required for method BayesA. Method BayesB requires the calculation of E(H) given that the locus is segregating. Since $4N_e m \ll 1$, the allele frequency distribution of a segregating locus is U-shaped and approximated by

$$f(p) = K/[p(1-p)]$$

(Crow and Kimura 1970), where p = allele frequency, and the constant $K = 0.5/\ln(2N_e - 1)$. Using this distribution, the expected heterozygosity is

$$E(H|s=1) = \int_0^1 2p(1-p)f(p)\,dp = 1/\ln(2N_e - 1) = 0.1889,$$

where $s = 1$ indicates that the locus is segregating. Hence, $E(\sigma_{gi}^2|s=1) = 0.1889 \times 0.102 = 0.019$.

The prior distribution of method BayesB also involves the probability that the QTL is not segregating, π. Because $E(H) = E(H|s=1) \times \text{Prob}(s=1)$, where $\text{Prob}(s=1) = 1 - \pi$, we have

$$\pi = 1 - E(H)/E(H|s=1) = 1 - 0.0099/0.1889 = 0.947.$$

Next we need the variance of $(\sigma_{gi}^2|s=1)$, which is approximated by

$$V(\sigma_{gi}^2|s=1) = [E(a^2)]^2 V(H|s=1) + [E(H|s=1)]^2 V(a^2).$$

$V(a^2)$ is obtained from the gamma(1.66, 0.4) distribution and equals 0.08. Further,

$$V(H|s=1) = E(H^2|s=1) - [E(H|s=1)]^2,$$

where $E(H^2|s=1) = 2K/3$, which is obtained by a similar integration as that of $E(H|s=1)$. This gives $V(H|s=1) = E(H|s=1)[1/3 - E(H|s=1)] = 0.0273$. Substitution of these terms in the above equation for $V(\sigma_{gi}^2|s=1)$ yields $V(\sigma_{gi}^2|s=1) = (0.102)^2 \times 0.0273 + (0.1889)^2 \times 0.08 = 0.00315$.

Finally we need the variance of σ_{gi}^2 unconditional on the segregation status of the QTL,

$$V(\sigma_{gi}^2) = E_s[V(\sigma_{gi}^2|s)] + V_s[E(\sigma_{gi}^2|s)],$$

where $E_s[]$ ($V_s[]$) denotes taking expectation (variance) over the segregation status, s. Because $V(\sigma_{gi}^2|s=0) = 0$ if the locus is not segregating, we have

$$E_s[V(\sigma_{gi}^2|s)] = (1-\pi)\,V(\sigma_{gi}^2|s=1) = 0.0001670,$$

where $(1 - \pi)$ is the probability that $s = 1$. Since the segregation status follows a binomial distribution with probability π yielding $E(\sigma_{gi}^2|s=0) = 0$, and probability $(1 - \pi)$ yielding $E(\sigma_{gi}^2|s=1)$, we have

$$V_s[E(\sigma_{gi}^2|s)] = \pi(1-\pi)\,[E(\sigma_{gi}^2|s=1)]^2 = 4.98 \times 10^{-7}.$$

Hence, the unconditional variance of σ_{gi}^2 is $V(\sigma_{gi}^2) = 0.0001670 + 4.98 \times 10^{-7} = 0.0001675$.

The inverted chi-square distribution as prior distribution: The aim here is to find an inverted chi-square distribution $x \sim \chi^{-2}(\nu, S)$ with parameters ν and S such that the mean and variance of x equal that of σ_{gi}^2 (BayesA) or $\sigma_{gi}^2|s=1$ (BayesB). From the inverted chi-square distribution $E(x) = S/(\nu - 2)$ and $[\text{CV}(x)]^2 = V(x)/[E(x)]^2 = 2/(\nu - 4)$, where CV( ) denotes coefficient of variance. Substituting the mean and variances of σ_{gi}^2 and $\sigma_{gi}^2|s=1$ that were obtained in the previous section of the appendix and backsolving for ν and S yields the prior distribution $\sigma_{gi}^2 \sim \chi^{-2}(4.012; 0.0020)$, which was used for method BayesA, and the prior distribution $\sigma_{gi}^2|s=1 \sim \chi^{-2}(4.2339; 0.0429)$, which was used for method BayesB.

VanRaden (2008)

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The problem

With SNP chips about to reach the market, it was necessary to define how to compute genomic predictions on a national scale. VanRaden compared linear and nonlinear methods and showed how to build the genomic relationship matrix, the G matrix, from genotypes centered by allele frequencies. With this matrix, genomic BLUP uses the same mixed model equations, simply replacing A with G.

Points to note while reading

The formula for the G matrix in this paper is the most widely used in the world. Note the equivalence between estimating marker effects and predicting breeding values with G, and the discussion of the allele frequencies used to center the genotypes. This detail will reappear in single step and in metafounders.

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Efficient Methods to Compute Genomic Predictions

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ABSTRACT

Efficient methods for processing genomic data were developed to increase reliability of estimated breeding values and to estimate thousands of marker effects simultaneously. Algorithms were derived and computer programs tested with simulated data for 2,967 bulls and 50,000 markers distributed randomly across 30 chromosomes. Estimation of genomic inbreeding coefficients required accurate estimates of allele frequencies in the base population. Linear model predictions of breeding values were computed by 3 equivalent methods: 1) iteration for individual allele effects followed by summation across loci to obtain estimated breeding values, 2) selection index including a genomic relationship matrix, and 3) mixed model equations including the inverse of genomic relationships. A blend of first- and second-order Jacobi iteration using 2 separate relaxation factors converged well for allele frequencies and effects. Reliability of predicted net merit for young bulls was 63% compared with 32% using the traditional relationship matrix. Nonlinear predictions were also computed using iteration on data and nonlinear regression on marker deviations; an additional (about 3%) gain in reliability for young bulls increased average reliability to 66%. Computing times increased linearly with number of genotypes. Estimation of allele frequencies required 2 processor days, and genomic predictions required <1 d per trait, and traits were processed in parallel. Information from genotyping was equivalent to about 20 daughters with phenotypic records. Actual gains may differ because the simulation did not account for linkage disequilibrium in the base population or selection in subsequent generations.
Key words: genomic selection, mixed model, computer program, relationship matrix

INTRODUCTION

Genomic selection increases the rate of genetic improvement and reduces cost of progeny testing by allowing breeders to preselect animals that inherited chromosome segments of greater merit (Meuwissen et al., 2001; Schaeffer, 2006). Single nucleotide polymorphism (**SNP**) markers can now cover the genome with high density and are inexpensive to obtain. Evaluations based on SNP genotypes can be computed as soon as DNA can be obtained, which allows selection in both sexes early in life. Application of genomic selection to dairy cattle has just begun (de Roos et al., 2007; van der Beek, 2007; Guillaume et al., 2008). Potential methods and strategies were compared by Meuwissen (2007).

Computer algorithms and programs are needed to incorporate genomic data into genetic evaluations and to process the rapidly expanding numbers of SNP genotypes. Previous algorithms for including markers often fit effects individually rather than simultaneously or fit additional polygenic effects because marker coverage of the genome was not yet complete (de Roos et al., 2007). Iterative algorithms such as Gauss-Seidel and preconditioned conjugate gradient can be used to estimate allele effects (Legarra and Misztal, 2008), but fewer numerical problems may result from direct inversion of variance matrices or mixed model equations (Lee and van der Werf, 2006). Genomic relationships can be included in multitrait derivative-free REML programs (Zhang et al., 2007).

Objectives of this research were 1) to develop computer methods to include genomic data in predictions, 2) to apply the methods to simulated data for actual Holstein and Jersey pedigrees, and 3) to estimate gains in reliability from genomic predictions.

MATERIALS AND METHODS

Predictions were computed by linear and nonlinear systems of equations. The linear predictions assumed that all markers contributed equally to genetic variation (no major genes). The nonlinear (Bayesian) predictions assumed that the prior distribution of marker or QTL effects was not normal. Genetic variance may not be equal across chromosomes or markers because, for example, major genes may exist on some chromosomes. The data vector in both linear and nonlinear predictions was modeled as a linear function of the unknown effects, but solutions for the unknown effects in the nonlinear predictions were nonlinear functions of the data vector. Nonlinear predictions may be better than

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the best linear predictions when data are not normally distributed (Henderson, 1963), in this case because of major genes.

Linear predictions were computed by selection index and by 2 equivalent sets of mixed model equations. One model estimated individual marker effects first and then summed those as in Meuwissen et al. (2001). The other model included a genomic relationship matrix in place of the traditional additive genetic relationship matrix (Garrick, 2007; VanRaden, 2007; Zhang et al., 2007). Regressions were on genotypes rather than haplotypes because haplotyping would increase computation time with little or no gain in accuracy at high marker numbers (Calus et al., 2008). Advantages from nonlinear predictions and from haplotyping were expected to be small because the allele effects simulated were not large.

Simulation

Data were simulated so that the experimental design and analysis methods could be assessed before actual genotyping was conducted. The simulated data provided a test for the computer programs, algorithms, time required, and reliability of predictions. The simulated genotypes differ from actual genotypes primarily in that all loci in the base population were in Hardy-Weinberg equilibrium and that selection was not practiced in the following generations, which could affect reported correlations and other statistics. Also, the distribution of QTL effects used in the simulation may not match the biology of actual traits.

Marker and QTL inheritance was simulated for 50,000 biallelic markers and 100 biallelic QTL on 30 equal-length chromosomes. Markers and QTL were randomly scattered across the genome by assigning the position of each by uniform distribution. None of the markers directly affected any trait; instead, the QTL had effects that were simulated with a normal distribution. Variance of QTL effects thus followed a chi-square distribution, with the largest effect accounting for about 10% of genetic variance. Allele frequencies were uniformly distributed between 0 and 1.

Predictions were tested with simulated genotypes for 2,967 Holstein bulls and 766 Jersey bulls. The Holstein bulls included 1,885 bulls born from 1995 through 1997, 290 ancestor bulls included in computing predictions, and 792 younger bulls born from 2001 through 2002 for testing predictions. The Jersey bulls included 563 older bulls to compute predictions plus 203 younger bulls to test predictions. Genotypes for all known ancestors born since 1950 also were simulated in age order for a total of 23,105 Holstein and 7,737 Jersey cows and bulls. Alleles in the earliest generation were in Hardy-Weinberg equilibrium. For subsequent generations, maternal or paternal chromosome segments were inherited with a mean of 1.7 crossovers simulated from Poisson distribution and located with uniform distribution across the chromosome. For actual data, mean crossovers per chromosome probably are closer to 1.0.

Genomic predictions can be obtained by combining traditional genetic evaluation results with genotypic data instead of reprocessing all phenotypic and pedigree data. Variables analogous to deregressed evaluations or daughter yield deviations (**DYD**) were simulated by adding an independent error to the simulated true breeding value. To mimic actual data as closely as possible, that DYD error variance was calculated from published USDA reliabilities for net merit and from reliabilities for parent averages from an actual set of bulls that are being genotyped by USDA (Van Tassell et al., 2007).

Reliability of net merit for older bulls was obtained from May 2003 evaluations when the younger bulls to be predicted were just 1 to 2 yr old. Parent averages were compared with genomic predictions for ability to predict either true breeding values of young bulls or simulated DYD with the same reliability as August 2007 evaluations. Observed reliability of genomic predictions was obtained by squaring the correlations of estimated with true breeding value. Regressions of true breeding value and of DYD on genomic predictions also were calculated. When DYD was the dependent variable, regressions were weighted by reliability from daughters, which was computed as total daughter equivalents minus daughter equivalents from parent average (VanRaden and Wiggans, 1991).

Genomic data sets include some missing genotypes and incorrect genotypes. Fractions of each were arbitrarily set to 1%. The number of Holstein genotypes generated was about a billion (23,105 animals in pedigree file × 50,000 SNP), but only 150 million (2,967 genotyped bulls × 50,000 SNP) were stored and used for genomic prediction. Seven replicates were formed by generating 7 independent sets of QTL effects and environmental errors. For the Jersey simulation, 10 replicates were formed. Time required for the Holstein simulation was only about 10 min, but initial memory usage was high at 8 gigabytes. Memory in the simulation program was reduced to <1 gigabyte by processing each chromosome separately and reusing the memory.

Genomic Relationships and Inbreeding

Let $\mathbf{M}$ be the matrix that specifies which marker alleles each individual inherited. Dimensions of $\mathbf{M}$ are the number of individuals (n) by the number of loci (m).

Equations can include marker information using $n \times n$ matrix $\mathbf{MM}'$ or $m \times m$ matrix $\mathbf{M}'\mathbf{M}$ (Legarra and Misztal, 2008). If elements of $\mathbf{M}$ are set to −1, 0, and 1 for the homozygote, heterozygote, and other homozygote, respectively, diagonals of $\mathbf{MM}'$ count the number of homozygous loci for each individual, and off-diagonals measure the number of alleles shared by relatives. In contrast, diagonals of $\mathbf{M}'\mathbf{M}$ count the number of homozygous individuals for each locus, and off-diagonals measure the number of times alleles at different loci were inherited by the same individual.

Let the frequency of the second allele at locus i be p_i, and let $\mathbf{P}$ contain allele frequencies expressed as a difference from 0.5 and multiplied by 2, such that column i of $\mathbf{P}$ is $2(p_i - 0.5)$. Subtraction of $\mathbf{P}$ from $\mathbf{M}$ gives $\mathbf{Z}$, which sets mean values of the allele effects to 0. Allele frequencies in $\mathbf{P}$ should be from the unselected base population rather than those that occur after selection or inbreeding. An earlier or later base population can lead to greater or fewer relationships and to more or less inbreeding. Subtraction of $\mathbf{P}$ gives more credit to rare alleles than to common alleles when calculating genomic relationships. Also, the genomic inbreeding coefficient is greater if the individual is homozygous for rare alleles than if homozygous for common alleles.

Genomic relationship matrix $\mathbf{G}$ can be obtained by at least 3 methods. The first uses the formula $\mathbf{G} = \frac{\mathbf{ZZ}'}{2\sum p_i(1-p_i)}$. Division by $2\sum p_i(1-p_i)$ scales $\mathbf{G}$ to be analogous to the numerator relationship matrix $\mathbf{A}$. The genomic inbreeding coefficient for individual j is simply $G_{jj} - 1$, and genomic relationships between individuals j and k, which are analogous to the relationship coefficients of Wright (1922), are obtained by dividing elements G_{jk} by square roots of diagonals G_{jj} and G_{kk}.

The second method for obtaining $\mathbf{G}$ weights markers by reciprocals of their expected variance instead of summing expectations across loci and then dividing: $\mathbf{G} = \mathbf{ZDZ}'$, where $\mathbf{D}$ is diagonal with $D_{ii} = \frac{1}{m[2p_i(1-p_i)]}$. That formula was proposed for human genetic studies (Leutenegger et al., 2003; Amin et al., 2007).

The third method for obtaining $\mathbf{G}$ does not require allele frequencies and instead adjusts for mean homozygosity by regressing $\mathbf{MM}'$ on $\mathbf{A}$ to obtain $\mathbf{G}$ using the model

$$\mathbf{MM}' = g_0\mathbf{11}' + g_1\mathbf{A} + \mathbf{E},$$

where g_0 is the intercept and g_1 is the slope. Matrix $\mathbf{E}$ includes differences of true from expected fractions of DNA in common plus measurement error because the full DNA sequences were not available and a subset of markers was genotyped instead. The regression was fit with $\mathbf{MM}'$ as dependent and $\mathbf{A}$ as independent variable instead of vice versa because $\mathbf{A}$ is the expected value of $\mathbf{G}$, not vice versa. However, that variable assignment required reversing the calculations after fitting the regression using the formula

$$\mathbf{G} = \frac{\mathbf{MM}' - g_0(\mathbf{11}')}{g_1}.$$

Estimates of g_0 and g_1 may not be obvious because the dependent and independent variables are matrices rather than vectors, but equations can be written with summation notation as

$$\begin{bmatrix} n^2 & \sum_j\sum_k A_{jk} \\ \sum_j\sum_k A_{jk} & \sum_j\sum_k A_{jk}^2 \end{bmatrix} \begin{bmatrix} g_0 \\ g_1 \end{bmatrix} = \begin{bmatrix} \sum_j\sum_k (MM')_{jk} \\ \sum_j\sum_k (MM')_{jk}A_{jk} \end{bmatrix}.$$

Matrix $\mathbf{G}$ is positive semidefinite with the first 2 methods but can be singular if numbers of loci are limited or if 2 individuals have identical genotypes; $\mathbf{G}$ must be singular if $m < n$. Identical twins or clones can cause singularity even in $\mathbf{A}$. An improved, nonsingular matrix $\mathbf{G}_w$ can be obtained as the weighted (w) mean, $w\mathbf{G} + (1 - w)\mathbf{A}$ if numbers of markers are limited and $\mathbf{A}$ is nonsingular. Weights are reciprocals of the error variance in measuring true fractions of alleles shared.

Elements of $\mathbf{A}$ for noninbred full-sibs are 0.5 and have error variance of $(0.05)^2$ for predicting true fractions of shared DNA. An earlier study (VanRaden, 2007) assumed greater numbers of crossovers and incorrectly reported a standard deviation of 0.035 instead of 0.05. Exact distributions of shared alleles can be calculated (Stam, 1980). Within any diploid species, the percentage of DNA identical by descent for full-sibs is approximately $50 \pm \frac{50}{\sqrt{2(L+c)}}$, where L is the length of the genome in Morgans (about 30 for cattle) and c is the number of chromosome pairs (30 for cattle).

Elements of $\mathbf{G}$ have error variance of $\frac{0.125}{m}$. If minor allele frequencies are much less than 0.5, $4\sum p_i(1-p_i)$ can be substituted for m to reflect the information from markers more precisely. From the preceding algebra,

$$w = \frac{0.05^2}{\left(0.05^2 + \frac{0.125}{m}\right)},$$

and **G** should get more weight than **A** if $m > 50$ and nearly all the weight (>0.99) if $m > 5{,}000$. Most calculations in this study used **G** instead of $\mathbf{G}_w$ because m was 50,000. To test if any weight on **A** might be beneficial, analyses including $\mathbf{G}_w$ with $w = 0.90$, 0.95, and 0.98 were compared with those with $w = 1$.

Linear Predictions

If each individual is measured once for a trait and the inheritance of all alleles is known, then data vector **y** can be modeled as

$$\mathbf{y} = \mathbf{Xb} + \mathbf{Zu} + \mathbf{e},$$

where **Xb** is the mean and **e** is a random error vector with variance of $\mathbf{R}\sigma_e^2$. Matrix **R** is diagonal with elements $R_{ii} = \dfrac{1}{R_{dau}} - 1$, where R_{dau} is the bull's reliability from daughters with parent information excluded. Vector **u** contains the additive genetic effects that correspond to allele substitution effects for each marker. The sum **Zu** over all marker loci is assumed to equal the vector of breeding values (**a**).

Although the genotypic coefficient matrix has been labeled **X** in some recent studies (e.g., Meuwissen, 2007; Legarra and Misztal, 2008), the original labels of Henderson (1963) were retained with **Xb** for fixed effects (those with flat priors) and **Zu** for random effects (those with informative priors). With repeated records or other more complicated models, another incidence matrix to assign individuals to records would be required instead of the identity matrix implied here. In the current study, **y** included 1 daughter deviation variable for each bull.

Three different approaches that provide equivalent predictions were used to evaluate genotyped individuals with and without phenotypes. Mixed model estimates of **u** ($\hat{\mathbf{u}}$) were solved by iteration on data (Schaeffer and Kennedy, 1986); in this case genotypic data. Scalar λ is defined as the ratio σ_e^2 / σ_u^2, which equals the sum across marker loci $2\sum p_i(1-p_i)$ times the ratio σ_e^2 / σ_a^2, where σ_a^2 is total genetic variance. When phenotypic records are processed directly, λ is the ratio of error to additive genetic variance as usual. In this study, **R** accounts for heritability and differences in daughter numbers in this study so that σ_e^2 / σ_a^2 simplifies to 1. The EBV ($\hat{\mathbf{a}}$) were obtained as $\mathbf{Z}\hat{\mathbf{u}}$, and the resulting equations were

$$\hat{\mathbf{a}} = \mathbf{Z}(\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{I}\boldsymbol{\lambda})^{-1}\mathbf{Z}'\mathbf{R}^{-1}(\mathbf{y} - \mathbf{X}\hat{\mathbf{b}}). \quad [1]$$

The identity matrix **I** results from an assumption that marker effects in a large, randomly mating, unselected base population are uncorrelated. That assumption is true for the simulated QTL effects, but the markers themselves have only indirect effects by tracing the inheritance of linked QTL. Marker and QTL alleles may not be in equilibrium in the base population, and marker regressions are fit across rather than within families.

Selection index equations predict $\hat{\mathbf{a}}$ directly using genomic relationship matrix **G**, which is computed as $\dfrac{\mathbf{ZZ}'}{2\sum p_i(1-p_i)}$. Selection index equations are constructed as the covariance of **y** and **a** multiplied by the inverse of the variance of **y** multiplied by deviation of **y** from $\mathbf{X}\hat{\mathbf{b}}$ or

$$\hat{\mathbf{a}} = \mathbf{G}\left[\mathbf{G} + \mathbf{R}\left(\frac{\sigma_e^2}{\sigma_a^2}\right)\right]^{-1}(\mathbf{y} - \mathbf{X}\hat{\mathbf{b}}). \quad [2]$$

Estimates of $\hat{\mathbf{u}}$ could also be obtained if needed using the selection index equations by substituting $\mathbf{Z}'$ for the leftmost **G** in [2] and then dividing by $2\sum p_i(1-p_i)$. This shows that $\hat{\mathbf{a}}$ is the sum $\mathbf{Z}\hat{\mathbf{u}}$ over all alleles that the individual inherited. Selection index and mixed model equations provide the same estimates of $\hat{\mathbf{a}}$ if the same estimates of $\mathbf{X}\hat{\mathbf{b}}$ are used (Henderson, 1963). Thus, [1] and [2] should be identical in many genomic analyses because DYD or deregressed evaluations are the data source and the fixed effects already have been removed.

A third solution strategy presented by Garrick (2007) could be more efficient than selection index because **G** can be inverted just once and then additional traits with differing heritability or **R** processed using iteration:

$$\hat{\mathbf{a}} = \left[\mathbf{R}^{-1} + \mathbf{G}^{-1}\left(\frac{\sigma_e^2}{\sigma_a^2}\right)\right]^{-1}\mathbf{R}^{-1}(\mathbf{y} - \mathbf{X}\hat{\mathbf{b}}). \quad [3]$$

Matrix **G** may be singular, for example, if the number of markers does not exceed the number of individuals genotyped.

A main goal is to predict merit for individuals from genotypes before their phenotypes are measured, but

different procedures were required when [1], [2], or [3] were used. When [1] was used, predictions for younger individuals were computed as $\mathbf{Z}_2\hat{\mathbf{u}}$, with $\mathbf{Z}_2$ constructed from their genotypes. When [2] was used, predictions were computed by replacing the leftmost $\mathbf{G}$ in [2] by the genomic covariance matrix $\mathbf{C}$ constructed as $\frac{\mathbf{Z}_2\mathbf{Z}'}{2\sum p_i(1-p_i)}$. When [3] was used, predictions were computed by $\mathbf{C}\mathbf{G}^{-1}\hat{\mathbf{a}}$, which is a genomic regression on EBV of individuals with phenotypes.

Expected reliabilities for $\hat{\mathbf{a}}$ were computed from the matrices in [2] and [3] but were not computed from [1] because the 50,000 × 50,000 dense matrix was too large to invert. From [2], reliabilities were obtained from diagonals of

$$\mathbf{G}\left[\mathbf{G}+\mathbf{R}\left(\frac{\sigma_e^2}{\sigma_a^2}\right)\right]^{-1}\mathbf{G}$$

for individuals with phenotypes and from

$$\mathbf{C}\left[\mathbf{G}+\mathbf{R}\left(\frac{\sigma_e^2}{\sigma_a^2}\right)\right]^{-1}\mathbf{C}'$$

for those without. From [3], reliabilities were obtained similarly from

$$\mathrm{G}-\left[\mathbf{R}^{-1}+\mathbf{G}^{-1}\left(\frac{\sigma_e^2}{\sigma_a^2}\right)\right]^{-1}\left[\frac{\sigma_e^2}{\sigma_a^2}\right]$$

and from

$$\mathbf{C}\mathbf{G}^{-1}\mathbf{C}'-\mathbf{C}\mathbf{G}^{-1}\left[\mathbf{R}^{-1}+\mathbf{G}^{-1}\left(\frac{\sigma_e^2}{\sigma_a^2}\right)\right]^{-1}\mathbf{G}^{-1}\mathbf{C}'\left[\frac{\sigma_e^2}{\sigma_a^2}\right].$$

Genetic bases often set the mean evaluations of some recent group of cows to 0. Estimation of the genetic base is accomplished by 2 equivalent formulas that depend on whether equations are solved by inversion or iteration. With inversion, inverse of the variance of $\mathbf{y}$ ($\mathbf{V}^{-1}$) is available so that the mean of $\hat{\mathbf{b}}$ can be obtained directly as $(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}(\mathbf{X}'\mathbf{V}^{-1}\mathbf{y})$, assuming that those equations are full rank. With iteration, updated EBV are available in each round so that the mean of $\mathbf{y}$ can be estimated iteratively as $\left(\mathbf{X}'\mathbf{R}^{-1}\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{R}^{-1}\left(\mathbf{y}-\hat{\mathbf{a}}\right)$. With those algorithms, genomic predictions $\hat{\mathbf{a}}$ are solved as deviations from a genetic base, and then the genetic base is added back to the predictions.

Nonlinear Predictions

For nonlinear predictions, a genetic variance component can be estimated for each marker, marker bracket, or haplotype (Meuwissen et al., 2001). Alternatively, markers with smaller effects can be regressed further toward 0 and markers with larger effects regressed less to account for the nonnormal prior distribution. Let marker deviations ($\hat{\mathbf{d}}$) be defined as

$$\hat{\mathbf{d}}=\left[\mathrm{diag}\left(\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z}\right)\right]^{-1}\left[\mathbf{Z}'\mathbf{R}^{-1}\left(\mathbf{y}-\mathbf{X}\hat{\mathbf{b}}-\mathbf{Z}\hat{\mathbf{u}}\right)\right]+\hat{\mathbf{u}}. \quad [4]$$

Addition of $\hat{\mathbf{u}}$ is required because subtraction of $\mathbf{Z}\hat{\mathbf{u}}$ in [4] removes not only off-diagonal terms in $\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z}$ but also the diagonal. Linear model solutions are obtained using linear regression on $\hat{\mathbf{d}}$:

$$\hat{\mathbf{u}}=\left[\mathrm{diag}\left(\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z}\right)+\mathbf{I}\lambda\right]^{-1}\mathrm{diag}\left(\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z}\right)\hat{\mathbf{d}}.$$

The optimal regression is nonlinear if distribution of $\hat{\mathbf{d}}$ is not normal. Of the total number of markers (m), only a subset (q) may be associated with QTL. If marker-QTL associations are not known a priori, distribution of $\hat{\mathbf{d}}$ is a mixture of 2 distributions. Elements of $\hat{\mathbf{d}}$ have a $1-\frac{q}{m}$ prior probability of containing only error variance and a $\frac{q}{m}$ prior probability of containing both error and QTL variance. Two normal density functions (f_{err} and $f_{QTL+err}$) were evaluated for each marker deviation in $\hat{\mathbf{d}}$ in each round of iteration to obtain converged posterior probabilities. To simplify calculations, marker solutions were first standardized by dividing $\hat{\mathbf{u}}$ by their standard deviations $\left(\sqrt{\frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{m}}\right)$. Then, f_{err} was the standard normal density with a mean of 0 and variance of 1. Function $f_{QTL+err}$ had greater variance calculated as m/q, the reciprocal of the proportion of loci with effects. For extreme marker effects, computation of normal densities resulted in underflow, which was avoided by setting error density to zero for marker deviations of >15 standard deviations.

Nonlinear and linear regressions on SNP marker deviations are shown in Figure 1. Nonlinear predic-

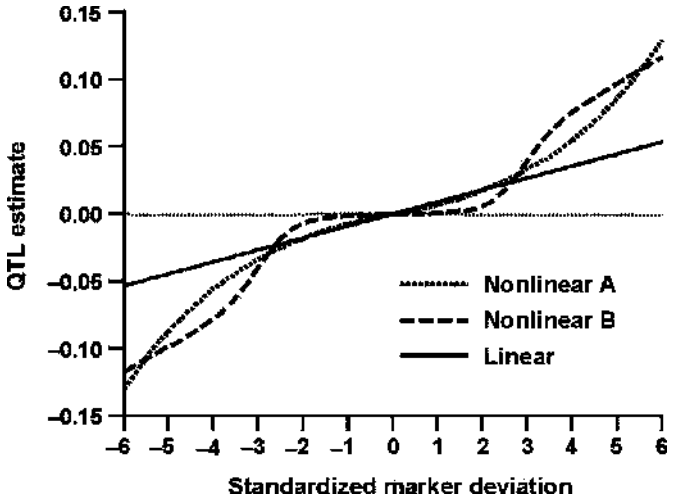


Figure 1. Nonlinear and linear regressions for marker allele effects.

tion A assumed that all markers had effects but with a simple, heavy-tailed distribution generated from a normal variable divided by $1.25^{\text{abs}(s-2)}$, where s is the number of standard deviations from the mean and 1.25 determines departure from normality. Instead of using a constant λ for all markers as in [1], nonlinear prediction A used individual λ_i for each locus, computed as $\lambda_i = \lambda/1.25^{\text{abs}(s-2)}$.

Nonlinear prediction B used:

$$\lambda_i = \lambda\left[\frac{q}{m} + \left(1 - \frac{q}{m}\right)\left(\frac{f_{err}}{f_{QTL+err}}\right)\right]. \quad [5]$$

Several markers may be needed for each actual QTL; therefore, the optimal choice for q was about 700 instead of the 100 QTL simulated. Nonlinear predictions A and B are analogous but not identical to Bayesian A and B methods of Meuwissen et al. (2001), and other prior distributions could fit actual data better. In this research, QTL effects were simulated with a normal distribution, and marker effects were estimated by iteration using an equation analogous to [1] but substituting λ_i from equation [5].

Computation

Allele frequencies in the base (founder) population were estimated with a linear model that solves for gene content of nongenotyped ancestors and descendants using pedigrees (Gengler et al., 2007). The known genotypes are treated as data, and the unknown genotypes of relatives are estimated using the inverse of the traditional relationship matrix and standard mixed model equations. Calculations are easy but not ideal because linear algebra is used instead of nonlinear probabilities. Equations without groups for unknown parents were presented by Gengler et al. (2007), and they also described and tested equations that included several groups. The current research included only 1 unknown-parent group because pedigrees were extremely complete and the goal was to estimate frequencies for the same population as the inbreeding base (1960).

Simple allele frequencies were also obtained as means of only the known genotypes. In the simple frequency estimates, summation was only over nonmissing genotypes so that missing genotypes did not contribute. Genomic predictions and inbreeding coefficients were computed using the simple estimate, the base population estimate, and also the true base frequencies to determine how sensitive the results were to different allele frequencies. Frequency estimation can bias the genomic inbreeding coefficients (Leutenegger et al., 2003).

Jacobi iteration was used in this study, but other methods may converge faster. First-order Jacobi iteration performed reasonably well with small data sets but converged too slowly with large data sets and unbalanced allele frequencies. A blend of first- and second-order Jacobi iteration performed better but required setting 2 relaxation factors.

Let the solution before relaxation from the current round be labeled $\mathbf{u}_c$, and let solutions after relaxation from the previous 2 rounds be $\mathbf{u}_1$ and $\mathbf{u}_2$. The solution after relaxation in the current round $\mathbf{u}_0$ was then computed as

$$\mathbf{u}_0 = \mathbf{u}_1 + relax_1(\mathbf{u}_c - \mathbf{u}_1) + relax_2(\mathbf{u}_1 - \mathbf{u}_2),$$

where $relax_1$ and $relax_2$ are relaxation factors. Optimal relaxation factors for this blend of first- and second-order Jacobi iteration were obtained by trial and error. The algorithms to estimate allele frequencies and allele effects both used this iterative technique, but the optimum relaxation factors differed.

Efficiency was greatly increased by summing elements of $\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z}\hat{\mathbf{u}}$ once per round of iteration and obtaining the off-diagonal sums for each row as

$$\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z}\hat{\mathbf{u}} - \text{diag}\left(\mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z}\right)\hat{\mathbf{u}}. \quad [6]$$

As genotypes are obtained, rows of $\mathbf{Z}$ are multiplied by $\hat{\mathbf{u}}$. Then, right-hand sides for each marker [elements of $\mathbf{Z}'\mathbf{R}^{-1}\left(\mathbf{y} - \mathbf{X}\hat{\mathbf{b}}\right)$] are adjusted for off-diagonal elements by subtracting the diagonal from the sum of all elements as in [6]. That procedure reduced computing

costs from quadratic with number of markers (Meuwissen et al., 2001) to linear. Similar computing strategies were discovered independently by Janss and de Jong (1999) and Legarra and Misztal (2008). As compared with the iterative methods of Legarra and Misztal (2008), the present algorithm requires much less memory by reading **Z** once per iteration instead of loading **Z** into memory, uses estimated allele frequencies instead of 0.5 to set coefficients of **Z**, accounts for missing genotypes, weights observations by including **R**, and is extended to nonlinear models.

Missing genotypes could be set to the gene content estimated from the algorithm of Gengler et al. (2007) or from a similar procedure. That approach would require recoding the data with fractional values instead of the simpler 0, 1, and 2 for known values and 5 to indicate unknown. Because ≤1% of genotypes were expected to be missing, simple genotype coding was retained for this study, but coefficients of **Z** were set to 0, which substitutes the mean allele frequency of the population for missing values. Other elements of **Z** were standardized to account for each animal's proportion of missing genotypes. That standardization was accomplished by replacing **Z** with **WZ**, where **W** is diagonal with

$$\mathbf{W}jj = \sqrt{\frac{\sum p_i(1-p_i) \text{ over all loci}}{\sum p_i(1-p_i) \text{ over only nonmissing loci}}}.$$

Elements of **WZ** for the 3 genotypes were then $-2p_i$, $1 - 2p_i$, and $2 - 2p_i$, each divided by $\mathbf{W}_{jj}$. That adjustment gave genomic relationship matrix $\frac{\mathbf{WZZ'W}}{\sum 2p_i(1-p_i)}$ an expected value of **A**, the traditional relationship matrix, when some genotypes were missing. Equations reduce to those of VanRaden (2007) when no genotypes are missing.

RESULTS AND DISCUSSION

Genomic predictions from several methods were all affordable, and timing tests on simulated data for 50,000 markers revealed that predictions for several traits of 3,000 bulls can be computed within a week. Genomic predictions offered much greater reliability for young animals than did traditional parent averages, even for traits affected by 100 QTL with unknown location.

Estimation of allele frequencies in the base population and gene content for 23,105 Holsteins in the pedigree file required about 400 iterations to converge to 5 digits of accuracy. Total time was 2 processor days, but actual clock time was reduced by processing loci on separate chromosomes in parallel. Time required for frequency estimation was proportional to the length of the pedigree file and number of markers. Optimal relaxation factors for the blended Jacobi iteration were 0.60 for the first factor and 0.88 for the second factor when used in the algorithm of Gengler et al. (2007). Simple frequency estimates required cycling through the known genotypes just once and <1 min of processing.

True allele frequencies were correlated with estimated frequencies in the base population by 0.98, but only by 0.94 with simple frequency estimates from the current population. The 2 estimates were correlated with each other by 0.97. Mean for base frequency estimates was 0.50 with a standard deviation of 0.27 compared with a mean of 0.49 and standard deviation of 0.30 for simple frequency estimates, both of which compared well with the mean of 0.50 and standard deviation of 0.29 for true allele frequencies simulated as uniformly distributed between 0 and 1. The largest difference between base frequency and simple frequency estimates was 0.41.

Genomic inbreeding coefficients from the first method for obtaining **G**, which weights summed allele effects by the sum of their expected variance, were more precise when base frequency rather than simple frequency estimates were used. Correlations between pedigree inbreeding coefficients from **A** and genomic inbreeding coefficients from **G** for the older Holstein bulls were 0.74 using true allele frequencies and 0.68 using estimates of base frequencies, but only 0.12 using simple allele frequency estimates. Corresponding correlations for the younger bulls were 0.66, 0.63, and 0.40. Genomic inbreeding coefficients were biased downward using either estimator of frequency. The mean of 7% was reasonable using true frequencies but was −4% with base frequency estimates and −2% with simple frequency estimates compared with 5% for traditional inbreeding coefficients of Holsteins from pedigree. Complete statistics including standard deviations and Jersey inbreeding coefficients are in Tables 1 and 2.

Inbreeding coefficients in **G** were less precise if **G** was estimated by weighting marker loci by reciprocals of their expected variance (second method) and more precise if **G** was obtained by adjusting for mean homozygosity by regression on **A** (third method) compared with the first method. Correlations of genomic inbreeding coefficients from the second method with pedigree inbreeding coefficients were about 0.06 lower than corresponding correlations for the first method when base allele frequencies were used. For the third method, which does not require allele frequencies, corresponding correlations were about 0.06 greater than for the first method. The first and third method had

Table 1. Correlations of pedigree with genomic inbreeding (F) coefficients computed with true or estimated allele frequencies

		Allele frequencies used in genomic F		
Breed	Birth year	True[1]	Base[2]	Simple[3]
Holstein	<2000	0.66	0.63	0.40
	2001 to 2002	0.74	0.68	0.12
Jersey	<2000	0.78	0.66	−0.26
	2001 to 2002	0.81	0.75	−0.28

[1]True frequencies in the base population.

[2]Estimated frequencies in the base population.

[3]Simple frequencies estimated by counting alleles in genotyped bulls.

nearly equal correlations when the first method used true rather than estimated base frequencies.

Subtraction of allele frequencies and weighting by variance may have theoretical appeal, but results from simulation indicated that simple counts of allele sharing and allele homozygosity can provide accurate measures of relationship and inbreeding. Comparisons using real data could give different results for the 3 methods because alleles in the base population may not be in equilibrium.

Construction of the genomic relationship matrix required 9 h for Holsteins and 1.5 h for Jerseys. Time increased by number of markers multiplied by number of bulls squared. Inversion of the matrix required about 12 min for Holsteins and 20 s for Jerseys. Time increased by number of bulls cubed. With that algorithm, traits with similar reliability such as milk, fat, and protein could be processed together with almost no additional cost.

Iteration on genotypic data took 1,000 rounds and 16 h to reach convergence for just 1 trait, but the cost for that algorithm did not increase as quickly as the cost of forming and inverting **G**. Time increased by number of iterations multiplied by number of markers multiplied by number of bulls multiplied by number of traits. In practice, clock times can be decreased by processing different traits in parallel and by using previously converged estimates of allele effects as starting values when additional genotypes are added. Rereading the genotypes increased time by 10% but decreased memory by >90% compared with storing the genotypes in memory. Optimal relaxation factors for the blended Jacobi iteration were 0.0002 for the first factor and 0.90 for the second factor. The first factor had greater optimums when numbers of markers were lower.

Predictions from iteration on data were correlated by 0.997 with those from direct inversion, which indicated agreement of [1] and [2]. With smaller data sets and more iteration, perfect correlations were achieved among predictions from [1], [2], and [3].

Reliability comparisons are in Table 3 and are means across replicates. Reliability averaged 0.66 for nonlinear predictions and 0.63 for linear predictions versus 0.32 for parent average for net merit of young Holstein bulls. Corresponding accuracies of selection obtained as square roots of those values were 0.81, 0.79, and 0.56. Thus, linear genomic predictions had reliabilities that were 0.31 greater than reliability for parent average for the younger Holstein bulls and 0.19 greater for the younger Jersey bulls. Simulated gains from genomic selection are much larger than estimates of Guillaume et al. (2008) because many more markers are now available.

Nonlinear predictions had a slight advantage over linear predictions (Table 3) of 0.04 for young Holstein bulls and 0.01 for young Jersey bulls. Reliability obtained from direct inversion of linear mixed model equations averaged 0.58, which is in reasonable agreement with the 0.63 obtained from the squared correlation of linear predictions with true breeding values. Reliability of older bulls and ancestors averaged 0.91 for predictions from both linear and nonlinear genomic models versus 0.90 from use of the traditional animal-model relationship matrix. Gains in reliability for the

Table 2. Means and SD of pedigree inbreeding (F) and genomic inbreeding coefficients computed with true or estimated allele frequencies

				Genomic F (%) using allele frequencies					
		Pedigree F (%)		True[1]		Base[2]		Simple[3]	
Breed	Birth year	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Holstein	<2000	4.6	2.4	6.7	3.9	−4.4	3.4	−2.1	5.4
	2001 to 2002	5.3	1.8	7.6	3.5	−3.6	3.0	−1.9	3.5
Jersey	<2000	5.1	3.4	7.2	4.8	0.0	4.0	−3.3	8.4
	2001 to 2002	7.1	2.7	9.2	4.3	2.1	3.7	−1.3	6.3

[1]True frequencies in the base population.

[2]Estimated frequencies in the base population.

[3]Simple frequencies estimated by counting alleles in genotyped bulls.

Table 3. Reliabilities of genomic predictions computed with true or estimated allele frequencies

		Reliability					
			Linear genomic model with frequencies				
Breed	Birth year	Animal model	True[1]	Base[2]	Simple[3]	0.5	Nonlinear model[2]
Holstein	<2000	0.896	0.915	0.913	0.912	0.905	0.913
	2001 to 2002	0.319	0.628	0.628	0.626	0.620	0.663
Jersey	<2000	0.797	0.817	0.817	0.817	0.814	0.816
	2001 to 2002	0.163	0.354	0.354	0.355	0.315	0.368

[1]True frequencies in the base population.
[2]Estimated frequencies in the base population.
[3]Simple frequencies estimated by counting alleles in genotyped bulls.

older bulls were small (about 0.01) because their reliabilities already were high and because the largest of the simulated QTL did not have very large effects. Gains from actual data could differ because of more or fewer QTL with non-normal distribution.

Reliability of predictions increased slightly for Holsteins when estimates of base allele frequency replaced simple estimates and nearly matched levels obtained when true allele frequencies were used. With iteration and coefficients of **Z** set to 0.5 and −0.5 as in Legarra and Misztal (2008), the system of equations converged more quickly, but predictions were less accurate. Realized reliabilities were about 0.007 greater for both younger and older Holstein bulls when simple estimates of allele frequency were used than when coefficients of 0.5 were used; corresponding gains for Jerseys were 0.040 and 0.003. With inversion, accuracy of predictions was similar to the use of true allele frequencies if elements of **G** were rescaled using

$$\frac{\mathbf{MM}' - g_0(\mathbf{11}')}{g_1}.$$

Thus, rescaling of **G** or of variance ratios seems useful, but predictions are not identical to those obtained by subtracting allele frequencies because elements of **G** are correlated by <1.

Matrix **G** was always nonsingular in this study because m was much larger than n, allowing use of equation [3], which requires $\mathbf{G}^{-1}$. Reliability of predictions increased very slightly when $\mathbf{G}_w$ replaced **G**, with maximum increase of only 0.0002 for $w = 0.95$. Reliability was lower for $w = 0.90$ than for $w = 1$. Thus, analyses may benefit slightly by including a small weight (5%) on traditional relationships.

Linear and nonlinear genomic predictions for younger Holstein bulls were correlated by 0.97 and had nearly equal means. The standard deviation was about 1.07 times larger for nonlinear predictions, which corresponds to the 0.04 greater reliability. Regressions of true breeding values on linear predictions were 0.99 ± 0.05 and on nonlinear predictions were 0.95 ± 0.05, only slightly less than the desired value of 1.0. Regressions of DYD on genomic predictions were similar, but with slightly larger standard error (0.06). Advantages of using base rather than simple estimates of allele frequency were small. Those extra gains in reliability were <0.01 for Holsteins and Jerseys.

Gains in reliability were converted to daughter equivalents by assuming a heritability of 20%, which is approximately the weighted mean of heritabilities of traits included in the net merit index (VanRaden and Multi-State Project S-1008, 2006). Total daughter equivalents equal daughter equivalents from parents, progeny, and own records (if available) plus additional daughter equivalents from genotyping. For both young and old bulls, information gained from genotyping was equivalent to including records from about 20 additional daughters. If similar gains occur with actual data, genomic evaluations should be released for young animals and for bulls with daughters and for cows with records instead of the traditional evaluations computed from only pedigree and phenotypic data. New methods are needed to explain genomic predictions (VanRaden and Tooker, 2007) because few scientists or breeders yet have the training or experience to understand how the new technology works.

During the review process for this study, the programs tested with simulated data were applied to actual genotypes of the Holstein bulls, and genomic predictions of merit were provided to North American breeders in April 2008. Primary advantages of simulated data were that true breeding values were known and statistical methods could be compared before actual data became available.

CONCLUSIONS

Computational methods for including genotypic data were developed and tested using simulation. Genomic

inbreeding coefficients required accurate estimates of allele frequencies in the base population; predictions of genetic merit were much less sensitive to allele frequency estimates. Genomic inbreeding and relationship coefficients that did not require allele frequencies were also obtained from simple counts of homozygous loci and alleles shared. Scaling these to match pedigree inbreeding and relationships was achieved by regression.

Three equivalent linear predictions were derived, each with different computational advantages. A nonlinear prediction produced slightly greater correlations with true breeding values and was solved using iteration on genotypic data and nonlinear regression on marker deviations. A blend of first- and second-order Jacobi iteration achieved reasonable convergence for estimating allele frequencies and effects, but other algorithms could be faster. Iterative algorithms are preferred because time increases linearly with number of genotypes. However, calculation of individual reliabilities required inverting mixed model equations that included genomic relationships. Calculation of predictions and reliabilities may require using more than one set of equations.

Tests on simulated data indicated that reliability for young animals could be >60% versus 32% from parent average. Genotypic data may add information worth about 20 daughter equivalents, and benefits should increase over time as more relatives are genotyped. However, actual genomic reliabilities may be affected by linkage disequilibrium in the base population and by subsequent selection, neither of which were simulated.

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6

Single step and the H matrix

2009 to 2020

In the first genomic evaluations, only part of the animals were genotyped and evaluations were carried out in several steps, with adjustments that were difficult to calibrate. The proposed solution was to combine pedigree relationships and genomic relationships into a single matrix, the H matrix, which allows genotyped and nongenotyped animals to be evaluated jointly. This block brings together the definition of the H matrix, the simplified form of its inverse, the independent derivation by Christensen and Lund, the APY algorithm for large populations and metafounders, which address the compatibility between the A and G matrices. The block closes with a review of the current state of genomic evaluation.

Legarra, Aguilar and Misztal (2009)

Misztal, Legarra and Aguilar (2009)

Aguilar et al. (2010)

Christensen and Lund (2010)

Misztal, Legarra and Aguilar (2014)

Legarra et al. (2015)

Misztal, Lourenco and Legarra (2020)

Legarra, Aguilar and Misztal (2009)

Legarra, A., Aguilar, I., and Misztal, I. 2009. A relationship matrix including full pedigree and genomic information. Journal of Dairy Science, vol. 92, no. 9, pp. 4656 to 4663.

The problem

The first genomic evaluations were carried out in several steps. First the traditional evaluation was run, then pseudophenotypes were computed for the genotyped animals, and only then was the genomic prediction made, with loss of information and the need for adjustments. Legarra, Aguilar and Misztal showed how to build a relationship matrix that combines pedigree and genotypes. The genomic information of the genotyped animals is propagated to their nongenotyped relatives through the pedigree. This is the H matrix.

Points to note while reading

Note the derivation, which conditions the breeding values of the nongenotyped animals on those of the genotyped animals. Note also that the paper presents the H matrix, but not yet the simple form of its inverse, which appears in the work of Aguilar and coworkers in 2010.

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A relationship matrix including full pedigree and genomic information

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ABSTRACT

Dense molecular markers are being used in genetic evaluation for parts of the population. This requires a two-step procedure where pseudo-data (for instance, daughter yield deviations) are computed from full records and pedigree data and later used for genomic evaluation. This results in bias and loss of information. One way to incorporate the genomic information into a full genetic evaluation is by modifying the numerator relationship matrix. A naive proposal is to substitute the relationships of genotyped animals with the genomic relationship matrix. However, this results in incoherencies because the genomic relationship matrix includes information on relationships among ancestors and descendants. In other words, using the pedigree-derived covariance between genotyped and ungenotyped individuals, with the pretense that genomic information does not exist, leads to inconsistencies. It is proposed to condition the genetic value of ungenotyped animals on the genetic value of genotyped animals via the selection index (e.g., pedigree information), and then use the genomic relationship matrix for the latter. This results in a joint distribution of genotyped and ungenotyped genetic values, with a pedigree-genomic relationship matrix **H**. In this matrix, genomic information is transmitted to the covariances among all ungenotyped individuals. The matrix is (semi)positive definite by construction, which is not the case for the naive approach. Numerical examples and alternative expressions are discussed. Matrix **H** is suitable for iteration on data algorithms that multiply a vector times a matrix, such as preconditioned conjugated gradients.
Key words: genetic evaluation, genomic selection, relationship matrix, mixed model

INTRODUCTION

Availability of dense molecular markers of type SNP has lead to the recent introduction of the so-called genome-wide or genomic selection evaluation models. Most such models are based on variants of simultaneous genome-wide association analysis, in which marker or haplotype effects (**a**) are estimated. Differences among methods are mostly on the a priori distribution of **a** (e.g., Meuwissen et al., 2001; Gianola et al., 2006).

Although these methods are very promising for animal breeding, genotyping is not feasible for an entire population because of its high cost or logistical constraints (i.e., culled, slaughtered, or foreign animals). This is of importance, for example, for foreign bulls for which no genotyping is possible. Animals that are genotyped include prospective and old males, and possibly prospective mothers of future candidates (e.g., embryo transfer dams).

As not all animals can be genotyped, a 2- or 3-step procedure has to be followed; first, a regular genetic evaluation is run; then, corrected phenotypes or pseudo-data are used in the second step, where the marker-assisted selection model is effectively applied (Guillaume et al., 2008; VanRaden et al., 2009). These phenotypes are daughter yield deviations (**DYD**) and yield deviations (**YD**) for dairy cattle.

After computation of pseudo-data, genomic or marker-assisted predictions can be obtained by either simultaneously fitting polygenic and QTL effects (Guillaume et al., 2008), or by computing the genomic prediction and combining it with estimated breeding values from the animal model (VanRaden et al., 2009). Genomic predictions can be obtained either by estimating **a** effects caused by markers or by using mixed model equations with a genomic relationship matrix **G** (VanRaden, 2008). This assumes that a priori marker effects are normally distributed with a common variance. Although the assumption is arguable, positing a more complicated prior distribution resulted in little gain in practice (VanRaden et al., 2009). On the other hand, the genomic relationship matrix is simple to interpret and handle.

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Advantages of the multistage system include no change to the regular evaluations and simple steps for predicting genomic values for young genotyped animals. Disadvantages include weighting parameters, such as variance components (Guillaume et al., 2008) or selection index coefficients (VanRaden et al., 2009), and loss of information. Furthermore, the extension to multiple traits is not obvious and tracing back anomalies in a two-step procedure might become very complicated.

As for the loss of information, several problems exist in the use of DYD and YD. These problems are weights (caused by different amount of information in the original data set), bias (caused by selection, for example), accuracy (for animals in small herds), and collinearity (for example, the YD of two cows in the same herd). As for the bias, if genomic selection is used, the expectation of Mendelian sampling in selected animals is not zero (Party and Ducrocq, 2009).

These problems may offset the benefit of marker-assisted selection, particularly for cows (Neuner et al., 2008, 2009). Also, in other species (sheep, swine, beef cattle) or traits (e.g., maternal traits, calving ease) DYD are more difficult to compute or even to define, or they might be poorly estimated—for example, if the contemporary groups are small.

One simplification of the current strategy would be to perform a joint evaluation using all phenotypic, pedigree, and genomic information. A possibility is to impute markers in ungenotyped animals via marker and pedigree information (i.e., linkage analysis) and estimate marker effects once imputation is done. However, in order to get a "best" predictor (in the sense of Henderson (1984), i.e., the conditional expectation), the incertitude in marker imputation, which is very high for most ungenotyped individuals, has to be accounted for via integration over the posterior distribution of marker imputations and marker effects. This can be achieved for example by peeling or Markov chain Monte Carlo (Abraham et al., 2007). However, this is unfeasible for a data set of even medium size when there are many loci or when many markers are missing, and particularly in the presence of loops, which are common in livestock pedigrees.

Another possibility is to use the same methodology as in the current evaluation (i.e., Henderson's mixed model equations) except that the relationship matrix **A** needs to be modified to include the genomic information. The purpose of this study is to provide such a relationship matrix, based on transmissions from genotyped animals to their offspring, or selection indexes from genotyped to ungenotyped animals. This will blend complementary information from recorded pedigree and molecular markers. Computational methods for such a modified numerator relationship matrix, even if complex, can be found in Misztal et al. (2009).

METHODS

Covariance Matrix of Breeding Values Including Genomic Information

Let $\mathbf{u}$ be a vector of genetic effects. Under a polygenic infinitesimal model of inheritance, $\mathrm{Var}(\mathbf{u}) = \mathbf{A}\sigma_u^2$, where $\mathbf{A}$ is the numerator relationship matrix based on pedigree. Consider three types of animals in $\mathbf{u}$: 1) ungenotyped ancestors with breeding values $\mathbf{u}_1$; 2) genotyped animals, with breeding values $\mathbf{u}_2$ (no ancestor is genotyped and phantom parents can be generated if necessary); and 3) ungenotyped animals with breeding values $\mathbf{u}_3$, which might descend from either one of the three types of animals. A particular case is one in which ungenotyped animals are ancestors and progeny of genotyped animals—for example, a bull dam daughter of another bull. They are arbitrarily put in group 1. Then $\mathbf{A}$ can be partitioned as follows:

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} & \mathbf{A}_{13} \\ \mathbf{A}_{21} & \mathbf{A}_{22} & \mathbf{A}_{23} \\ \mathbf{A}_{33} & \mathbf{A}_{32} & \mathbf{A}_{33} \end{bmatrix}.$$

Let $\mathbf{u}_2 = \mathbf{Za}$, $\mathbf{Z}$ being an incidence matrix and $\mathbf{a}$ the effects of markers. Matrix $\mathbf{Z}$ is centered by allele frequencies (VanRaden, 2008). Then $\mathrm{Var}(\mathbf{u}_2) = \mathbf{ZZ}'\sigma_a^2 = \frac{\mathbf{ZZ}'}{k}\sigma_u^2 = \mathbf{G}\sigma_u^2$, where k is twice the sum of heterozygosities of the markers (VanRaden, 2008).

In some implementations, matrix $\mathbf{G}$ can be seen as an "improved" matrix of relationships (Amin et al., 2007). Villanueva et al. (2005) and Visscher et al. (2006) propose to use a realized matrix of transmissions from parents to offspring in the data, averaging across all positions in the genome; this proposal is impractical in a general manner as genotypes are needed over entire families. VanRaden (2008) discussed how the expectation of $\mathbf{G}$ above is $\mathbf{A}$, the regular numerator relationship matrix, and that $\mathbf{G}$ represents observed, rather than average, relationships. Therefore, it accounts for Mendelian samplings (i.e., it can distinguish full-sibs) and unknown or far relationships. The gain by using $\mathbf{G}$ has been shown (González-Recio et al., 2008; Legarra et al., 2008; VanRaden et al., 2009). In principle, the additive variance using $\mathbf{G}$ is identical to that using $\mathbf{A}$ (Habier et al., 2007).

There is no need for $\mathbf{G}$ to have a particular genetic interpretation in terms of relationships. For example, using the reproducing kernel Hilbert spaces equations of González-Recio et al. (2008), $\mathrm{Var}(\mathbf{u}_2) = \mathbf{K}\sigma^2_\alpha$, where $\mathbf{K}$ is a matrix with "distances" among individuals. Matrix $\mathbf{K}$ can be scaled to $\mathbf{K}^*$ so that $\mathrm{Var}(\mathbf{u}_2) = \mathbf{K}^*\sigma^2_u$ by equating the expectation for the sum of squares of $\mathbf{u}_2$ in the data following polygenic and the reproducing kernel Hilbert spaces models. The expected sum of squares is, respectively (polygenic vs. reproducible kernel Hilbert spaces): $E(\mathbf{u}_2'\mathbf{u}_2) = tr(\mathbf{A}_{22})\sigma^2_u$ and $E(\mathbf{u}_2'\mathbf{u}_2) = tr(\mathbf{K})\sigma^2_\alpha = tr(\mathbf{K}^*)\sigma^2_u$, where tr is the trace operator. In absence of inbreeding, $tr(\mathbf{A}_{22}) = 1$ and thus $\mathbf{K}^* = \mathbf{K}/tr(\mathbf{K})$. Note that matrix $\mathbf{K}$ is also centered, and pseudo-inbreeding coefficients can be extracted from the diagonal of $\mathbf{K}^*$. Of course, by using a reproducing kernel Hilbert spaces model any "genealogical" intuition is lost.

In the following, and for simplicity of notation, it will be assumed that $\sigma^2_u = 1$.

Plug-in G. A simple way to use $\mathbf{G}$ is to plug it into $\mathbf{A}$; this results in the following modified $\mathbf{A}$:

$$\mathbf{A}_g = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} & \mathbf{A}_{13} \\ \mathbf{A}_{21} & \mathbf{G} & \mathbf{A}_{23} \\ \mathbf{A}_{33} & \mathbf{A}_{32} & \mathbf{A}_{33} \end{bmatrix} = \mathbf{A} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & \mathbf{G}-\mathbf{A}_{22} & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad [1]$$

where $\mathbf{A}_{22}$ has been simply replaced by $\mathbf{G}$. A proposal by Gianola and De los Campos (2008) to come up with predictions of ungenotyped animals from predictions of genotyped ones is to use $\mathbf{A}_{12}\mathbf{G}^{-1}\hat{\mathbf{u}}_2$. Their proposal reduces thus to a selection index by making the assumption that the covariance among individuals is described by $\mathbf{A}_g$.

Matrix $\mathbf{A}_g$ is simple to use but not properly constructed. The use of $\mathbf{G}$ potentially modifies covariances in ancestors and descendants of genotyped animals. For example, assume two full-sibs in the genotyped animals whose genomic relationship is 0.6. By using $\mathbf{A}_g$, it is assumed that average relationship among their daughters is 0.25, whereas in fact it is 0.3.

It can be verified by small numerical examples that $\mathbf{A}_g$ is indefinite (i.e., some eigenvalues are negative and some positive); the reason is that it is not based in an underlying linear model leading to a matrix crossproduct of the type **T'T** plus a diagonal matrix, like the numerator relationship matrix (Quaas, 1988) or the marker-assisted BLUP (Fernando and Grossman, 1989). Therefore, the statistical background is ill-defined (Searle, 1971; Harville, 1976). A correct statistical inference can only be made if the covariance matrix is positive or semi-positive definite. Matrix $\mathbf{A}_g$ might lead to correct inferences if the matrix is reasonable and numerical errors are not big. This ought to be checked by simulations.

Modification for Progeny. For this different approach, consider the descendants of genotyped animals. Following the decomposition of $\mathbf{A}$ (i.e., Quaas, 1988), let $\mathbf{P}$ be a matrix containing expected transmissions from ancestors to offspring, that is, with values of 0.5 in the son-dam and son-sire cells. Then $\mathbf{u} = \mathbf{P}\mathbf{u} + \boldsymbol{\varphi}$, where $\boldsymbol{\varphi}$ is a vector of Mendelian samplings and founder effects (Quaas, 1988). The variance of $\boldsymbol{\varphi}$ is indicated as $\mathbf{D}$.

In this particular group of animals:

$$\mathbf{u}_3 = \begin{bmatrix} \mathbf{P}_{31} & \mathbf{P}_{32} & \mathbf{P}_{33} \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \\ \mathbf{u}_3 \end{bmatrix} + \boldsymbol{\varphi}_3.$$

Then

$$\mathbf{u}_3 = \mathbf{T}_{33}(\mathbf{P}_{32}\mathbf{u}_2 + \mathbf{P}_{31}\mathbf{u}_1 + \boldsymbol{\varphi}_3),$$

which can be seen as a regression equation, and where $\mathbf{T}_{33} = (\mathbf{I} - \mathbf{P}_{33})^{-1}$ (Quaas, 1988).

Then

$$\mathrm{Var}(\mathbf{u}_3) = \mathbf{T}_{33}(\mathbf{P}_{32}\mathbf{G}\mathbf{P}_{32}' + \mathbf{P}_{31}\mathbf{A}_{11}\mathbf{P}_{31}' + \mathbf{P}_{32}\mathbf{A}_{21}\mathbf{P}_{31}' + \mathbf{P}_{31}\mathbf{A}_{12}\mathbf{P}_{32}' + \mathbf{D}_3)\mathbf{T}_{33}'$$

$$\mathrm{Cov}(\mathbf{u}_3, \mathbf{u}_1) = \mathbf{T}_{33}(\mathbf{P}_{31}\mathbf{A}_{11} + \mathbf{P}_{32}\mathbf{A}_{21}) = \mathbf{A}_{31}$$

$$\mathrm{Cov}(\mathbf{u}_3, \mathbf{u}_2) = \mathbf{T}_{33}(\mathbf{P}_{32}\mathbf{G} + \mathbf{P}_{31}\mathbf{A}_{12})$$

Then the covariance matrix becomes:

$$\mathbf{A}_p = \begin{bmatrix} \mathbf{A}_{11} & & \textit{symm} \\ \mathbf{A}_{21} & \mathbf{G} & \\ \mathbf{A}_{31} & \mathbf{T}_{33}\mathbf{P}_{32}\mathbf{G} + \mathbf{P}_{31}\mathbf{A}_{12} & \mathbf{T}_{33}(\mathbf{P}_{31}\mathbf{A}_{11}\mathbf{P}_{31}' + \mathbf{P}_{32}\mathbf{G}\mathbf{P}_{32}' + \mathbf{D}_3)\mathbf{T}_{33}' \end{bmatrix}. \quad [2]$$

Variance caused by Mendelian sampling in $\mathbf{D}_3$ is related to inbreeding in the founders $(\mathrm{Var}(\varphi) = (1/2 - (F_s + F_d)/4)\sigma^2_u)$, where F_s and F_d are inbreeding coefficients of sire and dam; this can be extracted from the diagonal of $\mathbf{G}$ if needed. Otherwise, $\mathbf{D}_3$ is the same as in classical methods. Assuming that $\mathbf{D}_3$ is equivalent in both cases (i.e., parents are not inbred), $\mathbf{A}_p$ can be formed by appropriately modifying $\mathbf{A}$:

$$\mathbf{A}_p = \mathbf{A} + \begin{bmatrix} 0 & 0 & symm \\ 0 & \mathbf{G} - \mathbf{A}_{22} & \\ 0 & \mathbf{T}_{33}\mathbf{P}_{32}\left(\mathbf{G} - \mathbf{A}_{22}\right) & \mathbf{T}_{33}\mathbf{P}_{31}\left(\mathbf{G} - \mathbf{A}_{22}\right)\mathbf{P}'_{31}\mathbf{T}'_{33} \end{bmatrix}.$$

Again, matrix $\mathbf{A}_{\mathrm{p}}$ might not be fully coherent (and indeed might be indefinite) because matrix $\mathbf{G}$ also includes information about the ancestors of genotyped animals. For example, two genotyped animals, say A and B, that have no relationship in the numerator relationship matrix $\mathbf{A}$ might show some relationship in $\mathbf{G}$, because of a common, unrecorded, ancestor. Thus, a relationship can be posited between the ancestors of A and B. Matrix $\mathbf{A}_{\mathrm{p}}$ would work if all founders were genotyped (e.g., in a nucleus scheme); in this case, the system is fully coherent. For practical purposes, $\mathbf{A}_{\mathrm{p}}$ might be reasonable because most information for sire evaluation is contained in the progeny, and not in the ancestors.

Modification for the Whole Pedigree. There is no distinction between ancestors or progeny of genotyped animals in this method; animals in 1 are ungenotyped, whereas animals in 2 are genotyped.

Then $\mathbf{A} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}$ with inverse $\mathbf{A}^{-1} = \begin{bmatrix} \mathbf{A}^{11} & \mathbf{A}^{12} \\ \mathbf{A}^{21} & \mathbf{A}^{22} \end{bmatrix}$.

Based on selection index theory and properties of the normal distribution, conditionally on pedigree (Sorensen and Gianola, 2002, p. 254; Gelman et al., 2004, p. 86), the distribution of breeding values of ungenotyped animals, conditioned on breeding values of genotyped animals, is:

$$p\left(\mathbf{u}_1 \middle| \mathbf{u}_2\right) = N\left(\mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{u}_2, \mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}\right) \qquad [3]$$

(which is the best predictor if we assume normality), or,

$$\mathbf{u}_1 = E\left(\mathbf{u}_1 \middle| \mathbf{u}_2\right) + \boldsymbol{\varepsilon} = \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{u}_2 + \boldsymbol{\varepsilon},$$
$$\mathrm{Var}\left(\boldsymbol{\varepsilon}\right) = \mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21} = \left(\mathbf{A}^{11}\right)^{-1}.$$

This can be seen just as a regression equation. Now substitute $\mathbf{u}_2 = \mathbf{Za}$. Then

$$\mathbf{u}_1 = \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{Za} + \boldsymbol{\varepsilon}$$

so that

$$\mathrm{Var}\left(\mathbf{u}_1\right) = \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{G}\mathbf{A}_{22}^{-1}\mathbf{A}_{21} + \mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}.$$

This can be reduced to

$$\mathrm{Var}\left(\mathbf{u}_1\right) = \mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\left(\mathbf{G} - \mathbf{A}_{22}\right)\mathbf{A}_{22}^{-1}\mathbf{A}_{21}$$

$$\mathrm{Var}\left(\mathbf{u}_2\right) = \mathbf{ZZ}'/k = \mathbf{G} \text{ and}$$

$$\mathrm{Cov}\left(\mathbf{u}_1, \mathbf{u}_2\right) = \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{G}.$$

Note that $\mathbf{A}_{12}\mathbf{A}_{22}^{-1} = -\left(\mathbf{A}^{11}\right)^{-1}\mathbf{A}^{12}$. This might be convenient for computation as $\mathbf{A}^{11}$ and $\mathbf{A}^{12}$ are sparse and simpler to create, following Henderson's rules, than $\mathbf{A}_{12}$ and $\mathbf{A}_{22}$.

Let us now call $\mathbf{H}$ the covariance matrix of breeding values including genomic information. This is:

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \mathbf{H}_{22} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\left(\mathbf{G} - \mathbf{A}_{22}\right)\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{G} \\ \mathbf{G}\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{G} \end{bmatrix}. \qquad [4]$$

Matrix $\mathbf{H}$ is identical to $\mathbf{A}_{\mathrm{p}}$ if all founders are genotyped, because in that case $\mathbf{A}_{12} = \mathbf{T}_1\mathbf{P}_{12}\mathbf{A}_{22}$. By construction, this matrix is semipositive or positive definite, which implies that the statistical background is sound (e.g., Harville, 1976). It is possible to come up with rules for inverting $\mathbf{H}$, in the lines of Wang et al. (1995). However, $\mathbf{H}^{-1}$ might be difficult to invert because full positive definiteness of $\mathbf{G}$ is not guaranteed and therefore their inverse (which is needed to get $\mathbf{H}^{-1}$) might not exist, or might be very ill-conditioned. Positive-definitiveness of $\mathbf{H}$ is not necessary for prediction (Harville, 1976; Henderson, 1984). Two alternative expressions for $\mathbf{H}$ that might be computationally convenient are:

$$\mathbf{H} = \begin{bmatrix} \left(\mathbf{A}^{11}\right)^{-1} + \left(\mathbf{A}^{11}\right)^{-1}\mathbf{A}^{12}\mathbf{G}\mathbf{A}^{21}\left(\mathbf{A}^{11}\right)^{-1} & -\left(\mathbf{A}^{11}\right)^{-1}\mathbf{A}^{12}\mathbf{G} \\ -\mathbf{G}\mathbf{A}^{21}\left(\mathbf{A}^{11}\right)^{-1} & \mathbf{G} \end{bmatrix} \qquad [5]$$

$$\mathbf{H} = \mathbf{A} + \begin{bmatrix} \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\left(\mathbf{G} - \mathbf{A}_{22}\right)\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\left(\mathbf{G} - \mathbf{A}_{22}\right) \\ \left(\mathbf{G} - \mathbf{A}_{22}\right)\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{G} - \mathbf{A}_{22} \end{bmatrix}$$
$$= \mathbf{A} + \begin{bmatrix} \mathbf{A}_{12}\mathbf{A}_{22}^{-1} & 0 \\ 0 & \mathbf{I} \end{bmatrix}\begin{bmatrix} \mathbf{I} \\ \mathbf{I} \end{bmatrix}\left[\mathbf{G} - \mathbf{A}_{22}\right]\begin{bmatrix} \mathbf{I} & \mathbf{I} \end{bmatrix}\begin{bmatrix} \mathbf{A}_{22}^{-1}\mathbf{A}_{21} & 0 \\ 0 & \mathbf{I} \end{bmatrix}. \qquad [6]$$

Computational Suggestions. An outline of some ideas for solving mixed model equations for big data sets will be shown here including matrix $\mathbf{H}$ (similar algorithms can be conceived for $\mathbf{A}_p$ and $\mathbf{A}_g$), whereas the companion paper by Misztal et al. (2009) gives more details and examples. Henderson (1984, 1985) gave expressions for the computation of the mixed model equations without use of the inverse of the relationship matrix. These expressions are valid for singular matrices (Harville, 1976), which might be the case for $\mathbf{G}$ as it was in our experience (unpublished). For the random effects the equation is: $\left[\mathbf{HZ}'\mathbf{R}^{-1}\mathbf{Z}+\mathbf{I}\right]\hat{\mathbf{u}} = \mathbf{W}\hat{\mathbf{u}} = \mathbf{HZ}'\mathbf{R}^{-1}\mathbf{y}$.

This equation can be solved, in methods such as preconditioned conjugated gradients, by repeatedly multiplying matrix $\mathbf{W}$ times the current guess of $\mathbf{u}$. This requires computing the product $\mathbf{Hq}$, where $\mathbf{q}$ is a vector. This is feasible using [6]. Whereas $\mathbf{G}$ is created explicitly, only $\mathbf{A}^{-1}$ can be created efficiently; $\mathbf{A}_{22}$ can be created from pedigree by computing single elements of the $\mathbf{A}$ matrix using recursive (Aguilar and Misztal, 2008) or indirect (Colleau, 2002) algorithms. For large data files, matrix $\mathbf{G}$ can be computed in parallel or even using iteration on data on genotype files. It will be assumed that $\mathbf{A}_{22}$ and $\mathbf{G}$ can be computed and stored in core. First, $\mathbf{Aq}$ can be computed by Colleau's (2002) indirect algorithm by reading twice the pedigree file without explicitly creating $\mathbf{A}$. This algorithm works by reading a pedigree twice. The other part is a product of the form $\mathbf{NQRSVq}$. This product can be computed as $\mathbf{N}(\mathbf{Q}(\mathbf{R}(\mathbf{S}(\mathbf{V}(\mathbf{q})))))$. The only difficult parts are the computations of $\mathbf{s} = \mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{t}_1$, where $\mathbf{t}_1$ is a vector of size equal to the number of ungenotyped animals, and its symmetric product of the form $\mathbf{A}_{12}\mathbf{A}_{22}^{-1}$. The product $\mathbf{p} = \mathbf{A}_{21}\mathbf{t}_1$ can be found as follows. Let be the product $\mathbf{A}\begin{bmatrix}\mathbf{t}_1\\ \mathbf{0}\end{bmatrix} = \begin{bmatrix}\mathbf{A}_{11} & \mathbf{A}_{12}\\ \mathbf{A}_{21} & \mathbf{A}_{22}\end{bmatrix}\begin{bmatrix}\mathbf{t}_1\\ \mathbf{0}\end{bmatrix} = \begin{bmatrix}\mathbf{A}_{11}\mathbf{t}_1\\ \mathbf{A}_{21}\mathbf{t}_1\end{bmatrix} = \begin{bmatrix}\mathbf{z}\\ \mathbf{y}\end{bmatrix}$, whose result $\mathbf{y}$ is needed. Now let $\mathbf{A}^*$ be the ordered relationship matrix (parents before offspring), and $\mathbf{x}$ a vector containing the reordered elements in $\mathbf{t}_1$ and zero otherwise (i.e., the values in $\mathbf{x}$ corresponding to animals in $\mathbf{A}_{22}$ are zero). Then, the product $\mathbf{A}^*\mathbf{x}$ can be computed by solving the system of equations $\mathbf{A}^{*-1}\mathbf{y}^* = \mathbf{x}$ by Colleau's algorithm and rearranging $\mathbf{y}^*$ into $\mathbf{z}$ and $\mathbf{y}$.

The product $\mathbf{s} = \mathbf{A}_{22}^{-1}\mathbf{p}$ can be computed directly if $\mathbf{A}_{22}^{-1}$ has been previously computed; or done by solving $\mathbf{A}_{22}\mathbf{s} = \mathbf{p}$ if it has not. Both operations have quadratic cost on the number of genotyped animals, say n. Even if $\mathbf{A}_{22}$ cannot be stored, solving $\mathbf{A}_{22}\mathbf{s} = \mathbf{p}$ can in principle be done by an iterative solver and repeated use of the Colleau's algorithm to compute the successive products $\mathbf{A}_{22}\mathbf{s}$. The opposite (multiplication followed by indirect algorithm) strategy can be applied in computing the product with $\mathbf{N}$. Product by $\mathbf{S}$ will involve n^2 operations. If $\mathbf{G}$ is smaller than $\mathbf{Z}$, products can be computed as $\mathbf{s} = \mathbf{Gp} = \mathbf{Z}(\mathbf{Z}'\mathbf{p})/k$ at a cost of $3nm$ (m being the number of markers). Overall, one iteration of the full algorithm involves reading the pedigree file 6 times, plus a number of operations being several times n^2 or $3nm$. For example, for 10 million animals in pedigree and $n = 10{,}000$ genotyped individuals, computing time per iteration will be roughly proportional to n^2. Thus, solving the mixed model equations may be feasible even for large pedigrees. More detailed explanations on the algorithms and preliminary studies of their performances can be found in the companion paper by Misztal et al. (2009).

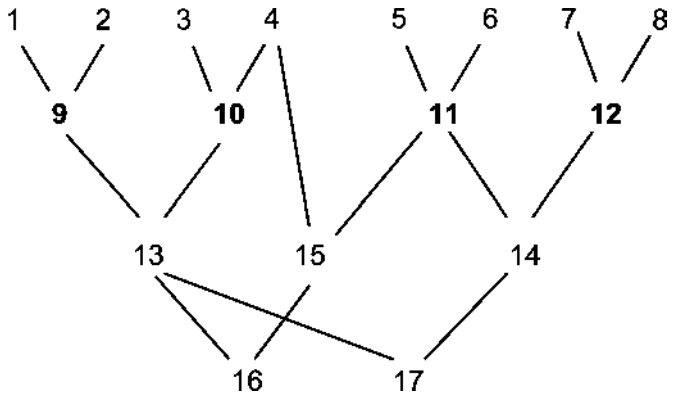


Figure 1. Example pedigree. Genotyped animals are in bold.

Example

Consider the pedigree in Figure 1. Animals 1 to 8 are unrelated founders, whereas animals 9 to 12 are genotyped. As an example, let $\mathbf{G}$ be a matrix with 1 on the diagonal and 0.7 otherwise (i.e., all animals are related although their founders are supposedly unrelated). The regular numerator relationship matrix $\mathbf{A}$ is in Table 1; only a slight modification is needed to get $\mathbf{A}_g$ (not shown). The modified $\mathbf{A}_p$, for progeny, is in Table 2, and the pedigree modified $\mathbf{H}$ is in Table 3. Even for this small example, $\mathbf{A}_p$ is indefinite, whereas $\mathbf{H}$ is positive definite.

It can be seen that in the latter, the relationships among genotyped individuals are projected backward and forward. The backward projection implies, for example, that parents of 9 and 10 are related, and 1 and 2 are not. In fact other possibilities exist (for example, that 2 and 3 were related but not 1 and 4), but the selection index gives a parsimonious solution. This is not the case in $\mathbf{A}_p$, where there is no backward projection. The nonexistence of this backward projection makes

Table 1. Numerator relationship matrix **A** for the pedigree in Figure 1[1]

1.00								0.50				0.25			0.13	0.13
	1.00							0.50				0.25			0.13	0.13
		1.00							0.50			0.25			0.13	0.13
			1.00						0.50			0.25		0.50	0.38	0.13
				1.00						0.50			0.25	0.25	0.13	0.13
					1.00					0.50			0.25	0.25	0.13	0.13
						1.00					0.50		0.25			0.13
							1.00				0.50		0.25			0.13
0.50	0.50							**1.00**				0.50			0.25	0.25
		0.50	0.50						**1.00**			0.50		0.25	0.38	0.25
				0.50	0.50					**1.00**			0.50	0.50	0.25	0.25
						0.50	0.50				**1.00**		0.50			0.25
0.25	0.25	0.25	0.25					0.50	0.50			1.00		0.13	0.56	0.50
				0.25	0.25	0.25	0.25			0.50	0.50		1.00	0.25	0.13	0.50
			0.50	0.25	0.25				0.25	0.50		0.13	0.25	1.00	0.56	0.19
0.13	0.13	0.13	0.38	0.13	0.13			0.25	0.38	0.25		0.56	0.13	0.56	1.06	0.34
0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.25	0.25	0.25	0.25	0.50	0.50	0.19	0.34	1.00

[1]Cells with 0 are empty to show the pattern. Coefficients for genotyped animals are in bold. Matrix $\mathbf{A}_g$ is obtained by setting the out-of-diagonal coefficients of genotyped animals to 0.7.

$\mathbf{A}_p$ for 1 to 12 indefinite, as the covariance structure it defines is ill-posed.

Also, in comparison to **A**, it can be seen that inbreeding coefficients appear in descendants of genotyped animals as these are related.

DISCUSSION

The system in [6] might also be expressed as if the overall genetic value was the sum of 2 different genetic values: the one in the infinitesimal model plus a difference whose covariance matrix is $\mathbf{G} - \mathbf{A}_{22}$. In the naive approach, this difference is not correctly accounted for in the relatives. If $\mathbf{G} = \mathbf{A}_{22}$ (which will not happen in practice), matrices **A** and **H** are identical as expected. Further, this shows that genetic variance in the population is the same on average (i.e., there is no artificial inflation). These are of course desirable properties.

The proposed matrix **H** is based on selection index principles or, equivalently, in assumptions of **A** being multivariate normal. Conditioning on breeding values of genotyped animals in [3] allowed us to develop a full multivariate distribution **H**. Thus, matrix **H** has been constructed from the joint density $p(\mathbf{u}_1,\mathbf{u}_2) = p(\mathbf{u}_1|\mathbf{u}_2)\, p(\mathbf{u}_2)$, where $p(\mathbf{u}_2)$ is obtained from genomic data. This distribution includes desirable aspects well known in genetic evaluation: the fact that sons inherit half their parents (as in the descendants of genotyped animals) and the notion of selection index (which is included in BLUP). So, these aspects are indeed used in the 2-step evaluation.

It is hard to envision other possibilities as it is not simple to come up with an underlying model and set up a probability distribution. For example, the "intuitive" expression $\hat{\mathbf{u}}_2 \mid \hat{\mathbf{u}}_1 = \mathbf{A}_{12}\mathbf{G}^{-1}\hat{\mathbf{u}}_1$ follows the logic of a se-

Table 2. Modified relationship matrix $\mathbf{A}_p$ including genomic information for genotyped animals and their progeny for the pedigree in Figure 1[1]

1.00								0.50				0.25			0.13	0.13
	1.00							0.50				0.25			0.13	0.13
		1.00							0.50			0.25			0.13	0.13
			1.00						0.50			0.25		0.50	0.38	0.13
				1.00						0.50			0.25	0.25	0.13	0.13
					1.00					0.50			0.25	0.25	0.13	0.13
						1.00					0.50		0.25			0.13
							1.00				0.50		0.25			0.13
0.50	0.50							**1.00**	**0.70**	**0.70**	**0.70**	0.85	0.70	0.35	0.60	0.78
		0.50	0.50					**0.70**	**1.00**	**0.70**	**0.70**	0.85	0.70	0.60	0.73	0.78
				0.50	0.50			**0.70**	**0.70**	**1.00**	**0.70**	0.70	0.85	0.50	0.60	0.78
						0.50	0.50	**0.70**	**0.70**	**0.70**	**1.00**	0.70	0.85	0.35	0.53	0.78
0.25	0.25	0.25	0.25					0.85	0.85	0.70	0.70	1.35	0.70	0.48	0.91	1.03
				0.25	0.25	0.25	0.25	0.70	0.70	0.85	0.85	0.70	1.35	0.43	0.56	1.03
			0.50	0.25	0.25			0.35	0.60	0.50	0.35	0.48	0.43	1.00	0.74	0.45
0.13	0.13	0.13	0.38	0.13	0.13			0.60	0.73	0.60	0.53	0.91	0.56	0.74	1.33	0.74
0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.78	0.78	0.78	0.78	1.03	1.03	0.45	0.74	1.53

[1]Cells with 0 are empty to show the pattern. Coefficients for genotyped animals are in bold.

Table 3. Modified relationship matrix **H** including genomic information for genotyped animals and all relatives for the pedigree in Figure 1[1]

1.00		0.18	0.18	0.18	0.18	0.18	0.18	0.50	0.35	0.35	0.35	0.43	0.35	0.26	0.34	0.39
	1.00	0.18	0.18	0.18	0.18	0.18	0.18	0.50	0.35	0.35	0.35	0.43	0.35	0.26	0.34	0.39
0.18	0.18	1.00		0.18	0.18	0.18	0.18	0.35	0.50	0.35	0.35	0.43	0.35	0.18	0.30	0.39
0.18	0.18		1.00	0.18	0.18	0.18	0.18	0.35	0.50	0.35	0.35	0.43	0.35	0.68	0.55	0.39
0.18	0.18	0.18	0.18	1.00		0.18	0.18	0.35	0.35	0.50	0.35	0.35	0.43	0.34	0.34	0.39
0.18	0.18	0.18	0.18		1.00	0.18	0.18	0.35	0.35	0.50	0.35	0.35	0.43	0.34	0.34	0.39
0.18	0.18	0.18	0.18	0.18	0.18	1.00		0.35	0.35	0.35	0.50	0.35	0.43	0.26	0.31	0.39
0.18	0.18	0.18	0.18	0.18	0.18		1.00	0.35	0.35	0.35	0.50	0.35	0.43	0.26	0.31	0.39
0.50	0.50	0.35	0.35	0.35	0.35	0.35	0.35	**1.00**	**0.70**	**0.70**	**0.70**	0.85	0.70	0.53	0.69	0.78
0.35	0.35	0.50	0.50	0.35	0.35	0.35	0.35	**0.70**	**1.00**	**0.70**	**0.70**	0.85	0.70	0.60	0.73	0.78
0.35	0.35	0.35	0.35	0.50	0.50	0.35	0.35	**0.70**	**0.70**	**1.00**	**0.70**	0.70	0.85	0.68	0.69	0.78
0.35	0.35	0.35	0.35	0.35	0.35	0.50	0.50	**0.70**	**0.70**	**0.70**	**1.00**	0.70	0.85	0.53	0.61	0.78
0.43	0.43	0.43	0.43	0.35	0.35	0.35	0.35	0.85	0.85	0.70	0.70	1.35	0.70	0.56	0.96	1.03
0.35	0.35	0.35	0.35	0.43	0.43	0.43	0.43	0.70	0.70	0.85	0.85	0.70	1.35	0.60	0.65	1.03
0.26	0.26	0.18	0.68	0.34	0.34	0.26	0.26	0.53	0.60	0.68	0.53	0.56	0.60	1.18	0.87	0.58
0.34	0.34	0.30	0.55	0.34	0.34	0.31	0.31	0.69	0.73	0.69	0.61	0.96	0.65	0.87	1.41	0.80
0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.78	0.78	0.78	0.78	1.03	1.03	0.58	0.80	1.53

[1]Cells with 0 are empty to show the pattern. Coefficients for genotyped animals are in bold.

lection index (or a multivariate normal distribution), but the covariances of $\mathbf{u}_1$ and $\mathbf{u}_2$ do not account for **G** as they should. It is not coherent to use **G** to derive $\text{Var}(\mathbf{u}_2)$ and not to derive $\text{Cov}(\mathbf{u}_1,\mathbf{u}_2)$. These covariances can be derived for descendants using the transmission vectors **P** and **T** as shown above, including **G** in the expression; however, it is more difficult to come up with a similar expression for ancestors. The selection index used as a conditional distribution overcomes this problem and accounts for **G** to generate the covariance of $\mathbf{u}_1$ and $\mathbf{u}_2$. This resulted in a parsimonious inclusion of all information (full pedigree and genomic relationships).

All of these assumptions are actually applied in the 2- or 3-step procedure for genomic selection mentioned previously, but as we discussed, information is lost by doing the steps procedure. A full relationship matrix would allow a joint evaluation and all the information would be accounted for automatically. We have also sketched how computations could be feasible in practice. Some aspects, like computation of reliabilities, deserve further research.

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Misztal, Legarra and Aguilar (2009)

> *Misztal, I., Legarra, A., and Aguilar, I. 2009. Computing procedures for genetic evaluation including phenotypic, full pedigree, and genomic information. Journal of Dairy Science, vol. 92, no. 9, pp. 4648 to 4655.*

The problem

Published in the same issue as the previous paper, this work deals with the computational side. The authors discuss how to include genomic information in evaluations that already used the animal model with millions of animals, without rebuilding the entire system. They present alternatives for solving the equations with the new relationship matrix and discuss the obstacles of each one.

Points to note while reading

Read together with the previous paper. Note the concern with keeping the structure of existing evaluations and the computational cost of each alternative. The limitations discussed here explain why the form of the inverse of H, presented the following year, was so important.

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Computing procedures for genetic evaluation including phenotypic, full pedigree, and genomic information

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ABSTRACT

Currently, genomic evaluations use multiple-step procedures, which are prone to biases and errors. A single-step procedure may be applicable when genomic predictions can be obtained by modifying the numerator relationship matrix $\mathbf{A}$ to $\mathbf{H} = \mathbf{A} + \mathbf{A}_{\Delta}$, where $\mathbf{A}_{\Delta}$ includes deviations from expected relationships. However, the traditional mixed model equations require $\mathbf{H}^{-1}$, which is usually difficult to obtain for large pedigrees. The computations with $\mathbf{H}$ are feasible when the mixed model equations are expressed in an alternate form that also applies for singular $\mathbf{H}$ and when those equations are solved by the conjugate gradient techniques. Then the only computations involving $\mathbf{H}$ are in the form of $\mathbf{Aq}$ or $\mathbf{A}_{\Delta}\mathbf{q}$, where $\mathbf{q}$ is a vector. The alternative equations have a nonsymmetric left-hand side. Computing $\mathbf{A}_{\Delta}\mathbf{q}$ is inexpensive when the number of nonzeros in $\mathbf{A}_{\Delta}$ is small, and the product $\mathbf{Aq}$ can be calculated efficiently in linear time using an indirect algorithm. Generalizations to more complicated models are proposed. The data included 10.2 million final scores on 6.2 million Holsteins and were analyzed by a repeatability model. Comparisons involved the regular and the alternative equations. The model for the second case included simulated $\mathbf{A}_{\Delta}$. Solutions were obtained by the preconditioned conjugate gradient algorithm, which works only with symmetric matrices, and by the bi-conjugate gradient stabilized algorithm, which also works with nonsymmetric matrices. The convergence rate associated with the nonsymmetric solvers was slightly better than that with the symmetric solver for the original equations, although the time per round was twice as much for the nonsymmetric solvers. The convergence rate associated with the alternative equations ranged from 2 times lower without $\mathbf{A}_{\Delta}$ to 3 times lower for the largest simulated $\mathbf{A}_{\Delta}$. When the information attributable to genomics can be expressed as modifications to the numerator relationship matrix, the proposed methodology may allow the upgrading of an existing evaluation to incorporate the genomic information.

Key words: best linear unbiased predictor, genomic selection, single nucleotide polymorphism, genetic evaluation

INTRODUCTION

Availability of dense molecular markers of type SNP led to the recent introduction of the genome-wide or genomic selection evaluation models. Those models are most often based on the simultaneous estimation of SNP marker effects $\mathbf{a}$. Differences among methods are mostly on the a priori distribution of marker effects $\mathbf{a}$ (Meuwissen et al., 2001; Gianola et al., 2006). Efficient procedures exist for the computation of $\mathbf{a}$, even for large data sets (Legarra and Misztal, 2008).

The genomic evaluation is currently implemented as a multistep procedure. For example, an implementation for US dairy cattle (VanRaden, 2008; VanRaden et al., 2009) requires 3 steps: a) regular evaluation by the animal model, b) estimation of genomic effects for a relatively small number of genotyped animals, and c) estimation of genomic breeding values by a selection index. The elements in the index include a parent average or PTA from step a), genomic solutions from step b), and a parent average or PTA computed based on genotyped ancestors. Weights in the index are functions of heritability and accuracy. The marker-assisted selection program in France simultaneously fits QTL and polygenic effects, with weights depending on associated variance components (Guillaume et al., 2008).

Advantages of the multistage procedure include no change to the regular evaluations and simple steps for predicting genomic values for young genotyped animals. Disadvantages are requirements for parameters in steps b) and c) such as prior variances and weights, and loss of accuracy and biases attributable to selection. Whereas the model in a) uses the information on all animals and can be multitrait, the model in b) is equivalent to a single-trait sire model for a highly selected set of sires. Incorrect parameters in b) and

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c) can result in unexpected changes for high-reliability bulls. Neuner et al. (2008) claimed that problems associated with the multistep procedure reduce its benefits, especially for cows.

VanRaden (2008) investigated 2 options for step b): "nonlinear," based on estimating effects attributable to SNP markers with a prior mixture distribution for those effects, and "linear," based on prior normal distribution for SNP markers. The latter is equivalent to using mixed model equations with a genomic relationship matrix. For most dairy traits, predictions based on the estimation of marker effects with nonlinear predictions were practically equivalent to linear predictions and thus to predictions with BLUP using a genomic relationship matrix (Cole et al., 2009; VanRaden et al., 2009). Therefore, using the genomic relationship matrix results in little or no loss of accuracy.

One way to simplify the multistep procedure is by incorporating the genomic information into step a), resulting in a single-step procedure. This could be accomplished by modifying the numerator relationship matrix $\mathbf{A}$ in that evaluation to include the genomic information. Such modifications are presented and discussed by Legarra et al. (2009) in a companion paper.

Assume that such a modification is known and that it involves relatively few elements of $\mathbf{A}$. The mixed model equations require $\mathbf{A}^{-1}$, which is very easy to create for large populations because of its sparsity and its special structure (Henderson, 1976). However, obtaining the inverse of the modified matrix is likely to be impossible in general for large populations. This is not only because the cost of inversion is high, but also because $\mathbf{A}$ is dense and thus too large to store for large pedigrees. Thus, an approach using a modified $\mathbf{A}$ is of little value unless a feasible computing approach is available. The purpose of this study is to develop an efficient computing strategy to obtain solutions to mixed model equations in which the numerator relationship matrix is modified by a known matrix accounting for the genomic information.

MATERIALS AND METHODS

Assume regular mixed model equations as used in a traditional genetic evaluation, for simplicity with only a single random effect:

$$\mathbf{y} = \mathbf{Xb} + \mathbf{Zu} + \mathbf{e},$$

where $\mathbf{y}$ is a vector of records, $\mathbf{b}$ is a vector of fixed effects, and $\mathbf{u}$ is a vector of animal effects. Under a polygenic infinitesimal model of inheritance, $\text{var}(\mathbf{u}) = \mathbf{A}\sigma_\alpha^2$, where $\mathbf{A}$ is the numerator relationship matrix based on pedigree. Furthermore, $\text{var}(\mathbf{e}) = \mathbf{I}\sigma_e^2$, and $\mathbf{X}$ and $\mathbf{Z}$ are appropriate incidence matrices.

Assume that the numerator relationship can be modified to account for genomic information:

$$\mathbf{H} = \mathbf{A} + \mathbf{A}_\Delta,$$

where $\mathbf{A}_\Delta$ is a matrix that can be stored explicitly, and $\mathbf{H}$ is the new modified matrix. In the simplest case, a genomic relationship matrix $\mathbf{G}$ replaces the numerator relationship matrix for the genotyped animals. Let indices 1 and 2 refer to ungenotyped and genotyped animals, respectively. Then

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix},$$

$$\mathbf{H} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{G} \end{bmatrix} = \mathbf{A} + \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{G} - \mathbf{A}_{22} \end{bmatrix},$$

and

$$\mathbf{A}_\Delta = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{G} - \mathbf{A}_{22} \end{bmatrix}.$$

Legarra et al. (2009) proposed several $\mathbf{H}$ based on the partition of animals into several groups, including ungenotyped and genotyped animals. Although their different $\mathbf{H}$ are more complex than in the simple case, most quantities can be computed efficiently without any steps involving large matrix multiplications. Therefore, for simplicity of presentations, the following computing formulas assume the simple case above.

Solving Algorithm

Assume that $\mathbf{G}$ and $\mathbf{A}_{22}$ are available. Temporarily assume that $\mathbf{H}$ is positive definite. The regular mixed model equations are

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{X} & \mathbf{Z}'\mathbf{Z} + \alpha\mathbf{H}^{-1} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{b}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{Z}'\mathbf{y} \end{bmatrix}$$

or

$$\mathbf{LHS}\ \mathbf{w} = \mathbf{RHS}$$

using the usual notation, where $\mathbf{LHS}$ and $\mathbf{RHS}$ are the left- and right-hand side, and $\mathbf{w} = \begin{bmatrix} \hat{\mathbf{b}}' & \hat{\mathbf{u}}' \end{bmatrix}'$.

Assume that the system of equations is solved using an algorithm that does not require the elements of **LHS** explicitly but only its product by a vector, say **LHS q**, as in the preconditioned conjugate gradient (**PCG**) iteration on data (Tsuruta et al., 2001). Then

$$\mathbf{LHS\ q} = \begin{bmatrix} \mathbf{X'Xq_1 + X'Zq_2} \\ \mathbf{Z'Xq_1 + Z'Zq_2 + \alpha H^{-1} q_2} \end{bmatrix} = \begin{bmatrix} \mathbf{c_1} \\ \mathbf{c_2 + c_3} \end{bmatrix},$$

with

$$\mathbf{q} = \begin{bmatrix} \mathbf{q_1} \\ \mathbf{q_2} \end{bmatrix};\ \mathbf{c_2} = \mathbf{Z'Xq_1 + Z'Zq_2};\ \mathbf{c_3} = \alpha \mathbf{H^{-1} q_2};\ \mathbf{RHS} = \begin{bmatrix} \mathbf{r_1} \\ \mathbf{r_2} \end{bmatrix}.$$

However, $\mathbf{H}^{-1}$ can be computed only for small populations; furthermore, **H** might be singular or close to singularity. Henderson (1984, 1985) and Harville (1976) described an unsymmetric set of mixed model equations in which only **H**, not necessarily of full rank, is required:

$$\begin{bmatrix} \mathbf{X'X} & \mathbf{X'Z} \\ \mathbf{HZ'X} & \mathbf{HZ'Z} + \alpha \mathbf{I} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{b}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X'y} \\ \mathbf{HZ'y} \end{bmatrix}$$

or

$$\mathbf{LHS_M w} = \mathbf{RHS_M}.$$

For that set,

$$\mathbf{LHS_M\ q} = \begin{bmatrix} \mathbf{X'Xq_1 + X'Zq_2} \\ \mathbf{HZ'Xq_1 + HZ'Zq_2 + \alpha q_2} \end{bmatrix} = \begin{bmatrix} \mathbf{c_1} \\ \mathbf{Hc_2 + \alpha q_2} \end{bmatrix} = \begin{bmatrix} \mathbf{c_1} \\ \mathbf{Ac_2 + A_\Delta c_2 + \alpha q_2} \end{bmatrix}$$

with

$$\mathbf{RHS_M} = \begin{bmatrix} \mathbf{r_1} \\ \mathbf{Hr_2} \end{bmatrix} = \begin{bmatrix} \mathbf{r_1} \\ \mathbf{Ar_2 + A_\Delta r_2} \end{bmatrix},$$

The new formulas do not include $\mathbf{H}^{-1}$ but include $\mathbf{A_\Delta c}_2$, $\mathbf{Ac}_2$, $\mathbf{A_\Delta r}_2$, and $\mathbf{Ar}_2$. For the simplistic **H**, the first term can be computed directly at a low cost. The second term can also be computed inexpensively following the algorithm by Colleau (2002; see Appendix A), which uses only the pedigree information and is completed in the amount of time proportional to the number of animals. The same algorithm also can be used to compute $\mathbf{Ar}_2$. Selected elements of **A** can be computed recursively, for example, by using the algorithm by Aguilar and Misztal (2008).

More Complicated Models

Assume a multiple trait model, possibly with effects such as random regression or maternal. The regular mixed model equations for such models can be presented as

$$\begin{bmatrix} \dots & \dots \\ \dots & \dots + \mathbf{G_{0a}^{-1}} \otimes \mathbf{H^{-1}} \end{bmatrix} \begin{bmatrix} \dots \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \dots \\ \mathbf{Z'R^{-1}y} \end{bmatrix},$$

where parts not listed (...) are due to effects other than $\hat{\mathbf{u}}$. By expanding the unsymmetric model by Henderson (1984) to multiple traits, the quantities needed for the iterations become

$$\mathbf{LHS_M q} = \begin{bmatrix} \mathbf{c_1} \\ \mathbf{G_0 \otimes Ac_2 + G_0 \otimes A_\Delta c_2 + Iq_2} \end{bmatrix}$$

with

$$\mathbf{RHS_M} = \begin{bmatrix} \mathbf{r_1} \\ \left(\mathbf{G_0 \otimes A + G_0 \otimes A_\Delta}\right)\mathbf{r_2} \end{bmatrix},$$

where quantities $\mathbf{c}_1$ and $\mathbf{r}_1$ are now associated with all effects other than the additive.

Nonsymmetric Solvers

The presented system of equations is nonsymmetric and the matrix **H** may be semipositive definite. The PCG algorithm (Barrett et al., 1994) is applicable only to symmetric systems of equations. Therefore, it is important to find a suitable conjugate-gradient type algorithm and ensure that it would converge even with a poorly conditioned **H**. Barrett et al. (1994) and Van der Vorst (2003) reviewed and presented several algorithms for solving the linear systems of equations. Based on their studies, the standard algorithm for solving sparse systems with nonsymmetric LHS is bi-conjugate gradient stabilized (**Bi-CGSTAB**; Van der Vorst, 1992; see Appendix B). This algorithm requires 2 **LHS** times a vector products per round as opposed to just one with PCG. When that product uses the majority of the computing time, Bi-CGSTAB is about twice as expensive as PCG per round of iteration.

Choice of Preconditioners

In initial tests (results not reported), Bi-CGSTAB converged very quickly with the unsymmetric equations for small models, but not for large ones. This was traced to large off-diagonal elements of the unsymmetric equations. The standard way in conjugate-gradient types

of algorithms to improve convergence is by choice of a preconditioner **M**, which approximates **LHS** but is easily invertible (Van der Vorst, 2003). Then the system of equations solved is equivalent to

$$\mathbf{M}^{-1}\mathbf{LHS}\ \mathbf{w} = \mathbf{M}^{-1}\mathbf{RHS},$$

which has better numerical properties than the original system. The preconditioner is never used explicitly, but only in multiplications with a vector.

Assuming a diagonal preconditioner,

$$\mathbf{M}^{-1} = \mathrm{diag}\left(\mathbf{LHS}\right)^{-1} = \begin{bmatrix} \mathbf{D}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_2 \end{bmatrix},$$

the **LHS** for regular equations after preconditioning is

$$\mathbf{M}^{-1}\ \mathbf{LHS} = \begin{bmatrix} \mathbf{D}_1\mathbf{X}'\mathbf{X} & \mathbf{D}_1\mathbf{X}'\mathbf{Z} \\ \mathbf{D}_2\mathbf{Z}'\mathbf{X} & \left[\mathbf{D}_2\left(\mathbf{Z}'\mathbf{Z} + \alpha\mathbf{A}^{-1}\right)\right] \end{bmatrix}.$$

The symmetry can be partially restored with a modified preconditioner:

$$\mathbf{M}_{\mathbf{M}}^{-1} = \begin{bmatrix} \mathbf{D}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_2\mathbf{A}^{-1} \end{bmatrix}.$$

Then

$$\begin{aligned} &\mathbf{M}_{\mathbf{M}}^{-1}\ \mathbf{LHS}_{\mathbf{M}} \\ &= \begin{bmatrix} \mathbf{D}_1\mathbf{X}'\mathbf{X} & \mathbf{D}_1\mathbf{X}'\mathbf{Z} \\ \mathbf{D}_2\left[\left(\mathbf{I} + \mathbf{A}^{-1}\mathbf{A}_\Delta\right)\mathbf{Z}'\mathbf{X}\right] & \mathbf{D}_2\left[\left(\mathbf{I} + \mathbf{A}^{-1}\mathbf{A}_\Delta\right)\mathbf{Z}'\mathbf{Z} + \alpha\mathbf{A}^{-1}\right] \end{bmatrix} \\ &= \mathbf{M}^{-1}\ \mathbf{LHS} + \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{D}_2\mathbf{A}^{-1}\mathbf{A}_\Delta\mathbf{Z}'\mathbf{X} & \mathbf{D}_2\mathbf{A}^{-1}\mathbf{A}_\Delta\mathbf{Z}'\mathbf{Z} \end{bmatrix}. \end{aligned}$$

When the genomic information is missing ($\mathbf{A}_\Delta = 0$), the preconditioned left-hand side of the unsymmetric system of equations is the same as with the preconditioned regular equations. With the genomic information, the off-diagonal elements are likely to be small for small $\mathbf{A}_\Delta$. The cost of the extra preconditioning is low because the product $\mathbf{D}_2\mathbf{A}^{-1}\mathbf{q}$, where $\mathbf{q}$ is a vector, can be done sequentially as $\mathbf{D}_2(\mathbf{A}^{-1}\mathbf{q})$.

Data

The data set included 10.5 million final scores on 6.2 million Holsteins as used for the recent genetic evaluation by the Holstein Association. Analyses were by a repeatability animal model. Two sets of mixed model equations were considered: regular and unsymmetric. For the second set, the genomic information was simulated for 5,000 randomly chosen animals as random numbers from the uniform distribution from 0 to b, where b was set to 0.0, 0.01, 0.03, and 0.05. For b = 0 there was no adjustment ($\mathbf{A}_\Delta = 0$). Only positive adjustments were included to avoid some elements of **H** being negative. Solving algorithms were PCG (for the regular equations only) and Bi-CGSTAB. The first algorithm used a diagonal preconditioner. The second algorithm used the modified preconditioner because no convergence was achieved with the diagonal preconditioner. In all cases, the stopping criterion was set at 10^{-12}. Computing was by the regular and modified program BLUP90IOD (Tsuruta et al., 2001) and was carried out on an Opteron system running at 3 GHz.

RESULTS AND DISCUSSION

The purpose of testing with the simulated genomic changes was to evaluate the computing feasibility of the method, and especially the robustness of the computing methodology. The results presented for the unsymmetric equations are only with the modified preconditioner. The Bi-CGSTAB diverged with the regular preconditioner and large data sets although it converged with small data sets. This is because products of **A** were very large for rows corresponding to popular bulls as all elements of **A** are positive; those products with $\mathbf{A}^{-1}$ are small because of cancellations; a contribution to a parent by a progeny in $\mathbf{A}^{-1}$ is proportional to [... 1.0 ... −0.5 ... −0.5 ...], which sums to 0.

Table 1 shows the number of rounds and computing time with PCG and Bi-CGSTAB for the regular and unsymmetric equations and with varying magnitudes of simulated changes. For the regular equations, Bi-CGSTAB was slightly faster but took twice the computing time (26 vs. 13 s). Figure 1 shows the convergence pattern for the regular equations. Whereas the pattern for PCG shows small fluctuations, the pattern for Bi-CGSTAB has more abrupt changes. Some differences in the number of rounds to convergence may be due to differences in the convergence criteria. However, the differences in solutions were very small (correlations >0.99999).

For the unsymmetric equations with no simulated changes, the number of rounds approximately doubled and the computing time increased by 30% (from 26 to 34 s). Adding small simulated changes (b = 0.01) increased the computing time per round by 10% (from 34 to 37 s) and slightly deteriorated convergence. The number of rounds increased by about 30% when changes were increased to b = 0.03 and again by 10% when

Table 1. The number of rounds (computing time per round in seconds) for different computing algorithms and different magnitudes of modification to the numerator relationship matrix

	Equation				
		Unsymmetric[2]			
Solving algorithm[1]	Regular	b = 0	b = 0.01	b = 0.03	b = 0.05
PCG	189 (13.1)	—	—	—	—
Bi-CGSTAB	166 (26.0)	318 (34.0)	369 (37.3)	477 (37.1)	520 (37.4)

[1]PCG = preconditioned conjugate gradient; Bi-CGSTAB = bi-conjugate gradient stabilized.
[2]Changes in relationships simulated from uniform (0, b) distribution for 5,000 randomly selected animals.

changes were increased by b = 0.05. Figure 2 shows the convergence pattern for the modified equations and b = 0.0 and 0.03. Much larger fluctuations than with the regular equations were observed, which may have been due to a more complex preconditioner. For a multiple-trait random regression model, Aguilar et al. (2008) observed much larger fluctuations in the convergence pattern with a block-diagonal preconditioner as compared with a diagonal one.

Additional computations will be necessary in practical applications of the method with the real genomic relationship matrix. For simple **H**, additional steps include the multiplications of $\mathbf{G}\text{-}\mathbf{A}_{22}$ and **A** by a vector. The last one can be done efficiently using the algorithm of Colleau (2002) in linear time (see Appendix A). The cost of this algorithm is equal to scanning the pedigree file twice and is small, especially with pedigrees in memory. Legarra et al. (2009) presented formulas for

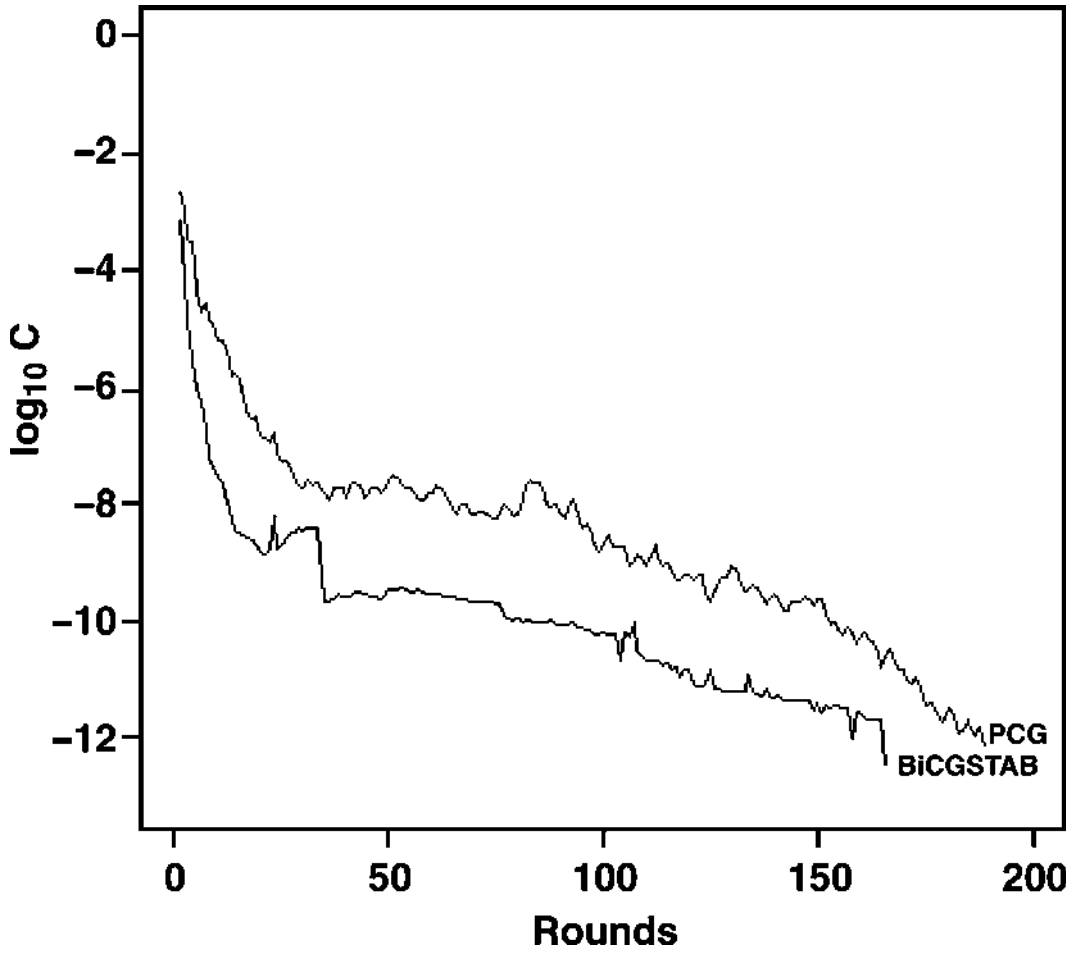


Figure 1. Convergence rate for the preconditioned conjugate gradient (PCG) and bi-conjugate gradient stabilized (Bi-CGSTAB) algorithms with the symmetric system of equations.

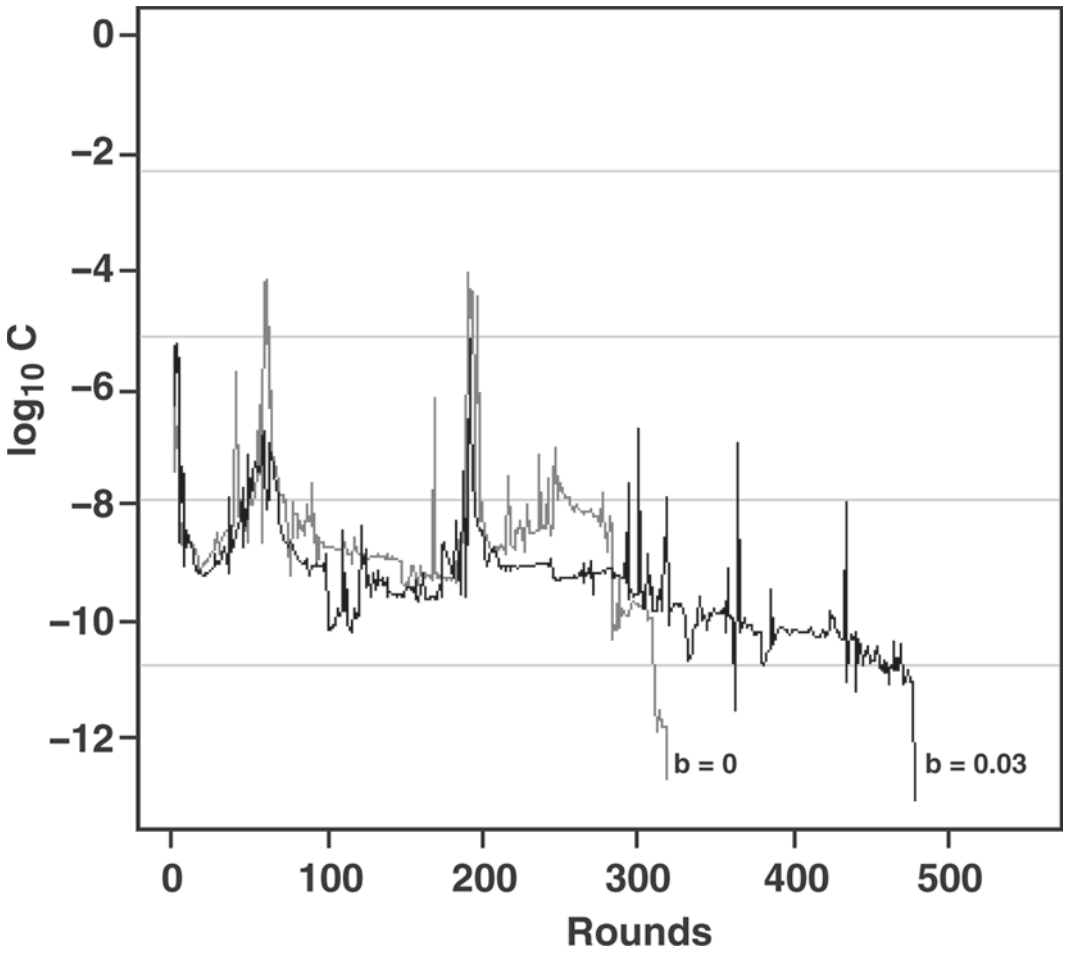


Figure 2. Convergence rate for the bi-conjugate gradient stabilized (Bi-CGSTAB) algorithm with the unsymmetric system of equations and either no (b = 0) or a middle level (b = 0.03) of simulated changes attributable to the genomic relationship matrix.

more realistic **H** and also computing details for a product of that **H** by a vector. With such a product, the only components that cannot be computed in linear but rather in quadratic time (matrix-vector multiplication) are those corresponding to **G** and possibly those due to $\mathbf{A}_{22}$. If $\mathbf{A}_{22}$ needs to be available explicitly, it can be computed by the method of Aguilar and Misztal (2008). When applied to 17 million Holsteins, that method calculated about 80,000 inbreeding coefficients/s. Assuming that computing one relationship costs no more than computing one inbreeding coefficient, on average, the computation of $\mathbf{A}_{22}$ for 20,000 animals would take 40 min. Alternatively, $\mathbf{A}_{22}$ can be computed by the repeated applications of the algorithm of Colleau (2002), in which the vector to multiply by would contain one 1.0 and zeros elsewhere.

When the number of genotyped animals is very high, say >50,000, storage and computations with matrix **G** and possibly $\mathbf{A}_{22}$ can be quite involving. A few choices may be applicable. First, some computations may easily be done in parallel. Current computers routinely include 4 processors (cores) per processor module, and computers with 4 modules are readily available. Second, some elements in $\mathbf{A}_{\Delta}$ may be very small or unimportant and could be neglected. Neglecting small elements in the computation of sparse inverse for the purpose of calculating accuracies reduced the computations by 50 times while retaining high precision (Thompson et al., 1994). Finally, genotypes of some animals may be unimportant and do not have to be included.

In summary, we have demonstrated that mixed model equations with small modifications to the numerator relationship matrix can be solved efficiently by conjugate-gradient type algorithms. Only a few modifications may be required for existing programs using the PCG algorithm.

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APPENDIX A

Below we show how to create a product $\mathbf{Aq}$, where $\mathbf{A}$ is the numerator relationship matrix and $\mathbf{q}$ is a vector. The recurrence equation for the additive effect is

$$\mathbf{a} = \mathbf{Pa} + \phi,$$

where $\mathbf{a}$ is a vector of animals ordered from oldest to youngest, ϕ is a diagonal matrix of Mendelian samplings, and $\mathbf{P}$ is a matrix relating animals to their parents; this matrix has at most 2 elements per row, both equal to 0.5. Following Quaas (1988),

$$\text{Var}(\mathbf{a}) = \mathbf{A} = (\mathbf{I} - \mathbf{P})^{-1}\mathbf{D}(\mathbf{I} - \mathbf{P})^{-1'},$$

where $\mathbf{D} = \text{var}(\phi)$. Colleau (2002) showed that the product of $\mathbf{A}$ by a vector, for example,

$$\mathbf{v} = \mathbf{Aq} = (\mathbf{I} - \mathbf{P})^{-1}\mathbf{D}(\mathbf{I} - \mathbf{P})^{-1'}\mathbf{q} = (\mathbf{I} - \mathbf{P})^{-1}\mathbf{D}[(\mathbf{I} - \mathbf{P})^{-1'}\mathbf{q}],$$

can be solved in linear time. In particular, quantities $\mathbf{r} = (\mathbf{I} - \mathbf{P})^{-1'}\mathbf{q}$ and $\mathbf{v} = (\mathbf{I} - \mathbf{P})^{-1}\mathbf{Dr}$ can be obtained by solving $(\mathbf{I} - \mathbf{P})'\mathbf{r} = \mathbf{q}$ and $(\mathbf{I} - \mathbf{P})\mathbf{v} = \mathbf{Dr}$, each one in a single sweep because $(\mathbf{I} - \mathbf{P})$ is triangular. The scalar formulas are

$$r_i = r_i + q_i;\ r_{si} = r_{si} + r_i/2;\ r_{di} = r_{di} + r_i/2;\ i = n, ..., 1$$

$$v_i = d_i r_i + (v_{si} + v_{di})/2,\ i = 1, ..., n,$$

where s_i and d_i are positions of the sire and dam of animal i, respectively.

The Colleau (2002) algorithm can be used to compute products of sections of matrices. For instance, the products below show how to compute $\mathbf{A}_{12}\mathbf{q}$, $\mathbf{A}_{22}\mathbf{q}$, $\mathbf{A}_{21}\mathbf{q}$, or $\mathbf{A}_{22}\mathbf{q}$:

$$\begin{bmatrix}\mathbf{A_{11}} & \mathbf{A_{12}}\\ \mathbf{A_{21}} & \mathbf{A_{22}}\end{bmatrix}\begin{bmatrix}\mathbf{q}\\ \mathbf{0}\end{bmatrix} = \begin{bmatrix}\mathbf{A_{11}q}\\ \mathbf{A_{21}q}\end{bmatrix},\ \begin{bmatrix}\mathbf{A_{11}} & \mathbf{A_{12}}\\ \mathbf{A_{21}} & \mathbf{A_{22}}\end{bmatrix}\begin{bmatrix}\mathbf{0}\\ \mathbf{q}\end{bmatrix} = \begin{bmatrix}\mathbf{A_{12}q}\\ \mathbf{A_{22}q}\end{bmatrix}.$$

APPENDIX B

The pseudo-program below implements the Bi-CG-STAB (Van der Vorst, 1992) for a system of equations $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{M}$ being a preconditioner. The major expenses in the algorithm are products of $\mathbf{A}$ by a vector, possibly followed by products of $\mathbf{M}^{-1}$, but only if $\mathbf{M}$ is of complex structure.

Compute $\mathbf{r}^{(0)} = \mathbf{b} - \mathbf{A}\mathbf{x}^{(0)}$ for some initial guess $\mathbf{x}^{(0)}$

Choose $\tilde{\mathbf{r}}$ (for example, $\tilde{\mathbf{r}} = \mathbf{r}^{(0)}$)

for i = 1, 2, …

$\rho_{i-1} = \tilde{\mathbf{r}}'\mathbf{r}^{(i-1)}$

if ρ_{i-1} = 0 method fails

if i = 1

$\mathbf{p}^{(i)} = \mathbf{r}^{(i-1)}$

else

$\beta_{i-1} = \left(\frac{\rho_{i-1}}{\rho_{i-2}}\right)\left(\frac{\alpha_{i-1}}{\omega_{i-1}}\right)$

$\mathbf{p}^{(i)} = \mathbf{r}^{(i-1)} + \beta_{i-1}\left(\mathbf{p}^{(i-1)} - \omega_{i-1}\boldsymbol{\upsilon}^{(i-1)}\right)$

endif

solve $\boldsymbol{M}^{-1}\hat{\mathbf{p}} = \mathbf{p}^{(i)}$

$\upsilon^{(i)} = \mathbf{A}\hat{\mathbf{p}}$

$\alpha_i = \frac{\rho_{i-1}}{\tilde{\mathbf{r}}'\boldsymbol{\upsilon}^{(1)}}$

$\mathbf{g} = \mathbf{r}^{(i-1)} - \alpha_i\boldsymbol{\upsilon}^{(i)}$

check norm of g; if small enough: set

$\mathbf{x}^{(i)} = \mathbf{x}^{(i-1)} + \alpha_i\hat{\mathbf{p}}$ and stop

solve $\mathbf{M}\hat{\mathbf{g}} = \mathbf{g}$

$\mathbf{t} = \mathbf{A}\hat{\mathbf{g}}$

$\omega_i = \frac{\mathbf{t}'\mathbf{g}}{\mathbf{t}'\mathbf{t}}$

$\mathbf{x}^{(i)} = \mathbf{x}^{(i-1)} + \alpha_i\hat{\mathbf{p}} + \omega_i\hat{\mathbf{g}}$

$\mathbf{r}^{(i)} = \mathbf{g} - \omega_i\mathbf{t}$

check for convergence; continue if necessary

for continuation it is necessary that $\omega_i \neq 0$

end

Aguilar et al. (2010)

Aguilar, I., Misztal, I., Johnson, D.L., Legarra, A., Tsuruta, S., and Lawlor, T.J. 2010. Hot topic: A unified approach to utilize phenotypic, full pedigree, and genomic information for genetic evaluation of Holstein final score. Journal of Dairy Science, vol. 93, no. 2, pp. 743 to 752.

The problem

This is the paper in which single step GBLUP was applied to real data. The authors show that the inverse of the H matrix has a very simple form. It equals the inverse of A plus a correction only in the block of genotyped animals, given by the difference between the inverse of G and the inverse of the block of A for those animals. As a result, it is enough to modify an existing animal model program. The method was tested on the evaluation of final score of Holstein cows in the United States.

Points to note while reading

Note the formula for the inverse of H and the comparison with the multiple step evaluation. Note also the blending of G with a portion of A, used to avoid numerical problems, and the questions of scale between G and A that would be addressed later.

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Hot topic: A unified approach to utilize phenotypic, full pedigree, and genomic information for genetic evaluation of Holstein final score[1]

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ABSTRACT

The first national single-step, full-information (phenotype, pedigree, and marker genotype) genetic evaluation was developed for final score of US Holsteins. Data included final scores recorded from 1955 to 2009 for 6,232,548 Holsteins cows. BovineSNP50 (Illumina, San Diego, CA) genotypes from the Cooperative Dairy DNA Repository (Beltsville, MD) were available for 6,508 bulls. Three analyses used a repeatability animal model as currently used for the national US evaluation. The first 2 analyses used final scores recorded up to 2004. The first analysis used only a pedigree-based relationship matrix. The second analysis used a relationship matrix based on both pedigree and genomic information (single-step approach). The third analysis used the complete data set and only the pedigree-based relationship matrix. The fourth analysis used predictions from the first analysis (final scores up to 2004 and only a pedigree-based relationship matrix) and prediction using a genomic based matrix to obtain genetic evaluation (multiple-step approach). Different allele frequencies were tested in construction of the genomic relationship matrix. Coefficients of determination between predictions of young bulls from parent average, single-step, and multiple-step approaches and their 2009 daughter deviations were 0.24, 0.37 to 0.41, and 0.40, respectively. The highest coefficient of determination for a single-step approach was observed when using a genomic relationship matrix with assumed allele frequencies of 0.5. Coefficients for regression of 2009 daughter deviations on parent-average, single-step, and multiple-step predictions were 0.76, 0.68 to 0.79, and 0.86, respectively, which indicated some inflation of predictions. The single-step regression coefficient could be increased up to 0.92 by scaling differences between the genomic and pedigree-based relationship matrices with little loss in accuracy of prediction. One complete evaluation took about 2 h of computing time and 2.7 gigabytes of memory. Computing times for single-step analyses were slightly longer (2%) than for pedigree-based analysis. A national single-step genetic evaluation with the pedigree relationship matrix augmented with genomic information provided genomic predictions with accuracy and bias comparable to multiple-step procedures and could account for any population or data structure. Advantages of single-step evaluations should increase in the future when animals are preselected on genotypes.
Key words: best linear unbiased predictor, genomic prediction, single nucleotide polymorphism, genetic evaluation

INTRODUCTION

Genomic evaluations are currently calculated with a multiple-step procedure (VanRaden, 2008; Hayes et al., 2009). A typical evaluation requires 1) traditional evaluation with an animal model, 2) extraction of pseudo-observations such as deregressed evaluations or daughter deviations (**DD**), 3) estimation of genomic effects for genotyped animals usually using simple sire models, and possibly 4) combining the genomic index with traditional parent averages (**PA**) and EBV (Hayes et al., 2009; VanRaden et al., 2009b). Those steps are dependent on many parameters and assumptions. For example, estimation of genomic effects has several options (Meuwissen et al., 2001; Gianola et al., 2006; VanRaden, 2008; de los Campos et al., 2009). The SNP marker effects can be estimated with different assumptions regarding the prior distribution of such effects. Genomic effects can also be estimated with a simple model that includes a genomic relationship matrix derived from genotypes and variances of the SNP marker effects (Nejati-Javaremi et al., 1997). Both methods are

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equivalent except for numerical properties (VanRaden, 2007).

Initially, genomic evaluation was tested with simulated data and a variety of assumptions (VanRaden, 2008). Experiences with actual data from dairy cattle (Hayes et al., 2009; VanRaden et al., 2009b) indicated that using a large number of markers with equal variance for all markers is appropriate for most traits. Limiting the number of SNP markers to only those with large effects resulted in reduced accuracy (Cole et al., 2009). However, little (if any) loss of accuracy occurred for most dairy cattle traits by assuming equal rather than different variance for each SNP marker (Cole et al., 2009; VanRaden et al., 2009b). Further, assuming equal variance allows the use of the same genomic relationship matrix for all traits.

Current experiences with genomic evaluations from the multiple-step procedure seem mixed. Genomic evaluations are more accurate than PA and approach the accuracy of evaluations for progeny-tested bulls, but they also seem inflated (VanRaden et al., 2009a). Although their inflation is lower than that of current PA, the potentially great utilization of top genomically evaluated young sires increases the importance of high accuracy and minimum bias. Inflation of genetic evaluations by genomic information causes top young bulls to have an unfair advantage over older progeny-tested bulls. Some of the problems with genomic evaluations may be caused by incorrect parameters and strong assumptions used in multiple-step procedures. However, effects of those parameters and assumptions are extremely difficult to verify, particularly in the presence of selection. An alternative explanation for the mixed results is that observed regressions and estimated reliabilities are biased downward by selective genotyping. A more serious problem is when pseudo-observations are poorly defined or of poor quality (e.g., for animals with small progeny numbers), which is often the case for monogastric species and for beef cattle.

Misztal et al. (2009) proposed a single-step evaluation in which the pedigree-based relationship matrix is augmented by contributions from the genomic relationship matrix. They also suggested a computing procedure based on a nonsymmetric system of mixed model equations that was suitable for millions of animals. Legarra et al. (2009) derived a joint relationship matrix based on pedigree and genomic relationships. Even though the matrix was expensive and complex to create, computations were feasible even for large data sets.

The single-step procedure provides a unified framework, eliminates several assumptions and parameters, and provides the opportunity to calculate more accurate genomic evaluations than the multiple-step procedures. The objective of this study was to use a single-step procedure for genomic evaluation in a national evaluation setting and compare its performance to a multiple-step procedure.

MATERIALS AND METHODS

Data

Data were US Holstein information for final score used for May 2009 official evaluations (Holstein Association USA, 2009). A total of 10,466,066 records were available for 6,232,548 cows. Pedigrees were available for 9,100,106 animals. Genotypes for 6,508 bulls were generated using the Illumina BovineSNP50 BeadChip (Illumina, San Diego, CA) and DNA from semen contributed by US and Canadian AI organizations to the Cooperative Dairy DNA Repository (Beltsville, MD); genotypes were provided by the Animal Improvement Programs Laboratory, Agricultural Research Service, USDA (Beltsville, MD).

Relationship Matrix with Pedigree and Genomic Information

Misztal et al. (2009) suggested that a numerator relationship matrix ($\mathbf{A}$) can be modified to a matrix ($\mathbf{H}$) that includes both pedigree-based relationships and differences between pedigree-based and genomic-based relationships ($\mathbf{A}_\Delta$): $\mathbf{H} = \mathbf{A} + \mathbf{A}_\Delta$. In their examples, they used

$$\mathbf{H} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{G} \end{bmatrix} = \mathbf{A} + \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{G} - \mathbf{A}_{22} \end{bmatrix},$$

where subscripts 1 and 2 represent ungenotyped and genotyped animals, respectively, and $\mathbf{G}$ is a genomic relationship matrix. In tests, such $\mathbf{H}$ did not work because off-diagonals of $\mathbf{H}$ were not functions of $\mathbf{G}$. Assume, for example, that no animal in $\mathbf{G}$ has records; then, according to $\mathbf{H}$, the predicted breeding value for genotyped animals ($\mathbf{u}_2$) would be $\mathbf{u}_2 \mid \mathbf{u}_1 = \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{u}_1$, where $\mathbf{u}_1$ is the predicted breeding value for ungenotyped animals, and $\mathbf{G}$ would have no role whatsoever.

Legarra et al. (2009) suggested deriving the joint density of $\mathbf{u}_1$ and $\mathbf{u}_2$ as $p(\mathbf{u}_1, \mathbf{u}_2) = p(\mathbf{u}_1 \mid \mathbf{u}_2)p(\mathbf{u}_2)$. The conditional distribution $p(\mathbf{u}_1 \mid \mathbf{u}_2)$ is based on pedigree through the selection index or multivariate normal properties; $p(\mathbf{u}_2)$ is based only on genomic information, possibly from genomic relationships. The covariance of the joint distribution of $\mathbf{u}_1$ and $\mathbf{u}_2$ is thus $\mathbf{H}$:

$$\mathbf{H} = \mathbf{A} + \begin{bmatrix} \mathbf{A}_{12}\mathbf{A}_{22}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{I} \\ \mathbf{I} \end{bmatrix} (\mathbf{G} - \mathbf{A}_{22}) \begin{bmatrix} \mathbf{I} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix},$$

which could be implemented in tests by using computing algorithms such as in Misztal et al. (2009) with only a few more computations per round of iteration than for traditional evaluations. Convergence was readily obtained for medium-sized data sets (up to 1 million); however, for larger data sets, convergence was strongly dependent on the type of **G** used.

An inverse of **H** that allows for drastically simpler computations (see Appendix A) is

$$\mathbf{H}^{-1} = \mathbf{A}^{-1} + \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{G}^{-1} - \mathbf{A}_{22}^{-1} \end{bmatrix},$$

where $\mathbf{A}_{22}^{-1}$ is the inverse of a pedigree-based relationship matrix for genotyped animals only. This expression has also been independently derived by Christensen and Lund (2009). However, the new formula introduces a small problem: **G** is usually singular and, therefore, cannot be inverted without additional steps.

Models and Analyses

A repeatability animal model was used for analysis as is currently done for US national evaluation of Holstein conformation traits (Holstein Association USA, 2009). The first 2 analyses used final scores through 2004 only. The first analysis (**Ped$_{04}$**) used only the pedigree-based relationship matrix; the second analysis (**PedGen1$_{04}$**) used relationships based on both pedigree and genomic information in a single-step approach. The third analysis (**Ped$_{09}$**) used the complete data set and only the pedigree-based relationship matrix. The fourth analysis (**PedGenM$_{04}$**) used predictions from Ped$_{04}$ and a multiple-step approach to obtain genomic predictions (**GP**) as described by VanRaden et al. (2009b). Options in the last analysis were genomic relationship matrix and base allele frequencies. Both PedGen1$_{04}$ and PedGenM$_{04}$ assumed equal variances per SNP marker effect.

"Raw" genomic relationships ($\mathbf{G}_b$) were created as

$$\mathbf{G}_b = \frac{\mathbf{ZZ}'}{k},$$

where **Z** is an incidence matrix for SNP effects with elements

$$z_{ij} = \begin{cases} 0 - 2p_j & \text{if homozygous 11} \\ 1 - 2p_j & \text{if heterozygous 12 or 21,} \\ 2 - 2p_j & \text{if homozygous 22} \end{cases}$$

for animal i and SNP j with allele frequency p_j. Several allele frequencies were used to center the matrix: 0.5, base population estimated by linear regression of gene content (Gengler et al., 2007), and current population. The scaling parameter k was defined as

$$k = 2\sum p_j(1 - p_j)$$

(VanRaden, 2008), which assumes a priori independence of SNP effects (Gianola et al., 2009).

Another scaling parameter has been proposed by Gianola et al. (2009) with

$$k = \left[(p_0 - q_0)^2 + 2\left(\frac{\sum p_j(1 - p_j)}{n} \right) \left(\frac{\alpha + \beta + 2}{\alpha + \beta} \right) \right] n,$$

where $p_0 = \alpha/(\alpha + \beta)$ is the expected allele frequency, $q_0 = (1 - p_0)$; α and β are parameters of the beta distribution fitting the base allelic frequency, and n is the number of SNP. That modification accounts for random ascertainment of SNP and their frequencies.

Matrices $\mathbf{G}_b$ were sometimes singular or close to singularity. To facilitate inversion, final analyses used a weighted **G** as proposed by VanRaden (2008): $\mathbf{G} = 0.95\mathbf{G}_b + 0.05\mathbf{A}_{22}$. The weights were not critical, and replacing them with 0.98 and 0.02 caused negligible differences.

Because GP could be scaled incorrectly, a series of analyses used $\mathbf{H}^{-1}$:

$$\mathbf{H}^{-1} = \mathbf{A}^{-1} + \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \lambda(\mathbf{G}^{-1} - \mathbf{A}_{22}^{-1}) \end{bmatrix},$$

where λ scales differences between genomic and pedigree-based information. More precisely (see Appendix B), λ sets the value of **G** in **H** to a new value ($\mathbf{G}^*$):

$$\mathbf{G}^* = \left[\lambda\mathbf{G}^{-1} + (1 - \lambda)\mathbf{A}_{22}^{-1} \right]^{-1},$$

thus blending genomic and pedigree information. For $\lambda = 1$, $\mathbf{G}^* = \mathbf{G}$; for $\lambda = 0$, $\mathbf{G}^* = \mathbf{A}_{22}$ and $\mathbf{H} = \mathbf{A}$. In fact, this corresponds to the following prior for genotyped animals:

$$p\left(u_2\right) = p\left(u_2 \mid \mathbf{G}, \lambda\right) p\left(u_2 \mid \mathbf{A}_{22}, \lambda\right)$$
$$= N\left(0, \mathbf{G} / \lambda\right) N\left(0, \mathbf{A}_{22} / \left(1-\lambda\right)\right).$$

Comparisons were based on the regressions

$$\mathrm{DD} = \mu + \delta \mathrm{EBV}_{04} + \mathrm{e}$$

and

$$\mathrm{EBV}_{09} = \mu + \delta \mathrm{EBV}_{04} + \mathrm{e},$$

where DD were deregressed evaluations (VanRaden et al., 2009b) from genotyped bulls without daughter records in 2004 but with daughter records in 2009 that were computed with complete final score data but without genomic information; EBV_{09} are breeding values based on final scores up to 2009 but without genomic information; μ is a mean; δ is a regression coefficient; EBV_{04} are breeding values based on final scores up to 2004; and e is residual error. Breeding values were calculated for 2 sets of genotyped bulls: 1) 2,575 young bulls with no daughter records in 2004 but with daughter records in 2009 and 2) 3,933 evaluated bulls with daughter records in 2004. The most accurate method for prediction for young bulls would have μ close to 0, δ close to 1, and R^2 as high as possible.

Both DD and EBV_{09} regressions were examined to allow more detailed comparison. Although DD computed through deregressed evaluations allow partial removal of the effect of PA, the removal is contingent on the accuracy of approximate reliabilities. Also, the goal of GP is not to predict DD but to predict future breeding values.

Software

Initial software for the construction of **G** and the multiple-step evaluation was provided by P. M. VanRaden (Animal Improvement Programs Laboratory, ARS, USDA, Beltsville, MD). Additional software for creating **G** was contributed by B. J. Hayes (Biosciences Research Division, Department of Primary Industries Victoria, Bundoora, Australia). Software refinement included rearrangements of code in Fortran 95 for efficient matrix multiplication, matrix inversion, and parallelization. Computation of $\mathbf{A}_{22}$ followed the formulas of Misztal et al. (2009), which used the algorithm of Colleau (2002). Genetic evaluation was performed by modified BLUP90IOD (Tsuruta et al., 2001; Misztal et al., 2002), which uses iteration on data with the preconditioned conjugate gradient algorithm.

Table 1. Coefficients of determination (R^2) and coefficients (δ) for regression of 2009 daughter deviations (DD) or corresponding estimated breeding values (EBV_{09}) for bulls progeny tested from 2005 through 2009 on 2004 predictions obtained by different algorithms

	DD		EBV_{09}	
Prediction method	R^2 (%)	δ	R^2 (%)	δ
Parent average	24	0.76	36	0.79
Multiple-step	40	0.86	50	0.82
Single-step[1]				
G5	41	0.76	49	0.70
GB	38	0.68	45	0.63
GC	37	0.71	45	0.66
GG – G5	41	0.79	50	0.73
GG – GB	38	0.77	46	0.71
GG – GC	39	0.79	46	0.73

[1]Assumed allele frequency of 0.5 (G5), base population (GB), current population (GC), or calculated as in [30] of Gianola et al. (2009) (GG).

RESULTS AND DISCUSSION

Precomputation of **G** and $\mathbf{A}_{22}$ took 650 s and 45 s, respectively, on an Opteron 64-bit processor with a clock speed of 3.02 GHz and a cache size of 1 MB, using one processor; their inversion took approximately 150 s. Time per 1 preconditioned conjugate gradient round for $\mathrm{PedGen1}_{04}$ was 13 s, which was 2% greater than 1 round for Ped_{04}. Convergence rates (not shown) for $\mathrm{PedGen1}_{04}$ and Ped_{04} were almost identical. A complete analysis with $\mathrm{PedGen1}_{04}$ took approximately 2 h. Memory requirement for precomputation of **G** was 2.7 GB.

Table 1 shows R^2 and δ for regression of 2009 DD and corresponding EBV_{09} on various 2004 predictions for young bulls. For PA, R^2 was 24% with δ of 0.76. The δ showed that PA overestimated the genetic evaluation with progeny included by 27%. For the multiple-step approach, R^2 increased to 40% and δ to 0.86. The increase in R^2 of 16% compared with PA R^2 was slightly higher than the increase of 13% reported by VanRaden et al. (2009b). VanRaden et al. (2009a) reported a regression coefficient of 0.74. Differences from the results of VanRaden et al. (2009a,b) were caused in part by slightly different data (theirs included Canadian evaluations but fewer genotypes and US records) and methodology details (e.g., different computation of approximate reliabilities).

For the single-step approaches (Table 1), R^2 for DD varied between 37 and 41%, and δ varied between 0.68 and 0.79 depending on **G**. The highest single-step increase in R^2 over prediction from PA was 1% higher than the multiple-step increase, which indicated that single-step breeding values were slightly more accurate than those by the multiple-step as implemented here.

Table 2. Coefficients of determination (R^2) and coefficients (δ) for regression of 2009 daughter deviations (DD) or corresponding breeding values (EBV_{09}) for bulls progeny tested from 2005 through 2009 on 2004 predictions from a single-step approach using an allele frequency of 0.5 and different relative variances for the genomic matrix (λ)

	DD		EBV_{09}	
λ	R^2 (%)	δ	R^2 (%)	δ
1.0	41	0.76	49	0.70
0.9	41	0.81	50	0.76
0.8	41	0.84	51	0.79
0.7	40	0.88	51	0.83
0.6	40	0.90	50	0.85
0.5	39	0.92	50	0.88
0.3	35	0.91	47	0.89

The best δ was 0.07 lower than the multiple-step δ, which indicated greater inflation of prediction for young bulls. The highest single-step R^2 and δ (least inflation) were for **G** based on equal allele frequencies with extra benefits from modifications by Gianola et al. (2009). For simplification, subsequent comparisons used the equal allele frequency **G** but without the modifications.

The R^2 values obtained using **G** matrix with equal allele frequency was greater compared with a **G** matrix created using base allele frequency. This was in the opposite direction to a similar study (VanRaden et al., 2008). In addition, the latter study reported correlations of 0.6 between genomic- and pedigree-based inbreeding coefficients, whereas a correlation of 0.2 using base allele frequencies was found in the current study. Further analyses need to be done to address such differences.

Results for EBV_{09} (Table 1) generally were similar to those for DD but with a slight advantage for the multiple-step approach. The δ indicated much greater inflation than for DD. Inflation on the EBV_{09} scale is important for producers because their comparisons are based on EBV and not on DD. It is debatable whether the results with EBV_{09} are valid in this case, because they contain information from PA. On the other hand, DD computed using approximated reliabilities may contain an extra noise.

Parent average was, in general, similar for runs with and without G. Thus, inflation higher than that in PA could be caused by too much indirect weight on genomic relationships. Inflation could be lowered by weighting $(\mathbf{G}^{-1} - \mathbf{A}_{22}^{-1})$ by λ (see Appendix B). Table 2 shows R^2 and δ for DD and EBV_{09} with such a weighting. As λ decreased from 1.0 to 0.5, R^2 gradually decreased for DD but had an interim maximum for EBV_{09}. At $\lambda = 0.7$, EBV_{09} R^2 increased to 51%, which was 1% better than for the multiple-step approach (Table 1); δ was also higher than for the multiple-step approach by 0.01. The δ can be increased to 0.92 with only a slight decrease in R^2. Because the primary interest of breeders is to identify animals with the highest genetic merit, a moderate reduction in bias (i.e., higher δ) would be preferred to a small increase in overall accuracy (R^2).

Accuracy of the single-step approach was dependent on the choice of **G** and the weighting placed on the difference between **G** and **A**. With the proper choice, accuracy of the single-step approach was superior to that of the multiple-step approach. One reason why the choice of **G** is critical is that genomic and pedigree relationship matrices should be compatible in both scale and structure. The importance of structure can be seen from the decomposition of the genomic breeding value in Appendix B. The weight of PA relative to genomic information depends on λ and even more on diagonals of $\mathbf{G}^{-1}$ and $\mathbf{A}_{22}^{-1}$. In general, the diagonal of $\mathbf{G}^{-1}$ depends on the genomic relationships and measures the amount of information provided to individual i by other animals.

The primary influence of the weighting factor (λ) appears to be related to the proportion of the additive variance explained by the genomic information (Appendix B). Snelling et al. (2009) found that different numbers of SNP genotypes used for the construction of **G** resulted in different decomposition of the additive variance between the genomic and polygenic effects. Genomic information from the best genotyped bulls would add relationship information for several animals and most likely result in higher additive variance. The Canadian official genomic evaluation system for Holsteins (Van Doormaal et al., 2009) assumes that only 80% of the additive variance is explained by the SNP information. Other factors behind the weighting factor may be related to final score as a trait in US Holsteins. For example, heritability based on records of grade animals is lower than with records on registered animals (Koduru, 2006). Other issues are preferential treatment of bull dams and the nature of final score, for which the definition changes over time (Tsuruta et al., 2005). Future studies with more traits and species will clarify the influence of the weighting factor as well as alternative weighting factors. Although our decomposition between the genomic and polygenic effects involved inverses of the respective matrices, it can also be done on the direct scale, by assuming that only part of the genetic variance is explained by the genomic information (Christensen and Lund, 2009).

What **G** should be is still undetermined. As implemented for this study, **G** was constructed so that linear effects were assumed for SNP genotypes while also collecting information about realized relationships (Van Raden, 2008). Other alternatives exist. For example, matrix **K** in González-Recio et al. (2008) included a similarity index across genotypes. Probabilities for

identity by descent can also be used and averaged across loci (Villanueva et al., 2005).

Use of regression coefficients to measure bias was described by Reverter et al. (1994) and forms the basis of Method R estimation of variance components. However, the use of δ to calibrate GP might be problematic. First, it relies on the same set of equations being used for old and recent evaluations, which was not true for this study; the "old" evaluation ($PedGen1_{04}$) used **H**, whereas the "recent" evaluation (Ped_{09}) used **A**. Second, as seen by experience with Method R, the estimated regression coefficient has large error and might be biased, especially with selection (Cantet et al., 2000; Schenkel and Schaeffer, 2000). On the other hand, little bias and very efficient computations were reported by Druet et al. (2001), who traced the bias to the use of fixed effects estimated from subsets of the data.

For this study, **G** was constructed with equal variances assumed for SNP marker effects. When variances are not equal; for example, as in Bayes-A or Bayes-B (Meuwissen et al., 2001), an equivalent **G** can be constructed by scaling contributions from different markers. Such construction requires precomputing those variances based on genotyped individuals and pseudo-data.

The generalization of the single-step approach to multiple traits is obvious when **G** is identical for each trait. However, separate **G** matrices for each trait may require single-trait analyses. For several traits, the benefits and simplicity of multiple-trait analysis using the same **G** may overcome the loss of accuracy from using less than the optimal **G** for each trait.

The single-step approach to evaluation as described in this study is easy to implement just by modifying the relationship matrix for current evaluations. Aside from simplification of genomic evaluation, the procedure is expected to improve evaluations for all ungenotyped animals. Updated PA and PTA for descendants of genotyped animals are possible using multiple-step methods with additional calculations (see http://aipl.arsusda.gov/reference/changes/eval0901.html). Advantages of single-step evaluations should increase in the future when animals have been pre-selected on genotypes. Traditional evaluations expect that Mendelian sampling averages zero, but in the future only animals with positive Mendelian sampling may receive phenotypes.

A substantial part of any current genomic selection is validation for young animals. In contrast, in BLUP based on pedigree information, such a validation is rarely performed and is implicitly replaced by variance component estimation, although some validation is performed indirectly for analyses used by Interbull multiple-trait across-country (MACE) evaluations (Interbull, 2001). With some assumptions, it is possible that the parameters of a single-step procedure are regular variance components plus weighting factors, either as proposed in this study or different. In such a case, the validation steps can be replaced by parameter estimation, greatly simplifying the use of the genomic information. Ways to estimate values of weighting factors by REML, Markov chain Monte Carlo, or other methods remain to be investigated.

CONCLUSIONS

Full genomic and pedigree evaluations by the single-step approach were as good as those obtained with the multiple-step approach in terms of accuracy and bias. Generalization for complex data structures or more complicated models are straightforward. Additional computational cost was small relative to pedigree evaluation. The highest accuracy was obtained with a scaled genomic relationship matrix created under the assumption of equal allele frequencies. The main advantages of the single-step approach are its simplicity and automatic weights for the various sources of information for the overall breeding value. Moreover, advantages of single-step evaluations should increase in the future when animals are preselected on genotypes.

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APPENDIX A

Let the inverse of the numerator relationship matrix ($\mathbf{A}$) be:

$$\mathbf{A}^{-1} = \begin{bmatrix} \mathbf{A}^{11} & \mathbf{A}^{12} \\ \mathbf{A}^{21} & \mathbf{A}^{22} \end{bmatrix},$$

where animals are partitioned into 2 groups with group 2 denoting genotyped animals.

To derive an inverse for the combined relationship matrix of Legarra et al. (2009), using the properties of the inverse of partitioned matrix, useful identities from $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ are

$$\mathbf{A}^{11}\mathbf{A}_{11} + \mathbf{A}^{12}\mathbf{A}_{21} = \mathbf{I}, \quad [A1]$$

$$\mathbf{A}^{21}\mathbf{A}_{12} + \mathbf{A}^{22}\mathbf{A}_{22} = \mathbf{I}, \quad [A2]$$

$$\mathbf{A}^{11}\mathbf{A}_{12} + \mathbf{A}^{12}\mathbf{A}_{22} = \mathbf{0}, \quad [A3]$$

$$\mathbf{A}^{21}\mathbf{A}_{11} + \mathbf{A}^{22}\mathbf{A}_{21} = \mathbf{0}, \text{ and} \quad [A4]$$

$$\left(\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}\right)^{-1} = \mathbf{A}^{11}. \quad [A5]$$

Using [A1] through [A4] and multiplying the whole-population matrix

$$\mathbf{H} = \mathbf{A} + \begin{bmatrix} \mathbf{A}_{12}\mathbf{A}_{22}^{-1}(\mathbf{G} - \mathbf{A}_{22})\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{A}_{12}\mathbf{A}_{22}^{-1}(\mathbf{G} - \mathbf{A}_{22}) \\ (\mathbf{G} - \mathbf{A}_{22})\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{G} - \mathbf{A}_{22} \end{bmatrix}$$

by

$$\mathbf{H}^{-1} = \mathbf{A}^{-1} + \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{G}^{-1} - \mathbf{A}_{22}^{-1} \end{bmatrix}$$

gives $\mathbf{H}^{-1}\mathbf{H} = \mathbf{I}$.

A direct approach to getting $\mathbf{H}^{-1}$ comes from the distribution function. Based on the conditional distribution

$$\mathbf{u}_1 \mid \mathbf{u}_2 \sim N\left(\mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{u}_2,\ \mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}\right)$$

and [A1] through [A5], the full distribution can be written as

$$\begin{aligned} p(\mathbf{u}_1, \mathbf{u}_2) &= p(\mathbf{u}_1, \mathbf{u}_2 \mid \mathbf{u}_2)p(\mathbf{u}_2) \\ &= p(\mathbf{u}_1 \mid \mathbf{u}_2)p(\mathbf{u}_2) \\ &\propto \exp[-0.5(\mathbf{u}_1 - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{u}_2)'\mathbf{A}^{11}(\mathbf{u}_1 - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{u}_2)]\exp[-0.5\mathbf{u}_2'\mathbf{G}^{-1}\mathbf{u}_2] \quad [A6] \\ &= \exp\left(-0.5\begin{bmatrix}\mathbf{u}_1' & \mathbf{u}_2'\end{bmatrix}\begin{bmatrix} \mathbf{A}^{11} & -\mathbf{A}^{11}\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{A}^{11} & \mathbf{G}^{-1} + \mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{A}^{11}\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \end{bmatrix}\begin{bmatrix}\mathbf{u}_1 \\ \mathbf{u}_2\end{bmatrix}\right) \\ &= \exp\left(-0.5\begin{bmatrix}\mathbf{u}_1' & \mathbf{u}_2'\end{bmatrix}\begin{bmatrix} \mathbf{A}^{11} & \mathbf{A}^{12} \\ \mathbf{A}^{21} & \mathbf{G}^{-1} + \mathbf{A}^{22} - \mathbf{A}_{22}^{-1} \end{bmatrix}\begin{bmatrix}\mathbf{u}_1 \\ \mathbf{u}_2\end{bmatrix}\right). \end{aligned}$$

The matrix in [A6] is the inverse of the variance matrix of the full distribution. Therefore,

$$\mathbf{H}^{-1} = \mathbf{A}^{-1} + \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{G}^{-1} - \mathbf{A}_{22}^{-1} \end{bmatrix}.$$

APPENDIX B

To illustrate the role of λ and decomposition of joint predictions in PA, genomic prediction (GP), and pedigree prediction from the subset of genotyped relatives (PP_{22}), consider $\mathbf{H}^{-1}$ after including λ:

$$\mathbf{H}^{-1} = \begin{bmatrix} \mathbf{A}^{11} & \mathbf{A}^{12} \\ \mathbf{A}^{21} & \mathbf{A}^{22} + \lambda(\mathbf{G}^{-1} + \mathbf{A}_{22}^{-1}) \end{bmatrix}. \quad [B1]$$

Denote $\mathbf{H}^{-1}$ as $\{h^{ij}\}$, $\mathbf{A}^{-1}$ as $\{a^{ij}\}$, $\mathbf{G}^{-1}$ as $\{g^{ij}\}$, and $\mathbf{A}_{22}^{-1}$ as $\{a_{22}^{ij}\}$. Consider the equation for breeding value u_i of individual i without records or progeny, in the spirit of VanRaden and Wiggans (1991); k indicates genotyped individuals (in $\mathbf{A}_{22}$), and j indicates all individuals (in $\mathbf{A}$):

$$\sum_j h^{ij} u_j = 0 \text{ and}$$

$$\lambda \sum_k g^{ik} u_k + (1-\lambda) \sum_k a_{22}^{ik} u_k + \sum_j a^{ij} u_j - \sum_k a_{22}^{ik} u_k = \lambda \sum_k g^{ik} u_k - \lambda \sum_k a_{22}^{ik} u_k + \sum_j a^{ij} u_j = 0.$$

Thus, for $\lambda = 0$, only contributions from pedigree relationships remain.

Consider more specifically young animal i without records or progeny. The equation with inbreeding ignored is

$$-u_s - u_d + 2u_i + \lambda \sum_j (g^{ij} - a_{22}^{ij}) u_j = 0,$$

where s and d correspond to sire and dam, respectively. Then,

$$\begin{aligned} u_i &= \frac{u_s + u_d + \lambda \sum_{j,j \neq i} (a_{22}^{ij} - g^{ij}) u_j}{2 + \lambda(g^{ii} - a_{22}^{ii})} \\ &= \left(\frac{u_s + u_d}{2} \right) \left(\frac{2}{2 + \lambda(g^{ii} - a_{22}^{ii})} \right) + \left(\frac{\lambda \sum_{j,j \neq i} a_{22}^{ij} u_j}{2 + \lambda(g^{ii} - a_{22}^{ii})} \right) - \left(\frac{\lambda \sum_{j,j \neq i} g^{ij} u_j}{2 + \lambda(g^{ii} - a_{22}^{ii})} \right) \\ &= 2(w) PA + \lambda(w) g^{ii} GP - \lambda(w) a_{22}^{ii} PP_{22}, \end{aligned} \quad [B2]$$

where

$$PA = \left(\frac{u_s + u_d}{2} \right),$$

$$GP = \frac{-\sum_{j,j \neq i} g^{ij} u^j}{g^{ii}}, \text{ and}$$

$$PP_{22} = \frac{-\sum_{j,j\neq i} a_{22}^{ij} u^j}{a_{22}^{ii}};$$

that is, parent average, genomic prediction, and subset pedigree prediction, with weights summing to 1. These are the same sources of information as in VanRaden et al. (2009b) except that they are estimated jointly. Note that PP_{22} might be different from PA because 1) both parents might not be genotyped and 2) only the subset of genotyped animals is considered if PP_{22} is computed independently (as in PedGenM$_{04}$). If $\lambda = 0$, only PA remains; if $\lambda = 1$, then weighting of the 3 sources of information depends on the elements $a^{ii} = 2$, g^{ii}, and a_{22}^{ii}, which measure precision of the 3 information sources relative to other breeding values. That approach is similar to the reliabilities used to combine the 3 information sources in VanRaden et al. (2009b).

Christensen and Lund (2010)

Christensen, O.F., and Lund, M.S. 2010. Genomic prediction when some animals are not genotyped. Genetics Selection Evolution, vol. 42, article 2.

The problem

At almost the same time as the University of Georgia group, Christensen and Lund independently arrived at the same combined relationship matrix and the same simple inverse. Their route was different. They started from a model for the genotypes and integrated over the unknown genotypes of the nongenotyped animals. They also discussed how to adjust G so that it is compatible with A.

Points to note while reading

Note how two different derivations lead to the same result, which reinforced the validity of the method. The concern with compatibility between G and A, that is, with the fact that the two matrices may have different bases, remains a central topic in single step.

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Genomic prediction when some animals are not genotyped

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Abstract

Background: The use of genomic selection in breeding programs may increase the rate of genetic improvement, reduce the generation time, and provide higher accuracy of estimated breeding values (EBVs). A number of different methods have been developed for genomic prediction of breeding values, but many of them assume that all animals have been genotyped. In practice, not all animals are genotyped, and the methods have to be adapted to this situation.

Results: In this paper we provide an extension of a linear mixed model method for genomic prediction to the situation with non-genotyped animals. The model specifies that a breeding value is the sum of a genomic and a polygenic genetic random effect, where genomic genetic random effects are correlated with a genomic relationship matrix constructed from markers and the polygenic genetic random effects are correlated with the usual relationship matrix. The extension of the model to non-genotyped animals is made by using the pedigree to derive an extension of the genomic relationship matrix to non-genotyped animals. As a result, in the extended model the estimated breeding values are obtained by blending the information used to compute traditional EBVs and the information used to compute purely genomic EBVs. Parameters in the model are estimated using average information REML and estimated breeding values are best linear unbiased predictions (BLUPs). The method is illustrated using a simulated data set.

Conclusions: The extension of the method to non-genotyped animals presented in this paper makes it possible to integrate all the genomic, pedigree and phenotype information into a one-step procedure for genomic prediction. Such a one-step procedure results in more accurate estimated breeding values and has the potential to become the standard tool for genomic prediction of breeding values in future practical evaluations in pig and cattle breeding.

Background

Genomic selection [1] has become the new paradigm in animal breeding programs using marker-assisted selection. It may increase the rate of genetic improvement, reduce the generation time, and provide higher accuracy of estimated breeding values (EBVs). Genomic prediction of breeding values can be based on a linear mixed model using matrix computations or a non-linear mixture type of model using Markov chain Monte Carlo (McMC) procedures. In this paper we provide a natural extension of a linear mixed model to the situation with non-genotyped animals.

A marker-based relationship matrix has been used by a number of authors, in particular VanRaden in [2] and [3], but also Gianola and van Kamm [4] in a dual formulation of their model. The types of genomic relationship matrices studied here are on the form

$$G(m) = (m - p)h(m - p)^{\mathrm{T}}, \quad (1)$$

as in VanRaden [3], but other types of genomic relationship matrices are discussed in the discussion section. In VanRaden [3] it is assumed that all animals are genotyped, which is unlikely to be a common scenario. In particular, in pig breeding it is probable that only boars or other selection candidates are genotyped, and in cattle breeding, traits being recorded for millions of animals it is very unlikely that all will be genotyped. We present an

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extension of matrix (1) in the situation where not all animals are genotyped. The approach presented here combines the relationship matrix (1) with a model for the markers. By marginalisation of the markers of non-genotyped animals a natural extension of (1) is obtained. The resulting extension of the genomic relationship matrix is the same as the one derived in Legarra et al. [5], but the details in the derivation are somewhat different and the derivation therefore sheds more light on this result.

To capture genetic variation not associated to the markers in a given SNP-panel, the model can also contain a polygenic genetic effect with the usual pedigree derived additive relationship matrix, as considered by [4,6] among others. The extension of the genomic relationship matrix to non-genotyped animals together with the addition of the polygenic effect provide a natural one-step procedure to blend the information from relatives and the genomic information into a combined genomically enhanced breeding value (GEBV). Genomic prediction with both a polygenic effect and with incomplete genotyping has been considered by a number of authors. Using a joint model for phenotypes and markers and using Bayesian inference, a general solution to sample missing markers in each McMC iteration has been suggested [4,7]. However, with a large number of SNP markers and many animals without genotypes such a solution seems computationally unfeasible in practice. In Gianola et al. [7] bivariate models are suggested, where the two traits are the traits of the genotyped and non-genotyped animals, respectively, and the genetic effect for a genotyped animal is the sum of a polygenic effect and a genomic effect whereas the genetic effect for a non-genotyped animal is just a polygenic effect (correlated with the polygenic effect of the genotyped animals). Since the model does not contain a genomic genetic effect for the non-genotyped animals, the phenotypic information from non-genotyped animals closely related to a given genotyped animal does not propagate properly into the estimate of the genomic genetic effect for this animal. Alternatively, the approach by Baruch and Weller [8] involves several steps, where first, expected genotypes are computed for non-genotyped animals, then marker effects are estimated (using expected genotypes for non-genotyped animals), phenotypes are adjusted by known or expected marker effects, and finally polygenic EBVs are computed from adjusted phenotypes. Although somewhat similar in idea to the approach taken here, the approach in [8] does not propagate any uncertainty from one step in the procedure to the next step, and the effects are not estimated simultaneously.

Methods

We assume that markers are summarised into a gene content matrix, m (m_{ij} = -1, when the SNP j of individual i is 11, m_{ij} = 0 for 12, and m_{ij} = 1 for 22), and we use capital letters M_{ij} to denote when the markers are random variables. For the genomic relationship matrix (1), the matrix p is the expectation of M, i.e. the entries in column j are $p_j = 2(\rho_j - 1/2)$ with ρ_j being the allele frequency of the second allele at loci j, and h is a diagonal matrix chosen such that $\mathrm{E}[G(M)] = A$, the usual pedigree derived additive relationship matrix. In VanRaden [3] three different genomic relationship matrices are presented, where the first two are on the form in (1), and here, we focus on the first one

$$G(m) = (m - p)(m - p)^{\mathrm{T}} / s \tag{2}$$

with $s = \sum_j 2\rho_j(1 - \rho_j)$.

The model is as follows

$$y = X\beta + Za + Zg + e, \tag{3}$$

where y is phenotype, X and Z are incidence matrices, β denotes fixed effects, e is error, $a \sim N(0, \sigma_a^2 A)$ is the polygenic genetic effect, and $g \sim N(0, \sigma_g^2 G^*(m^{obs}))$ is the genomic genetic effect. Here A is the usual pedigree derived additive relationship matrix, and $G^*(m^{obs})$ is the extension of (2) to be derived in the following section.

In the following sections, first, we derive the extension of the marker based relationship matrix, $G^*(m^{obs})$, and second, we study the variance-covariance matrix of the combined genetic effect $g + a$. Then procedures for parameter estimation using AI-REML, and breeding value estimation are presented. Finally, a simulation data set is described.

Genomic relationship matrix with a relationship of markers

Gengler et al. [9] suggested that missing genotypes could be modelled using the usual mixed model methodology with relationship matrix A. We now combine that idea with the genomic relationship matrix on the form (1). For simplicity, the derivation is made for the form (2), but it is straight-forward to generalise to (1) also.

The model for the genomic genetic effect is as follows

$$g \mid M \sim N(0, \sigma_g^2 G(M)), \quad \text{with} \quad G(M) = (M - p)(M - p)^{\mathrm{T}} / s,$$

where M is the gene content matrix. We assume that $\mathrm{E}[M_j] = 1p_j$, $\mathrm{Var}(M_j) = \nu_j A$, with A the usual relationship matrix, $\nu_j = 2\rho_j(1 - \rho_j)$, and $s = \sum_j \nu_j$. The covariances of M_j, and $M_{j'}$ for two different loci $j \neq j'$ are on the form $\mathrm{Cov}(M_j, M_{j'}) = \nu_{j,j'} A$ where the $\nu_{j,j'}$ s are unspecified since they are cancelling in the derivations that follow.

We split M into two sub-matrices containing the animals with observed genotypes and those without,

respectively,

$$M = \begin{bmatrix} M^{obs} \\ M^{miss} \end{bmatrix},$$

and in the following we distinguish between small letter m^{obs} (observed realisation of random variables M^{obs}) and capital letter M^{miss} (unobserved markers are random variables). In Appendix A, the mean vector and variance-covariance matrix of the conditional distribution $[g|m^{obs}]$ (with M^{miss} marginalised out) are shown to be

$$\mathrm{E}[g \mid m^{obs}] = 0, \quad \mathrm{Var}[g \mid m^{obs}] = \sigma_g^2 G^*(m^{obs}),$$

Where

$$G^*(m^{obs}) = \begin{bmatrix} G(m^{obs}) & G(m^{obs})A_{11}^{-1}A_{12} \\ A_{21}A_{11}^{-1}G(m^{obs}) & A_{21}A_{11}^{-1}G(m^{obs})A_{11}^{-1}A_{12} + A_{22} - A_{21}A_{11}^{-1}A_{12} \end{bmatrix}. \quad (4)$$

When all animals have been genotyped, $G^*(m^{obs}) = G(m^{obs})$, and when no animals have been genotyped, $G^*(m^{obs}) = A$, which makes the extension in (4) rather elegant. We assume that the distribution of $[g|m^{obs}]$ is multivariate normal, which for the non-genotyped animals is not strictly true, but an approximation.

The inverse of the genomic relationship matrix may be obtained from the inverse of A,

$$A^{-1} = \begin{bmatrix} A_{11}^{-1} + A_{11}^{-1}A_{12}(A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}(A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1} \\ -(A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1}A_{21}A_{11}^{-1} & (A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1} \end{bmatrix} (5)$$

Using some algebra, the inverse of the genomic relationship matrix becomes

$$\begin{aligned} &G^*(m^{obs})^{-1} \\ &= \begin{bmatrix} G(m^{obs})^{-1} + A_{11}^{-1}A_{12}(A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}(A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1} \\ -(A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1}A_{21}A_{11}^{-1} & (A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1} \end{bmatrix} \quad (6) \\ &= \begin{bmatrix} G(m^{obs})^{-1} - A_{11}^{-1} & 0 \\ 0 & 0 \end{bmatrix} + A^{-1}. \end{aligned}$$

Considering the terms in (6), because of the low dimension of $G(m^{obs})$ and A_{11} a direct inversion of these matrices should be possible for practical computations, and A^{-1} is a sparse matrix which can be computed directly without constructing A itself and using standard techniques. To compute A_{11} there might be cases where most of the A matrix has to be computed, potentially causing a memory storage problem.

Alternatively, $A_{11} = ((A^{-1})^{-1})_{11}$ may be computed using the formula (5) on A^{-1} and using sparse matrix computation. The formula (6) requires that $G(m^{obs})$ is invertible which may not actually be the case. In the next section this problem is automatically solved by combining the genomic genetic effect g with the polygenic effect a.

We also note that the determinant equals

$$\det(G^*(m^{obs})) = \det(G(m^{obs}))\det(A_{22} - A_{21}A_{11}^{-1}A_{12}),$$

where A_{22} - A_{21} $A_{11}^{-1}A_{12}$ is easily obtained from A^{-1}, and the determinant can be computed using sparse matrix computation.

The combined genetic effect

The combined genetic effect is the sum of the genomic genetic effect and the polygenic effect, $\tilde{g}$ = g + a, and using this notation the model (3) may now be written as

$$y = X\beta + Z\tilde{g} + e, \quad (7)$$

where $\tilde{g} \sim N(0, \sigma_g^2 G^*(m^{obs}) + \sigma_a^2 A)$. Introducing the notation $w = \sigma_a^2 / (\sigma_g^2 + \sigma_a^2)$ and $\sigma_{\tilde{g}}^2 = \sigma_g^2 + \sigma_a^2$, then

$$\tilde{g} \sim N(0, \sigma_g^2 \tilde{G}_w),$$

with $\tilde{G}_w$ = (1 - w)$G^*(m^{obs})$ + wA. Substituting (4) and rearranging the terms, we obtain

$$\tilde{G}_w = \begin{bmatrix} G_w & G_w A_{11}^{-1}A_{12} \\ A_{21}A_{11}^{-1}G_w & A_{21}A_{11}^{-1}G_w A_{11}^{-1}A_{12} + A_{22} - A_{21}A_{11}^{-1}A_{12} \end{bmatrix},$$

where

$$G_w = (1-w)G(m^{obs}) + wA_{11}.$$

The parameter w is interpreted as the relative weight on the polygenic effect, and it may be estimated from data as shown in the next section or be chosen to equal a small value.

Similar to the previous section the inverse equals

$$(\tilde{G}_w)^{-1} = \begin{bmatrix} G_w^{-1} - A_{11}^{-1} & 0 \\ 0 & 0 \end{bmatrix} + A^{-1}, \quad (8)$$

and here G_w is necessarily invertible when $w > 0$ (even when $G(m^{obs})$ is singular).

Variance component estimation

Here we consider parameter estimation using average information (AI)-REML based on the mixed model equations [10,11]

$$\begin{bmatrix} X^{\mathrm{T}}X & X^{\mathrm{T}}Z \\ Z^{\mathrm{T}}X & Z^{\mathrm{T}}Z + (\tilde{G}_w)^{-1}\lambda_{\tilde{g}}^2 \end{bmatrix} \begin{bmatrix} \beta \\ \tilde{g} \end{bmatrix} = \begin{bmatrix} X^{\mathrm{T}}y \\ Z^{\mathrm{T}}y \end{bmatrix}, \quad (9)$$

where $\lambda_{\tilde{g}} = (\sigma_{\tilde{g}}^2 / \sigma_e^2)^{-1}$. We will not enter into details, but just note that the sparse structure of the left hand side matrix in (9) is the cornerstone for the fast computation of the AI-matrix used in the numerical

maximisation of the REML likelihood. Considering the terms in this matrix, then $Z^{\mathrm{T}}Z$ is a sparse matrix, and from (4) we see that $\tilde{G}_w^{-1}$ has some sparse structure, although G_w^{-1} is a dense matrix. Depending on the proportion of animals genotyped it may in some cases not be necessarily advantageous to compute the AI-matrix using (9), but instead an AI-REML algorithm based on the inverse phenotypic variance-covariance matrix, $(\sigma_g^2\tilde{G}_w + \sigma_e^2 I)^{-1}$, could be used, see [12]. Here, we assume that the majority of animals are not genotyped and use the sparse structure of $G^*(m^{obs})^{-1}$ for AI-REML based on the mixed model equations.

The AI-REML method based on the mixed model equations is implemented in software DMU [13] and requires input in the form of the vector of phenotypes, the nonzero entries of $\tilde{G}_w^{-1}$ and the log-determinant log $(\det(\tilde{G}_w))$ = log$(\det(G_w))$ + log$(\det(A_{22}$ - $A_{21}A_{11}^{-1}A_{12}))$. For a given w the software provides estimates of σ_g^2 and σ_e^2, values of the REML log-likelihood at the maximum and (when required) BLUE solution $\hat{\beta}$ and BLUP solution $\hat{\tilde{g}}$. Here, the parameter w is estimated by using a grid of values, i.e. w = 0.01, 0.03, ..., 0.19, and computing the REML log-likelihood for each value. The resulting profile likelihood curve, log $\hat{L}(w)$, has a peak at the estimate $\hat{w}$, and a measure of the associated uncertainty is the interval $\{w|\log\ \hat{L}(w)\ >\ \log\ \hat{L}(\hat{w})\ -\ 3.84\}$ where 3.84 is the 95% quantile of a $\chi^2(1)$-distribution.

Breeding value estimation

Here we consider estimation (prediction) of breeding values. For animals included in the parameter estimation (animals with phenotypes, and some additional animals whose markers provide information about the unknown markers for non-genotyped animals with phenotypes), the GEBVs are the solution vector $\hat{\tilde{g}}$ to (9) with the parameter values being the estimated ones from the previous section. The software DMU provides these GEBVs and their precision.

For animals not included in the parameter estimation, then denoting this subset of animals by index 3 the GEBVs $\hat{\tilde{g}}_3$ are obtained by solving

$$\begin{bmatrix} X^{\mathrm{T}}X & X^{\mathrm{T}}Z_{all} \\ Z_{all}^{\mathrm{T}}X & Z_{all}^{\mathrm{T}}Z_{all} + (\tilde{G}_{all,w})^{-1}\lambda_{\tilde{g}}^2 \end{bmatrix} \begin{bmatrix} \beta \\ \tilde{g}_{all} \end{bmatrix} = \begin{bmatrix} X^{\mathrm{T}}y \\ Z^{\mathrm{T}}y \end{bmatrix},$$

where $\hat{\tilde{g}}_{all}^{\mathrm{T}} = (\hat{\tilde{g}}^{\mathrm{T}}, \hat{\tilde{g}}_3^{\mathrm{T}})$, Z_{all} and $\tilde{G}_{all,w}$ now contain all animals. Again software DMU provides these GEBVs and their precision.

For a scenario with a large number of genotyped animals whose marker information does not provide information for the parameter estimation, Appendix B presents a method for breeding value estimation where only part of the $\tilde{G}_{all,w}$ needs to be computed.

A simulated data set

The simulated data set is inspired by a pig nucleus breeding program, but is formulated in a simplified form. We assume, 10 chromosomes each 160 cM long, and a panel of p = 5000 equidistant SNP markers is used. It is assumed that 500 QTLs affect the phenotype, and the size of these effects is simulated from a Gamma (5.4, 0.42)-distribution. First, a base population consisting of 150 boars and 1500 sows is generated by assuming random mating for 50 generations in a population with an effective population size of 100. Then the following mating and selection scheme is followed for five generations. In each generation, 150 boars are mated with 1500 sows to produce 15000 offspring (half of them males). For the next generation, the 150 boars with the highest value of their own phenotype are selected, and 1500 sows are selected randomly. It is assumed that family records are available for all five generations, phenotypes of all boars available for all five generations (35000 records), and the selected boars in the last three generations are genotyped (450 animals). In addition, to estimate the allele frequencies required for the method, the 150 boars in the base population are genotyped (and the allele frequencies used are the estimated frequencies from these 150 boars). For prediction, it is assumed that 300 selection candidates (without phenotypes) for generation 6 are genotyped.

To evaluate the method advocated in this paper (one-step), two other methods are investigated. The first method (ped) computes traditional EBVs using the pedigree based relationship matrix (without using markers). The second method (two-step) is a two-step procedure similar to methods used in practical genomic selection [14,15] and is based on genotyped animals only using the model

$$y_{EBV} = \mu + \tilde{g} + e, \tag{10}$$

where y_{EBV} is the vector of traditional EBVs, and $\tilde{g} \sim N(0, \sigma_{\tilde{g}}^2 G_w)$ with $G_w = 0.99G(m^{obs}) + 0.01A_{11}$.

For the one-step method, the genotypes of the selection candidates provide information about the genotypes of their (non-genotyped) mothers and hence information about other non-genotyped animals further back in the pedigree. Therefore they also provide some information about the genotypes of the boars without offspring, and since these boars have phenotypes but not genotypes then the selection candidates should be included in the parameter estimation. However, to investigate how important it is to include these animals, a second analysis (one-step-2) is also performed where they are not included. Finally, to investigate the importance of obtaining the allele frequencies in the base population, the scenario where the boars in the base population

have not been genotyped is also studied. The use of three different allele frequencies are compared: 1) true allele frequencies (obtained from the 150 boars in the base population), 2) estimated allele frequencies for boars used in generation 3, 3) allele frequencies estimated using the approach by Gengler et al. [9].

Results

For the one-step method, the profile likelihood curve for w is shown in Figure 1. It is seen that the data do not support a large polygenic effect, with the estimate being about zero and the 5% confidence interval being about [0; 0.06]. For computational reasons, we decided to use $\hat{w} = 0.01$.

The parameter estimates and the correlation between GEBVs and true breeding values (BVs) are shown in Table 1. For comparison, the prediction using the pedigree based relationship matrix (ped method) and the genomic prediction using (10) based on genotyped animals (two-step) are also shown. We observe that the two methods using a marker-based relationship matrix perform better than the method using the pedigree based relationship matrix, but as expected the one-step method performs the best.

Column four in Table 1 shows the result obtained when ignoring the genotypes of the 300 selection candidates in the parameter estimation (one-step-2). Even though the parameter estimates are somewhat different betweeen one-step and one-step-2, only a minor difference in the correlation between GEBVs and the true breeding values is seen. Hence, for this data set this specific computational short-cut performs well. Finally, the results from the analyses where the boars in the base population are not genotyped show that the choice of allele frequencies is very important for parameter estimation. When using the true allele frequencies, $\hat{w} \approx 0$ is obtained, whereas when using allele frequencies estimated from the observed genotypes, $\hat{w} = 1$ is obtained for both methods estimating the allele frequencies. Since $\hat{w} = 1$ corresponds to the usual animal model, no further results from this comparison are shown here. We conclude that for this data set the parameter estimation is sensitive to the allele frequencies used in the one-step method.

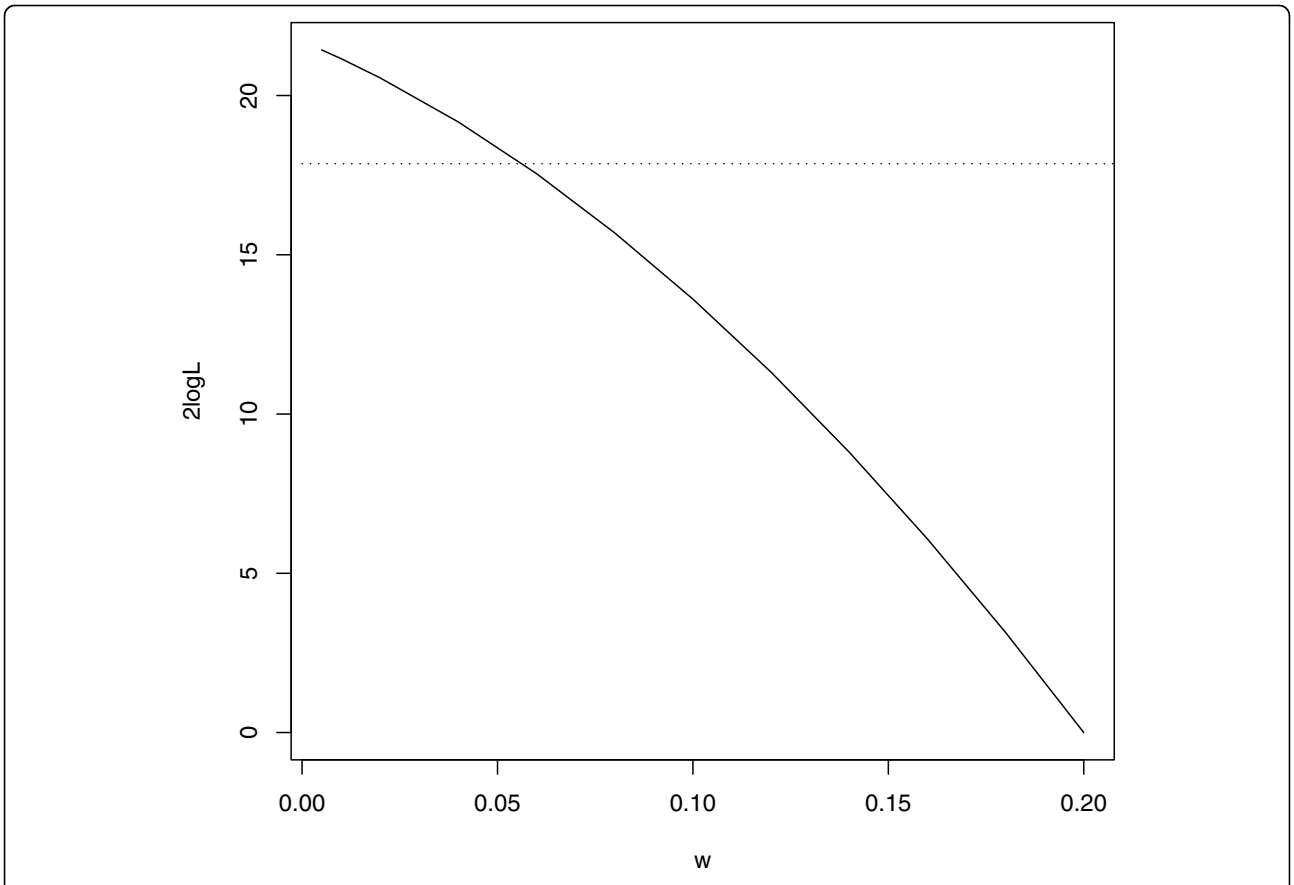


Figure 1 The profile log-likelihood curve for *w*. The dotted line corresponds to a the 95% quantile for a $\chi^2(1)$ distribution, and provides a 5% confidence interval of [0; 0.06] for w.

Table 1 Results from model with $\hat{w}$ = 0.01.

Method	$\hat{\sigma}^2_{\tilde{g}}$	$\hat{\sigma}^2_e$	Cor. true BV
one-step	4.16	16.22	0.6598
ped	5.03	15.80	0.3537
two-step	7.56	0.069	0.5869
one-step-2	5.98	15.58	0.6596

Method one-step is the method advocated in this paper, method ped uses the pedigree based relationship matrix, and method two-step is the genomic prediction method using only genotyped animals (note that parameter estimates for this method cannot be compared to parameter estimates from the other two methods). Finally, one-step-2 differs from one-step in that it ignores the markers of selection candidates in the parameter estimation. The right-most column shows the correlation between the estimated and the true breeding value (BV).

Discussion

For genomic prediction an extension of a linear mixed model to non-genotyped animals has been derived here. The extension of the method makes it possible to integrate in an optimal way the genomic, pedigree and phenotype information into a one-step procedure for breeding value estimation. Due to the simplicity of the method, the fact that it extends the traditional breeding value estimation method in a natural way, and the possibilities of handling large populations, such a one-step procedure has the potential to become the standard tool for genomic prediction of breeding values in practical pig or cattle evaluations in the future. The practical implementation of the approach uses an existing software DMU, and therefore the approach can be easily extended to other types of models implemented by that software, in particular multivariate analysis and generalised linear mixed models.

For such a one-step procedure to become the standard tool for computing GEBVs in practical pig or cattle evaluations, some technical issues of the method need further development. First, computing times necessary for the construction and the inversion of $G(m^{obs})$ are proportional to $n_1^2 p$ and n_1^3, respectively. These computations seem to be the computational bottle-necks for the method, and for a very large number of genotyped animals the method may not be feasible. Further research on efficient computation of $G(m^{obs})^{-1}$ seems necessary. Second, some computational short-cuts in the method could be imagined, as illustrated in our results by the good performance of the one-step method even when the marker information from selection candidates is ignored in the parameter estimation. Investigations by extensive simulation studies may reveal the benefits of other potential short-cuts. Third, the allele frequencies in the base population are considered known, or at least easily accessible. As illustrated in the results, the parameter estimation seems to be sensitive to the choice of these allele frequencies in a scenario with selection and where the base population itself has not been genotyped. To investigate whether the problems may be related to the strong selection on phenotype for the simulation data set, this analysis was repeated for a simulation with boars selected randomly. Here more sensible parameter estimates were obtained in the sense that $\hat{w} \approx 0$ when allele frequencies were estimated from observed genotypes. For practical dairy cattle evaluations, Misztal et al. [16] investigated the use of a number of different allele frequencies and obtained the best results by using ρ_j = 1/2 for all j but replacing $s = 2\Sigma_j \rho_j(1 - \rho_j) = p/2$ with a another scaling s which in practice was larger than $p/2$. Of course, whether that result is due to selection in this real data set is not known. Further research on the effect of selection and on how to handle appropriately the issue with allele frequencies is needed.

An assumption behind the genomic relationship matrix (2) is that all regions of the genome are equally important for the trait of interest. It is possible to instead use $G(m) \propto (m - p)h(m - p)^T$ where h is a diagonal matrix with known weights $h_{jj} = b_j^2$ with b_js being estimated SNP effects (estimated using for example a non-linear mixture type of model as in [1]). However, incorporating uncertainty on such estimated SNP effects into the method seems less straight-forward.

Considering other types of marker based relationship matrices, then

$$K(M)_{ii'} = \exp(-\sum_j (M_j^i - M_j^{i'})^2 / \phi), \quad (11)$$

with correlation parameter φ, corresponds to the method in [4] in it's dual formulation as a linear mixed model. For this choice of marker-based relationship matrix, the derivation of $K^*(m^{obs})$ = Var $[g|m^{obs}]$ is also possible, but as shown in Appendix C the form of the result differs from (4) in a number of ways. The implication is that using (4) and (6) with a marker based relationship matrix defined by (11) is possible, but lacks theoretical justification.

Appendix A

Here the mean and variances of the conditional distribution $[g \mid m^{obs}]$ (with M^{miss} marginalised out) are derived using formulas for conditional expectations, variances and covariances.

The mean vector

$$\mathrm{E}[g \mid m^{obs}] = \mathrm{E}[\mathrm{E}[g \mid m^{obs}, M^{miss}] \mid m^{obs}] = 0,$$

and the variance-covariance matrix

$$\mathrm{Var}[g \mid m^{obs}] = \mathrm{E}[Var[g \mid M^{miss}, m^{obs}] \mid m^{obs}] + \mathrm{Var}[\mathrm{E}[g \mid M^{miss}, m^{obs}] \mid \boldsymbol{m}^{obs}] = \sigma^2_{\tilde{g}} \mathrm{E}[G(M^{miss}, m^{obs}) \mid m^{obs}]$$
$$= \frac{\sigma^2_{\tilde{g}}}{s} \begin{bmatrix} (m^{obs} - p)(m^{obs} - p)^{\mathrm{T}} & (m^{obs} - p)(\mathrm{E}[M^{miss} \mid m^{obs}] - p)^{\mathrm{T}} \\ (\mathrm{E}[M^{miss} \mid m^{obs}] - p)(m^{obs} - p)^{\mathrm{T}} & (\mathrm{E}[M^{miss} \mid m^{obs}] - p)(\mathrm{E}[M^{miss} \mid m^{obs}] - p)^{\mathrm{T}} + \sum_j \mathrm{Var}[M_j^{miss} \mid m^{obs}] \end{bmatrix},$$

where

$$\mathrm{E}[M_j^{miss} \mid m^{obs}] = 1p_j + A_{21}A_{11}^{-1}(m_j^{obs} - 1p_j),$$

and

$$\mathrm{Var}[M_j^{miss} \mid m^{obs}] = \nu_j(A_{22} - A_{21}A_{11}^{-1}A_{12}),$$

with

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix},$$

and subdivision corresponding to (M^{obs}, M^{miss}). Using that $\Sigma_j \nu_j = s$, we obtain Var [$g \mid m^{obs}$] = $\sigma_g^2 G^*(m^{obs})$ where

$$G^*(m^{obs}) = \begin{bmatrix} G(m^{obs}) & G(m^{obs})A_{11}^{-1}A_{12} \\ A_{21}A_{11}^{-1}G(m^{obs}) & A_{21}A_{11}^{-1}G(m^{obs})A_{11}^{-1}A_{12} + A_{22} - A_{21}A_{11}^{-1}A_{12} \end{bmatrix}.$$

In the calculations above it is assumed that the conditional mean $\mathrm{E}[M_j^{miss} \mid m^{obs}] = \mathrm{E}[M_j^{miss} \mid m_j^{obs}]$ and the conditional variance-covariance $\mathrm{Var}[M_j^{miss} \mid m^{obs}] = \mathrm{Var}[M_j^{miss} \mid m_j^{obs}]$, and this is correct since

$$\mathrm{E}[M^{miss} \mid m^{obs}] = p + (I \otimes A_{21}A_{11}^{-1})(m^{obs} - p)$$

$$\mathrm{Var}[M^{miss} \mid m^{obs}] = V \otimes (A_{22} - A_{21}A_{11}^{-1}A_{12})$$

when Var(M) = $V \otimes A$.
In the main text we assume

$$g \mid m^{obs} \sim N(0, \sigma_g^2 G^*(m^{obs})),$$

where $G^*(m^{obs})$ is defined in (4). However, this is not strictly correct for a non-genotyped animal i where $g_i \mid X \sim N(0, X)$ with X here being a random variable with distribution $[\Sigma_j (M_{ij} - p_j)^2 | m^{obs}]$. This conditional distribution will never lead to a marginal normal distribution for g_i (the only exception is when X is a constant). The normal distribution of $g|m^{obs}$ is therefore only an approximation.

Appendix B

In some scenarios the number of genotyped animals not included in the parameter estimation may be large, for example if phenotypes are expensive to obtain and therefore only observed on a small subset of the population. To reduce the computational burden of creating the whole $G_{all}^*(m^{obs,other})$ for all animals, a procedure is presented where only a part of this matrix needs to be computed.

For genotyped animals used in the parameter estimation, let $\hat{\tilde{g}}_1$ be the corresponding sub-vector of $\hat{\tilde{g}}$. Estimated breeding values of other genotyped animals not included in the parameter estimation (denoting this subset of animals by index 3) are obtained by

$$\hat{\tilde{g}}_3 = [\tilde{G}_{w,31} \quad \tilde{G}_{w,32}](\tilde{G}_w)^{-1}\hat{\tilde{g}},$$

Where $\tilde{G}_{w,31} = (1-w)G_{31}^* + wA_{31}$, and $G_{31}^* = G_{all}^*(m^{obs}, m^{other})_{31}$ and $A_{31} = (A_{all})_{31}$ are sub-matrices of the full (containing all animals) genomic and polygenic relationship matrix, respectively. The matrices with index 32 are similarly defined. Since m^{other} does not influence M^{miss} directly,

$$\begin{bmatrix} G_{31}^* & G_{32}^* \end{bmatrix} = (1/s)(m^{other} - p)\left(\begin{bmatrix} m^{obs} \\ \mathrm{E}[M^{miss} \mid m^{obs}] \end{bmatrix} - p\right)^{\mathrm{T}} = G_{31}^*\begin{bmatrix} I & A_{11}^{-1}A_{12} \end{bmatrix}.$$

Considering the polygenic effect, then the assumption that m^{other} does not influence M^{miss} is equivalent to $A_{32} - A_{31}A_{11}^{-1}A_{12} = 0$. Using this relation we obtain

$$\begin{bmatrix} A_{31} & A_{32} \end{bmatrix} = A_{31}\begin{bmatrix} I & A_{11}^{-1}A_{12} \end{bmatrix}.$$

Hence,

$$\begin{bmatrix} \tilde{G}_{w,31} & \tilde{G}_{w,32} \end{bmatrix} = \tilde{G}_{w,31}\begin{bmatrix} I & A_{11}^{-1}A_{12} \end{bmatrix},$$

and therefore by using (8) and (5) the following form is obtained

$$\hat{\tilde{g}}_3 = \tilde{G}_{w,31}\begin{bmatrix} I & A_{11}^{-1}A_{12} \end{bmatrix}(\tilde{G}_w)^{-1}\hat{\tilde{g}} = \tilde{G}_{w,31}\begin{bmatrix} (G_w)^{-1} & 0 \end{bmatrix}\hat{\tilde{g}} = \tilde{G}_{\boldsymbol{w},31}(G_w)^{-1}\hat{\tilde{g}}_1. \quad (12)$$

This shows that the GEBVs of such genotyped animals only depend on $\hat{\tilde{g}}_1$. It also shows that only a part of the full genomic relationship matrix for genotyped animals is necessary to compute, since $G_{w,33} = (1 - w)G(m^{other}) + wA_{33}$ does not enter into (12).

In some cases the matrix A_{31} may be prohibitive to compute directly due to a large number of animals. In such a case, $\hat{\tilde{g}}_3 = (1-w)\hat{g}_3 + w\hat{a}_3$, where $\hat{g}_3 = G_{31}^*(G_w)^{-1}\hat{\tilde{g}}_1$ is computed directly and $\hat{a}_3 = A_{31}(G_w)^{-1}\tilde{g}_1$ may be obtained as the solution to the sparse system of equations

$$(A_{all})^{-1}\begin{bmatrix} \bar{a}_1 \\ \bar{a}_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} (G_w)^{-1}\tilde{g}_1 \\ 0 \\ 0 \end{bmatrix},$$

where $(A_{all})^{-1}$ is sparse and is computed directly, and $\bar{a}_1$ and $\bar{a}_2$ are dummy variables.

Appendix C

Here follows the derivation of the extension of the marker-based relationship matrix

$$K(M)_{ii'} = \exp(-\sum_j (M_j^i - M_j^{i'})^2 / \phi),$$

to non-genotyped animals.

The extension of the genomic relationship matrix is

$$K^*(m^{obs}) = \mathrm{Var}[g \mid m^{obs}] = \mathrm{E}[Var[g \mid M^{miss}, m^{obs}] \mid m^{obs}] + \mathrm{Var}[\mathrm{E}[g \mid M^{miss}, m^{obs}] \mid m^{obs}]$$
$$= \mathrm{E}[K(M^{miss}, m^{obs}) \mid m^{obs}] + 0 = \mathrm{E}[K(M^{miss}, m^{obs}) \mid m^{obs}].$$

As written in the discussion, the form of this matrix differs from (4) in a number of ways. First, all diagonal elements $K^*(m^{obs})_{ii} = 1$, and hence $K^*(m^{obs})$ does not simplify to the A matrix when no animals are genotyped. Second, the resulting matrix depends on the off-diagonal elements $v_{jj'}$ of V, since for non-genotyped animals i and i' the derivation

$$\mathrm{E}[K(m^{miss}, m^{obs}) \mid m^{obs}]_{ii'} = \prod_j \mathrm{E}[\exp(-M_j^i - M_j^{i'})^2 / \phi \mid m^{obs}]$$

requires that $M^1,\ldots, M^p$ are statistically independent (implying that V is a diagonal matrix). Third, the conditional expectation $\mathrm{E}[\exp(-M_j^i - M_j^{i'})^2 / \phi) \mid m^{obs}]$ depends on the distributional assumptions of the model for M, not just first and second moments. Fourth, assuming a multivariate normal distribution of M, then

$$\mathrm{E}[\exp(-M_j^i - M_j^{i'})^2 / \phi) \mid m^{obs}] = \exp(-\nu^2 / (1 + \tau^2)) / \sqrt{1 + \tau^2},$$

with $\nu = \mathrm{E}[(M_j^i - M_j^{i'}) / \sqrt{\phi} \mid m^{obs}]$ and $\tau^2 = \mathrm{Var}[(M_j^i - M_j^{i'}) / \sqrt{\phi} \mid m^{obs}]$ where these expectations and variances can be computed from the conditional expectations and variances given in Appendix A. The form $\exp(-\nu^2/(1 + \tau^2))/\sqrt{1 + \tau^2}$ with the variance τ^2 occurring in two places, implies that that the elements in $K^*(m^{obs})$ cannot be expressed in matrix form as in (4) but are on a more complicated form.

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Authors' contributions

OFC derived and implemented the methods, created and analysed the simulation study, and wrote the paper. MSL conceived the study, took part in discussions, and provided input to the writing of the paper. Both authors have read and approved the paper.

Competing interests

The authors declare that they have no competing interests.

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Misztal, Legarra and Aguilar (2014)

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The problem

Single step requires the inverse of G. Inverting G directly becomes infeasible when the number of genotyped animals exceeds a few hundred thousand. Misztal, Legarra and Aguilar showed that the inverse of G can be obtained recursively, as Henderson did with A, if the animals are divided into a core group and a noncore group. The noncore animals are conditioned on the core animals. This is the APY algorithm.

Points to note while reading

Note the analogy with Henderson's rules for the inverse of A, now applied to core animals in place of parents. The idea works because the genomic information of a population has limited dimensionality. APY is what makes it possible to run single step with millions of genotyped animals.

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Using recursion to compute the inverse of the genomic relationship matrix

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ABSTRACT

Computing the inverse of the genomic relationship matrix using recursion was investigated. A traditional algorithm to invert the numerator relationship matrix is based on the observation that the conditional expectation for an additive effect of 1 animal given the effects of all other animals depends on the effects of its sire and dam only, each with a coefficient of 0.5. With genomic relationships, such an expectation depends on all other genotyped animals, and the coefficients do not have any set value. For each animal, the coefficients plus the conditional variance can be called a genomic recursion. If such recursions are known, the mixed model equations can be solved without explicitly creating the inverse of the genomic relationship matrix. Several algorithms were developed to create genomic recursions. In an algorithm with sequential updates, genomic recursions are created animal by animal. That algorithm can also be used to update a known inverse of a genomic relationship matrix for additional genotypes. In an algorithm with forward updates, a newly computed recursion is immediately applied to update recursions for remaining animals. The computing costs for both algorithms depend on the sparsity pattern of the genomic recursions, but are lower or equal than for regular inversion. An algorithm for proven and young animals assumes that the genomic recursions for young animals contain coefficients only for proven animals. Such an algorithm generates exact genomic EBV in genomic BLUP and is an approximation in single-step genomic BLUP. That algorithm has a cubic cost for the number of proven animals and a linear cost for the number of young animals. The genomic recursions can provide new insight into genomic evaluation and possibly reduce costs of genetic predictions with extremely large numbers of genotypes.

Key words: genomic relationship matrix, recursion, genomic selection, single-step BLUP, preconditioned conjugate gradient (PCG) algorithm

INTRODUCTION

When only a fraction of animals are genotyped, a genomic relationship matrix **G** can be combined with a numerator relationship matrix **A** into a genomic-pedigree relationship matrix **H** (Legarra et al., 2009). Such a matrix is complicated, but has a simple inverse (Aguilar et al., 2010; Christensen and Lund, 2010). When the inverse of **H** is used with BLUP, the method is called single-step genomic BLUP (**ssGBLUP**). Advantages of ssGBLUP include simplicity of use (yet another BLUP), relatively high accuracy (Chen et al., 2011; Christensen et al., 2012; Gray et al., 2012), known and explicit control of biases because of different base populations in **A** and **G** as opposed to unknown properties of multistep methods (Tsuruta et al., 2011; Vitezica et al., 2011), and possible accounting for selection bias for genotyped animals (Patry and Ducrocq, 2011; VanRaden, 2012). Accuracy of ssGBLUP can be further improved by using a weighted **G** (Wang et al., 2012), which mimics Bayesian regressions.

The most expensive operation with ssGBLUP, as proposed by Aguilar et al. (2010) and Christensen and Lund (2010), is creating and then inverting **G**. Both operations have an approximately cubic cost with the number of genotypes. With efficient computing algorithms, both operations are feasible for up to 100,000 genotypes (Aguilar et al., 2011; Masuda and Suzuki, 2013). However, the US dairy industry has already collected over 400,000 Holstein genotypes; over 80% of genotypes are for animals without a BLUP evaluation, with a very slow increase in the number of genotypes for proven bulls (Council on Dairy Cattle Breeding, 2013).

Several approaches that do not require the inverse of **G** ($\mathbf{G}^{-1}$) have been proposed for ssGBLUP. Misztal et al. (2009) presented unsymmetric equations where only **H** was required. However, creating **H** directly is complicated. Legarra and Ducrocq (2012) presented

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different unsymmetric equations where inverses that were difficult to obtain were not required. Unsymmetric equations exhibited declining convergence with a larger number of genotypes (Aguilar et al., 2013), although they may be useful when a suitable preconditioner is found. Fernando et al. (2013) proposed a method where genotypes of nongenotyped animals were imputed and the final set of equations included SNP effects for all animals plus extra polygenic terms. However, the imputation is expensive; the volume of the imputed data are extremely large for big populations (up to dozens of millions of individuals for dairy cattle), and existing software is not applicable.

The cost of creating $\mathbf{A}$ by a tabular method (Emik and Terrill, 1949) is quadratic, and the cost to invert it directly is cubic. However, Henderson (1976) developed an algorithm based on recursion to obtain the inverse of $\mathbf{A}$ ($\mathbf{A}^{-1}$) directly at linear cost. Subsequently, $\mathbf{A}^{-1}$ can be computed for millions of animals in seconds. Faux et al. (2012) used Henderson's ideas and conditioned animals on a small number of relatives. However, the cost of their algorithm was higher than that by regular inversion. The purpose of our study was to determine whether recursion is useful in obtaining $\mathbf{G}^{-1}$ at a reasonable cost for a large number of genotypes.

MATERIALS AND METHODS

The method of Henderson (1976) to create $\mathbf{A}^{-1}$ directly depends on the recursion

$$u_{\mathrm{i}} = 0.5\left(u_{s_i} + u_{d_i}\right) + \varphi_i,$$

$$u_i = 0.5(u_{si} + u_{di}) + \varphi_i,$$

where u_i is the animal effect for animal i; s_i and d_i refer to the sire and dam of animal i, respectively; φ_i is Mendelian sampling; and founders of the pedigree are assumed to be unrelated. In matrix notation and with a genetic variance of σ_a^2 of 1 to simplify notation,

$$\mathbf{u} = \mathbf{P}\mathbf{u} + \Phi,\ \mathrm{var}(\Phi) = \mathbf{M},$$

and

$$\mathbf{A}^{-1} = (\mathbf{I} - \mathbf{P})'\mathbf{M}^{-1}(\mathbf{I} - \mathbf{P}) = \mathbf{T}'\mathbf{M}^{-1}\mathbf{T},$$

where $\mathbf{P}$ relates animals to parents; $\mathbf{T}$ is a triangular matrix if animals are ordered from oldest to youngest; and $\mathbf{M}$ is a diagonal matrix. Subsequently, $\mathbf{A}^{-1}$ can be created as a sum of outer products

$$\mathbf{A}^{-1} = \sum_i (\mathbf{t}'_{i,1:n}\,\mathbf{t}_{i,1:n}/m_i),$$

where $\mathbf{t}_{i,1:n}$ contains no more than 3 nonzero elements. Ignoring inbreeding, the value of m_i is (4 − number of known parents)/4. Henderson's rules are simple: when $\mathbf{u} \sim N(0, \mathbf{A}\sigma_a^2)$, animals are ordered from the oldest to the youngest, and all animals (including base animals) are included in the pedigree, $u_i \left| u_1, u_2, \ldots, u_{i-1}, u_{i+1}, \ldots, u_n = u_i \right| u_{s_i}, u_{d_i}$, or the conditional for an animal includes only its parents but not the rest of individuals in the pedigree. For instance, when only animals with records are included in $\mathbf{A}$ or older animals are conditioned on the younger animals, the conditional of u_i may involve more than 2 animals.

With genomic relationships, $\mathbf{u} \sim N(0, \mathbf{G}\sigma_a^2)$. The joint distribution of $u_1, \ldots, u_n$ can be written as

$$p(u_1, \ldots, u_n) = p(u_1)p(u_2|u_1)p(u_3|u_2, u_1)\ldots p(u_n|u_1, u_2, \ldots, u_{n-1}).$$

This decomposition is general and does not involve any particular ordering of individuals. Each of the conditional distributions can be written as

$$p\left(u_i|u_1, u_2, \ldots, u_{i-1}\right) \sim N \left[\mathbf{g}_{i,1:i-1}(\mathbf{G}_{1:i-1,1:i-1})^{-1}\mathbf{u}_{1:i-1}, g_{i,i} - \mathbf{g}_{i,1:i-1}(\mathbf{G}_{1:i-1,1:i-1})^{-1}\mathbf{g}'_{i,1:i-1}\right]$$

with $\mathbf{g}_{i,1:i-1}$ part of the ith row of $\mathbf{G}$, and with the following recursion equation:

$$u_i \mid u_1 \ldots u_{i-1} = \sum_{j=1}^{i-1} p_{ij}u_j + \varepsilon_i,$$

and

$$\mathbf{p}_{i,1:i-1} = \mathbf{g}_{i,1:i-1}(\mathbf{G}_{1:i-1:i-1})^{-1},\ \mathbf{M}_{i,i} = m_i = \mathrm{var}(\varepsilon_i) = \mathrm{g}_{i,i} - \mathbf{p}_{i,1:i-1}\mathbf{g}'_{i,1:i-1}. \quad [1]$$

Mimicking the developments of Henderson (1976) and Quaas (1988),

$$\mathbf{G}^{-1} = (\mathbf{I} - \mathbf{P})'\mathbf{M}^{-1}(\mathbf{I} - \mathbf{P}) = \mathbf{T}'\mathbf{M}^{-1}\mathbf{T},$$

where $\mathbf{T}$ is a triangular matrix as a result of the recursions of u_i on individuals $u_1 \ldots u_{i-1}$. Then $\mathbf{G}^{-1}$ can be created as a sum of outer products as

$$\mathbf{G}^{-1} = \sum_i (\mathbf{t}'_{i,1:n}\, \mathbf{t}_{i,1:n}/m_i).$$

The term $\{i, m_i, \mathbf{p}_{1:i-1,i}\}$ defines the genomic recursion for animal i, where m_i is the genomic Mendelian sampling and $\mathbf{p}_{1:i-1,i}$ are the genomic coefficients. Similarly to rules for $\mathbf{A}^{-1}$, $\mathbf{G}^{-1}$ can be constructed as a sum of contributions for each animal. However, $\mathbf{p}$ can have up to $n-1$ nonzero elements, and the elements of $\mathbf{p}$ generally are no longer 0 or 0.5. (See the Appendix for an example on regular and genomic recursions.)

The genomic recursions are a form of Cholesky decomposition:

$$\mathbf{G}^{-1} = \mathbf{T}'\mathbf{M}\mathbf{T} = \mathbf{U}\mathbf{D}\mathbf{U}',$$

where $\mathbf{U}$ is an upper triangular matrix and $\mathbf{D}$ is a diagonal matrix. However, the genomic recursions have a clear interpretation of what the elements of $\mathbf{U}$ and $\mathbf{D}$ are, which can be useful for insight into the genomic prediction [i.e., which animals contribute the most for a given animal's genomic EBV (**GEBV**)] and for computational refinements (progressive accumulation of elements in $\mathbf{G}^{-1}$, parallelization, and possible elimination of unimportant coefficients).

Efficient Computing Algorithms

Algorithm with Sequential Updates. The preceding algorithm is expensive, as it requires repeated inversion of sections of $\mathbf{G}$. The following algorithm creates those sections directly.

1. $GI_{n\times n} = 0; GI_{1:1,1:1} = 1/g_{1,1}$; $m_1 = g_{1,1}$.
2. $i = 2$.
3. $\mathbf{p}'_{i,1:i-1} = \mathbf{GI}_{1:i-1,1:i-1}\mathbf{g}_{1:i-1,i}$.
4. $m_i = g_{i,i} - \mathbf{p}_{i,1:i-1}\mathbf{g}_{1:i-1,i}$.
5. $$\mathbf{GI}_{1:i,1:i} = \mathbf{GI}_{1:i,1:i} + \begin{bmatrix} -\mathbf{p}'_{i,1:i-1} \\ 1 \end{bmatrix} \begin{bmatrix} -\mathbf{p}_{i,1:i-1} & 1 \end{bmatrix} / m_i.$$
6. $i = i + 1$; continue to 3 until $i > n$.

Step 3 calculates the genomic coefficients, step 4 calculates the genomic Mendelian sampling, and step 5 updates $\mathbf{G}^{-1}$ for the same animal. After the last step, $\mathbf{GI} = \mathbf{G}^{-1}$. This algorithm is equivalent to that of Sherman and Morrison (1950).

When the purpose of the algorithm is only to create genomic recursions [e.g., solving with a preconditioned conjugate gradient (**PCG**) algorithm directly], some steps can be modified:

1. $GI_{n\times n} = 0; GI_{1:1,1:1} = 1/g_{1,1}$; $m_1 = g_{1,1}$.
2. $i = 2$.
3. $$\mathbf{p}_{i,1:i-1} = \sum_{j=1}^{i-1} \left(\begin{bmatrix} -\mathbf{p}'_{j,1:j-1} \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \left(\begin{bmatrix} -\mathbf{p}_{j,1:j-1} & 1 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} \mathbf{g}_{1:j,i} \\ 0 \\ \vdots \\ 0 \end{bmatrix} / m_j \right) \right).$$
4. $m_i = g_{i,i} - \mathbf{p}_{i,1:i-1}\mathbf{g}_{1:i-1,i}$.
5. $i = i + 1$; continue to 3 until $i > n$.

Zeros were added above for conformity in matrix operations. In actual computing, operations on zeros are never done.

Algorithm with Forward Updates. The modified algorithm with sequential updates can be presented in a form where genomic coefficients for animal i are immediately applied to update recursions partially for the remaining animals (step 2):

1. $\mathbf{T} = \mathbf{I}$; $m_1 = g_{1,1}$; $i = 2$.
2. $\mathbf{t}_{j,1:i-1} = \mathbf{t}_{j,1:i-1} - \mathbf{t}_{i-1,1:i-1}(\mathbf{t}'_{i-1,1:i-1}\mathbf{g}'_{1:i-1,j})/m_{i-1}$, $j = i,n$.
3. $m_i = g_{i,i} + \mathbf{t}_{i,1:i-1}\mathbf{g}_{1:i-1,i}$.
4. $i = i + 1$; continue to 2 until $i > n$.

This algorithm is more suitable for vector or parallel computations.

Costs. Without additional optimization, both the sequential and forward update algorithms have cubic cost because they implement inversion for a general symmetric matrix. Costs can be reduced if many coefficients in $\mathbf{P}$ or $\mathbf{T}$ are so small that they can be set to zero with negligible effect on $\mathbf{G}^{-1}$, although the risk of propagation of numerical errors needs to be verified for large n.

In commercial analyses, the genomic relationship for each animal needs to be computed only once. If the population is expanded to include new animals, only genomic recursions for new animals need to be added. In such a case, the algorithm with sequential updates is equivalent to formulas for updates to $\mathbf{G}^{-1}$ when additional genotypes are available (Meyer et al., 2013).

Use of G^{-1} when Solving Large Systems of Equations. Solutions to mixed model equations that include $\mathbf{G}^{-1}$ can be computed without creating $\mathbf{G}^{-1}$ explicitly. Currently, a standard method to solve large BLUP equations is PCG iteration on data (Strandén and Lidauer, 1999; Tsuruta et al., 2001). For PCG computation, the left-hand side of the system of equa-

tions is not needed explicitly; however, each round of iteration requires a product of the left-hand side times a specified vector, say $\mathbf{r}$. The product $\mathbf{G}^{-1}\mathbf{r}$ can be computed as a series of vector-by-vector multiplications involving the genomic recursions,

$$\mathbf{G}^{-1}\mathbf{r} = \sum_i \left(\mathbf{t}'_{i,1:n}\,\mathbf{t}_{i,1:n}\mathbf{r}/m_i\right),$$

at computing cost $O(ns)$, where s is average length of $\mathbf{t}$. If $\mathbf{T}$ is dense, the cost is $O(n^2)$.

Algorithm when a Population Includes Proven and Young Animals

Assume that n initial animals have been proven (i.e., have phenotypes or progeny with phenotypes) and the remaining m are young. Then

$$u_i|u_1,u_2,\ldots,u_{i-1} = \sum_{j=1}^{n} p_{ij}u_j + \sum_{j=n+1}^{n+m} p_{ij}u_j + \varepsilon_i \quad \text{for } i > n.$$

In Bayesian regressions on SNP effects (e.g., Ridge regression BLUP and BLUP_SNP) and equivalent genomic BLUP, young animals do not provide any information to GEBV of other animals. This is not true for ssGBLUP because new genomic relationships may modify old pedigree relationships; however, the effect of genotypes of young animals on GEBV of other genotyped animals is likely small, especially if n is very large. Assuming that the contributions of young animals sum to 0,

$$u_i|u_1,u_2,\ldots,u_n = u_i|\mathbf{u}_{\text{proven}} = \sum_{j\in\text{proven}} p_{ij}u_j + \varepsilon_i \text{ for } i > n.$$

If p denotes proven animals, y denotes young animals, and $\mathbf{G} = \mathbf{Z}\mathbf{Z}'/q$ as in VanRaden (2008), then

$$\mathbf{P}_y = \mathbf{G}_{yp}\mathbf{G}_{pp}^{-1} = \mathbf{Z}_y\mathbf{Z}'_p/q\ \mathbf{G}_{pp}^{-1}$$

and

$$\text{var}(\varepsilon_i)/\sigma_u^2 \approx g_{ii} - \mathbf{G}_{ip}\mathbf{G}_{pp}^{-1}\mathbf{G}_{pi} = g_{ii} - \mathbf{z}_i'\mathbf{Z}'_p\mathbf{G}_{pp}^{-1}\mathbf{Z}_p\mathbf{z}_i/q^2 = m_i,$$

with the approximation because covariances among young animals were ignored. Finally,

$$\mathbf{G}^{-1} \approx \begin{bmatrix}\mathbf{G}_{pp}^{-1} & 0\\ 0 & 0\end{bmatrix} + \begin{bmatrix}-\mathbf{G}_{pp}^{-1}\mathbf{Z}_p\mathbf{Z}'_y/\text{q}\\ \mathbf{I}\end{bmatrix}\mathbf{M}^{-1}\begin{bmatrix}-\mathbf{Z}_y\mathbf{Z}'_p/\text{q}\,\mathbf{G}_{pp}^{-1} & \mathbf{I}\end{bmatrix}.$$

The last equation does not require the expensive operation of forming sections of $\mathbf{G}$ for young animals explicitly.

If n = number of proven animals, m = number of young animals, and ℓ = number of SNP, then

$$\text{cost}[\hat{\mathbf{u}}_y = \mathbf{Z}_y(\mathbf{Z}'_p/q)\mathbf{G}_{pp}^{-1}\hat{\mathbf{u}}_p] \sim n^3 + \ell(m+n);$$

$$\text{cost}(\mathbf{G}^{-1}\mathbf{r}) \sim n^2 + 2(\ell m + \ell n + n^2);$$

$$\text{cost}(\mathbf{G}_{pp}^{-1}) \sim n^3; \text{ and}$$

$$\text{cost}(M) \sim m\ell^2 \text{ if } \mathbf{Z}'_p\mathbf{G}_{pp}^{-1}\mathbf{Z}_p \text{ is precomputed at cost } \sim n^2\ell.$$

All costs are cubic with n, quadratic or linear with ℓ, and linear with the number of young animals. If n = 50,000, ℓ = 50,000, and PCG iteration completes in 300 rounds, then adding 450,000 young animals increases computing costs associated with $\mathbf{G}^{-1}$ by 11 times. With a regular algorithm for inversion, the costs increase 1,000 times.

Equivalent Formulas for Approximate Inverse of $\mathbf{A}_{22}$

Computing ssGBLUP also requires an inverse for $\mathbf{A}_{22}$ $\left(\mathbf{A}_{22}^{-1}\right)$, which is a submatrix of $\mathbf{A}$ for genotyped animals. Using notation analogous to that for $\mathbf{G}$,

$$\mathbf{A}_{22}^{-1} \approx \begin{bmatrix}\mathbf{A}_{pp}^{-1} & 0\\ 0 & 0\end{bmatrix} + \begin{bmatrix}-\mathbf{A}_{pp}^{-1}\mathbf{A}_{py}\\ \mathbf{I}\end{bmatrix}\mathbf{M}^{-1}\begin{bmatrix}-\mathbf{A}_{yp}\mathbf{A}_{pp}^{-1} & \mathbf{I}\end{bmatrix},$$

with

$$m_i \approx a_{22,ii} - \mathbf{a}_{ip}\mathbf{A}_{pp}^{-1}\mathbf{a}_{pi}.$$

The product of a section of $\mathbf{A}$ by a vector can be computed efficiently using an algorithm by Colleau (2002). In particular, that algorithm can be used to compute a large number of $\mathbf{a}_{ip}$ in parallel (Aguilar et al., 2011).

Alternately, the formulas for $\mathbf{A}_{22}^{-1}$ may be computed using the sequential update algorithm. This requires accessing elements 1 to i of $\mathbf{A}_{22}$ at a time, which can be done either by recursion or using Colleau's algorithm. When the depth of pedigree is limited, $\mathbf{A}_{22}^{-1}$ and $\mathbf{T}$ are relatively sparse (Faux and Gengler, 2013). Subsequently, the main cost of obtaining $\mathbf{A}_{22}^{-1}$ may be computing $\mathbf{A}_{22}$.

DISCUSSION

The algorithm for genomic selection based on genomic recursions offers an extra insight compared with algorithms based on estimation of SNP effects or based on **G** obtained through usual means. Methods based on estimation of SNP effects provide insight into the genetic architecture of traits. The genomic relationships allow for comparisons and, subsequently, quality control for pedigree-based relationships. They also aid in computing theoretical accuracy by inversion of the mixed model equations. The genomic recursions allow some insight into the flow of information from older to younger animals. Such perceptions may lead to modifications of recursion coefficients for a higher realized accuracy, because theoretical accuracies using traditional **G** are inflated (Su et al., 2012). Such modifications may include accounting for decay of genomic predictions over generations (Wolc et al., 2011) or for less information from imputed genotypes (Cleveland and Hickey, 2013). A great advantage of modified genomic recursions is that they result in at least semipositive definite $\mathbf{G}^{-1}$. The application of genomic recursions depends on their properties in practice. Although they are dense for small populations (not shown), their sparsity pattern with large populations is unknown.

If animals are sequentially added to a population, both the sequential and forward update algorithms provide a possibility to create genomic recursions that are computed once for every animal, which is functionally similar to ideas of Meyer et al. (2013). The genomic recursions are calculated as animals are registered. The costs of computations once the genomic recursions are known is reasonable because those pedigrees result in quadratic (or less if recursions are sparse) costs for a PCG iteration program.

The proposed algorithms were evaluated with a simulated data set of 1,500 genotypes and ssGBLUP. In the algorithm with sequential updates, setting very small elements (−0.001 to 0.001) in **P** to zero resulted in little sparsity. Setting larger elements (−0.01 to 0.01) to zero caused large errors in $\mathbf{G}^{-1}$due to accumulation of errors. However, this algorithm worked very well for $\mathbf{A}_{22}^{-1}$. When complete **P** was computed and its small elements were zeroed (−0.01 to 0.01), the accuracy of $\mathbf{G}^{-1}$ and GEBV were almost unaffected, but the sparsity level was moderate. The sparsity level increased to >60% when **G** was blended with 20% of $\mathbf{A}_{22}$. In all computations involving the algorithm for proven and young animals, the correlations of GEBV with those using the regular algorithm were >0.99. However, this testing is not extensive and its conclusions cannot by extended to real life data sets. Future, more detailed studies will address these points.

The algorithm based on decomposition for proven and young animals can reduce costs dramatically when the number of genotypes is >50,000 and potentially help with stability during iteration. In fact, evaluation with 1 million genotypes would be possible. Although GEBV computed with **G** from the proven-young algorithm are the same as with traditional **G** in GBLUP, which is not the case for ssGBLUP. In particular, genotypes of young animals can improve accuracy of EBV for their nongenotyped relatives, although the improvement is small (Christensen et al., 2012). Therefore, ssGBLUP computed using the proven-young algorithm is an approximation. With fewer coefficients in that algorithm, the numerical properties of mixed model equations are likely better. Aguilar et al. (2013) found that the convergence rate decreases with the addition of young animals. The proven-young algorithm is of interest only when the number of animals is large, (e.g., >50,000); otherwise, conventional algorithms are cost effective (Aguilar et al., 2011).

The increase in the number of animals with highly accurate evaluations (e.g., reliability of >95%) is fairly slow for dairy cattle; perhaps 1,000 such bulls are added yearly in the United States. It is unclear how the genomic accuracy of cows is affected if they are treated as young animals in the proven-young algorithm. Normally, the addition of genotypes of cows with records increases the accuracy of genomic predictions very little. For example, the mean accuracy across breeds and traits for New Zealand dairy cattle using about 7,000 male genotypes in the training population was approximately 0.70 (Harris et al., 2013). Adding 17,000 female genotypes increased that accuracy by approximately 0.02. Accuracies for cows treated as young with the proven-young algorithm in ssGBLUP may be close to that with exact **G**, but at a much lower cost.

CONCLUSIONS

The inverse of the genomic relationship matrix, which is needed for genomic evaluation, can be computed using genomic recursions. In general, the cost of obtaining those recursions depends on their sparsity pattern and needs to be determined experimentally. In a specific case where the genotyped population included proven and young animals, the approximate cost is cubic with the number of proven animals and linear with the number of young animals.

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APPENDIX

Example of Regular and Genomic Recursions

Consider 4 animals with the following pedigree:

Animal	Sire	Dam
1	0	0
2	0	0
3	0	2
4	1	2

.

The numerator relationship matrix is

$$\mathbf{A} = \begin{bmatrix} 1.00 & 0.00 & 0.00 & 0.50 \\ & 1.00 & 0.50 & 0.50 \\ & & 1.00 & 0.25 \\ symm & & & 1.00 \end{bmatrix}.$$

Using equation [1] yields the following recursions:

Animal	m		P	
1	1.00			
2	1.00	0.0		
3	0.75	0.0	0.5	
4	0.50	0.5	0.5	0.0

.

Those are the same values as in the Henderson algorithm, except that the values in t are shown explicitly. They point to the following recurrence equations (setting additive variance for simplicity to 1.0):

$$\begin{aligned} u_1 &= \varepsilon_1, \mathrm{var}(\varepsilon_1) = 1.00; \\ u_2 &= \varepsilon_2, \mathrm{var}(\varepsilon_2) = 1.00; \\ u_3 &= 0.50u_2 + \varepsilon_3, \mathrm{var}(\varepsilon_3) = 0.75; \text{ and} \\ u_4 &= 0.50u_1 + 0.50u_2 + \varepsilon_4, \mathrm{var}(\varepsilon_4) = 0.50. \end{aligned}$$

If the recursion is used in the reverse order from youngest to oldest animal, the recursions are

Animal	m			P	
1	0.667	0.000	−0.333	0.000	0.667
2	0.600	0.000	0.000	0.400	0.400
3	0.937	0.000	0.000	0.000	0.250
4	1.000	0.000	0.000	0.000	0.000

,

with the new recurrence equations

$$u_1 = -0.333u_2 + 0.667u_4 + \varepsilon_1, \text{var}(\varepsilon_1) = 0.667;$$
$$u_2 = 0.4u_3 + 0.4u_4 + \varepsilon_2, \text{var}(\varepsilon_2) = 0.600;$$
$$u_3 = 0.25u_4 + \varepsilon_3, \text{var}(\varepsilon_3) = 0.937; \text{ and}$$
$$u_4 = \varepsilon_4, \text{var}(\varepsilon_4) = 1.0.$$

The coefficients with reverse ordering are no longer 0.5, although some coefficients still equal zero. The new recurrence equations have no clear meaning although they result in the same $\mathbf{A}^{-1}$.

Now assume that the genomic relationships are slightly different:

$$\mathbf{G} = \begin{bmatrix} 1.02 & -0.04 & 0.06 & 0.51 \\ & 1.05 & 0.47 & 0.54 \\ & & 0.98 & 0.31 \\ symm & & & 0.97 \end{bmatrix}.$$

Using the same formulas yields results in the following genomic recursions:

Animal	m		P	
1	1.020			
2	1.048	−0.039		.
3	0.764	0.076	0.451	
4	0.414	0.518	0.517	0.036

The new recurrence equations are

$$u_1 = \varepsilon_1, \text{var}(\varepsilon_1) = 1.02;$$
$$u_2 = -0.039u_1 + \varepsilon_2, \text{var}(\varepsilon_2) = 1.048;$$
$$u_3 = 0.076u_1 + 0.451u_2 + \varepsilon_3, \text{var}(\varepsilon_3) = 0.764; \text{ and}$$
$$u_4 = 0.414u_1 + 0.518u_2 + 0.0306u_3 + \varepsilon_4, \text{var}(\varepsilon_4) = 0.414.$$

Example for an Algorithm Based on Decomposition for Young and Proven Animals

Assume 7 animals are genotyped for 10 SNP and have the following matrix of genotypes:

$$\mathbf{Z} = \begin{bmatrix} -1 & 0 & -1 & 0 & 1 & -1 & 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 & -1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & 1 & 0 & -1 & 0 & 1 & 1 & 1 & 1 \\ 0 & -1 & 0 & 0 & -1 & 1 & -1 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 1 & 0 & 1 & -1 & 1 & 1 \\ 0 & 1 & -1 & 0 & -1 & 0 & 0 & 1 & -1 & -1 \\ 1 & -1 & -1 & -1 & 0 & -1 & 1 & 0 & 0 & 1 \end{bmatrix}.$$

The genomic relationship matrix constructed according to VanRaden (2008) is

$$\mathbf{G} = \mathbf{ZZ}'/q = \begin{bmatrix} 0.795 & -0.318 & 0.000 & -0.477 & 0.636 & -0.159 & 0.318 \\ -0.318 & 0.955 & -0.159 & 0.318 & 0.000 & 0.636 & -0.477 \\ 0.000 & -0.159 & 1.114 & -0.159 & 0.159 & -0.159 & 0.000 \\ -0.477 & 0.318 & -0.159 & 0.795 & -0.477 & 0.159 & -0.318 \\ 0.636 & 0.000 & 0.159 & -0.477 & 1.273 & -0.477 & 0.159 \\ -0.159 & 0.636 & -0.159 & 0.159 & -0.477 & 0.955 & -0.159 \\ 0.318 & -0.477 & 0.000 & -0.318 & 0.159 & -0.159 & 1.114 \end{bmatrix},$$

where $q = 6.29$, which was chosen arbitrarily for a mean diagonal of 1.0. The inverse is

$$\mathbf{G}^{-1} = \begin{bmatrix} 12.229 & 14.726 & 1.704 & -2.121 & -12.225 & -12.902 & 2.114 \\ 14.726 & 23.208 & 2.269 & -4.877 & -17.428 & -19.874 & 3.996 \\ 1.704 & 2.269 & 1.191 & -0.200 & -1.817 & -1.834 & 0.426 \\ -2.121 & -4.877 & -0.200 & 3.199 & 3.930 & 4.208 & -0.530 \\ -12.225 & -17.428 & -1.817 & 3.930 & 14.774 & 15.553 & -2.742 \\ -12.902 & -19.874 & -1.834 & 4.208 & 15.553 & 18.379 & -3.225 \\ 2.114 & 3.996 & 0.426 & -0.530 & -2.742 & -3.225 & 1.786 \end{bmatrix}.$$

Assuming that observations are available for the first 5 animals,

$$\mathbf{y} = \begin{bmatrix} 31.856 & 46.657 & -6.941 & 34.636 & 51.571 & 0 & 0 \end{bmatrix},$$

the incidence matrix is

$$\mathbf{D} = \operatorname{diag}\left(\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 \end{bmatrix}\right).$$

Then the GBLUP equations are

$$(\mathbf{D} + \mathbf{G}^{-1})\hat{\mathbf{u}} = \mathbf{y},$$

with solutions

$$\hat{\mathbf{u}} = \begin{bmatrix} 10.962 & 23.830 & -5.688 & 7.958 & 29.040 & 4.893 & -9.151 \end{bmatrix}'.$$

For the algorithm that decomposes the animals into proven and young,

$$\mathbf{G}^{-1} = \begin{bmatrix} 3.154 & 0.837 & 0.429 & 0.858 & -1.309 & 0 & 0 \\ 0.837 & 1.505 & 0.242 & -0.414 & -0.604 & 0 & 0 \\ 0.429 & 0.242 & 0.999 & 0.202 & -0.264 & 0 & 0 \\ 0.858 & -0.414 & 0.202 & 2.200 & 0.371 & 0 & 0 \\ -1.309 & -0.604 & -0.264 & 0.371 & 1.612 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} +$$

$$\begin{bmatrix} -0.724 & -0.123 \\ -1.008 & 0.416 \\ -0.085 & 0.085 \\ 0.259 & 0.171 \\ 0.844 & -0.010 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0.080 & 0 \\ 0 & 0.820 \end{bmatrix}^{-1}$$

$$\begin{bmatrix} -0.724 & -1.008 & -0.085 & 0.259 & 0.844 & 1 & 0 \\ -0.123 & 0.416 & 0.085 & 0.171 & -0.010 & 0 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} 9.744 & 9.932 & 1.187 & -1.519 & -8.977 & -9.083 & -0.150 \\ 9.932 & 14.478 & 1.359 & -3.604 & -11.297 & -12.657 & 0.508 \\ 1.187 & 1.359 & 1.098 & -0.056 & -1.164 & -1.065 & 0.104 \\ -1.519 & -3.604 & -0.056 & 3.077 & 3.113 & 3.250 & 0.208 \\ -8.977 & -11.297 & -1.164 & 3.113 & 10.564 & 10.601 & -0.012 \\ -9.083 & -12.657 & -1.065 & 3.250 & 10.601 & 12.553 & 0 \\ -0.150 & 0.508 & 0.104 & 0.208 & -0.012 & 0 & 1.220 \end{bmatrix}.$$

Solutions from GBLUP with the new $\mathbf{G}^{-1}$ were identical.

Legarra et al. (2015)

Legarra, A., Christensen, O.F., Vitezica, Z.G., Aguilar, I., and Misztal, I. 2015. Ancestral relationships using metafounders. Finite ancestral populations and across population relationships. Genetics, vol. 200, no. 2, pp. 455 to 468.

The problem

The A matrix assumes that the founders of the pedigree are unrelated. The G matrix, in turn, measures relationships relative to the allele frequencies used in its construction. As a result, the two matrices may be on different scales, which causes bias in single step. Legarra and coworkers proposed treating base populations as metafounders, fictitious individuals with their own relationship and inbreeding. In this way A and G are placed on the same scale, even with several populations of origin.

Points to note while reading

Note the gamma parameter, which measures the average relationship within the base population, and how it is estimated from genotypes. Metafounders resolve at once the compatibility between A and G and the treatment of unknown parents, which was previously handled with genetic groups.

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Ancestral Relationships Using Metafounders: Finite Ancestral Populations and Across Population Relationships

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ABSTRACT Recent use of genomic (marker-based) relationships shows that relationships exist within and across base population (breeds or lines). However, current treatment of pedigree relationships is unable to consider relationships within or across base populations, although such relationships must exist due to finite size of the ancestral population and connections between populations. This complicates the conciliation of both approaches and, in particular, combining pedigree with genomic relationships. We present a coherent theoretical framework to consider base population in pedigree relationships. We suggest a conceptual framework that considers each ancestral population as a finite-sized pool of gametes. This generates across-individual relationships and contrasts with the classical view which each population is considered as an infinite, unrelated pool. Several ancestral populations may be connected and therefore related. Each ancestral population can be represented as a "metafounder," a pseudo-individual included as founder of the pedigree and similar to an "unknown parent group." Metafounders have self- and across relationships according to a set of parameters, which measure ancestral relationships, *i.e.*, homozygozities within populations and relationships across populations. These parameters can be estimated from existing pedigree and marker genotypes using maximum likelihood or a method based on summary statistics, for arbitrarily complex pedigrees. Equivalences of genetic variance and variance components between the classical and this new parameterization are shown. Segregation variance on crosses of populations is modeled. Efficient algorithms for computation of relationship matrices, their inverses, and inbreeding coefficients are presented. Use of metafounders leads to compatibility of genomic and pedigree relationship matrices and to simple computing algorithms. Examples and code are given.

KEYWORDS relationships; pedigree; genetic drift; base populations; marker genotypes; shared data resource; GenPred

POWELL *et al.* (2010) pointed out the conceptual conflict between identity-by-descent (IBD) relationships based on pedigree and identity-by-state (IBS) relationships based on marker genotypes. These are also known as pedigree and genomic (VanRaden 2008) relationships, respectively, and we use this terminology hereinafter. Whereas reference for pedigree relationships is formed by founders of the pedigree, reference for the genomic relationships is most often the current genotyped population (*e.g.*, Powell *et al.* 2010; Vitezica *et al.* 2011). Powell *et al.* (2010) showed that one can (at least conceptually) refer genomic relationship coefficients to the pedigree scale and vice versa. In the context of applied genetic evaluation of livestock, similar notions were introduced by VanRaden (2008) and Vitezica *et al.* (2011), explicitly modifying genomic relationships to refer to pedigree coefficients. However, an implicit assumption in these proposals is that the genotyped population has no pedigree structure, *e.g.*, no sib groups and only one generation (Christensen 2012), and the proposals are also difficult to extend to several base populations (Harris and Johnson 2010; Misztal *et al.* 2013; Makgahlela *et al.* 2014).

In addition, pedigree relationships have several problems. Pedigrees, which are incomplete by definition, end up in one or several base populations (lines or breeds). For instance, the pedigree of the Romane sheep synthetic breed traces back to

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two base populations, the Romanov and Berrichon du Cher breeds, and the pedigree of global Holstein cattle population can often be traced back to "European" and "North American" base populations. In another more complex case, pedigrees are incomplete for some categories of animals. For instance, in dairy sheep, the father of all males is known, but only 5–80% of females have a known father. To further complicate things, in the presence of selection, assuming that all unknown parents belong to the same base population and have the same genetic level is unfair since younger (or sometimes foreign) animals are selected and therefore "better" than the base population. If not properly accounted for, this structure in several base populations results in biases (*e.g.*, Ugarte *et al.* 1996; Misztal *et al.* 2013). Therefore, unknown fathers are assigned to different base populations, *e.g.*, depending on year of birth, country of origin, sex, or path of selection. Current practice of genetic evaluations assumes that individuals in the different base populations (typically known as genetic groups or unknown parent groups) have different *a priori* average values, and these values are estimated as fixed effects within the model (Thompson 1979; Quaas 1988). However, the quantitative genetics theory of unknown parent groups has not been much further developed. For instance, Kennedy (1991) pointed out that genetic groups are incorrectly assumed to be unrelated to each other and that reduction of variance due to drift and selection should be accounted for. With molecular markers, there are more and more examples of observed relationships across base populations that were *a priori* unrelated (Kijaas *et al.* 2009 ; Gibbs *et al.* 2009; Ter Braak *et al.* 2010 ; VanRaden *et al.* 2011).

The hypothesis of unrelatedness of founders in a base population implies that the base population is drawn from a very large ancestral population. Not only is this false but it also contradicts marker-based information (animals seemingly unrelated share alleles at markers). Although unrelatedness can simply be seen as an arbitrary starting point, we suggest that relaxing this hypothesis gives more flexibility to the models.

On the other hand, genomic measures of relationships are not dependent on knowledge of pedigree. Further, they are more accurate because they consider realized, not expected, relationships (VanRaden 2008; Hayes, *et al.* 2009; Hill and Weir 2011). Genomic relationships can be projected along the pedigree for animals with no genotypes (Legarra *et al.* 2009; Christensen and Lund 2010). The so-called single-step GBLUP (SSGBLUP) thus mixes pedigree and genomic relationships, and is becoming the *de facto* standard in genomic evaluations for livestock (*e.g.*, Legarra *et al.* 2014b). However, SSGBLUP requires genomic and pedigree relationship to refer to the same base. This base is however, hard to define. Genomic relationships of the current population change as more individuals are being included and are poorly defined if populations are structured (*i.e.*, in lines, breeds or origins) (*e.g.*, Harris and Johnson 2010). Defining a base is also difficult for pedigree relationships as pedigrees are incomplete and possibly end up in several base populations. An alternative is truncation of the pedigree, to have a more homogeneous base population (*e.g.*, Lourenco *et al.* 2014), but this is not always a feasible option. Furthermore, defining pedigree founders as unrelated is contradictory with results obtained if these individuals are genotyped. Christensen (2012) suggested taking for genomic relationships an arbitrary reference and an ideal population with 0.5 allele frequency at the markers, and referring pedigree relationships to this base population. By doing so, he showed that founders of the base population should become related, and this extra relatedness can be understood as an excess of identical-by-descent homozygosity. The approach can be understood as a marginalization with respect to uncertainty in allelic frequencies, and a stable definition of the genetic base across time and different populations is obtained. Extension of this method to several founder populations is, alas, not straightforward.

In this work, we present a theory to consider relationships within and across base, or founder, populations. This theory provides the tools, on the one hand, to generalize the "unknown parent groups" used in genetic evaluations and, on the other hand, to generalize Christensen's results, which conciliate pedigree and genomic relationships. The concepts developed in this work aim to be rather general and are based on pedigree considerations, but their use is of large interest in two cases: first, when combining genomic and pedigree relationships across individuals (as the SSGBLUP mentioned above) and, second, when considering several base populations simultaneously.

The outline of this article is as follows. First, we show that base populations with related individuals can be understood as issued from finite size ancestral populations. This, although not strictly necessary for practical purposes, gives a conceptual model and a genetic interpretation (Jacquard 1974). Second, such an ancestral population can be represented as a single pseudo-individual (a metafounder) with a particular self-relationship (a measure of homozygosity) and represents a pool of gametes. Several base populations can be represented as several, possibly related, metafounders. Metafounders are convenient because they simplify the representation and the algorithms for computing relationships and inbreeding. Finally, we show how parameters (ancestral homozygosities and relationships across populations) of ancestral populations can be estimated from the combined use of marker and pedigree data. Our work is an extension and unification of existing works by Jacquard (1969, 1974), VanRaden (1992), Aguilar and Misztal (2008), VanRaden *et al.* (2011), Colleau and Sargolzaei (2011), and Christensen (2012).

Theory

Relationships in a finite population

Relationships across base individuals: Let "ancestral" be the population from which founders of the pedigree are drawn and "base" population be the set of these pedigree founders

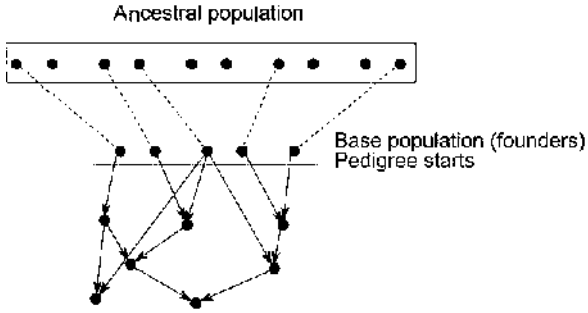


Figure 1 Ancestral and base population and pedigree.

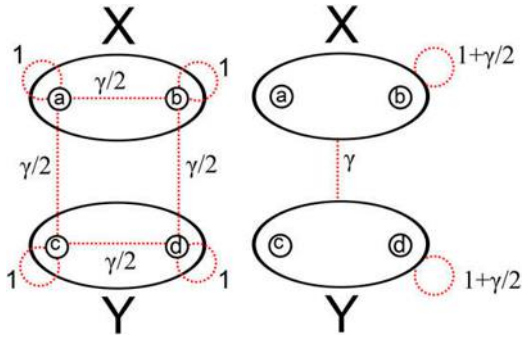


Figure 2 Relationships (red dashed line), in a related base population, of two individuals X and Y seen as gametes (left) or individuals (right). Individual X (Y) contains gametes a and b (c and d). Relationships across and within gametes are respectively $\boldsymbol{\gamma}/2$ and 1; relationships across and within individuals are $\boldsymbol{\gamma}$ and $1+\boldsymbol{\gamma}/2$.

(Figure 1). Typically, individuals in the base population are assumed to be drawn from a large, unrelated, ancestral population mating at random, so that the base population individuals will not be related. Jacquard (1969, 1974) considered relationships in a finite-size population, showing that inbreeding and relationships increase steadily. We redevelop his treatment in a simplified form. Pedigree founders in the base population are drawn at random, with replacement, from an ancestral finite monoecious population with effective size N_e, $2N_e$ gametes, "true" average breeding value μ, and genetic variance σ^2_u. In this ancestral population gametes are assumed to be independent (in a sense, the ancestral population becomes the new base). Imagine two gametes sampled with replacement from the finite ancestral population to form the base population. The second gamete will be identical to the first gamete $1/(2N_e)$ of the times. Therefore, the relationship coefficient (probability of identity by descent) between all pairs of gametes is $\gamma/2 = 1/2N_e$, and this relationship $\gamma/2$ can be understood as the correlation between gametes of Wright (1922). Jacquard (1974) used $\alpha = \gamma/2$ and called it the "inbreeding coefficient of a population."

Across-individual relationships in the base population are depicted in Figure 2. Consider diploid individuals *X* and *Y*. They are constituted by four gametes, a,b,c,d. These gametes have been drawn from a pool of gametes where the probability of being identical (by descent) is $\gamma/2$ across gametes and 1 with itself (Figure 2, left). Therefore, the coancestry coefficient between *X* and *Y* is the four-way average of probabilities of being identical for each possible pair of gametes, which sums to $\gamma/2$. Additive relationship between *X* and *Y* is twice the coancestry and therefore γ (Figure 2, right). Now consider individual *X*. The self-coancestry considers four ways of sampling alleles a and b (with replacement), and because $P(a \equiv b) = \gamma/2$, self-coancestry is equal to $1/2 + \gamma/4$, and therefore self-relationship is equal to $1 + \gamma/2$.

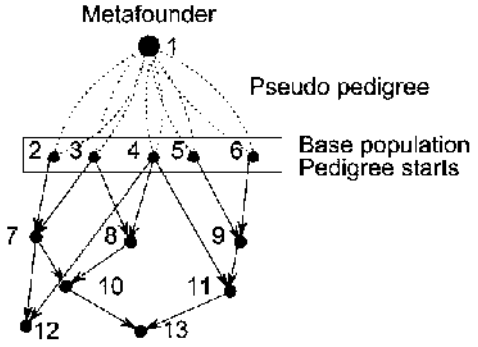
The base population has associated breeding values $\mathbf{u}_0$. From the developments above, the variance-covariance matrix of breeding values is $\mathrm{Var}(\mathbf{u}_0) = [\mathbf{I}(1-\gamma/2) + \mathbf{J}\gamma]\sigma^2_u$, where $\mathbf{I}$ is the identity matrix and $\mathbf{J}$ is a matrix of ones. This covariance structure was suggested by Christensen (2012) to correctly compare genomic relationships and pedigree relationships. Due to random sampling of a limited number of founders, the mean of the base population composed of n individuals ($\overline{u}_0 = \mathbf{1}'\mathbf{u}_0/n$) will drift around the mean of the ancestral population with variance $\mathrm{Var}(\overline{u}_0) = \gamma + (1-\gamma/2)/n$.

Pedigree relationships from related base populations: VanRaden (1992) (unaware of the work of Jacquard 1974) assigned nonzero relatedness to animals in the base populations to correctly estimate inbreeding when pedigree information is missing. The value assigned to this relatedness, which is equivalent to γ, was set to the average relatedness of contemporary individuals with known relationships. Lutaaya *et al.* (1999) showed that the classical algorithm for calculating inbreeding is very sensitive to even a small loss of pedigree while VanRaden (1992) algorithm is much better although not perfect. This idea was also applied by Aguilar and Misztal (2008), and Colleau and Sargolzaei (2011) used a closely related idea in a similar setting.

Obtaining pedigree relationships from related base populations is conceptually straightforward, can be done following the tabular rules (Emik and Terrill 1949), and leads to a matrix of additive relationships

$$\mathbf{A}^{\gamma} = \mathbf{A}\left(1 - \frac{\gamma}{2}\right) + \gamma\mathbf{J},$$

where $\mathbf{A}$ is the matrix with regular relationships and $\mathbf{J}$ a matrix of 1's. In Jacquard (1974, p. 169), this formula is presented using coancestries instead of relationships. However, algorithms for computation of inbreeding (*e.g.*, Quaas 1976; Meuwissen and Luo 1992), Henderson's (1976) sparse inverse of the pedigree relationships, and other algorithms (Colleau 2002) need to be modified to account for nonzero relatedness of founders (*e.g.*, Aguilar and Misztal 2008; Christensen 2012). These changes are rather complex and do not generalize well to the case of several base populations that are presented later. For this reason, and for its conceptual appeal, we have conceived the use of metafounders.



Pedigree		
1	0	0
2	1	1
3	1	1
4	1	1
5	1	1
6	1	1
7	2	3
8	3	4
9	5	6
10	7	8
11	4	6
12	7	4
13	10	11

Figure 3 Base population with a metafounder and corresponding pedigree.

Metafounder

We now introduce a different, but equivalent, representation of related base populations that allows a greater flexibility. This representation uses so-called metafounders.

Definition: The notion of metafounder comes as an extension of VanRaden (1992) method for estimation of across-breed relationships. Imagine a pseudo-individual who can be considered as, simultaneously, father and mother of all base animals (Figure 3). We call this pseudo-individual a metafounder. The metafounder in Figure 3 represents the ancestral population in Figure 1.

In Figure 3, the metafounder (individual 1) represents a finite-size pool of gametes, from which the gametes constituting individuals 2–6 (the base population) are drawn. Picking two gametes at random with replacement, these gametes have an across-gamete relationship of $\gamma/2$. Therefore, the metafounder can be considered as having a self-relationship of $a_{11} = \gamma$ and an individual inbreeding coefficient of $F_i = a_{11} - 1 = \gamma - 1$, which will usually be negative. Inbreeding means departure from Hardy–Weinberg equilibrium, and negative inbreeding represents excess of heterozygotes. Therefore, negative inbreeding means that in most cases two gametes are different, *i.e.*, the size of the pool is large, which is a tenable genetic hypothesis. For instance, considering $\gamma = 0$ (and therefore $F = -1$) means that the two gametes are always different (by descent) and unrelated, *i.e.*, the size of the pool is infinite, heterozygosity (by descent) is complete, and all individuals in the base population are unrelated. Considering $\gamma = 2$ (and $F = 1$) means that two gametes drawn at random are always identical, *i.e.*, the pool consists of one gamete, there is complete homozygosity, and all individuals in the base population are identical and completely inbred.

Algorithms for relationships and inbreeding with a single metafounder: With this representation using metafounders, regular rules for computation of relationships and inbreeding change only slightly. Consider the Emik and Terrill (1949) rules for computation of additive relationship coefficients. They start by assigning self-relationships of 1 to all animals in the base population and later two rules are used,

$$a_{ij} = 0.5(a_{dj} + a_{sj})$$

$$a_{ii} = 1 + 0.5(a_{sd}),$$

where d and s are the dam and sire of i, which must be younger than j. To include the metafounder, the only change is to set its self-relationship (a_{11} in the example) to γ. The Emik and Terrill rules do not otherwise need to be changed. For instance, for individual 2 in Figure 3, $a_{22} = 1 + 0.5a_{11} = 1 + \gamma/2$, and for individuals 1 and 2, $a_{12} = 0.5(a_{11} + a_{11}) = \gamma$. For individuals 2 and 3, $a_{23} = 0.5(a_{12} + a_{12}) = \gamma$. Therefore, assigning a metafounder with self-relationship γ is strictly equivalent to considering across-founder population relationships γ and founder self-relationships $1 + \gamma/2$. The recursive algorithms of Karigl (1981) and Aguilar and Misztal (2008) are versions of Emik and Terrill (1949) and therefore need no modification beyond setting a_{11} to γ. Using these rules, $\mathbf{A}^{\gamma}$ is easily created.

Consider Henderson's (1976) inverse of the relationship matrix **A**. This consists in a product on the form $\mathbf{A}^{-1} = \mathbf{L}^{-1}\mathbf{D}^{-1}\mathbf{L}^{-1'}$, where **D** is usually a diagonal matrix containing variances of the Mendelian sampling terms (deviation of an individual's breeding value from its parents' average) and $\mathbf{L}^{-1}$ contains ones in the diagonal and 0.5 coefficients linking parents to offspring. Elements of **D** are a function of inbreeding of the parents (see Thompson 1977 for the proof and Elzo (2008) for a detailed explanation). This reasoning applies equally well to the use of one metafounder. Thus, using pedigrees with a metafounder, all the information about covariance of gametes transmitted from base animals to their descendants is contained in the inbreeding of the base animals, and the algorithm of Henderson (1976) works without changes, provided (and this is important) that inbreeding for all individuals is computed previously. This is opposite to Christensen (2012), who had to devise modifications of the algorithm.

Inbreeding coefficients can be computed by Emik and Terrill (1949) or, equivalently, using recursion (Karigl 1981; Aguilar and Misztal 2008). However, efficient algorithms for computation of inbreeding use Henderson's (1976) decomposition of the numerator relationship matrix. These algorithms (*e.g.*, Quaas 1976; Meuwissen and Luo 1992) proceed by computing the variance of the Mendelian sampling term, D_{ii}. Meuwissen and Luo (1992) presented one rule,

$$D_{ii} = 0.5 - 0.25(F_s + F_d),$$

where, in the case of unknown ancestor, $s = 0$ (or $d = 0$), their programming set $F_0 = -1$. The same rule for computation of D_{ii} applies to the pedigree with one metafounder in Figure 3, by setting $F_1 = \gamma - 1$. In fact, the Meuwissen and Luo (1992) algorithm can be understood as having one

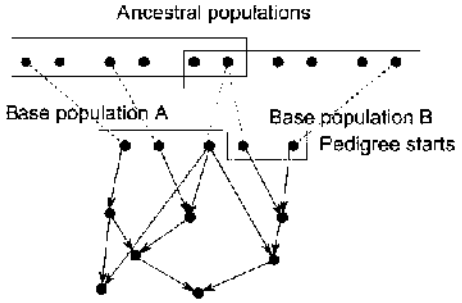


Figure 4 Several related base populations.

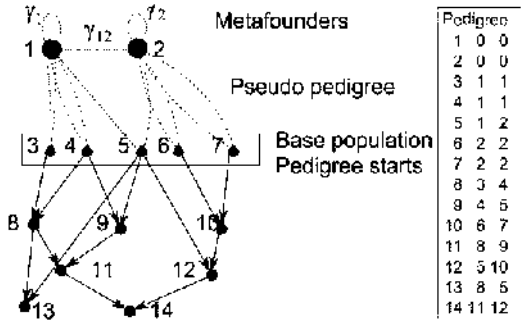


Figure 5 Population with two related metafounders 1 and 2, self-relationship coefficients $\boldsymbol{\gamma}^1$, $\boldsymbol{\gamma}^2$, and relationship coefficient $\boldsymbol{\gamma}^{1,2}$ and associated pedigree.

metafounder with $\gamma = 0$. Finally, the algorithm of Colleau (2002) for fast multiplication of matrix **A** with vector **x**, **Ax**, or extraction of elements of **A** also works.

Multiple base populations

Across-population relationships: An important case is the analysis of several populations at the same time, possibly with crosses. The conceptual model can easily be extended to several base populations, possibly with overlap as represented in Figure 4. In this case, we need to define within- and across-population relationships

$$\boldsymbol{\Gamma} = \begin{pmatrix} \gamma^{\mathrm{A}} & \gamma^{\mathrm{A,B}} \\ \text{symm} & \gamma^{\mathrm{B}} \\ & \cdots \end{pmatrix}.$$

This was suggested by VanRaden (1992) and used by VanRaden *et al.* (2011). The interpretation of the across-base population coefficients like $\gamma^{\mathrm{A,B}}$ is that the ancestor populations overlap, as seen in Figure 4. If population A is composed of n_{A} gametes, population B of n_{B} gametes, and they overlap to an extent of n_{AB} gametes (for instance, in Figure 4 these are 6, 6, and 2, respectively), then $\gamma^{\mathrm{A}} = 1/n_{\mathrm{A}}$, $\gamma^{\mathrm{B}} = 1/n_{\mathrm{B}}$, and $\gamma^{\mathrm{A,B}} = n_{\mathrm{AB}}/n_{\mathrm{A}}n_{\mathrm{B}}$. The last result can be explained as follows: $\gamma^{\mathrm{A,B}}$ is the probability that the gamete from A comes from the overlap $(n_{\mathrm{AB}}/n_{\mathrm{A}})$, times the probability that the gamete from B comes from the overlap $(n_{\mathrm{AB}}/n_{\mathrm{B}})$, times the probability that both gametes are actually the same, given that they come from the overlap $(1/n_{\mathrm{AB}})$. We allow values of γ^{A}, γ^{B}, and $\gamma^{\mathrm{A,B}}$ in a continuous range, even though the formulas only support values corresponding to integer values of n_{A}, n_{B} and n_{AB}. We also allow $\gamma^{\mathrm{A,B}}$ to potentially be negative, in order to consider the situation where populations have diverged due to selection in opposite directions. However, there is the restriction that the matrix $\boldsymbol{\Gamma}$ should be positive definite.

Metafounders: The consideration of each ancestral population as a metafounder is straightforward. Metafounders would be related by relationships

$$\boldsymbol{\Gamma} = \begin{pmatrix} \gamma^{\mathrm{A}} & \gamma^{\mathrm{A,B}} \\ & \gamma^{\mathrm{B}} \\ & \cdots \end{pmatrix}$$

(Figure 5). Actual numbers for the relationships within and across metafounders in $\boldsymbol{\Gamma}$ either can come from knowledge of the history of the populations (*i.e.*, they diverged so many generations ago) or can be inferred from genomic relationships; this is detailed later.

Algorithms for relationships and inbreeding with several metafounders: A pedigree with several metafounders defines a relationship matrix $\mathbf{A}^{\boldsymbol{\Gamma}}$. Algorithms for creation of this matrix are extensions of previous ones. To form $\mathbf{A}^{\boldsymbol{\Gamma}}$ using the tabular rules (Emik and Terrill 1949), the first step is to set $\boldsymbol{\Gamma}$ as relationships of the metafounders and then apply the regular rules. Rules for the inverse $\mathbf{A}^{\boldsymbol{\Gamma}-1}$ consist in, first, inverting $\boldsymbol{\Gamma}$ to create a small submatrix of $\boldsymbol{A}^{\boldsymbol{\Gamma}-1}$ and then using Henderson's rules (1976) with the elements D_{ii} for all individuals modified according to self-relationships of metafounders, as in the previous section. Using generalized inverses for inversion of $\boldsymbol{\Gamma}$ results in an algorithm that, for $\boldsymbol{\Gamma} = 0$, gives the same $\boldsymbol{A}^{\boldsymbol{\Gamma}-1}$ as with unknown parent groups as in Thompson (1979) or Quaas (1988). The reason for this is that the generalized inverse of $\boldsymbol{\Gamma} = 0$ is 0, and otherwise the rules for inversion and the values of D_{ii} are identical. This shows that metafounders are a generalization of unknown parent groups.

Computing D_{ii} involves computation of inbreeding coefficients, which can be done by recursion or modifying Meuwissen and Luo (1992). The Meuwissen and Luo (1992) algorithm goes up the ancestors of a given animal i and adds contributions $L_{ij}D_{jj}$ to the inbreeding coefficient of i; then animal j is deleted from the list of ancestors, and L_{ij} is set to zero. However, this does not work in the particular case of a crossbred individual issued directly from two related metafounders, *i.e.*, an F1 crossbred individual with unknown parents. This is a case that does sometimes exist, *e.g.*, in sheep and cattle. In this case, the contribution from the metafounders to A_{ii} is a sum over all metafounders $\sum_{k=1,\mathrm{nmf}}(\boldsymbol{L}_{i,k}\mathbf{K}_{:,k})^2$, where $\mathbf{K}_{:,k}$ is the kth column of $\mathbf{K}$, the lower triangular Cholesky decomposition of $\boldsymbol{\Gamma} = \mathbf{K}\mathbf{K}'$, and nmf is the number of metafounders. Therefore, in the case of several metafounders, their contributions need to be processed for simultaneously. The core modification for the Meuwissen and Luo code is

$\mathrm{K} = \mathrm{lower_choleski(gamma)}$

for (i in metafounders){ F(i) = 1 − gamma(i)}

for (i in (metafounders + 1, all animals)){

for (j in ancestors(i)){

...

if (j not in metafounders) then

Add $L_{ij}D_{jj}$ to A_{ii}

$L_{ij} = 0$

endif

}

Add $\sum\left(\left(\boldsymbol{L}_{\mathrm{i,1:nmf}}\mathbf{K}\right)^2\right)$ to A_{ii}

$L_{i,1:\mathrm{nmf}} = 0$

}

Finally, the algorithm of Colleau (2002) to efficiently compute products $\mathbf{A}^{\boldsymbol{\Gamma}}\mathbf{x}$ as $\mathbf{LDL'x}$ multiplies the result of $\boldsymbol{L}'x$ by $\boldsymbol{D}$, which has an upper diagonal block equal to $\boldsymbol{\Gamma}$ but that is diagonal otherwise. A complete code is furnished in Supporting Information, File S2.

Genetic variance considering related base populations

Single base population: The additive genetic variance is the variance of the breeding values of the set of individuals constituting a population. This definition does not involve a notion of (un)relatedness in itself. However, in the base population, these individuals are typically assumed unrelated, which simplifies the reasoning. A question is how to relate the genetic variance of a population modeled as "related" to the genetic variance of a population modeled as "unrelated." The breeding value is defined as relative to the average of the population. For this reason, any statistical model relating phenotypes to breeding values is forced to include an overall mean or an environmental effect confounded with it. A typical model for the phenotype can be written as

$$\mathbf{y} = 1\mu + \mathbf{u} + \mathbf{e}.$$

We follow the argument of Strandén and Christensen (2011), but for the sake of discussion, consider the mean as a random variable with variance σ^2_μ. The covariance of $\mathbf{y}$ is, for the classical model with unrelated base animals, $\mathrm{Var}(\mathbf{y}) = \mathbf{J}\sigma^2_\mu + \mathbf{A}\sigma^2_{u-\mathrm{unrelated}} + \mathbf{R}$, where $\mathrm{Var}(\mathbf{e}) = \mathbf{R}$. As for the new model with related base animals $\mathrm{Var}(\mathbf{y}) = \mathbf{J}\sigma^2_{\mu^*} + \mathbf{J}\gamma\sigma^2_{u-\mathrm{related}} + \mathbf{A}(1-\gamma/2)\sigma^2_{u-\mathrm{related}} + \mathbf{R}$. Two equivalent models (with equivalent likelihoods under multivariate normality) should have the same covariance for $\mathbf{y}$ and therefore

$$\sigma^2_{u-\mathrm{related}} = \frac{\sigma^2_{u-\mathrm{unrelated}}}{1-\gamma/2}$$

and $\sigma^2_\mu = \sigma^2_{\mu^*} + \gamma\sigma^2_{u-\mathrm{related}}$. In other words, the general across-individual covariance γ is absorbed by the overall mean (and it will be the case even if the mean is considered as a "fixed" effect; Strandén and Christensen 2011). An intuitive explanation is that, when sampling a finite number of animals from a population, animals will tend to be related and therefore the mean will drift from zero; but this drift of the mean will be accounted for by the general mean of the model. The expression above agrees with the numerical results in Christensen (2012).

This result looks puzzling because it suggests that an "inbred" population has higher genetic variance than a non-inbred one, but this is not actually the case. The parameter $\sigma^2_{u-\mathrm{related}}$ has to be interpreted as a parameter of the statistical linear model used for the analysis, and it cannot be interpreted as a genetic variance within the population (whereas $\sigma^2_{u-\mathrm{unrelated}}$ can be). In fact, the $\sigma^2_{u-\mathrm{related}}$ would be genetic variance in their hypothetical unrelated ancestral monoecious parents, and it would be reduced to $\sigma^2_{u-\mathrm{unrelated}}$ assuming a rate of inbreeding $\gamma/2$ from parents to offspring, as relatedness γ decreases the genetic variance within a population. Thus, the genetic variance *within* the population is always $\sigma^2_{u-\mathrm{unrelated}}$, and the variance component associated to the linear model is $\sigma^2_{u-\mathrm{related}}$. Along the same line, genetic gain in the "related" base population is not proportional to $\sigma^2_{u-\mathrm{related}}$ (because when selecting individuals, they will be related) but to $\sigma^2_{u-\mathrm{unrelated}}$.

Multiple base populations: The reasoning extends to the case with several populations but no crosses. For simplicity, we consider only two purebred populations. For breeds $b = \mathrm{A,B}$ the model for phenotypes is

$$\mathbf{y^b} = \mathbf{1}\mu^b + \mathbf{u^b},$$

where the variance–covariance matrix of the combined vector of breeding values is

$$\mathrm{var}\begin{pmatrix}\mathbf{u}^{\mathrm{A}}\\ \mathbf{u}^{\mathrm{B}}\end{pmatrix} = \sigma^2\begin{pmatrix}\mathbf{A}_{\mathrm{A,A}}\left(1-\frac{\gamma^{\mathrm{A}}}{2}\right) & 0\\ 0 & \mathbf{A}_{\mathrm{B,B}}\left(1-\frac{\gamma^{\mathrm{B}}}{2}\right)\end{pmatrix} + \sigma^2\begin{pmatrix}\gamma^{\mathrm{A}}\mathbf{J}_{\mathrm{AA}} & \gamma^{\mathrm{A,B}}\mathbf{J}_{\mathrm{AB}}\\ \gamma^{\mathrm{A,B}}\mathbf{J}_{\mathrm{BA}} & \gamma^{\mathrm{B}}\mathbf{J}_{\mathrm{BB}}\end{pmatrix},$$

with $\mathbf{A}_{b,b}$ being the relationship matrix of breed $b = A,B$ and $\mathbf{J}_{AA}$, $\mathbf{J}_{AB}$, $\mathbf{J}_{BA}, \mathbf{J}_{BB}$ being matrices consisting of 1's. Therefore, the vector of breeding values can be expressed as

$$\begin{pmatrix}\mathbf{u}^{\mathrm{A}}\\ \mathbf{u}^{\mathrm{B}}\end{pmatrix}=\begin{pmatrix}\mathbf{u}_u^{\mathrm{A}}\left(1-\frac{\gamma^{\mathrm{A}}}{2}\right)^{0.5}\\ \mathbf{u}_u^{\mathrm{B}}\left(1-\frac{\gamma^{\mathrm{B}}}{2}\right)^{0.5}\end{pmatrix}+\begin{pmatrix}\beta_{\mathrm{A}}1_{n_1}\\ \beta_{\mathrm{B}}1_{n_2}\end{pmatrix},$$

where subindex u on breeding values denotes that they are in the model with unrelated base populations, and

$$\begin{pmatrix}\beta_{\mathrm{A}}\\ \beta_{\mathrm{B}}\end{pmatrix}\sim N\left(\begin{pmatrix}0\\ 0\end{pmatrix},\sigma^2\begin{pmatrix}\gamma^{\mathrm{A}} & \gamma^{\mathrm{A,B}}\\ \gamma^{\mathrm{A,B}} & \gamma^{\mathrm{B}}\end{pmatrix}\right)$$

and assumed independence of breeding values $\mathbf{u}_u^{\mathrm{A}}$ and $\mathbf{u}_u^{\mathrm{B}}$. By an argument similar to above (*i.e.*, Stranden and Christensen 2011), the parameters β_{A} and β_{B} are absorbed into the two general mean parameters μ^{A} and μ^{B}, respectively. Therefore, the two models are equivalent in the sense that genetic variance parameters are just scaled by $(1-\gamma^b/2)$ and breeding values are just scaled and shifted. This model implies that phenotypes are separate by population and a mean (or distinct levels of fixed effects, *e.g.*, herds) has to be fit by population. The argument above is not difficult to generalize to any number of populations, as far as crosses that do not exist.

Multiple base populations with crossing: For crossbred populations the equivalence above does not hold because γ^{AB} enters into the covariances across individuals. A different approximate equivalence of variances can be constructed as follows. Assume a set of n base population individuals (n is assumed large) drawn from each of m populations. Let the genetic values of the across-breed base populations be $\mathbf{u}_0=\begin{pmatrix}\mathbf{u}_{\mathrm{A}0}\\ \mathbf{u}_{\mathrm{B}0}\end{pmatrix}$. The variance–covariance matrix is

$$\mathrm{Var}(\mathbf{u}_0)=\begin{pmatrix}1+\frac{\gamma^{\mathrm{A}}}{2} & \gamma^{\mathrm{A}} & \cdots & \gamma^{\mathrm{AB}} & \gamma^{\mathrm{AB}} & \cdots\\ \gamma^{\mathrm{A}} & 1+\frac{\gamma^{\mathrm{A}}}{2} & \cdots & \gamma^{\mathrm{AB}} & \gamma^{\mathrm{AB}} & \cdots\\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots\\ \gamma^{\mathrm{AB}} & \gamma^{\mathrm{AB}} & \cdots & 1+\frac{\gamma^{\mathrm{B}}}{2} & \gamma^{\mathrm{B}} & \cdots\\ \gamma^{\mathrm{AB}} & \gamma^{\mathrm{AB}} & \cdots & \gamma^{\mathrm{B}} & 1+\frac{\gamma^{\mathrm{B}}}{2} & \cdots\\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots\end{pmatrix}\sigma^2_{\mathrm{related}}.$$

The sample variance of $\mathbf{u}_0$, across all populations, is

$$S_u^2=\frac{\mathbf{u}_0{}'\mathbf{u}_0}{nm}-\overline{\mathbf{u}_0}^2,$$

which, for $\mathrm{Var}(\mathbf{u}_0)=\mathbf{K}\sigma_u^2$ (σ_u^2 is a parameter), has expectation (Searle 1982, p. 355)

$$E(S_u^2)=\left(\overline{\mathrm{diag}(\boldsymbol{K})}-\overline{\boldsymbol{K}}\right)\sigma_u^2.$$

In the classical parameterization (unrelated founders) $\mathbf{K}=\mathbf{I}$ and thus

$$E(S_u^2)=\left(\overline{\mathrm{diag}(\mathbf{K})}-\overline{\mathbf{K}}\right)\sigma_u^2=\left(1-\frac{1}{nm}\right)\sigma^2_{\mathrm{unrelated}},$$

which is equal to $\sigma^2_{\mathrm{unrelated}}$ if the population is reasonably large (a popular assumption) and therefore $\sigma_u^2=\sigma^2_{\mathrm{unrelated}}$ if founders are unrelated. This means that when the founders are unrelated, the genetic variance is, on expectation, equal to the variance component of the covariance structure.

Consider now the structure above for $\mathrm{Var}(\mathbf{u}_0)$. The two terms are equal to

$$\overline{\mathrm{diag}(\boldsymbol{K})}=1+\frac{\overline{\mathrm{diag}(\boldsymbol{\Gamma})}}{2}$$

$$\begin{aligned}\overline{\boldsymbol{K}}&=\frac{\left(n^2m^2\overline{\boldsymbol{\Gamma}}+nm-nm\left(\overline{\mathrm{diag}(\boldsymbol{\Gamma})}/2\right)\right)}{(nm)^2}\\&=\overline{\boldsymbol{\Gamma}}+\frac{1-\overline{\mathrm{diag}(\boldsymbol{\Gamma})}/2}{nm}\end{aligned}$$

$$\begin{aligned}E(S_u^2)&=\left(\overline{\mathrm{diag}(\boldsymbol{K})}-\overline{\boldsymbol{K}}\right)\sigma_u^2\\&=\left(1+\frac{\overline{\mathrm{diag}(\boldsymbol{\Gamma})}}{2}-\overline{\boldsymbol{\Gamma}}-\frac{1-\overline{\mathrm{diag}(\boldsymbol{\Gamma})}/2}{nm}\right)\sigma^2_{\mathrm{related}}\end{aligned}$$

in which we neglect the last term. This means that the genetic variance is, on expectation, equal to the variance component $\sigma^2_{\mathrm{related}}$ times a constant $(1+\overline{\mathrm{diag}(\boldsymbol{\Gamma})}/2-\overline{\boldsymbol{\Gamma}})$, which is <1. Equating these two expressions for $E(S_u^2)$ gives

$$\sigma^2_{\mathrm{related}}\approx\frac{\sigma^2_{\mathrm{unrelated}}}{\left(1+\overline{\mathrm{diag}(\boldsymbol{\Gamma})}/2-\overline{\boldsymbol{\Gamma}}\right)}.$$

This expression gives the previous result $\sigma^2_{\mathrm{related}}=\sigma^2_{\mathrm{unrelated}}/(1-\gamma/2)$ for a single population. Compared to the result in the previous subsection about multiple populations, this approximate equivalence is quite different. The result in the previous subsection is an equivalence between one genetic variance in a model with related base individuals and breed-specific genetic variances in a model with unrelated base individuals, whereas the result here is an approximate equivalence between two genetic variances, one being in a related base population and another being in an unrelated base population. This last expression $\sigma^2_{\mathrm{related}}\approx\sigma^2_{\mathrm{unrelated}}/\left[\left(1+\overline{\mathrm{diag}(\boldsymbol{\Gamma})}/2-\overline{\boldsymbol{\Gamma}}\right)\right]$ is more general because it can consider correctly crosses across individuals. The difference comes also because in the previous expression there were separate means for each population, something that is not required here.

Segregation variance: When crossing pure breeds, there is an increase of genetic variance due to the increase of heterozygosity of the QTL; for instance, if alternative alleles are fixed at each line. The additional variance in the F2 cross compared to the variance in the F1 cross is termed segregation variance (Lande 1981; Lo *et al.* 1993). This is typically

ignored in a classical framework, although methods exist (Lo *et al.* 1993; Garcia-Cortes and Toro 2006). This increase in the genetic variance can be considered using related metafounders, as we show here. Two individuals in an F1 population (assuming—in a pedigree sense—unrelated and non-inbred parents, and factorizing out $\sigma^2_{\text{related}}$) have

$$\text{Var}(\boldsymbol{u}_{\text{AB}}) = \begin{pmatrix} 1+\frac{\gamma^{\text{AB}}}{2} & \frac{\gamma^{\text{A}}}{4}+\frac{\gamma^{\text{B}}}{4}+\frac{\gamma^{\text{AB}}}{2} \\ \frac{\gamma^{\text{A}}}{4}+\frac{\gamma^{\text{B}}}{4}+\frac{\gamma^{\text{AB}}}{2} & 1+\frac{\gamma^{\text{AB}}}{2} \end{pmatrix}$$

whereas two individuals in an F2 population (parents in F1 above) have

$$\begin{aligned}\text{Var}(\mathbf{u}_{\text{AB}\times\text{AB}}) &= \begin{pmatrix} 1+\frac{\gamma^{m}}{4}+\frac{\gamma^{\text{AB}}}{4} & \frac{\gamma^{m}}{2}+\frac{\gamma^{\text{AB}}}{2} \\ \frac{\gamma^{m}}{2}+\frac{\gamma^{\text{AB}}}{2} & 1+\frac{\gamma^{m}}{4}+\frac{\gamma^{\text{AB}}}{4} \end{pmatrix} \\ &= \begin{pmatrix} 1+\frac{\gamma^{\text{F2}}}{2} & \gamma^{\text{F2}} \\ \gamma^{\text{F2}} & 1+\frac{\gamma^{\text{F2}}}{2} \end{pmatrix},\end{aligned}$$

with $\gamma^m = (\gamma^{\text{A}}+\gamma^{\text{B}})/2$ and $\gamma^{\text{F2}} = (\gamma^{\text{A}}+\gamma^{\text{B}})/4+\gamma^{\text{AB}}/2$. The γ^{F2} is transmitted forward and does not change in the F3, F4, etc. The genetic variance of such a population is thus $1-\gamma^{\text{F2}}/2 = 1-[(\gamma^{\text{A}}+\gamma^{\text{B}})/4-\gamma^{\text{AB}}/2]/2$. The variance–covariance matrix of two individuals in the F2 can be expressed as

$$\begin{aligned}\text{Var}(\mathbf{u}_{\text{AB}\times\text{AB}}) &= \text{Var}(\mathbf{u}_{\text{AB}}) \\ &\quad + \begin{pmatrix} (\gamma^{m}-\gamma^{\text{AB}})/4 & 0 \\ 0 & (\gamma^{m}-\gamma^{\text{AB}})/4 \end{pmatrix}\end{aligned}$$

showing that the segregation variance is $(\gamma^m-\gamma^{\text{AB}})/4$. Because $\boldsymbol{\Gamma}$ is positive definite, then this term must be ≥ 0. Slatkin and Lande (1994) showed that segregation variance is a function of within-loci squared differences of means at the two breeds, plus cross-products of differences across loci weighted by linkage. If γ^{AB} is estimated using markers as above, then it is implicitly assumed that genotypes at loci for the trait of interest have the same distribution across breeds and within the genome as marker genotypes. Reports of segregation variances in the livestock genetics literature are scarce (*e.g.*, Cardoso and Tempelman 2004; Munilla-Leguizamon and Cantet 2011), partly because of poor data sets, partly because of computational difficulties, and partly because the bulk of crossbred animals is in poultry and swine, where crosses do not go beyond F1 populations. So it is uncertain whether accounting for segregation variance is of any practical relevance.

Estimation of metafounders ancestral relationships from genomic data

Because the within- and across-founder relationships cannot be inferred from pedigree, we suggest estimating these relationships using molecular markers, referring them to a genetic base defined according to genomic relationships (Christensen 2012). The objective of this section is to obtain estimators of $\boldsymbol{\Gamma}$ based on two kinds of statistical inference: a method of maximum likelihood and a method of moments (roughly, make first- and second-order statistics of genomic and pedigree relationships comparable).

Maximum likelihood: Genomic information sheds light on relationships across breeds (Gibbs *et al.* 2009; Kijaas *et al.* 2009; VanRaden *et al.* 2011; Legarra *et al.* 2014a). Genomic relationships (VanRaden 2008; Hayes *et al.* 2009) are estimators of relatedness based on the observation of thousands of molecular markers, and typically matrix $\mathbf{G} = \mathbf{Z}\mathbf{Z}'/s$ is used, where $\mathbf{Z}$ contains centered genotypes and s is a measure of global heterozygosity, for instance, $s = 2\sum p_i q_i$, the total heterozygosity at the markers. This information can in principle be used to infer the $\boldsymbol{\Gamma}$ coefficients as follows. Marker genotypes follow Mendelian transmission, and therefore the covariance of genotypes of two individuals is determined by their relationship. Christensen (2012) used this to estimate γ in a single population. First, he integrated the likelihood over the unknown allelic frequencies, which results in using allelic frequencies of 0.5 as a reference ($\mathbf{Z}$ coded as $\{-1, 0, 1\}$). Assuming multivariate normality for $\mathbf{Z}$, the markers' genotype, the likelihood of observed genotypes conditional to γ and s is

$$\begin{aligned}\log p(\mathbf{Z}|\gamma, s) = \text{const} &- \frac{pn_2}{2}\log(s) - \frac{p}{2}\log\left(\left|\mathbf{A}^{\gamma}_{22}\right|\right) \\ &- \frac{p}{2s}\text{tr}\left(\mathbf{A}^{\gamma-1}_{22}\mathbf{Z}\mathbf{Z}'\right),\end{aligned}$$

where n_2 is the number of genotyped individuals and $\mathbf{A}^{\gamma}_{22}$ is the submatrix of $\mathbf{A}^{\gamma}$ corresponding to the genotyped individuals. The parameter s is a measure of heterozygosity in the genotyped population, and it is *not* equal to observed $2\sum p_i q_i$. The extension of this likelihood to multiple populations with different γ's in $\boldsymbol{\Gamma}$ is straightforward

$$\begin{aligned}\log p(\boldsymbol{Z}|\boldsymbol{\Gamma}, s) = \text{const} &- \frac{pn_2}{2}\log(s) - \frac{p}{2}\log\left(\left|\mathbf{A}^{\boldsymbol{\Gamma}}_{22}\right|\right) \\ &- \frac{p}{2s}tr\left(\mathbf{A}^{\boldsymbol{\Gamma}-1}_{22}\boldsymbol{Z}\boldsymbol{Z}'\right),\end{aligned}$$

where $\mathbf{A}^{\boldsymbol{\Gamma}}$ is the relationship matrix constructed with a given $\boldsymbol{\Gamma}$ matrix and $\mathbf{A}^{\boldsymbol{\Gamma}}_{22}$ is the submatrix corresponding to the genotyped individuals. This likelihood can be factorized by markers as

$$\begin{aligned}\log p(\mathbf{Z}|\boldsymbol{\Gamma}, s) = \text{const} &- \frac{pn_2}{2}\log(s) - \frac{p}{2}\log\left(\left|\mathbf{A}^{\boldsymbol{\Gamma}}_{22}\right|\right) \\ &- \frac{p}{2s}\sum_{i=1}^{nsnp}\mathbf{z}_i'\mathbf{A}^{\boldsymbol{\Gamma}-1}_{22}\mathbf{z}_i.\end{aligned}$$

The procedure can be completed by adding a prior distribution to $\boldsymbol{\Gamma}$ and using a Bayesian estimator instead of maximum likelihood. The prior distribution for $\boldsymbol{\Gamma}$ can be assigned

based on spatial or temporal distances; for instance, Latxa sheep founders in 1990 and 1992 should be closer than 1990 and 2000. However, in none of these forms of the likelihood can $\boldsymbol{\Gamma}$ be factorized out, and the maximization of the likelihood needs to be done by a search method such as Simplex or Monte Carlo methods. For this reason, we present a method based on summary statistics.

Method of moments based on summary statistics: This method matches summary statistics of across-individual and within-individual relationships in both $\mathbf{A}_{22}^{\boldsymbol{\Gamma}}$ (the matrix of extended pedigree-based relationships) and $\mathbf{G}$ (VanRaden *et al.* 2011; Vitezica *et al.* 2011; Christensen *et al.* 2012). This forces the equivalence between expected changes of the mean and variance under genetic drift (Vitezica *et al.* 2011; Christensen *et al.* 2012) for the populations described by either the pedigree or the genomic relationship matrices. For a set of n random variables $\mathbf{u}$ with variance–covariance matrix $\mathbf{K}$, the sample average $\overline{\mathbf{u}} = \mathbf{1}'\mathbf{u}/n$ has a variance $\mathrm{Var}(\overline{\mathbf{u}}) = \overline{\mathbf{K}}$, whereas the sample variance $S_u^2 = \mathbf{u}'\mathbf{u}/n - \overline{\mathbf{u}}^2$ has expectation $E(S_u^2) = \mathrm{tr}(\mathbf{K})/n - \overline{\mathbf{K}} = \overline{\mathrm{diag}(\mathbf{K})} - \overline{\mathbf{K}}$ (Searle, 1982, p. 355). The idea in the method is to force these two statistics of $\mathbf{K}$ ($\mathrm{Var}(\overline{\mathbf{u}})$ and $E(S_u^2)$) to be equivalent across both parameterizations ($\mathbf{K} = \mathbf{A}^{\boldsymbol{\Gamma}}$ and $\mathbf{K} = \mathbf{G}$). We consider three situations.

Single population: Two single unknowns need to be estimated: γ and s. Since $\boldsymbol{\gamma} = \mathbf{A}(1-\gamma/2) + \gamma\mathbf{J}$, the average of all elements is $\overline{\mathbf{A}_{22}^{\gamma}} = \overline{\mathbf{A}_{22}}(1-\gamma/2) + \gamma$, and the average of the diagonal is $\overline{\mathrm{diag}(\mathbf{A}_{22}^{\gamma})} = \overline{\mathrm{diag}(\mathbf{A}_{22})}(1+\gamma/2) + \gamma$, where $\mathbf{A}_{22}$ is the regular pedigree-relationship matrix for genotyped individuals. Therefore, a system of two equations needs to be set up,

$$\overline{\mathbf{A}_{22}}\left(1-\frac{\gamma}{2}\right) + \gamma = \overline{\mathbf{Z}\mathbf{Z}'}/s$$

$$\overline{\mathrm{diag}(\boldsymbol{A}_{22})}\left(1-\frac{\gamma}{2}\right) + \gamma = \overline{\mathrm{diag}(\mathbf{Z}\mathbf{Z}')}/s$$

with solutions

$$s = \frac{\overline{\mathrm{diag}(\mathbf{Z}\mathbf{Z}')}\left(1-\overline{\mathbf{A}_{22}}/2\right) - \overline{\mathbf{Z}\mathbf{Z}'}\left(1-\overline{\mathrm{diag}(\mathbf{A}_{22})}/2\right)}{\overline{\mathrm{diag}(\mathbf{A}_{22})} - \overline{\mathbf{A}_{22}}}$$

$$\gamma = \frac{\overline{\mathbf{Z}\mathbf{Z}'}/s - \overline{\mathbf{A}_{22}}}{1-\overline{\mathbf{A}_{22}}/2}.$$

These solutions have an interpretation in terms of measures of inbreeding in the population. In a population large enough and mating at random, inbreeding of the individuals is equal to half the relationships of their parents, $\overline{\mathrm{diag}(\mathbf{A}_{22})} = 1 + \overline{\mathbf{A}_{22}}/2 = 1 + \overline{F_{\mathrm{A}}}$ ($\overline{F_{\mathrm{A}}}$ is average pedigree inbreeding), and $\overline{\mathrm{diag}(\mathbf{Z}\mathbf{Z}')}/s = 1 + \overline{\mathbf{Z}\mathbf{Z}'}/2s = 1 + \overline{F_{\mathrm{G}}}$ ($\overline{F_{\mathrm{G}}}$ is average genomic inbreeding). Therefore, in this case $\gamma/2 = (\overline{F_{\mathrm{G}}} - \overline{F_{\mathrm{A}}})/(1-\overline{F_{\mathrm{A}}})$. This is basically the reverse of the expression derived by Vitezica *et al.* (2011), who adjusted $\mathbf{G}$ to match $\mathbf{A}$ and called $\gamma = \alpha$. The expression shows that γ is a correction for underestimation of inbreeding of $\mathbf{A}$ with respect to $\mathbf{G}$, following Wright's F coefficients theory. An advantage of the method is that it needs only statistics of the $\mathbf{A}_{22}$ and $\mathbf{G}$ matrices, which might be more available than full matrices.

Multiple pure populations: Assume that a sample from each pure breed is genotyped. Consider the purebred parts of $\mathbf{G}$ and $\mathbf{A}_{22}^{\boldsymbol{\Gamma}}$, for simplicity 2 breeds A and B:

$$\mathbf{G} = \begin{pmatrix} \mathbf{G}^{\mathrm{A,A}} & \mathbf{G}^{\mathrm{A,B}} \\ \mathbf{G}^{\mathrm{B,A}} & \mathbf{G}^{\mathrm{B,B}} \end{pmatrix} = \begin{pmatrix} \mathbf{Z}_{\mathrm{A}}\mathbf{Z}_{\mathrm{A}}' & \mathbf{Z}_{\mathrm{A}}\mathbf{Z}_{\mathrm{B}}' \\ \mathbf{Z}_{\mathrm{B}}\mathbf{Z}_{\mathrm{A}}' & \mathbf{Z}_{\mathrm{B}}\mathbf{Z}_{\mathrm{B}}' \end{pmatrix} \Big/ s$$

$$\begin{aligned} \boldsymbol{A}_{22}^{\boldsymbol{\Gamma}} &= \begin{pmatrix} \mathbf{A}_{22\mathrm{A,A}}^{\boldsymbol{\Gamma}} & \mathbf{A}_{22\mathrm{A,B}}^{\boldsymbol{\Gamma}} \\ \mathbf{A}_{22\mathrm{B,A}}^{\boldsymbol{\Gamma}} & \mathbf{A}_{22\mathrm{B,B}}^{\boldsymbol{\Gamma}} \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{A}_{22\mathrm{A,A}}\left(1-\frac{\gamma^{\mathrm{A}}}{2}\right) + \mathbf{J}\gamma^{\mathrm{A}} & \mathbf{J}\gamma^{\mathrm{A,B}} \\ \mathbf{J}\gamma^{\mathrm{A,B}} & \mathbf{A}_{22\mathrm{B,B}}\left(1-\frac{\gamma^{\mathrm{B}}}{2}\right) + \mathbf{J}\gamma^{\mathrm{B}} \end{pmatrix}. \end{aligned}$$

To meet the conditions of unbiasedness we need to force the equality of average diagonal and averages of $\mathbf{G}$ and $\mathbf{A}^{\boldsymbol{\Gamma}}$ and set up the four equations

$$\overline{\mathbf{A}_{22,\mathrm{A,A}}}\left(1-\frac{\gamma^{\mathrm{A}}}{2}\right) + \gamma^{\mathrm{A}} = \overline{\mathbf{Z}_{\mathbf{A}}\mathbf{Z}_{\mathbf{A}}'}/s$$

$$\overline{\mathbf{A}_{22,\mathrm{B,B}}}\left(1-\frac{\gamma^{\mathrm{B}}}{2}\right) + \gamma^{\mathrm{B}} = \overline{\mathbf{Z}_{\mathbf{B}}\mathbf{Z}_{\mathbf{B}}'}/s$$

$$\overline{\mathbf{A}_{22,\mathrm{A,B}}} + \gamma^{\mathrm{A,B}} = \overline{\mathbf{Z}_{\mathbf{A}}\mathbf{Z}_{\mathbf{B}}'}/s$$

$$\begin{aligned} &\overline{\mathrm{diag}(\mathbf{A}_{22,\mathrm{A,A}})}\left(1-\frac{\gamma^{\mathrm{A}}}{2}\right) + \gamma^{\mathrm{A}} + \overline{\mathrm{diag}(\mathbf{A}_{22,\mathrm{B,B}})}\left(1-\frac{\gamma^{\mathrm{B}}}{2}\right) + \gamma^{\mathrm{B}} \\ &\quad = \overline{\mathrm{diag}(\mathbf{Z}\mathbf{Z}')}/s. \end{aligned}$$

The solution is a generalization of the solutions for single populations. The scaling estimates for single populations are $s_{\mathrm{A}} = d_{\mathrm{A}}/m_{\mathrm{A}}$ and $s_{\mathrm{B}} = d_{\mathrm{B}}/m_{\mathrm{B}}$ with

$$\begin{aligned} d_{\mathrm{A}} = {} & \overline{\mathrm{diag}(\mathbf{Z}_{\mathbf{A}}\mathbf{Z}_{\mathbf{A}}')}\left(1-\overline{\mathbf{A}_{22,\mathrm{A,A}}}/2\right) \\ & - \overline{\mathbf{Z}_{\mathbf{A}}\mathbf{Z}_{\mathbf{A}}'}\left[1-\frac{\overline{\mathrm{diag}(\mathbf{A}_{22,\mathrm{A,A}})}}{2}\right] \end{aligned}$$

and $m_{\mathrm{A}} = \overline{\mathrm{diag}(\mathbf{A}_{22,\mathrm{A,A}})} - \overline{\mathbf{A}_{22,\mathrm{A,A}}}$, and d_{B} and m_{B} defined similarly. The solutions for two populations are

$$s = \frac{d_{\mathrm{A}}n_{\mathrm{A}} + d_{\mathrm{B}}n_{\mathrm{B}}}{m_{\mathrm{A}}n_{\mathrm{A}} + m_{\mathrm{B}}n_{\mathrm{B}}}$$

$$\gamma^{\mathrm{A}} = \frac{\overline{\mathbf{G}}_{\mathrm{A,A}} - \overline{\mathbf{A}}_{22\mathrm{A,A}}}{1-\overline{\mathbf{A}_{22,\mathrm{A,A}}}/2};\ \gamma^{\mathrm{B}} = \frac{\overline{\mathbf{G}}_{B,B} - \overline{\mathbf{A}}_{22\mathrm{B,B}}}{1-\overline{\mathbf{A}_{22,\mathrm{B,B}}}/2};\ \gamma^{\mathrm{A,B}} = \overline{\mathbf{G}}_{\mathrm{A,B}}$$

so that within-breed and across-breed average relationships agree. Assuming $\overline{\mathbf{A}_{22,A,A}}$ $\overline{\mathbf{A}_{22,B,B}}$ are close to zero, $\gamma^{A} = (\overline{\mathbf{G}}_{A,A} - \overline{\mathbf{A}}_{A,A})$, $\gamma^{B} = (\overline{\mathbf{G}}_{B,B} - \overline{\mathbf{A}}_{B,B})$, $\gamma^{A,B} = \overline{\mathbf{G}}_{A,B}$, which consist in setting $p = 0.5$ to construct the **G** matrices (VanRaden 2008) and then simply quantify average relationships across breeds. This is the simple method used by VanRaden *et al.* (2011), although they did not define scaling s as we have done. This reasoning can be extended to as many breeds as needed. Again, this method can be used from published statistics without access to raw data.

Populations with crosses: In some cases, pure populations may not be genotyped. For instance, Angus bulls may be mated to Limousine females and only the crossbreds and Angus genotyped. Another example is unknown parent groups (Quaas 1988), base populations that account for missing parentages. However, at some point descendants of these populations may be genotyped, and this information is usable. We propose an algorithm very similar to that of Harris and Johnson (2010). Let **Q** be a matrix containing in the i,j cell the expected fraction of metafounder j in the individual i (Quaas 1988). This matrix can be efficiently obtained using Colleau (2002), recursion, or tracing down the pedigree. The following identity, which is an extension of $\mathbf{A}^{\gamma} = \mathbf{A}(1-\gamma/2) + \gamma\mathbf{J}$, approximately holds

$$\mathbf{A}^{\boldsymbol{\Gamma}} \approx \mathbf{A}(\mathbf{I} - 0.5\ \mathrm{diag}(\mathbf{Q}\boldsymbol{\Gamma}\mathbf{Q}^{'})) + \mathbf{Q}\boldsymbol{\Gamma}\mathbf{Q}^{'}.$$

And therefore, $\mathbf{A}_{22}^{\boldsymbol{\Gamma}} \approx \mathbf{A}_{22}[\mathbf{I} - 0.5\ \mathrm{diag}(\mathbf{Q}_2\boldsymbol{\Gamma}\mathbf{Q}_2^{'})] + \mathbf{Q}_2\boldsymbol{\Gamma}\mathbf{Q}_2^{'}$. A linear model can be fit as $\mathbf{G} = \mathbf{A}_{22}^{\boldsymbol{\Gamma}} + \mathbf{E}$, where **E** is an error term and $\mathbf{Q}_2$ is the section of **Q** containing proportions of metafounders in genotyped individuals. We neglect the term $-0.5\ \mathrm{diag}(\mathbf{Q}_2\boldsymbol{\Gamma}\mathbf{Q}_2^{'})$, which is small with respect to the rest of elements and obtain a further approximation $\mathbf{G} = \mathbf{A}_{22} + \mathbf{Q}_2\boldsymbol{\Gamma}\mathbf{Q}_2^{'} + \mathbf{E}$, in which $\boldsymbol{\Gamma}$ is explicit. This expression can be linearized using the vec operator (Henderson and Searle 1979), and the least-squares estimator can be transformed back to a matrix form. This least-squares estimator of $\boldsymbol{\Gamma}$ is

$$\widehat{\boldsymbol{\Gamma}} = \left(\mathbf{Q}_2^{'}\mathbf{Q}_2\right)^{-1}\mathbf{Q}_2^{'}(\mathbf{G} - \mathbf{A}_{22})\mathbf{Q}_2\left(\mathbf{Q}_2^{'}\mathbf{Q}_2\right)^{-1}$$

using $\mathbf{G} = \mathbf{Z}\mathbf{Z}'/s$ and assuming that the value of s is known. If only pure population animals are genotyped, this is identical to the approximation of the estimator above for "pure populations." This solution for $\boldsymbol{\Gamma}$ is identical to the estimator proposed by Harris and Johnson (2010, Equations 13 and 14). As for s, one can use

$$s = \frac{\overline{\mathrm{diag}(\mathbf{Z}\mathbf{Z}^{'})} - \overline{\mathbf{Z}\mathbf{Z}^{'}}}{\overline{\mathrm{diag}(\mathbf{A}_{22}^{\boldsymbol{\Gamma}})} - \overline{\mathbf{A}_{22}^{\boldsymbol{\Gamma}}}},$$

where the approximation is used for $\mathbf{A}_{22}^{\boldsymbol{\Gamma}}$ [in this case including $-0.5\ \mathrm{diag}(\mathbf{Q}\boldsymbol{\Gamma}\mathbf{Q}^{'})$] such that $\overline{\mathrm{diag}(\mathbf{A}_{22}^{\boldsymbol{\Gamma}})}$ and $\overline{\mathbf{A}_{22}\boldsymbol{\Gamma}}$ are linear functions of $\boldsymbol{\Gamma}$. This system of two equations with two unknown is iterated until convergence. If there is little information for some metafounders (as is the case in ruminants), Bayesian estimation using a prior structure for $\boldsymbol{\Gamma}$ can be considered.

Combining pedigree relationships with metafounders and genomic relationships when not all individuals are genotyped

The SSGBLUP method for genomic evaluation (Aguilar *et al.* 2010; Christensen and Lund 2010; Legarra *et al.* 2014b) completes genomic information with pedigree-based information and in fact proceeds by correcting pedigree relationships in view of genomic relationships. Pedigree relationships are modified as (Legarra *et al.* 2009; Christensen and Lund 2010)

$$\mathbf{H} = \begin{pmatrix} \mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{A}_{22}^{-1}(\mathbf{G} - \mathbf{A}_{22})\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{G} \\ \mathbf{G}\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{G} \end{pmatrix},$$

where **H** is a matrix with relationships after including pedigree and genomic relationships, **G** is a matrix including genomic relationships for genotyped individuals ($\mathbf{u}_2$), which is projected upon relationships of ungenotyped animals ($\mathbf{u}_1$), **A** is the pedigree-based relationship matrix, and $\mathbf{A}_{22}$ is a relationship matrix across genotyped individuals. This joint matrix **H** can be understood as a linear imputation of genotypes over all nongenotyped individuals (Christensen and Lund 2010), considering also the uncertainty in the imputation. This covariance matrix is increasingly used in genomic predictions of genetic merit (Aguilar *et al.* 2010; Christensen *et al.* 2012) and also in QTL detection (Dikmen *et al.* 2013).

The algebraic development of matrix **H** assumes that base allelic frequencies are known or, equivalently, that mean and variance of the population do not change with time. This is notoriously false with small populations, deep pedigrees, or in presence of selection. Different adjustments had been suggested to modify genomic relationships so that their genetic base is the same as that of pedigree relationships (Vitezica *et al.* 2011; Christensen *et al.* 2012). This implicitly estimates the shift in breeding values (or allelic frequencies) from the pedigree base to the genotyped population (Vitezica *et al.* 2011). However, these adjustments do not consider the pedigree structure of the populations, and their generalizations to crosses of lines or breeds are neither completely satisfactory nor well understood (but see Harris and Johnson 2010; Makgahlela *et al.* 2014).

Christensen (2012) argued that, contrary to pedigree relationships, genomic relationships are independent of pedigree completeness and they should define the genetic base. He thus considered matching pedigree relationships to genomic relationships instead of the opposite. He showed that after marginalizing the allelic frequencies from the joint likelihood, the result was a related base population and suggested estimating γ and s using maximum likelihood. All our developments rely on this base and therefore, the extended pedigrees with metafounders do automatically conciliate marker and pedigree-based relationships, using estimates of $\boldsymbol{\Gamma}$ and s from markers. In particular, the inverse of the joint pedigree and markers relationship matrix is

$$\mathbf{H}^{\boldsymbol{\Gamma}-1} = \mathbf{A}^{\boldsymbol{\Gamma}-1} + \begin{pmatrix} 0 & 0 \\ 0 & \mathbf{G}^{-1} - \mathbf{A}_{22}^{\boldsymbol{\Gamma}-1} \end{pmatrix}.$$

This matrix can be fit into the mixed model equations of the SSGBLUP.

We have seen that the variance component assuming "related" founders is not the same as the genetic variance assuming "unrelated" founders; the latter is the one classically estimated and used. The most straightforward solution is to reestimate the variance using metafounders. Alternatively, to use current estimates of genetic variance in the implementation, the variance of the breeding values needs to be scaled to $\sigma^2_{\text{unrelated}}$. On expectation, the following equivalence holds:

$$\sigma^2_{\text{related}} = \sigma^2_{\text{unrelated}}/k, \quad \text{with } k = \left(1 + \frac{\overline{\text{diag}(\boldsymbol{\Gamma})}}{2} - \overline{\boldsymbol{\Gamma}}\right).$$

Thus

$$\text{Var}(\mathbf{u}) = \mathbf{H}^{\boldsymbol{\Gamma}}\sigma^2_{\text{related}} = \mathbf{H}^{\boldsymbol{\Gamma}}\frac{\sigma^2_{\text{unrelated}}}{k}$$

and

$$\text{Var}(\mathbf{u})^{-1} = k\mathbf{H}^{\boldsymbol{\Gamma}-1}\sigma^{-2}_{\text{unrelated}} = k\mathbf{A}^{\boldsymbol{\Gamma}-1}\sigma^{-2}_{\text{unrelated}} + \begin{pmatrix} 0 & 0 \\ 0 & k\mathbf{G}^{-1} - k\mathbf{A}_{22}^{\boldsymbol{\Gamma}-1} \end{pmatrix}\sigma^{-2}_{\text{unrelated}},$$

such that the inverse of the combined relationship matrix ($\mathbf{H}^{\boldsymbol{\Gamma}-1}$) can be multiplied by a single scalar, $k = (1 + \overline{\text{diag}(\boldsymbol{\Gamma})}/2 - \overline{\boldsymbol{\Gamma}})$.

Examples

Example 1: How pedigree relationships are modified

Consider the pedigree in Figure 4 and the relationships between the subset of individuals 8 (pure breed A), 10 (pure breed B) , and 14 (crossbred, 56% breed A and 44% breed B, grandson of 8 and of 10). Regular relationships ($\mathbf{A}_{\text{subset}}$) are

$$\mathbf{A}_{\text{subset}} = \begin{pmatrix} 1 & 0 & 0.31 \\ 0 & 1 & 0.25 \\ 0.31 & 0.25 & 1.06 \end{pmatrix}.$$

Consider now $\boldsymbol{\Gamma} = \begin{pmatrix} 0.1 & 0 \\ 0 & 0.2 \end{pmatrix}$. Then

$$\mathbf{A}_{\text{subset}} = \begin{pmatrix} 1.05 & 0 & 0.35 \\ 0 & 1.03 & 0.31 \\ 0.35 & 0.31 & 1.08 \end{pmatrix}.$$

All within-breed relationships have increased, because each base population is now assumed self-related. However, animals 8 and 10 are unrelated. Considering across-base population relationships in $\boldsymbol{\Gamma} = \begin{pmatrix} 0.1 & 0.05 \\ 0.05 & 0.2 \end{pmatrix}$ gives

$$\mathbf{A}_{\text{subset}} = \begin{pmatrix} 1.05 & 0.05 & 0.37 \\ 0.05 & 1.10 & 0.34 \\ 0.37 & 0.34 & 1.09 \end{pmatrix},$$

where the relationship between 8 and 10 appears, which in turn slightly increases the inbreeding coefficient of 14. To standardize to the genetic variance estimated assuming unrelated base individuals, $\mathbf{A}_{\text{subset}}$ must be divided by $(1 + \overline{\text{diag}(\boldsymbol{\Gamma})}/2 - \overline{\boldsymbol{\Gamma}}) = 0.975$.

Example 2: Interpretation of γ in a single population

Legarra *et al.* (2014a) used dairy sheep data (Manech Tête Rousse) for genomic prediction including 38,287 markers and 1295 rams. The relevant statistics are (observed) $2\sum p_i q_i = 14771$, $\overline{\text{diag}(\mathbf{ZZ}')} = 22798$, $\overline{\mathbf{ZZ}'} = 8654$, $\overline{\text{diag}(\mathbf{A}_{22})} = 1.011$, $\overline{\mathbf{A}_{22}} = 0.04$. Using the single population method above yields $\gamma = 0.434$, $s = 18602$. What do these numbers mean? They imply that heterozygosity of markers at the base population should have been $s = 18602$ (instead of observed 14771), to appropriately match the fact that the heterozygosity at the markers reduced from the base to the observed population, according to inbreeding observed in the pedigree. Based on this estimate, average genomic inbreeding is $1 - \overline{\text{diag}(\mathbf{ZZ}')}/s = 0.22$, which can be achieved with an effective size of the founder population $N_e = 1/0.43$ and therefore $\gamma = 0.43$. Although this effective size is very small, it refers to a reference with allelic frequencies equal to 0.5. This has to be taken as a reference point for the linear model and has no clear biological meaning.

Example 3: Numerical example of two breeds and crossbred individuals

VanRaden *et al.* (2011) estimated relationship coefficients across Jersey, Holstein, and Brown Swiss using 43,385 markers. Based on their published statistics and using the method based on summary statistics outlined above, we obtained an estimate of $\boldsymbol{\Gamma} = \begin{pmatrix} 0.55 & 0.48 \\ 0.48 & 0.77 \end{pmatrix}$ for Holstein and Jersey. Assuming the pedigree in Figure 5, we constructed $\mathbf{A}^{-1}$ using $\boldsymbol{\Gamma} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, which is equivalent to use of regular unknown parent group rules (Quaas 1988) and we also constructed $\mathbf{A}^{\Gamma-1}$ as described before with $\boldsymbol{\Gamma} = \begin{pmatrix} 0.55 & 0.48 \\ 0.48 & 0.77 \end{pmatrix}$; we scaled $\mathbf{A}^{\Gamma-1}$ to refer to the same regular genetic variance multiplying it by the constant $k = 1 + \overline{\text{diag}(\boldsymbol{\Gamma})}/2 - \overline{\boldsymbol{\Gamma}}$. Results are shown in Figure 6.

It can be observed that the sparsity pattern does not change, except for the nonnull values across metafounders. Also, the numbers do not change greatly but diagonal elements are higher because there is shrinkage associated to the metafounders, which is not the case for regular unknown parent groups.

Consider now the variance in the hypothetical crossed Holstein–Jersey individuals. The segregation variance is, by the

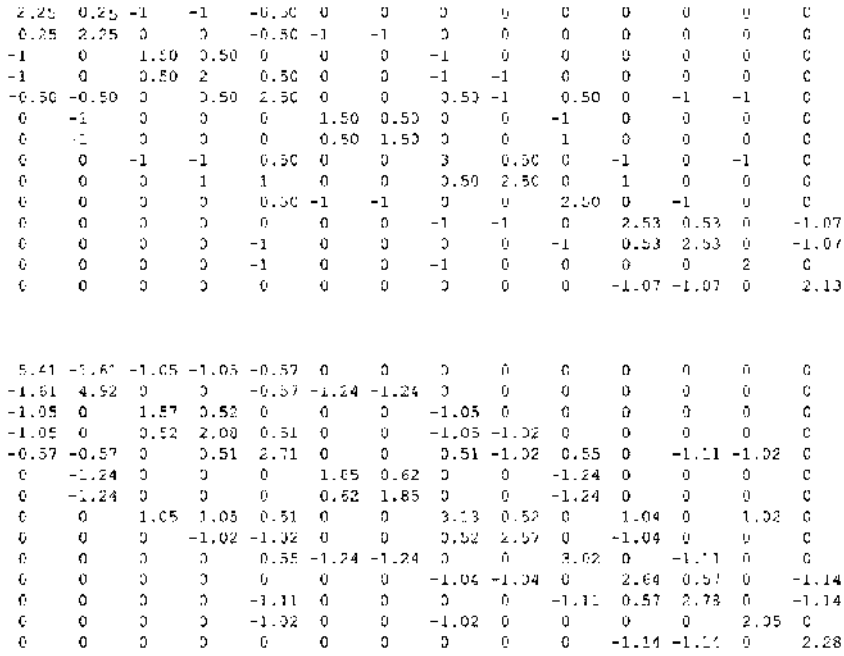


Figure 6 Inverse of the numerator relationship matrix with (up) unknown parent groups (equivalently, with $\mathbf{\Gamma}$= 0) or (down) with metafounders and Holstein–Jersey $\mathbf{\Gamma}$ coefficients () scaled to the same genetic variance. Pedigree as in Figure 5.

formula above, increased by $[(0.55 + 0.77)/2 - 0.48]/4 = 0.045$ compared to the variance in the F1.

Discussion

Conceptual developments

This work presents new conceptual developments for pedigree relationships, including ancestral relationships at the founders due to finite size of ancestral population and across-base population relationships due to overlapping. Such development is of conceptual interest *per se* (Kennedy 1991; VanRaden 1992; ter Braak 2010), but it is obliged for genomic evaluations integrating genotyped and nongenotyped individuals. In practice regular genetic evaluations including several base populations and their crosses assume that ancestral populations are of infinite size and unrelated. This leads to unsolved questions. For instance, assume three pure breeds A, B, C and all of their F1 crosses, all in the same environment. If breed A and B are more similar to each other than to breed C, does this need to be included in the genetic analysis? Another typical case is with ruminant population with missing parentages, which are modeled as animals entering from new base populations. These base populations will become gradually more inbred (VanRaden 1992) and they will drift from the oldest base population. Also, they will be related (*i.e.*, the unknown parent group "Holstein2004" will be more related to "Holstein2002" than to "Holstein1994"). All this can be conveniently modeled, estimated, and included in the genetic evaluations using metafounders. As genomic evaluation procedures are becoming more comprehensive, examples of these kind of problems are showing up in the animal breeding literature: Harris and Johnson (2010), Misztal *et al.* (2013), Makgahlela *et al.* (2014), Winkelman *et al.* (2015).

Metafounders and unknown parent groups

Metafounders are closely related to unknown parent groups or genetic groups (Thompson 1979; Quaas 1988). Genetic groups allow estimation of different genetic bases across the same population, which is necessary if the selection process is unknown (*i.e.*, importing animals or missing pedigrees). Genetic values of individuals in a genetic group model can be written as $\mathbf{u} = \mathbf{u}^* + \mathbf{Qg}$, where $\mathbf{g}$ has average values of the genetic groups. Genetic groups are usually considered as fixed but they can be conceived as random (Sullivan and Schaeffer 1994). For random $\mathbf{g}$ and $\mathrm{Var}(\mathbf{u}^*) = \mathbf{A}$ and $r(\mathbf{g}) = \mathbf{\Gamma}$, $\mathrm{Var}(\mathbf{u}^*) = \mathbf{A} + \mathbf{Q\Gamma Q}'$. This is similar to $\mathbf{A}^{\mathbf{\Gamma}}$, but does not correctly model crosses and overestimates inbreeding. As pointed out by Kennedy (1991) this traditional formulation of genetic groups did not consider inbreeding or drift. Our work can be seen as a generalization of genetic groups to include inbreeding, drift, and across-group relationships. This generalization overcomes the problems mentioned by Misztal *et al.* (2013), who realized that inclusion of unknown parent groups into single-step methods involved approximations in the setup of joint matrix $\mathbf{H}$.

Inclusion of finite size ancestral populations in genetic evaluation procedures has been largely neglected. Jacquard (1969, 1974) work on relationships in closed populations has been ignored. Independently, VanRaden (1992) made a first contribution to palliate the lack of genealogical information in cattle. He used inbreeding coefficients for unknown parent groups based on inbreeding of contemporaries; here we suggest using genomic information instead. Both ideas can possibly be merged.

A notion related to that of metafounders is *partial relationships* across pairs of individuals due to sharing alleles from some particular origin. This allows modeling the genetic value of an individual as a sum of genetic values from several breeds, and this is known as "splitting breeding values"

(Garcia-Cortes and Toro 2006). The relationship matrix with metafounders can be decomposed into such a structure, as explained in the Supporting Information, File S1.

Metafounders and pedigree and genomic relationships

The use of metafounders with $\boldsymbol{\Gamma}$ relationships allows a reconciliation of pedigree and genomic relationships and inbreeding (Powell *et al.* 2010; Vitezica *et al.* 2011). Homozygosity (or identity) can be considered as deviation from Hardy–Weinberg equilibrium (Wright 1922). These deviations cannot be easily measured because they depend on the assumed allelic frequencies, which change in time. Considering unrelated founders assumes that all founder alleles are different, which is not tenable in view of marker information. By assuming 0.5 allelic frequencies, the reference is constant and there are no ambiguities.

The fact that inbreeding automatically increases when considering metafounders may seem worrying. If the objective of quantifying inbreeding is to describe the incertitude *a priori* of inbred animals (*i.e.*, inbred animals tend to be more variable), this does not seem a concern. Use of pedigree inbreeding with metafounders to quantify inbreeding depression should not be problematic, for two reasons. The first is that adding a constant (roughly $\gamma/2$) to inbreeding will not change estimates. The second is that, due to purge, only "new" rate of inbreeding (ΔF) seems to have a measurable effect (*e.g.*, Hinrichs *et al.* 2007). Recent inbreeding could even be better estimated using metafounders (for instance, in incomplete pedigrees; VanRaden 1992).

Genomic relationships are based on markers, and commercial marker chips are often biased toward intermediate frequencies or toward specific breeds. For instance many markers conceived for *Bos taurus* are monomorphic in *Bos indicus* and their use will result in biased estimates of $\boldsymbol{\Gamma}$. For this reason, the approaches in this work should be considered with caution for such populations. Use of unbiased markers (*e.g.*, from sequence data or from random genotyping across the genome) will result in more accurate estimates of relationships across metafounders, if the populations are distant ones.

Genetic background across populations

Use of metafounders assumes a common genetic background across all base populations. This is typically accepted as true within breed, but breed itself is somewhat ill defined. Some genomic predictions across breeds assume identical genetic background (*i.e.*, Hayes *et al.* 2009; Harris and Johnson 2010). If the hypothesis of a homogeneous genetic background is not acceptable, for instance, in the case of genetic–environment interactions or scale effects, a genetic correlation model can be used (Wei and Vanderwerf 1994; Karoui *et al.* 2012).

Empirical checking

Practical performance of our model has to be ascertained with real data but we give an example of its interest. Winkelman *et al.* (2015), using a simplified single-step GBLUP, reported better performance of the Euclidean distance matrix relationship matrix (Gianola and van Kaam 2008) across breeds and their crosses, compared to **G** adjusted as in Harris and Johnson (2010). We have observed that numerically, **G** matrices based on EDM and **G** matrices based on 0.5 allelic frequencies tend to be similar (unpublished). It would seem that the appeal of the EDM relationship matrix is therefore its independence of within-breed allelic frequencies, as proposed by Christensen (2012). In this work, we have aimed at creating tools to make pedigree relationships compatible with this kind of **G** matrices.

Conclusion

We have defined the notion of metafounders, which can be understood as a limited pool of gametes from which the founders of the pedigree are drawn. Metafounders can also be understood as a generalization of unknown parent groups or genetic groups, which are essential in genetic evaluation of livestock. Use of metafounders makes it possible to analyze pedigreed populations allowing for relatedness within and across base populations, something that is desirable for genetic evaluations combining pedigree and genetic markers. Metafounders can account for extra segregation variances due to crosses of populations. Efficient algorithms exist for computation of relationship matrices and their inverses and inbreeding. Relationships across metafounders can be inferred from marker data. By doing so, compatibility of pedigree and genomic relationships is warranted by construction. This work provides new tools and concepts for genetic evaluation and management of populations.

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GENETICS

Supporting Information
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Ancestral Relationships Using Metafounders: Finite Ancestral Populations and Across Population Relationships

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FILE S1

Decomposition of relationships by population of origin and crosses (splitting breeding values).

García-Cortés and Toro (2006) suggested splitting crossbred relationships as sums of several relationship matrices, one by breed and segregation term. In our proposal, the overall relationship matrix can be decomposed in a similar manner, summing covariances across partial relationships matrices, each one by breed of origin or crosses of breeds. It can be shown that

$$\boldsymbol{A}^{\boldsymbol{\Gamma}} = \sum_{b=1}^{B} \left\{ \boldsymbol{A}^{(b)} \left(1 - \frac{\gamma_b}{2}\right) + \boldsymbol{C}^{(b)} \gamma_b \right\} + \sum_{b=1}^{B} \sum_{b'>b} \boldsymbol{C}_{ii'}^{(b,b')} \gamma_{bb'}$$

where $\boldsymbol{A}^{(b)}$ are partial relationship matrices for origin b and matrices $\boldsymbol{C}^{(b)}$ and $\boldsymbol{C}^{(b,b')}$ describe the covariance of breed fractions across individuals, and they are calculated using rules that are derived below.

Here, a recursive method for constructing $\boldsymbol{A}^{\boldsymbol{\Gamma}}$ is presented in complete generality. An additive relationship matrix should satisfy the following recursions

$$\boldsymbol{A}_{ii}^{\boldsymbol{\Gamma}} = 1 + \boldsymbol{A}_{f(i)m(i)}^{\boldsymbol{\Gamma}}/2$$

$$\boldsymbol{A}_{ii'}^{\boldsymbol{\Gamma}} = \left(\boldsymbol{A}_{f(i)i'}^{\boldsymbol{\Gamma}} + \boldsymbol{A}_{m(i)i'}^{\boldsymbol{\Gamma}}\right)/2$$

where $f(i)$ and $m(i)$ denote the two parents of individual i, and individual i' is not a direct descendant of individual i. Matrix $\boldsymbol{A}^{\boldsymbol{\Gamma}}$ is defined by base individuals in breed $b = 1, \ldots, p$ being related with relationship coefficient γ_b and inbred with inbreeding coefficient $\gamma_b / 2$, and by base individuals in different breeds $b \neq b'$ being related with relationship coefficient $\gamma_{bb'}$. This may for the base population be expressed as

$$A_{ii'}^{\Gamma} = \sum_{b} f_i^{\,b} A_{ii'}^{b} (1 - \gamma_b / 2) + \sum_{b} \sum_{b'} f_i^{\,b} f_{i'}^{\,b'} \gamma_{bb'},$$

where $f_i^{\,b}$ is the breed b proportion of individual i, and $A_{ii'}^{b}$ is the usual additive relationship when $i \neq i'$ and self-relationships when $i = i'$ between individuals in breed b (and zero when individuals are not breed b), and $\gamma_{bb} = \gamma_b$. The recursions required for an additive relationship matrix are satisfied for this expression as long as all animals are purebred. The recursions are also satisfied when individuals are crossbred with purebred parents with $A_{ii'}^{b}$ then denoting partial relationships (Garcia-Cortes and Toro, 2006), but they are not satisfied in general.

Here, we split $\mathbf{A}^{\boldsymbol{\Gamma}}$ into several components and derive how the recursions look like for the components. The formula above suggests that

$$A^{\Gamma}_{ii'} = \sum_b A^b_{ii'}(1-\gamma_b/2) + \sum_b C^b_{ii'}\gamma_b + \sum_b \sum_{b':b'>b} C^{b,b'}\gamma_{bb'},$$

where $\mathbf{A}^b$ the breed b specific partial relationship matrix (Garcia-Cortes and Toro, 2006) such that

$$A^b_{ii} = f^b_i + A^b_{f(i)m(i)}/2,$$

$$A^b_{ii'} = (A^b_{f(i)i'} + A^b_{m(i)i'})/2,$$

when individual i' is not a direct descendant of individual i. Inserting the suggested form of $\mathbf{A}^{\Gamma}$ into the recursive formulas of $\mathbf{A}^{\Gamma}$ we obtain (having used $1 = \sum_b f^b_i$) that diagonal elements should satisfy

$$A^b_{ii}(1-\gamma_b/2) + C^b_{ii}\gamma_b = f^b_i + (A^b_{f(i)m(i)}(1-\gamma_b/2) + C^b_{f(i)m(i)}\gamma_b)/2,$$

$$C^{b,b'}_{ii}\gamma_{bb'} = C^{b,b'}_{f(i)m(i)}\gamma_{bb'}/2,$$

from which we obtain (having used $f^b_i = f^b_i(1-\gamma_b/2) + f^b_i\gamma_b/2$) that $A^b_{ii} = f^b_i + A^b_{f(i)m(i)}/2$ is satisfied and

$$C^b_{ii} = (f^b_i + C^b_{f(i)m(i)})/2,$$

$$C^{b,b'}_{ii} = C^{b,b'}_{f(i)m(i)},$$

and off-diagonal elements should satisfy

$$A^b_{ii'} = (A^b_{f(i)i'} + A^b_{m(i)i'})/2,$$

$$C^b_{ii'} = (C^b_{f(i)i'} + C^b_{m(i)i'})/2,$$

$$C^{b,b'}_{ii'} = (C^{b,b'}_{f(i)i'} + C^{b,b'}_{m(i)i'})/2.$$

The description is completed by specifying that the recursions start by $A^b_{ii} = 1$, $A^b_{ii'} = 0$, $C^b_{ii} = C^b_{ii'} = 1$ when i and $i' \neq i$ are base animals of breed b, and $C^{b,b'}_{ii'} = 1$ when i and i' are base animals in different breeds b and b'.

We note that elements in matrices $\mathbf{A}^b$ and $\mathbf{C}^b$ are only non-zero for individuals which contain a breed b proportion. The matrix $\mathbf{C}^{b,b'}$ is complicated, but the rule is that off-diagonal elements are nonzero for pairs of individuals where one of them contain a breed b proportion and the other a b' proportion, and diagonal elements are non-zero for individuals where one of its parent contain a breed b and the other a breed b' proportion (i.e. certain crossbred animals).

FILE S2

Code for algorithms related to metafounders

File S2 is available for download as a compressed file (metafounders_code.tar.gz) at

www.genetics.org/lookup/suppl/doi:10.1534/genetics.115.177014/-/DC1

Misztal, Lourenco and Legarra (2020)

Misztal, I., Lourenco, D., and Legarra, A. 2020. Current status of genomic evaluation. Journal of Animal Science, vol. 98, no. 4, skaa101.

The problem

Just over ten years after the first official genomic evaluations, this review shows where the field stands. The authors discuss why single step GBLUP became the standard method, how APY makes it possible to work with millions of genotypes and which problems remain, such as bias in the evaluation of young animals, compatibility between A and G and the use of information from markers of large effect.

Points to note while reading

This text closes the collection and points to the questions that remain open. Almost all the concepts discussed here were built over the course of the previous papers, from Wright to Henderson and from VanRaden to metafounders.

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BOARD INVITED REVIEW

Current status of genomic evaluation

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Abstract

Early application of genomic selection relied on SNP estimation with phenotypes or de-regressed proofs (**DRP**). Chips of 50k SNP seemed sufficient for an accurate estimation of SNP effects. Genomic estimated breeding values (**GEBV**) were composed of an index with parent average, direct genomic value, and deduction of a parental index to eliminate double counting. Use of SNP selection or weighting increased accuracy with small data sets but had minimal to no impact with large data sets. Efforts to include potentially causative SNP derived from sequence data or high-density chips showed limited or no gain in accuracy. After the implementation of genomic selection, EBV by BLUP became biased because of genomic preselection and DRP computed based on EBV required adjustments, and the creation of DRP for females is hard and subject to double counting. Genomic selection was greatly simplified by single-step genomic BLUP (**ssGBLUP**). This method based on combining genomic and pedigree relationships automatically creates an index with all sources of information, can use any combination of male and female genotypes, and accounts for preselection. To avoid biases, especially under strong selection, ssGBLUP requires that pedigree and genomic relationships are compatible. Because the inversion of the genomic relationship matrix (**G**) becomes costly with more than 100k genotyped animals, large data computations in ssGBLUP were solved by exploiting limited dimensionality of genomic data due to limited effective population size. With such dimensionality ranging from 4k in chickens to about 15k in cattle, the inverse of **G** can be created directly (e.g., by the algorithm for proven and young) at a linear cost. Due to its simplicity and accuracy, ssGBLUP is routinely used for genomic selection by the major chicken, pig, and beef industries. Single step can be used to derive SNP effects for indirect prediction and for genome-wide association studies, including computations of the *P*-values. Alternative single-step formulations exist that use SNP effects for genotyped or for all animals. Although genomics is the new standard in breeding and genetics, there are still some problems that need to be solved. This involves new validation procedures that are unaffected by selection, parameter estimation that accounts for all the genomic data used in selection, and strategies to address reduction in genetic variances after genomic selection was implemented.

Key words: genomic evaluation, genomic selection, large data, single-step GBLUP

Introduction

Genomic selection is now widely practiced across the breeding and genetics industry. This is evident by large-scale genotyping using inexpensive SNP chips. As of November of 2019, genotypes were available for over 3 million U.S. Holsteins (https://queries.uscdcb.com/Genotype/cur_freq.html), over 700,000 for American Angus (S. P. Miller, American Angus Association, Saint Joseph, MO, personal communication), and over 100,000 animals per line for some pig and broiler breeding companies.

Generally, the beginning of genomic selection is attributed to a study by Meuwissen et al. (2001). They used simulated data to conduct analyses with a large number of equally spaced markers; no attempt was made to identify QTLs but

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Abbreviations

AF	allele frequencies
APY	algorithm for proven and young
BOO	breed of origin
DGV	direct genomic value
DRP	deregressed proofs
DYD	daughter yield deviation
G	genomic relationship matrix
GBLUP	genomic BLUP
GEBV	genomic estimated breeding values
GPU	graphical processing units
GWAS	genome-wide association studies
ICS	independent chromosome segments
LR	linear regression
PCG	preconditioned conjugate gradient
PEV	prediction error variance
ssBR	single-step Bayesian regression
ssGBLUP	single-step genomic BLUP
SVD	singular value decomposition
UPG	unknown parent groups

some markers were by chance to be closely linked to QTLs. Computations included haplotypes, and analyses were done by methods called BayesA and BayesB that assumed different distribution of haplotype effects. With 2,200 genotyped animals, they obtained prediction accuracies of 0.85. The accuracies were >0.7 after five generations without phenotyping or with only 500 genotyped animals. Such high accuracies with small data created high hopes in the animal breeding community.

The first large-scale genotyping was possible after the introduction of the SNP 50k bovine chip (Matukumalli et al., 2009), which provided an affordable and accurate technology for genotyping. Much subsequent work on the methodology of genomic selection focused on SNP effects and on the creation of a genomic relationship matrix (**G**) (VanRaden, 2008), a concept that allowed conceptual comparisons between pedigree-based and genome-based predictions. Methods using either SNP effects or genomic relationships led initially to field data analyses using a multistep methodology (VanRaden, 2008; VanRaden et al., 2009), where a regular genetic evaluation by pedigree BLUP (meant as non-genomic method throughout the paper) is followed by the extraction of pseudo-phenotypes of genotyped animals, a genomic analysis for genotyped animals, and the creation of an index combining results from BLUP and the genomic analysis (VanRaden et al., 2009). The multistep methodology is the natural choice when the genomic and pedigree/phenotypic data are owned by separate organizations.

When the genomic selection was introduced, the main focus was on testing models to increase accuracy, in particular increasing the accuracy of prediction by SNP selection (or differential weighting), assuming that it was possible to identify most pairs QTL-closest SNP from data. However, as the data grows bigger, gains with SNP selection become smaller or nonexistent (Karaman et al., 2016). Subsequently, most commercial evaluations do not use SNP selection.

The multistep method is relatively complicated and in its initial form relies on the existence of animals (bulls) with high accurate EBVs from pedigree information. It is also subject to double counting of the genomic information when both parents and progenies are genotyped. Because the genomic information can be expressed as genomic relationships (VanRaden, 2008), Misztal et al. (2009) proposed a single-step evaluation that enhanced the BLUP machinery with a relationship matrix that combines pedigree and genomic relationships. Subsequently, a pedigree-based BLUP with any model of analysis could support genomic models just by replacing the relationship matrix. A combined matrix was first shown by Legarra et al. (2009) and complete analysis using the so-called single-step genomic BLUP (**ssGBLUP**) was presented by Aguilar et al. (2010) and Christensen and Lund (2010). In the following studies, ssGBLUP was shown to be as accurate, or more, than multistep analyses.

Initially, the main focus of the single-step research was ensuring compatibility of genomic and pedigree information (Vitezica et al., 2011) because incompatibility creates biases, especially under strong selection. A later focus was extending ssGBLUP to larger numbers of genotyped animals (Legarra and Ducrocq, 2012; Fernando et al., 2014, Liu et al., 2014; Misztal et al., 2014a). Currently, ssGBLUP is the main tool for genomic evaluation in species other than dairy. If a population includes non-genotyped animals with phenotypes, the transition to some form of single step is unavoidable because BLUP, which is used to created pseudo-observations adopted in multistep, becomes biased by genomic preselection (Patry and Ducrocq, 2011b). The only alternative to ssGBLUP that has been explored is the use of segregation analysis to partially "infer" genotypes of the ancestors of genotyped animals, to later introduce this information in a refined ssGBLUP (Meuwissen et al., 2015). This strategy gave promising results but it is computationally complex and has not been pursued.

Advances in genotyping techniques are allowing sequence data to be generated at a lower cost; therefore, there is an interest to exploit these data (Georges et al., 2019). Sequence data can be used to identify recessive genes, targets for gene editing, and also potential causative SNP that can aid a genetic prediction across breeds or lines (Hayes and Daetwyler, 2019). However, gains with using the potential causative variants for genetic prediction appear to be limited (VanRaden et al., 2017; Fragomeni et al., 2019), with perhaps the exception of across-breed prediction (Moghaddar et al., 2019).

Although the rate of increase in genetic gain delivered by genomic selection can be over 100 % in some cases (García-Ruiz et al., 2016), issues have recently emerged. One important issue is the fast reduction in additive genetic variance and more undesirable genetic correlations between important traits (Hidalgo et al., 2020). This reduction is even more noticeable when genomic information is not used for variance components estimation. The same phenomenon may be responsible for the reported reduction of 33% in heritability in computations of genomic predictions for production yield traits to avoid bias in genomic estimated breeding values (**GEBV**; VanRaden et al., 2014).

From the onset of genomic selection, many ideas were proposed and usually tested by simulation, and many of these ideas were later applied to real data sets, first small then large. Many of these studies led to questions about various aspects of genomic selection, for example: Is it better to use haplotypes instead of SNP? What is the optimal number of SNP? Why there are discrepancies between simulation and field studies? Why is SNP selection less useful with large data? How can we use unknown parent groups (**UPG**) in genomic models? Is there any limit to genomic selection?. The purpose of this paper is to present and evaluate proposed ideas on genomic selection considering most up-to-date experiences with field data.

Exploring Genomic Selection Developments

Initial developments

Genomic selection is generally attributed to a study by Meuwissen et al. (2001) where simulated data for up to 2,200 phenotyped animals with genomic information expressed

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as 50k haplotypes. For prediction, haplotype effects were estimated by several methods including treating them as fixed effects, by haplotype-BLUP assuming haplotypes as normal random effects, by BayesA assuming a *t*-distribution of effects (allowing for large effects), and by BayesB assuming a mixture distribution where most haplotypes had null effects. The accuracy of predicting breeding values for the next generation was the highest using BayesB and reached 0.85. The persistence of accuracies over generations (without additional phenotypes) was excellent, decaying only to 0.72 after an additional five generations of data. Reducing the number of animals with phenotypes to 1,000 only slightly reduced the accuracy to 0.79. The study by Meuwissen et al. (2001) generated great excitement in the animal breeding community, showing the possibility of very high accuracy with small data. However, this turned out to be true because the simulation was unrealistic with a small genome and QTLs of large effect, and no selection. Muir (2007) showed low persistence of genomic predictions under selection and dependence of accuracy on population parameters.

Much of the work involving the methodology of genomic selection on a practical side was accomplished by VanRaden (2008) using SNP markers instead of haplotypes. Also, he showed the equivalence of BLUP with SNP effects to genomic BLUP (**GBLUP**) using **G**; where **G** = **ZZ**/*k*, with **Z** being the matrix of gene content, $k = 2\sum_{i=1}^{n\,SNP} p_i(1-p_i)$, and p_i the frequency of the *i*th SNP. While genomic and pedigree inbreeding were highly correlated (r = 0.68) using base allele frequencies (**AF**) but lowly using current AF (r = 0.12), any AF resulted in similar prediction accuracy. The SE for elements of **G** was inversely proportional to the square root of the number of markers. He stated that genomic relationships are due to shared alleles, and he related the distribution of such alleles to a study by Stam (1980). VanRaden (2008) findings on gene frequencies were validated by Strandén and Christensen (2011) who showed that in SNP, BLUP and GBLUP AF only affect the mean of predictions. The number of shared alleles, also known as independent chromosome segments (**ICS**), was used for approximating the accuracy of genomic selection based on the number of genotyped animals and heritability (Goddard, 2009); lower Ne means fewer segments to estimate and higher accuracy of genomic selection for the same population size.

Limited dimensionality of genomic information

Genomic prediction in farm animals is possible because of the small effective population size (Ne). Stretches of DNA from overrepresented ancestors (i.e., popular bulls) form relatively few segments called LD blocks (Muir, 2007), shared segments (VanRaden, 2008), or ICS (Goddard, 2009). While the segments are not easily identified and have fuzzy limits (i.e., they are broken at slightly different places across two sibs), they appear indirectly, for example, as singular **G** that needs to be blended to become full rank. The number of chromosome segments is usually quantified by the formula presented by Stam (1980) as 4 NeL, where L is genome length. In a simulated population, Pocrnic et al. (2016a) found that the accuracy of prediction using a recursion was maximized assuming 4 NeL segments. They also showed that for a large population the number of segments can be estimated as the number of the largest eigenvalues explaining 98% of the variation in **G**, with the remaining 2% interpreted as noise. Extension of their studies to farm animals (Pocrnic et al., 2016b) allowed to determine the number of segments, and indirectly Ne, and the optimal size of the SNP chip for several species (see Table 1).

The genomic prediction does not act on individual segments but on their clusters, where the four largest clusters could account for 10% of the genetic variation (Pocrnic et al., 2019a). Small data only allow to estimate only the largest eigenvalues (or clusters), but they explain a large portion of the genomic variation in **G**. Subsequently, moderate accuracy of genomic selection can be achieved with small data sets, and large data sets are needed for additional improvements. The same study explains why SNP selection improves accuracy in small but not in large data sets. Genomic selection works by estimating the effects of chromosome segments, and once nearly all are well estimated, the accuracy is high without SNP selection (Karaman et al., 2016) or weighting (Lourenco et al., 2017).

Estimation of haplotypes or SNP effects

If the DNA information is inherited as chromosome segments, it would be natural to base the estimation on haplotypes rather than on SNP effects. Using haplotypes would potentially account for epistasis within each block, as for instance, a segment of 5 SNP can be estimated as having 25 different SNP combinations, as opposed to only 5 SNP effects. In practice, the difference in accuracy in models with haplotypes and SNP effects is negligible (Cuyabano et al., 2015; Jónás et al., 2016). Problems using haplotypes are the need for a complex data preparation and arbitrary choices in their definition, poor estimates for rare haplotypes and the existence of spurious haplotypes due to genotyping errors.

Multistep genomic evaluation

A study by VanRaden et al. (2009) using field data sets established a mature multistep methodology for genomic selection in dairy cattle. The steps included running pedigree-based BLUP with the national database, creating pseudo-observations for genotyped animals (bulls) such as daughter yield deviation (**DYD**). These pseudo-observations are fit into a model estimating on SNP effects assuming normal (linear) or non-normal (nonlinear) distributions. Finally, genomic predictions for a genotyped animal or candidate to selection are obtained combining pedigree and genomic-based predictions into an index:

$$\mathrm{GEBV} = w_1\mathrm{PA} + w_2\mathrm{DGV} - w_3\mathrm{PI},$$

Table 1. Estimated number of chromosome segments, effective population size, and the optimal size of SNP chip following Pocrnic et al. (2016)

Species	Estimated number of segments	Estimated optimal size of SNP chip	Estimated effective population size
Broiler chicken	4.2k	50k	44
Pig	4.1k	49k	48
Angus cattle	10.6k	127k	113
Jersey cattle	11.5k	138k	101
Holstein cattle	14.0k	168k	149

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where GEBV is the genomic estimated breeding value, PA is parent average, **DGV** is direct genomic value, and **PI** is parental index created based on pedigree relationships for genotyped animals. Essentially, PI removes double counting of relationship information because part of PA is included in DGV, and weights w_1 to w_3 could be approximated from reliabilities of each component. Genomic prediction in the aforementioned paper was validated by forward prediction, where prediction for young bulls using truncated data were compared with pseudo-phenotypes of those bulls obtained by BLUP with complete data. That study also showed that increasing the reference size of the genotyped population has a higher impact on prediction accuracy than the number of SNP markers and that assuming non-normal SNP distribution has a positive effect only on traits with large effect QTL. Instead of creating DYD, which is time-consuming, pseudo-observation could be calculated as deregressed proofs (**DRP**) (VanRaden and Wiggans, 1991; Garrick et al., 2009; Wiggans et al., 2011).

Another version of a multistep method depended on using genomic predictions called molecular breeding value as a correlated trait (Kachman, 2008; MacNeil et al., 2010). In this version, GEBVs for genotyped animals based on previously estimated SNP effects were added as an extra trait with genetic correlation computed separately. Because genomic predictions indirectly include parent average, it was hard to account for double counting, and genetic trend was abnormally high even for early time periods when the genomic selection was not practiced (Lourenco et al., 2018).

Single-step genomic evaluations

The multistep methodology was well suited for scenarios where the phenotype and genomic data belong to separate organizations, and especially when most information can be condensed in a small number of animals to genotype. This includes dairy populations with a large number of average information bulls with many daughters. When a population includes both males and females, creating DRP free of double counting is hard, especially when genotyping includes parents and their progeny (Legarra et al., 2014). As the genomic information can be used to capture relationships, Misztal et al. (2009) proposed combining pedigree and genomic relationships into a combined relationship matrix. Subsequently, pedigree-only analyses could be converted to genomic analyses only by replacing the pedigree relationship matrix by the combined matrix, and the steps of construction pseudo-phenotypes and the index would no longer be needed. This combined matrix (**H**) was first presented by Legarra et al. (2009) who proposed to extend the genomic information to non-genotyped animals based on the joint distribution of breeding values of non-genotyped (u_1) and genotyped (u_2) animals:

$$\mathbf{H} = \begin{bmatrix} var(u_1) & cov(u_1, u_2) \\ cov(u_2, u_1) & var(u_2) \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{A}_{22}^{-1}(\mathbf{G} - \mathbf{A}_{22})\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{G} \\ \mathbf{G}\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{G} \end{bmatrix}$$

where subscripts 1 and 2 refer to non-genotyped and genotyped animals, respectively; **A** is the pedigree relationship matrix and **G** is the genomic relationship matrix or **G**. Christensen and Lund (2010) arrived to the same results, using the notion of predicting the genotype at non-genotyped individuals using pedigree information.

The inverse of **H** was presented by Aguilar et al. (2010) and Christensen and Lund (2010) as:

$$H^{-1} = A^{-1} + \begin{bmatrix} 0 & 0 \\ 0 & G^{-1} - A_{22}^{-1} \end{bmatrix}$$

In the analyses done by Aguilar et al. (2010), the reliability of the new method called ssGBLUP was as high or higher than in multistep.

Further research involving ssGBLUP was split into several directions. The first was compatibility between pedigree and genomic relationships, as incompatibility can generate biases or losses of accuracy under selection (Vitezica et al., 2011). The second was an extension to a very large number of genotyped animals as initial implementation was based on dense matrices (Aguilar et al., 2011), which restricted the number of genotyped animals to about 100k animals. Finally, there was an interest in accommodating SNP weighting via a weighted **G**, especially with potential causative SNP obtained from sequence data. The interest in single-step methods increased as the genomic selection was underway, because pedigree BLUP and, therefore, multistep methods were becoming biased due to genomic preselection (Patry and Ducrocq, 2011b), whereas ssGBLUP accounts for preselection.

Different single-step formulations

Several alternative single-step formulas were proposed. These included equations where **G** is not inverted (Legarra and Ducrocq, 2012), where SNP effects are estimated for the genotyped animals and a polygenic effect is fit for non-genotyped animals (Legarra and Ducrocq, 2012; Liu et al., 2014), and where SNP effects were fit for all animals using imputed genotypes (Fernando et al., 2014; Taskinen et al., 2017). The purpose of these formulas was to reduce computations with many genotyped animals. As opposed to a regular ssGBLUP, which can be applied to an existing BLUP software just by replacing the relationship matrix, SNP-based models require new programming. Meuwissen et al. (2014) proposed an alternative single-step approach by combining identical by descent and identical by state approaches.

Compatibility between genomic and pedigree relationships

An important issue in single-step methodology is the compatibility of genomic and pedigree relationships. While the genomic relationships indirectly account for all the ancestors but have an arbitrary scale depending on gene frequencies, the pedigree relationships have a well-defined scale but are limited by the depth and completeness of the pedigree. When pedigrees were complete up to a base population, scaling **G** for compatibility (same means for diagonals and off-diagonals) with the pedigree relationship matrix for genotyped animals ($\mathbf{A}_{22}$) improved accuracy and eliminated bias for a population under strong selection (Chen et al., 2011a; Vitezica et al., 2011). With no selection, the impact of scaling was minimal. Similar scaling could be accomplished automatically by using base population gene frequencies (Strandén and Christensen, 2011; Christensen, 2012), although finding those frequencies when the base population is not genotyped, for example, using the method of Gengler et al. (2008), can be time-consuming, and it suffers sometimes from a clear definition of base population as described below.

When the base populations are heterogeneous with missing pedigrees across generations, as is typical in ruminants, ssGBLUP may diverge or become biased, and the standard way to ensure convergence was by including a parameter ω as in Tsuruta et al. (2011):

$$H^{-1} = A^{-1} + \begin{bmatrix} 0 & 0 \\ 0 & G^{-1} - \omega A_{22}^{-1} \end{bmatrix}$$

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The parameter ω compensates for incomplete pedigree and incomplete accounting of inbreeding (Misztal et al., 2017). Incompleteness of pedigree in $\mathbf{A}_{22}$ can be minimized by truncating old data and pedigree (Misztal et al., 2013b) or by assigning nonzero inbreeding to unknown parents (VanRaden, 1992); old missing pedigree becomes irrelevant with truncation of data. Truncation to two generations of phenotypes and three generations of pedigree reduced bias without lowering accuracy (Lourenco et al., 2014; Howard et al., 2018).

In a single-step SNP-based model known as single-step Bayesian Regression (**ssBR**) developed by Fernando et al. (2014), the compatibility between genomic and pedigree information is provided partially by the use of fixed effects in a model for genotyped animals (Hsu et al., 2017), similar to Vitezica et al. (2011) where this effect is implicitly fit as random. This arises from the findings of Strandén and Christensen (2011) that solutions from SNP BLUP and GBLUP are independent of gene frequencies if the model includes a mean. However, the missing pedigree problem is present in all single-step formulations and it becomes more complex with several base populations as described below.

Missing pedigree and UPG

In several species, there is a need to define several populations. This is the case in ruminants with missing parents (whereas unknown parents of animals born in 2000 are better than unknown parents of animals born in 2016), and the case in pigs and birds (with several lines collapsing into one, and with 2-, 3-, and 4-way crosses). These base populations have different means due to selection and not considering them leads to strong biases.

In BLUP, the genetic merit of these different base populations is often modeled by genetic or UPG (Quaas and Pollak, 1981; Quaas, 1988; Westell et al., 1988). In ssGBLUP, when UPG are applied only to pedigree relationships (**A**) as follows:

$$\mathbf{H}^* = \mathbf{A}^* + \begin{bmatrix} 00 & 0 \\ 0\mathbf{G}^{-1} - \mathbf{A}_{22}^{-1} & 0 \\ 00 & 0 \end{bmatrix}$$

The convergence rate can be slow or no convergence may be reached (Tsuruta et al., 2014; Matilainen et al., 2016), partly because UPG were ignored in pedigree relationships for genotyped animals ($\mathbf{A}_{22}$). Indeed, construction of $\mathbf{A}_{22}$ implicitly assumes complete pedigrees. Misztal et al. (2013b) revised UPG equations to include groups also in the genomic portion of **H** based on Quaas–Pollak (QP) transformation:

$$\mathbf{H}^* = \mathbf{A}^* + \begin{bmatrix} 00 & 0 \\ 0\mathbf{G}^{-1} - \mathbf{A}_{22}^{-1} & -\left(\mathbf{G}^{-1} - \mathbf{A}_{22}^{-1}\right)\mathbf{Q}_2 \\ 0 & -\mathbf{Q}_2'\left(\mathbf{G}^{-1} - \mathbf{A}_{22}^{-1}\right) \mathbf{Q}_2'\left(\mathbf{G}^{-1} - \mathbf{A}_{22}^{-1}\right)\mathbf{Q}_2 \end{bmatrix}$$

When UPGs were applied to all components of **H** as above, convergence dramatically improved for a multitrait model in the Nordic dairy cattle population (Matilainen et al., 2016). Revised UPGs also worked well for the U.S. Holstein data up to 2014 (Misztal et al., 2017). However, using data updated to 2015, Masuda et al. (2018a), based on cross-validation, reported lower reliabilities using revised UPG than not using UPG at all. While most animals genotyped earlier were potentially elite, with complete pedigree, most genotyped animals after 2014 were commercial cows, often with incomplete pedigree and high pedigree error rate (Bradford et al., 2019a).

It is not clear whether the equations for UPG in ssGBLUP should be considered for **G** for a single breed as genomic relationships are not affected by missing pedigree, and, therefore, UPG are automatically accounted for. In other words, if all animals were genotyped, terms involving UPG should disappear from $\mathbf{H}^*$ above. Tsuruta et al. (2019a) found that removing **G** from the equation above improved accuracy and reduced bias. In GBLUP, using UPG for **G** did not increase accuracy for multi-breed populations (Plieschke et al., 2015).

Missing relationships also cause underestimation of inbreeding as animals with missing parents are automatically treated as not inbred. One solution is assigning nonzero inbreeding to missing parents (VanRaden, 1992; Lutaaya et al., 1999; Aguilar and Misztal, 2008). Such assignment improved convergence rate and bias in ssGBLUP in Holsteins (Misztal et al., 2017; Tsuruta et al., 2019a) although it only slightly affected the accuracy.

The concept of metafounders

Legarra et al. (2015) proposed to account for UPG while providing proper scaling by generalizing UPG to metafounders. In their approach, **G** would be derived using 0.5 AF as an "absolute reference" (Christensen, 2012), and **A** would be scaled for compatibility with **G** using relationships among and within metafounders, which are seen as pseudo-individuals. These relationships represent sizes and overlaps of the different base populations (Legarra et al., 2014). They can be estimated in such a way so that they account for scaling, unaccounted inbreeding, different genetic level (e.g., when using multi-breed animals or selected populations), and multiple breeds and crosses. Several methods were proposed to estimate the relationships, and, in practice, they imply estimating gene frequencies in the different base populations (Garcia-Baccino et al., 2017). In simulations and real data, the concept of metafounders delivered the least biased predictions (Garcia-Baccino et al., 2017; Meyer et al., 2018; Bradford et al., 2019b). When applied to dairy cattle, the relationships across metafounders could be well estimated only for metafounders associated with sufficient number of genotypes (S. Tsuruta, University of Georgia, Athens GA, personal communication). In dairy sheep, the use of metafounders reduces biases in predictions and instability of UPG estimates for small data sizes (F Macedo, INRAE, Toulouse, France, personal communication).

Evaluations of crossbred populations

Genomic evaluation of crossbred populations may be separated into two types, for specific crosses or for complex crosses. In pigs and chicken, there is an interest in using F1 and possibly three- to four-way crosses for the evaluation of purebreds on the commercial scale. In beef and dairy, the interest is to have a joint analysis of many breeds with complex crosses (e.g., "Kiwi" Jersey–Holstein crosses in New Zealand, 10+ breed crosses by International Genetic Solutions, and 50+ beef crosses by The Irish Cattle Breeding Federation). More recently, across-breed prediction with genomic data is not successful (Erbe et al., 2012; Kachman et al., 2013) because the breeds do not share the same chromosome segments. Also, the crossbreeds generate limited information if the amount of crossbred data is small and if they are progeny of very few parents (Pocrnic et al., 2019b). Genetic by environment interaction and purebred-crossbred correlations can be considered using multiple-trait models (Xiang et al., 2016; Vandenplas et al., 2017). With purebreds and defined crosses (F1), the genomic relationships can be adjusted

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separately for each breed combination using gene frequencies or other methods (Makgahlela et al., 2014; Lourenco et al., 2016) although the impact of such adjustment is small if the selection pressure is low. With many crosses, a simple approach is to ignore gene frequencies and have one set of SNP effects (Golden et al., 2018) or one **G** (Mäntysaari et al., 2017) for all breeds and breed combinations. Steyn et al. (2019) simulated five breeds using either shared or separate relationships. In the second case, the accuracy was compromised if the number of SNPs was reduced from 45k to 9k, and despite all breeds having identical QTLs, interbreed predictions had low accuracy. In U.S. dairy, SNP effects are estimated separately for each breed as otherwise the predictions would be based on the dominating breed—Holsteins (VanRaden et al., 2020); phenotypes of crossbreds are not used in the regular genetic evaluation of purebreds because of concerns of compromising the evaluation of purebreds. The most refined method for the F1 crossbreds is by phasing haplotypes in crossbreds originating from two parental lines and building a model with two H matrices, one per breed (sometimes called the breed of origin [**BOO**] model) (Christensen et al., 2014). Xiang et al. (2016) observed an increase in accuracy compared with fitting a single H matrix in an analysis of Landrace, Yorkshire, and crosses. The method becomes complex for more complex crosses as the origin of alleles in each crossbred is more difficult to establish.

The concept of metafounders provides a convenient solution to ssGBLUP applied to purebreds and crossbreds (Christensen et al., 2015; Xiang et al., 2017). In such a case, the relationship across breeds represent a distance from a common genetic origin (usually a small relationship, but potentially different across pairs of breeds), and the variances within breed reflect correct scaling separately for each breed and for all breeds simultaneously (Legarra et al., 2015). Xiang et al. (2017) fit this model treating each breed combination as a different trait to account for G × E and observed the same accuracy as in the BOO model of Xiang et al. (2016).

Modifying single step for large data sets

Single-step GBLUP requires explicit or implicit computations of $\mathbf{G}^{-1}$ and $\mathbf{A}_{22}^{-1}$. When created using dense matrix techniques (Aguilar et al., 2011), the practical limit is about 100k animals. This is because computations increase cubically and storage quadratically with the number of genotyped animals. Several strategies were proposed to overcome size limitations.

Indirect computations of A_{22}^{-1}

Matrix $\mathbf{A}_{22}^{-1}$ is dense and, therefore, cannot be created efficiently for a large number of genotyped animals. Henderson (1976) showed that the inverse of a submatrix of **A** could be obtained based on the rules for inversion of a partitioned matrix:

$$\mathbf{A}_{22}^{-1} = \mathbf{A}^{22} - (\mathbf{A}^{12})'\left(\mathbf{A}^{11}\right)^{-1}\mathbf{A}^{12}$$

When only a product of $\mathbf{A}_{22}^{-1}$ and a vector is required in the iteration process as in the preconditioned conjugate gradient (**PCG**) algorithm, that product can be calculated sequentially every round as follows (Masuda et al., 2017; Strandén et al., 2017):

$$\mathbf{A}_{22}^{-1}\mathbf{q} = [\mathbf{A}^{22} - (\mathbf{A}^{12})'\left(\mathbf{A}^{11}\right)^{-1}\mathbf{A}^{12}]\mathbf{q}$$

where the product:

$$s = \left(\mathbf{A}^{11}\right)^{-1}\mathbf{A}^{12}\mathbf{q}$$

is computed as a solution to:

$$\mathbf{A}^{11}s = \mathbf{A}^{12}\mathbf{q}$$

using sparse matrix techniques, in particular, because $\mathbf{A}^{11}$ is sparse and small. Masuda et al. (2017) found that, for a U.S. Holstein population, this algorithm required 2 min to set up and less than 1 s per round of multiplication.

Algorithm for proven and young

Because of small effective population size in farm animals, **G** has a rank of about 5k for pigs and chicken to about 15k for beef and dairy (Pocrnic et al., 2016b), indicating the existence of that many LD blocks or chromosome segments. Subsequently, the inverse of **G** can be obtained by recursion on a number of "core" animals equal to the rank of **G**, indirectly assuming that breeding values of N animals contain the same information as the effects of N chromosome segments. When animals are designated as core (c) or noncore (n), the inverse of **G** can be directly obtained as (Misztal, 2016):

$$\mathbf{G}_{\text{APY}}^{-1} = \begin{bmatrix}\mathbf{G}_{cc}^{-1}0\\ 0\quad 0\end{bmatrix} + \begin{bmatrix}-\mathbf{G}_{cc}^{-1}\mathbf{G}_{cn}\\ \mathbf{I}\end{bmatrix}\mathbf{M}^{-1}\left[-\mathbf{G}_{nc}\mathbf{G}_{cc}^{-1}\mathbf{I}\right]$$

where **M** is a diagonal matrix with elements:

$$m_i = g_{ii-}\mathbf{g}_{ic}\mathbf{G}_{cc}^{-1}\mathbf{g}_{ci}$$

where i refers to the *i*th genotyped, noncore animal. This method has almost a linear cost (computations and memory) with the number of animals (Fragomeni et al., 2015) and has been successfully applied to 2.3 M genotyped animals (Tsuruta et al., 2019b). The choice of core animals for recursion is not critical for accuracy when the number of core animals is sufficient but influences the convergence rate; the random choice is preferable (Bradford et al., 2017). Lately, Pocrnic et al. (2019a) found that accuracies obtained with N core animals are like those obtained with **G** ignoring all but the largest N eigenvalues. This explains why the accuracy with the algorithm for proven and young (**APY**) using 25% of the optimal number of core animals is almost the same, as 25% of important eigenvalues explain 90% of the genetic variation in **G**. In fact, the recursion acts not on individual chromosome segments but on their clusters.

Inverse by singular value decomposition

The inverse of **G** can be derived from the eigenvalue decomposition:

$$\mathbf{G} = \mathbf{UDU}'$$

where **U** is a matrix of eigenvectors and **D** is a matrix of eigenvalues. If all eigenvalues are positive, the inverse of **G** is:

$$\mathbf{G}^{-1} = \mathbf{U}'\mathbf{D}^{-1}\mathbf{U}$$

If **G** has small rank, only a small fraction of eigenvalues will be meaningful. Let $\mathbf{D}_t$ indicate a fraction of **D** with non-negligible eigenvalues, and let $\mathbf{U}_t$ be the corresponding eigenvectors. Then:

$$\mathbf{G}_t^{-1} = \mathbf{U}_t'\mathbf{D}_t^{-1}\mathbf{U}_t$$

While eigenvalue decomposition of **G** requires creating **G** explicitly and can be very expensive, a less expensive alternative, when there are more genotyped animals than SNP, is the

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singular value decomposition (**SVD**) of the matrix of SNP content (**Z**), where $\mathbf{Z} = \mathbf{UD}^{0.5}\mathbf{V}$. The SVD for a matrix of 720k animals by 60k SNP takes less than a day (Y. Masuda, University of Georgia, Athens, GA, personal communication). The SVD concept can be applied separately for each chromosome (Ødegård et al., 2018)

Inverse by the Woodbury formula

Mäntysaari et al. (2017) proposed an inverse of $\mathbf{G} = \mathbf{ZZ}' + \mathbf{I}\varepsilon$ based on the Woodbury formula to overcome computing challenges when the number of genotyped animals is greater than the number of SNP:

$$\mathbf{G}^{-1} = \frac{1}{\varepsilon}\mathbf{I} - \frac{1}{\varepsilon}\mathbf{Z}(\frac{1}{\varepsilon}\mathbf{Z}'\mathbf{Z} + \mathbf{I})^{-1}\mathbf{Z}'\frac{1}{\varepsilon}$$

where $\mathbf{Z}'\mathbf{Z}$ is the design matrix of SNP BLUP and **I** is an identity matrix with the same dimension. The formula is an exact inversion but is based on an arbitrary value of ε (i.e., 0.05**I**, $0.05\mathbf{A}_{22}$), without which **G** could not be full rank. The "Woodbury" $\mathbf{G}^{-1}$ is dense and is not used explicitly. Its use is only for PCG systems in which only a product of this matrix by a vector is desired, being reformulated as:

$$\mathbf{G}^{-1}\mathbf{q} = \frac{1}{\varepsilon}\left\{\mathbf{I} - \mathbf{Z}(\mathbf{Z}'\mathbf{Z} + \mathbf{I}\varepsilon)^{-1}\mathbf{Z}'\right\}$$
$$q = \frac{1}{\varepsilon}\left\{\mathbf{I} - \mathbf{Z}(\mathbf{UDU}')^{-1}\mathbf{Z}'\right\}$$
$$xq = \frac{1}{\varepsilon}\{\mathbf{I} - \mathbf{SS}'\}q,$$

with

$$\mathbf{S} = \mathbf{ZU}'\mathbf{D}^{-1/2}$$

Matrix **S** has dimensions equal to the number of animals by the number of SNP. In practice, the SNP BLUP design-matrix $\mathbf{Z}'\mathbf{Z}$ is not full rank, and one dimension can be reduced to the actual rank (5k to 15k for one breed) by truncating **U** and **D** to eliminate small eigenvalues.

Single-step Bayesian Regression

If 50k SNP are enough for predictions, an alternative idea was to impute genotypes of non-genotyped animals, resulting in the same 50k SNP effects to estimate regardless of the number of genotyped animals. Let $\mathbf{u}_2$, the vector of breeding values for genotyped animals be equal to **Za**, where **a** is a vector of SNP effects. Legarra et al. (2009) showed that the conditional distribution of breeding values for non-genotyped and genotyped animals has an expectation equal to $\mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{u}_2$. Replacing $\mathbf{u}_2$ by **Za**:

$$\mathbf{u}_1 = \mathrm{E}(\mathbf{u}_1 \mid \mathbf{u}_2) + \varepsilon = \mathbf{A}_{12}(\mathbf{A}_{22})^{-1}\mathbf{Za} + \varepsilon = \mathbf{Ta} + \varepsilon$$

where **T** can be called an imputation matrix for non-genotyped animals and ε can be called an imputation error. Then, the breeding values in an animal model can be replaced by:

$$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} T \\ Z \end{bmatrix}\mathbf{a} + \begin{bmatrix} \varepsilon \\ 0 \end{bmatrix}$$

Regardless of the number of animals, the number of "genomic" unknowns is equal to the number of SNP, although there is an additional uncorrelated effect ε with a simple relationship structure (Fernando et al., 2014). The model was reformulated for economy of memory by Fernando et al. (2016) who called it "hybrid model," although the same model had already been proposed by Legarra and Ducrocq (2012). As the imputation was expensive, and the model is conceived to use Gibbs sampling methods, the implementation of ssBR in the BOLT software used graphical processing units, that is, **GPU** (Garrick et al., 2018). Compared with ssGBLUP, ssBR allows the user to estimate SNP effects directly but the implementation of complex models (e.g., correlated maternal effects with multiple traits) is quite complex. The method of ssBR was used for a multi-breed evaluation done by the Simmental association for more than 10 breeds (Golden et al., 2018) but they decreased the number of SNPs from 50k to about 2.5k preselected SNP, contrary to all other species who abandoned the idea of preselecting markers because an optimal subset of markers may not be optimal a few generations later. A more general (and simple) formulation of the ssBR model was given by Taskinen et al. (2017).

Other approaches

Legarra and Ducrocq (2012) developed an assymetric method where **G** was not inverted, but the method did not scale up well. Also, both Legarra and Ducrocq (2012) and Liu et al. (2014) proposed methods that used SNP effects estimated for genotyped animals. Vandenplas et al (2019) showed that such models when solved by the PCG algorithm require a special preconditioner for convergence.

Preselection bias

Under genomic selection, BLUP becomes biased (Patry and Ducrocq, 2011a, 2011b) due to preselection on Mendelian sampling; for instance, only offspring that has received the "good" alleles from a sire gets to be recorded. This has an impact on multistep methods which use BLUP as a first step, because they will tend to penalize genomically selected animals and, therefore, to underestimate the genetic trend. The bias can be corrected for (Wiggans et al., 2011, 2012), but the corrections need to be reevaluated as genotyping increases. Single-step GBLUP is expected to be resistant to selection bias (VanRaden et al., 2012; Legarra et al., 2014) as it considers all available information jointly. Masuda et al. (2018b) ran evaluations with BLUP and ssGBLUP for production traits in U.S. Holsteins. They found that the trends for BLUP level off, when they should actually increase, whereas trends for ssGBLUP were consistent. Based on the work at UGA in dairy and in pigs (unpublished), typical trends for genotyped animals by BLUP and ssGBLUP indicating preselection in BLUP are shown in Figure 1. The preselection bias can intensify when more animals are genotyped.

Differences between trends from BLUP and ssGBLUP can be used indirectly as a measure of the effectiveness of the genomic section. If the trend by ssGBLUP is increasing and the trend by BLUP is lower, genomic selection is successful. If the trends by both methods are identical, genomic selection does not have an impact over the regular selection. If in an extreme case, the trends by ssGBLUP decrease, it means either poor implementation of genomic selection or a change in the selection objectives.

Validation of genomic predictions

Genomic evaluations are validated by realized accuracies or reliabilities computed from predictions based on incomplete data to predictions/phenotypes based on complete data —see review by Daetwyler et al. (2013) and Legarra and Reverter (2018). Several types of validations are currently applied, and each one is suitable for a different data structure. The k-fold cross-validation depends on splitting the population into *n* samples

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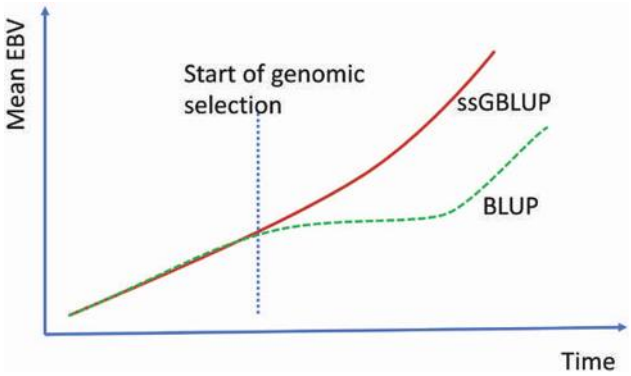


Figure 1. Trend of (G)EBV with ssGBLUP (solid) and BLUP (dashed) indicating preselection bias in BLUP.

and predicting phenotypes of one sample from the remaining samples (Saatchi et al., 2011). It is primarily used for small data sets when only one generation is genotyped or when other methods cannot be applied. As it follows from the decomposition of GEBV (VanRaden and Wright, 2013; Lourenco et al., 2015a), accuracies by clustering methods, such as the k-fold, depend on the algorithm for creating the clusters. In particular, BLUP may emerge as the best method if most animals in each cluster are progeny of the same ancestors.

The validation needs to consider that the breeder wants to predict the next generation from former ones, in other words, forward and not backward or "sideways." Thus, another validation method is based on a comparison of pseudo-observations of sires (DYD or DRP) with their (G)EBV obtained without their daughter's information (VanRaden et al., 2009). This validation type is only realistic for populations with sires that have large progeny sizes and phenotype recording is mainly on progeny, such as in dairy cattle. If pseudo-observations are computed by BLUP under genomic selection, this validation may be biased by preselection (Masuda et al., 2018b). If pseudo-observations are computed by ssGBLUP, the bias can be avoided but there is a danger of double counting of the genomic information, especially if progeny sizes are small. Yet another validation method that is called predictive ability or predictivity can be used when the validation animals have only their own records but not progeny (Legarra et al., 2008). It is based on correlations between GEBV obtained without a phenotype and the phenotype adjusted for fixed effects. However, it can only be computed for simple models and depends on the quality of adjustments (Legarra and Reverter, 2018). Accuracies based on validation are depressed by selection and, therefore, are lower than individual theoretical accuracies based on prediction error variance (**PEV**; Bijma, 2012; Lourenco et al., 2015a).

A completely different approach to validation was taken by Legarra and Reverter (2018) in a method called **LR**, which stands for linear regression. The method LR examines regressions and correlations of (G)EBV using complete and partial data sets while accounting for the relatedness of animals in the validation and additive variances under selection. The advantage of method LR is the ability to support any model and any data structure. For example, Bermann et al. (2020) were able to calculate the accuracy of evaluations for a threshold model. However, the method requires the additive variance for the validation population, which may be hard to estimate as typically these are a subset of selected animals. Without such a variance, only relative comparisons among methods are possible, although they are useful to rank methods.

Individual theoretical accuracies

Individual accuracies are published with (G)EBV as a measure of precision and they are based on true or approximated PEV derived from mixed model equations (Henderson, 1984). The PEV can be obtained either via efficient matrix inversion, for example, by REML with sparse matrix package YAMS (Masuda et al., 2015) or via Gibbs sampling (Tsuruta et al., 2017; Garrick et al., 2018). This is affordable for up to ~100K individuals genotyped. The last option can support larger data sets if the computation is by GPU (Garrick et al., 2018). For complex models and large populations, the computation of PEV is usually too expensive and approximations are used instead. With genomic information, the PEV for the ith animal can be approximated as (Misztal et al., 2013a):

$$\mathrm{PEV}_i \quad \sigma_e^2 \approx \frac{1}{\frac{\sigma_e^2}{\sigma_a^2} + d_i^r + d_i^p + d_i^g}$$

where d are contributions (in terms of effective daughters or observations) due to pedigrees (r), phenotypes (p), and genomic information (g), and σ_a^2 and σ_e^2 are additive and residual variances, respectively. Approximate contributions due to pedigree and phenotypic information were determined by earlier studies (Misztal and Wiggans, 1988; Meyer, 1989; VanRaden and Wiggans, 1991).

With the multistep SNP model, the contribution due to genomic information could be calculated by inversion for any number of genotyped animals (VanRaden et al., 2011; Liu et al., 2017b). To avoid double counting, the calculations exclude the genomic information that is already included in the pedigree information. In ssGBLUP, the genomic contribution can be calculated by combined differences between genomic and pedigree relationships (Misztal et al., 2013a). Edel et al. (2019) provided formulas for avoiding double counting in ssBR. Efficient computation of genomic accuracies for any model and data set is still a research topic but not a hot one because when the models are too large for direct inversion, genomic predictions are accurate enough for selection.

Genetic parameters under genomic selection

Plant breeders estimate variance components at each genetic evaluation, partly because they have several random effects (e.g., blocks) and partly because their data sets are small. In contrast, animal breeders tend to use either once-in-a-while estimates or to use pedigree-based estimates for genomic evaluation purposes, for example, as in VanRaden (2008). Genetic parameters can be estimated with genomic information using ssGBLUP and normal tools such as REML or Monte Carlo Markov Chain via Gibbs sampling. The use of the genomic information increases the costs of computations because the inverse of **G** is usually dense, whereas non-genomic mixed model equations are sparse. Masuda et al. (2014) developed a sparse matrix package that recognizes and processes dense blocks rapidly. A four-trait single-step AIREML model with 15k genotyped animals took less than 1 h with the new package (Masuda et al., 2015); the computations increase cubically with the number of genotyped animals and of traits.

Comparing genomic and pedigree-based estimates of variance components relies on the compatibility of genomic and pedigree information (Legarra, 2016). Without selection and with a complete pedigree, the estimates of variance components ignoring or using the genomic information are usually similar, although with the genomic information they have lower standard errors (Forni et al., 2011). Under strong

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selection, the estimates ignoring the genomic information are biased (Gao et al., 2019; Hidalgo et al., 2020). The computed bias due to preselection depends on the accuracy of modeling and intensity of selection. For example, the popular QMSim program for simulation of genomic data (Sargolzaei and Schenkel, 2009) only performs BLUP selection. If various types of single-step methods show different results despite being equivalent models, the actual variances are affected by small details in the models (e.g., Gao et al., 2019)

Before genomic selection, the genetic parameters were thought to be generally stable, but this was not studied in depth. Under genomic selection, there are indications of rapidly changing parameters, perhaps due to the Bulmer effect (Van Grevenhof et al., 2012; Hidalgo et al., 2020). For instance, bias for U.S. dairy genomic evaluations decreased when heritability was reduced to about 70%—50% of the original value (Wiggans et al., 2012; Misztal et al., 2017), which is an indicator of overestimated heritability. Hidalgo et al. (2020) used a Gibbs sampling approach to analyze the changes in genetic parameters for growth and fitness traits in pigs. To make the computations possible, analyses were done in time slices of 3 yr, and genotypes were restricted to parents and animals with records. Over time, heritabilities for growth were reduced by one half and the antagonistic genetic correlations between growth and fitness traits became almost twice as strong. Estimates without genomic information were quite different. The aforementioned study illustrated the tradeoffs in parameter estimation under genomic selection. Without genomic information, the estimates may be biased, and with all the genomic information available the computation are expensive. Cesarani et al. (2019) reported biased variance components estimates under genomic selection when the genomic information was truncated or too few generations were used. A modest compromise is to restrict genotypes only to those animals on which selection was more intense and to remove genotypes of all young animals and possibly of nonparents. Genetic parameter estimation with a large number of genotypes can be possible in GBLUP when the APY algorithm is applied. However, in ssGBLUP, $\mathbf{A}_{22}^{-1}$is relatively dense and using it in computations eliminates most of the gains due to using a sparse $\mathbf{G}_{APY}^{-1}$.

Stability of GEBV

Under BLUP, the evaluation of an animal depends nearly only on its phenotype, parents, and progeny. Therefore, EBV for animals with no new information are stable even if the accuracy is low (and PEV high). In genomic evaluations, all genotyped animals are connected through **G**. It means that information on new genotyped animals affects all the other genotyped animals, causing fluctuations. Changing core animals in the APY algorithm also causes fluctuations in GEBV even though the accuracy is not affected (Misztal et al., 2019). When short-term fluctuations are undesirable, for example, for merchandising, one solution is to use full model genomic prediction (by SSGBLUP or multistep methods) periodically (say once a month), compute SNP effects, and run interim (e.g., weekly) indirect predictions based on backsolved SNP effects. While with small data the indirect predictions can be inaccurate due to ignoring parent average, in large populations, the fraction of parent average in GEBV is small and indirect predictions have similar accuracy to complete predictions (Lourenco et al., 2015b; Garcia et al, 2020). To mitigate risk associated with potential rank changes of young bulls, semen from a team of bulls may be marketed instead of semen from individual bulls (e.g., https://www.dairynz.co.nz/animal/animal-evaluation/bull-team/).

Using sequence data for genomic predictions

As sequencing is becoming less expensive, there is an interest in exploiting sequence information in animal genetics. If all causative variants and their substitution effects could be identified, genomic prediction would be perfect (i.e., selection accuracy = 1.0). If those effects were conserved across breeds, accurate multi-breed evaluation would be possible (Goddard, 2017). But substitution effects may vary from breed to breed, even at the QTL level, due to gene–environment interaction and to nonadditive gene action (Duenk et al., 2020). Sequence data are available through selective sequencing of key animals across species (e.g., 1000 Bull Genomes Project; http://www.1000bullgenomes.com/; Hayes and Daetwyler, 2019) and imputation for the remaining animals (Ros-Freixedes et al., 2020). For a successful incorporation of potential causative SNP, they need to be very close to the actual causative SNP, and their a priori variance in a model need to be large as otherwise their value is strongly regressed toward 0 (Brøndum et al., 2015).

Practical results using sequence data from large data sets yielded mixed results. Some studies have found no improvement (Erbe et al., 2012) and some showed a small improvement, in particular Moghaddar et al. (2019) who found an increase in the accuracy of "distant" animals of ~0.10 using selected sequence variants. In a study that yielded up to 5% improvement in reliability across traits, VanRaden et al. (2017) partly used a bin concept, where they eliminated most of the SNP close to SNP with the largest effects. The bin concept, popular in plants, recognizes that QTLs are nested in chromosome segments and attempts to locate at most a few SNP per segment (Xu, 2013); fewer SNP reduce the impact of priors and reduce shrinkage of causative SNP. Fragomeni et al. (2017) showed that ssGBLUP can account for causative SNP if they have a large weight in a weighted **G**. In a study on stature in U.S. Holstein using the potential causative SNP identified by VanRaden et al. (2017), Fragomeni et al. (2019) found that the addition of potential causative SNP to the current SNP panel increased reliabilities in GBLUP but not in ssGBLUP, and reliabilities from ssGBLUP were the highest. Similar results were found in Belgian Blue cattle (J.L. Gualdrón-Duarte, University of Liege, Belgium, personal communication) and for health traits in dairy cattle (S. DeNise, Zoetis, Kalamazoo MO, personal communication). A few real, validated major genes (as identified by molecular genetics) explaining up to 10% of the genetic variance have been found and included in ssGBLUP evaluations, either as correlated traits (Legarra and Vitezica, 2015) or as weighting the G matrix appropriately. In general, any of these two strategies work and result in small, but less than expected, improvements on accuracy (Carillier-Jacquin et al., 2016; Teissier et al., 2018; Oget et al., 2019). Possibly, the causative SNP are already accounted for by the values of chromosome segments with large data. Some improvement with the causative SNP could be due to imperfect modeling by GBLUP with pseudo-data such as DRP or DYD instead of records.

There is a dilemma whether causative SNP with large effect, if found, should be used in selection programs for strongly selected traits. With long-term selection, most likely genes with positive effect for most traits are fixed or close to fixation, and genes that still have a large effect but are not fixed are likely to show undesirable pleiotropy. A chromosomal deletion in pigs increased growth but decreased fertility (Derks et al., 2018). Manhattan plots for mortality and milk yield, using a two-trait analysis, in U.S. Holstein showed the same peaks on chromosome 14 (Tsuruta et al., 2017). Georges et al. (2019) cite many studies indicating pleiotropy as a result of balancing

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selection, for example, where disruptive variants in genes increase muscularity but affect the viability of fitness. Negative effects of pleiotropy on low heritability traits may be hard to identify but can be important in the long run.

Balancing selection resulting in intermediate gene frequencies may be unlikely in cases where selection indices are utilized even with pleiotropy. While causative SNPs with large effects are not likely to be fixed after years of pedigree BLUP, the trend toward fixation will be faster with genomic selection. In the extreme, the fixation will negatively affect low heritability or sparsely recorded traits.

Genome-wide association studies

A standard tool for traditional genome-wide association studies (**GWAS**) is a model where one marker is analyzed at a time as fixed effect (Kennedy et al., 1992), for example, an efficient mixed-model association expedited—EMMAX (Kang et al., 2010). To reduce spurious signals due to a population structure, an animal effect using a pedigree or **G** is added to the model (Kennedy et al., 1992). Alternatively, many studies use Bayesian methods such as BayesB or BayesR with all SNP considered jointly, interpreting large signals as markers to nearby QTLs. While the former studies determine SNP significance using *P*-values, the latter usually estimate fractions of explained variance per segment of the genome, for example, 1 Mb.

Many studies, especially using small data, detect a large number of "large" markers, interpreting those as close to a QTL; however, the overlap of those markers across multiple populations or generations in a population under selection is minimal. This suggests that many detected associations are spurious (Fragomeni et al., 2014; Liu et al., 2017a). Studies using BayesB often show very high peaks, sometimes explaining >10% of the additive variance, especially with small data sets. As genomic selection with small data works on large clusters of chromosome segments (Pocrnic et al., 2019a), it is possible that some peaks may be tags to those clusters.

Many of these signals in GWAS are, therefore, probably false positives and can probably be explained by small data sets. If pedigree relationships are incomplete (e.g., ancestors not included), they would not account for population structure. In addition, *P*-values or False Discovery Rate are rarely reported.

Classical GWAS in EMMAX is conceived for a set of individuals that are genotyped and phenotyped. When genotyped animals have only records from progeny or other relatives, this method is only applicable in a multiple-step manner, that is, creating pseudo-phenotypes such as DRP or DYD as it was typically the case in dairy cattle (Boichard et al., 2003), but this is difficult to generalize to other species, where progeny sizes are smaller and many genotyped have phenotypes (e.g., weights) but not genotypes. However, Gualdrón-Duarte et al. (2014) and Bernal Rubio et al. (2016) showed the equivalence of *P*-values in GBLUP-based models with *P*-values in single-marker fixed regressions with a polygenic effect. Lu et al. (2018) extended the theory to ssGBLUP, and Aguilar et al. (2019) added this concept to the BLUPF90 package (Misztal et al., 2014b), with a successful implementation using 1 million birth weight phenotypes for American Angus, almost 2 million animals in the pedigrees, and 1,424 genotyped sires. The GWAS with *P*-values from ssGBLUP accounts for population structure, considers phenotypes from both genotyped and non-genotyped animals without additional steps, and allows for arbitrarily complex models. At this time, the method is limited to models where the left-hand side of the mixed model equations can be inverted, which sets a soft limit of perhaps ~100K genotyped animals.

Conclusions

Genomic selection methodology has been widely embraced by the animal breeding industry as evidenced by the scale of genotyping. The evaluation in most species except dairy cattle is by single-step methods, which consider all sources of information jointly, with methodology refined sufficiently to provide relatively unbiased evaluation for any data size, and easily accommodating causal genes. The dairy industry plans to move to single step are hampered by distributed ownership of phenotypic and genomic data. Most evaluations use <100K SNP chips without SNP selection or weighting, indirectly acknowledging that the prediction acts mostly on chromosome segments and less on markers of QTLs. Whether accurate determination of causative SNPs will lead to substantially increased accuracy of selection also across breeds is unclear. While the validations methods are less than perfect, they illustrate higher accuracy of evaluation with the genomic information. An important concern in long-term genomic selection may be a serious reduction of the additive variance that may limit future gains, especially given that the parameter estimation with the genomic information is difficult.

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Conflict of interest statement

The authors declare no real or perceived conflicts of interest.

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